Classical Electrodynamics

Problems and Worked Solutions — J. D. Jackson, 3rd ed.

Dark red text marks published errata and additional corrections or necessary qualifications explained in the corresponding solutions. The solutions follow Jackson's chapter conventions: SI units in Chapters 1–10 and Gaussian units in Chapters 11–16, with any problem-specific unit changes stated explicitly.

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1 Chapter 1. Introduction to Electrostatics

Problems

1.1 Use Gauss's theorem [and (1.21) if necessary] to prove the following:

(a) Any excess charge placed on a conductor must lie entirely on its surface. (A conductor by definition contains charges capable of moving freely under the action of applied electric fields.)

(b) A closed, hollow conductor shields its interior from fields due to charges outside, but does not shield its exterior from the fields due to charges placed inside it. For the exterior assertion, take the conductor to be initially neutral and isolated, with a nonzero net charge placed in its cavity.

(c) The electric field at the surface of a conductor is normal to the surface and has a magnitude $|\sigma|/\epsilon_0$, where σ is the charge density per unit area on the surface.

Solution.

We assume electrostatic equilibrium, a connected perfect conductor, and vacuum adjacent to its surface. Let $\mathbf{n}$ point from the metal into the vacuum.

(a) If the field in the metal were nonzero, its mobile charges would experience $q\mathbf{E}$ and move. Thus $\mathbf{E} = 0$ there. For every small volume wholly within the metal,

$$Q_V = \epsilon_0 \oint_{\partial V} \mathbf{E} \cdot d\mathbf{a} = 0.$$

Shrinking the volume gives $\rho = 0$ in the bulk. Excess charge can therefore reside only on the boundary. This statement concerns net excess charge, not the separate positive and negative microscopic charges making up the metal.

(b) The metal and hence the wall of an empty cavity are at one potential V_c. Put $u = \Phi - V_c$ in the cavity. Green's first identity gives

$$\int_{\text{cavity}} |\nabla u|^2 d^3x = \oint u\, \partial_n u\, da - \int u \nabla^2 u\, d^3x = 0,$$

since $u = 0$ on the wall and $\nabla^2 u = 0$ inside. Therefore $u = 0$ throughout: external charges produce no cavity field. This remains a shielding statement even if fixed charges already occupy the cavity: compare two solutions with the same interior charges and different exterior sources. Their difference is harmonic in the cavity and constant on its wall, so the same identity makes its gradient zero. The interior field therefore does not change. If a charge q is instead put in the cavity, a Gaussian surface in the metal enclosing the cavity gives $0 = q + Q_{\text{inner}}$, so $Q_{\text{inner}} = -q$. An initially neutral, isolated conductor must then have $Q_{\text{outer}} = q$. A Gaussian surface outside the whole system has flux q/ϵ_0, which is nonzero for $q \neq 0$. The qualification "isolated" matters: a grounded closed conductor can draw $-q$ from ground, leaving no exterior field when there are no exterior sources. A zero-net-charge interior distribution also need not give an exterior field.

(c) A narrow rectangular electrostatic loop straddling the surface has $\oint \mathbf{E}\cdot d\mathbf{l}=0$. Letting its height vanish gives $\mathbf{E}_{\text{out},t}=\mathbf{E}_{\text{in},t}=0$. For a pillbox of face area Δa,

$$(E_{\text{out},n}-E_{\text{in},n})\Delta a=\frac{\sigma\Delta a}{\epsilon_0}.$$

Since the interior field vanishes, $\boxed{\mathbf{E}_{\text{out}}=(\sigma/\epsilon_0)\mathbf{n}}$. Its magnitude is $|\sigma|/\epsilon_0$; the signed outward component is σ/ϵ_0.

1.2 The Dirac delta function in three dimensions can be taken as the improper limit as $\alpha\to 0$ of the Gaussian function

$$D(\alpha;\,x,y,z)=(2\pi)^{-3/2}\alpha^{-3}\exp\left[-\frac{1}{2\alpha^2}\left(x^2+y^2+z^2\right)\right]$$

Consider a general orthogonal coordinate system specified by the surfaces $u=$ constant, $v=$ constant, $w=$ constant, with length elements du/U, dv/V, dw/W in the three perpendicular directions. Show that

$$\delta(\mathbf{x}-\mathbf{x}')=\delta(u-u')\,\delta(v-v')\,\delta(w-w')\cdot UVW$$

by considering the limit of the Gaussian above. Note that as $\alpha\to 0$ only the infinitesimal length element need be used for the distance between the points in the exponent.

Solution.

Write $h_u=1/U$, $h_v=1/V$, and $h_w=1/W$. At a regular coordinate point $\mathbf{x}'$, the distance between neighboring points is

$$|\mathbf{x}-\mathbf{x}'|^2=h_u'^2(u-u')^2+h_v'^2(v-v')^2+h_w'^2(w-w')^2+O(|\Delta\mathbf{u}|^3).$$

Set $\xi_u=h_u'(u-u')/\alpha$, and similarly for v,w. The local volume measure is

$$d^3x=h_uh_vh_w\,du\,dv\,dw=\alpha^3[1+O(\alpha)]\,d\xi_u\,d\xi_v\,d\xi_w.$$

For any continuous test function f of compact support, the Gaussian integral therefore tends to

$$\begin{aligned}\lim_{\alpha\to0}\int D(\alpha;\mathbf{x}-\mathbf{x}')f(\mathbf{x})d^3x&=f(\mathbf{x}')(2\pi)^{-3/2}\int_{\mathbb{R}^3}e^{-(\xi_u^2+\xi_v^2+\xi_w^2)/2}d^3\xi\\&=f(\mathbf{x}').\end{aligned}$$

Equivalently, each one-dimensional Gaussian has width $\alpha/h_u'=\alpha U'$, so its normalization supplies a factor U'. Consequently

$$\boxed{\delta^3(\mathbf{x}-\mathbf{x}')=U'V'W'\delta(u-u')\delta(v-v')\delta(w-w').}$$

Unprimed UVW is equivalent because the deltas evaluate it at the primed point. Integrating this expression with $d^3x=du\,dv\,dw/(UVW)$ explicitly cancels the Jacobian. At coordinate singularities one uses a regular local chart or interprets the formula through this integration rule.

1.3 Using Dirac delta functions in the appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(\mathbf{x})$.

(a) In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R.

(b) In cylindrical coordinates, a charge λ per unit length uniformly distributed over a cylindrical surface of radius b.

(c) In cylindrical coordinates, a charge Q spread uniformly over a flat circular disc of negligible thickness and radius R.

(d) The same as part (c), but using spherical coordinates.

Solution.

The defining check is $\int \rho\, d^3x = Q$ (or charge per unit length λ). Let s denote cylindrical radius, to distinguish it from the charge density, and let H be the Heaviside function.

(a) The shell has surface density $Q/(4\pi R^2)$, hence

$$\boxed{\rho(r,\theta,\phi) = \frac{Q}{4\pi R^2}\delta(r-R).}$$

Indeed, $\int r^2\delta(r-R)dr = R^2$ and $\int d\Omega = 4\pi$.

(b) The charge per circumference is $\lambda/(2\pi b)$:

$$\boxed{\rho(s,\phi,z) = \frac{\lambda}{2\pi b}\delta(s-b).}$$

Integration with $s\,ds\,d\phi$ gives λ at every z.

(c) The surface density is $\sigma = Q/(\pi R^2)$, so

$$\boxed{\rho(s,\phi,z) = \sigma H(R-s)\delta(z).}$$

Its integral is $\sigma\int_0^R 2\pi s\, ds = Q$.

(d) Since $z = r\cos\theta$ and the disk lies at $\theta = \pi/2$,

$$\delta(r\cos\theta) = \frac{1}{r}\delta(\theta-\pi/2), \qquad r>0.$$

Thus a spherical-coordinate representation is

$$\boxed{\rho(r,\theta,\phi) = \frac{Q}{\pi R^2 r}H(R-r)\delta(\theta-\pi/2).}$$

Multiplication by $r^2\sin\theta\, dr\, d\theta\, d\phi$ leaves $\sigma r\, dr\, d\phi$, the correct disk measure. The apparent $1/r$ singularity is a coordinate factor and introduces no point charge at the origin.

1.4 Each of three charged spheres of radius a, one conducting, one having a uniform charge density within its volume, and one having a spherically symmetric charge density that varies radially as r^n $(n > -3)$, has a total charge Q. Use Gauss's theorem to obtain the electric fields both inside and outside each sphere. Sketch the behavior of the fields as a function of radius for the first two spheres, and for the third with $n = -2, +2$.

Solution.

Spherical symmetry gives $4\pi r^2 E_r = Q(< r)/\epsilon_0$. Put $E_a = Q/(4\pi\epsilon_0 a^2)$ and $s = r/a$ (the sketches take $Q > 0$). For the conductor, $Q(< r) = 0$ for $r < a$ and Q for $r > a$; therefore

$$E_r = \begin{cases} 0 & r < a, \\ E_a s^{-2} & r > a. \end{cases}$$

For uniform volume density $\rho_0 = 3Q/(4\pi a^3)$,

$$Q(< r) = 4\pi\rho_0 \int_0^r r'^2 dr' = Q(r/a)^3, \qquad E_r = E_a s \quad (r < a).$$

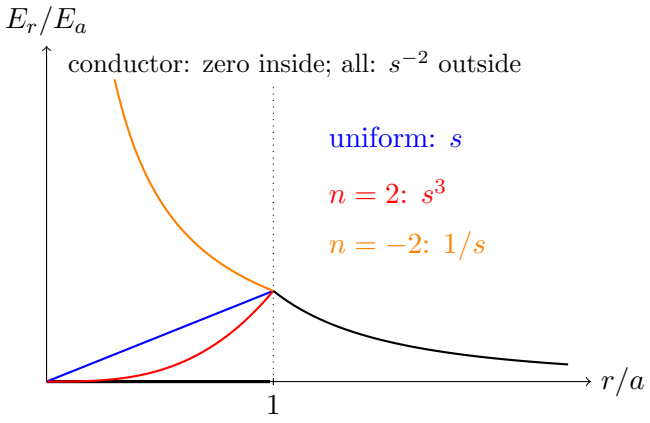
More generally, put $\rho = Kr^n H(a - r)$. Normalization and enclosed charge give

$$Q = \frac{4\pi K a^{n+3}}{n+3}, \qquad K = \frac{(n+3)Q}{4\pi a^{n+3}}, \qquad Q(< r) = Q(r/a)^{n+3}.$$

Hence

$$\boxed{E_r = E_a \begin{cases} s^{n+1} & 0 < r < a, \\ s^{-2} & r > a. \end{cases}}$$

For $n = -2$, the interior field is E_a/s; it diverges at the center but has vanishing enclosed charge as $r \to 0$, so there is no additional delta charge. For $n = 2$, it is $E_a s^3$. Only the conductor has a jump at $r = a$.



1.5 The time-averaged potential of a neutral hydrogen atom is given by

$$\Phi = \frac{q}{4\pi\epsilon_0}\frac{e^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right)$$

where q is the magnitude of the electronic charge, and $\alpha^{-1} = a_0/2$, a_0 being the Bohr radius. Find the distribution of charge (both continuous and discrete) that will give this potential and interpret your result physically.

Solution.

For $r > 0$, write $\Phi = (q/4\pi\epsilon_0) f(r)$ with $f = e^{-\alpha r}(1/r + \alpha/2)$. Direct differentiation gives

$$f' = -e^{-\alpha r}\left(\frac{1}{r^2} + \frac{\alpha}{r} + \frac{\alpha^2}{2}\right), \quad (r^2 f')' = \frac{\alpha^3 r^2}{2} e^{-\alpha r}.$$

Poisson's equation then yields the continuous part

$$\rho_c = -\epsilon_0 \nabla^2 \Phi = -\frac{q\alpha^3}{8\pi} e^{-\alpha r} \quad (r > 0).$$

The singular part is determined by a small Gaussian sphere:

$$\lim_{r\to 0} 4\pi\epsilon_0 r^2 (-\Phi') = q.$$

Thus

$$\boxed{\rho(\mathbf{r}) = q\delta^3(\mathbf{r}) - \frac{q\alpha^3}{8\pi} e^{-\alpha r}.}$$

Its continuous charge is $-q\alpha^3 \int_0^\infty r^2 e^{-\alpha r} dr/2 = -q$, since the integral is $2/\alpha^3$. With $\alpha = 2/a_0$, $\rho_c = -qe^{-2r/a_0}/(\pi a_0^3)$: a point proton plus the mean charge cloud of a hydrogen $1s$ electron. The total charge is zero, consistently with the exponentially screened large-r potential.

1.6 A simple capacitor is a device formed by two insulated conductors adjacent to each other. If equal and opposite charges are placed on the conductors, there will be a certain difference of potential between them. The ratio of the magnitude of the charge on one conductor to the magnitude of the potential difference is called the capacitance (in SI units it is measured in farads). Using Gauss's law, calculate the capacitance of

(a) two large, flat, conducting sheets of area A, separated by a small distance d;

(b) two concentric conducting spheres with radii a, b $(b > a)$;

(c) two concentric conducting cylinders of length L, large compared to their radii a, b $(b > a)$.

(d) What is the inner diameter of the outer conductor in an air-filled coaxial cable whose center conductor is a cylindrical wire of diameter 1 mm and whose capacitance is 3×10^{-11} F/m? 3×10^{-12} F/m?

Solution.

Take $Q > 0$ on the inner/positive conductor and define $V = \Phi_+ - \Phi_- > 0$. The electrostatic capacitance is $C = Q/V$.

(a) A Gaussian pillbox gives $E = Q/(\epsilon_0 A)$ in the gap. Consequently $V = \int_0^d E\, dz = Qd/(\epsilon_0 A)$, and $\boxed{C = \epsilon_0 A/d}$. Edge effects are neglected.

(b) A Gaussian sphere gives $E_r = Q/(4\pi\epsilon_0 r^2)$ for $a < r < b$. Therefore

$$V = \frac{Q}{4\pi\epsilon_0}\int_a^b \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right), \qquad \boxed{C = 4\pi\epsilon_0 \frac{ab}{b-a}}.$$

(c) For a coaxial Gaussian cylinder, $2\pi r L E_r = Q/\epsilon_0$, whence

$$V = \frac{Q}{2\pi\epsilon_0 L}\int_a^b \frac{dr}{r} = \frac{Q}{2\pi\epsilon_0 L}\ln\frac{b}{a}, \qquad \boxed{C = \frac{2\pi\epsilon_0 L}{\ln(b/a)}}.$$

End corrections are negligible when the length is much larger than the radii.

(d) With $C' = C/L$, $b/a = \exp(2\pi\epsilon_0/C')$. Since $2a = 1$ mm,

$$2b = (1\text{ mm})\exp(2\pi\epsilon_0/C') = \begin{cases} 6.388\text{ mm}, & C' = 3\times10^{-11}\text{ F/m}, \\ 1.1314\times10^{5}\text{ m}, & C' = 3\times10^{-12}\text{ F/m}. \end{cases}$$

Here air is approximated by vacuum, $\epsilon_0 \simeq 8.85419\times10^{-12}$ F/m. The second value is a formal coaxial result and would require an enormous outer cylinder and still greater length for the end-free approximation.

1.7 Two long, cylindrical conductors of radii a_1 and a_2 are parallel and separated by a distance d, which is large compared with either radius. Show that the capacitance per unit length is given approximately by

$$C \simeq \pi\epsilon_0\left(\ln\frac{d}{a}\right)^{-1}$$

where a is the geometrical mean of the two radii.

Approximately what gauge wire (state diameter in millimeters) would be necessary to make a two-wire transmission line with a capacitance of 1.2×10^{-11} F/m if the separation of the wires was 0.5 cm? 1.5 cm? 5.0 cm?

Solution.

Let the wire charges per unit length be $+\lambda$ and $-\lambda$. For a line charge, integrating $E_r = \lambda/(2\pi\epsilon_0 r)$ gives $\Phi(r_2)-\Phi(r_1) = -(\lambda/2\pi\epsilon_0)\ln(r_2/r_1)$. Superposing the two lines and evaluating at their surfaces, to leading order in a_i/d, gives

$$\Phi_1 = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{d}{a_1} + \Phi_*, \qquad \Phi_2 = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{d}{a_2} + \Phi_*.$$

Subtracting eliminates the arbitrary constant:

$$V = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{d^2}{a_1 a_2} = \frac{\lambda}{\pi\epsilon_0}\ln\frac{d}{\sqrt{a_1 a_2}}, \qquad \boxed{C' = \frac{\pi\epsilon_0}{\ln(d/a)}, \quad a = \sqrt{a_1 a_2}.}$$

For equal radii and $C' = 1.2\times10^{-11}$ F/m,

$$2a = 2d\exp\left(-\frac{\pi\epsilon_0}{C'}\right) = 0.19694\,d.$$

Thus the requested diameters for separations 0.5, 1.5, and 5.0 cm are, respectively,

$$\boxed{0.985,\ 2.954,\ 9.847\text{ mm}.}$$

These diameters specify the required wire size without choosing a national gauge convention. The ratio $a/d \simeq 0.0985$ also shows that the calculation is approximate rather than an exact finite-radius two-cylinder solution.

1.8

(a) For the three capacitor geometries in Problem 1.6 calculate the total electrostatic energy and express it alternatively in terms of the equal and opposite charges Q and $-Q$ placed on the conductors *and* the potential difference between them.

(b) Sketch the energy density of the electrostatic field in each case as a function of the appropriate linear coordinate.

Solution.

At an intermediate charge q, the voltage is q/C. The charging work is

$$U = \int_0^Q \frac{q}{C}\, dq = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV.$$

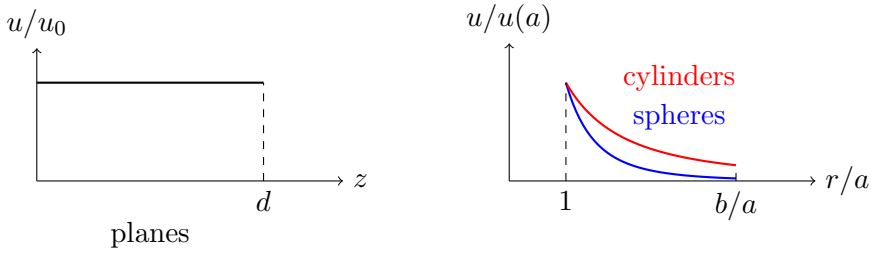
For the three geometries this gives

	$U(Q)$	$U(V)$
planes	$\dfrac{Q^2 d}{2\epsilon_0 A}$	$\dfrac{\epsilon_0 A V^2}{2d}$
spheres	$\dfrac{Q^2}{8\pi\epsilon_0}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)$	$\dfrac{2\pi\epsilon_0 abV^2}{b-a}$
cylinders	$\dfrac{Q^2}{4\pi\epsilon_0 L}\ln\dfrac{b}{a}$	$\dfrac{\pi\epsilon_0 LV^2}{\ln(b/a)}$

For part (b), the local energy density is $u = \epsilon_0 E^2/2$. Therefore

$$u = \begin{cases} Q^2/(2\epsilon_0 A^2), & 0 < z < d \quad \text{(planes)}, \\ Q^2/(32\pi^2\epsilon_0 r^4), & a < r < b \quad \text{(spheres)}, \\ Q^2/(8\pi^2\epsilon_0 L^2 r^2), & a < r < b \quad \text{(cylinders)}. \end{cases}$$

It is zero outside these gaps for the ideal equal-and-opposite capacitor charges. As an independent check, integrating the spherical density with $4\pi r^2 dr$ gives $Q^2(1/a - 1/b)/(8\pi\epsilon_0)$; integrating the cylindrical density with $2\pi r L dr$ gives $Q^2 \ln(b/a)/(4\pi\epsilon_0 L)$.



1.9 Calculate the attractive force between conductors in the parallel plate capacitor (Problem 1.6a) and the parallel cylinder capacitor (Problem 1.7) for

(a) fixed charges on each conductor;

(b) fixed potential difference between conductors.

Solution.

Let positive force mean increasing separation d. For fixed charge, $F_d = -\partial_d U$. For fixed voltage the source also exchanges energy: $dW_{\text{source}} = V\,dQ = V^2 dC$. Since $dU_{\text{field}} = \frac{1}{2}V^2 dC$, the mechanical work is $F_d\,dd = dW_{\text{source}} - dU_{\text{field}} = \frac{1}{2}V^2 dC$. Equivalently, differentiate $U_{\text{field}} - VQ = -\frac{1}{2}CV^2$.

(a) For the planes, $U = Q^2 d/(2\epsilon_0 A)$, so

$$\boxed{F_d = -\frac{Q^2}{2\epsilon_0 A}.}$$

For the wires use energy per length $U' = \lambda^2 \ln(d/a)/(2\pi\epsilon_0)$, obtaining

$$\boxed{\frac{F_d}{L} = -\frac{\lambda^2}{2\pi\epsilon_0 d}.}$$

(b) With $C = \epsilon_0 A/d$ for the planes,

$$\boxed{F_d = \tfrac{1}{2}V^2\frac{dC}{dd} = -\frac{\epsilon_0 A V^2}{2d^2}.}$$

For the wires,

$$\frac{dC'}{dd} = -\frac{\pi\epsilon_0}{d\ln^2(d/a)}, \qquad \boxed{\frac{F_d}{L} = -\frac{\pi\epsilon_0 V^2}{2d\ln^2(d/a)}.}$$

Substitution of $Q = CV$ or $\lambda = C'V$ recovers the fixed-charge formulas at the same instantaneous state. All signs are negative, hence attractive.

1.10 Prove the *mean value theorem*: For charge-free space the value of the electrostatic potential at any point is equal to the average of the potential over the surface of *any* sphere centered on that point.

Solution.

Let B_R be a ball wholly contained in a charge-free region, centered at $\mathbf{x}_0$, and let

$$M(R) = \frac{1}{4\pi}\int \Phi(\mathbf{x}_0 + R\hat{\mathbf{r}})d\Omega.$$

Differentiating at fixed solid angle and using $da = R^2 d\Omega$ gives

$$M'(R) = \frac{1}{4\pi R^2}\oint_{\partial B_R} \partial_n\Phi\, da = \frac{1}{4\pi R^2}\int_{B_R} \nabla^2\Phi\, d^3x = 0.$$

Thus M is independent of radius. Continuity implies $\lim_{R\to 0} M(R) = \Phi(\mathbf{x}_0)$, proving

$$\boxed{\Phi(\mathbf{x}_0) = \frac{1}{4\pi R^2}\oint_{\partial B_R} \Phi\, da.}$$

The entire ball must be free of charge and singularities, not just its surface.

1.11 Use Gauss's theorem to prove that at the surface of a curved charged conductor, the normal derivative of the electric field is given by

$$\frac{1}{E}\frac{\partial E}{\partial n} = -\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

where R_1 and R_2 are the principal radii of curvature of the surface.

Solution.

Choose a small patch of area dA and a tube whose sides follow the outward normals. At normal distance dn, its area is, to first order,

$$dA(dn) = dA\left(1 + \frac{dn}{R_1}\right)\left(1 + \frac{dn}{R_2}\right) = dA\left[1 + \left(\frac{1}{R_1} + \frac{1}{R_2}\right)dn\right] + O(dn^2).$$

The signed principal radii are positive on a convex surface with the outward normal convention. The field is normal at the conductor. On the side faces the tangential field starts from zero and is $O(dn)$, while their area is $O(dn)$. Side flux is therefore of second order in the tube height, so the first-order charge-free flux balance is

$$[E + \partial_n E\, dn]\, dA(dn) - E\, dA = 0 + O(dn^2).$$

Expanding, dividing by $dA\, dn$, and taking the limit gives

$$\partial_n E + E\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = 0, \qquad \boxed{\frac{\partial_n E}{E} = -\left(\frac{1}{R_1} + \frac{1}{R_2}\right).}$$

The divided form applies where $E \neq 0$; the undivided equation remains valid where it vanishes. A sphere has $E \propto r^{-2}$ and gives $-2/R$; a cylinder has $E \propto r^{-1}$ and gives $-1/R$, checking both curvature terms.

1.12 Prove *Green's reciprocation theorem*: If Φ is the potential due to a volume-charge density ρ within a volume V and a surface-charge density σ on the conducting surface S bounding the volume V, while Φ' is the potential due to another charge distribution ρ' and σ', then

$$\int_V \rho\Phi'\, d^3x + \int_S \sigma\Phi'\, da = \int_V \rho'\Phi\, d^3x + \int_S \sigma'\Phi\, da$$

Solution.

Use the normal $\mathbf{n}$ pointing out of the field volume and into each conductor. Since the electric field in the metal is zero, the conductor's surface charge is $\sigma = \epsilon_0\partial_n\Phi$, and similarly for primes. Green's second identity gives

$$\int_V (\Phi\nabla^2\Phi' - \Phi'\nabla^2\Phi)d^3x = \oint_S (\Phi\partial_n\Phi' - \Phi'\partial_n\Phi)da.$$

Insert $\nabla^2\Phi = -\rho/\epsilon_0$ and its primed counterpart:

$$\int_V (\rho\Phi' - \rho'\Phi)d^3x = \oint_S (\sigma'\Phi - \sigma\Phi')da.$$

Moving the surface terms to the respective sides proves

$$\boxed{\int_V \rho\Phi' d^3x + \oint_S \sigma\Phi' da = \int_V \rho'\Phi d^3x + \oint_S \sigma'\Phi da.}$$

If a boundary is at infinity, the identity assumes the usual decay that makes its surface contribution vanish; otherwise that boundary contribution must be retained.

1.13 Two infinite grounded parallel conducting planes are separated by a distance d. A point charge q is placed between the planes. Use the reciprocation theorem of Green to prove that the total induced charge on one of the planes is equal to $(-q)$ times the fractional perpendicular distance of the point charge from the other plane. (*Hint*: As your comparison electrostatic problem with the same surfaces choose one whose charge densities and potential are known and simple.)

Solution.

Take the planes at $z = 0$ and $z = d$, and put q at $z = z_0$. The actual plane potentials are zero. For the comparison problem hold the lower plane at V_* and the upper at zero, without volume charge. Its potential is

$$\Phi'(z) = V_*(1 - z/d), \qquad \rho' = 0.$$

In the reciprocity relation all right-hand terms vanish, because the actual potential on either conductor is zero. The left side is

$$qV_* \left(1 - \frac{z_0}{d}\right) + V_* Q_0 = 0.$$

Dividing by V_* gives $\boxed{Q_0 = -q(d - z_0)/d}$. Interchanging the reference potentials similarly gives $\boxed{Q_d = -qz_0/d}$. Their sum is $-q$, as also required by the Gaussian surface enclosing the charge and both induced distributions.

1.14 Consider the electrostatic Green functions of Section 1.10 for Dirichlet and Neumann boundary conditions on the surface S bounding the volume V. Apply Green's theorem (1.35) with integration variable $\mathbf{y}$ and $\phi = G(\mathbf{x}, \mathbf{y})$, $\psi = G(\mathbf{x}', \mathbf{y})$, with $\nabla_y^2 G(\mathbf{z}, \mathbf{y}) = -4\pi\delta(\mathbf{y} - \mathbf{z})$. Find an expression for the difference $[G(\mathbf{x}, \mathbf{x}') - G(\mathbf{x}', \mathbf{x})]$ in terms of an integral over the boundary surface S.

(a) For Dirichlet boundary conditions on the potential and the associated boundary condition on the Green function, show that $G_D(\mathbf{x}, \mathbf{x}')$ must be symmetric in $\mathbf{x}$ and $\mathbf{x}'$.

(b) For Neumann boundary conditions, use the boundary condition (1.45) for $G_N(\mathbf{x}, \mathbf{x}')$ to show that $G_N(\mathbf{x}, \mathbf{x}')$ is not symmetric in general, but that $G_N(\mathbf{x}, \mathbf{x}') - F(\mathbf{x})$ is symmetric in $\mathbf{x}$ and $\mathbf{x}'$, where

$$F(\mathbf{x}) = \frac{1}{S} \oint_S G_N(\mathbf{x}, \mathbf{y})\, da_y$$

(c) Show that the addition of $F(\mathbf{x})$ to the Green function does not affect the potential $\Phi(\mathbf{x})$. See problem 3.26 for an example of the Neumann Green function.

Solution.

Set $g(\mathbf{y}) = G(\mathbf{x}, \mathbf{y})$ and $h(\mathbf{y}) = G(\mathbf{x}', \mathbf{y})$. Green's identity yields

$$\begin{aligned} \oint_S (g\partial_n h - h\partial_n g) da &= \int_V (g\nabla_y^2 h - h\nabla_y^2 g) d^3y \\ &= -4\pi G(\mathbf{x}, \mathbf{x}') + 4\pi G(\mathbf{x}', \mathbf{x}). \end{aligned}$$

Thus

$$\boxed{G(\mathbf{x}, \mathbf{x}') - G(\mathbf{x}', \mathbf{x}) = \frac{1}{4\pi} \oint_S (h\partial_n g - g\partial_n h) da.}$$

(a) For G_D, both g and h vanish on S; the integral is zero.

(b) Jackson's normalization is $\partial_{n_y} G_N(\mathbf{z}, \mathbf{y}) = -4\pi/S$. Substitution gives

$$G_N(\mathbf{x}, \mathbf{x}') - G_N(\mathbf{x}', \mathbf{x}) = \frac{1}{S} \oint_S [g(\mathbf{y}) - h(\mathbf{y})] da = F(\mathbf{x}) - F(\mathbf{x}').$$

Rearrangement shows that $G_N(\mathbf{x}, \mathbf{x}') - F(\mathbf{x})$ is symmetric.

(c) The Neumann potential formula is

$$\Phi(\mathbf{x}) = \langle \Phi \rangle_S + \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{y}) G_N(\mathbf{x}, \mathbf{y}) d^3y + \frac{1}{4\pi} \oint_S G_N(\mathbf{x}, \mathbf{y}) \partial_n \Phi(\mathbf{y}) da.$$

Adding any function $f(\mathbf{x})$ independent of $\mathbf{y}$ changes this by

$$\Delta\Phi = \frac{f(\mathbf{x})}{4\pi} \left[\frac{1}{\epsilon_0} \int_V \rho \, d^3y + \oint_S \partial_n \Phi \, da \right] = 0,$$

because the divergence theorem and Poisson's equation give $\oint \partial_n \Phi = \int \nabla^2 \Phi = -\int \rho/\epsilon_0$. This applies to adding F as asked, or subtracting it to obtain the symmetric kernel. The independent additive constant $\langle \Phi \rangle_S$ still fixes the potential's gauge.

1.15 Prove *Thomson's theorem*: If a number of surfaces are fixed in position and a given total charge is placed on each surface, then the electrostatic energy in the region bounded by the surfaces is an absolute minimum when the charges are placed so that every surface is an equipotential, as happens when they are conductors.

Solution.

Let ρ_0 (including surface charges) be the equipotential distribution, with field $\mathbf{E}_0$ and potential Φ_0. Any other admissible redistribution is $\rho_0 + \delta\rho$, with field $\mathbf{E}_0 + \delta\mathbf{E}$, and with $\delta Q_i = 0$ on each surface. Expanding the field energy gives

$$U - U_0 = \epsilon_0 \int \mathbf{E}_0 \cdot \delta\mathbf{E} \, d^3x + \frac{\epsilon_0}{2} \int |\delta\mathbf{E}|^2 d^3x.$$

Integrating the cross term by parts over all field space, or treating surface charges as distributions, yields

$$\epsilon_0 \int \nabla\Phi_0 \cdot \nabla\delta\Phi \, d^3x = \int \Phi_0 \delta\rho \, d^3x = \sum_i \Phi_{0i} \delta Q_i = 0.$$

Consequently

$$\boxed{U - U_0 = \frac{\epsilon_0}{2} \int |\delta\mathbf{E}|^2 d^3x \geq 0.}$$

This is a global minimum, not just a stationary value. Equality requires no change of field. Fixed additional volume charges, if present, are left unchanged in the variation.

1.16 Prove the following theorem: If a number of conducting surfaces are fixed in position with a given total charge on each, the introduction of an uncharged, insulated conductor into the region bounded by the surfaces cannot increase the electrostatic energy. The decrease is strict if the field changes.

Solution.

Before inserting the new conductor, denote the equilibrium energy by U_0. In the enlarged set of possible charge distributions, consider the trial configuration that retains every old charge and places zero excess charge everywhere on the new conductor. It has the prescribed old charges and zero total new charge, so it is an admissible charge distribution for Thomson's minimization problem, although it is generally not an equilibrium distribution. Its field and energy, integrated also through the new body's volume, are the old ones: $U_{\text{trial}} = U_0$.

The new equilibrium minimizes the energy over this enlarged set. Therefore

$$\boxed{U_{\text{new}} \leq U_{\text{trial}} = U_0.}$$

More explicitly, applying the proof of Problem 1.15 with the new equilibrium as the reference gives

$$U_0 - U_{\text{new}} = \frac{\epsilon_0}{2} \int |\mathbf{E}_0 - \mathbf{E}_{\text{new}}|^2 d^3x \geq 0.$$

The inequality is strict if insertion changes the field. Equality is possible when the inserted body occupies a region that was already field-free. The old conductors are insulated with fixed charges; maintaining fixed potentials would involve additional work by voltage sources.

1.17 A volume V in vacuum is bounded by a surface S consisting of several separate conducting surfaces S_i. One conductor is held at *unit* potential and all the other conductors at zero potential.

(a) Show that the capacitance of the one conductor is given by

$$C = \epsilon_0 \int_V |\nabla\Phi|^2 \, d^3x$$

where $\Phi(\mathbf{x})$ is the solution for the potential.

(b) Show that the true capacitance C is always less than or equal to the quantity

$$C[\Psi] = \epsilon_0 \int_V |\nabla\Psi|^2 \, d^3x$$

where Ψ is any trial function satisfying the boundary conditions on the conductors. This is a variational principle for the capacitance that yields an *upper bound*.

Solution.

(a) There is no volume charge. With the normal directed out of V, Green's first identity gives

$$\int_V |\nabla\Phi|^2 d^3x = \oint_S \Phi \partial_n \Phi \, da = \int_{S_1} \partial_n \Phi \, da = Q_1/\epsilon_0.$$

Only S_1 contributes, because its potential is one and all others are zero. Since $Q_1 = C \times 1$, this proves $\boxed{C = \epsilon_0 \int_V |\nabla\Phi|^2 d^3x}$.

(b) Write a trial function as $\Psi = \Phi + \eta$ with $\eta = 0$ on every conductor. Then

$$C[\Psi] - C = 2\epsilon_0 \int_V \nabla\Phi \cdot \nabla\eta \, d^3x + \epsilon_0 \int_V |\nabla\eta|^2 d^3x.$$

The first integral is $\oint \eta \partial_n \Phi \, da - \int \eta \nabla^2 \Phi \, d^3x = 0$. Thus

$$\boxed{C[\Psi] - C = \epsilon_0 \int_V |\nabla\eta|^2 d^3x \geq 0.}$$

Trial functions need finite field energy and the specified boundary values; they need not solve Laplace's equation.

1.18 Consider the configuration of conductors of Problem 1.17, with all conductors except S_1 held at zero potential.

(a) Show that the potential $\Phi(\mathbf{x})$ anywhere in the volume V *and* on any of the surfaces S_i can be written

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}') G(\mathbf{x}, \mathbf{x}') \, da'$$

where $\sigma_1(\mathbf{x}')$ is the surface charge density on S_1 and $G(\mathbf{x}, \mathbf{x}')$ is the Green function potential for a point charge in the presence of all the surfaces that are held at zero potential (but with S_1 absent). Show also that the electrostatic energy is

$$W = \frac{1}{8\pi\epsilon_0} \oint_{S_1} da \oint_{S_1} da' \, \sigma_1(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \sigma_1(\mathbf{x}')$$

where the integrals are only over the surface S_1.

(b) Show that the variational expression

$$C^{-1}[\sigma] = \frac{\displaystyle\oint_{S_1} da \oint_{S_1} da' \, \sigma(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \sigma(\mathbf{x}')}{4\pi\epsilon_0 \left[\displaystyle\oint_{S_1} \sigma(\mathbf{x}) \, da \right]^2}$$

with an arbitrary integrable function $\sigma(\mathbf{x})$ defined on S_1, having nonzero total charge and finite electrostatic energy, is stationary for small variations of σ away from σ_1. Use Thomson's theorem to prove that the reciprocal of $C^{-1}[\sigma]$ gives a *lower bound* to the true capacitance of the conductor S_1.

Solution.

(a) Remove S_1 and use the Dirichlet Green function for the remaining grounded conductors. An element $dq = \sigma_1(\mathbf{x}')da'$ produces $d\Phi = G(\mathbf{x}, \mathbf{x}')dq/(4\pi\epsilon_0)$, including the response of those grounded conductors. Superposition gives

$$\boxed{\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}') G(\mathbf{x}, \mathbf{x}') da'.}$$

This is also the continuous single-layer potential at the surface itself. The electrostatic energy is $W = \frac{1}{2}\sum_i \int_{S_i} \sigma_i \Phi_i \, da$. All grounded surfaces have $\Phi_i = 0$, so substitution leaves

$$\boxed{W = \frac{1}{8\pi\epsilon_0} \oint_{S_1} da \oint_{S_1} da' \, \sigma_1(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \sigma_1(\mathbf{x}').}$$

(b) Write $Q[\sigma] = \int \sigma da$ and $B[\sigma] = \iint \sigma G \sigma \, da \, da'$. Then $C^{-1}[\sigma] = B/(4\pi\epsilon_0 Q^2) = 2W[\sigma]/Q^2$. Symmetry of G gives

$$\delta B = 2 \int \delta\sigma(\mathbf{x}) \left[\int G(\mathbf{x}, \mathbf{x}') \sigma_1(\mathbf{x}') da' \right] da = 8\pi\epsilon_0 V_1 \delta Q.$$

At equilibrium $B = 4\pi\epsilon_0 V_1 Q$, and hence

$$\delta C^{-1} = \frac{\delta B}{4\pi\epsilon_0 Q^2} - \frac{2B\delta Q}{4\pi\epsilon_0 Q^3} = 0.$$

Because this quotient is unchanged by rescaling σ, compare trial functions at the same nonzero charge Q. If $\delta\sigma = \sigma - \sigma_1$ and $\int \delta\sigma = 0$, the energy cross term is $V_1 \int \delta\sigma = 0$. The remaining term is

$$W[\sigma] - W[\sigma_1] = \frac{1}{8\pi\epsilon_0} \iint \delta\sigma G \delta\sigma = \frac{\epsilon_0}{2} \int |\nabla\delta\Phi|^2 d^3x \geq 0.$$

Thus $C^{-1}[\sigma] \geq C^{-1}$ and $\boxed{1/C^{-1}[\sigma] \leq C}$. The admissible trial density must have nonzero total charge and finite self-energy; integrability alone does not guarantee the latter. An infinite energy trial supplies only the trivial zero lower bound.

1.19 For the cylindrical capacitor of Problem 1.6c, evaluate the variational upper bound of Problem 1.17b with the naive trial function, $\Psi_1(\rho) = (b - \rho)/(b - a)$. Compare the variational result with the exact result for $b/a = 1.5$, 2, 3. Explain the trend of your results in terms of the functional form of Ψ_1. An improved trial function is treated by *Collin* (pp. 275–277).

Solution.

In the long-cylinder approximation, $\nabla\Psi_1 = -\widehat{\boldsymbol{\rho}}/(b - a)$, so

$$C[\Psi_1] = \epsilon_0 L \int_a^b \frac{2\pi\rho \, d\rho}{(b-a)^2} = \pi\epsilon_0 L \frac{b+a}{b-a}.$$

Divide by $C = 2\pi\epsilon_0 L/\ln(b/a)$ and set $t = b/a$:

$$\boxed{\frac{C[\Psi_1]}{C} = \frac{t+1}{2(t-1)} \ln t.}$$

The ratios at $t = 1.5, 2, 3$ are 1.013663, 1.039721, and 1.098612, i.e. upper-bound errors of 1.37%, 3.97%, and 9.86%. The exact potential is $\Phi = \ln(b/\rho)/\ln(b/a)$, whose gradient varies as $1/\rho$. A constant-gradient linear trial is close when the gap is narrow but cannot reproduce that variation across a wide annulus. In particular, putting $t = 1 + \delta$ gives

$$\frac{2+\delta}{2\delta} \left(\delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} - \cdots \right) = 1 + \frac{\delta^2}{12} + O(\delta^3),$$

which verifies approach to the exact result as the gap shrinks.

1.20 In estimating the capacitance of a given configuration of conductors, comparison with known capacitances is often helpful. Consider two configurations of n conductors in which the $(n-1)$ conductors held at zero potential are the same, but the one conductor whose capacitance we wish to know is different. In particular, let the conductor in one configuration have a closed surface S_1 and in the other configuration have surface S_1', with S_1' totally inside S_1.

(a) Use the extremum principle of Section 1.12 and the variational principle of Problem 1.17 to prove that the capacitance C' of the conductor with surface S_1' is less than or equal to the capacitance C of the conductor with surface S_1 that encloses S_1'.

(b) Set upper and lower limits for the capacitance of a conducting cube of side a. Compare your limits and also their average with the numerical value, $C \simeq 0.655(4\pi\epsilon_0 a)$.

(c) By how much do you estimate the capacitance per unit length of the two-wire system of Problem 1.7 will change (larger? smaller?) if *one* of the wires is replaced by a wire of square cross section whose side is equal to its diameter?

Solution.

(a) Normalize the larger conductor to unit potential, and call its exterior solution Φ. For the smaller-conductor problem choose the trial potential equal to Φ outside S_1 and equal to one throughout the space between S_1' and S_1. It is continuous and satisfies all the smaller problem's Dirichlet data. Its gradient vanishes in the added region, so its variational integral is exactly C. Problem 1.17 therefore gives $\boxed{C' \le C}$.

(b) The cube contains a sphere of radius $a/2$ and lies inside a sphere of radius $\sqrt{3}a/2$. Using $C_{\text{sphere}} = 4\pi\epsilon_0 R$ gives

$$\boxed{\frac{1}{2} \le \frac{C_{\text{cube}}}{4\pi\epsilon_0 a} \le \frac{\sqrt{3}}{2} = 0.866025.}$$

The average of the bounds is $(1+\sqrt{3})/4 = 0.683013$. Relative to the quoted 0.655, the lower bound is 23.66% low, the upper 32.22% high, and the average 4.28% high. The average is an estimate, not a variational bound.

(c) A square wire of side $2a_1$ contains the old round wire of radius a_1 and is contained in a round wire of radius $\sqrt{2}a_1$. The two-conductor capacitance is unchanged if the labels of the grounded and unit-potential wires are exchanged, so part (a) applies to replacing this wire. Writing $\ell = \ln[d/\sqrt{a_1 a_2}]$, the thin-wire formula gives

$$C_0' = \frac{\pi\epsilon_0}{\ell}, \qquad C_0' \le C'_{\text{square}} \lesssim \frac{\pi\epsilon_0}{\ell - \frac{1}{4}\ln 2}.$$

Thus the capacitance increases, with estimated fractional increase between zero and

$$\frac{(\ln 2)/4}{\ell - (\ln 2)/4}.$$

A simple midpoint estimate is half this interval. For the lines in 1.7, $\ell = \pi\epsilon_0/(1.2\times10^{-11}) \simeq 2.318$; the interval is 0 to 8.08% and its midpoint is about 4.04%. The monotonicity is exact, while these numerical percentages inherit the thin-wire approximation.

1.21 A two-dimensional potential problem consists of a unit square area ($0 \le x \le 1$, $0 \le y \le 1$) bounded by "surfaces" held at zero potential. Over the entire square there is a uniform charge density of unit strength (per unit length in z).

(a) Apply the variational principle (1.63) for the Poisson equation with the "variational" trial function $\Psi(x, y) = A \cdot x(1-x) \cdot y(1-y)$ to determine the best value of the constant A. [I use quotation marks around variational because there are no parameters to vary except the overall scale.]

(b) The exact (albeit series) solution for this problem is [see Problems 2.15 and 2.16]

$$4\pi\epsilon_0\Phi(x,y) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left\{1 - \frac{\cosh[(2m+1)\pi(y-\frac{1}{2})]}{\cosh[(2m+1)\pi/2]}\right\}$$

For $y = 0.25$ and $y = 0.5$, plot and compare the simple variational solution of part a with the exact solution as functions of x.

Solution.

(a) Put $f = x(1-x)y(1-y)$ and $\Psi = Af$. For unit charge density the functional (1.63) is

$$I[A] = \frac{1}{2}A^2\int_0^1\int_0^1 |\nabla f|^2 dxdy - \frac{A}{\epsilon_0}\int_0^1\int_0^1 f\, dxdy.$$

The elementary integrals are

$$\int_0^1 x(1-x)dx = \frac{1}{6}, \quad \int_0^1 x^2(1-x)^2dx = \frac{1}{30}, \quad \int_0^1 (1-2x)^2dx = \frac{1}{3}.$$

Consequently $\int f = 1/36$ and $\int |\nabla f|^2 = 2(1/3)(1/30) = 1/45$, giving

$$I[A] = \frac{A^2}{90} - \frac{A}{36\epsilon_0}, \qquad I'[A] = \frac{A}{45} - \frac{1}{36\epsilon_0} = 0.$$

Thus $\boxed{A = 5/(4\epsilon_0)}$, and $I'' = 1/45 > 0$. In the book's plotted normalization,

$$\boxed{u_{\text{trial}} = 4\pi\epsilon_0\Psi = 5\pi x(1-x)y(1-y).}$$

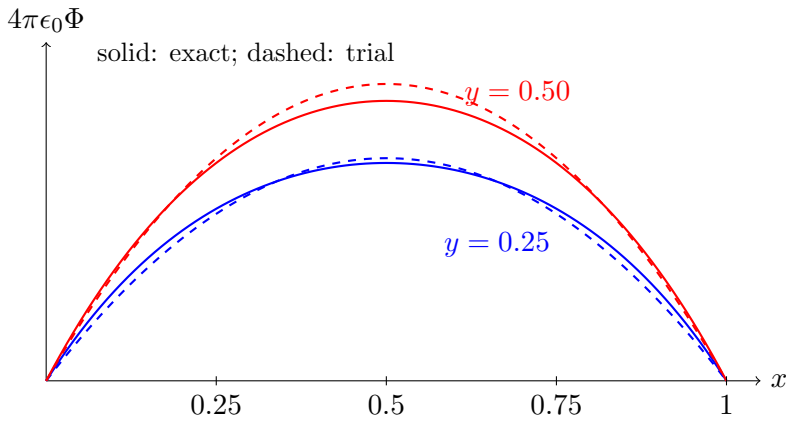
(b) Write $k_n = n\pi$ for positive odd n. The supplied exact series is

$$u = \frac{16}{\pi^2}\sum_{n\ \text{odd}>0}\frac{\sin k_n x}{n^3}\left[1 - \frac{\cosh(k_n(y-1/2))}{\cosh(k_n/2)}\right].$$

The boundary factors vanish at $x = 0, 1$ and $y = 0, 1$. Applying the Laplacian cancels the two derivatives of the hyperbolic term, leaving

$$\nabla^2 u = -16\sum_{n\ \text{odd}>0}\frac{\sin(n\pi x)}{n} = -4\pi,$$

where $1 = (4/\pi)\sum_{n\ \text{odd}>0}\sin(n\pi x)/n$ is the sine expansion of a constant on $0 < x < 1$. Thus the normalization indeed corresponds to $\rho = 1$. The comparison is



At $(1/2, 1/4)$ the exact and trial values are 0.720492 and $15\pi/64 = 0.736311$; at the center they are 0.925782 and $5\pi/16 = 0.981748$. The respective errors are about 2.20% and 6.04%. The variational principle minimizes the functional, not the pointwise error, so it does not make this potential a pointwise upper bound everywhere. The numerical series uses the stable ratio

$$\frac{\cosh(k(y-1/2))}{\cosh(k/2)} = \frac{e^{-ky} + e^{-k(1-y)}}{1 + e^{-k}},$$

which avoids overflow for large k.

1.22 Two-dimensional relaxation calculations commonly use sites on a square lattice with spacing $\Delta x = \Delta y = h$, and label the sites by (i, j), where i, j are integers and $x_i = ih + x_0$, $y_j = jh + y_0$. The value of the potential at (i, j) can be approximated by the average of the values at neighboring sites. [Recall the relevant theorem about harmonic functions.] But what average?

(a) If $F(x, y)$ is a well-behaved function in the neighborhood of the origin, but not necessarily harmonic, by explicit Taylor series expansions, show that the "cross" sum

$$S_c = F(h, 0) + F(0, h) + F(-h, 0) + F(0, -h)$$

can be expressed as

$$S_c = 4F(0, 0) + h^2\nabla^2 F + \frac{h^4}{12}\left(F_{xxxx} + F_{yyyy}\right) + O(h^6)$$

(b) Similarly, show that the "square" sum,

$$S_s = F(h, h) + F(-h, h) + F(-h, -h) + F(h, -h)$$

can be expressed as

$$S_s = 4F(0, 0) + 2h^2\nabla^2 F - \frac{h^4}{3}\left(F_{xxxx} + F_{yyyy}\right) + \frac{h^4}{2}\nabla^2(\nabla^2 F) + O(h^6)$$

Here F_{xxxx} is the fourth partial derivative of F with respect to x, evaluated at $x = 0$, $y = 0$, etc. If $\nabla^2 F = 0$, the averages $S_c/4$ and $S_s/4$ each give the value of $F(0, 0)$, correct to order h^3 inclusive. Note that an improvement can be obtained by forming the "improved" average,

$$\langle\langle F(0, 0)\rangle\rangle = \frac{1}{5}\left[S_c + \frac{1}{4}S_s\right]$$

where

$$\langle\langle F(0,0)\rangle\rangle = F(0,0) + \frac{3}{10}h^2\nabla^2 F + \frac{h^4}{40}\nabla^2(\nabla^2 F) + O(h^6)$$

If $\nabla^2 F = 0$, then the improved average gives $F(0,0)$, correct to order h^5 inclusive. For Poisson's equation, the charge density and its lowest order Laplacian can be inserted for the same accuracy.

Solution.

Write all derivatives below at the origin. The multivariate Taylor series is

$$F(x,y) = \sum_{p+q\le 5} \frac{x^p y^q}{p!q!}\partial_x^p \partial_y^q F + O((|x|+|y|)^6).$$

(a) Pairing opposite points cancels odd powers:

$$F(h,0) + F(-h,0) = 2F + h^2 F_{xx} + \frac{h^4}{12}F_{xxxx} + O(h^6).$$

The analogous y pair gives

$$\boxed{S_c = 4F + h^2\nabla^2 F + \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6).}$$

(b) In the four diagonal points, terms with either odd p or odd q cancel. The surviving terms through degree four are

$$S_s = 4F + 2h^2(F_{xx} + F_{yy}) + \frac{h^4}{6}(F_{xxxx} + F_{yyyy}) + h^4 F_{xxyy} + O(h^6).$$

Since $\nabla^4 F = F_{xxxx} + 2F_{xxyy} + F_{yyyy}$,

$$\boxed{S_s = 4F + 2h^2\nabla^2 F - \frac{h^4}{3}(F_{xxxx} + F_{yyyy}) + \frac{h^4}{2}\nabla^4 F + O(h^6).}$$

Now multiply S_c by $1/5$ and S_s by $1/20$. The fourth-derivative anisotropy cancels because $(1/5)(1/12) + (1/20)(-1/3) = 0$. The other coefficients are $4/5 + 4/20 = 1$, $1/5 + 2/20 = 3/10$, and $(1/20)(1/2) = 1/40$. Therefore

$$\boxed{\frac{4S_c + S_s}{20} = F + \frac{3}{10}h^2\nabla^2 F + \frac{h^4}{40}\nabla^4 F + O(h^6).}$$

If F is harmonic in a neighborhood, both displayed Laplacians vanish. For Poisson's equation $\nabla^2\Phi = -\rho/\epsilon_0$, instead rearrange to

$$\boxed{\Phi = \frac{4S_c + S_s}{20} + \frac{3h^2\rho}{10\epsilon_0} + \frac{h^4\nabla^2\rho}{40\epsilon_0} + O(h^6).}$$

This is a local consistency statement for smooth fields; corners or singular sources can reduce the convergence order of the full boundary-value solution.

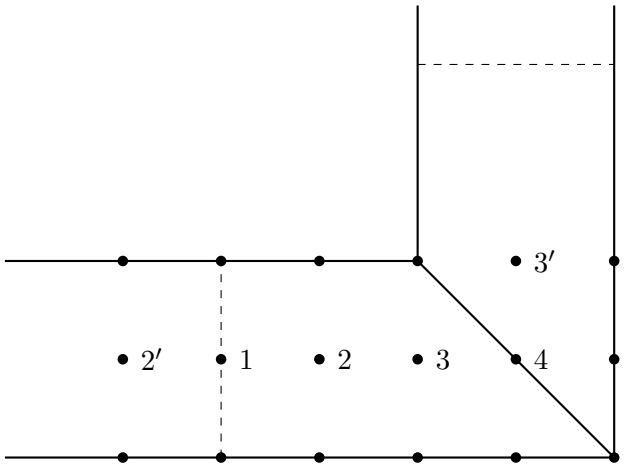
1.23 A transmission line consists of a long straight conductor with a hollow square region in its interior, with a square conductor of one-quarter the area of the hollow region centered in the empty space, with edges parallel to the inner sides of outer conductor. If the conductors are raised to different potentials, the potential and electric field in the space between them exhibit an eightfold symmetry; the basic unit is sketched in the accompanying figure. The efficacy of the relaxation method in determining the properties of the transmission line can be illustrated by a simple calculation.

(a) Using only the four interior points indicated in the figure, write down the relaxation equation for each point for the "cross" and the "improved" averaging schemes (defined in Problem 1.22) if the inner conductor has $\Phi = 100$ V and the outer has $\Phi = 0$. By performing either the relaxation iteration process or solving the set of algebraic equations for each scheme, find estimates for the potential at each of the four points for the two schemes.

(b) From the results of part a make the best estimate (or estimates) you can for the capacitance per unit length of the transmission line.

(c) (Optional) Using your favorite computational tools, repeat the relaxation calculation with half the lattice spacing (21 interior points) and compare.

Answer: $\Phi_1 = 48.87$ V, $\Phi_2 = 47.18$ V, $\Phi_3 = 38.34$ V, $\Phi_4 = 19.81$ V and $C' = 10.23\epsilon_0$ [textbook numerical reference; ϵ_0 already has units F/m, so no additional F/m multiplies it].



Problem 1.23

Solution.

Use a square annulus with outer walls $|x|, |y| \le 4h$ and inner conductor $|x|, |y| \le 2h$. By rotation/reflection, the four labeled values can be placed at $(0, 3h), (h, 3h), (2h, 3h), (3h, 3h)$; call them p_1, p_2, p_3, p_4.

(a) For the cross stencil the four axial-neighbor sums are, respectively, $2p_2 + 100$, $p_1 + p_3 + 100$, $p_2 + p_4 + 100$, and $2p_3$. Thus

$$4p_1 = 2p_2 + 100, \quad 4p_2 = p_1 + p_3 + 100, \quad 4p_3 = p_2 + p_4 + 100, \quad 4p_4 = 2p_3.$$

Eliminate in the order

$$p_4 = p_3/2, \qquad p_2 = \tfrac{7}{2}p_3 - 100, \qquad p_1 = 4p_2 - p_3 - 100 = 13p_3 - 500.$$

The first equation becomes $52p_3 - 2000 = 7p_3 - 100$, or $45p_3 = 1900$. Therefore

$$\boxed{(p_1, p_2, p_3, p_4) = \tfrac{1}{9}(440, 430, 380, 190)\ \text{V}.}$$

The diagonal sums for the improved stencil are 200, 200, $100+p_3$, and 100. Since $20p = 4S_c + S_s$, its equations are

$$20p_1 = 8p_2 + 600, \qquad 20p_2 = 4p_1 + 4p_3 + 600,$$
$$19p_3 = 4p_2 + 4p_4 + 500, \quad 20p_4 = 8p_3 + 100.$$

Now $p_2 = 5p_1/2 - 75$, $p_3 = 5p_2 - p_1 - 150 = 23p_1/2 - 525$, and $p_4 = 2p_3/5 + 5$. The third equation becomes $87p_3 = 50p_1 + 1100$, and hence $(1901/2)p_1 = 46775$. This gives

$$\boxed{(p_1, p_2, p_3, p_4) = \frac{1}{1901}(93550, 91300, 77800, 40625)\ \text{V}.}$$

Scheme	p_1	p_2	p_3	p_4
Cross	48.8889	47.7778	42.2222	21.1111
Improved	49.2109	48.0274	40.9258	21.3703
Quoted reference	48.87	47.18	38.34	19.81

The nine-point stencil reduces the corner-near error in p_3, but does not improve every value on this very coarse grid.

(b) Estimate the total charge from the smooth outer wall rather than the singular field at the inner corners. Define $g_j = E_n(jh, 4h)$, outward from the annulus toward the outer conductor. A one-sided quadratic derivative is

$$g_j \simeq \frac{4\Phi(jh, 3h) - \Phi(jh, 2h)}{2h}.$$

Here the field points toward the outer conductor, so its positive magnitude gives the magnitude of the negative outer charge. For $j = 0, \ldots, 4$,

$$(g_0, g_1, g_2, g_3, g_4) = \frac{1}{2h}(4p_1 - 100, 4p_2 - 100, 4p_3 - 100, 4p_4 - p_3, 0).$$

Use symmetry and composite Simpson integration on one half-side, then multiply by eight for the full perimeter:

$$\frac{C'}{\epsilon_0} \simeq \frac{8h}{3(100)}(g_0 + 4g_1 + 2g_2 + 4g_3 + g_4) = \frac{4p_1 + 16p_2 + 4p_3 + 16p_4}{75} - \frac{28}{3}.$$

The estimates are $\boxed{10.2222}$ (cross) and $\boxed{10.2788}$ (improved), compared with the quoted 10.23. The especially good coarse cross result is not a guaranteed accuracy bound.

A separate variational estimate follows by linearly interpolating the grid potential over right triangles. Each triangle contributes half the sum of squared potential differences on its horizontal and vertical legs to $\int |\nabla\Phi|^2 dA$. Summing triangles gives

$$\frac{C'_{\text{trial}}}{\epsilon_0} = \frac{1}{100^2} \sum_{\text{horizontal/vertical edges}} (\Delta\Phi)^2.$$

Boundary tangential edges carry zero differences. Problem 1.17 makes this an upper bound. The coarse cross and improved grids give 10.84444 and 10.85122, respectively. For the cross solution, discrete flux conservation also writes its energy as $4[500 - p_1 - 2p_2 - 2p_3]/100 = 10.84444$.

(c) With spacing $h/2$, keep coordinates measured in the original h. The 21 distinct sites satisfy $0 \le x \le y$, $y > 2$, and $y < 4$. At every site solve

$$4\Phi_{ij} - \sum_{\text{axial}} \Phi = 0 \quad \text{or} \quad 20\Phi_{ij} - 4\sum_{\text{axial}} \Phi - \sum_{\text{diagonal}} \Phi = 0,$$

replacing boundary values by 0 or 100 and reflected sites by their equal interior value. The results are:

$(x/h, y/h)$	Cross (V)	Improved (V)
$(0, 2.5)$	74.20689	74.28283
$(0.5, 2.5)$	73.95577	74.03723
$(1, 2.5)$	73.01509	73.10016
$(1.5, 2.5)$	70.62375	70.59951
$(2, 2.5)$	64.52396	63.42846
$(2.5, 2.5)$	47.62511	47.55550
$(0, 3)$	48.91600	49.00099
$(0.5, 3)$	48.60111	48.67741
$(1, 3)$	47.48085	47.50202
$(1.5, 3)$	44.95596	44.77078
$(2, 3)$	39.84699	39.29063
$(2.5, 3)$	30.72625	30.58736
$(3, 3)$	20.42497	20.40227
$(0, 3.5)$	24.25490	24.30382
$(0.5, 3.5)$	24.05181	24.08971
$(1, 3.5)$	23.35122	23.34158
$(1.5, 3.5)$	21.87224	21.75414
$(2, 3.5)$	19.18179	18.96016
$(2.5, 3.5)$	15.00793	14.91648
$(3, 3.5)$	10.12369	10.11165
$(3.5, 3.5)$	5.06184	5.06478

The outer-flux capacitance estimates become $10.2950\epsilon_0$ and $10.2633\epsilon_0$; the energy upper bounds become $10.46224\epsilon_0$ and $10.46574\epsilon_0$. Thus voltage errors and energy bounds improve, although the coarse cross flux estimate's accidental closeness does not persist under this first refinement. The script `supplementary/check_ch01.py` assembles the full annulus independently of the four-variable equations, checks residuals and symmetries, and also computes grids with 32, 64, and 128 intervals per side. For example, the 128-interval cross grid gives the energy bound $C'/\epsilon_0 \le 10.24757$ and the flux estimate 10.24500.

1.24 Consider solution of the two-dimensional Poisson equation problem of Problem 1.21, a unit square with zero potential on the boundary and a constant unit charge density in the interior, by the technique of relaxation. Choose $h = 0.25$ so that there are nine interior sites. Use symmetry to reduce the number of needed sites to three, at (0.25, 0.25), (0.5, 0.25), and (0.5, 0.5). With so few sites, it is easy to do the iterations with a block of paper and a pocket calculator, but suit yourself.

(a) Use the "improved grid" averaging of Problem 1.22 and the simple (Jacobi) iteration scheme, starting with $4\pi\epsilon_0\Phi = 1.0$ at all three interior sites. Do at least six iterations, preferably eight or ten.

(b) Repeat the iteration procedure with the same starting values, but using Gauss–Seidel iteration.

(c) Graph the two sets of results of each iteration versus iteration number and compare with the exact values, $4\pi\epsilon_0\Phi(0.25, 0.25) = 0.5691$, $4\pi\epsilon_0\Phi(0.5, 0.25) = 0.7205$, $4\pi\epsilon_0\Phi(0.5, 0.5) = 0.9258$. Comment on rate of convergence and final accuracy.

Solution.

Use $u = 4\pi\epsilon_0\Phi$; then $\nabla^2 u = -4\pi$. Let (a, b, c) denote the values at $(1/4, 1/4)$, $(1/2, 1/4)$, and $(1/2, 1/2)$. The source contribution in Problem 1.22 is $3h^2(4\pi)/10 = 3\pi/40$ for $h = 1/4$. The axial and diagonal sums are

	S_c	S_s
a	$2b$	c
b	$2a + c$	$2b$
c	$4b$	$4a$

Multiplying the averaging equation by 20 therefore gives

$$20a = 8b + c + s, \qquad 18b = 8a + 4c + s, \qquad 20c = 4a + 16b + s, \qquad s = \frac{3\pi}{2}.$$

Here the symmetry-related self term $2b$ has been moved to the diagonal before defining Jacobi and Gauss–Seidel on this three-variable system.

(a) Starting from $(a_0, b_0, c_0) = (1, 1, 1)$, Jacobi uses only the old values in each sweep:

$$a_{k+1} = \frac{8b_k + c_k + s}{20}, \quad b_{k+1} = \frac{8a_k + 4c_k + s}{18}, \quad c_{k+1} = \frac{4a_k + 16b_k + s}{20}.$$

For example, $a_1 = (9 + 3\pi/2)/20 = 0.685619$, $b_1 = (12 + 3\pi/2)/18 = 0.928466$, and $c_1 = (20 + 3\pi/2)/20 = 1.235619$.

(b) With update order a, b, c, Gauss–Seidel instead uses

$$a_{k+1} = \frac{8b_k + c_k + s}{20}, \quad b_{k+1} = \frac{8a_{k+1} + 4c_k + s}{18}, \quad c_{k+1} = \frac{4a_{k+1} + 16b_{k+1} + s}{20}.$$

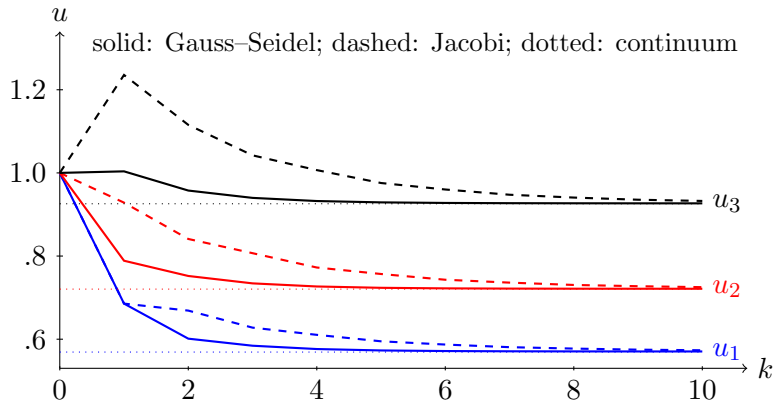
Thus its first sweep has the same a_1, but $b_1 = 0.788741$ and $c_1 = 1.003736$. The complete ten-sweep calculation is

k	Jacobi a_k	Jacobi b_k	Jacobi c_k	Gauss–Seidel a_k	Gauss–Seidel b_k	Gauss–Seidel c_k
0	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
1	0.685619	0.928466	1.235619	0.685619	0.788741	1.003736
2	0.668787	0.841101	1.115516	0.601303	0.752098	0.957558
3	0.627836	0.806930	1.042258	0.584336	0.734295	0.939923
4	0.610505	0.772450	1.006731	0.576334	0.726819	0.932342
5	0.594936	0.756853	0.975681	0.572964	0.723637	0.929122
6	0.587145	0.743033	0.960089	0.571530	0.722284	0.927753
7	0.580837	0.736106	0.947475	0.570921	0.721709	0.927171
8	0.577435	0.730499	0.940671	0.570662	0.721465	0.926924
9	0.574853	0.727475	0.935506	0.570552	0.721361	0.926818
10	0.573385	0.725180	0.932570	0.570505	0.721317	0.926774

(c) Solving the three equations directly gives

$$\boxed{(a_*, b_*, c_*) = \left(\frac{783\pi}{4312}, \frac{45\pi}{196}, \frac{159\pi}{539}\right) = (0.5704701, 0.7212840, 0.9267407).}$$

For explicit elimination, the third equation gives $c = a/5 + 4b/5 + s/20$; substituting it into the first and second gives $396a - 176b = 21s$ and $-176a + 296b = 24s$. Eliminating a then gives $b = 15s/98 = 45\pi/196$; back substitution produces a and c above. The continuum series gives $(0.5690826, 0.7204917, 0.9257815)$, so the final grid errors are $(0.0013874, 0.0007923, 0.0009592)$, or about $(0.244\%, 0.110\%, 0.104\%)$. Iterating further removes iteration error but cannot remove these discretization errors.



To quantify convergence, write $A = D + L + U$ for the coefficient matrix. Errors propagate by $B_J = -D^{-1}(L+U)$ and $B_{GS} = -(D+L)^{-1}U$. Their spectral radii are 0.648377 and 0.425149, respectively, explaining the faster Gauss–Seidel convergence for this update order. If one instead applies the unreduced nine-point averaging formula literally, the second Jacobi update is $b_{k+1} = (8a_k + 2b_k + 4c_k + s)/20$. That is another valid stationary iteration with different transient values and the same fixed point; it should not be mixed with the reduced-system iteration tabulated here.

2 Chapter 2. Boundary-Value Problems in Electrostatics: I

Problems

2.1 A point charge q is brought to a position a distance d away from an infinite plane conductor held at zero potential. Using the method of images, find:

(a) the surface-charge density induced on the plane, and plot it;

(b) the force between the plane and the charge by using Coulomb's law for the force between the charge and its image;

(c) the total force acting on the plane by integrating $\sigma^2/2\epsilon_0$ over the whole plane;

(d) the work necessary to remove the charge q from its position to infinity;

(e) the potential energy between the charge q and its image [compare the answer to part d and discuss].

(f) Find the answer to part d in electron volts for an electron originally one angstrom from the surface.

Solution.

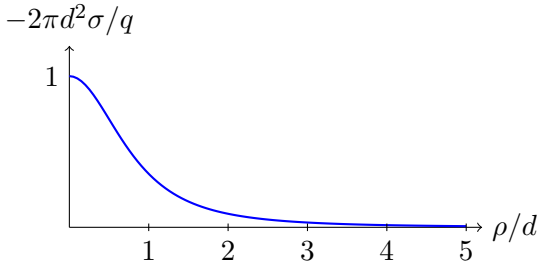
Put the plane at $z = 0$ and the real charge at $d\widehat{\mathbf{z}}$. All potentials below refer to the vacuum region $z > 0$; the image is $-q$ at $-d\widehat{\mathbf{z}}$. Thus

$$\Phi(\rho, z) = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{\sqrt{\rho^2 + (z-d)^2}} - \frac{1}{\sqrt{\rho^2 + (z+d)^2}}\right].$$

(a) The normal from the metal into the vacuum is $\widehat{\mathbf{z}}$, so $\sigma = -\epsilon_0 \partial_z \Phi|_{0^+}$. Differentiating the two terms gives respectively $d/(\rho^2 + d^2)^{3/2}$ and $d/(\rho^2 + d^2)^{3/2}$, whence

$$\boxed{\sigma(\rho) = -\frac{qd}{2\pi(\rho^2 + d^2)^{3/2}}.}$$

As a normalization check, $2\pi \int_0^\infty \sigma\rho\, d\rho = -qd[-(\rho^2 + d^2)^{-1/2}]_0^\infty = -q$. The following plot uses $u = \rho/d$ and $-2\pi d^2\sigma/q = (1 + u^2)^{-3/2}$ (the signed density is negative for $q > 0$).



(b) The separation of the charge and image is $2d$, hence

$$\boxed{\mathbf{F}_q = -\frac{q^2}{16\pi\epsilon_0 d^2}\hat{\mathbf{z}}.}$$

(c) The electrostatic pressure on the metal is toward the vacuum. Its integral is

$$\begin{aligned}\mathbf{F}_{\text{plane}} &= \hat{\mathbf{z}}\int_0^\infty \frac{q^2 d^2}{8\pi^2\epsilon_0(\rho^2+d^2)^3}\,2\pi\rho\,d\rho \\ &= \hat{\mathbf{z}}\frac{q^2d^2}{4\pi\epsilon_0}\left[-\frac{1}{4(\rho^2+d^2)^2}\right]_0^\infty = \frac{q^2}{16\pi\epsilon_0 d^2}\hat{\mathbf{z}}.\end{aligned}$$

This equals $-\mathbf{F}_q$. **(d)** In quasistatic removal the applied force opposes the attraction at every instantaneous height s, so

$$\boxed{W = \int_d^\infty \frac{q^2}{16\pi\epsilon_0 s^2}\,ds = \frac{q^2}{16\pi\epsilon_0 d}.}$$

(e) Two *independent* charges $q, -q$ a distance $2d$ apart have mutual energy $U_{q,q'} = -q^2/(8\pi\epsilon_0 d) = -2W$. The physical charge–conductor energy relative to infinite separation is instead

$$U = \frac{1}{2}q\Phi_{\text{induced}}(d) = -\frac{q^2}{16\pi\epsilon_0 d} = -W.$$

The factor $1/2$ follows by charging the real source from 0 to q: the induced potential is proportional to the charge already supplied, so $U = \int_0^q (q_1/q)\Phi_{\text{induced}}(q)\,dq_1$. An image is a representation of this induced charge, not an additional independently assembled physical source. **(f)** With $|q| = e$, $d = 10^{-10}\,\text{m}$, the conversion is

$$W[\text{eV}] = \frac{e}{16\pi\epsilon_0(10^{-10}\,\text{m})} \simeq \boxed{3.60\ \text{eV}}.$$

The numerical check uses the SI constants in `scipy.constants`.

2.2 Using the method of images, discuss the problem of a point charge q *inside* a hollow, grounded, conducting sphere of inner radius a. Find

(a) the potential inside the sphere;

(b) the induced surface-charge density;

(c) the magnitude and direction of the force acting on q.

(d) Is there any change in the solution if the sphere is kept at a fixed potential V? If the sphere has a total charge Q on its inner and outer surfaces?

Solution.

Let the charge be at $b\hat{\mathbf{z}}$, $0 < b < a$. Inversion in the sphere gives an image $q' = -qa/b$ at $b' = a^2/b > a$. On $r = a$, $|\mathbf{r} - b'\hat{\mathbf{z}}| = (a/b)|\mathbf{r} - b\hat{\mathbf{z}}|$, so their potentials cancel. **(a)** By uniqueness,

$$\boxed{\Phi(r,\theta) = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{\sqrt{r^2+b^2-2rb\cos\theta}} - \frac{a/b}{\sqrt{r^2+a^4/b^2-2ra^2\cos\theta/b}}\right], \quad r < a.}$$

The image lies outside the solution domain and creates no extra singularity in the cavity. **(b)** The metal-to-cavity normal is $-\hat{\mathbf{r}}$, so $\sigma = \epsilon_0 \partial_r \Phi|_{a^-}$. To simplify the derivative, write $D = \sqrt{a^2 + b^2 - 2ab\cos\theta}$ and use the distance relation:

$$\partial_r \Phi|_a = \frac{q}{4\pi\epsilon_0 D^3}\left[-(a - b\cos\theta) + \frac{b^2}{a^2}\left(a - \frac{a^2}{b}\cos\theta\right)\right].$$

Thus

$$\boxed{\sigma(\theta) = -\frac{q(a^2 - b^2)}{4\pi a(a^2 + b^2 - 2ab\cos\theta)^{3/2}}.}$$

Indeed, setting $u = \cos\theta$ gives

$$2\pi a^2 \int_{-1}^{1} \sigma(u)\, du = -\frac{qa(a^2 - b^2)}{2}\frac{1}{ab}\left(\frac{1}{a - b} - \frac{1}{a + b}\right) = -q.$$

(c) Only the image field acts on q after its own singular field is excluded. The distance is $b' - b = (a^2 - b^2)/b$, giving

$$\boxed{\mathbf{F} = \frac{q^2 ab}{4\pi\epsilon_0 (a^2 - b^2)^2}\hat{\mathbf{z}}.}$$

It is toward the nearer wall for either sign of q. For a centered charge the limits are $\Phi = q(1/r - 1/a)/(4\pi\epsilon_0)$, $\sigma = -q/(4\pi a^2)$, and $\mathbf{F} = 0$. **(d)** Holding the metal at V adds the constant V to the cavity potential; it leaves the cavity field, inner surface charge, and force unchanged. For an isolated metal with total charge Q, Gauss's law in the metal requires inner charge $-q$ and outer charge $Q + q$. The same cavity solution therefore holds up to a constant. That constant depends on the exterior geometry. For a concentric spherical outer surface of radius A, with no other external sources and zero potential at infinity, it is $(Q + q)/(4\pi\epsilon_0 A)$. For an ideal thin shell one sets $A = a$. Specifying only the inner radius does not specify this exterior capacitance.

2.3 A straight-line charge with constant linear charge density λ is located perpendicular to the x-y plane in the first quadrant at (x_0, y_0). The intersecting planes $x = 0$, $y \geq 0$ and $y = 0$, $x \geq 0$ are conducting boundary surfaces held at zero potential. Consider the potential, fields, and surface charges in the first quadrant.

(a) The well-known potential for an isolated line charge at (x_0, y_0) is $\Phi(x, y) = (\lambda/4\pi\epsilon_0)\ln(R^2/r^2)$, where $r^2 = (x - x_0)^2 + (y - y_0)^2$ and R is a constant. Determine the expression for the potential of the line charge in the presence of the intersecting planes. Verify explicitly that the potential and the tangential electric field vanish on the boundary surfaces.

(b) Determine the surface charge density σ on the plane $y = 0$, $x \geq 0$. Plot σ/λ versus x for $(x_0 = 2, y_0 = 1)$, $(x_0 = 1, y_0 = 1)$, and $(x_0 = 1, y_0 = 2)$.

(c) Show that the total charge (per unit length in z) on the plane $y = 0$, $x \geq 0$ is

$$Q_x = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{x_0}{y_0}\right)$$

What is the total charge on the plane $x = 0$?

(d) Show that far from the origin [$\rho \gg \rho_0$, where $\rho = \sqrt{x^2+y^2}$ and $\rho_0 = \sqrt{x_0^2+y_0^2}$] the leading term in the potential is

$$\Phi \to \Phi_{\text{asym}} = \frac{4\lambda}{\pi\epsilon_0}\frac{(x_0y_0)(xy)}{\rho^4}$$

Interpret.

Solution.

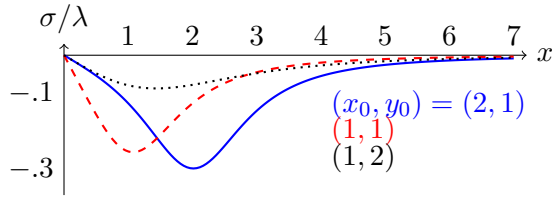
Define $D_{st} = (x - sx_0)^2 + (y - ty_0)^2$, with $s, t = \pm 1$. **(a)** Place line charges $st\lambda$ at (sx_0, ty_0). The arbitrary logarithmic reference lengths cancel because the four charges sum to zero, leaving

$$\boxed{\Phi = \frac{\lambda}{4\pi\epsilon_0}\ln\frac{D_{+-}D_{-+}}{D_{++}D_{--}}.}$$

At $y = 0$, $D_{++} = D_{+-}$ and $D_{-+} = D_{--}$, so $\Phi = 0$; at $x = 0$ the other two equalities give the same result. Differentiating these identities along either boundary gives $E_x(x, 0) = 0$ and $E_y(0, y) = 0$. All terms are independent of z, so $E_z = 0$ as well. **(b)** On $y = 0$ the outward metal normal is $+\hat{\mathbf{y}}$. Differentiating the logarithms before putting $y = 0$ gives

$$\boxed{\frac{\sigma(x)}{\lambda} = -\frac{y_0}{\pi}\left[\frac{1}{(x-x_0)^2+y_0^2} - \frac{1}{(x+x_0)^2+y_0^2}\right].}$$

It vanishes at the corner, is negative for $x > 0$ when $\lambda > 0$, and decays as $-4x_0y_0/(\pi x^3)$.



(c) Integrating each Lorentzian explicitly,

$$Q_x = -\frac{\lambda}{\pi}\left[\tan^{-1}\frac{x-x_0}{y_0} - \tan^{-1}\frac{x+x_0}{y_0}\right]_{x=0}^{\infty}$$
$$= \boxed{-\frac{2\lambda}{\pi}\tan^{-1}\frac{x_0}{y_0}}.$$

Interchange x_0, y_0 for the other plane: $Q_y = -2\lambda\tan^{-1}(y_0/x_0)/\pi$. Since both coordinates are positive, their inverse tangents add to $\pi/2$ and $Q_x + Q_y = -\lambda$. **(d)** Write $D_{st} = \rho^2 + B_{st}$, with $B_{st} = \rho_0^2 - 2sxx_0 - 2tyy_0$. In $\Phi = -\lambda\sum st\ln D_{st}/(4\pi\epsilon_0)$, the constant and first-order terms cancel. The needed quadratic sum is

$$\sum_{s,t} stB_{st}^2 = 32xx_0yy_0, \quad \sum_{s,t} st\ln D_{st} = -\frac{16xx_0yy_0}{\rho^4} + O(\rho_0^4/\rho^4).$$

Consequently

$$\boxed{\Phi \sim \frac{4\lambda x_0y_0xy}{\pi\epsilon_0\rho^4}.}$$

The images cancel both the net line charge and the dipole moment. This is the leading two-dimensional quadrupole, proportional to $\sin 2\phi/\rho^2$, with nodes on the grounded planes.

2.4 A point charge is placed a distance $d > R$ from the center of an equally charged, isolated, conducting sphere of radius R.

(a) Inside of what distance from the surface of the sphere is the point charge attracted rather than repelled by the charged sphere?

(b) What is the limiting value of the force of attraction when the point charge is located a distance $a\ (= d - R)$ from the surface of the sphere, if $a \ll R$?

(c) What are the results for parts a and b if the charge on the sphere is twice (half) as large as the point charge, but still the same sign?

[Answers: (a) $d/R - 1 = 0.6180(34)$, (b) $F = -q^2/(16\pi\epsilon_0 a^2)$, i.e., image force, (c) for $Q = 2q$, $d/R - 1 = 0.4276$; for $Q = q/2$, $d/R - 1 = 0.8823$. The answer for part b is the same.]

Solution.

Let $k = (4\pi\epsilon_0)^{-1}$ and write $Q = \alpha q$ for the sphere's charge. The image pair inside the sphere is

$$q' = -\frac{qR}{d}, \quad r' = \frac{R^2}{d}, \qquad q_0 = Q + \frac{qR}{d} \quad \text{at the center.}$$

The image q' cancels the angular variation on the surface and the central charge makes the total image charge Q. Taking positive force away from the center,

$$F = kq\left[\frac{Q + qR/d}{d^2} - \frac{qR/d}{(d - R^2/d)^2}\right] = \frac{kq^2}{R^2}\left[\frac{\alpha + 1/x}{x^2} - \frac{x}{(x^2-1)^2}\right], \quad x = \frac{d}{R}.$$

(a) Setting $F = 0$ is equivalent to

$$\alpha = H(x) := \frac{2x^2 - 1}{x(x^2-1)^2}.$$

For $x > 1$, $H'(x) = (-6x^4 + 3x^2 - 1)/[x^2(x^2-1)^3] < 0$; also $H(1^+) = \infty$ and $H(\infty) = 0$. There is exactly one root, with attraction below it. For $\alpha = 1$ the root is $x = (1 + \sqrt{5})/2$: using $x^2 = x + 1$ makes both sides of $x(x^2-1)^2 = 2x^2 - 1$ equal $2x + 1$. Thus

$$\boxed{0 < d - R < \frac{\sqrt{5} - 1}{2} R = 0.6180339887\ldots R.}$$

(b) Set $d = R + a$. Then $d - R^2/d = 2a + O(a^2/R)$ and $q' = -q + O(qa/R)$, whereas the central-charge force remains finite. Therefore

$$\boxed{F \sim -\frac{kq^2}{4a^2} = -\frac{q^2}{16\pi\epsilon_0 a^2} \quad (a/R \to 0).}$$

(c) Solving the same monotone equation $H(x) = \alpha$ gives

Q/q	x at zero force	$(d - R)/R$ at zero force
2	$1.427563934\ldots$	$0.427563934\ldots$
1/2	$1.882269146\ldots$	$0.882269146\ldots$

Attraction holds below these respective distances. The leading near-surface force in part (b) is independent of the finite value of Q/q because it comes from the nearby image.

2.5

(a) Show that the work done to remove the charge q from a distance $r > a$ to infinity against the force, Eq. (2.6), of a grounded conducting sphere is

$$W = \frac{q^2 a}{8\pi\epsilon_0 (r^2 - a^2)}$$

Relate this result to the electrostatic potential, Eq. (2.3), and the energy discussion of Section 1.11.

(b) Repeat the calculation of the work done to remove the charge q against the force, Eq. (2.9), of an isolated charged conducting sphere. Show that the work done is

$$W = \frac{1}{4\pi\epsilon_0}\left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right]$$

Relate the work to the electrostatic potential, Eq. (2.8), and the energy discussion of Section 1.11.

Solution.

Use $k = (4\pi\epsilon_0)^{-1}$ and take outward force as positive. **(a)** The grounded-sphere image at a source position $s > a$ is $-qa/s$ at a^2/s. Hence the force appearing in (2.6) is

$$F_g(s) = -k\frac{q^2 a/s}{(s - a^2/s)^2} = -k\frac{q^2 as}{(s^2 - a^2)^2}.$$

The quasistatic work is

$$W_g = -\int_r^\infty F_g(s)\, ds = kq^2 a\left[-\frac{1}{2(s^2 - a^2)}\right]_r^\infty = \boxed{\frac{q^2 a}{8\pi\epsilon_0 (r^2 - a^2)}}.$$

At the source the induced potential from the image construction of (2.3) is $\Phi_{\text{ind}} = -kqa/(r^2 - a^2)$. Therefore $U_g = \frac{1}{2}q\Phi_{\text{ind}} = -W_g$. The half factor is the work of building up the induced charge linearly with the real source, as in Section 1.11, with the divergent self-energy of the point charge omitted. **(b)** For an isolated sphere, add a central charge $Q + qa/s$ to the grounded image. Thus the force (2.9) is

$$F_Q(s) = -k\frac{q^2 as}{(s^2 - a^2)^2} + k\frac{qQ}{s^2} + k\frac{q^2 a}{s^3}.$$

Termwise integration gives

$$\begin{aligned} W_Q &= -\int_r^\infty F_Q(s)\, ds \\ &= \boxed{k\left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right]}. \end{aligned}$$

This work can be negative when repulsion supplies energy during removal. To see the energy content of (2.8), separate the potential at q into the pre-existing charge and the polarization:

$$\Phi_Q(r) = \frac{kQ}{r}, \qquad \Phi_{\text{pol}}(r) = kq\left[\frac{a}{r^2} - \frac{a}{r^2 - a^2}\right].$$

Relative to the infinitely separated system, the energy is

$$U_Q = q\Phi_Q + \frac{1}{2}q\Phi_{\rm pol} = -W_Q.$$

The pre-existing sphere charge Q has its full interaction factor; only the source-induced redistribution has the half factor. The constant self-energy $kQ^2/(2a)$ of the isolated charged sphere cancels between the initial and final configurations.

2.6 The electrostatic problem of a point charge q outside an isolated, charged conducting sphere is equivalent to that of three charges, the original and two others, one located at the center of the sphere and another ("the image charge") inside the now imaginary sphere, on the line joining the center and the original charge.

If the point charge and sphere are replaced by two conducting spheres of radii r_a and r_b, carrying total charges Q_a and Q_b, respectively, with centers separated by a distance $d > r_a + r_b$, there is an equivalence with an infinite set of charges within each sphere, one at the center and a set of images along the line joining the centers. The charges and their locations can be determined iteratively, starting with a charge $q_a(1)$ at the center of the first sphere and $q_b(1)$ correspondingly for the second sphere. The charge $q_b(1)$ has its image $q_a(2)$ within the first sphere and vice versa. Then the image charge within the first sphere induces another image within the second sphere, and so on. The sum of all the charges within each sphere must be scaled to be equal to Q_a or Q_b.

The electrostatic potential outside the spheres, the force between the spheres, etc. can be found by summing the contributions from all the charges.

(a) Show that the charges and their positions are determined iteratively by the relations,

$$\begin{aligned} q_a(j) &= -r_a q_b(j-1)/d_b(j-1), \qquad & x_a(j) &= r_a^2/d_b(j-1), \qquad & d_a(j) &= d - x_a(j) \\ q_b(j) &= -r_b q_a(j-1)/d_a(j-1), \qquad & x_b(j) &= r_b^2/d_a(j-1), \qquad & d_b(j) &= d - x_b(j) \end{aligned}$$

for $j = 2, 3, 4, \ldots$, with $d_a(1) = d_b(1) = d$, and $x_a(1) = x_b(1) = 0$.

(b) Find the image charges and their locations as well as the potentials on the spheres and force between them by means of a suitable computer program. [In computing the potential on each sphere, evaluate it in different places: e.g., in the equatorial plane and at the pole opposite the other sphere. This permits a check on the equipotential of the conductor and on the accuracy of computation.]

(c) As an example, show that for two equally charged spheres of the same radius R, the force between them when almost in contact is 0.6149 times the value that would be obtained if all the charge on each sphere were concentrated at its center. Show numerically and by explicit summation of the series that the capacitance of two identical conducting spheres in contact is $C/4\pi\epsilon_0 R = 1.3863\cdots$ $[= \ln 4]$.

Reference: J. A. Soules, *Am. J. Phys.* **58**, 1195 (1990).

Solution.

Use an axial coordinate X with centers at $0, d$. Distances x_a and x_b are measured from each center *toward* the other, so charges in sphere b have global coordinates $d - x_b$. **(a)** A charge $q_b(j-1)$ is a distance $d_b(j-1) = d - x_b(j-1) > r_a$ from center a. Sphere inversion therefore requires an image

$$q_a(j) = -\frac{r_a}{d_b(j-1)}q_b(j-1), \qquad x_a(j) = \frac{r_a^2}{d_b(j-1)}.$$

Interchanging a, b gives the other two recurrences in the question. Every external image is paired with its cancelling image in the other sphere. On sphere a these pairs have zero potential, leaving only the central charge's constant $kq_a(1)/r_a$; similarly on b. The two initial charges cannot in general be scaled independently after the iteration, because each subsequent chain crosses between the spheres.

(b) A reproducible implementation is the function `two_spheres` in the supplementary script `check_ch02.py`. It first constructs two unit-seed solutions, with initial central charges $(1, 0)$ and $(0, 1)$. If M_{ij} is the sum of the images in sphere i produced by seed j, solve

$$\begin{pmatrix} M_{aa} & M_{ab} \\ M_{ba} & M_{bb} \end{pmatrix} \begin{pmatrix} q_a(1) \\ q_b(1) \end{pmatrix} = \begin{pmatrix} Q_a \\ Q_b \end{pmatrix}.$$

In particular, writing $\Delta = M_{aa}M_{bb} - M_{ab}M_{ba}$,

$$q_a(1) = \frac{Q_a M_{bb} - Q_b M_{ab}}{\Delta}, \qquad q_b(1) = \frac{Q_b M_{aa} - Q_a M_{ba}}{\Delta}.$$

If the two stored unit-seed charges at a given image position are $c_{i,a}(j), c_{i,b}(j)$, its final charge is $q_i(j) = c_{i,a}(j)q_a(1) + c_{i,b}(j)q_b(1)$. Linearity then gives every image charge at the already computed positions. For a finite truncation the exterior potential and the outward force on sphere b are evaluated as

$$\Phi(X, \rho) = k \sum_j \left[\frac{q_a(j)}{\sqrt{(X - x_a(j))^2 + \rho^2}} + \frac{q_b(j)}{\sqrt{(X - d + x_b(j))^2 + \rho^2}} \right],$$

$$F_b = k \sum_{i,j} \frac{q_a(i) q_b(j)}{[d - x_a(i) - x_b(j)]^2}.$$

The latter follows by enclosing one sphere in a Maxwell-stress surface: its self-field gives zero net force, and the interaction with the other sphere can be replaced by the two sets of images. For a strictly positive gap, every successive image magnitude is bounded by the previous magnitude times

$$\eta = \max\left(\frac{r_a}{d - r_b}, \frac{r_b}{d - r_a} \right) < 1,$$

because each image remains inside its sphere. If c_N bounds the last retained charge in either unit-seed chain, each remaining tail is at most $c_N\eta/(1 - \eta)$. After combining the seeds this is multiplied by $|q_a(1)| + |q_b(1)|$. The code increases the image count until the unit-seed bound meets the requested tolerance and reports the resulting physical bound. Its checks repeat the calculation at a tighter tolerance, compare the resulting force and potentials, and compare each surface potential at the facing pole, rear pole, and equator with $V_i = kq_i(1)/r_i$. It exposes all charges and positions in the returned arrays and can write them to a CSV file using `--images PATH`. For example, $r_a = r_b = R$, $d = 3R$, $Q_a = Q_b = Q$ gives

$$\frac{4\pi\epsilon_0 RV}{Q} = 1.320647\ldots, \qquad \frac{F}{Q^2/(4\pi\epsilon_0 d^2)} = 0.849932\ldots.$$

The script prints higher-precision values and includes an unequal radius, unequal charge example to check the two-seed construction.

(c) For equal radii and equal charges put $d = 2R\cosh\alpha$, $\alpha > 0$. The recurrences are solved by

$$x_j = R\frac{\sinh[(j-1)\alpha]}{\sinh(j\alpha)}, \qquad q_j = (-1)^{j-1} q_1 \frac{\sinh\alpha}{\sinh(j\alpha)}.$$

For example, $d-x_{j-1} = R\sinh(j\alpha)/\sinh[(j-1)\alpha]$ by $2\cosh\alpha\sinh[(j-1)\alpha]-\sinh[(j-2)\alpha] = \sinh(j\alpha)$, which verifies both recurrences. Define

$$S(\alpha) = \frac{Q}{q_1} = \sinh\alpha \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{\sinh(j\alpha)}.$$

Then $V = kQ/(RS)$ and the capacitance of the pair at a common potential is

$$\frac{C}{4\pi\epsilon_0 R} = 2S(\alpha).$$

At contact the images become $x_j = R(j-1)/j$, $q_j = (-1)^{j-1}q_1/j$, and therefore

$$Q = q_1\left(1 - \frac{1}{2} + \frac{1}{3} - \cdots\right) = q_1 \ln 2, \qquad \boxed{\frac{C}{4\pi\epsilon_0 R} = 2\ln 2 = \ln 4}.$$

To differentiate near contact without differentiating a divergent termwise power expansion, first sum a geometric series:

$$S = 2\sinh\alpha \sum_{m=0}^{\infty} \frac{1}{e^{(2m+1)\alpha} + 1}.$$

For $f(t) = (e^t+1)^{-1}$, midpoint summation gives $h\sum_{m\geq 0} f((m+\frac{1}{2})h) = \int_0^\infty f(t)\,dt + h^2 f'(0)/24 + O(h^4)$. This follows by integrating the Taylor expansion on each interval: its midpoint error is $h^3 f''/24$, whose sum is $h^2\int_0^\infty f''/24 = -h^2 f'(0)/24$. With $h = 2\alpha$, $\int_0^\infty f = \ln 2$ and $f'(0) = -1/4$, we find

$$S = \ln 2 + \alpha^2\left(\frac{\ln 2}{6} - \frac{1}{24}\right) + O(\alpha^4).$$

At fixed charge on each sphere the total energy is $U = \frac{1}{2}(QV + QV) = kQ^2/(RS)$. Hence

$$F = -\frac{dU}{dd} = \frac{kQ^2}{RS^2}\frac{dS}{dd}, \qquad d = 2R + R\alpha^2 + O(\alpha^4).$$

Dividing by the center-charge force kQ^2/d^2 gives the limit

$$\boxed{\lim_{d\downarrow 2R} \frac{F}{kQ^2/d^2} = \frac{4\ln 2 - 1}{6(\ln 2)^2} = 0.6149018638\ldots.}$$

The printed value 0.6189 is inconsistent with this exact limit; the corrected value is 0.6149. For example, at $(d-2R)/R = 10^{-4}$ the image-force ratio is 0.6149340086, while the capacitance ratio is 1.3863091325, approaching $\ln 4 = 1.3862943611$. The supplementary code checks the image-force sum against both a Maxwell-stress integral and the energy derivative, and approaches both contact limits from positive gaps; it does not rely on a finite alternating sum at exact contact.

2.7 Consider a potential problem in the half-space defined by $z \geq 0$, with Dirichlet boundary conditions on the plane $z = 0$ (and at infinity).

(a) Write down the appropriate Green function $G(\mathbf{x}, \mathbf{x}')$.

(b) If the potential on the plane $z = 0$ is specified to be $\Phi = V$ *inside* a circle of radius a centered at the origin, and $\Phi = 0$ outside that circle, find an integral expression for the potential at the point P specified in terms of cylindrical coordinates (ρ, ϕ, z).

(c) Show that, along the axis of the circle ($\rho = 0$), the potential is given by

$$\Phi = V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right)$$

(d) Show that at large distances ($\rho^2 + z^2 \gg a^2$) the potential can be expanded in a power series in $(\rho^2 + z^2)^{-1}$, and that the leading terms are

$$\Phi = \frac{Va^2}{2}\frac{z}{(\rho^2 + z^2)^{3/2}}\left[1 - \frac{3a^2}{4(\rho^2 + z^2)} + \frac{5(3\rho^2 a^2 + a^4)}{8(\rho^2 + z^2)^2} + \cdots\right]$$

Verify that the results of parts c and d are consistent with each other in their common range of validity.

Solution.

Use the convention $\nabla'^2 G = -4\pi\delta^3(\mathbf{x} - \mathbf{x}')$. **(a)** Reflect the source in $z' = 0$:

$$\boxed{G(\mathbf{x}, \mathbf{x}') = \frac{1}{\sqrt{|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2 + (z - z')^2}} - \frac{1}{\sqrt{|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2 + (z + z')^2}}.}$$

It vanishes on the plane and at infinity. The outward normal of the upper half-space is $-\hat{\mathbf{z}}'$, giving $\partial_{n'} G|_0 = -2z/(|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2 + z^2)^{3/2}$. **(b)** The Dirichlet representation $\Phi = -(4\pi)^{-1}\int \Phi \partial_{n'} G \, dS'$ therefore yields

$$\boxed{\Phi(\rho, z) = \frac{Vz}{2\pi}\int_0^a s\, ds \int_0^{2\pi} \frac{d\varphi}{(\rho^2 + s^2 - 2\rho s\cos\varphi + z^2)^{3/2}}.}$$

The result is independent of the observation azimuth. **(c)** At $\rho = 0$ the angular integral is 2π, so

$$\Phi(0, z) = Vz\left[-\frac{1}{\sqrt{s^2 + z^2}}\right]_0^a = \boxed{V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right)}, \qquad z > 0.$$

(d) Put $L^2 = \rho^2 + z^2$ and $u = (s^2 - 2\rho s\cos\varphi)/L^2$. The binomial expansion is $(1 + u)^{-3/2} = 1 - 3u/2 + 15u^2/8 - 35u^3/16 + \cdots$. A disk average, $\langle f\rangle = (\pi a^2)^{-1}\int s\, ds\, d\varphi f$, gives

$$\langle u\rangle = \frac{a^2}{2L^2}, \qquad \langle u^2\rangle = \frac{a^4/3 + \rho^2 a^2}{L^4}.$$

Here $\langle s^2\rangle = a^2/2$, $\langle s^4\rangle = a^4/3$, and $\langle s^2\cos^2\varphi\rangle = a^2/4$. Substitution produces the displayed denominator expansion:

$$\Phi = \frac{Va^2 z}{2L^3}\left[1 - \frac{3a^2}{4L^2} + \frac{5(3\rho^2 a^2 + a^4)}{8L^4} + \cdots\right].$$

For clarity, terms grouped by *uniform far-distance order* require some further terms from the binomial series, since ρ can be comparable to L. With $s_\theta = \rho/L$, that ordering is

$$\Phi = \frac{Va^2 z}{2L^3}\left\{1 + \frac{a^2}{L^2}\left(-\frac{3}{4} + \frac{15}{8}s_\theta^2\right) + \frac{a^4}{L^4}\left(\frac{5}{8} - \frac{35}{8}s_\theta^2 + \frac{315}{64}s_\theta^4\right) + O(a^6/L^6)\right\}.$$

For example, the additional contributions at this order follow from $\langle u^3\rangle = (a^6/4 + 2\rho^2 a^4)/L^6$ and the $2\rho^4 a^4/L^8$ part of $\langle u^4\rangle$. On the axis the original bracket becomes $1 - 3a^2/(4z^2) + 5a^4/(8z^4) + \cdots$, which agrees term by term with $V[1 - (1 + a^2/z^2)^{-1/2}]$ from part (c).

2.8 A two-dimensional potential problem is defined by two straight parallel line charges separated by a distance R with equal and opposite linear charge densities λ and $-\lambda$.

(a) Show by direct construction that the surface of constant potential V is a circular cylinder (circle in the transverse dimensions) and find the coordinates of the axis of the cylinder and its radius in terms of R, λ, and V.

(b) Use the results of part a to show that the capacitance per unit length C of two right-circular cylindrical conductors, with radii a and b, separated by a distance $d > a + b$, is

$$C = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\dfrac{d^2 - a^2 - b^2}{2ab}\right)}$$

(c) Verify that the result for C agrees with the answer in Problem 1.7 in the appropriate limit and determine the next nonvanishing order correction in powers of a/d and b/d.

(d) Repeat the calculation of the capacitance per unit length for two cylinders inside each other ($d < |b - a|$). Check the result for concentric cylinders ($d = 0$).

Solution.

(a) Put $+\lambda$ at $(c, 0)$ and $-\lambda$ at $(-c, 0)$, where $c = R/2$, and choose zero potential at infinity. Then

$$\Phi = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{(x + c)^2 + y^2}{(x - c)^2 + y^2}.$$

Set $t = 2\pi\epsilon_0 V/\lambda$. The equipotential equation is $(x + c)^2 + y^2 = e^{2t}[(x - c)^2 + y^2]$. Dividing by $1 - e^{2t}$ and completing the square yields

$$\boxed{(x - c\coth t)^2 + y^2 = \frac{c^2}{\sinh^2 t}.}$$

Thus the cylinder axis passes through $(c\coth t, 0)$ and its radius is $c/|\sinh t|$. The exceptional level $V = 0$ is the plane $x = 0$ (the infinite-radius limit of the circles). **(b)** For two disjoint cylinders choose their surfaces to be $t = t_a > 0$ and $t = -t_b < 0$. Then

$$a = c\,\mathrm{csch}\, t_a, \quad b = c\,\mathrm{csch}\, t_b, \quad d = c(\coth t_a + \coth t_b).$$

Squaring the last relation and using $\coth^2 t - \mathrm{csch}^2 t = 1$ gives

$$\frac{d^2 - a^2 - b^2}{2ab} = \sinh t_a \sinh t_b + \cosh t_a \cosh t_b = \cosh(t_a + t_b).$$

Each cylinder encloses one of the two line charges, so Gauss's law gives charges per unit length $\pm\lambda$. Their voltage difference is $\Delta V = \lambda(t_a + t_b)/(2\pi\epsilon_0)$; therefore

$$\boxed{C = \frac{\lambda}{\Delta V} = \frac{2\pi\epsilon_0}{\cosh^{-1}[(d^2 - a^2 - b^2)/(2ab)]}.}$$

(c) Write $L = \ln[d^2/(ab)]$ and $S = a^2 + b^2$. For large X, $\cosh^{-1} X = \ln(2X) - 1/(4X^2) + O(X^{-4})$; substitution of $X = (d^2 - S)/(2ab)$ gives

$$\cosh^{-1} X = L - \frac{S}{d^2} - \frac{S^2/2 + a^2b^2}{d^4} + O(d^{-6}).$$

Thus the leading result is $C = \pi\epsilon_0/\ln[d/\sqrt{ab}]$, as in Problem 1.7, and the first nonzero relative correction is

$$\boxed{C = \frac{2\pi\epsilon_0}{L}\left[1 + \frac{a^2+b^2}{d^2 L} + O(d^{-4})\right].}$$

The length scales a, b are held fixed in this asymptotic notation. **(d)** For nested cylinders take two circles on the same branch, with $b > a$ and $t_a > t_b > 0$; then $d = c(\coth t_b - \coth t_a)$. In this case

$$a^2 + b^2 - d^2 = c^2[\operatorname{csch}^2 t_a + \operatorname{csch}^2 t_b - (\coth t_b - \coth t_a)^2] = 2c^2(\coth t_a \coth t_b - 1).$$

Dividing by $2ab = 2c^2/(\sinh t_a \sinh t_b)$ gives $\cosh t_a \cosh t_b - \sinh t_a \sinh t_b$. The voltage difference is now $\Delta V = \lambda(t_a - t_b)/(2\pi\epsilon_0)$, and hence

$$\cosh(t_a - t_b) = \frac{a^2+b^2-d^2}{2ab}, \qquad \boxed{C = \frac{2\pi\epsilon_0}{\cosh^{-1}[(a^2+b^2-d^2)/(2ab)]}}.$$

The formula is symmetric if a, b are interchanged. At $d = 0$, $(a^2+b^2)/(2ab) = \cosh\ln(b/a)$, giving the usual $C = 2\pi\epsilon_0/\ln(b/a)$ for concentric cylinders.

2.9 An insulated, spherical, conducting shell of radius a is in a uniform electric field E_0. If the sphere is cut into two hemispheres by a plane perpendicular to the field, find the force required to prevent the hemispheres from separating

(a) if the shell is uncharged;

(b) if the total charge on the shell is Q.

Solution.

Take the imposed field to be $E_0\hat{\mathbf{z}}$. With the hemispheres initially forming a sphere (the cut width tending to zero), the exterior and interior potentials are

$$\Phi_{\text{out}} = \frac{Q}{4\pi\epsilon_0 r} - E_0\left(r - \frac{a^3}{r^2}\right)\cos\theta, \qquad \Phi_{\text{in}} = \frac{Q}{4\pi\epsilon_0 a}.$$

The boundary is equipotential, the cavity field is zero, and

$$\sigma(\theta) = \frac{Q}{4\pi a^2} + 3\epsilon_0 E_0 \cos\theta =: s + 3\epsilon_0 E_0 \cos\theta.$$

The outward electrical pressure is $\sigma^2/(2\epsilon_0)$. For the northern hemisphere its z component integrates to

$$\begin{aligned} F_{N,z} &= \frac{\pi a^2}{\epsilon_0}\int_0^1 (s + 3\epsilon_0 E_0 u)^2 u\, du \\ &= \frac{\pi a^2}{\epsilon_0}\left(\frac{s^2}{2} + 2s\epsilon_0 E_0 + \frac{9}{4}\epsilon_0^2 E_0^2\right) \\ &= \frac{Q^2}{32\pi\epsilon_0 a^2} + \frac{QE_0}{2} + \frac{9}{4}\pi\epsilon_0 a^2 E_0^2. \end{aligned}$$

The southern hemisphere similarly has

$$F_{S,z} = \frac{QE_0}{2} - \left(\frac{Q^2}{32\pi\epsilon_0 a^2} + \frac{9}{4}\pi\epsilon_0 a^2 E_0^2\right).$$

Their sum is QE_0, as required for the force on the complete sphere. **(a)** When $Q = 0$, equal and opposite inward holding forces of magnitude

$$\boxed{F_{\text{hold}} = \frac{9}{4}\pi\epsilon_0 a^2 E_0^2}$$

prevent separation. **(b)** The force tending to separate the two equal hemispheres is half their force difference; the common term $QE_0/2$ produces translation. Thus their required mutual restraining force is

$$\boxed{F_{\text{separate}} = \frac{Q^2}{32\pi\epsilon_0 a^2} + \frac{9}{4}\pi\epsilon_0 a^2 E_0^2.}$$

If "holding" also means keeping each hemisphere at rest in the laboratory, the support forces must instead be $-F_{N,z}\widehat{\mathbf{z}}$ and $-F_{S,z}\widehat{\mathbf{z}}$ individually. Stating both forces distinguishes the separation load from the additional support required for the net force QE_0 on a charged shell.

2.10 A large parallel plate capacitor is made up of two plane conducting sheets with separation D, one of which has a small hemispherical boss of radius a on its inner surface ($D \gg a$). The conductor with the boss is kept at zero potential, and the other conductor is at a potential such that far from the boss the electric field between the plates is E_0.

(a) Calculate the surface-charge densities at an arbitrary point on the plane and on the boss, and sketch their behavior as a function of distance (or angle).

(b) Show that the total charge on the boss has the magnitude $3\pi\epsilon_0 E_0 a^2$.

(c) If, instead of the other conducting sheet at a different potential, a point charge q is placed directly above the hemispherical boss at a distance d from its center, show that the charge induced on the boss is

$$q' = -q\left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}}\right]$$

Solution.

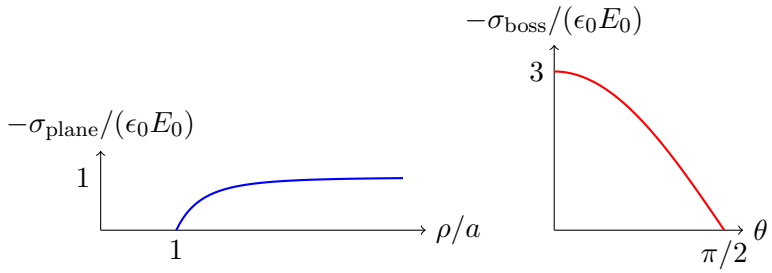
Take the grounded plane to be $z = 0$ and the boss to occupy $r = a$, $0 \le \theta \le \pi/2$. Choose the distant field to be $-E_0\widehat{\mathbf{z}}$, with $E_0 > 0$; reversing it reverses all surface-charge signs. In the approximation $a/D \to 0$ the other plate is represented by a uniform field. **(a)** The potential

$$\Phi = E_0\left(r - \frac{a^3}{r^2}\right)\cos\theta = E_0 z\left(1 - \frac{a^3}{r^3}\right)$$

vanishes both on the plane and on the boss and tends to $E_0 z$. Normal differentiation gives

$$\boxed{\sigma_{\text{plane}}(\rho) = -\epsilon_0 E_0\left(1 - \frac{a^3}{\rho^3}\right), \quad \rho \ge a; \qquad \sigma_{\text{boss}}(\theta) = -3\epsilon_0 E_0 \cos\theta.}$$

The plane density rises in magnitude from zero at the rim to the uniform value; on the boss its magnitude rises from zero at the rim to three times the uniform value at the top.



(b) Direct integration over the curved area gives

$$Q_{\text{boss}} = -3\epsilon_0 E_0 (2\pi a^2) \int_0^{\pi/2} \cos\theta \sin\theta \, d\theta = -3\pi\epsilon_0 E_0 a^2,$$

with the stated magnitude. **(c)** For a point charge at $d\widehat{\mathbf{z}}$, $d > a$, first reflect it in the plane, giving $-q$ at $-d\widehat{\mathbf{z}}$. Their sphere images are $-qa/d$ at $a^2\widehat{\mathbf{z}}/d$ and $+qa/d$ at $-a^2\widehat{\mathbf{z}}/d$. Each exterior/image pair vanishes on the sphere; the full set is odd under $z \mapsto -z$ and therefore vanishes on the plane as well. With $u = \cos\theta$, normal differentiation on the upper sphere gives

$$\sigma(u) = -\frac{q(d^2 - a^2)}{4\pi a} \left[\frac{1}{(d^2 + a^2 - 2adu)^{3/2}} - \frac{1}{(d^2 + a^2 + 2adu)^{3/2}} \right].$$

The elementary integral of the bracket from 0 to 1 is

$$\frac{1}{ad} \left(\frac{1}{d-a} + \frac{1}{d+a} - \frac{2}{\sqrt{d^2 + a^2}} \right).$$

Multiplying by $-qa(d^2 - a^2)/2$ yields

$$\boxed{Q_{\text{boss}} = -q \left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}} \right].}$$

It tends to $-q$ as the source reaches the top and tends to zero as $d \to \infty$, providing two useful checks.

2.11 A line charge with linear charge density τ is placed parallel to, and a distance R away from, the axis of a conducting cylinder of radius b held at fixed voltage such that the potential vanishes at infinity. Find

(a) the magnitude and position of the image charge(s);

(b) the potential at any point (expressed in polar coordinates with the origin at the axis of the cylinder and the direction from the origin to the line charge as the x axis), including the asymptotic form far from the cylinder;

(c) the induced surface-charge density, and plot it as a function of angle for $R/b = 2, 4$ in units of $\tau/2\pi b$;

(d) the force per unit length on the line charge.

Solution.

Assume $R > b$ and put the real line at $(R, 0)$ in the transverse plane. Let $c = b^2/R$. **(a)** The image is a single line of density $-\tau$ at $(c, 0)$. For a point on the cylinder,

$$R^2 + b^2 - 2Rb\cos\phi = \frac{R^2}{b^2}(b^2 + c^2 - 2bc\cos\phi).$$

Thus the ratio of the real-source and image distances is constant, which is precisely the equipotential condition. The total real plus image line charge is zero, permitting a finite potential at infinity. **(b)** With that limiting potential set to zero,

$$\boxed{\Phi(\rho,\phi) = \frac{\tau}{4\pi\epsilon_0}\ln\frac{\rho^2 + c^2 - 2\rho c\cos\phi}{\rho^2 + R^2 - 2\rho R\cos\phi}, \qquad \rho \geq b.}$$

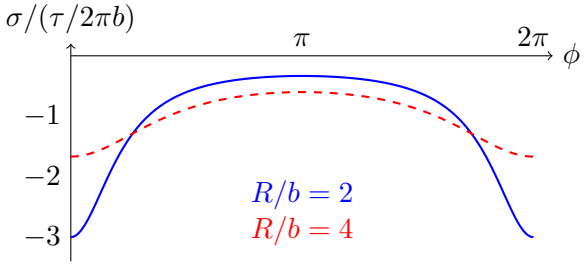
The conductor's voltage is consequently $V_b = \tau\ln(b/R)/(2\pi\epsilon_0)$. It is determined by the requirement at infinity: an additional central line charge could change V_b, but would produce an unbounded logarithmic potential at infinity. Expanding $\ln(1+u) = u + O(u^2)$ in the displayed solution gives

$$\boxed{\Phi \sim \frac{\tau(R - b^2/R)}{2\pi\epsilon_0\rho}\cos\phi.}$$

This is a two-dimensional dipole with moment per unit length $\tau(R - b^2/R)\hat{\mathbf{x}}$. **(c)** On the outside of the metal, $\sigma = -\epsilon_0\partial_\rho\Phi|_b$. Using the distance identity above to combine the two derivatives yields

$$\boxed{\sigma(\phi) = -\frac{\tau}{2\pi b}\frac{R^2 - b^2}{R^2 + b^2 - 2Rb\cos\phi}.}$$

The integral of $(1-t^2)/(1-2t\cos\phi+t^2)$ over a full circle is 2π for $|t| < 1$ (its nonconstant Fourier terms integrate to zero). With $t = b/R$ this checks that $b\int\sigma\,d\phi = -\tau$.



(d) The image field at the real line is $-\tau\hat{\mathbf{x}}/[2\pi\epsilon_0(R-c)]$. Thus

$$\boxed{\frac{\mathbf{F}}{L} = -\frac{\tau^2 R}{2\pi\epsilon_0(R^2 - b^2)}\hat{\mathbf{x}},}$$

an attraction for either sign of τ.

2.12 Starting with the series solution (2.71) for the two-dimensional potential problem with the potential specified on the surface of a cylinder of radius b, evaluate the coefficients formally, substitute them into the series, and sum it to obtain the potential *inside* the cylinder in the form of Poisson's integral:

$$\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b,\phi')\,\frac{b^2-\rho^2}{b^2+\rho^2-2b\rho\cos(\phi'-\phi)}\,d\phi'$$

What modification is necessary if the potential is desired in the region of space bounded by the cylinder and infinity?

Solution.

Write $f(\phi) = \Phi(b,\phi)$. The nonsingular interior specialization of (2.71) is

$$\Phi = A_0 + \sum_{n=1}^{\infty}\left(\frac{\rho}{b}\right)^n [A_n\cos n\phi + B_n \sin n\phi].$$

Orthogonality over $[0, 2\pi]$ determines

$$A_0 = \frac{1}{2\pi}\int_0^{2\pi} f(\phi')\,d\phi', \quad A_n = \frac{1}{\pi}\int_0^{2\pi} f(\phi')\cos n\phi'\,d\phi', \quad B_n = \frac{1}{\pi}\int_0^{2\pi} f(\phi')\sin n\phi'\,d\phi'.$$

Substitution and the cosine subtraction identity give, with $t = \rho/b < 1$,

$$\Phi = \frac{1}{2\pi}\int_0^{2\pi} f(\phi')\left[1 + 2\sum_{n=1}^{\infty} t^n \cos n(\phi-\phi')\right] d\phi'.$$

The series is absolutely convergent inside the circle. Summing the two geometric series explicitly,

$$1 + \frac{te^{i\Delta}}{1-te^{i\Delta}} + \frac{te^{-i\Delta}}{1-te^{-i\Delta}} = \frac{1-t^2}{1-2t\cos\Delta+t^2}.$$

Therefore

$$\boxed{\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} f(\phi')\frac{b^2-\rho^2}{b^2+\rho^2-2b\rho\cos(\phi-\phi')}\,d\phi'.}$$

For a solution bounded at infinity replace every $(\rho/b)^n$ by $(b/\rho)^n$. Equivalently the exterior kernel has numerator $\rho^2 - b^2$ and the same denominator. Its limit at infinity is the boundary average A_0; a zero potential at infinity is possible with a bounded solution only if $A_0 = 0$. If logarithmic growth is allowed, add $B_0\ln(\rho/b)$, which vanishes on the cylinder but corresponds to a net line charge. The boundary values alone do not fix B_0 without an exterior condition.

2.13

(a) Two halves of a long hollow conducting cylinder of inner radius b are separated by small lengthwise gaps on each side, and are kept at different potentials V_1 and V_2. Show that the potential inside is given by

$$\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{V_1-V_2}{\pi}\tan^{-1}\left(\frac{2b\rho}{b^2-\rho^2}\cos\phi\right)$$

where ϕ is measured from a plane perpendicular to the plane through the gap.

(b) Calculate the surface-charge density on each half of the cylinder.

Solution.

Choose the V_1 half to be $\cos\phi > 0$, and the V_2 half to be $\cos\phi < 0$. Define $\overline{V} = (V_1 + V_2)/2$ and $\Delta V = V_1 - V_2$. **(a)** The boundary function is $f(\phi) = \overline{V} + (\Delta V/2)\,\mathrm{sgn}(\cos\phi)$. Its sine coefficients vanish. Its cosine coefficients are zero for even n, and for $n = 2j+1$ are

$$A_{2j+1} = \frac{\Delta V}{\pi}\int_{-\pi/2}^{\pi/2} \cos[(2j+1)\phi]\,d\phi = \frac{2\Delta V}{\pi}\frac{(-1)^j}{2j+1}.$$

Thus, for $t = \rho/b < 1$,

$$\Phi - \overline{V} = \frac{2\Delta V}{\pi}\,\mathrm{Re}\sum_{j=0}^{\infty}\frac{(-1)^j(te^{i\phi})^{2j+1}}{2j+1} = \frac{2\Delta V}{\pi}\,\mathrm{Re}\tan^{-1}(te^{i\phi}).$$

From $\tan^{-1} w = (2i)^{-1}\ln[(1+iw)/(1-iw)]$, or the tangent addition formula applied to $w, \bar{w}$, one obtains

$$2\,\mathrm{Re}\tan^{-1}(te^{i\phi}) = \tan^{-1}\frac{2t\cos\phi}{1-t^2}.$$

Both sides vanish at $t = 0$ and remain on the same continuous branch for $t < 1$. Consequently

$$\boxed{\Phi = \overline{V} + \frac{\Delta V}{\pi}\tan^{-1}\frac{2b\rho\cos\phi}{b^2-\rho^2}.}$$

As $\rho \uparrow b$ the inverse tangent tends to $(\pi/2)\,\mathrm{sgn}(\cos\phi)$, as required. **(b)** The normal from the conductor into the cavity is $-\widehat{\boldsymbol{\rho}}$, so $\sigma_{\mathrm{inner}} = \epsilon_0\partial_\rho\Phi|_{b^-}$. In terms of t the derivative is

$$\partial_\rho\Phi = \frac{\Delta V}{\pi b}\frac{2(1+t^2)\cos\phi}{(1-t^2)^2 + 4t^2\cos^2\phi}.$$

Taking $t \to 1$ away from the gaps gives

$$\boxed{\sigma_{\mathrm{inner}}(\phi) = \frac{\epsilon_0(V_1 - V_2)}{\pi b\cos\phi}.}$$

For $V_1 > V_2$ this is positive on the V_1 half and negative on the V_2 half. It diverges as inverse distance to an ideal zero-width gap; a finite gap cuts off the logarithmic divergence of the charge on either half. The data for the cavity do not determine the charge on the exterior faces without exterior geometry and boundary data.

2.14 A variant of the preceding two-dimensional problem is a long hollow conducting cylinder of radius b that is divided into equal quarters, alternate segments being held at potential $+V$ and $-V$.

(a) Solve by means of the series solution (2.71) and show that the potential inside the cylinder is

$$\Phi(\rho,\phi) = \frac{4V}{\pi}\sum_{n=0}^{\infty}\left(\frac{\rho}{b}\right)^{4n+2}\frac{\sin[(4n+2)\phi]}{2n+1}$$

(b) Sum the series and show that

$$\Phi(\rho,\phi) = \frac{2V}{\pi}\tan^{-1}\left(\frac{2\rho^2 b^2\sin 2\phi}{b^4-\rho^4}\right)$$

(c) Sketch the field lines and equipotentials.

Solution.

Let the sectors $0 < \phi < \pi/2$ and $\pi < \phi < 3\pi/2$ be at $+V$; the other two are at $-V$. **(a)** The boundary function is $V\,\mathrm{sgn}(\sin 2\phi)$. It is odd in ϕ, invariant under $\phi \mapsto \phi + \pi$, and changes sign under $\phi \mapsto \phi + \pi/2$. These symmetries leave only sine modes with $m = 4n + 2$. Direct integration gives

$$B_m = \frac{1}{\pi}\int_0^{2\pi} V\,\mathrm{sgn}(\sin 2\phi)\sin m\phi\,d\phi = \frac{8V}{\pi m}, \quad m = 4n+2.$$

The regular radial solution for mode m is $(\rho/b)^m$, whence

$$\boxed{\Phi = \frac{4V}{\pi}\sum_{n=0}^{\infty}\frac{(\rho/b)^{4n+2}\sin[(4n+2)\phi]}{2n+1}.}$$

(b) Put $\zeta = (x + iy)/b$. The elementary identity $\operatorname{artanh} w = \sum_{n\ge0} w^{2n+1}/(2n+1) = \frac{1}{2}\ln[(1+w)/(1-w)]$ gives

$$\Phi = \frac{4V}{\pi}\,\mathrm{Im}\operatorname{artanh}(\zeta^2) = \frac{2V}{\pi}\arg\frac{1+\zeta^2}{1-\zeta^2}.$$

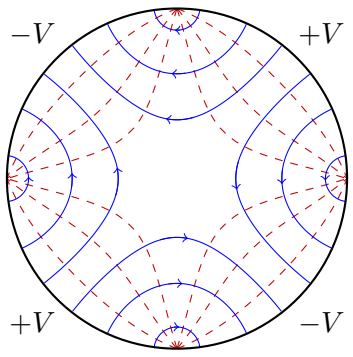
Multiplication by $1-\bar\zeta^2$ makes the real and imaginary parts of this quotient proportional to $1-|\zeta|^4$ and $2\,\mathrm{Im}\,\zeta^2$. The real part is positive for $|\zeta| < 1$, fixing the inverse tangent branch. Hence

$$\boxed{\Phi = \frac{2V}{\pi}\tan^{-1}\frac{2\rho^2 b^2\sin 2\phi}{b^4-\rho^4}.}$$

(c) An exact description of the two orthogonal curve families is obtained by writing $\operatorname{artanh}\zeta^2 = u + iv$. Equipotentials have $v = \pi\Phi/(4V)$ constant; field lines have u constant, with arrows toward decreasing v. In Cartesian form these conditions are

$$\frac{4b^2xy}{b^4-(x^2+y^2)^2} = \tan\frac{\pi\Phi}{2V}, \qquad \left|\frac{1+\zeta^2}{1-\zeta^2}\right| = e^{2u},$$

respectively. The supplementary plot is generated parametrically by $\zeta = \pm\sqrt{\tanh(u+iv)}$, so its curves follow the summed solution rather than a truncated Fourier series.



field lines
dashed: equipotentials

2.15

(a) Show that the Green function $G(x, y; x', y')$ appropriate for Dirichlet boundary conditions for a square two-dimensional region, $0 \le x \le 1$, $0 \le y \le 1$, has an expansion

$$G(x, y; x', y') = 2 \sum_{n=1}^{\infty} g_n(y, y') \sin(n\pi x) \sin(n\pi x')$$

where $g_n(y, y')$ satisfies

$$\left(\frac{\partial^2}{\partial y'^2} - n^2\pi^2 \right) g_n(y, y') = -4\pi\delta(y' - y) \quad \text{and} \quad g_n(y, 0) = g_n(y, 1) = 0$$

(b) Taking for $g_n(y, y')$ appropriate linear combinations of $\sinh(n\pi y')$ and $\cosh(n\pi y')$ in the two regions, $y' < y$ and $y' > y$, in accord with the boundary conditions and the discontinuity in slope required by the source delta function, show that the explicit form of G is

$$G(x, y; x', y') = 8 \sum_{n=1}^{\infty} \frac{1}{n \sinh(n\pi)} \sin(n\pi x) \sin(n\pi x') \sinh(n\pi y_<) \sinh[n\pi(1 - y_>)]$$

where $y_<$ ($y_>$) is the smaller (larger) of y and y'.

Solution.

(a) The sine basis on $0 < x' < 1$ obeys $\delta(x' - x) = 2 \sum_{n \ge 1} \sin(n\pi x) \sin(n\pi x')$ in the distributional sense. Expand

$$G = 2 \sum_{n \ge 1} g_n(y, y') \sin(n\pi x) \sin(n\pi x').$$

The factors already impose $G = 0$ at $x' = 0, 1$. Applying the primed Laplacian and comparing with $-4\pi\delta(x' - x)\delta(y' - y)$ gives, for each independent mode,

$$\left(\frac{d^2}{dy'^2} - n^2\pi^2 \right) g_n = -4\pi\delta(y' - y), \qquad g_n(y, 0) = g_n(y, 1) = 0.$$

(b) Let $k = n\pi$. On the two source-free intervals the boundary conditions select $\sinh(ky')$ and $\sinh[k(1 - y')]$. Continuity at $y' = y$ is imposed by writing

$$g_n = A \sinh(ky_<) \sinh[k(1 - y_>)].$$

Integrate its differential equation through $y' = y$: $\partial_{y'} g_n|_{y^+} - \partial_{y'} g_n|_{y^-} = -4\pi$. For the trial function the jump is

$$-Ak\{\sinh(ky) \cosh[k(1 - y)] + \cosh(ky) \sinh[k(1 - y)]\} = -Ak \sinh k.$$

Thus $A = 4\pi/(k \sinh k) = 4/[n \sinh(n\pi)]$, and

$$\boxed{G = 8 \sum_{n=1}^{\infty} \frac{\sin(n\pi x) \sin(n\pi x') \sinh(n\pi y_<) \sinh[n\pi(1 - y_>)]}{n \sinh(n\pi)}.}$$

The expression is symmetric under interchange of the observation and source points, and each term vanishes at all four walls.

2.16 A two-dimensional potential exists on a unit square area ($0 \le x \le 1$, $0 \le y \le 1$) bounded by "surfaces" held at zero potential. Over the entire square there is a uniform charge density of unit strength (per unit length in z). Using the Green function of Problem 2.15, show that the solution can be written as

$$\Phi(x,y) = \frac{4}{\pi^3 \epsilon_0} \sum_{m=0}^{\infty} \frac{\sin[(2m+1)\pi x]}{(2m+1)^3} \left\{ 1 - \frac{\cosh[(2m+1)\pi(y-\frac{1}{2})]}{\cosh[(2m+1)\pi/2]} \right\}$$

Solution.

The charge density is 1 in the stated units and all boundary terms vanish. Thus $\Phi(x,y) = (4\pi\epsilon_0)^{-1} \int_0^1 dx' \int_0^1 dy' G$. The x' integral is

$$\int_0^1 \sin(n\pi x')\, dx' = \frac{1-(-1)^n}{n\pi},$$

so only odd n survive. For $k = n\pi$, split the y' integral at $y' = y$ and integrate the two hyperbolic sines:

$$\begin{aligned} I_n &= \sinh[k(1-y)] \int_0^y \sinh(ky')\, dy' + \sinh(ky) \int_y^1 \sinh[k(1-y')]\, dy' \\ &= \frac{\sinh[k(1-y)](\cosh ky - 1) + \sinh(ky)(\cosh[k(1-y)] - 1)}{k} \\ &= \frac{\sinh k - \sinh(ky) - \sinh[k(1-y)]}{k} \\ &= \frac{\sinh k}{k} \left[1 - \frac{\cosh[k(y-1/2)]}{\cosh(k/2)} \right]. \end{aligned}$$

In the last line we used $\sinh(ky) + \sinh[k(1-y)] = 2\sinh(k/2)\cosh[k(y-1/2)]$. Combining $8/[n\sinh(n\pi)]$ from G, $2/(n\pi)$ from the first integral, and I_n, then setting $n = 2m+1$, gives

$$\boxed{\Phi(x,y) = \frac{4}{\pi^3 \epsilon_0} \sum_{m=0}^{\infty} \frac{\sin[(2m+1)\pi x]}{(2m+1)^3} \left[1 - \frac{\cosh[(2m+1)\pi(y-1/2)]}{\cosh[(2m+1)\pi/2]} \right].}$$

For an independent differential check, the Laplacian annihilates the product of sine and cosh, while the remaining term gives

$$\nabla^2 \Phi = -\frac{4}{\pi\epsilon_0} \sum_{n \text{ odd}} \frac{\sin(n\pi x)}{n} = -\frac{1}{\epsilon_0}, \quad 0 < x < 1,$$

since the sine series of the constant function 1 is $(4/\pi)\sum_{n \text{ odd}} \sin(n\pi x)/n$. All four boundary values also vanish.

2.17

(a) Construct the free-space Green function $G(x,y;x',y')$ for two-dimensional electrostatics by integrating $1/R$ with respect to $(z'-z)$ between the limits $\pm Z$, where Z is taken to be very large. Show that apart from an inessential constant, the Green function can be written alternately as

$$\begin{aligned} G(x,y;x',y') &= -\ln[(x-x')^2 + (y-y')^2] \\ &= -\ln[\rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi - \phi')] \end{aligned}$$

(b) Show explicitly by separation of variables in polar coordinates that the Green function can be expressed as a Fourier series in the azimuthal coordinate,

$$G = \frac{1}{2\pi} \sum_{-\infty}^{\infty} e^{im(\phi-\phi')} g_m(\rho, \rho')$$

where the radial Green functions satisfy

$$\frac{1}{\rho'} \frac{\partial}{\partial \rho'} \left(\rho' \frac{\partial g_m}{\partial \rho'} \right) - \frac{m^2}{\rho'^2} g_m = -4\pi \frac{\delta(\rho - \rho')}{\rho}$$

Note that $g_m(\rho, \rho')$ for fixed ρ is a different linear combination of the solutions of the homogeneous radial equation (2.68) for $\rho' < \rho$ and for $\rho' > \rho$, with a discontinuity of slope at $\rho' = \rho$ determined by the source delta function.

(c) Complete the solution and show that the free-space Green function has the expansion

$$G(\rho, \phi; \rho', \phi') = -\ln(\rho_>^2) + 2 \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{\rho_<}{\rho_>} \right)^m \cdot \cos[m(\phi - \phi')]$$

where $\rho_<$ ($\rho_>$) is the smaller (larger) of ρ and ρ'.

Solution.

(a) Let $s = |\boldsymbol{\rho} - \boldsymbol{\rho}'|$. Integration of the three-dimensional Green function gives

$$\int_{-Z}^{Z} \frac{d\zeta}{\sqrt{s^2 + \zeta^2}} = 2 \operatorname{arsinh} \frac{Z}{s} = 2 \ln \frac{Z + \sqrt{Z^2 + s^2}}{s} = 2 \ln(2Z) - \ln s^2 + O(s^2/Z^2).$$

Discarding the spatially constant first term leaves

$$\boxed{G = -\ln s^2 = -\ln[\rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')].}$$

A reference length inside the logarithm only changes the discarded constant. The normalization is checked by integrating the normal derivative around a small circle: $\partial_s G = -2/s$ gives flux $(-2/s)2\pi s = -4\pi$. **(b)** Expand the periodic angular delta function as $\delta_{2\pi}(\phi' - \phi) = (2\pi)^{-1} \sum_m e^{im(\phi-\phi')}$ and similarly put $G = (2\pi)^{-1} \sum_m e^{im(\phi-\phi')} g_m(\rho, \rho')$. Substitution into $\nabla'^2 G = -4\pi\delta(\rho' - \rho)\delta_{2\pi}(\phi' - \phi)/\rho$ yields

$$\frac{1}{\rho'} \frac{d}{d\rho'} \left(\rho' \frac{dg_m}{d\rho'} \right) - \frac{m^2}{\rho'^2} g_m = -4\pi \frac{\delta(\rho' - \rho)}{\rho}.$$

Away from the source the radial solutions are $\rho'^{\pm|m|}$ for $m \neq 0$, and $1, \ln \rho'$ for $m = 0$. Continuity and integration across the source require

$$[\rho' \partial_{\rho'} g_m]_{\rho^-}^{\rho^+} = -4\pi.$$

(c) Regularity at zero and decay of each nonzero angular mode at infinity select $g_m = A(\rho_</\rho_>)^{|m|}$. The derivative jump is $-2|m|A$, so $A = 2\pi/|m|$. For the zero mode, a constant inside and a

logarithm outside give $g_0 = -4\pi \ln \rho_>$, up to one common constant. The $+m$ and $-m$ contributions combine into cosines:

$$\boxed{G = -\ln \rho_>^2 + 2\sum_{m=1}^{\infty} \frac{1}{m}\left(\frac{\rho_<}{\rho_>}\right)^m \cos[m(\phi - \phi')].}$$

To verify the sum independently, use $-\ln(1 - te^{i\Delta}) = \sum_{m\geq 1} t^m e^{im\Delta}/m$ and its complex conjugate. Their sum is $-\ln(1 - 2t\cos\Delta + t^2)$, reproducing part (a). For unequal radii convergence is absolute; equal radii away from the source follow by the limit $t \uparrow 1$.

2.18

(a) By finding appropriate solutions of the radial equation in part b of Problem 2.17, find the Green function for the interior Dirichlet problem of a cylinder of radius b [$\rho' = b$: $G = 0$. See (1.40)]. First find the series expansion akin to the free-space Green function of Problem 2.17. Then show that it can be written in closed form as

$$G = \ln\left[\frac{\rho^2\rho'^2 + b^4 - 2\rho\rho' b^2\cos(\phi - \phi')}{b^2(\rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi - \phi'))}\right]$$

or

$$G = \ln\left[\frac{(b^2 - \rho^2)(b^2 - \rho'^2) + b^2\,|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{b^2\,|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}\right]$$

(b) Show that the solution of the Laplace equation with the potential given as $\Phi(b, \phi)$ on the cylinder can be expressed as Poisson's integral of Problem 2.12.

(c) What changes are necessary for the Green function for the *exterior* problem ($b < \rho < \infty$), for both the Fourier expansion and the closed form? [Note that the exterior Green function is not rigorously correct because it does not vanish for ρ or $\rho' \to \infty$. For situations in which the potential falls off fast enough as $\rho \to \infty$, no mistake is made in its use.]

Solution.

Put $\Delta = \phi - \phi'$ and use the normalization of Problem 2.17. **(a)** The interior radial solution regular at zero is ρ'^m; the combination that vanishes at b is $\rho'^{-m} - \rho'^m/b^{2m}$. For these two functions f, h,

$$\rho'(fh' - f'h) = -2m.$$

Thus $g_m = Af(\rho_<)h(\rho_>)$ has jump $[\rho'\partial_{\rho'} g_m] = -2mA = -4\pi$, so $A = 2\pi/m$. For $m = 0$, the regular inner solution is constant and the outer one is $\ln(b/\rho')$, whose weighted derivative is -1; the same jump sets its coefficient to 4π. These constants give

$$g_m = \frac{2\pi}{m}\left[\left(\frac{\rho_<}{\rho_>}\right)^m - \left(\frac{\rho\rho'}{b^2}\right)^m\right], \quad m > 0, \qquad g_0 = 4\pi \ln\frac{b}{\rho_>}.$$

Hence

$$G = 2\ln\frac{b}{\rho_>} + 2\sum_{m=1}^{\infty}\frac{\cos m\Delta}{m}\left[\left(\frac{\rho_<}{\rho_>}\right)^m - \left(\frac{\rho\rho'}{b^2}\right)^m\right].$$

Summing the logarithmic series with $D = \rho^2 + \rho'^2 - 2\rho\rho'\cos\Delta$ gives

$$G = \ln\frac{b^2}{\rho_>^2} - \ln\frac{D}{\rho_>^2} + \ln\left(1 - \frac{2\rho\rho'}{b^2}\cos\Delta + \frac{\rho^2\rho'^2}{b^4}\right)$$
$$= \boxed{\ln\frac{\rho^2\rho'^2 + b^4 - 2\rho\rho' b^2\cos\Delta}{b^2 D}}.$$

The numerator can equally be written $(b^2-\rho^2)(b^2-\rho'^2) + b^2 D$, proving the second form. Either form vanishes at $\rho' = b$ and is reciprocal. **(b)** Differentiating the closed form at $\rho' = b$ gives

$$\partial_{n'} G|_b = \partial_{\rho'} G\big|_b = -\frac{2(b^2-\rho^2)}{b[b^2+\rho^2-2b\rho\cos\Delta]}.$$

Inserting this in $\Phi = -(4\pi)^{-1}\int_0^{2\pi}\Phi(b,\phi')\partial_{n'}G\, b\, d\phi'$ is exactly the Poisson integral of Problem 2.12. **(c)** For the exterior, the two nonzero-mode solutions are $\rho'^m - b^{2m}/\rho'^m$ at the inner boundary and ρ'^{-m} at infinity. Their normalized combination gives

$$\boxed{G_{\text{ext}} = 2\ln\frac{\rho_<}{b} + 2\sum_{m=1}^{\infty}\frac{\cos m\Delta}{m}\left[\left(\frac{\rho_<}{\rho_>}\right)^m - \left(\frac{b^2}{\rho\rho'}\right)^m\right].}$$

The zero mode is now $g_0 = 4\pi\ln(\rho_</b)$. Summation produces *the same algebraic closed form* as in part (a), now evaluated for $\rho, \rho' > b$. For fixed ρ', its limit as $\rho\to\infty$ is $2\ln(\rho'/b)$, not zero. This is the two-dimensional zero-mode obstruction mentioned in the question. Green's formula remains valid for decaying potentials when the boundary integral at infinity vanishes; an arbitrary nondecaying exterior solution requires keeping that boundary term. On the inner wall the exterior-domain normal is $-\widehat{\boldsymbol{\rho}}'$; using it gives the exterior Poisson kernel $(\rho^2-b^2)/(b^2+\rho^2-2b\rho\cos\Delta)$.

2.19 Show that the two-dimensional Green function for Dirichlet boundary conditions for the annular region, $b \le \rho \le c$ (concentric cylinders) has the expansion

$$G = \frac{\ln(\rho_<^2/b^2)\,\ln(c^2/\rho_>^2)}{\ln(c^2/b^2)} + 2\sum_{m=1}^{\infty}\frac{\cos[m(\phi-\phi')]}{m[1-(b/c)^{2m}]}\left(\rho_<^m - b^{2m}/\rho_<^m\right)\left(1/\rho_>^m - \rho_>^m/c^{2m}\right)$$

Solution.

For a nonzero angular mode $m > 0$, choose the two radial solutions

$$f_m(r) = r^m - \frac{b^{2m}}{r^m}, \qquad h_m(r) = \frac{1}{r^m} - \frac{r^m}{c^{2m}}.$$

They vanish at $r = b, c$ respectively. Continuity across the source is enforced by $g_m = A_m f_m(\rho_<)h_m(\rho_>)$. To impose the source normalization, differentiate the two factors:

$$r[f_m h_m' - f_m' h_m] = -2m\,[1-(b/c)^{2m}].$$

Thus the jump of $\rho'\partial_{\rho'}g_m$ is $-2mA_m[1-(b/c)^{2m}] = -4\pi$, and

$$A_m = \frac{2\pi}{m[1-(b/c)^{2m}]}.$$

For $m = 0$ use $f_0(r) = \ln(r/b)$ and $h_0(r) = \ln(c/r)$. Their Wronskian is $r(f_0 h_0' - f_0' h_0) = -\ln(c/b)$, so

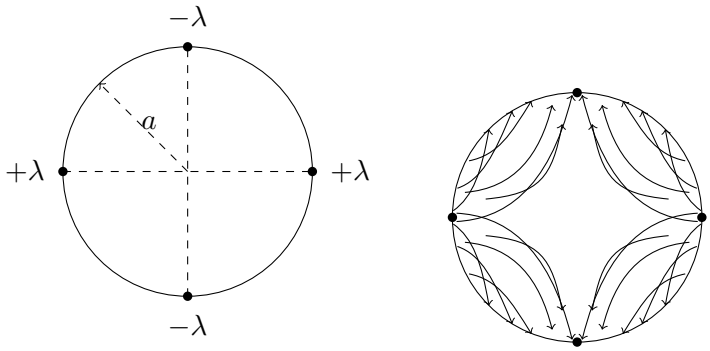
$$g_0 = \frac{4\pi}{\ln(c/b)} \ln\frac{\rho_<}{b} \ln\frac{c}{\rho_>}.$$

Inserting these modes into $G = (2\pi)^{-1}[g_0 + 2\sum_{m\geq 1} g_m \cos(m\Delta)]$ yields

$$\boxed{G = \frac{\ln(\rho_<^2/b^2)\ln(c^2/\rho_>^2)}{\ln(c^2/b^2)} + 2\sum_{m=1}^{\infty} \frac{\cos(m\Delta)}{m[1-(b/c)^{2m}]} \left(\rho_<^m - \frac{b^{2m}}{\rho_<^m}\right)\left(\frac{1}{\rho_>^m} - \frac{\rho_>^m}{c^{2m}}\right).}$$

Both boundary values, reciprocity, and the source jump can be checked separately in every angular mode. As $c \to \infty$ with other radii fixed, the zero mode tends to $2\ln(\rho_</b)$ and the nonzero modes become the exterior-cylinder series of Problem 2.18.

2.20 Two-dimensional electric quadrupole focusing fields for particle accelerators can be modeled by a set of four symmetrically placed line charges, with linear charge densities $\pm\lambda$, as shown in the left-hand figure (the right-hand figure shows the electric field lines).



Problem 2.20

The charge density in two dimensions can be expressed as

$$\sigma(\rho,\phi) = \frac{\lambda}{a}\sum_{n=0}^{3}(-1)^n\,\delta(\rho-a)\,\delta(\phi - n\pi/2)$$

(a) Using the Green function expansion from Problem 2.17c, show that the electrostatic potential is

$$\Phi(\rho,\phi) = \frac{\lambda}{\pi\epsilon_0}\sum_{k=0}^{\infty}\frac{1}{2k+1}\left(\frac{\rho_<}{\rho_>}\right)^{4k+2}\cos[(4k+2)\,\phi]$$

(b) Relate the solution of part a to the real part of the complex function

$$w(z) = \frac{2\lambda}{4\pi\epsilon_0}\ln\left[\frac{(z-ia)(z+ia)}{(z-a)(z+a)}\right]$$

where $z = x + iy = \rho e^{i\phi}$. Comment on the connection to Problem 2.3.

(c) Find expressions for the Cartesian components of the electric field near the origin, expressed in terms of x and y. Keep the $k = 0$ and $k = 1$ terms in the expansion. For $y = 0$ what is the relative magnitude of the $k = 1$ (2^6-pole) contribution to E_x compared to the $k = 0$ (2^2-pole or quadrupole) term?

Solution.

(a) The source integral has measure $\rho'\,d\rho'\,d\phi'$, so the factor λ/a in the density assigns line charge $(-1)^n\lambda$ to each point on $\rho' = a$. Their total charge is zero, eliminating the logarithmic zero mode of G. The angular sum is

$$\sum_{n=0}^{3}(-1)^n e^{-imn\pi/2} = \begin{cases} 4, & m = 4k+2, \\ 0, & \text{otherwise.} \end{cases}$$

Indeed it is the four-term geometric sum with ratio $-e^{-im\pi/2}$, which is zero unless the ratio is one. With $t = \min(\rho, a)/\max(\rho, a)$, the remaining Green-function terms give

$$\Phi = \frac{\lambda}{4\pi\epsilon_0} 2\sum_{k\geq 0}\frac{4t^{4k+2}}{4k+2}\cos[(4k+2)\phi] = \boxed{\frac{\lambda}{\pi\epsilon_0}\sum_{k\geq 0}\frac{t^{4k+2}}{2k+1}\cos[(4k+2)\phi].}$$

(b) Taking the real part of the stated analytic function uses $\operatorname{Re}\ln(z - z_j) = \ln|z - z_j|$ and gives

$$\operatorname{Re} w = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{|z - ia|\,|z + ia|}{|z - a|\,|z + a|},$$

which is the sum of the four line-charge potentials. For $|z| < a$,

$$\ln\frac{z^2 + a^2}{z^2 - a^2} = \text{pure imaginary constant} + \ln\frac{1 + z^2/a^2}{1 - z^2/a^2} = \text{constant} + 2\sum_{k\geq 0}\frac{(z/a)^{4k+2}}{2k+1}.$$

For $|z| > a$ expand in a^2/z^2 instead; its real part has the same cosines. These two expansions reproduce part (a). The four alternating line sources are the same image construction as in Problem 2.3 when $x_0 = y_0$, after rotating the axes by $\pi/4$ and choosing the corresponding length scale. Here the zero-potential planes are the diagonals between the charged axes. **(c)** Keeping $k = 0, 1$ inside the source circle gives

$$\Phi = \frac{\lambda}{\pi\epsilon_0}\left[\frac{x^2 - y^2}{a^2} + \frac{x^6 - 15x^4y^2 + 15x^2y^4 - y^6}{3a^6} + O(\rho^{10}/a^{10})\right].$$

Termwise Cartesian differentiation gives

$$\boxed{\begin{aligned} E_x &= -\frac{2\lambda}{\pi\epsilon_0 a^2}\left[x + \frac{x^5 - 10x^3y^2 + 5xy^4}{a^4}\right] + O\left(\frac{\lambda\rho^9}{\epsilon_0 a^{10}}\right), \\ E_y &= \frac{2\lambda}{\pi\epsilon_0 a^2}\left[y + \frac{5x^4y - 10x^2y^3 + y^5}{a^4}\right] + O\left(\frac{\lambda\rho^9}{\epsilon_0 a^{10}}\right). \end{aligned}}$$

On $y = 0$ their two E_x contributions have the same sign and

$$\boxed{\left|\frac{E_x^{(k=1)}}{E_x^{(k=0)}}\right| = (x/a)^4}$$

for $x \neq 0$, with limiting ratio zero at the origin.

2.21 Use Cauchy's theorem to derive the Poisson integral solution. Cauchy's theorem states that if $F(z)$ is analytic in a region R bounded by a closed curve C, then

$$\frac{1}{2\pi i}\oint_C \frac{F(z')\,dz'}{z'-z} = \begin{cases} F(z) & \text{if } z \text{ is inside } R \\ 0 & \text{outside} \end{cases}$$

Hint: You may wish to add an integral that vanishes (associated with the image point) to the integral for the point inside the circle.

Solution.

Let F be analytic in the disk $|\zeta| < b$, continuous on its boundary, and let $\Phi = \operatorname{Re} F$. Fix an interior point z and its exterior image $z_i = b^2/\bar{z}$. By Cauchy's formula,

$$F(z) = \frac{1}{2\pi i}\oint_{|\zeta|=b} F(\zeta)\left[\frac{1}{\zeta - z} - \frac{1}{\zeta - z_i}\right] d\zeta.$$

The second integral is zero because its pole is outside the disk. Put $\zeta = be^{i\phi'}$, $d\zeta = i\zeta\, d\phi'$. Since $\zeta\bar{\zeta} = b^2$ on the contour,

$$-\frac{\zeta}{\zeta - b^2/\bar{z}} = \frac{\bar{z}}{\bar{\zeta} - \bar{z}}.$$

Consequently the two kernels combine to a real quantity:

$$\begin{aligned} \frac{\zeta}{\zeta - z} + \frac{\bar{z}}{\bar{\zeta} - \bar{z}} &= \frac{\zeta(\bar{\zeta} - \bar{z}) + \bar{z}(\zeta - z)}{|\zeta - z|^2} \\ &= \frac{b^2 - |z|^2}{|\zeta - z|^2}. \end{aligned}$$

Taking real parts now removes the otherwise unknown harmonic conjugate of the boundary potential. With $z = \rho e^{i\phi}$,

$$\boxed{\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b,\phi')\frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi - \phi')}\, d\phi'.}$$

At $z = 0$ the image is understood as a limit at infinity and the kernel is one, giving the average boundary value. Every harmonic potential in the simply connected disk has a harmonic conjugate, so this establishes the Poisson formula for the stated Dirichlet problem; piecewise boundary data follow by taking interior limits.

2.22

(a) For the example of oppositely charged conducting hemispherical shells separated by a tiny gap, as shown in Figure 2.8, show that the interior potential ($r < a$) on the z axis is

$$\Phi_{\text{in}}(z) = V\,\frac{a}{z}\left[1 - \frac{(a^2 - z^2)}{a\sqrt{a^2 + z^2}}\right]$$

Find the first few terms of the expansion in powers of z and show that they agree with (2.27) with the appropriate substitutions.

(b) From the result of part a and (2.22), show that the axial electric field E_z (equal to E_r on the positive z axis) is

$$E_z(z) = \frac{Va^2}{(z^2+a^2)^{3/2}}\left(3+\frac{a^2}{z^2}\right)$$

for $z > a$, and

$$E_z(z) = -\frac{V}{a}\left[\frac{3+(a/z)^2}{(1+(z/a)^2)^{3/2}} - \frac{a^2}{z^2}\right]$$

for $|z| < a$. Show that the second form is well behaved at the origin, with the value $E_z(0) = -3V/2a$. Show that at $z = a$ (north pole inside) it has the value $-(\sqrt{2}-1)V/a$. Show that the radial field at the north pole outside has the value $\sqrt{2}\,V/a$.

(c) Make a sketch of the electric field lines, both inside and outside the conducting hemispheres, with directions indicated. Make a *plot* of the axial electric field E_z along the z axis from $z = -2a$ to $z = +2a$.

Solution.

The northern hemisphere is at $+V$ and the southern at $-V$. **(a)** The interior spherical Poisson kernel gives, on the axis,

$$\Phi_{\rm in}(z) = \frac{Va(a^2-z^2)}{2}\int_{-1}^{1}\frac{\operatorname{sgn} u\, du}{(a^2+z^2-2azu)^{3/2}}, \qquad |z|<a.$$

For $z \neq 0$, the integral is

$$\frac{1}{az}\left[\frac{1}{a-z}+\frac{1}{a+z}-\frac{2}{\sqrt{a^2+z^2}}\right].$$

Combining the first two fractions before multiplying the prefactor therefore gives

$$\boxed{\Phi_{\rm in}(z) = \frac{Va}{z}\left[1-\frac{a^2-z^2}{a\sqrt{a^2+z^2}}\right].}$$

Put $t = z/a$ and expand $(1+t^2)^{-1/2} = 1 - t^2/2 + 3t^4/8 - 5t^6/16 + 35t^8/128 + \cdots$. Multiplication by $1-t^2$ and division by t give

$$\frac{\Phi_{\rm in}}{V} = \frac{3}{2}t - \frac{7}{8}t^3 + \frac{11}{16}t^5 - \frac{75}{128}t^7 + O(t^9).$$

The interior and exterior harmonic expansions are related by Kelvin inversion, $\Phi_{\rm out}(r,\theta) = (a/r)\Phi_{\rm in}(a^2/r,\theta)$. Thus the exterior terms are $V[\frac{3}{2}(a/r)^2P_1(\cos\theta) - \frac{7}{8}(a/r)^4P_3(\cos\theta) + \cdots]$. On the positive axis $P_l(1) = 1$, so the replacements $(a/r)^{l+1} \mapsto (r/a)^l$ recover the interior coefficients above from (2.27).

(b) Differentiating the interior expression at fixed sign of z gives the axial component

$$\boxed{E_z^{\rm in}(z) = -\frac{V}{a}\left[\frac{3+(a/z)^2}{[1+(z/a)^2]^{3/2}} - \frac{a^2}{z^2}\right].}$$

Its apparently singular terms cancel. In fact differentiating the power series gives

$$E_z^{\rm in}(z) = \frac{V}{a}\left[-\frac{3}{2}+\frac{21}{8}t^2-\frac{55}{16}t^4+\frac{525}{128}t^6+O(t^8)\right].$$

The origin value is therefore $-3V/(2a)$. The exterior positive-axis potential (2.22), also obtained by the inversion above, is

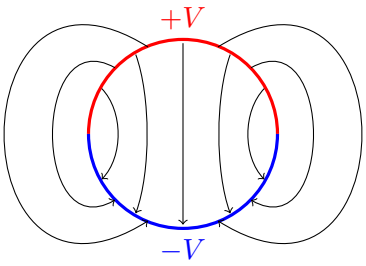
$$\Phi_{\text{out}}(z) = V\left[1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}}\right], \quad z > a.$$

Its derivative is

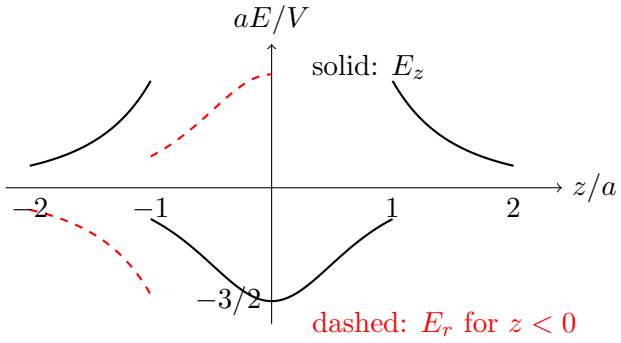
$$\boxed{E_z^{\text{out}}(z) = \frac{Va^2}{(z^2 + a^2)^{3/2}}\left(3 + \frac{a^2}{z^2}\right), \quad z > a.}$$

On the positive axis $E_z = E_r$. At the north pole the limiting values are $E_r(a^-) = -\left(\sqrt{2} - 1\right) V/a$ and $E_r(a^+) = \sqrt{2}V/a$, agreeing with the question. Their difference is $(2\sqrt{2} - 1)V/a$, the surface charge divided by ϵ_0.

(c) Antisymmetry of the potential, $\Phi(-z) = -\Phi(z)$, implies $E_z(-z) = E_z(z)$. The radial basis vector reverses on the negative axis, so strictly $E_r(z) = \text{sgn}(z)E_z(z)$ for $z \neq 0$; E_r has no single basis direction at the origin. The finite origin value is the continuous *axial* component; the statement uses corrected notation to make this distinction explicit. The plot shows that component together with the true radial component on the negative axis, where they differ. Fields jump at $z = \pm a$; there is no field value inside an ideal zero-thickness metal surface itself.



Meridional field-line sketch; arrows follow **E**.



The inside lines run from the northern to the southern inner face; the outside lines connect the outer faces around the shell. The sketch illustrates topology; the accompanying axial curves use the exact formulas above.

2.23 A hollow cube has conducting walls defined by six planes $x = 0$, $y = 0$, $z = 0$, and $x = a$, $y = a$, $z = a$. The walls $z = 0$ and $z = a$ are held at a constant potential V. The other four sides are at zero potential.

(a) Find the potential $\Phi(x, y, z)$ at any point inside the cube.

(b) Evaluate the potential at the center of the cube numerically, accurate to three significant figures. How many terms in the series is it necessary to keep in order to attain this accuracy? Compare your numerical result with the average value of the potential on the walls. See Problem 2.28.

(c) Find the surface-charge density on the surface $z = a$.

Solution.

(a) The four grounded side walls select $\sin(m\pi x/a)\sin(n\pi y/a)$. For a fixed pair the remaining separation equation is $Z'' - k_{mn}^2 Z = 0$, where $k_{mn} = \pi\sqrt{m^2+n^2}/a$. Equal values at $z = 0, a$ select $Z = \cosh[k_{mn}(z - a/2)]/\cosh(k_{mn}a/2)$. The double sine coefficients of a constant on a square are

$$\frac{4}{a^2}\int_0^a dx \int_0^a dy\, V \sin\frac{m\pi x}{a}\sin\frac{n\pi y}{a} = \frac{4V[1-(-1)^m][1-(-1)^n]}{mn\pi^2}.$$

Thus only odd m, n occur, and

$$\boxed{\Phi = \frac{16V}{\pi^2}\sum_{\substack{m,n\geq 1\\ m,n\ \text{odd}}} \frac{\sin(m\pi x/a)\sin(n\pi y/a)}{mn}\frac{\cosh[k_{mn}(z-a/2)]}{\cosh(k_{mn}a/2)}.}$$

At the edges the imposed boundary data have different one-sided limits; the interior solution and the values on open faces are well defined. **(b)** At the center, the two sines supply $(-1)^{(m+n-2)/2}$. Summing all pairs with $m, n = 1, 3, \ldots, 2N-1$ gives

N	number of pairs	$\Phi(a/2, a/2, a/2)/V$
1	1	0.347545849
2	4	0.332957715
3	9	0.333345107
4	16	0.333332938

The four-pair truncation already rounds to 0.333 to three significant figures; it contains only three distinct values after using $m \leftrightarrow n$ symmetry. There is also an exact check: by cube symmetry the harmonic response at the center to each unit-potential face is the same, and the sum of the six responses is one. Two faces at V therefore give

$$\boxed{\Phi_{\text{center}} = V/3,}$$

exactly the average wall potential $(V + V)/6$. This proves the accuracy of the reported truncation without inferring convergence from successive partial sums alone. **(c)** On the inner face $z = a$ the conductor-to-cavity normal is $-\widehat{\mathbf{z}}$, giving $\sigma = \epsilon_0 \partial_z \Phi|_{a^-}$. Differentiation yields

$$\boxed{\sigma(x,y) = \frac{16\epsilon_0 V}{\pi^2}\sum_{\substack{m,n\geq 1\\ m,n\ \text{odd}}} \frac{k_{mn}\tanh(k_{mn}a/2)}{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{a}.}$$

The derivative series is understood as its interior limit $z \uparrow a$, or an equivalent Abel limit, away from edges; charge crowding makes the ideal sharp-edge density singular. This determines the cavity-facing charge only, since no exterior boundary problem was specified.

2.24 In the two-dimensional region shown in Fig. 2.12, the angular functions appropriate for Dirichlet boundary conditions at $\phi = 0$ and $\phi = \beta$ are $\Phi(\phi) = A_m \sin(m\pi\phi/\beta)$. Show that the completeness relation for these functions is

$$\delta(\phi - \phi') = \frac{2}{\beta} \sum_{m=1}^{\infty} \sin(m\pi\phi/\beta)\, \sin(m\pi\phi'/\beta) \qquad \text{for } 0 < \phi,\, \phi' < \beta$$

Solution.

For positive integers m, n, product-to-sum integration gives

$$\int_0^\beta \sin\frac{m\pi\phi}{\beta} \sin\frac{n\pi\phi}{\beta}\, d\phi = \frac{\beta}{2}\delta_{mn}.$$

To show that the normalized modes are complete, take a smooth test function f with support in $(0, \beta)$, extend it oddly to $(-\beta, \beta)$, and periodically with period 2β. The ordinary Fourier expansion of this odd function contains only sines, with coefficients

$$f_m = \frac{2}{\beta} \int_0^\beta f(\phi') \sin\frac{m\pi\phi'}{\beta}\, d\phi'.$$

Substituting these coefficients back into the Fourier series yields

$$f(\phi) = \int_0^\beta \left[\frac{2}{\beta} \sum_{m=1}^{\infty} \sin\frac{m\pi\phi}{\beta} \sin\frac{m\pi\phi'}{\beta} \right] f(\phi')\, d\phi'.$$

The bracket therefore has precisely the action of the delta function on every such test function:

$$\boxed{\delta(\phi - \phi') = \frac{2}{\beta} \sum_{m=1}^{\infty} \sin\frac{m\pi\phi}{\beta} \sin\frac{m\pi\phi'}{\beta}.}$$

This is a distributional completeness relation for interior points, not a pointwise convergent numerical series for the delta function. The restriction away from the endpoints avoids the half-weight endpoint convention of Fourier boundary limits.

2.25 Two conducting planes at zero potential meet along the z axis, making an angle β between them, as in Fig. 2.12. A unit line charge parallel to the z axis is located between the planes at position (ρ', ϕ').

(a) Show that $(4\pi\epsilon_0)$ times the potential in the space between the planes, that is, the Dirichlet Green function $G(\rho, \phi; \rho', \phi')$, is given by the infinite series

$$G(\rho, \phi; \rho', \phi') = 4 \sum_{m=1}^{\infty} \frac{1}{m} \rho_<^{m\pi/\beta} \rho_>^{-m\pi/\beta} \sin(m\pi\phi/\beta)\, \sin(m\pi\phi'/\beta)$$

(b) By means of complex-variable techniques or other means, show that the series can be summed to give a closed form,

$$G(\rho, \phi; \rho', \phi') = \ln \left\{ \frac{(\rho)^{2\pi/\beta} + (\rho')^{2\pi/\beta} - 2(\rho\rho')^{\pi/\beta} \cos[\pi(\phi + \phi')/\beta]}{(\rho)^{2\pi/\beta} + (\rho')^{2\pi/\beta} - 2(\rho\rho')^{\pi/\beta} \cos[\pi(\phi - \phi')/\beta]} \right\}$$

(c) Verify that you obtain the familiar results when $\beta = \pi$ and $\beta = \pi/2$.

Solution.

(a) With $\nu_m = m\pi/\beta$, expand

$$G = \frac{2}{\beta} \sum_{m \geq 1} g_m(\rho, \rho') \sin(\nu_m \phi) \sin(\nu_m \phi').$$

Using Problem 2.24, the radial equation is

$$\frac{1}{\rho'}(\rho' g_m')' - \frac{\nu_m^2}{\rho'^2} g_m = -4\pi \frac{\delta(\rho' - \rho)}{\rho}.$$

Boundedness at zero and decay at infinity select $g_m = A_m \rho_<^{\nu_m} \rho_>^{-\nu_m}$. The radial derivative jump is $-2\nu_m A_m = -4\pi$, so $A_m = 2\pi/\nu_m$. Multiplying by the angular normalization $2/\beta$ gives

$$\boxed{G = 4 \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{\rho_<}{\rho_>} \right)^{m\pi/\beta} \sin \frac{m\pi\phi}{\beta} \sin \frac{m\pi\phi'}{\beta}.}$$

For unit line charge, $\Phi = G/(4\pi\epsilon_0)$. **(b)** Put $t = (\rho_</\rho_>)^{\pi/\beta}$, $A = \pi\phi/\beta$, and $B = \pi\phi'/\beta$. Since $2 \sin mA \sin mB = \cos m(A - B) - \cos m(A + B)$ and $\sum_{m \geq 1} t^m \cos m\theta / m = -\frac{1}{2} \ln(1 - 2t \cos\theta + t^2)$,

$$G = \ln \frac{1 - 2t \cos(A + B) + t^2}{1 - 2t \cos(A - B) + t^2}.$$

Multiplication of numerator and denominator by $\rho_>^{2\pi/\beta}$ yields exactly

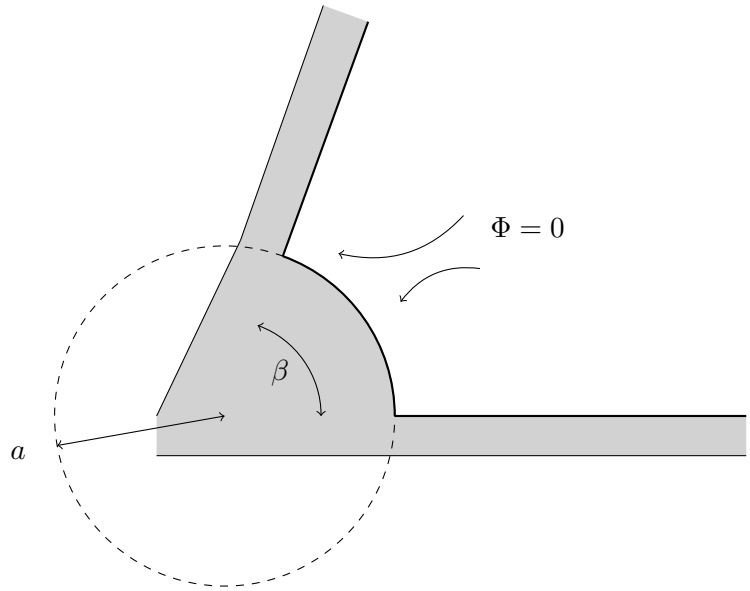
$$\boxed{G = \ln \frac{\rho^{2\pi/\beta} + \rho'^{2\pi/\beta} - 2(\rho\rho')^{\pi/\beta} \cos[\pi(\phi + \phi')/\beta]}{\rho^{2\pi/\beta} + \rho'^{2\pi/\beta} - 2(\rho\rho')^{\pi/\beta} \cos[\pi(\phi - \phi')/\beta]}.}$$

Equivalently the analytic mapping $w = z^{\pi/\beta}$ sends the wedge to the upper half-plane; the numerator is the squared distance to the reflected source in that plane. Near the physical source the analytic mapping is locally linear, so its scale cancels in the ratio and the singular term remains $-\ln|z - z'|^2$. **(c)** For $\beta = \pi$ the result is $G = \ln[|z - \bar{z}'|^2/|z - z'|^2]$, the real line and its negative mirror in a single plane. For $\beta = \pi/2$, factor the squares:

$$G = \ln \frac{|z^2 - \bar{z}'^2|^2}{|z^2 - z'^2|^2} = \ln \frac{|z - \bar{z}'|^2 |z + \bar{z}'|^2}{|z - z'|^2 |z + z'|^2}.$$

These are a positive source at z', a positive image at $-z'$, and negative images at $\bar{z}'$ and $-\bar{z}'$, exactly the four-line construction of Problem 2.3.

2.26 The two-dimensional region, $\rho \geq a$, $0 \leq \phi \leq \beta$, is bounded by conducting surfaces at $\phi = 0$, $\rho = a$, and $\phi = \beta$ held at zero potential, as indicated in the sketch. At large ρ the potential is determined by some configuration of charges and/or conductors at fixed potentials.



Problem 2.26

(a) Write down a solution for the potential $\Phi(\rho, \phi)$ that satisfies the boundary conditions for finite ρ.

(b) Keeping only the lowest nonvanishing terms, calculate the electric field components E_ρ and E_ϕ and also the surface-charge densities $\sigma(\rho, 0)$, $\sigma(\rho, \beta)$, and $\sigma(a, \phi)$ on the three boundary surfaces.

(c) Consider $\beta = \pi$ (a plane conductor with a half-cylinder of radius a on it). Show that far from the half-cylinder the lowest order terms of part b give a uniform electric field normal to the plane. Sketch the charge density on and in the neighborhood of the half-cylinder. For fixed electric field strength far from the plane, show that the total charge on the half-cylinder (actually charge per unit length in the z direction) is twice as large as would reside on a strip of width $2a$ in its absence. Show that the extra portion is drawn from regions of the plane nearby, so that the total charge on a strip of width large compared to a is the same whether the half-cylinder is there or not.

Solution.

(a) Separation of Laplace's equation with zero values on the two radial walls selects angular factors $\sin(\nu_m \phi)$, $\nu_m = m\pi/\beta$. The radial solutions are ρ^{ν_m} and $\rho^{-\nu_m}$. Imposing zero potential at $\rho = a$ fixes their relative coefficient:

$$\boxed{\Phi(\rho, \phi) = \sum_{m=1}^{\infty} A_m \left(\rho^{\nu_m} - a^{2\nu_m} \rho^{-\nu_m}\right) \sin(\nu_m \phi).}$$

The remaining A_m encode the remote boundary conditions. The series is used in the source-free region between the local conductor and those remote sources, within its region of convergence. **(b)** If $A_1 \neq 0$, set $\nu = \pi/\beta$, $A = A_1$ and retain that lowest mode. Direct differentiation gives

$$\boxed{\begin{aligned} E_\rho &= -A\nu(\rho^{\nu-1} + a^{2\nu}\rho^{-\nu-1}) \sin \nu\phi, \\ E_\phi &= -A\nu(\rho^{\nu-1} - a^{2\nu}\rho^{-\nu-1}) \cos \nu\phi. \end{aligned}}$$

The normals from the conductors into the free region are $+\widehat{\boldsymbol{\phi}}$ at $\phi = 0$, $-\widehat{\boldsymbol{\phi}}$ at $\phi = \beta$, and $+\widehat{\boldsymbol{\rho}}$ at $\rho = a$. Therefore

$$\boxed{\begin{aligned}\sigma(\rho,0) = \sigma(\rho,\beta) &= -\epsilon_0 A\nu(\rho^{\nu-1} - a^{2\nu}\rho^{-\nu-1}),\\ \sigma(a,\phi) &= -2\epsilon_0 A\nu a^{\nu-1}\sin\nu\phi.\end{aligned}}$$

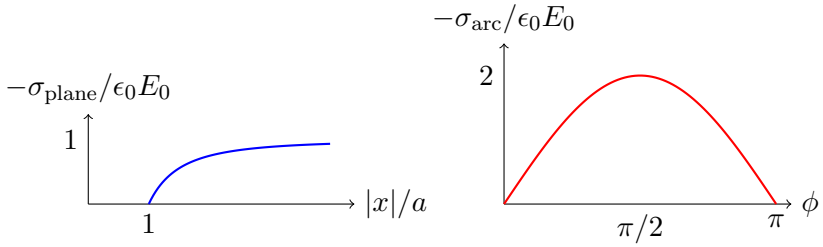
If symmetry makes $A_1 = 0$, use the first nonzero m instead; the second wall then has $\sigma(\rho,\beta) = \epsilon_0 A_m \nu_m(\rho^{\nu_m-1} - a^{2\nu_m}\rho^{-\nu_m-1})(-1)^m$. **(c)** For $\beta = \pi$, $\nu = 1$. Choosing $A = E_0 > 0$ gives

$$\Phi = E_0\left(\rho - \frac{a^2}{\rho}\right)\sin\phi = E_0 y\left(1 - \frac{a^2}{x^2+y^2}\right),$$

so far away $\mathbf{E} \to -E_0\widehat{\mathbf{y}}$. The densities are

$$\sigma_{\text{plane}}(x) = -\epsilon_0 E_0(1 - a^2/x^2), \quad |x| \geq a, \qquad \sigma_{\text{arc}}(\phi) = -2\epsilon_0 E_0 \sin\phi.$$

Both vanish where the arc meets the plane. The arc density is twice the planar magnitude at its top, and the plane approaches its undisturbed value monotonically away from the bump.



The arc charge per unit z length is

$$Q_{\text{arc}} = -2\epsilon_0 E_0 a\int_0^\pi \sin\phi\, d\phi = -4\epsilon_0 E_0 a,$$

twice the charge $-2\epsilon_0 E_0 a$ on the displaced flat strip. The deficit in charge magnitude on the remaining plane is

$$2\int_a^\infty [\sigma_{\text{plane}}(x) + \epsilon_0 E_0]\, dx = 2\epsilon_0 E_0 a^2\int_a^\infty \frac{dx}{x^2} = 2\epsilon_0 E_0 a,$$

exactly the negative of the extra arc charge. More precisely, the charge on the arc plus the plane portions $a < |x| < L$ is

$$Q_L = -2\epsilon_0 E_0 L - \frac{2\epsilon_0 E_0 a^2}{L}.$$

Compared with the flat value $-2\epsilon_0 E_0 L$, the relative change is $a^2/L^2 \to 0$ for a wide strip.

2.27 Consider the two-dimensional wedge-shaped region of Problem 2.26, with $\beta = 2\pi$. This corresponds to a semi-infinite thin sheet of conductor on the positive x axis from $x = a$ to infinity with a conducting cylinder of radius a fastened to its edge.

(a) Sketch the surface-charge densities on the cylinder and on the top and bottom of the sheet, using the lowest order solution.

(b) Calculate the total charge on the cylinder and compare with the total deficiency of charge on the sheet near the cylinder, that is, the total difference in charge for a finite compared with $a = 0$, assuming that the charge density far from the cylinder is the same.

Solution.

For $\beta = 2\pi$ the lowest angular index is $\nu = 1/2$. Let its coefficient A be positive; all signs reverse if $A < 0$. The potential near the cylinder is

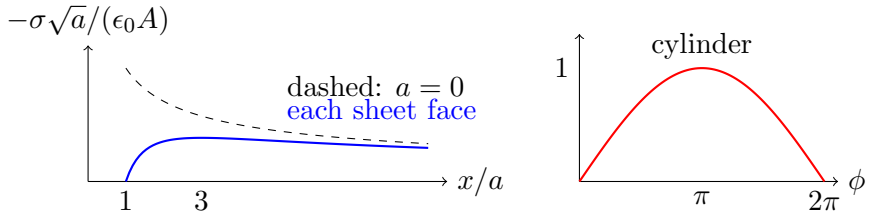
$$\Phi = A\left(\sqrt{\rho} - \frac{a}{\sqrt{\rho}}\right)\sin\frac{\phi}{2}.$$

(a) The formulas of Problem 2.26 give

$$\boxed{\sigma_{\text{cylinder}}(\phi) = -\frac{\epsilon_0 A}{\sqrt{a}}\sin\frac{\phi}{2}, \quad 0 < \phi < 2\pi,}$$

$$\boxed{\sigma_{\text{top}}(x) = \sigma_{\text{bottom}}(x) = -\frac{\epsilon_0 A}{2\sqrt{x}}\left(1 - \frac{a}{x}\right), \quad x > a.}$$

The density on the cylinder vanishes at the attachment point and is largest in magnitude at the leftmost point, $\phi = \pi$. Each sheet face starts from zero at $x = a$, reaches its largest magnitude at $x = 3a$ (differentiate $x^{-1/2} - ax^{-3/2}$), and then approaches the $a = 0$ edge law $-\epsilon_0 A/(2\sqrt{x})$. The normalized maximum sheet magnitude is $1/(3\sqrt{3})$ in units $\epsilon_0 A/\sqrt{a}$.



(b) Integration on the full circular surface gives

$$Q_c = -\epsilon_0 A\sqrt{a}\int_0^{2\pi}\sin\frac{\phi}{2}\,d\phi = \boxed{-4\epsilon_0 A\sqrt{a}}.$$

Although the total charge of an infinite sheet diverges, the difference between the two configurations is integrable. On the part $x > a$ that remains in both configurations,

$$\Delta Q_{x>a} = 2\int_a^\infty \frac{\epsilon_0 A a}{2x^{3/2}}\,dx = 2\epsilon_0 A\sqrt{a}.$$

The reference sheet also had the part $0 < x < a$, on both faces, which has now been replaced by the cylinder. Its removed charge is

$$Q^{(0)}_{0<x<a} = 2\int_0^a -\frac{\epsilon_0 A}{2\sqrt{x}}\,dx = -2\epsilon_0 A\sqrt{a}.$$

Thus the *total* sheet deficiency is $\Delta Q_{\text{sheet}} = 2\epsilon_0 A\sqrt{a} - (-2\epsilon_0 A\sqrt{a}) = 4\epsilon_0 A\sqrt{a} = -Q_c$. Omitting the removed strip would count only half the deficiency.

2.28 A closed volume is bounded by conducting surfaces that are the n sides of a regular polyhedron ($n = 4,\ 6,\ 8,\ 12,\ 20$). The n surfaces are at different potentials V_i, $i = 1, 2, \ldots, n$. Prove in the simplest way you can that the potential at the center of the polyhedron is the average of the potential on the n sides. This problem bears on Problem 2.23b, and has an interesting similarity to the result of Problem 1.10.

Solution.

Let $u_i(\mathbf{x})$ solve Laplace's equation with boundary value one on face i and zero on the other faces. By linearity and Dirichlet uniqueness,

$$\Phi(\mathbf{x}) = \sum_{i=1}^{n} V_i u_i(\mathbf{x}).$$

A symmetry of a regular polyhedron carries any face into any other while leaving its center $\mathbf{x}_c$ fixed. Applying that symmetry to a unit-face solution and using uniqueness therefore proves $u_i(\mathbf{x}_c) = u_j(\mathbf{x}_c)$ for all i, j. Moreover, $\sum_i u_i$ has boundary value one on every open face, so its unique interior solution is the constant one. At the center,

$$1 = \sum_i u_i(\mathbf{x}_c) = n u_i(\mathbf{x}_c).$$

It follows immediately that

$$\boxed{\Phi(\mathbf{x}_c) = \frac{1}{n} \sum_{i=1}^{n} V_i.}$$

Idealized voltage jumps on shared edges can be understood as limits of narrow insulating gaps; the interior harmonic solution is unaffected by the assignment of boundary values on the edges alone. The proof uses face-transitive symmetry, not a general assertion that the potential equals an area average on an arbitrary enclosing surface. For the cube of Problem 2.23, two of its six faces contribute, so the center potential is $V/3$.

2.29 For the Galerkin method on a two-dimensional square lattice with lattice spacing h, verify the relations (2.81) for the localized "pyramid" basis functions, $\phi_{ij}(x, y) = (1 - |x|/h)(1 - |y|/h)$, $|x| < h$, $|y| < h$, where x and y are measured from the site (i, j). In particular,

$$\int dx \int dy\ \phi_{i,j}(x, y) = h^2; \qquad \int dx \int dy\ \nabla\phi_{i,j} \cdot \nabla\phi_{i,j} = \frac{8}{3};$$
$$\int dx \int dy\ \nabla\phi_{i+1,j} \cdot \nabla\phi_{i,j} = -\frac{1}{3}; \qquad \int dx \int dy\ \nabla\phi_{i,j+1} \cdot \nabla\phi_{i,j} = -\frac{1}{3};$$
$$\int dx \int dy\ \nabla\phi_{i+1,j+1} \cdot \nabla\phi_{i,j} = -\frac{1}{3}$$

Solution.

Define a one-dimensional tent $f(x) = 1 - |x|/h$ for $|x| < h$, and zero otherwise. Then $\phi_{ij}(x, y) = f(x - ih) f(y - jh)$, and each required two-dimensional integral factors into one-dimensional integrals.

For a tent and its nearest translate, direct integration gives

$$
\begin{aligned}
I &= \int f(x)\,dx = 2\int_0^h (1-x/h)\,dx = h,\\
M_0 &= \int f^2\,dx = 2\int_0^h (1-x/h)^2\,dx = \frac{2h}{3},\\
M_1 &= \int f(x)f(x-h)\,dx = \int_0^h (1-x/h)(x/h)\,dx = \frac{h}{6},\\
K_0 &= \int [f'(x)]^2\,dx = \int_{-h}^h \frac{dx}{h^2} = \frac{2}{h},\\
K_1 &= \int f'(x)f'(x-h)\,dx = \int_0^h \left(-\frac{1}{h}\right)\left(\frac{1}{h}\right)dx = -\frac{1}{h}.
\end{aligned}
$$

Supports separated by two or more mesh spacings have zero overlap except on sets of measure zero. First,

$$\boxed{\int \phi_{ij}\,dx\,dy = I^2 = h^2.}$$

For any two sites, the gradient product has an x-derivative term and a y-derivative term. The coincident-site integral is therefore

$$\boxed{K_0M_0 + M_0K_0 = 2(2/h)(2h/3) = \frac{8}{3}.}$$

For a horizontal neighbor the result is

$$\boxed{K_1M_0 + M_1K_0 = -\frac{2}{3} + \frac{1}{3} = -\frac{1}{3},}$$

and a vertical neighbor interchanges the two terms with the same answer. For a diagonal neighbor both coordinates are translated:

$$\boxed{K_1M_1 + M_1K_1 = 2(-1/h)(h/6) = -\frac{1}{3}.}$$

The same values hold for all reflected nearest neighbors. Thus a row of the interior stiffness matrix has $8/3$ on its own site and $-1/3$ at each of eight neighboring sites; their sum is zero, as required for a constant potential to have zero gradient.

2.30 Using the results of Problem 2.29, apply the Galerkin method to the integral equivalent of the Poisson equation with zero potential on the boundary,

$$\int_V dx\,dy\,[\nabla\phi_{i,j}\cdot\nabla\psi - 4\pi\rho\,\phi_{i,j}] = 0 \quad \text{with} \quad \psi(x,y) = \sum_{i',j'=1}^{N} \psi_{i'j'}\,\phi_{i'j'}(x,y)$$

for the lattice of Problem 1.24, with its three independent lattice sites. Show that you get three coupled equations for the $\psi_{i,j}$ values (ψ_1,ψ_2,ψ_3) and solve to find the "Galerkin" approximations for the potential at these sites. Compare with the exact values and the results of the various iterations of Problem 1.24c. Comment.

$$[\psi = 4\pi\epsilon_0\Phi].$$

Solution.

Here $\rho = 1$, $h = 1/4$, and $\psi = 4\pi\epsilon_0\Phi$. Inserting the basis expansion into the weak equation and using Problem 2.29 gives

$$\frac{8}{3}\psi_{ij} - \frac{1}{3}\sum_{\text{eight neighbors}} \psi_{i'j'} = 4\pi h^2, \quad \text{or} \quad 8\psi_{ij} - \sum_{\text{eight neighbors}} \psi_{i'j'} = \frac{3\pi}{4}.$$

Boundary nodes have zero potential. Label the three independent interior values by their sites in Problem 1.24: $\psi_1 = \psi(1/4, 1/4)$, $\psi_2 = \psi(1/2, 1/4)$, $\psi_3 = \psi(1/2, 1/2)$. At a corner-type interior site the nonzero neighbors are $2\psi_2 + \psi_3$; at an edge-type site they are $2\psi_1 + 2\psi_2 + \psi_3$; at the center they are $4\psi_1 + 4\psi_2$. Thus

$$\boxed{\begin{aligned} 8\psi_1 - 2\psi_2 - \psi_3 &= 3\pi/4, \\ -2\psi_1 + 6\psi_2 - \psi_3 &= 3\pi/4, \\ -4\psi_1 - 4\psi_2 + 8\psi_3 &= 3\pi/4. \end{aligned}}$$

Subtracting the second equation from the first gives $10\psi_1 - 8\psi_2 = 0$, so $\psi_2 = 5\psi_1/4$. The first then gives $\psi_3 = 11\psi_1/2 - 3\pi/4$. Substitution into the third gives $35\psi_1 = 27\pi/4$. Consequently

$$\boxed{\psi_1 = \frac{27\pi}{140} = 0.605878583, \quad \psi_2 = \frac{27\pi}{112} = 0.757348229, \quad \psi_3 = \frac{87\pi}{280} = 0.976137717.}$$

For comparison, the improved finite-difference equations of Problem 1.24 are

$$\begin{pmatrix} 20 & -8 & -1 \\ -8 & 18 & -4 \\ -4 & -16 & 20 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} = \frac{3\pi}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

To exhibit the iterations, put $b_0 = 3\pi/2$. The reduced Jacobi updates are

$$\psi_1^{(r+1)} = \frac{b_0 + 8\psi_2^{(r)} + \psi_3^{(r)}}{20}, \quad \psi_2^{(r+1)} = \frac{b_0 + 8\psi_1^{(r)} + 4\psi_3^{(r)}}{18}, \quad \psi_3^{(r+1)} = \frac{b_0 + 4\psi_1^{(r)} + 16\psi_2^{(r)}}{20}.$$

Ordered Gauss–Seidel uses the just-computed $\psi_1^{(r+1)}$ in the second equation and both new $\psi_1^{(r+1)}, \psi_2^{(r+1)}$ in the third. Solving the three simultaneous equations, for example eliminating $\psi_3 = 20\psi_1 - 8\psi_2 - b_0$ from the first, gives

$$-88\psi_1 + 50\psi_2 = -3b_0, \qquad 396\psi_1 - 176\psi_2 = 21b_0.$$

Their solution is $(\psi_1, \psi_2, \psi_3) = (783\pi/4312, 45\pi/196, 159\pi/539)$. Starting from $(1, 1, 1)$ and applying Jacobi or ordered Gauss–Seidel to this reduced matrix (with diagonal 18 in its second row, as in Problem 1.24) gives the comparison below.

Method	ψ_1	ψ_2	ψ_3
Exact series, Problem 2.16	.569082647	.720491684	.925781529
Galerkin, $h = 1/4$	.605878583	.757348229	.976137717
Jacobi, iteration 2	.668786843	.841101243	1.115516182
Jacobi, iteration 6	.587144572	.743033367	.960088859
Jacobi, iteration 10	.573384927	.725179706	.932570355
Gauss–Seidel, iteration 2	.601302817	.752097624	.957558112
Gauss–Seidel, iteration 6	.571530449	.722284498	.927753137
Gauss–Seidel, iteration 10	.570504738	.721316714	.926773768
Improved stencil, converged	.570470095	.721284028	.926740690

The Galerkin values exceed the exact values by approximately 6.47%, 5.12%, and 5.44%, respectively, whereas the converged improved stencil errors are 0.244%, 0.110%, and 0.104%. This comparison separates the error in solving a discrete system from its spatial approximation error: more iterations converge to the stencil's fixed solution, not to the continuum solution. The Galerkin calculation is also a different discrete approximation, using piecewise bilinear tents and a variational integral; on this coarse grid it is less accurate at these nodes. This is not a general failure of finite elements: mesh refinement, or richer basis functions, changes that approximation. Galerkin stationarity controls the integrated field-energy error, not the sign or size of each nodal potential error.

3 Chapter 3. Boundary-Value Problems in Electrostatics: II

Problems

3.1 Two concentric spheres have radii a, b ($b > a$) and each is divided into two hemispheres by the same horizontal plane. The upper hemisphere of the inner sphere and the lower hemisphere of the outer sphere are maintained at potential V. The other hemispheres are at zero potential.

Determine the potential in the region $a \le r \le b$ as a series in Legendre polynomials. Include terms at least up to $l = 4$. Check your solution against known results in the limiting cases $b \to \infty$, and $a \to 0$.

Solution.

Put $\mu = \cos\theta$ and expand the boundary values as $V_a(\mu) = \sum_l v_l P_l(\mu)$ and $V_b(\mu) = \sum_l w_l P_l(\mu)$. Orthogonality gives

$$v_l = \frac{(2l+1)V}{2}\int_0^1 P_l(u)\,du, \qquad w_l = (-1)^l v_l.$$

For $l \ge 1$, integrate $(2l+1)P_l = (P_{l+1} - P_{l-1})'$ to obtain

$$v_l = \frac{V}{2}[P_{l-1}(0) - P_{l+1}(0)].$$

Thus $v_0 = V/2$, $v_1 = 3V/4$, $v_3 = -7V/16$, and all even $l > 0$ coefficients vanish. For each angular order, Laplace's equation gives $f_l(r) = \alpha_l r^l + \beta_l r^{-l-1}$. The two boundary equations, after multiplication by a^{l+1} and b^{l+1}, are

$$\alpha_l a^{2l+1} + \beta_l = a^{l+1} v_l, \qquad \alpha_l b^{2l+1} + \beta_l = b^{l+1} w_l.$$

Writing $D_l = b^{2l+1} - a^{2l+1}$, subtraction and back-substitution give

$$\alpha_l = \frac{b^{l+1}w_l - a^{l+1}v_l}{D_l}, \qquad \beta_l = \frac{a^{l+1}b^{2l+1}v_l - b^{l+1}a^{2l+1}w_l}{D_l}.$$

Thus $f_l = v_l A_l + w_l B_l$, where the two radial interpolants are

$$A_l(r) = \frac{a^{l+1}(b^{2l+1} - r^{2l+1})}{r^{l+1}(b^{2l+1} - a^{2l+1})}, \qquad B_l(r) = \frac{b^{l+1}(r^{2l+1} - a^{2l+1})}{r^{l+1}(b^{2l+1} - a^{2l+1})}.$$

Each is a linear combination of r^l and r^{-l-1}, and $(A_l(a), A_l(b)) = (1, 0)$, $(B_l(a), B_l(b)) = (0, 1)$. Consequently

$$\boxed{\Phi = \frac{V}{2} + \sum_{\substack{l \ge 1 \\ l \text{ odd}}} \frac{V}{2}[P_{l-1}(0) - P_{l+1}(0)][A_l(r) - B_l(r)]P_l(\mu).}$$

Through $l = 4$ this is

$$\Phi = \frac{V}{2} + \frac{3V}{4}(A_1 - B_1)\mu - \frac{7V}{16}(A_3 - B_3)\frac{5\mu^3 - 3\mu}{2} + \text{terms with } l \ge 5.$$

The $l = 0$ contribution is constant because $A_0 + B_0 = 1$. At a boundary jump the Legendre series takes the mean value $V/2$.

For fixed a, r, $b \to \infty$ gives $A_l \to (a/r)^{l+1}$ and $B_l \to 0$ for $l > 0$. The limit is therefore the exterior hemispherical solution with asymptotic potential $V/2$; the distant outer boundary retains this constant offset. In the limit $a \to 0$ at fixed r, b, $A_l \to 0$ and $B_l \to (r/b)^l$ for $l > 0$, giving the regular interior solution with the lower hemisphere of the sphere at V.

3.2 A spherical surface of radius R has charge uniformly distributed over its surface with a density $Q/4\pi R^2$, except for a spherical cap at the north pole, defined by the cone $\theta = \alpha$.

(a) Show that the potential inside the spherical surface can be expressed as

$$\Phi = \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \left[P_{l+1}(\cos\alpha) - P_{l-1}(\cos\alpha)\right] \frac{r^l}{R^{l+1}} P_l(\cos\theta)$$

where, for $l = 0$, $P_{l-1}(\cos\alpha) = -1$. What is the potential outside?

(b) Find the magnitude and the direction of the electric field at the origin.

(c) Discuss the limiting forms of the potential (part a) and electric field (part b) as the spherical cap becomes (1) very small, and (2) so large that the area with charge on it becomes a very small cap at the south pole.

Solution.

Write $c = \cos\alpha$; the charged portion is $-1 \le \mu' \le c$. The symbol Q specifies the original full-sphere density: the charge actually present is $Q_{\rm rem} = Q(1+c)/2$. Averaging the Coulomb expansion over the source azimuth gives

$$\int_0^{2\pi} P_l(\cos\gamma)\, d\phi' = 2\pi P_l(\mu) P_l(\mu').$$

(a) Hence

$$\Phi = \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} P_l(\mu) \frac{r_<^l}{r_>^{l+1}} \int_{-1}^{c} P_l(u)\, du, \quad r_< = \min(r,R), \quad r_> = \max(r,R).$$

For $l \ge 1$ the integral is $[P_{l+1}(c) - P_{l-1}(c)]/(2l+1)$ because the lower-end terms cancel. For $l = 0$ it is $1 + c$, reproduced by the stipulated $P_{-1} = -1$. This proves the displayed interior expression. The exterior expression is the same sum with r^l/R^{l+1} replaced by R^l/r^{l+1}. Continuity at $r = R$ follows term by term; its radial derivative has the jump $\partial_r\Phi_{\rm out} - \partial_r\Phi_{\rm in} = -\sigma/\epsilon_0$.

(b) Only $l = 1$ contributes to the gradient at the origin. Since $P_2(c) - P_0(c) = -3(1-c^2)/2$,

$$\Phi = \frac{Q(1+c)}{8\pi\epsilon_0 R} - \frac{Q\sin^2\alpha}{16\pi\epsilon_0 R^2} z + O(r^2), \qquad \boxed{\mathbf{E}(0) = \frac{Q\sin^2\alpha}{16\pi\epsilon_0 R^2}\,\hat{\mathbf{z}}.}$$

For positive Q the field points toward the missing north cap; reversing Q reverses its direction.

(c) For $\alpha \ll 1$, the missing charge is $\delta Q = Q(1-c)/2 = Q\alpha^2/4 + O(\alpha^4)$. Away from the shrinking cap its potential is that of a point charge at $R\hat{\mathbf{z}}$ to leading order. Thus

$$\Phi = \frac{Q}{4\pi\epsilon_0 \max(r,R)} - \frac{Q\alpha^2}{16\pi\epsilon_0 |\mathbf{r} - R\hat{\mathbf{z}}|} + O(\alpha^4), \qquad \mathbf{E}(0) = \frac{Q\alpha^2}{16\pi\epsilon_0 R^2}\hat{\mathbf{z}} + O(\alpha^4).$$

For $\beta = \pi - \alpha \ll 1$, only the south cap remains, with $Q_{\rm rem} = Q\beta^2/4 + O(\beta^4)$, and

$$\Phi = \frac{Q\beta^2}{16\pi\epsilon_0 |\mathbf{r} + R\hat{\mathbf{z}}|} + O(\beta^4), \qquad \mathbf{E}(0) = \frac{Q\beta^2}{16\pi\epsilon_0 R^2}\hat{\mathbf{z}} + O(\beta^4).$$

These expansions are not uniform at the cap itself. With the specified fixed density the remaining charge and field tend to zero; if instead its total charge is held fixed, the south-cap limit is an ordinary point charge of that fixed magnitude.

3.3 A thin, flat, conducting, circular disc of radius R is located in the x-y plane with its center at the origin, and is maintained at a fixed potential V. With the information that the charge density on a disc at fixed potential is proportional to $(R^2-\rho^2)^{-1/2}$, where ρ is the distance out from the center of the disc,

(a) show that for $r > R$ the potential is

$$\Phi(r,\theta,\phi) = \frac{2V}{\pi}\frac{R}{r}\sum_{l=0}^{\infty}\frac{(-1)^l}{2l+1}\left(\frac{R}{r}\right)^{2l}P_{2l}(\cos\theta)$$

(b) find the potential for $r < R$.

(c) What is the capacitance of the disc?

Solution.

Let $\sigma(\rho) = C_0/\sqrt{R^2-\rho^2}$ denote the total sheet charge density, including both faces. At the center,

$$V = \frac{1}{4\pi\epsilon_0}\int_0^R \frac{2\pi\rho\, C_0\, d\rho}{\rho\sqrt{R^2-\rho^2}} = \frac{\pi C_0}{4\epsilon_0}, \quad C_0 = \frac{4\epsilon_0 V}{\pi}.$$

(a) The azimuthal average of the Coulomb expansion yields

$$\Phi = \frac{C_0}{2\epsilon_0}\sum_{n=0}^{\infty}\frac{P_{2n}(0)P_{2n}(\cos\theta)}{r^{2n+1}}\int_0^R\frac{\rho^{2n+1}\,d\rho}{\sqrt{R^2-\rho^2}}.$$

Odd orders vanish. With $t = \rho^2/R^2$ the integral equals

$$\frac{R^{2n+1}}{2}B(n+1,\tfrac{1}{2}) = R^{2n+1}\frac{2^{2n}(n!)^2}{(2n+1)!}.$$

Multiplication by $P_{2n}(0) = (-1)^n(2n)!/[2^{2n}(n!)^2]$ gives $(-1)^n R^{2n+1}/(2n+1)$ and hence the required series.

(b) First compute the potential on the positive axis:

$$\Phi(0,z) = \frac{2V}{\pi}\int_0^R\frac{\rho\,d\rho}{\sqrt{R^2-\rho^2}\sqrt{\rho^2+z^2}} = \frac{2V}{\pi}\tan^{-1}\frac{R}{z} = V - \frac{2V}{\pi}\tan^{-1}\frac{z}{R}.$$

For example, $u = \sqrt{R^2-\rho^2}$ changes the integral to $\int_0^R du/\sqrt{R^2+z^2-u^2}$, giving the stated inverse tangent. In the upper half-ball expand the last expression in powers of z/R. The unique regular, axially symmetric harmonic extension of z^l is $r^l P_l(\cos\theta)$. Reflection in the disc then gives

$$\boxed{\Phi = V - \frac{2V}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\left(\frac{r}{R}\right)^{2n+1}P_{2n+1}(|\cos\theta|), \qquad r<R.}$$

On the disc all odd Legendre polynomials vanish at zero, so $\Phi = V$. The absolute value is essential: there is a normal-derivative jump at the charged sheet and no single harmonic power series through it.

(c) Integration of the sheet density gives

$$Q = 2\pi C_0\int_0^R\frac{\rho\,d\rho}{\sqrt{R^2-\rho^2}} = 8\epsilon_0 RV, \qquad \boxed{C = 8\epsilon_0 R.}$$

The exterior monopole term is correspondingly $Q/(4\pi\epsilon_0 r)$.

3.4 The surface of a hollow conducting sphere of inner radius a is divided into an *even number* of equal segments by a set of planes; their common line of intersection is the z axis and they are distributed uniformly in the angle ϕ. (The segments are like the skin on wedges of an apple, or the earth's surface between successive meridians of longitude.) The segments are kept at fixed potentials $\pm V$, alternately.

(a) Set up a series representation for the potential inside the sphere for the general case of $2n$ segments, and carry the calculation of the coefficients in the series far enough to determine exactly which coefficients are different from zero. For the nonvanishing terms, exhibit the coefficients as an integral over $\cos\theta$.

(b) For the special case of $n = 1$ (two hemispheres) determine explicitly the potential up to and including all terms with $l = 3$. By a coordinate transformation verify that this reduces to result (3.36) of Section 3.3.

Solution.

Choose the angular origin so that the boundary value is $V\ \mathrm{sgn}[\sin(n\phi)]$. A rotation about z covers any other placement of the segments. Its Fourier series is

$$V(\phi) = \frac{4V}{\pi}\sum_{s=0}^{\infty}\frac{\sin(m_s\phi)}{2s+1}, \qquad m_s = (2s+1)n.$$

(a) Use real associated Legendre functions with the Condon–Shortley phase, $P_l^m = (-1)^m(1-u^2)^{m/2}d^mP_l/du^m$, as in Jackson's Eq. (3.49), p. 108. Consequently $Y_{lm} = [(2l+1)(l-m)!/(4\pi(l+m)!)]^{1/2}P_l^m e^{im\phi}$ has no additional phase [Eq. (3.53)]. The regular solution and its coefficients are

$$\Phi = \sum_{s=0}^{\infty}\sum_{l=m_s}^{\infty} c_{lm_s}(r/a)^l P_l^{m_s}(\cos\theta)\sin(m_s\phi),$$

$$c_{lm} = \frac{2V(2l+1)}{\pi(2s+1)}\frac{(l-m)!}{(l+m)!}\int_{-1}^{1}P_l^m(u)\,du, \qquad m = (2s+1)n.$$

Here the factor $(2l+1)(l-m)!/[2(l+m)!]$ is the inverse associated-Legendre norm. All cosine modes, $m = 0$ modes, and all other azimuthal orders vanish. Parity gives $P_l^m(-u) = (-1)^{l+m}P_l^m(u)$, so $l+m$ must be even. To prove that there are no further zeros, put $I_{lm} = \int_{-1}^{1}P_l^m(u)\,du$. The associated-Legendre identities

$$(1-u^2)(P_l^m)' = -luP_l^m + (l+m)P_{l-1}^m, \qquad (2l+1)uP_l^m = (l-m+1)P_{l+1}^m + (l+m)P_{l-1}^m$$

give, after integration by parts in the first identity,

$$(l+2)\int_{-1}^{1}uP_l^m\,du = (l+m)I_{l-1,m}, \qquad I_{l+1,m} = \frac{(l-1)(l+m)}{(l+2)(l-m+1)}I_{l-1,m}.$$

The boundary term vanishes because $(1-u^2)P_l^m \to 0$. Relabeling $l = L+1$ yields

$$I_{L+2,m} = \frac{L(L+m+1)}{(L+3)(L-m+2)}I_{L,m}.$$

The starting value follows from $P_m^m = (-1)^m(2m-1)!!(1-u^2)^{m/2}$:

$$I_{mm} = (-1)^m(2m-1)!!\,B(1/2, 1+m/2) \neq 0.$$

Every multiplier in the recurrence is positive for $L \geq m \geq 1$; hence every $l = m, m+2, \ldots$ survives. Iterating the recurrence and using $\Gamma(x+1) = x\Gamma(x)$ also gives the closed form

$$\int_{-1}^{1} P_l^m(u)\, du = (-1)^m 2^{m-1} m\, \frac{\Gamma(l/2)\Gamma((l+m+1)/2)}{\Gamma((l-m+2)/2)\Gamma((l+3)/2)},$$

whose gamma factors are finite and positive. This also specifies unambiguously which coefficients in the integral formula survive.

(b) For $n = 1$, the required integrals are

$$\int P_1^1\, du = -\frac{\pi}{2}, \qquad \int P_3^1\, du = -\frac{3\pi}{16}, \qquad \int P_3^3\, du = -\frac{45\pi}{8},$$

where all integrals run from -1 to 1. Therefore, through $l = 3$,

$$\Phi = -\frac{3V}{2}\frac{r}{a}P_1^1(\cos\theta)\sin\phi - \left(\frac{r}{a}\right)^3 \left[\frac{7V}{32}P_3^1(\cos\theta)\sin\phi + \frac{7V}{192}P_3^3(\cos\theta)\sin 3\phi\right] + O((r/a)^5).$$

The new polar axis may be chosen along $+y$, so $\cos\Theta = y/r = \sin\theta\sin\phi$. Substitution of $P_1^1 = -\sin\theta$, $P_3^1 = -\frac{3}{2}(5\cos^2\theta - 1)\sin\theta$, and $P_3^3 = -15\sin^3\theta$, with $\sin 3\phi = 3\sin\phi - 4\sin^3\phi$, reduces this to

$$\Phi = V\left[\frac{3}{2}\frac{r}{a}P_1(\cos\Theta) - \frac{7}{8}\left(\frac{r}{a}\right)^3 P_3(\cos\Theta)\right] + O((r/a)^5),$$

which is Eq. (3.36) in the rotated coordinates.

3.5 A hollow sphere of inner radius a has the potential specified on its surface to be $\Phi = V(\theta, \phi)$. Prove the equivalence of the two forms of solution for the potential inside the sphere:

(a) $\Phi(\mathbf{x}) = \dfrac{a(a^2 - r^2)}{4\pi} \displaystyle\int \frac{V(\theta', \phi')}{(r^2 + a^2 - 2ar\cos\gamma)^{3/2}}\, d\Omega'$

where $\cos\gamma = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi - \phi')$.

(b) $\Phi(\mathbf{x}) = \displaystyle\sum_{l=0}^{\infty}\sum_{m=-l}^{l} A_{lm}\left(\frac{r}{a}\right)^l Y_{lm}(\theta, \phi)$

where $A_{lm} = \int d\Omega'\, Y_{lm}^*(\theta', \phi')V(\theta', \phi')$.

Solution.

Let $t = r/a < 1$ and $u = \cos\gamma$. The Legendre generating function is $F(t, u) = (1 - 2tu + t^2)^{-1/2} = \sum_{l\geq 0} t^l P_l(u)$. Differentiate within its convergent power series:

$$F + 2t\partial_t F = \frac{1 - t^2}{(1 - 2tu + t^2)^{3/2}} = \sum_{l=0}^{\infty}(2l+1)t^l P_l(u).$$

Consequently the kernel in (a) is

$$\frac{a(a^2 - r^2)}{4\pi(a^2 + r^2 - 2ar\cos\gamma)^{3/2}} = \sum_{l=0}^{\infty}\frac{2l+1}{4\pi}(r/a)^l P_l(\cos\gamma).$$

Use the addition theorem

$$\frac{2l+1}{4\pi}P_l(\cos\gamma) = \sum_{m=-l}^{l} Y_{lm}(\theta,\phi)Y_{lm}^*(\theta',\phi').$$

Inserting it into (a), interchanging the integral and the uniformly convergent series on any closed ball $r \le r_0 < a$, and grouping the source integral gives exactly (b) with its stated A_{lm}. Conversely, insert those coefficients into (b) and sum the same kernel to recover (a). For an integrable boundary value this proves the interior equivalence; for continuous boundary data the limiting Poisson integral takes the assigned surface value. Constant data provide a normalization check: only $l = 0$ remains and $\Phi = V$.

3.6 Two point charges q and $-q$ are located on the z axis at $z = +a$ and $z = -a$, respectively.

(a) Find the electrostatic potential as an expansion in spherical harmonics and powers of r for both $r > a$ and $r < a$.

(b) Keeping the product $qa \equiv p/2$ constant, take the limit of $a \to 0$ and find the potential for $r \neq 0$. This is by definition a dipole along the z axis and its potential.

(c) Suppose now that the dipole of part b is surrounded by a *grounded* spherical shell of radius b concentric with the origin. By linear superposition find the potential everywhere inside the shell.

Solution.

(a) The direct potential is

$$\Phi = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\mathbf{r} - a\hat{\mathbf{z}}|} - \frac{1}{|\mathbf{r} + a\hat{\mathbf{z}}|}\right).$$

In the Coulomb expansion the two source directions contribute $P_l(\cos\theta)$ and $P_l(-\cos\theta) = (-1)^l P_l(\cos\theta)$. Even orders cancel, giving

$$\boxed{\Phi = \frac{q}{2\pi\epsilon_0}\sum_{j=0}^{\infty} P_{2j+1}(\cos\theta)\begin{cases} a^{2j+1}/r^{2j+2}, & r > a, \\ r^{2j+1}/a^{2j+2}, & r < a. \end{cases}}$$

This is a spherical-harmonic expansion because $P_l(\cos\theta) = \sqrt{4\pi/(2l+1)}Y_{l0}$; all $m \neq 0$ coefficients vanish. The source points themselves are excluded.

(b) With $2qa = p$ fixed, the $l = 1$ coefficient remains finite and every higher term has an extra factor a^{2j}. At each fixed $r > 0$,

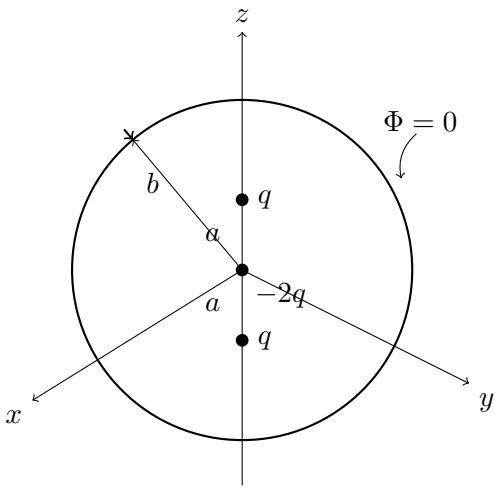
$$\boxed{\Phi_{\text{dip}} = \frac{p\cos\theta}{4\pi\epsilon_0 r^2}.}$$

(c) A regular $l = 1$ harmonic $Ar\cos\theta$ can cancel the dipole's surface value. The equation $p/(4\pi\epsilon_0 b^2) + Ab = 0$ fixes $A = -p/(4\pi\epsilon_0 b^3)$. Thus

$$\boxed{\Phi = \frac{p\cos\theta}{4\pi\epsilon_0}\left(\frac{1}{r^2} - \frac{r}{b^3}\right), \qquad 0 < r < b.}$$

The added field is the uniform field $+p\hat{\mathbf{z}}/(4\pi\epsilon_0 b^3)$; it is regular at the source and makes the total potential vanish at $r = b$.

3.7 Three point charges (q, $-2q$, q) are located in a straight line with separation a and with the middle charge ($-2q$) at the origin of a grounded conducting spherical shell of radius b, as indicated in the sketch.



Problem 3.7

(a) Write down the potential of the three charges in the absence of the grounded sphere. Find the limiting form of the potential as $a \to 0$, but the product $qa^2 = Q$ remains finite. Write this latter answer in spherical coordinates.

(b) The presence of the grounded sphere of radius b alters the potential for $r < b$. The added potential can be viewed as caused by the surface-charge density induced on the inner surface at $r = b$ or by image charges located at $r > b$. Use linear superposition to satisfy the boundary conditions and find the potential everywhere inside the sphere for $r < a$ and $r > a$. Show that in the limit $a \to 0$,

$$\Phi(r, \theta, \phi) \to \frac{Q}{2\pi\epsilon_0 r^3}\left(1 - \frac{r^5}{b^5}\right) P_2(\cos\theta)$$

Solution.

(a) Before inserting the conductor,

$$\Phi_{\text{free}} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{r^2 + a^2 - 2ar\cos\theta}} + \frac{1}{\sqrt{r^2 + a^2 + 2ar\cos\theta}} - \frac{2}{r}\right).$$

For $r > a$ the $l = 0$ terms cancel and odd terms cancel by reflection:

$$\Phi_{\text{free}} = \frac{q}{2\pi\epsilon_0}\sum_{j=1}^{\infty}\frac{a^{2j}}{r^{2j+1}}P_{2j}(\cos\theta).$$

Therefore, for fixed $r > 0$ and $qa^2 = Q$,

$$\Phi_{\text{free}} \longrightarrow \frac{Q}{2\pi\epsilon_0 r^3}P_2(\cos\theta) = \frac{Q(3\cos^2\theta - 1)}{4\pi\epsilon_0 r^3}.$$

(b) Assume $a < b$. The regular harmonic correction that cancels the preceding free-space series at b is

$$\Phi_{\text{ind}} = -\frac{q}{2\pi\epsilon_0}\sum_{j=1}^{\infty}\frac{a^{2j}r^{2j}}{b^{4j+1}}P_{2j}(\cos\theta).$$

It is valid throughout $r < b$. Explicitly the full result is

$$\boxed{\Phi = \frac{q}{2\pi\epsilon_0}\sum_{j=1}^{\infty} a^{2j}\left(\frac{1}{r^{2j+1}} - \frac{r^{2j}}{b^{4j+1}}\right)P_{2j}(\cos\theta), \quad a < r < b,}$$

and

$$\boxed{\Phi = \frac{q}{2\pi\epsilon_0}\left[\frac{1}{a} - \frac{1}{r} + \sum_{j=1}^{\infty} r^{2j}\left(\frac{1}{a^{2j+1}} - \frac{a^{2j}}{b^{4j+1}}\right)P_{2j}(\cos\theta)\right], \quad 0 < r < a.}$$

An independent image construction gives image charges $-qb/a$ at $z = \pm b^2/a$ and the constant $2q/(4\pi\epsilon_0 b)$ for the central charge. The monopoles of the two images cancel that constant, leaving precisely Φ_{ind}. In the quadrupole limit only $j = 1$ survives, and

$$\boxed{\Phi \longrightarrow \frac{Q}{2\pi\epsilon_0 r^3}\left(1 - \frac{r^5}{b^5}\right)P_2(\cos\theta).}$$

All expressions exclude the point-charge singularities; the series at $r = a$ can be understood by a limit away from the sources.

3.8 There is a puzzling aspect of the solution (3.136) for the potential inside a grounded sphere with a uniformly charged wire along a diameter. Very close to the wire (i.e., for $\rho = r\sin\theta \ll b$), the potential should be that of a uniformly charged wire, namely, $\Phi = (Q/4\pi\epsilon_0 b)\ln(b/\rho) + \Phi_0$. The solution (3.136) does not explicitly have this behavior.

(a) Show by use of the Legendre differential equation (3.10) and some integration by parts, that $\ln(\text{cosec}\,\theta)$ has the appropriate expansion in spherical harmonics to permit the solution (3.136) to be written in the alternative form,

$$\Phi(\mathbf{x}) = \frac{Q}{4\pi\epsilon_0 b}\left\{\ln\left(\frac{2b}{r\sin\theta}\right) - 1 - \sum_{j=1}^{\infty}\frac{4j+1}{2j(2j+1)}\left(\frac{r}{b}\right)^{2j}P_{2j}(\cos\theta)\right\}$$

in which the expected behavior near the wire is manifest. Give an interpretation of the constant term $\Phi_0 = -Q/4\pi\epsilon_0 b$. Note that in this form, for any $r/b < 1$ the Legendre polynomial series is rapidly convergent at all angles.

(b) Show by use of the expansion (3.38) that

$$\frac{1}{2}\left(\frac{1}{\sin(\theta/2)} + \frac{1}{\cos(\theta/2)}\right) = 2\sum_{j=0}^{\infty}P_{2j}(\cos\theta)$$

and that therefore the charge density on the inner surface of the sphere, Eq. (3.137), can be expressed alternatively as

$$\sigma(\theta) = -\frac{Q}{4\pi b^2}\left\{\frac{1}{2}\left(\frac{1}{\sin(\theta/2)} + \frac{1}{\cos(\theta/2)}\right) - \sum_{j=0}^{\infty}\frac{1}{2j+1}P_{2j}(\cos\theta)\right\}$$

The (integrable) singular behavior at $\theta = 0$ and $\theta = \pi$ is now exhibited explicitly. The series provides corrections in $\ln(1/\theta)$ as $\theta \to 0$.

Solution.

(a) Put $x = \cos\theta$ and $f(x) = -\frac{1}{2}\ln(1-x^2)$. Its constant Legendre coefficient is

$$\frac{1}{2}\int_{-1}^{1} f(x)\,dx = 1 - \ln 2.$$

For $l > 0$, use $[(1-x^2)P_l']' = -l(l+1)P_l$ and integrate by parts. The boundary term $f(1-x^2)P_l'$ vanishes, although f diverges:

$$\begin{aligned} l(l+1)\int_{-1}^{1} fP_l\,dx &= \int_{-1}^{1} f'(1-x^2)P_l'\,dx = \int_{-1}^{1} xP_l'\,dx \\ &= [xP_l]_{-1}^{1} - \int_{-1}^{1} P_l\,dx = 1 + (-1)^l. \end{aligned}$$

Thus, for $0 < \theta < \pi$,

$$\ln\csc\theta = 1 - \ln 2 + \sum_{j=1}^{\infty} \frac{4j+1}{2j(2j+1)} P_{2j}(\cos\theta).$$

Equation (3.136), checked in the text, is

$$\Phi = \frac{Q}{4\pi\epsilon_0 b}\left[\ln\frac{b}{r} + \sum_{j=1}^{\infty}\frac{4j+1}{2j(2j+1)}\{1-(r/b)^{2j}\}P_{2j}(\cos\theta)\right].$$

Replacing its r-independent series by the logarithm proves the requested alternative expression.

The constant $-Q/(4\pi\epsilon_0 b)$ is the induced potential at the center: each line element dq at z' has an image $-b\,dq/|z'|$ at radius $b^2/|z'|$, contributing $-dq/(4\pi\epsilon_0 b)$ there. Their sum is the stated constant. Near the center the bare finite-wire potential is $[Q/(4\pi\epsilon_0 b)]\ln(2b/\rho)$, explaining the $2b$ in the logarithm. If one instead writes $\ln(b/\rho)$, its additive constant must also include $[Q/(4\pi\epsilon_0 b)]\ln 2$. At noncentral positions the regular induced series contributes too; one should not interpret the displayed constant as the entire position-independent remainder everywhere along the wire.

(b) The generating function, first with $0 < t < 1$, gives

$$2\sum_{j=0}^{\infty} t^{2j}P_{2j}(x) = \frac{1}{\sqrt{1-2tx+t^2}} + \frac{1}{\sqrt{1+2tx+t^2}}.$$

Taking $t \uparrow 1$ away from the poles gives

$$2\sum_{j=0}^{\infty} P_{2j}(\cos\theta) = \frac{1}{2}\left(\csc\frac{\theta}{2} + \sec\frac{\theta}{2}\right).$$

The radial derivative of (3.136), with the inward-facing conductor normal accounted for, is

$$\sigma = \epsilon_0\partial_r\Phi\big|_{b^-} = -\frac{Q}{4\pi b^2}\sum_{j=0}^{\infty}\frac{4j+1}{2j+1}P_{2j}(\cos\theta).$$

Use $(4j+1)/(2j+1) = 2 - 1/(2j+1)$ to obtain the stated result. Near a pole its leading term is proportional to $-1/\theta$; the surface measure $\sin\theta\,d\theta$ makes it integrable. The correction series equals $\int_0^1 \sum_j t^{2j}P_{2j}(\cos\theta)\,dt$; the region $t \simeq 1$ has integrand proportional to $[(1-t)^2+\theta^2]^{-1/2}$ and therefore contributes a logarithm. Only the $j = 0$ term survives the full surface integral, giving the necessary check $\int \sigma\,dS = -Q$.

3.9 A hollow right circular cylinder of radius b has its axis coincident with the z axis and its ends at $z = 0$ and $z = L$. The potential on the end faces is zero, while the potential on the cylindrical surface is given as $V(\phi, z)$. Using the appropriate separation of variables in cylindrical coordinates, find a series solution for the potential anywhere inside the cylinder.

Solution.

Write $\Phi = R(\rho)F(\phi)Z(z)$ in Laplace's equation. Periodicity requires $F = e^{im\phi}$, $m \in \mathbb{Z}$; zero potentials at both ends require $Z = \sin(k_n z)$, $k_n = n\pi/L$, $n \geq 1$. The remaining radial equation is

$$R'' + \frac{1}{\rho}R' - \left(k_n^2 + \frac{m^2}{\rho^2}\right) R = 0.$$

Its solutions are $I_{|m|}(k_n\rho)$ and $K_{|m|}(k_n\rho)$. Regularity at the axis excludes K, so normalize each radial function to unity at b. The result is

$$\boxed{\Phi(\rho, \phi, z) = \sum_{m=-\infty}^{\infty} \sum_{n=1}^{\infty} v_{mn} \frac{I_{|m|}(k_n\rho)}{I_{|m|}(k_n b)} e^{im\phi} \sin(k_n z),}$$

$$v_{mn} = \frac{1}{\pi L} \int_0^{2\pi} d\phi' \int_0^L dz'\, V(\phi', z') e^{-im\phi'} \sin(k_n z').$$

The normalization follows from $\int_0^{2\pi} e^{i(m-m')\phi} d\phi = 2\pi\delta_{mm'}$ and $\int_0^L \sin(k_n z) \sin(k_{n'} z) dz = L\delta_{nn'}/2$. For real boundary data $v_{-m,n} = v_{mn}^*$, so the potential is real. The series solves the interior problem even if the assigned side and end values disagree at the circular rims; those boundary jumps are understood as one-sided limits.

3.10 For the cylinder in Problem 3.9 the cylindrical surface is made of two equal half-cylinders, one at potential V and the other at potential $-V$, so that

$$V(\phi, z) = \begin{cases} V & \text{for } -\pi/2 < \phi < \pi/2 \\ -V & \text{for } \pi/2 < \phi < 3\pi/2 \end{cases}$$

(a) Find the potential inside the cylinder.

(b) Assuming $L \gg b$, consider the potential at $z = L/2$ as a function of ρ and ϕ and compare it with two-dimensional Problem 2.13.

Solution.

(a) The two needed Fourier expansions are

$$\operatorname{sgn}(\cos\phi) = \frac{4}{\pi} \sum_{s=0}^{\infty} \frac{(-1)^s}{2s+1} \cos[(2s+1)\phi], \qquad 1 = \frac{4}{\pi} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{1}{n} \sin\frac{n\pi z}{L} \quad (0 < z < L).$$

Substitution in Problem 3.9 gives

$$\boxed{\Phi = \frac{16V}{\pi^2} \sum_{s=0}^{\infty} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{(-1)^s}{(2s+1)n} \frac{I_{2s+1}(n\pi\rho/L)}{I_{2s+1}(n\pi b/L)} \cos[(2s+1)\phi] \sin\frac{n\pi z}{L}.}$$

At the side it gives $V\,\mathrm{sgn}(\cos\phi)$, and each term vanishes at both ends.

(b) Far from the end faces the two-dimensional solution is

$$\Phi_\infty = \frac{4V}{\pi}\sum_{s=0}^{\infty}\frac{(-1)^s}{2s+1}(\rho/b)^{2s+1}\cos[(2s+1)\phi] = \frac{4V}{\pi}\,\mathrm{Re}\tan^{-1}\left(\frac{\rho}{b}e^{i\phi}\right).$$

For $\rho < b$ a manifestly real form is

$$\boxed{\Phi_\infty = \frac{2V}{\pi}\tan^{-1}\left(\frac{2(\rho/b)\cos\phi}{1-(\rho/b)^2}\right).}$$

To justify the long-cylinder limit without interchanging nonuniform sums, set $u = \Phi - \Phi_\infty$. This harmonic correction is zero on the side and equals $-\Phi_\infty$ on both end faces. Expand its end data in transverse Dirichlet modes $J_m(x_{mn}\rho/b)e^{im\phi}$, where $J_m(x_{mn}) = 0$. Each mode has axial factor

$$\frac{\cosh[x_{mn}(z-L/2)/b]}{\cosh[x_{mn}L/(2b)]}.$$

At $z = L/2$ all such corrections decrease exponentially with L/b. The smallest allowed azimuthal order is $m = 1$; hence the leading possible correction is of order $e^{-x_{11}L/(2b)}$ in the interior. The surviving limit is exactly the two-dimensional hemisplit-circle potential in Problem 2.13.

3.11 A modified Bessel–Fourier series on the interval $0 \le \rho \le a$ for an arbitrary function $f(\rho)$ can be based on the "homogeneous" boundary conditions:

$$\text{At } \rho = 0, \qquad \rho\,J_\nu(k\rho)\,\frac{dJ_\nu(k'\rho)}{d\rho} = 0$$

$$\text{At } \rho = a, \qquad \frac{d}{d\rho}\ln[J_\nu(k\rho)] = -\frac{\lambda}{a} \qquad (\lambda \text{ real})$$

The first condition restricts ν. The second condition yields eigenvalues $k = y_{\nu n}/a$, where $y_{\nu n}$ is the nth positive root of $x\,dJ_\nu(x)/dx + \lambda J_\nu(x) = 0$.

(a) Show that the Bessel functions of different eigenvalues are orthogonal in the usual way.

(b) Find the normalization integral and show that for a regular real order $\nu \ge 0$ and $\lambda > -\nu$, an arbitrary function $f(\rho)$ can be expanded on the interval in the modified Bessel–Fourier series

$$f(\rho) = \sum_{n=1}^{\infty} A_n J_\nu\left(\frac{y_{\nu n}\rho}{a}\right)$$

with the coefficients A_n given by

$$A_n = \frac{2}{a^2}\left[\left(1-\frac{\nu^2}{y_{\nu n}^2}\right)J_\nu^2(y_{\nu n}) + \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}}\right)^2\right]^{-1}\int_0^a f(\rho)\,\rho\,J_\nu\left(\frac{y_{\nu n}\rho}{a}\right)d\rho$$

The dependence on λ is implicit in this form, but the square bracket has alternative forms:

$$\begin{aligned}\left[\left(1-\frac{\nu^2}{y_{\nu n}^2}\right)J_\nu^2(y_{\nu n}) + \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}}\right)^2\right] &= \left(1+\frac{\lambda^2-\nu^2}{y_{\nu n}^2}\right)J_\nu^2(y_{\nu n}) \\ &= \left(1+\frac{y_{\nu n}^2-\nu^2}{\lambda^2}\right)\left[\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}}\right]^2 \\ &= \left[J_\nu^2(y_{\nu n}) - J_{\nu-1}(y_{\nu n})J_{\nu+1}(y_{\nu n})\right]\end{aligned}$$

For $\lambda \to \infty$ we recover the result of (3.96) and (3.97). The choice $\lambda = 0$ is another simple alternative.

For $\lambda \le -\nu$, determine and add the zero or negative eigenvalue mode needed to complete the positive-root expansion. In particular, include the constant mode for $\nu = \lambda = 0$. This qualification corrects the positive-root completeness claim; the original problem allows arbitrary real λ.

Solution.

Take a real order $\nu \ge 0$ and the regular radial solution at the origin. For $\nu > 0$, $\rho J_\nu(k\rho)\partial_\rho J_\nu(k'\rho) = O(\rho^{2\nu}) \to 0$; for $\nu = 0$ the derivative starts at order ρ, also giving zero. A general real negative noninteger order does not obey the stated endpoint condition.

(a) Let $u = J_\nu(k\rho)$, $v = J_\nu(k'\rho)$. Subtract their Sturm–Liouville equations to obtain

$$(k^2 - k'^2)\int_0^a \rho uv\, d\rho = [\rho(uv' - vu')]_0^a.$$

The lower endpoint vanishes as above; at a both functions satisfy $u' = -\lambda u/a$, $v' = -\lambda v/a$, so the upper endpoint vanishes as well. Distinct eigenvalues are therefore orthogonal in the weight ρ.

(b) The Bessel equation directly implies

$$\frac{d}{dx}\left\{\frac{1}{2}\left[(x^2-\nu^2)J_\nu^2(x) + x^2 J_\nu'^2(x)\right]\right\} = xJ_\nu^2(x).$$

The expression in braces vanishes at $x = 0$ for the regular branch. With $x = y\rho/a$, its integral gives

$$N_y = \int_0^a \rho J_\nu^2(y\rho/a)\, d\rho = \frac{a^2}{2}\left[\left(1 - \frac{\nu^2}{y^2}\right)J_\nu^2(y) + J_\nu'^2(y)\right].$$

Projection gives $A_n = N_{y_{\nu n}}^{-1}\int_0^a f(\rho)\rho J_\nu(y_{\nu n}\rho/a)d\rho$, exactly the coefficient displayed in the question. At an eigenvalue $J_\nu'(y) = -\lambda J_\nu(y)/y$, which proves the first two alternative forms (the one dividing by λ^2 is used only for $\lambda \ne 0$). Finally

$$J_{\nu-1} = \frac{\nu}{y}J_\nu + J_\nu', \qquad J_{\nu+1} = \frac{\nu}{y}J_\nu - J_\nu'$$

show that the bracket is $J_\nu^2 - J_{\nu-1}J_{\nu+1}$. As $\lambda \to +\infty$ the positive roots approach the zeros of J_ν and $N_y \to a^2 J_{\nu+1}^2(y)/2$.

Completeness qualification for the stated "λ real." The positive-root series is complete by itself only for $\lambda > -\nu$. This is the reason for the correction to the statement. Indeed, write a regular radial function as $u = \rho^\nu h$; integration by parts gives the exact quadratic form

$$\int_0^a \left(\rho u'^2 + \frac{\nu^2}{\rho}u^2\right)d\rho + \lambda u(a)^2 = \int_0^a \rho^{2\nu+1}h'^2 d\rho + (\nu+\lambda)u(a)^2.$$

At $\lambda = -\nu$ there is a zero mode $u_0 = (\rho/a)^\nu$, with $N_0 = a^2/(2\nu+2)$, and its projection must be added. In particular, $\nu = \lambda = 0$ requires the constant mode; a positive-root Neumann series alone cannot represent a constant. For $\lambda < -\nu$ there is instead one negative eigenvalue $k^2 = -\chi^2/a^2$, whose regular eigenfunction is $I_\nu(\chi\rho/a)$ and whose root satisfies $\chi I_\nu'(\chi) + \lambda I_\nu(\chi) = 0$. To establish uniqueness, write the positive-term series

$$I_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1)}\sum_{k=0}^{\infty}\frac{(x^2/4)^k}{k!(\nu+1)_k},$$

where $(\nu+1)_k = \Gamma(\nu+k+1)/\Gamma(\nu+1)$. Divide each summand by their sum to form weights w_k. Logarithmic differentiation then gives

$$\frac{xI'_\nu}{I_\nu} = \nu + 2\sum_k k w_k, \qquad \frac{d}{d\log x}\left(\frac{xI'_\nu}{I_\nu}\right) = 4\left[\sum_k k^2 w_k - \left(\sum_k k w_k\right)^2\right] > 0.$$

The small-x series gives the limiting value ν, whereas $I_\nu(x) \sim e^x/\sqrt{2\pi x}$ gives divergence as $x \to \infty$. There is therefore exactly one root when $-\lambda > \nu$. Its normalization can be evaluated by differentiating the modified-Bessel equation:

$$N_\chi = \int_0^a \rho I_\nu^2(\chi\rho/a)d\rho = \frac{a^2}{2}\left[\left(1+\frac{\nu^2}{\chi^2}\right)I_\nu^2(\chi) - I_\nu'^2(\chi)\right].$$

Add $I_\nu(\chi\rho/a)N_\chi^{-1}\int_0^a f(s)I_\nu(\chi s/a)s\,ds$ to the positive-root expansion. These additions restore the arbitrary-function claim for all real λ without changing the positive-root formulas.

3.12 An infinite, thin, plane sheet of conducting material has a circular hole of radius a cut in it. A thin, flat disc of the same material and slightly smaller radius lies in the plane, filling the hole, but separated from the sheet by a very narrow insulating ring. The disc is maintained at a fixed potential V, while the infinite sheet is kept at zero potential.

(a) Using appropriate cylindrical coordinates, find an integral expression involving Bessel functions for the potential at any point *above* the plane.

(b) Show that the potential a perpendicular distance z above the *center* of the disc is

$$\Phi_0(z) = V\left(1 - \frac{z}{\sqrt{a^2+z^2}}\right)$$

(c) Show that the potential a perpendicular distance z above the *edge* of the disc is

$$\Phi_a(z) = \frac{V}{2}\left[1 - \frac{kz}{\pi a}K(k)\right]$$

where $k = 2a/(z^2+4a^2)^{1/2}$, and $K(k)$ is the complete elliptic integral of the first kind.

Solution.

The insulating gap is taken in the zero-width idealization, so the boundary value at $z = 0$ is $V\,H(a-\rho)$. **(a)** An axially symmetric decaying harmonic function in $z > 0$ is $\Phi = \int_0^\infty A(k)J_0(k\rho)e^{-kz}dk$. Hankel inversion determines

$$A(k) = k\int_0^\infty \rho\Phi(\rho,0)J_0(k\rho)\,d\rho = Vk\int_0^a \rho J_0(k\rho)\,d\rho = VaJ_1(ka).$$

Therefore

$$\boxed{\Phi(\rho,z) = Va\int_0^\infty J_1(ka)J_0(k\rho)e^{-kz}\,dk.}$$

(b) On the axis, use $aJ_1(ka) = -\partial_k J_0(ka)$. Integration by parts, together with $\int_0^\infty e^{-kz}J_0(ka)dk = (z^2+a^2)^{-1/2}$, gives

$$\Phi(0,z) = V\left[1 - z\int_0^\infty e^{-kz}J_0(ka)dk\right] = V\left(1 - \frac{z}{\sqrt{z^2+a^2}}\right).$$

(c) At $\rho = a$, $2aJ_1(ka)J_0(ka) = -\partial_k[J_0^2(ka)]$, so

$$\Phi(a,z) = \frac{V}{2}\left[1 - z\int_0^\infty e^{-kz}J_0^2(ka)dk\right].$$

The Bessel addition theorem, averaged over angle, yields

$$J_0^2(ka) = \frac{1}{\pi}\int_0^\pi J_0\left(2ka\sin\frac{\psi}{2}\right)d\psi.$$

Perform the k integral first and put $u = (\pi - \psi)/2$:

$$\begin{aligned}\int_0^\infty e^{-kz}J_0^2(ka)dk &= \frac{1}{\pi}\int_0^\pi \frac{d\psi}{\sqrt{z^2 + 4a^2\sin^2(\psi/2)}} \\ &= \frac{2}{\pi\sqrt{z^2+4a^2}}\int_0^{\pi/2}\frac{du}{\sqrt{1-\kappa^2\sin^2 u}} = \frac{\kappa}{\pi a}K(\kappa),\end{aligned}$$

where $\kappa = 2a/\sqrt{z^2+4a^2}$. This proves the stated result. It approaches $V/2$ as $z \downarrow 0$ at the edge; on the axis the limit is V. Both far-field expressions begin with $Va^2/(2z^2)$.

3.13 Solve for the potential in Problem 3.1, using the appropriate Green function obtained in the text, and verify that the answer obtained in this way agrees with the direct solution from the differential equation.

Solution.

Use the convention $\nabla'^2 G_D = -4\pi\delta(\mathbf{r}-\mathbf{r}')$. The Dirichlet Green function between two concentric spheres, Eq. (3.125), can be written

$$G_D = \sum_{l=0}^\infty g_l(r,r')P_l(\cos\gamma), \quad g_l = \frac{(r_<^l - a^{2l+1}r_<^{-l-1})(r_>^{-l-1} - r_>^l/b^{2l+1})}{1-(a/b)^{2l+1}}.$$

Both boundary values vanish; the jump $[r'^2\partial_{r'}g_l]_{r^-}^{r^+} = -(2l+1)$ fixes the Coulomb normalization. Direct differentiation on the source boundary gives

$$\partial_{r'}g_l(r,a) = \frac{2l+1}{a^2}A_l(r), \qquad \partial_{r'}g_l(r,b) = -\frac{2l+1}{b^2}B_l(r),$$

where A_l, B_l are the radial functions in Problem 3.1. The outward normal of the annular volume is $-\hat{\mathbf{r}}'$ on the inner sphere and $+\hat{\mathbf{r}}'$ on the outer sphere. Green's boundary formula therefore reads

$$\Phi = \frac{a^2}{4\pi}\int V_a(\Omega')\partial_{r'}G_D(r,a)\,d\Omega' - \frac{b^2}{4\pi}\int V_b(\Omega')\partial_{r'}G_D(r,b)\,d\Omega'.$$

For axially symmetric data, $\int_0^{2\pi}P_l(\cos\gamma)d\phi' = 2\pi P_l(\mu)P_l(\mu')$. Consequently

$$\Phi = \sum_l [A_l(r)v_l + B_l(r)w_l]P_l(\mu), \quad v_l = \frac{2l+1}{2}\int_{-1}^1 V_a(u)P_l(u)du,$$

with the analogous definition of w_l. For the complementary hemispheres $w_l = (-1)^l v_l$, $v_0 = V/2$, and $v_l = V[P_{l-1}(0) - P_{l+1}(0)]/2$ for $l > 0$. Thus the Green-function solution is term by term the solution of Problem 3.1, including its monopole and limiting cases.

3.14 A line charge of length $2d$ with a total charge Q has a linear charge density varying as $(d^2 - z^2)$, where z is the distance from the midpoint. A grounded, conducting, spherical shell of inner radius $b > d$ is centered at the midpoint of the line charge.

(a) Find the potential everywhere inside the spherical shell as an expansion in Legendre polynomials.

(b) Calculate the surface-charge density induced on the shell.

(c) Discuss your answers to parts a and b in the limit that $d \ll b$.

Solution.

Normalize the line density first:

$$\lambda(s) = \lambda_0(d^2 - s^2), \quad |s| < d, \qquad Q = \lambda_0 \int_{-d}^{d} (d^2 - s^2)ds = \frac{4}{3}\lambda_0 d^3, \quad \lambda_0 = \frac{3Q}{4d^3}.$$

The even multipole moments are

$$M_l = \int_{-d}^{d} \lambda(s) s^l ds = \frac{3Qd^l}{(l+1)(l+3)} \quad (l \text{ even}); \qquad M_l = 0 \quad (l \text{ odd}).$$

(a) A source at $s\hat{\mathbf{z}}$ inside a grounded sphere has radial Green coefficient $r_<^l/r_>^{l+1} - r^l|s|^l/b^{2l+1}$. Pairing the two halves of the line gives

$$\boxed{\Phi = \frac{1}{4\pi\epsilon_0} \sum_{\substack{l \geq 0 \\ l \text{ even}}} \left[C_l(r) - \frac{M_l r^l}{b^{2l+1}} \right] P_l(\cos\theta).}$$

For $d < r < b$, $C_l = M_l/r^{l+1}$. For $0 < r < d$, split the line integral at $|s| = r$ to obtain

$$C_l = 2\lambda_0 \left[\frac{1}{r^{l+1}} \int_0^r (d^2 - s^2) s^l ds + r^l \int_r^d (d^2 - s^2) s^{-l-1} ds \right].$$

Evaluation, including the exceptional logarithmic cases, gives

$$C_0 = 2\lambda_0 \left[d^2 \ln\frac{d}{r} + \frac{d^2}{2} + \frac{r^2}{6} \right],$$

$$C_2 = 2\lambda_0 \left[\frac{5d^2}{6} - \frac{7r^2}{10} - r^2 \ln\frac{d}{r} \right],$$

$$C_l = 2\lambda_0 \left[\frac{d^2}{l+1} - \frac{r^2}{l+3} + \frac{d^2}{l}\left(1 - \frac{r^l}{d^l}\right) - \frac{r^l(d^{2-l} - r^{2-l})}{2-l} \right], \quad l = 4, 6, \ldots.$$

These expressions match M_l/d^{l+1} at $r = d$. The actual charged line is a logarithmic singularity; "everywhere" means all points outside that source, with its divergence specified.

(b) At the inner conducting surface $\sigma = \epsilon_0 \partial_r \Phi(b^-)$, so

$$\boxed{\sigma(\theta) = -\frac{3Q}{4\pi b^2} \sum_{j=0}^{\infty} \frac{4j+1}{(2j+1)(2j+3)} (d/b)^{2j} P_{2j}(\cos\theta).}$$

Its integral is $-Q$, since only $j = 0$ has a nonzero spherical average.

(c) For $d/b \ll 1$ the induced density is nearly uniform:

$$\sigma = -\frac{Q}{4\pi b^2}\left[1 + (d/b)^2 P_2(\cos\theta) + O((d/b)^4)\right].$$

At observation points with $r \gg d$,

$$\Phi = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{r} - \frac{1}{b} + \frac{d^2}{5}\left(\frac{1}{r^3} - \frac{r^2}{b^5}\right)P_2(\cos\theta) + O\big(d^4/r^5 + d^4 r^4/b^9\big)\right].$$

Thus the leading term is a central point charge plus its grounding constant. This point-charge approximation is not uniform within $O(d)$ of the shrinking line.

3.15 Consider the following "spherical cow" model of a battery connected to an external circuit. A sphere of radius a and conductivity σ is embedded in a uniform medium of conductivity σ'. Inside the sphere there is a uniform (chemical) force in the z direction acting on the charge carriers; its strength as an effective electric field entering Ohm's law is F. In the steady state, electric fields exist inside and outside the sphere and surface charge resides on its surface.

(a) Find the electric field (in addition to F) and current density everywhere in space. Determine the surface-charge density and show that the electric dipole moment of the sphere is $p = 4\pi\epsilon_0\sigma a^3 F/(\sigma + 2\sigma')$.

(b) Show that the total current flowing out through the upper hemisphere of the sphere is

$$I = \frac{2\sigma\sigma'}{\sigma + 2\sigma'} \cdot \pi a^2 F$$

Calculate the total power dissipation outside the sphere. Using the lumped circuit relations, $P = I^2 R_e = I V_e$, find the effective external resistance R_e and voltage V_e.

(c) Find the power dissipated within the sphere and deduce the effective internal resistance R_i and voltage V_i.

(d) Define the total voltage through the relation, $V_t = (R_e + R_i)I$ and show that $V_t = 4aF/3$, as well as $V_e + V_i = V_t$. Show that IV_t is the power supplied by the "chemical" force.

Reference: W. M. Saslow, *Am. J. Phys.* **62**, 495–501 (1994).

Solution.

Assume the permittivity is ϵ_0 on both sides and no field is applied at infinity. Put $D = \sigma + 2\sigma'$ and $A = \sigma F/D$. **(a)** In each uniform medium, steady charge conservation $\nabla\cdot\mathbf{J} = 0$, Ohm's law, and $\nabla\cdot(F\hat{\mathbf{z}}) = 0$ imply $\nabla^2\Phi = 0$. The normal component of the driving term on the sphere is $F\cos\theta$, so the boundary matching selects only $l = 1$. Regularity at the origin and decay at infinity then give

$$\Phi_{\rm in} = Ar\cos\theta, \qquad \Phi_{\rm out} = Aa^3\cos\theta/r^2.$$

Continuity of potential fixes the relative coefficient. Ohm's law is $\mathbf{J} = \sigma(\mathbf{E} + F\hat{\mathbf{z}})$ inside and $\mathbf{J} = \sigma'\mathbf{E}$ outside. At the surface its normal continuity requires $\sigma(F - A)\cos\theta = 2\sigma' A\cos\theta$, giving A above. Therefore

$$\mathbf{E}_{\rm in} = -A\hat{\mathbf{z}}, \qquad \mathbf{J}_{\rm in} = 2\sigma' A\hat{\mathbf{z}},$$

$$\mathbf{E}_{\rm out} = \frac{Aa^3}{r^3}(2\cos\theta\,\hat{\mathbf{r}} + \sin\theta\,\hat{\boldsymbol{\theta}}), \qquad \mathbf{J}_{\rm out} = \sigma'\mathbf{E}_{\rm out}.$$

The normal field jump gives $\sigma_s = \epsilon_0(E_{r,\text{out}} - E_{r,\text{in}}) = 3\epsilon_0 A\cos\theta$. Integration of $z\sigma_s$ over the sphere yields

$$p = 3\epsilon_0 A a^3 \int \cos^2\theta \, d\Omega = 4\pi\epsilon_0 A a^3 = \frac{4\pi\epsilon_0 \sigma a^3 F}{D}.$$

(b) Integrating J_r over the upper hemisphere gives

$$I = 2\sigma' A a^2 \, 2\pi \int_0^{\pi/2} \cos\theta \sin\theta \, d\theta = 2\pi\sigma' A a^2 = \frac{2\sigma\sigma'}{D}\pi a^2 F.$$

The exterior dissipation is

$$\begin{aligned} P_e &= \sigma' A^2 a^6 \int_a^\infty r^{-4} dr \int (4\cos^2\theta + \sin^2\theta) d\Omega \\ &= \frac{8\pi}{3}\sigma' A^2 a^3. \end{aligned}$$

Consequently

$$\boxed{R_e = \frac{P_e}{I^2} = \frac{2}{3\pi\sigma' a}, \qquad V_e = \frac{P_e}{I} = \frac{4Aa}{3}.}$$

(c) Joule heating inside is J^2/σ, not σE^2, because the chemical force also drives the current:

$$P_i = \frac{(2\sigma' A)^2}{\sigma}\frac{4\pi a^3}{3} = \frac{16\pi\sigma'^2 A^2 a^3}{3\sigma}.$$

Thus

$$\boxed{R_i = \frac{4}{3\pi\sigma a}, \qquad V_i = \frac{8\sigma' A a}{3\sigma}.}$$

(d) The voltage sum is

$$V_e + V_i = \frac{4Aa}{3}\left(1 + \frac{2\sigma'}{\sigma}\right) = \frac{4aF}{3} = V_t = (R_e + R_i)I.$$

The chemical work per unit time is

$$P_{\text{chem}} = \int_{r<a} \mathbf{J} \cdot F\hat{\mathbf{z}} \, d^3 r = 2\sigma' A F \frac{4\pi a^3}{3} = I\frac{4aF}{3} = P_e + P_i.$$

The resistance formulas assume positive conductivities; open-circuit limits are obtained from these expressions by continuity.

3.16

(a) Starting from the Bessel differential equation and appropriate limiting procedures, verify the generalization of (3.108),

$$\frac{1}{k}\,\delta(k - k') = \int_0^\infty \rho \, J_\nu(k\rho) J_\nu(k'\rho) \, d\rho$$

or equivalently that

$$\frac{1}{\rho}\,\delta(\rho - \rho') = \int_0^\infty k \, J_\nu(k\rho) J_\nu(k\rho') \, dk$$

where $\text{Re}(\nu) > -1$.

(b) Obtain the following expansion:

$$\frac{1}{|\mathbf{x}-\mathbf{x}'|} = \sum_{m=-\infty}^{\infty} \int_0^\infty dk\, e^{im(\phi-\phi')} J_m(k\rho) J_m(k\rho') e^{-k(z_>-z_<)}$$

(c) By appropriate limiting procedures prove the following expansions:

$$\frac{1}{\sqrt{\rho^2+z^2}} = \int_0^\infty e^{-k|z|} J_0(k\rho)\, dk$$

$$J_0\Big(k\sqrt{\rho^2+\rho'^2-2\rho\rho'\cos\phi}\Big) = \sum_{m=-\infty}^{\infty} e^{im\phi} J_m(k\rho) J_m(k\rho')$$

$$e^{ik\rho\cos\phi} = \sum_{m=-\infty}^{\infty} i^m e^{im\phi} J_m(k\rho)$$

(d) From the last result obtain an integral representation of the Bessel function:

$$J_m(x) = \frac{1}{2\pi i^m} \int_0^{2\pi} e^{ix\cos\phi - im\phi}\, d\phi$$

Compare the standard integral representations.

Solution.

(a) Interpret the identities as distributions for positive k, k', ρ, ρ', not as ordinary absolutely convergent integrals. The Bessel equations give the finite-radius identity

$$\int_0^R \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho = \frac{R[J_\nu(kR)k'J'_\nu(k'R) - J_\nu(k'R)kJ'_\nu(kR)]}{k^2-k'^2}.$$

The combined lower Wronskian vanishes for $\mathrm{Re}\,\nu > -1$: its leading powers cancel and the remainder is $O(\rho^{2\nu+2})$. For real positive k, k', insert

$$J_\nu(kR) = \sqrt{\frac{2}{\pi kR}} \cos(kR - \tfrac{\pi\nu}{2} - \tfrac{\pi}{4}) + O(R^{-3/2}).$$

The slowly oscillating part of the quotient is

$$\frac{1}{\pi\sqrt{kk'}} \frac{\sin[(k-k')R]}{k-k'}.$$

Its integral against a smooth test function supported away from $k' = 0$ tends to that function at $k' = k$, divided by k. The term oscillating with $k + k'$ has no delta support on the positive axis and tends weakly to zero. This proves $\int_0^\infty \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho = \delta(k-k')/k$. Exchange the names of the integration and spectral variables to obtain the second identity. For complex order the same argument uses analytic continuation of the origin-admissible solutions; there is no complex conjugation in the bilinear identity displayed in the problem.

(b) The cylindrical delta function is

$$\delta^3(\mathbf{x}-\mathbf{x}') = \frac{1}{2\pi} \sum_m e^{im(\phi-\phi')} \int_0^\infty k\, dk\, J_m(k\rho) J_m(k\rho') \delta(z-z').$$

Applying the Laplacian to a separated term leaves $(\partial_z^2 - k^2)f$. Since

$$(\partial_z^2 - k^2)e^{-k|z-z'|} = -2k\delta(z-z'),$$

comparison with $\nabla^2 G = -4\pi\delta^3$ gives

$$G = \sum_m \int_0^\infty dk\, e^{im(\phi-\phi')} J_m(k\rho) J_m(k\rho') e^{-k|z-z'|}.$$

It decays at infinity, has the Coulomb singularity, and is therefore $1/|\mathbf{x}-\mathbf{x}'|$.

(c) First put $\rho' = 0$, $z' = 0$; since $J_m(0) = 0$ for $m \neq 0$ and $J_0(0) = 1$, this gives the first Laplace-transform identity. Next let $s^2 = \rho^2 + \rho'^2 - 2\rho\rho'\cos\phi$ and $h = |z - z'| > 0$. The first identity expresses $1/\sqrt{s^2+h^2}$ as the Laplace transform of $J_0(ks)$, whereas (b) expresses it as the transform of $\sum_m e^{im\phi} J_m(k\rho) J_m(k\rho')$. Uniqueness of the Laplace transform gives the second identity. A direct limiting derivation of the plane-wave identity is also useful. Take $\rho' = R \to \infty$ in this addition theorem. Then $s = R - \rho\cos\phi + O(R^{-1})$. Using the large-argument Bessel asymptotic on both sides and comparing the coefficients of $e^{-i(kR-\pi/4)}$ gives

$$e^{ik\rho\cos\phi} = \sum_{m=-\infty}^{\infty} i^m e^{im\phi} J_m(k\rho).$$

Comparison of the other exponential gives its conjugate. One may first keep a finite Fourier truncation and then use the rapid decay of $J_m(k\rho)$ at fixed $k\rho$ to justify the limit.

(d) Multiply that Fourier series by $e^{-im\phi}$ and integrate over one period. Orthogonality gives exactly

$$J_m(x) = \frac{1}{2\pi i^m} \int_0^{2\pi} e^{ix\cos\phi - im\phi} d\phi.$$

With $\phi = t - \pi/2$, the factors $i^{-m} e^{im\pi/2}$ cancel. Then $t \mapsto -t$ and periodicity give $J_m(x) = (2\pi)^{-1} \int_{-\pi}^{\pi} e^{i(mt - x\sin t)} dt$. Pairing t and $-t$ yields the familiar real representation

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(mt - x\sin t) dt \qquad (m \in \mathbb{Z}).$$

3.17 The Dirichlet Green function for the unbounded space between the planes at $z = 0$ and $z = L$ allows discussion of a point charge or a distribution of charge between parallel conducting planes held at zero potential.

(a) Using cylindrical coordinates show that one form of the Green function is

$$G(\mathbf{x},\mathbf{x}') = \frac{4}{L} \sum_{n=1}^{\infty} \sum_{m=-\infty}^{\infty} e^{im(\phi-\phi')} \sin\Big(\frac{n\pi z}{L}\Big) \sin\Big(\frac{n\pi z'}{L}\Big) I_m\Big(\frac{n\pi}{L}\rho_<\Big) K_m\Big(\frac{n\pi}{L}\rho_>\Big)$$

(b) Show that an alternative form of the Green function is

$$G(\mathbf{x},\mathbf{x}') = 2 \sum_{m=-\infty}^{\infty} \int_0^\infty dk\; e^{im(\phi-\phi')} J_m(k\rho) J_m(k\rho') \frac{\sinh(kz_<)\,\sinh[k(L-z_>)]}{\sinh(kL)}$$

Solution.

Throughout, $\nabla^2 G = -4\pi\delta^3(\mathbf{x}-\mathbf{x}')$; G vanishes on each plane and decays at transverse infinity. **(a)** Expand the angular and axial delta functions:

$$\delta(\phi-\phi') = \frac{1}{2\pi}\sum_m e^{im(\phi-\phi')}, \qquad \delta(z-z') = \frac{2}{L}\sum_{n=1}^{\infty}\sin(\kappa_n z)\sin(\kappa_n z'), \quad \kappa_n = \frac{n\pi}{L}.$$

For each m, n the radial Green equation is

$$\left[\frac{1}{\rho}\partial_\rho(\rho\partial_\rho) - \frac{m^2}{\rho^2} - \kappa_n^2\right] g = -\frac{\delta(\rho-\rho')}{\rho}.$$

The solution finite at the axis and decaying at infinity is $g = I_m(\kappa_n\rho_<)K_m(\kappa_n\rho_>)$. Indeed the modified-Bessel Wronskian gives $[\rho\partial_\rho g]_{\rho'-}^{\rho'+} = -1$. Multiplication by the delta-function normalization therefore yields

$$G = \frac{4}{L}\sum_{n=1}^{\infty}\sum_{m=-\infty}^{\infty} e^{im(\phi-\phi')}\sin(\kappa_n z)\sin(\kappa_n z')I_m(\kappa_n\rho_<)K_m(\kappa_n\rho_>),$$

proving (a).

(b) Instead use the radial completeness relation of Problem 3.16 and leave the axial equation unsummed. The two Dirichlet axial solutions are $u(z) = \sinh(kz)$ and $v(z) = \sinh[k(L-z)]$, with Wronskian $uv' - u'v = -k\sinh(kL)$. Thus

$$h_k(z,z') = \frac{\sinh(kz_<)\sinh[k(L-z_>)]}{\sinh(kL)}, \qquad [\partial_z h_k]_{z'-}^{z'+} = -k.$$

Since the radial delta expansion contains $k\,dk$, the required coefficient is 2:

$$G = 2\sum_m \int_0^\infty dk\, e^{im(\phi-\phi')}J_m(k\rho)J_m(k\rho')h_k(z,z').$$

This proves the second form and its normalization. Reciprocity is explicit in both representations. Expanding h_k in sine modes,

$$h_k = \frac{2k}{L}\sum_{n=1}^{\infty}\frac{\sin(\kappa_n z)\sin(\kappa_n z')}{k^2+\kappa_n^2},$$

also shows directly how the two representations are related.

3.18 The configuration of Problem 3.12 is modified by placing a conducting plane held at zero potential parallel to and a distance L away from the plane with the disc insert in it. For definiteness put the grounded plane at $z = 0$ and the other plane with the center of the disc on the z axis at $z = L$.

(a) Show that the potential between the planes can be written in cylindrical coordinates (z, ρ, ϕ) as

$$\Phi(z,\rho) = V\int_0^\infty d\lambda\; J_1(\lambda)J_0(\lambda\rho/a)\,\frac{\sinh(\lambda z/a)}{\sinh(\lambda L/a)}$$

(b) Show that in the limit $a\to\infty$ with z, ρ, L fixed the solution of part a reduces to the expected result. Viewing your result as the lowest order answer in an expansion in powers of a^{-1}, consider the question of corrections to the lowest order expression if a is *large* compared to ρ and L, but not infinite. Are there difficulties? Can you obtain an explicit *estimate* of the corrections?

(c) Consider the limit of $L \to \infty$ with $(L - z)$, a and ρ fixed and show that the results of Problem 3.12 are recovered. What about corrections for $L \gg a$, but not $L \to \infty$?

Solution.

(a) The upper boundary is $VH(a - \rho)$ and the lower boundary is zero. Hankel expansion and axial separation give

$$\Phi = Va \int_0^\infty dk\, J_1(ka) J_0(k\rho) \frac{\sinh(kz)}{\sinh(kL)}.$$

The coefficient $VaJ_1(ka)$ follows from $kV \int_0^a \rho J_0(k\rho) d\rho = VaJ_1(ka)$, as in Problem 3.12. Putting $\lambda = ka$ proves the specified expression.

(b) For a disc covering the whole upper plane, uniqueness gives $\Phi = Vz/L$. One can obtain both this limit and its error without a nonuniform expansion of the Hankel integral. Subtract from Vz/L the Poisson integral of the missing exterior of the disc. Denote that subtracted potential by Φ_{ext}. For $\kappa_n = n\pi/L$, the outward derivative at $z' = L$ of the Green function in Problem 3.17 contains $\partial_{z'} \sin(\kappa_n z') = \kappa_n (-1)^n$. The Dirichlet formula therefore gives, for $\rho < a$,

$$\frac{\Phi_{\text{ext}}}{V} = -\frac{2}{L} \sum_{n=1}^\infty (-1)^n \kappa_n \sin(\kappa_n z) I_0(\kappa_n \rho) \int_a^\infty \rho' K_0(\kappa_n \rho')\, d\rho'.$$

Here angular integration supplies the factor 2π and leaves the $m = 0$ cylindrical mode. Since $\Phi = Vz/L - \Phi_{\text{ext}}$, this gives

$$\boxed{\frac{\Phi}{V} = \frac{z}{L} - \frac{2a}{L} \sum_{n=1}^\infty (-1)^{n+1} \sin \frac{n\pi z}{L} I_0\Big(\frac{n\pi\rho}{L}\Big) K_1\Big(\frac{n\pi a}{L}\Big).}$$

For clarity, the radial step in this derivation is

$$\int_a^\infty \rho' K_0(\kappa_n \rho') d\rho' = \frac{a}{\kappa_n} K_1(\kappa_n a), \quad \int_0^{2\pi} K_0(\kappa_n |\boldsymbol{\rho} - \boldsymbol{\rho}'|) d\phi' = 2\pi I_0(\kappa_n \rho) K_0(\kappa_n \rho').$$

With ρ, z, L fixed and $a/L \gg 1$, the leading correction is

$$\frac{\Phi}{V} - \frac{z}{L} = -\sqrt{\frac{2a}{L}}\, e^{-\pi a/L} I_0(\pi\rho/L) \sin(\pi z/L) \left[1 + \frac{3L}{8\pi a} + O((L/a)^2)\right] + O(\sqrt{a/L}\, e^{-2\pi a/L}).$$

There are no nonzero algebraic corrections in powers of a^{-1} at fixed interior position: the leading effect is exponentially small. Expanding $J_0(\lambda\rho/a)$ and the hyperbolic ratio under the original integral would require divergent ordinary moments such as $\int_0^\infty \lambda^2 J_1(\lambda) d\lambda$; it misses this nonuniform effect. Near the rim the relevant distance is $a - \rho$, and the approximation requires $a - \rho \gg L$.

(c) Put $h = L - z$ and hold h, a, ρ fixed. Exactly,

$$\frac{\sinh[k(L-h)]}{\sinh(kL)} = e^{-kh} + \sum_{j=1}^\infty \left[e^{-k(2jL+h)} - e^{-k(2jL-h)}\right].$$

If $F(\rho, s)$ denotes the half-space answer of Problem 3.12 at height s, then

$$\Phi = F(\rho, h) + \sum_{j=1}^\infty [F(\rho, 2jL + h) - F(\rho, 2jL - h)].$$

Thus $L \to \infty$ recovers that problem. For $L \gg a, \rho, h$, expand $J_1(ka) = ka/2 + O(k^3a^3)$ and $J_0(k\rho) = 1 + O(k^2\rho^2)$ in the far-away image terms: $F(\rho, s) = Va^2/(2s^2) + O(Va^2(a^2+\rho^2)/s^4)$. The difference of the pair at $2jL$ is $-Va^2h/(4j^3L^3) + O(Va^2h(a^2+\rho^2+h^2)/(jL)^5)$, so

$$\boxed{\Phi = F(\rho, h) - \frac{Va^2h}{4L^3}\zeta(3) + O\left(\frac{Va^2h(a^2+\rho^2+h^2)}{L^5}\right).}$$

For merely $L \gg a$ with h not small compared with L, the exact image series, rather than this additional expansion in h/L, provides the correction.

3.19 Consider a point charge q between two infinite parallel conducting planes held at zero potential. Let the planes be located at $z = 0$ and $z = L$ in a cylindrical coordinate system, with the charge on the z axis at $z = z_0$, $0 < z_0 < L$. Use Green's reciprocation theorem of Problem 1.12 with problem 3.18 as the comparison problem.

(a) Show that the amount of induced charge on the plate at $z = L$ inside a circle of radius a whose center is on the z axis is given by

$$Q_L(a) = -\frac{q}{V}\,\Phi(z_0, 0)$$

where $\Phi(z_0, 0)$ is the potential of Problem 3.18 evaluated at $z = z_0$, $\rho = 0$. Find the *total* charge induced on the upper plate. Compare with the solution (in method and answer) of Problem 1.13.

(b) Show that the induced charge *density* on the upper plate can be written as

$$\sigma(\rho) = -\frac{q}{2\pi}\int_0^\infty dk\,\frac{\sinh(kz_0)}{\sinh(kL)}\,kJ_0(k\rho)$$

This integral can be expressed (see, e.g., *Gradshteyn and Ryzhik*, p. 728, formula 6.666) as an infinite series involving the modified Bessel functions $K_0(n\pi\rho/L)$, showing that at large radial distances the induced charge density falls off as $(\rho)^{-1/2}e^{-\pi\rho/L}$.

(c) Show that the charge density at $\rho = 0$ can be written as the series

$$\sigma(0) = -\frac{q}{2\pi L^2}\sum_{n>0,\ \text{odd}}\left[(n - z_0/L)^{-2} - (n + z_0/L)^{-2}\right]$$

Solution.

(a) Use configuration A for the point charge and grounded plates, and configuration B for the voltage-V disc boundary of Problem 3.18, with no volume charge. Reciprocity includes both volume and conductor charges. In $\int \rho_A\Phi_B = \int \rho_B\Phi_A$, the right side is zero because $\Phi_A = 0$ on every conductor. The left side is $q\Phi_B(z_0, 0) + VQ_L(a)$, all other conductor terms being zero. Therefore

$$\boxed{Q_L(a) = -q\Phi_B(z_0, 0)/V.}$$

As $a \to \infty$, $\Phi_B \to Vz_0/L$, so $Q_L = -qz_0/L$ and $Q_0 = -q(1 - z_0/L)$. This is the same result as Problem 1.13: reciprocity with a whole plate held at unit voltage gives a comparison potential z/L. No detailed induced-charge distribution is needed for the total.

(b) Set $\rho = 0$ in Problem 3.18 before rescaling its integration variable:

$$Q_L(a) = -qa \int_0^\infty J_1(ka) \frac{\sinh(kz_0)}{\sinh(kL)} dk.$$

Since $d[aJ_1(ka)]/da = kaJ_0(ka)$ and $dQ_L/da = 2\pi a\sigma(a)$,

$$\boxed{\sigma(\rho) = -\frac{q}{2\pi} \int_0^\infty kJ_0(k\rho) \frac{\sinh(kz_0)}{\sinh(kL)} dk.}$$

Alternatively, its $m = 0$ sine modes give the source potential

$$\Phi_A(\rho, z) = \frac{q}{\pi\epsilon_0 L} \sum_{n=1}^\infty \sin(\kappa_n z_0) \sin(\kappa_n z) K_0(\kappa_n \rho), \qquad \kappa_n = n\pi/L.$$

The normal from the upper conductor into the gap is $-\hat{\mathbf{z}}$, so $\sigma = \epsilon_0 \partial_z \Phi_A|_{L^-}$. Using $\partial_z \sin(\kappa_n z)|_L = \kappa_n(-1)^n$ gives

$$\sigma(\rho) = \frac{q}{L^2} \sum_{n=1}^\infty (-1)^n n \sin(n\pi z_0/L) K_0(n\pi\rho/L), \qquad \rho > 0.$$

For large ρ, the $n = 1$ term dominates and

$$\sigma(\rho) \sim -\frac{q}{L^2} \sin(\pi z_0/L) \sqrt{\frac{L}{2\rho}} e^{-\pi\rho/L},$$

using $K_0(x) \sim \sqrt{\pi/(2x)} e^{-x}$.

(c) For $k > 0$ expand the ratio into decaying exponentials:

$$\frac{\sinh(kz_0)}{\sinh(kL)} = \sum_{j=0}^\infty \left[e^{-k[(2j+1)L - z_0]} - e^{-k[(2j+1)L + z_0]} \right].$$

At $\rho = 0$, $J_0(0) = 1$ and $\int_0^\infty ke^{-ck} dk = 1/c^2$. Integrating each positive pair therefore gives exactly

$$\boxed{\sigma(0) = -\frac{q}{2\pi L^2} \sum_{\substack{n>0 \\ n \text{ odd}}} [(n - z_0/L)^{-2} - (n + z_0/L)^{-2}].}$$

Each pair decreases as n^{-3}, so this representation is convergent and well suited to axis evaluation.

3.20

(a) From the results of Problem 3.17 or from first principles show that the potential of a point charge q between two infinite parallel conducting planes held at zero potential can be written as

$$\Phi(z, \rho) = \frac{q}{\pi\epsilon_0 L} \sum_{n=1}^\infty \sin\Big(\frac{n\pi z_0}{L}\Big) \sin\Big(\frac{n\pi z}{L}\Big) K_0\Big(\frac{n\pi\rho}{L}\Big)$$

where the planes are at $z = 0$ and $z = L$ and the charge is on the z axis at the point $z = z_0$.

(b) Calculate the induced surface-charge densities $\sigma_0(\rho)$ and $\sigma_L(\rho)$ on the lower and upper plates. The result for $\sigma_L(\rho)$ is

$$\sigma_L(\rho) = \frac{q}{L^2}\sum_{n=1}^{\infty}(-1)^n n \sin\Big(\frac{n\pi z_0}{L}\Big) K_0\Big(\frac{n\pi\rho}{L}\Big)$$

Discuss the connection of this expression with that of Problem 3.19b and 3.19c.

(c) From the answer in part b, calculate the total charge Q_L on the plate at $z = L$. By summing the Fourier series or by other means of comparison, check your answer against the known expression of Problem 1.13 [C. Y. Fong and C. Kittel, *Am. J. Phys.* **35**, 1091 (1967).]

Solution.

(a) Put $\rho' = 0$, $z' = z_0$ in Problem 3.17(a). Since $I_m(0) = 0$ for $m \neq 0$ and $I_0(0) = 1$, only $m = 0$ survives. Multiplication by $q/(4\pi\epsilon_0)$ gives

$$\boxed{\Phi(z,\rho) = \frac{q}{\pi\epsilon_0 L}\sum_{n=1}^{\infty}\sin(n\pi z_0/L)\sin(n\pi z/L)K_0(n\pi\rho/L).}$$

This is the potential *of* the point charge; it diverges at the point charge itself. The convergent K_0 representation is most convenient for $\rho > 0$.

(b) The conductor normal into the gap is $+\hat{\mathbf{z}}$ on the lower plate and $-\hat{\mathbf{z}}$ on the upper plate. Thus $\sigma_0 = -\epsilon_0\partial_z\Phi(0^+)$ and $\sigma_L = +\epsilon_0\partial_z\Phi(L^-)$, giving

$$\sigma_0(\rho) = -\frac{q}{L^2}\sum_{n=1}^{\infty} n\sin(n\pi z_0/L)K_0(n\pi\rho/L),$$
$$\sigma_L(\rho) = \frac{q}{L^2}\sum_{n=1}^{\infty}(-1)^n n\sin(n\pi z_0/L)K_0(n\pi\rho/L).$$

To compare with Problem 3.19, the sine expansion of $f(z) = \sinh(kz)/\sinh(kL)$ has coefficient

$$\frac{2}{L}\int_0^L f(z)\sin(\kappa_n z)dz = \frac{2(-1)^{n+1}\kappa_n}{L(k^2+\kappa_n^2)}.$$

Use $\int_0^\infty kJ_0(k\rho)/(k^2+\kappa_n^2)dk = K_0(\kappa_n\rho)$ in the integral of Problem 3.19(b); it gives precisely σ_L above. One must not set $\rho = 0$ term by term in this K_0 series: each term is singular there. Sum first, or use the exponentially convergent k integral and its inverse-square series from 3.19(c). The actual axis density is finite for $0 < z_0 < L$.

(c) Since $\int_0^\infty \rho K_0(\kappa_n\rho)d\rho = 1/\kappa_n^2$,

$$Q_L = \frac{2q}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin(nx), \qquad x = \pi z_0/L.$$

This interchange can be regularized first with an Abel factor t^n, then $t \uparrow 1$. For $0 < x < \pi$,

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}e^{inx} = \ln(1+e^{ix}), \qquad \operatorname{Im}\ln(1+e^{ix}) = x/2.$$

It follows that $Q_L = -qx/\pi = -qz_0/L$, agreeing with reciprocity and Problem 1.13. Similarly $\sum_n \sin(nx)/n = (\pi - x)/2$ gives $Q_0 = -q(1 - z_0/L)$, and their sum is $-q$.

3.21

(a) By using the Green function of Problem 3.17b in the limit $L \to \infty$, show that the capacitance of a flat, thin, circular, conducting disc of radius R located parallel to, and a distance d above, a grounded conducting plane is given by

$$\frac{4\pi\epsilon_0}{C} = \int_0^\infty dk\,(1 - e^{-2kd})\,\frac{\left[\int_0^R \rho J_0(k\rho)\sigma(\rho)\,d\rho\right]^2}{\left[\int_0^R \rho\sigma(\rho)\,d\rho\right]^2}$$

where $\sigma(\rho)$ is the charge density on the disc.

(b) Use the expression in part a as a variational or stationary principle for C^{-1} with the approximation that $\sigma(\rho) =$ constant. Show explicitly that you obtain the correct limiting value for C^{-1} as $d \ll R$. Determine an approximate value of C^{-1} for an isolated disc ($d \gg R$) and evaluate the ratio of it to the exact result, $4\pi\epsilon_0/C = (\pi/2)R^{-1}$.

(c) As a better trial form for $\sigma(\rho)$ consider a linear combination of a constant and $(R^2 - \rho^2)^{-1/2}$, the latter being the correct form for an isolated disc.

For part b the following integrals may be of use:

$$\int_0^\infty dt\left[\frac{J_1(t)}{t}\right]^2 = \frac{4}{3\pi}, \qquad \int_0^\infty \frac{dt\,J_1^2(t)}{t} = \frac{1}{2}$$

Solution.

Let σ be the total charge per disc area (both faces), and set $S(k) = \int_0^R \rho\sigma(\rho)J_0(k\rho)d\rho$, $Q = 2\pi S(0)$. **(a)** In the half-space limit of Problem 3.17(b), at two points of height d,

$$G = \sum_m \int_0^\infty dk\, e^{im(\phi-\phi')}J_m(k\rho)J_m(k\rho')(1 - e^{-2kd}).$$

Both angular integrations in the electrostatic energy select $m = 0$:

$$U = \frac{1}{8\pi\epsilon_0}\iint \sigma G\sigma'\,dS\,dS' = \frac{\pi}{2\epsilon_0}\int_0^\infty (1 - e^{-2kd})S(k)^2 dk.$$

At equilibrium $U = Q^2/(2C)$, so division gives the required formula. At fixed total charge this positive energy is stationary and minimal when the disc potential is constant: its first variation is $\delta U = \int \Phi\,\delta\sigma\,dS = V\delta Q = 0$. Thus a trial density gives an upper bound on C^{-1}.

(b) A uniform density has $S(k)/S(0) = 2J_1(kR)/(kR)$. Put $t = kR$ and $\eta = d/R$:

$$\left(\frac{4\pi\epsilon_0}{C}\right)_{\text{unif}} = \frac{4}{R}\int_0^\infty (1 - e^{-2\eta t})\frac{J_1^2(t)}{t^2}dt.$$

For $\eta \downarrow 0$, the bound $(1 - e^{-2\eta t})/\eta \le 2t$ justifies dominated convergence using the second supplied integral. The result is $4d/R^2$, or $C \sim \epsilon_0\pi R^2/d$, the parallel-plate value. For an isolated disc the first supplied integral gives

$$\left(\frac{4\pi\epsilon_0}{C}\right)_{\text{unif}} \longrightarrow \frac{16}{3\pi R}, \qquad \frac{(C^{-1})_{\text{unif}}}{(C^{-1})_{\text{exact}}} = \frac{32}{3\pi^2} = 1.080759\ldots.$$

(c) Choose charge-normalized basis densities and one real variational parameter c:

$$\sigma_c(\rho) = \frac{Q(1-c)}{\pi R^2} + \frac{Qc}{2\pi R\sqrt{R^2-\rho^2}}.$$

Each basis density carries total charge Q; hence $S(k)/S(0) = (1-c)f_0(t) + cf_1(t)$ with $f_0 = 2J_1(t)/t$, $f_1 = \sin t/t$. For the second transform, put $\rho = R\sin\vartheta$ and expand J_0:

$$\begin{aligned}\int_0^R \frac{\rho J_0(k\rho)}{\sqrt{R^2-\rho^2}}d\rho &= R\sum_{j=0}^{\infty}\frac{(-1)^j(kR)^{2j}}{2^{2j}(j!)^2}\int_0^{\pi/2}\sin^{2j+1}\vartheta\,d\vartheta \\ &= R\sum_{j=0}^{\infty}\frac{(-1)^j(kR)^{2j}}{(2j+1)!} = \frac{\sin(kR)}{k}.\end{aligned}$$

The beta integral in the middle step is $2^{2j}(j!)^2/(2j+1)!$. Define

$$M_{ij}(\eta) = \int_0^{\infty}(1-e^{-2\eta t})f_i(t)f_j(t)dt.$$

The trial inverse capacitance is the explicit quadratic

$$\frac{1}{R}[(1-c)^2M_{00} + 2c(1-c)M_{01} + c^2M_{11}].$$

Differentiating with respect to c gives

$$\boxed{c_* = \frac{M_{00}-M_{01}}{M_{00}-2M_{01}+M_{11}}, \qquad \left(\frac{4\pi\epsilon_0}{C}\right)_{\text{trial}} = \frac{M_{00}M_{11}-M_{01}^2}{R(M_{00}-2M_{01}+M_{11})}.}$$

The denominator is the positive integral of $(1-e^{-2\eta t})(f_0-f_1)^2$. These one-dimensional integrals fully determine the approximation for any $d/R > 0$. Useful formulas for stable numerical evaluation are

$$M_{11} = \tan^{-1}\eta + \frac{\eta}{2}\ln(1+\eta^{-2}), \quad M_{00} = \frac{16}{3\pi} - \int_0^{\infty}e^{-2\eta t}f_0^2dt, \quad M_{01} = \frac{\pi}{2} - \int_0^{\infty}e^{-2\eta t}f_0f_1dt.$$

For the first formula, differentiate under the convergent integral:

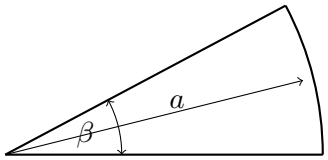
$$M_{11}'(\eta) = 2\int_0^{\infty}e^{-2\eta t}\frac{\sin^2 t}{t}dt = \frac{1}{2}\log(1+\eta^{-2}).$$

Here $\sin^2 t = (1-\cos 2t)/2$ and $\int_0^{\infty}e^{-at}(1-\cos bt)dt/t = \frac{1}{2}\log(1+b^2/a^2)$, obtained by integrating ∂_b from $b=0$. Integration in η, with $M_{11}(0)=0$, yields the displayed $\arctan\eta + (\eta/2)\log(1+\eta^{-2})$. For the last constant, the exact isolated-disc density produces a constant potential, so its mutual energy with any equal-charge trial equals its self-energy; hence $M_{01}(\infty) = M_{11}(\infty) = \pi/2$. Alternatively it follows by inserting the disc Hankel transform in the constant surface-potential integral. For example, the supplementary quadrature gives

d/R	c_*	$R(4\pi\epsilon_0/C)_{\text{unif}}$	$R(4\pi\epsilon_0/C)_{\text{trial}}$
0.05	0.289107	0.175277	0.170416
0.10	0.427071	0.318671	0.303972
0.50	0.839454	0.948980	0.862857
1.00	0.952640	1.246310	1.131688
5.00	0.999362	1.598147	1.471452

As $\eta\to\infty$, $c_*\to 1$ and the exact isolated result is recovered. As $\eta\to 0$, $M_{00} = 4\eta + o(\eta)$, $M_{01} = 4\eta + o(\eta)$, whereas $M_{11} = \eta[1+\ln(1/\eta)] + o(\eta)$; consequently $c_*\to 0$ and the leading parallel-plate result is retained.

3.22 The geometry of a two-dimensional potential problem is defined in polar coordinates by the surfaces $\phi = 0$, $\phi = \beta$, and $\rho = a$, as indicated in the sketch.



Problem 3.22

Using separation of variables in polar coordinates, show that the Green function can be written as

$$G(\rho,\phi;\rho',\phi') = \sum_{m=1}^{\infty} \frac{4}{m}\,\rho_<^{m\pi/\beta}\left(\frac{1}{\rho_>^{m\pi/\beta}} - \frac{\rho_>^{m\pi/\beta}}{a^{2m\pi/\beta}}\right)\sin\!\left(\frac{m\pi\phi}{\beta}\right)\sin\!\left(\frac{m\pi\phi'}{\beta}\right)$$

Problem 2.25 may be of use.

Solution.

Use the two-dimensional convention $\nabla^2 G = -4\pi\delta(\rho-\rho')\delta(\phi-\phi')/\rho$. On the interval $0 < \phi < \beta$ the normalized delta expansion is

$$\delta(\phi-\phi') = \frac{2}{\beta}\sum_{m=1}^{\infty}\sin(\nu_m\phi)\sin(\nu_m\phi'), \qquad \nu_m = m\pi/\beta.$$

Write $G = \sum_m g_m(\rho,\rho')\sin(\nu_m\phi)\sin(\nu_m\phi')$. The radial equation is

$$\left[\frac{1}{\rho}\frac{d}{d\rho}\rho\frac{d}{d\rho} - \frac{\nu_m^2}{\rho^2}\right] g_m = -\frac{8\pi}{\beta}\frac{\delta(\rho-\rho')}{\rho}.$$

Its homogeneous solutions are ρ^{ν_m} and $\rho^{-\nu_m}$. Regularity at the apex and the zero value at a require

$$g_m = C_m\rho_<^{\nu_m}(\rho_>^{-\nu_m} - \rho_>^{\nu_m}/a^{2\nu_m}).$$

This is continuous at ρ'; its derivative jump is $[\rho\partial_\rho g_m]_-^+ = -2\nu_m C_m$. Integrating the radial equation across the source then fixes $C_m = 4\pi/(\beta\nu_m) = 4/m$, proving the required formula. Both radial faces and the arc have zero potential, the apex solution is regular, and interchange of source and observation coordinates leaves the result invariant.

3.23 A point charge q is located at the point (ρ',ϕ',z') inside a grounded cylindrical box defined by the surfaces $z = 0$, $z = L$, $\rho = a$. Show that the potential inside the box can be expressed in the

following alternative forms:

$$\Phi(\mathbf{x},\mathbf{x}') = \frac{q}{\pi\epsilon_0 a}\sum_{m=-\infty}^{\infty}\sum_{n=1}^{\infty}\frac{e^{im(\phi-\phi')}J_m\Big(\frac{x_{mn}\rho}{a}\Big)J_m\Big(\frac{x_{mn}\rho'}{a}\Big)}{x_{mn}J_{m+1}^2(x_{mn})\sinh\Big(\frac{x_{mn}L}{a}\Big)} \times \sinh\Big[\frac{x_{mn}}{a}z_<\Big]\sinh\Big[\frac{x_{mn}}{a}(L-z_>)\Big]$$

$$\Phi(\mathbf{x},\mathbf{x}') = \frac{q}{\pi\epsilon_0 L}\sum_{m=-\infty}^{\infty}\sum_{n=1}^{\infty}e^{im(\phi-\phi')}\sin\Big(\frac{n\pi z}{L}\Big)\sin\Big(\frac{n\pi z'}{L}\Big)\frac{I_m\Big(\frac{n\pi\rho_<}{L}\Big)}{I_m\Big(\frac{n\pi a}{L}\Big)} \times\Big[I_m\Big(\frac{n\pi a}{L}\Big)K_m\Big(\frac{n\pi\rho_>}{L}\Big)-K_m\Big(\frac{n\pi a}{L}\Big)I_m\Big(\frac{n\pi\rho_>}{L}\Big)\Big]$$

$$\Phi(\mathbf{x},\mathbf{x}') = \frac{2q}{\pi\epsilon_0 L a^2}\sum_{m=-\infty}^{\infty}\sum_{k=1}^{\infty}\sum_{n=1}^{\infty}\frac{e^{im(\phi-\phi')}\sin\Big(\frac{k\pi z}{L}\Big)\sin\Big(\frac{k\pi z'}{L}\Big)J_m\Big(\frac{x_{mn}\rho}{a}\Big)J_m\Big(\frac{x_{mn}\rho'}{a}\Big)}{\left[\Big(\frac{x_{mn}}{a}\Big)^2+\Big(\frac{k\pi}{L}\Big)^2\right]J_{m+1}^2(x_{mn})}$$

Discuss the relation of the last expansion (with its extra summation) to the other two.

Solution.

Use $\Phi = qG_D/(4\pi\epsilon_0)$ and denote the positive zeros of J_m by x_{mn} (for negative m, use the corresponding zeros of $J_{|m|}$). The radial norm and completeness relation are

$$\int_0^a \rho J_m(x_{mn}\rho/a)J_m(x_{mn'}\rho/a)d\rho = \frac{a^2}{2}J_{m+1}^2(x_{mn})\delta_{nn'},$$

$$\frac{\delta(\rho-\rho')}{\rho} = \frac{2}{a^2}\sum_{n=1}^{\infty}\frac{J_m(x_{mn}\rho/a)J_m(x_{mn}\rho'/a)}{J_{m+1}^2(x_{mn})}.$$

The sign in $J_{-m} = (-1)^m J_m$ cancels in the products and norms.

First use this radial basis and angular Fourier modes. For $\alpha = x_{mn}/a$, the axial inverse of $-\partial_z^2+\alpha^2$ is

$$h_\alpha(z,z') = \frac{\sinh(\alpha z_<)\sinh[\alpha(L-z_>)]}{\alpha\sinh(\alpha L)}, \qquad [\partial_z h_\alpha]_-^+ = -1.$$

The source normalization gives

$$G_D = \frac{4}{a^2}\sum_{m,n}\frac{e^{im(\phi-\phi')}J_m(x_{mn}\rho/a)J_m(x_{mn}\rho'/a)}{J_{m+1}^2(x_{mn})}\,h_{x_{mn}/a}(z,z').$$

Multiplying by $q/(4\pi\epsilon_0)$ and using $\alpha = x_{mn}/a$ is exactly the first displayed potential.

Alternatively expand in axial sine modes with $\kappa = n\pi/L$. The radial function satisfying both regularity and the wall condition is

$$R_m(\rho,\rho';\kappa) = I_m(\kappa\rho_<)\left[K_m(\kappa\rho_>)-\frac{K_m(\kappa a)}{I_m(\kappa a)}I_m(\kappa\rho_>)\right].$$

Its derivative jump is $[\rho\partial_\rho R_m]_-^+ = -1$, by the I_m, K_m Wronskian. Therefore

$$G_D = \frac{4}{L}\sum_{m,n} e^{im(\phi-\phi')}\sin(\kappa z)\sin(\kappa z')R_m(\rho,\rho';\kappa),$$

which, after rearranging the bracket, gives the second potential.

Finally expand in both sets of eigenfunctions. Their full Laplacian eigenvalue is $\alpha^2+\kappa^2$, so inversion gives

$$G_D = \frac{8}{La^2}\sum_{m,k,n} \frac{e^{im(\phi-\phi')}\sin(k\pi z/L)\sin(k\pi z'/L)J_m(x_{mn}\rho/a)J_m(x_{mn}\rho'/a)}{[(x_{mn}/a)^2+(k\pi/L)^2]J_{m+1}^2(x_{mn})}.$$

This is the third expression with its correct factor after multiplying by $q/(4\pi\epsilon_0)$. The extra sum has simply left both one-dimensional inversions in spectral form. Explicitly,

$$\frac{2}{L}\sum_{k=1}^{\infty}\frac{\sin(k\pi z/L)\sin(k\pi z'/L)}{\alpha^2+(k\pi/L)^2} = h_\alpha(z,z'),$$

reduces the third expression to the first; the radial resolvent

$$\frac{2}{a^2}\sum_n \frac{J_m(x_{mn}\rho/a)J_m(x_{mn}\rho'/a)}{[(x_{mn}/a)^2+\kappa^2]J_{m+1}^2(x_{mn})} = R_m(\rho,\rho';\kappa)$$

reduces it to the second. All three have the same boundary values, source jump, and reciprocity, so uniqueness also establishes their equivalence.

3.24 The walls of the conducting cylindrical box of Problem 3.23 are all at zero potential, except for a disc in the upper end, defined by $\rho = b < a$, at potential V.

(a) Using the various forms of the Green function obtained in Problem 3.23, find three expansions for the potential inside the cylinder.

(b) For each series, calculate numerically the ratio of the potential at $\rho = 0$, $z = L/2$ to the potential of the disc, assuming $b = L/4 = a/2$. Try to obtain at least two-significant-figure accuracy. Is one series less rapidly convergent than the others? Why?

(*Abramowitz and Stegun* have tables; *Mathematica* has Bessel functions, as does the software of *Press et al.*)

Solution.

(a) Only the upper disc contributes to the Dirichlet boundary formula,

$$\Phi = -\frac{V}{4\pi}\int_0^{2\pi} d\phi' \int_0^b \rho' d\rho'\, \partial_{z'} G_D(\rho,\phi,z;\rho',\phi',L).$$

The angular integral selects $m = 0$. Put $x_n = x_{0n}$, $\alpha_n = x_n/a$, and $\kappa_k = k\pi/L$. In the first form of Problem 3.23, the upper-end derivative of h_{α_n} is $-\sinh(\alpha_n z)/\sinh(\alpha_n L)$. Using $\int_0^b \rho' J_0(\alpha_n\rho')d\rho' = bJ_1(\alpha_n b)/\alpha_n$ gives the first expansion:

$$\boxed{\frac{\Phi}{V} = \frac{2b}{a}\sum_{n=1}^{\infty}\frac{J_1(x_n b/a)}{x_n J_1^2(x_n)}J_0(x_n\rho/a)\frac{\sinh(x_n z/a)}{\sinh(x_n L/a)}.} \qquad \text{(A)}$$

For the second Green function define $c_k = K_0(\kappa_k a)/I_0(\kappa_k a)$ and

$$H_k(\rho) = \int_0^b \rho' I_0(\kappa_k \rho_<)[K_0(\kappa_k \rho_>) - c_k I_0(\kappa_k \rho_>)]d\rho'.$$

The primitive integrals are $\int \rho I_0(\kappa\rho)d\rho = \rho I_1(\kappa\rho)/\kappa$ and $\int \rho K_0(\kappa\rho)d\rho = -\rho K_1(\kappa\rho)/\kappa$. Splitting at $\rho' = \rho$ and using $I_0(x)K_1(x) + I_1(x)K_0(x) = 1/x$ yields

$$H_k(\rho) = \begin{cases} \kappa_k^{-2} - \dfrac{b}{\kappa_k} I_0(\kappa_k \rho)[K_1(\kappa_k b) + c_k I_1(\kappa_k b)], & \rho \le b, \\ \dfrac{b}{\kappa_k} I_1(\kappa_k b)[K_0(\kappa_k \rho) - c_k I_0(\kappa_k \rho)], & b \le \rho < a. \end{cases}$$

The second expansion is therefore

$$\boxed{\frac{\Phi}{V} = \frac{2}{L}\sum_{k=1}^{\infty}(-1)^{k+1}\kappa_k \sin(\kappa_k z) H_k(\rho).} \tag{B}$$

The third Green function gives directly

$$\boxed{\frac{\Phi}{V} = \frac{4b}{La}\sum_{n=1}^{\infty}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}\kappa_k \sin(\kappa_k z) J_0(x_n \rho/a) J_1(x_n b/a)}{x_n J_1^2(x_n)(\alpha_n^2 + \kappa_k^2)}.} \tag{C}$$

As a boundary limit these are interpreted in the usual Fourier/Abel sense at jumps. For strictly interior points they give the same harmonic solution.

(b) Choose units $a = 1$, so $b = 1/2$, $L = 2$, $z = 1$. The factor in (A) is $\sinh x_n / \sinh 2x_n = 1/(2\cosh x_n)$. Its partial sums are

N	1	2	5	10
Φ_A/V	0.06893268	0.07158219	0.07152937	0.07152937289

For (B) the raw sums and an accelerated evaluation are

K	11	51	101	501
Φ_B/V	0.04518810	0.06541029	0.07464976	0.07216345

The slow part can be summed exactly: $(2/\pi)\sum_{k\ge1}(-1)^{k+1}\sin(k\pi z/L)/k = z/L$. Subtracting this $1/k$ tail from (B), summing the remaining Bessel terms, and adding z/L gives 0.07152937289 (51 terms suffice).

For (C), rectangular sums over $(N, K) = (20, 51), (100, 501), (200, 2001)$ give respectively 0.06595877, 0.07212779, and 0.07168131. An independent acceleration is Euler's transformation of the alternating odd-k sum at $z = L/2$. If it is $\sum_{j\ge0}(-1)^j u_j$, use $\sum_{p\ge0}\Delta^p u_0/2^{p+1}$ with $\Delta u_j = u_j - u_{j+1}$. Taking $N = 20$ radial modes and 32 odd axial terms gives 0.071529372893; increasing to $(40, 64)$ gives 0.071529372888. Thus every representation yields

$$\boxed{\Phi(0, L/2)/V = 0.07152937289\ldots \simeq 0.072.}$$

Series (A) has exponential suppression across the distance from the upper boundary. The unaccelerated axial sine series (B) and (C) retain a $1/k$ tail because their basis functions vanish at the end where the imposed boundary value jumps to a nonzero value. The extra unsummed spectrum in (C) adds work without accelerating that tail. The supplementary code reproduces all three calculations.

3.25 Consider the surface-charge densities for the problem of Section 3.13 of the conducting plane with a circular hole of radius a.

(a) Show that the surface-charge densities on the top and bottom of the plane for $\rho \geq a$ are

$$\sigma_+(\rho) = -\epsilon_0 E_0 + \Delta\sigma(\rho)$$
$$\sigma_-(\rho) = \epsilon_0 E_1 + \Delta\sigma(\rho)$$

where

$$\Delta\sigma(\rho) = -\epsilon_0 \frac{(E_0 - E_1)}{\pi}\left[\frac{a}{\sqrt{\rho^2 - a^2}} - \sin^{-1}\left(\frac{a}{\rho}\right)\right]$$

How does $\Delta\sigma(\rho)$ behave for large ρ? Is $\Delta\sigma(\rho)$, defined in terms of $\Phi^{(1)}$, zero for $\rho < a$? Explain.

(b) Show by direct integration that

$$\lim_{R\to\infty}\left[2\pi\int_a^R d\rho\ \rho(\sigma_+ + \sigma_-) + 2\pi\epsilon_0\int_0^R d\rho\ \rho(E_0 - E_1)\right] = 0$$

Interpret.

Solution.

Put $\Delta E = E_0 - E_1$. Section 3.13 gives $\Phi = E_0 z + \Phi^{(1)}$ above and $\Phi = E_1 z + \Phi^{(1)}$ below, where the even correction has the transform

$$\Phi^{(1)} = \frac{\Delta E}{\pi}\int_0^\infty \left[\frac{\sin(ka)}{k^2} - \frac{a\cos(ka)}{k}\right] e^{-k|z|} J_0(k\rho)dk.$$

(a) On the upper conducting face $\sigma_+ = -\epsilon_0 \partial_z \Phi|_{0^+}$, so its correction is

$$\Delta\sigma = \frac{\epsilon_0 \Delta E}{\pi}\left[\int_0^\infty \frac{\sin(ka)}{k} J_0(k\rho)dk - a\int_0^\infty \cos(ka) J_0(k\rho)dk\right].$$

For $\rho > a$, take the real part of $\int_0^\infty e^{-(s-ia)k} J_0(k\rho)dk = [(s-ia)^2 + \rho^2]^{-1/2}$ as $s \downarrow 0$ to get $(\rho^2 - a^2)^{-1/2}$. Integration with respect to a from zero gives the first integral $\sin^{-1}(a/\rho)$. Hence the stated expression for $\Delta\sigma$ follows. Evenness of $\Phi^{(1)}$ makes the correction on the bottom face the same, whereas the background contribution has the opposite normal sign. Thus $\sigma_+ = -\epsilon_0 E_0 + \Delta\sigma$ and $\sigma_- = \epsilon_0 E_1 + \Delta\sigma$. At large ρ, set $t = a/\rho$ and expand:

$$\frac{t}{\sqrt{1-t^2}} - \sin^{-1} t = \frac{t^3}{3} + \frac{3t^5}{10} + O(t^7), \quad \Delta\sigma = -\frac{\epsilon_0 \Delta E\, a^3}{3\pi\rho^3} + O(\rho^{-5}).$$

Inside the hole the second transform is zero while the first is $\pi/2$. Therefore the correction *defined from* $\Phi^{(1)}$ is $\Delta\sigma = \epsilon_0 \Delta E/2$ per face, not zero. It is a fictitious correction to the intact-plane background: the combined correction $2\Delta\sigma$ cancels its background sheet charge $-\epsilon_0\Delta E$ in the hole. There is no physical surface charge where the material is absent.

(b) A direct primitive gives

$$I(R) = \int_a^R \rho\left[\frac{a}{\sqrt{\rho^2 - a^2}} - \sin^{-1}\frac{a}{\rho}\right] d\rho$$
$$= \frac{a}{2}\sqrt{R^2 - a^2} - \frac{R^2}{2}\sin^{-1}(a/R) + \frac{\pi a^2}{4} \longrightarrow \frac{\pi a^2}{4}.$$

The quantity in the question equals

$$\pi\epsilon_0\Delta E\, a^2 - 4\epsilon_0\Delta E I(R) = 2\epsilon_0\Delta E\left[R^2\sin^{-1}(a/R) - a\sqrt{R^2-a^2}\right] = \frac{4\epsilon_0\Delta E\, a^3}{3R} + O(R^{-3}),$$

and therefore tends to zero. The missing intact-plane charge in the hole is exactly compensated by the redistribution on the remaining conductor. Equivalently, the total charge associated with $\Phi^{(1)}$ is zero, consistent with its absence of a monopole far field.

3.26 Consider the Green function appropriate for Neumann boundary conditions for the volume V between the concentric spherical surfaces defined by $r = a$ and $r = b$, $a < b$. To be able to use (1.46) for the potential, impose the simple constraint (1.45). Use an expansion in spherical harmonics of the form,

$$G(\mathbf{x},\mathbf{x}') = \sum_{l=0}^{\infty} g_l(r,r')P_l(\cos\gamma)$$

where $g_l(r,r') = r_<^l/r_>^{l+1} + f_l(r,r')$.

(a) Show that for $l > 0$, the radial Green function has the symmetric form

$$g_l(r,r') = \frac{r_<^l}{r_>^{l+1}} + \frac{1}{(b^{2l+1}-a^{2l+1})}\left[\frac{l+1}{l}(rr')^l + \frac{l}{l+1}\frac{(ab)^{2l+1}}{(rr')^{l+1}} + a^{2l+1}\left(\frac{r^l}{r'^{l+1}} + \frac{r'^l}{r^{l+1}}\right)\right]$$

(b) Show that for $l = 0$

$$g_0(r,r') = \frac{1}{r_>} - \left(\frac{a^2}{a^2+b^2}\right)\frac{1}{r'} + f(r)$$

where $f(r)$ is arbitrary. Show explicitly in (1.46) that answers for the potential $\Phi(\mathbf{x})$ are independent of $f(r)$.

[The arbitrariness in the Neumann Green function can be removed by symmetrizing g_0 in r and r' with a suitable choice of $f(r)$.]

Solution.

The Neumann convention of Eqs. (1.45)–(1.46) is

$$\nabla'^2 G_N = -4\pi\delta^3(\mathbf{r}-\mathbf{r}'), \qquad \partial_{n'}G_N = -\frac{4\pi}{S} = -\frac{1}{a^2+b^2}, \quad S = 4\pi(a^2+b^2).$$

The inner outward normal is $-\hat{\mathbf{r}}'$ and the outer normal is $+\hat{\mathbf{r}}'$.

(a) For $l > 0$ the boundary derivative must vanish. Write

$$f_l = A_l(r)r'^l + B_l(r)r'^{-l-1}.$$

At $r' = a < r$, differentiating the Coulomb term and f_l gives

$$A_l - \frac{l+1}{l}\frac{B_l}{a^{2l+1}} = -r^{-l-1}.$$

At $r' = b > r$ the analogous condition is

$$\frac{l}{l+1} A_l b^{2l+1} - B_l = r^l.$$

Eliminating A_l, B_l, with $D_l = b^{2l+1} - a^{2l+1}$, yields

$$A_l = \frac{(l+1)r^l/l + a^{2l+1} r^{-l-1}}{D_l}, \qquad B_l = \frac{a^{2l+1} r^l + [l/(l+1)](ab)^{2l+1} r^{-l-1}}{D_l}.$$

Insertion into f_l gives

$$\boxed{g_l = \frac{r_<^l}{r_>^{l+1}} + \frac{1}{D_l}\left[\frac{l+1}{l}(rr')^l + \frac{l}{l+1}\frac{(ab)^{2l+1}}{(rr')^{l+1}} + a^{2l+1}\left(\frac{r^l}{r'^{l+1}} + \frac{r'^l}{r^{l+1}}\right)\right].}$$

It is explicitly symmetric and has the same source derivative jump as the Coulomb radial term; the added terms are harmonic in the source coordinate.

(b) Put $c = a^2/(a^2+b^2)$. For $l = 0$ the proposed form

$$g_0 = \frac{1}{r_>} - \frac{c}{r'} + f(r)$$

is harmonic in r' away from the source. Its inner radial derivative is $c/a^2 = 1/(a^2+b^2)$ and its outer derivative is $(-1+c)/b^2 = -1/(a^2+b^2)$, exactly the required normal derivatives. The function $f(r)$ is arbitrary because the defining differential equation and boundary constraint act on the primed argument. Choosing $f(r) = -c/r + C$ makes g_0 symmetric.

To verify that this freedom is immaterial, write the Neumann representation with ϱ denoting the volume charge density:

$$\Phi(\mathbf{r}) - \langle\Phi\rangle_S = \frac{1}{4\pi\epsilon_0}\int_V \varrho(\mathbf{r}') G_N \, d^3r' + \frac{1}{4\pi}\oint_S G_N \partial_{n'}\Phi \, dS', \quad \langle\Phi\rangle_S = \frac{1}{S}\oint_S \Phi \, dS.$$

Changing f by $\delta f(r)$ changes its right side by

$$\frac{\delta f(r)}{4\pi}\left[\frac{Q_V}{\epsilon_0} + \oint_S \partial_n\Phi \, dS\right] = 0,$$

because Gauss's law/Poisson's equation gives $\oint_S \partial_n\Phi \, dS = -Q_V/\epsilon_0$. This is also the compatibility condition for the prescribed Neumann data. Only the arbitrary additive constant in Φ remains.

3.27 Apply the Neumann Green function of Problem 3.26 to the situation in which the normal electric field is $E_r = -E_0 \cos\theta$ at the outer surface ($r = b$) and is $E_r = 0$ on the inner surface ($r = a$).

(a) Show that the electrostatic potential inside the volume V is

$$\Phi(\mathbf{x}) = E_0 \frac{r\cos\theta}{1-p^3}\left(1 + \frac{a^3}{2r^3}\right)$$

where $p = a/b$. Find the components of the electric field,

$$E_r(r,\theta) = -E_0 \frac{\cos\theta}{1-p^3}\left(1 - \frac{a^3}{r^3}\right), \qquad E_\theta(r,\theta) = E_0 \frac{\sin\theta}{1-p^3}\left(1 + \frac{a^3}{2r^3}\right)$$

(b) Calculate the Cartesian or cylindrical components of the field, E_z and E_ρ, and make a sketch or computer plot of the lines of electric force for a typical case of $p = 0.5$.

Solution.

(a) There is no volume charge. On $r = b$, $\partial_n\Phi = \partial_r\Phi = -E_r = E_0\cos\theta$; on $r = a$ the derivative is zero. Choose the additive constant so that the surface average of Φ is zero. The Neumann formula is then

$$\Phi = \frac{E_0 b^2}{4\pi}\int G_N(\mathbf{r}, b\hat{\mathbf{r}}')\cos\theta'\,d\Omega'.$$

The angular addition theorem gives $\int P_l(\cos\gamma)\cos\theta'\,d\Omega' = (4\pi/3)\delta_{l1}\cos\theta$. From Problem 3.26,

$$g_1(r,b) = \frac{3}{b^2(1-a^3/b^3)}\left(r + \frac{a^3}{2r^2}\right).$$

Thus, with $p = a/b$,

$$\boxed{\Phi = \frac{E_0}{1-p^3}\left(r + \frac{a^3}{2r^2}\right)\cos\theta.}$$

Differentiating, $E_r = -\partial_r\Phi$ and $E_\theta = -r^{-1}\partial_\theta\Phi$, gives exactly

$$E_r = -\frac{E_0}{1-p^3}(1 - a^3/r^3)\cos\theta, \qquad E_\theta = \frac{E_0}{1-p^3}(1 + a^3/(2r^3))\sin\theta.$$

In particular $E_r(a) = 0$ and $E_r(b) = -E_0\cos\theta$, checking both Neumann conditions.

(b) Resolve the unit vectors along and perpendicular to z: $E_z = E_r\cos\theta - E_\theta\sin\theta$, $E_\rho = E_r\sin\theta + E_\theta\cos\theta$. If $K = E_0/(1-p^3)$, this gives

$$\boxed{E_z = -K\left[1 - \frac{a^3}{2r^3}(3\cos^2\theta - 1)\right], \qquad E_\rho = \frac{3Ka^3}{2r^3}\sin\theta\cos\theta.}$$

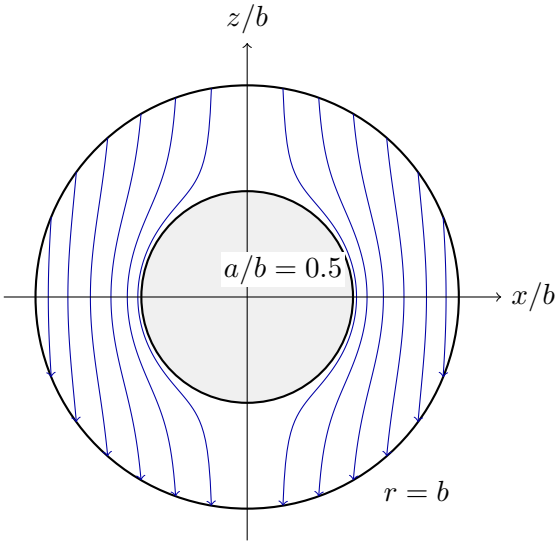
Equivalently $E_\rho = 3Ka^3\rho z/(2r^5)$ and $E_z = -K[1 - a^3(2z^2 - \rho^2)/(2r^5)]$. A useful axisymmetric flux function is

$$\Psi = -\frac{K}{2}\left(r^2 - \frac{a^3}{r}\right)\sin^2\theta, \quad E_r = \frac{\partial_\theta\Psi}{r^2\sin\theta}, \quad E_\theta = -\frac{\partial_r\Psi}{r\sin\theta}.$$

Hence field lines have constant

$$\boxed{\rho^2\left[1 - \frac{a^3}{(\rho^2+z^2)^{3/2}}\right].}$$

This exact first integral was used to generate the following meridional section for $p = 0.5$ and $E_0 > 0$. Arrows run from the upper outer hemisphere to the lower one; the inner sphere is a zero-normal-flux surface. Rotation about z gives the full field.

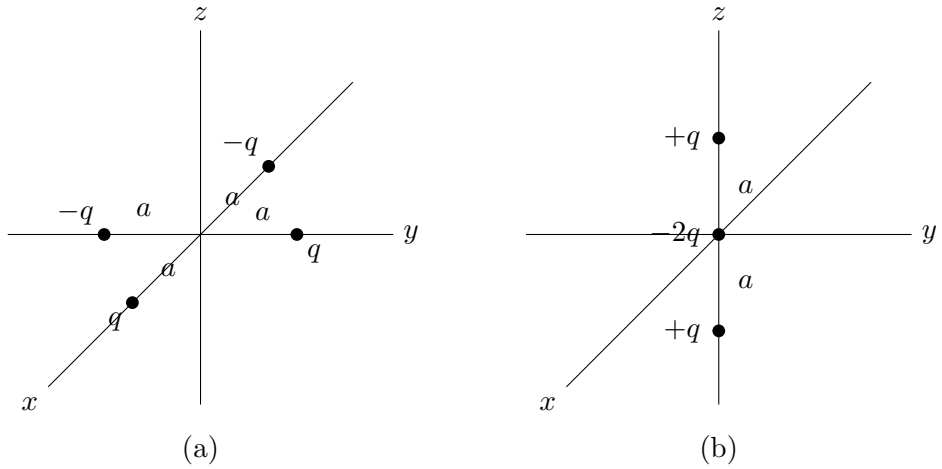


The supplementary check differentiates both the potential and the flux function, tests both boundaries, and verifies the plotted curves against this first integral.

4 Chapter 4. Multipoles, Electrostatics of Macroscopic Media, Dielectrics

Problems

4.1 Calculate the multipole moments q_{lm} of the charge distributions shown as parts a and b. Try to obtain results for the nonvanishing moments valid for all l, but in each case find the first *two* sets of nonvanishing moments at the very least.



Problem 4.1

(c) For the charge distribution of the second set b write down the multipole expansion for the potential. Keeping only the lowest-order term in the expansion, plot the potential in the x-y plane as a function of distance from the origin for distances greater than a.

(d) Calculate directly from Coulomb's law the exact potential for b in the x-y plane. Plot it as a function of distance and compare with the result found in part c.

Divide out the asymptotic form in parts c and d to see the behavior at large distances more clearly.

Solution.

We use Jackson's (4.1), (4.3), and (4.7):

$$q_{lm} = \int r^l Y^*_{lm} \rho \, d^3r, \qquad \Phi = \frac{1}{4\pi\epsilon_0} \sum_{l,m} \frac{4\pi}{2l+1} \frac{q_{lm} Y_{lm}}{r^{l+1}}, \qquad q_{l,-m} = (-1)^m q^*_{lm}.$$

Here $Y_{lm} = N_{lm} P_l^m(\cos\theta) e^{im\phi}$ for $m \geq 0$, $N_{lm} = [(2l+1)(l-m)!/(4\pi(l+m)!)]^{1/2}$, and P_l^m includes the Condon–Shortley factor $(-1)^m$.

(a) The positive x axis points down and left in the drawing. The charges are therefore $+q$ at $a\hat{\mathbf{x}}, a\hat{\mathbf{y}}$ and $-q$ at $-a\hat{\mathbf{x}}, -a\hat{\mathbf{y}}$. Substitution gives

$$\begin{aligned} q_{lm} &= qa^l N_{lm} P_l^m(0)[1 + e^{-im\pi/2} - e^{-im\pi} - e^{-3im\pi/2}] \\ &= qa^l N_{lm} P_l^m(0)[1 - (-1)^m][1 + e^{-im\pi/2}]. \end{aligned}$$

Even l vanish by odd inversion symmetry; $P_l^m(0) = 0$ when $l + m$ is odd. To evaluate the nonzero values, use Rodrigues' polynomial and $P_l^m(x) = (-1)^m(1-x^2)^{m/2}d^m P_l/dx^m$:

$$P_l(x) = 2^{-l}\sum_{j=0}^{\lfloor l/2\rfloor}\frac{(-1)^j(2l-2j)!}{j!(l-j)!(l-2j)!}\,x^{l-2j}.$$

At $x = 0$, only the term with $l - 2j = m$ survives the m derivatives. For even $l + m$ this gives

$$P_l^m(0) = (-1)^{(l+m)/2}\frac{(l+m)!}{2^l[(l-m)/2]![(l+m)/2]!} = (-1)^{(l+m)/2}\frac{(l+m-1)!!}{(l-m)!!}.$$

The factor $1-(-1)^m$ in the charge sum also requires odd m. For odd l and positive odd $m \le l$ the complete answer is

$$\boxed{q_{lm} = 2qa^l N_{lm}(-1)^{(l+m)/2}\frac{(l+m-1)!!}{(l-m)!!}(1+e^{-im\pi/2}).}$$

Negative m follow by conjugation, and all remaining moments vanish. The first two nonzero sets are

$$q_{11} = -qa\sqrt{\frac{3}{2\pi}}(1-i), \qquad q_{1,-1} = -q_{11}^*,$$

$$q_{31} = qa^3\sqrt{\frac{21}{16\pi}}(1-i), \quad q_{33} = -qa^3\sqrt{\frac{35}{16\pi}}(1+i), \quad q_{3,-1} = -q_{31}^*, \quad q_{3,-3} = -q_{33}^*.$$

As a check, $\mathbf{p} = 2qa(\hat{\mathbf{x}} + \hat{\mathbf{y}})$ reproduces q_{11} through Jackson's (4.5).

(b) Axial symmetry permits only $m = 0$. The central charge contributes only to $l = 0$, cancelling the other two charges. Since $Y_{l0}(0) = \sqrt{(2l+1)/(4\pi)}$ and $Y_{l0}(\pi) = (-1)^l Y_{l0}(0)$,

$$\boxed{q_{2n,0} = 2qa^{2n}\sqrt{\frac{4n+1}{4\pi}}, \quad n \ge 1.}$$

Thus $q_{20} = qa^2\sqrt{5/\pi}$ and $q_{40} = 3qa^4/\sqrt{\pi}$ are the first two sets; all other moments vanish.

(c) Substitution into the potential, using $Y_{l0} = \sqrt{(2l+1)/(4\pi)}P_l$, gives

$$\boxed{\Phi(r,\theta) = \frac{2q}{4\pi\epsilon_0 r}\sum_{n=1}^{\infty}\left(\frac{a}{r}\right)^{2n}P_{2n}(\cos\theta), \quad r > a.}$$

In the x–y plane let $s = \sqrt{x^2+y^2}$. As $P_2(0) = -1/2$, $\Phi_2(s) = -qa^2/(4\pi\epsilon_0 s^3)$.

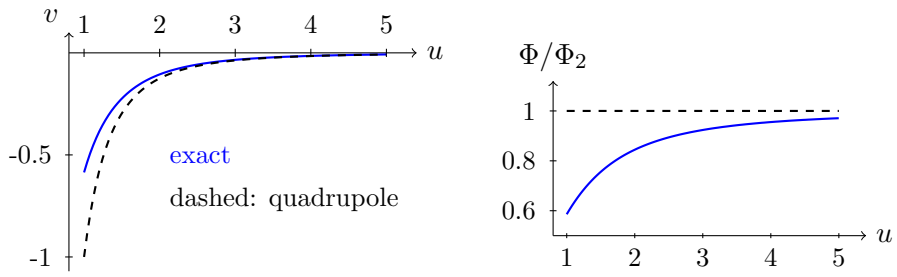
(d) The distances to the three charges are $\sqrt{s^2+a^2}, s, \sqrt{s^2+a^2}$. Hence

$$\boxed{\Phi_{\text{exact}}(s) = \frac{2q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{s^2+a^2}} - \frac{1}{s}\right).}$$

Set $u = s/a$ and $v = 4\pi\epsilon_0 a\Phi/q$. Then $v_2 = -u^{-3}$ and $v_{\text{exact}} = 2[(u^2+1)^{-1/2} - u^{-1}]$. Dividing by the common asymptotic potential and rationalizing gives

$$\begin{aligned}\frac{\Phi_{\text{exact}}}{\Phi_2} &= 2u^2[1-(1+u^{-2})^{-1/2}] \\ &= \frac{2}{\sqrt{1+u^{-2}}[1+\sqrt{1+u^{-2}}]} = 1 - \frac{3}{4u^2} + \frac{5}{8u^4} + O(u^{-6}).\end{aligned}$$

The quadrupole term overestimates the magnitude. The ratio increases from $2-\sqrt{2}$ at $u = 1$ to 1 at infinity.



For example the ratios at $u = 2, 3, 5$ are $0.84458, 0.92370, 0.97097$.

4.2 A point dipole with dipole moment $\mathbf{p}$ is located at the point $\mathbf{x}_0$. From the properties of the derivative of a Dirac delta function, show that for calculation of the potential Φ or the energy of a dipole in an external field, the dipole can be described by an effective charge density

$$\rho_{\text{eff}}(\mathbf{x}) = -\mathbf{p} \cdot \nabla \delta(\mathbf{x} - \mathbf{x}_0)$$

Solution.

The defining distributional identity is

$$\int f(\mathbf{x}) \partial_i \delta(\mathbf{x} - \mathbf{x}_0)\, d^3x = -\partial_i f(\mathbf{x}_0).$$

Thus $\int \rho_{\text{eff}} f\, d^3x = \mathbf{p} \cdot \nabla f(\mathbf{x}_0)$. For an observation point $\mathbf{r} \neq \mathbf{x}_0$, put $f(\mathbf{x}) = 1/|\mathbf{r} - \mathbf{x}|$. The derivative is with respect to the source coordinate, so

$$\nabla_{\mathbf{x}} \frac{1}{|\mathbf{r} - \mathbf{x}|} = \frac{\mathbf{r} - \mathbf{x}}{|\mathbf{r} - \mathbf{x}|^3}, \qquad \boxed{\Phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot (\mathbf{r} - \mathbf{x}_0)}{|\mathbf{r} - \mathbf{x}_0|^3}.}$$

For the energy choose $f = \Phi^{(0)}$, smooth at $\mathbf{x}_0$. Since this is interaction with distinct external sources, there is no factor $1/2$:

$$\boxed{W = \int \rho_{\text{eff}} \Phi^{(0)}\, d^3x = \mathbf{p} \cdot \nabla \Phi^{(0)}(\mathbf{x}_0) = -\mathbf{p} \cdot \mathbf{E}^{(0)}(\mathbf{x}_0).}$$

Self-energy is excluded. Taking $f = 1$ checks zero total charge; taking $f = x_i$ checks $\int x_i \rho_{\text{eff}}\, d^3x = p_i$. For a further direct construction, place $\pm q$ a displacement $\pm\mathbf{d}/2$ from $\mathbf{x}_0$. Taylor expansion gives

$$q[\delta(\mathbf{x} - \mathbf{x}_0 - \mathbf{d}/2) - \delta(\mathbf{x} - \mathbf{x}_0 + \mathbf{d}/2)] = -q\mathbf{d} \cdot \nabla \delta(\mathbf{x} - \mathbf{x}_0) + O(qd^3).$$

Taking $\mathbf{d} \to 0$ with $q\mathbf{d} = \mathbf{p}$ fixed gives the same density and fixes its sign.

4.3 The lth term in the multipole expansion (4.1) of the potential is specified by the $(2l + 1)$ multipole moments q_{lm}. On the other hand, the Cartesian multipole moments,

$$Q^{(l)}_{\alpha\beta\gamma} = \int \rho(\mathbf{x})\, x^\alpha y^\beta z^\gamma\, d^3x$$

with α, β, γ nonnegative integers subject to the constraint $\alpha + \beta + \gamma = l$, are $(l + 1)(l + 2)/2$ in number. Thus, for $l > 1$ there are more Cartesian multipole moments than seem necessary to describe the term in the potential whose radial dependence is r^{-l-1}.

Show that while the q_{lm} transform under rotations as irreducible spherical tensors of rank l, the Cartesian multipole moments correspond to reducible spherical tensors of ranks l, $l-2$, $l-4$, …, $l_{\min}$, where $l_{\min}=0$ or 1 for l even or odd, respectively. Check that the number of different tensorial components adds up to the total number of Cartesian tensors. Why are only the q_{lm} needed in the expansion (4.1)?

Solution.

Package the Cartesian moments into the symmetric rank-l tensor $M_{i_1\cdots i_l}=\int \rho x_{i_1}\cdots x_{i_l}\,d^3x$. Its independent components correspond to homogeneous polynomials of degree l, whose monomials are $x^\alpha y^\beta z^\gamma$ with $\alpha+\beta+\gamma=l$. There are $\binom{l+2}{2}$ such monomials.

A homogeneous polynomial has the unique harmonic decomposition

$$F_l(\mathbf{x})=H_l(\mathbf{x})+r^2H_{l-2}(\mathbf{x})+r^4H_{l-4}(\mathbf{x})+\cdots, \qquad \nabla^2H_k=0,$$

where each H_k is homogeneous of degree k. Here is why traces can be removed recursively. For such a harmonic H_k, Euler's identity $\mathbf{x}\cdot\nabla H_k=kH_k$ gives

$$\begin{aligned}\nabla^2(r^{2s}H_k)&=(\nabla^2r^{2s})H_k+2\nabla r^{2s}\cdot\nabla H_k\\&=[2s(2s+1)+4sk]r^{2s-2}H_k=2s(2k+2s+1)r^{2s-2}H_k.\end{aligned}$$

The coefficient is nonzero for $s\geq 1$. Decompose ∇^2F_l recursively and subtract the corresponding $r^{2s}H_k$ terms; the remainder is harmonic. The same nonzero coefficients establish uniqueness. In tensor language H_l is the symmetric traceless part, while each factor r^2 corresponds to a Kronecker delta, or trace.

The degree-k harmonic polynomials have the basis $\mathcal{Y}_{km}=r^kY_{km}$, $-k\leq m\leq k$. Rotations preserve degree and harmonicity and mix only these $2k+1$ components. Moreover $L^2Y_{km}=k(k+1)Y_{km}$, and $L_\pm$ connect all m in that set; this is the irreducible rank-k representation. Therefore the Cartesian moments contain the ranks

$$\boxed{l,\ l-2,\ l-4,\ldots,l_{\min},}$$

whereas the integrals $q_{lm}=\int\rho\mathcal{Y}^*_{lm}\,d^3x$ select just the irreducible rank-l part. The lower-rank pieces of $M^{(l)}$ involve radial weights r^{2s}; they are not generally the ordinary moments of order $l-2s$.

For $l=2n$, the number of components is

$$\sum_{k=0}^{n}(4k+1)=(n+1)(2n+1)=\frac{(l+1)(l+2)}{2};$$

for $l=2n+1$ it is

$$\sum_{k=0}^{n}(4k+3)=(n+1)(2n+3)=\frac{(l+1)(l+2)}{2}.$$

Finally, Taylor expansion of the Coulomb kernel yields

$$\Phi_l(\mathbf{r})=\frac{1}{4\pi\epsilon_0}\frac{(-1)^l}{l!}M_{i_1\cdots i_l}\partial_{i_1}\cdots\partial_{i_l}\frac{1}{r}.$$

Any trace in M contracts two derivatives into $\nabla^2(1/r)=0$ at exterior points. Only its symmetric traceless part contributes, and that part is fully specified by the $2l+1$ q_{lm}. This explains why the extra Cartesian components do not appear in (4.1).

4.4

(a) Prove the following theorem: For an arbitrary charge distribution $\rho(\mathbf{x})$ the values of the $(2l+1)$ moments of the first nonvanishing multipole are independent of the origin of the coordinate axes, but the values of all higher multipole moments do in general depend on the choice of origin. (The different moments q_{lm} for fixed l depend, of course, on the orientation of the axes.)

(b) A charge distribution has multipole moments q, $\mathbf{p}$, Q_{ij}, ... with respect to one set of coordinate axes, and moments q', $\mathbf{p}'$, Q'_{ij}, ... with respect to another set whose axes are parallel to the first, but whose origin is located at the point $\mathbf{R} = (X, Y, Z)$ relative to the first. Determine explicitly the connections between the monopole, dipole, and quadrupole moments in the two coordinate frames.

(c) If $q \neq 0$, can $\mathbf{R}$ be found so that $\mathbf{p}' = 0$? If $q \neq 0$, $\mathbf{p} \neq 0$, or at least $\mathbf{p} \neq 0$, can $\mathbf{R}$ be found so that $Q'_{ij} = 0$?

Solution.

(a) Keep the axes parallel and move the origin to $\mathbf{R}$, so that $\mathbf{x}' = \mathbf{x} - \mathbf{R}$. For the regular solid harmonic $\mathcal{Y}_{lm} = r^l Y_{lm}$, the finite Taylor expansion is

$$\mathcal{Y}^*_{lm}(\mathbf{x} - \mathbf{R}) = \sum_{k=0}^{l} \frac{(-\mathbf{R} \cdot \nabla)^k}{k!} \mathcal{Y}^*_{lm}(\mathbf{x}).$$

Each derivative on the right is a homogeneous polynomial of degree $l - k$ and is harmonic, since differentiation commutes with ∇^2. It is consequently a linear combination of the $\mathcal{Y}_{l-k,m'}$. Multiplying by ρ and integrating shows that

$$q'_{lm} = q_{lm} + \text{linear combinations of moments of orders } 0, \ldots, l-1.$$

If all moments below order L vanish, then $q'_{Lm} = q_{Lm}$ and all moments below L remain zero. Higher moments contain terms involving the nonzero order-L moments, so in general they change under translation. Particular components may be unchanged for particular translations; the theorem does not assert otherwise. Rotations still mix the m components within each order.

(b) With $q = \int \rho \, d^3x$, $p_i = \int \rho x_i \, d^3x$, and $Q_{ij} = \int \rho(3x_i x_j - r^2\delta_{ij}) \, d^3x$,

$$\boxed{q' = q, \qquad p'_i = p_i - qR_i.}$$

For the quadrupole, expand its integrand before integrating:

$$\begin{aligned} 3(x_i - R_i)(x_j - R_j) - |\mathbf{x} - \mathbf{R}|^2\delta_{ij} = {} & 3x_i x_j - r^2\delta_{ij} - 3(R_i x_j + R_j x_i) \\ & + 2(\mathbf{R} \cdot \mathbf{x})\delta_{ij} + 3R_i R_j - R^2\delta_{ij}. \end{aligned}$$

Thus

$$\boxed{Q'_{ij} = Q_{ij} - 3(R_i p_j + R_j p_i) + 2\mathbf{R} \cdot \mathbf{p} \, \delta_{ij} + q(3R_i R_j - R^2\delta_{ij}).}$$

The trace remains zero, as it must.

(c) If $q \neq 0$, the unique dipole-free origin is $\boxed{\mathbf{R} = \mathbf{p}/q}$. A quadrupole-free origin generally does not exist. To give the precise restriction, define the quadrupole at this dipole-free origin,

$$Q^c_{ij} = Q_{ij} - \frac{3p_i p_j - p^2\delta_{ij}}{q}, \qquad \mathbf{s} = \mathbf{R} - \frac{\mathbf{p}}{q}.$$

Completing the square in the transformation law gives

$$Q'_{ij} = Q^c_{ij} + q(3s_i s_j - s^2\delta_{ij}).$$

The matrix $3\mathbf{s}\mathbf{s}^{\mathsf{T}} - s^2\mathbf{I}$ has eigenvalue $2s^2$ along $\mathbf{s}$ and $-s^2$ along each perpendicular direction; this identifies the possible spectra of its negative. Therefore $Q' = 0$ is possible exactly when $Q^c = 0$ (choose $\mathbf{s} = 0$), or when Q^c has eigenvalues $(-2\lambda, \lambda, \lambda)$ with $\lambda/q > 0$. The direction of $\mathbf{s}$ must be the first principal axis, and $s^2 = \lambda/q$. A generic traceless symmetric tensor has three distinct eigenvalues and fails this condition. For example, equal pairs of charges $+e$ at $\pm a\hat{\mathbf{x}}$ and $+2e$ at $\pm a\hat{\mathbf{y}}$ have $q = 6e$, $\mathbf{p} = 0$, and $Q^c = 6ea^2\,\mathrm{diag}(0, 1, -1)$, which cannot be removed by translation. Describing this same distribution from another origin makes $\mathbf{p} \neq 0$ without removing the obstruction.

If $q = 0$ but $\mathbf{p} \neq 0$, choose axes with $\mathbf{p} = p\hat{\mathbf{z}}$. The six symmetric components transform as

$$Q'_{xx} = Q_{xx} + 2pR_z, \quad Q'_{yy} = Q_{yy} + 2pR_z, \quad Q'_{zz} = Q_{zz} - 4pR_z,$$
$$Q'_{xy} = Q_{xy}, \quad Q'_{xz} = Q_{xz} - 3pR_x, \quad Q'_{yz} = Q_{yz} - 3pR_y.$$

Thus $Q' = 0$ requires $Q_{xy} = 0$ and $Q_{xx} = Q_{yy}$; when these conditions hold, choose

$$(R_x, R_y, R_z) = \left(\frac{Q_{xz}}{3p}, \frac{Q_{yz}}{3p}, -\frac{Q_{xx}}{2p}\right).$$

If both q and $\mathbf{p}$ vanish, the quadrupole is invariant and can be zero in a translated frame only if it was zero already.

4.5 A localized charge density $\rho(x, y, z)$ is placed in an external electrostatic field described by a potential $\Phi^{(0)}(x, y, z)$. The external potential varies slowly in space over the region where the charge density is different from zero. Assume that the external sources lie outside a neighborhood of this region, so that $\nabla^2\Phi^{(0)} = 0$ there, as in the derivation of (4.23)–(4.24).

(a) From first principles calculate the total *force* acting on the charge distribution as an expansion in multipole moments times derivatives of the electric field, up to and including the quadrupole moments. Show that the force is

$$\mathbf{F} = q\mathbf{E}^{(0)}(0) + \{\nabla[\mathbf{p}\cdot\mathbf{E}^{(0)}(\mathbf{x})]\}_0 + \left\{\nabla\left[\frac{1}{6}\sum_{j,k} Q_{jk}\frac{\partial E^{(0)}_j}{\partial x_k}(\mathbf{x})\right]\right\}_0 + \cdots$$

Compare this to the expansion (4.24) of the *energy* W. Note that (4.24) is a number—it is not a function of $\mathbf{x}$ that can be differentiated! What is its connection to $\mathbf{F}$?

(b) Repeat the calculation of part a for the total *torque*. For simplicity, evaluate only one Cartesian component of the torque, say N_1. Show that this component is

$$N_1 = [\mathbf{p}\times\mathbf{E}^{(0)}(0)]_1 + \frac{1}{3}\left[\frac{\partial}{\partial x_3}\left(\sum_j Q_{2j}E^{(0)}_j\right) - \frac{\partial}{\partial x_2}\left(\sum_j Q_{3j}E^{(0)}_j\right)\right]_0 + \cdots$$

Solution.

As in Jackson's derivation of (4.23)–(4.24), assume that the external sources lie outside a neighborhood containing the localized distribution. Then $\nabla\cdot\mathbf{E}^{(0)} = 0$ and $\nabla\times\mathbf{E}^{(0)} = 0$ there.

This source-free condition is needed for the stated force formula with only traceless moments. All derivatives below are of the external field.

(a) Start with the force on each charge element:

$$F_i = \int \rho(\mathbf{x}) E_i^{(0)}(\mathbf{x})\, d^3x.$$

Taylor-expand the field around the origin, integrate term by term, and define $M_{jk} = \int \rho x_j x_k\, d^3x$ and $S = \int \rho r^2\, d^3x$:

$$F_i = qE_i^{(0)}(0) + p_j \partial_j E_i^{(0)}(0) + \frac{1}{2} M_{jk} \partial_j \partial_k E_i^{(0)}(0) + \cdots,$$
$$M_{jk} = \frac{1}{3}(Q_{jk} + S\delta_{jk}).$$

The trace term is $S\nabla^2 E_i^{(0)}/6 = 0$, because $\nabla^2 \mathbf{E}^{(0)} = \nabla(\nabla \cdot \mathbf{E}^{(0)}) = 0$. Electrostatics also gives $\partial_j E_i^{(0)} = \partial_i E_j^{(0)}$, so

$$p_j \partial_j E_i^{(0)} = \partial_i (p_j E_j^{(0)}),$$
$$Q_{jk} \partial_j \partial_k E_i^{(0)} = \partial_i (Q_{jk} \partial_k E_j^{(0)}).$$

These identities give exactly

$$\boxed{\mathbf{F} = q\mathbf{E}^{(0)}(0) + \nabla(\mathbf{p} \cdot \mathbf{E}^{(0)})\Big|_0 + \frac{1}{6} \nabla(Q_{jk} \partial_k E_j^{(0)})\Big|_0 + \cdots.}$$

Repeated Cartesian indices are summed. If external charge overlaps the region, the additional term $S\nabla(\nabla \cdot \mathbf{E}^{(0)})/6$ must be retained; the displayed problem formula alone is then insufficient.

To connect force and energy, let $\mathbf{R}$ be a *physical rigid translation* of the charge distribution and let $\mathbf{u}$ denote coordinates fixed in it. Its interaction energy is the function

$$W(\mathbf{R}) = \int \rho(\mathbf{u}) \Phi^{(0)}(\mathbf{R} + \mathbf{u})\, d^3u$$
$$= q\Phi^{(0)}(\mathbf{R}) - p_j E_j^{(0)}(\mathbf{R}) - \frac{1}{6} Q_{jk} \partial_k E_j^{(0)}(\mathbf{R}) + \cdots.$$

The moments are held fixed while differentiating, and $\mathbf{F} = -\nabla_{\mathbf{R}} W(\mathbf{R})|_{\mathbf{R}=0}$. Equation (4.24) is the value $W(0)$, not a spatial function to which one may directly apply $\nabla_{\mathbf{x}}$.

(b) The torque about the chosen origin is

$$N_1 = \int \rho(x_2 E_3^{(0)} - x_3 E_2^{(0)})\, d^3x.$$

Keeping the constant and linear terms of the field produces

$$N_1 = p_2 E_3^{(0)}(0) - p_3 E_2^{(0)}(0) + M_{2j} \partial_j E_3^{(0)}(0) - M_{3j} \partial_j E_2^{(0)}(0) + \cdots$$
$$= [\mathbf{p} \times \mathbf{E}^{(0)}(0)]_1 + \frac{1}{3}[Q_{2j} \partial_j E_3^{(0)} - Q_{3j} \partial_j E_2^{(0)}]_0 + \frac{S}{3}[\partial_2 E_3^{(0)} - \partial_3 E_2^{(0)}]_0 + \cdots.$$

The last term vanishes by zero curl. Interchanging the derivative and field indices in the preceding term then gives

$$\boxed{N_1 = [\mathbf{p} \times \mathbf{E}^{(0)}(0)]_1 + \frac{1}{3}[\partial_3(Q_{2j} E_j^{(0)}) - \partial_2(Q_{3j} E_j^{(0)})]_0 + \cdots.}$$

This torque is about the origin; about a different point it changes by the usual moment of the total force.

4.6 A nucleus with quadrupole moment Q finds itself in a cylindrically symmetric electric field with a gradient $(\partial E_z/\partial z)_0$ along the z axis at the position of the nucleus. Take the nuclear symmetry axis along z and the external field to be source-free at the nucleus; these conditions specify the aligned quadrupole energy below.

(a) Show that the energy of quadrupole interaction is

$$W = -\frac{e}{4} Q \left(\frac{\partial E_z}{\partial z}\right)_0$$

(b) If it is known that $Q = 2 \times 10^{-28}$ m^2 and that W/h is 10 MHz, where h is Planck's constant, calculate $(\partial E_z/\partial z)_0$ in units of $e/4\pi\epsilon_0 a_0^3$, where $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.529 \times 10^{-10}$ m is the Bohr radius in hydrogen.

(c) Nuclear charge distributions can be approximated by a constant charge density throughout a spheroidal volume of semimajor axis a and semiminor axis b. Calculate the quadrupole moment of such a nucleus, assuming that the total charge is Ze. Given that Eu^{153} $(Z = 63)$ has a quadrupole moment $Q = 2.5 \times 10^{-28}$ m^2 and a mean radius

$$R = (a + b)/2 = 7 \times 10^{-15} \text{ m}$$

determine the fractional difference in radius $(a - b)/R$.

Solution.

(a) Jackson's nuclear convention (4.25) is $Q = Q_{zz}/e$, with dimensions of area. Take the nuclear symmetry axis along z, as in that definition. Axial symmetry and tracelessness give

$$Q_{zz} = eQ, \qquad Q_{xx} = Q_{yy} = -eQ/2, \qquad Q_{ij} = 0 \ (i \neq j).$$

At a source-free point of an axially symmetric external field, let $G = (\partial_z E_z)_0$. Then $\partial_x E_x = \partial_y E_y$ and $\partial_x E_x + \partial_y E_y + \partial_z E_z = 0$, so each transverse entry is $-G/2$. Thus

$$W_Q = -\frac{1}{6} Q_{ij} \partial_j E_i = -\frac{1}{6}\left[eQG + 2\left(-\frac{eQ}{2}\right)\left(-\frac{G}{2}\right)\right] = \boxed{-\frac{eQG}{4}}.$$

For a nucleus whose symmetry axis makes angle β with the field axis, write $Q_{ij} = eQ(3\hat{u}_i\hat{u}_j - \delta_{ij})/2$, where $\hat{\mathbf{u}} \cdot \hat{\mathbf{z}} = \cos\beta$. The traceless gradient tensor is $G_{ij} = G(3\hat{z}_i\hat{z}_j - \delta_{ij})/2$. Thus

$$-\frac{1}{6} Q_{ij} G_{ij} = -\frac{eQG}{24}(9\cos^2\beta - 3) = -\frac{eQG}{4} P_2(\cos\beta).$$

The stated formula uses the aligned convention $\beta = 0$.

(b) With $W/h = +10^7$ s^{-1} and $Q > 0$,

$$G = -\frac{4h(10^7 \text{ s}^{-1})}{e(2 \times 10^{-28} \text{ m}^2)} = -8.2713 \times 10^{20} \text{ V m}^{-2}.$$

Using the problem's rounded $a_0 = 0.529 \times 10^{-10}$ m,

$$\frac{e}{4\pi\epsilon_0 a_0^3} = 9.7271 \times 10^{21} \text{ V m}^{-2}, \qquad \boxed{\frac{G}{e/(4\pi\epsilon_0 a_0^3)} = -0.08503.}$$

If the measured 10 MHz denotes only the magnitude of a splitting, only $|G|$ is determined and the ratio is 0.08503 in magnitude.

(c) For a prolate spheroid put the semiaxis a along z and the equal semiaxes b along x, y. Its density is $\rho = 3Ze/(4\pi ab^2)$. Transform the ellipsoid to the unit ball by $x = bu$, $y = bv$, $z = aw$. Spherical symmetry on that ball gives

$$\int_{u^2+v^2+w^2\le 1} w^2\, d^3u = \frac{1}{3}\int_0^1 4\pi t^4\, dt = \frac{4\pi}{15}.$$

Since $d^3x = ab^2 d^3u$,

$$\int \rho z^2\, d^3x = \frac{Zea^2}{5}, \qquad \int \rho x^2\, d^3x = \int \rho y^2\, d^3x = \frac{Zeb^2}{5}.$$

It follows that

$$\boxed{Q = \frac{1}{e}\int \rho(2z^2 - x^2 - y^2)\, d^3x = \frac{2Z}{5}(a^2 - b^2).}$$

For an oblate spheroid the axial semiaxis is the smaller one, and the quadrupole has the opposite sign. Here $Q > 0$ selects the prolate case. Using $a + b = 2R$, with no small-deformation approximation,

$$Q = \frac{4Z}{5}R(a-b), \qquad \boxed{\frac{a-b}{R} = \frac{5Q}{4ZR^2} = \frac{5(2.5\times 10^{-28})}{4(63)(7\times 10^{-15})^2} = 0.10123.}$$

Thus $a \simeq 7.354 \times 10^{-15}$ m and $b \simeq 6.646 \times 10^{-15}$ m in this uniform-density model.

4.7 A localized distribution of charge has a charge density

$$\rho(\mathbf{r}) = \frac{1}{64\pi} r^2 e^{-r} \sin^2\theta$$

(a) Make a multipole expansion of the potential due to this charge density and determine all the nonvanishing multipole moments. Write down the potential at large distances as a finite expansion in Legendre polynomials.

(b) Determine the potential explicitly at any point in space, and show that near the origin, correct to r^2 inclusive,

$$\Phi(\mathbf{r}) \simeq \frac{1}{4\pi\epsilon_0}\left[\frac{1}{4} - \frac{r^2}{120}P_2(\cos\theta)\right]$$

(c) If there exists at the origin a nucleus with a quadrupole moment $Q = 10^{-28}$ m^2, with its symmetry axis along z, determine the magnitude of the interaction energy, assuming that the unit of charge in $\rho(\mathbf{r})$ above is the electronic charge and the unit of length is the hydrogen Bohr radius $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.529 \times 10^{-10}$ m. Express your answer as a frequency by dividing by Planck's constant h.

The charge density in this problem is that for the $m = \pm 1$ states of the $2p$ level in hydrogen, while the quadrupole interaction is of the same order as found in molecules.

Solution.

(a) Use $\sin^2\theta = \frac{2}{3}[1 - P_2(\cos\theta)]$ to write

$$\rho = \frac{r^2 e^{-r}}{96\pi}[1 - P_2(\cos\theta)].$$

Azimuthal integration eliminates $m \neq 0$, and Legendre orthogonality eliminates $l \neq 0, 2$. Since $\int_0^\infty r^n e^{-r}\, dr = n!$ and $\int P_l^2\, d\Omega = 4\pi/(2l+1)$, the total charge and the Legendre quadrupole integral are

$$q = \frac{4\pi}{96\pi}\, 4! = 1, \qquad \int \rho r^2 P_2\, d^3r = -\frac{1}{96\pi}\frac{4\pi}{5}\, 6! = -6.$$

Therefore

$$\boxed{q_{00} = \frac{1}{\sqrt{4\pi}}, \qquad q_{20} = -6\sqrt{\frac{5}{4\pi}} = -3\sqrt{\frac{5}{\pi}},}$$

with all other moments zero. The large-distance expansion is

$$\boxed{\Phi \sim \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r} - \frac{6}{r^3}P_2(\cos\theta)\right].}$$

The density is exponentially localized but is not compactly supported. Consequently this finite algebraic expansion is asymptotic, with exponentially small corrections at finite r; it is not the exact potential throughout an exterior charge-free region.

(b) Split the Coulomb integral into source radii $s < r$ and $s > r$, using the kernel $r_<^l / r_>^{l+1}$. The same angular orthogonality gives

$$4\pi\epsilon_0\Phi = A(r) + B(r)P_2(\cos\theta),$$

$$A(r) = \frac{1}{24}\left[\frac{1}{r}\int_0^r s^4 e^{-s}\, ds + \int_r^\infty s^3 e^{-s}\, ds\right],$$
$$B(r) = -\frac{1}{120}\left[\frac{1}{r^3}\int_0^r s^6 e^{-s}\, ds + r^2\int_r^\infty s e^{-s}\, ds\right].$$

Repeated integration by parts gives

$$\int_r^\infty s^n e^{-s}\, ds = n! e^{-r}\sum_{k=0}^n \frac{r^k}{k!}, \qquad \int_0^r s^n e^{-s}\, ds = n!\left[1 - e^{-r}\sum_{k=0}^n \frac{r^k}{k!}\right].$$

Substituting and collecting equal powers yields the explicit answer

$$\boxed{A(r)} = \boxed{\frac{1}{r} - e^{-r}\left(\frac{1}{r} + \frac{3}{4} + \frac{r}{4} + \frac{r^2}{24}\right),}$$
$$\boxed{B(r)} = \boxed{-\frac{6}{r^3} + e^{-r}\left(\frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{r^2}{24}\right).}$$

These expressions apply for $r > 0$; their apparent singularities are removable at the origin. The integral forms show the cancellations without subtracting divergent terms. For small r, the needed integrals are

$$\int_0^r s^4 e^{-s}\, ds = \frac{r^5}{5} + O(r^6), \qquad \int_r^\infty s^3 e^{-s}\, ds = 6 - \frac{r^4}{4} + O(r^5),$$
$$\int_0^r s^6 e^{-s}\, ds = \frac{r^7}{7} + O(r^8), \qquad \int_r^\infty s e^{-s}\, ds = 1 - \frac{r^2}{2} + O(r^3).$$

Consequently

$$A(r) = \frac{1}{24}\left[6 + \left(\frac{1}{5} - \frac{1}{4}\right) r^4 + O(r^5)\right] = \frac{1}{4} - \frac{r^4}{480} + O(r^5),$$
$$B(r) = -\frac{1}{120}\left[r^2 + \left(\frac{1}{7} - \frac{1}{2}\right) r^4 + O(r^5)\right] = -\frac{r^2}{120} + \frac{r^4}{336} + O(r^5),$$

so

$$\boxed{\Phi = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{4} - \frac{r^2}{120} P_2(\cos\theta) + O(r^4)\right].}$$

For example, differentiating the exact radial functions checks Poisson's equation directly:

$$A'' + \frac{2}{r}A' = -\frac{r^2 e^{-r}}{24}, \qquad B'' + \frac{2}{r}B' - \frac{6}{r^2}B = +\frac{r^2 e^{-r}}{24}.$$

(c) Restore physical length $\mathbf{R} = a_0\mathbf{r}$ and charge q_e (equal to $-e$ for an electron). The quadratic physical potential is

$$\Phi_Q(\mathbf{R}) = -\frac{q_e}{4\pi\epsilon_0 a_0^3}\frac{3Z^2 - R^2}{240}.$$

Since $3Z^2 - R^2 = 2Z^2 - X^2 - Y^2$,

$$\left(\frac{\partial E_Z}{\partial Z}\right)_0 = -\partial_Z^2 \Phi_Q = \frac{q_e}{4\pi\epsilon_0 a_0^3}\frac{1}{60}.$$

For the nuclear quadrupole convention and alignment of Problem 4.6,

$$|W_Q| = \frac{e^2|Q|}{240(4\pi\epsilon_0)a_0^3}, \qquad \boxed{\frac{|W_Q|}{h} = 9.8000 \times 10^5 \text{ Hz} \simeq 0.980 \text{ MHz}.}$$

The sign reverses if the sign of the charge distribution is reversed; for an electron and $Q > 0$, this aligned quadrupole contribution is positive.

4.8 A very long, right circular, cylindrical shell of dielectric constant ϵ/ϵ_0 and inner and outer radii a and b, respectively, is placed in a previously uniform electric field E_0 with its axis perpendicular to the field. The medium inside and outside the cylinder has a dielectric constant of unity.

(a) Determine the potential and electric field in the three regions, neglecting end effects.

(b) Sketch the lines of force for a typical case of $b \simeq 2a$.

(c) Discuss the limiting forms of your solution appropriate for a solid dielectric cylinder in a uniform field, and a cylindrical cavity in a uniform dielectric.

Solution.

(a) Let $\kappa = \epsilon/\epsilon_0 > 0$, use cylindrical coordinates (s, ϕ, z), and take the applied field along x, so $x = s\cos\phi$. Translation invariance along z and the $\cos\phi$ boundary condition leave only the cylindrical harmonic of order one. The solutions regular at $s = 0$ and asymptotic to $-E_0 x$ are

$$\Phi = -E_0\cos\phi \begin{cases} As, & s < a, \\ Bs + C/s, & a < s < b, \\ s - D/s, & s > b. \end{cases}$$

There is no free surface charge at either dielectric boundary. Continuity of Φ and of $D_s = -\epsilon\partial_s\Phi$ gives four equations:

$$A = B + C/a^2, \qquad A = \kappa(B - C/a^2),$$
$$B + C/b^2 = 1 - D/b^2, \qquad \kappa(B - C/b^2) = 1 + D/b^2.$$

The first pair gives $C = a^2B(\kappa-1)/(\kappa+1)$ and $A = 2\kappa B/(\kappa+1)$. Adding the second pair eliminates D:

$$(\kappa+1)B - (\kappa-1)C/b^2 = 2.$$

Define $\eta = a^2/b^2$ and $\Delta = (\kappa+1)^2 - (\kappa-1)^2\eta$. Substitution gives

$$\boxed{A = \frac{4\kappa}{\Delta}, \quad B = \frac{2(\kappa+1)}{\Delta}, \quad C = \frac{2(\kappa-1)a^2}{\Delta}, \quad D = \frac{(\kappa^2-1)(b^2-a^2)}{\Delta}.}$$

The field is $\mathbf{E} = E_s\hat{\mathbf{s}} + E_\phi\hat{\boldsymbol{\phi}}$, with $E_z = 0$ and

$$\boxed{\frac{E_s}{E_0} = \cos\phi \begin{cases} A, & s < a, \\ B - C/s^2, & a < s < b, \\ 1 + D/s^2, & s > b, \end{cases} \qquad \frac{E_\phi}{E_0} = -\sin\phi \begin{cases} A, & s < a, \\ B + C/s^2, & a < s < b, \\ 1 - D/s^2, & s > b. \end{cases}}$$

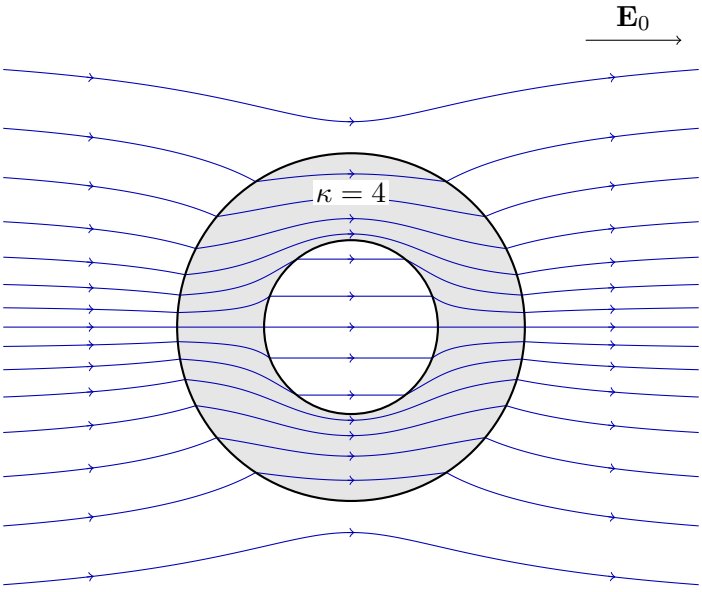
Thus the field in the hollow core is uniform, $\mathbf{E} = AE_0\hat{\mathbf{x}}$. For $\kappa = 1$, $A = B = 1$ and $C = D = 0$, recovering a uniform vacuum field.

(b) The field lines are contours of a stream function. In each homogeneous region introduce the stream function for $\mathbf{D}/(\epsilon_0 E_0)$:

$$\Psi(s,\phi) = \sin\phi \begin{cases} As, & s < a, \\ \kappa(Bs - C/s), & a < s < b, \\ s + D/s, & s > b. \end{cases}$$

It obeys $D_s/(\epsilon_0 E_0) = s^{-1}\partial_\phi\Psi$ and $D_\phi/(\epsilon_0 E_0) = -\partial_s\Psi$. Hence $\mathbf{E}\cdot\nabla\Psi = 0$, and contours of Ψ are field lines. The boundary equations above make Ψ continuous across both circles. The direction changes there in accordance with continuous E_ϕ and ϵE_s.

The plot uses $b = 2a$, $\kappa = 4$: $A = 64/91$, $B = 40/91$, $C/a^2 = 24/91$, $D/a^2 = 180/91$. The shaded annulus is the dielectric, and arrows point along the field.



$b = 2a$, coordinates in units of a

(c) For a solid cylinder set $a \to 0$, keeping b fixed. The shell becomes the entire interior, with

$$\boxed{\Phi_{\text{in}} = -\frac{2E_0}{\kappa+1} s\cos\phi, \qquad \Phi_{\text{out}} = -E_0\left(s - \frac{\kappa-1}{\kappa+1}\frac{b^2}{s}\right)\cos\phi.}$$

For an isolated cylindrical cavity of radius a in an infinite dielectric, denote the prescribed field deep in the dielectric by E_∞. The cavity and dielectric potentials are

$$\boxed{\Phi_{\text{cavity}} = -\frac{2\kappa}{\kappa+1} E_\infty s\cos\phi, \qquad \Phi_{\text{diel}} = -E_\infty\left(s + \frac{\kappa-1}{\kappa+1}\frac{a^2}{s}\right)\cos\phi.}$$

To see this limit in the shell solution, first let $b \to \infty$ at fixed s, a. Then $B \to 2/(\kappa+1)$ and $A \to 4\kappa/(\kappa+1)^2$, so the bulk dielectric field is $E_\infty = 2E_0/(\kappa+1)$. Expressing the result in terms of this bulk field yields the boxed cavity formulas. Keeping the distant *vacuum* field E_0 fixed instead would retain this additional outer-interface screening factor.

4.9 A point charge q is located in free space a distance d from the center of a dielectric sphere of radius a ($a < d$) and dielectric constant ϵ/ϵ_0.

(a) Find the potential at all points in space as an expansion in spherical harmonics.

(b) Calculate the rectangular components of the electric field *near* the center of the sphere.

(c) Verify that, in the limit $\epsilon/\epsilon_0 \to \infty$, your result is the same as that for the conducting sphere.

Solution.

(a) Choose the polar axis through the charge, so its position is $d\hat{\mathbf{z}}$, and set $\kappa = \epsilon/\epsilon_0$. For $r < d$ its vacuum potential has the expansion

$$\frac{q}{4\pi\epsilon_0|\mathbf{r} - d\hat{\mathbf{z}}|} = \frac{q}{4\pi\epsilon_0}\sum_{l=0}^{\infty}\frac{r^l}{d^{l+1}}P_l(\cos\theta).$$

Write the interior and exterior solutions as

$$\Phi_{\rm in} = \frac{q}{4\pi\epsilon_0}\sum_{l=0}^{\infty} A_l \frac{r^l}{d^{l+1}} P_l(\cos\theta),$$
$$\Phi_{\rm out} = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{|\mathbf{r}-d\hat{\mathbf{z}}|} + \sum_{l=0}^{\infty} B_l \frac{a^{2l+1}}{d^{l+1}r^{l+1}} P_l(\cos\theta)\right].$$

At $r=a$, continuity of potential gives $A_l = 1 + B_l$. Continuity of the normal displacement gives $\kappa l A_l = l - (l+1)B_l$. Therefore

$$[(\kappa+1)l+1]B_l = -(\kappa-1)l, \qquad \boxed{A_l = \frac{2l+1}{(\kappa+1)l+1}, \quad B_l = -\frac{(\kappa-1)l}{(\kappa+1)l+1}.}$$

These coefficients in the preceding two formulas determine the potential everywhere except at the point charge itself. In particular $A_0 = 1$ and $B_0 = 0$: the induced polarization has zero net charge.

These are spherical-harmonic expansions because $P_l(\cos\theta) = \sqrt{4\pi/(2l+1)}Y_{l0}(\theta,\phi)$. For a source in a general direction Ω_d, replace each P_l by

$$P_l(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}}) = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{lm}(\Omega)Y_{lm}^*(\Omega_d).$$

If an expansion of the direct term is desired also outside d, use

$$\frac{1}{|\mathbf{r}-\mathbf{d}|} = \sum_{l=0}^{\infty}\frac{r_<^l}{r_>^{l+1}}P_l(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}}), \quad r_< = \min(r,d), \quad r_> = \max(r,d).$$

The exact Coulomb term avoids convergence qualifications on the sphere $r=d$, while the induced series converges throughout $r \geq a$ since $a < d$.

(b) Keep $l = 0, 1, 2$ and use $rP_1 = z$, $r^2P_2 = (2z^2 - x^2 - y^2)/2$. Near the center,

$$\Phi_{\rm in} = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{d} + \frac{3z}{(\kappa+2)d^2} + \frac{5(2z^2-x^2-y^2)}{2(2\kappa+3)d^3} + O\left(\frac{r^3}{d^4}\right)\right].$$

Taking the negative Cartesian gradient gives

$$\boxed{E_x} = \boxed{\frac{q}{4\pi\epsilon_0}\frac{5x}{(2\kappa+3)d^3} + O\left(\frac{qr^2}{4\pi\epsilon_0 d^4}\right)},$$
$$\boxed{E_y} = \boxed{\frac{q}{4\pi\epsilon_0}\frac{5y}{(2\kappa+3)d^3} + O\left(\frac{qr^2}{4\pi\epsilon_0 d^4}\right)},$$
$$\boxed{E_z} = \boxed{-\frac{q}{4\pi\epsilon_0}\left[\frac{3}{(\kappa+2)d^2} + \frac{10z}{(2\kappa+3)d^3}\right] + O\left(\frac{qr^2}{4\pi\epsilon_0 d^4}\right)}.$$

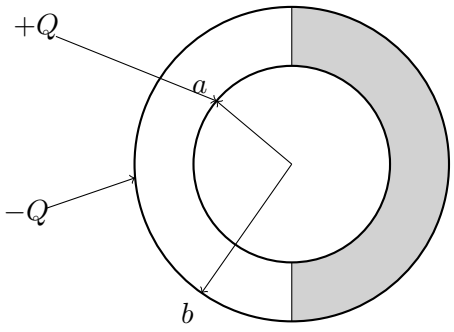
In particular the central field is the applied vacuum field at the center times $3/(\kappa+2)$.

(c) As $\kappa \to \infty$, $A_l \to 0$ for $l \geq 1$, $A_0 = 1$, and $B_l \to -1$ for $l \geq 1$ while $B_0 = 0$. Thus the sphere is equipotential at $q/(4\pi\epsilon_0 d)$. Set $d' = a^2/d$. The exterior induced series sums to

$$-\frac{q}{4\pi\epsilon_0}\sum_{l=1}^{\infty}\frac{a^{2l+1}}{d^{l+1}r^{l+1}}P_l = \frac{1}{4\pi\epsilon_0}\left[-\frac{qa/d}{|\mathbf{r}-d'\hat{\mathbf{z}}|} + \frac{qa/d}{r}\right].$$

These are the image charge $-qa/d$ at d' and compensating charge $+qa/d$ at the center. Their total is zero. The limit is therefore an *isolated neutral* conducting sphere, rather than a grounded sphere; omitting the compensating central image would give the wrong monopole and boundary potential for this dielectric problem.

4.10 Two concentric conducting spheres of inner and outer radii a and b, respectively, carry charges $\pm Q$. The empty space between the spheres is half-filled by a hemispherical shell of dielectric (of dielectric constant ϵ/ϵ_0), as shown in the figure.



Problem 4.10

(a) Find the electric field everywhere between the spheres.

(b) Calculate the surface-charge distribution on the inner sphere.

(c) Calculate the polarization-charge density induced on the surface of the dielectric at $r = a$.

Solution.

(a) Let the inner conductor carry $+Q$ and the outer one $-Q$, as in the textbook figure. Seek a radial field $\mathbf{E} = K\hat{\mathbf{r}}/r^2$. Its potential between the conductors, with the outer one set to zero, is

$$\Phi(r) = K\left(\frac{1}{r} - \frac{1}{b}\right).$$

Both spherical conductors are equipotentials. On the plane separating the two hemispheres, this radial field is tangential, so E_t is continuous and $D_n = 0$ on both sides. Thus all interface conditions are satisfied. A Gaussian sphere of radius $a < r < b$ gives

$$Q = \oint \mathbf{D}\cdot d\mathbf{a} = \epsilon_0 \frac{K}{r^2}(2\pi r^2) + \epsilon \frac{K}{r^2}(2\pi r^2) = 2\pi(\epsilon + \epsilon_0)K.$$

Consequently, by uniqueness for positive permittivities,

$$\boxed{\mathbf{E}(r) = \frac{Q}{2\pi(\epsilon + \epsilon_0)r^2}\hat{\mathbf{r}}, \qquad a < r < b.}$$

For $\epsilon = \epsilon_0$ this reduces to the usual spherical-capacitor field $Q/(4\pi\epsilon_0 r^2)$.

(b) The free surface charge on the inner conductor is $\sigma_f = D_r(a^+)$, since the field inside it is zero. Hence

$$\boxed{\sigma_f = \begin{cases} \dfrac{\epsilon Q}{2\pi(\epsilon + \epsilon_0)a^2}, & \text{hemisphere touching dielectric,} \\ \dfrac{\epsilon_0 Q}{2\pi(\epsilon + \epsilon_0)a^2}, & \text{hemisphere touching vacuum.} \end{cases}}$$

The two hemispheres carry $Q\epsilon/(\epsilon+\epsilon_0)$ and $Q\epsilon_0/(\epsilon+\epsilon_0)$, respectively, which sum to Q. The finite jump at the equator does not require an additional line charge in this solution.

(c) In the dielectric $\mathbf{P} = (\epsilon - \epsilon_0)\mathbf{E}$. Its outward unit normal at its inner surface is $-\hat{\mathbf{r}}$, so

$$\boxed{\sigma_{\rm pol}(a) = \mathbf{P}\cdot(-\hat{\mathbf{r}}) = -\frac{(\epsilon-\epsilon_0)Q}{2\pi(\epsilon+\epsilon_0)a^2}.}$$

This applies only on the dielectric hemisphere. As checks, $\rho_{\rm pol} = -\nabla\cdot\mathbf{P} = 0$ within the dielectric, the flat interface has no polarization charge because $\mathbf{P}$ is tangential, and the bound charge at $r = b$ has opposite total charge to that at $r = a$.

4.11 The following data on the variation of dielectric constant with pressure are taken from the *Smithsonian Physical Tables*, 9th ed., p. 424:

Air at 292 K

Pressure (atm)	ϵ/ϵ_0	
20	1.0108	Relative density of air as a function of
40	1.0218	pressure is given in *AIP Handbook*,
60	1.0333	[3rd ed., McGraw-Hill, New York
80	1.0439	(1972), p. 4-165].
100	1.0548	

Pentane (C_5H_{12}) at 303 K

Pressure (atm)	**Density (g/cm^3)**	ϵ/ϵ_0
1	0.613	1.82
10^3	0.701	1.96
4×10^3	0.796	2.12
8×10^3	0.865	2.24
12×10^3	0.907	2.33

Test the Clausius–Mossotti relation between dielectric constants and density for air and pentane in the ranges tabulated. Does it hold exactly? Approximately? If approximately, discuss fractional variations in density and $(\epsilon/\epsilon_0 - 1)$. For pentane, compare the Clausius–Mossotti relation to the cruder relation, $(\epsilon/\epsilon_0 - 1) \propto$ density.

Solution.

Put $\kappa = \epsilon/\epsilon_0$ and $F = (\kappa-1)/(\kappa+2)$. For fixed molecular mass and polarizability, Clausius–Mossotti says

$$F = \frac{N\gamma_{\rm mol}}{3} = K\rho, \qquad \boxed{K = \frac{\kappa-1}{(\kappa+2)\rho} = \text{constant}.}$$

It is F, rather than κ or $\kappa - 1$ itself, that should be proportional to mass density.

Air. The pressure table alone does not specify real-gas densities. To make the density input explicit, use the primary air tables of Hilsenrath et al., *NBS Circular 564* (1955), Table 2-1, p. 31, for the compressibility factor $Z_g = PV/(nRT)$.[1] The relevant rows are

[1] The source is available at `https://nvlpubs.nist.gov/nistpubs/Legacy/circ/nbscircular564.pdf`. The input rows, and the independent density rows from Table 2-2, p. 37, are transcribed in `supplementary/ch04-source/air_reference.csv`. This supplies a checked density source in place of the AIP reference cited in the problem.

P (atm)	10	40	70	100
Z_g(290 K)	0.99623	0.98782	0.98420	0.98600
Z_g(300 K)	0.99717	0.99135	0.99000	0.99330

Linearly interpolate in temperature, $Z_g(292, P) = 0.8Z_g(290, P) + 0.2Z_g(300, P)$. For the intervening pressures use the cubic interpolating polynomial through the four pressure entries:

$$Z_g(P) = \sum_{i=1}^{4} Z_g(P_i) \prod_{j \neq i} \frac{P - P_j}{P_i - P_j}.$$

Since the temperature and composition are fixed, $\rho \propto P/Z_g$, so the reference density cancels:

$$\frac{\rho(P)}{\rho(20)} = \frac{PZ_g(20)}{20Z_g(P)}, \qquad \frac{K(P)}{K(20)} = \frac{F(P)/F(20)}{\rho(P)/\rho(20)}.$$

This gives

P (atm)	Z_g	$\rho/\rho(20)$	10^3F	$K/K(20)$
20	0.99330	1.00000	3.58709	1.00000
40	0.98853	2.00965	7.21424	1.00076
60	0.98586	3.02264	10.97814	1.01251
80	0.98545	4.03185	14.42229	0.99721
100	0.98746	5.02955	17.93898	0.99432

The ratio is approximately constant: its peak-to-peak spread is about 1.8% of its 20-atm value. Linear instead of cubic pressure interpolation changes the normalized ratios by less than 0.05%, so the percent-level conclusion does not depend on this interpolation choice. These are approximate density-based comparisons, not a claim of exact agreement. The dielectric data have no stated uncertainties, so their scatter alone should not be assigned a definite microscopic cause.

Across the range,

$$\frac{\Delta\rho}{\rho_{20}} \simeq 4.0295 \quad (403.0\%), \qquad \frac{\Delta(\kappa - 1)}{\kappa_{20} - 1} = \frac{0.0548 - 0.0108}{0.0108} = 4.0741 \quad (407.4\%).$$

Thus the density and susceptibility increase by almost the same fraction even though each changes by roughly a factor of five. This is expected because $F = (\kappa - 1)/[3 + (\kappa - 1)] \simeq (\kappa - 1)/3$ for this gas. If ideal-gas densities were used instead, F/P would range from 1.7935×10^{-4} to 1.8297×10^{-4} atm^{-1}, again giving approximate agreement but omitting the real-gas correction.

Pentane. Here the actual densities are given. The two proposed constant ratios are $K = F/\rho$ and the cruder $J = (\kappa - 1)/\rho$, both in cm^3/g:

P (atm)	K	K/K_1	J	J/J_1
1	0.350179	1.00000	1.337684	1.00000
1000	0.345826	0.98757	1.369472	1.02376
4000	0.341513	0.97525	1.407035	1.05184
8000	0.338096	0.96549	1.433526	1.07165
12000	0.338654	0.96709	1.466373	1.09620

Clausius–Mossotti is again approximate: K varies by about 3.45% relative to its initial value, whereas J rises by 9.62%. It is therefore the better of the two proportionalities for these data, but neither is exact. The endpoint fractional changes are

$$\frac{\Delta\rho}{\rho_1} = \frac{0.907 - 0.613}{0.613} = 0.47961, \qquad \frac{\Delta(\kappa-1)}{\kappa_1 - 1} = \frac{1.33 - 0.82}{0.82} = 0.62195.$$

Density rises 47.96%, while susceptibility rises 62.20%. The difference is consistent in direction with the local-field correction: differentiating $F = (\kappa-1)/(\kappa+2) = K\rho$ at constant K gives

$$\frac{d(\kappa-1)}{\kappa-1} = \frac{\kappa+2}{3}\frac{d\rho}{\rho}.$$

The multiplier is close to one for air, but is appreciably larger for pentane. The tabulated residual variation of K measures the remaining failure of a strictly density-independent molecular-polarizability model.

4.12 Water vapor is a polar gas whose dielectric constant exhibits an appreciable temperature dependence. The following table gives experimental data on this effect. Assuming that water vapor obeys the ideal gas law, calculate the molecular polarizability as a function of inverse temperature and plot it. From the slope of the curve, deduce a value for the permanent dipole moment of the H_2O molecule (express the dipole moment in coulomb-meters).

T(K)	**Pressure (cm Hg)**	$\boldsymbol{(\epsilon/\epsilon_0 - 1) \times 10^5}$
393	56.49	400.2
423	60.93	371.7
453	65.34	348.8
483	69.75	328.7

Solution.

Use Jackson's convention $\langle \mathbf{p}_{\rm mol} \rangle = \epsilon_0 \gamma_{\rm mol} \mathbf{E}_{\rm local}$, so the molecular polarizability $\gamma_{\rm mol}$ has units of volume. Here k denotes Boltzmann's constant, as in (4.81)–(4.82). The alternative SI convention is $\alpha = \epsilon_0 \gamma_{\rm mol}$, with units $\mathrm{C\,m^2/V}$. The ideal gas law gives $N = P/(kT)$. Applying the Clausius–Mossotti relation (4.70) to each row, with $\delta = \epsilon/\epsilon_0 - 1$, yields

$$\boxed{\gamma_{\rm mol}(T) = \frac{3kT}{P}\frac{\delta}{3+\delta}, \qquad \alpha(T) = \epsilon_0 \gamma_{\rm mol}(T).}$$

Here P must be in pascals: 1 cm Hg = 101325/76 Pa. The resulting values are

T (K)	$10^3/T$ ($\mathrm{K^{-1}}$)	$\gamma_{\rm mol}/(10^{-28}\ \mathrm{m^3})$	$\alpha/(10^{-39}\ \mathrm{C\,m^2/V})$
393	2.54453	2.87938	2.54946
423	2.36407	2.66897	2.36316
453	2.20751	2.50133	2.21473
483	2.07039	2.35455	2.08476

For completeness, the inverse-temperature law follows by averaging the permanent dipoles. If p_0 is the fixed molecular dipole magnitude and u its direction cosine with the local field, its orientation energy is $-p_0 E u$. In a weak field $p_0 E \ll kT$,

$$\begin{aligned}\langle p_{\parallel} \rangle &= p_0 \frac{\int_{-1}^{1} u \exp[p_0 E u/(kT)]\, du}{\int_{-1}^{1} \exp[p_0 E u/(kT)]\, du} \\ &= p_0 \frac{(p_0 E/kT)\int_{-1}^{1} u^2\, du + O(E^3)}{2 + O(E^2)} = \frac{p_0^2 E}{3kT} + O(E^3).\end{aligned}$$

Adding a temperature-independent induced polarizability gives

$$\gamma_{\mathrm{mol}}(T) = \gamma_i + \frac{p_0^2}{3\epsilon_0 k}\frac{1}{T}.$$

An unweighted straight-line fit $\gamma_{\mathrm{mol}} = A + B/T$ is obtained from $x_i = 1/T_i$, $y_i = \gamma_{\mathrm{mol}}(T_i)$ by

$$B = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad A = \bar{y} - B\bar{x}.$$

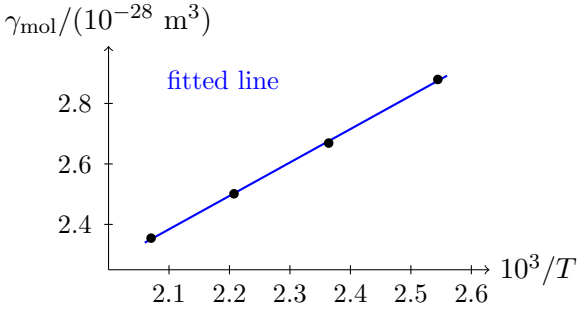
The four rows give

$$A = 6.3875 \times 10^{-30}\ \mathrm{m}^3, \qquad B = 1.10475 \times 10^{-25}\ \mathrm{m}^3\,\mathrm{K}.$$

Thus the requested dipole magnitude is

$$\boxed{p_0 = \sqrt{3\epsilon_0 k B} = 6.365 \times 10^{-30}\ \mathrm{C\,m} \simeq 6.4 \times 10^{-30}\ \mathrm{C\,m}.}$$

The rounding of the data does not warrant interpreting the last digits as an experimental precision. The largest deviation of a tabulated polarizability from this fitted line is 0.25%.



Neglecting the small local-field correction would replace $3\delta/(3+\delta)$ by δ; retaining it as above is consistent with the chapter's definition of molecular polarizability.

4.13 Two long, coaxial, cylindrical conducting surfaces of radii a and b are lowered vertically into a liquid dielectric. If the liquid rises an average height h between the electrodes when a potential difference V is established between them, show that the susceptibility of the liquid is

$$\chi_e = \frac{(b^2 - a^2)\rho g h\, \ln(b/a)}{\epsilon_0 V^2}$$

where ρ is the density of the liquid, g is the acceleration due to gravity, and the susceptibility of air is neglected.

Solution.

Neglect end effects, surface tension, and changes of liquid density, and take the external reservoir to be large enough that its level is fixed. Write $\epsilon = \epsilon_0(1+\chi_e)$ and $\Lambda = \ln(b/a)$. A homogeneous length L of the coaxial capacitor has radial field $E_r = V/(r\Lambda)$ and capacitance

$$C = \frac{2\pi\epsilon L}{\Lambda}.$$

Raising the dielectric by dh replaces a length dh of air, so

$$\frac{dC}{dh} = \frac{2\pi(\epsilon-\epsilon_0)}{\Lambda} = \frac{2\pi\epsilon_0\chi_e}{\Lambda}.$$

At fixed voltage, the battery supplies $V\,dQ = V^2 dC$, while the field energy increases by $d(\frac{1}{2}CV^2) = \frac{1}{2}V^2 dC$. Their difference is available for mechanical work, hence

$$F_{\text{electric}}\,dh = \frac{1}{2}V^2 dC, \qquad F_{\text{electric}} = \frac{\pi\epsilon_0\chi_e V^2}{\Lambda}.$$

Equivalently, the mechanical potential at fixed V includes $-CV^2/2$, not $+CV^2/2$.

The excess volume of liquid between the electrodes is $\pi(b^2-a^2)h$, where h is the area-averaged rise. Its weight, after subtracting the hydrostatic support at the external free-surface level, is

$$F_g = \rho g \pi (b^2-a^2)h.$$

Force balance gives

$$\rho g\pi(b^2-a^2)h = \frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)}, \qquad \boxed{\chi_e = \frac{(b^2-a^2)\rho g h \ln(b/a)}{\epsilon_0 V^2}.}$$

For a flat column the same result follows by minimizing $\rho g\pi(b^2-a^2)h^2/2 - C(h)V^2/2$. The total electric force can also be checked by integrating the leading local dielectric pressure over the annular cross section:

$$\int_a^b \frac{1}{2}(\epsilon-\epsilon_0)\left[\frac{V}{r\ln(b/a)}\right]^2 2\pi r\,dr = \frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)}.$$

Thus use of the average rise is consistent with the varying radial field. The susceptibility is positive for a liquid that rises, and all dimensions cancel in the boxed expression.

5 Chapter 5. Magnetostatics, Faraday's Law, Quasi-Static Fields Problems

5.1 Starting with the differential expression

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi}\, d\mathbf{l}' \times \frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}$$

for the magnetic induction at the point P with coordinate $\mathbf{x}$ produced by an increment of current $I\, d\mathbf{l}'$ at $\mathbf{x}'$, show explicitly that for a closed loop carrying a current I the magnetic induction at P is

$$\mathbf{B} = \frac{\mu_0 I}{4\pi}\nabla\Omega$$

where Ω is the solid angle subtended by the loop at the point P. This corresponds to a magnetic scalar potential, $\Phi_M = -\mu_0 I\Omega/4\pi$. The sign convention for the solid angle is that Ω is positive if the point P views the "inner" side of the surface spanning the loop, that is, if a unit normal $\mathbf{n}$ to the surface is defined by the direction of current flow via the right-hand rule, Ω is positive if $\mathbf{n}$ points *away* from the point P, and negative otherwise. This is the same convention as in Section 1.6 for the electric dipole layer.

Solution.

Let $G = |\mathbf{x}-\mathbf{x}'|^{-1}$ and orient a spanning surface S by the right-hand rule for the current. The specified solid angle is

$$\Omega(\mathbf{x}) = \int_S \frac{\mathbf{n}'\cdot(\mathbf{x}'-\mathbf{x})}{|\mathbf{x}-\mathbf{x}'|^3} da' = \int_S n'_j \partial_j G\, da'.$$

Here unprimed derivatives act on the observation point, and $\partial'_j G = -\partial_j G$. Put $\mathbf{C} = \oint d\mathbf{l}' \times \nabla' G$. Componentwise Stokes's theorem gives

$$\begin{aligned} C_i &= \epsilon_{ijk}\epsilon_{jlm}\int_S n'_l \partial'_m \partial'_k G\, da' \\ &= \int_S [n'_k \partial'_i \partial'_k G - n'_i \nabla'^2 G] da' = \partial_i \int_S n'_k \partial_k G\, da' = \partial_i \Omega. \end{aligned}$$

The delta term $\nabla'^2 G = -4\pi\delta^3(\mathbf{x}'-\mathbf{x})$ is absent when the observation point is off the chosen spanning surface. Since $\nabla' G = (\mathbf{x}-\mathbf{x}')/|\mathbf{x}-\mathbf{x}'|^3$, Biot–Savart now gives

$$\boxed{\mathbf{B} = \frac{\mu_0 I}{4\pi}\nabla\Omega, \qquad \Phi_M = -\frac{\mu_0 I}{4\pi}\Omega, \quad \mathbf{B} = -\nabla\Phi_M.}$$

For $\mathbf{H}$ rather than $\mathbf{B}$, divide this scalar potential by μ_0. The solid angle changes by 4π on crossing a spanning surface. One must use a continuous local branch before differentiating; a distributional derivative of a discontinuous branch would add a spurious sheet term. This multivaluedness is necessary because a path linking the wire has $\oint \mathbf{H}\cdot d\mathbf{l} = I$, despite the existence of a local scalar potential away from the current.

5.2 A long, right cylindrical, ideal solenoid of arbitrary cross section is created by stacking a large number of identical current-carrying loops one above the other, with N coils per unit length and each loop carrying a current I. [In practice such a solenoid could be wound on a mandrel machined to the arbitrary cross section. After the coil was made rigid (e.g., with epoxy), the mandrel would be withdrawn.]

(a) In the approximation that the solenoidal coil is an ideal current sheet and infinitely long, use Problem 5.1 to establish that at any point inside the coil the magnetic field is axial and equal to

$$H = NI$$

and that $H = 0$ for any point outside the coil.

(b) For a realistic solenoid of circular cross section of radius a ($Na \gg 1$), but still infinite in length, show that the "smoothed" magnetic field just outside the solenoid (averaged axially over several turns) is not zero, but is the same in magnitude and direction as that of a single wire on the axis carrying a current I, even if $Na \to \infty$. Compare fields inside and out.

Solution.

(a) Orient the loops counterclockwise as seen from $+z$. Let D be their cross section and $u = z' - z$ the height of a source loop above the observation point. Its principal solid angle is

$$\Omega_0(u) = \int_D \frac{u\, d^2 s'}{[|\mathbf{s} - \mathbf{s}'|^2 + u^2]^{3/2}}.$$

It is odd in u, tends to zero at $u = \pm\infty$, and, for $\mathbf{s}$ strictly inside D, tends to $\pm 2\pi$ as $u \to 0^{\pm}$. The last limit follows by integrating over a small disk centered at $\mathbf{s}$:

$$\int_0^{\delta} \frac{2\pi u\, s\, ds}{(s^2 + u^2)^{3/2}} = 2\pi \left[\operatorname{sgn} u - \frac{u}{\sqrt{\delta^2 + u^2}} \right].$$

Outside D there is no jump.

For interior observation points choose the continuous local branch $\widetilde{\Omega} = \Omega_0(z' - z) + 4\pi H(z - z')$; outside use Ω_0. The sum of the loop potentials for a finite stack is $-NI \int \widetilde{\Omega}\, dz'/(4\pi)$ for the $\mathbf{H}$ potential. As both ends recede to infinity, derivatives of the principal-angle integral vanish: its transverse contributions cancel by oddness, and its axial end terms tend to zero. Inside, the added branch contributes

$$\nabla \int 4\pi H(z - z')\, dz' = 4\pi \widehat{\mathbf{z}}.$$

There is no such term outside. Problem 5.1 therefore gives

$$\boxed{\mathbf{H} = NI\, \widehat{\mathbf{z}} \quad (\mathbf{s} \in D), \qquad \mathbf{H} = 0 \quad (\mathbf{s} \notin \overline{D}).}$$

This result is independent of the shape of D. The jump agrees with $\mathbf{n} \times (\mathbf{H}_{\text{out}} - \mathbf{H}_{\text{in}}) = \mathbf{K}$, where $\mathbf{K} = NI\, \widehat{\mathbf{z}} \times \mathbf{n}$.

(b) A continuous helical wire advances by one pitch $1/N$ per turn. Axial averaging gives two components of sheet current:

$$K_\phi = NI, \qquad 2\pi a K_z = I, \qquad K_z = \frac{I}{2\pi a}.$$

The second equality follows because a plane perpendicular to the axis cuts the single winding once and carries net axial current I. The azimuthal sheet produces the field in part (a). An Amperian circle for the axial sheet gives $2\pi\rho H_\phi = I$ outside, and zero inside. Thus, neglecting the spatial ripple between turns,

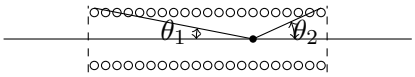
$$\boxed{\mathbf{H}_{\rm in} = NI\,\widehat{\mathbf{z}}, \qquad \mathbf{H}_{\rm out} = \frac{I}{2\pi\rho}\widehat{\boldsymbol{\phi}}.}$$

At $\rho = a^+$, $H_{\rm out}/H_{\rm in} = 1/(2\pi Na)$. For fixed I, the exterior field does not vanish as N increases, although its ratio to the interior field does. A return lead can alter this exterior field; the result is for the isolated helical winding.

5.3 A right-circular solenoid of finite length L and radius a has N turns per unit length and carries a current I. Show that the magnetic induction on the cylinder axis in the limit $NL \to \infty$ is

$$B_z = \frac{\mu_0 NI}{2}\,(\cos\theta_1 + \cos\theta_2)$$

where the angles are defined in the figure.



Problem 5.3

Solution.

First derive the axial field of one loop. Parameterize it by

$$\mathbf{x}' = (a\cos\phi', a\sin\phi', z'), \qquad d\mathbf{l}' = a(-\sin\phi', \cos\phi', 0)d\phi'.$$

Then the axial component of $d\mathbf{l}' \times (\mathbf{x} - \mathbf{x}')$ at $(0, 0, z)$ is $a^2 d\phi'$. The transverse components integrate to zero, so

$$dB_z = \frac{\mu_0 (NI\,dz')a^2}{2[a^2 + (z - z')^2]^{3/2}}.$$

Choose the ends at $\pm L/2$ and integrate:

$$\begin{aligned} B_z(z) &= \frac{\mu_0 NI}{2}\int_{-L/2}^{L/2} \frac{a^2\,dz'}{[a^2 + (z - z')^2]^{3/2}} \\ &= \frac{\mu_0 NI}{2}\left[\frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z - L/2)^2}}\right]. \end{aligned}$$

The antiderivative used here is $u/\sqrt{a^2 + u^2}$, whose derivative is $a^2/(a^2 + u^2)^{3/2}$. For a point between the ends,

$$\cos\theta_1 = \frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}}, \qquad \cos\theta_2 = \frac{L/2 - z}{\sqrt{a^2 + (L/2 - z)^2}},$$

which proves the stated result. The algebraic formula also applies outside the solenoid, with the corresponding signed cosines. The continuum limit replaces the discrete turns by a sheet of current NI; a long solenoid gives $B_z \to \mu_0 NI$ deep inside.

5.4 A magnetic induction **B** in a current-free region in a uniform medium is cylindrically symmetric with components $B_z(\rho, z)$ and $B_\rho(\rho, z)$ and with a known $B_z(0, z)$ on the axis of symmetry. The magnitude of the axial field varies slowly in z.

(a) Show that near the axis the axial and radial components of magnetic induction are approximately

$$B_z(\rho, z) \simeq B_z(0, z) - \left(\frac{\rho^2}{4}\right)\left[\frac{\partial^2 B_z(0, z)}{\partial z^2}\right] + \cdots$$
$$B_\rho(\rho, z) \simeq -\left(\frac{\rho}{2}\right)\left[\frac{\partial B_z(0, z)}{\partial z}\right] + \left(\frac{\rho^3}{16}\right)\left[\frac{\partial^3 B_z(0, z)}{\partial z^3}\right] + \cdots$$

(b) What are the magnitudes of the neglected terms, or equivalently what is the criterion defining "near" the axis?

Solution.

(a) Smooth axial symmetry requires expansions

$$B_z(\rho, z) = \sum_{n=0}^{\infty} a_n(z)\rho^{2n}, \qquad B_\rho(\rho, z) = \sum_{n=0}^{\infty} b_n(z)\rho^{2n+1}.$$

The divergence equation and the azimuthal curl equation are

$$\rho^{-1}\partial_\rho(\rho B_\rho) + \partial_z B_z = 0, \qquad \partial_z B_\rho - \partial_\rho B_z = 0.$$

Equating powers of ρ gives

$$(2n + 2)b_n + a_n' = 0, \qquad b_n' = 2(n + 1)a_{n+1}.$$

Combining these, $a_{n+1} = -a_n''/[4(n + 1)^2]$. Writing $f(z) = B_z(0, z) = a_0(z)$, repeated substitution yields

$$a_n = \frac{(-1)^n f^{(2n)}}{2^{2n}(n!)^2}, \qquad b_n = \frac{(-1)^{n+1} f^{(2n+1)}}{2^{2n+1} n!(n + 1)!}.$$

In particular,

$$\boxed{B_z = f - \frac{\rho^2}{4}f'' + \frac{\rho^4}{64}f^{(4)} + \cdots,} \qquad \boxed{B_\rho = -\frac{\rho}{2}f' + \frac{\rho^3}{16}f''' - \frac{\rho^5}{384}f^{(5)} + \cdots.}$$

(b) The first omitted terms in the question are $\rho^4 f^{(4)}/64$ and $-\rho^5 f^{(5)}/384$. If derivatives have a common variation length ℓ, $|f^{(n)}| \sim |f|/\ell^n$, the expansion parameter is $\rho/\ell \ll 1$. More precisely, each omitted term must be small relative to the accuracy required of the retained sum. Ratios to f' or f'' alone are inappropriate where those derivatives happen to vanish. The series is local and its radius of convergence cannot extend through a current sheet or another field singularity.

5.5

(a) Use the results of Problems 5.4 and 5.3 to find the axial and radial components of magnetic induction in the *central region* ($|z| \ll L/2$) of a long uniform solenoid of radius a and ends at $z = \pm L/2$, including the value of B_z just inside the coil ($\rho = a^-$).

(b) Use Ampère's law to show that the longitudinal magnetic induction just outside the coil is approximately

$$B_z(\rho = a^+, z) \simeq -\left(\frac{2\mu_0 NIa^2}{L^2}\right)\left(1 + \frac{12z^2}{L^2} - \frac{9a^2}{L^2} + \cdots\right)$$

For $L \gg a$, the field outside is negligible compared to inside. How does this axial component compare in size to the azimuthal component of Problem 5.2b?

(c) Show that at the end of the solenoid the magnetic induction near the axis has components

$$B_z \simeq \frac{\mu_0 NI}{2}, \qquad B_\rho \simeq \pm\frac{\mu_0 NI}{4}\left(\frac{\rho}{a}\right)$$

Solution.

Put $\ell = L/2$ and $B_s = \mu_0 NI$. We consider $a, \rho, |z| \ll \ell$. The exact on-axis result of 5.3 is

$$\frac{f(z)}{B_s} = \frac{1}{2}\left[\frac{\ell + z}{\sqrt{a^2 + (\ell + z)^2}} + \frac{\ell - z}{\sqrt{a^2 + (\ell - z)^2}}\right].$$

Use $(1 + x)^{-1/2} = 1 - x/2 + 3x^2/8 + \cdots$ and then expand in z/ℓ:

$$\frac{f}{B_s} = 1 - \frac{a^2}{2\ell^2} - \frac{3a^2z^2}{2\ell^4} + \frac{3a^4}{8\ell^4} + O\left(\frac{a^2z^4 + a^4z^2 + a^6}{\ell^6}\right).$$

(a) Differentiating this axis function and using 5.4 gives

$$\boxed{\frac{B_z}{B_s} = 1 - \frac{a^2}{2\ell^2} - \frac{3a^2z^2}{2\ell^4} + \frac{3a^4}{8\ell^4} + \frac{3a^2\rho^2}{4\ell^4} + \cdots, \qquad \frac{B_\rho}{B_s} = \frac{3a^2\rho z}{2\ell^4} + \cdots.}$$

For example, the next terms in B_ρ/B_s are $5a^2\rho z^3/\ell^6 - 15a^4\rho z/(4\ell^6) - 15a^2\rho^3 z/(4\ell^6)$. At the inner wall,

$$B_z(a^-, z) = B_s\left[1 - \frac{2a^2}{L^2} - \frac{24a^2z^2}{L^4} + \frac{18a^4}{L^4} + \cdots\right].$$

(b) A narrow rectangular Ampere loop straddling the winding gives $H_z(a^-) - H_z(a^+) = NI$. Subtracting B_s from the preceding expression therefore gives

$$\boxed{B_z(a^+, z) = -\frac{2\mu_0 NIa^2}{L^2}\left[1 + \frac{12z^2}{L^2} - \frac{9a^2}{L^2} + \cdots\right].}$$

It is an axial return field. For an actual helical winding, $B_\phi(a^+) = \mu_0 I/(2\pi a)$, so near the center

$$\frac{|B_z(a^+)|}{|B_\phi(a^+)|} \simeq \frac{4\pi N a^3}{L^2}.$$

Thus either exterior component can dominate, depending on N, a, L; both can be small compared with the main interior field.

(c) At the upper end $z = \ell$,

$$f(\ell) = \frac{B_s}{2}\frac{L}{\sqrt{L^2 + a^2}} \simeq \frac{B_s}{2}, \qquad f'(\ell) = \frac{B_s}{2}\left[\frac{a^2}{(L^2 + a^2)^{3/2}} - \frac{1}{a}\right] \simeq -\frac{B_s}{2a}.$$

For $\rho \ll a$ and $L \gg a$, 5.4 then gives $B_\rho \simeq B_s\rho/(4a)$. At the lower end the derivative changes sign. Consequently

$$\boxed{B_z \simeq B_s/2, \qquad B_\rho \simeq \pm B_s\rho/(4a)}$$

at the upper/lower ends, respectively.

5.6 A cylindrical conductor of radius a has a hole of radius b bored parallel to, and centered a distance d from, the cylinder axis ($d + b < a$). The current density is uniform throughout the remaining metal of the cylinder and is parallel to the axis. Use Ampère's law and principle of linear superposition to find the magnitude and the direction of the magnetic-flux density in the hole.

Solution.

Let the axis of the original cylinder be the origin of transverse coordinates, and let the hole center have displacement $\mathbf{d}$. Take $\mathbf{J} = J\widehat{\mathbf{z}}$ in the remaining metal. For a uniformly filled cylinder, an Amperian circle gives $2\pi\rho B_\phi = \mu_0 J\pi\rho^2$, or

$$\mathbf{B}_{\text{full}}(\boldsymbol{\rho}) = \frac{\mu_0}{2}\mathbf{J}\times\boldsymbol{\rho}.$$

Represent the hole by superposing a cylinder with density $-\mathbf{J}$ and local coordinate $\boldsymbol{\rho}-\mathbf{d}$. Every point in the hole is inside both cylinders, so

$$\boxed{\mathbf{B}_{\text{hole}} = \frac{\mu_0}{2}\left[\mathbf{J}\times\boldsymbol{\rho} - \mathbf{J}\times(\boldsymbol{\rho}-\mathbf{d})\right] = \frac{\mu_0}{2}\mathbf{J}\times\mathbf{d}.}$$

The field is uniform, of magnitude $\mu_0 Jd/2$, perpendicular both to the current and to the displacement toward the hole. If the total current is I, then $J = I/[\pi(a^2-b^2)]$. A centered hole gives zero field, providing a symmetry check.

5.7 A compact circular coil of radius a, carrying a current I (perhaps N turns, each with current I/N), lies in the x-y plane with its center at the origin.

(a) By elementary means [Eq. (5.4)] find the magnetic induction at any point on the z axis.

(b) An identical coil with the same magnitude and sense of the current is located on the same axis, parallel to, and a distance b above, the first coil. With the coordinate origin relocated at the point midway between the centers of the two coils, determine the magnetic induction on the axis near the origin as an expansion in powers of z, up to z^4 inclusive:

$$B_z = \left(\frac{\mu_0 I a^2}{d^3}\right)\left[1 + \frac{3(b^2-a^2)z^2}{2d^4} + \frac{15(b^4-6b^2a^2+2a^4)z^4}{16d^8} + \cdots\right]$$

where $d^2 = a^2 + b^2/4$.

(c) Show that, off-axis near the origin, the axial and radial components, correct to second order in the coordinates, take the form

$$B_z = \sigma_0 + \sigma_2\left(z^2 - \frac{\rho^2}{2}\right); \qquad B_\rho = -\sigma_2 z\rho$$

(d) For the two coils in part b show that the magnetic induction on the z axis for large $|z|$ is given by the expansion in inverse odd powers of $|z|$ obtained from the small z expansion of part b by the formal substitution, $d \to |z|$.

(e) If $b = a$, the two coils are known as a pair of Helmholtz coils. For this choice of geometry the second terms in the expansions of parts b and d are absent ($\sigma_2 = 0$ in part c). The field near the origin is then very uniform. What is the maximum permitted value of $|z|/a$ if the axial field is to be uniform to one part in 10^4, one part in 10^2?

Solution.

(a) The element calculation in 5.3 gives

$$\boxed{B_z(0,z)=\frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}},\qquad B_x=B_y=0.}$$

Here I is the total ampere-turns specified in the question.

(b) Put the centers at $z=\pm b/2$ and $d^2=a^2+b^2/4$. The exact sum is

$$B_z=\frac{\mu_0 I a^2}{2d^3}\left[\left(1+\frac{z^2-bz}{d^2}\right)^{-3/2}+\left(1+\frac{z^2+bz}{d^2}\right)^{-3/2}\right].$$

The binomial coefficients needed are $1,-3/2,15/8,-35/16,315/128$. After averaging the two signs, the coefficients of z^2 and z^4 are

$$-\frac{3}{2d^2}+\frac{15b^2}{8d^4}=\frac{3(b^2-a^2)}{2d^4},$$

$$\frac{15}{8d^4}-\frac{105b^2}{16d^6}+\frac{315b^4}{128d^8}=\frac{15(b^4-6a^2b^2+2a^4)}{16d^8}.$$

The odd powers cancel, proving the stated series.

(c) Write its first terms as $f(z)=\sigma_0+\sigma_2 z^2+\cdots$, where

$$\sigma_0=\frac{\mu_0 I a^2}{d^3},\qquad \sigma_2=\sigma_0\frac{3(b^2-a^2)}{2d^4}.$$

Then $f'=2\sigma_2 z+\cdots$ and $f''=2\sigma_2+\cdots$. Substitution into 5.4 gives

$$\boxed{B_z=\sigma_0+\sigma_2(z^2-\rho^2/2)+O(r^4),\qquad B_\rho=-\sigma_2 z\rho+O(r^4).}$$

(d) For $|z|\gg d$, factor $|z|^3$ from the exact denominators:

$$B_z=\frac{\mu_0 I a^2}{2|z|^3}\sum_{\pm}\left(1\pm\frac{b}{|z|}+\frac{d^2}{z^2}\right)^{-3/2}.$$

The same binomial calculation yields

$$\boxed{B_z=\frac{\mu_0 I a^2}{|z|^3}\left[1+\frac{3(b^2-a^2)}{2z^2}+\frac{15(b^4-6a^2b^2+2a^4)}{16z^4}+\cdots\right].}$$

This is the stated formal replacement $d\mapsto|z|$ in part (b), including its prefactor. It does not replace a or b.

(e) At $b=a$, $d^2=5a^2/4$ and

$$\frac{B_z(z)}{B_z(0)}=1-\frac{144}{125}(z/a)^4+O((z/a)^6).$$

The fourth-order estimate for relative tolerance δ is $|z|/a\le(125\delta/144)^{1/4}$, namely 0.09652 for 10^{-4} and 0.30524 for 10^{-2}. To determine the actual edge of the central uniform interval, solve with the exact field:

$$\frac{(5/4)^{3/2}}{2}\left\{[1+(u-1/2)^2]^{-3/2}+[1+(u+1/2)^2]^{-3/2}\right\}=1-\delta,\qquad u=|z|/a.$$

The exact central-interval limits are $\boxed{|z|/a\le 0.0967727}$ and $\boxed{|z|/a\le 0.3137460}$, respectively. Substitution into the exact expression gives field ratios 0.9999 and 0.9900.

5.8 A localized cylindrically symmetric current distribution is such that the current flows only in the azimuthal direction; the current density is a function only of r and θ (or ρ and z): $\mathbf{J} = \hat{\boldsymbol{\phi}} J(r,\theta)$. The distribution is "hollow" in the sense that there is a current-free region near the origin, as well as outside.

(a) Show that the magnetic field can be derived from the azimuthal component of the vector potential, with a multipole expansion

$$A_\phi(r,\theta) = -\frac{\mu_0}{4\pi} \sum_L m_L r^L P_L^1(\cos\theta)$$

in the interior and

$$A_\phi(r,\theta) = -\frac{\mu_0}{4\pi} \sum_L \mu_L r^{-L-1} P_L^1(\cos\theta)$$

outside the current distribution.

(b) Show that the internal and external multipole moments are

$$m_L = -\frac{1}{L(L+1)} \int d^3x\, r^{-L-1} P_L^1(\cos\theta)\, J(r,\theta)$$

and

$$\mu_L = -\frac{1}{L(L+1)} \int d^3x\, r^L P_L^1(\cos\theta)\, J(r,\theta)$$

Solution.

(a) Choose the observation azimuth to be zero. Reflection in the x–z plane cancels the radial part of $\mathbf{A} = (\mu_0/4\pi)\int \mathbf{J}(\mathbf{x}')/|\mathbf{x}-\mathbf{x}'|\, d^3x'$. There is no z component, and

$$A_\phi = \frac{\mu_0}{4\pi} \int \frac{J(r',\theta')\cos\phi'}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'.$$

Expand the Coulomb kernel as $\sum_{L\ge 0} r_<^L r_>^{-L-1} P_L(\cos\gamma)$. The addition theorem in real azimuthal form is

$$P_L(\cos\gamma) = P_L(\cos\theta)P_L(\cos\theta') + 2\sum_{m=1}^{L} \frac{(L-m)!}{(L+m)!} P_L^m(\cos\theta) P_L^m(\cos\theta')\cos(m\phi').$$

Multiplication by $\cos\phi'$ and integration over $0 \le \phi' < 2\pi$ kills every term except $m = 1$, whose cosine integral is π. Thus

$$A_\phi = \frac{\mu_0}{4\pi} \sum_{L=1}^{\infty} \frac{P_L^1(\cos\theta)}{L(L+1)} \int d^3x'\, \frac{r_<^L}{r_>^{L+1}} P_L^1(\cos\theta') J(r',\theta').$$

(b) In a source-free ball containing the origin, $r < r'$ for all source points, so factor out r^L. In the exterior of a sphere containing all currents, $r > r'$ and one factors out r^{-L-1}. Define

$$\boxed{m_L = -\frac{1}{L(L+1)} \int r'^{-L-1} P_L^1(\cos\theta') J\, d^3x', \qquad \mu_L = -\frac{1}{L(L+1)} \int r'^{L} P_L^1(\cos\theta') J\, d^3x'.}$$

Then

$$\boxed{A_{\phi,\mathrm{in}} = -\frac{\mu_0}{4\pi}\sum_{L\geq 1} m_L r^L P_L^1, \qquad A_{\phi,\mathrm{out}} = -\frac{\mu_0}{4\pi}\sum_{L\geq 1} \mu_L r^{-L-1} P_L^1.}$$

The moments use the Condon–Shortley convention $P_L^1(\cos\theta) = \partial_\theta P_L(\cos\theta)$. Finally, $B_r = (r\sin\theta)^{-1}\partial_\theta(\sin\theta A_\phi)$ and $B_\theta = -r^{-1}\partial_r(rA_\phi)$, with $B_\phi = 0$. These convergence domains are spherical source-free regions; an arbitrary point merely outside the current support need not belong to the convergence domain of either expansion.

5.9 The two circular coils of radius a and separation b of Problem 5.7 can be described in cylindrical coordinates by the current density

$$\mathbf{J} = \hat{\boldsymbol{\phi}} I\delta(\rho - a)[\delta(z - b/2) + \delta(z + b/2)]$$

(a) Using the formalism of Problem 5.8, calculate the internal and external multipole moments for $L = 1, \ldots, 5$.

(b) Using the internal multipole expansion of Problem 5.8, write down explicitly an expression for B_z on the z axis and relate it to the answer of Problem 5.7b.

Solution.

(a) Let $d = \sqrt{a^2 + b^2/4}$ and $c = b/(2d)$. At the two source rings, $r' = d$ and $\cos\theta' = \pm c$. The cylindrical volume element supplies $\int J\, d^3x' = 2\pi aI$ for each ring. Because $P_L^1(-c) = (-1)^{L+1}P_L^1(c)$, even L vanish and odd L give

$$m_L = -\frac{4\pi aI}{L(L+1)d^{L+1}}P_L^1(c), \qquad \mu_L = d^{2L+1}m_L.$$

Use

$$P_1^1(c) = -\sqrt{1-c^2}, \quad P_3^1(c) = -\tfrac{3}{2}(5c^2-1)\sqrt{1-c^2}, \quad P_5^1(c) = -\tfrac{15}{8}(21c^4 - 14c^2 + 1)\sqrt{1-c^2},$$

with $\sqrt{1-c^2} = a/d$. The explicit moments are

L	m_L	μ_L
1	$2\pi Ia^2/d^3$	$2\pi Ia^2$
2	0	0
3	$\pi Ia^2(b^2-a^2)/(2d^7)$	$\pi Ia^2(b^2-a^2)/2$
4	0	0
5	$\pi Ia^2(b^4 - 6a^2b^2 + 2a^4)/(8d^{11})$	$\pi Ia^2(b^4 - 6a^2b^2 + 2a^4)/8$

(b) The identity $\partial_\theta(\sin\theta P_L^1) = -L(L+1)\sin\theta P_L$ gives

$$B_{r,\mathrm{in}} = \frac{\mu_0}{4\pi}\sum_{L\geq 1} L(L+1)m_L r^{L-1}P_L(\cos\theta).$$

On the positive z axis $P_L(1) = 1$ and $B_z = B_r$, hence

$$B_z(0,z) = \frac{\mu_0}{4\pi}[2m_1 + 12m_3z^2 + 30m_5z^4 + \cdots].$$

Inserting the table reproduces every coefficient in 5.7(b). Reflection symmetry extends this even polynomial to negative z.

5.10 A circular current loop of radius a carrying a current I lies in the x-y plane with its center at the origin.

(a) Show that the only nonvanishing component of the vector potential is

$$A_\phi(\rho,z)=\frac{\mu_0 Ia}{\pi}\int_0^\infty dk\,\cos kz\,I_1(k\rho_<)K_1(k\rho_>)$$

where $\rho_<$ ($\rho_>$) is the smaller (larger) of a and ρ.

(b) Show that an alternative expression for A_ϕ is

$$A_\phi(\rho,z)=\frac{\mu_0 Ia}{2}\int_0^\infty dk\,e^{-k|z|}J_1(ka)J_1(k\rho)$$

(c) Write down integral expressions for the components of magnetic induction, using the expressions of parts a and b. Evaluate explicitly the components of $\mathbf{B}$ on the z axis by performing the necessary integrations.

Solution.

The scalar cylindrical Green expansions needed here follow by separating the Fourier variable along z, or by separating radial Bessel modes:

$$\frac{1}{|\mathbf{x}-\mathbf{x}'|}=\frac{2}{\pi}\sum_{m=-\infty}^{\infty}e^{im(\phi-\phi')}\int_0^\infty\cos[k(z-z')]I_m(k\rho_<)K_m(k\rho_>)\,dk,$$
$$\frac{1}{|\mathbf{x}-\mathbf{x}'|}=\sum_{m=-\infty}^{\infty}e^{im(\phi-\phi')}\int_0^\infty e^{-k|z-z'|}J_m(k\rho)J_m(k\rho')\,dk.$$

For completeness, the first normalization follows from $\int_0^\infty\cos(kz)K_0(ks)\,dk=\pi/[2\sqrt{s^2+z^2}]$ and the cylindrical addition theorem for K_0. The second follows from $\int_0^\infty e^{-k|z|}J_0(ks)\,dk=1/\sqrt{s^2+z^2}$ and the addition theorem for J_0.

(a),(b) The current loop has $d\mathbf{l}'=a\widehat{\boldsymbol{\phi}}'\,d\phi'$. Projecting onto the observation $\widehat{\boldsymbol{\phi}}$ supplies $\cos(\phi-\phi')$. Its angular integral is π for each of $m=\pm1$ and zero for all other m. Consequently the first kernel gives

$$\boxed{A_\phi=\frac{\mu_0 Ia}{\pi}\int_0^\infty\cos(kz)I_1(k\rho_<)K_1(k\rho_>)\,dk,}$$

and the second gives

$$\boxed{A_\phi=\frac{\mu_0 Ia}{2}\int_0^\infty e^{-k|z|}J_1(ka)J_1(k\rho)\,dk.}$$

All other components vanish by the same reflection symmetry as in 5.8.

(c) The curl equations are $B_\rho=-\partial_z A_\phi$ and $B_z=\rho^{-1}\partial_\rho(\rho A_\phi)$. Using $(xI_1)'=xI_0$, $(xK_1)'=-xK_0$, and $(xJ_1)'=xJ_0$, one obtains

$$B_\rho=\frac{\mu_0 Ia}{\pi}\int_0^\infty k\sin(kz)I_1(k\rho_<)K_1(k\rho_>)\,dk,$$

$$B_z=\frac{\mu_0 Ia}{\pi}\int_0^\infty k\cos(kz)\begin{cases}I_0(k\rho)K_1(ka), & \rho<a,\\ -I_1(ka)K_0(k\rho), & \rho>a\end{cases}dk,$$

or, for $z \neq 0$,

$$\boxed{\begin{aligned} B_\rho &= \frac{\mu_0 I a}{2}\,\mathrm{sgn}(z)\int_0^\infty k e^{-k|z|} J_1(ka) J_1(k\rho)\,dk, \\ B_z &= \frac{\mu_0 I a}{2}\int_0^\infty k e^{-k|z|} J_1(ka) J_0(k\rho)\,dk. \end{aligned}}$$

At $z = 0$ off the wire, take the limiting fields; symmetry gives $B_\rho = 0$. On the axis $J_1(0) = 0$ and $J_0(0) = 1$. Differentiate the elementary Laplace integral $\int_0^\infty e^{-sk} J_0(ak)\,dk = (s^2 + a^2)^{-1/2}$ with respect to a:

$$\int_0^\infty k e^{-sk} J_1(ak)\,dk = \frac{a}{(s^2 + a^2)^{3/2}}.$$

This gives $\boxed{B_\rho(0, z) = 0,\ B_z(0, z) = \mu_0 I a^2/[2(a^2 + z^2)^{3/2}]}$. The modified-Bessel representation independently gives the same result: differentiate $\int_0^\infty \cos(kz) K_0(ka)\,dk = \pi/[2\sqrt{a^2 + z^2}]$ with respect to a, producing the required integral of $kK_1(ka)$.

5.11 A circular loop of wire carrying a current I is located with its center at the origin of coordinates and the normal to its plane having spherical angles θ_0, ϕ_0. There is an applied magnetic field, $B_x = B_0(1 + \beta y)$ and $B_y = B_0(1 + \beta x)$.

(a) Calculate the force acting on the loop without making any approximations. Compare your result with the approximate result (5.69). Comment.

(b) Calculate the torque in lowest order. Can you deduce anything about the higher order contributions? Do they vanish for the circular loop? What about for other shapes?

Solution.

Let the radius be a, the area be $S = \pi a^2$, and

$$\mathbf{m} = IS(\sin\theta_0 \cos\phi_0, \sin\theta_0 \sin\phi_0, \cos\theta_0).$$

(a) Starting with $\mathbf{F} = I \oint d\mathbf{l} \times \mathbf{B}$ and writing $B_k = B_{k0} + G_{kj} x_j$, the constant term vanishes. Stokes's theorem gives $\oint x_j\,dx_i = \epsilon_{jil} S n_l$. In indices, the remaining term is

$$F_i = I\epsilon_{ijk} G_{k\ell} \oint x_\ell\,dx_j = \epsilon_{ijk}\epsilon_{\ell jn} G_{k\ell} m_n = m_k G_{ki} - m_i G_{kk}.$$

Here $\epsilon_{ijk}\epsilon_{\ell jn} = \delta_{i\ell}\delta_{kn} - \delta_{in}\delta_{k\ell}$ and $G_{kk} = \nabla \cdot \mathbf{B} = 0$. Therefore $F_i = m_j \partial_i B_j$. Here $\partial_y B_x = \partial_x B_y = \beta B_0$, and thus

$$\boxed{\mathbf{F} = \beta B_0 (m_y, m_x, 0) = \beta B_0 I \pi a^2 \sin\theta_0 (\sin\phi_0, \cos\phi_0, 0).}$$

This is exactly $\nabla(\mathbf{m} \cdot \mathbf{B})$, with no small-loop approximation: a linear applied field has no derivatives that could couple to higher force multipoles.

(b) To dipole order,

$$\boxed{\mathbf{N} = \mathbf{m} \times \mathbf{B}(0) = B_0(-m_z, m_z, m_x - m_y).}$$

One can also see why this is exact for the centered circle. The torque integrand is $\mathbf{x} \times [I\,d\mathbf{l} \times \mathbf{B}]$. Its correction from the linear part of $\mathbf{B}$ is odd under $(\mathbf{x}, d\mathbf{l}) \mapsto (-\mathbf{x}, -d\mathbf{l})$, so opposite points cancel. Terms beyond this correction are absent because the field is linear.

For a general planar loop with area centroid $\overline{\mathbf{x}} = S^{-1}\int_S \mathbf{x}\,da$, an explicit surface form of the torque is

$$\mathbf{N} = I\int_S [\mathbf{n}\times\mathbf{B}(\mathbf{x}) + \mathbf{x}\times\nabla(\mathbf{n}\cdot\mathbf{B})]\,da.$$

This follows by dividing the spanning surface into small loops; all interior edges cancel, leaving the original boundary loop. For the present linear field it becomes

$$\mathbf{N} = \mathbf{m}\times[\mathbf{B}(0) + G\overline{\mathbf{x}}] + \overline{\mathbf{x}}\times(G^T\mathbf{m}).$$

Thus a general shape can have a correction about a specified origin. It vanishes for every planar shape when the origin is its area centroid, not only for circles or shapes with inversion symmetry.

5.12 Two concentric circular loops of radii a, b and currents I, I', respectively ($b < a$), have an angle α between their planes. Show that the torque on one of the loops is about the line of intersection of the two planes containing the loops and has the signed component in the direction of increasing α

$$N_\alpha = \frac{\mu_0\pi II'b^2}{2a}\sum_{n=0}^{\infty}\frac{(n+1)}{(2n+1)}\left[\frac{\Gamma(n+\frac{3}{2})}{\Gamma(n+2)\Gamma(\frac{3}{2})}\right]^2\left(\frac{b}{a}\right)^{2n}P^1_{2n+1}(\cos\alpha)$$

where $P^1_l(\cos\alpha)$ is an associated Legendre polynomial. Determine the sense of the torque for α an acute angle and the currents in the same (opposite) directions.

Solution.

Take the larger loop's normal along z and let α be the angle through which the smaller normal is rotated about the line of intersection. We use $P^1_l(\cos\alpha) = dP_l(\cos\alpha)/d\alpha$. For the single larger loop, the interior moment from 5.8 is

$$m_l = -\frac{2\pi I}{l(l+1)a^l}P^1_l(0), \qquad l = 1,3,5,\ldots.$$

Its scalar potential for $\mathbf{H}$ is

$$\Phi_H = -\frac{1}{4\pi}\sum_l (l+1)m_l r^l P_l(\cos\theta),$$

as direct radial differentiation verifies against the curl of A_ϕ.

Introduce coordinates with z' normal to the smaller loop. Azimuthal averaging about that axis, using the addition theorem, replaces $P_l(\cos\theta)$ by $P_l(\cos\alpha)P_l(\cos\theta')$. On its spanning disk $z' = 0$,

$$\partial_{z'}[r^l P_l(z'/r)]\Big|_{z'=0} = r^{l-1}P'_l(0) = -r^{l-1}P^1_l(0).$$

Integrating $B_{z'} = -\mu_0\partial_{z'}\Phi_H$ over the disk gives its flux:

$$\begin{aligned} F &= -\frac{\mu_0}{4\pi}\sum_l (l+1)m_l P^1_l(0)P_l(\cos\alpha)\int_0^b 2\pi r^l\,dr \\ &= -\frac{\mu_0}{2}\sum_l m_l P^1_l(0)b^{l+1}P_l(\cos\alpha). \end{aligned}$$

Thus

$$M(\alpha) = F/I = \mu_0 \pi \sum_{l \text{ odd}} \frac{[P_l^1(0)]^2}{l(l+1)} \frac{b^{l+1}}{a^l} P_l(\cos\alpha).$$

At fixed currents the torque component that increases α is $N_\alpha = II'\, dM/d\alpha$. Put $l = 2n+1$ and use

$$P_{2n+1}^1(0) = (-1)^{n+1} \frac{(2n+1)!!}{(2n)!!} = (-1)^{n+1}(n+1) \frac{\Gamma(n+3/2)}{\Gamma(n+2)\Gamma(3/2)}.$$

Then

$$\boxed{N_\alpha = \frac{\mu_0 \pi I I' b^2}{2a} \sum_{n=0}^{\infty} \frac{n+1}{2n+1} \left[\frac{\Gamma(n+3/2)}{\Gamma(n+2)\Gamma(3/2)}\right]^2 \left(\frac{b}{a}\right)^{2n} P_{2n+1}^1(\cos\alpha).}$$

The mechanical interaction potential is $-II'M$, which fixes this sign. The displayed expression is a signed component, not an always-positive magnitude; the magnitude is $|N_\alpha|$.

For $0 < \alpha < \pi/2$, like-directed currents tend to decrease α, and opposite currents tend to increase it. To establish the sign beyond the dipole limit, put $c = \cos\alpha > 0$ and parameterize the smaller ring. Its cylindrical radius satisfies $\rho^2 = b^2(\cos^2\varphi + c^2 \sin^2\varphi)$, and $\mathbf{A} \cdot d\mathbf{l} = b^2 c (A_\phi/\rho) d\varphi$. On the sphere $r = b < a$, the Biot–Savart expression for A_ϕ/ρ has a convergent expansion in nonnegative powers of ρ^2 with positive coefficients: expand $[a^2 + b^2 - 2a\rho\cos\varphi']^{-1/2}$ and integrate its product with $\cos\varphi'$. Only positive even cosine integrals survive. Hence $M(c) = c$ times a positive, increasing function of c^2, so $dM/d\alpha = -\sin\alpha\, dM/dc < 0$. Rotational symmetry leaves only a torque about the line of intersection.

5.13 A sphere of radius a carries a uniform surface-charge distribution σ. The sphere is rotated about a diameter with constant angular velocity ω. Find the vector potential and magnetic-flux density both inside and outside the sphere.

Solution.

Take $\boldsymbol{\omega} = \omega \widehat{\mathbf{z}}$. The moving surface charge is the surface current $\mathbf{K} = \sigma(\boldsymbol{\omega} \times \mathbf{r}) = \sigma\omega a \sin\theta\, \widehat{\boldsymbol{\phi}}$. It has only the $l = 1$ angular dependence, so write

$$A_\phi = \begin{cases} Cr\sin\theta, & r < a, \\ Dr^{-2}\sin\theta, & r > a. \end{cases}$$

Continuity of A_ϕ gives $D = Ca^3$. Taking its curl gives

$$\mathbf{B}_{\text{in}} = 2C(\cos\theta\, \widehat{\mathbf{r}} - \sin\theta\, \widehat{\boldsymbol{\theta}}), \quad \mathbf{B}_{\text{out}} = \frac{D}{r^3}(2\cos\theta\, \widehat{\mathbf{r}} + \sin\theta\, \widehat{\boldsymbol{\theta}}).$$

The normal component is continuous, while the sheet-current condition is $B_{\theta,\text{out}} - B_{\theta,\text{in}} = \mu_0 K_\phi$. Thus $3C\sin\theta = \mu_0 \sigma\omega a \sin\theta$, giving

$$\boxed{A_\phi = \frac{\mu_0 \sigma \omega a}{3} \begin{cases} r\sin\theta, & r < a, \\ a^3 r^{-2}\sin\theta, & r > a, \end{cases}}$$

$$\boxed{\mathbf{B}_{\text{in}} = \frac{2\mu_0\sigma\omega a}{3}\widehat{\mathbf{z}}, \qquad \mathbf{B}_{\text{out}} = \frac{\mu_0\sigma\omega a^4}{3r^3}(2\cos\theta\, \widehat{\mathbf{r}} + \sin\theta\, \widehat{\boldsymbol{\theta}}).}$$

The external moment is $\mathbf{m} = (4\pi\sigma\omega a^4/3)\widehat{\mathbf{z}} = (Q\omega a^2/3)\widehat{\mathbf{z}}$. Independently, $\mathbf{m} = \frac{1}{2}\int \mathbf{r} \times \mathbf{K}\, da$ gives $m_z = (\sigma\omega a^2/2)\int \sin^2\theta\, da = 4\pi\sigma\omega a^4/3$.

5.14 A long, hollow, right circular cylinder of inner (outer) radius a (b), and of relative permeability μ_r, is placed in a region of initially uniform magnetic-flux density $\mathbf{B}_0$ at right angles to the field. Find the flux density at all points in space, and sketch the logarithm of the ratio of the magnitudes of $\mathbf{B}$ on the cylinder axis to $\mathbf{B}_0$ as a function of $\log_{10}\mu_r$ for $a^2/b^2 = 0.5,\ 0.1$. Neglect end effects.

Solution.

Let $\kappa = \mu_r > 0$, $\eta = a^2/b^2$, and put the original field along x. The magnetic scalar potential for $\mathbf{H}$ obeys the same boundary equations as the dielectric-shell potential of Problem 4.8: H_ϕ and μH_ρ are continuous. Define

$$\Delta = (\kappa+1)^2-(\kappa-1)^2\eta, \qquad A = \frac{4\kappa}{\Delta}, \quad B = \frac{2(\kappa+1)}{\Delta}, \quad C = \frac{2(\kappa-1)a^2}{\Delta}, \quad D = \frac{(\kappa^2-1)(b^2-a^2)}{\Delta}.$$

In detail, the trial potential $\Phi_H = -(B_0/\mu_0)\cos\phi\{A\rho,\ B\rho + C/\rho,\ \rho - D/\rho\}$ gives the four interface equations

$$A = B + C/a^2, \quad A = \kappa(B - C/a^2), \quad B + C/b^2 = 1 - D/b^2, \quad \kappa(B - C/b^2) = 1 + D/b^2.$$

The first pair gives $C = a^2(\kappa-1)B/(\kappa+1)$ and $A = 2\kappa B/(\kappa+1)$. Adding the last two equations eliminates D:

$$(\kappa+1)B - (\kappa-1)C/b^2 = 2, \qquad \frac{(\kappa+1)^2 - (\kappa-1)^2\eta}{\kappa+1}B = 2.$$

Thus $B = 2(\kappa+1)/\Delta$, which gives A, C above. Finally

$$\frac{D}{b^2} = 1 - B - \frac{C}{b^2} = \frac{\Delta - 2(\kappa+1) - 2(\kappa-1)\eta}{\Delta} = \frac{(\kappa^2-1)(1-\eta)}{\Delta}.$$

The requested induction components are

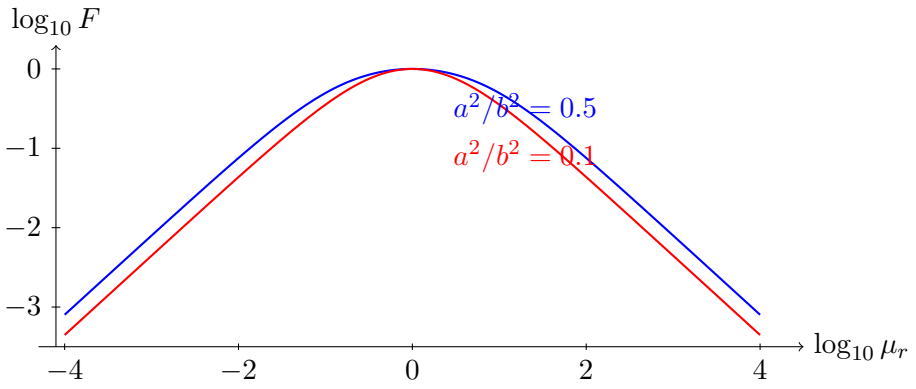
$$\boxed{\frac{B_\rho}{B_0} = \cos\phi \begin{cases} A, & \rho < a, \\ \kappa(B - C/\rho^2), & a < \rho < b, \\ 1 + D/\rho^2, & \rho > b, \end{cases}}$$

$$\boxed{\frac{B_\phi}{B_0} = -\sin\phi \begin{cases} A, & \rho < a, \\ \kappa(B + C/\rho^2), & a < \rho < b, \\ 1 - D/\rho^2, & \rho > b, \end{cases} \qquad B_z = 0.}$$

In particular the core field is uniform, with

$$F = \frac{|\mathbf{B}_{\rm core}|}{B_0} = \frac{4\kappa}{(1-\eta)(\kappa^2+1) + 2(1+\eta)\kappa}.$$

The logarithmic curves below use $x = \log_{10}\kappa$ and $y = \log_{10} F$. They are symmetric under $x \mapsto -x$, since $F(\kappa) = F(1/\kappa)$. Both reach $F = 1$ at $\kappa = 1$; at large κ, $F \sim 4/[(1-\eta)\kappa]$, while at small κ, $F \sim 4\kappa/(1-\eta)$.



Thus a permeable shell and an ideal diamagnetic shell can both shield the cavity, by different limiting boundary conditions.

5.15 Consider two long, straight wires, parallel to the z axis, spaced a distance d apart and carrying currents I in opposite directions. Describe the magnetic field $\mathbf{H}$ in terms of a magnetic scalar potential Φ_M, with $\mathbf{H} = -\nabla\Phi_M$.

(a) If the wires are parallel to the z axis with positions, $x = \pm d/2$, $y = 0$, show that in the limit of small spacing, the potential is approximately that of a two-dimensional dipole,

$$\Phi_M \simeq -\frac{Id\,\sin\phi}{2\pi\rho} + O(d^2/\rho^2)$$

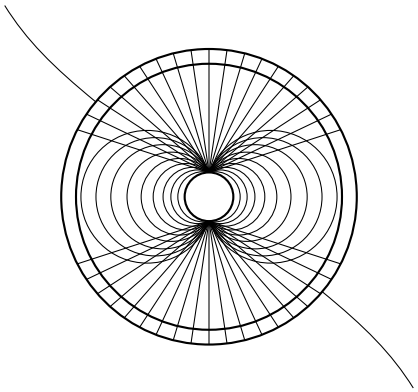
where ρ and ϕ are the usual polar coordinates.

(b) The closely spaced wires are now centered in a hollow right circular cylinder of steel, of inner (outer) radius a (b) and magnetic permeability $\mu = \mu_r\mu_0$. Determine the magnetic scalar potential in the three regions, $0 < \rho < a$, $a < \rho < b$, and $\rho > b$. Show that the field outside the steel cylinder is a two-dimensional dipole field, as in part a, but with a strength reduced by the factor

$$F = \frac{4\mu_r b^2}{(\mu_r+1)^2 b^2 - (\mu_r-1)^2 a^2}$$

Relate your result to Problem 5.14.

(c) Assuming that $\mu_r \gg 1$, and $b = a + t$, where the thickness $t \ll b$, write down an approximate expression for F and determine its numerical value for $\mu_r = 200$ (typical of steel at 20 G), $b = 1.25$ cm, $t = 3$ mm. The shielding effect is relevant for reduction of stray fields in residential and commercial 60 Hz, 110 or 220 V wiring. The figure illustrates the shielding effect for $a/b = 0.9$, $\mu_r = 100$.



Problem 5.15

Solution.

(a) Put $+I\widehat{\mathbf{z}}$ at $x = d/2$ and $-I\widehat{\mathbf{z}}$ at $x = -d/2$. A straight wire has $\mathbf{H} = (I/2\pi)\nabla\phi$, so a local scalar potential for the pair is

$$\Phi_H = \frac{I}{2\pi}\{\arg(x + iy + d/2) - \arg(x + iy - d/2)\}.$$

For $\rho \gg d$, expand the logarithms whose imaginary parts give the arguments:

$$\ln(w + d/2) - \ln(w - d/2) = \frac{d}{w} + \frac{d^3}{12w^3} + \cdots, \qquad w = \rho e^{i\phi}.$$

Consequently

$$\boxed{\Phi_H = -\frac{Id}{2\pi\rho}\sin\phi + O\!\left(I\frac{d^3}{\rho^3}\right).}$$

The next term is actually odd in d and of third order; the second-order remainder quoted in the question is a looser estimate.

(b) Work to this dipole order, so $d \ll a$. Set $p = Id/(2\pi)$, $\kappa = \mu_r$, $\eta = a^2/b^2$, and $\Delta = (\kappa+1)^2 - (\kappa-1)^2\eta$. Write

$$\Phi_H = -p\sin\phi \begin{cases} \rho^{-1} + A\rho, & 0 < \rho < a, \\ B\rho^{-1} + C\rho, & a < \rho < b, \\ D\rho^{-1}, & \rho > b. \end{cases}$$

Continuity of potential and $\mu\partial_\rho\Phi_H$ gives

$$1 + Aa^2 = B + Ca^2, \quad Aa^2 - 1 = \kappa(Ca^2 - B), \quad B + Cb^2 = D, \quad \kappa(Cb^2 - B) = -D.$$

The last pair gives $C = (\kappa-1)B/[(\kappa+1)b^2]$ and $D = 2\kappa B/(\kappa+1)$. The first pair then implies $(\kappa+1)B - (\kappa-1)Ca^2 = 2$. Hence

$$\boxed{A = -\frac{(\kappa^2-1)(1-\eta)}{\Delta a^2}, \quad B = \frac{2(\kappa+1)}{\Delta}, \quad C = \frac{2(\kappa-1)}{\Delta b^2}, \quad D = \frac{4\kappa}{\Delta} = F.}$$

The exterior is the original two-dimensional dipole multiplied by F. This is exactly the core-field transmission factor of 5.14, consistent with reciprocity between a source inside and an applied field

outside. The singular central dipole represents the two wires only at distances large compared with d.

(c) It is useful to rearrange the exact factor before expanding:

$$F^{-1} = 1 + \frac{(\kappa - 1)^2}{4\kappa}(1 - a^2/b^2).$$

For $t = b - a \ll b$, $1 - a^2/b^2 = 2t/b - t^2/b^2$, so for $\kappa \gg 1$

$$\boxed{F \simeq \frac{1}{1 + \kappa t/(2b)}.}$$

If also $\kappa t/(2b) \gg 1$, this further reduces to $2b/(\kappa t)$. With the specified numbers these estimates are 0.0400 and 0.04167, respectively. The exact factor is 0.04564, because $t/b = 0.24$ is only moderately small. The exact suppression is about a factor of 21.9; the leading thin-shell estimate gives a factor of 25.

5.16 A circular loop of wire of radius a and negligible thickness carries a current I. The loop is centered in a spherical cavity of radius $b > a$ in a large block of soft iron. Assume that the relative permeability of the iron is effectively infinite and that of the medium in the cavity, unity.

(a) In the approximation of $b \gg a$, show that the magnetic field at the center of the loop is augmented by a factor $(1 + a^3/2b^3)$ by the presence of the iron.

(b) What is the radius of the "image" current loop (carrying the same current) that simulates the leading, nearly uniform reaction field of the iron for $r < b$, in the same $b \gg a$ approximation? Check whether a single such loop can reproduce the complete reaction field exactly.

Solution.

(a) For $b \gg a$, the loop's field at the cavity wall is dominated by its dipole moment $m = I\pi a^2$. The direct scalar potential for $\mathbf{H}$ there is $m\cos\theta/(4\pi r^2)$. Infinite permeability outside makes tangential $\mathbf{H}$ vanish at $r = b$, so the wall must be an equipotential. The regular $l = 1$ reaction potential that cancels its angular variation is

$$\Phi_{H,\text{react}} = -\frac{m}{4\pi b^3} r\cos\theta.$$

Taking $-\mu_0\nabla$ gives the uniform reaction induction $\mathbf{B}_{\text{react}} = \mu_0 m\hat{\mathbf{z}}/(4\pi b^3)$. Adding the exact free-loop center field $\mu_0 I/(2a)$ gives

$$\boxed{B_{\text{center}} \simeq \frac{\mu_0 I}{2a}\left(1 + \frac{a^3}{2b^3}\right).}$$

Higher regular multipoles do not contribute at the exact center, so this center value is in fact unchanged by retaining the other loop multipoles; the dipole approximation is needed for the statement that the reaction field is uniform throughout the cavity.

(b) A coaxial image loop carrying the same current I has center field $\mu_0 I/(2R_{\text{im}})$. Matching the leading uniform reaction field gives

$$\frac{\mu_0 I}{2R_{\text{im}}} = \frac{\mu_0 I a^2}{4b^3}, \qquad \boxed{R_{\text{im}} = \frac{2b^3}{a^2}.}$$

Since $R_{\rm im} \gg b$ when $b \gg a$, this image produces an almost uniform field over the cavity.

It is not an exact image for all $r < b$ at arbitrary a/b. Here is a direct check of that necessary qualification. For an odd l, the real-loop exterior A_ϕ is proportional to $\mu_l r^{-l-1} P_l^1$, while the regular reaction term is proportional to $m_l^R r^l P_l^1$. The condition $B_\theta(b) = 0$ requires

$$\frac{m_l^R}{\mu_l} = \frac{l}{l+1} b^{-2l-1}.$$

A single same-current image loop of radius R instead has, from 5.8, $m_l^R/\mu_l = R^{-l} a^{-l-1}$. The $l = 1$ condition demands $R = 2b^3/a^2$, whereas $l = 3$ demands $R^3 = 4b^7/(3a^4)$. These cannot both hold for $b > a$: their combination would require $b^2/a^2 = 1/6$. Thus the equal-current image-loop statement must be understood in the leading uniform-field approximation, as made explicit above.

5.17 A current distribution $\mathbf{J}(\mathbf{x})$ exists in a medium of unit relative permeability adjacent to a semi-infinite slab of material having relative permeability μ_r and filling the half-space, $z < 0$.

(a) Show that for $z > 0$ the magnetic induction can be calculated by replacing the medium of permeability μ_r by an image current distribution, $\mathbf{J}^*$, with components,

$$\left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_x(x, y, -z), \qquad \left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_y(x, y, -z), \qquad -\left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_z(x, y, -z)$$

(b) Show that for $z < 0$ the magnetic induction appears to be due to a current distribution $[2\mu_r/(\mu_r + 1)]\mathbf{J}$ in a medium of unit relative permeability.

Solution.

Let $\mathcal{R} = \mathrm{diag}(1, 1, -1)$ reflect a vector across the interface, and let $\mathbf{B}^{(0)}$ be the field of the original current computed in vacuum everywhere. Under spatial reflection a current is a polar vector but its magnetic field is an axial vector. Direct substitution in Biot–Savart gives

$$\mathbf{J}_{\rm ref}(\mathbf{x}) = \mathcal{R}\mathbf{J}(\mathcal{R}\mathbf{x}), \qquad \mathbf{B}_{\rm ref}(\mathbf{x}) = -\mathcal{R}\mathbf{B}^{(0)}(\mathcal{R}\mathbf{x}).$$

In particular, at $z = 0$ its tangential field reverses sign and its normal field keeps its sign.

Take the upper field to be $\mathbf{B}^{(0)} + \chi \mathbf{B}_{\rm ref}$ and the lower field to be $T\mathbf{B}^{(0)}$. At the interface, continuity of B_n and H_t requires

$$1 + \chi = T, \qquad 1 - \chi = T/\mu_r.$$

Solving gives $\chi = (\mu_r - 1)/(\mu_r + 1)$ and $T = 2\mu_r/(\mu_r + 1)$. Therefore

$$\boxed{\mathbf{J}^*(x, y, z) = \frac{\mu_r - 1}{\mu_r + 1}(J_x(x, y, -z), J_y(x, y, -z), -J_z(x, y, -z))}$$

is the image distribution for part (a), and

$$\boxed{\mathbf{B}_{z<0} = \frac{2\mu_r}{\mu_r + 1}\mathbf{B}^{(0)}}$$

proves part (b). Both candidate fields are divergence-free and satisfy the correct source-free curl equations in their respective regions, the interface conditions, and decay at infinity; uniqueness completes the construction. At $\mu_r = 1$ the image disappears and $T = 1$.

5.18 A circular loop of wire having a radius a and carrying a current I is located in vacuum with its center a distance d away from a semi-infinite slab of permeability μ. Find the force acting on the loop when

(a) the plane of the loop is parallel to the face of the slab,

(b) the plane of the loop is perpendicular to the face of the slab.

(c) Determine the limiting form of your answer to parts a and b when $d \gg a$. Can you obtain these limiting values in some simple and direct way?

Solution.

Choose the slab as $z < 0$, the center as $(0, 0, d)$, and write $\chi = (\mu - \mu_0)/(\mu + \mu_0)$. For each orientation the image is centered at $z = -d$. Let $s = 2d$. In computing the Lorentz force, the real loop moves in the field of an image at its instantaneous position; one does not differentiate that position a second time. Equivalently, the mechanical image interaction energy carries a factor $1/2$.

(a) A loop parallel to the interface has a same-sense image with current χI. For coaxial equal loops (derived in 5.28),

$$M_{\parallel}(s) = \mu_0 a \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right], \qquad k = \frac{2a}{\sqrt{4a^2 + s^2}}.$$

The force is purely normal, $F_z = \chi I^2 M'_{\parallel}(s)$. Using $K' = E/[k(1-k^2)] - K/k$, $E' = (E-K)/k$, and $dk/ds = -ks/(4a^2 + s^2)$ gives

$$\boxed{F_z = -\frac{\mu_0 \chi I^2 s k}{4a} \left[\frac{2 - k^2}{1 - k^2} E(k) - 2K(k) \right]_{s=2d}}.$$

This holds for any $d > 0$ in the ideal filament model.

(b) A perpendicular loop must have $d > a$ to lie wholly in vacuum. Its image is coplanar with it and carries opposite-sense current $-\chi I$. Define the positive coefficients

$$D_n = \frac{[(2n+2)!]^2}{2^{2n+2} n! (n+2)! [(n+1)!]^2}, \qquad E_n = D_n \frac{(2n+2)!}{2^{2n+2} [(n+1)!]^2}.$$

The convergent coplanar mutual inductance derived in 5.34 is $M_{\perp}(s) = -\mu_0 \pi a \sum_{n \geq 0} E_n (a/s)^{2n+3}$. Thus the exact force throughout the allowed domain $s = 2d > 2a$ is

$$\boxed{F_z = -\chi I^2 M'_{\perp}(2d) = -\mu_0 \pi \chi I^2 \sum_{n=0}^{\infty} (2n+3) E_n \left(\frac{a}{2d} \right)^{2n+4}}.$$

All coefficients are positive, so the direction is toward a more permeable slab and away from a less permeable one. The two in-plane transverse force components cancel by symmetry.

(c) Expanding the preceding results, or using $D_0 = 1/2$ and $E_0 = 1/4$, gives

$$\boxed{F_{z,\parallel} \sim -\frac{3\mu_0 \pi \chi I^2 a^4}{32 d^4}, \qquad F_{z,\perp} \sim -\frac{3\mu_0 \pi \chi I^2 a^4}{64 d^4}}.$$

For a direct dipole check, use $m = I\pi a^2$ and separation s. Parallel-to-interface loops have axial moments $m, \chi m$: $U_{\text{pair}} = -2\mu_0 \chi m^2/(4\pi s^3)$. Perpendicular loops have transverse moments $m, -\chi m$: $U_{\text{pair}} = -\mu_0 \chi m^2/(4\pi s^3)$. The force on the real dipole is $-\partial_s U_{\text{pair}}$. Putting $s = 2d$ gives the two boxed limits, with the first twice the second.

5.19 A magnetically "hard" material is in the shape of a right circular cylinder of length L and radius a. The cylinder has a permanent magnetization M_0, uniform throughout its volume and parallel to its axis.

(a) Determine the magnetic field $\mathbf{H}$ and magnetic induction $\mathbf{B}$ at all points on the axis of the cylinder, both inside and outside.

(b) Plot the ratios $\mathbf{B}/\mu_0 M_0$ and $\mathbf{H}/M_0$ on the axis as functions of z for $L/a = 5$.

Solution.

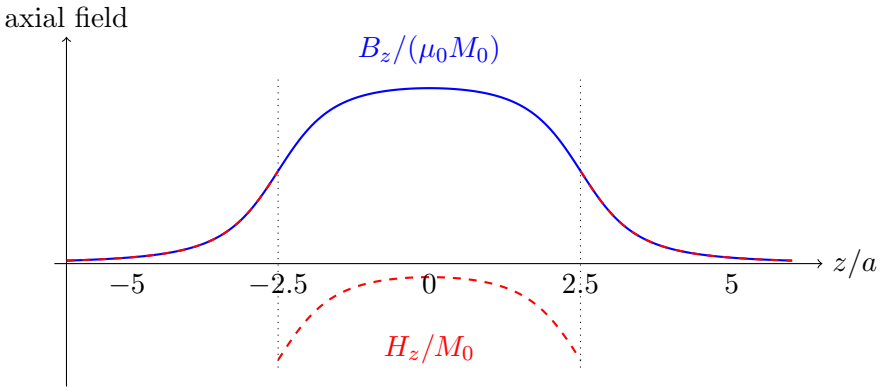
(a) Let the cylinder occupy $-L/2 < z < L/2$, with $\mathbf{M} = M_0\hat{\mathbf{z}}$. Its volume bound current vanishes, and its curved surface carries $\mathbf{K}_M = \mathbf{M} \times \hat{\boldsymbol{\rho}} = M_0\hat{\boldsymbol{\phi}}$. Thus its induction is that of the ideal finite solenoid in 5.3, with NI replaced by M_0. Define

$$f(z) = \frac{1}{2}\left[\frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z - L/2)^2}}\right].$$

The answers on the entire axis, with one-sided limits at the end faces, are

$$\boxed{\mathbf{B}(z) = \mu_0 M_0 f(z)\hat{\mathbf{z}}, \qquad \mathbf{H}(z) = M_0[f(z) - \mathbf{1}_{\{|z|<L/2\}}]\hat{\mathbf{z}}.}$$

The subtraction follows from $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$. The axial induction is continuous across either face, while H_z jumps by the magnetic pole-sheet strength M_0. **(b)** For $L/a = 5$, at the center $f(0) = 2.5/\sqrt{7.25} = 0.928477$, so $H_z(0)/M_0 = -0.0715233$. At either end $f = \frac{1}{2}(5/\sqrt{26}) = 0.490290$.



The red interior field opposes the magnetization; outside, $\mathbf{B}/\mu_0 = \mathbf{H}$ as required in vacuum.

5.20

(a) Starting from the force equation (5.12) and the fact that a magnetization $\mathbf{M}$ inside a volume V bounded by a surface S is equivalent to a volume current density $\mathbf{J}_M = (\nabla \times \mathbf{M})$ and a surface current density $(\mathbf{M} \times \mathbf{n})$, show that in the absence of macroscopic conduction currents the total magnetic force on the body can be written

$$\mathbf{F} = -\int_V (\nabla \cdot \mathbf{M})\mathbf{B}_e \, d^3x + \int_S (\mathbf{M} \cdot \mathbf{n})\mathbf{B}_e \, da$$

where $\mathbf{B}_e$ is the applied magnetic induction (not including that of the body in question). The force is now expressed in terms of the effective charge densities ρ_M and σ_M. If the distribution of magnetization is not discontinuous, the surface can be at infinity and the force given by just the volume integral.

(b) A sphere of radius R with uniform magnetization has its center at the origin of coordinates and its direction of magnetization making spherical angles θ_0, ϕ_0. If the external magnetic field is the same as in Problem 5.11, use the expression of part a to evaluate the components of the force acting on the sphere.

Solution.

(a) The external sources are outside the body, so $\nabla \times \mathbf{B}_e = 0$ and $\nabla \cdot \mathbf{B}_e = 0$ in its neighborhood. Start from both bound-current contributions:

$$\mathbf{F} = \int_V (\nabla \times \mathbf{M}) \times \mathbf{B}_e \, d^3x + \oint_S (\mathbf{M} \times \mathbf{n}) \times \mathbf{B}_e \, da.$$

The volume integrand has components $B_{e,j}\partial_j M_i - B_{e,j}\partial_i M_j$. Integrating both derivatives by parts gives the boundary term $M_i(\mathbf{B}_e \cdot \mathbf{n}) - n_i(\mathbf{M} \cdot \mathbf{B}_e)$ and the volume term $M_j \partial_i B_{e,j} - M_i \partial_j B_{e,j}$. The surface-current term cancels the boundary term exactly. Zero divergence removes its second volume term, leaving

$$F_i = \int_V M_j \partial_i B_{e,j} \, d^3x.$$

Zero curl allows $\partial_i B_{e,j} = \partial_j B_{e,i}$. A final integration by parts then gives

$$\boxed{\mathbf{F} = -\int_V (\nabla \cdot \mathbf{M}) \mathbf{B}_e \, d^3x + \oint_S (\mathbf{M} \cdot \mathbf{n}) \mathbf{B}_e \, da.}$$

These are effective pole densities $\rho_M = -\nabla \cdot \mathbf{M}$ and $\sigma_M = \mathbf{M} \cdot \mathbf{n}$. If imposed external current overlaps V, zero curl is unavailable: the additional force term is $\int_V \mathbf{M} \times (\nabla \times \mathbf{B}_e) d^3x$.

(b) For uniform $\mathbf{M}$ the volume pole density is zero. On the sphere $\sigma_M = M_j n_j$, and the constant part of the external field integrates to zero because $\int \mathbf{n} \, da = 0$. For its linear part, $x_k = R n_k$ and $\int n_j n_k \, da = (4\pi R^2/3)\delta_{jk}$, so

$$F_i = R \, \partial_k B_{e,i} \, M_j \int n_j n_k \, da = \frac{4\pi R^3}{3} M_j \partial_j B_{e,i}.$$

With the field of 5.11 this gives

$$\boxed{\mathbf{F} = \frac{4\pi R^3}{3} \beta B_0 M \sin\theta_0 (\sin\phi_0, \cos\phi_0, 0).}$$

It equals the loop result with $\mathbf{m} = (4\pi R^3/3)\mathbf{M}$, as expected for an affine field.

5.21 A magnetostatic field is due entirely to a localized distribution of permanent magnetization.

(a) Show that

$$\int \mathbf{B} \cdot \mathbf{H} \, d^3x = 0$$

provided the integral is taken over all space.

(b) From the potential energy (5.72) of a dipole in an external field, show that for a continuous distribution of permanent magnetization the magnetostatic energy can be written

$$W = \frac{\mu_0}{2} \int \mathbf{H} \cdot \mathbf{H} \, d^3x = -\frac{\mu_0}{2} \int \mathbf{M} \cdot \mathbf{H} \, d^3x$$

apart from an additive constant, which is independent of the orientation or position of the various constituent magnetized bodies.

Solution.

(a) With no free currents, $\mathbf{H} = -\nabla\Phi_H$ throughout space. Therefore

$$\int \mathbf{B}\cdot\mathbf{H}\, d^3x = -\int \nabla\cdot(\Phi_H\mathbf{B}) d^3x + \int \Phi_H \nabla\cdot\mathbf{B}\, d^3x = 0.$$

The volume term vanishes by $\nabla\cdot\mathbf{B} = 0$. For a localized magnetization, $\Phi_H = O(r^{-2})$ and $B = O(r^{-3})$, so the surface term at infinity is $O(r^{-3})$ and vanishes. Internal interfaces cancel when all space is included.

(b) The pair interaction energy of permanent dipoles is $-\mathbf{m}\cdot\mathbf{B}_e$. Summing it over the distribution and dividing by two to count each pair once gives $-\frac{1}{2}\int \mathbf{M}\cdot\mathbf{B}\, d^3x$, apart from local self-energy terms. Since $\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M})$,

$$-\frac{1}{2}\int \mathbf{M}\cdot\mathbf{B} = -\frac{\mu_0}{2}\int \mathbf{M}\cdot\mathbf{H} - \frac{\mu_0}{2}\int M^2.$$

The last term and the local self-energy constants are unchanged by translations or rotations of permanently magnetized bodies. They may be absorbed into the additive constant specified in the question. Part (a) also gives

$$0 = \mu_0\int (\mathbf{H}+\mathbf{M})\cdot\mathbf{H}\, d^3x, \quad \text{hence} \quad \int H^2 = -\int \mathbf{M}\cdot\mathbf{H}.$$

Consequently the position- and orientation-dependent energy can be written

$$\boxed{W = \frac{\mu_0}{2}\int H^2 d^3x = -\frac{\mu_0}{2}\int \mathbf{M}\cdot\mathbf{H}\, d^3x + \text{constant}.}$$

The moment magnitudes are fixed here; changing them would change the discarded intrinsic energy and would be a different thermodynamic problem.

5.22 Show that in general a long, straight bar of uniform cross-sectional area A with uniform lengthwise magnetization M, when placed with its flat end against an infinitely permeable flat surface, adheres with a force given approximately by

$$F \simeq \frac{\mu_0}{2} AM^2$$

Relate your discussion to the electrostatic considerations in Section 1.11.

Solution.

Resolve contact as the limit of a narrow vacuum gap. The end face has effective magnetic surface charge of magnitude M. Infinite permeability can be represented by an image magnet extending the real bar through the plane; the facing image pole has opposite sign. For a bar long compared with its transverse size, the far ends make negligible contributions in the gap. Each facing pole sheet supplies $H = M/2$, so the total gap field is $H = M$, $B = \mu_0 M$.

The force on the real face is its pole strength MA times the *image* induction $\mu_0 M/2$, giving

$$\boxed{F \simeq \frac{\mu_0}{2} AM^2.}$$

The same result follows from the normal vacuum magnetic stress $B^2/(2\mu_0)$ times A. In the electrostatic capacitor the corresponding pressure is $\epsilon_0 E^2/2 = \sigma^2/(2\epsilon_0)$. Using the field of both sheets as the acting external field would incorrectly double the force; the face does not exert a net force on itself.

5.23 A right circular cylinder of length L and radius a has a uniform lengthwise magnetization M.

(a) Show that, when it is placed with its flat end against an infinitely permeable plane surface, it adheres with a force

$$F = 2\mu_0 a L M^2 \left[\frac{K(k) - E(k)}{k} - \frac{K(k_1) - E(k_1)}{k_1} \right]$$

where

$$k = \frac{2a}{\sqrt{4a^2 + L^2}}, \qquad k_1 = \frac{a}{\sqrt{a^2 + L^2}}$$

(b) Find the limiting form for the force if $L \gg a$.

Solution.

(a) Use magnetic pole disks $\pm M$ at the two end faces and the image cylinder across the infinitely permeable plane. Let $f(s) > 0$ be the attraction of two oppositely charged disks of radius a, density magnitudes M, and axial separation s. Biot–Savart's equivalent pole force is

$$f(s) = \frac{\mu_0 M^2}{4\pi} \int_D d^2x \int_D d^2x' \frac{s}{[s^2 + |\mathbf{x} - \mathbf{x}'|^2]^{3/2}}.$$

The overlap area of disks displaced transversely by t is

$$C(t) = 2a^2 \cos^{-1}(t/2a) - \frac{t}{2}\sqrt{4a^2 - t^2}, \quad 0 \le t \le 2a, \qquad C'(t) = -\sqrt{4a^2 - t^2}.$$

Integrating by the displacement variable, then by parts, yields

$$\begin{aligned} f(s) &= \frac{\mu_0 M^2}{2} \int_0^{2a} C(t) \frac{st\,dt}{(s^2 + t^2)^{3/2}} \\ &= \frac{\mu_0 M^2}{2} \left[\pi a^2 - s \int_0^{2a} \frac{\sqrt{4a^2 - t^2}}{\sqrt{s^2 + t^2}} dt \right]. \end{aligned}$$

For the remaining integral put $t = 2a\cos u$ and $k_s = 2a/\sqrt{4a^2 + s^2}$. It becomes

$$\frac{4a^2}{\sqrt{4a^2 + s^2}} \int_0^{\pi/2} \frac{\sin^2 u\, du}{\sqrt{1 - k_s^2 \sin^2 u}} = \frac{2a}{k_s}[K(k_s) - E(k_s)].$$

Thus $f(s) = \mu_0 M^2 \{\pi a^2/2 - as[K(k_s) - E(k_s)]/k_s\}$. In particular $f(0^+) = \mu_0 M^2 \pi a^2/2$.

The four real-image pairs give attraction $f(0)$ and $f(2L)$, and two repulsions $f(L)$. Hence

$$F = f(0) - 2f(L) + f(2L) = \boxed{2\mu_0 a L M^2 \left[\frac{K(k) - E(k)}{k} - \frac{K(k_1) - E(k_1)}{k_1} \right]},$$

with $k = k_L$ and $k_1 = k_{2L}$ exactly as in the question.

(b) At small k, $K = \frac{\pi}{2}(1 + k^2/4 + \cdots)$ and $E = \frac{\pi}{2}(1 - k^2/4 + \cdots)$, so $(K - E)/k = \pi k/4 + O(k^3)$. When $L \gg a$, $k \simeq 2a/L$ and $k_1 \simeq a/L$, giving

$$\boxed{F \longrightarrow \frac{\mu_0 \pi a^2 M^2}{2}.}$$

This agrees with 5.22 for $A = \pi a^2$.

5.24

(a) For the perfectly conducting plane of Section 5.13 with the circular hole in it and the asymptotically uniform tangential magnetic field $\mathbf{H}_0$ on one side, calculate the added tangential magnetic field $\mathbf{H}^{(1)}$ on the side of the plane with $\mathbf{H}_0$. Show that its components for $\rho > a$ are

$$H_x^{(1)} = \frac{2H_0a^3}{\pi}\frac{xy}{\rho^4\sqrt{\rho^2-a^2}}$$

$$H_y^{(1)} = \frac{2H_0a^3}{\pi}\frac{y^2}{\rho^4\sqrt{\rho^2-a^2}} + \frac{H_0}{\pi}\left[\frac{a}{\rho}\sqrt{1-\frac{a^2}{\rho^2}} - \sin^{-1}\left(\frac{a}{\rho}\right)\right]$$

(b) Sketch the lines of surface current flow in the neighborhood of the hole on both sides of the plane.

Solution.

Take $\mathbf{H}_0 = H_0\widehat{\mathbf{y}}$ on the $z > 0$ side. The scalar potentials in Section 5.13 are $\Phi_+ = -H_0y + \Phi_1$ and $\Phi_- = -\Phi_1$, with

$$\Phi_1 = \frac{2H_0a^2}{\pi}\sin\phi\int_0^\infty j_1(ka)e^{-k|z|}J_1(k\rho)\,dk.$$

We evaluate its plane value rather than leaving a Bessel integral. The identity $a^2j_1(ka) = \int_0^a t\sin(kt)\,dt$, followed by integration by parts in k, gives, for $t < \rho$,

$$\int_0^\infty \sin(kt)J_1(k\rho)\,dk = \frac{t}{\rho}\int_0^\infty \cos(kt)J_0(k\rho)\,dk = \frac{t}{\rho\sqrt{\rho^2-t^2}}.$$

These oscillatory integrals are understood as limits with positive exponential damping, consistent with $z \to 0^+$. Consequently, for $\rho > a$,

$$\int_0^\infty j_1(ka)J_1(k\rho)\,dk = \frac{1}{a^2\rho}\int_0^a \frac{t^2\,dt}{\sqrt{\rho^2-t^2}} = \frac{\rho}{2a^2}f(\rho),$$

where $f(\rho) = \sin^{-1}(a/\rho) - (a/\rho)\sqrt{1-a^2/\rho^2}$. Thus $\Phi_1(x,y,0) = H_0yf(\rho)/\pi$.

(a) Differentiate explicitly:

$$f'(\rho) = -\frac{2a^3}{\rho^3\sqrt{\rho^2-a^2}}, \qquad \partial_x\rho = x/\rho, \quad \partial_y\rho = y/\rho.$$

Therefore

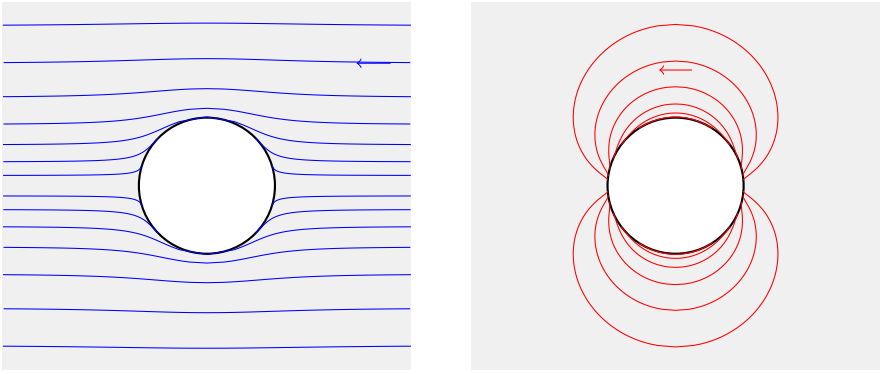
$$\boxed{H_x^{(1)} = -\frac{H_0}{\pi}yf'(\rho)\frac{x}{\rho} = \frac{2H_0a^3}{\pi}\frac{xy}{\rho^4\sqrt{\rho^2-a^2}},}$$

$$\boxed{H_y^{(1)} = -\frac{H_0}{\pi}\left[f(\rho) + \frac{y^2}{\rho}f'(\rho)\right] = \frac{2H_0a^3y^2}{\pi\rho^4\sqrt{\rho^2-a^2}} + \frac{H_0}{\pi}\left[\frac{a}{\rho}\sqrt{1-\frac{a^2}{\rho^2}} - \sin^{-1}\frac{a}{\rho}\right].}$$

(b) On the upper face the outward normal from the conductor is $+\widehat{\mathbf{z}}$ and on the lower face it is $-\widehat{\mathbf{z}}$. Hence

$$\mathbf{K}_+ = -H_0\widehat{\mathbf{x}} + \widehat{\mathbf{z}}\times\mathbf{H}^{(1)}, \qquad \mathbf{K}_- = \widehat{\mathbf{z}}\times\mathbf{H}^{(1)}.$$

If $\psi_+ = -y + yf/\pi$ and $\psi_- = yf/\pi$, both currents can be written $\mathbf{K}_\pm/H_0 = (\partial_y\psi_\pm, -\partial_x\psi_\pm)$. Their streamlines are therefore contours of the respective ψ. At the rim $f(a) = \pi/2$; the currents on the two faces join around the edge, whereas there is no conducting sheet in the open disk.



upper face: applied field lower face

Far from the hole the upper current is uniform toward $-x$, and the lower current tends to zero. The square-root edge divergence is the ideal zero-thickness conductor's edge field.

5.25 A flat right rectangular loop carrying a constant current I_1 is placed near a long straight wire carrying a current I_2. The loop is oriented so that its center is a perpendicular distance d from the wire; the sides of length a are parallel to the wire and the sides of length b make an angle α with the plane containing the wire and the loop's center. The direction of the current I_1 is the same as that of I_2 in the side of the rectangle nearest the wire.

(a) Show that the interaction magnetic energy

$$W_{12} = \int \mathbf{J}_1 \cdot \mathbf{A}_2 \, d^3x = I_1 F_2$$

(where F_2 is the magnetic flux from I_2 linking the rectangular circuit carrying I_1), is

$$W_{12} = \frac{\mu_0 I_1 I_2 a}{4\pi} \ln\left[\frac{4d^2 + b^2 + 4d\,b\cos\alpha}{4d^2 + b^2 - 4d\,b\cos\alpha}\right]$$

(b) Calculate the force between the loop and the wire for fixed currents.

(c) Repeat the calculation for a circular loop of radius a, whose plane is parallel to the wire and makes an angle α with respect to the plane containing the center of the loop and the wire. Show that the interaction energy is

$$W_{12} = \mu_0 I_1 I_2\, d \cdot \mathrm{Re}\left\{e^{i\alpha} - \sqrt{e^{2i\alpha} - a^2/d^2}\right\}$$

Find the force.

(d) For both loops, show that when $d \gg a$,b the interaction energy reduces to $W_{12} \simeq \mathbf{m} \cdot \mathbf{B}$, where $\mathbf{m}$ is the magnetic moment of the loop. Explain the sign.

Solution.

Put the long wire on the z axis with current $I_2\widehat{\mathbf{z}}$. The loop center is $(d, 0, 0)$; define the transverse in-plane direction $\mathbf{u} = (\cos\alpha, \sin\alpha, 0)$. A convenient vector potential is $\mathbf{A}_2 = -(\mu_0 I_2/2\pi)\ln\rho\,\widehat{\mathbf{z}}$. Its additive constant cancels around a closed loop.

(a) The two long rectangular sides have transverse positions $\mathbf{r}_\pm = d\hat{\mathbf{x}} \pm (b/2)\mathbf{u}$ and distances $r_\pm^2 = d^2 + b^2/4 \pm db\cos\alpha$. The near side carries $+I_1\hat{\mathbf{z}}$ and the far side the opposite current. The transverse sides have $\mathbf{A}_2 \cdot d\mathbf{l} = 0$. Thus

$$W_{12} = I_1 \oint \mathbf{A}_2 \cdot d\mathbf{l} = I_1 a[A_{2z}(r_-) - A_{2z}(r_+)]$$
$$= \frac{\mu_0 I_1 I_2 a}{2\pi} \ln\frac{r_+}{r_-} = \boxed{\frac{\mu_0 I_1 I_2 a}{4\pi} \ln\frac{4d^2 + b^2 + 4db\cos\alpha}{4d^2 + b^2 - 4db\cos\alpha}.}$$

Stokes's theorem identifies $\oint \mathbf{A}_2 \cdot d\mathbf{l} = F_2$, proving $W_{12} = I_1 F_2$ as well.

(b) At fixed currents the force is $+\nabla W_{12}$, with the physical loop orientation held fixed. Directly adding the forces on the two long sides gives the complete vector answer

$$\boxed{\mathbf{F} = \frac{\mu_0 I_1 I_2 a}{2\pi}\left(\frac{\mathbf{r}_+}{r_+^2} - \frac{\mathbf{r}_-}{r_-^2}\right), \qquad F_z = 0.}$$

The two transverse sides have equal and opposite z forces. Explicitly,

$$F_x = \frac{\mu_0 I_1 I_2 a}{2\pi}\left[\frac{d + (b/2)\cos\alpha}{r_+^2} - \frac{d - (b/2)\cos\alpha}{r_-^2}\right],$$
$$F_y = \frac{\mu_0 I_1 I_2 ab\sin\alpha}{4\pi}\left(\frac{1}{r_+^2} + \frac{1}{r_-^2}\right).$$

For $\alpha = 0$ and $d > b/2$, this is an attraction toward the wire.

(c) Parameterize the circular loop by $\mathbf{r} = d\hat{\mathbf{x}} + a\cos\theta\,\mathbf{u} - a\sin\theta\,\hat{\mathbf{z}}$, so the nearer side again carries current in $+z$. Then

$$W_{12} = \frac{\mu_0 I_1 I_2 a}{2\pi}\,\mathrm{Re}\int_0^{2\pi} \cos\theta \ln(d + ae^{i\alpha}\cos\theta)\,d\theta.$$

For $t = ae^{i\alpha}$, integrate by parts and use the elementary integral of $(d + t\cos\theta)^{-1}$:

$$\int_0^{2\pi} \cos\theta\ln(d + t\cos\theta)d\theta = t\int_0^{2\pi} \frac{\sin^2\theta}{d + t\cos\theta}d\theta = \frac{2\pi}{t}\left(d - \sqrt{d^2 - t^2}\right).$$

Conjugating inside the real part and simplifying gives

$$\boxed{W_{12} = \mu_0 I_1 I_2\,\mathrm{Re}\left[de^{i\alpha} - \sqrt{d^2e^{2i\alpha} - a^2}\right].}$$

Choose the square-root branch that approaches $de^{i\alpha}$ as $a/d \to 0$, and continue it without crossing a wire-loop intersection. This branch specification is essential; a principal square root used without regard to orientation can give the wrong flux.

At fixed orientation in space, displacement along x differentiates d at fixed α, whereas displacement along y changes the angle relative to the radial plane by $d\alpha/dy = -1/d$. Define

$$Z = e^{i\alpha} - \frac{de^{2i\alpha}}{\sqrt{d^2e^{2i\alpha} - a^2}}.$$

The force is therefore

$$\boxed{F_x = \mu_0 I_1 I_2\,\mathrm{Re}\,Z, \qquad F_y = \mu_0 I_1 I_2\,\mathrm{Im}\,Z, \qquad F_z = 0.}$$

Indeed $F_x = \partial_d W_{12}$ and $F_y = -d^{-1}\partial_\alpha W_{12}$.

(d) For the rectangle, $\ln[(1+\varepsilon)/(1-\varepsilon)] = 2\varepsilon + O(\varepsilon^3)$ gives $W_{12} \simeq \mu_0 I_1 I_2 ab\cos\alpha/(2\pi d)$. For the circle, expand the square root to get $W_{12} \simeq \mu_0 I_1 I_2 a^2 \cos\alpha/(2d)$. In both cases $\mathbf{m} = I_1 S(\widehat{\mathbf{z}} \times \mathbf{u})$ and $\mathbf{B}(d) = \mu_0 I_2 \widehat{\mathbf{y}}/(2\pi d)$, so these are $+\mathbf{m}\cdot\mathbf{B}$. This is the cross term in the *field energy*. Current sources exchange twice its change when mutual inductance changes at fixed currents. The mechanical potential is its negative, yielding the familiar force and dipole mechanical energy $-\mathbf{m}\cdot\mathbf{B}$.

5.26 A two-wire transmission line consists of a pair of nonpermeable parallel wires of radii a and b separated by a distance $d > a+b$. A current flows down one wire and back the other. It is uniformly distributed over the cross section of each wire. Show that the self-inductance per unit length is

$$L = \frac{\mu_0}{4\pi}\left[1 + 2\ln\left(\frac{d^2}{ab}\right)\right]$$

Solution.

Let the axial densities be $J_1 = I/(\pi a^2)$ and $J_2 = -I/(\pi b^2)$. In two dimensions

$$A_z(\mathbf{x}) = -\frac{\mu_0}{2\pi}\int J_z(\mathbf{x}')\ln|\mathbf{x}-\mathbf{x}'|\,d^2x'.$$

The arbitrary length inside the logarithm cancels because the total current is zero. With $W' = \frac{1}{2}\int J_z A_z\, d^2x$, the energy per length is determined by three mean logarithmic distances.

For a unit-normalized uniform disk of radius a, its logarithmic potential inside is

$$g_a(r) = \frac{1}{\pi a^2}\int_{D_a} \ln|\mathbf{r}-\mathbf{r}'|\,d^2r' = \ln a - \frac{1}{2} + \frac{r^2}{2a^2}.$$

To verify it, use $\nabla^2 \ln|\mathbf{r}-\mathbf{r}'| = 2\pi\delta^2$: $r^{-1}(rg_a')' = 2/a^2$, so regularity gives $g_a' = r/a^2$. Continuity with the exterior $g_a = \ln r$ fixes the constant. Disk averaging, with $\langle r^2\rangle = a^2/2$, then gives $\langle \ln|\mathbf{r}-\mathbf{r}'|\rangle = \ln a - 1/4$. The second disk similarly gives $\ln b - 1/4$. For disjoint disks the logarithmic kernel is harmonic in either disk separately. Applying the mean-value theorem twice makes the cross average exactly $\ln d$.

Therefore

$$W' = -\frac{\mu_0 I^2}{4\pi}\left[(\ln a - \tfrac{1}{4}) + (\ln b - \tfrac{1}{4}) - 2\ln d\right] = \frac{\mu_0 I^2}{4\pi}\left[\ln\frac{d^2}{ab} + \frac{1}{2}\right].$$

Since $W' = L'I^2/2$,

$$\boxed{L' = \frac{\mu_0}{4\pi}\left[1 + 2\ln\frac{d^2}{ab}\right].}$$

Within the stipulated uniform-current model this is exact for $d > a+b$, not just for widely separated wires. The constant term is the sum of the two internal inductances, $\mu_0/(8\pi)$ per wire.

5.27 A circuit consists of a long thin conducting shell of radius a and a parallel return wire of radius b on axis inside. If the current is assumed distributed uniformly throughout the cross section of the wire, calculate the self-inductance per unit length. What is the self-inductance if the inner conductor is a thin hollow tube?

Solution.

Use I along the inner conductor and $-I$ in the thin outer shell. Ampere's law gives

$$B_\phi(r) = \begin{cases} \mu_0 I r/(2\pi b^2), & 0 < r < b, \\ \mu_0 I/(2\pi r), & b < r < a, \\ 0, & r > a. \end{cases}$$

The energy per unit length is

$$\begin{aligned} W' &= \int_0^b \frac{B_\phi^2}{2\mu_0} 2\pi r\, dr + \int_b^a \frac{B_\phi^2}{2\mu_0} 2\pi r\, dr \\ &= \frac{\mu_0 I^2}{4\pi b^4}\frac{b^4}{4} + \frac{\mu_0 I^2}{4\pi}\ln\frac{a}{b} = \frac{\mu_0 I^2}{16\pi} + \frac{\mu_0 I^2}{4\pi}\ln\frac{a}{b}. \end{aligned}$$

Thus

$$\boxed{L' = \frac{\mu_0}{2\pi}\ln\frac{a}{b} + \frac{\mu_0}{8\pi}.}$$

If the inner conductor is a thin hollow tube, its current is a surface sheet and $B = 0$ for $r < b$. The internal term disappears: $\boxed{L'_{\text{tube}} = \mu_0 \ln(a/b)/(2\pi)}$. The infinitely thin outer shell contributes no internal energy.

5.28 Show that the mutual inductance of two circular coaxial loops in a homogeneous medium of permeability μ is

$$M_{12} = \mu\sqrt{ab}\left[\left(\frac{2}{k} - k\right)K(k) - \frac{2}{k}\,E(k)\right]$$

where

$$k^2 = \frac{4ab}{(a+b)^2 + d^2}$$

and a, b are the radii of the loops, d is the distance between their centers, and K and E are the complete elliptic integrals.

Find the limiting value when $d \ll a$, b and $a \simeq b$.

Solution.

The Neumann formula is $M = \mu(4\pi)^{-1}\oint\oint d\mathbf{l}_1 \cdot d\mathbf{l}_2/R$. For coaxial circles, the integrand depends only on the azimuthal difference:

$$M = \frac{\mu ab}{2}\int_0^{2\pi} \frac{\cos\varphi\, d\varphi}{\sqrt{a^2 + b^2 + d^2 - 2ab\cos\varphi}}.$$

Put $D = \sqrt{(a+b)^2 + d^2}$ and $k^2 = 4ab/D^2$. Set $\varphi = 2t$. Then $\cos\varphi = 2\cos^2 t - 1$ and the denominator is $D\sqrt{1 - k^2\cos^2 t}$. The integrand is symmetric about $t = \pi/2$, so the original integral over $0 < \varphi < 2\pi$ becomes

$$\frac{4}{D}\int_0^{\pi/2} \frac{2\cos^2 t - 1}{\sqrt{1 - k^2\cos^2 t}}\,dt.$$

Now set $u = \pi/2 - t$. The result is

$$\frac{4}{D}\int_0^{\pi/2} \frac{2\sin^2 u - 1}{\sqrt{1 - k^2\sin^2 u}}\,du = \frac{4}{D}\left[\frac{2(K-E)}{k^2} - K\right].$$

Here the definitions of K and E give the needed reduction:

$$K - E = \int_0^{\pi/2} \frac{1 - (1 - k^2 \sin^2 u)}{\sqrt{1 - k^2 \sin^2 u}} du = k^2 \int_0^{\pi/2} \frac{\sin^2 u}{\sqrt{1 - k^2 \sin^2 u}} du.$$

Since $2ab/D = k\sqrt{ab}$, this gives

$$\boxed{M = \mu\sqrt{ab}\left[\left(\frac{2}{k} - k\right) K(k) - \frac{2}{k} E(k)\right].}$$

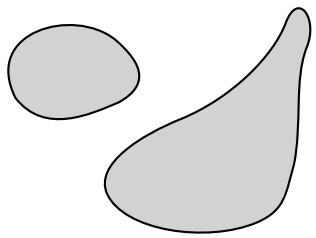
Here the arguments are elliptic moduli; numerical libraries often take the parameter k^2 instead.

For nearly coincident loops set $\delta = \sqrt{(a-b)^2 + d^2}$, so $k' = \sqrt{1 - k^2} = \delta/D \ll 1$. Using $K(k) = \ln(4/k') + O(k'^2 \ln k')$ and $E(k) = 1 + O(k'^2 \ln k')$ gives

$$\boxed{M \simeq \mu\sqrt{ab}\left[\ln \frac{4\sqrt{(a+b)^2 + d^2}}{\sqrt{(a-b)^2 + d^2}} - 2\right] \simeq \mu a \left[\ln \frac{8a}{\sqrt{(a-b)^2 + d^2}} - 2\right].}$$

For equal radii this is $\mu a[\ln(8a/|d|) - 2]$. The logarithmic divergence at exact coincidence belongs to the filament model; finite wire radius supplies the cutoff for self-inductance.

5.29 The figure represents a transmission line consisting of two, parallel *perfect* conductors of arbitrary, but constant, cross section. Current flows down one conductor and returns via the other.



Problem 5.29

Show that the product of the inductance per unit length L and the capacitance per unit length C is

$$LC = \mu\epsilon$$

where μ and ϵ are the permeability and the permittivity of the medium surrounding the conductors. (See the discussion about magnetic fields near perfect conductors at the beginning of Section 5.13.)

Solution.

Use the perfect-conductor magnetic boundary condition stated in Section 5.13: no magnetic field in the conductors and no trapped flux. Let the electrostatic transverse field for voltage difference V be $\mathbf{E}$, with charges per length $\pm\lambda$, so $\lambda = C'V$. In the homogeneous exterior medium, $\mathbf{E}$ is both curl-free and divergence-free, and normal to each conductor.

The rotated field $\mathbf{H} = \gamma\hat{\mathbf{z}} \times \mathbf{E}$ is consequently curl-free and divergence-free in the transverse exterior region, and tangent to the conductor surfaces. On a contour around one wire,

$$\oint \mathbf{H} \cdot d\mathbf{l} = \gamma \oint \mathbf{E} \cdot \mathbf{n}\, dl = \gamma \frac{\lambda}{\epsilon} = I.$$

Thus $\gamma = \epsilon I/\lambda$. The energy relation $\frac{\epsilon}{2}\int E^2 d^2x = C'V^2/2$ then gives

$$W'_m = \frac{\mu}{2}\int H^2 d^2x = \frac{\mu}{\epsilon}\left(\frac{\epsilon I}{C'V}\right)^2 \frac{C'V^2}{2} = \frac{\mu\epsilon I^2}{2C'}.$$

Comparing with $W'_m = L'I^2/2$ proves $\boxed{L'C' = \mu\epsilon}$. This result concerns the external inductance of perfect conductors. An internal magnetic field, such as that of a uniform-current wire in 5.26, adds internal inductance and does not satisfy this same product.

5.30

(a) Show that a surface current density $K(\phi) = I\cos\phi/2R$ flowing in the axial direction on a right circular cylindrical surface of radius R produces inside the cylinder a uniform magnetic induction $B_0 = \mu_0 I/4R$ in a direction perpendicular to the cylinder axis. Show that the field outside is that of a two-dimensional dipole.

(b) Calculate the total magnetostatic field energy per unit length. How is it divided inside and outside the cylinder?

(c) What is the inductance per unit length of the system, viewed as a long circuit with current flowing up one side of the cylinder and back the other?

Answer: $L = \pi\mu_0/8$.

Solution.

(a) Set $K_z = K_0\cos\phi$, $K_0 = I/(2R)$. The axial vector potential solves the two-dimensional Laplace equation on either side of the sheet. Regularity, decay of the field, and continuity give

$$A_z = C\cos\phi\begin{cases}\rho, & \rho < R,\\ R^2/\rho, & \rho > R.\end{cases}$$

Since $B_\rho = \rho^{-1}\partial_\phi A_z$ and $B_\phi = -\partial_\rho A_z$,

$$\mathbf{B}_{\rm in} = -C\widehat{\mathbf{y}}, \qquad (B_\rho, B_\phi)_{\rm out} = \frac{CR^2}{\rho^2}(-\sin\phi, \cos\phi).$$

The jump $B_{\phi,\rm out} - B_{\phi,\rm in} = \mu_0 K_z$ gives $2C\cos\phi = \mu_0 I\cos\phi/(2R)$. Therefore $\boxed{C = B_0 = \mu_0 I/(4R)}$, and the exterior field is the two-dimensional dipole field displayed above.

(b) The energy per length inside is $W'_{\rm in} = \pi R^2 B_0^2/(2\mu_0)$. Outside the magnitude is B_0R^2/ρ^2, so

$$W'_{\rm out} = \frac{1}{2\mu_0}\int_R^\infty \frac{B_0^2R^4}{\rho^4}2\pi\rho\,d\rho = \frac{\pi R^2 B_0^2}{2\mu_0} = W'_{\rm in}.$$

Thus $\boxed{W'_{\rm total} = \pi\mu_0 I^2/16}$, divided equally.

(c) The positive axial current integrates to $\int_{-\pi/2}^{\pi/2} K_z R\,d\phi = I$; the other half carries $-I$. Accordingly the circuit current is precisely the I in the density, and $\boxed{L' = 2W'_{\rm total}/I^2 = \pi\mu_0/8}$.

5.31 An accelerator bending magnet consists of N turns of superconducting cable whose current configuration can be described approximately by the axial current density

$$J_z(\rho,\phi) = \left(\frac{NI}{2R}\right)\cos\phi\ \delta(\rho - R)$$

The right circular current cylinder is centered on the axis of a hollow iron cylinder of inner radius R' ($R' > R$). The relative dimensions (R, R' a few centimeters and a magnet length of several meters) permit the use of a two-dimensional approximation, at least away from the ends of the magnet. Assume that the relative permeability of the iron can be taken as infinite. [Then the outer radius of the iron is irrelevant.]

(a) Show that the magnetic field inside the current sheath is perpendicular to the axis of the cylinder in the direction defined by $\phi = \pm\pi/2$ and has the magnitude

$$B_0 = \left(\frac{\mu_0 NI}{4R}\right)\left[1 + \frac{R^2}{R'^2}\right]$$

(b) Show that the magnetic energy inside $r = R$ is augmented (and that outside diminished) relative to the values in the absence of the iron. (Compare part b of Problem 5.30.)

(c) Show that the inductance per unit length is

$$\frac{dL}{dz} = \left(\frac{\pi\mu_0 N^2}{8}\right)\left[1 + \frac{R^2}{R'^2}\right]$$

Solution.

Let $c_0 = \mu_0 NI/(4R)$ and $\eta = R^2/R'^2$. Write

$$A_z = \cos\phi \begin{cases} C\rho, & \rho < R, \\ D\rho + E/\rho, & R < \rho < R'. \end{cases}$$

At the inner face of infinitely permeable iron, tangential H_ϕ vanishes, so $D - E/R'^2 = 0$. At the current sheath, continuity gives $C = D + E/R^2$ and the derivative jump gives $2E/R^2 = \mu_0 NI/(2R)$. These equations yield $E = c_0 R^2$, $D = c_0\eta$, and $C = c_0(1+\eta)$.

(a) As in 5.30 the core field is along $-y$ for the displayed positive-current convention and has magnitude

$$\boxed{B_0 = c_0(1+\eta) = \frac{\mu_0 NI}{4R}\left(1 + \frac{R^2}{R'^2}\right).}$$

Reversing the current reverses its direction.

(b) Put $W_0' = \pi R^2 c_0^2/(2\mu_0)$, the energy in each region for the corresponding free-space sheath. Inside, $W_{\text{in}}' = W_0'(1+\eta)^2$. In the gap, $B^2 = D^2 + E^2/\rho^4 - 2DE\cos(2\phi)/\rho^2$. Angular integration removes the last term, giving

$$\begin{aligned} W_{\text{out}}' &= \frac{\pi}{\mu_0}\int_R^{R'} (D^2\rho + E^2/\rho^3)\,d\rho \\ &= \frac{\pi}{2\mu_0}[D^2(R'^2 - R^2) + E^2(R^{-2} - R'^{-2})] = \boxed{W_0'(1-\eta^2)}. \end{aligned}$$

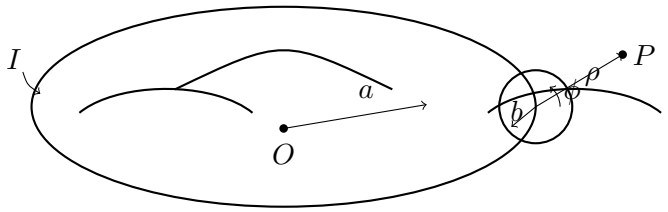
The iron contributes zero energy in the infinite-permeability limit. Thus the interior energy increases and the exterior energy decreases.

(c) Their sum is $2W_0'(1+\eta)$. Dividing twice this energy by the turn current squared gives

$$\boxed{L' = \frac{\pi\mu_0 N^2}{8}(1+\eta).}$$

The factor is N^2, because I is the current in each turn whereas the sheet strength is proportional to NI.

5.32 A circular loop of mean radius a is made of wire having a circular cross section of radius b, with $b \ll a$. The sketch shows the relevant dimensions and coordinates for this problem.



Problem 5.32

(a) Using (5.37), the expression for the vector potential of a filamentary circular loop, and appropriate approximations for the elliptic integrals, show that the vector potential at the point P near the wire is approximately

$$A_\phi = (\mu_0 I/2\pi)[\ln(8a/\rho) - 2]$$

where ρ is the transverse coordinate shown in the figure and corrections are of order $(\rho/a)\cos\phi$ and $(\rho/a)^2$.

(b) Since the vector potential of part a is, apart from a constant, just that outside a straight circular wire carrying a current I, determine the vector potential inside the wire ($\rho < b$) in the same approximation by requiring continuity of A_ϕ and its radial derivative at $\rho = b$, assuming that the current is uniform in density inside the wire:

$$A_\phi = (\mu_0 I/4\pi)(1 - \rho^2/b^2) + (\mu_0 I/2\pi)[\ln(8a/b) - 2], \qquad \rho < b$$

(c) Use (5.149) to find the magnetic energy, hence the self-inductance,

$$L = \mu_0 a[\ln(8a/b) - 7/4]$$

Are the corrections of order b/a or $(b/a)^2$? What is the change in L if the current is assumed to flow only on the surface of the wire (as occurs at high frequencies when the skin depth is small compared to b)?

Solution.

Distinguish the large cylindrical radius s about the loop axis from the small local distance ρ from the wire center. Near the wire, $s = a + \rho\cos\phi$ and $z = \rho\sin\phi$.

(a) From the circular-loop potential, or 5.28 divided by the circumference $2\pi s$,

$$A_\varphi = \frac{\mu_0 I}{\pi k}\sqrt{\frac{a}{s}}\left[\left(1-\frac{k^2}{2}\right)K(k) - E(k)\right], \quad k^2 = \frac{4as}{(a+s)^2+z^2}.$$

The complementary modulus is $k' = \rho/\sqrt{(a+s)^2+z^2} = \rho/(2a)[1+O(\rho/a)]$. Use $K = \ln(4/k') + O(k'^2 \ln k')$ and $E = 1 + O(k'^2 \ln k')$, with $s \simeq a$, $k \simeq 1$:

$$\boxed{A_\varphi \simeq \frac{\mu_0 I}{2\pi}\left[\ln\frac{8a}{\rho} - 2\right].}$$

The first geometrical correction is proportional to $(\rho/a)\cos\phi$ (with logarithmic coefficients), followed by second-order terms. The local angle ϕ in the figure is not the azimuth φ specifying the direction of this vector potential.

(b) Locally the straight-wire equation is $\rho^{-1}(\rho A'_\varphi)' = -\mu_0 I/(\pi b^2)$. Integrate once and discard the singular $1/\rho$ derivative at the center: $A'_\varphi = -\mu_0 I\rho/(2\pi b^2)$. Matching its value to the exterior potential at $\rho = b$ gives

$$\boxed{A_\varphi = \frac{\mu_0 I}{4\pi}\left(1-\frac{\rho^2}{b^2}\right) + \frac{\mu_0 I}{2\pi}\left[\ln\frac{8a}{b} - 2\right], \quad \rho < b.}$$

Its derivative at b is $-\mu_0 I/(2\pi b)$, also matching outside.

(c) To this order the volume element is $2\pi a\, d^2\rho$, and $J = I/(\pi b^2)$. Since $\langle\rho^2\rangle = b^2/2$ in a uniform disk,

$$\langle A_\varphi\rangle = \frac{\mu_0 I}{8\pi} + \frac{\mu_0 I}{2\pi}\left[\ln\frac{8a}{b} - 2\right] = \frac{\mu_0 I}{2\pi}\left[\ln\frac{8a}{b} - \frac{7}{4}\right].$$

Therefore $W = \frac{1}{2}(2\pi a) I\langle A_\varphi\rangle$ and

$$\boxed{L = \mu_0 a\left[\ln\frac{8a}{b} - \frac{7}{4}\right].}$$

Every first-order curvature term contains $\cos\phi$ and averages to zero, including the first correction to the toroidal volume element. The surviving corrections start at second order in b/a, possibly multiplied by $\ln(a/b)$. For surface current the local internal field and its internal energy vanish; use $A_\varphi(b)$ in the energy integral instead. Then

$$\boxed{L_{\text{surface}} = \mu_0 a\left[\ln\frac{8a}{b} - 2\right] = L - \frac{\mu_0 a}{4}.}$$

5.33 Consider two current loops (as in Fig. 5.3) whose orientation in space is fixed, but whose relative separation can be changed. Let O_1 and O_2 be origins in the two loops, fixed relative to each loop, and $\mathbf{x}_1$ and $\mathbf{x}_2$ be coordinates of elements $d\mathbf{l}_1$ and $d\mathbf{l}_2$, respectively, of the loops referred to the respective origins. Let $\mathbf{R}$ be the relative coordinate of the origins, directed from loop 2 to loop 1.

(a) Starting from (5.10), the expression for the force between the loops, show that it can be written

$$\mathbf{F}_{12} = I_1 I_2 \nabla_R M_{12}(\mathbf{R})$$

where M_{12} is the mutual inductance of the loops,

$$M_{12}(\mathbf{R}) = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|}$$

and it is assumed that the orientation of the loops does not change with $\mathbf{R}$.

(b) Show that the mutual inductance, viewed as a function of $\mathbf{R}$ in any open region where the two filaments do not intersect, is a solution of the Laplace equation,

$$\nabla_R^2 M_{12}(\mathbf{R}) = 0$$

The importance of this result is that the uniqueness of solutions of the Laplace equation allows the exploitation of the properties of such solutions, provided a solution can be found for a particular value of $\mathbf{R}$.

Solution.

(a) Put $\mathbf{s} = \mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}$. The magnetic force on loop 1 is

$$\mathbf{F}_{12} = \frac{\mu_0 I_1 I_2}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \times (d\mathbf{l}_2 \times \mathbf{s})}{s^3}.$$

Expanding the triple product gives $d\mathbf{l}_2(d\mathbf{l}_1 \cdot \mathbf{s}) - \mathbf{s}(d\mathbf{l}_1 \cdot d\mathbf{l}_2)$. For fixed loop-2 element, $d\mathbf{l}_1 \cdot \mathbf{s}/s^3 = -d(1/s)$, whose integral around closed loop 1 is zero. The remaining term is $(d\mathbf{l}_1 \cdot d\mathbf{l}_2)\nabla_R(1/s)$. The loop orientations and their local coordinates are independent of $\mathbf{R}$, so the derivative can be taken outside:

$$\boxed{\mathbf{F}_{12} = I_1 I_2 \nabla_R M_{12}, \qquad M_{12} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{|\mathbf{x}_1 - \mathbf{x}_2 + \mathbf{R}|}.}$$

(b) For any open set of separations in which the filaments are disjoint, $s > 0$ and $\nabla_R^2(1/s) = 0$ throughout the integration. Consequently $\boxed{\nabla_R^2 M_{12} = 0}$ in that set. This qualification matters: at intersecting-loop separations, the distributional identity instead reads

$$\nabla_R^2 M_{12} = -\mu_0 \oint \oint d\mathbf{l}_1 \cdot d\mathbf{l}_2 \, \delta^3(\mathbf{R} + \mathbf{x}_1 - \mathbf{x}_2).$$

Uniqueness and harmonic continuation are therefore used only within a current-free region of separation space.

5.34 Two identical circular loops of radius a are initially located a distance R apart on a common axis perpendicular to their planes.

(a) From the expression $W_{12} = \int d^3x \, \mathbf{J}_1 \cdot \mathbf{A}_2$ and the result for A_ϕ from Problem 5.10b, show that the mutual inductance of the loops is

$$M_{12} = \mu_0 \pi a^2 \int_0^\infty dk \, e^{-kR} J_1^2(ka)$$

(b) Show that for $R > 2a$, M_{12} has the expansion,

$$M_{12} = \frac{\mu_0 \pi a}{2} \left[\left(\frac{a}{R}\right)^3 - 3\left(\frac{a}{R}\right)^5 + \frac{75}{8}\left(\frac{a}{R}\right)^7 + \cdots \right]$$

(c) Use the techniques of Section 3.3 for solutions of the Laplace equation to show that the mutual inductance for two coplanar identical circular loops of radius a whose centers are separated by a distance $R > 2a$ is

$$M_{12} = -\frac{\mu_0 \pi a}{4}\left[\left(\frac{a}{R}\right)^3 + \frac{9}{4}\left(\frac{a}{R}\right)^5 + \frac{375}{64}\left(\frac{a}{R}\right)^7 + \cdots\right]$$

(d) Calculate the forces between the loops in the common axis and coplanar configurations. Relate the answers to those of Problem 5.18.

Solution.

(a) The azimuthal potential of loop 1 on loop 2 is constant. Multiplying the result in 5.10(b) at $\rho = a, z = R$ by $2\pi a I_2$ gives $W_{12} = I_1 I_2 M_{12}$ with

$$\boxed{M_\parallel(R) = \mu_0 \pi a^2 \int_0^\infty e^{-kR} J_1(ka)^2\, dk.}$$

(b) The power series $J_1(x) = \sum_{j\geq 0}(-1)^j (x/2)^{2j+1}/[j!(j+1)!]$ gives

$$J_1(x)^2 = \frac{x^2}{4} - \frac{x^4}{16} + \frac{5x^6}{768} + \cdots.$$

More generally its coefficient of $(-1)^n (x/2)^{2n+2}$ is

$$\sum_{j=0}^{n} \frac{1}{j!(j+1)!(n-j)!(n-j+1)!} = \frac{(2n+2)!}{n!(n+2)![(n+1)!]^2}.$$

This follows by factoring out $1/[n!(n+2)!]$ and applying the binomial convolution to $\sum_j \binom{n}{j}\binom{n+2}{j+1}$. Integrating each term with $\int_0^\infty k^m e^{-kR} dk = m!/R^{m+1}$ gives

$$M_\parallel(R) = \mu_0 \pi a \sum_{n=0}^{\infty} (-1)^n D_n (a/R)^{2n+3}, \quad D_n = \frac{[(2n+2)!]^2}{2^{2n+2} n!(n+2)![(n+1)!]^2}.$$

The ratio test gives convergence for $R > 2a$. For $n = 0, 1, 2$, $D_n = 1/2, 3/2, 75/16$, yielding

$$\boxed{M_\parallel = \frac{\mu_0 \pi a}{2}\left[(a/R)^3 - 3(a/R)^5 + \frac{75}{8}(a/R)^7 + \cdots\right].}$$

(c) With parallel loop normals fixed, rotational symmetry makes $M(\mathbf{R})$ axisymmetric in separation space. By 5.33 it is harmonic for $R > 2a$ and decays at infinity. The exterior harmonic with axial value R^{-2n-3} is $R^{-2n-3} P_{2n+2}(\cos\Theta)$. Thus replace each term by this angular form. At $\Theta = \pi/2$,

$$P_{2n+2}(0) = (-1)^{n+1} \frac{(2n+2)!}{2^{2n+2}[(n+1)!]^2},$$

so all terms in the coplanar mutual inductance are negative:

$$\boxed{M_\perp = -\frac{\mu_0 \pi a}{4}\left[(a/R)^3 + \frac{9}{4}(a/R)^5 + \frac{375}{64}(a/R)^7 + \cdots\right].}$$

(d) At fixed currents, $F_R = I_1 I_2\, dM/dR$. For positive $I_1 I_2$,

$$\boxed{F_{R,\parallel} = -\frac{3\mu_0 \pi I_1 I_2}{2}(a/R)^4 \left[1 - 5(a/R)^2 + \frac{175}{8}(a/R)^4 + \cdots\right],}$$

$$\boxed{F_{R,\perp} = +\frac{3\mu_0 \pi I_1 I_2}{4}(a/R)^4 \left[1 + \frac{15}{4}(a/R)^2 + \frac{875}{64}(a/R)^4 + \cdots\right].}$$

Coaxial like-current loops attract; coplanar like-current loops repel. Reverse one current and both forces reverse. For 5.18 use $R = 2d$ and $I_2 = \chi I_1$ in the axial case, but $I_2 = -\chi I_1$ in the coplanar case. Both then attract a more-permeable slab, with precisely the two dipole limits found there.

5.35 An insulated coil is wound on the surface of a sphere of radius a in such a way as to produce a uniform magnetic induction B_0 in the z direction inside the sphere and dipole field outside the sphere. The medium inside and outside the sphere has a uniform conductivity σ and permeability μ.

(a) Find the necessary surface current density $\mathbf{K}$ and show that the vector potential describing the magnetic field has only an azimuthal component, given by

$$A_\phi = \frac{B_0 a^2}{2}\frac{r_<}{r_>^2}\sin\theta$$

where $r_<$ ($r_>$) is the smaller (larger) of r and a.

(b) At $t = 0$ the current in the coil is cut off. [The coil's presence may be ignored from now on.] With the neglect of Maxwell's displacement current, the decay of the magnetic field is described by the diffusion equation, (5.160). Using a Laplace transform and a spherical Bessel function expansion (3.113), show that the vector potential at times $t > 0$ is given by

$$A_\phi = \frac{3B_0 a}{\pi}\sin\theta \int_0^\infty e^{-\nu t k^2} j_1(k) j_1\left(\frac{kr}{a}\right) dk$$

where $\nu = 1/\mu\sigma a^2$ is a characteristic decay rate and $j_1(x)$ is the spherical Bessel function of order one. Show that the magnetic field at the center of the sphere can be written explicitly in terms of the error function $\Phi(x)$ as

$$B_z(0,t) = B_0\left[\Phi\left(\frac{1}{\sqrt{4\nu t}}\right) - \frac{1}{\sqrt{\pi\nu t}}\exp\left(-\frac{1}{4\nu t}\right)\right]$$

(c) Show that the total magnetic energy at time $t > 0$ can be written

$$W_{\rm m} = \frac{6B_0^2 a^3}{\mu}\int_0^\infty e^{-2\nu t k^2}[j_1(k)]^2\, dk$$

Show that at long times ($\nu t \gg 1$) the magnetic energy decays asymptotically as

$$W_{\rm m} \to \frac{\sqrt{2\pi} B_0^2 a^3}{24\mu(\nu t)^{3/2}}$$

(d) Find a corresponding expression for the asymptotic form of the vector potential (at fixed r, θ and $\nu t \to \infty$) and show that it decays as $(\nu t)^{-3/2}$ as well. Since the energy is quadratic in the field strength, there seems to be a puzzle here. Show by numerical or analytic means that the behavior of the magnetic field at time t is such that, for distances small compared to $R = a(\nu t)^{1/2} \gg a$, the field is uniform with strength $(B_0/6\pi^{1/2})\,(\nu t)^{-3/2}$, and for distances large compared to R, the field is essentially the original dipole field. Explain physically.

Solution.

(a) Write $A_\phi = f(r)\sin\theta$. A uniform interior field requires $f = B_0 r/2$. Continuity with the decaying exterior dipole requires $f = B_0 a^3/(2r^2)$ outside. Thus

$$\boxed{A_\phi = \frac{B_0 a^2}{2}\frac{r_<}{r_>^2}\sin\theta.}$$

Its curl gives $B_{r,\mathrm{in}} = B_0\cos\theta$, $B_{\theta,\mathrm{in}} = -B_0\sin\theta$, and $B_{r,\mathrm{out}} = B_0(a/r)^3\cos\theta$, $B_{\theta,\mathrm{out}} = (B_0/2)(a/r)^3\sin\theta$. The tangential jump is $3B_0\sin\theta/2$, so

$$\boxed{\mathbf{K} = \frac{3B_0}{2\mu}\sin\theta\,\widehat{\boldsymbol{\phi}}.}$$

(b) Put $x = r/a$, $\tau = \nu t$, $\nu = (\mu\sigma a^2)^{-1}$. The vector diffusion equation reduces to

$$\partial_t f = \nu\mathcal{L}_x f, \qquad \mathcal{L}_x = \frac{d^2}{dx^2} + \frac{2}{x}\frac{d}{dx} - \frac{2}{x^2}.$$

Initially $f = (B_0 a/2)g(x)$ with $g = x$ for $x < 1$ and $g = x^{-2}$ for $x > 1$. Since g is continuous and its first derivative jumps from 1 to -2, $\mathcal{L}_x g = -3\delta(x-1)$. The spherical Bessel functions satisfy $\mathcal{L}_x j_1(kx) = -k^2 j_1(kx)$ and

$$\int_0^\infty x^2 j_1(kx) j_1(qx)dx = \frac{\pi}{2k^2}\delta(k-q).$$

The radial operator is self-adjoint with weight x^2: integrating $u\mathcal{L}_x v - v\mathcal{L}_x u$ gives $[x^2(uv' - u'v)]_0^\infty$. Apply this to $u = j_1(kx)$ and $v = g(x)$, with a vanishing positive convergence factor at infinity if needed for the oscillatory integral. The endpoint term vanishes, leaving the derivative jump at $x = 1$. Hence the initial transform satisfies

$$-k^2\int_0^\infty x^2 g(x) j_1(kx)dx = \int_0^\infty x^2 j_1(kx)\mathcal{L}_x g(x)dx = -3j_1(k).$$

Thus $\widetilde{g}(k) = 3j_1(k)/k^2$. Laplace transformation in time gives $(p + \nu k^2)\overline{\widetilde{f}}(k,p) = (B_0 a/2)\widetilde{g}(k)$. Inverting the Laplace transform yields $e^{-\nu k^2 t}$; inverting the spherical Bessel transform, with measure $2k^2 dk/\pi$, then gives

$$\boxed{A_\phi = \frac{3B_0 a}{\pi}\sin\theta\int_0^\infty e^{-\tau k^2} j_1(k) j_1(kx)dk.}$$

Near the center $j_1(kx) = kx/3 + O(x^3)$ and a potential $A_\phi = Cr\sin\theta$ has $B_z = 2C$. Hence

$$B_z(0,t) = \frac{2B_0}{\pi}\int_0^\infty e^{-\tau k^2}\left(\frac{\sin k}{k} - \cos k\right)dk.$$

The cosine Gaussian integral is $\sqrt{\pi}\,e^{-u^2/(4\tau)}/(2\sqrt{\tau})$. Integrating this identity from $u=0$ to $u=1$ gives $\int e^{-\tau k^2}\sin k\,dk/k = (\pi/2)\,\mathrm{erf}(1/(2\sqrt{\tau}))$. Therefore

$$\boxed{B_z(0,t) = B_0\left[\mathrm{erf}\,\frac{1}{2\sqrt{\tau}} - \frac{e^{-1/(4\tau)}}{\sqrt{\pi\tau}}\right].}$$

The question's Φ denotes this error function, not an electrostatic potential. The expression tends to B_0 as $t\downarrow 0$.

(c) Integration by parts gives $\int B^2 d^3x = \int \mathbf{A}\cdot(-\nabla^2\mathbf{A})d^3x$. The angular integral is $\int \sin^2\theta\,d\Omega = 8\pi/3$, so

$$W_m = \frac{4\pi}{3\mu}\int_0^\infty r^2 f(-\mathcal{L}_r f)\,dr.$$

Insert the two Bessel integrals for f and $-\mathcal{L}_r f$. The second supplies k^2/a^2; orthogonality in physical radius gives $\pi a^3\delta(k-q)/(2k^2)$. The prefactors reduce to $(4\pi/3\mu)(3B_0a/\pi)^2(\pi a/2) = 6B_0^2a^3/\mu$:

$$\boxed{W_m = \frac{6B_0^2a^3}{\mu}\int_0^\infty e^{-2\tau k^2} j_1(k)^2 dk.}$$

At $\tau \gg 1$, only $k = O(\tau^{-1/2})$ contributes. Use $j_1(k) = k/3 + O(k^3)$ and $\int_0^\infty k^2 e^{-2\tau k^2}dk = \sqrt{\pi}/[4(2\tau)^{3/2}]$:

$$\boxed{W_m \sim \frac{\sqrt{2\pi}B_0^2a^3}{24\mu\,\tau^{3/2}}.}$$

As a normalization check, the initial energy is the sum of the uniform interior contribution $2\pi B_0^2a^3/(3\mu)$ and the exterior dipole contribution $\pi B_0^2a^3/(3\mu)$, namely $\pi B_0^2a^3/\mu$. The integral formula agrees, since $\int_0^\infty j_1(k)^2dk = \pi/6$.

(d) At fixed r, expand both Bessel factors:

$$A_\phi \sim \frac{3B_0a}{\pi}\frac{x}{9}\sin\theta\int_0^\infty k^2 e^{-\tau k^2}dk = \boxed{\frac{B_0 r\sin\theta}{12\sqrt{\pi}}\tau^{-3/2}.}$$

To describe the changing spatial scale, keep $x/\sqrt{\tau}$ finite and expand only $j_1(k)$. The remaining integral can be evaluated:

$$A_\phi \sim \frac{B_0a^3}{2r^2}\mathcal{F}(s)\sin\theta, \qquad s = \frac{r}{2a\sqrt{\tau}}, \quad \mathcal{F}(s) = \mathrm{erf}\,s - \frac{2s}{\sqrt{\pi}}e^{-s^2}.$$

Indeed the integral of $ke^{-\tau k^2}j_1(kx)$ is $\pi\mathcal{F}(x/(2\sqrt{\tau}))/(2x^2)$, by the same sine and cosine integrals used for the center field. Taking the curl gives

$$B_r \sim \frac{B_0a^3}{r^3}\mathcal{F}(s)\cos\theta, \qquad B_\theta \sim \frac{B_0a^3}{2r^3}[\mathcal{F}(s) - s\mathcal{F}'(s)]\sin\theta.$$

For $s \ll 1$, $\mathcal{F}(s) = 4s^3/(3\sqrt{\pi}) + O(s^5)$, so the field is uniform to leading order, with $\boxed{B_{\mathrm{core}} \sim B_0\tau^{-3/2}/(6\sqrt{\pi})}$. For $s \gg 1$, $\mathcal{F}\to 1$ and $\mathcal{F}'\to 0$, giving the original dipole field.

There is no conflict between field and total-energy decay: the nearly uniform region grows to radius $R = O(a\sqrt{\tau})$. Its energy scales as $B_{\mathrm{core}}^2R^3 \sim \tau^{-3}\tau^{3/2}$. The exterior dipole energy beyond R scales as R^{-3}, the same $\tau^{-3/2}$. Magnetic diffusion changes a progressively larger volume; it does not multiply a fixed spatial field shape by one time-dependent amplitude.

5.36 The time-varying magnetic field for $t > 0$ in Problem 5.35 induces an electric field and causes current to flow.

(a) What components of electric field exist? Determine integral expressions for the components of the electric field and find a simple explicit form of the current density $\mathbf{J} = \sigma\mathbf{E}$ at $t = 0^+$. Compare your result with the current density of Problem 5.35a. Find the asymptotic behavior of the electric fields in time.

(b) With Ohm's law and the electric fields found in part a, show that the total power dissipated in the resistive medium can be written

$$P = \frac{12B_0^2 a^3 \nu}{\mu} \int_0^\infty e^{-2\nu t k^2} [k j_1(k)]^2 \, dk$$

Note that the power is the negative time derivative of the *magnetic* energy, W_m.

(c) Because of Ohm's law, the total *electric* energy is $W_\mathrm{e} = \epsilon_0 P/2\sigma$. The total energy is the sum of W_e and W_m; its time derivative should be the negative of the power dissipation. Show that the neglect of the energy in the electric field is the same order of approximation as neglect of the displacement current in the equations governing the magnetic field.

Solution.

(a) There are no imposed electric charges or scalar-potential gradients in this axially symmetric diffusion problem. In the divergence-free gauge, $\mathbf{E} = -\partial_t \mathbf{A}$, so $E_r = E_\theta = 0$ and

$$\boxed{E_\phi(r,\theta,t) = \frac{3B_0 a\nu}{\pi} \sin\theta \int_0^\infty k^2 e^{-\tau k^2} j_1(k) j_1(kr/a) dk.}$$

At $t = 0^+$ use $\partial_t f = \nu \mathcal{L}_x f$ and the initial $\mathcal{L}_x g = -3\delta(x-1)$ found in 5.35:

$$E_\phi(0^+) = \frac{3B_0 a\nu}{2} \sin\theta \, \delta(r/a - 1) = \frac{3B_0 a^2 \nu}{2} \sin\theta \, \delta(r-a).$$

Since $\sigma a^2 \nu = 1/\mu$,

$$\boxed{\mathbf{J}(0^+) = \frac{3B_0}{2\mu} \sin\theta \, \delta(r-a) \widehat{\boldsymbol{\phi}}.}$$

This induced sheet current initially replaces exactly the switched-off coil current in 5.35(a). It is a distributional limit: at every positive time diffusion smooths the sheet, and the instantaneous switch has a singular initial dissipation in the ideal diffusion model. For fixed r at late times, differentiate the asymptotic potential:

$$\boxed{E_\phi \sim \frac{B_0 \nu r \sin\theta}{8\sqrt{\pi}} \tau^{-5/2}.}$$

This limit is not uniform over radii of order $a\sqrt{\tau}$ or larger.

(b) Ohmic dissipation is $P = \sigma \int E_\phi^2 d^3x$. The angular integral is again $8\pi/3$, and the physical radial orthogonality integral is $\pi a^3 \delta(k-q)/(2k^2)$. Consequently

$$P = \sigma \left(\frac{3B_0 a\nu}{\pi} \right)^2 \frac{8\pi}{3} \frac{\pi a^3}{2} \int_0^\infty k^2 e^{-2\tau k^2} j_1(k)^2 dk.$$

Using $\sigma a^2 \nu = 1/\mu$ reduces this to

$$\boxed{P = \frac{12 B_0^2 a^3 \nu}{\mu} \int_0^\infty e^{-2\tau k^2} [k j_1(k)]^2 dk = -\frac{dW_m}{dt}.}$$

The equality follows independently by differentiating the energy integral in 5.35(c) for $t > 0$.

(c) Because $\mathbf{J} = \sigma \mathbf{E}$, $P = \sigma \int E^2 d^3x$, whereas $W_e = (\epsilon_0/2) \int E^2 d^3x$. Thus $W_e = \epsilon_0 P/(2\sigma)$ as stated. If the fields vary on a time scale T, the displacement-to-conduction current ratio is

$$\frac{|\epsilon_0 \partial_t \mathbf{E}|}{|\sigma \mathbf{E}|} = O\left(\frac{\epsilon_0}{\sigma T}\right).$$

In the energy balance, the discarded term has the same relative order:

$$\frac{|dW_e/dt|}{P} = \frac{\epsilon_0}{2\sigma} \left| \frac{d \ln P}{dt} \right| = O\left(\frac{\epsilon_0}{\sigma T}\right).$$

For example, at late times $W_m \propto t^{-3/2}$ and $P = (3/2) W_m/t \propto t^{-5/2}$, so $W_e/W_m = 3\epsilon_0/(4\sigma t)$ and $|dW_e/dt|/P = 5\epsilon_0/(4\sigma t)$. Thus retaining electric energy while omitting displacement current would mix approximation orders. For times comparable to ϵ_0/σ, the neglected current and electric energy must both be restored; the ideal $t = 0^+$ sheet limit is not a claim that the quasistatic model resolves arbitrarily fast switching.

6 Chapter 6. Maxwell Equations, Macroscopic Electromagnetism, Conservation Laws

Problems

6.1 In three dimensions the solution to the wave equation (6.32) for a point source in space and time (a light flash at $t' = 0$, $\mathbf{x}' = 0$) is a spherical shell disturbance of radius $R = ct$, namely the Green function $G^{(+)}$ (6.44). It may be initially surprising that in one or two dimensions, the disturbance possesses a "wake," even though the source is a "point" in space and time. The solutions for fewer dimensions than three can be found by superposition in the superfluous dimension(s), to eliminate dependence on such variable(s). For example, a flashing line source of uniform amplitude is equivalent to a point source in two dimensions.

(a) Starting with the retarded solution to the three-dimensional wave equation (6.47), show that the source $f(\mathbf{x}', t') = \delta(x')\delta(y')\delta(t')$, equivalent to a $t = 0$ point source at the origin in two spatial dimensions, produces a two-dimensional wave,

$$\Psi(x, y, t) = \frac{2c\Theta(ct - \rho)}{\sqrt{c^2t^2 - \rho^2}}$$

where $\rho^2 = x^2 + y^2$ and $\Theta(\xi)$ is the unit step function [$\Theta(\xi) = 0$ (1) if $\xi < 0$ (> 0).]

(b) Show that a "sheet" source, equivalent to a point pulsed source at the origin in one space dimension, produces a one-dimensional wave proportional to

$$\Psi(x, t) = 2\pi c\,\Theta(ct - |x|)$$

Solution.

The normalization in (6.32) is $(\nabla^2 - c^{-2}\partial_t^2)\Psi = -4\pi f$; thus (6.47), with no incident homogeneous wave, gives

$$\Psi(\mathbf{x}, t) = \int \frac{f(\mathbf{x}', t - |\mathbf{x} - \mathbf{x}'|/c)}{|\mathbf{x} - \mathbf{x}'|}\, d^3x'.$$

(a) Translation invariance along z permits setting the observation point's z coordinate to zero. Put $R = (\rho^2 + z'^2)^{1/2}$. Then

$$\Psi = \int_{-\infty}^{\infty} \frac{\delta(t - R/c)}{R}\, dz'.$$

There are no roots for $ct < \rho$. For $ct > \rho$ the two roots are $z'_\pm = \pm\sqrt{c^2t^2 - \rho^2}$, and $|d(t - R/c)/dz'| = |z'|/(cR)$. Hence each root contributes $c/\sqrt{c^2t^2 - \rho^2}$, proving

$$\boxed{\Psi = \frac{2c\Theta(ct - \rho)}{\sqrt{c^2t^2 - \rho^2}}.}$$

The inverse square root at the wavefront is understood as a locally integrable distribution, not as a finite pointwise value on the front.

(b) For $f = \delta(x')\delta(t')$, use polar coordinates s, φ in the source $y'z'$ plane and $R = (x^2 + s^2)^{1/2}$:

$$\Psi = 2\pi \int_0^{\infty} \frac{s\, ds}{\sqrt{x^2 + s^2}} \delta\left(t - \frac{\sqrt{x^2 + s^2}}{c}\right) = 2\pi \int_{|x|}^{\infty} \delta(t - R/c)\, dR = \boxed{2\pi c\Theta(ct - |x|)}.$$

For a fixed observation point, progressively more distant portions of the line or sheet contribute after the earliest arrival. Their superposition explains the wake without introducing a propagation speed other than c.

6.2 The charge and current densities for a single point charge q can be written formally as

$$\rho(\mathbf{x}', t') = q\delta[\mathbf{x}' - \mathbf{r}(t')]; \qquad \mathbf{J}(\mathbf{x}', t') = q\mathbf{v}(t')\delta[\mathbf{x}' - \mathbf{r}(t')]$$

where $\mathbf{r}(t')$ is the charge's position at time t' and $\mathbf{v}(t')$ is its velocity. In evaluating expressions involving the retarded time, one must put $t' = t_{\rm ret} = t - R(t')/c$, where $\mathbf{R} = \mathbf{x} - \mathbf{r}(t')$ (but $\mathbf{R} = \mathbf{x} - \mathbf{x}'(t')$ inside the delta functions). Assume a smooth subluminal trajectory and an observation event off the worldline with a finite retarded intersection. Subluminal motion alone ensures uniqueness, but does not ensure that such an intersection exists.

(a) As a preliminary to deriving the Heaviside–Feynman expressions for the electric and magnetic fields of a point charge, show that

$$\int d^3x' \; \delta[\mathbf{x}' - \mathbf{r}(t_{\rm ret})] = \frac{1}{\kappa}$$

where $\kappa = 1 - \mathbf{v} \cdot \hat{\mathbf{R}}/c$. Note that κ is evaluated at the retarded time.

(b) Starting with the Jefimenko generalizations of the Coulomb and Biot–Savart laws, use the expressions for the charge and current densities for a point charge and the result of part a to obtain the Heaviside–Feynman expressions for the electric and magnetic fields of a point charge,

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret} + \frac{\partial}{c\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret} - \frac{\partial}{c^2\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret}\right\}$$

and

$$\mathbf{B} = \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret} + \frac{\partial}{c\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}\right\}$$

(c) In our notation Feynman's expression for the electric field is

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret} + \frac{[R]_{\rm ret}}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret} + \frac{\partial^2}{c^2\partial t^2}\left[\hat{\mathbf{R}}\right]_{\rm ret}\right\}$$

while Heaviside's expression for the magnetic field is

$$\mathbf{B} = \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa^2 R^2}\right]_{\rm ret} + \frac{1}{c[R]_{\rm ret}}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret}\right\}$$

Show the equivalence of the two sets of expressions for the fields.

References: O. Heaviside, *Electromagnetic Theory*, Vol. 3 (1912), p. 464, Eq. (214). R. P. Feynman, *The Feynman Lectures in Physics*, Vol. 1 (1963), Chapter 28, Eq. (28.3).

Solution.

Assume a smooth subluminal trajectory and an observation event off the worldline whose past light cone intersects the trajectory at a finite time. Write $s = t_{\rm ret}$, $\mathbf{n} = \hat{\mathbf{R}}$, $v_n = \mathbf{v} \cdot \mathbf{n}$, and $\kappa = 1 - v_n/c > 0$. Quantities below are evaluated at s unless stated otherwise.

(a) In the spatial integral the retarded time is $s(\mathbf{x}') = t - |\mathbf{x} - \mathbf{x}'|/c$, so $\partial s/\partial x'_j = n_j/c$. The derivative matrix of the delta function's argument is

$$\frac{\partial[x'_i - r_i(s(\mathbf{x}'))]}{\partial x'_j} = \delta_{ij} - \frac{v_i n_j}{c}, \qquad \det\left(\mathbf{1} - \frac{\mathbf{v}\mathbf{n}^{\mathsf{T}}}{c}\right) = 1 - \frac{\mathbf{n}\cdot\mathbf{v}}{c} = \kappa.$$

The last equality follows by the rank-one determinant formula. The function $s + R(s)/c$ has derivative $\kappa > 0$, so the assumed retarded root is unique. Subluminal motion alone does not guarantee existence: in units $c = 1$, the trajectory $\mathbf{r}(s) = \sqrt{1+s^2}\,\hat{\mathbf{x}}$ has $|\mathbf{v}| < 1$, but an observer at $\mathbf{x} = 0$, $t = 0$ would require $s + \sqrt{1+s^2} = 0$, which has no finite solution. On the stated domain the delta-function rule therefore gives

$$\int F(\mathbf{x}')\delta^{(3)}[\mathbf{x}' - \mathbf{r}(s(\mathbf{x}'))]\, d^3x' = \left[\frac{F}{\kappa}\right]_{\text{ret}}.$$

Setting $F = 1$ proves the assertion. Omitting this Jacobian would incorrectly replace a retarded charge distribution by an instantaneous one.

(b) Equations (6.55)–(6.57), with $R = |\mathbf{x} - \mathbf{x}'|$ inside the integrals, read

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\left\{\int \frac{\mathbf{n}[\rho]}{R^2}\, d^3x' + \frac{1}{c}\partial_t \int \frac{\mathbf{n}[\rho]}{R}\, d^3x' - \frac{1}{c^2}\partial_t \int \frac{[\mathbf{J}]}{R}\, d^3x'\right\},$$
$$\mathbf{B} = \frac{\mu_0}{4\pi}\left\{\int \frac{[\mathbf{J}]\times\mathbf{n}}{R^2}\, d^3x' + \frac{1}{c}\partial_t \int \frac{[\mathbf{J}]\times\mathbf{n}}{R}\, d^3x'\right\}.$$

At fixed integration point, $\partial_t[\rho] = [\partial_{t'}\rho]$; this justifies taking the derivatives outside. Applying part (a) to each integral yields

$$\boxed{\mathbf{E}} = \boxed{\frac{q}{4\pi\epsilon_0}\left[\frac{\mathbf{n}}{\kappa R^2} + \frac{1}{c}\partial_t\left(\frac{\mathbf{n}}{\kappa R}\right) - \frac{1}{c^2}\partial_t\left(\frac{\mathbf{v}}{\kappa R}\right)\right]},$$
$$\boxed{\mathbf{B}} = \boxed{\frac{\mu_0 q}{4\pi}\left[\frac{\mathbf{v}\times\mathbf{n}}{\kappa R^2} + \frac{1}{c}\partial_t\left(\frac{\mathbf{v}\times\mathbf{n}}{\kappa R}\right)\right]}.$$

These derivatives act on the entire retarded expression, including κ.

(c) Differentiating $t = s + R(s)/c$ at fixed observer gives

$$\frac{ds}{dt} = \frac{1}{\kappa}, \quad \dot{R} = -\frac{v_n}{\kappa}, \quad \dot{\mathbf{n}} = \frac{-\mathbf{v} + v_n\mathbf{n}}{\kappa R}, \quad \frac{1}{\kappa} = 1 - \frac{\dot{R}}{c}.$$

Here dots denote observation-time derivatives, not source-time derivatives. In particular,

$$\frac{\mathbf{n}}{\kappa R} = \frac{\mathbf{n}}{R} - \frac{\mathbf{n}\dot{R}}{cR}, \qquad \frac{\mathbf{v}}{\kappa R} = -\dot{\mathbf{n}} - \frac{\mathbf{n}\dot{R}}{R}.$$

Substitute these into the electric field of part (b). The two derivatives of $\mathbf{n}\dot{R}/R$ cancel, leaving

$$\begin{aligned}\frac{4\pi\epsilon_0}{q}\mathbf{E} &= \frac{\mathbf{n}}{R^2} - \frac{\mathbf{n}\dot{R}}{cR^2} + \frac{1}{c}\partial_t\left(\frac{\mathbf{n}}{R}\right) + \frac{\ddot{\mathbf{n}}}{c^2} \\ &= \frac{\mathbf{n}}{R^2} + \frac{R}{c}\partial_t\left(\frac{\mathbf{n}}{R^2}\right) + \frac{\ddot{\mathbf{n}}}{c^2}.\end{aligned}$$

For the magnetic field put $\mathbf{u} = \mathbf{v}\times\mathbf{n}$ and expand only the derivative of $1/R$:

$$\frac{\mathbf{u}}{\kappa R^2} + \frac{1}{c}\partial_t\frac{\mathbf{u}}{\kappa R} = \frac{\mathbf{u}}{\kappa R^2}\left(1 - \frac{\dot{R}}{c}\right) + \frac{1}{cR}\partial_t\frac{\mathbf{u}}{\kappa} = \frac{\mathbf{u}}{\kappa^2 R^2} + \frac{1}{cR}\partial_t\frac{\mathbf{u}}{\kappa}.$$

These are precisely Feynman's electric and Heaviside's magnetic formulas.

6.3 The homogeneous diffusion equation (5.160) for the vector potential for quasi-static fields in unbounded conducting media has a solution to the initial value problem of the form,

$$\mathbf{A}(\mathbf{x},t) = \int d^3x'\, G(\mathbf{x}-\mathbf{x}',t)\mathbf{A}(\mathbf{x}',0)$$

where $\mathbf{A}(\mathbf{x}',0)$ describes the initial field configuration and G is an appropriate kernel.

(a) Solve the initial value problem by use of a three-dimensional Fourier transform in space for $\mathbf{A}(\mathbf{x},t)$. With the usual assumptions on interchange of orders of integration, show that the Green function has the Fourier representation,

$$G(\mathbf{x}-\mathbf{x}',t) = \frac{1}{(2\pi)^3}\int d^3k\; e^{-k^2t/\mu\sigma}e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}$$

and it is assumed that $t > 0$.

(b) By introducing a Fourier decomposition in both space and time, and performing the frequency integral in the complex ω plane to recover the result of part a, show that $G(\mathbf{x}-\mathbf{x}',t)$ is the diffusion Green function that satisfies the inhomogeneous equation,

$$\frac{\partial G}{\partial t} - \frac{1}{\mu\sigma}\nabla^2 G = \delta^{(3)}(\mathbf{x}-\mathbf{x}')\delta(t)$$

and vanishes for $t < 0$.

(c) Show that if σ is uniform throughout all space, the Green function is

$$G(\mathbf{x},t;\mathbf{x}',0) = \Theta(t)\Big(\frac{\mu\sigma}{4\pi t}\Big)^{3/2}\exp\Big(\frac{-\mu\sigma|\mathbf{x}-\mathbf{x}'|^2}{4t}\Big)$$

(d) Suppose that at time $t' = 0$, the initial vector potential $\mathbf{A}(\mathbf{x}',0)$ is nonvanishing only in a localized region of linear extent d around the origin. The time dependence of the fields is observed at a point P far from the origin, i.e., $|\mathbf{x}| = r \gg d$. Show that there are three regimes of time, $0 < t \ll T_1$, $T_1 \ll t \ll T_2$, and $t \gg T_2$. Give plausible definitions of T_1 and T_2, and describe qualitatively the time dependence at P. Show that in the last regime, the vector potential is proportional to the volume integral of $\mathbf{A}(\mathbf{x}',0)$ times $t^{-3/2}$, assuming that integral exists. Relate your discussion to those of Section 5.18.B and Problems 5.35 and 5.36.

Solution.

Put $D = 1/(\mu\sigma) > 0$, with constant μ, σ. The equation checked in Section 5.18, (5.160), is $\partial_t\mathbf{A} = D\nabla^2\mathbf{A}$. Each Cartesian component obeys the same scalar equation.

(a) Use $\widetilde{\mathbf{A}}(\mathbf{k},t) = \int e^{-i\mathbf{k}\cdot\mathbf{x}}\mathbf{A}(\mathbf{x},t)\,d^3x$. Fourier transformation and the initial condition give

$$\partial_t\widetilde{\mathbf{A}} = -Dk^2\widetilde{\mathbf{A}}, \qquad \widetilde{\mathbf{A}}(\mathbf{k},t) = e^{-Dk^2t}\widetilde{\mathbf{A}}(\mathbf{k},0).$$

Inserting the initial Fourier integral into the inverse transform gives

$$\mathbf{A}(\mathbf{x},t) = \int d^3x'\,\mathbf{A}(\mathbf{x}',0)\frac{1}{(2\pi)^3}\int d^3k\, e^{-Dk^2t+i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}.$$

The inner integral is the required kernel for $t > 0$.

(b) With the temporal convention $e^{-i\omega t}$, the causal inhomogeneous Green function has representation

$$G(\mathbf{R},t)=\int\frac{d^3k\,d\omega}{(2\pi)^4}\frac{e^{i\mathbf{k}\cdot\mathbf{R}-i\omega t}}{Dk^2-i\omega}.$$

The pole is at $\omega=-iDk^2$. For $t>0$ close the contour in the lower half-plane, clockwise: the residue ie^{-Dk^2t} gives e^{-Dk^2t}. For $t<0$ the upper contour contains no pole. The $k=0$ pole has the retarded limiting prescription. Consequently $G=\Theta(t)G_{t>0}$. Acting with $\partial_t-D\nabla^2$ multiplies the Fourier integrand by $Dk^2-i\omega$, proving the delta-function equation. Equivalently, its time derivative supplies $\delta(t)G(\mathbf{R},0^+)=\delta(t)\delta^{(3)}(\mathbf{R})$.

(c) Completing the square in each of the three Gaussian integrals,

$$\int_{-\infty}^{\infty}e^{-Dtk_j^2+ik_jR_j}\,dk_j=\sqrt{\frac{\pi}{Dt}}e^{-R_j^2/(4Dt)},$$

yields

$$\boxed{G(\mathbf{x},t;\mathbf{x}',0)=\frac{\Theta(t)}{(4\pi Dt)^{3/2}}\exp\left[-\frac{|\mathbf{x}-\mathbf{x}'|^2}{4Dt}\right].}$$

It has unit spatial integral for every $t>0$, tends to a spatial delta function as $t\downarrow 0$, and satisfies the diffusion equation away from $t=0$.

(d) Suppose $|\mathbf{x}'|\lesssim d\ll r$. Factor the convolution as

$$\mathbf{A}(\mathbf{x},t)=\frac{e^{-r^2/(4Dt)}}{(4\pi Dt)^{3/2}}\int\mathbf{A}(\mathbf{x}',0)\exp\left[\frac{\mathbf{x}\cdot\mathbf{x}'}{2Dt}-\frac{r'^2}{4Dt}\right]d^3x'.$$

Useful definitions, up to order-one factors, are $T_1=rd/(2D)$ and $T_2=r^2/(4D)$, with $T_1\ll T_2$. For $t\ll T_1$, the response is exponentially small, of order $\exp[-(r-d)^2/(4Dt)]$, and can depend strongly on which part of the initial region is nearest the observer. For $T_1\ll t\ll T_2$, the exponential inside the integral is nearly one: the region acts as a point source, but the response still contains the small factor $e^{-r^2/(4Dt)}$. Near $t\sim T_2$ it becomes appreciable; the monopole kernel itself peaks at $t=r^2/(6D)$. Finally, for $t\gg T_2$,

$$\boxed{\mathbf{A}(\mathbf{x},t)\sim\frac{t^{-3/2}}{(4\pi D)^{3/2}}\int\mathbf{A}(\mathbf{x}',0)\,d^3x'.}$$

If that integral vanishes, this coefficient is zero and higher moments of the initial data determine the faster decay. In particular, a smooth, compactly supported, divergence-free Coulomb-gauge potential has $\int A_i\,d^3x=\int\nabla\cdot(x_i\mathbf{A})\,d^3x=0$. Thus the displayed asymptotic statement need not have a nonzero coefficient for every physical initial configuration.

The length reached by diffusion is $\sqrt{Dt}$, as in Section 5.18.B's current sheets; in that effectively one-dimensional geometry the basic kernel scales as $t^{-1/2}$. Problems 5.35–5.36 instead begin with a potential extending to infinity as a dipole potential. Their fixed-position $t^{-3/2}$ field behavior and expanding affected region $r\sim\sqrt{Dt}$ explain why a total energy can decay with a different power from the square of a field at a fixed point. The compact-support moment argument above must not be applied to those noncompact initial data without checking convergence. The arbitrarily early, exponentially small diffusion tail is part of the quasistatic approximation; it is not an exact relativistic signal.

6.4 A uniformly magnetized and conducting sphere of radius R and total magnetic moment $m = 4\pi MR^3/3$ rotates about its magnetization axis with angular speed ω. In the steady state no current flows in the conductor. The motion is nonrelativistic; the sphere has no excess charge on it.

(a) By considering Ohm's law in the moving conductor, show that the motion induces an electric field and a uniform volume charge density in the conductor, $\rho = -m\omega/\pi c^2 R^3$.

(b) Because the sphere is electrically neutral, there is no monopole electric field outside. Use symmetry arguments to show that the lowest possible electric multipolarity is quadrupole. Show that only a quadrupole field exists outside and that the quadrupole moment tensor has nonvanishing components, $Q_{33} = -4m\omega R^2/3c^2$, $Q_{11} = Q_{22} = -Q_{33}/2$.

(c) By considering the radial electric fields inside and outside the sphere, show that the necessary surface-charge density $\sigma(\theta)$ is

$$\sigma(\theta) = \frac{1}{4\pi R^2} \cdot \frac{4m\omega}{3c^2} \cdot \left[1 - \frac{5}{2} P_2(\cos\theta)\right]$$

(d) The rotating sphere serves as a unipolar induction device if a stationary circuit is attached by a slip ring to the pole and a sliding contact to the equator. Show that the line integral of the electric field from the equator contact to the pole contact (by any path) is $\mathcal{E} = \mu_0 m\omega/4\pi R$.

[See Landau and Lifshitz, *Electrodynamics of Continuous Media*, p. 221, for an alternative discussion of this electromotive force.]

Solution.

Take the common magnetization and rotation axis to be z, with positive $m\omega$ for parallel vectors. The uniformly magnetized sphere has $\mathbf{B}_0 = (2\mu_0 M/3)\hat{\mathbf{z}} = \mu_0 m\hat{\mathbf{z}}/(2\pi R^3)$ inside. Work through first order in $\omega R/c$; fields of the induced rotating charge change this magnetic field only at the next relevant order.

(a) To this order Ohm's law in the local moving conductor is $\mathbf{J}_{\text{cond}} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}_0) = 0$. With cylindrical radius s and $\mathbf{v} = \omega s\hat{\boldsymbol{\phi}}$,

$$\mathbf{E}_{\text{in}} = -\omega B_0 s\hat{\mathbf{s}}, \quad \Phi_{\text{in}} = \frac{1}{2}\omega B_0 s^2 + C, \quad \rho = \epsilon_0 \nabla \cdot \mathbf{E}_{\text{in}} = -2\epsilon_0 \omega B_0 = \boxed{-\frac{m\omega}{\pi c^2 R^3}}.$$

Here "no current" means no conduction current relative to the sphere; the induced charge does have a higher-order convection current.

(b) Axial symmetry excludes $m \neq 0$, inversion symmetry excludes odd l, and neutrality excludes the exterior $l = 0$ term. Moreover,

$$\Phi_{\text{in}} = \frac{\omega B_0 r^2}{3}[1 - P_2(\cos\theta)] + C.$$

The surface value contains only $l = 0, 2$. Uniqueness and decay at infinity therefore imply an exterior quadrupole alone. Continuity and neutrality fix

$$C = -\frac{\omega B_0 R^2}{3}, \qquad \Phi_{\text{out}} = -\frac{\omega B_0 R^5}{3r^3} P_2(\cos\theta).$$

For an axial traceless quadrupole, $\Phi_Q = Q_{33} P_2/(8\pi\epsilon_0 r^3)$. Thus

$$\boxed{Q_{33} = -\frac{4m\omega R^2}{3c^2}, \qquad Q_{11} = Q_{22} = -\frac{1}{2}Q_{33}.}$$

The volume charge is uniform and contributes no traceless quadrupole; this moment resides in the anisotropic surface charge.

(c) At $r = R$,

$$E_{r,\text{in}} = -\frac{2\omega B_0 R}{3}(1 - P_2), \qquad E_{r,\text{out}} = -\omega B_0 R P_2.$$

Gauss's boundary condition consequently gives

$$\sigma = \epsilon_0(E_{r,\text{out}} - E_{r,\text{in}}) = \frac{\epsilon_0 \omega B_0 R}{3}(2 - 5P_2) = \boxed{\frac{1}{4\pi R^2}\frac{4m\omega}{3c^2}\left(1 - \frac{5}{2}P_2\right)}.$$

Its integral is $4m\omega/(3c^2)$, canceling the volume charge $\rho(4\pi R^3/3) = -4m\omega/(3c^2)$.

(d) The stationary laboratory fields are electrostatic, so the integral between the contacts is independent of path:

$$\int_{\text{equator}}^{\text{pole}} \mathbf{E} \cdot d\mathbf{l} = \Phi_{\text{equator}} - \Phi_{\text{pole}} = \frac{1}{2}\omega B_0 R^2 = \boxed{\frac{\mu_0 m\omega}{4\pi R}}.$$

The motional force on charges in the rotating conductor sustains this terminal voltage; a wholly stationary electrostatic closed-path integral still vanishes.

6.5 A localized electric charge distribution produces an electrostatic field, $\mathbf{E} = -\nabla\Phi$. Into this field is placed a small localized time-independent current density $\mathbf{J}(\mathbf{x})$, which generates a magnetic field $\mathbf{H}$.

(a) Show that the momentum of these electromagnetic fields, (6.117), can be transformed to

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2}\int \Phi\mathbf{J}\, d^3x$$

provided the product $\Phi\mathbf{H}$ falls off rapidly enough at large distances. How rapidly is "rapidly enough"?

(b) Assuming that the current distribution is localized to a region small compared to the scale of variation of the electric field, expand the electrostatic potential in a Taylor series and show that

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2}\mathbf{E}(0) \times \mathbf{m}$$

where $\mathbf{E}(0)$ is the electric field at the current distribution and $\mathbf{m}$ is the magnetic moment, (5.54), caused by the current.

(c) Suppose the current distribution is placed instead in a *uniform* electric field $\mathbf{E}_0$ (filling all space). Interpret the conditionally convergent all-space field integrals as limits over concentric spheres. Show that, no matter how complicated is the localized $\mathbf{J}$, the result in part a is augmented by a surface integral contribution from infinity equal to minus one-third of the result of part b, yielding

$$\mathbf{P}_{\text{field}} = \frac{2}{3c^2}\mathbf{E}_0 \times \mathbf{m}$$

Compare this result with that obtained by working directly with (6.117) and the considerations at the end of Section 5.6.

Solution.

(a) For a finite volume, $\nabla \times \mathbf{H} = \mathbf{J}$ and $\nabla \times (\Phi \mathbf{H}) = -\mathbf{E} \times \mathbf{H} + \Phi \mathbf{J}$. Using the surface theorem for a curl gives

$$\mathbf{P}_V = \frac{1}{c^2} \int_V \mathbf{E} \times \mathbf{H} \, d^3x = \frac{1}{c^2} \left[\int_V \Phi \mathbf{J} \, d^3x - \oint_{\partial V} \Phi \, \mathbf{n} \times \mathbf{H} \, da \right].$$

For spherical surfaces, the sufficient uniform decay condition $\Phi \mathbf{H} = o(r^{-2})$ makes the surface term vanish. More generally it suffices that its angular integral multiplied by r^2 tend to zero. Localized electric and stationary current sources give $\Phi = O(r^{-1})$ and $H = O(r^{-3})$, and hence meet this condition.

(b) Localized steady current obeys $\nabla \cdot \mathbf{J} = 0$. Integration of $\partial_j(x_i J_j)$ gives $\int J_i = 0$; integration of $\partial_k(x_i x_j J_k)$ gives

$$\int (x_i J_j + x_j J_i) \, d^3x = 0, \qquad \int x_i J_j \, d^3x = \epsilon_{ijk} m_k,$$

where $m_k = \frac{1}{2} \epsilon_{kij} \int x_i J_j \, d^3x$. Expand $\Phi = \Phi(0) - x_i E_i(0) + \cdots$. The constant term integrates to zero and the linear term gives

$$(P_{\text{field}})_j = -\frac{1}{c^2} E_i(0) \epsilon_{ijk} m_k + \cdots, \qquad \boxed{\mathbf{P}_{\text{field}} = \frac{\mathbf{E}(0) \times \mathbf{m}}{c^2}}$$

to leading order in source size divided by the field-variation length.

(c) Now $\Phi = -\mathbf{E}_0 \cdot \mathbf{x}$ and the surface term cannot be discarded. On a large sphere,

$$\mathbf{H} = \frac{3\mathbf{n}(\mathbf{m} \cdot \mathbf{n}) - \mathbf{m}}{4\pi r^3} + O(r^{-4}), \qquad \oint \Phi \mathbf{n} \times \mathbf{H} \, da = \frac{1}{4\pi} \int (\mathbf{E}_0 \cdot \mathbf{n})(\mathbf{n} \times \mathbf{m}) \, d\Omega = \frac{1}{3} \mathbf{E}_0 \times \mathbf{m}.$$

We used $\int n_i n_j \, d\Omega = 4\pi \delta_{ij}/3$. The volume term in part (a) is exactly $\mathbf{E}_0 \times \mathbf{m}$ for any localized current, so

$$\boxed{\mathbf{P}_{\text{field}} = \frac{2}{3c^2} \mathbf{E}_0 \times \mathbf{m}}.$$

Directly, $\mathbf{A} = \mu_0 \mathbf{m} \times \mathbf{n}/(4\pi r^2) + O(r^{-3})$ implies

$$\int \mathbf{B} \, d^3x = \lim_{R \to \infty} \oint \mathbf{n} \times \mathbf{A} \, da = \frac{\mu_0}{4\pi} \int [\mathbf{m} - \mathbf{n}(\mathbf{n} \cdot \mathbf{m})] \, d\Omega = \frac{2\mu_0}{3} \mathbf{m}.$$

Then $\epsilon_0 \mathbf{E}_0 \times \int \mathbf{B}$ gives the same result, including the interior field emphasized in Section 5.6. The all-space uniform-field integral is conditionally convergent; the specified result uses concentric spherical exhaustion. Other asymptotic boundary shapes need not give the same integral. A localized physical source of an almost uniform electric field instead has the vanishing surface term of part (a).

6.6

(a) Consider a circular toroidal coil of mean radius a and N turns, with a small uniform cross section of area A (both height and width small compared to a). The toroid has a current I flowing in it and there is a point charge Q located at its center. Calculate all the components of field momentum of the system; show that the component along the axis of the toroid is

$$(\mathbf{P}_{\text{field}})_z \approx \pm \frac{\mu_0 Q I N A}{4\pi a^2}$$

where the sign depends on the sense of the current flow in the coil. Assume that the electric field of the charge penetrates unimpeded into the region of nonvanishing magnetic field, as would happen for a toroid that is actually a set of N small nonconducting tubes inside which ionized gas moves to provide the current flow.

Check that the answer conforms to the approximation of Problem 6.5b.

(b) If $Q = 10^{-6}$ C ($\approx 6 \times 10^{12}$ electronic charges), $I = 1.0$ A, $N = 2000$, $A = 10^{-4}$ m^2, $a = 0.1$ m, find the electric field at the toroid in volts per meter, the magnetic induction in tesla, and the electromagnetic momentum in newton-seconds. Compare with the momentum of a 10 μg insect flying at a speed of 0.1 m/s.

[Note that the system of charge and toroid is at rest. Its *total* momentum must vanish. There must therefore be a canceling "hidden" mechanical momentum—see Problem 12.8.]

Solution.

(a) Let $\eta = \pm 1$ specify $\mathbf{B} = \eta B_\phi \hat{\boldsymbol{\phi}}$. In cylindrical coordinates (s, ϕ, z) inside an ideal closely wound toroid,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0} \frac{s\hat{\mathbf{s}} + z\hat{\mathbf{z}}}{(s^2 + z^2)^{3/2}}, \qquad B_\phi = \frac{\mu_0 NI}{2\pi s}.$$

Thus $\epsilon_0 \mathbf{E} \times \mathbf{B} = \eta \epsilon_0 B_\phi (E_s \hat{\mathbf{z}} - E_z \hat{\mathbf{s}})$. The radial part integrates to zero around ϕ even without reflection symmetry of the cross section; hence $P_x = P_y = 0$. The axial integral is

$$P_z = \eta \frac{\mu_0 QNI}{4\pi} \int_A \frac{s\, ds\, dz}{(s^2 + z^2)^{3/2}} \simeq \boxed{\eta \frac{\mu_0 QINA}{4\pi a^2}}.$$

The approximation replaces s by a and z by zero across the small cross section. Each turn separately is a dipole $\mathbf{m}_{\text{turn}} = \eta IA\hat{\boldsymbol{\phi}}$ in its local electric field $\mathbf{E} \simeq Q\hat{\mathbf{s}}/(4\pi\epsilon_0 a^2)$. Summing $c^{-2}\mathbf{E} \times \mathbf{m}_{\text{turn}}$ over the N turns reproduces the result. The entire toroid has zero net dipole moment, so part 6.5(b) cannot be applied once to the whole toroid: its applied electric field is not uniform on that scale.

(b) Substitution of the specified SI values gives

$$\boxed{E = 8.988 \times 10^5 \text{ V m}^{-1}, \quad B = 4.000 \times 10^{-3} \text{ T}, \quad |\mathbf{P}_{\text{field}}| = 2.000 \times 10^{-12} \text{ N s}.}$$

The insect's mass is $10\,\mu\text{g} = 10^{-8}$ kg, so its momentum is 10^{-9} N s, about 500 times larger. These are field momenta only; the stationary charge–toroid system has compensating mechanical momentum as specified in the question.

6.7 The microscopic current $\mathbf{j}(\mathbf{x}, t)$ can be written as

$$\mathbf{j}(\mathbf{x}, t) = \sum_j q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{x}_j(t))$$

where the point charge q_j is located at the point $\mathbf{x}_j(t)$ and has velocity $\mathbf{v}_j = d\mathbf{x}_j(t)/dt$. Just as for the charge density, this current can be broken up into a "free" (conduction) electron contribution and a bound (molecular) current contribution.

Following the averaging procedures of Section 6.6 *and* assuming *nonrelativistic* addition of velocities, consider the averaged current, $\langle \mathbf{j}(\mathbf{x}, t) \rangle$.

(a) Show that the averaged current can be written in the form of (6.96) with the definitions (6.92), (6.97), and (6.98).

(b) Show that for a medium whose internal molecular velocities can be neglected, but which is in bulk motion (i.e., $\mathbf{v}_n = \mathbf{v}$ for all n),

$$\frac{1}{\mu_0}\mathbf{B} - \mathbf{H} = \mathbf{M} + (\mathbf{D} - \epsilon_0\mathbf{E}) \times \mathbf{v}$$

This shows that a moving polarization ($\mathbf{P}$) produces an effective magnetization density.

Hints for part a: Consider quantities like $(d\mathbf{p}_n/dt)$, $(dQ'^{(n)}_{\alpha\beta}/dt)$ and see what they look like. Also note that

$$\frac{df}{dt}(\mathbf{x} - \mathbf{x}_n(t)) = -\mathbf{v}_n \cdot \nabla f(\mathbf{x} - \mathbf{x}_n(t))$$

Solution.

(a) Use the averaging function $f_n = f(\mathbf{x} - \mathbf{x}_n)$ of Section 6.6. For one molecule suppress n, write its internal coordinates and velocities as $\boldsymbol{\xi}_j, \mathbf{u}_j$, and its center velocity as $\mathbf{v}$. Define

$$q = \sum_j q_j, \quad p_\alpha = \sum_j q_j \xi_{j\alpha}, \quad S_{\alpha\beta} = \sum_j q_j \xi_{j\alpha}\xi_{j\beta} = \tfrac{1}{3}Q'_{\alpha\beta}, \quad U_{\alpha\beta} = \sum_j q_j u_{j\alpha}\xi_{j\beta}.$$

The factor 3 in Q' is Jackson's actual convention (6.77); Q' is not traceless. Repeated Cartesian indices below are summed. Taylor expansion of (6.94) gives the terms displayed in (6.96):

$$\langle j_\alpha \rangle_n = q v_\alpha f + \dot{p}_\alpha f - (v_\alpha p_\beta + U_{\alpha\beta})\partial_\beta f + \frac{1}{2} v_\alpha S_{\beta\gamma}\partial_\beta\partial_\gamma f + \cdots .$$

The omitted internal-current second spatial moments belong to higher multipoles in the ellipsis of (6.96). The identities needed to rearrange the retained terms are

$$\dot{S}_{\alpha\beta} = U_{\alpha\beta} + U_{\beta\alpha}, \quad \epsilon_{\alpha\beta\gamma} m_\gamma = \tfrac{1}{2}(U_{\beta\alpha} - U_{\alpha\beta}), \quad \partial_t f = -v_\gamma \partial_\gamma f.$$

In particular,

$$\partial_t \left(p_\alpha f - \tfrac{1}{2} S_{\alpha\beta}\partial_\beta f \right) = \dot{p}_\alpha f - p_\alpha v_\beta \partial_\beta f - \tfrac{1}{2}(U_{\alpha\beta} + U_{\beta\alpha})\partial_\beta f + \tfrac{1}{2} S_{\alpha\beta} v_\gamma \partial_\beta \partial_\gamma f.$$

Adding $\epsilon_{\alpha\beta\gamma}\partial_\beta(m_\gamma f)$ changes the internal-velocity term to $-U_{\alpha\beta}\partial_\beta f$. The remaining convection terms are exactly

$$\partial_\beta[(p_\alpha v_\beta - p_\beta v_\alpha) f] - \frac{1}{6}\partial_\beta\partial_\gamma[(Q'_{\alpha\beta} v_\gamma - Q'_{\gamma\beta} v_\alpha) f].$$

Sum over molecules and restore the free-particle current. With

$$J_\alpha = \left\langle \sum_{j\,\text{free}} q_j v_{j\alpha} \delta(\mathbf{x} - \mathbf{x}_j) + \sum_n q_n v_{n\alpha} \delta(\mathbf{x} - \mathbf{x}_n) \right\rangle,$$

$$M_\alpha = \left\langle \sum_n m_{n\alpha} \delta(\mathbf{x} - \mathbf{x}_n) \right\rangle,$$

$$D_\alpha - \epsilon_0 E_\alpha = P_\alpha - \partial_\beta Q_{\alpha\beta}, \quad P_\alpha = \langle \sum_n p_{n\alpha}\delta_n \rangle, \quad Q_{\alpha\beta} = \tfrac{1}{6}\langle \sum_n Q'^{(n)}_{\alpha\beta}\delta_n \rangle,$$

one obtains explicitly

$$\begin{aligned}\langle j_\alpha\rangle &= J_\alpha + \partial_t(D_\alpha - \epsilon_0 E_\alpha) + \epsilon_{\alpha\beta\gamma}\partial_\beta M_\gamma \\ &\quad + \partial_\beta \left\langle \sum_n (p_{n\alpha} v_{n\beta} - p_{n\beta} v_{n\alpha})\delta_n \right\rangle \\ &\quad - \frac{1}{6}\partial_\beta\partial_\gamma \left\langle \sum_n (Q'^{(n)}_{\alpha\beta} v_{n\gamma} - Q'^{(n)}_{\gamma\beta} v_{n\alpha})\delta_n \right\rangle + \cdots .\end{aligned}$$

This is (6.96), with precisely (6.92), (6.97), and (6.98).

(b) For spatially uniform common $\mathbf{v}$, the two last lines are curls. Their effective magnetization is

$$\mathbf{M}_{\rm conv} = \mathbf{P}\times\mathbf{v} - (\nabla\cdot\overleftrightarrow{Q})\times\mathbf{v} = (\mathbf{D}-\epsilon_0\mathbf{E})\times\mathbf{v}, \qquad (\nabla\cdot Q)_\alpha = \partial_\beta Q_{\alpha\beta}.$$

Insert $\langle\mathbf{j}\rangle = \mathbf{J} + \partial_t(\mathbf{D}-\epsilon_0\mathbf{E}) + \nabla\times(\mathbf{M}+\mathbf{M}_{\rm conv})$ into the averaged microscopic Ampère equation. Rearrangement gives

$$\nabla\times\mathbf{H} = \mathbf{J} + \partial_t\mathbf{D}, \qquad \boxed{\frac{\mathbf{B}}{\mu_0} - \mathbf{H} = \mathbf{M} + (\mathbf{D}-\epsilon_0\mathbf{E})\times\mathbf{v}}.$$

If all internal motion and intrinsic moments vanish, $\mathbf{M}=0$; retaining it allows the intrinsic moments mentioned after (6.98).

6.8 A dielectric sphere of dielectric constant ϵ and radius a is located at the origin. There is a uniform applied electric field E_0 in the x direction. The sphere rotates with an angular velocity ω about the z axis. Show that there is a magnetic field $\mathbf{H} = -\nabla\Phi_M$, where

$$\Phi_M = \frac{3}{5}\left(\frac{\epsilon-\epsilon_0}{\epsilon+2\epsilon_0}\right)\epsilon_0 E_0\omega\left(\frac{a}{r_>}\right)^5 \cdot xz$$

where $r_>$ is the larger of r and a. The motion is nonrelativistic.

You may use the results of Section 4.4 for the dielectric sphere in an applied field.

Solution.

Work to first order in $\omega a/c$, assume a lossless isotropic dielectric with its induced polarization following the applied field, and neglect magnetic susceptibility. Section 4.4 gives

$$\mathbf{E}_{\rm in} = \frac{3\epsilon_0}{\epsilon+2\epsilon_0}E_0\hat{\mathbf{x}}, \qquad \mathbf{P} = P\hat{\mathbf{x}}, \qquad P = 3\epsilon_0\frac{\epsilon-\epsilon_0}{\epsilon+2\epsilon_0}E_0.$$

The velocity $\mathbf{v} = \omega(-y,x,0)$ therefore produces the effective magnetization, to dipole order in the molecular expansion,

$$\mathbf{M}_{\rm eff} = \mathbf{P}\times\mathbf{v} = P\omega x\hat{\mathbf{z}} \quad (r<a), \qquad \mathbf{M}_{\rm eff} = 0 \quad (r>a).$$

The macroscopic polarization is stationary in laboratory coordinates. There is no free current and $\partial_t\mathbf{D}=0$, so $\nabla\times\mathbf{H}=0$ and $\mathbf{H}=-\nabla\Phi_M$. Since $\nabla\cdot\mathbf{M}_{\rm eff}=0$ in the interior, the only magnetic scalar-potential source is the surface density

$$\sigma_M = \mathbf{M}_{\rm eff}\cdot\mathbf{n} = P\omega\frac{xz}{a} = P\omega a\, n_x n_z.$$

The angular function $n_x n_z$ is a pure $l = 2$ harmonic. The regular interior and decaying exterior solutions continuous at a can thus be written $\Phi_{M,\text{in}} = Cxz$ and $\Phi_{M,\text{out}} = Ca^5 xz/r^5$. At the surface,

$$H_{r,\text{out}} - H_{r,\text{in}} = -\partial_r \Phi_{M,\text{out}} + \partial_r \Phi_{M,\text{in}} = (3+2)Ca\, n_x n_z = \sigma_M.$$

Hence $C = P\omega/5$ and

$$\boxed{\Phi_M = \frac{3}{5}\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0}\epsilon_0 E_0 \omega \left(\frac{a}{r_>}\right)^5 xz.}$$

For example, $\mathbf{H}_{\text{in}} = -(P\omega/5)(z\hat{\mathbf{x}} + x\hat{\mathbf{z}})$. The induction is $\mu_0(\mathbf{H} + \mathbf{M}_{\text{eff}})$ inside and $\mu_0 \mathbf{H}$ outside; it has continuous normal component. The result vanishes for $\epsilon = \epsilon_0$ or $\omega = 0$, as required.

6.9 Discuss the conservation of energy and linear momentum for a macroscopic system of sources and electromagnetic fields in a uniform, isotropic medium described by a permittivity ϵ and a permeability μ. Show that in a straightforward calculation the energy density, Poynting vector, field-momentum density, and Maxwell stress tensor are given by the Minkowski expressions,

$$\begin{aligned} u &= \frac{1}{2}\left(\epsilon E^2 + \mu H^2\right) \\ \mathbf{S} &= \mathbf{E} \times \mathbf{H} \\ \mathbf{g} &= \mu\epsilon \mathbf{E} \times \mathbf{H} \\ T_{ij} &= \left[\epsilon E_i E_j + \mu H_i H_j - \tfrac{1}{2}\delta_{ij}(\epsilon E^2 + \mu H^2)\right] \end{aligned}$$

What modifications arise if ϵ and μ are functions of position?

Solution.

Assume instantaneous linear constitutive laws $\mathbf{D} = \epsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$, real positive time-independent ϵ, μ, and let $\rho, \mathbf{J}$ be the macroscopic free sources. These assumptions exclude temporal dispersion and absorption already represented by complex constitutive coefficients.

Dot $\nabla \times \mathbf{H} = \mathbf{J} + \partial_t \mathbf{D}$ with $\mathbf{E}$ and $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ with $\mathbf{H}$. Their difference, using the divergence of a cross product, gives

$$\partial_t\left(\tfrac{1}{2}\epsilon E^2 + \tfrac{1}{2}\mu H^2\right) + \nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{J} \cdot \mathbf{E}.$$

Thus u and $\mathbf{S}$ have exactly the stated forms. Mechanical work and heat receive $\mathbf{J} \cdot \mathbf{E}$, so the integrated total energy changes by $-\oint \mathbf{S} \cdot \mathbf{n}\, da$ for a fixed closed system.

For momentum, eliminate the free sources from $\mathbf{f} = \rho\mathbf{E} + \mathbf{J} \times \mathbf{B}$:

$$\begin{aligned} \mathbf{f} + \partial_t(\mathbf{D} \times \mathbf{B}) = \epsilon[\mathbf{E}(\nabla \cdot \mathbf{E}) - \mathbf{E} \times (\nabla \times \mathbf{E})] \\ + \mu[\mathbf{H}(\nabla \cdot \mathbf{H}) - \mathbf{H} \times (\nabla \times \mathbf{H})]. \end{aligned}$$

For any vector $\mathbf{F}$, $F_i \partial_j F_j - [\mathbf{F} \times (\nabla \times \mathbf{F})]_i = \partial_j(F_i F_j - \tfrac{1}{2}\delta_{ij} F^2)$. Consequently

$$\boxed{\partial_t g_i + f_i = \partial_j T_{ij}, \quad \mathbf{g} = \mathbf{D} \times \mathbf{B} = \epsilon\mu \mathbf{E} \times \mathbf{H}, \quad T_{ij} = \epsilon E_i E_j + \mu H_i H_j - \delta_{ij} u.}$$

The outward momentum flux is $-T_{ij} n_j$. Including the mechanical momentum delivered by $\mathbf{f}$ gives $d(\mathbf{P}_{\text{field}} + \mathbf{P}_{\text{mech}})/dt = \oint \overleftrightarrow{T} \cdot \mathbf{n}\, da$, when material does not cross the boundary.

For spatially varying but time-independent $\epsilon(\mathbf{x}), \mu(\mathbf{x})$, the energy calculation is unchanged. Differentiating the stress coefficients in the momentum calculation instead gives

$$\partial_j T_{ij} - \partial_t g_i = \rho E_i + (\mathbf{J} \times \mathbf{B})_i - \frac{1}{2}E^2 \partial_i \epsilon - \frac{1}{2}H^2 \partial_i \mu.$$

Thus the material force density acquires $-\frac{1}{2}E^2\nabla\epsilon - \frac{1}{2}H^2\nabla\mu$. At sharp interfaces one uses the traction jump of T rather than multiplying ambiguous one-sided field values by a delta function. This is the Minkowski macroscopic momentum partition requested here; it does not identify the free-source Lorentz term alone with every microscopic force in matter.

6.10 With the same assumptions as in Problem 6.9 discuss the conservation of angular momentum. Show that the differential and integral forms of the conservation law are

$$\frac{\partial}{\partial t}\left(\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}\right) + \nabla \cdot \overleftrightarrow{\mathbf{M}} = 0$$

and

$$\frac{d}{dt}\int_V \left(\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}\right) d^3x + \int_S \mathbf{n} \cdot \overleftrightarrow{\mathbf{M}}\, da = 0$$

where the field angular-momentum density is

$$\mathcal{L}_{\text{field}} = \mathbf{x} \times \mathbf{g} = \mu\epsilon\, \mathbf{x} \times (\mathbf{E} \times \mathbf{H})$$

and the flux of angular momentum is described by the tensor

$$\overleftrightarrow{\mathbf{M}} = \overleftrightarrow{\mathbf{T}} \times \mathbf{x}$$

Note: Here we have used the dyadic notation for M_{ij} and T_{ij}. A double-headed arrow conveys a fairly obvious meaning. For example, $\mathbf{n} \cdot \overleftrightarrow{\mathbf{M}}$ is a vector whose jth component is $\sum_i n_i M_{ij}$. The second-rank $\overleftrightarrow{\mathbf{M}}$ can be written as a third-rank tensor, $M_{ijk} = T_{ij}x_k - T_{ik}x_j$. But in the indices j and k it is antisymmetric and so has only three independent elements. Including the index i, M_{ijk} therefore has nine components and can be written as a pseudotensor of the second rank, as above.

Solution.

Use the stress and momentum of Problem 6.9, and write the matter's local rate of electromagnetic momentum acquisition as $\partial_t p_{\text{mech},i} = f_i$ (no material transport across the control volume). Then $\partial_t(p_{\text{mech},i} + g_i) = \partial_k T_{ki}$, where T is symmetric. Cross with $\mathbf{x}$ and define

$$\mathcal{L}_{\text{field},j} = \epsilon_{j\ell m} x_\ell g_m, \qquad M_{ij} = \epsilon_{jk\ell} T_{ik} x_\ell.$$

This component convention is precisely $\overleftrightarrow{M} = \overleftrightarrow{T} \times \mathbf{x}$. Its divergence is

$$\begin{aligned}\partial_i M_{ij} &= \epsilon_{jk\ell} x_\ell \partial_i T_{ik} + \epsilon_{jki} T_{ik} \\ &= -\epsilon_{j\ell k} x_\ell \partial_i T_{ik},\end{aligned}$$

because the contraction of a symmetric stress with an antisymmetric Levi-Civita symbol vanishes. Hence

$$\boxed{\partial_t(\boldsymbol{\mathcal{L}}_{\text{mech}} + \boldsymbol{\mathcal{L}}_{\text{field}}) + \nabla \cdot \overleftrightarrow{M} = 0, \qquad \boldsymbol{\mathcal{L}}_{\text{field}} = \epsilon\mu\, \mathbf{x} \times (\mathbf{E} \times \mathbf{H}).}$$

Integrating over a fixed V and applying the divergence theorem yields

$$\boxed{\frac{d}{dt}\int_V \left(\boldsymbol{\mathcal{L}}_{\text{mech}} + \boldsymbol{\mathcal{L}}_{\text{field}}\right) d^3x + \oint_S \mathbf{n} \cdot \overleftrightarrow{M}\, da = 0.}$$

Indeed $\mathbf{n} \cdot M = (\mathbf{n} \cdot T) \times \mathbf{x}$ is the outward angular-momentum flux, the negative of the torque traction. Spatially varying scalar ϵ, μ leave T symmetric; the additional material torque is $\mathbf{x} \times [-E^2\nabla\epsilon/2 - H^2\nabla\mu/2]$. If matter flows across the boundary or carries mechanical stress, its angular-momentum flux must also be included. The formula in the question uses the same fixed-matter convention as (6.110)–(6.122).

6.11 A transverse plane wave is incident normally in vacuum on a perfectly absorbing flat screen.

(a) From the law of conservation of linear momentum, show that the pressure (called radiation pressure) exerted on the screen is equal to the field energy per unit volume in the wave.

(b) In the neighborhood of the earth the flux of electromagnetic energy from the sun is approximately 1.4 kW/m^2. If an interplanetary "sailplane" had a sail of mass 1 g/m^2 of area and negligible other weight, what would be its maximum acceleration in meters per second squared due to the solar radiation pressure? How does this compare with the acceleration due to the solar "wind" (corpuscular radiation)?

Solution.

(a) Let the incident wave travel along $+z$. In vacuum, $B = E/c$, $u = \epsilon_0 E^2$, $S_z = cu$, $g_z = u/c$, and the stress has $T_{zz} = -u$. The momentum incident per unit area per unit time is $cg_z = u = S_z/c$. A perfectly absorbing screen has no outgoing momentum flux, so its time-averaged pressure is

$$\boxed{p_{\text{abs}} = \langle u \rangle = \mathcal{I}/c},$$

where $\mathcal{I} = \langle S_z \rangle$ is the intensity. A control-volume momentum balance across the screen gives the same result.

(b) For mass per area $\Sigma = 10^{-3}\ \text{kg}\,\text{m}^{-2}$ and normal incidence at the stated $\mathcal{I} = 1400\ \text{W}\,\text{m}^{-2}$,

$$p_{\text{abs}} = 4.670 \times 10^{-6}\ \text{Pa}, \qquad \boxed{a_{\text{abs}} = \frac{\mathcal{I}}{c\Sigma} = 4.670 \times 10^{-3}\ \text{m}\,\text{s}^{-2}.}$$

This is the maximum over orientations for the absorbing sail of part (a). If "maximum" also allows choosing its optical properties, a perfectly reflecting sail at normal incidence reverses the photon momentum and gives

$$\boxed{a_{\text{reflect}} = \frac{2\mathcal{I}}{c\Sigma} = 9.340 \times 10^{-3}\ \text{m}\,\text{s}^{-2}.}$$

These are the radiation contributions at the specified solar flux; no payload or solar-gravity subtraction is included.

A solar-wind comparison needs density and speed, neither specified in the problem. As an explicit representative estimate, take proton density $n_p = 5\ \text{cm}^{-3}$ and speed $v = 440\ \text{km}\,\text{s}^{-1}$, the values given by E. Moebius in NASA/GSFC's *Cosmicopia*, "Solar Wind Electricity?" (October 2006). [2] For complete absorption of the protons,

$$p_{\text{wind}} = n_p m_p v^2 = 1.619 \times 10^{-9}\ \text{Pa}, \qquad \boxed{a_{\text{wind}} = 1.619 \times 10^{-6}\ \text{m}\,\text{s}^{-2}.}$$

It is 3.47×10^{-4} of the absorbing-radiation acceleration (about 2.88×10^3 times smaller). Perfect reflection doubles the particle result too. Composition, solar activity, and the sail's particle stopping fraction alter this estimate; the specified numbers make the comparison reproducible.

[2] `https://science.gsfc.nasa.gov/attic/cosmicopia.gsfc.nasa.gov/qa_sun.html`.

6.12 Consider the definition of the admittance $Y = G - iB$ of a two-terminal linear passive network in terms of field quantities by means of the complex Poynting theorem of Section 6.9.

(a) By considering the complex conjugate of (6.134) obtain general expressions for the conductance G and susceptance B for the general case including radiation loss.

(b) Show that at low frequencies the expressions equivalent to (6.139) and (6.140) are

$$G \simeq \frac{1}{|V_i|^2} \int_V \sigma |\mathbf{E}|^2 \, d^3x$$
$$B \simeq -\frac{4\omega}{|V_i|^2} \int_V (w_m - w_e) \, d^3x$$

Solution.

(a) Use the book's $e^{-i\omega t}$ convention. Define

$$C_J = \int_V \mathbf{E} \cdot \mathbf{J}^* \, d^3x, \quad \Delta W = \int_V (w_m - w_e) \, d^3x, \quad P_S = \int_{S - S_i} \tfrac{1}{2} (\mathbf{E} \times \mathbf{H}^*) \cdot \mathbf{n} \, da.$$

The energy densities may be complex when constitutive losses are included. Equation (6.136), obtained from (6.134) by isolating the input port, is

$$\tfrac{1}{2} I_i^* V_i = \tfrac{1}{2} C_J - 2i\omega \Delta W + P_S.$$

Complex conjugation is essential because $I_i = Y V_i$:

$$Y |V_i|^2 = C_J^* + 4i\omega \Delta W^* + 2P_S^*.$$

With $Y = G - iB$, taking real and imaginary parts yields the general formulas

$$\boxed{G} = \boxed{\frac{\operatorname{Re} C_J + 4\omega \operatorname{Im} \Delta W + 2 \operatorname{Re} P_S}{|V_i|^2}},$$
$$\boxed{B} = \boxed{\frac{\operatorname{Im} C_J - 4\omega \operatorname{Re} \Delta W + 2 \operatorname{Im} P_S}{|V_i|^2}}.$$

If the non-port boundary is in the outgoing radiation zone at infinity, P_S is real and nonnegative: its contribution $2P_S/|V_i|^2$ is the radiation conductance. On a finite boundary its imaginary part must also be retained. Material loss must be counted either in C_J or in the constitutive coefficients, without double counting.

(b) At low frequency neglect radiation and take the only appreciable dissipation to be $\mathbf{J} = \sigma \mathbf{E}$, with real σ. Then $C_J = \int \sigma |\mathbf{E}|^2$ is real and w_e, w_m are real to the retained accuracy. Thus

$$\boxed{G \simeq \frac{\int_V \sigma |\mathbf{E}|^2 \, d^3x}{|V_i|^2}, \qquad B \simeq -\frac{4\omega}{|V_i|^2} \int_V (w_m - w_e) \, d^3x.}$$

A capacitor gives $B = \omega C > 0$ and $Y = -i\omega C$, verifying the susceptance sign for this time convention.

6.13 A parallel plate capacitor is formed of two flat rectangular perfectly conducting sheets of dimensions a and b separated by a distance d small compared to a or b. Current is fed in and taken out uniformly along the adjacent edges of length b. With the input current and voltage defined at this end of the capacitor, calculate the input impedance or admittance using the field concepts of Section 6.9.

(a) Calculate the electric and magnetic fields in the capacitor correct to second order in powers of the frequency, but neglecting fringing fields.

(b) Show that the expansion of the reactance (6.140) in powers of the frequency to an appropriate order is the same as that obtained for a lumped circuit consisting of a capacitance $C = \epsilon_0 ab/d$ in series with an inductance $L = \mu_0 ad/3b$.

Solution.

Choose $x = 0$ at the input and $x = a$ at the far edge, the plate normal along z, and fields independent of y, z within the gap. Neglect fringing and use $e^{-i\omega t}$, $k = \omega/c$. Define voltage so that $V_i = dE_z(0)$ and input current $I_i = -bH_y(0)$; these are the passive port signs when positive plate charge produces positive E_z.

(a) Maxwell's curl equations reduce to

$$-E_z' = i\omega\mu_0 H_y, \qquad H_y' = -i\omega\epsilon_0 E_z, \qquad E_z'' + k^2 E_z = 0.$$

No current passes beyond $x = a$, so $H_y(a) = 0$ and $E_z'(a) = 0$. The exact one-dimensional solution normalized to the total input current is

$$E_z(x) = \frac{I_i k}{-i\omega\epsilon_0 b}\frac{\cos[k(a-x)]}{\sin ka}, \qquad H_y(x) = -\frac{I_i}{b}\frac{\sin[k(a-x)]}{\sin ka}.$$

All other components vanish in this approximation. Expanding the spatial profiles through relative order k^2 gives

$$\boxed{E_z(x)} = \boxed{\frac{I_i}{-i\omega\epsilon_0 ab}\left[1 + k^2\left(\frac{a^2}{6} - \frac{(a-x)^2}{2}\right) + O(k^4a^4)\right]},$$

$$\boxed{H_y(x)} = \boxed{-\frac{I_i}{b}\frac{a-x}{a}\left[1 + \frac{k^2}{6}\{a^2 - (a-x)^2\} + O(k^4a^4)\right]}.$$

These expressions also specify the absolute orders: at fixed current, E starts at ω^{-1} and H at ω^0. For fixed voltage, $E_z = (V_i/d)[1 + k^2\{a^2 - (a-x)^2\}/2 + O(k^4)]$ and $H_y = i\omega\epsilon_0(V_i/d)(a-x) + O(\omega^3)$.

(b) The average of $a^2/6 - (a-x)^2/2$ over the plate is zero. Explicitly,

$$\int_0^a \left[\frac{a^2}{6} - \frac{(a-x)^2}{2}\right] dx = \frac{a^3}{6} - \frac{1}{2}\frac{a^3}{3} = 0, \qquad \int_0^a \left(\frac{a-x}{a}\right)^2 dx = \frac{a}{3}.$$

Therefore, at fixed I_i, the energy integrals needed through the inductive term in X are

$$W_e = \frac{\epsilon_0 bd}{4}\int_0^a |E_z|^2 dx = \frac{|I_i|^2 d}{4\epsilon_0 ab\omega^2}[1 + O(k^4a^4)],$$

$$W_m = \frac{\mu_0 bd}{4}\int_0^a |H_y|^2 dx = \frac{\mu_0 |I_i|^2 ad}{12b}[1 + O(k^2a^2)].$$

Equation (6.140) then gives

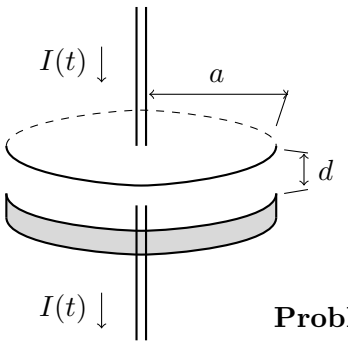
$$X = \frac{4\omega(W_m - W_e)}{|I_i|^2} = -\frac{1}{\omega C} + \omega L + O(\omega^3), \qquad \boxed{C = \frac{\epsilon_0 ab}{d}, \qquad L = \frac{\mu_0 ad}{3b}}.$$

The port solution provides an independent exact check:

$$Z = \frac{V_i}{I_i} = i\sqrt{\frac{\mu_0}{\epsilon_0}}\frac{d}{b}\cot ka = \frac{i}{\omega C} - i\omega L + O(\omega^3), \qquad Y = Z^{-1}.$$

Thus $Z = -iX$ is the stated lossless series LC impedance, within the small-ka and negligible-fringing approximations.

6.14 An ideal circular parallel plate capacitor of radius a and plate separation $d \ll a$ is connected to a current source by axial leads, as shown in the sketch. The current in the wire is $I(t) = I_0 \cos \omega t$.



Problem 6.14

(a) Calculate the electric and magnetic fields between the plates to second order in powers of the frequency (or wave number), neglecting the effects of fringing fields.

(b) Calculate the volume integrals of w_e and w_m that enter the definition of the reactance X, (6.140), through second relative order in $\omega a/c$. (For absolute order ω^2 at fixed current, the electric energy also needs the relative $(\omega a/c)^4/192$ correction derived below.) Show that in terms of the input current I_i, defined by $I_i = -i\omega Q$, where Q is the *total charge* on one plate, these energies are

$$\int w_e \, d^3x = \frac{1}{4\pi\epsilon_0}\frac{|I_i|^2 d}{\omega^2 a^2}, \qquad \int w_m \, d^3x = \frac{\mu_0}{4\pi}\frac{|I_i|^2 d}{8}\left(1 + \frac{\omega^2 a^2}{12c^2}\right)$$

(c) Show that the equivalent series circuit has $C \simeq \pi\epsilon_0 a^2/d$, $L \simeq \mu_0 d/8\pi$, and that an estimate for the resonant frequency of the system is $\omega_{\rm res} \simeq 2\sqrt{2}\, c/a$. Compare with the first root of $J_0(x)$.

Solution.

Use cylindrical radius s and the phasor convention $e^{-i\omega t}$. Let $Q = \epsilon_0 \int E_z \, da$ be the signed plate charge and $I_i = -i\omega Q$. The axial leads end on their respective plates; there is no conducting connection through the gap (as in the source sketch on printed p. 289).

(a) Neglecting fringing, the fields are $E_z(s), H_\phi(s)$. The curl equations give

$$-\frac{dE_z}{ds} = i\omega\mu_0 H_\phi, \qquad \frac{1}{s}\frac{d}{ds}(sH_\phi) = -i\omega\epsilon_0 E_z, \qquad \frac{1}{s}(sE_z')' + k^2 E_z = 0.$$

Regularity at $s = 0$ selects $E_z = AJ_0(ks)$. Since $\int_0^a sJ_0(ks)ds = aJ_1(ka)/k$, charge normalization fixes

$$\boxed{E_z(s) = \frac{I_i k}{-i\omega\, 2\pi\epsilon_0 a J_1(ka)} J_0(ks), \qquad H_\phi(s) = \frac{I_i}{2\pi a J_1(ka)} J_1(ks).}$$

Here $B_\phi = \mu_0 H_\phi$, and the actual fields are the real parts of the phasors times $e^{-i\omega t}$. For $ka \ll 1$,

$$E_z = \frac{I_i}{-i\omega\pi\epsilon_0 a^2}\left[1 + k^2\left(\frac{a^2}{8} - \frac{s^2}{4}\right) + O(k^4a^4)\right],$$
$$H_\phi = \frac{I_i s}{2\pi a^2}\left[1 + \frac{k^2}{8}(a^2 - s^2) + O(k^4a^4)\right].$$

In particular $2\pi a H_\phi(a) = I_i$ exactly; the field at the rim contains the total displacement current.

(b) With $w_e = \epsilon_0|E|^2/4$ and $w_m = \mu_0|H|^2/4$, $\langle s^2\rangle_{\text{disk}} = a^2/2$ makes the relative k^2 correction to W_e vanish. For the magnetic integral,

$$\int_0^a s^3\left[1 + \frac{k^2}{4}(a^2 - s^2)\right] ds = \frac{a^4}{4}\left(1 + \frac{k^2a^2}{12}\right).$$

Thus the two displayed leading-order results in the question are

$$W_e = \frac{|I_i|^2 d}{4\pi\epsilon_0\omega^2 a^2}[1 + O((ka)^4)], \qquad W_m = \frac{\mu_0|I_i|^2 d}{32\pi}\left[1 + \frac{(ka)^2}{12} + O((ka)^4)\right].$$

There is an ordering subtlety in the wording "to second order in ω": at fixed I_i, the leading electric energy is $O(\omega^{-2})$. For its *absolute* $O(\omega^2)$ term one must retain the relative k^4 correction. The exact Bessel solution settles this without extending the approximate Maxwell calculation. Writing $z = ka$ and taking Bessel functions below at z,

$$\int_0^a sJ_\nu(ks)^2\, ds = \frac{a^2}{2}\left[J_\nu(z)^2 - J_{\nu-1}(z)J_{\nu+1}(z)\right].$$

For $\nu = 0$, $J_{-1} = -J_1$ makes the bracket $J_0^2 + J_1^2$; for $\nu = 1$ it is $J_1^2 - J_0J_2$. The identity follows by differentiating its right-hand side, using $2J'_\nu = J_{\nu-1} - J_{\nu+1}$ and $2\nu J_\nu/z = J_{\nu-1} + J_{\nu+1}$. Substitution of the exact field amplitudes then gives

$$W_e = \frac{|I_i|^2 d}{4\pi\epsilon_0\omega^2 a^2}\frac{z^2(J_0^2 + J_1^2)}{4J_1^2} = \frac{|I_i|^2 d}{4\pi\epsilon_0\omega^2 a^2}\left[1 + \frac{z^4}{192} + O(z^6)\right],$$
$$W_m = \frac{\mu_0|I_i|^2 d}{16\pi}\left[1 - \frac{J_0J_2}{J_1^2}\right].$$

The $z^4/192$ term is absent from the book's displayed W_e; it is necessary if both energies are demanded through absolute order ω^2, but it does not affect the requested leading L, C.

(c) From $X = 4\omega(W_m - W_e)/|I_i|^2$,

$$X = -\frac{1}{\omega C} + \omega L + O(\omega^3), \qquad \boxed{C = \frac{\pi\epsilon_0 a^2}{d}, \quad L = \frac{\mu_0 d}{8\pi}, \quad \omega_{LC} = \frac{1}{\sqrt{LC}} = 2\sqrt{2}\,\frac{c}{a}.}$$

The exact energy difference, using $J_0 + J_2 = 2J_1/z$, gives

$$X = -\frac{\mu_0\omega d}{2\pi}\frac{J_0(z)}{zJ_1(z)}.$$

Its first zero is $z = j_{0,1} = 2.4048255577$, whereas $2\sqrt{2} = 2.8284271247$: the lumped estimate is 17.61% too high. This is expected because its putative resonance has ka of order one, outside the small-frequency expansion. Both results neglect edge and lead effects; they describe the idealized gap field model.

6.15 If a conductor or semiconductor has current flowing in it because of an applied electric field, and a transverse magnetic field is applied, there develops a component of electric field in the direction orthogonal to both the applied electric field (direction of current flow) and the magnetic field, resulting in a voltage difference between the sides of the conductor. This phenomenon is known as the *Hall effect*.

(a) Use the known properties of electromagnetic fields under rotations and spatial reflections and the assumption of Taylor series expansions around zero magnetic field strength to show that for an isotropic medium the generalization of Ohm's law, correct to second order in the magnetic field, must have the form

$$\mathbf{E} = \rho_0 \mathbf{J} + R(\mathbf{H} \times \mathbf{J}) + \beta_1 H^2 \mathbf{J} + \beta_2 (\mathbf{H} \cdot \mathbf{J})\mathbf{H}$$

where ρ_0 is the resistivity in the absence of the magnetic field and R is called the Hall coefficient.

(b) What about the requirements of time reversal invariance?

Solution.

(a) Linear response to current means $E_i = \rho_{ij}(\mathbf{H}) J_j$. For an isotropic medium the rank-two tensors that can be formed through second order in the axial vector $\mathbf{H}$ are

$$\rho_{ij} = \rho_0 \delta_{ij} + R\epsilon_{ikj} H_k + \beta_1 H_k H_k \delta_{ij} + \beta_2 H_i H_j + O(H^3).$$

To see completeness, the zero-order isotropic tensor is δ_{ij}; the only rank-three isotropic pseudotensor is ϵ_{ikj}; and the three rank-four delta products, contracted with $H_k H_l$, reduce to $H^2\delta_{ij}$ and $H_i H_j$. Under inversion $\mathbf{E}, \mathbf{J}$ are polar and $\mathbf{H}$ is axial, so all these terms transform as required. Consequently

$$\boxed{\mathbf{E} = \rho_0 \mathbf{J} + R(\mathbf{H} \times \mathbf{J}) + \beta_1 H^2 \mathbf{J} + \beta_2 (\mathbf{H} \cdot \mathbf{J})\mathbf{H} + O(H^3 J).}$$

The Hall part is antisymmetric and does no Joule work:

$$\mathbf{J} \cdot \mathbf{E} = (\rho_0 + \beta_1 H^2) J_\perp^2 + [\rho_0 + (\beta_1 + \beta_2) H^2] J_\parallel^2.$$

Passivity requires these two coefficients to be nonnegative over the field range in which the expansion is used.

(b) Under time reversal $\mathbf{E}$ is even, whereas $\mathbf{J}$ and $\mathbf{H}$ are odd. Thus the Hall term is even, as required for a reversible force balance, while the longitudinal dissipative terms are odd. A resistive macroscopic law with a fixed sign of dissipation is not itself invariant under reversal of a nonequilibrium history: heat flow and the reservoir state must be reversed too. It would therefore be incorrect to set ρ_0, β_1, β_2 to zero by applying time reversal directly to Ohm's dissipative law.

Microscopic time-reversal symmetry in equilibrium instead implies the Onsager–Casimir condition $\rho_{ij}(\mathbf{H}) = \rho_{ji}(-\mathbf{H})$, provided the magnetic field is the only time-reversal-odd bias and there is no spontaneous magnetic order. The even-in-H symmetric terms and the odd-in-H antisymmetric Hall term above all satisfy this condition. It imposes no further relation among the four coefficients.

6.16

(a) Calculate the force in newtons acting on a Dirac monopole of the minimum magnetic charge located a distance 0.5 Å from and in the median plane of a magnetic dipole with dipole moment equal to one nuclear magneton ($e\hbar/2m_p$).

(b) Compare the force in part a with atomic forces such as the direct electrostatic force between charges (at the same separation), the spin-orbit force, the hyperfine interaction. Comment on the question of binding of magnetic monopoles to nuclei with magnetic moments. Assume that the monopole mass is at least that of a proton.

Reference: D. Sivers, *Phys. Rev.* **D2**, 2048 (1970).

Solution.

(a) In the SI convention of (6.153), $eg_D = 2\pi\hbar$ and the static force on a monopole is $g_D\mathbf{H} = g_D\mathbf{B}/\mu_0$. For $\mathbf{m} = \mu_N\hat{\mathbf{z}}$, $\mu_N = e\hbar/(2m_p)$, the *median-plane* dipole field is $\mathbf{B} = -\mu_0\mu_N\hat{\mathbf{z}}/(4\pi r^3)$. It is half the magnitude of the axial dipole field. Therefore

$$\mathbf{F} = -\frac{g_D\mu_N}{4\pi r^3}\hat{\mathbf{z}} = -\frac{\hbar^2}{4m_p r^3}\hat{\mathbf{z}}, \qquad \boxed{|\mathbf{F}| = 1.330 \times 10^{-11}\ \text{N} \quad (r = 0.5\ \text{Å}).}$$

Here $g_D = h/e = 4.13567 \times 10^{-15}$ Wb; a negative monopole reverses the force.

(b) The Coulomb force at the same separation is

$$F_C = \frac{e^2}{4\pi\epsilon_0 r^2} = 9.228 \times 10^{-8}\ \text{N}, \qquad \frac{|F|}{F_C} = \frac{1}{4}\frac{m_e}{m_p}\frac{a_0}{r} = 1.441 \times 10^{-4}.$$

Thus the monopole force is about 6940 times smaller than the direct atomic electric force. Atomic spin-orbit forces have the characteristic relativistic scale $\alpha^2 F_C \simeq 4.91 \times 10^{-12}$ N at $r \sim a_0$ (angular and state-dependent coefficients are not specified). The monopole force is about 2.7 times this scale. For a concrete hyperfine comparison, two magnetic moments of magnitudes μ_B and μ_N have dipolar energy scale $\mu_0\mu_B\mu_N/(4\pi r^3)$ and radial force scale

$$F_{\text{hf}} \sim \frac{3\mu_0\mu_B\mu_N}{4\pi r^4} = 2.25 \times 10^{-15}\ \text{N}.$$

The monopole force is roughly 5900 times this scale. These last two comparisons are scales, not a claim that every atomic level has a nonzero spin-orbit or hyperfine force.

The dipole–monopole potential is anisotropic:

$$U(r,\theta) = \frac{g_D\mu_N}{4\pi r^2}\cos\theta = \frac{\hbar^2}{4m_p r^2}\cos\theta.$$

Here $\mathbf{r}$ points from the fixed dipole to the monopole. The energy is negative on the $\cos\theta < 0$ side and becomes strong at short distance. Its characteristic magnitude at the stated atomic distance is $|U|_{\text{scale}} = 4.15 \times 10^{-3}$ eV, but at a nuclear cutoff r_N the same point-dipole estimate grows as r_N^{-2}. A localization kinetic-energy scale for relative mass m_{red} is $\hbar^2/(2m_{\text{red}}r^2)$, so their ratio for a moment of one nuclear magneton is $m_{\text{red}}/(2m_p)$. A heavy monopole and a heavy nucleus can therefore make the attraction competitive with localization energy; electron binding is much less favored by its small reduced mass. Binding to suitable magnetic nuclei is consequently plausible despite the weak force at atomic distances. A universal binding energy or a proof of binding for every nucleus does not follow from this estimate: spin, electric charge–monopole angular momentum, the actual nuclear moment, and the finite nuclear core are required. The mass assumption alone does not supply those data. In particular the attractive inverse square potential has no intrinsic length scale, so a physical short-distance model is indispensable for a binding energy.

6.17

(a) For a particle possessing both electric and magnetic charges, show that the generalization of the Lorentz force in vacuum is

$$\mathbf{F} = q_{\mathrm{e}}\mathbf{E} + q_{\mathrm{m}}\mathbf{B}/\mu_0 + q_{\mathrm{e}}\mathbf{v}\times\mathbf{B} - q_{\mathrm{m}}\mathbf{v}\times\epsilon_0\mathbf{E}$$

(b) Show that this expression for the force is invariant under a duality transformation of both fields and charges, (6.151) and (6.152).

(c) Show that the Dirac quantization condition, (6.153), is generalized for two particles possessing electric and magnetic charges e_1, g_1 and e_2, g_2, respectively, to

$$\frac{e_1 g_2 - e_2 g_1}{\hbar} = 2\pi n$$

and that the relation is invariant under a duality transformation of the charges.

Solution.

(a) Duality maps the electric force $q_e(\mathbf{E}+\mathbf{v}\times\mathbf{B})$ to a magnetic-charge force. In vacuum put $Z_0 = \sqrt{\mu_0/\epsilon_0} = \mu_0 c$, and use a quarter-turn of the normalized field pair $(\mathbf{E}, c\mathbf{B})$. The corresponding magnetic force is $q_m(\mathbf{B}/\mu_0 - \epsilon_0\mathbf{v}\times\mathbf{E})$. Superposition therefore gives

$$\boxed{\mathbf{F} = q_e\mathbf{E} + q_m\mathbf{B}/\mu_0 + q_e\mathbf{v}\times\mathbf{B} - q_m\epsilon_0\mathbf{v}\times\mathbf{E}.}$$

This also has the required power $\mathbf{F}\cdot\mathbf{v} = q_e\mathbf{E}\cdot\mathbf{v} + q_m\mathbf{H}\cdot\mathbf{v}$; both velocity cross-product terms do no work.

(b) Express (6.151)–(6.152) as a common rotation by an arbitrary angle χ (changing χ to $-\chi$ exchanges the book's primed/unprimed convention):

$$\begin{pmatrix}\mathbf{E}'\\ c\mathbf{B}'\end{pmatrix} = \begin{pmatrix}\cos\chi & \sin\chi\\ -\sin\chi & \cos\chi\end{pmatrix}\begin{pmatrix}\mathbf{E}\\ c\mathbf{B}\end{pmatrix}, \qquad \begin{pmatrix}q_e'\\ q_m'/Z_0\end{pmatrix} = \begin{pmatrix}\cos\chi & \sin\chi\\ -\sin\chi & \cos\chi\end{pmatrix}\begin{pmatrix}q_e\\ q_m/Z_0\end{pmatrix}.$$

With $a = q_e$, $b = q_m/Z_0$, and $\mathbf{F}_1 = \mathbf{E}$, $\mathbf{F}_2 = c\mathbf{B}$, the force is

$$\mathbf{F} = a\mathbf{F}_1 + b\mathbf{F}_2 + \frac{\mathbf{v}}{c}\times(a\mathbf{F}_2 - b\mathbf{F}_1).$$

The first combination is the scalar product of two pairs; the second is their antisymmetric product. Both are preserved by a common rotation of determinant one, proving force invariance including all factors of c, Z_0.

(c) Rotate particle 1 to a purely electric charge. If it is nonzero, choose

$$e_1' = \sqrt{e_1^2 + (g_1/Z_0)^2}, \qquad g_1' = 0.$$

The determinant of the two charge vectors is invariant, so $e_1' g_2' = e_1 g_2 - e_2 g_1$. Applying the electric–magnetic Dirac condition in that frame gives

$$\boxed{e_1 g_2 - e_2 g_1 = 2\pi n\hbar, \qquad n\in\mathbb{Z}.}$$

If particle 1 carries neither charge the left side vanishes and $n = 0$. Equivalently, the two-source field angular momentum has coefficient $(e_1 g_2 - e_2 g_1)/(4\pi)$, whose allowed half-integral multiples of $\hbar$ give the same condition. The determinant argument proves its invariance under every common duality transformation.

6.18 Consider the Dirac expression

$$\mathbf{A}(\mathbf{x}) = \frac{g}{4\pi}\int_L \frac{d\mathbf{l}' \times (\mathbf{x}-\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|^3}$$

for the vector potential of a magnetic monopole and its associated string L. Suppose for definiteness that the monopole is located at the origin and the string along the negative z axis.

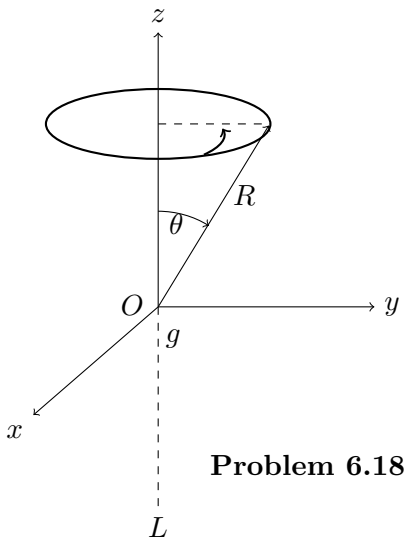
(a) Calculate $\mathbf{A}$ explicitly and show that in spherical coordinates it has components

$$A_r = 0, \qquad A_\theta = 0, \qquad A_\phi = \frac{g(1-\cos\theta)}{4\pi r \sin\theta} = \left(\frac{g}{4\pi r}\right)\tan\left(\frac{\theta}{2}\right)$$

(b) Verify that $\mathbf{B} = \nabla \times \mathbf{A}$ is the Coulomb-like field of a point charge, except perhaps at $\theta = \pi$.

(c) With the $\mathbf{B}$ determined in part b, evaluate the total magnetic flux passing through the circular loop of radius $R\sin\theta$ shown in the figure. Consider $\theta < \pi/2$ and $\theta > \pi/2$ separately, but always calculate the upward flux.

(d) From $\oint \mathbf{A}\cdot d\mathbf{l}$ around the loop, determine the total magnetic flux through the loop. Compare the result with that found in part c. Show that they are equal for $0 < \theta < \pi/2$, but have a *constant* difference for $\pi/2 < \theta < \pi$. Interpret this difference.



Problem 6.18

Solution.

(a) Parameterize the string by $\mathbf{x}' = z'\hat{\mathbf{z}}$, $-\infty < z' \le 0$, with $d\mathbf{l}' = \hat{\mathbf{z}}\,dz'$. Its numerator is $d\mathbf{l}' \times (\mathbf{x}-\mathbf{x}') = s\hat{\boldsymbol{\phi}}\,dz'$, where $s = r\sin\theta$. Thus

$$\begin{aligned} A_\phi &= \frac{g}{4\pi}\int_{-\infty}^{0} \frac{s\,dz'}{[s^2+(z-z')^2]^{3/2}} \\ &= \frac{g}{4\pi s}\left(1 - \frac{z}{\sqrt{s^2+z^2}}\right) = \boxed{\frac{g}{4\pi r}\frac{1-\cos\theta}{\sin\theta} = \frac{g}{4\pi r}\tan\frac{\theta}{2}}. \end{aligned}$$

The other components vanish. This potential is regular on the positive axis and singular on the negative axis.

(b) Away from that string,

$$B_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\phi) = \frac{g}{4\pi r^2}, \qquad B_\theta = -\frac{1}{r}\partial_r(rA_\phi) = 0, \qquad B_\phi = 0.$$

Thus the ordinary curl is the monopole field $\boxed{\mathbf{B} = g\hat{\mathbf{r}}/(4\pi r^2)}$ off the string and origin.

(c) The horizontal disk lies at $z_0 = R\cos\theta$, with radius $s_0 = R\sin\theta$. For $z_0 \neq 0$ its upward monopole flux is

$$\Phi^{\uparrow}_{\text{mono}} = \frac{g}{2}\int_0^{s_0} \frac{z_0 s\, ds}{(s^2+z_0^2)^{3/2}} = \frac{g}{2}\left[\text{sgn}(z_0) - \frac{z_0}{R}\right]$$
$$= \boxed{\begin{cases} \frac{g}{2}(1-\cos\theta), & 0<\theta<\pi/2, \\ -\frac{g}{2}(1+\cos\theta), & \pi/2<\theta<\pi. \end{cases}}$$

The disk at $\theta = \pi/2$ passes through the monopole and requires a singular-surface prescription; it is excluded from these two cases.

(d) Counterclockwise circulation viewed from above gives

$$\oint \mathbf{A}\cdot d\mathbf{l} = 2\pi R\sin\theta\, A_\phi(R,\theta) = \boxed{\frac{g}{2}(1-\cos\theta)}.$$

It equals the flux in part (c) for the upper disk, but exceeds that flux by g for the lower disk. The distributional curl includes a return-flux string:

$$\nabla\times\mathbf{A} = \frac{g}{4\pi r^2}\hat{\mathbf{r}} + g\delta(x)\delta(y)\Theta(-z)\hat{\mathbf{z}}.$$

Its upward flux through every disk cutting the negative axis is g. Its divergence is $-g\delta^{(3)}(\mathbf{x})$, which cancels the monopole field's divergence, as every globally defined distributional curl must. Excluding the string gives the isolated monopole field; retaining it restores Stokes's theorem for disks crossing the string.

6.19

(a) Apply space inversion to the monopole vector potential of Problem 6.18 and show that the vector potential becomes

$$A'_\phi = -g\frac{(1+\cos\theta)}{4\pi r\,\sin\theta} = -\frac{g}{4\pi r}\cot\left(\frac{\theta}{2}\right)$$

with the other components vanishing. Show explicitly that its curl gives the magnetic field of a magnetic monopole, except perhaps at $\theta = 0$. [Remember the space-inversion properties of the magnetic charge!]

(b) Show that the difference, $\delta\mathbf{A} = \mathbf{A}' - \mathbf{A}$, can be expressed as the gradient of a scalar function, indicating that the original and space-inverted vector potentials differ by a gauge transformation.

(c) Interpret the gauge function in terms of Fig. 6.9. [*Hint*: Choose the contour C to be a rectangle lying in a plane containing the z axis, with three sides at infinity.]

Solution.

(a) A vector potential is a polar vector, so under active parity $\mathbf{A}^P(\mathbf{x}) = -\mathbf{A}(-\mathbf{x})$, whereas magnetic charge is a pseudoscalar, $g^P = -g$. At $-\mathbf{x}$, the angles are $(\pi - \theta, \phi + \pi)$ and $\hat{\boldsymbol{\phi}}(\phi + \pi) = -\hat{\boldsymbol{\phi}}(\phi)$. Applying these facts to 6.18 gives

$$\mathbf{A}^P(\mathbf{x}) = \frac{g(1+\cos\theta)}{4\pi r \sin\theta}\hat{\boldsymbol{\phi}} = -\frac{g^P(1+\cos\theta)}{4\pi r\sin\theta}\hat{\boldsymbol{\phi}}.$$

Now denote the *transformed* monopole charge by g, as in the question. The potential of that charge with its string on the positive axis is

$$\boxed{A'_\phi = -\frac{g(1+\cos\theta)}{4\pi r\sin\theta} = -\frac{g}{4\pi r}\cot\frac{\theta}{2}, \qquad A'_r = A'_\theta = 0.}$$

Direct differentiation gives $B'_r = (r\sin\theta)^{-1}\partial_\theta(\sin\theta A'_\phi) = g/(4\pi r^2)$ and $B'_\theta = B'_\phi = 0$ away from the positive-axis string. Keeping the original pre-parity charge label instead gives a monopole of charge $-g$, consistently with its pseudoscalar character.

(b) Comparing potentials of the *same* charge g on their common nonsingular domain,

$$\delta\mathbf{A} = -\frac{g}{2\pi r\sin\theta}\hat{\boldsymbol{\phi}} = \nabla\Lambda, \qquad \boxed{\Lambda = -\frac{g}{2\pi}\phi + \text{constant}.}$$

Since $\nabla\phi = \hat{\boldsymbol{\phi}}/(r\sin\theta)$, this is a gauge transformation on a patch with a chosen azimuthal cut. The scalar is not globally single-valued: $\phi \mapsto \phi + 2\pi$ changes it by $-g$. The quantum phase $\exp(ie\Lambda/\hbar)$ is nevertheless single-valued when $eg = 2\pi n\hbar$.

(c) In Fig. 6.9 the contour is $C = L' - L$. For strings on opposite halves of the z axis, close this contour with three sides of a rectangle at infinity in a meridian plane. Its spanning surface becomes a half-plane ending on the axis. On a unit sphere centered at the observer, a change $d\phi$ sweeps out a spherical lune. Its unsigned area is

$$\int_\phi^{\phi+d\phi} d\varphi \int_0^\pi \sin\vartheta\, d\vartheta = 2\, d\phi.$$

The contour orientation $C = L' - L$ assigns the negative sign, so the oriented solid angle changes by $-2\, d\phi$. Thus, on a chosen branch,

$$\Omega_C = -2\phi + \text{constant}, \qquad \Lambda = \frac{g}{4\pi}\Omega_C.$$

This is exactly Jackson's (6.162). Crossing the half-plane changes the solid-angle branch by 4π, or the gauge function by g in magnitude; that jump represents relocation of the unobservable string.

6.20 An example of the preservation of causality and finite speed of propagation in spite of the use of the Coulomb gauge is afforded by a dipole source that is flashed on and off at $t = 0$. The effective charge and current densities are

$$\rho(\mathbf{x}, t) = \delta(x)\delta(y)\delta'(z)\delta(t)$$
$$J_z(\mathbf{x}, t) = -\delta(x)\delta(y)\delta(z)\delta'(t)$$

where a prime means differentiation with respect to the argument. This dipole is of unit strength and it points in the negative z direction.

(a) Show that the instantaneous Coulomb potential (6.23) is

$$\Phi(\mathbf{x},t) = -\frac{1}{4\pi\epsilon_0}\,\delta(t)\,\frac{z}{r^3}$$

(b) Show that the *transverse* current $\mathbf{J}_t$ is

$$\mathbf{J}_t(\mathbf{x},t) = -\delta'(t)\left[\tfrac{2}{3}\boldsymbol{\epsilon}_3\delta(\mathbf{x}) - \frac{\boldsymbol{\epsilon}_3}{4\pi r^3} + \frac{3}{4\pi r^3}\,\mathbf{n}(\boldsymbol{\epsilon}_3\cdot\mathbf{n})\right]$$

where the factor of 2/3 multiplying the delta function comes from treating the gradient of z/r^3 according to (4.20).

(c) Show that the electric and magnetic fields are causal and that the electric field components are

$$E_x(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\frac{c}{r}\left[-\delta''(r-ct) + \frac{3}{r}\delta'(r-ct) - \frac{3}{r^2}\delta(r-ct)\right]\sin\theta\cos\theta\cos\phi$$

E_y is the same as E_x, with $\cos\phi$ replaced by $\sin\phi$, and

$$E_z(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\frac{c}{r}\left[\sin^2\theta\,\delta''(r-ct) + (3\cos^2\theta - 1)\cdot\left(\frac{\delta'(r-ct)}{r} - \frac{\delta(r-ct)}{r^2}\right)\right]$$

Hint: While the answer in part b displays the transverse current explicitly, the less explicit form

$$\mathbf{J}_t(\mathbf{x},t) = -\delta'(t)\left[\boldsymbol{\epsilon}_3\delta(\mathbf{x}) + \frac{1}{4\pi}\nabla\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right]$$

can be used with (6.47) to calculate the vector potential and the fields for part c. An alternative method is to use the Fourier transforms in time of $\mathbf{J}_t$ and $\mathbf{A}$, the Green function (6.40) and its spherical wave expansion from Chapter 9.

Solution.

All fields below are distributions, and the explicit spherical formulas are for $r > 0$. Let $K = 1/(4\pi\epsilon_0)$ and $\mathbf{e}_z = \boldsymbol{\epsilon}_3$.

(a) Integrating the derivative of the source delta function gives

$$\Phi_C = K\delta(t)\int\frac{\delta(x')\delta(y')\delta'(z')}{|\mathbf{x}-\mathbf{x}'|}\,d^3x' = -K\delta(t)\,\partial_{z'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\bigg|_0 = \boxed{-K\delta(t)\frac{z}{r^3}}.$$

(b) Charge conservation and $\nabla^2\Phi_C = -\rho/\epsilon_0$ show that the longitudinal current is $\mathbf{J}_\ell = \epsilon_0\nabla\partial_t\Phi_C$. Consequently

$$\mathbf{J}_t = \mathbf{J} - \mathbf{J}_\ell = -\delta'(t)\left[\mathbf{e}_z\delta^{(3)}(\mathbf{x}) + \frac{1}{4\pi}\nabla\partial_z\frac{1}{r}\right].$$

The distributional derivative checked in (4.20) is

$$\partial_i\partial_j\frac{1}{r} = \frac{3n_in_j - \delta_{ij}}{r^3} - \frac{4\pi}{3}\delta_{ij}\delta^{(3)}(\mathbf{x}),$$

where the r^{-3} expression uses spherical principal-value exclusion. It changes the contact term from 1 to 2/3 and yields

$$\boxed{\mathbf{J}_t = -\delta'(t)\left[\frac{2}{3}\mathbf{e}_z\delta^{(3)}(\mathbf{x}) - \frac{\mathbf{e}_z}{4\pi r^3} + \frac{3\mathbf{n}n_z}{4\pi r^3}\right].}$$

Taking a divergence of the less explicit form gives zero, since $\nabla^2(1/r) = -4\pi\delta^{(3)}(\mathbf{x})$.

(c) One can display the instantaneous cancellation without integrating the spatially nonlocal $\mathbf{J}_t$ directly. The source is an electric dipole $\mathbf{p}(t) = -\mathbf{e}_z\delta(t)$. Applying the retarded potential integral (6.47) to the source deltas gives the Lorenz-gauge potentials, with $\tau = t - r/c$,

$$\Phi_L = -Kn_z\left[\frac{\delta(\tau)}{r^2} + \frac{\delta'(\tau)}{cr}\right], \qquad \mathbf{A}_L = -\frac{K}{c^2 r}\mathbf{e}_z\delta'(\tau).$$

For example, the second term in Φ_L is the derivative of the retarded time when the source-coordinate derivative acts on the delta function. Define

$$\chi = Kn_z\left[\frac{\Theta(t) - \Theta(\tau)}{r^2} - \frac{\delta(\tau)}{cr}\right], \qquad \mathbf{A}_C = \mathbf{A}_L + \nabla\chi.$$

Direct differentiation gives $\partial_t\chi = \Phi_L - \Phi_C$. For the spatial check write $\chi = Kn_zF$, $H = \Theta(t) - \Theta(\tau)$, and $F = H/r^2 - \delta(\tau)/(cr)$. Using $\partial_r H = \delta(\tau)/c$ gives

$$F' = -\frac{2H}{r^3} + \frac{2\delta(\tau)}{cr^2} + \frac{\delta'(\tau)}{c^2 r}, \qquad F'' + \frac{2F'}{r} - \frac{2F}{r^2} = -\frac{\delta'(\tau)}{c^2r^2} - \frac{\delta''(\tau)}{c^3 r}.$$

The last expression times Kn_z is $-\nabla\cdot\mathbf{A}_L$. Thus $\nabla^2\chi = -\nabla\cdot\mathbf{A}_L$, so $\mathbf{A}_C$ is the Coulomb-gauge potential; both statements include the source distributions. Therefore

$$-\nabla\Phi_C - \partial_t\mathbf{A}_C = -\nabla\Phi_C - \partial_t\mathbf{A}_L - \nabla(\Phi_L - \Phi_C) = -\nabla\Phi_L - \partial_t\mathbf{A}_L.$$

The instantaneous Coulomb term cancels explicitly. Differentiating the retarded potentials at $r > 0$ yields

$$\mathbf{E} = K\left[(\mathbf{e}_z - 3\mathbf{n}n_z)\left(\frac{\delta(\tau)}{r^3} + \frac{\delta'(\tau)}{cr^2}\right) + (\mathbf{e}_z - \mathbf{n}n_z)\frac{\delta''(\tau)}{c^2 r}\right],$$
$$\mathbf{B} = -\frac{\mu_0}{4\pi}(\mathbf{e}_z\times\mathbf{n})\left[\frac{\delta'(\tau)}{r^2} + \frac{\delta''(\tau)}{cr}\right].$$

Set $u = r - ct$. The distribution identities $\delta(\tau) = c\delta(u)$, $\delta'(\tau) = -c^2\delta'(u)$, and $\delta''(\tau) = c^3\delta''(u)$ give

$$\boxed{E_x} = \boxed{\frac{Kc}{r}n_xn_z\left[-\delta''(u) + \frac{3}{r}\delta'(u) - \frac{3}{r^2}\delta(u)\right]},$$
$$E_y = \frac{Kc}{r}n_yn_z\left[-\delta''(u) + \frac{3}{r}\delta'(u) - \frac{3}{r^2}\delta(u)\right],$$
$$\boxed{E_z} = \boxed{\frac{Kc}{r}\left[(1-n_z^2)\delta''(u) + (3n_z^2-1)\left(\frac{\delta'(u)}{r} - \frac{\delta(u)}{r^2}\right)\right]},$$
$$\mathbf{B} = \frac{\mu_0 c^2}{4\pi r}\sin\theta\left[\frac{\delta'(u)}{r} - \delta''(u)\right]\hat{\boldsymbol{\phi}}.$$

Substituting $n_x = \sin\theta\cos\phi$, $n_y = \sin\theta\sin\phi$, $n_z = \cos\theta$ proves all requested components. Every field at $r > 0$ has support only on $r = ct$, $t > 0$; the nonlocal Coulomb-gauge potentials have produced a causal flash with finite speed c.

6.21 An electric dipole of dipole moment $\mathbf{p}$, fixed in direction and magnitude, is located at a position $\mathbf{r}_0(t)$ with respect to the origin. Its velocity $\mathbf{v} = d\mathbf{r}_0/dt$ is nonrelativistic.

(a) Show that the dipole's charge and current densities can be expressed formally as

$$\rho(\mathbf{x}, t) = -(\mathbf{p} \cdot \nabla)\delta(\mathbf{x} - \mathbf{r}_0(t)); \qquad \mathbf{J}(\mathbf{x}, t) = -\mathbf{v}(\mathbf{p} \cdot \nabla)\delta(\mathbf{x} - \mathbf{r}_0(t))$$

(b) Show that the off-center moving dipole gives rise to a magnetic dipole field and an electric quadrupole field in addition to an electric dipole field, with moments

$$\mathbf{m} = \tfrac{1}{2}\mathbf{p} \times \mathbf{v}$$

and

$$Q_{ij} = 3(x_{0i}p_j + x_{0j}p_i) - 2\mathbf{r}_0 \cdot \mathbf{p}\,\delta_{ij}$$

[There are, of course, still higher moments.]

(c) Show that the quasi-static electric quadrupole field is

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^4}\left[15\mathbf{n}(\mathbf{n} \cdot \mathbf{r}_0)(\mathbf{n} \cdot \mathbf{p}) - 3\mathbf{r}_0(\mathbf{n} \cdot \mathbf{p}) - 3\mathbf{p}(\mathbf{n} \cdot \mathbf{r}_0) - 3\mathbf{n}(\mathbf{r}_0 \cdot \mathbf{p})\right]$$

where $\mathbf{n}$ is a unit vector in the radial direction.

Solution.

Take $\mathbf{p}$ to be constant in magnitude as well as direction. If only its direction were fixed while its magnitude varied, the displayed current would need an additional $\dot{\mathbf{p}}\delta(\mathbf{x} - \mathbf{r}_0)$. The following derivation makes this assumption explicit.

(a) Represent the dipole by $+q$ at $\mathbf{r}_0 + \mathbf{d}/2$ and $-q$ at $\mathbf{r}_0 - \mathbf{d}/2$, translating with the common velocity $\mathbf{v}$, and let $\mathbf{p} = q\mathbf{d}$ be fixed as $d \to 0$. Expansion of the two deltas gives

$$\rho = q[\delta(\mathbf{x} - \mathbf{r}_0 - \mathbf{d}/2) - \delta(\mathbf{x} - \mathbf{r}_0 + \mathbf{d}/2)] \longrightarrow \boxed{-\mathbf{p} \cdot \nabla\delta(\mathbf{x} - \mathbf{r}_0)}, \qquad \boxed{\mathbf{J} = \mathbf{v}\rho}.$$

Since $\partial_t\delta(\mathbf{x} - \mathbf{r}_0) = -\mathbf{v} \cdot \nabla\delta$, these expressions satisfy $\partial_t\rho + \nabla \cdot \mathbf{J} = 0$.

(b) For any smooth test function F,

$$\int F\rho\, d^3x = p_k\partial_k F(\mathbf{r}_0).$$

Taking $F = 1, x_i$ gives zero net charge and electric moment $\mathbf{p}$. For the magnetic moment,

$$m_i = \frac{1}{2}\epsilon_{ijk}\int x_j v_k \rho\, d^3x = \frac{1}{2}\epsilon_{ijk}p_j v_k, \qquad \boxed{\mathbf{m} = \tfrac{1}{2}\mathbf{p} \times \mathbf{v}}.$$

For the electric quadrupole use $F = 3x_ix_j - r^2\delta_{ij}$:

$$\boxed{Q_{ij} = p_k\partial_k(3x_ix_j - r^2\delta_{ij})\big|_{\mathbf{r}_0} = 3(x_{0i}p_j + x_{0j}p_i) - 2(\mathbf{r}_0 \cdot \mathbf{p})\delta_{ij}}.$$

Its trace vanishes. A moving off-center dipole generally has further moments; neither formula assumes their absence.

(c) The quadrupole term in the far-zone, quasistatic potential is

$$\Phi_Q = \frac{1}{4\pi\epsilon_0}\frac{Q_{ij}n_in_j}{2r^3} = \frac{1}{4\pi\epsilon_0}\left[\frac{3(\mathbf{r}_0 \cdot \mathbf{x})(\mathbf{p} \cdot \mathbf{x})}{r^5} - \frac{\mathbf{r}_0 \cdot \mathbf{p}}{r^3}\right].$$

Use $\nabla r^{-s} = -s\mathbf{n}r^{-s-1}$ and the product rule for the quadratic numerator. Its negative gradient is

$$\boxed{\mathbf{E}_Q = \frac{1}{4\pi\epsilon_0 r^4}\left[15\mathbf{n}(\mathbf{n}\cdot\mathbf{r}_0)(\mathbf{n}\cdot\mathbf{p}) - 3\mathbf{r}_0(\mathbf{n}\cdot\mathbf{p}) - 3\mathbf{p}(\mathbf{n}\cdot\mathbf{r}_0) - 3\mathbf{n}(\mathbf{r}_0\cdot\mathbf{p})\right].}$$

This is the r^{-4} term of the field expanded about the origin, valid for $r \gg |\mathbf{r}_0|$; it is not the entire translated dipole field.

6.22

(a) For the off-center, slowly moving, electric dipole of Problem 6.21, show that the leading term for $r \gg |\mathbf{r}_0|$ in the quasi-static vector potential produced by the current flow associated with the dipole motion is

$$\mathbf{A}(\mathbf{x},t) = \frac{\mu_0\mathbf{v}(\mathbf{n}\cdot\mathbf{p})}{4\pi r^2} = \frac{\mu_0}{4\pi}\left[\frac{1}{2}\frac{(\mathbf{p}\times\mathbf{v})\times\mathbf{x}}{r^3} + \frac{1}{2}\frac{[\mathbf{p}(\mathbf{x}\cdot\mathbf{v}) + \mathbf{v}(\mathbf{x}\cdot\mathbf{p})]}{r^3}\right]$$

where the first term of the second form (antisymmetric in $\mathbf{v}$ and $\mathbf{p}$) is the vector potential of the magnetic dipole whose moment is given in Problem 6.21. The added term is symmetric in $\mathbf{v}$ and $\mathbf{p}$.

(b) Show that the magnetic field of the symmetric term is

$$\mathbf{B}_{\text{sym}} = -\frac{3\mu_0}{8\pi r^3}\,\mathbf{n}\times[\mathbf{p}(\mathbf{v}\cdot\mathbf{n}) + \mathbf{v}(\mathbf{p}\cdot\mathbf{n})]$$

(c) By calculating its curl, show that $\mathbf{B}_{\text{sym}}$ is consistent with being the quasi-static magnetic field associated with the electric quadrupole field of Problem 6.21c.

(d) Show that the total magnetic field (computed from the first form of the vector potential, i.e., the sum of $\mathbf{B}_{\text{sym}}$ and the magnetic *dipole* field) is

$$\mathbf{B} = \frac{\mu_0}{4\pi}\,\mathbf{v}\times\frac{[3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}]}{r^3}$$

Comment.

Solution.

Keep the leading far-zone term in $|\mathbf{r}_0|/r$, and first order in v/c, for the constant dipole moment of 6.21. A quasistatic observation also requires retardation across the source and observer distance to be small on the variation time scale.

(a) Direct integration of the derivative source gives the instantaneous potential before expanding in r_0/r:

$$\mathbf{A}(\mathbf{x},t) = \frac{\mu_0\mathbf{v}}{4\pi}\frac{\mathbf{p}\cdot(\mathbf{x}-\mathbf{r}_0)}{|\mathbf{x}-\mathbf{r}_0|^3}.$$

Its leading term is $\mu_0\mathbf{v}(\mathbf{p}\cdot\mathbf{x})/(4\pi r^3)$, as stated. The triple-product identity $(\mathbf{p}\times\mathbf{v})\times\mathbf{x} = \mathbf{v}(\mathbf{p}\cdot\mathbf{x}) - \mathbf{p}(\mathbf{v}\cdot\mathbf{x})$ splits it exactly into

$$\boxed{\mathbf{A} = \frac{\mu_0}{8\pi r^3}\left[(\mathbf{p}\times\mathbf{v})\times\mathbf{x} + \mathbf{p}(\mathbf{v}\cdot\mathbf{x}) + \mathbf{v}(\mathbf{p}\cdot\mathbf{x})\right].}$$

The first term is the potential of $\mathbf{m} = \mathbf{p} \times \mathbf{v}/2$.

(b) Let $\mathbf{C} = \mathbf{p}(\mathbf{v}\cdot\mathbf{x}) + \mathbf{v}(\mathbf{p}\cdot\mathbf{x})$. Its curl vanishes, since the two constant-vector cross products cancel. Therefore

$$\mathbf{B}_{\rm sym} = \frac{\mu_0}{8\pi}\nabla(r^{-3}) \times \mathbf{C} = \boxed{-\frac{3\mu_0}{8\pi r^3}\mathbf{n} \times [\mathbf{p}(\mathbf{v}\cdot\mathbf{n}) + \mathbf{v}(\mathbf{p}\cdot\mathbf{n})]}.$$

(c) Component differentiation using $\partial_i n_j = (\delta_{ij} - n_i n_j)/r$ gives

$$\nabla \times \mathbf{B}_{\rm sym} = \frac{\mu_0}{4\pi r^4}\left[15\mathbf{n}(\mathbf{n}\cdot\mathbf{v})(\mathbf{n}\cdot\mathbf{p}) - 3\mathbf{v}(\mathbf{n}\cdot\mathbf{p}) - 3\mathbf{p}(\mathbf{n}\cdot\mathbf{v}) - 3\mathbf{n}(\mathbf{v}\cdot\mathbf{p})\right].$$

For an explicit way to organize the differentiation, use $\nabla \times (\nabla \times \mathbf{A}_{\rm sym}) = \nabla(\nabla\cdot\mathbf{A}_{\rm sym}) - \nabla^2\mathbf{A}_{\rm sym}$; each x_j/r^3 is harmonic for $r > 0$, so the Laplacian vanishes, and

$$\nabla\cdot\mathbf{A}_{\rm sym} = \frac{\mu_0}{4\pi}\left[\frac{\mathbf{p}\cdot\mathbf{v}}{r^3} - \frac{3(\mathbf{p}\cdot\mathbf{x})(\mathbf{v}\cdot\mathbf{x})}{r^5}\right].$$

Its gradient is the displayed curl. Taking ∂_t of 6.21(c) replaces $\mathbf{r}_0$ by $\mathbf{v}$, hence

$$\boxed{\nabla \times \mathbf{B}_{\rm sym} = \mu_0\epsilon_0\partial_t\mathbf{E}_Q.}$$

This is exactly the source-free Ampère–Maxwell equation at the retained multipole order. The magnetic dipole field has zero curl for $r > 0$ and therefore cannot supply this displacement current by itself.

(d) Curl the unsplit potential:

$$\mathbf{B} = \frac{\mu_0}{4\pi}\nabla\left(\frac{\mathbf{p}\cdot\mathbf{x}}{r^3}\right) \times \mathbf{v} = \boxed{\frac{\mu_0}{4\pi r^3}\mathbf{v} \times [3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}]}.$$

It equals $\mathbf{v} \times \mathbf{E}_{\rm dip}/c^2$, the first-order magnetic field generated by translating an electric dipole. The magnetic dipole moment alone gives only part of it; the symmetric potential and its electric-quadrupole displacement current are essential for the full field. The exact instantaneous expression uses $\mathbf{x} - \mathbf{r}_0$ instead of $\mathbf{x}$, as shown in part (a).

6.23 The wave equations (6.168) for the Hertz vectors contain arbitrary source terms involving the functions $\mathbf{V}$ and ξ. Consider the gauge transformations

$$\mathbf{\Pi}'_{\rm e} = \mathbf{\Pi}_{\rm e} + \mu_0\nabla \times \mathbf{G} - \nabla g; \qquad \mathbf{\Pi}'_{\rm m} = \mathbf{\Pi}_{\rm m} - \mu\frac{\partial\mathbf{G}}{\partial t}$$

where $\mathbf{G}$ and g are well-behaved functions of space and time.

(a) Show that, if $\mathbf{G}$ and g satisfy the wave equations

$$\left(\mu\epsilon\frac{\partial^2}{\partial t^2} - \nabla^2\right)\begin{Bmatrix}\mathbf{G}\\ g\end{Bmatrix} = \begin{Bmatrix}\frac{1}{\mu}(\mathbf{V} + \nabla\xi)\\ 0\end{Bmatrix}$$

then the new polarization potentials $\mathbf{\Pi}'_{\rm e}$ and $\mathbf{\Pi}'_{\rm m}$ satisfy (6.168) with vanishing $\mathbf{V}$ and ξ.

(b) Show that the gauge transformation on the Hertz vectors is equivalent to a gauge transformation on $\mathbf{A}$ and Φ. What is the gauge function Λ of (6.19) in terms of $\mathbf{G}$ and g?

Solution.

Let $\mathcal{D} = \mu\epsilon\partial_t^2 - \nabla^2$, with constant ϵ, μ. The actual equations (6.168), including their signs, are

$$\mathcal{D}\mathbf{\Pi}_e = \mathbf{P}_{\text{ext}} - \frac{\mu_0}{\mu}\nabla\times\mathbf{V}, \qquad \mathcal{D}\mathbf{\Pi}_m = \mathbf{M}_{\text{ext}} + \partial_t\mathbf{V} + \nabla\partial_t\xi.$$

(a) Constant-coefficient derivatives commute with $\mathcal{D}$. Apply it to the proposed transformed vectors and insert $\mathcal{D}\mathbf{G} = (\mathbf{V} + \nabla\xi)/\mu$, $\mathcal{D}g = 0$:

$$\mathcal{D}\mathbf{\Pi}'_e = \mathbf{P}_{\text{ext}} - \frac{\mu_0}{\mu}\nabla\times\mathbf{V} + \frac{\mu_0}{\mu}\nabla\times(\mathbf{V}+\nabla\xi) - \nabla\mathcal{D}g = \boxed{\mathbf{P}_{\text{ext}}},$$
$$\mathcal{D}\mathbf{\Pi}'_m = \mathbf{M}_{\text{ext}} + \partial_t(\mathbf{V}+\nabla\xi) - \partial_t(\mathbf{V}+\nabla\xi) = \boxed{\mathbf{M}_{\text{ext}}}.$$

These are (6.168) with both arbitrary source functions set to zero.

(b) Equation (6.166) defines

$$\mathbf{A} = \mu\partial_t\mathbf{\Pi}_e + \mu_0\nabla\times\mathbf{\Pi}_m, \qquad \Phi = -\epsilon^{-1}\nabla\cdot\mathbf{\Pi}_e.$$

The $\mathbf{G}$ contributions cancel identically in $\mathbf{A}$, while its divergence contribution to Φ is zero. The remaining changes are

$$\delta\mathbf{A} = -\mu\nabla\partial_t g, \qquad \delta\Phi = \epsilon^{-1}\nabla^2 g = \mu\partial_t^2 g,$$

where the last equality uses $\mathcal{D}g = 0$. Thus

$$\boxed{\Lambda = -\mu\partial_t g, \qquad \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \qquad \Phi' = \Phi - \partial_t\Lambda.}$$

The arbitrary vector $\mathbf{G}$ changes the Hertz representation without changing even $\mathbf{A}, \Phi$; the scalar g generates their residual Lorenz-gauge transformation. Both leave $\mathbf{E}, \mathbf{B}$ unchanged.

6.24 A current distribution $\mathbf{J}(\mathbf{x}, t)$ localized near the origin varies slowly in time.

(a) Use the Jefimenko expressions (6.55) and (6.56) for the retarded fields to evaluate the quasi-static fields far from the current distribution. Assuming that there are no electric multipole moments and retaining only the magnetic dipole contributions, show that the magnetic and electric fields at the point ($\mathbf{x} = \hat{\mathbf{r}}r$, t) to first order in an expansion in successive time derivatives are

$$\mathbf{B} = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left(1 + \frac{r}{c}\frac{\partial}{\partial t}\right)[3(\mathbf{m}(t-r/c)\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}(t-r/c)]$$
$$\mathbf{E} = \frac{\mu_0}{4\pi}\frac{1}{r^2}\,\hat{\mathbf{r}}\times\frac{\partial\mathbf{m}(t-r/c)}{\partial t}$$

(b) The construction and excitation of an infinite, straight, right circular solenoid of radius a, with N turns per unit length, are such that its current $I(t)$ is the same everywhere along its length and is changed very slowly in time. Show that the fields far from the solenoid are approximately

$$\mathbf{B} \approx 0 + O\left(\frac{\partial^2 I}{\partial t^2}\right)$$
$$\mathbf{E} \approx -\frac{\mu_0}{2}\frac{Na^2}{\rho}\frac{\partial I(t-\rho/c)}{\partial t}\,\hat{\boldsymbol{\phi}}$$

where ρ is the perpendicular distance from the axis, provided $\max(|dI/dt/I|) \ll c/\rho$. The derivative expansion also requires control of higher derivatives, or a low-frequency spectrum; the displayed relative first-derivative bound alone is insufficient. A long solenoid with a time-varying current has an electric field outside it, in contrast to the static situation. Verify that Faraday's law is satisfied.

Solution.

(a) With no time-dependent charge multipoles, the localized magnetic current has $\nabla' \cdot \mathbf{J} = 0$, $\int \mathbf{J}\, d^3x' = 0$, and

$$\int x'_j J_i \, d^3x' = \epsilon_{jik} m_k(t), \qquad \mathbf{m} = \tfrac{1}{2} \int \mathbf{x}' \times \mathbf{J} \, d^3x'.$$

These follow by integrating divergences of $x'_i \mathbf{J}$ and $x'_i x'_j \mathbf{J}$. To use Jefimenko's formulas directly, observe that at fixed source point

$$\nabla \times \frac{\mathbf{J}(\mathbf{x}', t - R/c)}{R} = \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR} \right]_{\text{ret}} \times \hat{\mathbf{R}}.$$

This follows from $\nabla R = \hat{\mathbf{R}}$ and $\nabla(t - R/c) = -\hat{\mathbf{R}}/c$, and is exactly the integrand of (6.56). The absence of electric source multipoles leaves only the $-\dot{\mathbf{J}}/(c^2 R)$ term of (6.55). Thus one auxiliary integral $\mathbf{A} = (\mu_0/4\pi) \int [\mathbf{J}]_{\text{ret}}/R \, d^3x'$ generates both Jefimenko fields by a curl and a negative time derivative. Expand this retarded source integral in x'/r; with $t_r = t - r/c$,

$$\frac{\mathbf{J}(\mathbf{x}', t - R/c)}{R} = \frac{\mathbf{J}(\mathbf{x}', t_r)}{r} + (\mathbf{n} \cdot \mathbf{x}') \left[\frac{\mathbf{J}(\mathbf{x}', t_r)}{r^2} + \frac{\dot{\mathbf{J}}(\mathbf{x}', t_r)}{cr} \right] + \cdots .$$

The zeroth moment is zero, and the first moment gives

$$\mathbf{A} = \frac{\mu_0}{4\pi} \left[\frac{\mathbf{m}(t_r) \times \mathbf{n}}{r^2} + \frac{\dot{\mathbf{m}}(t_r) \times \mathbf{n}}{cr} \right].$$

This same expansion in the $-\dot{\mathbf{J}}/(c^2 R)$ term of (6.55) and the two current terms of (6.56) yields respectively $-\partial_t \mathbf{A}$ and $\nabla \times \mathbf{A}$; these operations retain exactly the source moments displayed above. Curling while keeping $\nabla t_r = -\mathbf{n}/c$ uses the two intermediate identities

$$\nabla \times \frac{\mathbf{m}(t_r) \times \mathbf{n}}{r^2} = \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{r^3} + \frac{\mathbf{n}(\mathbf{n} \cdot \dot{\mathbf{m}}) - \dot{\mathbf{m}}}{cr^2},$$
$$\nabla \times \frac{\dot{\mathbf{m}}(t_r) \times \mathbf{n}}{cr} = \frac{2\mathbf{n}(\mathbf{n} \cdot \dot{\mathbf{m}})}{cr^2} + \frac{\mathbf{n}(\mathbf{n} \cdot \ddot{\mathbf{m}}) - \ddot{\mathbf{m}}}{c^2 r}.$$

Adding these and taking $-\partial_t \mathbf{A}$ gives

$$\mathbf{B} = \frac{\mu_0}{4\pi} \left\{ \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{r^3} + \frac{3\mathbf{n}(\mathbf{n} \cdot \dot{\mathbf{m}}) - \dot{\mathbf{m}}}{cr^2} + \frac{\mathbf{n} \times (\mathbf{n} \times \ddot{\mathbf{m}})}{c^2 r} \right\},$$
$$\mathbf{E} = \frac{\mu_0}{4\pi} \left[\frac{\mathbf{n} \times \dot{\mathbf{m}}}{r^2} + \frac{\mathbf{n} \times \ddot{\mathbf{m}}}{cr} \right].$$

All moments here are at t_r. Dropping second and higher time derivatives proves both stated formulas. This is a far-source multipole expansion (r much larger than source size) within the quasistatic zone (r/c small compared with the variation time), not an omission of the retardation in the displayed first-derivative terms.

(b) Approximate each solenoid slice dz' as an axial magnetic dipole $d\mathbf{m} = N\pi a^2 I(t)\, dz'\hat{\mathbf{z}}$, valid for $\rho \gg a$. Put the observer at $z = 0$ and $R = (\rho^2 + z'^2)^{1/2}$. Then $\mathbf{n} \times \hat{\mathbf{z}} = -(\rho/R)\hat{\boldsymbol{\phi}}$, so

$$E_\phi = -\frac{\mu_0 N a^2 \rho}{4} \int_{-\infty}^{\infty} \frac{\dot{I}(t - R/c)}{R^3}\, dz'.$$

To first derivative order, replace $\dot{I}(t - R/c)$ by $\dot{I}(t - \rho/c)$. The integral is $2/\rho^2$, giving

$$\boxed{\mathbf{E} \simeq -\frac{\mu_0 N a^2}{2\rho} \dot{I}(t - \rho/c)\hat{\boldsymbol{\phi}}.}$$

For the magnetic field, odd radial contributions cancel between z' and $-z'$. The static axial part is proportional to

$$\int_{-\infty}^{\infty} \left(\frac{3z'^2}{R^5} - \frac{1}{R^3} \right) dz' = -\int_{-\infty}^{\infty} \frac{d}{dz'} \left(\frac{z'}{R^3} \right) dz' = 0.$$

In each retained dipole term, expanding $I(t - R/c) = I(t) - (R/c)\dot{I}(t) + \cdots$ makes the first retardation correction cancel the explicit $(R/c)\partial_t$ contribution in part (a). Thus

$$\boxed{\mathbf{B}_{\text{outside}} = 0 \quad \text{through first time-derivative order.}}$$

Faraday's integral law gives a direct check. To this accuracy the interior flux is $\Phi_B = \mu_0 N I(t)\pi a^2$, and

$$\oint \mathbf{E} \cdot d\mathbf{l} = 2\pi\rho E_\phi = -\mu_0 N \pi a^2 \dot{I}(t - \rho/c) = -\dot{\Phi}_B + O(\ddot{I}).$$

Locally outside, ρE_ϕ is constant to first derivative order, so $\nabla \times \mathbf{E} = 0$ at that order as well.

For a truly infinite solenoid the higher-order expansion is not uniform at arbitrarily large $|z'|$. The exact harmonic outgoing cylindrical wave has exterior B_z proportional to $\omega^2 \log(\omega\rho/c)$ at small frequency. Thus the book's $O(\partial_t^2 I)$ denotes omitted derivative order, possibly with logarithms, not a rigorous uniform power-series bound. Its first-order result is valid for smooth slowly varying currents with bounded higher derivatives or a narrow low-frequency spectrum; $|\dot{I}/I| \ll c/\rho$ alone should be understood as that quasistatic scale condition, not applied at zeros of a sinusoidal current. To see why a first-derivative bound is insufficient by itself, introduce a fixed time T and let $I(t) = I_0[1 + \varepsilon^3 \sin(t/(\varepsilon^2 T))]$, with $I_0 > 0$ and $0 < \varepsilon \ll 1$. Then

$$\max \left| \frac{\dot{I}}{I} \right| \le \frac{\varepsilon}{T(1 - \varepsilon^3)}, \qquad \max |\ddot{I}| = \frac{I_0}{\varepsilon T^2}.$$

The first expression tends to zero while the second diverges. A controlled derivative expansion therefore also bounds the higher derivatives, or restricts all appreciable Fourier components to $|\omega|\rho/c \ll 1$.

6.25

(a) Starting with the Lorentz force expression (6.114), show that in the dipole approximation the force acting on a neutral atom at rest can be expressed as

$$\frac{d\mathbf{P}_{\text{atom}}}{dt} = (\mathbf{d} \cdot \nabla)\mathbf{E} + \dot{\mathbf{d}} \times \mathbf{B}$$

where $\mathbf{d}$ is the atomic dipole moment and $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields at the site of the atom.

(b) For a uniform plane wave of frequency ω in a nonmagnetic tenuous dielectric medium with index of refraction $n(\omega)$, show that the time rate of change of mechanical momentum per unit volume $\mathbf{g}_{\text{mech}}$ accompanying the electromagnetic momentum $\mathbf{g}_{\text{em}}$ (6.118) of the wave is

$$\frac{d\mathbf{g}_{\text{mech}}}{dt} = \frac{1}{2}\left(n^2 - 1\right)\frac{d\mathbf{g}_{\text{em}}}{dt}$$

[see Peierls (loc. cit.) for corrections for dense media and non-uniform waves.]

Solution.

(a) Let $\mathbf{R}$ be the fixed atomic center and $\boldsymbol{\xi}_a$ the charge coordinates relative to it. Expand the external fields across the atom in the microscopic Lorentz force:

$$\mathbf{F} = \sum_a q_a \left[\mathbf{E}(\mathbf{R}) + (\boldsymbol{\xi}_a \cdot \nabla)\mathbf{E}(\mathbf{R}) + \dot{\boldsymbol{\xi}}_a \times \mathbf{B}(\mathbf{R})\right] + \cdots .$$

Neutrality removes the first term. Define $\mathbf{d} = \sum_a q_a \boldsymbol{\xi}_a$ and $\dot{\mathbf{d}} = \sum_a q_a \dot{\boldsymbol{\xi}}_a$; the retained electric and magnetic terms therefore give

$$\boxed{\frac{d\mathbf{P}_{\text{atom}}}{dt} = (\mathbf{d} \cdot \nabla)\mathbf{E} + \dot{\mathbf{d}} \times \mathbf{B}.}$$

The omitted terms contain electric quadrupoles, intrinsic magnetic moments, and higher field gradients. Forces internal to the neutral atom cancel in its total mechanical momentum balance.

(b) For a transparent tenuous nonmagnetic medium, let N be the atomic number density and $\alpha(\omega)$ the real electric polarizability at the single frequency in the question. Neglecting local-field corrections,

$$\mathbf{d} = \alpha\mathbf{E}, \qquad N\alpha = \epsilon_0[n^2(\omega) - 1].$$

For a transverse plane wave propagating along $\hat{\mathbf{k}}$, $\mathbf{B} = (n/c)\hat{\mathbf{k}} \times \mathbf{E}$ and spatial dependence is only along $\hat{\mathbf{k}}$. Hence $(\mathbf{d} \cdot \nabla)\mathbf{E} = 0$, while the vector triple product gives

$$\frac{d\mathbf{g}_{\text{mech}}}{dt} = N\dot{\mathbf{d}} \times \mathbf{B} = \frac{N\alpha n}{c}\hat{\mathbf{k}}(\dot{\mathbf{E}} \cdot \mathbf{E}) = \frac{N\alpha n}{2c}\hat{\mathbf{k}}\,\partial_t E^2.$$

The electromagnetic momentum density in (6.118) is the vacuum-field (or Abraham) expression, not the Minkowski quantity of 6.9:

$$\mathbf{g}_{\text{em}} = \epsilon_0 \mathbf{E} \times \mathbf{B} = \frac{\epsilon_0 n}{c} E^2 \hat{\mathbf{k}}.$$

Combining the last two equations proves

$$\boxed{\frac{d\mathbf{g}_{\text{mech}}}{dt} = \frac{1}{2}(n^2 - 1)\frac{d\mathbf{g}_{\text{em}}}{dt}.}$$

This is an instantaneous relation for the monochromatic lossless response; its cycle average has zero derivative. A momentum assigned to a finite pulse by integrating through its turn-on requires the envelope, dispersion, and boundary forces to be included. In particular the single-frequency relation $\mathbf{d} = \alpha(\omega)\mathbf{E}$ cannot be used as an instantaneous constitutive law for an arbitrary broadband pulse. These are the limitations behind the problem's restriction to a tenuous, uniform plane wave.

Note of explanation: The reader may be startled to find (in all but the earliest printings) the association of Danish physicist Ludvig V. Lorenz's name instead of Dutch physicist Hendrik A.

Lorentz's with the relation (6.14) between the scalar and vector potentials. Yet it is a fact that in 1867 Lorenz, in a paper entitled "On the identity of the vibrations of light with electrical currents," (op. cit.) exploited the retarded solutions for the potentials, derived (6.14) and equations equivalent to wave equations for the electric field, and discussed the characteristics of light propagation in conductors and transparent media, contemporaneously with Maxwell. H. A. Lorentz has ample recognition in physics terminology without the mis-attribution of (6.14) to him (by others, beginning around 1900). As Van Bladel[3] observes, it is up to textbook authors to accord Lorenz his due.[4]

[3]J. Van Bladel, IEEE Antennas and Propagation Magazine **33**, No. 2, 69 (April 1991).

[4]An earlier author who deplored the lack of recognition of Lorenz's contributions is A. O'Rahilly, *Electromagnetic Theory*, Dover Publications, New York (1965) [originally published as *Electromagnetics*, Longman Green and Cork University Press (1938)], footnote, p. 184.

7 Chapter 7. Plane Electromagnetic Waves and Wave Propagation Problems

7.1 For each set of Stokes parameters given below deduce the amplitude of the electric field, up to an overall phase, in both linear polarization and circular polarization bases and make an accurate drawing similar to Fig. 7.4 showing the lengths of the axes of one of the ellipses and its orientation.

(a) $s_0 = 3, \quad s_1 = -1, \quad s_2 = 2, \quad s_3 = -2;$

(b) $s_0 = 25, \quad s_1 = 0, \quad s_2 = 24, \quad s_3 = 7.$

Solution.

Use the conventions of (7.27)–(7.28), with physical fields the real part of amplitudes times $e^{ikz-i\omega t}$:

$$s_0 = |E_x|^2 + |E_y|^2, \quad s_1 = |E_x|^2 - |E_y|^2, \quad s_2 = 2\,\mathrm{Re}(E_x^* E_y), \quad s_3 = 2\,\mathrm{Im}(E_x^* E_y).$$

Both sets obey $s_0^2 = s_1^2 + s_2^2 + s_3^2$, so each represents a single fully polarized wave. Choose the overall phase to make $E_x = \sqrt{(s_0+s_1)/2} > 0$. Then

$$E_y = \frac{s_2 + i s_3}{2E_x}, \qquad E_+ = \frac{E_x - iE_y}{\sqrt{2}}, \qquad E_- = \frac{E_x + iE_y}{\sqrt{2}},$$

where $\boldsymbol{\epsilon}_\pm = (\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$. The real ellipse is generated by $\mathbf{E}(t) = \mathrm{Re}\,\mathbf{E}_0 \cos\omega t + \mathrm{Im}\,\mathbf{E}_0 \sin\omega t$ at $z = 0$. The squared semiaxes are the eigenvalues of

$$\mathrm{Re}(\mathbf{E}_0 \mathbf{E}_0^\dagger) = \frac{1}{2}\begin{pmatrix} s_0 + s_1 & s_2 \\ s_2 & s_0 - s_1 \end{pmatrix}.$$

Thus

$$a^2, b^2 = \frac{s_0 \pm \sqrt{s_1^2 + s_2^2}}{2}, \qquad \psi = \frac{1}{2}\,\mathrm{atan2}(s_2, s_1),$$

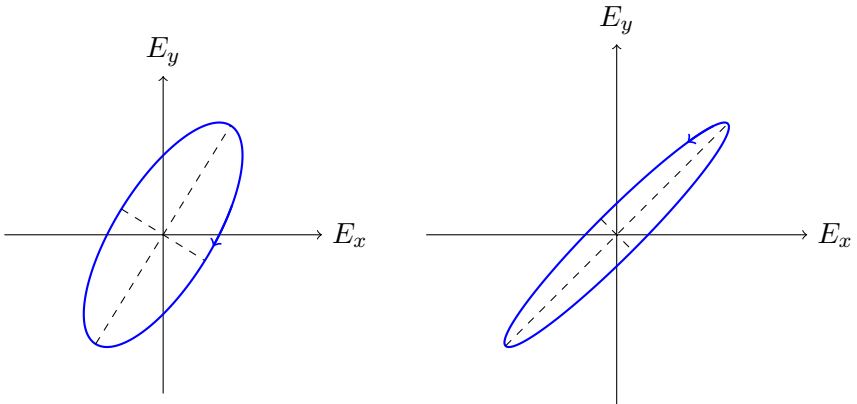
where $a \geq b$ and ψ is the major-axis angle from $+x$. **(a)** Substitution gives

$$\boxed{(E_x, E_y) = (1, 1-i), \qquad (E_+, E_-) = \frac{1}{\sqrt{2}}(-i, 2+i).}$$

Here $a = (1+\sqrt{5})/2 = 1.618034\ldots$, $b = (\sqrt{5}-1)/2 = 0.618034\ldots$, and $\psi = 58.282526°$. The full axis lengths are $2a, 2b$. At $t = 0$ the field is $(1, 1)$ and its derivative is $(0, -\omega)$, fixing the clockwise sense in an x-right, y-up drawing. **(b)** The corresponding amplitudes are

$$\boxed{(E_x, E_y) = \left(\frac{5}{\sqrt{2}}, \frac{24+7i}{5\sqrt{2}}\right), \qquad (E_+, E_-) = \left(\frac{16-12i}{5}, \frac{9+12i}{5}\right).}$$

Their circular intensities are 16 and 9, giving $s_3 = 7$. The ellipse has $a = 7/\sqrt{2}$, $b = 1/\sqrt{2}$, and $\psi = 45°$; its full axis lengths are $7\sqrt{2}$ and $\sqrt{2}$. Its traversal is counterclockwise. The plots use the actual amplitudes and label the semiaxes; a common scale is not imposed on the two separate panels.

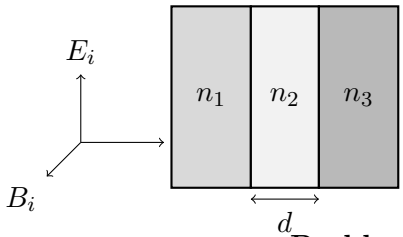


(a) $a = (1+\sqrt{5})/2,\ b = (\sqrt{5}-1)/2$
$\psi = 58.2825°$

(b) $a = 7/\sqrt{2},\ b = 1/\sqrt{2}$
$\psi = 45°$

7.2 A plane wave is incident on a layered interface as shown in the figure. The indices of refraction of the three nonpermeable media are n_1, n_2, n_3. The thickness of the intermediate layer is d. Each of the other media is semi-infinite.

(a) Calculate the transmission and reflection coefficients (ratios of transmitted and reflected Poynting's flux to the incident flux), and sketch their behavior as a function of frequency for $n_1 = 1$, $n_2 = 2$, $n_3 = 3$; $n_1 = 3$, $n_2 = 2$, $n_3 = 1$; and $n_1 = 2$, $n_2 = 4$, $n_3 = 1$.



Problem 7.2

(b) The medium n_1 is part of an optical system (e.g., a lens); medium n_3 is air ($n_3 = 1$). It is desired to put an optical coating (medium n_2) on the surface so that there is no reflected wave for a frequency ω_0. What thickness d and index of refraction n_2 are necessary?

Solution.

The drawing specifies normal incidence. Take $e^{-i\omega t}$ time dependence, and reference incident/reflected amplitudes to the first face and the transmitted amplitude to the second face. Define

$$r_{ij} = \frac{n_i - n_j}{n_i + n_j}, \quad t_{ij} = \frac{2n_i}{n_i + n_j}, \quad \delta = \frac{n_2 \omega d}{c}.$$

At either face continuity of E_t, H_t gives $E_i + E_r = E_t$ and $n_i(E_i - E_r) = n_j E_t$, proving these single-face coefficients. **(a)** A round trip inside the film multiplies an amplitude by $r_{21} r_{23} e^{2i\delta} = -r_{12} r_{23} e^{2i\delta}$. Summing all transmitted trips therefore gives

$$t = \frac{t_{12} t_{23} e^{i\delta}}{1 + r_{12} r_{23} e^{2i\delta}}.$$

The direct reflection plus the waves returned through the first face gives

$$r = r_{12} + \frac{t_{12}t_{21}r_{23}e^{2i\delta}}{1 + r_{12}r_{23}e^{2i\delta}} = \frac{r_{12} + r_{23}e^{2i\delta}}{1 + r_{12}r_{23}e^{2i\delta}},$$

where $t_{12}t_{21} = 1 - r_{12}^2$ was used. Since the flux for a traveling electric amplitude is $n|E|^2/(2Z_0)$,

$$\boxed{R = |r|^2, \qquad T = \frac{n_3}{n_1}|t|^2.}$$

A useful real form, obtained by multiplying the denominator by $e^{-i\delta}$, is

$$\boxed{T = \frac{4n_1n_3}{(n_1+n_3)^2\cos^2\delta + (n_2 + n_1n_3/n_2)^2\sin^2\delta}, \qquad R = 1 - T.}$$

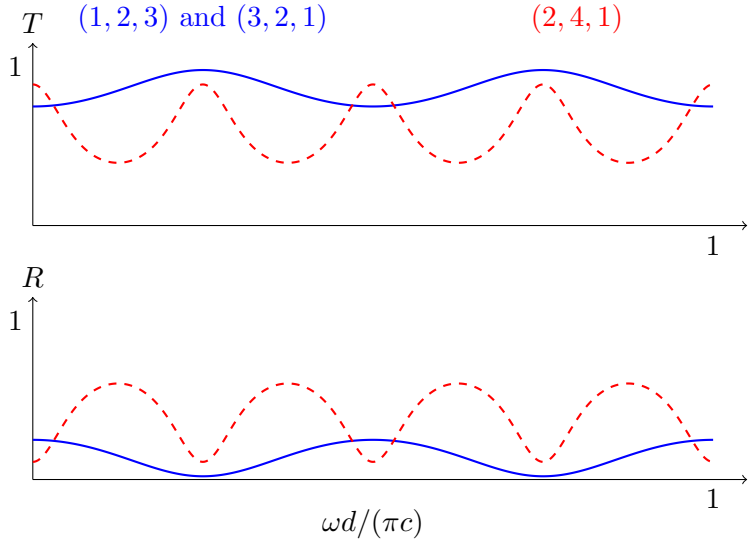
Indeed the numerator of R is the same denominator minus $4n_1n_3$, namely $(n_1 - n_3)^2\cos^2\delta + (n_2 - n_1n_3/n_2)^2\sin^2\delta$. For $(n_1, n_2, n_3) = (1, 2, 3)$ and $(3, 2, 1)$ the flux curves coincide:

$$T = \frac{12}{16\cos^2\delta + (49/4)\sin^2\delta}, \qquad \frac{3}{4} \le T \le \frac{48}{49}.$$

For $(2, 4, 1)$,

$$T = \frac{8}{9\cos^2\delta + (81/4)\sin^2\delta}, \qquad \frac{32}{81} \le T \le \frac{8}{9}.$$

The plot uses a common frequency coordinate $\omega d/(\pi c)$, so the third film has twice as many fringes as the first two.



(b) For a nontrivial mismatch $n_1 \neq n_3$, the two nonnegative terms in the numerator of R vanish together only if $\cos\delta = 0$ and $n_2^2 = n_1n_3$. With $n_3 = 1$ this gives

$$\boxed{n_2 = \sqrt{n_1}, \qquad d = \frac{(2m+1)\pi c}{2n_2\omega_0} = \frac{(2m+1)\lambda_0}{4n_2}, \quad m = 0, 1, \ldots.}$$

The thinnest coating is a quarter-wave optical thickness. If the outer media already match, $n_1 = n_3$, integer half-wave layers also transmit perfectly, and a coating with the same index is unnecessary.

7.3 Two plane semi-infinite slabs of the same uniform, isotropic, nonpermeable, lossless dielectric with index of refraction n are parallel and separated by an air gap ($n = 1$) of width d. A plane electromagnetic wave of frequency ω is incident on the gap from one of the slabs with angle of incidence i. For linear polarization *both* parallel to *and* perpendicular to the plane of incidence,

(a) calculate the ratio of power transmitted into the second slab to the incident power and the ratio of reflected to incident power;

(b) for i greater than the critical angle for total internal reflection, sketch the ratio of transmitted power to incident power as a function of d measured in units of wavelength in the gap.

Solution.

Let $k_0 = \omega/c$, $\beta = n\cos i$, and $q = \sqrt{1 - n^2 \sin^2 i}$, with the square root chosen real positive or positive imaginary. The tangential wave number $k_x = nk_0 \sin i$ is conserved at both faces. Use the tangential electric amplitude to define the dimensionless admittance $Y = Z_0 H_t / E_t$; its forward-wave values are

	dielectric slab Y_1	air gap Y_2
$\perp$ (TE)	β	q
$\parallel$ (TM)	n^2/β	$1/q$

For example, TE gives $H_x = -k_z E_y/(\mu_0\omega)$, whereas TM uses $E_x = k_z H_y/(\epsilon\omega)$; the sign used in Y is the one yielding positive normal flux for a forward wave. **(a)** The film calculation of Problem 7.2 applies with n_j replaced by the corresponding admittances, with equal outer admittances and phase $\delta = k_0 q d$. The transmitted tangential amplitude is

$$t = \left[\cos\delta - \frac{i}{2}\left(\frac{Y_1}{Y_2} + \frac{Y_2}{Y_1}\right)\sin\delta\right]^{-1}.$$

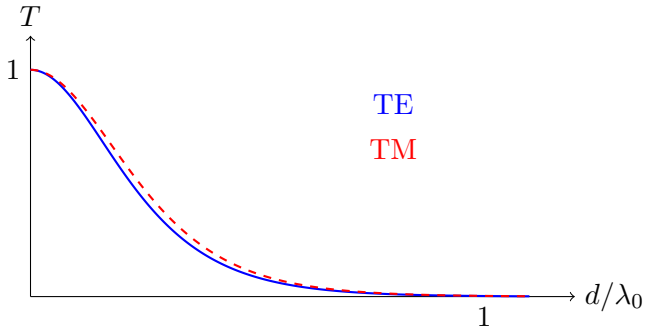
The two outer media are identical, so $T = |t|^2$ and $R = 1 - T$. When $i < i_c = \sin^{-1}(1/n)$, both admittances are real and

$$\boxed{T = \left[1 + \frac{(Y_1^2 - Y_2^2)^2}{4Y_1^2 Y_2^2}\sin^2(k_0 q d)\right]^{-1}, \quad R = 1 - T.}$$

This includes perfect transmission at each gap resonance. At the TM Brewster angle the admittances match and $T = 1$ for every d. **(b)** For $i > i_c$ write $q = i\kappa$, $\kappa = \sqrt{n^2 \sin^2 i - 1}$. Put $\eta = \kappa$ for TE and $\eta = 1/\kappa$ for TM, so $Y_2 = \pm i\eta$. Substituting $\sin(iu) = i\sinh u$ into the preceding expression gives

$$\boxed{T = \left[1 + \frac{(Y_1^2 + \eta^2)^2}{4Y_1^2 \eta^2}\sinh^2(k_0 \kappa d)\right]^{-1}, \quad R = 1 - T.}$$

Thus $T(0) = 1$ and T decreases monotonically to zero; for a wide gap it is proportional to $e^{-2k_0\kappa d}$. A single isolated interface reflects totally, but a second dielectric within the penetration distance receives power through the evanescent gap. An evanescent normal wave has no real normal wavelength. In the plot, "wavelength in the gap" means the air reference wavelength $\lambda_0 = 2\pi/k_0$ at this frequency. The illustrative choice is $n = 1.5$, $i = 50^\circ > i_c$.



At the critical angle the finite limits, obtained using $\sin(k_0 qd) \sim k_0 qd$, are $T_\perp = [1 + (\beta k_0 d/2)^2]^{-1}$ and $T_\parallel = [1 + (k_0 d/2Y_1)^2]^{-1}$ with $Y_1 = n^2/\beta$.

7.4 A plane-polarized electromagnetic wave of frequency ω in free space is incident normally on the flat surface of a nonpermeable medium of conductivity σ and dielectric constant ϵ.

(a) Calculate the amplitude and phase of the reflected wave relative to the incident wave for arbitrary σ and ϵ.

(b) Discuss the limiting cases of a very poor and a very good conductor, and show that for a good conductor the reflection coefficient (ratio of reflected to incident intensity) is approximately

$$R \simeq 1 - 2\,\frac{\omega}{c}\,\delta$$

where δ is the skin depth.

Solution.

Put the interface at $z = 0$, with the material in $z > 0$, and use $e^{-i\omega t}$. Ohm's law can be absorbed into $\epsilon_{\text{eff}} = \epsilon + i\sigma/\omega$. **(a)** Define the passive refractive index

$$N = u + iv = \sqrt{\frac{\epsilon}{\epsilon_0} + i\frac{\sigma}{\epsilon_0 \omega}}, \quad u, v \geq 0.$$

Writing $e_r = \epsilon/\epsilon_0$ and $s = \sigma/(\epsilon_0\omega)$, comparison of real and imaginary parts of N^2 gives

$$u = \sqrt{\frac{\sqrt{e_r^2 + s^2} + e_r}{2}}, \qquad v = \sqrt{\frac{\sqrt{e_r^2 + s^2} - e_r}{2}}.$$

Tangential boundary continuity gives $1 + r = t$ and $1 - r = Nt$. Solving and rationalizing the complex ratio gives

$$r = \frac{1 - N}{1 + N} = \frac{1 - u^2 - v^2 - 2iv}{(1 + u)^2 + v^2}.$$

Therefore the requested relative amplitude and phase are

$$\boxed{|r| = \sqrt{\frac{(1-u)^2 + v^2}{(1+u)^2 + v^2}}, \qquad \arg r = \operatorname{atan2}(-2v, 1 - u^2 - v^2).}$$

The phase is immaterial when $r = 0$, and otherwise is defined modulo 2π. This formula also handles real negative ϵ on the passive square-root branch. **(b)** For a poor conductor with positive ϵ and $\sigma \ll \omega\epsilon$, write $n_d = \sqrt{\epsilon/\epsilon_0}$. Then

$$N = n_d + \frac{is}{2n_d} + O(s^2), \qquad r = \frac{1-n_d}{1+n_d} - \frac{is}{n_d(1+n_d)^2} + O(s^2).$$

Thus the lossless Fresnel intensity is unchanged at first order, while the phase changes. In the matched case $n_d = 1$ the leading reflection is $r = -is/4$ and $R = s^2/16 + O(s^4)$. For a good conductor, $\sigma \gg \omega\epsilon$, set $\delta = \sqrt{2/(\mu_0\sigma\omega)}$ and $p = \omega\delta/c \ll 1$. Then $N \simeq (1+i)/p$ and

$$r = -1 + \frac{2}{1+N} = -1 + (1-i)p + O(p^2).$$

Squaring the modulus and retaining the first correction yields

$$\boxed{R = 1 - 2p + O(p^2) = 1 - 2\frac{\omega}{c}\delta + O(p^2).}$$

The reflected field has nearly the same magnitude and the opposite sign as the incident field; on the phase branch approaching $-\pi$, $\arg r = -\pi + p + O(p^2)$.

7.5 A plane polarized electromagnetic wave $\mathbf{E} = \mathbf{E}_i e^{i\mathbf{k}\cdot\mathbf{x} - i\omega t}$ is incident normally on a flat uniform sheet of an *excellent* conductor ($\sigma \gg \omega\epsilon_0$) having a thickness D. Assuming that in space and in the conducting sheet $\mu/\mu_0 = \epsilon/\epsilon_0 = 1$, discuss the reflection and transmission of the incident wave.

(a) Show that the amplitudes of the reflected and transmitted waves, correct to the first order in $(\epsilon_0\omega/\sigma)^{1/2}$, are:

$$\frac{E_r}{E_i} = \frac{-(1-e^{-2\lambda})}{(1-e^{-2\lambda}) + \gamma(1+e^{-2\lambda})}$$
$$\frac{E_t}{E_i} = \frac{2\gamma e^{-\lambda}}{(1-e^{-2\lambda}) + \gamma(1+e^{-2\lambda})}$$

where

$$\gamma = \sqrt{\frac{2\epsilon_0\omega}{\sigma}}\,(1-i) = \frac{\omega\delta}{c}\,(1-i)$$
$$\lambda = (1-i)D/\delta$$

and $\delta = \sqrt{2/\omega\mu\sigma}$ is the penetration depth.

(b) Verify that for zero thickness and infinite thickness you obtain the proper limiting results.

(c) Show that, except for sheets of very small thickness, the transmission coefficient is

$$T = \frac{8(\mathrm{Re}\,\gamma)^2 e^{-2D/\delta}}{1 - 2e^{-2D/\delta}\cos(2D/\delta) + e^{-4D/\delta}}$$

Sketch $\log T$ as a function of (D/δ), assuming $\mathrm{Re}\,\gamma = 10^{-2}$.

Define "very small thickness."

Solution.

Reference E_t to the back face of the sheet, and put $k_0 = \omega/c$, $N = \sqrt{1 + i\sigma/(\epsilon_0\omega)}$. **(a)** Solving the two-interface problem exactly, as in Problem 7.2, gives

$$r = \frac{(1-N^2)(1-e^{2iNk_0D})}{(1+N)^2 - (1-N)^2 e^{2iNk_0D}}, \qquad t = \frac{4Ne^{iNk_0D}}{(1+N)^2 - (1-N)^2 e^{2iNk_0D}}.$$

For an excellent conductor, $N \simeq (1+i)/(k_0\delta)$, $2/N \simeq \gamma = k_0\delta(1-i)$, and $-iNk_0D \simeq \lambda = (1-i)D/\delta$. Divide numerator and denominator by N^2. The denominator becomes

$$(1 - e^{-2\lambda}) + \gamma(1 + e^{-2\lambda}) + \frac{\gamma^2}{4}(1 - e^{-2\lambda}).$$

The reflection numerator is $-(1-\gamma^2/4)(1-e^{-2\lambda})$ and the transmission numerator is $2\gamma e^{-\lambda}$. Dropping the quadratic corrections, while retaining the linear term in the denominator, gives the uniform thin-sheet approximation

$$\boxed{r = -\frac{1 - e^{-2\lambda}}{1 - e^{-2\lambda} + \gamma(1 + e^{-2\lambda})}, \qquad t = \frac{2\gamma e^{-\lambda}}{1 - e^{-2\lambda} + \gamma(1 + e^{-2\lambda})}.}$$

Keeping that denominator term is essential when the thickness also tends to zero. The approximation to the propagation exponent is understood at fixed D/δ in the good-conductor limit. **(b)** At $D = 0$, numerator and denominator give $r = 0, t = 1$. As $D/\delta \to \infty$, $r = -1/(1+\gamma) = -1 + \gamma + O(\gamma^2)$ and $t \to 0$, agreeing with the half-space result of Problem 7.4. For a thin but finite sheet, expand $e^{-2\lambda} = 1 - 2\lambda + O(\lambda^2)$ to obtain

$$r \simeq -\frac{\lambda}{\lambda + \gamma}, \qquad t \simeq \frac{\gamma}{\lambda + \gamma}.$$

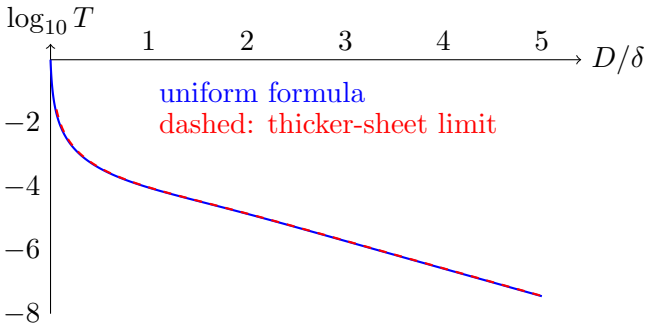
Since $\lambda/\gamma = Z_0\sigma D/2$, these are the resistive-sheet formulas $r = -g/(2+g)$ and $t = 2/(2+g)$ with $g = Z_0\sigma D$. **(c)** Neglect the γ term in the denominator when $|\tanh\lambda| \gg |\gamma|$. Then

$$T = |t|^2 \simeq \frac{4|\gamma|^2 e^{-2D/\delta}}{|1 - e^{-2(1-i)D/\delta}|^2} = \boxed{\frac{8(\mathrm{Re}\,\gamma)^2 e^{-2D/\delta}}{1 - 2e^{-2D/\delta}\cos(2D/\delta) + e^{-4D/\delta}}}.$$

In the thin limit the condition is

$$\boxed{D \gg k_0\delta^2 = \frac{2}{Z_0\sigma}.}$$

Thus "very small" here means $D \lesssim 2/(Z_0\sigma)$, a much smaller length than δ. Applying the simplified expression there would spuriously make T diverge as $D \to 0$. The plot for $\mathrm{Re}\,\gamma = 10^{-2}$ shows the uniform formula through that region and the simplified formula where valid. For $x = D/\delta \gg 1$, its straight-line asymptote is $\log_{10} T = \log_{10}(8 \times 10^{-4}) - 2x/\ln 10$.



7.6 A plane wave of frequency ω is incident normally from vacuum on a semi-infinite slab of material with a *complex* index of refraction $n(\omega)$ $[n^2(\omega) = \epsilon(\omega)/\epsilon_0]$.

(a) Show that the ratio of reflected power to incident power is

$$R = \left| \frac{1 - n(\omega)}{1 + n(\omega)} \right|^2$$

while the ratio of power transmitted into the medium to the incident power is

$$T = \frac{4 \,\mathrm{Re}\, n(\omega)}{|1 + n(\omega)|^2}$$

(b) Evaluate $\mathrm{Re}[i\omega(\mathbf{E} \cdot \mathbf{D}^* - \mathbf{B} \cdot \mathbf{H}^*)/2]$ as a function of (x, y, z). Show that this rate of change of energy per unit volume accounts for the relative transmitted power T.

(c) For a conductor, with $n^2 = 1 + i(\sigma/\omega\epsilon_0)$, σ real, write out the results of parts a and b in the limit $\epsilon_0\omega \ll \sigma$. Express your answer in terms of δ as much as possible. Calculate $\frac{1}{2}\mathrm{Re}(\mathbf{J}^* \cdot \mathbf{E})$ and compare with the result of part b. Do both enter the complex form of Poynting's theorem?

Solution.

Take the material in $z > 0$, $n = u + iv$ with $u, v \geq 0$, and $k_0 = \omega/c$. For $\mathbf{E}_i = E_i \mathbf{e}_x$ the transmitted fields are

$$\mathbf{E} = \frac{2E_i}{1+n} \mathbf{e}_x e^{ik_0 nz}, \qquad \mathbf{H} = \frac{n}{Z_0} \frac{2E_i}{1+n} \mathbf{e}_y e^{ik_0 nz}.$$

(a) The same boundary equations $1 + r = t$, $1 - r = nt$ give $r = (1 - n)/(1 + n)$ and $t = 2/(1 + n)$. The averaged incident flux is $|E_i|^2/(2Z_0)$, while the transmitted flux at the surface is $u|tE_i|^2/(2Z_0)$. Therefore

$$\boxed{R = \left| \frac{1-n}{1+n} \right|^2, \qquad T = \frac{4 \,\mathrm{Re}\, n}{|1+n|^2}.}$$

Expansion of the two squared moduli proves $R+T = 1$ at the interface. Here T measures flux entering the material, not the flux surviving to infinite depth. **(b)** The magnetic term $\mathbf{B} \cdot \mathbf{H}^* = \mu_0 |\mathbf{H}|^2$ is real. With $\epsilon = \epsilon_0 n^2$ one finds

$$\begin{aligned} q(z) &= \mathrm{Re}\, \frac{i\omega}{2} (\mathbf{E} \cdot \mathbf{D}^* - \mathbf{B} \cdot \mathbf{H}^*) \\ &= \frac{\omega}{2} \,\mathrm{Im}\, \epsilon \, |\mathbf{E}|^2 = \boxed{\frac{4\omega\epsilon_0 uv |E_i|^2}{|1+n|^2} e^{-2k_0 vz}}. \end{aligned}$$

There is no dependence on x, y. The flux at depth z is $S_z(z) = 2u|E_i|^2 e^{-2k_0 vz}/[Z_0|1 + n|^2]$; using $1/Z_0 = \epsilon_0 c$ verifies $q = -dS_z/dz$. For $v > 0$, integrating to infinity gives $\int_0^\infty q \, dz = S_z(0) = T|E_i|^2/(2Z_0)$. In the lossless case $v = 0$, the dissipation vanishes and the entering flux instead propagates to infinity; it must not be equated to a vanishing absorption integral. **(c)** For a good conductor write $p = k_0\delta \ll 1$. Then

$$n \simeq \frac{1+i}{p}, \quad t \simeq (1-i)p, \quad R \simeq 1 - 2p, \quad T \simeq 2p,$$

and

$$\boxed{q(z) \simeq \sigma p^2 |E_i|^2 e^{-2z/\delta} = 2\epsilon_0 \omega |E_i|^2 e^{-2z/\delta}.}$$

The equality uses $\sigma\delta^2 = 2/(\mu_0\omega)$. Ohm's law gives $\frac{1}{2}\,\mathrm{Re}(\mathbf{J}^* \cdot \mathbf{E}) = \sigma|\mathbf{E}|^2/2$, precisely the same heat production. It enters Poynting's theorem once: either keep an explicit conduction current with a real background permittivity, or absorb it into $\epsilon + i\sigma/\omega$. Including both descriptions of the same current would double-count the loss.

7.7 A ribbon beam of plane-polarized radiation of wavelength λ is totally reflected internally at a plane boundary between two nonpermeable media with indices of refraction n and n' ($n' < n$). As discussed in Section 7.4, the ratio of the reflected to incident amplitudes is a complex number of modulus unity, $E_0''/E_0 = \exp[i\phi(i, i_0)]$ for the angle of incidence $i > i_0$, where $\sin i_0 = n'/n$.

(a) Show that for a "monochromatic" ribbon beam of radiation in the z direction with an electric field amplitude, $E(x)e^{i(kz-\omega t)}$, where $E(x)$ is smooth and finite in transverse extent (but many wavelengths broad), the lowest order approximation in terms of plane waves is

$$\mathbf{E}(x, z, t) = \boldsymbol{\epsilon} \int d\kappa\; A(\kappa) e^{i\kappa x + ikz - i\omega t}$$

where $\boldsymbol{\epsilon}$ is a polarization vector, and $A(\kappa)$ is the Fourier transform of $E(x)$, with support in κ around $\kappa = 0$ small compared to k. The finite beam consists of plane waves with a small range of angles of incidence, centered around the geometrical optics value.

(b) Consider the reflected beam and show that for $i > i_0$ the electric field can be expressed approximately as

$$\mathbf{E}''(x, z, t) = \boldsymbol{\epsilon}'' E(x'' - \delta x)\, \exp[i\mathbf{k}'' \cdot \mathbf{x} - i\omega t + i\phi(i, i_0)]$$

where $\boldsymbol{\epsilon}''$ is a polarization vector, x'' is the x coordinate perpendicular to $\mathbf{k}''$, the reflected wave vector, and $\delta x = -(1/k)[d\phi(i, i_0)/di]$.

(c) With the Fresnel expressions of Section 7.3 for the phases $\phi(i, i_0)$ for the two states of plane polarization, show that the lateral displacements of reflected beams with respect to the geometric optics position are

$$D_\perp = \frac{\lambda}{\pi} \frac{\sin i}{(\sin^2 i - \sin^2 i_0)^{1/2}} \qquad \text{and} \qquad D_\parallel = D_\perp \cdot \frac{\sin^2 i_0}{[\sin^2 i - \cos^2 i \cdot \sin^2 i_0]}$$

The displacement is known as the *Goos–Hänchen effect* (op. cit.).

Solution.

Let $k = 2\pi/\lambda$ be the wave number in the incident medium; λ is correspondingly its wavelength. **(a)** Define $A(\kappa) = (2\pi)^{-1} \int E(x) e^{-i\kappa x}\, dx$. The problem writes the inverse integral without a prefactor, so this A is Jackson's unitary Fourier amplitude [Eqs. (7.80)–(7.81), p. 323] divided by $\sqrt{2\pi}$. The exact scalar angular spectrum in a homogeneous medium is

$$\int d\kappa\, A(\kappa) e^{i\kappa x + i\sqrt{k^2 - \kappa^2}\, z - i\omega t}.$$

A smooth beam of width $w \gg \lambda$ has its important spectrum at $|\kappa| \sim 1/w \ll k$. Expanding $\sqrt{k^2 - \kappa^2} = k - \kappa^2/(2k) + \cdots$ and keeping the lowest term gives the expression in the question.

The polarization vector can likewise be replaced by its central value to this order. The ignored quadratic phase describes diffraction and is small over propagation distances $|z| \ll k/(\Delta\kappa)^2$. The angular change of an individual component is $\Delta i \simeq \kappa/k$. **(b)** The reflected spectrum is multiplied by $e^{i\phi(i+\kappa/k)}$. Linearization of that phase gives

$$e^{i\phi(i+\kappa/k)} \simeq e^{i\phi(i)} e^{i\kappa\phi'(i)/k}.$$

After extracting the central reflected plane-wave phase, its transverse spectrum is therefore

$$\int A(\kappa) e^{i\kappa[x''+\phi'(i)/k]}\, d\kappa = E\big(x'' + \phi'(i)/k\big).$$

Thus

$$\boxed{\mathbf{E}'' \simeq \boldsymbol{\epsilon}'' E(x'' - \delta x) e^{i\mathbf{k}''\cdot\mathbf{x} - i\omega t + i\phi(i)}, \qquad \delta x = -\frac{1}{k}\frac{d\phi}{di}.}$$

This approximation requires the reflection phase to be approximately linear over the beam's angular spread. **(c)** Put $m = n'/n = \sin i_0$, $h = \sqrt{\sin^2 i - m^2}$, and $c_i = \cos i$. The two total-reflection phase factors, up to constant phases irrelevant to the shift, are

$$r_\perp = \frac{c_i - ih}{c_i + ih}, \qquad r_\parallel = \frac{m^2 c_i - ih}{m^2 c_i + ih}.$$

Hence $\phi_\perp = -2\tan^{-1}(h/c_i)$ and $\phi_\parallel = -2\tan^{-1}[h/(m^2 c_i)]$. Since $h' = \sin i \cos i/h$ and $h^2 + c_i^2 = 1 - m^2$,

$$\phi'_\perp = -\frac{2\sin i}{h}, \qquad \frac{\phi'_\parallel}{\phi'_\perp} = \frac{m^2(1-m^2)}{h^2 + m^4 c_i^2} = \frac{m^2}{\sin^2 i - m^2\cos^2 i}.$$

Substitution in part (b) gives the stated transverse displacements:

$$\boxed{D_\perp = \frac{\lambda}{\pi}\frac{\sin i}{\sqrt{\sin^2 i - \sin^2 i_0}}, \quad D_\parallel = D_\perp \frac{\sin^2 i_0}{\sin^2 i - \cos^2 i \sin^2 i_0}.}$$

Here $D = \delta x$ is measured perpendicular to the reflected beam. The corresponding displacement of the beam's footprint *along the interface* is $D/\cos i$. Close to i_0 the formal derivative diverges, but the linear phase approximation fails when the angular spread is comparable to $i - i_0$; a finite beam then requires the unexpanded angular-spectrum integral.

7.8 A monochromatic plane wave of frequency ω is incident normally on a stack of layers of various thicknesses t_j and lossless indices of refraction n_j. Inside the stack, the wave has both forward and backward moving components. The change in the wave through any interface and also from one side of a layer to the other can be described by means of 2×2 transfer matrices. If the electric field is written as

$$E = E_+ e^{ikx} + E_- e^{-ikx}$$

in each layer, the transfer matrix equation $E' = TE$ is explicitly

$$\begin{pmatrix} E'_+ \\ E'_- \end{pmatrix} = \begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \begin{pmatrix} E_+ \\ E_- \end{pmatrix}$$

(a) Show that the transfer matrix for propagation inside, but across, a layer of index of refraction n_j and thickness t_j is

$$T_{\text{layer}}(n_j, t_j) = \begin{pmatrix} e^{ik_j t_j} & 0 \\ 0 & e^{-ik_j t_j} \end{pmatrix} = I\cos(k_j t_j) + i\sigma_3 \sin(k_j t_j)$$

where $k_j = n_j\omega/c$, I is the unit matrix, and σ_k are the Pauli spin matrices of quantum mechanics. Show that the inverse matrix is T^*.

(b) Show that the transfer matrix to cross an interface from n_1 ($x < x_0$) to n_2 ($x > x_0$) is

$$T_{\text{interface}}(2,1) = \frac{1}{2}\begin{pmatrix} n+1 & -(n-1) \\ -(n-1) & n+1 \end{pmatrix} = I\,\frac{(n+1)}{2} - \sigma_1 \frac{(n-1)}{2}$$

where $n = n_1/n_2$.

(c) Show that for a complete stack, the incident, reflected, and transmitted waves are related by

$$E_{\text{trans}} = \frac{\det(T)}{t_{22}}\,E_{\text{inc}}, \qquad E_{\text{refl}} = -\frac{t_{21}}{t_{22}}\,E_{\text{inc}}$$

where t_{ij} are the elements of T, the product of the forward-going transfer matrices, including from the material filling space on the incident side into the first layer and from the last layer into the medium filling the space on the transmitted side.

Solution.

Amplitudes in a transfer vector are evaluated at a reference plane; this choice absorbs the local traveling-wave phases at that plane. If one instead uses fixed global exponential coordinates, the same relations acquire the corresponding reference-plane phase factors. **(a)** Across a uniform thickness t_j, the forward wave acquires $e^{ik_j t_j}$ and the backward wave acquires $e^{-ik_j t_j}$. Thus

$$\boxed{T_{\text{layer}} = \begin{pmatrix} e^{ik_j t_j} & 0 \\ 0 & e^{-ik_j t_j} \end{pmatrix} = I\cos(k_j t_j) + i\sigma_3 \sin(k_j t_j).}$$

Here $\sigma_3 = \text{diag}(1, -1)$. For real $k_j t_j$ its inverse is its componentwise complex conjugate, since each diagonal factor is then of unit modulus. **(b)** At an interface let $n = n_1/n_2$. Continuity of the tangential electric and magnetic fields gives

$$E_{2+} + E_{2-} = E_{1+} + E_{1-}, \qquad E_{2+} - E_{2-} = n(E_{1+} - E_{1-}).$$

Adding and subtracting these equations proves

$$\boxed{T_{\text{interface}}(2,1) = \frac{1}{2}\begin{pmatrix} n+1 & 1-n \\ 1-n & n+1 \end{pmatrix} = \frac{n+1}{2}I - \frac{n-1}{2}\sigma_1.}$$

Its determinant is n, and its inverse is the interface matrix with 1 and 2 interchanged. **(c)** Multiply matrices in the order encountered, the first one acting being on the right. With no wave incident from the transmitted side,

$$\begin{pmatrix} E_{\text{trans}} \\ 0 \end{pmatrix} = \begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \begin{pmatrix} E_{\text{inc}} \\ E_{\text{refl}} \end{pmatrix}.$$

The second row gives $E_{\rm refl} = -t_{21}E_{\rm inc}/t_{22}$. Inserting it in the first row gives

$$\boxed{E_{\rm trans} = \frac{t_{11}t_{22} - t_{12}t_{21}}{t_{22}}E_{\rm inc} = \frac{\det T}{t_{22}}E_{\rm inc}, \qquad E_{\rm refl} = -\frac{t_{21}}{t_{22}}E_{\rm inc}.}$$

The layer determinants are one and the interface determinants telescope, so $\det T = n_{\rm inc}/n_{\rm out}$. Flux conservation provides an independent check: $T^\dagger\sigma_3 T = (n_{\rm inc}/n_{\rm out})\sigma_3$. Consequently the amplitudes above obey $|r|^2 + (n_{\rm out}/n_{\rm inc})|t|^2 = 1$ for a lossless stack.

7.9 A stack of optical elements consists of N layers with index of refraction n and thickness t_1, separated by air gaps ($n_2 = 1$) of thickness t_2. A monochromatic plane wave is incident normally. With appropriate thicknesses, a modest number of layers can cause almost total reflection of a given range of frequencies, even for normal n values (e.g., $1.3 < n < 1.8$).

(a) Show the transfer matrix for the stack is $T_{\rm stack} = T^N(I\cos\alpha_2 - i\sigma_3\sin\alpha_2)$, where $\alpha_2 = \omega t_2/c$, and the single air gap plus foil transfer matrix is

$$\begin{aligned} T = (1/4n)\{&[(n+1)^2\cos(\alpha_1+\alpha_2) - (n-1)^2\cos(\alpha_1-\alpha_2)]I \\ &+ i\sigma_3[(n+1)^2\sin(\alpha_1+\alpha_2) + (n-1)^2\sin(\alpha_1-\alpha_2)] \\ &+ 2\sigma_1(n^2-1)\sin\alpha_1\sin\alpha_2 \\ &- 2\sigma_2(n^2-1)\sin\alpha_1\cos\alpha_2\} \end{aligned}$$

with $\alpha_1 = n\omega t_1/c$.

(b) If all the layers (both air gaps and foils) have optical thicknesses of a quarter-wavelength of the incident wave, show that

$$T = -\exp(-\lambda\sigma_1), \qquad \text{where } \lambda = \ln(n)$$

is (roughly) the amplitude "decay constant" per layer. Show that the fractional transmitted intensity is

$$\frac{|E_{\rm trans}|^2}{|E_{\rm inc}|^2} = \operatorname{sech}^2[N\ln(n)] = \frac{4n^{2N}}{(n^{2N}+1)^2} \to 4\,\exp[-N\ln(n^2)]$$

The asymptotic form holds for $n^{2N} \gg 1$.

Solution.

Let $P_j = I\cos\alpha_j + i\sigma_3\sin\alpha_j$, and write $I(b,a)$ for the interface matrix from index a to b. A single foil surrounded by air has transfer matrix

$$F = I(1,n)P_1 I(n,1) = I\cos\alpha_1 + iA\sigma_3\sin\alpha_1 - B\sigma_2\sin\alpha_1,$$

where $A = (n^2+1)/(2n)$ and $B = (n^2-1)/(2n)$. For example, the nontrivial multiplication follows from

$$I(1,n)\sigma_3 I(n,1) = A\sigma_3 + iB\sigma_2,$$

using $\sigma_3\sigma_1 = i\sigma_2$, $\sigma_1\sigma_3 = -i\sigma_2$ and $\sigma_1\sigma_3\sigma_1 = -\sigma_3$. **(a)** Define the cell to be a gap followed by a foil, so $T = FP_2$. Then

$$\begin{aligned} T = &(\cos\alpha_1\cos\alpha_2 - A\sin\alpha_1\sin\alpha_2)I \\ &+ i(\cos\alpha_1\sin\alpha_2 + A\sin\alpha_1\cos\alpha_2)\sigma_3 \\ &+ B\sin\alpha_1\sin\alpha_2\sigma_1 - B\sin\alpha_1\cos\alpha_2\sigma_2. \end{aligned}$$

Product-to-sum identities turn the first two coefficients into

$$\frac{(n+1)^2\cos(\alpha_1+\alpha_2)-(n-1)^2\cos(\alpha_1-\alpha_2)}{4n},$$

$$\frac{(n+1)^2\sin(\alpha_1+\alpha_2)+(n-1)^2\sin(\alpha_1-\alpha_2)}{4n},$$

respectively, while the last two are exactly the coefficients printed in the question. A stack with N foils and $N-1$ gaps is $FP_2FP_2\cdots F=(FP_2)^NP_2^{-1}$; hence

$$\boxed{T_{\text{stack}}=T^N(I\cos\alpha_2-i\sigma_3\sin\alpha_2).}$$

(b) With $\alpha_1=\alpha_2=\pi/2$, $T=-AI+B\sigma_1$. If $\lambda=\ln n$, then $A=\cosh\lambda$ and $B=\sinh\lambda$. Since $\sigma_1^2=I$,

$$\boxed{T=-e^{-\lambda\sigma_1},\quad T^N=(-1)^N[I\cosh(N\lambda)-\sigma_1\sinh(N\lambda)].}$$

Multiplication by $P_2^{-1}=-i\sigma_3$ only changes the phases of its columns. The magnitude of t_{22} for the full stack is $\cosh(N\lambda)$ and its determinant is one. The incident and transmitted media are both air, so

$$\boxed{\frac{|E_{\text{trans}}|^2}{|E_{\text{inc}}|^2}=\operatorname{sech}^2(N\ln n)=\frac{4n^{2N}}{(n^{2N}+1)^2}\sim 4e^{-N\ln(n^2)}.}$$

Thus the decaying cell eigenvalue has magnitude $e^{-\lambda}$; intensity transmission decreases exponentially with the number of quarter-wave foils, with prefactors fixed by the end interfaces.

7.10 An arbitrary optical element of length L is placed in a uniform nonabsorbing medium with index of refraction $n(\omega)$ with its front face at $x=0$ and its back face at $x=L$. If a monochromatic plane wave of frequency ω with amplitude $\psi_{\text{inc}}(\omega,x,t)=\exp[ik(\omega)x-i\omega t]$ is incident on the front face of the element, the transmitted wave amplitude is $\psi_{\text{trans}}(\omega,x,t)=T(\omega)\exp[ik(\omega)(x-L)-i\omega t]$, where the relative transmission amplitude $T(\omega)=\tau(\omega)\exp[i\phi(\omega)]$ is a complex quantity of magnitude $\tau(\omega)$ and phase $\phi(\omega)$.

A plane wave of radiation $\psi_{\text{inc}}(x,t)$, consisting of a coherent superposition of different frequencies centered around $\omega=\omega_0$, with support $A(\omega)$ narrow on the scale of variation of $\tau(\omega)$, $\phi(\omega)$ and $k(\omega)$, is incident on the optical element. Show that the transmitted wave for $x>L$ is approximately

$$\psi_{\text{trans}}(x,t)\approx\tau(\omega_0)e^{i\beta}\psi_{\text{inc}}(x',t')$$

where β is a constant phase and $x'=x-L$, $t'=t-T$. The transit or group delay time (sometimes attributed in another context to E. P. Wigner) is $T=[d\phi(\omega)/d\omega]_{\omega=\omega_0}$. If $cT<L$, some authors speak of superluminal propagation through the element. Discuss.

Solution.

Write the incident packet as $\psi_{\text{inc}}(x,t)=(2\pi)^{-1/2}\int A(\omega)e^{ik(\omega)x-i\omega t}\,d\omega$. Each frequency is transmitted independently, giving

$$\psi_{\text{trans}}(x,t)=\frac{1}{\sqrt{2\pi}}\int A(\omega)\tau(\omega)e^{i\phi(\omega)+ik(\omega)(x-L)-i\omega t}\,d\omega.$$

Across its narrow spectrum approximate $\tau(\omega) \simeq \tau_0$ and $\phi(\omega) \simeq \phi_0 + T(\omega - \omega_0)$, where $T = \phi'(\omega_0)$. Combining terms in the exponent then gives

$$\psi_{\text{trans}}(x,t) \simeq \frac{\tau_0 e^{i(\phi_0 - \omega_0 T)}}{\sqrt{2\pi}} \int A(\omega) e^{ik(\omega)(x-L) - i\omega(t-T)}\, d\omega$$
$$= \boxed{\tau_0 e^{i\beta} \psi_{\text{inc}}(x - L, t - T), \qquad \beta = \phi_0 - \omega_0 T.}$$

The conditions include $|\tau'/\tau_0|\Delta\omega \ll 1$ and $|\phi''|(\Delta\omega)^2 \ll 1$; otherwise the packet is distorted, and a single translation does not describe it. A linear expansion of $k(\omega)$ identifies the usual group motion of its envelope in the surrounding medium, but is not needed for the algebraic shift relation itself.

The quantity T measures the shift of a narrow-band envelope or its peak. It is not in general the arrival time of a new signal front. A frequency-selective element can reshape the leading tail and thereby advance the transmitted peak, giving $cT < L$ or even a negative T. A sharp front contains frequencies outside this narrow-band approximation. For a causal electromagnetic material whose high-frequency index tends to one, the high-frequency propagator fixes front propagation at no more than c, consistently with the causal response discussed in Section 7.10. Thus $cT < L$ by itself does not establish faster-than-light transport of new information; the approximation used to infer it cannot describe the required front.

7.11 A simple example of the transit time of the preceding problem is afforded by a slab of lossless dielectric of thickness d and index of refraction n in vacuum.

(a) For a plane wave incident normally, show that the magnitude of the transmitted amplitude is

$$|\tau(\omega)| = \frac{4n}{\sqrt{[(n+1)^2 - (n-1)^2\cos(2z)]^2 + [(n-1)^2\sin(2z)]^2}}$$

while its phase is

$$\phi(\omega) = z + \arctan\left[\frac{(n-1)^2\sin(2z)}{(n+1)^2 - (n-1)^2\cos(2z)}\right]$$

where $z = n\omega d/c$.

(b) Neglecting dispersion, show that for $z \to 0$ and $z = \pi$, $|\tau| = 1.0$ and $cT/d = (n^2+1)/2$, while for $z = \pi/2$ (quarter-wave plate), $|\tau| = 2n/(n^2+1)$ and $cT/d = 2n^2/(n^2+1)$. Show also that cT/d, averaged from $z = 0$ to any integer multiple of $\pi/2$, is $\langle cT/d\rangle = n$. Does this result tell you anything about what you might expect for the observed transit time of a long wave train ($\Delta\omega/\omega \ll 1$) through a piece of window glass? Explain.

(c) Calculate numerically and plot the results as functions of z for the magnitude of the transmission amplitude, its phase, and the transit time in units of d/c for $n = 1.5$ and $n = 2.0$.

Solution.

(a) The two-interface result for a slab in vacuum is

$$t(\omega) = \frac{4ne^{iz}}{(n+1)^2 - (n-1)^2 e^{2iz}}, \qquad z = \frac{n\omega d}{c}.$$

The denominator has real and imaginary parts $(n+1)^2 - (n-1)^2\cos 2z$ and $-(n-1)^2\sin 2z$. Taking its modulus and subtracting its phase proves the two expressions in the question. Its real part is always positive for $n > 0$, so the continuous, unwrapped phase is

$$\phi = z + \tan^{-1}\frac{(n-1)^2\sin 2z}{(n+1)^2 - (n-1)^2\cos 2z}.$$

It is useful to divide numerator and denominator by $4ne^{iz}$:

$$t = \frac{1}{\cos z - iA\sin z}, \quad A = \frac{n^2+1}{2n}, \qquad |t|^2 = \frac{1}{\cos^2 z + A^2 \sin^2 z}.$$

(b) Differentiation of the argument of $\cos z + iA\sin z$ gives

$$\frac{d\phi}{dz} = \frac{A\cos^2 z + A\sin^2 z}{\cos^2 z + A^2\sin^2 z} = \frac{A}{\cos^2 z + A^2 \sin^2 z}.$$

For nondispersive n, $dz/d\omega = nd/c$, hence

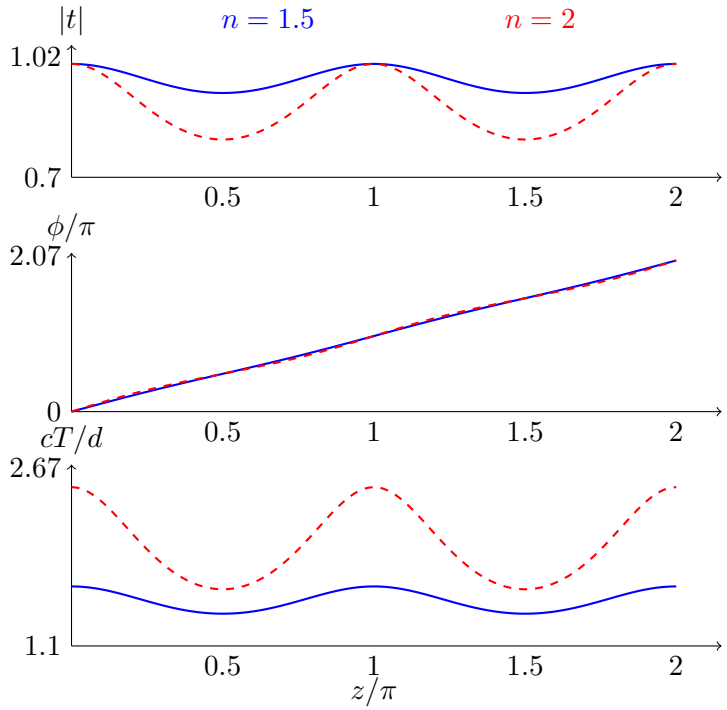
$$\boxed{\frac{cT}{d} = \frac{nA}{\cos^2 z + A^2 \sin^2 z} = nA|t|^2.}$$

At $z = 0, \pi$, $|t| = 1$ and $cT/d = nA = (n^2+1)/2$; at $z = \pi/2$, $|t| = 1/A = 2n/(n^2+1)$ and $cT/d = n/A = 2n^2/(n^2+1)$. On a quarter-wave interval with endpoints at consecutive multiples of $\pi/2$, substitute $u = \tan z$ (or use symmetry for the next interval) to find

$$\int_0^{\pi/2} \frac{A\,dz}{\cos^2 z + A^2 \sin^2 z} = \int_0^{\infty} \frac{A\,du}{1 + A^2 u^2} = \frac{\pi}{2}.$$

Consequently $\langle cT/d \rangle = n$ from $z = 0$ to any integer multiple of $\pi/2$, or over any whole fringe period π. An interval of length $\pi/2$ starting at an arbitrary phase need not have that average; the quarter-wave endpoints matter.

This unweighted average over optical phase is not a universal measurement of a packet's transit time. A sufficiently narrow spectrum, $nd\Delta\omega/c \ll 1$, samples the delay at its carrier phase, anywhere between the two limits above. Even a long train with $\Delta\omega/\omega \ll 1$ can cover many fringes in a thick window because $z \gg 1$. In that regime the single-delay approximation can fail and the separate multiply reflected pulses, coherence, and detector weighting must be considered. The direct transmitted pulse travels in time nd/c; the displayed average explains that scale but does not fix every measured peak or energy-centroid delay. **(c)** The plotted curves use the exact amplitude, unwrapped phase, and analytic derivative, not numerical differentiation of a wrapped arctangent. For $n = 1.5$, the amplitude ranges from $12/13$ to 1 and cT/d from $18/13$ to $13/8$. For $n = 2$, the ranges are $4/5$ to 1 and $8/5$ to $5/2$. The phase increases by π over each period of the amplitude.



7.12 The time dependence of electrical disturbances in good conductors is governed by the frequency-dependent conductivity (7.58). Consider longitudinal electric fields in a conductor, using Ohm's law, the continuity equation, and the differential form of Coulomb's law.

(a) Show that the time-Fourier-transformed charge density satisfies the equation

$$[\sigma(\omega) - i\omega\epsilon_0]\rho(\mathbf{x},\omega) = 0$$

(b) Using the representation $\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$, where $\sigma_0 = \epsilon_0\omega_p^2\tau$ and τ is a damping time, show that in the approximation $\omega_p\tau \gg 1$ any initial disturbance will oscillate with the plasma frequency and decay in amplitude with a decay constant $\lambda = 1/2\tau$. Note that if you use $\sigma(\omega) \simeq \sigma(0) = \sigma_0$ in part a, you will find no oscillations and extremely rapid damping with the (wrong) decay constant $\lambda_w = \sigma_0/\epsilon_0$.

Reference: W. M. Saslow and G. Wilkinson, *Am. J. Phys.* **39**, 1244 (1971).

Solution.

Use the convention $e^{-i\omega t}$ and take a homogeneous conductor. **(a)** The three relevant equations are

$$-i\omega\rho + \nabla\cdot\mathbf{J} = 0, \qquad \mathbf{J} = \sigma(\omega)\mathbf{E}, \qquad \nabla\cdot\mathbf{E} = \rho/\epsilon_0.$$

Their elimination gives $[-i\omega + \sigma(\omega)/\epsilon_0]\rho = 0$, or

$$\boxed{[\sigma(\omega) - i\omega\epsilon_0]\rho(\mathbf{x},\omega) = 0.}$$

This is the free longitudinal relaxation condition, with no imposed external charge source. It applies pointwise to each spatial mode. **(b)** Substitution of the Drude conductivity and multiplication by $1 - i\omega\tau$ yield

$$\left(\omega_p^2 - \omega^2 - \frac{i\omega}{\tau}\right)\rho = 0, \qquad \omega_\pm = \pm\sqrt{\omega_p^2 - \frac{1}{4\tau^2}} - \frac{i}{2\tau}.$$

Equivalently the time-domain equation is

$$\ddot{\rho} + \tau^{-1}\dot{\rho} + \omega_p^2 \rho = 0.$$

Writing $\Omega = \sqrt{\omega_p^2 - 1/(4\tau^2)}$, its solution for general initial charge and initial current divergence is

$$\boxed{\rho(\mathbf{x},t) = e^{-t/(2\tau)}\left[\rho(\mathbf{x},0)\cos\Omega t + \frac{\dot{\rho}(\mathbf{x},0) + \rho(\mathbf{x},0)/(2\tau)}{\Omega}\sin\Omega t\right].}$$

For $\omega_p\tau \gg 1$, $\Omega \simeq \omega_p$ and the amplitude decay constant is $1/(2\tau)$. In contrast, replacing $\sigma(\omega)$ by its static value before solving gives $\dot{\rho} = -(\sigma_0/\epsilon_0)\rho$ and $\lambda_w = \omega_p^2\tau$. That approximation requires $|\omega\tau| \ll 1$, whereas the actual modes have $|\omega\tau| \simeq \omega_p\tau \gg 1$. Even its predicted decay rate would give $\lambda_w\tau = (\omega_p\tau)^2 \gg 1$, outside its own range of validity.

7.13 A stylized model of the ionosphere is a medium described by the dielectric constant (7.59). Consider the earth with such a medium beginning suddenly at a height h and extending to infinity. For waves with polarization both perpendicular to the plane of incidence (from a horizontal antenna) and in the plane of incidence (from a vertical antenna),

(a) show from Fresnel's equations for reflection and refraction that for $\omega > \omega_p$ there is a range of angles of incidence for which reflection is not total, but for larger angles there is total reflection back toward the earth.

(b) A radio amateur operating at a wavelength of 21 meters in the early evening finds that she can receive distant stations located more than 1000 km away, but none closer. Assuming that the signals are being reflected from the F layer of the ionosphere at an effective height of 300 km, calculate the electron density. Compare with the known maximum and minimum F layer densities of $\sim 2 \times 10^{12}$ m^{-3} in the daytime and $\sim$ (2-4) $\times 10^{11}$ m^{-3} at night.

Solution.

(a) In the collisionless model (7.59), $n^2 = \epsilon/\epsilon_0 = 1 - \omega_p^2/\omega^2$, with $\mu = \mu_0$. For $\omega > \omega_p$, $0 < n < 1$. Conservation of the tangential wave number gives the normal transmitted wave number

$$k_z' = \frac{\omega}{c}q, \qquad q = \sqrt{n^2 - \sin^2 i} = \sqrt{\cos^2 i - \omega_p^2/\omega^2},$$

where the branch has $q \geq 0$ or $\mathrm{Im}\, q > 0$. The reflection coefficients, with fixed tangential electric-field reference directions, have magnitudes given by

$$r_\perp = \frac{\cos i - q}{\cos i + q}, \qquad r_\parallel = \frac{q - n^2\cos i}{q + n^2\cos i}.$$

For the parallel wave $Y_1 = 1/\cos i$ and $Y_2 = n^2/q$; continuity gives $1 + r = t$ and $Y_1(1 - r) = Y_2 t$, hence the displayed $r_\parallel = (Y_1 - Y_2)/(Y_1 + Y_2)$. A reflected polarization vector opposite to the fixed tangential direction, as in the usual Fresnel basis, changes its sign but not its magnitude. When q is real there is a transmitted flux and generally $|r| < 1$; for the parallel polarization there is also a Brewster zero. At

$$\boxed{\sin i_c = n, \qquad \cos i_c = \omega_p/\omega,}$$

q vanishes. For $i > i_c$, write $q = i\kappa$ with real $\kappa > 0$. Both Fresnel coefficients are ratios of complex conjugates, so $|r_\perp| = |r_\parallel| = 1$ and the field above the interface is evanescent. For $\omega \leq \omega_p$ there is no propagating transmitted wave at any incidence angle.

(b) First use the locally flat-earth approximation. The shortest one-hop ground range $D = 1000\,\text{km}$ corresponds to the critical ray, so

$$\tan i_c = \frac{D}{2h} = \frac{5}{3}, \qquad \frac{\omega_p}{\omega} = \frac{2h}{\sqrt{D^2+4h^2}} = \frac{3}{\sqrt{34}}.$$

With $\omega = 2\pi c/(21\,\text{m})$ and $\omega_p^2 = n_e e^2/(\epsilon_0 m_e)$,

$$\boxed{\omega_p = 4.6149\times10^7\,\text{s}^{-1}, \qquad n_e = \frac{\epsilon_0 m_e}{e^2}\left(\frac{2\pi c}{21\,\text{m}}\right)^2\frac{9}{34} = 6.69\times10^{11}\,\text{m}^{-3}.}$$

This is below the stated daytime maximum and above the stated nighttime range, consistent with an intermediate early-evening value in this simplified model. Both polarizations have the same critical angle and give the same inferred density.

Earth curvature is a modest systematic correction to this estimate. If a spherical earth of radius R_E is used, with D the ground arc length, put $\vartheta = D/(2R_E)$. At the reflection point,

$$\cos i_c = \frac{h + R_E(1-\cos\vartheta)}{\sqrt{R_E^2\sin^2\vartheta + [h + R_E(1-\cos\vartheta)]^2}}.$$

For the explicitly chosen radius $R_E = 6370\,\text{km}$ this becomes 0.53898 and gives $n_e = 7.34\times10^{11}\,\text{m}^{-3}$. The problem specifies no earth radius; the flat result above is the usual plane-layer estimate, while this formula states the geometry needed for a spherical refinement.

7.14 A simple model of propagation of radio waves in the earth's atmosphere or ionosphere consists of a flat earth at $z = 0$ and a nonuniform medium with $\epsilon = \epsilon(z)$ for $z > 0$. Consider the Maxwell equations under the assumption that the fields are independent of y and can be written as functions of z times $e^{i(kx-\omega t)}$.

(a) Show that the wave equation governing the propagation for $z > 0$ is

$$\frac{d^2F}{dz^2} + q^2(z)F = 0$$

where

$$q^2(z) = \omega^2\mu_0\epsilon(z) - k^2$$

and $F = E_y$ for *horizontal* polarization, and

$$q^2(z) = \omega^2\mu_0\epsilon(z) + \frac{1}{2\epsilon}\frac{d^2\epsilon}{dz^2} - \frac{3}{4\epsilon^2}\left(\frac{d\epsilon}{dz}\right)^2 - k^2$$

with $F = \sqrt{\epsilon/\epsilon_0}\,E_z$ for *vertical* polarization when $k \neq 0$. At $k = 0$, use a transverse electric component and the horizontal-polarization equation.

(b) Use the WKB approximation to treat the propagation of waves directed vertically into the ionosphere ($k = 0$), assuming that the dielectric constant is given by (7.59) with a plasma frequency $\omega_p(z)$ governed by an electron density like that shown in Fig. 7.11. Verify that the qualitative arguments in Section 7.6 hold, with departures in detail only for $\omega \sim \omega_{p,\max}$.

(c) Using the WKB results of part b and the concepts of the propagation of a pulse from Section 7.8, define an effective height of the ionosphere $h'(\omega)$ by calculating the time T for a pulse of dominant frequency ω to travel up and be reflected back ($h' \equiv cT/2$). [The WKB approximation is discussed in most books on quantum mechanics.]

Solution.

Assume $\mu = \mu_0$ and use the stated $e^{ikx-i\omega t}$ dependence. **(a)** For horizontal polarization, set $\mathbf{E} = E_y(z)\mathbf{e}_y$. Faraday's law gives $H_x = -E_y'/(i\omega\mu_0)$ and $H_z = kE_y/(\omega\mu_0)$. The y component of Ampère's law then gives

$$E_y'' + [\omega^2\mu_0\epsilon(z) - k^2]E_y = 0.$$

For vertical polarization set $\mathbf{H} = H_y(z)\mathbf{e}_y$. Ampère's law gives

$$E_x = \frac{H_y'}{i\omega\epsilon}, \qquad E_z = -\frac{kH_y}{\omega\epsilon}.$$

Substitution into $E_x' - ikE_z = i\omega\mu_0 H_y$ yields

$$H_y'' - \frac{\epsilon'}{\epsilon}H_y' + [\omega^2\mu_0\epsilon - k^2]H_y = 0.$$

Put $H_y = \sqrt{\epsilon}\,F$ and differentiate both factors. The coefficient of F' cancels, leaving

$$\boxed{F'' + \left[\omega^2\mu_0\epsilon - k^2 + \frac{\epsilon''}{2\epsilon} - \frac{3\epsilon'^2}{4\epsilon^2}\right]F = 0.}$$

For $k \neq 0$, $\sqrt{\epsilon/\epsilon_0}E_z = -kF/(\omega\sqrt{\epsilon_0})$, so this is precisely the stated choice of dependent variable up to a constant. At strictly normal incidence $E_z = 0$; that variable cannot describe a nonzero wave. One should instead choose either transverse electric component, which obeys the horizontal-polarization equation. This also avoids a spurious singular description at a zero of ϵ.

(b) At $k = 0$ the physical transverse equation has

$$q^2(z) = \frac{\omega^2 - \omega_p^2(z)}{c^2}.$$

Away from turning points, the WKB solutions in an allowed region are

$$F(z) \simeq \frac{C_+}{\sqrt{q(z)}}e^{i\int^z q(s)\,ds} + \frac{C_-}{\sqrt{q(z)}}e^{-i\int^z q(s)\,ds}, \qquad |q'| \ll q^2.$$

In a forbidden region put $q = i\kappa$; the corresponding solutions are $\kappa^{-1/2}\exp(\pm\int\kappa\,dz)$. For $\omega < \omega_{p,\max}$, the rising density produces a turning point z_t with $\omega_p(z_t) = \omega$. Near a simple turning point write $q^2(z) \simeq -\alpha(z - z_t)$, with $\alpha > 0$, and put $\xi = \alpha^{1/3}(z - z_t)$. Then the equation becomes $F_{\xi\xi} - \xi F = 0$. Its decaying solution $\mathrm{Ai}(\xi)$ has the allowed-side asymptote

$$\mathrm{Ai}(-X) \sim \frac{1}{\sqrt{\pi}X^{1/4}}\sin\left(\frac{2}{3}X^{3/2} + \frac{\pi}{4}\right), \qquad X > 0.$$

Here $\frac{2}{3}X^{3/2} = \int_z^{z_t} q(s)\,ds$ to the linearized order. Writing the sine as two exponentials gives equal incoming and reflected amplitudes and thus equal flux. Their phase ratio at $z = 0$ is $-i\exp[2i\int_0^{z_t} q\,dz]$, which will give the delay in part (c). If the density decreases again above its

maximum there is a second turning point z_2; a thick forbidden layer transmits only a fraction of order

$$\mathcal{T} \sim \exp\left[-2\int_{z_t}^{z_2} \frac{\sqrt{\omega_p^2(z)-\omega^2}}{c}\,dz\right].$$

For $\omega > \omega_{p,\max}$, q^2 is positive throughout and a smooth layer transmits with small WKB reflection. Close to the maximum the turning points merge, or the minimum positive q becomes too small for the WKB inequality; partial reflection and transmission then require a uniform turning-point treatment. These are the qualifications to the reflection/cutoff picture of Section 7.6. The required slow density variation is part of the WKB model; a sharp layer would instead require Fresnel matching.

(c) Refer the reflected wave to $z = 0$. For an isolated simple turning point its phase is

$$\phi_r(\omega) = 2\int_0^{z_t(\omega)} q(z,\omega)\,dz - \frac{\pi}{2}.$$

The narrow-band group delay is $T = d\phi_r/d\omega$. Since $q(z_t,\omega) = 0$, differentiating the endpoint gives no term. Thus

$$\boxed{T(\omega) = \frac{2}{c}\int_0^{z_t(\omega)} \frac{dz}{\sqrt{1-\omega_p^2(z)/\omega^2}}, \qquad h'(\omega) = \int_0^{z_t(\omega)} \frac{dz}{\sqrt{1-\omega_p^2(z)/\omega^2}}.}$$

The virtual height h' exceeds the physical turning height because the group velocity $c\sqrt{1-\omega_p^2/\omega^2}$ is below c. The square-root singularity is integrable at a simple turning point. Near a smooth density maximum the growing delay and coalescing turning points demand the same uniform treatment as in part (b). A frequency with no appreciable reflected pulse has no reflection height defined by this prescription.

7.15 The partially ionized interstellar medium (mostly hydrogen) responds to optical frequencies as an electronic plasma in a weak magnetic field. The broad-spectrum pulses from a pulsar allow determination of some average properties of the interstellar medium (e.g., mean electron density and mean magnetic field). The treatment of an electronic plasma in a magnetic field of Section 7.6 is pertinent.

(a) Ignoring the weak magnetic field and assuming that $\max(\omega_p) \ll \omega$, show that c times the transit time of a pulse of mean frequency ω from a pulsar a distance R away is

$$ct(\omega) \approx R + \frac{e^2}{2\epsilon_0 m_\mathrm{e}\omega^2}\int n_\mathrm{e}(z)\,dz$$

where $n_\mathrm{e}(z)$ is the electron density along the path of the light.

(b) The presence of the magnetic field causes a rotation of the plane of linear polarization (Faraday effect). Show that to lowest order in the magnetic field, the polarized light from the pulsar has its polarization rotated through an angle $\delta\theta(\omega)$:

$$\delta\theta(\omega) \approx -\frac{e^3}{2\epsilon_0 c m_\mathrm{e}^2\omega^2}\int n_\mathrm{e}(z)B_\parallel(z)\,dz$$

where $B_\parallel(z)$ is the component of $\mathbf{B}$ parallel to the path of the light. The signed angle here is positive from $\mathbf{e}_x$ toward $-\mathbf{e}_y$ in a right-handed basis with $\mathbf{e}_z$ along propagation; the opposite azimuth convention reverses the sign.

(c) Assuming you had an independent measure of the pulsar distance R, what observations would you make in order to infer $\langle n_{\rm e}\rangle$ and $\langle B_\parallel\rangle$? What assumptions, if any, about the polarization are necessary?

Solution.

Take z along propagation, neglect collisions and ion motion, and assume the density and magnetic field change slowly on a wavelength. **(a)** From $k = c^{-1}\sqrt{\omega^2 - \omega_p^2(z)}$,

$$\frac{dt}{dz} = \frac{\partial k}{\partial \omega} = \frac{1}{c\sqrt{1-\omega_p^2(z)/\omega^2}} = \frac{1}{c}\left[1 + \frac{\omega_p^2(z)}{2\omega^2} + \cdots\right].$$

Using $\omega_p^2(z) = n_e(z)e^2/(\epsilon_0 m_e)$ and integrating along the path gives

$$\boxed{ct(\omega) = R + \frac{e^2}{2\epsilon_0 m_e \omega^2}\int_0^R n_e(z)\,dz + O(\omega^{-4}).}$$

(b) Choose a right-handed transverse basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ and define $\boldsymbol{\epsilon}_\pm = (\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$. The helicity conventions of (7.67) give, to first order in $\omega_B = eB_\parallel/m_e$ and leading order in ω_p^2/ω^2,

$$n_\pm \simeq 1 - \frac{\omega_p^2}{2\omega^2} \mp \frac{\omega_p^2\omega_B}{2\omega^3}, \qquad k_+ - k_- = -\frac{\omega_p^2\omega_B}{c\omega^2}.$$

A wave initially along $\mathbf{e}_x$ is the equal superposition of these two helicities. After propagation its Jones vector, apart from a common phase, is

$$\mathbf{e}_x \cos\frac{\phi_+ - \phi_-}{2} - \mathbf{e}_y \sin\frac{\phi_+ - \phi_-}{2}, \qquad \phi_\pm = \int_0^R k_\pm(z)\,dz.$$

Consequently the angle convention specified in the amended question, $\delta\theta = (\phi_+ - \phi_-)/2$, gives

$$\boxed{\delta\theta = -\frac{e^3}{2\epsilon_0 c m_e^2 \omega^2}\int_0^R n_e(z)B_\parallel(z)\,dz.}$$

If polarization azimuth is instead measured positively from $\mathbf{e}_x$ toward $\mathbf{e}_y$, its change is $\delta\psi = -\delta\theta$. Stating the angle convention is essential for a signed Faraday formula; the magnitude is independent of it.

(c) Measure pulse arrival times in several frequency bands and fit the coefficient of ω^{-2}. A common unknown emission time is removed by taking differences of arrival times. The fitted column density $\mathcal{N}_e = \int n_e\,dz$ gives

$$\langle n_e\rangle = \mathcal{N}_e/R.$$

Measure the linear-polarization angle in the same bands, unwrap its π ambiguity, and fit its ω^{-2} coefficient. This gives $\mathcal{R} = \int n_e B_\parallel\,dz$, and hence

$$\boxed{\langle B_\parallel\rangle_{n_e} = \frac{\mathcal{R}}{\mathcal{N}_e}.}$$

This is an electron-density-weighted magnetic field, not generally the ordinary length average. Interpreting it as the latter requires nearly uniform density or an appropriate absence of correlations. The source must have measurable linear polarization; its intrinsic angle must be independent of frequency over the band, or otherwise known well enough to remove its frequency dependence. Similarly, intrinsic frequency-dependent pulse emission delays must be negligible or modeled. Foreground contributions other than the path being inferred must also be included in those fitted columns.

7.16 Plane waves propagate in a homogeneous, nonpermeable, but *anisotropic* dielectric. Assume a transparent reciprocal dielectric, with a real symmetric positive tensor ϵ_{ij}. If coordinate axes are chosen as the principal axes, the components of displacement along these axes are related to the electric-field components by $D_i = \epsilon_i E_i$ $(i = 1, 2, 3)$, where ϵ_i are the eigenvalues of the matrix ϵ_{ij}.

(a) Show that plane waves with frequency ω and wave vector $\mathbf{k}$ must satisfy

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{E}) + \mu_0 \omega^2 \mathbf{D} = 0$$

(b) Show that for a given wave vector $\mathbf{k} = k\mathbf{n}$ there are two polarization modes, whose phase velocities $v = \omega/k$ may coincide. Their velocities satisfy the Fresnel equation in its denominator-cleared polynomial form; for nonzero denominators this is

$$\sum_{i=1}^{3} \frac{n_i^2}{v^2 - v_i^2} = 0$$

where $v_i = 1/\sqrt{\mu_0 \epsilon_i}$ is called a principal velocity, and n_i is the component of $\mathbf{n}$ along the ith principal axis.

(c) Show that $\mathbf{D}_a \cdot \mathbf{D}_b = 0$ for distinct phase velocities, and that orthogonal displacements can be chosen in the degenerate case, where $\mathbf{D}_a$, $\mathbf{D}_b$ are the displacements associated with the two modes of propagation.

Solution.

The real, symmetric, positive dielectric tensor specified in the amended statement is needed for real principal velocities and the ordinary orthogonality asserted here. Write $V = \mathrm{diag}(v_1^2, v_2^2, v_3^2)$, so $\mathbf{E} = \mu_0 V \mathbf{D}$. **(a)** The Fourier-space Maxwell equations are

$$\mathbf{k} \times \mathbf{E} = \omega \mathbf{B}, \qquad \mathbf{k} \times \mathbf{B} = -\mu_0 \omega \mathbf{D}.$$

Taking one more cross product in the first equation gives

$$\boxed{\mathbf{k} \times (\mathbf{k} \times \mathbf{E}) + \mu_0 \omega^2 \mathbf{D} = 0.}$$

Its scalar product with $\mathbf{k}$ gives $\mathbf{n} \cdot \mathbf{D} = 0$ for a nonzero frequency. **(b)** With $P = I - \mathbf{n}\mathbf{n}^{\mathsf{T}}$, the vector equation becomes the two-dimensional eigenvalue problem

$$v^2 \mathbf{D} = PV\mathbf{D} = PVP\mathbf{D}, \qquad \mathbf{D} \in \mathbf{n}^{\perp}.$$

The restriction of the real symmetric operator PVP to this plane has two positive eigenvalues, counted with multiplicity. Componentwise,

$$(v^2 - v_i^2) D_i = -n_i S, \qquad S = \mathbf{n} \cdot V\mathbf{D}.$$

If $S \neq 0$ and none of the denominators vanish, imposing $\sum_i n_i D_i = 0$ proves the rational Fresnel equation. The form valid also at exceptional directions is its denominator-cleared quadratic:

$$\boxed{\sum_{i=1}^{3} n_i^2 \prod_{j \neq i} (v^2 - v_j^2) = 0.}$$

For example, for $\mathbf{n} = (1, 0, 0)$ its roots are v_2^2 and v_3^2, with displacements along axes 2 and 3. Neither appears as a root of the unqualified expression $1/(v^2 - v_1^2) = 0$. This explicit principal-axis example verifies why the polynomial qualification in the statement is necessary. Eigenvalues may coincide, for example in an isotropic material or on an optic axis; "two different phase velocities" is not valid in those cases. **(c)** Symmetry of PVP gives

$$v_a^2 \mathbf{D}_b \cdot \mathbf{D}_a = \mathbf{D}_b \cdot PVP\mathbf{D}_a = \mathbf{D}_a \cdot PVP\mathbf{D}_b = v_b^2 \mathbf{D}_a \cdot \mathbf{D}_b.$$

Thus $(v_a^2 - v_b^2)\mathbf{D}_a \cdot \mathbf{D}_b = 0$. For distinct eigenvalues the two displacements are necessarily orthogonal; at a repeated eigenvalue an orthonormal basis of the two-dimensional eigenspace may be chosen. The electric fields need not be orthogonal, since applying V does not in general preserve angles.

7.17 Consider the problem of dispersion and waves in an electronic plasma when a uniform external magnetic induction $\mathbf{B}_0$ is present, as in Section 7.6.

(a) Show that in general the susceptibility tensor $\chi_{jk}(\omega)$ defined through $D_j = \sum_k \epsilon_{jk} E_k$, and $\epsilon_{jk} = \epsilon_0(\delta_{jk} + \chi_{jk})$, is

$$\chi_{jk} = -\frac{\omega_p^2}{\omega^2(\omega^2 - \omega_B^2)} \left[\omega^2 \delta_{jk} - \omega_B^2 b_j b_k - i\omega\omega_B \epsilon_{jkl} b_l\right]$$

where $\mathbf{b}$ is a unit vector in the direction of $\mathbf{B}_0$.

(b) By straightforward diagonalization of the dielectric tensor ϵ_{jk} or by an airtight argument based on the approach and results of Section 7.6, find the eigenvalues ϵ_j, $j = 1, 2, 3$.

(c) A plane wave (ω, $\mathbf{k} = k\mathbf{n}$) must satisfy the vector equation of Problem 7.16a. Show that in terms of χ_{jk} the electric field and wave number must satisfy the three homogeneous equations,

$$(1 - \xi)E_j + \xi n_j(\mathbf{n} \cdot \mathbf{E}) + \sum_k \chi_{jk} E_k = 0, \qquad j = 1, 2, 3$$

where $\xi = (ck/\omega)^2$. Keeping only first-order terms in an expansion of χ_{jk} in powers of ω_B/ω, show that the effective dielectric constant for propagation of the plane wave is

$$\epsilon_\pm/\epsilon_0 \approx 1 - \frac{\omega_p^2}{\omega^2} \mp \frac{\omega_p^2 \omega_B \mathbf{b} \cdot \mathbf{n}}{\omega^3}$$

for positive and negative helicity waves.

Solution.

Let $e > 0$ be the magnitude of the electron charge, $\omega_B = eB_0/m_e$, and $\omega_p^2 = n_e e^2/(\epsilon_0 m_e)$. Neglect damping, ion motion, and terms quadratic in the wave fields. **(a)** Define the linear operator $C\mathbf{a} = \mathbf{b} \times \mathbf{a}$. The electron equation $m_e \ddot{\mathbf{x}} = -e(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}_0)$ becomes

$$(\omega^2 I - i\omega\omega_B C)\mathbf{x} = \frac{e}{m_e}\mathbf{E}.$$

Write $P_\parallel = \mathbf{b}\mathbf{b}^\mathsf{T}$ and $P_\perp = I - P_\parallel$. Since $CP_\parallel = 0$ and $C^2 = -P_\perp$, direct multiplication gives

$$(\omega^2 I - i\omega\omega_B C)^{-1} = \frac{P_\parallel}{\omega^2} + \frac{P_\perp + i(\omega_B/\omega)C}{\omega^2 - \omega_B^2}.$$

The polarization is $\mathbf{P} = -n_e e\mathbf{x}$, so

$$\chi = -\omega_p^2 \left[\frac{P_\parallel}{\omega^2} + \frac{P_\perp + i(\omega_B/\omega)C}{\omega^2 - \omega_B^2} \right].$$

Using $C_{jk} = -\epsilon_{jkl} b_l$ and combining denominators proves

$$\boxed{\chi_{jk} = -\frac{\omega_p^2}{\omega^2(\omega^2 - \omega_B^2)} [\omega^2 \delta_{jk} - \omega_B^2 b_j b_k - i\omega\omega_B \epsilon_{jkl} b_l].}$$

(b) The vector $\mathbf{b}$ has C eigenvalue zero. In a right-handed basis about $\mathbf{b}$, the vectors $\boldsymbol{\epsilon}_\pm = (\mathbf{e}_1 \pm i\mathbf{e}_2)/\sqrt{2}$ have C eigenvalues $\mp i$. The three dielectric eigenvalues are therefore

$$\boxed{\frac{\epsilon_\parallel}{\epsilon_0} = 1 - \frac{\omega_p^2}{\omega^2}, \qquad \frac{\epsilon_\pm}{\epsilon_0} = 1 - \frac{\omega_p^2}{\omega(\omega \mp \omega_B)}.}$$

These are material eigenvalues; for arbitrary propagation direction the electromagnetic normal modes also have to satisfy Maxwell's longitudinal constraint. **(c)** Divide the equation in Problem 7.16(a) by ω^2/c^2 and substitute $\mathbf{D} = \epsilon_0(I + \chi)\mathbf{E}$:

$$\boxed{(1 - \xi)E_j + \xi n_j (\mathbf{n} \cdot \mathbf{E}) + \sum_k \chi_{jk} E_k = 0, \qquad \xi = (ck/\omega)^2.}$$

To first order in ω_B/ω,

$$\chi = -\frac{\omega_p^2}{\omega^2} I - i\frac{\omega_p^2 \omega_B}{\omega^3} C + O(\omega_B^2).$$

Project onto the plane perpendicular to $\mathbf{n}$. In that plane $P_n C P_n = (\mathbf{b} \cdot \mathbf{n}) C_n$, where $C_n \mathbf{a} = \mathbf{n} \times \mathbf{a}$. Its circular eigenvectors have eigenvalues $\mp i$, so the two transverse equations give

$$\boxed{\xi_\pm \equiv \epsilon_{\mathrm{eff},\pm}/\epsilon_0 = 1 - \frac{\omega_p^2}{\omega^2} \mp \frac{\omega_p^2 \omega_B (\mathbf{b} \cdot \mathbf{n})}{\omega^3} + O(\omega_B^2).}$$

There is in general a longitudinal electric component of first order in ω_B, fixed by $\mathbf{n} \cdot \mathbf{D} = 0$. Its feedback into the transverse eigenvalues starts at second order. This expansion is not uniform at the unmagnetized plasma resonance $1 - \omega_p^2/\omega^2 = 0$, where eliminating that longitudinal component involves a small denominator.

7.18 Magnetohydrodynamic waves can occur in a compressible, nonviscous, perfectly conducting fluid in a uniform static magnetic induction $\mathbf{B}_0$. If the propagation direction is not parallel or perpendicular to $\mathbf{B}_0$, the waves are not separated into purely longitudinal (magnetosonic) or transverse (Alfvén) waves. Let the angle between the propagation direction $\mathbf{k}$ and the field $\mathbf{B}_0$ be θ.

(a) Show that there are three different waves with phase velocities given by

$$u_1^2 = (v_A \cos\theta)^2$$
$$u_{2,3}^2 = \tfrac{1}{2}(s^2 + v_A^2) \pm \tfrac{1}{2}[(s^2 + v_A^2)^2 - 4s^2 v_A^2 \cos^2\theta]^{1/2}$$

where s is the sound velocity in the fluid, and $v_A = (B_0^2/\mu\rho_0)^{1/2}$ is the Alfvén velocity.

(b) Find the velocity eigenvectors for the three different waves, and prove that the first (Alfvén) wave is always transverse, while the other two are neither longitudinal nor transverse.

(c) Evaluate the phase velocities and eigenvectors of the mixed waves in the approximation that $v_A \gg s$. Show that for one wave the only appreciable component of velocity is parallel to the magnetic field, while for the other the only component is perpendicular to the field and in the plane containing $\mathbf{k}$ and $\mathbf{B}_0$.

Solution.

Let $\mathbf{B}_0 = B_0\mathbf{e}_z$ and $\mathbf{k} = k(\sin\theta, 0, \cos\theta)$, with equilibrium density ρ_0 and $\delta p = s^2\delta\rho$. Write all perturbations as $e^{i\mathbf{k}\cdot\mathbf{x}-i\omega t}$. **(a)** The linearized continuity, induction, and momentum equations of ideal MHD are

$$\begin{aligned}\rho_1 &= \frac{\rho_0}{\omega}\mathbf{k}\cdot\mathbf{v},\\ \mathbf{B}_1 &= \frac{(\mathbf{k}\cdot\mathbf{v})\mathbf{B}_0 - (\mathbf{k}\cdot\mathbf{B}_0)\mathbf{v}}{\omega},\\ -\omega\rho_0\mathbf{v} &= -\mathbf{k}s^2\rho_1 + \frac{1}{\mu}(\mathbf{k}\times\mathbf{B}_1)\times\mathbf{B}_0.\end{aligned}$$

Writing $s_\theta = \sin\theta$, $c_\theta = \cos\theta$, the induction formula has the explicit components

$$\mathbf{B}_1 = \frac{kB_0}{\omega}(-c_\theta v_x, -c_\theta v_y, s_\theta v_x).$$

Since $(\mathbf{k}\times\mathbf{B}_1)\times\mathbf{B}_0 = \mathbf{B}_1(\mathbf{k}\cdot\mathbf{B}_0) - \mathbf{k}(\mathbf{B}_1\cdot\mathbf{B}_0)$, its components are

$$-\frac{k^2B_0^2}{\omega}(v_x, c_\theta^2 v_y, 0).$$

The pressure force is proportional to $(s_\theta, 0, c_\theta)(s_\theta v_x + c_\theta v_z)$. Substitute these into the momentum equation, multiply by $-\omega/(k^2\rho_0)$, and write $u = \omega/k$, $v_A^2 = B_0^2/(\mu\rho_0)$. The result is

$$u^2\begin{pmatrix}v_x\\ v_y\\ v_z\end{pmatrix} = \begin{pmatrix}v_A^2 + s^2\sin^2\theta & 0 & s^2\sin\theta\cos\theta\\ 0 & v_A^2\cos^2\theta & 0\\ s^2\sin\theta\cos\theta & 0 & s^2\cos^2\theta\end{pmatrix}\begin{pmatrix}v_x\\ v_y\\ v_z\end{pmatrix}.$$

The y eigenvalue is $u_1^2 = v_A^2\cos^2\theta$. The xz block has trace $s^2 + v_A^2$ and determinant $s^2v_A^2\cos^2\theta$. Its characteristic equation is

$$u^4 - (s^2 + v_A^2)u^2 + s^2v_A^2\cos^2\theta = 0,$$

which gives

$$\boxed{u_{2,3}^2 = \frac{s^2 + v_A^2}{2} \pm \frac{1}{2}\sqrt{(s^2 + v_A^2)^2 - 4s^2v_A^2\cos^2\theta}.}$$

(b) The Alfvén polarization is $\mathbf{v}_1 \parallel \mathbf{e}_y \parallel \mathbf{B}_0\times\mathbf{k}$, hence $\mathbf{k}\cdot\mathbf{v}_1 = 0$ and $\rho_1 = 0$. For either mixed root, an unnormalized eigenvector is

$$\boxed{\mathbf{v}_{2,3} \;\propto\; \big(u_{2,3}^2 - s^2\cos^2\theta,\; 0,\; s^2\sin\theta\cos\theta\big).}$$

For the generic oblique case in the question, with $s, v_A > 0$, these vectors are nonzero and can be normalized by their Euclidean lengths. To test a putative longitudinal mode, apply the xz matrix to

$(\sin\theta, \cos\theta)$; the pressure part is parallel to this vector but the magnetic part is $(v_A^2 \sin\theta, 0)$, which is not. For a putative transverse vector $(\cos\theta, -\sin\theta)$, the pressure part vanishes and the magnetic result $(v_A^2 \cos\theta, 0)$ is again not parallel. Thus neither mixed mode is longitudinal or transverse for $\sin\theta\cos\theta \neq 0$. The excluded parallel/perpendicular directions and vanishing sound or magnetic speeds supply the familiar special cases. **(c)** Put $\eta = s^2/v_A^2 \ll 1$. Expansion of the quadratic roots and their eigenvectors gives

$$u_{\text{fast}}^2 = v_A^2 + s^2 \sin^2\theta + O(s^4/v_A^2), \qquad \mathbf{v}_{\text{fast}} \propto (1, 0, \eta \sin\theta\cos\theta) + O(\eta^2),$$
$$u_{\text{slow}}^2 = s^2 \cos^2\theta[1 - \eta \sin^2\theta + O(\eta^2)], \qquad \mathbf{v}_{\text{slow}} \propto (-\eta \sin\theta\cos\theta, 0, 1) + O(\eta^2).$$

Thus the fast speed is $v_A[1 + \eta \sin^2\theta/2 + O(\eta^2)]$ and the slow speed is $s|\cos\theta|[1 - \eta \sin^2\theta/2 + O(\eta^2)]$. The fast fluid motion is predominantly perpendicular to the field in the $\mathbf{k}, \mathbf{B}_0$ plane, and the slow motion is predominantly parallel to the field, as requested.

7.19 An approximately monochromatic plane wave packet in one dimension has the instantaneous form, $u(x, 0) = f(x)\, e^{ik_0 x}$, with $f(x)$ the modulation envelope. For each of the forms $f(x)$ below, calculate the wave-number spectrum $|A(k)|^2$ of the packet, sketch $|u(x, 0)|^2$ and $|A(k)|^2$, evaluate explicitly the rms deviations from the means Δx and Δk (defined in terms of the intensities $|u(x, 0)|^2$ and $|A(k)|^2$), and test inequality (7.82).

(a) $f(x) = Ne^{-\alpha|x|/2}$

(b) $f(x) = Ne^{-\alpha^2 x^2/4}$

(c) $f(x) = \begin{cases} N(1 - \alpha|x|) & \text{for } \alpha|x| < 1 \\ 0 & \text{for } \alpha|x| > 1 \end{cases}$

(d) $f(x) = \begin{cases} N & \text{for } |x| < a \\ 0 & \text{for } |x| > a \end{cases}$

Solution.

Choose the unitary transform

$$A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i(k-k_0)x}\, dx, \qquad \int |f|^2 dx = \int |A|^2 dk = 1.$$

This fixes N without affecting the requested normalized variances. Set $q = k - k_0$ and $\operatorname{sinc} u = \sin u/u$, with its continuous value at zero. Each spatial density is even, so $\langle x \rangle = 0$; for (a)–(c) the spectral mean exists and is $\langle k \rangle = k_0$. **(a)** Normalization gives $N^2 = \alpha/2$. The cosine integral $2\int_0^\infty e^{-\alpha x/2} \cos(qx) dx = \alpha/(q^2 + \alpha^2/4)$ gives

$$A(k) = \frac{N\alpha}{\sqrt{2\pi}(q^2 + \alpha^2/4)}, \qquad \boxed{|A(k)|^2 = \frac{\alpha^3}{4\pi(q^2 + \alpha^2/4)^2}.}$$

Using $\int_0^\infty x^2 e^{-\alpha x} dx = 2/\alpha^3$ gives $\Delta x^2 = 2/\alpha^2$. Alternatively evaluating the spectral integral, or using Parseval for the square-integrable weak derivative,

$$\Delta k^2 = \int |f'(x)|^2 dx = \frac{\alpha^2}{4}.$$

Thus $\boxed{\Delta x = \sqrt{2}/\alpha, \quad \Delta k = \alpha/2, \quad \Delta x \Delta k = 1/\sqrt{2} > 1/2}$. **(b)** Here $N^2 = \alpha/\sqrt{2\pi}$. Completing the square in the Gaussian Fourier integral uses

$$-\frac{\alpha^2 x^2}{4} - iqx = -\frac{\alpha^2}{4}\left(x + \frac{2iq}{\alpha^2}\right)^2 - \frac{q^2}{\alpha^2}.$$

The Gaussian integral on the shifted contour is $2\sqrt{\pi}/\alpha$ (the integrand is entire and its connecting tails vanish). Therefore

$$A(k) = \frac{\sqrt{2}N}{\alpha} e^{-q^2/\alpha^2}, \qquad \boxed{|A(k)|^2 = \frac{\sqrt{2/\pi}}{\alpha} e^{-2q^2/\alpha^2}.}$$

The spatial and spectral Gaussian variances are respectively $1/\alpha^2$ and $\alpha^2/4$, so

$$\boxed{\Delta x = 1/\alpha, \qquad \Delta k = \alpha/2, \qquad \Delta x \Delta k = 1/2.}$$

This Gaussian saturates (7.82). **(c)** The integrals $2N^2 \int_0^{1/\alpha} (1-\alpha x)^2 dx = 1$ and $2N^2 \int_0^{1/\alpha} x^2 (1-\alpha x)^2 dx$ give $N^2 = 3\alpha/2$ and $\Delta x^2 = 1/(10\alpha^2)$. Two integrations of the elementary cosine integral give

$$A(k) = \frac{2N\alpha}{\sqrt{2\pi} q^2}[1 - \cos(q/\alpha)] = \frac{N}{\alpha\sqrt{2\pi}} \operatorname{sinc}^2\!\Big(\frac{q}{2\alpha}\Big),$$

$$\boxed{|A(k)|^2 = \frac{3}{4\pi\alpha} \operatorname{sinc}^4\!\Big(\frac{q}{2\alpha}\Big).}$$

The envelope is continuous at its endpoints. Its weak first derivative is $\mp N\alpha$ on the two halves of its support and zero elsewhere, so Parseval gives $\Delta k^2 = 2N^2\alpha^2/\alpha = 3\alpha^2$. Therefore

$$\boxed{\Delta x = \frac{1}{\sqrt{10}\alpha}, \qquad \Delta k = \sqrt{3}\alpha, \qquad \Delta x \Delta k = \sqrt{3/10} > 1/2.}$$

(d) Now $N^2 = 1/(2a)$, and direct integration gives

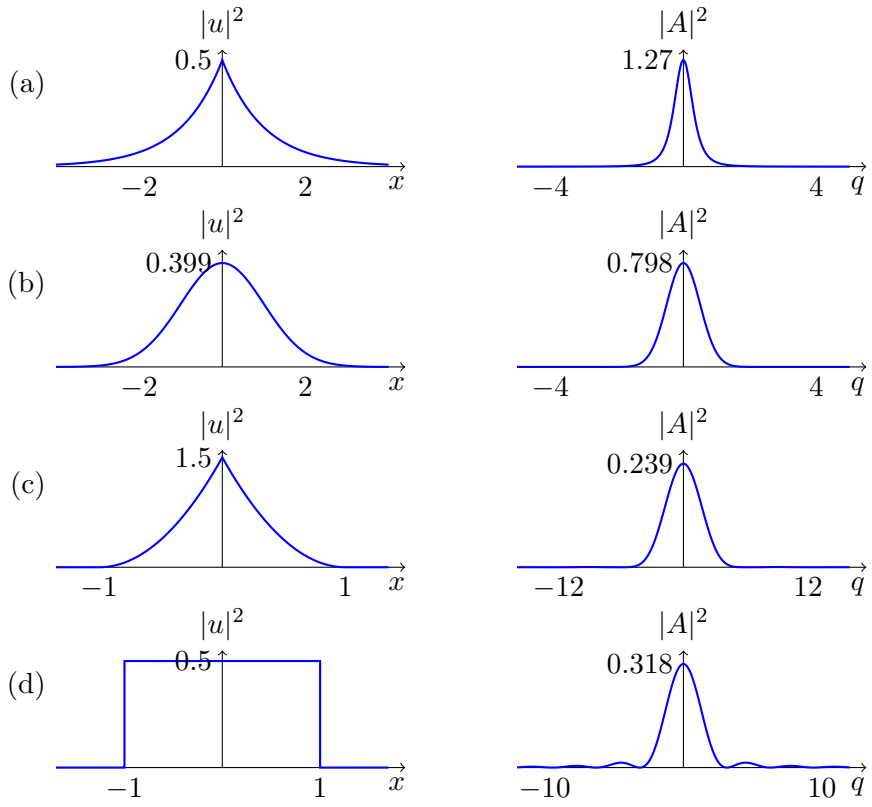
$$A(k) = N\sqrt{\frac{2}{\pi}} \frac{\sin(qa)}{q}, \qquad \boxed{|A(k)|^2 = \frac{a}{\pi} \operatorname{sinc}^2(qa), \qquad \Delta x = \frac{a}{\sqrt{3}}.}$$

The spectral tail is too slow for a finite rms width:

$$\int_{-K}^{K} q^2 |A|^2 dq = \frac{1}{\pi a}\left[K - \frac{\sin(2Ka)}{2a}\right] \longrightarrow \infty.$$

Indeed the absolute first spectral moment also diverges logarithmically; k_0 is its symmetry center and principal-value mean, not an ordinary finite mean. Thus there is no finite Δk in the usual variance sense; every symmetric cutoff yields an unbounded rms width as the cutoff grows. This case does not contradict (7.82), whose finite-width hypotheses fail. The discontinuities of the rectangle produce this tail, while the continuous triangle has a finite derivative norm.

The following sketches use $\alpha = 1$ in (a)–(c), $a = 1$ in (d), and $q = k - k_0$. Each plotted density is normalized; the displayed finite spectral interval includes the central lobes and representative tails.



7.20 A homogeneous, isotropic, nonpermeable dielectric is characterized by an index of refraction $n(\omega)$, which is in general complex in order to describe absorptive processes.

(a) Show that the general solution for plane waves in one dimension can be written

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \; e^{-i\omega t}[A(\omega)e^{i(\omega/c)n(\omega)x} + B(\omega)e^{-i(\omega/c)n(\omega)x}]$$

where $u(x,t)$ is a component of $\mathbf{E}$ or $\mathbf{B}$.

(b) Assuming a real material-response kernel and the causal branch of n that approaches $+1$ at high frequency, show that $n(-\omega) = n^*(\omega)$ and state the conditions on A and B for $u(x,t)$ to be real.

(c) Show that, if $u(0,t)$ and $\partial u(0,t)/\partial x$ are the boundary values of u and its derivative at $x = 0$, the coefficients $A(\omega)$ and $B(\omega)$ are

$$\begin{Bmatrix} A(\omega) \\ B(\omega) \end{Bmatrix} = \frac{1}{2}\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{i\omega t} \left[u(0,t) \mp \frac{ic}{\omega n(\omega)} \frac{\partial u}{\partial x}(0,t) \right]$$

Solution.

(a) Temporal Fourier transformation in a homogeneous medium turns the wave equation into

$$\frac{d^2U(x,\omega)}{dx^2} + k^2(\omega)U(x,\omega) = 0, \qquad k(\omega) = \frac{\omega}{c}n(\omega).$$

For $k \neq 0$ its two independent solutions give

$$U(x,\omega) = A(\omega)e^{ik(\omega)x} + B(\omega)e^{-ik(\omega)x}.$$

Inverse Fourier transformation is exactly the integral in part (a). The amplitudes and radiation or boundary conditions select the physically appropriate growing/decaying directions in an absorbing medium; both solutions are required for the local general solution. At a zero of k, the independent solutions are 1 and x and are obtained as limits of suitable combinations of the two exponentials. **(b)** For a real temporal material-response kernel, $\epsilon(-\omega) = \epsilon^*(\omega)$ on the real frequency axis. Choose the causal square-root branch of $n^2 = \epsilon/\epsilon_0$ that approaches $+1$ at high frequency. That branch then satisfies

$$\boxed{n(-\omega) = n^*(\omega), \qquad k(-\omega) = -k^*(\omega).}$$

Indeed, the complex conjugate of $e^{ik(\omega)x - i\omega t}$ is the negative-frequency wave with this wave number. Reality of u requires $U(x,-\omega) = U^*(x,\omega)$ and is achieved by $A(-\omega) = A^*(\omega)$ and $B(-\omega) = B^*(\omega)$. A single real field which fails to excite some frequencies cannot by itself determine a material's index at those frequencies; this is why the physical real-response and branch assumptions have been made explicit in the statement. **(c)** Let $U_0(\omega)$ and $U_0'(\omega)$ be the unitary time transforms of the two boundary data. Evaluating the solution and its derivative at zero gives

$$A + B = U_0, \qquad ik(A - B) = U_0',$$

so

$$\boxed{\begin{Bmatrix} A(\omega) \\ B(\omega) \end{Bmatrix} = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega t} \left[u(0,t) \mp \frac{ic}{\omega n(\omega)} \frac{\partial u}{\partial x}(0,t) \right] dt.}$$

At $k = 0$ use instead $U(x) = U_0 + xU_0'$; the apparent $1/k$ singularity is a degeneracy of the exponential basis, not of that boundary-value solution.

7.21 Consider the nonlocal (in time) connection between $\mathbf{D}$ and $\mathbf{E}$,

$$\mathbf{D}(\mathbf{x},t) = \epsilon_0 \left\{ \mathbf{E}(\mathbf{x},t) + \int d\tau \; G(\tau)\mathbf{E}(\mathbf{x},t-\tau) \right\}$$

with the $G(\tau)$ appropriate for the single-resonance model,

$$\epsilon(\omega)/\epsilon_0 = 1 + \omega_p^2(\omega_0^2 - \omega^2 - i\gamma\omega)^{-1}$$

(a) Convert the nonlocal connection between $\mathbf{D}$ and $\mathbf{E}$ into an instantaneous relation involving derivatives of $\mathbf{E}$ with respect to time by expanding the electric field in the integral in a Taylor series in τ. Evaluate the integrals over $G(\tau)$ explicitly up to at least $\partial^2\mathbf{E}/\partial t^2$.

(b) Show that the series obtained in part a can be obtained formally by converting the frequency-representation relation, $\mathbf{D}(\mathbf{x},\omega) = \epsilon(\omega)\mathbf{E}(\mathbf{x},\omega)$ into a space-time relation,

$$\mathbf{D}(\mathbf{x},t) = \epsilon\left(i\frac{\partial}{\partial t}\right)\mathbf{E}(\mathbf{x},t)$$

where the variable ω in $\epsilon(\omega)$ is replaced by $\omega \to i(\partial/\partial t)$.

Solution.

(a) Assume $\omega_0 > 0$, so a Taylor expansion about zero frequency exists. The causal one-resonance kernel (7.110) is

$$G(\tau) = \theta(\tau)\frac{\omega_p^2}{\Omega}e^{-\gamma\tau/2}\sin(\Omega\tau), \qquad \Omega^2 = \omega_0^2 - \gamma^2/4.$$

For an overdamped oscillator the same expression is continued to imaginary Ω, and at $\Omega = 0$ the ratio becomes τ. For $\gamma > 0$, define $M_j = \int_0^\infty \tau^j G(\tau)d\tau$. Starting from $\int_0^\infty e^{-s\tau}\sin(\Omega\tau)d\tau = \Omega/(s^2+\Omega^2)$, differentiation gives

$$\begin{aligned}
M_0 &= \frac{\omega_p^2}{\omega_0^2}, \\
M_1 &= \left.\frac{2s\omega_p^2}{(s^2+\Omega^2)^2}\right|_{s=\gamma/2} = \frac{\gamma\omega_p^2}{\omega_0^4}, \\
M_2 &= \left.\frac{2\omega_p^2(3s^2-\Omega^2)}{(s^2+\Omega^2)^3}\right|_{s=\gamma/2} = \frac{2\omega_p^2(\gamma^2-\omega_0^2)}{\omega_0^6}.
\end{aligned}$$

Inserting $\mathbf{E}(t-\tau) = \mathbf{E}(t) - \tau\dot{\mathbf{E}}(t) + \tau^2\ddot{\mathbf{E}}(t)/2 + \cdots$ therefore gives

$$\boxed{\frac{\mathbf{D}}{\epsilon_0} = \left(1 + \frac{\omega_p^2}{\omega_0^2}\right)\mathbf{E} - \frac{\gamma\omega_p^2}{\omega_0^4}\dot{\mathbf{E}} + \frac{\omega_p^2(\gamma^2-\omega_0^2)}{\omega_0^6}\ddot{\mathbf{E}} + \cdots.}$$

The negative second moment for weak damping is allowed because G is an oscillating response kernel, not a probability density. This is a slow-field derivative expansion; arbitrary rapidly varying fields do not justify interchanging a Taylor series and the full memory integral. Undamped results can be read as Abel limits $\gamma \downarrow 0$. **(b)** Direct expansion of the frequency response gives

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{\omega_p^2}{\omega_0^2} + i\frac{\gamma\omega_p^2}{\omega_0^4}\omega + \frac{\omega_p^2(\omega_0^2-\gamma^2)}{\omega_0^6}\omega^2 + O(\omega^3).$$

Since $i\partial_t e^{-i\omega t} = \omega e^{-i\omega t}$, replacing ω by $i\partial_t$ reproduces every term above:

$$\boxed{\mathbf{D}(t) = \epsilon(i\partial_t)\mathbf{E}(t).}$$

As an exact local differential statement, with causal initial conditions selecting the inverse operator, the same model is

$$(\partial_t^2 + \gamma\partial_t + \omega_0^2)\left(\frac{\mathbf{D}}{\epsilon_0} - \mathbf{E}\right) = \omega_p^2\mathbf{E}.$$

This last form specifies the response even when the low-frequency series is unsuitable.

7.22 Use the Kramers–Kronig relation (7.120) to calculate the real part of $\epsilon(\omega)$, given the imaginary part of $\epsilon(\omega)$ for positive ω as

(a) $\text{Im}\,\epsilon/\epsilon_0 = \lambda[\theta(\omega-\omega_1) - \theta(\omega-\omega_2)], \qquad \omega_2 > \omega_1 > 0$

(b) $\text{Im}\,\epsilon/\epsilon_0 = \dfrac{\lambda\gamma\omega}{(\omega_0^2-\omega^2)^2 + \gamma^2\omega^2}$

In each case sketch the behavior of Im $\epsilon(\omega)$ and the result for Re $\epsilon(\omega)$ as functions of ω. Comment on the reasons for similarities or differences of your results as compared with the curves in Fig. 7.8. The step function is $\theta(x) = 0$, $x < 0$ and $\theta(x) = 1$, $x > 0$.

Solution.

Write $\chi' = \mathrm{Re}(\epsilon/\epsilon_0) - 1$ and $\chi'' = \mathrm{Im}(\epsilon/\epsilon_0)$. Equation (7.120), with the high-frequency limit $\epsilon/\epsilon_0 \to 1$, is

$$\chi'(\omega) = \frac{2}{\pi} P \int_0^\infty \frac{x\chi''(x)}{x^2 - \omega^2}\, dx.$$

(a) Only $\omega_1 < x < \omega_2$ contributes. Since the antiderivative of $x/(x^2 - \omega^2)$ is $\frac{1}{2}\ln|x^2 - \omega^2|$, the principal value gives

$$\boxed{\frac{\mathrm{Re}\,\epsilon}{\epsilon_0} = 1 + \frac{\lambda}{\pi}\ln\left|\frac{\omega_2^2 - \omega^2}{\omega_1^2 - \omega^2}\right|.}$$

It rises to $+\infty$ at the lower absorption edge and falls to $-\infty$ at the upper one. At zero frequency it is $1 + (2\lambda/\pi)\ln(\omega_2/\omega_1)$, while at high frequency it is $1 - \lambda(\omega_2^2 - \omega_1^2)/(\pi\omega^2) + O(\omega^{-4})$. These logarithmic edge singularities arise from the imposed perfectly sharp jumps in absorption. Smoothing the jumps rounds them.

(b) An explicit evaluation is possible without assuming the real part in advance. Put $a = \omega_0^2$ and $D(x) = (a - x^2)^2 + \gamma^2 x^2$. Partial fractions give

$$\frac{x^2}{D(x)(x^2 - \omega^2)} = \frac{1}{D(\omega)}\left[\frac{\omega^2}{x^2 - \omega^2} + \frac{a^2 - \omega^2 x^2}{D(x)}\right].$$

For $\omega > 0$, $P\int_0^\infty dx/(x^2 - \omega^2) = 0$. Define $J_j = \int_0^\infty x^j/D(x)\, dx$. The substitution $x = a/y$ gives $J_2 = aJ_0$. Also

$$\frac{d}{dx}\tan^{-1}\frac{x^2 - a}{\gamma x} = \frac{\gamma(x^2 + a)}{D(x)},$$

whose integral from zero to infinity is π. Consequently

$$J_2 = \frac{\pi}{2\gamma}, \qquad J_0 = \frac{\pi}{2a\gamma}.$$

Substituting these in the Kramers–Kronig integral gives

$$\chi' = \frac{2\lambda\gamma}{\pi D(\omega)}(a^2 J_0 - \omega^2 J_2) = \frac{\lambda(a - \omega^2)}{D(\omega)},$$

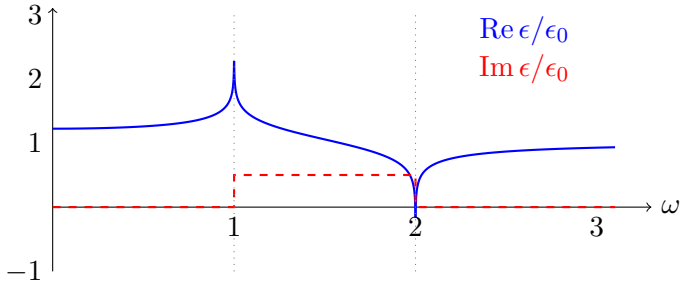
so

$$\boxed{\frac{\mathrm{Re}\,\epsilon}{\epsilon_0} = 1 + \frac{\lambda(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}.}$$

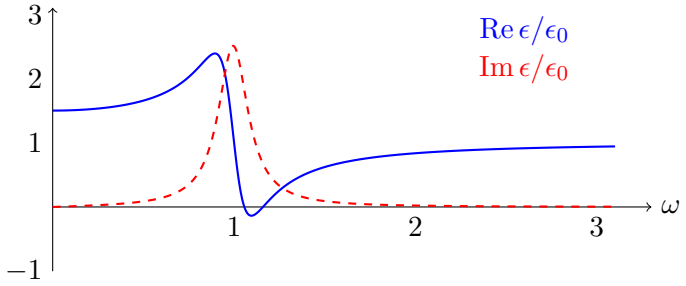
The zero-frequency value is $1 + \lambda/\omega_0^2$, and the high-frequency behavior is $1 - \lambda/\omega^2 + O(\omega^{-4})$. The imaginary part is a smooth absorption peak and the real part has the familiar dispersive rise and fall of Fig. 7.8. In fact this is that same single damped-oscillator response, with λ in place of its oscillator strength. The band model in (a) has the same causal link between absorption and dispersion but an entirely different sharp-edged absorption spectrum.

The sketches use $(\omega_1, \omega_2, \lambda) = (1, 2, 0.5)$ for (a) and $(\omega_0, \gamma, \lambda) = (1, 0.2, 0.5)$ for (b), in consistent frequency units. Dashed vertical lines mark the divergent edges in (a); the branches are deliberately not joined across them.

(a) sharp absorption band



(b) damped oscillator



7.23 Discuss the extension of the Kramers–Kronig relations (7.120) for a medium with a static electrical conductivity σ. Show that the first equation in (7.120) is unchanged, but that the second is changed into

$$\operatorname{Im}\epsilon(\omega) = \frac{\sigma}{\omega} - \frac{2\omega}{\pi} P \int_0^\infty \frac{[\operatorname{Re}\epsilon(\omega') - \epsilon_0]}{\omega'^2 - \omega^2}\, d\omega'$$

[*Hint*: Consider $\epsilon(\omega) - i\sigma/\omega$ as analytic for $\operatorname{Im}\omega \geq 0$.]

Solution.

A static conductivity contributes $i\sigma/\omega$ to the dielectric function with the convention $e^{-i\omega t}$. Subtract this pole:

$$h(\omega) = \epsilon(\omega) - \frac{i\sigma}{\omega}.$$

Under the analyticity assumption in the hint, h obeys the ordinary Kramers–Kronig relations with limiting value ϵ_0. For nonzero real ω, $\operatorname{Re} h = \operatorname{Re}\epsilon$ and $\operatorname{Im} h = \operatorname{Im}\epsilon - \sigma/\omega$. The real-part relation becomes

$$\operatorname{Re}\epsilon - \epsilon_0 = \frac{2}{\pi} P \int_0^\infty \frac{x \operatorname{Im}\epsilon(x) - \sigma}{x^2 - \omega^2}\, dx.$$

The subtracted term vanishes because

$$P \int_0^\infty \frac{dx}{x^2 - \omega^2} = \left[\frac{1}{2\omega} \ln \left| \frac{x - \omega}{x + \omega} \right| \right]_0^\infty = 0,$$

where the logarithmic terms on the two sides of the pole cancel. Thus the first equation of (7.120) is unchanged for $\omega \neq 0$. The imaginary-part relation for h is instead

$$\operatorname{Im}\epsilon - \frac{\sigma}{\omega} = -\frac{2\omega}{\pi} P \int_0^\infty \frac{\operatorname{Re}\epsilon(x) - \epsilon_0}{x^2 - \omega^2}\, dx,$$

or

$$\boxed{\operatorname{Im}\epsilon(\omega) = \frac{\sigma}{\omega} - \frac{2\omega}{\pi} P\int_0^\infty \frac{\operatorname{Re}\epsilon(x) - \epsilon_0}{x^2 - \omega^2}\, dx.}$$

The value at $\omega = 0$ must be interpreted through the pole subtraction or a causal limiting prescription, not by substitution into the nonzero-frequency principal-value formula.

7.24

(a) Use the relation (7.113) and the analyticity of $\epsilon(\omega)/\epsilon_0$ for $\operatorname{Im}\omega \geq 0$ to prove that on the positive imaginary axis $\epsilon(\omega)/\epsilon_0$ is real and monotonically decreasing away from the origin toward unity as $\omega \to i\infty$, provided $\operatorname{Im}\epsilon \geq 0$ for real positive frequencies. Assume that (7.114) holds for real ω.

(b) With the assumption that $\operatorname{Im}\epsilon$ vanishes for finite real ω only at $\omega = 0$, show that $\epsilon(\omega)$ has no zeros in the upper half-ω-plane.

(c) Write down a Kramers–Kronig relation for $\epsilon_0/\epsilon(\omega)$ and deduce a sum rule similar to (7.122), but as an integral over $\operatorname{Im}[\epsilon_0/\epsilon(\omega)]$.

(d) With the one-resonance model (7.107) for $\epsilon(\omega)$ determine $\operatorname{Im}\epsilon(\omega)$ and $\operatorname{Im}[1/\epsilon(\omega)]$ and verify explicitly that the sum rules (7.122) and part c are satisfied.

Solution.

Let $g(z) = \epsilon(z)/\epsilon_0$ and $g''(x) = \operatorname{Im} g(x)$ on the positive real axis. Cauchy's representation is Eq. (7.115), p. 333:

$$g(z) - 1 = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{g(x) - 1}{x - z}\, dx, \qquad \operatorname{Im} z > 0.$$

The reflected function $g(z^*)^* - 1$ is analytic in the lower half-plane. Closing its real-axis integral there encloses no pole of $1/(x - z)$, so $\int_{-\infty}^{\infty}[g(x)^* - 1]/(x - z)\, dx = 0$. Subtracting this zero integral from the preceding expression gives

$$g(z) - 1 = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{g''(x)}{x - z}\, dx.$$

The reality relation (7.113), p. 332, gives $g(-x) = g^*(x)$, hence $g''(-x) = -g''(x)$. Pairing the positive and negative frequency contributions uses

$$\frac{1}{x - z} + \frac{1}{x + z} = \frac{2x}{x^2 - z^2}$$

and gives

$$g(z) = 1 + \frac{2}{\pi}\int_0^\infty \frac{x g''(x)}{x^2 - z^2}\, dx, \qquad \operatorname{Im} z > 0. \tag{*}$$

The large-frequency falloff (7.114) supplies convergence; the vanishing upper and lower semicircle integrals use the causal high-frequency behavior discussed immediately before Eq. (7.115). Following Jackson's definition (7.121), p. 335, denote its leading coefficient by $\omega_p^2 = \lim_{|z|\to\infty} z^2[1 - g(z)]$. Thus $g(z) = 1 - \omega_p^2/z^2 + o(z^{-2})$; the falloff orders alone do not fix the numerical value of ω_p^2. **(a)** For $z = iy$, $y > 0$, equation (*) gives

$$g(iy) = 1 + \frac{2}{\pi}\int_0^\infty \frac{x g''(x)}{x^2 + y^2}\, dx \geq 1, \qquad \frac{dg(iy)}{dy} = -\frac{4y}{\pi}\int_0^\infty \frac{x g''(x)}{(x^2 + y^2)^2}\, dx \leq 0.$$

Both quantities are real. Unless the medium has no absorption at all, the derivative is strictly negative. The integral tends to zero as $y \to \infty$, by the finite spectral moment implicit in (7.114), so $g(iy)$ decreases toward unity. **(b)** Put $z = u + iy$. The imaginary part of (*) is

$$\operatorname{Im} g(u+iy) = \frac{4uy}{\pi} \int_0^\infty \frac{x g''(x)}{(x^2 - u^2 + y^2)^2 + 4u^2y^2}\, dx.$$

Under the stated positive absorption assumption, its sign is the sign of u for $u \neq 0$; hence g cannot vanish there. On $u = 0$, part (a) gives $g \geq 1$, so there is no zero anywhere in the open upper half-plane. On the real axis g'' excludes a zero at any nonzero frequency, while the zero-frequency limit of (*) is positive or divergent. This also states the boundary qualifications needed when applying the reciprocal response. **(c)** Since $f(z) = 1/g(z) = \epsilon_0/\epsilon(z)$ has no upper half-plane pole and tends to one, it satisfies

$$\boxed{\operatorname{Re} f(\omega) - 1 = \frac{2}{\pi} P \int_0^\infty \frac{x \operatorname{Im} f(x)}{x^2 - \omega^2}\, dx,}$$

with the companion relation $\operatorname{Im} f(\omega) = -(2\omega/\pi) P \int_0^\infty [\operatorname{Re} f(x) - 1]/(x^2 - \omega^2)\, dx$. Its large-frequency expansion is $f(z) = 1 + \omega_p^2/z^2 + \cdots$. For example take $z = iy$ and multiply the dispersion integral by y^2. Since $y^2/(x^2 + y^2) \to 1$, the convergent first spectral moment permits passage to the limit. The expansion $f(iy) - 1 = -\omega_p^2/y^2 + \cdots$ then gives

$$\boxed{\frac{2}{\pi} \int_0^\infty x \operatorname{Im} \frac{\epsilon_0}{\epsilon(x)}\, dx = -\omega_p^2.}$$

The negative sign agrees with $\operatorname{Im}(1/g) = -g''/|g|^2$ for a passive medium. **(d)** For the one-resonance response set $\omega_L^2 = \omega_0^2 + \omega_p^2$. Direct division gives

$$\operatorname{Im} \epsilon(\omega) = \frac{\epsilon_0 \omega_p^2 \gamma \omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2},$$
$$\operatorname{Im} \frac{1}{\epsilon(\omega)} = -\frac{\omega_p^2 \gamma \omega/\epsilon_0}{(\omega_L^2 - \omega^2)^2 + \gamma^2 \omega^2}.$$

The imaginary part of the reciprocal has its loss feature at the longitudinal frequency ω_L, rather than at ω_0. The integral evaluated in Problem 7.22(b) states, for any $a > 0$,

$$\int_0^\infty \frac{\gamma \omega^2\, d\omega}{(a^2 - \omega^2)^2 + \gamma^2 \omega^2} = \frac{\pi}{2}.$$

Using it once with $a = \omega_0$ and once with $a = \omega_L$ proves explicitly

$$\frac{2}{\pi} \int_0^\infty \omega \frac{\operatorname{Im} \epsilon}{\epsilon_0}\, d\omega = \omega_p^2, \qquad \frac{2}{\pi} \int_0^\infty \omega \operatorname{Im} \frac{\epsilon_0}{\epsilon}\, d\omega = -\omega_p^2.$$

7.25 Equation (7.67) is an expression for the square of the index of refraction for waves propagating along field lines through a plasma in a uniform external magnetic field. Using this as a model for propagation in the magnetosphere, consider the arrival of a whistler signal (actually the Brillouin precursor and subsequently of Section 7.11).

(a) Make a reasonably careful sketch of $c\,dk/d\omega$, where $k = \omega n(\omega)/c$, for the positive helicity wave, assuming $\omega_p/\omega_B \geq 1$. Indicate the interval where $c\,dk/d\omega$ is imaginary, but do not try to sketch it there!

(b) Show that on the interval, $0 < \omega < \omega_B$, the minimum of $c\,dk/d\omega$ occurs at $\omega/\omega_B \simeq \frac{1}{4}$, provided $\omega_p/\omega_B \geq 1$. Find approximate expressions for $c\,dk/d\omega$ for ω near zero and for ω near ω_B.

(c) By means of the method of stationary phase and the general structure of the solution to Problem 7.20a, show that the arrival of a whistler is signaled by a rising and falling frequency as a function of time, the falling frequency component being the source of the name.

(d) (Optional) Consider the form of the signal in the Brillouin precursor. Show that it consists of a modulated waveform of frequency $\omega_* \simeq \omega_B/4$, where ω_* is the exact group-delay minimum whose envelope is the Airy integral. This then evolves into a signal beating with the two frequencies of part c.

Solution.

Use the positive-helicity sign in (7.67), namely $n^2 = 1 - \omega_p^2/[\omega(\omega - \omega_B)]$, for propagation parallel to $\mathbf{B}_0$. Define

$$x = \omega/\omega_B, \qquad p = \omega_p/\omega_B \geq 1, \qquad K(x) = \frac{ck}{\omega_B} = \sqrt{f(x)}, \qquad f(x) = x^2 + \frac{p^2 x}{1-x}.$$

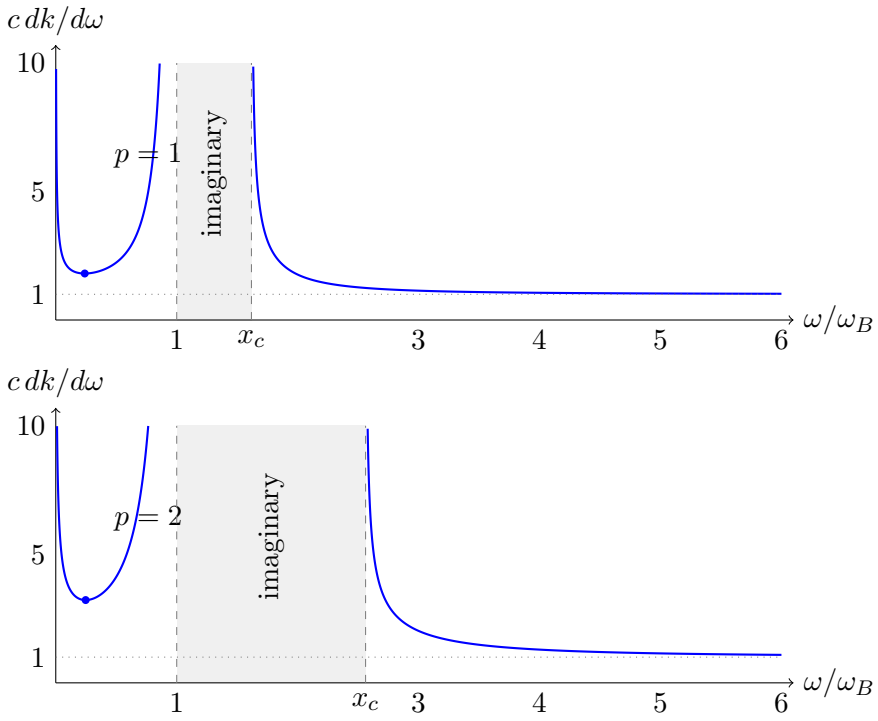
(a) In either propagating band choose the positive real root for K. The dimensionless group delay per length is

$$\boxed{n_g(x) \equiv c\frac{dk}{d\omega} = \frac{2x + p^2/(1-x)^2}{2\sqrt{x^2 + p^2 x/(1-x)}}.}$$

The lower band is $0 < x < 1$. The stop band is

$$1 < x < x_c, \qquad x_c = \frac{1 + \sqrt{1+4p^2}}{2},$$

where k and $dk/d\omega$ are imaginary in this lossless model. For $x > x_c$, n_g starts at $+\infty$ and decreases toward one. To check monotonicity, its derivative has the sign of $p^2(4x-1) + 4x^3(1-x)$. Since $p^2 \leq x(x-1)$ in this band, that expression is at most $-x(x-1)(2x-1)^2 < 0$. The sketch shows both real bands for representative $p \geq 1$; no real curve is drawn in the stop band.



(b) Differentiating $K' = f'/(2\sqrt{f})$ shows that its minimum satisfies $2ff'' - f'^2 = 0$. Substitution and cancellation give

$$\boxed{p^2(4x_* - 1) + 4x_*^3(1 - x_*) = 0.}$$

There is one root in $(0, 1/4)$: the left side is negative at zero, positive at $1/4$, and its derivative is positive on this interval. It remains positive for $x \geq 1/4$. For example, $x_* = 0.2395469,\ 0.2471584,\ 0.2495336$ for $p = 1, 2, 5$. Thus $x_* \simeq 1/4$ is good even for $p = 1$, but is not an exact identity at finite p. Neglecting x^2 against $p^2 x/(1 - x)$ gives

$$K \simeq p\sqrt{\frac{x}{1-x}}, \qquad n_g \simeq \frac{p}{2\sqrt{x}(1-x)^{3/2}}, \qquad n_{g,\min} \simeq \frac{8p}{3\sqrt{3}}.$$

The exact formula also gives the endpoint asymptotes

$$\boxed{n_g \sim \frac{p}{2\sqrt{x}} \quad (x \downarrow 0), \qquad n_g \sim \frac{p}{2(1-x)^{3/2}} \quad (x \uparrow 1).}$$

The discarded-to-retained ratio in f is $x(1 - x)/p^2 \leq 1/(4p^2)$, which explains the useful but approximate quarter-frequency result. **(c)** For a path length L in this uniform model, write the positive-frequency signal using the unitary temporal Fourier normalization of Eq. (7.104) and Problem 7.20:

$$\psi(L, t) = \frac{1}{\sqrt{2\pi}} \int A(\omega) e^{i[k(\omega)L - \omega t]}\, d\omega.$$

A stationary phase frequency obeys $t = Lk'(\omega) = (L/c)n_g(\omega)$. The two low-band stationary points first merge at $t_* = Ln_g(\omega_*)/c$. For $t > t_*$ they separate: the lower frequency falls toward zero, and the upper one rises toward ω_B. Near an isolated stationary point set $\omega = \omega_j + \nu$. Its phase is $k_j L - \omega_j t + \frac{1}{2} L k_j'' \nu^2 + O(\nu^3)$. The regulated Gaussian integral is

$$\int_{-\infty}^{\infty} e^{iLk_j''\nu^2/2}\, d\nu = \sqrt{\frac{2\pi}{L|k_j''|}}\, e^{i\pi\,\mathrm{sgn}(k_j'')/4}.$$

Multiplying by the unitary Fourier prefactor yields, well away from the merger,

$$\psi \simeq \sum_{j=1}^{2} A(\omega_j) \frac{1}{\sqrt{L|k''(\omega_j)|}} \exp\left\{ i[k(\omega_j)L - \omega_j t] + i\frac{\pi}{4} \operatorname{sgn} k''(\omega_j) \right\}.$$

The endpoint formulas imply the late-time tracks

$$\boxed{\omega_{\rm low} \sim \frac{\omega_p^2 L^2}{4c^2 \omega_B t^2}, \qquad \omega_{\rm high} \sim \omega_B \left[1 - \left(\frac{pL}{2ct} \right)^{2/3} \right].}$$

A sufficiently broad incident spectrum must populate both tracks; the descending component produces the familiar falling whistle. The stop band and any absorption modify amplitudes but not this stationary-phase mechanism within the stated model. **(d)** At the merger $k''(\omega_*) = 0$ and $k'''(\omega_*) > 0$. With $\Omega = \omega - \omega_*$, expand

$$k(\omega)L - \omega t = k_* L - \omega_* t + (t_* - t)\Omega + \frac{L k_*'''}{6} \Omega^3 + \cdots .$$

On the narrow frequency scale of this coalescence take $A(\omega) \simeq A_*$ and put $a = (L k_*'''/2)^{1/3}$. The Airy integral $\int_{-\infty}^{\infty} e^{i(u^3/3 + yu)} du = 2\pi \, \mathrm{Ai}(y)$ then gives the uniform leading approximation

$$\boxed{\psi(L,t) \simeq \frac{\sqrt{2\pi} A_*}{a} e^{i(k_* L - \omega_* t)} \, \mathrm{Ai}\!\left(\frac{t_* - t}{a} \right).}$$

In the plasma-dominated approximation, $\omega_* \simeq \omega_B/4$ and $k_*''' \simeq 256p/(9\sqrt{3}\, c\omega_B^2)$. Thus the Airy envelope modulates a carrier near $\omega_B/4$. For $t > t_*$ the Airy function oscillates, and its two saddle phases have instantaneous frequencies

$$\omega_* \pm \sqrt{\frac{2(t - t_*)}{L k_*'''}},$$

matching the two branches of part (c) near their merger. This is the whistler/Brillouin onset, not the earliest causal signal front; the high-frequency band supports an earlier precursor approaching vacuum transit time. The marked approximation in part (d) is required by the exact minimum equation above.

7.26 A charged particle (charge Ze) moves at constant velocity $\mathbf{v}$ through a medium described by a dielectric function $\epsilon(\mathbf{q}, \omega)/\epsilon_0$ or, equivalently, by a conductivity function $\sigma(\mathbf{q}, \omega) = i\omega[\epsilon_0 - \epsilon(\mathbf{q}, \omega)]$. It is desired to calculate the energy loss per unit time by the moving particle in terms of the dielectric function $\epsilon(\mathbf{q}, \omega)$ in the approximation that the electric field is the negative gradient of the potential and current flow obeys Ohm's law, $\mathbf{J}(\mathbf{q}, \omega) = \sigma(\mathbf{q}, \omega)\mathbf{E}(\mathbf{q}, \omega)$.

(a) Show that with suitable normalization, the Fourier transform of the particle's charge density is

$$\rho(\mathbf{q}, \omega) = \frac{Ze}{(2\pi)^3} \, \delta(\omega - \mathbf{q} \cdot \mathbf{v})$$

(b) Show that the Fourier components of the scalar potential are

$$\phi(\mathbf{q}, \omega) = \frac{\rho(\mathbf{q}, \omega)}{q^2 \, \epsilon(\mathbf{q}, \omega)}$$

(c) Starting from $-dW/dt = \int \mathbf{J}\cdot\mathbf{E}\, d^3x$, where W is the particle energy, show that the energy loss per unit time can be written as

$$-\frac{dW}{dt} = -\frac{Z^2e^2}{4\pi^3}\int \frac{d^3q}{q^2}\int_0^\infty d\omega\,\omega\,\mathrm{Im}\left[\frac{1}{\epsilon(\mathbf{q},\omega)}\right]\delta(\omega-\mathbf{q}\cdot\mathbf{v})$$

[This shows that $-\mathrm{Im}[1/\epsilon(\mathbf{q},\omega)]$ is related to energy loss and provides, by studying characteristic energy losses in thin foils, information on $\epsilon(\mathbf{q},\omega)$ for solids.]

Solution.

Use the four-dimensional Fourier pair

$$f(\mathbf{x},t) = \int d^3q\, d\omega\, f(\mathbf{q},\omega)e^{i\mathbf{q}\cdot\mathbf{x}-i\omega t}, \qquad f(\mathbf{q},\omega) = \frac{1}{(2\pi)^4}\int d^3x\, dt\, f(\mathbf{x},t)e^{-i\mathbf{q}\cdot\mathbf{x}+i\omega t}.$$

Assume a homogeneous stationary medium with a real response kernel, so $\epsilon(-\mathbf{q},-\omega)=\epsilon^*(\mathbf{q},\omega)$.
(a) Inserting $\rho(\mathbf{x},t) = Ze\delta^3(\mathbf{x}-\mathbf{v}t)$ gives

$$\rho(\mathbf{q},\omega) = \frac{Ze}{(2\pi)^4}\int_{-\infty}^{\infty} e^{i(\omega-\mathbf{q}\cdot\mathbf{v})t}dt = \boxed{\frac{Ze}{(2\pi)^3}\delta(\omega-\mathbf{q}\cdot\mathbf{v}).}$$

(b) With $\mathbf{E} = -i\mathbf{q}\phi$ and $i\mathbf{q}\cdot\mathbf{D} = \rho$, the longitudinal dielectric relation gives $q^2\epsilon(\mathbf{q},\omega)\phi = \rho$, hence

$$\boxed{\phi(\mathbf{q},\omega) = \frac{\rho(\mathbf{q},\omega)}{q^2\epsilon(\mathbf{q},\omega)}.}$$

(c) Integrate the frequency delta function first. If $\omega_q = \mathbf{q}\cdot\mathbf{v}$, the field is

$$\mathbf{E}(\mathbf{x},t) = \frac{Ze}{(2\pi)^3}\int d^3q\,\frac{-i\mathbf{q}\,e^{i\mathbf{q}\cdot(\mathbf{x}-\mathbf{v}t)}}{q^2\epsilon(\mathbf{q},\omega_q)}.$$

The current has the same spatial Fourier amplitude multiplied by $\sigma(\mathbf{q},\omega_q)$. Spatial Parseval, pairing $\mathbf{q}$ with $-\mathbf{q}$, therefore gives the real dissipated power

$$P_{\mathrm{med}} = \int \mathbf{J}\cdot\mathbf{E}\, d^3x = \frac{Z^2e^2}{(2\pi)^3}\int\frac{d^3q}{q^2}\frac{\mathrm{Re}\,\sigma(\mathbf{q},\omega_q)}{|\epsilon(\mathbf{q},\omega_q)|^2}.$$

This calculation uses no square of a frequency delta function. From $\sigma = i\omega(\epsilon_0-\epsilon)$,

$$\mathrm{Re}\,\sigma = \omega\,\mathrm{Im}\,\epsilon, \qquad \frac{\mathrm{Re}\,\sigma}{|\epsilon|^2} = -\omega\,\mathrm{Im}\,\frac{1}{\epsilon}.$$

Thus $P_{\mathrm{med}} = -(Z^2e^2/(2\pi)^3)\int d^3q\,\omega_q\,\mathrm{Im}[1/\epsilon(\mathbf{q},\omega_q)]/q^2$. The integrand is even under $\mathbf{q}\to-\mathbf{q}$ when its frequency is also reversed. Restricting the frequency to positive values consequently supplies a factor two:

$$\boxed{-\frac{dW}{dt} = P_{\mathrm{med}} = -\frac{Z^2e^2}{4\pi^3}\int\frac{d^3q}{q^2}\int_0^\infty d\omega\,\omega\,\mathrm{Im}\left[\frac{1}{\epsilon(\mathbf{q},\omega)}\right]\delta(\omega-\mathbf{q}\cdot\mathbf{v}).}$$

Here W is the particle energy; an external agent maintaining constant speed supplies this same positive power. For a passive medium $\mathrm{Im}\,\epsilon > 0$ at positive frequency, whereas $\mathrm{Im}(1/\epsilon) < 0$. This direct sign test and the Parseval derivation verify the two marked sign corrections to the printed part (c): its original positive right-hand side would give negative loss. The nonnegative loss function is $-\mathrm{Im}(1/\epsilon)$ (often normalized by ϵ_0). The formula assumes the material response supplies a convergent wave-number integral, or that the continuum model has an appropriate microscopic cutoff; no universal finite loss follows from a purely local dielectric at arbitrarily large q.

7.27 The angular momentum of a distribution of electromagnetic fields in vacuum is given by

$$\mathbf{L} = \frac{1}{\mu_0 c^2} \int d^3x \, \mathbf{x} \times (\mathbf{E} \times \mathbf{B})$$

where the integration is over all space.

(a) For source-free radiation fields produced a finite time in the past (and so localized to a finite region of space) show that, provided the magnetic field is eliminated in favor of the vector potential $\mathbf{A}$, the angular momentum can be written in the form

$$\mathbf{L} = \frac{1}{\mu_0 c^2} \int d^3x \left[\mathbf{E} \times \mathbf{A} + \sum_{j=1}^{3} E_j (\mathbf{x} \times \nabla) A_j \right]$$

The first term is sometimes identified with the "spin" of the photon and the second with its "orbital" angular momentum because of the presence of the angular momentum operator $\mathbf{L}_{\text{op}} = -i(\mathbf{x} \times \nabla)$.

(b) Consider an expansion of the vector potential in the radiation gauge in terms of plane waves:

$$\mathbf{A}(\mathbf{x}, t) = \sum_{\lambda} \int \frac{d^3k}{(2\pi)^3} \left[\boldsymbol{\epsilon}_\lambda(\mathbf{k}) a_\lambda(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x} - i\omega t} + \text{c.c.} \right]$$

The polarization vectors $\boldsymbol{\epsilon}_\lambda(\mathbf{k})$ are conveniently chosen as the positive and negative helicity vectors $\boldsymbol{\epsilon}_\pm = (1/\sqrt{2})(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)$ where $\boldsymbol{\epsilon}_1$ and $\boldsymbol{\epsilon}_2$ are real orthogonal vectors in the plane whose positive normal is in the direction of $\mathbf{k}$.

Show that the time average of the first (spin) term of $\mathbf{L}$ can be written as

$$\mathbf{L}_{\text{spin}} = \frac{2}{\mu_0 c} \int \frac{d^3k}{(2\pi)^3} \mathbf{k} [|a_+(\mathbf{k})|^2 - |a_-(\mathbf{k})|^2]$$

Can the term "spin" angular momentum be justified from this expression? Calculate the energy of the field in terms of the plane wave expansion of $\mathbf{A}$ and compare.

Solution.

(a) Since $\mathbf{B} = \nabla \times \mathbf{A}$,

$$(\mathbf{E} \times \mathbf{B})_i = \sum_j E_j \partial_i A_j - \sum_j E_j \partial_j A_i.$$

The first term immediately gives the orbital expression after multiplication by $\epsilon_{abi} x_b$. Integrating the second by parts gives, componentwise,

$$-\int \epsilon_{abi} x_b E_j \partial_j A_i \, d^3x = \int \epsilon_{aji} E_j A_i \, d^3x + \int \epsilon_{abi} x_b A_i \nabla \cdot \mathbf{E} \, d^3x,$$

where localization removes the surface term. Consequently the *general* identity is

$$\mathbf{L} = \epsilon_0 \int \left[\mathbf{E} \times \mathbf{A} + \sum_j E_j (\mathbf{x} \times \nabla) A_j \right] d^3x + \int \rho (\mathbf{x} \times \mathbf{A}) \, d^3x.$$

For source-free radiation $\rho = 0$, it reduces to the requested

$$\boxed{\mathbf{L} = \frac{1}{\mu_0 c^2} \int \left[\mathbf{E} \times \mathbf{A} + \sum_j E_j (\mathbf{x} \times \nabla) A_j \right] d^3x.}$$

Localization removes a boundary term but does not by itself remove the charge-density term; this is the reason for the marked source-free qualification in the statement. The split depends on gauge, whereas the original total field angular momentum does not. **(b)** In the radiation gauge for free waves, $\mathbf{E} = -\partial_t \mathbf{A}$ and $\omega = ck$. Spatial integration pairs equal wave vectors. Time averaging removes terms with two positive or two negative frequencies. The remaining product in the spin integral is

$$2i\omega \sum_{\lambda\lambda'} a_\lambda a^*_{\lambda'} \boldsymbol{\epsilon}_\lambda \times \boldsymbol{\epsilon}^*_{\lambda'}.$$

The circular vectors obey

$$\boldsymbol{\epsilon}_+ \times \boldsymbol{\epsilon}^*_+ = -i\widehat{\mathbf{k}}, \qquad \boldsymbol{\epsilon}_- \times \boldsymbol{\epsilon}^*_- = +i\widehat{\mathbf{k}}, \qquad \boldsymbol{\epsilon}_+ \times \boldsymbol{\epsilon}^*_- = 0.$$

It follows that

$$\boxed{\mathbf{L}_{\text{spin}} = \frac{2}{\mu_0 c} \int \frac{d^3k}{(2\pi)^3} \, \mathbf{k} \, [|a_+(\mathbf{k})|^2 - |a_-(\mathbf{k})|^2].}$$

For the energy, the two surviving conjugate products give

$$\int \langle \mathbf{E}^2 \rangle d^3x = 2 \int \frac{d^3k}{(2\pi)^3} \, \omega^2 \sum_\lambda |a_\lambda|^2, \qquad \int \langle \mathbf{B}^2 \rangle d^3x = 2 \int \frac{d^3k}{(2\pi)^3} \, k^2 \sum_\lambda |a_\lambda|^2.$$

The second identity uses $|\mathbf{k} \times \boldsymbol{\epsilon}_\lambda|^2 = k^2$. Multiplying by $\epsilon_0/2$ and $1/(2\mu_0)$ shows that the two energy contributions are equal. Adding them gives

$$\boxed{U = 2\epsilon_0 \int \frac{d^3k}{(2\pi)^3} \, \omega^2 (|a_+|^2 + |a_-|^2) = \frac{2}{\mu_0} \int \frac{d^3k}{(2\pi)^3} \, k^2 (|a_+|^2 + |a_-|^2).}$$

For one wave vector and one helicity, $\mathbf{L}_{\text{spin}}/U = \pm\widehat{\mathbf{k}}/\omega$. Assigning energy $\hbar\omega$ per photon therefore assigns $\pm\hbar\widehat{\mathbf{k}}$ per photon. This intrinsic helicity, independent of a transverse orbital profile, justifies the spin terminology in the fixed radiation gauge. There are two transverse helicities; this argument introduces no longitudinal photon state.

7.28 A circularly polarized plane wave moving in the z direction has a finite extent in the x and y directions. Assuming that the amplitude modulation is slowly varying (the wave is many wavelengths broad), show that the electric and magnetic fields are given approximately by

$$\mathbf{E}(x, y, z, t) \simeq \left[E_0(x, y)(\mathbf{e}_1 \pm i\mathbf{e}_2) + \frac{i}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \mathbf{e}_3 \right] e^{ikz - i\omega t}$$
$$\mathbf{B} \simeq \mp i\sqrt{\mu\epsilon} \, \mathbf{E}$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ are unit vectors in the x, y, z directions.

Solution.

Write the helicity sign as $s = \pm 1$ and suppress the common factor $e^{ikz-i\omega t}$. At leading order the transverse field is $\mathbf{E}_\perp = E_0(\mathbf{e}_x + is\mathbf{e}_y)$. For a homogeneous source-free medium, $\nabla \cdot \mathbf{E} = 0$ requires

$$\partial_x E_0 + is\partial_y E_0 + ikE_z = 0, \qquad \boxed{E_z = \frac{i}{k}(\partial_x + is\partial_y)E_0.}$$

Let w denote the scale of envelope variation. This component is of relative order $(kw)^{-1}$. Faraday's law is $\mathbf{B} = (i\omega)^{-1}\nabla \times \mathbf{E}$. The transverse components, dropping second transverse derivatives of E_0, are

$$B_x = -is\frac{k}{\omega}E_0 + O((kw)^{-2}E_0 k/\omega), \qquad B_y = \frac{k}{\omega}E_0 + O((kw)^{-2}E_0 k/\omega).$$

The longitudinal component is

$$B_z = \frac{1}{i\omega}(is\partial_x E_0 - \partial_y E_0) = \frac{s\partial_x E_0 + i\partial_y E_0}{\omega} = -is\frac{k}{\omega}E_z.$$

Since $k/\omega = \sqrt{\mu\epsilon}$, these equations give

$$\boxed{\mathbf{E} \simeq \left[E_0(\mathbf{e}_x + is\mathbf{e}_y) + \frac{i}{k}(\partial_x + is\partial_y)E_0\,\mathbf{e}_z\right]e^{ikz-i\omega t}, \qquad \mathbf{B} \simeq -is\sqrt{\mu\epsilon}\,\mathbf{E}.}$$

The errors in the transverse magnetic field and the wave equation are of relative order $(kw)^{-2}$. A finite beam cannot have exactly unchanging transverse shape for all z; the displayed expression neglects diffraction over distances small compared with kw^2. For a propagating lossless beam the expansion assumes real k and $kw \gg 1$.

7.29 For the circularly polarized wave of Problem 7.28 with $E_0(x, y)$ a real function of x and y, calculate the time-averaged component of angular momentum parallel to the direction of propagation. Show that the ratio of this component of angular momentum to the energy of the wave in vacuum is

$$\frac{L_3}{U} = \pm\omega^{-1}$$

Interpret this result in terms of quanta of radiation (photons). Show that for a cylindrically symmetric, finite plane wave, the transverse components of angular momentum vanish.

Solution.

Take vacuum, put $s = \pm 1$, and set $f(x, y) = E_0(x, y)$ real and $I = f^2$. With the phasor convention of Problem 7.28,

$$\mathbf{E} = (f, isf, (if_x - sf_y)/k), \qquad \mathbf{B} = -is\mathbf{E}/c$$

to the indicated paraxial order. A direct cross product gives

$$\langle \mathbf{S} \rangle = \frac{1}{2\mu_0}\,\mathrm{Re}(\mathbf{E} \times \mathbf{B}^*) = \frac{1}{\mu_0 c}\left[I\mathbf{e}_z + \frac{s}{2k}(I_y\mathbf{e}_x - I_x\mathbf{e}_y)\right].$$

Therefore the momentum density is

$$\mathbf{g} = \frac{\langle \mathbf{S} \rangle}{c^2} = \frac{\epsilon_0}{c}\left[I\mathbf{e}_z + \frac{s}{2k}(I_y\mathbf{e}_x - I_x\mathbf{e}_y)\right].$$

The small transverse flow, although one order smaller in the paraxial parameter, provides the axial angular momentum:

$$L_3 = \int (x g_y - y g_x) d^3x = -\frac{s\epsilon_0}{2kc} \int (x I_x + y I_y) d^3x.$$

For a finite beam whose transverse intensity decays sufficiently fast, integration by parts gives $\int x I_x \, dxdy = -\int I \, dxdy$ and the same result for $y I_y$. Consequently

$$L_3 = \frac{s\epsilon_0}{kc} \int I \, d^3x.$$

The leading energy density is $\epsilon_0 |\mathbf{E}|^2/4 + |\mathbf{B}|^2/(4\mu_0) = \epsilon_0 I$, so

$$\boxed{\frac{L_3}{U} = \frac{s}{kc} = \pm\frac{1}{\omega}.}$$

For an indefinitely long monochromatic beam both volume integrals are infinite; the ratio is understood per unit longitudinal length, or for a long finite packet with a common slowly varying longitudinal envelope. Corrections beyond the approximation of Problem 7.28 are of relative order $(kw)^{-2}$. A photon of energy $\hbar\omega$ thus carries axial angular momentum $s\hbar$, agreeing with the helicity interpretation of Problem 7.27.

For a cylindrically symmetric beam centered on the z axis, $\int x I \, dxdy = \int y I \, dxdy = 0$. Also $\int g_x \, dxdy = \int g_y \, dxdy = 0$ because they are total transverse derivatives. Hence

$$L_x = \int (y g_z - z g_y) d^3x = 0, \qquad L_y = \int (z g_x - x g_z) d^3x = 0.$$

These cancellations hold at each z for a symmetric finite packet; the transverse angular momentum vanishes about its central axis.

7.30 Starting with the expression for the total energy of an arbitrary superposition of plane electromagnetic waves (7.8, 7.11) in otherwise empty space, show that the total number of photons (defined for each plane wave of wave vector $\mathbf{k}$ and polarization $\boldsymbol{\epsilon}$ as its energy divided by $\hbar ck$) is given by the double integral

$$N = \frac{\epsilon_0}{4\pi^2 \hbar c} \int d^3x \int d^3x' \left[\frac{\mathbf{E}(\mathbf{x}, t) \cdot \mathbf{E}(\mathbf{x}', t) + c^2 \, \mathbf{B}(\mathbf{x}, t) \cdot \mathbf{B}(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|^2} \right]$$

Solution.

At a fixed time use the three-dimensional version of Jackson's unitary Fourier pair [Eqs. (7.80)–(7.81)]:

$$\mathbf{E}(\mathbf{x}) = \frac{1}{(2\pi)^{3/2}} \int d^3k \, \widetilde{\mathbf{E}}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \widetilde{\mathbf{E}}(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int d^3x \, \mathbf{E}(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}.$$

and similarly for $\mathbf{B}$. Reality relates the two signs of $\mathbf{k}$. Parseval gives the total energy

$$U = \frac{\epsilon_0}{2} \int d^3k \left(|\widetilde{\mathbf{E}}(\mathbf{k})|^2 + c^2 |\widetilde{\mathbf{B}}(\mathbf{k})|^2 \right).$$

For free radiation each mode at wave-number magnitude k has frequency ck. Dividing its spectral energy by $\hbar ck$ gives

$$N = \frac{\epsilon_0}{2\hbar c} \int \frac{d^3k}{k} \left(|\widetilde{\mathbf{E}}|^2 + c^2 |\widetilde{\mathbf{B}}|^2 \right).$$

Expanding each squared Fourier amplitude into its two spatial integrals supplies a factor $(2\pi)^{-3}$. To transform the $1/k$ kernel back to space, insert a regulator $e^{-\eta k}$, $\eta > 0$, and put $R = |\mathbf{x} - \mathbf{x}'|$. Angular integration gives

$$\begin{aligned} K_\eta(R) &= \int \frac{d^3k}{(2\pi)^3} \frac{e^{-\eta k + i\mathbf{k}\cdot\mathbf{R}}}{k} \\ &= \frac{1}{2\pi^2 R} \int_0^\infty e^{-\eta k} \sin(kR)\, dk = \frac{1}{2\pi^2 (R^2 + \eta^2)}. \end{aligned}$$

Thus $K(R) = \lim_{\eta \downarrow 0} K_\eta(R) = 1/(2\pi^2 R^2)$ in the regulated Fourier sense. Expanding the squared Fourier amplitudes as double spatial integrals now yields

$$\boxed{N = \frac{\epsilon_0}{4\pi^2 \hbar c} \int d^3x\, d^3x' \, \frac{\mathbf{E}(\mathbf{x}, t) \cdot \mathbf{E}(\mathbf{x}', t) + c^2 \mathbf{B}(\mathbf{x}, t) \cdot \mathbf{B}(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|^2}.}$$

The diagonal singularity is locally integrable for smooth fields because the three-dimensional volume element supplies $R^2 dR$. The fields must also have sufficient decay, and their spectrum must have a finite $1/k$-weighted energy, for a finite photon number; finite energy alone does not exclude an infrared divergence of that number. The expression uses only physical fields and is independent of gauge. Its nonlocal spatial kernel reflects the mode-dependent division by frequency.

8 Chapter 8. Waveguides, Resonant Cavities, and Optical Fibers

Problems

8.1 Consider the electric and magnetic fields in the surface region of an excellent conductor in the approximation of Section 8.1, where the skin depth is very small compared to the radii of curvature of the surface or the scale of significant spatial variation of the fields just outside.

(a) For a single-frequency component, show that the magnetic field $\mathbf{H}$ and the current density $\mathbf{J}$ are such that $\mathbf{f}$, the time-averaged force per unit area at the surface from the conduction current, is given by

$$\mathbf{f} = -\mathbf{n}\,\frac{\mu_c}{4}\,|\mathbf{H}_\parallel|^2$$

where $\mathbf{H}_\parallel$ is the peak parallel component of magnetic field at the surface, μ_c is the magnetic permeability of the conductor, and $\mathbf{n}$ is the outward normal at the surface.

(b) If the magnetic permeability μ outside the surface is different from μ_c, is there an additional magnetic force per unit area? What about electric forces?

(c) Assume that the fields are a superposition of different frequencies (all high enough that the approximations still hold). Show that the time-averaged force takes the same form as in part a with $|\mathbf{H}_\parallel|^2$ replaced by $2\langle|\mathbf{H}_\parallel|^2\rangle$, where the angle brackets $\langle\cdots\rangle$ mean time average.

Solution.

Use peak phasors with time dependence $e^{-i\omega t}$ and measure $u \geq 0$ inward from the surface. Locally the conductor is a half-space, and the leading skin-layer fields are

$$\mathbf{H}(u) = \mathbf{H}_\parallel e^{-(1-i)u/\delta}, \qquad \mathbf{J} = \nabla\times\mathbf{H} = \frac{1-i}{\delta}\,\mathbf{n}\times\mathbf{H}, \quad \delta = \sqrt{\frac{2}{\omega\mu_c\sigma}}.$$

Here $\nabla = -\mathbf{n}\partial_u$ to this order and displacement current is negligible compared with $\sigma\mathbf{E}$.

(a) The time average of the conduction-current force density is

$$\langle \mathbf{J}\times\mathbf{B}\rangle = \frac{\mu_c}{2}\,\mathrm{Re}(\mathbf{J}\times\mathbf{H}^*) = -\frac{\mu_c}{2\delta}\,\mathbf{n}|\mathbf{H}_\parallel|^2 e^{-2u/\delta},$$

where $(\mathbf{n}\times\mathbf{H})\times\mathbf{H}^* = -\mathbf{n}|\mathbf{H}|^2$ since $\mathbf{n}\cdot\mathbf{H} = 0$. Integration through the layer gives

$$\boxed{\mathbf{f} = \int_0^\infty \langle \mathbf{J}\times\mathbf{B}\rangle du = -\frac{\mu_c}{4}|\mathbf{H}_\parallel|^2\mathbf{n}.}$$

The result is a magnetic pressure directed into the conductor.

(b) This is only the force from the conduction current. In the macroscopic Maxwell-stress description, the permeability jump also produces the magnetic traction

$$\mathbf{f}_\mu = \frac{\mu_c-\mu}{4}|\mathbf{H}_\parallel|^2\mathbf{n}.$$

One way to see the sign and magnitude is to integrate the magnetic material-force term $-\frac{1}{2}H^2\nabla\mu$ across the interface: $\int \nabla\mu\, du_{\text{outward}} = (\mu-\mu_c)\mathbf{n}$ and $\langle H^2\rangle = |H_\parallel|^2/2$. Adding the two contributions leaves the exterior magnetic traction $-\mu|H_\parallel|^2\mathbf{n}/4$. The surface charge supplies an outward electric

traction $+\epsilon|E_\perp|^2\mathbf{n}/4$, to leading order in the excellent-conductor approximation. Thus the leading total macroscopic electromagnetic traction is

$$\frac{1}{4}\left(\epsilon|E_\perp|^2 - \mu|H_\parallel|^2\right)\mathbf{n}.$$

This uses linear, isotropic media and the stated surface approximation; mechanical stresses of the material are separate from this electromagnetic traction. Tangential electric fields and internal normal electric fields supply higher-order corrections.

(c) For distinct frequencies, products oscillating at their sum or difference average to zero over a sufficiently long time. Combine components of the same frequency into a single phasor first. For frequency-independent μ_c in the range considered, summing (a) gives

$$\mathbf{f} = -\frac{\mu_c}{4}\mathbf{n}\sum_\omega |\mathbf{H}_\omega|^2 = -\frac{\mu_c}{2}\mathbf{n}\langle|\mathbf{H}_\parallel(t)|^2\rangle.$$

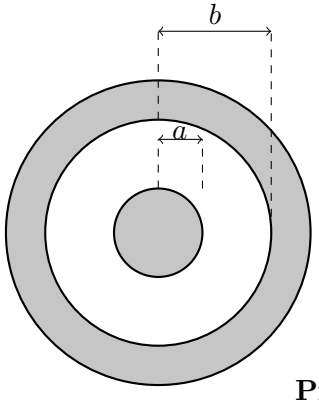
This is the claimed replacement. If permeability is dispersive, the first expression instead contains $\mu_c(\omega)$ in each frequency contribution; it cannot in general be factored out.

8.2 A transmission line consisting of two concentric circular cylinders of metal with conductivity σ and skin depth δ, as shown, is filled with a uniform lossless dielectric (μ, ϵ). A TEM mode is propagated along this line. Section 8.1 applies.

(a) Show that the time-averaged power flow along the line is

$$P = \sqrt{\frac{\mu}{\epsilon}}\,\pi a^2\,|H_0|^2\,\ln\left(\frac{b}{a}\right)$$

where H_0 is the peak value of the azimuthal magnetic field at the surface of the inner conductor.



Problem 8.2

(b) Show that the transmitted power is attenuated along the line as

$$P(z) = P_0 e^{-2\gamma z}$$

where

$$\gamma = \frac{1}{2\sigma\delta}\sqrt{\frac{\epsilon}{\mu}}\,\frac{\left(\frac{1}{a}+\frac{1}{b}\right)}{\ln\left(\frac{b}{a}\right)}$$

(c) The characteristic impedance Z_0 of the line is defined as the ratio of the voltage between the cylinders to the axial current flowing in one of them at any position z. Show that for this line

$$Z_0 = \frac{1}{2\pi}\sqrt{\frac{\mu}{\epsilon}}\,\ln\!\left(\frac{b}{a}\right)$$

(d) Show that the series resistance and inductance per unit length of the line are

$$R = \frac{1}{2\pi\sigma\delta}\left(\frac{1}{a}+\frac{1}{b}\right)$$
$$L = \left\{\frac{\mu}{2\pi}\ln\!\left(\frac{b}{a}\right) + \frac{\mu_c\delta}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right)\right\}$$

where μ_c is the permeability of the conductor. The correction to the inductance comes from the penetration of the flux into the conductors by a distance of order δ.

Solution.

Here b is the radius of the *inner surface* of the outer conductor. Set $\eta = \sqrt{\mu/\epsilon}$ and $R_s = 1/(\sigma\delta)$, the surface resistance. To leading order in skin depth, the TEM fields in $a < \rho < b$ are

$$H_\phi(\rho) = \frac{aH_0}{\rho}, \qquad E_\rho = \eta H_\phi, \qquad k = \omega\sqrt{\mu\epsilon}.$$

Ampere's law fixes $I = 2\pi a H_0$, while the radial electrostatic solution and the plane-wave relation fix E_ρ.

(a) With peak phasors the mean axial Poynting flux is $\eta|H_\phi|^2/2$. Therefore

$$P = \frac{\eta}{2}\int_a^b \frac{a^2|H_0|^2}{\rho^2}2\pi\rho\,d\rho = \boxed{\eta\pi a^2|H_0|^2\ln(b/a)}.$$

(b) The dissipated power per unit length on both surfaces is

$$P'_\ell = \frac{R_s}{2}\oint |H_\parallel|^2 dl = \pi R_s a^2|H_0|^2\left(\frac{1}{a}+\frac{1}{b}\right).$$

Conservation of energy requires $dP/dz = -P'_\ell = -2\gamma P$. Dividing by the preceding expression for P gives

$$\gamma = \frac{R_s}{2\eta}\frac{a^{-1}+b^{-1}}{\ln(b/a)}, \qquad P(z) = P(0)e^{-2\gamma z}.$$

(c) The voltage magnitude is $V = \int_a^b E_\rho d\rho = \eta a H_0\ln(b/a)$. Consequently

$$\boxed{Z_0 = V/I = \frac{\eta}{2\pi}\ln(b/a).}$$

It is the leading, real characteristic impedance; conductor losses produce small complex corrections to the exact impedance.

(d) From $P'_\ell = R|I|^2/2$,

$$\boxed{R = \frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right).}$$

The average magnetic energy per unit length in the dielectric is

$$U_{m,\text{ext}} = \frac{\mu}{4}\int_a^b |H_\phi|^2\, 2\pi\rho d\rho = \frac{\mu\pi a^2|H_0|^2}{2}\ln(b/a).$$

The penetrating field decays as $e^{-u/\delta}$ in amplitude, giving

$$U_{m,\text{skin}} = \frac{\mu_c}{4}\oint dl\,|H_\parallel|^2\int_0^\infty e^{-2u/\delta}du = \frac{\mu_c\delta\pi a^2|H_0|^2}{4}\left(\frac{1}{a}+\frac{1}{b}\right).$$

Using $U_m = L|I|^2/4$ then gives

$$\boxed{L = \frac{\mu}{2\pi}\ln(b/a) + \frac{\mu_c\delta}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right).}$$

As a check, $\omega L_{\text{skin}} = R$ because $\omega\mu_c\delta/2 = 1/(\sigma\delta)$, the equality of the resistive and reactive parts of the good-conductor surface impedance.

8.3

(a) A transmission line consists of two identical thin strips of metal, shown in cross section in the sketch. Assuming that $b \gg a$, discuss the propagation of a TEM mode on this line, repeating the derivations of Problem 8.2. Show that

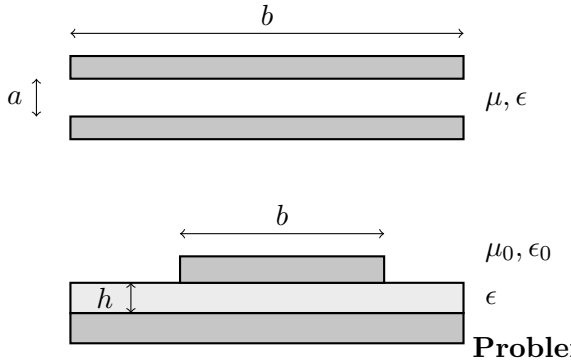
$$P = \frac{ab}{2}\sqrt{\frac{\mu}{\epsilon}}\,|H_0|^2$$

$$\gamma = \frac{1}{a\sigma\delta}\sqrt{\frac{\epsilon}{\mu}}$$

$$Z_0 = \sqrt{\frac{\mu}{\epsilon}}\left(\frac{a}{b}\right)$$

$$R = \frac{2}{\sigma\delta b}$$

$$L = \left(\frac{\mu a + \mu_c\delta}{b}\right)$$



Problem 8.3

where the symbols on the left have the same meanings as in Problem 8.2.

(b) The lower half of the figure shows the cross section of a microstrip line with a strip of width b mounted on a dielectric substrate of thickness h and dielectric constant ϵ, all on a ground plane. What differences occur here compared to part a if $b \gg h$? If $b \ll h$?

Solution.

(a) Neglect fringing when $b \gg a$ and assume each metal strip is thick compared with its skin depth. In the gap the fields are uniform, with $E_0 = \eta H_0$, where $\eta = \sqrt{\mu/\epsilon}$. The voltage and current are $V = aE_0$ and $I = bH_0$. The two inner faces carry opposite axial currents; the exterior field is negligible at this order. Thus

$$P = \frac{1}{2}ab\eta|H_0|^2, \qquad Z_0 = \frac{V}{I} = \eta\frac{a}{b}.$$

The two active surfaces dissipate $P'_\ell = (R_s/2)(2b|H_0|^2) = R_s b|H_0|^2$. Comparison with $P'_\ell = 2\gamma P = R|I|^2/2$ gives

$$\boxed{\gamma = \frac{R_s}{\eta a} = \frac{1}{a\sigma\delta}\sqrt{\frac{\epsilon}{\mu}}, \qquad R = \frac{2R_s}{b} = \frac{2}{\sigma\delta b}.}$$

The external magnetic energy is $\mu ab|H_0|^2/4$. The energy in both skin layers is $(\mu_c/4)(2b|H_0|^2)(\delta/2) = \mu_c\delta b|H_0|^2/4$. Equating their sum to $L|I|^2/4$ yields

$$\boxed{L = (\mu a + \mu_c\delta)/b.}$$

These are all the stated results, with energy velocity $1/\sqrt{\mu\epsilon}$ to leading order. "Thin strips" means thin geometrically, not thinner than the penetration depth.

(b) For $b \gg h$, nearly all of the electric and magnetic energy lies between the strip and the ground plane. The preceding parallel-plate formulas apply to leading order with $a = h$ and the substrate permittivity and permeability; fields at the two strip edges provide relative corrections that vanish as $h/b \to 0$. For finite b/h the air and substrate have different wave speeds, so the mode is generally only approximately TEM.

For $b \ll h$ the fields spread transversely over distances of order h, both in the substrate and in the air. There is no uniform-field plate region. In the electrically small, quasi-static limit the line is described by $Z_0 = \sqrt{L'/C'}$ and $v = 1/\sqrt{L'C'}$, where C' is obtained from the two-dielectric electrostatic problem and L' from the corresponding magnetostatic problem. For nonmagnetic media the effective permittivity lies between ϵ_0 and ϵ; in the very narrow-strip limit its leading value is $(\epsilon + \epsilon_0)/2$, because the local field around a strip on the interface occupies two half-spaces. To see both this average and the logarithmic dependence, let λ be charge per unit length and consider $b \ll \rho \ll h$. The two semicircles in a Gaussian cylinder give

$$\pi\rho(\epsilon + \epsilon_0)E_\rho \simeq \lambda, \qquad V \simeq \frac{\lambda}{\pi(\epsilon + \epsilon_0)}\mathcal{L}, \qquad \mathcal{L} = \ln(h/b) + O(1).$$

The additive constant depends on the near-strip and ground-plane geometry. Ampere's law gives $H_\phi \simeq I/(2\pi\rho)$ in the same intermediate region. Integrating its magnetic energy gives

$$\frac{\mu_0}{4}\int_b^h \left|\frac{I}{2\pi\rho}\right|^2 2\pi\rho\, d\rho \simeq \frac{\mu_0|I|^2}{8\pi}\mathcal{L} = \frac{L'|I|^2}{4}.$$

Thus $C' \simeq \pi(\epsilon + \epsilon_0)/\mathcal{L}$, $L' \simeq \mu_0\mathcal{L}/(2\pi)$, and

$$v \simeq \frac{1}{\sqrt{\mu_0(\epsilon + \epsilon_0)/2}}, \qquad Z_0 \simeq \frac{\mathcal{L}}{2\pi}\sqrt{\frac{2\mu_0}{\epsilon + \epsilon_0}}.$$

The impedance therefore grows logarithmically with h/b. The strongly nonuniform strip and return currents also require the surface-loss integral with their actual distributions; the wide-plate resistance and attenuation formulas no longer apply. At wavelengths comparable with the transverse dimensions, the mode has longitudinal field components and geometric dispersion, so even the quasi-TEM circuit description needs correction.

8.4 Transverse electric and magnetic waves are propagated along a hollow, right circular cylinder with inner radius R and conductivity σ.

(a) Find the cutoff frequencies of the various TE and TM modes. Determine numerically the lowest cutoff frequency (the dominant mode) in terms of the tube radius and the ratio of cutoff frequencies of the next four higher modes to that of the dominant mode. For this part assume that the conductivity of the cylinder is infinite.

(b) Calculate the attenuation constants of the waveguide as a function of frequency for the lowest two distinct modes and plot them as a function of frequency.

Solution.

Let $v = 1/\sqrt{\mu\epsilon}$ and $\eta = \sqrt{\mu/\epsilon}$ for the lossless filling (vacuum gives $v = c$). Write the fields as $e^{ikz-i\omega t}$, and put $\gamma^2 = \omega^2/v^2 - k^2$. Here γ is the transverse eigenvalue parameter of (8.34), and β below is the attenuation constant of Section 8.5. We use $H_z = B_z/\mu$ when evaluating conductor losses; the constant factor μ does not change the scalar eigenvalue problem.

(a) Regular longitudinal fields have radial dependence $J_m(\gamma\rho)$ and angular dependence $\cos m\phi$ or $\sin m\phi$. A TM mode has $E_z = 0$ on the wall, while a TE mode has $\partial_\rho H_z = 0$ there. Thus

$$\boxed{\omega_{c,mn}^{\mathrm{TM}} = \frac{v}{R}x_{mn}, \quad J_m(x_{mn}) = 0; \qquad \omega_{c,mn}^{\mathrm{TE}} = \frac{v}{R}x'_{mn}, \quad J'_m(x'_{mn}) = 0.}$$

Use positive roots only; the constant Neumann scalar mode does not produce a nonzero TE electromagnetic field in a hollow guide. For $m > 0$ each family has two angular polarizations. The first families, with their cutoffs divided by that of TE_{11}, are

family	$\omega_c R/v$	$\omega_c/\omega_{c,11}^{\mathrm{TE}}$
TE_{11}	1.841183781	1
TM_{01}	2.404825558	1.306130101
TE_{21}	3.054236928	1.658844141
TE_{01}	3.831705970	2.081109995
TM_{11}	3.831705970	2.081109995
TE_{31}	4.201188941	2.281786850

The middle two families have the same cutoff. Thus the table gives both the next four families and, if "higher" is counted by distinct frequency, the next four distinct frequencies. The dominant cutoff is $\omega_c = 1.841183781v/R$, or $f_c = 0.293033v/R$.

(b) Let $R_s = 1/(\sigma\delta) = \sqrt{\omega\mu_c/(2\sigma)}$ and denote amplitude attenuation by β, so power decays as $e^{-2\beta z}$. The loss formula is

$$\beta = \frac{R_s}{4P}\oint |H_\parallel|^2 dl.$$

For clarity, the field and normalization steps behind this formula are as follows. Maxwell's equations, with the phase convention above, give the transverse fields in (8.26):

$$\mathbf{E}_t = \frac{i}{\gamma^2}(k\nabla_t E_z - \omega\mu\hat{\mathbf{z}} \times \nabla_t H_z),$$
$$\mathbf{H}_t = \frac{i}{\gamma^2}(k\nabla_t H_z + \omega\epsilon\hat{\mathbf{z}} \times \nabla_t E_z).$$

For either scalar ψ, integration by parts and its Dirichlet or Neumann wall condition imply

$$\int |\nabla_t \psi|^2 dA = -\int \psi^* \nabla_t^2 \psi \, dA + \oint \psi^* \partial_n \psi \, dl = \gamma^2 N, \qquad N = \int |\psi|^2 dA.$$

Inserting the transverse fields into $P = \frac{1}{2}\,\mathrm{Re}\int (\mathbf{E}_t \times \mathbf{H}_t^*) \cdot \hat{\mathbf{z}}\, dA$ then supplies the power formulas below. For a TM field $\psi = E_z$ with $N = \int |\psi|^2 dA$,

$$P = \frac{\omega\epsilon k}{2\gamma^2} N, \qquad \oint |H_\parallel|^2 dl = \frac{\omega^2\epsilon^2}{\gamma^4} \oint |\partial_n \psi|^2 dl.$$

The radial normalization follows directly from Bessel's equation:

$$\frac{d}{dx}\left\{x^2 J_m'(x)^2 + (x^2 - m^2) J_m(x)^2\right\} = 2x J_m(x)^2.$$

Its integral from 0 to $\chi = \gamma R$ gives

$$\int_0^R \rho J_m(\gamma\rho)^2 d\rho = \frac{R^2}{2}\left[J_m'(\chi)^2 + \left(1 - \frac{m^2}{\chi^2}\right) J_m(\chi)^2\right].$$

At a TM zero, the angular factor cancels between the area integral and the boundary integral $R\gamma^2 J_m'(\chi)^2$ times that factor. This gives $\oint |\partial_n \psi|^2 dl = 2\gamma^2 N/R$. Hence

$$\boxed{\beta_{\mathrm{TM}} = \frac{R_s}{\eta R\sqrt{1 - (\omega_c/\omega)^2}}.}$$

For TE modes put $\psi = H_z$. At the wall the longitudinal field and the tangential transverse field both contribute:

$$P = \frac{\omega\mu k}{2\gamma^2} N, \qquad \oint |H_\parallel|^2 dl = \oint \left(|\psi|^2 + \frac{k^2}{\gamma^4}|\partial_s \psi|^2\right) dl.$$

For $\chi = x'_{mn}$ the Bessel norm and the angular derivative give

$$\frac{\oint |\psi|^2 dl}{N} = \frac{2}{R(1 - m^2/\chi^2)}, \qquad \oint |\partial_s \psi|^2 dl = \frac{m^2}{R^2} \oint |\psi|^2 dl.$$

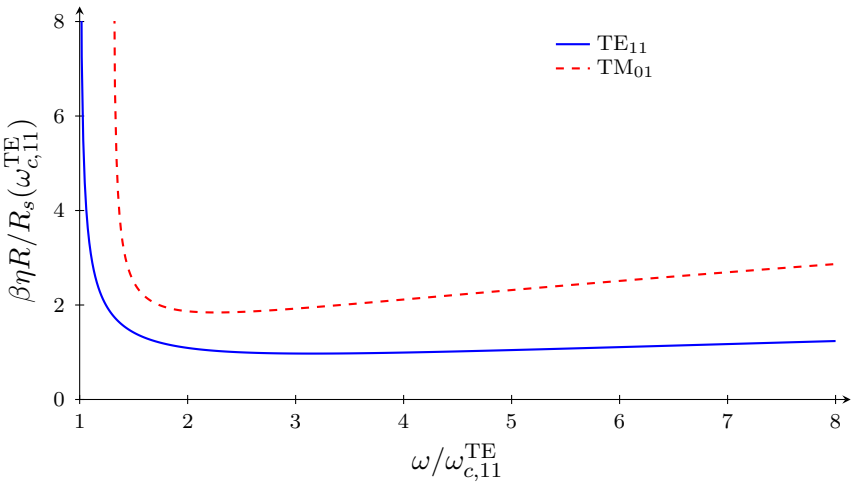
Substitute $k^2 = (\omega/v)^2 - \chi^2/R^2$. To display the simplification, put $t = \omega_c/\omega$; the dimensionless factor is

$$\frac{t^2}{1 - m^2/\chi^2}\left[1 + \frac{m^2}{\chi^2}\frac{1 - t^2}{t^2}\right] = t^2 + \frac{m^2}{\chi^2 - m^2}.$$

Therefore

$$\boxed{\beta_{\mathrm{TE},mn} = \frac{R_s}{\eta R\sqrt{1 - (\omega_c/\omega)^2}}\left[\left(\frac{\omega_c}{\omega}\right)^2 + \frac{m^2}{\chi^2 - m^2}\right].}$$

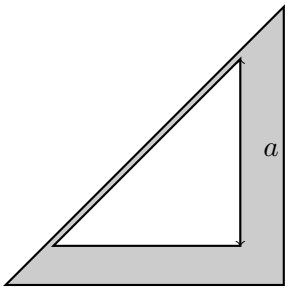
Use $m = 1$, $\chi = 1.841183781$ for the lowest TE mode, and the TM formula with $\omega_c R/v = 2.404825558$ for the next distinct mode. The plot retains $R_s \propto \sqrt{\omega}$; the normalization uses its value at the dominant cutoff.



The perturbative formulas apply above cutoff when the attenuation is small compared with k. Their divergences arbitrarily close to cutoff mark a failure of that approximation, not an infinite physical loss. Below cutoff there is already evanescent decay in a perfect guide.

8.5 A waveguide is constructed so that the cross section of the guide forms a right triangle with sides of length a, a, $\sqrt{2}a$, as shown. The medium inside has $\mu_r = \epsilon_r = 1$.

(a) Assuming infinite conductivity for the walls, determine the possible modes of propagation and their cutoff frequencies.



Problem 8.5

(b) For the lowest modes of each type calculate the attenuation constant, assuming that the walls have large, but finite, conductivity. Compare the result with that for a square guide of side a made from the same material.

Solution.

Choose the triangle $0 < y < x < a$, one half of the square $0 < x, y < a$. Reflection across $x = y$ converts the triangular boundary conditions into a parity condition in the square.

(a) TM fields vanish on all three sides. Antisymmetric combinations of the square's Dirichlet modes give

$$E_z = E_0 \left[\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{a} - \sin\frac{n\pi x}{a}\sin\frac{m\pi y}{a}\right], \quad m > n \geq 1.$$

TE fields have zero normal derivative. The required symmetric Neumann combinations are

$$H_z = H_0 \left[\cos\frac{m\pi x}{a}\cos\frac{n\pi y}{a} + \cos\frac{n\pi x}{a}\cos\frac{m\pi y}{a}\right], \quad m \geq n \geq 0, \quad (m,n) \neq (0,0).$$

For $m = n$ the redundant factor of two can be absorbed into H_0. Reflection gives a complete set, and for both types

$$\boxed{\omega_{c,mn} = \frac{\pi c}{a}\sqrt{m^2+n^2}, \qquad k^2 = (\omega/c)^2 - (\pi/a)^2(m^2+n^2).}$$

There is no TEM mode in this simply connected hollow cross section. The lowest TE and TM fields are respectively $\cos(\pi x/a) + \cos(\pi y/a)$ and the antisymmetric $(2,1)$ combination; their cutoffs are $\pi c/a$ and $\sqrt{5}\pi c/a$.

(b) Set $R_s = 1/(\sigma\delta)$, $\eta_0 = \sqrt{\mu_0/\epsilon_0}$, $t = \omega_c/\omega$ and $s = \sqrt{1-t^2}$ for the mode in question. For TM modes the general surface-loss expression is

$$\beta_{\mathrm{TM}} = \frac{R_s \omega \epsilon_0}{2k\gamma^2} \frac{\oint |\partial_n \psi|^2 dl}{\int |\psi|^2 dA}.$$

For the lowest TM mode of unit longitudinal amplitude, ψ^2 is symmetric across the diagonal. Orthogonality of the two sine products over the full square therefore gives $N = \frac{1}{2}[2(a/2)^2] = a^2/4$. The bottom and right normal derivatives, up to an irrelevant overall sign, are

$$\frac{\pi}{a}\left[\sin\frac{2\pi x}{a} - 2\sin\frac{\pi x}{a}\right], \qquad \frac{\pi}{a}\left[2\sin\frac{\pi y}{a} + \sin\frac{2\pi y}{a}\right].$$

Their squared integrals are each $(\pi/a)^2(a/2)(1+4) = 5\pi^2/(2a)$. On the diagonal, put $x = y = u$, use $dl = \sqrt{2}\,du$, and find $|\partial_n\psi|^2 = (8\pi^2/a^2)\sin^6(\pi u/a)$. Since $\int_0^\pi \sin^6 v\,dv = 5\pi/16$, the diagonal contributes $5\sqrt{2}\pi^2/(2a)$. Consequently

$$\boxed{\beta^{\triangle}_{\mathrm{TM},21} = \frac{R_s}{\eta_0 a\sqrt{1-(\omega_{c,21}/\omega)^2}}(2+\sqrt{2}).}$$

For the lowest TE field $\psi = \cos(\pi x/a) + \cos(\pi y/a)$, $\gamma = \pi/a$ and direct integrals on the same three sides give

$$N = \frac{a^2}{2}, \qquad S_0 = \oint \psi^2 dl = a(3+2\sqrt{2}), \qquad S_1 = \oint |\partial_s\psi|^2 dl = \frac{\pi^2}{a}(1+\sqrt{2}).$$

For example, the bottom and right contributions to S_0 are each $3a/2$, while the diagonal gives $2\sqrt{2}a$; their contributions to S_1 are $\pi^2/(2a), \pi^2/(2a), \sqrt{2}\pi^2/a$. Use

$$\beta_{\mathrm{TE}} = \frac{R_s\gamma^2}{2\omega\mu_0 kN}\left(S_0 + \frac{k^2}{\gamma^4}S_1\right)$$

to obtain

$$\boxed{\beta^{\triangle}_{\mathrm{TE},10} = \frac{R_s}{\eta_0 a s}\left[(1+\sqrt{2}) + (2+\sqrt{2})t^2\right].}$$

For comparison, the square of side a has

$$\beta^{\square}_{\mathrm{TE},10} = \frac{R_s}{\eta_0 a s}(1+2t^2), \quad \omega^{\square}_{c,10} = \pi c/a, \qquad \beta^{\square}_{\mathrm{TM},11} = \frac{2R_s}{\eta_0 a s}, \quad \omega^{\square}_{c,11} = \sqrt{2}\pi c/a.$$

The TE cutoffs are the same, so their attenuation ratio at fixed frequency is $[(1+\sqrt{2}) + (2+\sqrt{2})t^2]/(1+2t^2)$. For TM the square has the lower cutoff. At a fixed frequency above both cutoffs, the ratio is $(1+1/\sqrt{2})s_{\square}/s_{\triangle}$.

8.6 A resonant cavity of copper consists of a hollow, right circular cylinder of inner radius R and length L, with flat end faces.

(a) Determine the resonant frequencies of the cavity for all types of waves. With $(1/\sqrt{\mu\epsilon}\,R)$ as a unit of frequency, plot the lowest four resonant frequencies of each type as a function of R/L for $0 < R/L < 2$. Does the same mode have the lowest frequency for all R/L?

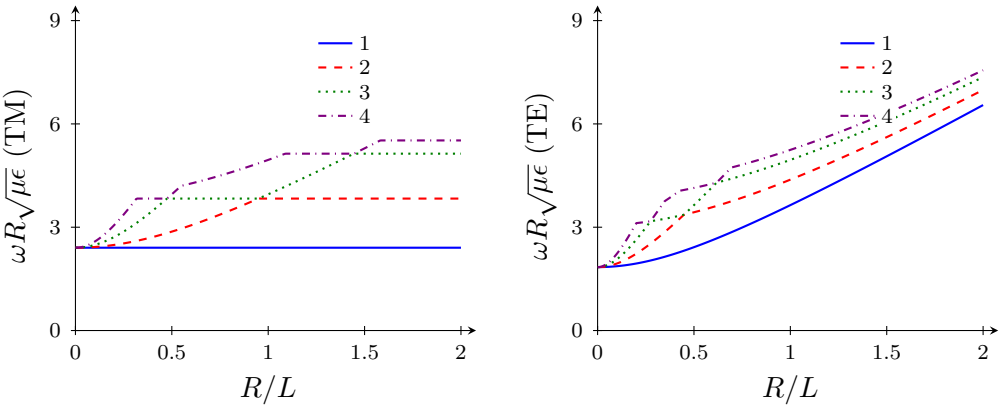
(b) If $R = 2$ cm, $L = 3$ cm, and the cavity is made of pure copper, what is the numerical value of Q for the lowest resonant mode?

Solution.

(a) Separation in z imposes different axial standing waves: $E_z = \psi(\rho,\phi)\cos(p\pi z/L)$ for TM, with $p = 0, 1, \ldots$, and $H_z = \psi(\rho,\phi)\sin(p\pi z/L)$ for TE, with $p = 1, 2, \ldots$. At an end plate the tangential electric field must vanish. For TM fields it is proportional to $\partial_z E_z$, so a cosine standing wave satisfies both end conditions. For TE fields it is proportional to H_z, so a sine is required. The TM solution with $p = 0$ remains nonzero; the analogous TE sine would vanish identically. The side-wall conditions are those of Problem 8.4, so

$$\boxed{\omega^{\mathrm{TM}}_{mnp} = \frac{1}{\sqrt{\mu\epsilon}R}\sqrt{x_{mn}^2 + (p\pi R/L)^2}, \qquad \omega^{\mathrm{TE}}_{mnp} = \frac{1}{\sqrt{\mu\epsilon}R}\sqrt{x'^2_{mn} + (p\pi R/L)^2}.}$$

Here $m = 0, 1, \ldots$, $n = 1, 2, \ldots$ and only positive radial roots are retained. Every $m > 0$ family has two angular polarizations. The plots show the four smallest frequencies at each aspect ratio, counting a polarization pair once. The ordering can change as axial and radial families cross, so the plotted curves are the ordered eigenvalues, not four fixed mode labels.



The lowest TM frequency is always TM_{010}, $\omega R\sqrt{\mu\epsilon} = 2.404825558$; the lowest TE frequency is always TE_{111}, $\sqrt{1.841183781^2 + \pi^2(R/L)^2}$. Their crossing occurs at

$$\frac{R}{L} = \frac{\sqrt{2.404825558^2 - 1.841183781^2}}{\pi} = 0.492427374.$$

Thus a long cavity has TE_{111} as its lowest mode, whereas a shorter cavity has TM_{010}.

(b) For $R/L = 2/3$ the lowest mode is TM_{010}. Write $x = x_{01}$ and $H_\phi = H_1 J_1(x\rho/R)$; the fields are independent of z. At resonance the average electric and magnetic energies are equal, so

$$Q = \frac{\omega U}{P_\ell} = \frac{\omega\mu\int_V |H|^2 dV}{R_s \oint_S |H_\parallel|^2 dS}.$$

The radial Bessel integral, using $J_0(x) = 0$, is $\int_0^R \rho J_1^2(x\rho/R)d\rho = R^2 J_1^2(x)/2$. Indeed, $J_1' = J_0 - J_1/x$ gives $J_1'(x) = -J_1(x)/x$ at this root; inserting it into the Bessel norm identity derived in Problem 8.4 cancels the two terms proportional to J_1^2/x^2. Consequently the volume integral is $\pi R^2 L|H_1|^2 J_1^2(x)$, while the side wall and the two end plates contribute respectively $2\pi RL|H_1|^2 J_1^2(x)$ and $2\pi R^2|H_1|^2 J_1^2(x)$. Therefore

$$\boxed{Q = \frac{\omega\mu}{R_s}\frac{RL}{2(R+L)} = \frac{\mu}{\mu_c}\frac{RL}{\delta(R+L)}.}$$

For the numerical estimate take vacuum inside, $\mu_c = \mu_0$, and room-temperature copper represented by $\sigma = 5.80 \times 10^7\ \mathrm{S\,m^{-1}}$ (100% IACS at 20°C, NBS *Copper Wire Tables*, Circular 31, 2nd ed., 1914, p. 9). With $R = 0.020$ m and $L = 0.030$ m,

$$\omega = 3.60474 \times 10^{10}\ \mathrm{s^{-1}}, \quad f = 5.73713\ \mathrm{GHz}, \quad \delta = 8.72487 \times 10^{-7}\ \mathrm{m}, \quad \boxed{Q \simeq 1.38 \times 10^4.}$$

The material estimate is explicit: $Q = 1.37538 \times 10^4 \sqrt{\sigma/(5.80 \times 10^7\ \mathrm{S\,m^{-1}})}$ under these geometric and permeability assumptions. A more precise value for "pure copper" requires its temperature and conductivity.

8.7 A resonant cavity consists of the empty space between two perfectly conducting, concentric spherical shells, the smaller having an outer radius a and the larger an inner radius b. As shown in Section 8.9, the azimuthal magnetic field has a radial dependence given by spherical Bessel functions, $j_l(kr)$ and $n_l(kr)$, where $k = \omega/c$.

(a) Write down the transcendental equation for the characteristic frequencies of the cavity for arbitrary l.

(b) For $l = 1$ use the explicit forms of the spherical Bessel functions to show that the characteristic frequencies are given by

$$\frac{\tan kh}{kh} = \frac{\left(k^2 + \dfrac{1}{ab}\right)}{k^2 + ab\left(k^2 - \dfrac{1}{a^2}\right)\left(k^2 - \dfrac{1}{b^2}\right)}$$

where $h = b - a$.

(c) For $h/a \ll 1$, verify that the result of part b yields the frequency found in Section 8.9, and find the first order correction in h/a. [The result of part b seems to have been derived first by J. J. Thomson and published in his book *Recent Researches in Electricity and Magnetism*, Oxford Clarendon Press, 1893, pp. 373 ff.]

Solution.

(a) For the electric-multipole (TM) family used in Section 8.9, put $u_l(r) = r[Aj_l(kr) + Bn_l(kr)]$. The radial equation and the conducting boundary conditions are

$$u_l'' + \left[k^2 - \frac{l(l+1)}{r^2}\right] u_l = 0, \qquad u_l'(a) = u_l'(b) = 0.$$

The tangential electric field is proportional to u_l'. Eliminating A, B gives

$$\boxed{[j_l(ka) + kaj_l'(ka)][n_l(kb) + kbn_l'(kb)] - [n_l(ka) + kan_l'(ka)][j_l(kb) + kbj_l'(kb)] = 0.}$$

Take $l \geq 1$ and nonzero k; the $2l+1$ azimuthal orientations are degenerate. For completeness the cavity also has a magnetic-multipole (TE) family, whose tangential electric field vanishes when $u_l(a) = u_l(b) = 0$. Its equation is $j_l(ka)n_l(kb) - n_l(ka)j_l(kb) = 0$. Parts (b)–(c) concern the first, Schumann-resonance family.

(b) For $l = 1$,

$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \qquad n_1(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x}.$$

Define $A_r = k^2 - r^{-2}$ and $B_r = k/r$. Direct differentiation gives

$$\frac{d}{dr}[rj_1(kr)] = k^{-2}[A_r \sin(kr) + B_r \cos(kr)], \quad \frac{d}{dr}[rn_1(kr)] = k^{-2}[-A_r \cos(kr) + B_r \sin(kr)].$$

Their determinant reduces by angle-subtraction identities to

$$(A_a A_b + B_a B_b)\sin(kh) + (A_a B_b - B_a A_b)\cos(kh) = 0.$$

Now

$$A_a B_b - B_a A_b = -\frac{kh}{ab}\left(k^2 + \frac{1}{ab}\right), \quad ab(A_a A_b + B_a B_b) = k^2 + ab(k^2 - a^{-2})(k^2 - b^{-2}).$$

Division gives exactly the stated formula for $\tan(kh)/(kh)$. At poles or zeros of quantities used in that division, the determinant form is the safe equation. In particular the formal $k = 0$ solution of the divided equation is spurious: a static monopole field is not a nonzero $l = 1$ oscillatory mode.

(c) For the lowest radial mode and $h/a \ll 1$, the Neumann eigenfunction is constant at zeroth order. Integrating the radial equation across the gap eliminates u_l'' by the two boundary conditions and gives the exact identity for the positive lowest mode

$$k^2 = l(l+1)\frac{\int_a^{a+h} u_l(r) r^{-2}\, dr}{\int_a^{a+h} u_l(r)\, dr}.$$

Normalize u_l to have a fixed mean. Its first-order change then has zero integral, so its contribution multiplied by the zeroth-order constant $r^{-2} = a^{-2} + O(h/a^3)$ cancels from this ratio. To first order in h/a its eigenvalue is therefore the average centrifugal potential:

$$k^2 = \frac{l(l+1)}{h}\int_a^{a+h} \frac{dr}{r^2} + O(h^2/a^4) = \frac{l(l+1)}{a(a+h)} + O(h^2/a^4).$$

Equivalently this follows by expanding the determinant in (b); changes in the almost constant eigenfunction affect no first-order term. For $l = 1$,

$$\boxed{\omega = \frac{\sqrt{2}c}{a}\left[1 - \frac{h}{2a} + O((h/a)^2)\right].}$$

The leading term is Eq. (8.105), and the first correction replaces the earth's radius by the mid-gap radius to this order. Higher radial modes instead have $k \sim n\pi/h$ and are not described by this low-frequency expansion.

8.8 For the Schumann resonances of Section 8.9 calculate the Q values on the assumption that the earth has a conductivity σ_e and the ionosphere has a conductivity σ_i, with corresponding skin depths δ_e and δ_i.

(a) Show that to lowest order in h/a the Q value is given by $Q = Nh/(\delta_e + \delta_i)$ and determine the numerical factor N for all l.

(b) For the lowest Schumann resonance evaluate the Q value assuming $\sigma_e = 0.1$ $(\Omega\text{m})^{-1}$, $\sigma_i = 10^{-5}$ $(\Omega\text{m})^{-1}$, $h = 10^2$ km.

(c) Discuss the validity of the approximations used in part a for the range of parameters used in part b.

Solution.

(a) In a thin cavity the lowest radial Schumann modes have almost constant $u_l = rH_\phi$ across the gap, so the tangential magnetic field changes only by relative order h/a. Let $\mathcal{H} = \int |H_\parallel(a,\Omega)|^2 a^2 d\Omega$. The mean total stored energy, using equality of electric and magnetic energies, and the wall dissipation are

$$U = \frac{\mu_0}{2} h\mathcal{H}[1 + O(h/a)], \qquad P_\ell = \frac{R_{s,e} + R_{s,i}}{2}\mathcal{H}[1 + O(h/a)].$$

Here both conductors are taken as nonmagnetic and $R_{s,j} = 1/(\sigma_j \delta_j) = \omega\mu_0\delta_j/2$. The common angular integral cancels for every $l \geq 1$:

$$\boxed{Q = \frac{\omega\mu_0 h}{R_{s,e} + R_{s,i}} = \frac{2h}{\delta_e + \delta_i}, \qquad N = 2.}$$

There is implicit l dependence through the frequency entering the skin depths, $\omega_l \simeq c\sqrt{l(l+1)}/a$.

(b) Use $a = 6.371 \times 10^6$ m and the ideal lowest frequency $\omega_1 = 66.5469$ s^{-1} ($f_1 = 10.5913$ Hz). The specified conductivities give

$$\delta_e = \sqrt{\frac{2}{\omega_1\mu_0\sigma_e}} = 4.89042 \times 10^2 \text{ m}, \qquad \delta_i = 4.89042 \times 10^4 \text{ m}.$$

For $h = 10^5$ m,

$$\boxed{Q \simeq \frac{2 \times 10^5}{489.042 + 48904.2} = 4.04914.}$$

This is the result of the specified homogeneous-conductor model, using its ideal resonance frequency to evaluate the losses.

(c) The geometrical thin-gap parameter is $h/a = 0.0157$, and even $\delta_i/a = 0.00768$ is small, so curvature corrections to the local skin-layer treatment are modest. Displacement current is also negligible: $\omega\epsilon_0/\sigma_i \simeq 5.9 \times 10^{-5}$, and its ratio is smaller still in the earth. However, $\delta_i/h \simeq 0.489$ and Q is only about four. The use of unperturbed cavity fields to calculate a small damping correction is therefore not quantitatively controlled by a very small loss parameter. The first-order frequency shift of magnitude $\omega/(2Q)$ is already about 12%; an accurate answer requires solving with the finite surface impedances, and a realistic ionospheric profile may require more than a homogeneous half-space.

8.9 A hollow volume V containing a uniform isotropic linear medium (ϵ, μ) is bounded by a perfectly conducting closed surface S (which may have more than one disconnected part). A harmonic electric field inside the cavity satisfies the vector Helmholtz equation,

$$\nabla \times (\nabla \times \mathbf{E}) = k^2 \mathbf{E} \qquad \text{with } k^2 = \omega^2 \mu \epsilon$$

The boundary condition is $\mathbf{n} \times \mathbf{E} = 0$ (and $\mathbf{n} \cdot \mathbf{B} = 0$) on S.

(a) Show that

$$k^2 = \frac{\displaystyle\int_V \mathbf{E}^* \cdot [\nabla \times (\nabla \times \mathbf{E})]\, d^3x}{\displaystyle\int_V \mathbf{E}^* \cdot \mathbf{E}\, d^3x}$$

is a variational principle for the eigenvalue k^2 in the sense that a change of $\mathbf{E} \to \mathbf{E} + \delta\mathbf{E}$, where both $\mathbf{E}$ and $\delta\mathbf{E}$ satisfy the boundary conditions on S, leads to only second-order changes in k^2.

(b) Apply the variational principle to the TM_{010} mode of a right cylindrical cavity of radius R and length d, using the trial longitudinal electric field $E_z = E_0 \cos(\pi\rho/2R)$ [no variational parameters]. Show that the estimate of the eigenvalue is

$$kR = \frac{\pi}{2}\sqrt{\frac{\pi^2 + 4}{\pi^2 - 4}}$$

Compare numerically with the known eigenvalue, the root x_{01} of $J_0(x)$.

(c) Repeat the calculation of part b with $E_z = E_0[1 + \alpha(\rho/R)^2 - (1+\alpha)(\rho/R)^4]$, where α is a variational parameter. Show that for this trial function the best estimate is

$$kR = \left[80 \cdot \left(\frac{17 - 2\sqrt{34}}{68 + \sqrt{34}}\right)\right]^{1/2}$$

How much better is this truly variational calculation than part b?

Solution.

(a) Write $\mathcal{A} = \nabla \times \nabla\times$ and $\langle \mathbf{F}, \mathbf{G} \rangle = \int_V \mathbf{F}^* \cdot \mathbf{G}\, dV$. Integration by parts gives

$$\langle \mathbf{F}, \mathcal{A}\mathbf{G} \rangle = \int_V (\nabla \times \mathbf{F})^* \cdot (\nabla \times \mathbf{G}) dV - \oint_S \mathbf{n} \cdot [\mathbf{F}^* \times (\nabla \times \mathbf{G})] dS.$$

The boundary term vanishes if $\mathbf{n} \times \mathbf{F} = 0$, so $\mathcal{A}$ is symmetric on the admissible cavity fields. For $\mathcal{A}\mathbf{E} = k^2\mathbf{E}$ and $N = \langle \mathbf{E}, \mathbf{E} \rangle$, expansion of the Rayleigh quotient $\mathcal{R}$ gives

$$\delta\mathcal{R} = \frac{2\,\mathrm{Re}\langle \delta\mathbf{E}, (\mathcal{A} - k^2)\mathbf{E} \rangle}{N} = 0.$$

More explicitly the exact difference for a finite admissible variation is

$$\mathcal{R}[\mathbf{E} + \delta\mathbf{E}] - k^2 = \frac{\langle \delta\mathbf{E}, (\mathcal{A} - k^2)\delta\mathbf{E} \rangle}{\langle \mathbf{E} + \delta\mathbf{E}, \mathbf{E} + \delta\mathbf{E} \rangle},$$

which is quadratic at leading order. For upper bounds to positive Maxwell eigenvalues use divergence-free trial fields, and for an excited mode also restrict to its invariant symmetry subspace or remove

overlaps with lower modes. The trials below are z-independent longitudinal fields and therefore divergence-free.

(b) Put $u = \rho/R$ and $f(u) = \cos(\pi u/2)$. The common axial and angular factors cancel, leaving

$$k^2R^2 = \frac{\int_0^1 u[f'(u)]^2 du}{\int_0^1 u f(u)^2 du}.$$

Using $\cos^2 x = (1+\cos 2x)/2$ and integration by parts,

$$\int_0^1 u\cos^2\frac{\pi u}{2}du = \frac{1}{4} - \frac{1}{\pi^2}, \qquad \int_0^1 u\sin^2\frac{\pi u}{2}du = \frac{1}{4} + \frac{1}{\pi^2}.$$

Thus

$$kR = \frac{\pi}{2}\sqrt{\frac{\pi^2+4}{\pi^2-4}} = 2.414612798.$$

The exact value $x_{01} = 2.404825558$ is smaller by 0.40698% when the error is measured relative to the exact frequency.

(c) For $f = 1 + \alpha u^2 - (1+\alpha)u^4$, expand the polynomials and integrate their powers of u:

$$\int_0^1 u f^2 du = \frac{\alpha^2+7\alpha+16}{60}, \qquad \int_0^1 u(f')^2 du = \frac{\alpha^2+4\alpha+6}{3}.$$

Therefore

$$\mathcal{R}(\alpha)R^2 = 20\frac{\alpha^2+4\alpha+6}{\alpha^2+7\alpha+16}, \quad \frac{d(\mathcal{R}R^2)}{d\alpha} = 20\frac{3\alpha^2+20\alpha+22}{(\alpha^2+7\alpha+16)^2}.$$

The minimizing stationary point is $\alpha = (-10+\sqrt{34})/3$; the other root gives the larger stationary quotient. Substitution yields

$$\boxed{kR = \left[80\frac{17-2\sqrt{34}}{68+\sqrt{34}}\right]^{1/2} = 2.405021472.}$$

The relative error is 0.0081467%, about fifty times smaller than the error in (b). Both results lie above the exact fundamental TM eigenvalue, as the variational principle requires in this sector.

8.10 Use the variational principle of Problem 8.9 in terms of the electric field **E** to find an estimate of the eigenvalue for k^2 for the TE_{111} mode in a right circular cylinder cavity of radius R and length d with perfectly conducting walls. Use as a trial function $B_z = B_0(\rho/R)(1-\rho/2R)\cos\phi\sin(\pi z/d)$. [This function satisfies the boundary conditions of $B_z = 0$ at $z = 0$ and $z = d$, and $\partial B_z/\partial\rho = 0$ at $\rho = R$.]

(a) First show that the variational principle can be reexpressed as

$$k^2 = \frac{\displaystyle\int_V (\nabla\times\mathbf{E}^*)\cdot(\nabla\times\mathbf{E})\, d^3x}{\displaystyle\int_V \mathbf{E}^*\cdot\mathbf{E}\, d^3x}$$

(b) Show that the (transverse) components of the (trial) electric field are

$$E_\rho = E_0(1-\rho/2R)\,\sin\phi\,\sin(\pi z/d); \qquad E_\phi = E_0(1-\rho/R)\,\cos\phi\,\sin(\pi z/d)$$

(c) Calculate the curl of $\mathbf{E}$ and show that the approximation for k^2 is

$$k^2 = \frac{18}{5R^2} + \frac{\pi^2}{d^2}$$

Compare with the exact result. For small enough d/R, this mode has a larger eigenvalue than the TM_{010} mode. Why should the present variational estimate be at all reliable?

(d) The original variational expression in Problem 8.9 has an equivalent integrand in the numerator, $\mathbf{E}^* \cdot [\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}]$. Discuss the relative merits of this integrand compared with the square of the curl of $\mathbf{E}$ in part a for the present problem.

Solution.

Put $u = \rho/R$ and $q = \pi/d$. **(a)** The integration-by-parts identity in Problem 8.9 has zero surface term because $\mathbf{n} \times \mathbf{E} = 0$. Consequently its numerator is $\int |\nabla \times \mathbf{E}|^2 dV$, which gives the stated curl-squared quotient. This form requires only square-integrable first derivatives of the trial field.

(b) The transverse TE electric field is proportional to $\hat{\mathbf{z}} \times \nabla_t B_z$. For the proposed B_z,

$$-\frac{1}{\rho}\partial_\phi B_z = \frac{B_0}{R}(1 - u/2)\sin\phi \sin qz, \qquad \partial_\rho B_z = \frac{B_0}{R}(1-u)\cos\phi\sin qz.$$

Absorb the common proportionality constant into E_0 to obtain exactly the two trial components in the question. They satisfy $E_\phi(R) = 0$, vanish at both end faces, and have $E_z = 0$. Moreover

$$\nabla \cdot \mathbf{E} = \frac{E_0 \sin\phi \sin qz}{\rho}[(1-u) - (1-u)] = 0.$$

At the axis the leading cylindrical components combine into the finite Cartesian vector $E_0 \hat{\mathbf{y}} \sin qz$.

(c) The curl is

$$\begin{aligned}
(\nabla \times \mathbf{E})_\rho &= -qE_0(1-u)\cos\phi\cos qz, \\
(\nabla \times \mathbf{E})_\phi &= qE_0(1-u/2)\sin\phi\cos qz, \\
(\nabla \times \mathbf{E})_z &= -\frac{3E_0}{2R}\cos\phi\sin qz.
\end{aligned}$$

The field norm follows from $\int_0^{2\pi} \sin^2\phi\, d\phi = \int_0^{2\pi} \cos^2\phi\, d\phi = \pi$ and the corresponding axial integrals $d/2$:

$$N = \frac{\pi d R^2 |E_0|^2}{2}\int_0^1 u[(1-u/2)^2 + (1-u)^2]du = \frac{5\pi d R^2 |E_0|^2}{32}.$$

The transverse curl contributes $q^2 N$ and its longitudinal component contributes $9\pi d|E_0|^2/16$. Therefore

$$\boxed{k^2 = \frac{18}{5R^2} + \frac{\pi^2}{d^2}.}$$

The exact result is $(x'_{11}/R)^2 + (\pi/d)^2$, with $x'^2_{11} = 3.389957717$. The trial replaces this transverse number by 3.6, a 6.20% error in that term, corresponding to at most 3.05% error in frequency and a smaller error as d/R shrinks.

Reliability does not require TE_{111} to be the absolute lowest cavity mode. The trial belongs to the $m = 1$, first axial TE sector. Its transverse field is a rotated gradient, whereas a TM transverse field is a gradient of a Dirichlet scalar. Their cross integral is a boundary integral with that scalar

zero on the wall, and vanishes. Angular and axial orthogonality remove the other TE sectors. Hence this is a Rayleigh upper bound to the lowest mode in the appropriate invariant sector even when a TM mode lies below it.

(d) The alternatives are mathematically equivalent if the vector Laplacian, including derivatives of cylindrical unit vectors, is evaluated correctly and all integrable axis terms are retained. The present field is continuous at the axis but its first derivatives are direction-dependent there; second derivatives contain $1/\rho$ terms. In fact the cylindrical vector Laplacian gives

$$(\nabla^2\mathbf{E})_\rho = -q^2E_\rho - \frac{3E_0}{2R\rho}\sin\phi\sin qz, \qquad (\nabla^2\mathbf{E})_\phi = -q^2E_\phi.$$

The extra contribution to $-\int \mathbf{E}^*\cdot\nabla^2\mathbf{E}\,dV$ is

$$\frac{3|E_0|^2}{2R}\frac{\pi d}{2}\int_0^R\left(1-\frac{\rho}{2R}\right)d\rho = \frac{9\pi d|E_0|^2}{16},$$

as in (c). These terms are integrable with the area element $\rho\,d\rho\,d\phi$, but make $\mathbf{E}^*\cdot[\nabla\nabla\cdot\mathbf{E}-\nabla^2\mathbf{E}]$ more cumbersome and cancellation-prone. Excising a small cylinder around the axis and then shrinking it gives a vanishing additional boundary term, so no singular contribution is lost by the curl-squared form. It is the simpler positive integral for this trial.

8.11 Apply the variational method of Problem 8.9 to estimate the resonant frequency of the lowest TM mode in a "breadbox" cavity with perfectly conducting walls, of length d in the z direction, radius R for the curved quarter-circle "front" of the breadbox, and the "bottom" and "back" of the box defined by the plane segments ($y=0$, $0<x<R$) and ($x=0$, $0<y<R$), respectively. Use the trial function,

$$E_z = E_0(\rho/R)^\nu(1-\rho/R)\sin 2\phi$$

for the only component of electric field present. This function gives vanishing tangential component of $\mathbf{E}$ on the boundary surfaces; the index ν is a variational parameter. Show that

$$k^2R^2 = \frac{(\nu+2)(2\nu+3)(\nu^2+\nu+4)}{\nu(2\nu+1)}$$

Minimize with respect to ν to find the best estimate of kR from the given trial function. Compare with the exact answer, $kR = 5.13562$, the first root of $J_2(x)$.

Solution.

The lowest TM cavity mode is independent of z, so the axial length and field amplitude cancel in the Rayleigh quotient. Set $u=\rho/R$ and $f=u^\nu(1-u)$; finite gradient energy at the apex requires $\nu>0$. Angular integration over $0<\phi<\pi/2$ gives the same factor $\pi/4$ for $\sin^2 2\phi$ and $\cos^2 2\phi$. Consequently

$$k^2R^2 = \frac{\int_0^1[u(f')^2+4f^2/u]du}{\int_0^1 uf^2du}.$$

Use $f' = \nu u^{\nu-1}-(\nu+1)u^\nu$ to evaluate the numerator:

$$D = \frac{\nu^2+4}{2\nu} - \frac{2\nu(\nu+1)+8}{2\nu+1} + \frac{(\nu+1)^2+4}{2\nu+2},$$
$$N = \frac{1}{2\nu+2} - \frac{2}{2\nu+3} + \frac{1}{2\nu+4}.$$

Combining these fractions gives exactly

$$\boxed{k^2R^2 = \frac{(\nu+2)(2\nu+3)(\nu^2+\nu+4)}{\nu(2\nu+1)}.}$$

Its derivative has positive denominator and numerator

$$8\nu^5 + 24\nu^4 + 18\nu^3 - 47\nu^2 - 96\nu - 24.$$

There is exactly one positive root by Descartes' rule of signs. The quotient diverges as $\nu \downarrow 0$ and as $\nu \to \infty$, so that root is the global minimum. Numerical root finding gives

$$\nu_* = 1.56757838, \qquad \boxed{kR = 5.20544971.}$$

The exact eigenfunction is $J_2(x_{21}\rho/R)\sin 2\phi$ with $x_{21} = 5.13562230$. The trial frequency is 1.36% high. The noninteger optimum is admissible in the finite-energy variational space, although it does not reproduce the exact analytic ρ^2 behavior at the corner; a richer trial space could improve that shape.

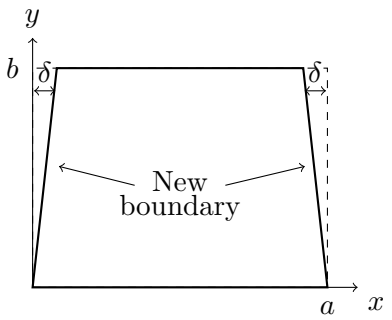
8.12 A waveguide with lossless dielectric inside and perfectly conducting walls has a cross-sectional contour C that departs slightly from a comparison contour C_0 whose fields are known. The difference in boundaries is described by $\delta(x, y)$, the length measured from C_0 to C along the normal to C_0 at the boundary point (x, y). The derivative $d\delta/ds$ along the boundary is higher order in small quantities.

(a) If the eigenvalue parameters and solutions for C and C_0 are (γ^2, ψ) and (γ_0^2, ψ_0), respectively, without degeneracy, show that to first order in δ

$$\gamma^2 - \gamma_0^2 = -\frac{\oint_{C_0} \delta(x,y)\left[\left|\dfrac{\partial\psi_0}{\partial n}\right|^2 - \psi_0^* \dfrac{\partial^2\psi_0}{\partial n^2}\right] dl}{\displaystyle\int_{S_0} |\psi_0|^2 \, dx\, dy}$$

where only the first (second) term in the numerator occurs for TM (TE) modes. [*Hint*: Follow the same general approach as used in Section 8.6 for the effects of finite conductivity.]

(b) Determine the perturbed value of γ^2 for the lowest TE and TM modes ($\mathrm{TE}_{1,0}$ and $\mathrm{TM}_{1,1}$) in a rectangular guide if the change in shape is as shown in the accompanying sketch.



Problem 8.12

Solution.

Take the normal outward from the guide cross section and let a positive $\delta(s)$ expand the domain. Write $\psi = \psi_0 + \delta\psi$ and $\gamma^2 = \gamma_0^2 + \delta\gamma^2$. **(a)** Subtract the two transverse Helmholtz equations and use Green's identity on the original domain:

$$\delta\gamma^2 \int_{S_0} |\psi_0|^2 dA = -\oint_{C_0} [\psi_0^* \partial_n \delta\psi - \delta\psi\, \partial_n \psi_0^*]\, dl + O(\delta^2).$$

For TM, the displaced Dirichlet condition is $\delta\psi = -\delta\, \partial_n \psi_0$ on C_0. Since $\psi_0 = 0$ there, substitution gives

$$\delta\gamma^2 = -\frac{\oint \delta |\partial_n \psi_0|^2 dl}{N}, \qquad N = \int |\psi_0|^2 dA.$$

For TE, the displaced outward normal is $\mathbf{n} - \delta'(s)\mathbf{t}$ to first order. Expansion of the Neumann condition therefore gives

$$\partial_n \delta\psi = -\delta \partial_n^2 \psi_0 + \delta' \partial_s \psi_0.$$

Under the problem's assumption that δ' is higher order, omit the last term. Then $\delta\gamma^2 = \oint \delta\psi_0^* \partial_n^2 \psi_0 dl/N$, which together with the TM result proves the displayed formula. For a general deformation the slope term must be retained. An integration by parts around the contour, using $\partial_n^2\psi_0 = -\gamma_0^2\psi_0 - \partial_s^2\psi_0$ on a Neumann boundary, gives the more general form

$$\boxed{\delta\gamma_{\mathrm{TE}}^2 = \frac{\oint \delta[|\partial_s \psi_0|^2 - \gamma_0^2 |\psi_0|^2] dl}{N}.}$$

This qualification matters for the ellipse in Problem 8.13.

(b) Let the positive symbol δ in the sketch denote the inward displacement at either top corner. The two side boundaries move by the outward-normal amount $\delta_n(y) = -\delta y/b$; the horizontal boundaries do not move. Assume $a > b$ so that the stated TE_{10} mode is nondegenerate. Its unperturbed field is $\psi_0 = \cos(\pi x/a)$, with $N = ab/2$. On the sloping sides $\partial_s\psi_0 = 0$ and $\psi_0 \partial_n^2 \psi_0 = -\pi^2/a^2$, so the slope correction vanishes even though δ_n' is first order. Thus

$$\delta\gamma_{\mathrm{TE}}^2 = \frac{2\int_0^b (-\delta y/b)(-\pi^2/a^2) dy}{ab/2} = \frac{2\pi^2\delta}{a^3}.$$

For TM_{11}, $\psi_0 = \sin(\pi x/a)\sin(\pi y/b)$ and $N = ab/4$. On either side $(\partial_n\psi_0)^2 = (\pi/a)^2 \sin^2(\pi y/b)$. Using $\int_0^b y \sin^2(\pi y/b) dy = b^2/4$ gives the same shift. Therefore

$$\boxed{\gamma_{\mathrm{TE}}^2 = \frac{\pi^2}{a^2}\left(1 + \frac{2\delta}{a}\right) + O(\delta^2), \qquad \gamma_{\mathrm{TM}}^2 = \frac{\pi^2}{a^2} + \frac{\pi^2}{b^2} + \frac{2\pi^2\delta}{a^3} + O(\delta^2).}$$

The positive shifts agree with the reduction in mean width from a to $a - \delta$. For a square, the degenerate TE pair must instead be treated together as in Problem 8.13.

8.13 To treat perturbations if there is a degeneracy of modes in guides or cavities under ideal conditions, one must use degenerate-state perturbation theory. Consider the two-dimensional (waveguide) situation in which there is an N-fold degeneracy in the ideal circumstances (of perfect conductivity or chosen shape of cross section), with no other nearby modes. There are N linearly independent solutions $\psi_0^{(i)}$, chosen to be orthogonal, to the transverse wave equation, $(\nabla_t^2 + \gamma_0^2)\psi_0^{(i)} = 0$, $i = 1, 2, \ldots, N$. In response to the perturbation, the degeneracy is in general lifted. There is a set of perturbed eigenvalues, γ_k^2, with associated eigenmodes, ψ_k, which can be expanded (in lowest order) in terms of the N unperturbed eigenmodes: $\psi_k = \sum_i a_i \psi_0^{(i)}$.

(a) Show that the generalization of (8.68) for finite conductivity (and the corresponding expression in Problem 8.12 on distortion of the shape of a waveguide) is the set of algebraic equations,

$$\sum_{i=1}^{N} [(\gamma^2 - \gamma_0^2) N_j \delta_{ji} + \Delta_{ji}] a_i = 0 \qquad (j = 1, 2, \ldots, N)$$

where

$$N_j = \int_A |\psi_0^{(j)}|^2 \, da \qquad \text{and} \qquad \Delta_{ji} = f \oint_C \frac{\partial \psi_0^{(j)*}}{\partial n} \frac{\partial \psi_0^{(i)}}{\partial n} \, dl$$

for finite conductivity of TM modes (as in Eq. (8.68)), and

$$\Delta_{ji} = \oint_C \delta \left[\partial_n \psi_0^{(j)*} \partial_n \psi_0^{(i)} - \psi_0^{(j)*} \partial_n^2 \psi_0^{(i)} \right] dl + \oint_C \frac{d\delta}{ds} \psi_0^{(j)*} \partial_s \psi_0^{(i)} \, dl$$

for distortion of the boundary shape. The extra term accounts for rotation of the normal. It vanishes for TM modes and may be neglected for TE modes only under the higher-order slope assumption of Problem 8.12. That assumption does not apply to the ellipse in (b).

(b) The lowest mode in a circular guide of radius R is the twofold degenerate TE_{11} mode, with fields given by

$$\psi^{(\pm)} = B_z = B_0 J_1(\gamma_0 \rho) \exp(\pm i\phi) \exp(ikz - i\omega t)$$

The eigenvalue parameter is $\gamma_0 = 1.841/R$, corresponding to the first root of $dJ_1(x)/dx$. Suppose that the circular waveguide is distorted along its length into an elliptical shape with semimajor and semiminor axes, $a = R + \Delta R$, $b = R - \Delta R$, respectively. To first order in $\Delta R/R$, the area and circumference of the guide remain unchanged. Show that the degeneracy is lifted by the distortion and that to first order in $\Delta R/R$, $\gamma_1^2 = \gamma_0^2(1 + \lambda \Delta R/R)$ and $\gamma_2^2 = \gamma_0^2(1 - \lambda \Delta R/R)$. Determine the numerical value of λ and find the eigenmodes as linear combinations of $\psi^{(\pm)}$. Explain physically why the eigenmodes turn out as they do.

Solution.

(a) Multiply the perturbed wave equation by $\psi_0^{(j)*}$, subtract its unperturbed adjoint equation, and integrate over the cross section. Orthogonality makes the volume term $(\gamma^2 - \gamma_0^2) N_j a_j$. Linearizing the boundary condition supplies a boundary term linear in every a_i. For the TM surface-impedance condition of Eq. (8.65), with its normal directed *into* the guide, the boundary value is $f \partial_n \psi_0$. Equivalently, with the outward normal used here it is $-f \partial_n \psi_0$. Green's identity therefore gives

$$\sum_i [(\gamma^2 - \gamma_0^2) N_j \delta_{ji} + \Delta_{ji}] a_i = 0, \qquad \Delta_{ji} = f \oint \partial_n \psi_0^{(j)*} \partial_n \psi_0^{(i)} dl.$$

Normal reversal has no effect on the product of the two derivatives. This quoted finite-conductivity formula is the TM formula; setting Neumann derivatives to zero does not describe TE conductor loss. For TE the surface-impedance perturbation gives instead

$$\Delta_{ji}^{\mathrm{TE}} = (1 + i) \frac{R_s \gamma_0^2}{\omega \mu} \oint \left[\psi_0^{(j)*} \psi_0^{(i)} + \frac{k^2}{\gamma_0^4} \partial_s \psi_0^{(j)*} \partial_s \psi_0^{(i)} \right] dl.$$

For a diagonal matrix element this yields $\delta\gamma^2 = -2(1 + i)k\beta$, with β obtained by the surface-loss calculation in Problem 8.4. To see the off-diagonal terms, take $\psi = B_z$ as in (8.36) and in part (b), and form an arbitrary superposition $\psi = \sum_i a_i \psi_i$. Its wall-current norm is the quadratic form

$$\oint |\mathbf{H}_\parallel|^2 dl = \frac{1}{\mu^2} \sum_{j,i} a_j^* a_i \oint \left[\psi_j^* \psi_i + \frac{k^2}{\gamma_0^4} (\partial_s \psi_j^*)(\partial_s \psi_i) \right] dl.$$

Here $\|$ means tangential to the conducting wall. Its longitudinal and contour-tangent magnetic-field components are orthogonal, giving the two terms. The factor $1/\mu^2$ converts $\mathbf{B}$ to $\mathbf{H}$. The modal power is $(\omega k/2\mu\gamma_0^2)\sum_i N_i|a_i|^2$. Equating $2\beta P$ to $(R_s/2)\oint|\mathbf{H}_\||^2 dl$ fixes the dissipative quadratic form; the equal reactive and resistive parts of a good conductor supply its factor $1+i$. Since this holds for every set of complex a_i, polarization of the quadratic form fixes every matrix element as displayed. If TE and TM families are mutually degenerate, both sets of fields and their mutual surface-current overlaps must be included.

For a displaced boundary the Dirichlet calculation gives $\Delta_{ji} = \oint \delta\partial_n\psi_j^*\partial_n\psi_i dl$. The general Neumann calculation in Problem 8.12 gives

$$\begin{aligned}\Delta_{ji}^{\mathrm{TE}} &= -\oint \delta\psi_j^*\partial_n^2\psi_i dl + \oint \delta'\psi_j^*\partial_s\psi_i dl \\ &= \oint \delta\left[\gamma_0^2\psi_j^*\psi_i - \partial_s\psi_j^*\partial_s\psi_i\right] dl.\end{aligned}$$

The second line follows by integrating the tangential second derivative by parts. It is Hermitian for a real shape perturbation. When δ' is of higher order, the first line reduces to the shape formula originally quoted. For finite-conductivity TM modes and either shape problem, nontrivial coefficients require

$$\det[(\gamma^2-\gamma_0^2)\mathbf{N}+\mathbf{\Delta}] = 0.$$

Equivalently, diagonalize $\mathbf{N}^{-1/2}\mathbf{\Delta}\mathbf{N}^{-1/2}$; its negative eigenvalues are the first-order shifts of γ^2.

(b) Put $x = x'_{11} = 1.841183781$ and suppress the common z,t factors. To first order the ellipse is the radial deformation $\delta(\phi) = \Delta R\cos 2\phi$. Its tangential derivative is $\delta' = -2(\Delta R/R)\sin 2\phi$, of the same perturbative order; it cannot be discarded. In the basis $\psi_\pm = B_0 J_1(\gamma_0\rho)e^{\pm i\phi}$, both norms are

$$N = \pi R^2|B_0 J_1(x)|^2(1-x^{-2}).$$

The diagonal shape matrix elements vanish because the integrals of $\cos 2\phi$ vanish. For the off-diagonal element, $\psi_+^*\psi_- = |B_0J_1(x)|^2e^{-2i\phi}$ and $\partial_s\psi_+^*\partial_s\psi_- = -R^{-2}\psi_+^*\psi_-$. Since $\int_0^{2\pi}\cos 2\phi\, e^{-2i\phi}d\phi = \pi$,

$$\Delta_{+-} = \Delta_{-+} = \pi R\Delta R|B_0J_1(x)|^2(\gamma_0^2+R^{-2}).$$

The combinations $(\psi_+ + \psi_-)/\sqrt{2} \propto \cos\phi$ and $(\psi_+ - \psi_-)/(i\sqrt{2}) \propto \sin\phi$ diagonalize this matrix. Their shifts are

$$\boxed{\gamma_{\cos}^2 = \gamma_0^2\left(1-\lambda\frac{\Delta R}{R}\right), \qquad \gamma_{\sin}^2 = \gamma_0^2\left(1+\lambda\frac{\Delta R}{R}\right), \quad \lambda = \frac{x^2+1}{x^2-1} = 1.836834889.}$$

The reflection axes of the ellipse select standing angular patterns; a traveling $e^{\pm i\phi}$ pattern no longer has a single eigenvalue. The $\cos\phi$ longitudinal magnetic field varies mainly across the longer x direction and has the lower transverse wave number; its electric field is predominantly along y. The other polarization has the opposite ordering.

As an independent check, map the ellipse to the unit disk by $x = (1+\varepsilon)X$, $y = (1-\varepsilon)Y$. The Neumann Rayleigh quotient changes by

$$\delta\gamma^2 = \frac{2\varepsilon\int(|\partial_Y\psi|^2 - |\partial_X\psi|^2)dA}{\int|\psi|^2 dA}.$$

For $J_1(x\rho/R)\cos\phi$, direct radial and angular integration gives $-1.836834889\,\varepsilon\gamma_0^2$, agreeing with the complete boundary perturbation. Omitting normal rotation would instead give $\lambda = 1$, which fails this independent check.

8.14 Consider an optical fiber with a graded index of refraction for rays confined to the x-z plane, $n(x) = n(0)\,\mathrm{sech}(\alpha x)$. The fiber has large enough transverse dimensions (x) to contain all rays of interest, which are evidently symmetric about $x = 0$. The invariant $\bar{n} = n(x_{\max}) = n(0)\cos\theta(0)$.

(a) Solve the eikonal equations for the transverse coordinate $x(z)$ of the ray and show that

$$\alpha x = \sinh^{-1}[\sinh(\alpha x_{\max})\,\sin(\alpha z)]$$

where the origin in z is chosen when the ray has $x = 0$. Sketch rays over one half-period for "launch angles" $\theta(0) = \pi/6$, $\pi/4$, and $\pi/3$.

(b) Find the half-period Z of the ray. Does it depend on $\bar{n}$?

(c) Show that the optical path length for a half period, $L_{\text{opt}} = \int n(x)\,ds$ is $L_{\text{opt}} = n(0)Z$. Comment on the effectiveness of this particular grading of the index.

Hint: In the computation of L_{opt}, a useful change of variables is $\sinh(\alpha x) = \sinh(\alpha x_{\max})\sin t$. The resulting integral can be done by contour integration:

$$\int_0^{\pi/2} \frac{d\theta}{1 + a^2\sin^2\theta} = \frac{\pi}{2\sqrt{1+a^2}}$$

Solution.

(a) Write $n_0 = n(0)$, $\theta_0 = \theta(0)$ and take $0 \le \theta_0 < \pi/2$. The eikonal equation $d(n\hat{\mathbf{t}})/ds = \nabla n$ has a conserved z component,

$$n\frac{dz}{ds} = \bar{n} = n_0\cos\theta_0.$$

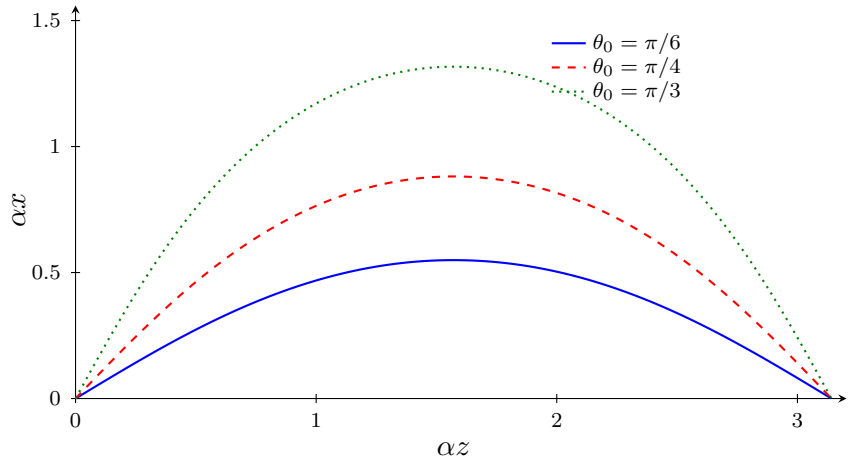
Thus $dx/dz = \pm\sqrt{n^2/\bar{n}^2 - 1}$. With $A = \tan\theta_0 = \sinh(\alpha x_{\max})$ and $y = \sinh(\alpha x)$ this becomes

$$\left(\frac{dy}{dz}\right)^2 = \alpha^2\left[A^2 - y^2\right].$$

Starting at $x(0) = 0$ with positive slope gives $y = A\sin\alpha z$; continuing smoothly through the turning point gives

$$\boxed{x(z) = \alpha^{-1}\,\mathrm{arsinh}[\tan\theta_0\sin(\alpha z)].}$$

The three requested rays are plotted below. Their different heights are $\alpha x_{\max} = \mathrm{arsinh}(\tan\theta_0)$.



(b) The next crossing of the axis is at

$$\boxed{Z = \pi/\alpha,}$$

independent of $\bar{n}$. The axial ray is the limiting case of this family; its transverse oscillation amplitude is zero.

(c) For the optical length use $ds/dz = n/\bar{n}$:

$$\begin{aligned} L_{\text{opt}} &= \frac{n_0^2}{\alpha\bar{n}} \int_0^{\pi} \frac{dt}{1 + A^2 \sin^2 t} \\ &= \frac{n_0^2}{\alpha\bar{n}} \frac{\pi}{\sqrt{1 + A^2}} = \boxed{\frac{\pi n_0}{\alpha} = n_0 Z}. \end{aligned}$$

For example the half-range integral follows immediately by $u = \tan t$: $\int_0^{\infty} du/[1 + (1 + A^2)u^2] = \pi/(2\sqrt{1 + A^2})$. The longer geometrical paths at larger launch angles are exactly compensated by their lower refractive indices. In geometrical optics, with nondispersive $n(x)$, all these rays have the same transit time $n_0 Z/c$ and no differential delay over a half-period. Material dispersion and corrections outside the ray approximation are not included in this conclusion.

8.15 Discuss the TE and TM modes in the dielectric slab waveguide of Section 8.11.A as a boundary-value problem.

(a) Show that (8.123) emerges as the determining relation for both even and odd modes (in x) and that even or odd p goes with the evenness or oddness of the mode, defined by the symmetry in x of the transverse fields.

(b) Show that the eigenvalues of ξ for the TE modes are given approximately by

$$\xi \approx \frac{(p+1)\pi}{2(V+1)} \left[1 - \frac{(p+1)^2\pi^2}{24(V+1)^3}\right]$$

The lowest order result is accurate for $V \gg 1$ and small p. Check the accuracy of the full expression against solution of (8.123) by Newton's method for $V = 1, 2, 3$.

(c) Calculate the power flow in the z direction (per unit length in the y direction) within the core ($|x| < a$) and in the cladding ($|x| > a$) for the even TE modes and show that the fractions are

$$F_{\text{core}} = \frac{1}{S}\left[1 + \frac{\sin(2V\xi)}{2V\xi}\right] \qquad \text{and} \qquad F_{\text{clad}} = \frac{1}{S}\left[\frac{\cos^2(V\xi)}{V\sqrt{1-\xi^2}}\right]$$

where

$$S = \left[1 + \frac{\sin(2V\xi)}{2V\xi}\right] + \left[\frac{\cos^2(V\xi)}{V\sqrt{1-\xi^2}}\right]$$

where ξ is the root of (8.123) for the pth mode. Find corresponding expressions for the odd TE modes.

Solution.

(a) Assume isotropic, lossless, nonmagnetic media, with no variation in y. Set $k_0 = \omega/c$ and define

$$q^2 = n_1^2 k_0^2 - k_z^2, \quad \beta^2 = k_z^2 - n_2^2 k_0^2, \quad u = qa = V\xi, \quad w = \beta a = V\sqrt{1-\xi^2}.$$

Here $u, w > 0$ for a bound mode and $u^2 + w^2 = V^2$. For TE polarization take $E_y = E_0 f(x) e^{ik_z z - i\omega t}$. In the core, $f = \cos qx$ for even modes or $f = \sin qx$ for odd modes. Outside, match to $f(a)e^{-\beta(x-a)}$ at the right surface and to its parity reflection at the left surface. Continuity of E_y and $H_z = \partial_x E_y/(i\omega\mu_0)$ requires continuity of f, f'. Therefore

$$u \tan u = w \quad \text{(even TE)}, \qquad -u \cot u = w \quad \text{(odd TE)}.$$

For TM polarization replace E_y by H_y. Continuity of H_y and $E_z = -\partial_x H_y/(i\omega\epsilon)$ requires continuity of f and f'/ϵ, giving the same equations with $w \mapsto (n_1^2/n_2^2)w$. The combined result is

$$\boxed{\tan(V\xi - p\pi/2) = f\,\frac{\sqrt{1-\xi^2}}{\xi}, \quad f = 1 \text{ (TE)}, \quad f = \frac{n_1^2}{n_2^2} \text{ (TM)}.}$$

The root belonging to p lies in $p\pi/2 < u < (p+1)\pi/2$; even p is an even transverse field, and odd p is odd. The longitudinal component has the opposite parity, which explains why specifying transverse parity matters. A bound mode exists when $V > p\pi/2$, with no positive threshold for $p = 0$.

(b) For TE modes the phase equation can be written without tangent poles: since $w/u = \sqrt{1-\xi^2}/\xi$ and $0 < \xi < 1$, $\arctan(w/u) = \arccos\xi = \pi/2 - \arcsin\xi$. Thus

$$V\xi + \arcsin\xi = (p+1)\pi/2.$$

Expanding $\arcsin\xi = \xi + \xi^3/6 + O(\xi^5)$ and putting $A_p = (p+1)\pi/[2(V+1)]$ first on the cubic term gives

$$\xi = A_p - \frac{A_p^3}{6(V+1)} + \cdots = \boxed{\frac{(p+1)\pi}{2(V+1)}\left[1 - \frac{(p+1)^2\pi^2}{24(V+1)^3}\right].}$$

A stable Newton implementation uses $t = \arccos\xi$ and $F(t) = V\cos t - t - p\pi/2$, with

$$t_{j+1} = t_j + \frac{V\cos t_j - t_j - p\pi/2}{V\sin t_j + 1}.$$

The supplementary program retains a bracket $0 < t < \pi/2$ and bisects if a Newton step leaves it. It gives all bound TE modes at the requested values:

V	p	Newton ξ	Approximate ξ	Relative error (%)
1	0	0.7390851332	0.7450254073	0.8037
2	0	0.5149332647	0.5156239102	0.1341
2	1	0.9477471335	0.9833986280	3.7617
3	0	0.3900403167	0.3901757844	0.0347
3	1	0.7596208867	0.7652117854	0.7360

The larger error near threshold is expected because ξ is then near one, outside the small-ξ expansion.

(c) The local longitudinal power density is $S_z = k_z|E_y|^2/(2\omega\mu_0)$. Let $P_* = k_z|E_0|^2 a/(2\omega\mu_0)$. For even modes direct integration gives

$$P_{\text{core}} = P_* \int_{-1}^{1} \cos^2(ut)\,dt = P_*\left[1 + \frac{\sin 2u}{2u}\right],$$

$$P_{\text{clad}} = \frac{k_z|E_0|^2}{2\omega\mu_0} 2\cos^2 u \int_0^\infty e^{-2\beta x}\,dx = P_* \frac{\cos^2 u}{w}.$$

Dividing by their sum gives precisely the stated fractions. For odd modes define

$$S_o = 1 - \frac{\sin 2u}{2u} + \frac{\sin^2 u}{w}.$$

Then

$$\boxed{F_{\text{core}}^{(o)} = \frac{1 - \sin(2u)/(2u)}{S_o}, \qquad F_{\text{clad}}^{(o)} = \frac{\sin^2 u/w}{S_o}.}$$

The two claddings are both included. Their fraction tends to one at a mode's threshold ($w \downarrow 0$), and to zero for a fixed mode as $V \to \infty$.

8.16 The longitudinal phase velocity in the dielectric slab waveguide of the preceding problem is $v_p = \omega/k_z = c/(n_1 \cos\theta_p)$. Intermodal dispersion occurs because the dielectric media have dispersion and also because the group velocity differs intrinsically for different modes.

(a) Making the approximation that the dielectrics' dispersion can be neglected, show that the group velocity $v_g = d\omega/dk_z$ for the TE_p mode is

$$v_g = \frac{c \cos\theta_p}{n_1} \left[\frac{1 + \beta_p a}{\cos^2 \theta_p + \beta_p a} \right]$$

where θ_p is the eigenangle of the pth mode ($\cos\theta_p = \sqrt{1 - 2\Delta\xi_p^2}$) and β_p is given by (8.124). Interpret the departure from $v_p v_g = c^2/n_1^2$ (as in metallic waveguides; $\beta_p \to \infty$) in terms of the Goos–Hänchen effect and ray-like propagation at the simple phase speed c/n_1. [*Hint*: Write the eigenvalue relation (8.121) in terms of the independent variable ω and the dependent variable k_z and differentiate with respect to ω.]

(b) Write a program to evaluate v_g versus V/V_t, where $V_t = p\pi/2$ is the threshold frequency variable for the pth mode. Make a plot of v_g/c for $n_1 = 1.5$, $n_2 = 1.0$ for $p = 0, 1, 2, \ldots, 6$ as a function of $V/V_t(p = 1)$ on the range $(0, 10)$.

(c) Relate the results of part b to the optical path length difference for the step-index fiber shown in Fig. 8.12b. Can you generate a plot from the results for $v_g(p)$ at fixed V for $n_1 = 1.01$, $n_2 = 1.0$ to compare with the "classical" ray result?

Solution.

(a) Keep n_1, n_2 independent of frequency, as stipulated. Put $W = \omega a/c$, $K = k_z a$, $u = qa$, $w = \beta a$ and $C = \cos\theta_p = K/(n_1 W)$. The TE phase relation is

$$u = \arctan(w/u) + p\pi/2.$$

Its differential is

$$(V^2 + w)\, du = u\, dw,$$

whereas differentiation of $u^2 = n_1^2 W^2 - K^2$, $w^2 = K^2 - n_2^2 W^2$ gives

$$u\, du = n_1^2 W\, dW - K\, dK, \qquad w\, dw = K\, dK - n_2^2 W\, dW.$$

Multiplying the differential phase equation by uw and inserting these two differentials gives

$$w(V^2 + w)(n_1^2 W\, dW - K\, dK) = u^2 (K\, dK - n_2^2 W\, dW).$$

The coefficient of $K\,dK$ is $V^2(1+w)$. Also $n_1^2w^2+n_2^2u^2=(n_1^2-n_2^2)K^2=V^2K^2/W^2$. Collecting the dW terms and cancelling V^2 therefore yields

$$K\frac{dK}{dW}=\frac{n_1^2Ww+K^2/W}{1+w},\qquad \frac{dK}{dW}=n_1\frac{C^2+w}{C(1+w)}.$$

Since $v_g/c=dW/dK$, the result is

$$\boxed{v_g=\frac{cC}{n_1}\frac{1+w}{C^2+w}.}$$

In particular

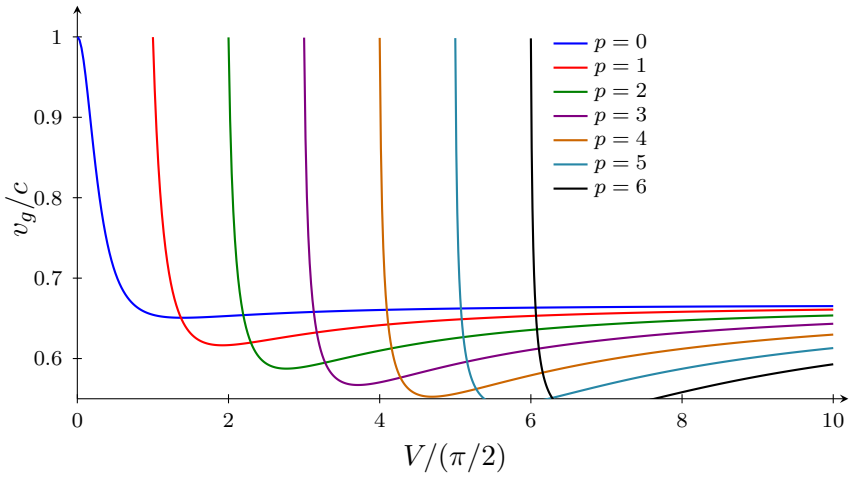
$$v_pv_g=\frac{c^2}{n_1^2}\frac{1+w}{C^2+w}>\frac{c^2}{n_1^2}$$

for a nonaxial, finitely confined mode. A simple zigzag ray with speed c/n_1 would have $v_{g,z}=cC/n_1$. The reflection phase varies with ω and k_z, producing a Goos–Hänchen displacement. For TE reflection $\phi=-2\arctan(\beta/q)$, its displacement along the interface is

$$-\left.\frac{\partial\phi}{\partial k_z}\right|_{\omega}=\frac{2k_z}{q\beta}.$$

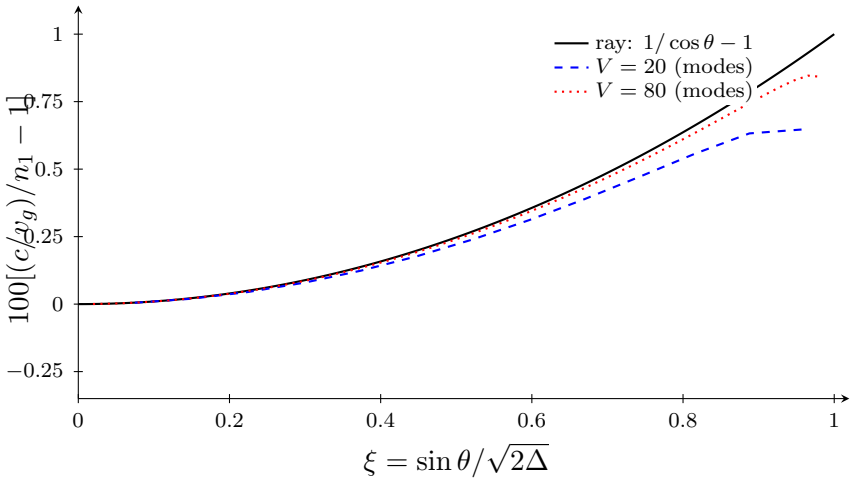
The wave packet thus advances along the interface as well as along the core ray segments. The associated cladding field carries part of the energy in the lower-index medium. These effects account for the factor $(1+w)/(C^2+w)>1$; neglecting the reflection-phase derivative would miss it. As $w\to\infty$ the core ray expression is recovered. At a threshold $w\to 0$, $C\to n_2/n_1$ and $v_g\to c/n_2$, consistently with the energy being mostly outside.

(b) The program solves the phase relation independently for each p, then evaluates the boxed expression. Since $V_t(0)=0$, the common horizontal scale is $V_t(1)=\pi/2$, as specified in the plot request. Each higher branch starts at $V/(\pi/2)=p$.



All curves tend to c/n_1 for large V at fixed p. They need not be monotone: the transition from cladding-dominated propagation to the oblique core ray can produce a minimum.

(c) For a length L the optical length inferred from delay is $c\tau=cL/v_g$, while the simple ray expression is $n_1L/\cos\theta$. The following comparison uses $n_1=1.01$, $n_2=1$; each colored curve joins the discrete modes at a fixed V.



The vertical coordinate is the percent excess delay relative to $n_1 L/c$. Away from the weakly confined highest modes, the points approach $100(1/\cos\theta - 1)$ as V increases. This is the step-index optical-path difference of Fig. 8.12b. Near threshold the delay is reduced by cladding propagation, so applying the core-only ray formula there overestimates it. Even in this nondispersive model different modes have different delays; material dispersion would add a further frequency dependence.

8.17 Consider the propagating modes in a cylindrical optical fiber waveguide of radius a with a step index of refraction, n_1 in the core ($\rho < a$) and $n_2 < n_1$ in the cladding ($\rho > a$). Assume that the fields vary as $e^{im\phi + ik_z z - i\omega t}$. For bound modes, the fields in the core (cladding) are proportional to ordinary (modified) Bessel functions $J_\nu(K_\nu)$ with appropriate values of ν and argument, as in (8.128).

(a) Show that for $m \neq 0$ the eigenvalue relation for the transverse parameter γ_{mn} (and β_{mn}) is

$$\left(\frac{n_1^2}{\gamma}\frac{J_m'}{J_m} + \frac{n_2^2}{\beta}\frac{K_m'}{K_m}\right)\left(\frac{1}{\gamma}\frac{J_m'}{J_m} + \frac{1}{\beta}\frac{K_m'}{K_m}\right) = \frac{m^2}{a^2}\left(\frac{n_1^2}{\gamma^2} + \frac{n_2^2}{\beta^2}\right)\left(\frac{1}{\gamma^2} + \frac{1}{\beta^2}\right)$$

where $\gamma^2 = n_1^2\omega^2/c^2 - k_z^2$ and $\beta^2 = k_z^2 - n_2^2\omega^2/c^2$, while primes indicate derivatives with respect to the argument, and the argument of J_m (K_m) is γa (βa). The first subscript on γ is the azimuthal index m; the second designates the nth root of the eigenvalue equation for fixed m.

(b) Determine the eigenvalue equation for the $m = 0$ modes (TE and TM) and show that the lowest "cutoff" frequency corresponds to $V = 2.405$, the first root of $J_0(x)$, where "cutoff" is the frequency below which the guide radiates rather than confines.

(c) Show that the lowest HE mode (HE_{11}) has no cutoff frequency and that for $V \ll 1$ the decay parameter $\beta a \approx Ae^{-B/V^2}$. Find A and B in terms of n_1 and n_2.

Solution.

(a) Take $\mu = \mu_0$, $\epsilon_i = n_i^2\epsilon_0$ and put $k_0 = \omega/c$. At $\rho = a$ let the continuous longitudinal field amplitudes be E and H. In each region Maxwell's equations give

$$\mathbf{E}_t = \frac{i}{g^2}(k_z\nabla_t E_z - \omega\mu_0\hat{\mathbf{z}}\times\nabla_t H_z), \quad \mathbf{H}_t = \frac{i}{g^2}(k_z\nabla_t H_z + \omega\epsilon\hat{\mathbf{z}}\times\nabla_t E_z),$$

with $g^2=\gamma^2$ inside and $g^2=-\beta^2$ outside. The radial logarithmic derivatives of the longitudinal fields are $\gamma J_m'/J_m$ and $\beta K_m'/K_m$ respectively. Define

$$D=\frac{J_m'}{\gamma J_m}+\frac{K_m'}{\beta K_m},\quad F=\frac{n_1^2 J_m'}{\gamma J_m}+\frac{n_2^2 K_m'}{\beta K_m},\quad C_m=\frac{mk_z}{a}(\gamma^{-2}+\beta^{-2}).$$

Continuity of E_ϕ, H_ϕ becomes the two homogeneous equations

$$C_m E+i\omega\mu_0 DH=0,\qquad i\omega\epsilon_0 FE-C_m H=0.$$

A nonzero solution requires

$$FD=\frac{k_z^2}{k_0^2}\frac{m^2}{a^2}(\gamma^{-2}+\beta^{-2})^2.$$

The identity

$$\frac{n_1^2}{\gamma^2}+\frac{n_2^2}{\beta^2}=\frac{k_z^2}{k_0^2}(\gamma^{-2}+\beta^{-2})$$

follows by substituting the definitions of γ,β and converts this determinant to the equation in part (a). The normal D,B conditions then follow from the tangential Maxwell equations.

(b) For $m=0$ the two equations decouple. With $u=\gamma a$, $w=\beta a$, $u^2+w^2=V^2$, they are

$$\boxed{\frac{J_0'(u)}{uJ_0(u)}+\frac{K_0'(w)}{wK_0(w)}=0\quad\text{(TE)},}$$

$$\boxed{n_1^2\frac{J_0'(u)}{uJ_0(u)}+n_2^2\frac{K_0'(w)}{wK_0(w)}=0\quad\text{(TM)}.}$$

At loss of confinement, $w\downarrow 0$ and $K_0'(w)/(wK_0(w))\sim -1/[w^2(-\log(w/2)-\gamma_E)]$ diverges negatively. Cancellation requires the positive pole of the J_0 term, with u approaching a zero of J_0 from above. The lowest threshold is therefore

$$\boxed{V_t=x_{01}=2.4048255577\ldots.}$$

(c) For the fundamental hybrid mode put $m=1$, and abbreviate $S=n_1^2+n_2^2$, $A_u=J_1'(u)/(uJ_1(u))$, $B_w=K_1'(w)/(wK_1(w))$. The determinant in dimensionless form is

$$(n_1^2A_u+n_2^2B_w)(A_u+B_w)=(n_1^2/u^2+n_2^2/w^2)(1/u^2+1/w^2).$$

For $w\ll u\ll 1$, start with

$$J_1(u)=\frac{u}{2}-\frac{u^3}{16}+\frac{u^5}{384}+O(u^7),\qquad K_1(w)=\frac{1}{w}+\frac{w}{2}\left[\log(w/2)+\gamma_E-\frac{1}{2}\right]+O(w^3\log w).$$

Differentiating and dividing these series gives

$$B_w=-w^{-2}-L+O(w^2L^2),\quad L=-\log(w/2)-\gamma_E,\qquad A_u=u^{-2}-\tfrac{1}{4}-\tfrac{u^2}{96}+O(u^4).$$

The w^{-4} terms cancel identically. The w^{-2} coefficients on the left and right are $2n_2^2L-SA_u$ and S/u^2. Equating them gives

$$2n_2^2L=S(A_u+u^{-2})=S\left(\frac{2}{u^2}-\frac{1}{4}-\frac{u^2}{96}+\cdots\right).$$

As $u^2 = V^2 - w^2$ and w is exponentially small, this becomes

$$\log(w/2) = -\gamma_E - \frac{S}{n_2^2 V^2} + \frac{S}{8n_2^2} + O(V^2).$$

Exponentiating yields

$$\boxed{\beta a \sim A\exp(-B/V^2), \qquad A = 2\exp\left[-\gamma_E + \frac{n_1^2+n_2^2}{8n_2^2}\right], \quad B = \frac{n_1^2+n_2^2}{n_2^2}.}$$

The relative correction to this leading expression is $O(V^2)$. There is a positive but exponentially small decay constant at arbitrarily small positive V: the HE_{11} branch has no positive cutoff. Its very long cladding tail should not be confused with a well-confined ray. The supplementary code solves a cancellation-free form of the determinant in $\log w$ and checks this limit; for $n_1 = 1.5, n_2 = 1, V = 0.3$ it finds $\log w = -35.58739754$, versus -35.58892959 from the asymptote.

8.18

(a) From the use of Green's theorem in two dimensions show that the TM and TE modes in a waveguide defined by the boundary-value problems (8.34) and (8.36) are orthogonal in the sense that

$$\int_A E_{z\lambda} E_{z\mu}\, da = 0 \qquad \text{for } \lambda \neq \mu$$

for TM modes, and a corresponding relation for H_z for TE modes.

(b) Prove that the relations (8.131)–(8.134) form a consistent set of normalization conditions for the fields, including the circumstances when λ is a TM mode and μ is a TE mode.

Solution.

(a) Use a uniform real ϵ, μ and a perfectly conducting boundary. Choose real scalar eigenfunctions of $-\nabla_t^2$, Dirichlet for TM and Neumann for TE. For two scalar eigenfunctions f_λ, f_μ, Green's identity is

$$(\gamma_\lambda^2 - \gamma_\mu^2)\int_A f_\lambda f_\mu\, da = \oint_{\partial A} (f_\lambda \partial_n f_\mu - f_\mu \partial_n f_\lambda)\, dl = 0.$$

Thus different eigenvalues are orthogonal. Within a degenerate eigenspace choose an orthogonal basis; the eigenvalue equation alone does not make arbitrary degenerate linear combinations orthogonal. This establishes the stated relation for E_z and its TE counterpart for H_z. The constant Neumann scalar is excluded because it produces no TE field.

(b) Let D_λ, N_λ be Dirichlet and nonconstant Neumann scalars normalized to $\int_A f_\lambda f_\mu\, da = \delta_{\lambda\mu}$. Integration by parts gives

$$\int_A \nabla_t f_\lambda \cdot \nabla_t f_\mu\, da = \gamma_\mu^2 \delta_{\lambda\mu}.$$

The normalized transverse electric fields can therefore be chosen as

$$\mathbf{E}_\lambda^{\mathrm{TM}} = \frac{\nabla_t D_\lambda}{\gamma_\lambda}, \qquad \mathbf{E}_\lambda^{\mathrm{TE}} = -\frac{\widehat{\mathbf{z}} \times \nabla_t N_\lambda}{\gamma_\lambda}.$$

Within either family their inner product is $\delta_{\lambda\mu}$. For a mixed pair, $\nabla_t \cdot (\widehat{\mathbf{z}} \times \nabla_t N) = 0$ and $D = 0$ on the wall imply

$$\int_A \nabla_t D \cdot (\widehat{\mathbf{z}} \times \nabla_t N)\, da = \oint_{\partial A} D(\widehat{\mathbf{z}} \times \nabla_t N) \cdot \mathbf{n}\, dl = 0.$$

This proves (8.131) for all pairs, including accidental degeneracy between TE and TM families.

With $k_\lambda^2 = \omega^2\mu\epsilon - \gamma_\lambda^2$,

$$\mathbf{H}_\lambda = \frac{\widehat{\mathbf{z}} \times \mathbf{E}_\lambda}{Z_\lambda}, \qquad Z_{\mathrm{TM}} = \frac{k_\lambda}{\omega\epsilon}, \quad Z_{\mathrm{TE}} = \frac{\omega\mu}{k_\lambda},$$

so the remaining transverse bilinear integrals follow directly:

$$\int_A \mathbf{H}_\lambda \cdot \mathbf{H}_\mu \, da = \frac{\delta_{\lambda\mu}}{Z_\lambda^2}, \qquad \frac{1}{2}\int_A (\mathbf{E}_\lambda \times \mathbf{H}_\mu) \cdot \widehat{\mathbf{z}} \, da = \frac{\delta_{\lambda\mu}}{2Z_\lambda}.$$

Finally $\nabla \cdot \mathbf{E} = \nabla \cdot \mathbf{H} = 0$ gives

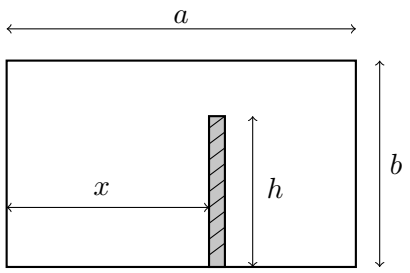
$$E_{z\lambda}^{\mathrm{TM}} = -\frac{i\gamma_\lambda}{k_\lambda} D_\lambda, \qquad H_{z\lambda}^{\mathrm{TE}} = -\frac{i\gamma_\lambda}{k_\lambda Z_\lambda} N_\lambda,$$

and consequently

$$\boxed{\int_A E_{z\lambda} E_{z\mu} \, da = -\frac{\gamma_\lambda^2}{k_\lambda^2}\delta_{\lambda\mu}, \qquad \int_A H_{z\lambda} H_{z\mu} \, da = -\frac{\gamma_\lambda^2}{k_\lambda^2 Z_\lambda^2}\delta_{\lambda\mu}.}$$

These are exactly the signs and factors of (8.134): the textbook uses bilinear products, without complex conjugation, and longitudinal fields are imaginary when the transverse electric field is real. For a propagating mode Z is positive and the power of amplitude A is $|A|^2/(2Z)$. For an evanescent mode the bilinear equations remain algebraic normalization identities, while the actual mean power uses $\frac{1}{2}\,\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*)$ and is zero for a single decaying mode. Exactly at cutoff, $k = 0$, these field conventions must be interpreted by a rescaled limit rather than by division by zero.

8.19 The figure shows a cross-sectional view of an infinitely long rectangular waveguide with the center conductor of a coaxial line extending vertically a distance h into its interior at $z = 0$. The current along the probe oscillates sinusoidally in time with frequency ω, and its variation in space can be approximated as $I(y) = I_0 \sin[(\omega/c)(h - y)]$. The thickness of the probe can be neglected. The frequency is such that only the TE_{10} mode can propagate in the guide.



Problem 8.19

(a) Calculate the amplitudes for excitation of both TE and TM modes for all (m, n) and show how the amplitudes depend on m and n for $m, n \gg 1$ for a fixed frequency ω.

(b) For the propagating mode show that the power radiated in the positive z direction is

$$P = \frac{\mu c^2 I_0^2}{\omega k a b} \sin^2\left(\frac{\pi X}{a}\right) \sin^4\left(\frac{\omega h}{2c}\right)$$

with an equal amount in the opposite direction. Here k is the wave number for the TE_{10} mode.

(c) Discuss the modifications that occur if the guide, instead of running off to infinity in both directions, is terminated with a perfectly conducting surface at $z = L$. What values of L will maximize the power flow for a fixed current I_0? What is the radiation resistance of the probe (defined as the ratio of power flow to one-half the square of the current at the base of the probe) at maximum?

Solution.

(a) Use X for the horizontal probe position marked x in the sketch, and $k_0 = \omega/c$. The prescribed current is

$$\mathbf{J} = \widehat{\mathbf{y}} I_0 \sin[k_0(h-y)]\delta(x-X)\delta(z), \qquad 0 < y < h.$$

Normalize the transverse electric fields to unit squared integral, as in (8.131). Put

$$q_m = m\pi/a, \quad q_n = n\pi/b, \quad \gamma_{mn}^2 = q_m^2 + q_n^2, \quad k_{mn} = \sqrt{k_0^2 - \gamma_{mn}^2},$$

choosing $k > 0$ for propagation and $\operatorname{Im} k > 0$ for decay. With $N_{mn} = \sqrt{(2-\delta_{m0})(2-\delta_{n0})/(ab)}$, the components needed in the source integral are

$$E_{y,mn}^{\mathrm{TM}} = N_{mn}\frac{q_n}{\gamma_{mn}}\sin(q_m x)\cos(q_n y), \qquad E_{y,mn}^{\mathrm{TE}} = N_{mn}\frac{q_m}{\gamma_{mn}}\sin(q_m x)\cos(q_n y).$$

TM requires $m, n \geq 1$; TE permits either index to be zero but excludes $(0,0)$. These signs agree with (8.135)–(8.136). The product-to-sum identity gives the elementary source integral:

$$F_n = \int_0^h \sin[k_0(h-y)]\cos(q_n y)\,dy = \frac{\cos(q_n h) - \cos(k_0 h)}{2}\left(\frac{1}{k_0 - q_n} + \frac{1}{k_0 + q_n}\right).$$

Combining the fractions,

$$F_n = \frac{k_0[\cos(q_n h) - \cos(k_0 h)]}{k_0^2 - q_n^2}.$$

When $q_n = k_0$, its continuous limit is $F_n = (h/2)\sin(k_0 h)$, so the apparent denominator singularity is removable.

Reciprocity, (8.146), gives $A_\lambda^{(\pm)} = -(Z_\lambda/2)\int \mathbf{J}\cdot\mathbf{E}_\lambda^{(\mp)} d^3x$. The transverse source at $z = 0$ therefore excites equal amplitudes in both directions:

$$\boxed{A_{mn}^{\mathrm{TM},\pm} = -\frac{I_0 Z_{mn}^{\mathrm{TM}}}{2} N_{mn}\frac{q_n}{\gamma_{mn}}\sin(q_m X)F_n, \qquad Z_{mn}^{\mathrm{TM}} = \frac{k_{mn}}{\omega\epsilon},}$$

$$\boxed{A_{mn}^{\mathrm{TE},\pm} = -\frac{I_0 Z_{mn}^{\mathrm{TE}}}{2} N_{mn}\frac{q_m}{\gamma_{mn}}\sin(q_m X)F_n, \qquad Z_{mn}^{\mathrm{TE}} = \frac{\omega\mu}{k_{mn}}.}$$

In particular all $m = 0$ TE amplitudes vanish. At fixed ω and $m, n \gg 1$, $k \sim i\gamma$ and $F_n = O(n^{-2})$. Keeping the oscillating trigonometric factors understood,

$$A_{mn}^{\mathrm{TM}} = O(n^{-1}), \qquad A_{mn}^{\mathrm{TE}} = O\left(\frac{m}{(m^2+n^2)n^2}\right).$$

When both indices grow at a fixed ratio these are $O(M^{-1})$ and $O(M^{-3})$, respectively. Some coefficients vanish faster or exactly by symmetry. These are source-plane amplitudes; at $z \neq 0$ the high modes also contain $e^{-\gamma|z|}$.

(b) For TE_{10}, $F_0 = 2\sin^2(k_0 h/2)/k_0$, $N_{10} = \sqrt{2/(ab)}$, so

$$A_{10}^{\pm} = -\frac{ZI_0}{k_0}\sqrt{\frac{2}{ab}}\sin\frac{\pi X}{a}\sin^2\frac{k_0 h}{2}.$$

Since the power in either outgoing wave is $|A_{10}|^2/(2Z)$,

$$\boxed{P = \frac{ZI_0^2}{abk_0^2}\sin^2\frac{\pi X}{a}\sin^4\frac{k_0 h}{2} = \frac{\mu c^2 I_0^2}{\omega kab}\sin^2\frac{\pi X}{a}\sin^4\frac{\omega h}{2c}.}$$

Here I_0 is a peak phasor amplitude; replace I_0^2 by $|I_0|^2$ if its phase has not been chosen real.

(c) A short at $z = L > 0$ reflects transverse electric fields with coefficient -1. Equivalently add a negative image probe at $z = 2L$. The outgoing amplitude at $z < 0$ for each mode is

$$A_{mn}^{\text{left}} = A_{mn}^{-}(1 - e^{2ik_{mn}L}).$$

In $0 < z < L$ the field is a standing-wave combination with zero tangential electric field at the short. The same image construction includes the evanescent near fields; for those modes e^{2ikL} is real and decaying. Only the propagating mode transports power far to the left, giving

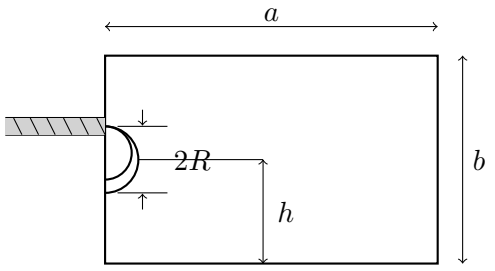
$$P_{\text{left}} = 4P\sin^2 kL, \qquad \boxed{L = (2j+1)\pi/(2k), \quad j = 0, 1, \ldots}$$

for maxima at fixed prescribed current. The maximum is $4P$, compared with total power $2P$ in the unterminated guide. With $I_{\text{base}} = I_0 \sin k_0 h$, the radiation resistance there is

$$\boxed{R_{\text{rad,max}} = \frac{4P}{|I_{\text{base}}|^2/2} = \frac{2Z}{abk_0^2}\sin^2\frac{\pi X}{a}\tan^2\frac{k_0 h}{2}.}$$

This definition presumes nonzero base current; a prescribed profile with zero base current cannot be assigned a finite resistance by that ratio. A physical source of fixed voltage would also have its current changed by the reflected wave, whereas the problem holds I_0 fixed.

8.20 An infinitely long rectangular waveguide has a coaxial line terminating in the short side of the guide with the thin central conductor forming a semicircular loop of radius R whose center is a height h above the floor of the guide, as shown in the accompanying cross-sectional view. The half-loop is in the plane $z = 0$ and its radius R is sufficiently small that the current can be taken as having a constant value I_0 everywhere on the loop.



Problem 8.20

(a) Prove that to the extent that the current is constant around the half-loop, the TM modes are not excited. Give a physical explanation of this lack of excitation.

(b) Determine the amplitude for the lowest TE mode in the guide and show that its value is independent of the height h.

(c) Show that the power radiated in either direction in the lowest TE mode is

$$P = \frac{I_0^2}{16} Z \frac{a}{b} \left(\frac{\pi R}{a}\right)^4$$

where Z is the wave impedance of the TE_{10} mode. Here assume $R \ll a, b$.

Solution.

(a) The mode coefficient for the filament is

$$A_\lambda^\pm = -\frac{Z_\lambda I_0}{2} \int_{\mathrm{arc}} \mathbf{E}_{t\lambda} \cdot d\boldsymbol{\ell}.$$

For TM modes $\mathbf{E}_t = \nabla_t D/\gamma$, where D vanishes on the metal boundary. Consequently

$$\int_{\mathrm{arc}} \mathbf{E}_t^{\mathrm{TM}} \cdot d\boldsymbol{\ell} = \frac{D(\mathrm{end}) - D(\mathrm{start})}{\gamma} = 0.$$

This proves the absence of every TM mode, including evanescent ones, without a small-R expansion. The constant current together with its return in the wall forms a solenoidal transverse loop. It has magnetic circulation and no varying line charge along the arc, so its transverse overlap with a gradient field is zero. If the current varied around the loop, integration by parts would leave a term involving that variation and TM excitation could occur.

(b) Choose the direction of current to increase ϑ in

$$x = R\cos\vartheta, \qquad y = h + R\sin\vartheta, \qquad -\pi/2 \le \vartheta \le \pi/2.$$

For TE_{10}, $E_y = \sqrt{2/(ab)}\sin(\pi x/a)$ and $E_x = 0$. Its overlap is independent of h:

$$\begin{aligned}\int_{\mathrm{arc}} \mathbf{E}_t \cdot d\boldsymbol{\ell} &= \sqrt{\frac{2}{ab}} R \int_{-\pi/2}^{\pi/2} \cos\vartheta \sin\left(\frac{\pi R}{a}\cos\vartheta\right) d\vartheta \\ &= \sqrt{\frac{2}{ab}}\, \pi R J_1(\pi R/a).\end{aligned}$$

The last identity follows by differentiating $\int_{-\pi/2}^{\pi/2} \cos(t\cos\vartheta) d\vartheta = \pi J_0(t)$. Thus, for constant current, the exact coefficient is

$$\boxed{A_{10}^\pm = -\frac{ZI_0}{2}\sqrt{\frac{2}{ab}}\pi R J_1(\pi R/a).}$$

Reversing the loop orientation reverses its sign but not its power. For $R \ll a, b$, $J_1(t) = t/2 + O(t^3)$, and therefore

$$A_{10}^\pm = -ZI_0\sqrt{\frac{2}{ab}}\frac{\pi^2 R^2}{4a}[1 + O(R^2/a^2)].$$

(c) The power in either direction is

$$\boxed{P = \frac{|A_{10}|^2}{2Z} = \frac{I_0^2}{16} Z \frac{a}{b}\left(\frac{\pi R}{a}\right)^4 [1 + O(R^2/a^2)].}$$

The exact constant-current result before this expansion is $P = ZI_0^2\pi^2 R^2 J_1^2(\pi R/a)/(4ab)$. The independence of h follows from the absence of any y dependence in the TE_{10} field, provided the entire loop remains inside the guide.

8.21 A hollow metallic waveguide with a distortion in the form of a localized bend or increase in cross section can support nonpropagating ("bound state") configurations of fields in the vicinity of the distortion. Consider a rectangular guide that has its distortion confined to a plane, as shown in the figure, and TE_{10} as its lowest propagating mode, with perpendicular electric field $E_\perp = \psi$. On either side of the distortion the guide is straight and of width a. Without distortion, $\psi = E_0 \sin(\pi y/a)\exp(\pm ikz)$, where $k^2 = (\omega/c)^2 - (\pi/a)^2$. The distortion is described by a curvature $\kappa(s) = 1/R(s)$ and a width $w(s)$. Locally the element of area in the plane is $dA = h(s,t)\, ds\, dt$, where s is the length along the guide wall and t the transverse coordinate, as shown in the figure, and $h(s,t) = 1 - \kappa(s)t$. In terms of s and t the Laplacian is

$$\nabla^2\psi = \frac{1}{h}\frac{\partial}{\partial t}\left(h\frac{\partial\psi}{\partial t}\right) + \frac{1}{h}\frac{\partial}{\partial s}\left(\frac{1}{h}\frac{\partial\psi}{\partial s}\right)$$

If the distortions are very small and change slowly in s on the scale of the width a, an ansatz for the solution is

$$\psi(s,t) = \frac{u(s)}{\sqrt{h(s,t)}}\cdot\sin\left[\frac{\pi t}{w(s)}\right]$$

[The factor in the denominator is equivalent to the factor $\rho^{-1/2}$ familiar from Bessel functions that converts the radial part of the Laplacian in polar coordinates to a simple second partial derivative (plus an additional term without derivatives).]

(a) Show that substitution of the ansatz into the two-dimensional wave equation, $(\nabla^2+\omega^2/c^2)\psi = 0$, leads to the equation for $u(s)$,

$$\frac{d^2u}{ds^2} + [k^2 - v(s)]u = 0 \qquad \text{with } v(s) = \pi^2\left(\frac{1}{w^2(s)} - \frac{1}{a^2}\right) - \frac{1}{4}\kappa^2(s)$$

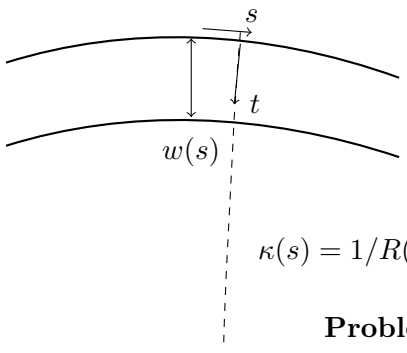
if small terms are neglected. Interpret $v(s)$ in analogy with the Schrödinger equation in one dimension.

(b) If the distortion is in the form of a bend through an angle θ with constant radius of curvature $R \gg a$, show that for $\theta a/R \ll 1$ and $|\theta| \ll 1$ there is a "bound state" at frequency ω_0 where

$$\omega_0^2 \approx \left(\frac{\pi c}{a}\right)^2\left[1 - \left(\frac{\theta a}{8\pi R}\right)^2\right]$$

The additional small-angle assumption is needed for the stated leading coefficient; the finite-angle bound state is treated in the solution.

References: J. Goldstone and R. L. Jaffe, *Phys. Rev. B* **45**, 14100 (1992); J. P. Carini, J. T. Londergan, K. Mullen, and D. P. Murdock, *Phys. Rev. B* **48**, 4503 (1993).



Problem 8.21

Solution.

(a) Put $f(s,t) = u(s)\sin[\pi t/w(s)]$ and $\psi = h^{-1/2}f$. At fixed s, $h_t = -\kappa$, and direct differentiation gives

$$\frac{1}{h}\partial_t(h\partial_t\psi) = h^{-1/2}\left(\partial_t^2 f + \frac{\kappa^2}{4h^2}f\right).$$

For $|\kappa|a \ll 1$, slowly varying width and curvature, and a longitudinal amplitude varying on scales large compared with a, the longitudinal part can be expanded explicitly as

$$\frac{1}{h}\partial_s(h^{-1}\partial_s\psi) = h^{-1/2}\left[\frac{f_{ss}}{h^2} - \frac{2h_s f_s}{h^3} + \left(\frac{5h_s^2}{4h^4} - \frac{h_{ss}}{2h^3}\right)f\right].$$

Here $h_s = -t\kappa'$ and $h_{ss} = -t\kappa''$. Writing $\vartheta = \pi t/w(s)$ also gives

$$f_{ss} = u''\sin\vartheta + 2u'\vartheta_s\cos\vartheta + u\vartheta_{ss}\cos\vartheta - u\vartheta_s^2\sin\vartheta, \quad \vartheta_s = -\frac{\pi t w'}{w^2}, \quad \vartheta_{ss} = -\frac{\pi t w''}{w^2} + \frac{2\pi t(w')^2}{w^3}.$$

Discarding these width and curvature derivatives, and replacing h^{-2} by 1 in the slow longitudinal derivative term, leaves $h^{-1/2}u''\sin(\pi t/w)$. These are precisely the stated adiabatic and small-curvature approximations. Using $\partial_t^2 f = -(\pi/w)^2 f$ and $h^{-2} = 1 + O(\kappa a)$ in the curvature term gives

$$u'' + \left[\frac{\omega^2}{c^2} - \frac{\pi^2}{w^2(s)} + \frac{\kappa^2(s)}{4}\right]u = 0.$$

Separating the straight-guide cutoff gives the stated equation,

$$\boxed{-u'' + v(s)u = k^2 u, \qquad v(s) = \pi^2\left[\frac{1}{w^2(s)} - \frac{1}{a^2}\right] - \frac{\kappa^2(s)}{4}.}$$

It is a one-dimensional Schrödinger eigenproblem with energy k^2. A widening makes the first term attractive; curvature makes the second term attractive regardless of the direction of bending. A localized negative potential in one dimension supports a bound ground state. Here $k^2 = -q^2 < 0$ means fields decay in the straight sections, with frequency below $\pi c/a$; it does not mean a negative electromagnetic frequency squared.

(b) For a bend of constant radius and unchanged width, the reduced potential is a square well of depth $1/(4R^2)$ and length $\ell = R|\theta|$. Smooth end transitions much shorter than R and than ℓ, but longer than a, give the same leading well limit while preserving the slow-variation assumption of (a). This smooth-well interpretation requires $a \ll \ell$ as well as $a \ll R$. With the origin at its center the even ground-state solution is proportional to $\cos(bs)$ inside and $e^{-q(|s|-\ell/2)}$ outside, where

$$b^2 = \frac{1}{4R^2} - q^2, \qquad \boxed{q = b\tan(b\ell/2).}$$

For a weak, short well, $b\ell/2 \ll 1$. This requires $|\theta| \ll 1$, independently of a/R. Expanding the tangent then gives

$$q = \frac{\ell}{8R^2}[1 + O(\theta^2)] = \frac{|\theta|}{8R}[1 + O(\theta^2)].$$

Since $\omega_0^2/c^2 = (\pi/a)^2 - q^2$, the requested leading small-angle result is

$$\boxed{\omega_0^2 \simeq \left(\frac{\pi c}{a}\right)^2\left[1 - \left(\frac{\theta a}{8\pi R}\right)^2\right].}$$

The retained relative shift is small when $|\theta|a/R \ll 1$, but that condition alone does not justify its coefficient. This is the reason for the red correction to part (b). For finite angles the full reduced-model answer remains available: let $x \in (0, \pi/2)$ solve

$$\frac{x}{\cos x} = \frac{|\theta|}{4}, \qquad qR = \tfrac{1}{2}\sin x, \qquad \omega_0^2 = \left(\frac{\pi c}{a}\right)^2 - \frac{c^2 \sin^2 x}{4R^2}.$$

For example $\theta = \pi$ gives $qR = 0.2958470340$, whereas $|\theta|/8 = 0.3926990817$. Taking R/a arbitrarily large satisfies the originally printed $\theta a/R \ll 1$ without removing this discrepancy. Thus the small-angle requirement is mathematical, not merely a preference about how tightly the guide should bend.

9 Chapter 9. Radiating Systems, Multipole Fields and Radiation Problems

9.1 A common textbook example of a radiating system (see Problem 9.2) is a configuration of charges fixed relative to each other but in rotation. The charge density is obviously a function of time, but it is not in the form of (9.1).

(a) Show that for rotating charges one alternative is to calculate *real* time-dependent multipole moments using $\rho(\mathbf{x}, t)$ directly and then compute the multipole moments for a given harmonic frequency with the convention of (9.1) by inspection or Fourier decomposition of the time-dependent moments. Note that care must be taken when calculating $q_{lm}(t)$ to form linear combinations that are real before making the connection.

(b) Consider a charge density $\rho(\mathbf{x}, t)$ that is periodic in time with period $T = 2\pi/\omega_0$. By making a Fourier *series* expansion, show that it can be written as

$$\rho(\mathbf{x}, t) = \rho_0(\mathbf{x}) + \sum_{n=1}^{\infty} \mathrm{Re}[2\rho_n(\mathbf{x}) e^{-in\omega_0 t}]$$

where

$$\rho_n(\mathbf{x}) = \frac{1}{T} \int_0^T \rho(\mathbf{x}, t) e^{in\omega_0 t}\, dt$$

This shows explicitly how to establish connection with (9.1).

(c) For a single charge q rotating about the origin in the x-y plane in a circle of radius R at constant angular speed ω_0, calculate the $l = 0$ and $l = 1$ multipole moments by the methods of parts a and b and compare. In method b express the charge density $\rho_n(\mathbf{x})$ in cylindrical coordinates. Are there higher multipoles, for example, quadrupole? At what frequencies?

Solution.

Use $Y_{lm} = N_{lm} P_l^m(\cos\theta) e^{im\phi}$ with the Condon–Shortley phase, with $N_{lm} = [(2l+1)(l-m)!/(4\pi(l+m)!)]^{1/2}$. We use $q_{lm}(t) = \int r^l Y_{lm}^* \rho(\mathbf{x}, t) d^3x$ for the instantaneous complex spherical moment, and Jackson's Q_{lm} for its electric source phasor at a specified positive frequency. A phasor below means the coefficient inside $\mathrm{Re}[\cdots e^{-i\Omega t}]$, not a two-sided Fourier coefficient.

(a) Multipole formation is linear. Thus for every real Cartesian moment $F(t)$, Fourier expansion commutes with the spatial integral defining F. In particular, expand p_x, p_y, p_z or the five real components of Q_{ij}, and then convert their phasors to spherical components. A complex quantity such as $q_{11}(t) = -\sqrt{3/(8\pi)}[p_x(t) - ip_y(t)]$ is not itself a real observable; taking its real part before this conversion loses information.

(b) Orthogonality of $e^{in\omega_0 t}$ gives

$$\rho = \sum_{n=-\infty}^{\infty} \rho_n e^{-in\omega_0 t}, \qquad \rho_n = T^{-1} \int_0^T \rho e^{in\omega_0 t} dt.$$

Reality gives $\rho_{-n} = \rho_n^*$, so pairing n with $-n$ yields exactly $\rho_0 + \sum_{n\geq 1} \mathrm{Re}[2\rho_n e^{-in\omega_0 t}]$. Consequently the positive-frequency source phasor is $2\rho_n$.

(c) Write cylindrical radius as s and use the periodic delta $\delta_{2\pi}$. Then

$$\rho = \frac{q}{s}\delta(s-R)\delta(z)\delta_{2\pi}(\phi-\omega_0 t), \qquad \boxed{\rho_n = \frac{q}{2\pi s}\delta(s-R)\delta(z)e^{in\phi}.}$$

Directly, the real moments are

$$q_{00} = \frac{q}{\sqrt{4\pi}}, \quad \mathbf{p}(t) = qR(\hat{\mathbf{x}}\cos\omega_0 t + \hat{\mathbf{y}}\sin\omega_0 t), \quad q_{10} = 0.$$

Thus $\mathbf{p}_{\omega_0} = qR(\hat{\mathbf{x}} + i\hat{\mathbf{y}})$, $(Q_{11})_{\omega_0} = -2qR\sqrt{3/(8\pi)}$, and $(Q_{1,-1})_{\omega_0} = 0$. Alternatively direct integration of ρ_n gives

$$\int r^l Y^*_{lm}\rho_n\, d^3x = qR^l Y^*_{lm}(\pi/2, 0)\delta_{mn}.$$

Doubling this for $n > 0$ reproduces the preceding phasors. Before taking a phasor, the complex time-dependent moments themselves are $q_{11}(t) = -qR\sqrt{3/(8\pi)}e^{-i\omega_0 t}$ and $q_{1,-1}(t) = qR\sqrt{3/(8\pi)}e^{i\omega_0 t}$; this displays the factor-of-two trap in part (a).

All l, m for which $P_l^m(0) \neq 0$ occur: $l + m$ must be even. The $m = 0$ terms are static, while $|m| > 0$ terms oscillate at $|m|\omega_0$. For example $q_{20} = -qR^2\sqrt{5/(16\pi)}$ is static and $q_{22} = qR^2\sqrt{15/(32\pi)}e^{-2i\omega_0 t}$, with its reality-related $m = -2$ partner, supplies electric quadrupole radiation at $2\omega_0$.

9.2 A radiating quadrupole consists of a square of side a with charges $\pm q$ at alternate corners. The square rotates with angular velocity ω about an axis normal to the plane of the square and through its center. Calculate the quadrupole moments, the radiation fields, the angular distribution of radiation, and the total radiated power, all in the long-wavelength approximation. What is the frequency of the radiation?

Solution.

Choose the positive charges initially at $\pm R\hat{\mathbf{x}}$ and the negative charges at $\pm R\hat{\mathbf{y}}$, where $R = a/\sqrt{2}$. A change of the initial orientation changes only phases. The net charge, electric dipole and magnetic dipole vanish. With $\Omega = 2\omega$, direct summation of $Q_{ij} = \sum_s q_s(3x_{si}x_{sj} - r_s^2\delta_{ij})$ gives

$$Q(t) = 3qa^2\begin{pmatrix}\cos\Omega t & \sin\Omega t & 0\\ \sin\Omega t & -\cos\Omega t & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad \mathcal{Q} = 3qa^2\begin{pmatrix}1 & i & 0\\ i & -1 & 0\\ 0 & 0 & 0\end{pmatrix}.$$

Thus the positive-frequency spherical moment is only $Q_{22} = qa^2\sqrt{15/(2\pi)}$. Put $K = \Omega/c$ and $\mathbf{v} = \mathcal{Q}\mathbf{n}$. The radiation fields are

$$\boxed{\mathbf{E} = -\frac{iK^3}{24\pi\epsilon_0}[\mathbf{v} - \mathbf{n}(\mathbf{n}\cdot\mathbf{v})]\frac{e^{iKr}}{r}, \qquad \mathbf{H} = \frac{1}{Z_0}\mathbf{n}\times\mathbf{E}.}$$

These follow by expanding the retarded vector potential to first order in the source coordinate and using continuity: its symmetric part is $\mathbf{A}_Q = -\mu_0 cK^2\mathcal{Q}\mathbf{n}\, e^{iKr}/(24\pi r)$; $\mathbf{E}_T = i\Omega\mathbf{A}_T$. Explicitly, if $A = 3qa^2$, $\mathbf{v} = A\sin\theta e^{i\phi}(\hat{\mathbf{x}} + i\hat{\mathbf{y}})$, so $|\mathbf{v}|^2 = 2A^2\sin^2\theta$ and $|\mathbf{n}\cdot\mathbf{v}|^2 = A^2\sin^4\theta$. Therefore

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{cq^2a^4K^6}{128\pi^2\epsilon_0}(1-\cos^4\theta).}$$

Since $2\pi \int_{-1}^{1}(1-u^4)du = 16\pi/5$,

$$\boxed{\overline{P} = \frac{cq^2a^4K^6}{40\pi\epsilon_0} = \frac{8q^2a^4\omega^6}{5\pi\epsilon_0 c^5}, \qquad \Omega = 2\omega.}$$

The physical fields are the real parts after multiplication by $e^{-i\Omega t}$; the square returns to the same charge configuration after half a revolution.

9.3 Two halves of a spherical metallic shell of radius R and infinite conductivity are separated by a very small insulating gap. An alternating potential is applied between the two halves of the sphere so that the potentials are $\pm V\cos\omega t$. In the long-wavelength limit, find the radiation fields, the angular distribution of radiated power, and the total radiated power from the sphere.

Solution.

Take the gap in the equatorial plane. In the quasistatic exterior solution write $\Phi(R,\theta) = V\ \mathrm{sgn}(\cos\theta)$ for its phasor. Its $l=1$ Legendre coefficient is

$$A_1 = \frac{3}{2}\int_{-1}^{1} Vu\ \mathrm{sgn}\,u\,du = \frac{3}{2}V.$$

Thus $\Phi_1 = (3VR^2/2r^2)\cos\theta$ and comparison with $\Phi_p = p\cos\theta/(4\pi\epsilon_0 r^2)$ gives $\mathbf{p} = 6\pi\epsilon_0 VR^2\hat{\mathbf{z}}$. Higher odd electric multipoles are smaller in radiated amplitude by successive powers of $(kR)^2$. The leading radiation is

$$E_\theta = -\frac{3}{2}Vk^2R^2\sin\theta\frac{e^{ikr}}{r}, \qquad H_\phi = E_\theta/Z_0,$$

with all other radiation components zero. Consequently

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{9}{8}\epsilon_0 cV^2k^4R^4\sin^2\theta, \qquad \overline{P} = 3\pi\epsilon_0 cV^2k^4R^4.}$$

Here the voltage difference is $2V$, not V, and $\int\sin^2\theta\,d\Omega = 8\pi/3$.

9.4 Apply the approach of Problem 9.1b to the current and magnetization densities of the particle of charge q rotating about the origin in the x-y plane in a circle of radius R at constant angular speed ω_0. The motion is such that $\omega_0 R \ll c$.

(a) Find $(J_x)_n$, $(J_y)_n$, and $(J_z)_n$ in terms of cylindrical coordinates for all n. Also determine the components of the orbital "magnetization," $(\mathbf{x}\times\mathbf{J}_n)/2$, and its divergence [which plays the role of a magnetic charge density for magnetic multipoles, as in M_{lm} (9.172)].

(b) What long-wavelength magnetic multipoles (l,m) occur and at what frequencies? [Remember that the multipole order l does not necessarily equal the harmonic number n.]

(c) Use linear superposition to generalize your argument to the four charges rotating in Problem 9.2 at radius $R = a/\sqrt{2}$. What harmonics occur, and what magnetic multipoles at each harmonic? Is there a magnetic multipole contribution at the $E2$ frequency of Problem 9.2? Is it significant relative to the $E2$ radiation?

Solution.

(a) The particle velocity at its position is $\omega_0 s\hat{\boldsymbol{\phi}}$. Multiplication of the Fourier density in 9.1 therefore gives, for every integer n,

$$\mathbf{J}_n = \frac{q\omega_0}{2\pi}\delta(s-R)\delta(z)e^{in\phi}\hat{\boldsymbol{\phi}}.$$

Its Cartesian components are this scalar times $(-\sin\phi, \cos\phi, 0)$. The apparent extra harmonics in these components are not extra temporal frequencies; the cylindrical basis itself depends on position. On the support $z = 0$,

$$\mathcal{M}_n = \tfrac{1}{2}\mathbf{x}\times\mathbf{J}_n = \frac{q\omega_0 s}{4\pi}\delta(s-R)\delta(z)e^{in\phi}\hat{\mathbf{z}}, \quad \nabla\cdot\mathcal{M}_n = \frac{q\omega_0 s}{4\pi}\delta(s-R)\delta'(z)e^{in\phi}.$$

The vanishing distribution $z\delta(z)$ in the transverse component makes no further contribution.

(b) The orbital moment convention of (9.172) is $M_{lm} = -(2/(l+1))\int r^l Y_{lm}^* \nabla\cdot\mathcal{M}\, d^3x$. Integrating by parts in z and then integrating ϕ gives the *Fourier coefficient*

$$\boxed{(M_{lm})_n = \frac{q\omega_0 R^{l+1}}{l+1} N_{lm}(P_l^m)'(0)\delta_{mn}.}$$

In obtaining this expression use $\partial_z[r^l P_l^m(z/r)]_{z=0} = R^{l-1}(P_l^m)'(0)$. It is nonzero only when $l+m$ is odd. For $n > 0$ the phasor is twice this expression: the allowed magnetic multipoles are $m = n$, $l = n+1, n+3, \ldots$, at frequency $n\omega_0$. The $n = 0$ moments have odd l and are static; in particular $m_z = q\omega_0 R^2/2$. Negative-frequency components are fixed by reality.

(c) Superposition adds the phase factor

$$\sum_{j=0}^{3}(-1)^j e^{-in\pi j/2} = \begin{cases} 4, & n = 2 \pmod 4, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the positive harmonics are $n = 2, 6, 10, \ldots$; at each one the magnetic orders are $l = n+1, n+3, \ldots$, $m = n$. There is an $M3$ contribution at the $E2$ frequency $2\omega_0$. For example the leading phasors satisfy

$$\frac{M_{32}}{Q_{22}} = \frac{\sqrt{7}}{4}\omega_0 R^2.$$

The general powers are proportional to $k^{2l+2}[(l+1)/l]|Q_{lm}|^2/[(2l+1)!!]^2$, with $Q \mapsto M/c$ for magnetic radiation. At $k = 2\omega_0/c$ their ratio is

$$\boxed{\frac{P_{M3}}{P_{E2}} = \frac{k^2}{c^2}\frac{8}{9}\frac{1}{49}\left|\frac{M_{32}}{Q_{22}}\right|^2 = \frac{2}{63}\left(\frac{\omega_0 R}{c}\right)^4.}$$

It is negligible in the stated nonrelativistic limit, despite occurring at the same frequency.

9.5

(a) Show that for harmonic time variation at frequency ω the electric dipole scalar and vector potentials in the Lorenz gauge and the long-wavelength limit are

$$\Phi(\mathbf{x}) = \frac{e^{ikr}}{4\pi\epsilon_0 r^2}\,\mathbf{n}\cdot\mathbf{p}(1-ikr)$$

$$\mathbf{A}(\mathbf{x}) = -i\,\frac{\mu_0\omega}{4\pi}\frac{e^{ikr}}{r}\,\mathbf{p} \qquad \text{[this is (9.16)]}$$

where $k = \omega/c$, $\mathbf{n}$ is a unit vector in the radial direction, $\mathbf{p}$ is the dipole moment (9.17), and the time dependence $e^{-i\omega t}$ is understood.

(b) Calculate the electric and magnetic fields *from the potentials* and show that they are given by (9.18).

Solution.

(a) Expand the outgoing Helmholtz Green function for a localized source:

$$\frac{e^{ik|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|} = \frac{e^{ikr}}{r} - \mathbf{x}'\cdot\nabla\frac{e^{ikr}}{r} + \cdots .$$

The electric dipole part of the scalar potential is consequently $\Phi_p = -(4\pi\epsilon_0)^{-1}\mathbf{p}\cdot\nabla(e^{ikr}/r)$. Since $\nabla(e^{ikr}/r) = \mathbf{n}e^{ikr}(ik/r - 1/r^2)$, this is the stated scalar potential. Integrating $\nabla'\cdot\mathbf{J} = i\omega\rho$ against x_i' gives $\int J_i\, d^3x' = -i\omega p_i$; the zeroth spatial moment of the vector potential is therefore $\mathbf{A} = -i\mu_0\omega\mathbf{p}\, e^{ikr}/(4\pi r)$. A time-independent net charge, if present, has its separate Coulomb field.

(b) Define $f(r) = e^{ikr}/r$. Then

$$\mathbf{B} = \nabla\times\mathbf{A} = -\frac{i\mu_0\omega}{4\pi}f'\mathbf{n}\times\mathbf{p} = \frac{\mu_0 c}{4\pi}e^{ikr}\left(\frac{k^2}{r} + \frac{ik}{r^2}\right)\mathbf{n}\times\mathbf{p}.$$

For the electric field use $\partial_i\partial_j f = (f'' - f'/r)n_i n_j + (f'/r)\delta_{ij}$ and

$$f' = e^{ikr}(ik/r - 1/r^2), \qquad f'' = e^{ikr}(-k^2/r - 2ik/r^2 + 2/r^3).$$

Then $\mathbf{E} = -\nabla\Phi + i\omega\mathbf{A}$ gives

$$\boxed{\mathbf{E} = \frac{e^{ikr}}{4\pi\epsilon_0}\left\{\frac{k^2}{r}[(\mathbf{n}\times\mathbf{p})\times\mathbf{n}] + \left(\frac{1}{r^3} - \frac{ik}{r^2}\right)[3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}]\right\},}$$

$$\boxed{\mathbf{H} = \frac{ck^2}{4\pi}\frac{e^{ikr}}{r}\left(1 + \frac{i}{kr}\right)\mathbf{n}\times\mathbf{p}.}$$

These retain the electrostatic, induction and radiation fields while keeping only the electric dipole source moment.

9.6

(a) Starting from the general expression (9.2) for $\mathbf{A}$ and the corresponding expression for Φ, expand both $R = |\mathbf{x}-\mathbf{x}'|$ and $t' = t - R/c$ to first order in $|\mathbf{x}'|/r$ to obtain the electric dipole potentials for arbitrary time variation

$$\Phi(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^2}\,\mathbf{n}\cdot\mathbf{p}_{\rm ret} + \frac{1}{cr}\,\mathbf{n}\cdot\frac{\partial\mathbf{p}_{\rm ret}}{\partial t}\right]$$

$$\mathbf{A}(\mathbf{x},t) = \frac{\mu_0}{4\pi r}\frac{\partial\mathbf{p}_{\rm ret}}{\partial t}$$

where $\mathbf{p}_{\rm ret} = \mathbf{p}(t' = t - r/c)$ is the dipole moment evaluated at the retarded time measured from the origin.

(b) Calculate the dipole electric and magnetic fields directly from these potentials and show that

$$\mathbf{B}(\mathbf{x},t) = \frac{\mu_0}{4\pi}\left[-\frac{1}{r^2}\,\mathbf{n}\times\frac{\partial\mathbf{p}_{\rm ret}}{\partial t} - \frac{1}{cr}\,\mathbf{n}\times\frac{\partial^2\mathbf{p}_{\rm ret}}{\partial t^2}\right]$$

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\left\{\left(1 + \frac{r}{c}\frac{\partial}{\partial t}\right)\left[\frac{3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}_{\rm ret}) - \mathbf{p}_{\rm ret}}{r^3}\right] + \frac{1}{c^2 r}\,\mathbf{n}\times\left(\mathbf{n}\times\frac{\partial^2\mathbf{p}_{\rm ret}}{\partial t^2}\right)\right\}$$

(c) Show explicitly how you can go back and forth between these results and the harmonic fields of (9.18) by the substitutions $-i\omega \leftrightarrow \partial/\partial t$ and $\mathbf{p}e^{ikr-i\omega t} \leftrightarrow \mathbf{p}_{\rm ret}(t')$.

Solution.

(a) Put $t_r = t - r/c$. To first order in the source coordinate,

$$R = r - \mathbf{n}\cdot\mathbf{x}' + \cdots, \quad R^{-1} = r^{-1} + r^{-2}\mathbf{n}\cdot\mathbf{x}' + \cdots, \quad \rho(\mathbf{x}', t - R/c) = \rho(\mathbf{x}', t_r) + \frac{\mathbf{n}\cdot\mathbf{x}'}{c}\dot\rho(\mathbf{x}', t_r) + \cdots.$$

Integrate their product in the retarded scalar potential. Its dipole terms are $\mathbf{n}\cdot\mathbf{p}/r^2 + \mathbf{n}\cdot\dot{\mathbf{p}}/(cr)$, divided by $4\pi\epsilon_0$. The omitted monopole is the constant total charge divided by $4\pi\epsilon_0 r$. Integration of the continuity equation times x_i' gives $\int J_i\, d^3x' = \dot p_i$, so the dipole vector potential is $\mu_0\dot{\mathbf{p}}(t_r)/(4\pi r)$ as claimed. This source expansion also requires the source light-crossing time to be short compared with its variation time; r itself need not be in the radiation zone.

(b) All time derivatives below are at t_r. The identities

$$\partial_i t_r = -n_i/c, \qquad \partial_i n_j = (\delta_{ij} - n_i n_j)/r$$

ensure that spatial differentiation includes retardation. In particular

$$\nabla\frac{\mathbf{n}\cdot\mathbf{p}}{r^2} = \frac{\mathbf{p} - 3\mathbf{n}(\mathbf{n}\cdot\mathbf{p})}{r^3} - \frac{\mathbf{n}(\mathbf{n}\cdot\dot{\mathbf{p}})}{cr^2},$$

$$\nabla\frac{\mathbf{n}\cdot\dot{\mathbf{p}}}{cr} = \frac{\dot{\mathbf{p}} - 2\mathbf{n}(\mathbf{n}\cdot\dot{\mathbf{p}})}{cr^2} - \frac{\mathbf{n}(\mathbf{n}\cdot\ddot{\mathbf{p}})}{c^2 r}.$$

Adding $-\partial_t\mathbf{A} = -\ddot{\mathbf{p}}/(4\pi\epsilon_0 c^2 r)$ to minus these gradients gives

$$\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0}\left[\frac{3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}}{r^3} + \frac{3\mathbf{n}(\mathbf{n}\cdot\dot{\mathbf{p}}) - \dot{\mathbf{p}}}{cr^2} + \frac{\mathbf{n}\times(\mathbf{n}\times\ddot{\mathbf{p}})}{c^2 r}\right].}$$

Curling $\dot{\mathbf{p}}(t_r)/r$ gives separately the derivative of $1/r$ and that of its retarded argument:

$$\boxed{\mathbf{B} = -\frac{\mu_0}{4\pi}\left[\frac{\mathbf{n}\times\dot{\mathbf{p}}}{r^2} + \frac{\mathbf{n}\times\ddot{\mathbf{p}}}{cr}\right].}$$

This also confirms the two corrected radial powers in the question.

(c) For $\mathbf{p}(t_r) = \mathbf{p}_0 e^{-i\omega(t-r/c)}$, $\dot{\mathbf{p}} = -i\omega\mathbf{p}$ and $\ddot{\mathbf{p}} = -\omega^2\mathbf{p}$. Substitution in the last two expressions gives the two boxed harmonic fields in 9.5 after removing $e^{-i\omega t}$. Conversely each power of $-i\omega$ in those fields becomes a time derivative under Fourier superposition. This applies to arbitrary sufficiently differentiable source histories, not only a single sinusoid.

9.7

(a) By means of Fourier superposition of different frequencies or equivalent means, show for a real electric dipole $\mathbf{p}(t)$ that the instantaneous radiated power per unit solid angle at a distance r from the dipole in a direction $\mathbf{n}$ is

$$\frac{dP(t)}{d\Omega} = \frac{Z_0}{16\pi^2 c^2}\left|\left[\mathbf{n}\times\frac{d^2\mathbf{p}}{dt'^2}(t')\right]\times\mathbf{n}\right|^2$$

where $t' = t - r/c$ is the retarded time. For a magnetic dipole $\mathbf{m}(t)$, substitute $(1/c)\ddot{\mathbf{m}}\times\mathbf{n}$ for $(\mathbf{n}\times\ddot{\mathbf{p}})\times\mathbf{n}$.

(b) Show similarly for a real quadrupole tensor $Q_{\alpha\beta}(t)$ given by (9.41) with a real charge density $\rho(\mathbf{x},t)$ that the instantaneous radiated power per unit solid angle is

$$\frac{dP(t)}{d\Omega} = \frac{Z_0}{576\pi^2 c^4}\left|\left[\mathbf{n}\times\frac{d^3\mathbf{Q}}{dt'^3}(\mathbf{n},t')\right]\times\mathbf{n}\right|^2$$

where $\mathbf{Q}(\mathbf{n},t)$ is defined by (9.43).

Solution.

(a) The r^{-1} electric field in 9.6 is $\mathbf{E}_{\rm rad} = [\mathbf{n}\times(\mathbf{n}\times\ddot{\mathbf{p}})]/(4\pi\epsilon_0 c^2 r)$ and $\mathbf{H}_{\rm rad} = \mathbf{n}\times\mathbf{E}_{\rm rad}/Z_0$. For real instantaneous fields there is no factor $1/2$ in the Poynting vector. Hence $dP/d\Omega = r^2|\mathbf{E}_{\rm rad}|^2/Z_0$, and $1/(\epsilon_0^2 c^4 Z_0) = Z_0/c^2$ gives the stated coefficient. Electric–magnetic duality gives the replacement $(\mathbf{n}\times\ddot{\mathbf{p}})\times\mathbf{n} \mapsto \ddot{\mathbf{m}}\times\mathbf{n}/c$.

(b) The symmetric first spatial moment of the retarded vector potential is

$$(A_Q)_i = \frac{\mu_0}{24\pi c r} n_j \ddot{Q}_{ij}(t_r) + \text{a longitudinal trace term.}$$

To see the coefficient, continuity implies $\int (x_i J_j + x_j J_i)d^3x = \partial_t \int x_i x_j \rho\, d^3x$; the symmetric part is one half of this, and extracting the traceless quadrupole supplies another factor $1/3$. The retarded spatial expansion supplies $n_j\partial_t/c$. Taking $-\partial_t \mathbf{A}_T$ therefore gives

$$\mathbf{E}_Q = -\frac{1}{24\pi\epsilon_0 c^3 r}\{\mathbf{Q}^{(3)}(\mathbf{n},t_r) - \mathbf{n}[\mathbf{n}\cdot\mathbf{Q}^{(3)}(\mathbf{n},t_r)]\}.$$

Squaring, multiplying by r^2/Z_0, and using $1/(\epsilon_0^2 c^6 Z_0) = Z_0/c^4$ proves the second formula. For a harmonic phasor, time averaging introduces the additional factor $1/2$ and replaces ∂_t^3 by $(-i\omega)^3$, agreeing with 9.2.

9.8

(a) Show that a classical oscillating electric dipole $\mathbf{p}$ with fields given by (9.18) radiates electromagnetic angular momentum to infinity at the rate

$$\frac{d\mathbf{L}}{dt} = \frac{k^3}{12\pi\epsilon_0}\,\mathrm{Im}[\mathbf{p}^*\times\mathbf{p}]$$

(b) What is the ratio of angular momentum radiated to energy radiated? Interpret.

(c) For a charge e rotating in the x-y plane at radius a and angular speed ω, show that there is only a z component of radiated angular momentum with magnitude $dL_z/dt = e^2k^3a^2/6\pi\epsilon_0$. What about a charge oscillating along the z axis?

(d) What are the results corresponding to parts a and b for magnetic dipole radiation?

Hint: The electromagnetic angular momentum density comes from more than the transverse (radiation zone) components of the fields.

Solution.

(a) Use the outward angular-momentum flux through a sphere, $-r^3\int \mathbf{n}\times(\langle T\rangle\cdot\mathbf{n})d\Omega$, where T is the Maxwell stress tensor. Since $B_r = 0$ for an electric dipole, its nonzero contribution is

$$\dot{\mathbf{L}} = -\frac{\epsilon_0 r^3}{2}\,\mathrm{Re}\int E_r^*\,\mathbf{n}\times\mathbf{E}_T\,d\Omega.$$

At large r retain the induction term in the radial field:

$$E_r = -\frac{2ik}{4\pi\epsilon_0 r^2}(\mathbf{n}\cdot\mathbf{p})e^{ikr} + O(r^{-3}), \qquad \mathbf{n}\times\mathbf{E}_T = \frac{k^2 e^{ikr}}{4\pi\epsilon_0 r}\mathbf{n}\times\mathbf{p} + O(r^{-2}).$$

The r^3 prefactor cancels these radial powers. Finally, $\int(\mathbf{n}\cdot\mathbf{p}^*)\mathbf{n}\,d\Omega = (4\pi/3)\mathbf{p}^*$ gives

$$\boxed{\dot{\mathbf{L}} = \frac{k^3}{12\pi\epsilon_0}\,\mathrm{Im}(\mathbf{p}^*\times\mathbf{p}).}$$

Dropping the induction field would incorrectly give zero.

(b) Integrating the dipole angular power gives $\overline{P} = ck^4|\mathbf{p}|^2/(12\pi\epsilon_0)$, so

$$\boxed{\frac{\dot{\mathbf{L}}}{\overline{P}} = \frac{1}{\omega}\frac{\mathrm{Im}(\mathbf{p}^*\times\mathbf{p})}{|\mathbf{p}|^2}.}$$

For $\mathbf{p} = \mathbf{u} + i\mathbf{v}$ with real vectors, $\mathrm{Im}(\mathbf{p}^*\times\mathbf{p}) = 2\mathbf{u}\times\mathbf{v}$. Its magnitude is at most $|\mathbf{u}|^2 + |\mathbf{v}|^2$; equality holds for circular polarization. A pure circular dipole carries angular momentum $\pm\hbar$ along its rotation axis per photon of energy $\hbar\omega$; linear polarization has zero net angular momentum.

(c) For positive rotation in the xy plane, $\mathbf{p} = ea(\hat{\mathbf{x}} + i\hat{\mathbf{y}})$, so $\mathrm{Im}(\mathbf{p}^*\times\mathbf{p}) = 2e^2a^2\hat{\mathbf{z}}$. This gives $\dot{L}_z = e^2k^3a^2/(6\pi\epsilon_0)$ and no transverse components. For $\mathbf{p} = p_0\hat{\mathbf{z}}$ all components vanish, although the radiated energy is nonzero.

(d) Duality replaces $\mathbf{p}$ by $\mathbf{m}/c$:

$$\boxed{\dot{\mathbf{L}}_M = \frac{\mu_0 k^3}{12\pi}\,\mathrm{Im}(\mathbf{m}^*\times\mathbf{m}), \quad \overline{P}_M = \frac{\mu_0 ck^4}{12\pi}|\mathbf{m}|^2.}$$

The ratio is the expression in (b) with $\mathbf{p}\mapsto\mathbf{m}$.

9.9

(a) From the electric dipole fields with general time dependence of Problem 9.6, show that the total power and the total rate of radiation of angular momentum through a sphere at large radius r and time t are

$$P(t) = \frac{1}{6\pi\epsilon_0 c^3}\left(\frac{\partial^2\mathbf{p}_{\mathrm{ret}}}{\partial t^2}\right)^2$$

$$\frac{d\mathbf{L}_{em}}{dt} = \frac{1}{6\pi\epsilon_0 c^3}\left(\frac{\partial\mathbf{p}_{\mathrm{ret}}}{\partial t}\times\frac{\partial^2\mathbf{p}_{\mathrm{ret}}}{\partial t^2}\right)$$

where the dipole moment $\mathbf{p}$ is evaluated at the retarded time $t' = t - r/c$.

(b) The dipole moment is caused by a particle of mass m and charge e moving nonrelativistically in a fixed central potential $V(r)$. Show that the radiated power and angular momentum for such a particle can be written as

$$P(t) = \frac{\tau}{m}\left(\frac{dV}{dr}\right)^2$$

$$\frac{d\mathbf{L}_{em}}{dt} = \frac{\tau}{m}\left(\frac{dV}{r\,dr}\right)\mathbf{L}$$

where $\tau = e^2/6\pi\epsilon_0 mc^3$ ($= 2e^2/3mc^3$ in Gaussian units) is a characteristic time, $\mathbf{L}$ is the particle's angular momentum, and the right-hand sides are evaluated at the retarded time. Relate these results to those from the Abraham–Lorentz equation for radiation damping [Section 16.2].

(c) Suppose the charged particle is an electron in a hydrogen atom. Show that the inverse time defined by the ratio of the rate of angular momentum radiated to the particle's angular momentum is of the order of $\alpha^4 c/a_0$, where $\alpha = e^2/4\pi\epsilon_0\hbar c \approx 1/137$ is the fine structure constant and a_0 is the Bohr radius. How does this inverse time compare to the observed rate of radiation in hydrogen atoms?

(d) Relate the expressions in parts a and b to those for harmonic time dependence in Problem 9.8.

Solution.

(a) From 9.6 the far electric field has magnitude squared $[|\ddot{\mathbf{p}}|^2 - (\mathbf{n}\cdot\ddot{\mathbf{p}})^2]/(16\pi^2\epsilon_0^2 c^4 r^2)$. The identity $\int n_i n_j d\Omega = (4\pi/3)\delta_{ij}$ gives

$$\boxed{P = \frac{|\ddot{\mathbf{p}}|^2}{6\pi\epsilon_0 c^3}.}$$

For angular momentum use the same stress flux as 9.8 without a time average. Now $E_r = 2\mathbf{n}\cdot\dot{\mathbf{p}}/(4\pi\epsilon_0 c r^2) + O(r^{-3})$ and $\mathbf{n}\times\mathbf{E}_T = -\mathbf{n}\times\ddot{\mathbf{p}}/(4\pi\epsilon_0 c^2 r)$. Thus

$$\dot{\mathbf{L}}_{em} = \frac{1}{8\pi^2\epsilon_0 c^3}\int (\mathbf{n}\cdot\dot{\mathbf{p}})\mathbf{n}\times\ddot{\mathbf{p}}\, d\Omega = \boxed{\frac{\dot{\mathbf{p}}\times\ddot{\mathbf{p}}}{6\pi\epsilon_0 c^3}}.$$

All quantities on the right are evaluated at $t - r/c$.

(b) With $\mathbf{p} = e\mathbf{r}$ and $m\ddot{\mathbf{r}} = -V'(r)\hat{\mathbf{r}}$,

$$P = m\tau|\dddot{\mathbf{r}}|^2 = \frac{\tau}{m}[V'(r)]^2, \quad m\tau\ddot{\mathbf{r}}\times\dddot{\mathbf{r}} = \frac{\tau V'(r)}{r}\mathbf{r}\times\dot{\mathbf{r}} = \frac{\tau}{m}\frac{V'(r)}{r}\mathbf{L}.$$

These expressions use the unperturbed central-force orbit to first order in radiation damping. The Abraham–Lorentz force is $m\tau\,\ddot{\dddot{\mathbf{r}}}$. Its work and torque obey

$$\dot{E}_{\text{mech}} = m\tau\mathbf{v}\cdot\dot{\mathbf{a}} = \frac{d}{dt}(m\tau\mathbf{v}\cdot\mathbf{a}) - m\tau a^2,$$

$$\dot{\mathbf{L}}_{\text{mech}} = m\tau\mathbf{r}\times\dot{\mathbf{a}} = \frac{d}{dt}(m\tau\mathbf{r}\times\mathbf{a}) - m\tau\mathbf{v}\times\mathbf{a}.$$

The last terms are minus the radiated energy and angular momentum. The total derivatives describe reversible exchange with the near field; they average to zero on a periodic orbit. For the zeroth-order central acceleration $\mathbf{r}\times\mathbf{a} = 0$ already.

(c) For hydrogen $V = -\alpha\hbar c/r$, $a_0 = \hbar/(mc\alpha)$ and $\tau = 2\alpha\hbar/(3mc^2)$. At $r \sim a_0$,

$$\frac{|\dot{\mathbf{L}}_{em}|}{|\mathbf{L}|} \sim \frac{\tau\alpha\hbar c}{m a_0^3} = \boxed{\frac{2}{3}\frac{\alpha^4 c}{a_0}}.$$

The scale $\alpha^4 c/a_0 = 1.6065\times10^{10}\ \text{s}^{-1}$ is larger than the hydrogen $2p \to 1s$ rate $6.265\times10^8\ \text{s}^{-1}$ by about 26; the classical coefficient $2/3$ reduces this ratio to about 17. This is a scale estimate, not a classical prediction of a stationary atomic state's lifetime. The evaluated transition rate is tabulated by

Wiese and Fuhr, *J. Phys. Chem. Ref. Data* **38**, 565 (2009), Table 2, p. 572, DOI: 10.1063/1.3077727; the source and numerical constants are recorded in `supplementary/ch09-source/`.

(d) For a phasor, $\langle|\ddot{\mathbf{p}}|^2\rangle = \omega^4|\mathbf{p}|^2/2$ and

$$\langle \dot{\mathbf{p}} \times \ddot{\mathbf{p}} \rangle = \tfrac{1}{2}\,\mathrm{Re}[(-i\omega\mathbf{p}) \times (-\omega^2\mathbf{p}^*)] = \frac{\omega^3}{2}\,\mathrm{Im}(\mathbf{p}^* \times \mathbf{p}).$$

Substitution reproduces both harmonic results in 9.8.

9.10 The transitional charge and current densities for the radiative transition from the $m = 0$, $2p$ state in hydrogen to the $1s$ ground state are, in the notation of (9.1) and with the neglect of spin,

$$\rho(r,\theta,\phi,t) = \frac{2e}{\sqrt{6}\,a_0^4} \cdot r e^{-3r/2a_0} Y_{00} Y_{10} e^{-i\omega_0 t}$$

$$\mathbf{J}(r,\theta,\phi,t) = \frac{-iv_0}{2}\left(\frac{\hat{\mathbf{r}}}{2} + \frac{a_0}{z}\,\hat{\mathbf{z}}\right)\rho(r,\theta,\phi,t)$$

where $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.529 \times 10^{-10}$ m is the Bohr radius, $\omega_0 = 3e^2/32\pi\epsilon_0\hbar a_0$ is the frequency difference of the levels, and $v_0 = e^2/4\pi\epsilon_0\hbar = \alpha c \approx c/137$ is the Bohr orbit speed.

(a) Show that the effective transitional (orbital) "magnetization" is

$$\text{“}\mathcal{M}\text{”}(r,\theta,\phi,t) = -i\,\frac{\alpha c a_0}{4}\tan\theta(\hat{\mathbf{x}}\sin\phi - \hat{\mathbf{y}}\cos\phi)\cdot\rho(r,\theta,\phi,t)$$

Calculate $\nabla\cdot$“$\mathcal{M}$” and evaluate all the nonvanishing radiation multipoles in the long-wavelength limit.

(b) In the electric dipole approximation calculate the total time-averaged power radiated. Express your answer in units of $(\hbar\omega_0)\cdot(\alpha^4 c/a_0)$, where $\alpha = e^2/4\pi\epsilon_0\hbar c$ is the fine structure constant.

(c) Interpreting the classically calculated power as the photon energy $(\hbar\omega_0)$ times the transition probability, evaluate numerically the transition probability in units of reciprocal seconds.

(d) If, instead of the semiclassical charge density used above, the electron in the $2p$ state was described by a *circular* comparison orbit of radius $2a_0$, rotating with the transitional frequency ω_0, what would the radiated power be? Express your answer in the same units as in part b and evaluate the ratio of the two powers numerically.

Solution.

Write $\rho(\mathbf{x},t) = \rho_\omega(\mathbf{x})e^{-i\omega_0 t}$ and $v_0 = \alpha c$. The apparent $1/z$ in the current causes no singularity: $Y_{10} \propto z/r$, so $\rho_\omega/z \propto e^{-3r/(2a_0)}$ has a finite continuous extension to $z = 0$. At the origin this radial function has the usual Coulomb cusp; continuity here does not imply differentiability there.

(a) The radial part of $\mathbf{J}$ has zero cross product with $\mathbf{x}$. Its axial part gives

$$\mathcal{M} = \tfrac{1}{2}\mathbf{x}\times\mathbf{J} = -\frac{iv_0 a_0}{4z}\mathbf{x}\times\hat{\mathbf{z}}\,\rho = -\frac{i\alpha c a_0}{4}\tan\theta(\hat{\mathbf{x}}\sin\phi - \hat{\mathbf{y}}\cos\phi)\rho.$$

This is purely azimuthal and independent of ϕ in spherical components; hence $\nabla\cdot\mathcal{M} = 0$ distributionally as well as away from the axis. All orbital magnetic moments vanish. Orthogonality of Y_{lm}

in $\rho \propto Y_{10}$ leaves only Q_{10} in the leading electric moments. Equivalently the only dipole component is

$$p_z = \frac{2e}{\sqrt{6}a_0^4}\left(\int Y_{00}Y_{10}\cos\theta\, d\Omega\right)\int_0^\infty r^4 e^{-3r/(2a_0)}dr$$
$$= \frac{2e}{\sqrt{18}a_0^4}\frac{4!}{[3/(2a_0)]^5} = \boxed{\frac{256\sqrt{2}}{243}ea_0}, \qquad Q_{10} = \sqrt{\frac{3}{4\pi}}p_z.$$

There is no intrinsic magnetization in the specified spinless model.

(b) With $\omega_0 = 3\alpha c/(8a_0)$,

$$\overline{P} = \frac{\omega_0^4 p_z^2}{12\pi\epsilon_0 c^3}, \qquad \frac{\overline{P}}{(\hbar\omega_0)(\alpha^4 c/a_0)} = \frac{1}{3}\left(\frac{3}{8}\right)^3\left(\frac{256\sqrt{2}}{243}\right)^2 = \boxed{\frac{256}{6561} = 0.03901844.}$$

The factor 2 in the supplied charge phasor is essential to this result.

(c) Using $a_0 = 5.29177210544 \times 10^{-11}$ m and $\alpha = 7.2973525643 \times 10^{-3}$ gives

$$\boxed{\Gamma = \frac{\overline{P}}{\hbar\omega_0} = \frac{256}{6561}\frac{\alpha^4 c}{a_0} = 6.2683 \times 10^8 \text{ s}^{-1}.}$$

This is the nonrelativistic infinite-nuclear-mass result. It is close to the evaluated physical-hydrogen rate quoted in 9.9; equality to all shown digits is not implied.

(d) The specified circular model has $\mathbf{p} = 2ea_0(\hat{\mathbf{x}} + i\hat{\mathbf{y}})$ and hence $|\mathbf{p}|^2 = 8e^2a_0^2$, rather than the squared amplitude of one linear component. Therefore

$$\boxed{\frac{\overline{P}_{\text{circ}}}{(\hbar\omega_0)(\alpha^4 c/a_0)} = \frac{9}{64}, \qquad \frac{\overline{P}_{\text{circ}}}{\overline{P}_{2p\to 1s}} = \frac{59049}{16384} = 3.60406494.}$$

The stipulated radius and frequency define this comparison model; they are not a self-consistent $n = 2$ Coulomb Bohr orbit, whose radius is $4a_0$.

9.11 Three charges are located along the z axis, a charge $+2q$ at the origin, and charges $-q$ at $z = \pm a\cos\omega t$. Determine the lowest nonvanishing multipole moments, the angular distribution of radiation, and the total power radiated. Assume that $ka \ll 1$.

Solution.

The total charge and electric dipole vanish at every time. The particles move along the line through the origin, so $\mathbf{x} \times \mathbf{J} = 0$ and there are no magnetic multipoles. Put $s = a\cos\omega t$. The first nonzero electric tensor is

$$Q(t) = qs^2\,\text{diag}(2, 2, -4) = qa^2[1 + \cos(2\omega t)]\,\text{diag}(1, 1, -2).$$

The constant part does not radiate. The oscillating phasor at $\Omega = 2\omega$ is $\mathcal{Q} = qa^2\,\text{diag}(1, 1, -2)$; equivalently $Q_{20} = -qa^2\sqrt{5/(4\pi)}$. With $K = 2\omega/c$,

$$|\mathcal{Q}\mathbf{n}|^2 - |\mathbf{n}\cdot\mathcal{Q}\mathbf{n}|^2 = 9q^2a^4\sin^2\theta\cos^2\theta.$$

For clarity this follows by writing $\mathcal{Q}\mathbf{n} = qa^2(\mathbf{n} - 3\hat{\mathbf{z}}\cos\theta)$ and removing its component along $\mathbf{n}$. Thus

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{cq^2a^4K^6}{128\pi^2\epsilon_0}\sin^2\theta\cos^2\theta, \qquad \overline{P} = \frac{cq^2a^4K^6}{240\pi\epsilon_0} = \frac{4q^2a^4\omega^6}{15\pi\epsilon_0 c^5}.}$$

Here $\int \sin^2\theta\cos^2\theta\, d\Omega = 8\pi/15$. If instantaneous rather than averaged power is desired, multiply these angular and total results by $2\sin^2[2\omega(t - r/c)]$, since the third derivative of the quadrupole is proportional to that sine.

9.12 An almost spherical surface defined by

$$R(\theta) = R_0[1 + \beta P_2(\cos\theta)]$$

has inside of it a uniform volume distribution of charge totaling Q. The small parameter β varies harmonically in time at frequency ω. This corresponds to surface waves on a sphere. Keeping only lowest order terms in β and making the long-wavelength approximation, calculate the nonvanishing multipole moments, the angular distribution of radiation, and the total power radiated. Assume an axisymmetric meridional charge flow realizing the surface deformation, with no independently imposed circulating current.

Solution.

Let $\beta(t) = \mathrm{Re}(\beta_0 e^{-i\omega t})$. The volume is $\int R(\theta)^3 d\Omega/3 = 4\pi R_0^3/3 + O(\beta^2)$ because $\int P_2 d\Omega = 0$. Thus $\rho = 3Q/(4\pi R_0^3) + O(\beta^2)$. For a spherical multipole, performing the radial integral first gives

$$q_{lm} = \frac{\rho R_0^{l+3}}{l+3} \int Y_{lm}^*[1 + (l+3)\beta P_2] d\Omega + O(\beta^2).$$

Orthogonality leaves the constant monopole $q_{00} = Q/\sqrt{4\pi}$ and

$$\boxed{Q_{20} = \frac{3QR_0^2\beta_0}{\sqrt{20\pi}}, \qquad \mathcal{Q} = \frac{3QR_0^2\beta_0}{5}\,\mathrm{diag}(-1,-1,2).}$$

There is no dipole. The surface deformation is taken to carry its charge without an added circulating current. One explicit flow consistent with the uniform density to first order is

$$\mathbf{v} = \dot\beta(-x/2, -y/2, z), \qquad \nabla\cdot\mathbf{v} = 0, \qquad \mathbf{v}\cdot\hat{\mathbf{r}} = \dot\beta r P_2(\cos\theta).$$

Its normal surface speed is the required $\partial_t R$ at this order. For this axisymmetric meridional current, $\mathbf{r}\times\mathbf{J}$ is azimuthal with zero divergence, including at the moving surface; all leading magnetic moments therefore vanish. Charge density alone would leave an arbitrary additional divergence-free current undetermined. As in 9.11,

$$|\mathcal{Q}\mathbf{n}|^2 - |\mathbf{n}\cdot\mathcal{Q}\mathbf{n}|^2 = \frac{81}{25}Q^2R_0^4|\beta_0|^2\sin^2\theta\cos^2\theta.$$

Consequently, with $k = \omega/c$,

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{9ck^6Q^2R_0^4|\beta_0|^2}{3200\pi^2\epsilon_0}\sin^2\theta\cos^2\theta, \quad \overline{P} = \frac{3ck^6Q^2R_0^4|\beta_0|^2}{2000\pi\epsilon_0}.}$$

The time-varying moment is linear in the shape amplitude and the power is quadratic; the monopole and the static spherical part cannot radiate.

9.13 The uniform charge density of Problem 9.12 is replaced by a uniform density of intrinsic magnetization parallel to the z axis and having total magnetic moment M. With the same approximations as above calculate the nonvanishing radiation multipole moments, the angular distribution of radiation, and the total power radiated. For the total leading radiated power, also retain the finite-wavelength correction to the magnetic dipole, which contributes at the same order as the magnetic octupole.

Solution.

Use the prescribed-magnetization model, with no additional electric polarization or conduction current: $\boldsymbol{\mathcal{M}} = 3M\hat{\mathbf{z}}/(4\pi R_0^3)$ inside the moving surface, zero outside, to first order in β. The associated current is $\nabla \times \boldsymbol{\mathcal{M}}$, including its surface term. Since $\nabla \cdot (\mathbf{r} \times \boldsymbol{\mathcal{M}}) = 0$ by axial symmetry, there are no electric radiation moments, including the intrinsic-magnetization contribution Q'_{lm} of (9.170).

The leading magnetic moments follow by integration by parts:

$$M'_{l0} = \int \boldsymbol{\mathcal{M}} \cdot \nabla(r^l Y_{l0})\, d^3x, \qquad \partial_z[r^l P_l(\cos\theta)] = l r^{l-1} P_{l-1}(\cos\theta).$$

The undeformed volume gives only $M'_{10} = \sqrt{3/(4\pi)}M$, which is constant. The boundary displacement adds

$$\delta M'_{l0} = \frac{3M}{4\pi R_0^3}\, l\sqrt{\frac{2l+1}{4\pi}}\beta R_0^{l+2} \int P_{l-1}P_2\, d\Omega.$$

Only $l = 3$ survives, with phasor

$$\boxed{M'_{30} = \frac{9}{5}\sqrt{\frac{7}{4\pi}} M R_0^2 \beta_0.}$$

If one keeps only the leading term in each static moment, the radiating multipole is therefore $M3$. Its coefficient is $a_M(3,0) = ik^5\sqrt{4/3}\, M'_{30}/105$. Equivalently its far field is

$$H_\theta^{(3)} = \frac{Mk^4 R_0^2 \beta_0}{100\pi} \sin\theta P'_3(\cos\theta)\frac{e^{ikr}}{r}, \qquad E_\phi^{(3)} = -Z_0 H_\theta^{(3)}.$$

Thus that isolated multipole has

$$\frac{d\overline{P}_3}{d\Omega} = \frac{9Z_0 M^2 k^8 R_0^4 |\beta_0|^2}{80000\pi^2} \sin^2\theta(5\cos^2\theta - 1)^2, \quad \overline{P}_3 = \frac{3Z_0 M^2 k^8 R_0^4 |\beta_0|^2}{8750\pi}.$$

There is an essential qualification to a *total leading-power* answer: the finite-wavelength correction to $M1$ is of the same order as this $M3$ field. It must be retained even though the static total moment is constant. This can be checked without any multipole convention. The spatial Fourier transform of the magnetization is

$$\widetilde{\boldsymbol{\mathcal{M}}}(\mathbf{n}) = \hat{\mathbf{z}} \int \mathcal{M}_z e^{-ik\mathbf{n}\cdot\mathbf{r}} d^3r.$$

Its time-dependent second moments are

$$\delta\langle x^2\rangle = \delta\langle y^2\rangle = -\frac{\beta_0 R_0^2}{5}, \qquad \delta\langle z^2\rangle = \frac{2\beta_0 R_0^2}{5}.$$

The constant Fourier term is M, so its oscillating part starts at

$$\delta\widetilde{\mathcal{M}}_z = -\frac{k^2 M}{2} n_i n_j \delta\langle x_i x_j\rangle = -\frac{Mk^2 R_0^2 \beta_0}{5} P_2(\cos\theta).$$

For example each second moment follows by integrating $r_i r_j$ over the displaced shell $R_0 < r < R_0[1 + \beta P_2]$; the angular integrals are $\int u^2 P_2 d\Omega = 8\pi/15$ and $\int n_x^2 P_2 d\Omega = -4\pi/15$. Since $\widetilde{\mathbf{J}} = ik\mathbf{n} \times \widetilde{\boldsymbol{\mathcal{M}}}$, the complete leading far field is

$$\boxed{H_\theta = \frac{Mk^4 R_0^2 \beta_0}{20\pi} \sin\theta P_2(\cos\theta)\frac{e^{ikr}}{r}, \qquad E_\phi = -Z_0 H_\theta.}$$

The identity $5P_2 = P_3' - 1$ separates this into the preceding $M3$ field and an $M1$ field with effective dynamic dipole $m_{\text{eff}} = Mk^2R_0^2\beta_0/25$. The latter contributes $\overline{P}_1 = Z_0M^2k^8R_0^4|\beta_0|^2/(7500\pi)$. The angular interference need not vanish pointwise, but its integral does by multipole orthogonality. The consistent total result is

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{Z_0M^2k^8R_0^4|\beta_0|^2}{800\pi^2}\sin^2\theta P_2(\cos\theta)^2, \qquad \overline{P} = \frac{Z_0M^2k^8R_0^4|\beta_0|^2}{2100\pi}.}$$

Here $\int \sin^2\theta P_2^2 d\Omega = 8\pi/21$. The distinction between the isolated leading static $M3$ approximation and the full leading radiation is the reason for the qualification added to the problem statement.

9.14 An antenna consists of a circular loop of wire of radius a located in the x-y plane with its center at the origin. The current in the wire is

$$I = I_0\cos\omega t = \text{Re}\, I_0 e^{-i\omega t}$$

(a) Find the expressions for $\mathbf{E}$, $\mathbf{H}$ in the radiation zone without approximations as to the magnitude of ka. Determine the power radiated per unit solid angle.

(b) What is the lowest nonvanishing multipole moment ($Q_{l,m}$ or $M_{l,m}$)? Evaluate this moment in the limit $ka \ll 1$.

Solution.

(a) In the radiation zone use the exact source transform

$$\mathbf{A} = \frac{\mu_0 I_0 e^{ikr}}{4\pi r}\oint d\boldsymbol{\ell}' e^{-ik\mathbf{n}\cdot\mathbf{r}'}.$$

First choose the observation azimuth zero. Then $\mathbf{n}\cdot\mathbf{r}' = a\sin\theta\cos\phi'$ and $d\boldsymbol{\ell}' = a(-\sin\phi'\hat{\mathbf{x}} + \cos\phi'\hat{\mathbf{y}})d\phi'$. The x integral is odd, while differentiating $\int_0^{2\pi} e^{-iu\cos\phi'}d\phi' = 2\pi J_0(u)$ gives $\int\cos\phi' e^{-iu\cos\phi'}d\phi' = -2\pi i J_1(u)$. Rotation to general azimuth therefore gives

$$A_\phi = -\frac{i\mu_0 I_0 a}{2}J_1(ka\sin\theta)\frac{e^{ikr}}{r}.$$

The source has no oscillating charge, and the far field is transverse:

$$\boxed{E_\phi = \frac{Z_0 k I_0 a}{2}J_1(ka\sin\theta)\frac{e^{ikr}}{r}, \qquad H_\theta = -\frac{kI_0 a}{2}J_1(ka\sin\theta)\frac{e^{ikr}}{r}.}$$

All other r^{-1} components vanish. No small-ka approximation was made, so

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{Z_0k^2|I_0|^2a^2}{8}J_1(ka\sin\theta)^2.}$$

(b) The first moment is $\mathbf{m} = \frac{1}{2}I_0\oint\mathbf{r}'\times d\boldsymbol{\ell}' = \pi a^2 I_0\hat{\mathbf{z}}$, so

$$\boxed{M_{10} = \sqrt{\frac{3}{4\pi}}\,\pi a^2 I_0.}$$

All electric moments vanish; axial symmetry leaves $m = 0$, and the magnetic orders are odd. Since $J_1(u) = u/2 + O(u^3)$, the field reduces to $E_\phi = Z_0k^2m\sin\theta\, e^{ikr}/(4\pi r)$ and $\overline{P} = Z_0k^4|m|^2/(12\pi)$, the magnetic dipole result.

9.15 Two fixed electric dipoles of dipole moment p are located in the x-y plane a distance $2a$ apart, their axes parallel and perpendicular to the plane, but their moments directed oppositely. The dipoles rotate with constant angular speed ω about a z axis located halfway between them. The motion is nonrelativistic ($\omega a/c \ll 1$).

(a) Find the lowest nonvanishing multipole moments.

(b) Show that the magnetic field in the radiation zone is, apart from an overall phase factor,

$$\mathbf{H} = \frac{cpa}{2\pi}\, k^3[(\hat{\mathbf{x}} + i\hat{\mathbf{y}})\cos\theta - \hat{\mathbf{z}}\sin\theta\, e^{i\phi}]\cos\theta\, \frac{e^{ikr}}{r}$$

(c) Show that the angular distribution of the radiation is proportional to $(\cos^2\theta + \cos^4\theta)$ and the total time-averaged power radiated is

$$P = \frac{4}{15\pi\epsilon_0}\, ck^6p^2a^2$$

Hint: Problem 6.21 is relevant.

Solution.

Let $\mathbf{u}(t) = \hat{\mathbf{x}}\cos\omega t + \hat{\mathbf{y}}\sin\omega t$. The positions are $\pm a\mathbf{u}$, the fixed dipole vectors are $\pm p\hat{\mathbf{z}}$, and their velocities are $\pm a\omega\hat{\mathbf{z}} \times \mathbf{u}$.

(a) The net electric dipole is zero. Applying 6.21 to each moving dipole gives

$$\mathbf{m}(t) = \tfrac{1}{2}\sum_s \mathbf{p}_s \times \mathbf{v}_s = -pa\omega\mathbf{u}(t), \qquad Q_{ij}(t) = 6pa(u_i\delta_{jz} + \delta_{iz}u_j).$$

Thus, with $\mathbf{e} = \hat{\mathbf{x}} + i\hat{\mathbf{y}}$,

$$\boxed{\mathbf{m} = -pa\omega\mathbf{e}, \qquad \mathcal{Q} = 6pa(\mathbf{e}\hat{\mathbf{z}} + \hat{\mathbf{z}}\mathbf{e}).}$$

These are $M1$ and $E2$ at frequency ω. The magnetic moment is smaller by one power of velocity, but magnetic dipole radiation has one less power of source size than electric quadrupole radiation; both fields are the same order here.

(b) The two magnetic radiation fields are

$$\mathbf{H}_M = -\frac{cpak^3}{4\pi}[\mathbf{e} - \mathbf{n}(\mathbf{n}\cdot\mathbf{e})]\frac{e^{ikr}}{r},$$

$$\mathbf{H}_Q = -\frac{icpak^3}{4\pi}[\cos\theta\,\mathbf{n}\times\mathbf{e} + (\mathbf{n}\cdot\mathbf{e})\mathbf{n}\times\hat{\mathbf{z}}]\frac{e^{ikr}}{r}.$$

Use $\mathbf{n}\cdot\mathbf{e} = \sin\theta e^{i\phi}$ and

$$\mathbf{e} = e^{i\phi}(\sin\theta\,\mathbf{n} + \cos\theta\,\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\phi}}), \qquad \hat{\mathbf{z}} = \cos\theta\,\mathbf{n} - \sin\theta\,\hat{\boldsymbol{\theta}}.$$

Writing $C = cpak^3e^{ikr}/(4\pi r)$, this gives $\mathbf{H}_M = -Ce^{i\phi}(\cos\theta\,\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\phi}})$ and $\mathbf{H}_Q = -Ce^{i\phi}[\cos\theta\,\hat{\boldsymbol{\theta}} + i(2\cos^2\theta - 1)\hat{\boldsymbol{\phi}}]$. Their sum is

$$\boxed{\mathbf{H} = -\frac{cpak^3}{2\pi}[\mathbf{e}\cos\theta - \hat{\mathbf{z}}\sin\theta e^{i\phi}]\cos\theta\frac{e^{ikr}}{r}.}$$

The minus sign is the irrelevant overall phase allowed in the question. For example the bracket equals $e^{i\phi}(\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\phi}}\cos\theta)$, which also makes transversality explicit.

(c) That bracket has squared norm $1 + \cos^2\theta$, so

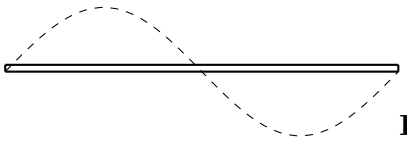
$$\frac{d\overline{P}}{d\Omega} = \frac{cp^2a^2k^6}{8\pi^2\epsilon_0}(\cos^2\theta + \cos^4\theta).$$

Integrating $2\pi\int_{-1}^{1}(u^2+u^4)du = 32\pi/15$ gives

$$\boxed{\overline{P} = \frac{4cp^2a^2k^6}{15\pi\epsilon_0}.}$$

The two multipole powers add after angular integration, but their interference is required to obtain the correct angular pattern.

9.16 A thin linear antenna of length d is excited in such a way that the sinusoidal current makes a full wavelength of oscillation as shown in the figure.



Problem 9.16

(a) Calculate exactly the power radiated per unit solid angle and plot the angular distribution of radiation.

(b) Determine the total power radiated and find a numerical value for the radiation resistance.

Solution.

Use z along the wire and choose the overall current phase so that $I(z) = I_0\sin(kz)$ on $-d/2 < z < d/2$. A full spatial wavelength means $kd = 2\pi$, with $k = \omega/c$. I_0 is the *maximum* current amplitude, not the current at the central node, which is zero.

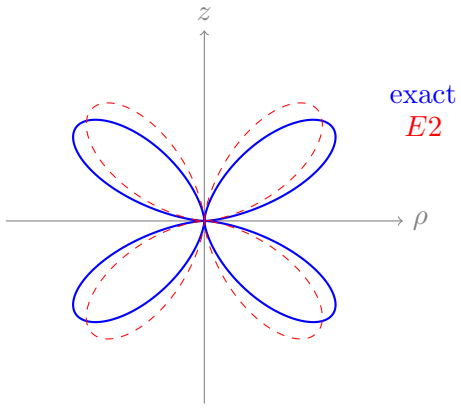
(a) With $u = \cos\theta$, the exact radiation source integral is

$$\begin{aligned} F(u) &= \int_{-\pi/k}^{\pi/k} I_0\sin(kz)e^{-ikuz}dz \\ &= -\frac{iI_0}{k}\left[\frac{\sin[\pi(1-u)]}{1-u} - \frac{\sin[\pi(1+u)]}{1+u}\right] = -\frac{2iI_0}{k}\frac{\sin\pi u}{1-u^2}. \end{aligned}$$

The far vector potential is $\mu_0 F(u)\hat{\mathbf{z}}e^{ikr}/(4\pi r)$. Its transverse electric field has magnitude $Z_0 k|F|\sin\theta/(4\pi r)$, giving

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{Z_0|I_0|^2}{8\pi^2}\frac{\sin^2(\pi\cos\theta)}{\sin^2\theta}.}$$

The apparent axial singularities are removable: the expression tends to zero as θ^2 at either pole. It also vanishes in the equatorial plane. The polar plot shows the normalized intensity in a meridian; the dashed curve is the normalized $E2$ pattern used in 9.17.



(b) Integration gives

$$\overline{P} = \frac{Z_0 |I_0|^2}{4\pi} \int_{-1}^{1} \frac{\sin^2 \pi u}{1-u^2}\, du.$$

By symmetry and partial fractions, the last integral is

$$\int_{-1}^{1} \frac{\sin^2 \pi u}{1-u}\, du = \frac{1}{2} \int_0^{4\pi} \frac{1-\cos t}{t} dt = \frac{1}{2}[\gamma_E + \ln(4\pi) - \mathrm{Ci}(4\pi)].$$

Therefore, defining resistance by $\overline{P} = R_{\mathrm{rad}} |I_0|^2/2$,

$$\boxed{R_{\mathrm{rad}} = \frac{Z_0}{4\pi}[\gamma_E + \ln(4\pi) - \mathrm{Ci}(4\pi)] = 93.3661\ \Omega.}$$

This is resistance referred to a current antinode. A feed resistance referred to the zero central current is not finite in this idealized prescribed-current model.

9.17 Treat the linear antenna of Problem 9.16 by the multipole expansion method.

(a) Calculate the multipole moments (electric dipole, magnetic dipole, and electric quadrupole) exactly and in the long-wavelength approximation.

(b) Compare the shape of the angular distribution of radiated power for the lowest nonvanishing multipole with the exact distribution of Problem 9.16.

(c) Determine the total power radiated for the lowest multipole and the corresponding radiation resistance using both multipole moments from part a. Compare with Problem 9.16b. Is there a paradox here?

Solution.

(a) The source is purely radial along its axis, so $\mathbf{r} \times \mathbf{J} = 0$: all magnetic moments vanish exactly. Because $I(-z) = -I(z)$, continuity gives an even charge density, $\rho = I'(z)\delta(x)\delta(y)/(i\omega)$; the current vanishes at the endpoints, so there are no endpoint delta charges. The electric dipole vanishes by parity, both statically and in the exact dynamic expansion. The leading electric order is $l = 2, m = 0$.

For the static, long-wavelength tensor,

$$Q_{zz} = \frac{2}{i\omega} \int_{-\pi/k}^{\pi/k} z^2 I'(z) dz = -\frac{4}{i\omega} \int z I(z) dz = \frac{8\pi i I_0}{\omega k^2}, \qquad Q_{xx} = Q_{yy} = -Q_{zz}/2.$$

Hence $Q_{20}^{\rm LW} = \sqrt{5/(16\pi)}\,Q_{zz}$. To calculate the dynamic moment without a long-wavelength expansion, write $N_l = \sqrt{(2l+1)/(4\pi)}$ and use (9.167). For even l it yields

$$a_E(l,0) = \frac{2kN_l}{\sqrt{l(l+1)}} \int_0^{\pi/k} \{-I'[rj_l(kr)]' + k^2 r I j_l(kr)\}\, dr.$$

Integration by parts leaves

$$\frac{2kN_l}{\sqrt{l(l+1)}} \left\{ -[I' r j_l]_0^{\pi/k} + \int_0^{\pi/k} r j_l (I'' + k^2 I) dr \right\}.$$

The integral is zero for the specified sinusoid, the lower endpoint vanishes, and $I'(\pi/k) = -kI_0$. Thus

$$\boxed{a_E(l,0) = kI_0 \sqrt{\frac{\pi(2l+1)}{l(l+1)}}\, j_l(\pi) \quad (l = 2, 4, \ldots).}$$

In particular $j_2(\pi) = 3/\pi^2$. Defining an exact effective quadrupole by $a_E(2,0) = ck^4\sqrt{3/2}\,Q_{20}^{\rm eff}/(15i)$ gives

$$\boxed{Q_{20}^{\rm eff} = \frac{15}{2\pi^2} Q_{20}^{\rm LW}.}$$

It is this frequency-dependent moment, not the static integral alone, that reproduces the entire $l = 2$ component at finite kd.

(b) Either quadrupole approximation has normalized intensity proportional to $\sin^2\theta\cos^2\theta$; its maxima are at $\theta = 45°, 135°$. The exact pattern is the solid curve in 9.16. Its maxima satisfy $\pi\cot(\pi u) + u/(1-u^2) = 0$, giving $|u| \simeq 0.588981$, or $\theta \simeq 53.915°, 126.085°$. The nodes agree, but higher even electric multipoles change the lobes.

(c) Orthogonality of vector spherical harmonics gives $P_l = Z_0 |a_E(l,0)|^2/(2k^2)$. Therefore

$$\boxed{R_{E2}^{\rm exact} = \frac{15 Z_0}{2\pi^3} = 91.1260\ \Omega, \qquad R_{E2}^{\rm LW} = \frac{2\pi Z_0}{15} = 157.8044\ \Omega.}$$

The exact $E2$ accounts for 97.601% of the 93.3661 Ω total; the remaining power is in $l = 4, 6, \ldots$. There is no paradox in the long-wavelength estimate exceeding the full power: $kd = 2\pi$ is not small, so that source-size approximation is uncontrolled. Multipole *order* truncation using the exact coefficient remains accurate. In particular, inserting $j_2(x) \simeq x^2/15$ into the boundary-only formula above is not a legitimate way to derive the static moment: its cancellation used the full Bessel differential equation and $kd = 2\pi$ before the approximation.

9.18 A qualitative understanding of the result for the reactance of a short antenna whose radiation fields are described by the electric dipole fields of Section 9.2 can be achieved by considering the idealized dipole fields (9.18).

(a) Show that the integral over all angles at fixed distance r of $\epsilon_0|\mathbf{E}|^2 - \mu_0|\mathbf{H}|^2$ is

$$\int \left[\epsilon_0|\mathbf{E}|^2 - \mu_0|\mathbf{H}|^2\right] d\Omega = \frac{1}{2\pi\epsilon_0}\frac{|\mathbf{p}|^2}{r^6}$$

(b) Using (6.140) for the reactance, show that the contribution X_a to the reactance from fields at distances $r > a$ is

$$X_a = -\frac{\omega}{6\pi\epsilon_0}\frac{|\mathbf{p}|^2}{|I_i|^2}\frac{1}{a^3}$$

where I_i is the input current.

(c) For the short center-fed antenna of Section 9.2 show that $X_a \simeq -d^2/24\pi\epsilon_0\omega a^3$, corresponding to an effective capacitance $24\pi\epsilon_0 a^3/d^2$. With $a = d/2$, X_a gives only a small fraction of the total negative reactance of a short antenna. The fields close to the antenna, obviously not dipole in character, contribute heavily. For calculations of reactances of short antennas, see the book by *Schelkunoff and Friis*.

Solution.

(a) Put $\mathbf{p}_T = \mathbf{p} - \mathbf{n}(\mathbf{n}\cdot\mathbf{p})$, $p_n = \mathbf{n}\cdot\mathbf{p}$. The exact dipole fields give

$$E_r = \frac{2p_n e^{ikr}}{4\pi\epsilon_0}(r^{-3} - ikr^{-2}), \quad \mathbf{E}_T = \frac{\mathbf{p}_T e^{ikr}}{4\pi\epsilon_0}(k^2 r^{-1} + ikr^{-2} - r^{-3}),$$

$$\mathbf{H} = \frac{ce^{ikr}}{4\pi}(k^2 r^{-1} + ikr^{-2})\mathbf{n}\times\mathbf{p}.$$

Since $\mu_0 c^2 = 1/\epsilon_0$, subtraction of the transverse squared coefficients leaves $r^{-6} - 2k^2 r^{-4}$, while the radial electric term contributes $4|p_n|^2(r^{-6} + k^2 r^{-4})$. Use $\int |p_n|^2 d\Omega = 4\pi|\mathbf{p}|^2/3$ and $\int |\mathbf{p}_T|^2 d\Omega = 8\pi|\mathbf{p}|^2/3$. The k^2 terms cancel, and the r^{-6} coefficient becomes

$$\boxed{\int (\epsilon_0|\mathbf{E}|^2 - \mu_0|\mathbf{H}|^2)d\Omega = \frac{|\mathbf{p}|^2}{2\pi\epsilon_0 r^6}.}$$

(b) With the reactance convention in (6.140),

$$X_a = -\frac{\omega}{|I_i|^2}\int_a^\infty r^2 dr \int (\epsilon_0|\mathbf{E}|^2 - \mu_0|\mathbf{H}|^2)d\Omega.$$

As $\int_a^\infty r^{-4}dr = 1/(3a^3)$, this yields $X_a = -\omega|\mathbf{p}|^2/(6\pi\epsilon_0|I_i|^2 a^3)$. It is negative because the remaining near-field energy is predominantly electric in this convention.

(c) The short center-fed wire has approximately triangular current $I(z) = I_i(1-2|z|/d)$, $|z| < d/2$. Continuity gives $-i\omega p_z = \int I(z)dz = I_i d/2$, and thus $|p_z|^2/|I_i|^2 = d^2/(4\omega^2)$. Substitution gives

$$\boxed{X_a \simeq -\frac{d^2}{24\pi\epsilon_0\omega a^3}, \qquad C_a = \frac{24\pi\epsilon_0 a^3}{d^2}.}$$

For $a = d/2$, $C_a = 3\pi\epsilon_0 d$. This exterior-field estimate excludes the strong fields near the wire and gap; it does not determine the full input reactance of a finite-radius antenna.

9.19 Consider the excitation of a waveguide in Problem 8.19 from the point of view of multipole moments of the source.

(a) For the linear probe antenna calculate the multipole moment components of $\mathbf{p}$, $\mathbf{m}$, $Q_{\alpha\beta}$, $Q^M_{\alpha\beta}$ that enter (9.69).

(b) Calculate the amplitudes for excitation of the $TE_{1,0}$ mode and evaluate the power flow. Compare the multipole expansion result with the answer given in Problem 8.19b. Discuss the reasons for agreement or disagreement. What about the comparison for excitation of other modes?

Solution.

Use the local origin at the probe's base $(X,0,0)$ and measure y along it. The mode fields in the expansion are therefore evaluated at that base. This avoids representing the harmless offset X by extra translated moments. Write $k_0=\omega/c$, $q=k_0h$, and define

$$S_j=\int_0^h y^j I_0\sin[k_0(h-y)]dy.$$

The first two integrals, by elementary integration by parts, are

$$S_0=\frac{I_0}{k_0}(1-\cos q),\qquad S_1=I_0\left(\frac{h}{k_0}-\frac{\sin q}{k_0^2}\right).$$

(a) Continuity, including injection at the base, gives $\rho=\partial_y[I(y)\mathbf{1}_{(0,h)}]/(i\omega)$ times the two transverse deltas. Integrating its derivative against y and y^2 gives

$$\boxed{\mathbf{p}=\frac{iS_0}{\omega}\hat{\mathbf{y}},\quad \mathbf{m}=0,\quad Q=\frac{2iS_1}{\omega}\operatorname{diag}(-1,2,-1),\quad Q^M=0.}$$

The magnetic moments vanish because $\mathbf{r}\parallel\mathbf{J}$ in these local coordinates. The charge at the base has zero moment but makes the total oscillating charge zero. For reference, translating the origin back to the guide corner adds $Q_{xy}=Q_{yx}=3Xp_y$ and $m_z=XS_0/2$; such translated moments must be used together with a Taylor expansion about that same corner.

(b) Normalize the transverse mode to unit squared integral, as in 8.19. For TE_{10},

$$E_y(X,y,0)=\sqrt{\frac{2}{ab}}\sin\frac{\pi X}{a},\qquad Z=\frac{\omega\mu}{k},\quad k^2=k_0^2-(\pi/a)^2.$$

The quadrupole contraction can be reduced explicitly:

$$\sum_{ij}Q_{ij}\partial_jE_i=\frac{2iS_1}{\omega}(-\partial_xE_x+2\partial_yE_y-\partial_zE_z)=\frac{6iS_1}{\omega}\partial_yE_y=0.$$

The second equality uses $\nabla\cdot\mathbf{E}=0$ and the last uses the mode's independence of y. Equation (9.69) consequently gives

$$A_{10}^{\pm}=\frac{i\omega Z}{2}\mathbf{p}\cdot\mathbf{E}_{10}^{\mp}=-\frac{ZI_0}{k_0}\sqrt{\frac{2}{ab}}\sin\frac{\pi X}{a}\sin^2\frac{k_0h}{2}.$$

Its one-direction power is

$$\boxed{P_+=P_-=\frac{ZI_0^2}{abk_0^2}\sin^2\frac{\pi X}{a}\sin^4\frac{k_0h}{2},}$$

exactly the answer in 8.19. This agreement does not require $k_0h\ll1$: only E_y enters the source overlap, and it is constant all along the probe, so the exact integral is already encoded in $\mathbf{p}$.

For a mode with $n>0$, $E_y\propto\cos(q_ny)$, $q_n=n\pi/b$, and 8.19 contains the different overlap

$$I_0\frac{k_0[\cos(q_nh)-\cos(k_0h)]}{k_0^2-q_n^2}=S_0-\frac{q_n^2}{2}S_2+\cdots,\quad S_2=I_0\left[\frac{h^2}{k_0}-\frac{2(1-\cos q)}{k_0^3}\right].$$

The quadrupole again gives zero because $\partial_yE_y=0$ at the base; the first omitted coupling is electric octupole order. A dipole approximation is reliable when the overlap samples $q_nh\ll1$, apart from cancellations of S_0, and not uniformly for high transverse orders. For $q_n=k_0$ the exact overlap has the removable limit $I_0h\sin(k_0h)/2$. Modes with E_y identically zero are not excited at any order by this ideal vertical probe.

9.20

(a) Verify by direct calculation that the *static* tangential electric field (3.186) in a circular opening in a flat conducting plane, when inserted into the defining equation (9.72) for the electric dipole moment $\mathbf{p}_{\text{eff}}$, leads to the expression (9.75).

(b) Determine the value of $i\mu\omega\mathbf{m}_{\text{eff}}$ given by (9.72) with the static electric field in part a.

(c) Use the *static* normal magnetic field (5.132) for the corresponding magnetic boundary problem with a circular opening to compute via (9.74) the magnetic dipole moment $\mathbf{m}_{\text{eff}}$ and compare with (9.75).

(d) Comment on the differences between the results of parts b and c and the use of the definitions (9.72) in a consistent fashion. [See Section 9 of the article, Diffraction Theory, by C. J. B. Bouwkamp in *Reports on Progress in Physics*, Vol. 17, ed. A. C. Strickland, The Physical Society, London (1954).]

Solution.

Take the aperture normal into the region $z > 0$ to be $\mathbf{n} = \hat{\mathbf{z}}$ and its radius to be a. In the convention of the static electrostatic solution (3.186), the asymptotic field in this region is $\mathbf{E}_0 = -E_0\hat{\mathbf{z}}$; with zero field on the other side,

$$\mathbf{E}_{\tan}^{(0)} = \frac{E_0}{\pi}\frac{\boldsymbol{s}}{\sqrt{a^2 - s^2}}, \quad 0 < s < a.$$

If both sides have imposed fields, replace E_0 by the corresponding scalar difference. This sign convention is important in comparing with (9.75).

(a) The required radial integral is

$$\int_0^a \frac{s^3}{\sqrt{a^2 - s^2}} ds = a^3 \int_0^{\pi/2} \sin^3 u \, du = \frac{2a^3}{3}.$$

Hence

$$\mathbf{p}_{\text{eff}} = \epsilon_0\hat{\mathbf{z}} \int \boldsymbol{s} \cdot \mathbf{E}_{\tan}^{(0)} d^2s = \frac{4\epsilon_0 a^3 E_0}{3}\hat{\mathbf{z}} = \boxed{-\frac{4\epsilon_0 a^3}{3}\mathbf{E}_0}.$$

(b) The other definition gives

$$i\mu\omega\mathbf{m}_{\text{eff}} = 2\int \mathbf{n} \times \mathbf{E}_{\tan}^{(0)} d^2s = 0,$$

because the angular integral of $\hat{\boldsymbol{\phi}}$ is zero. This calculation uses only the electrostatic part of the aperture field.

(c) For an imposed $\mathbf{H}_0 = H_0\hat{\mathbf{y}}$ in the same upper region, the static magnetic solution (5.132) is $H_z = 2H_0 s \sin\phi/[\pi\sqrt{a^2 - s^2}]$. Thus

$$\mathbf{m}_{\text{eff}} = 2\int \boldsymbol{s} H_z d^2s = \frac{4H_0}{\pi}\frac{2a^3}{3}\int_0^{2\pi} (\hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi)\sin\phi \, d\phi = \boxed{\frac{8a^3}{3}\mathbf{H}_0}.$$

Reversing the observation side reverses the effective moments for fixed aperture fields, consistently with the normal in their definitions.

(d) There is no inconsistency between the two exact definitions. For a harmonic Maxwell field, integration by parts in the aperture and $\nabla_t \times \mathbf{E}_t = i\omega\mu H_n$ give

$$\int \mathbf{n} \times \mathbf{E}_t \, d^2s = \int \boldsymbol{s}(\nabla_t \times \mathbf{E}_t)_n \, d^2s = i\omega\mu \int \boldsymbol{s} H_n \, d^2s.$$

The boundary term is zero because the electric field tangent to the rim is zero; the integrable inverse-square-root edge singularity does not alter it. In a low-frequency expansion $\mathbf{E}_t = \mathbf{E}_t^{(0)} + \omega\mathbf{E}_t^{(1)} + \cdots$, the first term determines $\mathbf{p}_{\text{eff}}$, but its curl and its integrated cross product vanish. The explicit $1/\omega$ in the magnetic definition promotes $\omega\mathbf{E}_t^{(1)}$ to a finite leading term. One must retain that inductive electric field, or equivalently use the static magnetic solution in (9.74). The electric and magnetic static boundary problems specify independent leading excitations; setting the former equal to the entire finite-frequency field loses the latter.

9.21 The fields representing a transverse magnetic wave propagating in a cylindrical waveguide of radius R are:

$$\begin{aligned} E_z &= J_m(\gamma r) e^{im\phi} e^{i\beta z - i\omega t}, \qquad & H_z &= 0 \\ E_\phi &= \frac{-m\beta}{\gamma^2}\frac{E_z}{r}, & H_r &= -\frac{k}{Z_0\beta} E_\phi \\ E_r &= \frac{i\beta}{\gamma^2}\frac{\partial E_z}{\partial r}, & H_\phi &= \frac{k}{Z_0\beta} E_r \end{aligned}$$

where m is the index specifying the angular dependence, β is the propagation constant, $\gamma^2 = k^2 - \beta^2$ ($k = \omega/c$), where γ is such that $J_m(\gamma R) = 0$. Calculate the ratio of the z component of the electromagnetic angular momentum to the energy in the field. It may be advantageous to perform some integrations by parts, and to use the differential equation satisfied by E_z, to simplify your calculations.

Solution.

All energies and angular momenta below are time averages per unit length. Let $\psi = J_m(\gamma r)e^{im\phi}$ and $N = \int_{r<R} |\psi|^2 dA$. The transverse Helmholtz equation and $\psi(R) = 0$ imply, by integration by parts with the vanishing boundary term,

$$\int |\nabla_t \psi|^2 dA = -\int \psi^* \nabla_t^2 \psi \, dA = \gamma^2 N.$$

Consequently the field expressions in the question give

$$\int |\mathbf{E}_t|^2 dA = \frac{\beta^2}{\gamma^2} N, \qquad \int |\mathbf{H}_t|^2 dA = \frac{k^2}{Z_0^2\gamma^2} N.$$

Since $\mu_0/Z_0^2 = \epsilon_0$ and $k^2 = \beta^2 + \gamma^2$,

$$\frac{U}{L} = \frac{1}{4}\int (\epsilon_0|\mathbf{E}|^2 + \mu_0|\mathbf{H}|^2) dA = \frac{\epsilon_0 N}{4}\left[1 + \frac{\beta^2 + k^2}{\gamma^2}\right] = \frac{\epsilon_0 k^2 N}{2\gamma^2}.$$

The angular momentum density is $\mathbf{r} \times \text{Re}(\mathbf{E} \times \mathbf{H}^*)/(2c^2)$. Here $(\mathbf{E} \times \mathbf{H}^*)_\phi = E_z H_r^*$ and $H_r = km\psi/(Z_0\gamma^2 r)$. Therefore

$$\frac{L_z}{L} = \frac{1}{2c^2}\int r \, \text{Re}(E_z H_r^*) dA = \frac{mkN}{2c^2 Z_0 \gamma^2}.$$

Dividing and using $\epsilon_0 Z_0 = 1/c$ gives

$$\boxed{\frac{L_z}{U} = \frac{m}{ck} = \frac{m}{\omega}.}$$

The sign is that of the helical angular phase $e^{im\phi}$. A real standing angular pattern such as $\cos m\phi$ combines equal $+m$ and $-m$ waves and has zero net L_z, despite the same energy.

9.22 A spherical hole of radius a in a conducting medium can serve as an electromagnetic resonant cavity.

(a) Assuming infinite conductivity, determine the transcendental equations for the characteristic frequencies ω_{lm} of the cavity for TE and TM modes.

(b) Calculate numerical values for the wavelength λ_{lm} in units of the radius a for the four lowest modes for TE and TM waves.

(c) Calculate explicitly the electric and magnetic fields inside the cavity for the lowest TE and lowest TM mode.

Solution.

(a) A regular spherical wave has radial function $j_l(kr)$, $l \geq 1$. With normalized vector harmonic $\mathbf{X}_{lm} = \mathbf{L}Y_{lm}/\sqrt{l(l+1)}$, where $\mathbf{L} = -i\mathbf{r} \times \nabla$ as in (9.101), TE modes have $\mathbf{E} \propto j_l(kr)\mathbf{X}_{lm}$ and TM modes have $\mathbf{H} \propto j_l(kr)\mathbf{X}_{lm}$. Curling the latter gives a tangential electric field proportional to $[xj_l(x)]'/x$. The perfectly conducting boundary therefore requires

$$\boxed{j_l(x_{ln}^{\mathrm{TE}}) = 0, \qquad [xj_l(x)]'_{x=x_{ln}^{\mathrm{TM}}} = 0, \qquad \omega_{ln} = cx_{ln}/a.}$$

Here $n = 1, 2, \ldots$ counts positive radial roots; it is distinct from the azimuthal index $m = -l, \ldots, l$. Every listed frequency has $2l + 1$ angular degeneracy. There is no nonzero-frequency $l = 0$ electromagnetic cavity mode.

(b) Root bracketing and refinement give the four lowest *distinct frequencies* in each family:

Family	(l, n)	$x = \omega a/c$	$\lambda/a = 2\pi/x$
TE	$(1, 1)$	4.493409458	1.398311319
TE	$(2, 1)$	5.763459197	1.090176072
TE	$(3, 1)$	6.987932001	0.899148032
TE	$(1, 2)$	7.725251837	0.813330807
TM	$(1, 1)$	2.743707270	2.290034865
TM	$(2, 1)$	3.870238580	1.623462011
TM	$(3, 1)$	4.973420351	1.263352957
TM	$(4, 1)$	6.061949363	1.036495842

The supplementary calculation searches several l and radial roots before sorting, so the TE fourth entry is not incorrectly identified with $(4, 1)$.

(c) Choose the $m = 0$ member; rotations and complex linear combinations produce the other members of each lowest $l = 1$ triplet. Let $x = kr$, $j = j_1(x)$, and $D = [xj_1(x)]'$. An explicit TE field with arbitrary amplitude E_* is

$$\boxed{E_\phi = E_* j \sin\theta, \quad H_r = \frac{2E_*}{i\omega\mu_0 r} j\cos\theta, \quad H_\theta = -\frac{E_*}{i\omega\mu_0 r} D\sin\theta,}$$

with other components zero and $ka = 4.493409458$. These follow directly from $(\nabla \times \mathbf{E})_r = (r\sin\theta)^{-1}\partial_\theta(\sin\theta E_\phi)$ and $(\nabla \times \mathbf{E})_\theta = -r^{-1}\partial_r(rE_\phi)$, using $\nabla \times \mathbf{E} = i\omega\mu_0\mathbf{H}$. For the TM mode, choose an arbitrary H_* and use $\nabla \times \mathbf{H} = -i\omega\epsilon_0\mathbf{E}$:

$$\boxed{H_\phi = H_* j\sin\theta, \quad E_r = -\frac{2H_*}{i\omega\epsilon_0 r} j\cos\theta, \quad E_\theta = \frac{H_*}{i\omega\epsilon_0 r} D\sin\theta,}$$

with $ka = 2.743707270$. The limits at the origin are finite because $j_1(x) \sim x/3$ and $D \sim 2x/3$. The respective conditions $j(ka) = 0$ and $D(ka) = 0$ make the wall's tangential electric field vanish.

9.23 The spherical resonant cavity of Problem 9.22 has nonpermeable walls of large, but finite, conductivity. In the approximation that the skin depth δ is small compared to the cavity radius a, show that the Q of the cavity, defined by equation (8.86), is given by

$$Q = \frac{a}{\delta} \qquad \text{for all TE modes}$$

$$Q = \frac{a}{\delta}\left(1 - \frac{l(l+1)}{x_{lm}^2}\right) \qquad \text{for TM modes}$$

where $x_{lm} = (a/c)\omega_{lm}$ for TM modes.

Solution.

For a good nonmagnetic conductor, $R_s = 1/(\sigma\delta) = \omega\mu_0\delta/2$. The leading wall dissipation is $P_{\text{loss}} = (R_s/2)\int_{r=a} |\mathbf{H}_t|^2 dA$. Electric and magnetic stored energies are equal in a lossless cavity, so its total time-averaged energy is $U = (\mu_0/2)\int |\mathbf{H}|^2 dV$. Thus

$$Q = \frac{\omega U}{P_{\text{loss}}} = \frac{2}{\delta}\frac{\int |\mathbf{H}|^2 dV}{\int_{r=a} |\mathbf{H}_t|^2 dA}.$$

Let $x = ka$ and $I_l = \int_0^a r^2 j_l(kr)^2 dr$. Multiplying the spherical Bessel equation by its solution, or differentiating its adjacent-order recurrences, gives

$$I_l = \frac{a^3}{2}[j_l(x)^2 - j_{l-1}(x)j_{l+1}(x)].$$

To verify the integral rather than assume it, put $j = j_l(x)$ and use $j_{l-1} = j' + (l+1)j/x$, $j_{l+1} = lj/x - j'$. The proposed dimensionless antiderivative is

$$F(x) = \frac{x^3}{2}(j^2 + j'^2) + \frac{x^2}{2}jj' - \frac{l(l+1)x}{2}j^2.$$

Differentiation gives

$$F' = \frac{3x^2}{2}(j^2 + j'^2) + x^3(jj' + j'j'') + xjj' + \frac{x^2}{2}(j'^2 + jj'') - \frac{l(l+1)}{2}j^2 - l(l+1)xjj'.$$

Substituting $j'' = -2j'/x - [1 - l(l+1)/x^2]j$ cancels all jj' and j'^2 terms and leaves $F' = x^2 j^2$. Integrating from zero to ka and dividing by k^3 proves the identity; its lower endpoint vanishes since $l \geq 1$.

For TM modes $\mathbf{H} = H_* j_l(kr)\mathbf{X}_{lm}$, with $\int |\mathbf{X}_{lm}|^2 d\Omega = 1$, so the ratio of integrals is $I_l/[a^2 j_l(x)^2]$. The condition $j_l + x j_l' = 0$ and the recurrences give $j_{l-1} = l j_l/x$, $j_{l+1} = (l+1) j_l/x$. Hence

$$\boxed{Q_{\rm TM} = \frac{a}{\delta}\left[1 - \frac{l(l+1)}{x^2}\right].}$$

For TE modes take $\mathbf{E} = E_* j_l \mathbf{X}_{lm}$. After curling, the angularly integrated magnetic norm is proportional to $\int_0^a \{l(l+1) j_l^2 + [(r j_l)']^2\} dr$. For the integration by parts, set $y(r) = r j_l(kr)$. Its equation is $y'' + [k^2 - l(l+1)/r^2]y = 0$, so

$$\int_0^a (y')^2 dr = [yy']_0^a - \int_0^a yy'' dr = [yy']_0^a + k^2 \int_0^a y^2 dr - l(l+1)\int_0^a \frac{y^2}{r^2} dr.$$

The last term cancels the other term in the magnetic norm. The result is $k^2 I_l + [r j_l (r j_l)']_0^a = k^2 I_l$ at the TE root $j_l(x) = 0$. The surface norm has the same common factor times $[x j_l'(x)]^2$. At this root $j_{l-1} = j_l'$ and $j_{l+1} = -j_l'$, so $I_l = a^3 j_l'^2/2$. Consequently

$$\boxed{Q_{\rm TE} = \frac{2}{\delta}\frac{k^2 a^3 j_l'^2/2}{x^2 j_l'^2} = \frac{a}{\delta}.}$$

The skin depth is evaluated at the mode frequency. Corrections to these expressions are beyond leading order in δ/a; the lossless mode fields suffice at this order.

9.24 Discuss the normal modes of oscillation of a perfectly conducting solid sphere of radius a in free space. (This problem was solved by J. J. Thomson in the 1880s.)

(a) Determine the characteristic equations for the eigenfrequencies for TE and TM modes of oscillation. Show that the roots for ω always have a negative imaginary part, assuming a time dependence of $e^{-i\omega t}$.

(b) Calculate the eigenfrequencies for the $l = 1$ and $l = 2$ TE and TM modes. Tabulate the wavelength (defined in terms of the real part of the frequency) in units of the radius a and the decay time (defined as the time taken for the *energy* to fall to e^{-1} of its initial value) in units of the transit time (a/c) for each of the modes.

Solution.

(a) Exterior natural oscillations obey an outgoing condition, so replace the regular j_l of 9.22 by $h_l^{(1)}$. With $x = \omega a/c$,

$$\boxed{h_l^{(1)}(x) = 0 \quad \text{(TE)}, \qquad [x h_l^{(1)}(x)]' = 0 \quad \text{(TM)}, \qquad l \geq 1.}$$

These are radiative resonances, not the real eigenfrequencies of a closed cavity. To exclude $\mathrm{Im}\,\omega > 0$, note that an outgoing field would then decay exponentially at infinity. Integrating $\nabla \cdot (\mathbf{E} \times \mathbf{H}^*) = i\omega\mu_0 |\mathbf{H}|^2 - i\omega^* \epsilon_0 |\mathbf{E}|^2$ over the exterior and taking the real part gives

$$0 = -\,\mathrm{Im}\,\omega \int (\mu_0 |\mathbf{H}|^2 + \epsilon_0 |\mathbf{E}|^2) dV.$$

The PEC surface has no normal energy flow and the outer surface term vanishes, so a nonzero mode is impossible in that half-plane. For real nonzero ω, outgoing radiation instead has strictly positive

power at infinity, again incompatible with a source-free PEC boundary. Thus all nonzero radiative roots have $\operatorname{Im}\omega < 0$. Static exterior charge or magnetic fields at $\omega = 0$ are separate boundary-value solutions, not these oscillatory radiation modes.

(b) The explicit outgoing functions are

$$h_1^{(1)}(x) = -\frac{e^{ix}}{x^2}(x+i), \qquad h_2^{(1)}(x) = \frac{ie^{ix}}{x^3}(x^2+3ix-3).$$

Thus TE roots are $x = -i$ for $l = 1$ and $x = \pm\sqrt{3}/2 - 3i/2$ for $l = 2$. Differentiating xh_l gives the TM polynomials

$$l = 1: \quad x^2 + ix - 1 = 0, \qquad l = 2: \quad x^3 + 3ix^2 - 6x - 6i = 0.$$

The first gives $x = \pm\sqrt{3}/2 - i/2$. For the cubic put $x = -iy$; it becomes $y^3 - 3y^2 + 6y - 6 = 0$. With

$$b = 1 + \sqrt[3]{1+\sqrt{2}} + \sqrt[3]{1-\sqrt{2}} = 1.596071638,$$

its roots in x are

$$x = -ib, \qquad x = \pm\frac{1}{2}\sqrt{3b^2 - 6b + 15} - \frac{i}{2}(3-b) = \pm 1.807339494 - 0.701964181i.$$

For each root the local field amplitude varies as $e^{\operatorname{Im}\omega t}$ and its energy as $e^{2\operatorname{Im}\omega t}$. Therefore $\tau_E/(a/c) = -1/(2\operatorname{Im} x)$ and $\lambda/a = 2\pi/|\operatorname{Re} x|$ when the real part is nonzero:

Family	$x = \omega a/c$	λ/a	$\tau_E/(a/c)$
TE, $l = 1$	$-i$	∞ (no oscillation)	0.500000
TM, $l = 1$	$\pm 0.866025404 - 0.5i$	7.255197457	1.000000
TE, $l = 2$	$\pm 0.866025404 - 1.5i$	7.255197457	0.333333
TM, $l = 2$	$-1.596071638i$	∞ (no oscillation)	0.313269
TM, $l = 2$	$\pm 1.807339494 - 0.701964181i$	3.476483	0.712287

Each row also has the usual $2l + 1$ angular degeneracy. Negative-real-part partners supply the reality-related time dependence, rather than new positive-frequency resonances. The outgoing resonance functions grow spatially at infinity for these complex frequencies, so a single such mode is not a finite-total-energy field on all of space at one time. The stated decay time describes the ringing field in a fixed bounded exterior region; a causal physical signal includes its propagating front.

10 Chapter 10. Scattering and Diffraction

Problems

10.1

(a) Show that for arbitrary initial polarization, the scattering cross section of a perfectly conducting sphere of radius a, summed over outgoing polarizations, is given in the long-wavelength limit by

$$\frac{d\sigma}{d\Omega}(\boldsymbol{\epsilon}_0, \mathbf{n}_0, \mathbf{n}) = k^4 a^6 \left[\frac{5}{4} - |\boldsymbol{\epsilon}_0 \cdot \mathbf{n}|^2 - \frac{1}{4}|\mathbf{n} \cdot (\mathbf{n}_0 \times \boldsymbol{\epsilon}_0)|^2 - \mathbf{n}_0 \cdot \mathbf{n}\right]$$

where $\mathbf{n}_0$ and $\mathbf{n}$ are the directions of the incident and scattered radiations, respectively, while $\boldsymbol{\epsilon}_0$ is the (perhaps complex) unit polarization vector of the incident radiation ($\boldsymbol{\epsilon}_0^* \cdot \boldsymbol{\epsilon}_0 = 1$; $\mathbf{n}_0 \cdot \boldsymbol{\epsilon}_0 = 0$).

(b) If the incident radiation is linearly polarized, show that the cross section is

$$\frac{d\sigma}{d\Omega}(\boldsymbol{\epsilon}_0, \mathbf{n}_0, \mathbf{n}) = k^4 a^6 \left[\frac{5}{8}(1 + \cos^2\theta) - \cos\theta - \frac{3}{8}\sin^2\theta\, \cos 2\phi\right]$$

where $\mathbf{n} \cdot \mathbf{n}_0 = \cos\theta$ and the azimuthal angle ϕ is measured from the direction of the linear polarization.

(c) What is the ratio of scattered intensities at $\theta = \pi/2$, $\phi = 0$ and $\theta = \pi/2$, $\phi = \pi/2$? Explain physically in terms of the induced multipoles and their radiation patterns.

Solution.

Use phasors with time dependence $e^{-i\omega t}$, and put $\mathbf{h} = \mathbf{n}_0 \times \boldsymbol{\epsilon}_0$. Jackson's induced moments (10.12)–(10.14) are

$$\mathbf{p} = 4\pi\epsilon_0 a^3 E_0 \boldsymbol{\epsilon}_0, \qquad \frac{\mathbf{m}}{c} = -2\pi\epsilon_0 a^3 E_0 \mathbf{h}.$$

The electric dipole field is proportional to $\mathbf{p} - \mathbf{n}(\mathbf{n} \cdot \mathbf{p})$ and the magnetic dipole field to $-\mathbf{n} \times \mathbf{m}/c$. Thus the normalized outgoing amplitude is

$$\mathbf{f} = k^2 a^3 \left[\boldsymbol{\epsilon}_0 - \mathbf{n}(\mathbf{n} \cdot \boldsymbol{\epsilon}_0) + \frac{1}{2}\mathbf{n} \times \mathbf{h}\right].$$

(a) Summing over any two orthonormal outgoing transverse polarizations gives $\sum_s |\boldsymbol{\epsilon}_s^* \cdot \mathbf{f}|^2 = \mathbf{f}^* \cdot \mathbf{f}$. Now

$$|\boldsymbol{\epsilon}_0 - \mathbf{n}(\mathbf{n} \cdot \boldsymbol{\epsilon}_0)|^2 = 1 - |\mathbf{n} \cdot \boldsymbol{\epsilon}_0|^2, \qquad |\mathbf{n} \times \mathbf{h}|^2 = 1 - |\mathbf{n} \cdot \mathbf{h}|^2,$$

and the electric–magnetic cross term follows from

$$\boldsymbol{\epsilon}_0^* \cdot (\mathbf{n} \times \mathbf{h}) = \boldsymbol{\epsilon}_0^* \cdot [\mathbf{n}_0(\mathbf{n} \cdot \boldsymbol{\epsilon}_0) - \boldsymbol{\epsilon}_0(\mathbf{n} \cdot \mathbf{n}_0)] = -\mathbf{n} \cdot \mathbf{n}_0.$$

Consequently

$$\boxed{\frac{d\sigma}{d\Omega} = k^4 a^6 \left[\frac{5}{4} - |\mathbf{n} \cdot \boldsymbol{\epsilon}_0|^2 - \frac{1}{4}|\mathbf{n} \cdot \mathbf{h}|^2 - \mathbf{n} \cdot \mathbf{n}_0\right].}$$

(b) Choose $\mathbf{n}_0 = \widehat{\mathbf{z}}$ and $\boldsymbol{\epsilon}_0 = \widehat{\mathbf{x}}$, so $\mathbf{h} = \widehat{\mathbf{y}}$. The bracket becomes $5/4 - \sin^2\theta\cos^2\phi - \sin^2\theta\sin^2\phi/4 - \cos\theta$. Using $\cos^2\phi = (1+\cos 2\phi)/2$ gives

$$\frac{d\sigma}{d\Omega} = k^4 a^6 \left[\frac{5}{8}(1+\cos^2\theta) - \cos\theta - \frac{3}{8}\sin^2\theta\cos 2\phi\right].$$

Its integral is $10\pi k^4 a^6/3$, independent of the incident polarization.

(c) At $\theta = \pi/2$ the two values are $k^4a^6/4$ and k^4a^6, respectively, hence $I_{\phi=0}/I_{\phi=\pi/2} = 1/4$. Along $\widehat{\mathbf{x}}$ the electric dipole does not radiate and only the magnetic dipole contributes; along $\widehat{\mathbf{y}}$ the magnetic dipole does not radiate and only the electric dipole contributes. Their effective radiation moments obey $|\mathbf{m}|/c = |\mathbf{p}|/2$, so their isolated intensities have ratio 1/4.

10.2 Electromagnetic radiation with elliptic polarization, described (in the notation of Section 7.2) by the polarization vector,

$$\boldsymbol{\epsilon} = \frac{1}{\sqrt{1+r^2}}\left(\boldsymbol{\epsilon}_+ + re^{i\alpha}\boldsymbol{\epsilon}_-\right)$$

is scattered by a perfectly conducting sphere of radius a. Generalize the amplitude in the scattering cross section (10.71), which applies for $r = 0$ or $r = \infty$, and calculate the cross section for scattering in the long-wavelength limit. Show that

$$\frac{d\sigma}{d\Omega} = k^4 a^6 \left[\frac{5}{8}\left(1+\cos^2\theta\right) - \cos\theta - \frac{3}{4}\left(\frac{r}{1+r^2}\right)\sin^2\theta\,\cos(2\phi-\alpha)\right]$$

Compare with Problem 10.1.

Solution.

Let $\boldsymbol{\epsilon}_\pm = (\widehat{\mathbf{x}} \pm i\widehat{\mathbf{y}})/\sqrt{2}$ and $\mathbf{n}_0 = \widehat{\mathbf{z}}$. Linearity of the fields, rather than addition of intensities, gives

$$\mathbf{f} = \frac{k^2a^3}{\sqrt{1+r^2}}(\mathbf{A}_+ + re^{i\alpha}\mathbf{A}_-), \qquad \mathbf{A}_\pm = \boldsymbol{\epsilon}_\pm - \mathbf{n}(\mathbf{n}\cdot\boldsymbol{\epsilon}_\pm) + \frac{1}{2}\mathbf{n}\times(\widehat{\mathbf{z}}\times\boldsymbol{\epsilon}_\pm).$$

In the spherical-harmonic notation of (10.71) the same expression is

$$\mathbf{f} = \sqrt{\frac{2\pi}{3}}\,k^2a^3\frac{(\mathbf{X}_{11} - 2i\mathbf{n}\times\mathbf{X}_{11}) + re^{i\alpha}(\mathbf{X}_{1,-1} + 2i\mathbf{n}\times\mathbf{X}_{1,-1})}{\sqrt{1+r^2}},$$

up to an irrelevant common phase. The relative phase is fixed by the circular vectors specified above. Directly,

$$|\mathbf{n}\cdot\boldsymbol{\epsilon}|^2 = \frac{\sin^2\theta}{2}\left[1 + \frac{2r}{1+r^2}\cos(2\phi-\alpha)\right], \quad |\mathbf{n}\cdot(\widehat{\mathbf{z}}\times\boldsymbol{\epsilon})|^2 = \frac{\sin^2\theta}{2}\left[1 - \frac{2r}{1+r^2}\cos(2\phi-\alpha)\right].$$

Substitution into 10.1 gives

$$\boxed{\frac{d\sigma}{d\Omega} = k^4a^6\left[\frac{5}{8}(1+\cos^2\theta) - \cos\theta - \frac{3r}{4(1+r^2)}\sin^2\theta\cos(2\phi-\alpha)\right].}$$

For $r = 0$ or $r \to \infty$ the azimuthal term vanishes, reproducing (10.71). For $r = 1$ the polarization is linear, along azimuth $\alpha/2$ up to a common phase; this is 10.1(b) with $\phi \mapsto \phi - \alpha/2$. The total is $10\pi k^4a^6/3$.

10.3 A solid uniform sphere of radius R and conductivity σ acts as a scatterer of a plane-wave beam of unpolarized radiation of frequency ω, with $\omega R/c \ll 1$. The conductivity is large enough that the skin depth δ is small compared to R.

(a) Justify and use a magnetostatic scalar potential to determine the magnetic field around the sphere, assuming the conductivity is infinite. (Remember that $\omega \neq 0$.)

(b) Use the technique of Section 8.1 to determine the absorption cross section of the sphere. Show that it varies as $(\omega)^{1/2}$ provided σ is independent of frequency.

Solution.

Take the sphere to be nonmagnetic, with exterior permeability μ_0. **(a)** In a region of size $R \ll c/\omega$, the incident magnetic field is nearly uniform. To leading quasistatic order the exterior field is curl-free and divergence-free. Infinite conductivity at nonzero frequency forces the oscillating field inside to vanish: it is not the static problem of a sphere placed in a preexisting penetrating flux. Choose the polar axis along $\mathbf{H}_i = H_i\widehat{\mathbf{z}}$ and write

$$\Psi_m = -H_i r\cos\vartheta + \frac{A\cos\vartheta}{r^2}, \qquad \mathbf{H} = -\boldsymbol{\nabla}\Psi_m.$$

The boundary condition $H_r(R) = 0$ gives $A = -H_i R^3/2$. Hence

$$H_r = H_i\left(1 - \frac{R^3}{r^3}\right)\cos\vartheta, \qquad H_\vartheta = -H_i\left(1 + \frac{R^3}{2r^3}\right)\sin\vartheta.$$

In particular $\mathbf{m} = -2\pi R^3\mathbf{H}_i$ and $H_\vartheta(R) = -3H_i\sin\vartheta/2$.

(b) The planar skin-layer solution of Section 8.1 applies locally when $\delta = \sqrt{2/(\mu_0\sigma\omega)} \ll R$. Its dissipated time-averaged power per area is $\frac{1}{2}R_s|\mathbf{H}_t|^2$, where $R_s = 1/(\sigma\delta) = \mu_0\omega\delta/2$. Therefore

$$P_{\text{abs}} = \frac{R_s}{2}\frac{9}{4}|H_i|^2R^2\int\sin^2\vartheta\,d\Omega = 3\pi R_s R^2|H_i|^2.$$

Divide by the incident flux $Z_0|H_i|^2/2$:

$$\boxed{\sigma_{\text{abs}} = 6\pi R^2\frac{R_s}{Z_0} = 3\pi k\delta R^2 = 3\pi R^2\sqrt{\frac{2\epsilon_0\omega}{\sigma}}.}$$

The result applies to each incident polarization and hence to unpolarized radiation. Its $\sqrt{\omega}$ behavior assumes frequency-independent σ and a range where both $kR \ll 1$ and $\delta/R \ll 1$ hold. Electric-field corrections to the surface current enter at higher order in kR.

10.4 An unpolarized wave of frequency $\omega = ck$ is scattered by a *slightly* lossy uniform isotropic dielectric sphere of radius R much smaller than a wavelength. The sphere is characterized by an ordinary real dielectric constant ϵ_r and a real conductivity σ. The parameters are such that the skin depth δ is very *large* compared to the radius R.

(a) Calculate the differential and total *scattering* cross sections.

(b) Show that the absorption cross section is

$$\sigma_{\text{abs}} = 12\pi R^2\frac{(RZ_0\sigma)}{(\epsilon_r+2)^2 + (Z_0\sigma/k)^2}$$

(c) From part a write down the forward scattering amplitude and use the optical theorem to evaluate the total cross section. Compare your answer with the sum of the scattering and absorption cross sections from parts a and b. Comment.

Solution.

Put

$$\widetilde{\epsilon}_r = \epsilon_r + is, \qquad s = \frac{\sigma}{\epsilon_0\omega} = \frac{Z_0\sigma}{k}, \qquad A = \frac{\widetilde{\epsilon}_r - 1}{\widetilde{\epsilon}_r + 2}, \qquad D = (\epsilon_r + 2)^2 + s^2.$$

Here $\mu = \mu_0$, and the electrostatic internal-field approximation requires $|k\sqrt{\widetilde{\epsilon}_r}\, R| \ll 1$ as well as $kR \ll 1$. For an ordinary dielectric this follows from the stated long wavelength and weak skin effect. To match the quasistatic fields, align the polar axis with $\mathbf{E}_i$ and write

$$\Phi_{\rm in} = -C_{\rm in} E_i r\cos\theta, \qquad \Phi_{\rm out} = -E_i r\cos\theta + C_{\rm out} E_i R^3 \frac{\cos\theta}{r^2}.$$

Potential and normal-displacement continuity require

$$C_{\rm in} = 1 - C_{\rm out}, \qquad \widetilde{\epsilon}_r C_{\rm in} = 1 + 2C_{\rm out}.$$

Thus $C_{\rm out} = A$ and $C_{\rm in} = 3/(\widetilde{\epsilon}_r + 2)$, giving

$$\mathbf{E}_{\rm in} = \frac{3}{\widetilde{\epsilon}_r + 2}\mathbf{E}_i, \qquad \mathbf{p} = 4\pi\epsilon_0 R^3 A\, \mathbf{E}_i.$$

(a) The electric dipole amplitude is $\mathbf{f} = k^2 R^3 A[\boldsymbol{\epsilon}_0 - \mathbf{n}(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)]$. Summing outgoing and averaging incident polarizations yields

$$\boxed{\frac{d\sigma_{\rm sc}}{d\Omega} = k^4 R^6 \frac{(\epsilon_r - 1)^2 + s^2}{D}\frac{1 + \cos^2\theta}{2}, \qquad \sigma_{\rm sc} = \frac{8\pi}{3} k^4 R^6 \frac{(\epsilon_r - 1)^2 + s^2}{D}.}$$

(b) Joule heating is

$$P_{\rm abs} = \frac{\sigma}{2}|\mathbf{E}_{\rm in}|^2 \frac{4\pi R^3}{3} = \frac{6\pi\sigma R^3 |E_i|^2}{D}.$$

Dividing by $|E_i|^2/(2Z_0)$ gives

$$\boxed{\sigma_{\rm abs} = \frac{12\pi R^3 Z_0 \sigma}{D}.}$$

(c) The forward amplitude in the original polarization is $f(0) = k^2 R^3 A$. Since $\operatorname{Im} A = 3s/D$, the optical theorem gives

$$\frac{4\pi}{k}\operatorname{Im} f(0) = \frac{12\pi R^3 Z_0 \sigma}{D} = \sigma_{\rm abs}.$$

It misses the positive scattering term. This is the approximation-order issue discussed after (10.149)–(10.151): a real leading polarizability does scatter, although its leading forward amplitude has no imaginary part. For example, the radiation-reaction completion of the dipole response is

$$A_{\rm rad} = \frac{A}{1 - \frac{2i}{3}k^3 R^3 A}.$$

Writing $b = 2k^3 R^3/3$, one checks

$$\operatorname{Im} A_{\rm rad} = \frac{\operatorname{Im} A + b|A|^2}{|1 - ibA|^2}, \qquad |A_{\rm rad}|^2 = \frac{|A|^2}{|1 - ibA|^2}.$$

Thus extinction equals absorption plus scattering at consistent dipole order. Dynamic depolarization also corrects the real part at relative order $(kR)^2$; the radiative imaginary part is still necessary.

10.5 The scattering by the dielectric sphere of Problem 10.4 was treated as purely electric dipole scattering. This is adequate unless it happens that the real dielectric constant ϵ/ϵ_0 is very large. In these circumstances a magnetic dipole contribution, even though higher order in kR, may be important. Retain the internal quasistatic condition $|k\sqrt{\tilde{\epsilon}_r}R| \ll 1$. In (a) the displayed moment is its conduction-current part; in (b) the displayed cross section retains material absorption, with radiative extinction to be added when comparable.

(a) Show that the changing magnetic flux of the incident wave induces an azimuthal current flow in the sphere and produces a magnetic dipole moment,

$$\mathbf{m} = \frac{i4\pi\sigma Z_0}{k\mu_0}\,(kR)^2\,\frac{R^3}{30}\,\mathbf{B}_{\rm inc}$$

(b) Show that application of the optical theorem to the coherent sum of the electric and magnetic dipole contributions leads to a total cross section,

$$\sigma_t = 12\pi R^2 (RZ_0\sigma)\left[\frac{1}{(\epsilon_r+2)^2 + (Z_0\sigma/k)^2} + \frac{1}{90}\,(kR)^2\right]$$

(Compare Landau and Lifshitz, *Electrodynamics of Continuous Media*, p. 323).

Solution.

The required approximation is $|k\sqrt{\tilde{\epsilon}_r}R| \ll 1$, not merely $kR \ll 1$ and $\delta \gg R$. In this limit the internal magnetic field can be replaced by its spatially uniform incident value.

(a) Faraday's law, $\boldsymbol{\nabla}\times\mathbf{E} = i\omega\mathbf{B}_i$, is satisfied by

$$\mathbf{E}_{\rm rot} = \frac{i\omega}{2}\mathbf{B}_i \times \mathbf{r}.$$

For $\mathbf{B}_i$ along z, this is an azimuthal field $E_\varphi = i\omega B_i\rho/2$, giving $\mathbf{J}_{\rm cond} = i\sigma\omega\,\mathbf{B}_i\times\mathbf{r}/2$. Using $\int_{r<R} r_i r_j\, d^3r = (4\pi R^5/15)\delta_{ij}$,

$$\begin{aligned}\mathbf{m}_{\rm cond} &= \frac{1}{2}\int \mathbf{r}\times\mathbf{J}_{\rm cond}\, d^3r \\ &= \frac{i\sigma\omega}{4}\int [\mathbf{B}_i r^2 - \mathbf{r}(\mathbf{r}\cdot\mathbf{B}_i)]\, d^3r = \frac{2\pi i}{15}\sigma\omega R^5\mathbf{B}_i \\ &= \boxed{\frac{i4\pi\sigma Z_0}{k\mu_0}(kR)^2\frac{R^3}{30}\mathbf{B}_i}.\end{aligned}$$

This is the *conduction-current* moment requested. A dielectric also has the polarization current $-i\omega\epsilon_0(\epsilon_r-1)\mathbf{E}$. Together the two give, to the same order,

$$\mathbf{m} = \frac{2\pi}{15\mu_0}(\tilde{\epsilon}_r - 1)k^2R^5\mathbf{B}_i.$$

Its real part affects scattering but not leading material absorption.

(b) Let $\mathbf{h} = \mathbf{n}_0\times\boldsymbol{\epsilon}_0$ and write $\mathbf{m}/c = 4\pi\epsilon_0 R^3 E_i B\,\mathbf{h}$. Then

$$B = \frac{(\tilde{\epsilon}_r-1)(kR)^2}{30}, \qquad \mathbf{f} = k^2R^3\left\{A[\boldsymbol{\epsilon}_0 - \mathbf{n}(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)] - B\,\mathbf{n}\times\mathbf{h}\right\}.$$

Forward, $-\mathbf{n}_0 \times \mathbf{h} = \boldsymbol{\epsilon}_0$, so $f(0) = k^2R^3(A+B)$. Since $\operatorname{Im} B = s(kR)^2/30$, the optical theorem at the retained absorptive order gives

$$\boxed{\sigma_t^{(\text{material})} = 12\pi R^2 (RZ_0\sigma)\left[\frac{1}{(\epsilon_r+2)^2+s^2} + \frac{(kR)^2}{90}\right].}$$

An independent check of the magnetic term is direct Joule heating:

$$P_m = \frac{\sigma\omega^2|B_i|^2}{8}\int \rho^2\, d^3r = \frac{\pi\sigma\omega^2R^5|B_i|^2}{15}, \quad \sigma_{\text{abs},m} = \frac{2\pi}{15}R^3Z_0\sigma(kR)^2.$$

The electric term is suppressed by $|\widetilde{\epsilon}_r + 2|^{-2}$; this explains the importance of magnetic absorption at large dielectric constant. As in 10.4(c), the displayed total omits radiative extinction: the dipole scattering term $8\pi k^4R^6(|A|^2+|B|^2)/3$ must be added when comparable, with consistent radiative corrections to the moments. The displayed expression is not the complete extinction at $\sigma = 0$.

10.6

(a) Show that for the scattered wave (10.57) the *normalized* scattering amplitude (10.138) is

$$\mathbf{f} = \frac{1}{ik}\sqrt{\frac{\pi}{2}}\sum_l \sqrt{2l+1}\,[\alpha_\pm(l)\mathbf{X}_{l,\pm1} \pm i\beta_\pm(l)\mathbf{n}\times\mathbf{X}_{l,\pm1}]$$

where the polarization vector of the incident wave is $(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)/\sqrt{2}$.

(b) Deduce an expression for the total cross section of σ_t from the optical theorem (10.139) and the above expression for $\mathbf{f}$.

Solution.

(a) In (10.57), insert

$$h_l^{(1)}(kr) \sim (-i)^{l+1}\frac{e^{ikr}}{kr}, \qquad \frac{1}{k}\boldsymbol{\nabla}\times(h_l^{(1)}\mathbf{X}_{lm}) \sim ih_l^{(1)}\mathbf{n}\times\mathbf{X}_{lm}.$$

The factor $i^l(-i)^{l+1} = -i$ is independent of l. The incident electric vector in (10.55) has norm $\sqrt{2}$; normalization supplies the additional $1/\sqrt{2}$. The coefficient of E_ie^{ikr}/r is therefore

$$\boxed{\mathbf{f} = \frac{1}{ik}\sqrt{\frac{\pi}{2}}\sum_{l=1}^{\infty}\sqrt{2l+1}[\alpha_\pm(l)\mathbf{X}_{l,\pm1} \pm i\beta_\pm(l)\mathbf{n}\times\mathbf{X}_{l,\pm1}].}$$

(b) At the north pole, using the spherical harmonics with Condon–Shortley phase and $\mathbf{X}_{lm} = \mathbf{L}Y_{lm}/\sqrt{l(l+1)}$,

$$\mathbf{X}_{l,\pm1}(0,\phi) = \sqrt{\frac{2l+1}{8\pi}}\,\boldsymbol{\epsilon}_\pm, \qquad \widehat{\mathbf{z}}\times\boldsymbol{\epsilon}_\pm = \mp i\boldsymbol{\epsilon}_\pm.$$

These limits follow, for example, from $Y_{l1} \sim -\sqrt{(2l+1)l(l+1)/(16\pi)}\,\theta e^{i\phi}$. Consequently

$$\boldsymbol{\epsilon}_\pm^*\cdot\mathbf{f}(0) = \frac{1}{4ik}\sum_l(2l+1)[\alpha_\pm(l)+\beta_\pm(l)],$$

and the optical theorem (10.139) gives

$$\boxed{\sigma_t = -\frac{\pi}{k^2}\sum_l (2l+1)\,\mathrm{Re}[\alpha_\pm(l)+\beta_\pm(l)].}$$

For a rotationally invariant sphere the helicities have identical coefficients. Adding (10.61)'s scattering and absorption expressions gives the same answer, since $2-|1+\alpha|^2-|1+\beta|^2+|\alpha|^2+|\beta|^2 = -2\,\mathrm{Re}(\alpha+\beta)$.

10.7 Discuss the scattering of a plane wave of electromagnetic radiation by a nonpermeable, dielectric sphere of radius a and dielectric constant ϵ_r.

(a) By finding the fields inside the sphere and matching to the incident plus scattered wave outside the sphere, determine without any restriction on ka the multipole coefficients in the scattered wave. Define suitable phase shifts for the problem.

(b) Consider the long-wavelength limit ($ka \ll 1$) and determine explicitly the differential and total scattering cross sections. Compare your results with those of Section 10.1.B.

(c) In the limit $\epsilon_r \to \infty$ compare your results to those for the perfectly conducting sphere.

Solution.

Let $x = ka$, $\nu = \sqrt{\epsilon_r}$, $y = \nu x$, with permeability μ_0 on both sides. For an absorbing sphere choose $\mathrm{Im}\,\nu \geq 0$. Introduce Riccati functions

$$\psi_l(z) = z j_l(z), \quad \chi_l(z) = z n_l(z), \quad \xi_l(z) = z h_l^{(1)}(z) = \psi_l(z) + i\chi_l(z).$$

A prime denotes differentiation with respect to the entire argument.

(a) The regular interior fields are combinations of $j_l(\nu kr)\mathbf{X}_{lm}$ and $(\nu k)^{-1}\boldsymbol{\nabla}\times[j_l(\nu kr)\mathbf{X}_{lm}]$. The exterior radial functions are $j_l(kr)+\alpha_l h_l^{(1)}(kr)/2$ for the magnetic multipole and $j_l(kr)+\beta_l h_l^{(1)}(kr)/2$ for the electric multipole. Set $\alpha_l = -2b_l$, $\beta_l = -2a_l$. Tangential E, H continuity gives, with normalized interior amplitudes t_M, t_E,

$$\begin{aligned} \psi_l(x) - b_l\xi_l(x) &= t_M\psi_l(y), \quad & \psi_l'(x) - b_l\xi_l'(x) &= \nu t_M\psi_l'(y), \\ \psi_l(x) - a_l\xi_l(x) &= \nu t_E\psi_l(y), \quad & \psi_l'(x) - a_l\xi_l'(x) &= t_E\psi_l'(y). \end{aligned}$$

Elimination yields the coefficients at arbitrary size:

$$\boxed{b_l = \frac{\psi_l(y)\psi_l'(x) - \nu\psi_l(x)\psi_l'(y)}{\psi_l(y)\xi_l'(x) - \nu\xi_l(x)\psi_l'(y)}, \qquad a_l = \frac{\nu\psi_l(y)\psi_l'(x) - \psi_l(x)\psi_l'(y)}{\nu\psi_l(y)\xi_l'(x) - \xi_l(x)\psi_l'(y)}.}$$

Either pair determines the interior amplitude; keeping the pair instead of dividing only by $\psi_l(y)$ also handles its zeros. For clarity, define $\mathbf{M}^{\mathrm{in}}_{lm} = j_l(\nu kr)\mathbf{X}_{lm}$ and $\mathbf{N}^{\mathrm{in}}_{lm} = (\nu k)^{-1}\nabla\times\mathbf{M}^{\mathrm{in}}_{lm}$. The normalized circular incident wave has $d_l = i^l\sqrt{2\pi(2l+1)}$ in (10.55). Its complete internal field is

$$\mathbf{E}_{\mathrm{in}} = E_i\sum_{l=1}^{\infty} d_l\nu[t_M(l)\mathbf{M}^{\mathrm{in}}_{l,\pm1} \pm t_E(l)\mathbf{N}^{\mathrm{in}}_{l,\pm1}], \qquad \mathbf{B}_{\mathrm{in}} = \frac{\nabla\times\mathbf{E}_{\mathrm{in}}}{i\omega}.$$

For real positive ϵ_r, let N_M, D_M be the numerator of b_l and the same expression with exterior ψ_l replaced by χ_l. Then $b_l = N_M/(N_M + iD_M)$, and suitable phase shifts are

$$\tan\delta_l = N_M/D_M, \quad 1+\alpha_l = e^{2i\delta_l}; \qquad \tan\delta_l' = N_E/D_E, \quad 1+\beta_l = e^{2i\delta_l'}.$$

Continuous phase branches cover zeros of D. Absorption makes the phase shifts complex. The cross sections are

$$\sigma_{\rm sc} = \frac{2\pi}{k^2}\sum_l (2l+1)(|a_l|^2+|b_l|^2), \quad \sigma_t = \frac{2\pi}{k^2}\sum_l (2l+1)\,{\rm Re}(a_l+b_l).$$

(b) Expand $j_l(z) = z^l/(2l+1)!! + O(z^{l+2})$ and $n_l(z) = -(2l-1)!!/z^{l+1} + O(z^{-l+1})$. For fixed ϵ_r and thus $|y| \ll 1$,

$$a_1 = -\frac{2i}{3}\frac{\epsilon_r - 1}{\epsilon_r + 2}x^3 + O(x^5), \qquad b_1 = -\frac{i}{45}(\epsilon_r - 1)x^5 + O(x^7).$$

Higher electric multipoles start at x^{2l+1}. For unpolarized incidence,

$$\boxed{\frac{d\sigma}{d\Omega} = k^4a^6\left|\frac{\epsilon_r-1}{\epsilon_r+2}\right|^2\frac{1+\cos^2\theta}{2}, \quad \sigma_{\rm sc} = \frac{8\pi}{3}k^4a^6\left|\frac{\epsilon_r-1}{\epsilon_r+2}\right|^2.}$$

This agrees with (10.6)–(10.11). For fixed initial polarization the amplitude is the transverse projection of $k^2a^3(\epsilon_r-1)\boldsymbol{\epsilon}_0/(\epsilon_r+2)$.

(c) The perfect-conductor coefficients are

$$b_l^{\rm PEC} = \psi_l(x)/\xi_l(x), \qquad a_l^{\rm PEC} = \psi_l'(x)/\xi_l'(x),$$

equivalent to (10.68). They are approached if $|\nu| \to \infty$ with sufficient absorption to suppress internal standing-wave resonances (for example ${\rm Im}\,\nu\,x \to \infty$). A strictly real $\epsilon_r \to +\infty$ has no pointwise limit along all sequences: at any zero $\psi_l(\nu x) = 0$, the exact electric coefficient is $\psi_l(x)/\xi_l(x)$, rather than the PEC derivative ratio; such zeros occur for arbitrarily large ν. Away from the shrinking resonance intervals the PEC expressions result.

The small-x expansion in (b) is not uniform in ϵ_r. Taking $\epsilon_r \to \infty$ in that electric-only approximation gives $8\pi k^4a^6/3$, whereas a PEC sphere has magnetic moment $\mathbf{m} = -2\pi a^3\mathbf{H}_i$ and total $10\pi k^4a^6/3$. The condition $|\nu x| \ll 1$ has failed before the conductor limit; interchanging these limits loses the magnetic response.

10.8 Consider the scattering of a plane wave by a nonpermeable sphere of radius a and very good, but not perfect, conductivity following the spherical multipole field approach of Section 10.4. Assume that $ka \ll 1$ and that the skin depth $\delta < a$. The local surface-impedance approximation is controlled for $\delta \ll a$; the comparison at $\delta = a$ in (d) is a formal extrapolation of that model.

(a) Show from the analysis of Section 8.1 that

$$\frac{Z_s}{Z_0} = \frac{k\delta}{2}(1-i)$$

(b) In the long-wavelength limit, show that for $l = 1$ the coefficients $\alpha_\pm(l)$ and $\beta_\pm(l)$ in (10.65) are

$$\alpha_\pm(1) \simeq -\frac{2i}{3}(ka)^3\left[\frac{\left(1-\dfrac{\delta}{a}\right) - i\,\dfrac{\delta}{a}}{\left(1+\dfrac{\delta}{2a}\right) + i\,\dfrac{\delta}{2a}}\right]$$

$$\beta_\pm(1) \simeq \frac{4i}{3}(ka)^3$$

(c) Write out explicitly the differential scattering cross section, correct to *first* order in δ/a and lowest order in ka.

(d) Using (10.61), evaluate the absorption cross section. Show that to first order in δ it is $\sigma_{\rm abs} \simeq 3\pi(k\delta)a^2$. How different is the value if $\delta = a$?

Solution.

Put $x = ka$ and $d = \delta/a$. The local surface-impedance treatment requires $d \ll 1$ for controlled accuracy; a rational function of d does not extend that boundary condition to skin depths comparable with a.

(a) The decaying planar skin-layer solution is proportional to $\exp[-(1-i)s/\delta]$, where s measures depth. Faraday's law gives

$$Z_s = \frac{1-i}{\sigma\delta} = \frac{\mu_0\omega\delta}{2}(1-i), \qquad \boxed{\frac{Z_s}{Z_0} = \frac{k\delta}{2}(1-i).}$$

(b) Put $z_s = Z_s/Z_0$, $\psi_l(x) = x j_l(x)$, and $\xi_l(x) = x h_l^{(1)}(x)$. Multiplying the first of (10.65) by x gives

$$\psi_l + \frac{\alpha_l}{2}\xi_l = iz_s\left(\psi_l' + \frac{\alpha_l}{2}\xi_l'\right), \qquad \alpha_l = -2\frac{\psi_l - iz_s\psi_l'}{\xi_l - iz_s\xi_l'}.$$

Primes here denote derivatives with respect to x. For $x \ll 1$, write $D_l = (2l-1)!!$, so $(2l+1)!! = (2l+1)D_l$. The needed terms are

$$\psi_l \simeq \frac{x^{l+1}}{(2l+1)D_l}, \quad \psi_l' \simeq \frac{(l+1)x^l}{(2l+1)D_l}, \quad \xi_l \simeq -iD_l x^{-l}, \quad \xi_l' \simeq ilD_l x^{-l-1}.$$

Consequently the numerator is $x^l[x - i(l+1)z_s]/[(2l+1)D_l]$, while the denominator is $-iD_l x^{-l-1}(x + ilz_s)$. Their ratio gives

$$\alpha_l \simeq -\frac{2ix^{2l+1}}{(2l+1)[(2l-1)!!]^2}\frac{x - i(l+1)Z_s/Z_0}{x + ilZ_s/Z_0}.$$

For $l = 1$, substitute $Z_s/Z_0 = xd(1-i)/2$:

$$\boxed{\alpha_1 = -\frac{2i}{3}x^3F(d), \quad F(d) = \frac{1-d-id}{1+d/2+id/2}, \qquad \beta_1 = \frac{4i}{3}x^3 + O(x^5).}$$

For β the second boundary equation replaces z_s by $1/z_s$. At $l = 1$ the ratio becomes

$$\frac{x - 2i/z_s}{x + i/z_s} = \frac{xz_s - 2i}{xz_s + i} = -2 + O(x^2 d),$$

which gives the displayed β_1. Neither coefficient depends on helicity.

(c) The electric dipole retains its PEC value; the magnetic dipole is multiplied by F. For unpolarized incidence,

$$\frac{d\sigma}{d\Omega} = k^4a^6\left[\frac{1 + |F|^2/4}{2}(1 + \cos^2\theta) - \mathrm{Re}\, F\cos\theta\right].$$

Since $F = 1 - \frac{3}{2}(1+i)d + O(d^2)$,

$$\boxed{\frac{d\sigma}{d\Omega} = k^4 a^6 \left[\frac{5}{8}(1+\cos^2\theta) - \cos\theta + d\left\{-\frac{3}{8}(1+\cos^2\theta) + \frac{3}{2}\cos\theta\right\} + O(d^2, x^2)\right].}$$

For a specified polarization the answer is $k^4 a^6 |\boldsymbol{\epsilon}_0 - \mathbf{n}(\mathbf{n}\cdot\boldsymbol{\epsilon}_0) + (F/2)\mathbf{n}\times(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)|^2$.

(d) Equation (10.61) gives

$$\sigma_{\rm abs} = \frac{\pi}{2k^2}\sum_l (2l+1)[-2\,\mathrm{Re}(\alpha_l+\beta_l) - |\alpha_l|^2 - |\beta_l|^2].$$

At the leading material-loss order $x^3 d$ only $\mathrm{Re}\,\alpha_1$ is needed. The higher x^6 scattering terms must be accompanied by the radiative x^6 real parts of the coefficients; discarding both consistently avoids a spurious negative loss at $d = 0$. Since

$$\mathrm{Im}\,F = -\frac{3d/2}{1+d+d^2/2}, \qquad \mathrm{Re}\,\alpha_1 = -\frac{x^3 d}{1+d+d^2/2},$$

the impedance model gives

$$\boxed{\sigma_{\rm abs} \simeq \frac{3\pi k\delta a^2}{1+d+d^2/2} = 3\pi k\delta a^2[1+O(d)].}$$

Formally setting $d = 1$ gives 2/5 of the first-order extrapolation, a 60% reduction. This comparison belongs to the rational model; the planar impedance approximation is not controlled at $\delta = a$. The linear result agrees with 10.3.

10.9 In the scattering of light by a gas very near the critical point the scattered light is observed to be "whiter" (i.e., its spectrum is less predominantly peaked toward the blue) than far from the critical point. Show that this can be understood by the fact that the volumes of the density fluctuations become large enough that Rayleigh's law fails to hold. In particular, consider the lowest order approximation to the scattering by a uniform dielectric sphere of radius a whose dielectric constant ϵ_r differs only slightly from unity.

(a) Show that for $ka \gg 1$, the differential cross section is sharply peaked in the forward direction and the total scattering cross section is approximately

$$\sigma \simeq \frac{\pi}{2}(ka)^2|\epsilon_r - 1|^2 a^2$$

with a k^2, rather than k^4, dependence on frequency.

(b) Show that for arbitrary ka the total cross section to lowest order in $(\epsilon_r - 1)$ is the expression given in part a, multiplied by the function

$$F(z) = 1 + 5z^{-2} - \tfrac{7}{2}z^{-4}(1-\cos 2z) - z^{-3}\sin 2z - 4(z^{-2}-z^{-4})\int_0^{2z}\frac{1-\cos t}{t}\,dt$$

where $z = 2ka$. [This result is due to Lord Rayleigh, 1914.]

Solution.

Set $\Delta = \epsilon_r - 1$. In the first Born approximation (10.31), the induced polarization is $\epsilon_0 \Delta \mathbf{E}_i$ throughout the sphere. Its Fourier transform is

$$\int_{r<a} e^{i\mathbf{q}\cdot\mathbf{r}}\, d^3r = 4\pi \frac{\sin(qa) - qa\cos(qa)}{q^3}, \qquad q = 2k\sin(\theta/2).$$

Thus, with $u = qa$,

$$\frac{d\sigma}{d\Omega} = k^4 a^6 |\Delta|^2 \frac{1+\cos^2\theta}{2} \left[\frac{\sin u - u\cos u}{u^3}\right]^2.$$

This is the coefficient of $|\Delta|^2$ at fixed ka. For a uniform approximation at large ka, the phase perturbation across the sphere must also be small: $ka|\Delta| \ll 1$. Small contrast alone does not permit ka to grow without bound.

(a) The form factor is largest for $u \lesssim 1$, hence $\theta \lesssim 1/(ka)$ at large ka. In this forward cone set $(1+\cos^2\theta)/2 \simeq 1$ and $d\Omega \simeq 2\pi u\, du/(ka)^2$:

$$\sigma \simeq 2\pi k^2 a^4 |\Delta|^2 \int_0^\infty \frac{(\sin u - u\cos u)^2}{u^5}\, du = \boxed{\frac{\pi}{2} k^2 a^4 |\Delta|^2}.$$

The finite integral $A_0(z)$ evaluated explicitly below tends to $1/4$ as $z \to \infty$, which supplies the displayed normalization. Compared with Rayleigh's k^4 law, the forward peak keeps its k^4 height but its solid angle shrinks as k^{-2}. Large correlated density fluctuations consequently produce less preferential enhancement of blue light.

(b) Put $z = 2ka$. Since $\cos\theta = 1 - 2u^2/z^2$ and $d\Omega = 8\pi u\, du/z^2$, exactly within Born theory,

$$\sigma = \frac{\pi}{2} k^2 a^4 |\Delta|^2 F(z), \qquad F(z) = 4\int_0^z \left(1 - \frac{2u^2}{z^2} + \frac{2u^4}{z^4}\right) \frac{(\sin u - u\cos u)^2}{u^5}\, du.$$

For an explicit reduction define $A_j(z) = \int_0^z (\sin u - u\cos u)^2 u^{j-5}\, du$ for $j = 0, 2, 4$. Then $F = 4A_0 - 8A_2/z^2 + 8A_4/z^4$. Expand the square into $\frac{1}{2}(1-\cos 2u) - u\sin 2u + \frac{1}{2}u^2(1+\cos 2u)$, integrate the negative powers by parts, and retain the finite lower-end limits before combining terms. In terms of $H(z) = \gamma + \ln(2z) - \mathrm{Ci}(2z)$, the intermediate integrals are

$$\begin{aligned}
A_0 &= \frac{1}{4} - \frac{1}{4z^2} + \frac{\sin 2z}{4z^3} - \frac{1-\cos 2z}{8z^4}, \\
A_2 &= \frac{H}{2} + \frac{\sin 2z}{2z} - \frac{1-\cos 2z}{4z^2} - \frac{1}{2}, \\
A_4 &= \frac{z^2}{4} + \frac{z\sin 2z}{4} + \frac{H}{2} + \frac{5}{8}(\cos 2z - 1).
\end{aligned}$$

Differentiating each returns its defining integrand, and each tends to zero at the lower endpoint. Combining them yields

$$\boxed{\begin{aligned} F(z) = 1 + \frac{5}{z^2} - \frac{7(1-\cos 2z)}{2z^4} - \frac{\sin 2z}{z^3} \\ - 4\left(\frac{1}{z^2} - \frac{1}{z^4}\right)[\gamma + \ln(2z) - \mathrm{Ci}(2z)]. \end{aligned}}$$

Here $\gamma + \ln(2z) - \mathrm{Ci}(2z) = \int_0^{2z} (1-\cos t)\, dt/t$. The expression has a removable singularity at zero; the integral form makes positivity and the limit transparent:

$$F(z) = \frac{4z^2}{27} + O(z^4), \quad \sigma \longrightarrow \frac{8\pi}{27} k^4 a^6 |\Delta|^2, \qquad F(z) \longrightarrow 1 \quad (z \to \infty).$$

The small-z value agrees with the weak-contrast limit of 10.7(b).

10.10 The aperture or apertures in a perfectly conducting plane screen can be viewed as the location of effective sources that produce radiation (the diffracted fields). An aperture whose dimensions are small compared with a wavelength acts as a source of dipole radiation with the contributions of other multipoles being negligible.

(a) Beginning with (10.101) show that the effective electric and magnetic dipole moments can be expressed in terms of integrals of the tangential electric field in the aperture as follows:

$$\mathbf{p} = \epsilon \mathbf{n} \int (\mathbf{x} \cdot \mathbf{E}_{\tan})\, da \tag{9.72}$$

$$\mathbf{m} = \frac{2}{i\omega\mu} \int (\mathbf{n} \times \mathbf{E}_{\tan})\, da$$

where $\mathbf{E}_{\tan}$ is the *exact* tangential electric field in the aperture, $\mathbf{n}$ is the normal to the plane screen, directed into the region of interest, and the integration is over the area of the openings.

(b) Show that the expression for the magnetic moment can be transformed into

$$\mathbf{m} = \frac{2}{\mu} \int \mathbf{x}(\mathbf{n} \cdot \mathbf{B})\, da \tag{9.74}$$

Be careful about possible contributions from the edge of the aperture where some components of the fields are singular if the screen is infinitesimally thick.

Solution.

Let $\mathbf{N}$ be the fixed aperture normal (the problem's $\mathbf{n}$) and $\mathbf{n} = \mathbf{r}/r$ the observation direction. In a homogeneous medium let $v = 1/\sqrt{\epsilon\mu}$ and $k = \omega/v$. The radiation limit of Smythe's exact relation is

$$\mathbf{E}(\mathbf{r}) = \frac{ike^{ikr}}{2\pi r}\, \mathbf{n} \times \int_A (\mathbf{N} \times \mathbf{E}_t) e^{-ik\mathbf{n}\cdot\mathbf{x}'}\, da'.$$

(a) Define $\mathbf{K} = \int_A \mathbf{N} \times \mathbf{E}_t\, da$ and $T_{ij} = \int_A x_i E_{t,j}\, da$, with i, j in the aperture plane. The zeroth term of the exponential gives $ike^{ikr}\mathbf{n} \times \mathbf{K}/(2\pi r)$. Compare this with the magnetic dipole field $-k^2 e^{ikr}\mathbf{n} \times \mathbf{m}/(4\pi\epsilon v r)$:

$$\boxed{\mathbf{m} = \frac{2}{i\omega\mu}\mathbf{K}.}$$

The next term is

$$\mathbf{E}^{(1)} = \frac{k^2 e^{ikr}}{2\pi r}\, \mathbf{n} \times \left[\mathbf{N} \times (T^{\mathsf{T}}\mathbf{n})\right].$$

Its electric dipole part is supplied by the two-dimensional trace $T_{ij} \mapsto \frac{1}{2}(\operatorname{tr} T)(\delta_{ij} - N_i N_j)$, and equals

$$\frac{k^2 e^{ikr}}{4\pi r}(\operatorname{tr} T)[\mathbf{N} - \mathbf{n}(\mathbf{n} \cdot \mathbf{N})].$$

The remaining trace-free tensor is of higher multipole type. Comparison with the electric dipole radiation field gives

$$\boxed{\mathbf{p} = \epsilon \mathbf{N}\ \operatorname{tr} T = \epsilon \mathbf{N} \int_A \mathbf{x}' \cdot \mathbf{E}_t\, da'.}$$

For a quasistatic aperture field $\mathbf{E}_t = -\boldsymbol{\nabla}_t \Phi$, with $\Phi = 0$ on its rim, integration by parts gives $T_{ij} = \delta_{ij} \int_A \Phi\, da$, so the trace selection is especially explicit. The formulas using the exact field

retain the leading dipole terms; higher powers of the aperture size times k generate retardation and higher multipoles.

(b) Set $\mathbf{N} = \widehat{\mathbf{z}}$ and use $i\omega B_z = \partial_x E_y - \partial_y E_x$. Integration by parts gives

$$i\omega \int_A (x, y, 0) B_z \, da = \int_A (-E_y, E_x, 0) \, da + \oint_{\partial A} (x, y, 0)(E_y \nu_x - E_x \nu_y) \, ds.$$

Here $\boldsymbol{\nu}$ is the outward in-plane normal and $E_y \nu_x - E_x \nu_y = \mathbf{E}_t \cdot \mathbf{t}$ is the component *parallel* to the rim. Regulate the integral by a contour a distance η inside a smooth edge. For the finite-energy thin-screen solution, the component perpendicular to the rim can diverge as $\eta^{-1/2}$, but the parallel component vanishes as $\eta^{1/2}$. Thus the line term vanishes, rather than inheriting the singularity of the other component. Taking the limit and substituting in part (a) gives

$$\boxed{\mathbf{m} = \frac{2}{\mu} \int_A \mathbf{x}' (\mathbf{N} \cdot \mathbf{B}) \, da'.}$$

This argument also identifies precisely which edge behavior is needed; one may not simply assume that every electric-field component is bounded.

10.11 A perfectly conducting flat screen occupies half of the x-y plane (i.e., $x < 0$). A plane wave of intensity I_0 and wave number k is incident along the z axis from the region $z < 0$. Discuss the values of the diffracted fields in the plane parallel to the x-y plane defined by $z = Z > 0$. Let the coordinates of the observation point be $(X, 0, Z)$.

(a) Show that, for the usual scalar Kirchhoff approximation and in the limit $Z \gg X$ and $\sqrt{kZ} \gg 1$, the diffracted field is

$$\psi(X, 0, Z, t) \simeq I_0^{1/2} e^{ikZ - i\omega t} \left(\frac{1+i}{2i} \right) \sqrt{\frac{2}{\pi}} \int_{-\xi}^{\infty} e^{it^2} \, dt$$

where $\xi = (k/2Z)^{1/2} X$.

(b) Show that the intensity can be written

$$I = |\psi|^2 = \frac{I_0}{2} \left[(C(\xi) + \tfrac{1}{2})^2 + (S(\xi) + \tfrac{1}{2})^2 \right]$$

where $C(\xi)$ and $S(\xi)$ are the so-called Fresnel integrals. Determine the asymptotic behavior of I for ξ large and positive (illuminated region) and ξ large and negative (shadow region). What is the value of I at $X = 0$? Make a sketch of I as a function of X for fixed Z.

(c) Use the vector formula (10.101) to obtain a result equivalent to that of part a. Compare the two expressions.

Solution.

The open part of the plane is $x' > 0$. Normalize the scalar amplitude so that its squared modulus is intensity. **(a)** For $|X| \ll Z$ and $kZ \gg 1$ the obliquity factor is unity, the slowly varying denominator is Z, and

$$R = |\mathbf{r} - \mathbf{r}'| \simeq Z + \frac{(x' - X)^2 + y'^2}{2Z}.$$

The scalar Kirchhoff integral becomes

$$\psi \simeq \frac{k\sqrt{I_0}\, e^{ikZ-i\omega t}}{2\pi i Z} \int_0^\infty e^{ik(x'-X)^2/(2Z)} dx' \int_{-\infty}^\infty e^{iky'^2/(2Z)} dy'.$$

The second integral is $\sqrt{2\pi Z/k}\, e^{i\pi/4}$. With $u = \sqrt{k/(2Z)}(x'-X)$ and $\xi = \sqrt{k/(2Z)}X$,

$$\boxed{\psi = \sqrt{I_0} e^{ikZ-i\omega t} A(\xi), \qquad A(\xi) = \frac{1-i}{\sqrt{2\pi}} \int_{-\xi}^\infty e^{iu^2}\, du.}$$

This coefficient is precisely the one in the question. Convergence can be defined by a vanishing positive Gaussian damping.

(b) With the question's argument convention define

$$C(\xi) = \sqrt{\frac{2}{\pi}} \int_0^\xi \cos u^2\, du, \qquad S(\xi) = \sqrt{\frac{2}{\pi}} \int_0^\xi \sin u^2\, du.$$

These equal the more usual Fresnel functions at argument $\sqrt{2/\pi}\,\xi$. Both tend to $1/2$ at positive infinity. Thus

$$A = \frac{1-i}{2}[(C+\tfrac12) + i(S+\tfrac12)], \qquad \boxed{\frac{I}{I_0} = \frac12[(C+\tfrac12)^2 + (S+\tfrac12)^2].}$$

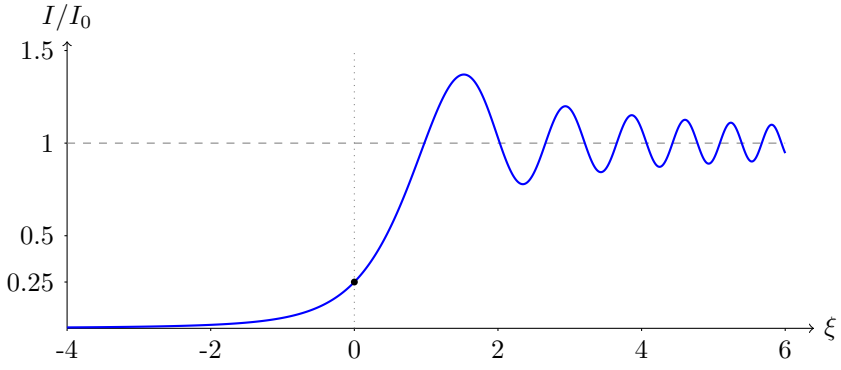
Integration by parts in the tail gives

$$\int_\xi^\infty e^{iu^2} du = e^{i\xi^2}\left[\frac{i}{2\xi} + \frac{1}{4\xi^3} + O(\xi^{-5})\right] \quad (\xi > 0).$$

Consequently

$$\frac{I}{I_0} = \begin{cases} 1 - \dfrac{\cos(\xi^2+\pi/4)}{\sqrt{\pi}\,\xi} + \dfrac{1}{4\pi\xi^2} + O(\xi^{-3}), & \xi \to +\infty, \\ \dfrac{1}{4\pi\xi^2} + O(|\xi|^{-6}), & \xi \to -\infty. \end{cases}$$

At the geometrical edge $A(0) = 1/2$, so $I(0) = I_0/4$. The scale on the plot is $X = \sqrt{2Z/k}\,\xi$.



(c) Set the aperture field equal to $E_0\boldsymbol{\epsilon}$ on $x' > 0$, where $\boldsymbol{\epsilon}\cdot\hat{\mathbf{z}} = 0$. In (10.101) the leading curl acts on e^{ikZ}: $\boldsymbol{\nabla}\times \to ik\hat{\mathbf{z}}\times$. Since $\hat{\mathbf{z}}\times(\hat{\mathbf{z}}\times\boldsymbol{\epsilon}) = -\boldsymbol{\epsilon}$, the coefficient is $-ik/(2\pi Z) = k/(2\pi i Z)$, the same as in (a). Hence

$$\mathbf{E} = E_0 e^{ikZ-i\omega t} A(\xi)\boldsymbol{\epsilon} + O((kZ)^{-1/2}E_0), \qquad \mathbf{B} \simeq \hat{\mathbf{z}}\times\mathbf{E}/c.$$

For a component polarized along x, transversality also requires $E_z = (i/k)\partial_X E_x$ at the next paraxial order; an incident y component has no such leading longitudinal correction. The scalar and vector intensities agree to the retained order, while the vector relation supplies the polarization and these smaller corrections.

10.12 A linearly polarized plane wave of amplitude E_0 and wave number k is incident on a circular opening of radius a in an otherwise perfectly conducting flat screen. The incident wave vector makes an angle α with the normal to the screen. The polarization vector is perpendicular to the plane of incidence.

(a) Calculate the diffracted fields and the power per unit solid angle transmitted through the opening, using the vector Smythe–Kirchhoff formula (10.101) with the assumption that the tangential electric field in the opening is the unperturbed incident field.

(b) Compare your result in part a with the standard scalar Kirchhoff approximation and with the result in Section 10.9 for the polarization vector in the plane of incidence.

Solution.

Take the aperture normal along z, the incident wave vector in the xz plane, and $\mathbf{E}_i = E_0\widehat{\mathbf{y}}e^{ik(x\sin\alpha + z\cos\alpha)}$. Define

$$\xi^2 = (\sin\theta\cos\phi - \sin\alpha)^2 + \sin^2\theta\sin^2\phi, \qquad D(u) = \frac{2J_1(u)}{u}, \quad D(0) = 1.$$

(a) The aperture Fourier transform is

$$\int_A e^{-ik[(n_x - \sin\alpha)x' + n_y y']}\, da' = 2\pi\int_0^a \rho J_0(k\xi\rho)\, d\rho = \pi a^2 D(ka\xi).$$

Because $\widehat{\mathbf{z}} \times \mathbf{E}_i = -E_0\widehat{\mathbf{x}}$, (10.109) gives

$$\boxed{\mathbf{E}_\perp = -\frac{ika^2E_0}{2r}e^{ikr - i\omega t}D(ka\xi)\,\mathbf{n}\times\widehat{\mathbf{x}}, \qquad \mathbf{B}_\perp = \frac{1}{c}\mathbf{n}\times\mathbf{E}_\perp.}$$

Therefore, writing $C_0 = |E_0|^2k^2a^4/(8Z_0)$,

$$\boxed{\frac{dP_\perp}{d\Omega} = C_0(1 - \sin^2\theta\cos^2\phi)D(ka\xi)^2.}$$

Equivalently $C_0 = P_0(ka)^2/(4\pi\cos\alpha)$, where $P_0 = |E_0|^2\pi a^2\cos\alpha/(2Z_0)$ is the incident power on the hole.

(b) The scalar Kirchhoff result and the in-plane polarization result (10.114), at the same E_0, are

$$\frac{dP_{\text{scalar}}}{d\Omega} = C_0\left(\frac{\cos\theta + \cos\alpha}{2}\right)^2 D^2, \qquad \frac{dP_\parallel}{d\Omega} = C_0\cos^2\alpha(1 - \sin^2\theta\sin^2\phi)D^2.$$

All contain the same diffraction factor. The scalar expression cannot distinguish polarizations; the vector factors are the radiation patterns of the corresponding aperture magnetic currents. In the incident direction $(\theta, \phi) = (\alpha, 0)$ all three factors equal $\cos^2\alpha$. At normal incidence the two vector patterns rotate into one another under $\phi \mapsto \phi + \pi/2$. When $ka \gg 1$ their common narrow main beam makes the approximations nearly coincide.

10.13 Discuss the diffraction of a plane wave by a circular hole of radius a, following Section 10.9, but using a vector Kirchhoff approximation based on (10.90) instead of the Smythe formula (10.101).

(a) Show that the diffracted electric field in this approximation differs from (10.112) in two ways, first, that $\cos\alpha$ is replaced by $(\cos\theta+\cos\alpha)/2$, and second, by the addition of a term proportional to $(\mathbf{k}\times\boldsymbol{\epsilon}_3)$. Compare with the obliquity factors $\mathcal{O}$ of the scalar theory.

(b) Evaluate the ratio of the scattered power for this vector Kirchhoff approximation to that of (10.114) for the conditions shown in Fig. 10.14. Sketch the two angular distributions.

Solution.

Use the geometry and $D(ka\xi)$ of 10.12, now with $\mathbf{E}_i/E_0=\widehat{\mathbf{x}}\cos\alpha-\widehat{\mathbf{z}}\sin\alpha$ and $c\mathbf{B}_i/E_0=\widehat{\mathbf{y}}$. **(a)** Before making a Kirchhoff replacement, project (10.90) onto the transverse radiation field, equivalently use (10.92). The aperture electric and magnetic terms then give

$$\begin{aligned}\mathbf{E}_K&=\frac{ikE_0e^{ikr}}{4\pi r}\,\pi a^2D\,\{-\widehat{\mathbf{x}}+n_x\mathbf{n}+\cos\alpha\,\mathbf{n}\times\widehat{\mathbf{y}}\}\\&=\boxed{\frac{iE_0a^2e^{ikr}}{2r}D\left[\frac{\cos\theta+\cos\alpha}{2}\,\mathbf{k}\times\widehat{\mathbf{y}}-\frac{\sin\theta\sin\phi}{2}\,\mathbf{k}\times\widehat{\mathbf{z}}\right].}\end{aligned}$$

The identity used is $-\widehat{\mathbf{x}}+n_x\mathbf{n}=\mathbf{n}\times(n_z\widehat{\mathbf{y}}-n_y\widehat{\mathbf{z}})$. Thus the coefficient $\cos\alpha$ of the Smythe expression becomes $(\cos\theta+\cos\alpha)/2$, accompanied by the stated extra term. The former is the scalar Kirchhoff obliquity factor; the scalar formula cannot represent the second, polarization-dependent term.

There is a useful consistency subtlety. Directly replacing all three fields in (10.90) by discontinuously truncated incident values, without first taking the transverse part, leaves an unphysical radial term proportional to $(\sin\alpha-n_x)\mathbf{n}$. The exact boundary data cancel it. Equation (10.92) implements that exact transversality before the approximation and specifies the radiation prescription used here.

(b) The squared vector factor simplifies to

$$|(\cos\theta+\cos\alpha)\mathbf{n}\times\widehat{\mathbf{y}}-n_y\mathbf{n}\times\widehat{\mathbf{z}}|^2=(\cos\theta+\cos\alpha)^2+\sin^2\alpha\,n_y^2.$$

Consequently

$$\boxed{\frac{(dP/d\Omega)_K}{(dP/d\Omega)_S}=\frac{(\cos\theta+\cos\alpha)^2+\sin^2\alpha\sin^2\theta\sin^2\phi}{4\cos^2\alpha(1-\sin^2\theta\sin^2\phi)}.}$$

At common diffraction zeros this denotes the ratio of the smooth envelopes. Figure 10.14 has $\alpha=\pi/4$, $ka=\pi$ (one wavelength in diameter). In its E plane, $\phi=0$ with signed angle $-\pi/2\le\theta\le\pi/2$,

$$R_E=\tfrac{1}{2}(\cos\theta+1/\sqrt{2})^2.$$

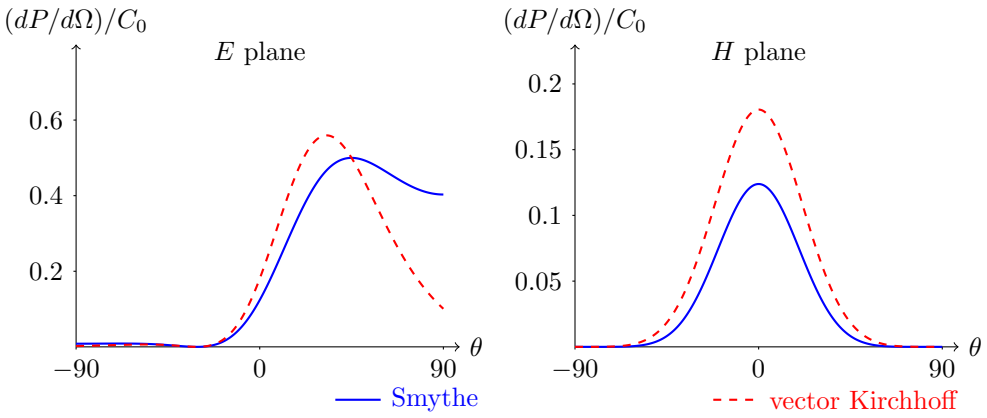
In its perpendicular H plane, $\phi=\pi/2$,

$$R_H=\frac{(\cos\theta+1/\sqrt{2})^2+\frac{1}{2}\sin^2\theta}{2\cos^2\theta}.$$

For example,

θ	0°	30°	45°	60°
R_E	1.4571	1.2374	1.0000	0.7286
R_H	1.4571	1.7332	2.2500	3.6642

At grazing observation $R_E=1/4$; R_H diverges because the Smythe H-plane intensity vanishes, while the Kirchhoff one remains finite. The curves below use the common normalization C_0 from 10.12 and include the diffraction factor, not just these envelopes.



10.14 A rectangular opening with sides of length a and $b \geq a$ defined by $x = \pm(a/2)$, $y = \pm(b/2)$ exists in a flat, perfectly conducting plane sheet filling the x-y plane. A plane wave is normally incident with its polarization vector making an angle β with the long edges of the opening.

(a) Calculate the diffracted fields and power per unit solid angle with the vector Smythe–Kirchhoff relation (10.109), assuming that the tangential electric field in the opening is the incident unperturbed field.

(b) Calculate the corresponding result of the scalar Kirchhoff approximation.

(c) For $b = a$, $\beta = 45°$, $ka = 4\pi$, compute the vector and scalar approximations to the diffracted power per unit solid angle as a function of the angle θ for $\phi = 0$. Plot a graph showing a comparison between the two results.

Solution.

Let $\mathbf{e} = \sin\beta\,\hat{\mathbf{x}} + \cos\beta\,\hat{\mathbf{y}}$, since the long edges run along y, and define $\operatorname{sinc} u = \sin u/u$, continuously equal to one at zero. The rectangular aperture transform is

$$\mathcal{A}(\theta,\phi) = ab\,\operatorname{sinc}\left(\frac{ka}{2}\sin\theta\cos\phi\right)\operatorname{sinc}\left(\frac{kb}{2}\sin\theta\sin\phi\right).$$

(a) In (10.109),

$$\boxed{\mathbf{E}_v = \frac{ikE_0 e^{ikr-i\omega t}}{2\pi r}\mathcal{A}\,\mathbf{n}\times(\hat{\mathbf{z}}\times\mathbf{e}), \qquad \mathbf{B}_v = \mathbf{n}\times\mathbf{E}_v/c.}$$

The angular distribution is therefore

$$\boxed{\frac{dP_v}{d\Omega} = \frac{|E_0|^2k^2|\mathcal{A}|^2}{8\pi^2 Z_0}\left[\cos^2\theta + \sin^2\theta\sin^2(\phi+\beta)\right].}$$

The factor follows from $|\mathbf{n}\times(\hat{\mathbf{z}}\times\mathbf{e})|^2 = \cos^2\theta + (\mathbf{n}\cdot\mathbf{e})^2$.

(b) The scalar field has the same aperture transform, multiplied by $(1+\cos\theta)/2$ instead of the vector factor, hence

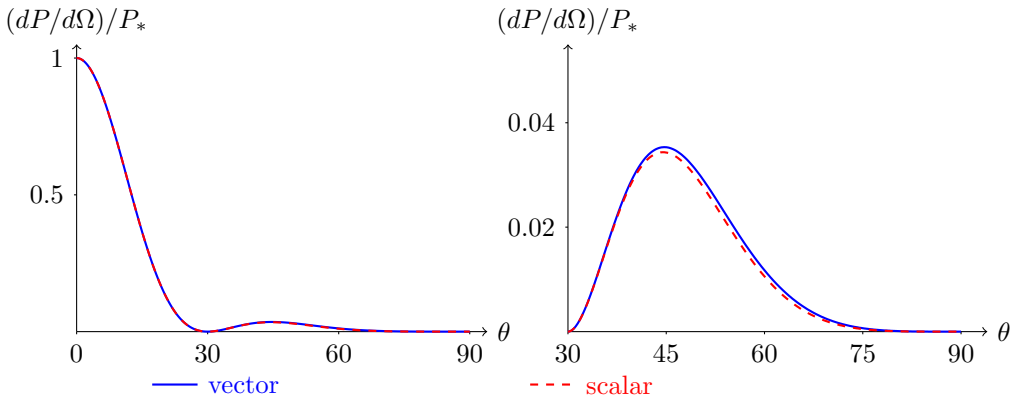
$$\boxed{\frac{dP_s}{d\Omega} = \frac{|E_0|^2k^2|\mathcal{A}|^2}{8\pi^2 Z_0}\left(\frac{1+\cos\theta}{2}\right)^2.}$$

Both give the same intensity on axis; the scalar result is independent of β.

(c) For the parameters specified, normalize to the common axial value $P_* = |E_0|^2 k^2 a^4/(8\pi^2 Z_0)$. Then

$$\boxed{\frac{1}{P_*}\frac{dP_v}{d\Omega} = \frac{1+\cos^2\theta}{2}\,\mathrm{sinc}^2(2\pi\sin\theta), \qquad \frac{1}{P_*}\frac{dP_s}{d\Omega} = \frac{(1+\cos\theta)^2}{4}\,\mathrm{sinc}^2(2\pi\sin\theta).}$$

The common zeros occur at $\theta = 30°, 90°$. Their difference is the nonnegative quantity $\frac{1}{4}(1-\cos\theta)^2\,\mathrm{sinc}^2(2\pi\sin\theta)$. At $\theta = 45°, 60°$ the vector values are $0.03530, 0.01174$ and the scalar values $0.03429, 0.01057$.



10.15 A cylindrical coaxial transmission line of inner radius a and outer radius b has its axis along the negative z axis. Both inner and outer conductors end at $z = 0$, and the outer one is connected to an infinite plane copper flange occupying the whole x-y plane (except for the annulus of inner radius a and outer radius b around the origin). The transmission line is excited at frequency ω in its dominant TEM mode, with the peak voltage between the cylinders being V. Use the vector Smythe–Kirchhoff approximation to discuss the radiated fields, the angular distribution of radiation, and the total power radiated.

Solution.

Assume vacuum in the coax and neglect the finite surface resistance of the copper flange. The prescribed Smythe–Kirchhoff approximation uses the unperturbed TEM aperture field

$$\mathbf{E}_t(\rho, 0) = \frac{V}{\rho\ln(b/a)}\widehat{\boldsymbol{\rho}}, \qquad a < \rho < b.$$

Here V is a *peak phasor voltage*; time-averaged powers contain $1/2$. Put $\ell = \ln(b/a)$ and $q = k\sin\theta$. Choosing the observation azimuth as zero during the integral,

$$\begin{aligned}\int_A \widehat{\mathbf{z}} \times \mathbf{E}_t\, e^{-i\mathbf{k}_t\cdot\mathbf{r}'}\,da' &= \frac{V}{\ell}\int_a^b d\rho \int_0^{2\pi} \widehat{\boldsymbol{\varphi}}' e^{-iq\rho\cos\varphi'}\,d\varphi' \\ &= -\frac{2\pi i V}{q\ell}[J_0(qa) - J_0(qb)]\widehat{\boldsymbol{\varphi}}.\end{aligned}$$

Since $\mathbf{n} \times \widehat{\boldsymbol{\varphi}} = -\widehat{\boldsymbol{\theta}}$, the radiation fields are

$$\boxed{E_\theta = -\frac{V e^{ikr - i\omega t}}{r\ell}\,\frac{J_0(ka\sin\theta) - J_0(kb\sin\theta)}{\sin\theta}, \qquad B_\varphi = E_\theta/c.}$$

All other leading $1/r$ components vanish. The apparent axial singularity is removable: the numerator is $k^2(b^2-a^2)\sin^2\theta/4+O(\sin^4\theta)$, so $E_\theta=0$ on axis. The polarization is meridional and the intensity is azimuthally symmetric:

$$\boxed{\frac{dP}{d\Omega}=\frac{|V|^2}{2Z_0\ell^2}\frac{[J_0(ka\sin\theta)-J_0(kb\sin\theta)]^2}{\sin^2\theta}.}$$

Only the upper hemisphere is radiated into, giving the convergent one-dimensional formula, valid at arbitrary ka, kb within this model,

$$\boxed{P=\frac{\pi|V|^2}{Z_0\ell^2}\int_0^{\pi/2}\frac{[J_0(ka\sin\theta)-J_0(kb\sin\theta)]^2}{\sin\theta}\,d\theta.}$$

For $kb\ll 1$,

$$P\simeq\frac{\pi|V|^2k^4(b^2-a^2)^2}{24Z_0\ell^2},\qquad \frac{dP}{d\Omega}\simeq\frac{|V|^2k^4(b^2-a^2)^2}{32Z_0\ell^2}\sin^2\theta.$$

This is the pattern of a normal electric dipole; the zeroth aperture magnetic moment vanishes by rotational symmetry.

The incident TEM power for the same voltage is $P_{\rm TEM}=|V|^2/(2Z_c)=\pi|V|^2/(Z_0\ell)$, with $Z_c=Z_0\ell/(2\pi)$. Thus the displayed integral divided by ℓ is the approximate transmission fraction. At large ka, kb with fixed b/a, set $u=k\sin\theta$ in its forward-dominated contribution and use

$$\int_0^\infty[J_0(au)-J_0(bu)]^2\,\frac{du}{u}=\ln(b/a)$$

to obtain $P/P_{\rm TEM}\to 1$. To verify this last integral, call its left side $I(a,b)$. Hankel inversion of a disk indicator gives $\int_0^\infty J_0(au)J_1(bu)\,du=1/b$ for $a<b$, while $\int_0^\infty J_0(bu)J_1(bu)\,du=1/(2b)$ follows by integrating $-\frac{1}{2}\,d[J_0(bu)^2]/du$. Therefore

$$\partial_b I=2\int_0^\infty[J_0(au)-J_0(bu)]J_1(bu)\,du=\frac{1}{b},\qquad I(a,a)=0,$$

which proves $I=\ln(b/a)$ (oscillatory integrals can be Abel regulated). At small apertures the actual reflected TEM wave changes the aperture field; prescribing the unperturbed field does not determine the true input reflection coefficient.

10.16

(a) Show from (10.125) that the integral of the shadow scattering differential cross section, summed over outgoing polarizations, can be written in the short-wavelength limit as

$$\sigma_{\rm sh}=\int d^2x_\perp\int d^2x'_\perp\cdot\frac{1}{4\pi^2}\int e^{i(\mathbf{x}_\perp-\mathbf{x}'_\perp)\cdot\mathbf{k}_\perp}\,d^2k_\perp$$

and therefore is equal to the projected area of the scatterer, independent of its detailed shape.

(b) Apply the optical theorem to the "shadow" amplitude (10.125) to obtain the total cross section under the assumption that in the forward direction the contribution from the illuminated side of the scatterer is negligible in comparison.

Solution.

Let $\chi_A(\mathbf{x}_\perp)$ be the indicator of the projected shadow. Equation (10.125), divided by incident amplitude, reads in its narrow forward cone

$$\mathbf{f}_{\rm sh} \simeq \frac{ik}{2\pi}\boldsymbol{\epsilon}_0 \int_A e^{-i\mathbf{k}_\perp\cdot\mathbf{x}_\perp} d^2x_\perp .$$

(a) In that cone the outgoing polarization sum tends to $|\boldsymbol{\epsilon}_0|^2 = 1$, and $d\Omega \simeq d^2k_\perp/k^2$. The transverse wave-number domain can be extended to the entire plane at the retained short-wave order. Expanding the modulus squared gives

$$\sigma_{\rm sh} \simeq \int_A d^2x_\perp \int_A d^2x'_\perp \frac{1}{4\pi^2}\int_{\mathbb{R}^2} e^{i(\mathbf{x}_\perp-\mathbf{x}'_\perp)\cdot\mathbf{k}_\perp}\, d^2k_\perp .$$

The inner Fourier integral is $\delta^{(2)}(\mathbf{x}_\perp - \mathbf{x}'_\perp)$, or equivalently Parseval's theorem gives $\int |\chi_A|^2 d^2x_\perp$. Hence

$$\boxed{\sigma_{\rm sh} = A_{\rm proj} \quad \text{to leading short-wave order.}}$$

Only the projected area enters, including any holes in that projection.

(b) At $\mathbf{k}_\perp = 0$, $f_{\rm sh}(0) = ikA_{\rm proj}/(2\pi)$ in the original polarization. Under the stated dominance assumption the optical theorem gives

$$\boxed{\sigma_t = \frac{4\pi}{k}\,\mathrm{Im}\, f(0) \simeq 2A_{\rm proj}.}$$

One projected area is the shadow diffraction contribution. The other is removed through reflection and/or absorption on the illuminated surface; its division depends on the material. This is compatible with the area result in (a), since (a) is only one contribution to extinction.

10.17

(a) Using the approximate amplitudes of Section 10.10, show that, for a linearly polarized plane wave of wave number k incident on a perfectly conducting sphere of radius a in the limit of large ka, the differential scattering cross section in the E plane ($\boldsymbol{\epsilon}_0$, $\mathbf{k}_0$, and $\mathbf{k}$ coplanar) is

$$\frac{d\sigma}{d\Omega}\,(E\text{ plane}) = \frac{a^2}{4}\Big[4\cot^2\theta\, J_1^2(ka\sin\theta) + 1 - 4\cot\theta\, J_1(ka\sin\theta)\sin\left(2\,ka\,\sin\frac{\theta}{2}\right)\Big]$$

and in the H plane ($\boldsymbol{\epsilon}_0$ perpendicular to $\mathbf{k}_0$ and $\mathbf{k}$) is

$$\frac{d\sigma}{d\Omega}\,(H\text{ plane}) = \frac{a^2}{4}\Big[4\,\mathrm{cosec}^2\theta\, J_1^2(ka\sin\theta) + 1 + 4\,\mathrm{cosec}\,\theta\, J_1(ka\sin\theta)\sin\left(2\,ka\,\sin\frac{\theta}{2}\right)\Big]$$

(The dashed curve in Fig. 10.16 is the average of these two expressions.)

(b) Look up the exact calculations in *King and Wu* (Appendix) or *Bowman, Senior and Uslenghi* (pp. 402–405). Are the *qualitative* aspects of the interference between the diffractive and reflective amplitudes exhibited in part a in agreement with the exact results? What about quantitative agreement?

Solution.

Put $x = ka$, $\chi = 2x\sin(\theta/2)$ and $g = J_1(x\sin\theta)/\sin\theta$, interpreted continuously at a forward axis. The amplitudes here have the incident electric amplitude divided out.

(a) The shadow amplitude (10.127) is $iag\,\boldsymbol{\epsilon}_0$, with projection onto the outgoing transverse polarization understood. The reflected amplitude (10.132) is $(a/2)e^{-i\chi}\boldsymbol{\epsilon}_r$, where $\boldsymbol{\epsilon}_r = -\boldsymbol{\epsilon}_0 + 2(\mathbf{n}_s \cdot \boldsymbol{\epsilon}_0)\mathbf{n}_s$ and $\mathbf{n}_s \parallel \mathbf{n} - \mathbf{n}_0$.

For observation in the xz plane, $\mathbf{n}_s = (\cos(\theta/2), 0, -\sin(\theta/2))$. An incident x polarization has $\boldsymbol{\epsilon}_r = \widehat{\boldsymbol{\theta}}$ and $\widehat{\boldsymbol{\theta}} \cdot \hat{\mathbf{x}} = \cos\theta$. An incident y polarization has $\boldsymbol{\epsilon}_r = -\hat{\mathbf{y}}$. Thus

$$f_E = \frac{a}{2}[e^{-i\chi} + 2ig\cos\theta], \qquad f_H = \frac{a}{2}[-e^{-i\chi} + 2ig].$$

Squaring the coherent sums, including their relative phases, gives

$$\boxed{\frac{d\sigma_E}{d\Omega} = \frac{a^2}{4}[1 + 4g^2\cos^2\theta - 4g\cos\theta\sin\chi],} \qquad \boxed{\frac{d\sigma_H}{d\Omega} = \frac{a^2}{4}[1 + 4g^2 + 4g\sin\chi].}$$

These are the requested cotangent and cosecant expressions. Their average is the unpolarized approximation in Fig. 10.16. The stationary-phase reflected amplitude requires $2ka\sin(\theta/2) \gg 1$; the shadow expression was derived for a narrow forward cone. Extending both formulas over all angles is the plotted approximation, and is not a uniform large-ka expansion, particularly near backward scattering.

(b) Jackson's Fig. 10.16, printed p. 500, explicitly reproduces the *King–Wu* exact result at $ka = 10$. It shows much stronger dips below the geometrical-optics level than the approximate unpolarized curve. The discussion on printed p. 499 identifies stronger interference in the E plane than in the H plane. For a quantitative comparison in the separate planes, evaluate the exact conducting-sphere series (10.57), (10.68), without a high-frequency approximation:

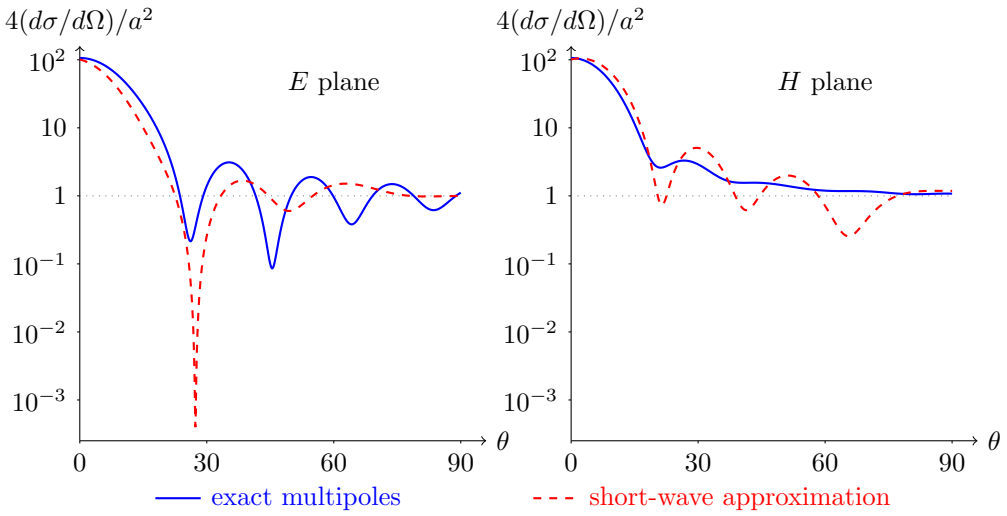
$$a_l = \frac{\psi_l'(x)}{\xi_l'(x)}, \qquad b_l = \frac{\psi_l(x)}{\xi_l(x)}, \qquad f_{E,H} = \frac{i}{k}S_{E,H},$$

$$S_E = \sum_{l=1}^{\infty} \frac{2l+1}{l(l+1)}(a_l\tau_l + b_l\pi_l),$$

$$S_H = \sum_{l=1}^{\infty} \frac{2l+1}{l(l+1)}(a_l\pi_l + b_l\tau_l).$$

Here $\pi_l = P_l'(\cos\theta)$, $\tau_l = l\cos\theta\,\pi_l - (l+1)\pi_{l-1}$, with $\pi_0 = 0, \pi_1 = 1$. For numerical generation use $\pi_{l+1} = [(2l+1)\cos\theta\,\pi_l - (l+1)\pi_{l-1}]/l$. This recurrence remains finite at the axes. At $x = 10$, increasing the sum from $l_{\max} = 30$ to 40 does not change the reported digits. The table and curves use $4(d\sigma/d\Omega)/a^2$:

θ	exact E	approx. E	exact H	approx. H
0°	106.358	101.000	106.358	101.000
10°	51.808	31.267	41.859	58.719
20°	5.668	2.491	2.754	1.348
30°	1.398	0.258	2.900	5.060
45°	0.0992	0.9361	1.551	1.094
60°	0.9579	1.4389	1.193	0.7108
90°	1.1133	1.0000	1.0773	1.1814



The coherent two-amplitude picture correctly predicts oscillations about the reflection level and a polarization-dependent interference sign. Quantitative agreement at $ka = 10$ is poor for the locations and depths of the minima; at 45° the E-plane approximation is about 9.4 times the exact value. The exact H pattern settles toward the reflection level with much smaller oscillations than the exact E pattern. These numerical comparisons use the converged exact series; the historical graph inspected is the King–Wu reproduction in Jackson, rather than a claim to have accessed the original Appendix pages.

10.18 Discuss the diffraction due to a small, circular hole of radius a in a flat, perfectly conducting sheet, assuming that $ka \ll 1$.

(a) If the fields near the screen on the incident side are normal $\mathbf{E}_0 e^{-i\omega t}$ and tangential $\mathbf{B}_0 e^{-i\omega t}$, show that the diffracted electric field in the Fraunhofer zone is

$$\mathbf{E} = \frac{e^{ikr-i\omega t}}{3\pi r} k^2 a^3 \left[2c \frac{\mathbf{k}}{k} \times \mathbf{B}_0 + \frac{\mathbf{k}}{k} \times \left(\mathbf{E}_0 \times \frac{\mathbf{k}}{k} \right) \right]$$

where $\mathbf{k}$ is the wave vector in the direction of observation.

(b) Determine the angular distribution of the diffracted radiation and show that the total power transmitted through the hole is

$$P = \frac{2}{27\pi Z_0} k^4 a^6 (4c^2 B_0^2 + E_0^2)$$

Solution.

The fields $\mathbf{E}_0, \mathbf{B}_0$ mean the fields on the incident side of the *closed* screen, locally away from the small perturbation caused by the hole. They need not equal the amplitudes of an incident traveling wave alone.

(a) The circular-hole static polarizabilities (9.75), printed p. 424, have opposite signs as viewed from the two sides of the screen. On the transmitted side the effective moments are

$$\mathbf{p} = \frac{4}{3}\epsilon_0 a^3 \mathbf{E}_0, \qquad \mathbf{m} = -\frac{8}{3\mu_0} a^3 \mathbf{B}_0.$$

Substituting them into the electric and magnetic dipole radiation fields (10.2) gives

$$\mathbf{E} = \frac{k^2 e^{ikr-i\omega t}}{4\pi\epsilon_0 r}\left[\mathbf{p} - \mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \frac{\mathbf{n}\times\mathbf{m}}{c}\right]$$
$$= \boxed{\frac{k^2 a^3 e^{ikr-i\omega t}}{3\pi r}[2c\,\mathbf{n}\times\mathbf{B}_0 + \mathbf{E}_0 - \mathbf{n}(\mathbf{n}\cdot\mathbf{E}_0)]}.$$

Here $1/(\mu_0\epsilon_0 c) = c$ and $\mathbf{E}_0 - \mathbf{n}(\mathbf{n}\cdot\mathbf{E}_0) = \mathbf{n}\times(\mathbf{E}_0\times\mathbf{n})$. Also $\mathbf{B}_{\rm rad} = \mathbf{n}\times\mathbf{E}/c$.

(b) For possibly complex phasors the angular distribution is

$$\boxed{\frac{dP}{d\Omega} = \frac{k^4 a^6}{18\pi^2 Z_0}\left[4c^2\big(|\mathbf{B}_0|^2 - |\mathbf{n}\cdot\mathbf{B}_0|^2\big) + |\mathbf{E}_0|^2 - |\mathbf{n}\cdot\mathbf{E}_0|^2 + 4c\ \mathrm{Re}\{\mathbf{E}_0^*\cdot(\mathbf{n}\times\mathbf{B}_0)\}\right].}$$

Integrate over the transmitted hemisphere. Its second moments are $\int n_i n_j\, d\Omega = (2\pi/3)\delta_{ij}$. Thus each isolated dipole pattern integrates to $4\pi/3$ times its squared moment. Since $\mathbf{E}_0$ is normal and $\mathbf{B}_0$ tangential, the interference term is proportional to n_x or n_y; its azimuthal integral vanishes. Therefore

$$\boxed{P = \frac{2k^4 a^6}{27\pi Z_0}(4c^2|\mathbf{B}_0|^2 + |\mathbf{E}_0|^2).}$$

For real amplitudes this is the expression stated. It is half the full-space dipole power, since the aperture emits here into one side only.

10.19 Specialize the discussion of Problem 10.18 to the diffraction of a plane wave by the small, circular hole. Treat the general case of oblique incidence at an angle α to the normal, with polarization in and perpendicular to the plane of incidence.

(a) Calculate the angular distributions of the diffracted radiation and compare them to the vector Smythe–Kirchhoff approximation results of Section 10.9 and Problem 10.12 in the limit $ka \ll 1$.

(b) For the conditions of Fig. 10.14 (but for $ka \ll 1$) compute the diffraction intensity in the plane of incidence and compare the relative values with the solid curve in Fig. 10.14. (Use a protractor and a ruler to read off the values from Fig. 10.14 at several angles.)

(c) Show that the transmission coefficient [defined above (10.116)] for the two states of polarization are

$$T_\parallel = \frac{64}{27\pi^2}\,(ka)^4\left(\frac{4+\sin^2\alpha}{4\cos\alpha}\right)$$
$$T_\perp = \frac{64}{27\pi^2}\,(ka)^4\cos\alpha$$

Note that these transmission coefficients are a factor $(ka)^2$ smaller than those given by the vector Smythe–Kirchhoff approximation in the same limit.

Solution.

Let the transmitted side be $z > 0$, let incidence be in the xz plane, and set $s = \sin\alpha$, $c_\alpha = \cos\alpha$. Use a common incident traveling-wave electric amplitude E_i.

(a) For in-plane (TM) polarization, $\mathbf{E}_i = E_i(c_\alpha, 0, -s)$ and $c\mathbf{B}_i = E_i\widehat{\mathbf{y}}$. Adding the reflected wave at a closed PEC plane cancels tangential E and doubles normal E and tangential B:

$$\mathbf{E}_0 = -2E_i s\,\widehat{\mathbf{z}}, \qquad c\mathbf{B}_0 = 2E_i\widehat{\mathbf{y}}.$$

For perpendicular (TE) polarization, $\mathbf{E}_i = E_i\widehat{\mathbf{y}}$ and $c\mathbf{B}_i = E_i(-c_\alpha, 0, s)$, hence

$$\mathbf{E}_0 = 0, \qquad c\mathbf{B}_0 = -2E_i c_\alpha\,\widehat{\mathbf{x}}.$$

Insert these into 10.18. With $C_B = 2|E_i|^2k^4a^6/(9\pi^2Z_0)$,

$$\boxed{\begin{aligned}\frac{dP_\parallel}{d\Omega} &= C_B[4(1-n_y^2) + s^2(1-n_z^2) - 4sn_x],\\ \frac{dP_\perp}{d\Omega} &= 4C_Bc_\alpha^2(1-n_x^2).\end{aligned}}$$

The TM interference term follows from $(\mathbf{n}\times\widehat{\mathbf{y}})\cdot[\widehat{\mathbf{z}} - n_z\mathbf{n}] = n_x$. This step fixes its sign.

For comparison, set $D(ka\xi) \to 1$ in the Smythe–Kirchhoff results of 10.12:

$$\left(\frac{dP_\parallel}{d\Omega}\right)_S = C_0c_\alpha^2(1-n_y^2), \qquad \left(\frac{dP_\perp}{d\Omega}\right)_S = C_0(1-n_x^2), \quad C_0 = \frac{|E_i|^2k^2a^4}{8Z_0}.$$

The exact leading small-hole result scales as k^4a^6, while the unperturbed-aperture model scales as k^2a^4. The TE shapes agree, with a different amplitude; the TM result also includes a normal electric dipole and its interference with the magnetic dipole, which changes the shape.

(b) In the E plane let θ be signed, so $\mathbf{n} = (\sin\theta, 0, \cos\theta)$. At $\alpha = 45°$,

$$\frac{dP_\parallel}{d\Omega} = C_B(2 - \sin\theta/\sqrt{2})^2.$$

Normalize each curve to its own value at $\theta = 45°$. The source figure was measured using rays from the aperture origin on a 400-dpi rendering of Jackson's printed p. 495; radial length is intensity in this polar diagram. The retained crop and measured radial intervals are in `supplementary/ch10-source`.

θ	small hole $ka \ll 1$	Fig. 10.14 solid, ruler	Smythe formula $ka = \pi$
0°	1.778	0.25	0.248
15°	1.467	0.58	0.596
30°	1.205	0.89	0.899
45°	1.000	1.00	1.000
60°	0.856	0.94	0.939
75°	0.771	0.84	0.846

The ruler readings have approximately 0.02–0.03 uncertainty from line thickness and locating the origin. The final column is an independent evaluation of $D(\pi|\sin\theta - 1/\sqrt{2}|)^2$ and checks the interpretation of the printed polar graph. A small hole radiates broadly, with greater TM intensity toward negative signed angles, whereas the wavelength-size hole in Fig. 10.14 has its main lobe near the incident direction. Merely shrinking ka in the unperturbed-aperture field model would give a constant E-plane envelope and also misses this dipole interference.

(c) The incident power intercepted by the aperture is $P_0 = |E_i|^2\pi a^2c_\alpha/(2Z_0)$. The hemispherical integrals from 10.18 give

$$P_\parallel = \frac{8|E_i|^2k^4a^6}{27\pi Z_0}(4+s^2), \qquad P_\perp = \frac{32|E_i|^2k^4a^6}{27\pi Z_0}c_\alpha^2.$$

Dividing each by P_0 yields

$$\boxed{T_\parallel = \frac{64}{27\pi^2}(ka)^4\frac{4+\sin^2\alpha}{4\cos\alpha}, \qquad T_\perp = \frac{64}{27\pi^2}(ka)^4\cos\alpha.}$$

For comparison $T_{\parallel,S} = (ka)^2 c_\alpha/3$ and $T_{\perp,S} = (ka)^2/(3c_\alpha)$; at fixed nongrazing angle, the exact coefficients are smaller by an additional order $(ka)^2$. The apparent divergence of the TM coefficient at grazing incidence comes from normalizing by $P_0 \propto \cos\alpha$, not divergent radiated power; the limiting grazing setup requires care with illumination.

10.20 A suspension of transparent fibers in a clear liquid is modeled as a collection of scatterers, each being a right circular cylinder of radius a and length L of uniform dielectric material whose electric susceptibility differs from the surrounding medium by a small fractional amount $\delta\epsilon/\epsilon$.

(a) Show that using the amplitude to first order in $\delta\epsilon/\epsilon$, the scattering cross section per scatterer for unpolarized radiation of wave number k is

$$\frac{d\sigma}{d\Omega} = \left|\frac{\delta\epsilon}{\epsilon}\right|^2 \frac{k^4a^4L^2}{32}(1+\cos^2\theta)\left|\frac{2J_1(q_\perp a)}{q_\perp a}\cdot\frac{\sin(q_\parallel L/2)}{q_\parallel L/2}\right|^2$$

where $J_1(z)$ is the Bessel function of order unity and $q_\parallel$ ($q_\perp$) is the component of the wave number transfer parallel (perpendicular) to the cylinder axis.

(b) In the limit of very slender cylinders ($ka \ll 1$), show that the scattering cross section, averaged over the orientation of the cylinder (appropriate for an ensemble of randomly oriented fibers), is

$$\left\langle\frac{d\sigma}{d\Omega}\right\rangle = \left|\frac{\delta\epsilon}{\epsilon}\right|^2 \frac{k^4a^4L^2}{32}(1+\cos^2\theta)\left[\frac{2}{qL}\,\mathrm{Si}(qL) - \left(\frac{\sin(qL/2)}{qL/2}\right)^2\right]$$

where $\mathrm{Si}(x) = \int_0^x[(\sin x)/x]\,dx$ is the sine integral (*Abramowitz and Stegun*, p. 231) and $q^2 = 2k^2(1-\cos\theta)$.

(c) Plot the square-bracketed quantity in part b as a function of q^2L^2 on the range $(0, 100)$. Verify that the cross section is the expected one when $kL \ll 1$ and show that when $kL \gg 1$ (but $ka \ll 1$) the total scattering cross section is approximately

$$\sigma_{\mathrm{scatt}} \approx \frac{11\pi^2}{60}\left|\frac{\delta\epsilon}{\epsilon}\right|^2 k^3a^4L\left\{1+O\left[\frac{1}{kL},\frac{\ln(kL)}{kL}\right]\right\}$$

Comment on the frequency dependence.

Solution.

Write $\eta = \delta\epsilon/\epsilon$, let $\hat{\mathbf{u}}$ be the fiber axis, and use the host-medium wave number k. The first Born approximation uses the incident field in the excess polarization. Besides $|\eta| \ll 1$, its use for long objects requires the accumulated internal phase perturbation to remain small along relevant propagation paths.

(a) With the cylinder centered at the origin, separate the volume Fourier transform into axial and transverse factors:

$$\begin{aligned}\mathcal{V}(\mathbf{q}) &= \int_{-L/2}^{L/2} e^{iq_\parallel z}\,dz \int_0^a \rho\,d\rho \int_0^{2\pi} e^{iq_\perp\rho\cos\varphi}\,d\varphi \\ &= \pi a^2 L\,\frac{2J_1(q_\perp a)}{q_\perp a}\frac{\sin(q_\parallel L/2)}{q_\parallel L/2}.\end{aligned}$$

The normalized polarized amplitude (10.31) is $f_{\epsilon'\epsilon} = k^2\eta\mathcal{V}(\boldsymbol{\epsilon}'^{*}\cdot\boldsymbol{\epsilon})/(4\pi)$. The sum over outgoing and average over incident polarizations is $(1+\cos^2\theta)/2$. Therefore

$$\boxed{\frac{d\sigma}{d\Omega} = |\eta|^2\frac{k^4a^4L^2}{32}(1+\cos^2\theta)\left|\frac{2J_1(q_\perp a)}{q_\perp a}\frac{\sin(q_\parallel L/2)}{q_\parallel L/2}\right|^2.}$$

It is the *amplitude* that is first order in η; the leading cross section is second order.

(b) If $ka \ll 1$, then $q_\perp a \le 2ka \ll 1$ and the Bessel factor is one to relative order $(ka)^2$. For an isotropic axis orientation, $\mu = \widehat{\mathbf{q}}\cdot\widehat{\mathbf{u}}$ is uniform on $[-1,1]$. With $u = qL$ the remaining average is

$$H(u) = \frac{1}{2}\int_{-1}^{1}\left[\frac{\sin(u\mu/2)}{u\mu/2}\right]^2 d\mu = \frac{2}{u}\int_0^{u/2}\frac{\sin^2 t}{t^2}\,dt.$$

Integrating by parts,

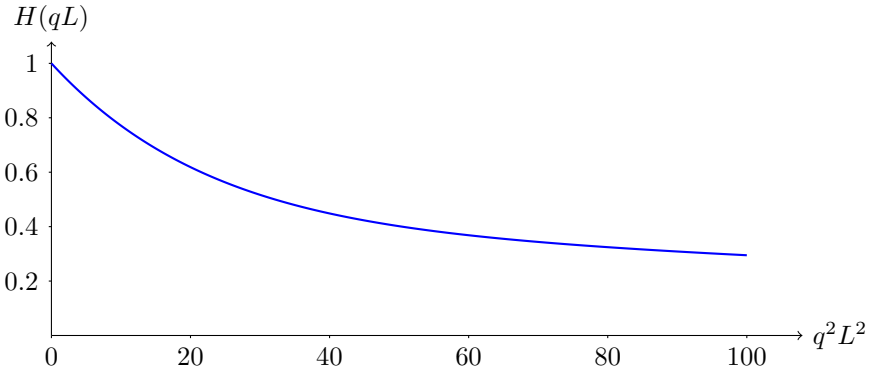
$$\int_0^v \frac{\sin^2 t}{t^2}\,dt = -\frac{\sin^2 v}{v} + \int_0^v \frac{\sin 2t}{t}\,dt = \mathrm{Si}(2v) - \frac{\sin^2 v}{v}.$$

Consequently

$$\boxed{H(u) = \frac{2\,\mathrm{Si}(u)}{u} - \left[\frac{\sin(u/2)}{u/2}\right]^2,}$$

which multiplies the prefactor in (a). Kinematics gives $q^2 = |\mathbf{k}_0 - \mathbf{k}|^2 = 2k^2(1-\cos\theta)$.

(c) The plot uses $u^2 = q^2L^2$ on its horizontal axis.



Expanding the orientation integral term by term gives

$$H(u) = 1 - \frac{u^2}{36} + \frac{u^4}{1800} + O(u^6).$$

Thus for $kL \ll 1$ the object has Born electric dipole moment $\epsilon\eta\pi a^2 L\,\mathbf{E}_i$ and

$$\boxed{\sigma_{\mathrm{sc}} = \frac{\pi}{6}|\eta|^2k^4a^4L^2[1+O((kL)^2,(ka)^2)]}.$$

Indeed $\int(1+\cos^2\theta)d\Omega = 16\pi/3$ supplies this coefficient.

For $u \gg 1$, integration by parts in the sine-integral tail gives $H(u) = \pi/u - 2/u^2 + O(u^{-3})$; the leading oscillatory u^{-2} terms cancel between the two terms defining H. Inserting $\pi/(qL)$ into the total angular integral yields

$$\sigma_{\mathrm{sc}} \simeq \frac{\pi^2}{16}|\eta|^2 k^3 a^4 L \int_{-1}^{1} \frac{1+\mu^2}{\sqrt{2(1-\mu)}}\, d\mu,$$

$$\int_{-1}^{1} \frac{1+\mu^2}{\sqrt{2(1-\mu)}}\, d\mu = 2\int_0^1 [1+(1-2t^2)^2]\, dt = 4 - \frac{8}{3} + \frac{8}{5} = \frac{44}{15},$$

where $t = \sqrt{(1-\mu)/2}$. Therefore

$$\boxed{\sigma_{\mathrm{sc}} \simeq \frac{11\pi^2}{60}|\eta|^2 k^3 a^4 L.}$$

The forward cap $qL \lesssim 1$ contributes a relative $O((kL)^{-1})$ correction; the u^{-2} term integrated outside that cap gives $O(\ln(kL)/(kL))$. These explain the stated error orders. For frequency-independent contrast and host index the long, thin random fibers have a k^3 law, between the point-particle k^4 law and the large weak-contrast sphere's k^2 law in 10.9.

11 Chapter 11. Special Theory of Relativity

Problems

Conventions for the solutions. This chapter of the source uses Gaussian electromagnetic units. We use $g_{\alpha\beta} = \mathrm{diag}(1,-1,-1,-1)$, $x^0 = ct$, $F^{0i} = -E_i$, and $F^{ij} = -\varepsilon_{ijk}B_k$, as in Eqs. (11.69), (11.75), and (11.137), pp. 542, 556. Spatial subscripts on $\mathbf{E},\mathbf{B}$ are ordinary Cartesian components, not covariant four-vector indices. The dual has $(\mathbf{E},\mathbf{B}) \mapsto (\mathbf{B},-\mathbf{E})$ [Eq. (11.140)]; macroscopic ϵ,μ are dimensionless in these Gaussian constitutive laws [Eq. (11.145), p. 557]. For the particle-kinematics Problems 11.19–11.26, masses and momenta are expressed in energy units, equivalently setting $c = 1$; factors of c are restored when converting a distance or a time.

11.1 Two equivalent inertial frames K and K' are such that K' moves in the positive x direction with speed v as seen from K. The spatial coordinate axes in K' are parallel to those in K and the two origins are coincident at times $t = t' = 0$.

(a) Show that the isotropy and homogeneity of space-time and equivalence of different inertial frames (first postulate of relativity) require that the most general transformation between the space-time coordinates (x,y,z,t) and (x',y',z',t') is the *linear* transformation,

$$x' = f(v^2)x - vf(v^2)t; \qquad t' = g(v^2)t - vh(v^2)x; \qquad y' = y; \qquad z' = z$$

and the inverse,

$$x = f(v^2)x' + vf(v^2)t'; \qquad t = g(v^2)t' + vh(v^2)x'; \qquad y = y'; \qquad z = z'$$

where f, g, and h are functions of v^2, the structures of the x' and x equations are determined by the definition of the inertial frames in relative motion, and the signs in the inverse equation are a reflection of the reversal of roles of the two frames.

(b) Show that consistency of the initial transformation and its inverse requires

$$f = g \qquad \text{and} \qquad f^2 - v^2 fh = 1$$

(c) If a physical entity has speed u' parallel to the x' axis in K', show that its speed u parallel to the x axis in K is

$$u = \frac{u' + v}{1 + vu'(h/f)}$$

Using the second postulate $2'$ (universal limiting speed C), show that $h = f/C^2$ is required and that the Lorentz transformation of the coordinates results. The universal limiting speed C is to be determined from experiment.

Solution.

(a) Let X denote the four coordinate components, and let $T_v(X)$ map them between the two frames. Homogeneity means that the transformed displacement between two events depends only on their displacement: $T_v(X + A) - T_v(X) = T_v(A)$ after choosing coincident origins. Thus T_v is additive. Continuity of the coordinate map extends additivity from rational to real multiples, so it is linear. Rotational symmetry about the relative-velocity axis forbids mixing of (x,t) with the transverse coordinates. With parallel transverse axes their transformation is $y' = a(v)y$, $z' = a(v)z$.

The inverse has factor $a(-v)$, so $a(v)a(-v) = 1$. Isotropy under reversal of the x direction makes $a(-v) = a(v)$; continuity from $a(0) = 1$ consequently gives $a = 1$.

Write the remaining linear equations as $x' = Ax + Bt$, $t' = Cx + Dt$. The origin of K' follows $x = vt$, and hence $B = -vA$. Under simultaneous reversal of the x axes, v changes sign while x' and x change sign and t', t do not. Therefore A, D are even functions of v, and C is odd. Set $A = f(v^2)$, $D = g(v^2)$, $C = -vh(v^2)$:

$$\boxed{x' = f(x - vt), \qquad t' = gt - vhx, \qquad y' = y, \quad z' = z.}$$

The relativity principle says the inverse is the same rule with $v \to -v$, giving exactly the inverse in the question. The usual shared orientation of clocks and rods fixes the branch continuous with the identity at $v = 0$. **(b)** Substitute the primed equations into the inverse:

$$x = f[fx - vft + v(gt - vhx)] = (f^2 - v^2fh)x + vf(g - f)t,$$
$$t = g(gt - vhx) + vh(fx - vft) = (g^2 - v^2fh)t + vh(f - g)x.$$

For these to hold for every x, t, and for nonzero v, the coefficient of t in the first is zero, so $f = g$; the coefficient of x is one, so

$$\boxed{f = g, \qquad f^2 - v^2fh = 1.}$$

Continuity includes $v = 0$. **(c)** Along the moving entity's path $dx' = u'\,dt'$. Differentiating the inverse and taking their ratio gives

$$u = \frac{f(dx' + v\,dt')}{f\,dt' + vh\,dx'} = \boxed{\frac{u' + v}{1 + vu'h/f}.}$$

The map is continuous and monotonic on the physical velocity interval. Equivalence of frames requires its universal upper endpoint $u' = C$ to map to $u = C$ (one may take a limit from below if no massive entity attains C). Thus

$$C(1 + vCh/f) = C + v, \qquad \boxed{h = f/C^2}.$$

Substitution into part (b) gives $f^2(1 - v^2/C^2) = 1$. On the identity branch,

$$\boxed{f = \frac{1}{\sqrt{1 - v^2/C^2}}, \qquad x' = f(x - vt), \quad t' = f(t - vx/C^2).}$$

These are the Lorentz transformations with invariant limiting speed C; identifying C with the vacuum light speed is experimental input, as emphasized in the question.

11.2 Consider three inertial frames and coordinates $K(x, t)$, $K'(x', t')$, and $K''(x'', t'')$. Frame K' moves in the x direction with speed v_1 relative to K; frame K'' moves with speed v_2 relative to K', and speed v_3 relative to K. By considering the group property of the transformations of Problem 11.1 (including the results of parts a and b), $(x'', t'') \to (x', t') \to (x, t)$ and $(x'', t'') \to (x, t)$ directly, show that $|h(v^2)/f(v^2)|$ is a universal constant with the dimensions of an inverse speed squared.

This approach obtains the Lorentz transformation without reference to electromagnetism or the second postulate, but requires experiment to show that $h/f > 0$.

Reference: Y. P. Terletskii, *Paradoxes in the Theory of Relativity*, Plenum Press, New York (1968), pp. 17–25.

Solution.

Use the inverse transformation of Problem 11.1 and write $\kappa(v) = h(v^2)/f(v^2)$. On the (x,t) column it is

$$M(v) = f(v^2)\begin{pmatrix} 1 & v \\ \kappa(v)v & 1 \end{pmatrix}.$$

Two successive transformations give

$$M(v_1)M(v_2) = f_1 f_2 \begin{pmatrix} 1+\kappa_2 v_1 v_2 & v_1+v_2 \\ \kappa_1 v_1 + \kappa_2 v_2 & 1+\kappa_1 v_1 v_2 \end{pmatrix}.$$

By the group property this must have the form $M(v_3)$, whose two diagonal entries are equal. For $v_1 v_2 \neq 0$ this forces $\kappa_1 = \kappa_2$. Since the two velocities are arbitrary, and continuity extends the result to zero, the signed ratio itself is universal:

$$\boxed{h(v^2)/f(v^2) = \kappa = \text{constant}, \qquad [\kappa] = [\text{speed}]^{-2}.}$$

Comparison of the upper right and diagonal entries then gives

$$v_3 = \frac{v_1+v_2}{1+\kappa v_1 v_2}, \qquad f_3 = f_1 f_2 (1+\kappa v_1 v_2), \qquad f(v^2) = (1-\kappa v^2)^{-1/2}.$$

For $\kappa > 0$ put $\kappa = 1/C^2$ and obtain Lorentz kinematics. For $\kappa = 0$, $f = 1$ and the transformations are Galilean. A negative κ leads locally to a different, rotation-like kinematics with no finite invariant positive speed. The algebra alone has not measured the sign or the magnitude of κ; experiment selects the positive value $1/c^2$. In particular, proving a universal magnitude does not by itself exclude the zero, Galilean case.

11.3 Show explicitly that two successive Lorentz transformations in the same direction are equivalent to a single Lorentz transformation with a velocity

$$v = \frac{v_1+v_2}{1+(v_1 v_2/c^2)}$$

This is an alternative way to derive the parallel-velocity addition law.

Solution.

Write $\beta_i = v_i/c$ and $\gamma_i = (1-\beta_i^2)^{-1/2}$. On (ct,x) a boost from a frame to one moving in the positive x direction is

$$\Lambda(\beta_i) = \gamma_i \begin{pmatrix} 1 & -\beta_i \\ -\beta_i & 1 \end{pmatrix}.$$

The two-step transformation is

$$\Lambda(\beta_2)\Lambda(\beta_1) = \gamma_1\gamma_2 \begin{pmatrix} 1+\beta_1\beta_2 & -(\beta_1+\beta_2) \\ -(\beta_1+\beta_2) & 1+\beta_1\beta_2 \end{pmatrix}.$$

Factoring out its common diagonal coefficient identifies

$$\beta = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2}, \qquad \gamma = \gamma_1\gamma_2(1+\beta_1\beta_2).$$

To check that this factor is the required Lorentz gamma, compute

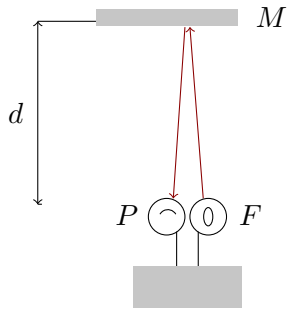
$$1-\beta^2 = \frac{(1+\beta_1\beta_2)^2-(\beta_1+\beta_2)^2}{(1+\beta_1\beta_2)^2} = \frac{(1-\beta_1^2)(1-\beta_2^2)}{(1+\beta_1\beta_2)^2}.$$

The transverse coordinates are unchanged by both transformations, so this proves equality of the full four-dimensional maps and gives

$$\boxed{v = \frac{v_1 + v_2}{1 + v_1 v_2/c^2}.}$$

Equivalently, writing $\beta_i = \tanh\zeta_i$ makes the product a boost of rapidity $\zeta_1 + \zeta_2$; the same result is the hyperbolic-tangent addition formula.

11.4 A possible clock is shown in the figure. It consists of a flashtube F and a photocell P shielded so that each views only the mirror M, located a distance d away, and mounted rigidly with respect to the flashtube-photocell assembly. The electronic innards of the box are such that when the photocell responds to a light flash from the mirror, the flashtube is triggered with a negligible delay and emits a short flash toward the mirror. The clock thus "ticks" once every $(2d/c)$ seconds when at rest.



Problem 11.4

(a) Suppose that the clock moves with a uniform velocity v, perpendicular to the line from PF to M, relative to an observer. Using the second postulate of relativity, show by explicit geometrical or algebraic construction that the observer sees the relativistic time dilatation as the clock moves by.

(b) Suppose that the clock moves with a velocity v parallel to the line from PF to M. Verify that here, too, the clock is observed to tick more slowly, by the same time dilatation factor.

Solution.

The arrows are restored to the source figure's $F \to M \to P$ direction (Jackson, p. 569). Treat the flashtube and photocell as effectively at one point, as implicit in the stipulated rest period $\Delta\tau = 2d/c$. The measured period below is a coordinate-time interval between successive emissions in the observer's inertial frame; travel-time changes from clock to observer would additionally produce a Doppler shift of received flashes. **(a)** Let $T_\uparrow$ be the outward travel time with the clock moving transversely at speed v. The mirror advances by $vT_\uparrow$, so the light ray has length $\sqrt{d^2 + v^2T_\uparrow^2}$. Constancy of light speed gives

$$c^2T_\uparrow^2 = d^2 + v^2T_\uparrow^2, \qquad T_\uparrow = \frac{d}{\sqrt{c^2 - v^2}} = \frac{\gamma d}{c}.$$

The return leg has the same geometry and duration. Therefore

$$\boxed{\Delta t = 2\gamma d/c = \gamma\Delta\tau.}$$

(b) Now the clock's longitudinal separation in the observer's frame is $\ell = d/\gamma$. Choose the mirror ahead of the emitting assembly in the direction of motion. On the outward leg the light catches a receding mirror: $cT_\uparrow = \ell + vT_\uparrow$, giving $T_\uparrow = \ell/(c-v)$. On the return leg the emitting assembly approaches the light, giving $T_\downarrow = \ell/(c+v)$. Thus

$$\Delta t = \frac{\ell}{c-v} + \frac{\ell}{c+v} = \frac{2c\ell}{c^2-v^2} = \boxed{\frac{2\gamma d}{c} = \gamma \Delta\tau.}$$

The contraction of the longitudinal mirror separation is essential: using its rest length directly in these laboratory travel equations would count one gamma factor too many.

11.5 A coordinate system K' moves with a velocity $\mathbf{v}$ relative to another system K. In K' a particle has a velocity $\mathbf{u}'$ and an acceleration $\mathbf{a}'$. Find the Lorentz transformation law for accelerations, and show that in the system K the components of acceleration parallel and perpendicular to $\mathbf{v}$ are

$$\mathbf{a}_\parallel = \frac{\left(1-\dfrac{v^2}{c^2}\right)^{3/2}}{\left(1+\dfrac{\mathbf{v}\cdot\mathbf{u}'}{c^2}\right)^3}\,\mathbf{a}'_\parallel$$

$$\mathbf{a}_\perp = \frac{\left(1-\dfrac{v^2}{c^2}\right)}{\left(1+\dfrac{\mathbf{v}\cdot\mathbf{u}'}{c^2}\right)^3}\left(\mathbf{a}'_\perp + \frac{\mathbf{v}}{c^2}\times(\mathbf{a}'\times\mathbf{u}')\right)$$

Solution.

Let $s = 1 + \mathbf{v}\cdot\mathbf{u}'/c^2$ and $\gamma = (1-v^2/c^2)^{-1/2}$, with parallel/perpendicular projections relative to the fixed boost velocity $\mathbf{v}$. The inverse coordinate transformation and velocity-addition law give

$$dt = \gamma s\, dt', \qquad \mathbf{u}_\parallel = \frac{\mathbf{u}'_\parallel + \mathbf{v}}{s}, \qquad \mathbf{u}_\perp = \frac{\mathbf{u}'_\perp}{\gamma s}, \qquad \frac{ds}{dt'} = \frac{\mathbf{v}\cdot\mathbf{a}'}{c^2}.$$

Differentiate the first velocity quotient and divide by dt/dt':

$$\mathbf{a}_\parallel = \frac{s\mathbf{a}'_\parallel - (\mathbf{u}'_\parallel + \mathbf{v})(\mathbf{v}\cdot\mathbf{a}')/c^2}{\gamma s^3}.$$

All vectors in this numerator are parallel to $\mathbf{v}$. Writing their signed scalar components along that axis reduces its coefficient to $a'_\parallel[1 + vu'_\parallel/c^2 - v(u'_\parallel + v)/c^2] = a'_\parallel(1 - v^2/c^2)$. Hence

$$\boxed{\mathbf{a}_\parallel = \frac{\mathbf{a}'_\parallel}{\gamma^3 s^3}.}$$

For the perpendicular quotient the same operations give

$$\mathbf{a}_\perp = \frac{s\mathbf{a}'_\perp - \mathbf{u}'_\perp(\mathbf{v}\cdot\mathbf{a}')/c^2}{\gamma^2 s^3}.$$

The triple-product identity gives

$$\mathbf{v}\times(\mathbf{a}'\times\mathbf{u}') = \mathbf{a}'(\mathbf{v}\cdot\mathbf{u}') - \mathbf{u}'(\mathbf{v}\cdot\mathbf{a}').$$

Its parallel part is zero; its perpendicular part is $\mathbf{a}'_\perp(\mathbf{v}\cdot\mathbf{u}') - \mathbf{u}'_\perp(\mathbf{v}\cdot\mathbf{a}')$. Consequently

$$\boxed{\mathbf{a}_\perp = \frac{\mathbf{a}'_\perp + \mathbf{v}\times(\mathbf{a}'\times\mathbf{u}')/c^2}{\gamma^2 s^3}.}$$

These are the requested formulas. As a check, for a particle instantaneously at rest in K', $s=1$, and the familiar proper-to-lab relations are $a_\parallel = a'_\parallel/\gamma^3$ and $a_\perp = a'_\perp/\gamma^2$.

11.6 Assume that a rocket ship leaves the earth in the year 2100. One of a set of twins born in 2080 remains on earth; the other rides in the rocket. The rocket ship is so constructed that it has an acceleration g in its own rest frame (this makes the occupants feel at home). It accelerates in a straight-line path for 5 years (by its own clocks), decelerates at the same rate for 5 more years, turns around, accelerates for 5 years, decelerates for 5 years, and lands on earth. The twin in the rocket is 40 years old.

(a) What year is it on earth?

(b) How far away from the earth did the rocket ship travel?

Solution.

Treat the earth as an inertial frame, use flat spacetime, and take g to be the constant magnitude of the rocket's proper acceleration. For one-dimensional motion introduce rapidity η:

$$v = c\tanh\eta, \qquad \frac{dt}{d\tau} = \cosh\eta, \qquad \frac{dx}{d\tau} = c\sinh\eta.$$

The longitudinal result of Problem 11.5 gives $dv/dt = g/\gamma^3$; combining it with $dv/d\eta = c\,\mathrm{sech}^2\,\eta$ and $dt/d\tau = \gamma$ gives $d\eta/d\tau = g/c$ during outward acceleration. Starting at rest and integrating,

$$\eta = \frac{g\tau}{c}, \qquad t(\tau) = \frac{c}{g}\sinh\frac{g\tau}{c}, \qquad x(\tau) = \frac{c^2}{g}\left(\cosh\frac{g\tau}{c} - 1\right).$$

Let $\tau_0 = 5$ years and $\eta_0 = g\tau_0/c$. In the second leg the rapidity decreases from η_0 to zero:

$$\Delta t_2 = \frac{c}{g}\int_0^{\eta_0}\cosh\eta\,d\eta = \frac{c}{g}\sinh\eta_0, \qquad \Delta x_2 = \frac{c^2}{g}\int_0^{\eta_0}\sinh\eta\,d\eta = \frac{c^2}{g}(\cosh\eta_0 - 1).$$

The two return legs have these same time increments and the negatives of these displacements. The rocket is momentarily at rest at the most distant point; no instantaneous velocity reversal is required. Thus

$$\boxed{t_{\text{earth}} = \frac{4c}{g}\sinh\eta_0, \qquad x_{\max} = \frac{2c^2}{g}(\cosh\eta_0 - 1).}$$

Using $g = 9.80665\,\mathrm{m\,s^{-2}}$ and a Julian year of 365.25 days gives $\eta_0 = 5.1614764$. **(a)** The elapsed earth time is 337.9189 years, so arrival is in approximately the year

$$\boxed{2438 \quad (2100 + 337.9189).}$$

The earth twin is then about 357.9 years old; the rocket twin has aged the stipulated 20 years and is 40. **(b)** The maximum separation, reached halfway through the proper-time journey, is

$$\boxed{x_{\max} = 167.0331 \text{ light-years} = 1.58026\times10^{15}\,\text{km}.}$$

The precise fractions of a year depend on the specified value of g and the convention for the duration of a year; the formulas above retain both explicitly.

11.7 In the reference frame K two very evenly matched sprinters are lined up a distance d apart on the y axis for a race parallel to the x axis. Two starters, one beside each man, will fire their starting pistols at slightly different times, giving a handicap to the better of the two runners. The time difference in K is T.

(a) For what range of time differences will there be a reference frame K' in which there is no handicap, and for what range of time differences is there a frame K' in which there is a true (not apparent) handicap?

(b) Determine explicitly the Lorentz transformation to the frame K' appropriate for each of the two possibilities in part a, finding the velocity of K' relative to K and the space-time positions of each sprinter in K'.

Solution.

Choose the first start event as $(t, x, y, z) = (0, 0, 0, 0)$ and the second as $(T, 0, d, 0)$, taking $d > 0$ and allowing either sign of T. Their invariant interval is $s^2 = c^2T^2 - d^2$. A boost of signed speed V along y gives

$$t' = \Gamma(t - Vy/c^2), \qquad y' = \Gamma(y - Vt), \qquad x' = x, \quad z' = z, \quad \Gamma = (1 - V^2/c^2)^{-1/2}.$$

The first event stays at the origin, while the second is

$$(t'_2, x'_2, y'_2, z'_2) = \big(\Gamma(T - Vd/c^2), 0, \Gamma(d - VT), 0\big).$$

(a) and (b), spacelike separation. If $|cT| < d$, choose

$$\boxed{V = c^2T/d, \qquad (t'_2, x'_2, y'_2, z'_2) = (0, 0, \sqrt{d^2 - c^2T^2}, 0).}$$

Both starts are simultaneous in this frame, so the time offset can be removed. This includes $T = 0$, for which $V = 0$. **(a) and (b), timelike separation.** If $|cT| > d$, choose

$$\boxed{V = d/T, \qquad (t'_2, x'_2, y'_2, z'_2) = (\operatorname{sgn} T\sqrt{T^2 - d^2/c^2}, 0, 0, 0).}$$

The two starts now occur at the same spatial coordinate, separated by a nonzero proper time. Their ordering cannot be reversed by an inertial transformation, so this is a genuine temporal handicap. At the limiting cases $|cT| = d$ the separation is lightlike: no boost with $|V| < c$ makes the distinct events simultaneous or colocated, and their temporal ordering is still fixed. The formal speeds above would be $\pm c$ and do not define an inertial frame.

These are the requested spacetime positions of the start events. For completeness, a sprinter still waiting at the fixed laboratory coordinate $y = y_i$ has world line $y'_i = y_i/\Gamma - Vt'$. After starting, if its laboratory running speed is u_i along x, velocity transformation gives $u'_{ix} = u_i/\Gamma$, $u'_{iy} = -V$. Together with the corresponding start event these specify its subsequent uniform-running world line without assuming numerical speeds not supplied in the question.

11.8

(a) Use the relativistic velocity addition law and the invariance of phase to discuss the Fizeau experiments on the velocity of propagation of light in moving liquids. Show that for liquid

flow at a speed v parallel or antiparallel to the path of the light the speed of the light, as observed in the laboratory, is given to first order in v by

$$u = \frac{c}{n(\omega)} \pm v \left(1 - \frac{1}{n^2} + \frac{\omega}{n} \frac{dn(\omega)}{d\omega} \right)$$

where ω is the frequency of the light in the laboratory (in the liquid and outside it) and $n(\omega)$ is the index of refraction of the liquid. Because of the extinction theorem, it is assumed that the light travels with speed $u' = c/n(\omega')$ relative to the moving liquid.

(b) Consult the paper of W. M. Macek, J. R. Schneider, and R. M. Salamon [*J. Appl. Phys.* **35**, 2556 (1964)] and discuss the status of the Fizeau experiments.

Solution.

(a) Let $s = +1$ for flow along the light and $s = -1$ for opposite flow; choose the light propagation direction as $+x$ in each case. Invariance of the phase $kx - \omega t = k'x' - \omega' t'$ under a boost of speed sv gives the wave four-vector law

$$\omega' = \gamma_v(\omega - svk), \qquad k' = \gamma_v(k - sv\omega/c^2).$$

Taking the ratio of the inverse relations, $\omega = \gamma_v(\omega' + svk')$ and $k = \gamma_v(k' + sv\omega'/c^2)$, gives the collinear *phase*-velocity addition rule

$$u = \frac{\omega}{k} = \frac{u' + sv}{1 + svu'/c^2}, \qquad u' = \frac{c}{n(\omega')}.$$

This derivation explains why the rule applies to the wave phase here; it is not an assumption about a photon trajectory in matter. To first order in v/c, substitute the zero-flow wave number $k = n(\omega)\omega/c$ into the frequency shift:

$$\omega' - \omega = -s\frac{v}{c} n(\omega)\omega + O(\omega v^2/c^2).$$

Consequently the rest-frame index and phase speed are

$$n(\omega') = n(\omega) - s\frac{v}{c}\omega n(\omega)\frac{dn}{d\omega} + O(v^2/c^2),$$
$$u' = \frac{c}{n} + sv\frac{\omega}{n}\frac{dn}{d\omega} + O(v^2/c).$$

Expanding the velocity denominator supplies a distinct correction:

$$u = (u' + sv)(1 - svu'/c^2) + O(v^2/c) = u' + sv(1 - u'^2/c^2) + O(v^2/c).$$

The factor multiplying v needs only the zero-order value $u' = c/n$. Hence

$$\boxed{u = \frac{c}{n(\omega)} \pm v \left[1 - \frac{1}{n(\omega)^2} + \frac{\omega}{n(\omega)} \frac{dn}{d\omega} \right] + O(v^2/c).}$$

The laboratory apparatus and liquid interfaces are stationary, so the laboratory frequency is fixed across the interfaces. The Doppler shift belongs to the material rest frame. Omitting it would lose the dispersion term. In terms of vacuum wavelength $\lambda = 2\pi c/\omega$, the coefficient is equivalently $f = 1 - n^{-2} - (\lambda/n)\, dn/d\lambda$.

For a traversed laboratory length ℓ, the directional phase accumulations $\phi_\pm = \omega\ell/u_\pm$ obey

$$\phi_- - \phi_+ = \frac{2\omega\ell v}{c^2} n^2 f + O(v^2/c^2) = \frac{2\omega\ell v}{c^2}\left(n^2 - 1 + \omega n \frac{dn}{d\omega}\right) + O(v^2/c^2).$$

Thus interference measures a small differential phase, without having to measure the absolute propagation speed. Reversing the flow reverses this signal. Nondispersive vacuum has $n = 1$ and $f = 0$, a useful limiting check. **(b)** The cited article is W. M. Macek, J. R. Schneider, and R. M. Salamon, "Measurement of Fresnel Drag with the Ring Laser," *J. Appl. Phys.* **35**, 2556–2557 (1964), DOI 10.1063/1.1702908. The ring-laser extension measures the two oppositely circulating resonances. Its relation to part (a) follows by imposing the resonance condition $\phi_\pm(\omega_\pm) = 2\pi N$. If ω_0 is their common frequency at zero flow and $\delta\phi_\pm$ their flow-induced phase changes, linearization gives

$$\phi_0'(\omega_0)\delta\omega_\pm + \delta\phi_\pm = 0, \qquad |\omega_+ - \omega_-| = \frac{|\delta\phi_+ - \delta\phi_-|}{\phi_0'(\omega_0)}.$$

Defining the cavity group optical length $L_g = c\phi_0'(\omega_0)$, this predicts, for a single moving segment of length ℓ,

$$|\nu_+ - \nu_-| = \frac{2\ell|v|}{\lambda L_g}\left|n^2 - 1 + \omega n \frac{dn}{d\omega}\right|.$$

The optical phase difference is thereby converted into a beat frequency. Material dispersion enters both the numerator and the round-trip phase derivative; treating the latter as a geometric length requires negligible cavity dispersion.

For experimentally checked status, Bilger and Zavodny's primary 1972 report describes a triangular ring laser at $0.6328\,\mu$m with fused silica and finds drag consistent, within its stated errors, with the theory including dispersion. A later direct water-flow experiment by Lahaye, Labastie, and Mathevet (2012) also reports agreement with the relativistic phase shift and identifies the earlier ring-laser work as extending the test to solids, liquids, and gases. These results support the coefficient in part (a), including a separately testable dispersive correction; "drag" denotes this propagation effect and does not require a material aether. [5]

11.9 An infinitesimal Lorentz transformation and its inverse can be written as

$$x'^\alpha = (g^{\alpha\beta} + \epsilon^{\alpha\beta})x_\beta$$
$$x^\alpha = (g^{\alpha\beta} + \epsilon'^{\alpha\beta})x'_\beta$$

where $\epsilon^{\alpha\beta}$ and $\epsilon'^{\alpha\beta}$ are infinitesimal.

(a) Show from the definition of the inverse that $\epsilon'^{\alpha\beta} = -\epsilon^{\alpha\beta}$.

(b) Show from the preservation of the norm that $\epsilon^{\alpha\beta} = -\epsilon^{\beta\alpha}$.

(c) By writing the transformation in terms of contravariant components on both sides of the equation, show that $\epsilon^{\alpha\beta}$ is equivalent to the matrix L (11.93).

[5] Primary reports checked: H. R. Bilger and A. T. Zavodny, *Phys. Rev. A* **5**, 591 (1972), publisher abstract, `https://doi.org/10.1103/PhysRevA.5.591`; T. Lahaye, P. Labastie, and R. Mathevet, *Am. J. Phys.* **80**, 497–505 (2012), `https://doi.org/10.1119/1.3690117`. The 1964 article's publisher record was checked, but its full text was inaccessible during this preparation. Thus no numerical precision, raw data, or dispersion measurement is attributed here to that inaccessible paper; the experimental assessment uses the accessible primary reports above.

Solution.

Use $g = \mathrm{diag}(1,-1,-1,-1)$ and define the mixed matrix $L^\alpha{}_\beta = \epsilon^{\alpha\gamma} g_{\gamma\beta}$, so $x'^\alpha = (\delta^\alpha{}_\beta + L^\alpha{}_\beta)x^\beta$. **(a)** Composing the inverse and forward maps gives $(I+L')(I+L) = I + L' + L + O(\epsilon^2)$. The identity requirement is $L' = -L$. Multiplication by $g^{\beta\gamma}$ raises the second index and proves

$$\boxed{\epsilon'^{\alpha\beta} = -\epsilon^{\alpha\beta}}$$

to the retained order. **(b)** Norm preservation is $(I+L)^{\mathsf{T}} g (I+L) = g$, hence $L^{\mathsf{T}} g + gL = 0$. Equivalently, direct substitution in $x'^\alpha x'_\alpha = x^\alpha x_\alpha$ gives

$$0 = 2\epsilon^{\alpha\beta} x_\alpha x_\beta = (\epsilon^{\alpha\beta} + \epsilon^{\beta\alpha}) x_\alpha x_\beta$$

for every x. Its symmetric coefficient must vanish:

$$\boxed{\epsilon^{\alpha\beta} = -\epsilon^{\beta\alpha}.}$$

(c) Lowering the second index changes the signs of the spatial columns, so $L^0{}_i = L^i{}_0 = -\epsilon^{0i}$ and $L^i{}_j = -\epsilon^{ij}$. Set $\epsilon^{0i} = \zeta_i$ and $\epsilon^{ij} = -\varepsilon_{ijk}\vartheta_k$. The explicit map is

$$L = \begin{pmatrix} 0 & -\zeta_1 & -\zeta_2 & -\zeta_3 \\ -\zeta_1 & 0 & \vartheta_3 & -\vartheta_2 \\ -\zeta_2 & -\vartheta_3 & 0 & \vartheta_1 \\ -\zeta_3 & \vartheta_2 & -\vartheta_1 & 0 \end{pmatrix} = \boxed{-\boldsymbol{\vartheta}\cdot\mathbf{S} - \boldsymbol{\zeta}\cdot\mathbf{K}.}$$

This is (11.93) to infinitesimal order. Antisymmetry of the both-upper-index $\epsilon^{\alpha\beta}$ is thus consistent with the symmetric time-space block of the mixed matrix L; they have different index placements.

11.10

(a) For the Lorentz boost and rotation matrices $\mathbf{K}$ and $\mathbf{S}$ show that

$$(\boldsymbol{\epsilon}\cdot\mathbf{S})^3 = -\boldsymbol{\epsilon}\cdot\mathbf{S}$$
$$(\boldsymbol{\epsilon}'\cdot\mathbf{K})^3 = \boldsymbol{\epsilon}'\cdot\mathbf{K}$$

where $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'$ are any real unit 3-vectors.

(b) Use the results of part a to show that

$$\exp(-\zeta\hat{\boldsymbol{\beta}}\cdot\mathbf{K}) = I - \hat{\boldsymbol{\beta}}\cdot\mathbf{K}\sinh\zeta + (\hat{\boldsymbol{\beta}}\cdot\mathbf{K})^2[\cosh\zeta - 1]$$

Solution.

The matrices (11.91) act on a four-vector $(a^0, \mathbf{a})$ as

$$S(\mathbf{e})(a^0,\mathbf{a}) = (0, \mathbf{e}\times\mathbf{a}), \qquad K(\mathbf{e})(a^0,\mathbf{a}) = (\mathbf{e}\cdot\mathbf{a}, a^0\mathbf{e}),$$

where $S(\mathbf{e}) = \mathbf{e}\cdot\mathbf{S}$ and similarly for K. **(a)** For a unit $\mathbf{e}$, applying the first map twice gives

$$S^2(\mathbf{e})(a^0,\mathbf{a}) = (0, \mathbf{e}(\mathbf{e}\cdot\mathbf{a}) - \mathbf{a}).$$

Taking another cross product with $\mathbf{e}$ annihilates the first term and changes the second to $-\mathbf{e}\times\mathbf{a}$. Since this holds on every four-vector, $\boxed{S^3(\mathbf{e}) = -S(\mathbf{e})}$. For the boost map,

$$K^2(\mathbf{e})(a^0,\mathbf{a}) = (a^0, \mathbf{e}(\mathbf{e}\cdot\mathbf{a})),$$

and the next application returns $(\mathbf{e}\cdot\mathbf{a}, a^0\mathbf{e})$. Thus $\boxed{K^3(\mathbf{e}) = K(\mathbf{e})}$. **(b)** Put $K = \widehat{\boldsymbol{\beta}}\cdot\mathbf{K}$. For every positive integer j, the preceding identity gives $K^{2j} = K^2$ and $K^{2j+1} = K$. Separating odd and even powers in the exponential series gives

$$\begin{aligned} e^{-\zeta K} &= I - K\sum_{j=0}^{\infty}\frac{\zeta^{2j+1}}{(2j+1)!} + K^2\sum_{j=1}^{\infty}\frac{\zeta^{2j}}{(2j)!} \\ &= \boxed{I - K\sinh\zeta + K^2(\cosh\zeta - 1).} \end{aligned}$$

The identity includes the transverse subspace, on which K vanishes and the transformation is exactly the identity.

11.11 Two Lorentz transformations differ by an infinitesimal amount. In the notation of Section 11.7 they are represented by $A_1 = e^L$, $A_2 = e^{L+\delta L}$. Without using explicit matrix representations show that *to first order in* δL the Lorentz transformation $A = A_2A_1^{-1}$ can be written as

$$A = I + \delta L + \frac{1}{2!}[L,\delta L] + \frac{1}{3!}[L,[L,\delta L]] + \frac{1}{4!}[L,[L,[L,\delta L]]] + \cdots$$

Hint: The early terms can be found by brute force, but alternatively consider the Taylor series expansion in λ of the operator $A(\lambda) = e^{\lambda(L+\delta L)}e^{-\lambda L}$ and then put $\lambda = 1$.

Solution.

Define $A(\lambda) = e^{\lambda(L+\delta L)}e^{-\lambda L}$ and differentiate without commuting L and δL:

$$\frac{dA}{d\lambda} = (L+\delta L)A - AL = [L,A] + \delta L\, A, \qquad A(0) = I.$$

Write $A = I + B + O(\delta L^2)$ with B first order. Its equation is

$$B' = [L,B] + \delta L, \qquad B(0) = 0.$$

To solve it, conjugate by $e^{-\lambda L}$:

$$\frac{d}{d\lambda}(e^{-\lambda L}Be^{\lambda L}) = e^{-\lambda L}\delta L e^{\lambda L}.$$

Integrating and changing the integration variable gives

$$B(\lambda) = \int_0^{\lambda} e^{sL}\delta L e^{-sL}\, ds.$$

The conjugated operator has derivatives $\frac{d^n}{ds^n}(e^{sL}\delta L e^{-sL})|_{s=0} = \mathrm{ad}_L^n(\delta L)$, where $\mathrm{ad}_L X = [L,X]$. Thus

$$B(\lambda) = \sum_{n=0}^{\infty}\frac{\lambda^{n+1}}{(n+1)!}\,\mathrm{ad}_L^n(\delta L).$$

Putting $\lambda = 1$ proves, with no matrix representation,

$$\boxed{A = I + \delta L + \frac{[L,\delta L]}{2!} + \frac{[L,[L,\delta L]]}{3!} + \frac{[L,[L,[L,\delta L]]]}{4!} + \cdots + O(\delta L^2).}$$

Here L need not be small. The expansion is only in the variation δL; replacing the full answer by $I + \delta L$ would lose terms of that same first order whenever the generators do not commute.

11.12 Apply the result of Problem 11.11 to a purely algebraic derivation of (11.116) on Thomas precession.

(a) With

$$L = -\frac{\boldsymbol{\beta}\cdot\mathbf{K}(\tanh^{-1}\beta)}{\beta}$$

$$L+\delta L = -\frac{(\boldsymbol{\beta}+\delta\boldsymbol{\beta}_\parallel+\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{K}(\tanh^{-1}\beta')}{\beta'}$$

where $\beta' = \sqrt{(\boldsymbol{\beta}+\delta\boldsymbol{\beta}_\parallel)^2+(\delta\boldsymbol{\beta}_\perp)^2}$, show that

$$\delta L = -\gamma^2\delta\boldsymbol{\beta}_\parallel\cdot\mathbf{K} - \frac{\delta\boldsymbol{\beta}_\perp\cdot\mathbf{K}(\tanh^{-1}\beta)}{\beta}$$

(b) Using the commutation relations for $\mathbf{K}$ and $\mathbf{S}$, show that

$$C_1 = [L,\delta L] = -\left(\frac{\tanh^{-1}\beta}{\beta}\right)^2(\boldsymbol{\beta}\times\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{S}$$
$$C_2 = [L,C_1] = (\tanh^{-1}\beta)^2\,\delta L_\perp$$
$$C_3 = [L,C_2] = (\tanh^{-1}\beta)^2 C_1$$
$$C_4 = [L,C_3] = (\tanh^{-1}\beta)^4\,\delta L_\perp$$

where $\delta L_\perp$ is the term in δL involving $\delta\boldsymbol{\beta}_\perp$.

(c) Sum the series of terms for $A_T = A_2A_1^{-1}$ to obtain

$$A_T = I - (\gamma^2\delta\boldsymbol{\beta}_\parallel+\gamma\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{K} - \frac{\gamma^2}{\gamma+1}(\boldsymbol{\beta}\times\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{S}$$

correct to first order in $\delta\boldsymbol{\beta}$. [See D. Shelupsky, *Am. J. Phys.* **35**, 650 (1967).]

Solution.

Put $\eta = \tanh^{-1}\beta$ and $\mathbf{n} = \boldsymbol{\beta}/\beta$, so $L = -\eta\mathbf{n}\cdot\mathbf{K}$. **(a)** To first order, the magnitude changes by $\delta\beta = \mathbf{n}\cdot\delta\boldsymbol{\beta}_\parallel$, whereas the direction changes by $\delta\mathbf{n} = \delta\boldsymbol{\beta}_\perp/\beta$. Since $d\eta/d\beta = (1-\beta^2)^{-1} = \gamma^2$,

$$\delta(\eta\mathbf{n}) = \gamma^2\delta\boldsymbol{\beta}_\parallel + \frac{\eta}{\beta}\delta\boldsymbol{\beta}_\perp,$$

and hence

$$\boxed{\delta L = -\gamma^2\delta\boldsymbol{\beta}_\parallel\cdot\mathbf{K} - \frac{\eta}{\beta}\delta\boldsymbol{\beta}_\perp\cdot\mathbf{K}.}$$

(b) Equation (11.99) implies $[\mathbf{a}\cdot\mathbf{K},\mathbf{b}\cdot\mathbf{K}] = -(\mathbf{a}\times\mathbf{b})\cdot\mathbf{S}$ and $[\mathbf{a}\cdot\mathbf{K},\mathbf{b}\cdot\mathbf{S}] = (\mathbf{a}\times\mathbf{b})\cdot\mathbf{K}$. The parallel part commutes with L, so

$$C_1 = -\frac{\eta^2}{\beta^2}(\boldsymbol{\beta}\times\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{S}.$$

One further commutator uses $\boldsymbol{\beta}\times(\boldsymbol{\beta}\times\delta\boldsymbol{\beta}_\perp) = -\beta^2\delta\boldsymbol{\beta}_\perp$:

$$C_2 = -\frac{\eta^3}{\beta}\delta\boldsymbol{\beta}_\perp\cdot\mathbf{K} = \eta^2\delta L_\perp.$$

The next two immediately give $C_3 = \eta^2 C_1$ and $C_4 = \eta^4 \delta L_\perp$, as stated. The induction continues alternating the same two operators. **(c)** Inserting them in Problem 11.11 groups the series as

$$\begin{aligned} A_T &= I + \delta L_\parallel + \delta L_\perp \left(1 + \frac{\eta^2}{3!} + \frac{\eta^4}{5!} + \cdots\right) \\ &\quad + C_1 \left(\frac{1}{2!} + \frac{\eta^2}{4!} + \frac{\eta^4}{6!} + \cdots\right) \\ &= I + \delta L_\parallel + \frac{\sinh\eta}{\eta}\delta L_\perp + \frac{\cosh\eta - 1}{\eta^2} C_1. \end{aligned}$$

Use $\sinh\eta = \gamma\beta$, $\cosh\eta = \gamma$, and $(\gamma-1)/\beta^2 = \gamma^2/(\gamma+1)$ to obtain

$$\boxed{A_T = I - (\gamma^2 \delta\boldsymbol{\beta}_\parallel + \gamma \delta\boldsymbol{\beta}_\perp)\cdot \mathbf{K} - \frac{\gamma^2}{\gamma+1}(\boldsymbol{\beta}\times\delta\boldsymbol{\beta}_\perp)\cdot\mathbf{S} + O(\delta\boldsymbol{\beta}^2).}$$

The rotation term is the infinitesimal Thomas rotation in the coordinate-transformation convention of (11.116). All apparent $1/\beta$ factors have regular limits as $\beta \to 0$; in particular the rotation starts at $-\frac{1}{2}(\boldsymbol{\beta}\times\delta\boldsymbol{\beta})\cdot\mathbf{S}$.

11.13 An infinitely long straight wire of negligible cross-sectional area is at rest and has a uniform linear charge density q_0 in the inertial frame K'. The frame K' (and the wire) move with a velocity $\mathbf{v}$ parallel to the direction of the wire with respect to the laboratory frame K.

(a) Write down the electric and magnetic fields in cylindrical coordinates in the rest frame of the wire. Using the Lorentz transformation properties of the fields, find the components of the electric and magnetic fields in the laboratory.

(b) What are the charge and current densities associated with the wire in its rest frame? In the laboratory?

(c) From the laboratory charge and current densities, calculate directly the electric and magnetic fields in the laboratory. Compare with the results of part a.

Solution.

Take the wire along z and $\mathbf{v} = v\hat{\mathbf{z}}$. **(a)** A Gaussian cylinder of radius ρ and length L in the rest frame gives $2\pi\rho L E'_\rho = 4\pi q_0 L$. There is no current in that frame, so

$$\mathbf{E}' = \frac{2q_0}{\rho}\hat{\boldsymbol{\rho}}, \qquad \mathbf{B}' = 0.$$

The inverse of the field transformation (11.149) has $\boldsymbol{\beta} \to -\boldsymbol{\beta}$. Since the electric field is transverse and $\rho' = \rho$, it gives

$$\boxed{E_\rho = \frac{2\gamma q_0}{\rho}, \qquad B_\phi = \frac{2\gamma\beta q_0}{\rho},}$$

with every other component zero. In particular $\hat{\mathbf{z}}\times\hat{\boldsymbol{\rho}} = \hat{\boldsymbol{\phi}}$ fixes the sign of B_ϕ. **(b)** To avoid confusing charge density with the cylindrical radius, denote volume charge density by ρ_e. In Cartesian transverse coordinates the distributions are

$$\rho'_e = q_0\delta(x')\delta(y'), \quad \mathbf{J}' = 0.$$

Transform the four-current $(c\rho'_e, \mathbf{0})$ by the inverse boost. The transverse delta functions are unchanged, yielding

$$\boxed{\rho_e = \gamma q_0 \delta(x)\delta(y), \qquad \mathbf{J} = \gamma q_0 v \delta(x)\delta(y)\hat{\mathbf{z}}.}$$

Equivalently the laboratory line charge and current are $\lambda = \gamma q_0$ and $I = \lambda v$. **(c)** The densities are time independent. Gauss's law gives $E_\rho(2\pi\rho L) = 4\pi\lambda L$, while the Gaussian-unit Ampère law gives $B_\phi(2\pi\rho) = 4\pi I/c$. Consequently $E_\rho = 2\lambda/\rho$ and $B_\phi = 2I/(c\rho)$, which reproduce both transformed fields, including their directions.

11.14

(a) Express the Lorentz scalars $F^{\alpha\beta}F_{\alpha\beta}$, $\mathcal{F}^{\alpha\beta}F_{\alpha\beta}$, and $\mathcal{F}^{\alpha\beta}\mathcal{F}_{\alpha\beta}$ in terms of $\mathbf{E}$ and $\mathbf{B}$. Are there any other invariants quadratic in the field strengths $\mathbf{E}$ and $\mathbf{B}$?

(b) Is it possible to have an electromagnetic field that appears as a purely electric field in one inertial frame and as a purely magnetic field in some other inertial frame? What are the criteria imposed on $\mathbf{E}$ and $\mathbf{B}$ such that there is an inertial frame in which there is no electric field?

(c) For macroscopic media, $\mathbf{E}$, $\mathbf{B}$ form the field tensor $F^{\alpha\beta}$ and $\mathbf{D}$, $\mathbf{H}$ the tensor $G^{\alpha\beta}$. What further invariants can be formed? What are their explicit expressions in terms of the 3-vector fields?

Solution.

(a) Lowering both indices changes the signs of the $0i$ entries but leaves the spatial entries unchanged. Antisymmetry counts each independent entry twice, so

$$\begin{aligned}
F^{\alpha\beta}F_{\alpha\beta} &= 2(-E_x^2 - E_y^2 - E_z^2 + B_x^2 + B_y^2 + B_z^2) = \boxed{2(B^2 - E^2)}, \\
\mathcal{F}^{\alpha\beta}F_{\alpha\beta} &= 2[-\mathbf{B}\cdot\mathbf{E} + (-\mathbf{E})\cdot\mathbf{B}] = \boxed{-4\mathbf{E}\cdot\mathbf{B}}, \\
\mathcal{F}^{\alpha\beta}\mathcal{F}_{\alpha\beta} &= \boxed{2(E^2 - B^2)}.
\end{aligned}$$

Here (11.140) substitutes $(\mathbf{E}, \mathbf{B}) \mapsto (\mathbf{B}, -\mathbf{E})$ to form the dual. Thus there are only two independent quadratic invariants. Indeed a rotational scalar quadratic in the two vectors is a linear combination of E^2, B^2, and $\mathbf{E}\cdot\mathbf{B}$; invariance under a boost requires the first two coefficients to be opposite. The dot product is a pseudoscalar under a spatial reflection, but a scalar under proper Lorentz transformations. **(b)** A nonzero purely electric field has $B^2 - E^2 < 0$; a purely magnetic field has $B^2 - E^2 > 0$. The invariant cannot change sign, so the proposed conversion is impossible unless the entire field is zero. For a frame with $\mathbf{E}' = 0$, the necessary conditions for a nonzero field are

$$\boxed{\mathbf{E}\cdot\mathbf{B} = 0, \qquad B^2 > E^2.}$$

They are also sufficient. Choose the frame velocity

$$\boldsymbol{\beta} = \frac{\mathbf{E}\times\mathbf{B}}{B^2}, \qquad |\boldsymbol{\beta}| = E/B < 1.$$

Then $\boldsymbol{\beta}\cdot\mathbf{E} = 0$ and $\boldsymbol{\beta}\times\mathbf{B} = -\mathbf{E}$, so (11.149) gives $\mathbf{E}' = 0$. These are pointwise criteria: for a nonuniform field to be purely magnetic throughout a region, one constant boost must eliminate $\mathbf{E}$ at every point of that region. Independent pointwise choices do not establish such a global frame.

The remaining field has magnitude $\sqrt{B^2 - E^2}$. A nonzero null field, with $\mathbf{E}\cdot\mathbf{B} = 0$ and $E = B$, would require a light-speed boost and is not an exception. The identically zero field is a separate trivial case. **(c)** Forming the same contractions with G, whose entries are $(\mathbf{D}, \mathbf{H})$, gives the four additional independent quadratic invariants

$$\begin{aligned} G^{\alpha\beta}G_{\alpha\beta} &= 2(H^2 - D^2), \\ \mathcal{G}^{\alpha\beta}G_{\alpha\beta} &= -4\mathbf{D}\cdot\mathbf{H}, \\ F^{\alpha\beta}G_{\alpha\beta} &= 2(\mathbf{B}\cdot\mathbf{H} - \mathbf{E}\cdot\mathbf{D}), \\ \mathcal{F}^{\alpha\beta}G_{\alpha\beta} = F^{\alpha\beta}\mathcal{G}_{\alpha\beta} &= -2(\mathbf{B}\cdot\mathbf{D} + \mathbf{E}\cdot\mathbf{H}). \end{aligned}$$

Dualizing both factors supplies no new quantity: $\mathcal{G}{:}\mathcal{G} = -G{:}G$ and $\mathcal{F}{:}\mathcal{G} = -F{:}G$, where the colon denotes the full index contraction. Constitutive relations can impose further relations among these quantities for a particular medium.

11.15 In a certain reference frame a static, uniform, electric field E_0 is parallel to the x axis, and a static, uniform, magnetic induction $B_0 = 2E_0$ lies in the x-y plane, making an angle θ with the axis. Determine the relative velocity of a reference frame in which the electric and magnetic fields are parallel. What are the fields in that frame for $\theta \ll 1$ and $\theta \to (\pi/2)$?

Solution.

Write $s = \sin\theta$ and $h = \cos\theta$, with $0 \le \theta \le \pi$ for the angle between the two initial fields. Choose a boost perpendicular to the initial fields, $\boldsymbol{\beta} = b\hat{\mathbf{z}}$. Then $\mathbf{E} = E_0\hat{\mathbf{x}}$ and $\mathbf{B} = 2E_0(h\hat{\mathbf{x}} + s\hat{\mathbf{y}})$ give

$$\mathbf{E}' = \gamma E_0[(1 - 2bs)\hat{\mathbf{x}} + 2bh\hat{\mathbf{y}}], \quad \mathbf{B}' = \gamma E_0[2h\hat{\mathbf{x}} + (2s - b)\hat{\mathbf{y}}].$$

Their cross product vanishes if

$$(1 - 2bs)(2s - b) - 4bh^2 = 2s(1 + b^2) - 5b = 0.$$

The root continuous with $b = 0$ at $s = 0$ and less than unity is

$$\boxed{\mathbf{v} = c\frac{4\sin\theta}{5 + \sqrt{25 - 16\sin^2\theta}}\hat{\mathbf{z}}.}$$

For $s \neq 0$ the other root is its reciprocal, and is not a physical boost. The displayed transformed fields with this b also specify their common direction. This frame is not unique: a subsequent boost along the common field direction preserves parallelism.

For $\theta \ll 1$ the quadratic gives $b = 2\theta/5 + O(\theta^3)$, $\gamma = 1 + 2\theta^2/25 + O(\theta^4)$, so, to first order,

$$\boxed{\mathbf{E}' = E_0(\hat{\mathbf{x}} + \tfrac{4}{5}\theta\hat{\mathbf{y}}) + O(E_0\theta^2), \qquad \mathbf{B}' = 2E_0(\hat{\mathbf{x}} + \tfrac{4}{5}\theta\hat{\mathbf{y}}) + O(\theta^2).}$$

The fields tilt through $4\theta/5$ at this order. Their exact magnitudes can also be found without further component algebra. The invariants give $B'^2 - E'^2 = 3E_0^2$ and $E'B' = 2E_0^2|\cos\theta|$ once their cross product vanishes. For $0 \le \theta < \pi/2$ they point in the same direction; for $\pi/2 < \theta \le \pi$ they are antiparallel. If "parallel" is meant to exclude antiparallel directions, that latter range has no solution, because the negative invariant $\mathbf{E}\cdot\mathbf{B}$ cannot become positive. The displayed boost makes the fields collinear throughout the range. Solving the quadratic for E'^2 gives

$$E'^2 = \frac{E_0^2}{2}[\sqrt{9 + 16\cos^2\theta} - 3], \qquad B'^2 = \frac{E_0^2}{2}[\sqrt{9 + 16\cos^2\theta} + 3].$$

Thus $E' = E_0(1 - 2\theta^2/5 + O(\theta^4))$ and $B' = 2E_0(1 - \theta^2/10 + O(\theta^4))$. As $\theta \to \pi/2$, $b \to 1/2$, $\gamma \to 2/\sqrt{3}$, and the component formulas give

$$\boxed{\mathbf{E}' \to 0, \qquad \mathbf{B}' \to \sqrt{3}E_0\hat{\mathbf{y}}.}$$

At the endpoint "parallel" therefore means a vanishing cross product; the zero electric field has no direction.

11.16 In the rest frame of a conducting medium the current density satisfies Ohm's law, $\mathbf{J}' = \sigma\mathbf{E}'$, where σ is the conductivity and primes denote quantities in the rest frame.

(a) Taking into account the possibility of convection current as well as conduction current, show that the covariant generalization of Ohm's law is

$$J^\alpha - \frac{1}{c^2}(U_\beta J^\beta)U^\alpha = \frac{\sigma}{c}F^{\alpha\beta}U_\beta$$

where U^α is the 4-velocity of the medium.

(b) Show that if the medium has a velocity $\mathbf{v} = c\boldsymbol{\beta}$ with respect to some inertial frame that the 3-vector current in that frame is

$$\mathbf{J} = \gamma\sigma[\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B} - \boldsymbol{\beta}(\boldsymbol{\beta} \cdot \mathbf{E})] + \rho\mathbf{v}$$

where ρ is the charge density observed in that frame.

(c) If the medium is uncharged in its rest frame ($\rho' = 0$), what is the charge density and the expression for $\mathbf{J}$ in the frame of part b? This is the relativistic generalization, in the Gaussian units of this chapter, of the equation $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}/c)$ (compare the SI form on p. 320).

Solution.

(a) The proper charge density is the scalar $\rho_0 = U_\beta J^\beta/c^2$. Decompose the four-current into a convective part $\rho_0 U^\alpha$ and an orthogonal conduction part $j^\alpha = J^\alpha - \rho_0 U^\alpha$, for which $U_\alpha j^\alpha = 0$. In the medium's rest frame, $U'^\alpha = (c, \mathbf{0})$, $\rho_0 = \rho'$, and $j'^\alpha = (0, \mathbf{J}')$. Since $F'^{i0} = E'_i$,

$$\frac{\sigma}{c}F'^{\alpha\beta}U'_\beta = (0, \sigma\mathbf{E}') = j'^\alpha.$$

Equality of these four-vectors in one frame implies equality in all frames:

$$\boxed{J^\alpha - \frac{U_\beta J^\beta}{c^2}U^\alpha = \frac{\sigma}{c}F^{\alpha\beta}U_\beta.}$$

The right side is automatically orthogonal to U because F is antisymmetric. Here σ is the scalar conductivity defined by a local, isotropic rest-frame Ohm law. **(b)** With $U^\alpha = \gamma(c, \mathbf{v})$, lower indices give $U_\alpha = \gamma(c, -\mathbf{v})$. The temporal and spatial components of the covariant equation are respectively

$$\rho - \gamma\rho_0 = \frac{\gamma\sigma}{c}\boldsymbol{\beta} \cdot \mathbf{E}, \qquad \mathbf{J} - \gamma\rho_0\mathbf{v} = \gamma\sigma(\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}).$$

Substitute $\gamma\rho_0 = \rho - (\gamma\sigma/c)\boldsymbol{\beta} \cdot \mathbf{E}$ into the second equation:

$$\boxed{\mathbf{J} = \rho\mathbf{v} + \gamma\sigma[\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B} - \boldsymbol{\beta}(\boldsymbol{\beta} \cdot \mathbf{E})].}$$

(c) If $\rho' = \rho_0 = 0$, the two component equations instead directly give

$$\boxed{\rho = \frac{\gamma\sigma}{c}\boldsymbol{\beta}\cdot\mathbf{E}, \qquad \mathbf{J} = \gamma\sigma(\mathbf{E} + \boldsymbol{\beta}\times\mathbf{B}).}$$

A neutral conductor need not remain locally neutral in a frame where its conduction current moves. At small v/c, the current reduces to $\sigma(\mathbf{E} + \mathbf{v}\times\mathbf{B}/c)$. The factor $1/c$ in the corrected statement is required by the Gaussian units used in this chapter; the reference on printed p. 320 is in the earlier SI-unit part of the book.

11.17 The electric and magnetic fields (11.152) of a charge in uniform motion can be obtained from Coulomb's law in the charge's rest frame and the fact that the field strength $F^{\alpha\beta}$ is an antisymmetric tensor of rank 2 without considering *explicitly* the Lorentz transformation. The idea is the following. For a charge in uniform motion the only relevant variables are the charge's 4-velocity U^α and the relative coordinate $X^\alpha = x_p^\alpha - x_q^\alpha$, where x_p^α and x_q^α are the 4-vector coordinates of the observation point and the charge, respectively. The only ordinary antisymmetric tensor formed without a Levi-Civita tensor is $(X^\alpha U^\beta - X^\beta U^\alpha)$. The dual tensor is excluded by the absence of a magnetic field in the charge rest frame. Thus the electromagnetic field $F^{\alpha\beta}$ must be this tensor multiplied by some scalar function of the possible scalar products, $X_\alpha X^\alpha$, $X_\alpha U^\alpha$, $U_\alpha U^\alpha$.

(a) For the geometry of Fig. 11.8 the coordinates of P and q at a common time in K can be written $x_p^\alpha = (ct, \mathbf{b})$, $x_q^\alpha = (ct, \mathbf{v}t)$, with $\mathbf{b}\cdot\mathbf{v} = 0$. By considering the general form of $F^{\alpha\beta}$ in the rest frame of the charge, show that

$$F^{\alpha\beta} = \frac{q}{c}\frac{(X^\alpha U^\beta - X^\beta U^\alpha)}{\left[\dfrac{1}{c^2}(U_\alpha X^\alpha)^2 - X_\alpha X^\alpha\right]^{3/2}}$$

Verify that this yields the expressions (11.152) in the inertial frame K.

(b) Repeat the calculation, using as the starting point the common-time coordinates in the rest frame, $x_p'^\alpha = (ct', \mathbf{b} - \mathbf{v}t')$ and $x_q'^\alpha = (ct', 0)$. Show that

$$F^{\alpha\beta} = \frac{q}{c}\frac{(Y^\alpha U^\beta - Y^\beta U^\alpha)}{(-Y_\alpha Y^\alpha)^{3/2}}$$

where $Y'^\alpha = x_p'^\alpha - x_q'^\alpha$. Verify that the fields are the same as in part a. Note that to obtain the results of (11.152) it is necessary to use the time t of the observation point P in K as the time parameter.

(c) Finally, consider the coordinate $x_p^\alpha = (ct, \mathbf{b})$ and the "retarded-time" coordinate $x_q^\alpha = [ct - R, \boldsymbol{\beta}(ct - R)]$ where R is the distance between P and q at the retarded time. Define the difference as $Z^\alpha = [R, \mathbf{b} - \boldsymbol{\beta}(ct - R)]$. Show that in terms of Z^α and U^α the field is

$$F^{\alpha\beta} = \frac{q}{c}\frac{(Z^\alpha U^\beta - Z^\beta U^\alpha)}{\left(\dfrac{1}{c}U_\alpha Z^\alpha\right)^3}$$

Solution.

Let $X \wedge U$ denote $X^\alpha U^\beta - X^\beta U^\alpha$. A dual tensor constructed with the Levi-Civita tensor is also algebraically possible, but its coefficient must vanish: in the rest frame it would supply a radial

magnetic field, absent from Coulomb's solution. Thus the ordinary tensor specified in the question is the relevant one. **(a)** In the rest frame, $U' = (c, \mathbf{0})$, and

$$S^2 \equiv \frac{(U\cdot X)^2}{c^2} - X^2 = (X'^0)^2 - [(X'^0)^2 - |\mathbf{X}'|^2] = |\mathbf{X}'|^2.$$

Also $(X' \wedge U')^{i0} = cX'^i$. Matching $F'^{i0} = qX'^i/|\mathbf{X}'|^3$ fixes both the coefficient and its radial dependence, independently of the rest-frame time separation. Hence

$$\boxed{F^{\alpha\beta} = \frac{q}{c}\frac{(X\wedge U)^{\alpha\beta}}{S^3}.}$$

For simultaneous laboratory events, $X = (0, \mathbf{r})$, $\mathbf{r} = \mathbf{b} - \mathbf{v}t$, and $U = \gamma(c, \mathbf{v})$. Since $\mathbf{b}\cdot\mathbf{v} = 0$,

$$U\cdot X = \gamma v^2 t, \qquad X^2 = -b^2 - v^2t^2, \quad S^2 = b^2 + \gamma^2 v^2 t^2.$$

Reading the tensor entries gives

$$\boxed{\mathbf{E} = \frac{q\gamma(\mathbf{b}-\mathbf{v}t)}{(b^2+\gamma^2v^2t^2)^{3/2}}, \qquad \mathbf{B} = \boldsymbol{\beta}\times\mathbf{E}.}$$

In particular its longitudinal component is $-q\gamma vt/(b^2+\gamma^2v^2t^2)^{3/2}$ and its transverse component is $q\gamma\mathbf{b}/(b^2+\gamma^2v^2t^2)^{3/2}$, which are (11.152). **(b)** Now $Y' = (0, \mathbf{b} - \mathbf{v}t')$, so $U'\cdot Y' = 0$ and $S^2 = -Y^2$. At the observation event $\mathbf{x}_p = \mathbf{b}$, the time transformation is $t' = \gamma t$. Boosting Y' back to the laboratory therefore gives

$$Y^0 = -\gamma^2 v^2 t/c, \qquad \mathbf{Y} = \mathbf{b} - \gamma^2\mathbf{v}t.$$

Equivalently, without components,

$$Y = X - \frac{U\cdot X}{c^2}U, \qquad Y\wedge U = X\wedge U, \qquad -Y^2 = \frac{(U\cdot X)^2}{c^2} - X^2.$$

Consequently

$$\boxed{F^{\alpha\beta} = \frac{q}{c}\frac{(Y\wedge U)^{\alpha\beta}}{(-Y^2)^{3/2}}}$$

has precisely the numerator and denominator of part (a). For example $Y^i c\gamma - Y^0\gamma v_i = c\gamma(b_i - v_i t)$, displaying explicitly the otherwise important cancellation of the extra factors of γ. **(c)** In the retarded construction, $Z = (R, \mathbf{r} + \boldsymbol{\beta}R) = X + [R/(\gamma c)]U$. Thus $Z\wedge U = X\wedge U$. The retarded separation is null, $Z^2 = 0$, and the scalar S^2 is unchanged by adding any multiple of U to X. Therefore $S^2 = (U\cdot Z/c)^2$. The positive, retarded root fixes the sign: writing $\mathbf{r} + \boldsymbol{\beta}R = R\mathbf{n}$,

$$\frac{U\cdot Z}{c} = \gamma R(1 - \boldsymbol{\beta}\cdot\mathbf{n}) > 0 \quad (|\boldsymbol{\beta}| < 1).$$

It follows that

$$\boxed{F^{\alpha\beta} = \frac{q}{c}\frac{(Z\wedge U)^{\alpha\beta}}{(U\cdot Z/c)^3}.}$$

The three choices select different events on the same straight charge world line, while leaving its perpendicular separation from the observation event unchanged.

11.18 The electric and magnetic fields of a particle of charge q moving in a straight line with speed $v = \beta c$, given by (11.152), become more and more concentrated as $\beta \to 1$, as is indicated in Fig. 11.9. Choose axes so that the charge moves along the z axis in the positive direction, passing the origin at $t = 0$. Let the spatial coordinates of the observation point be (x, y, z) and define the transverse vector $\mathbf{r}_\perp$, with components x and y. Consider the fields and the source in the limit of $\beta = 1$.

(a) Show that the fields can be written as

$$\mathbf{E} = 2q\,\frac{\mathbf{r}_\perp}{r_\perp^2}\,\delta(ct - z); \qquad \mathbf{B} = 2q\,\frac{\hat{\mathbf{v}} \times \mathbf{r}_\perp}{r_\perp^2}\,\delta(ct - z)$$

where $\hat{\mathbf{v}}$ is a unit vector in the direction of the particle's velocity.

(b) Show by substitution into the Maxwell equations that these fields are consistent with a 4-vector source density,

$$J^\alpha = qcv^\alpha \delta^{(2)}(\mathbf{r}_\perp)\delta(ct - z)$$

where the 4-vector $v^\alpha = (1, \hat{\mathbf{v}})$.

(c) Show that the fields of part a are derivable from either of the following 4-vector potentials,

$$A^0 = A^z = -2q\delta(ct - z)\,\ln(\lambda r_\perp); \qquad \mathbf{A}_\perp = 0$$

or

$$A^0 = 0 = A^z; \qquad \mathbf{A}_\perp = -2q\Theta(ct - z)\,\nabla_\perp \ln(\lambda r_\perp)$$

where λ is an irrelevant parameter setting the scale of the logarithm.

Show that the two potentials differ by a gauge transformation and find the gauge function, χ.

Reference: R. Jackiw, D. Kabat, and M. Ortiz, *Phys. Lett.* B **277**, 148 (1992).

Solution.

Interpret the limit distributionally; it describes a formal lightlike source, not a massive particle accelerated to c. **(a)** With $s = z - vt$, the finite-speed fields are

$$\mathbf{E}_\perp = \frac{q\gamma \mathbf{r}_\perp}{(r_\perp^2 + \gamma^2 s^2)^{3/2}}, \qquad E_z = \frac{q\gamma s}{(r_\perp^2 + \gamma^2 s^2)^{3/2}}, \qquad \mathbf{B} = \beta\hat{\mathbf{z}} \times \mathbf{E}_\perp.$$

For fixed $r_\perp > 0$, test the first kernel against a smooth compactly supported function $f(z)$ and set $u = \gamma(z - vt)$:

$$\int \frac{\gamma f(z)\,dz}{(r_\perp^2 + \gamma^2(z - vt)^2)^{3/2}} = \int \frac{f(vt + u/\gamma)\,du}{(r_\perp^2 + u^2)^{3/2}} \longrightarrow \frac{2f(ct)}{r_\perp^2}.$$

The last integral follows from the primitive $u/[r_\perp^2\sqrt{r_\perp^2 + u^2}]$. For the longitudinal kernel the same change of variable gives

$$\left|\int \frac{\gamma(z - vt)f(z)\,dz}{(r_\perp^2 + \gamma^2(z - vt)^2)^{3/2}}\right| \le \frac{\|f\|_\infty}{\gamma}\int_{-\infty}^{\infty} \frac{|u|\,du}{(r_\perp^2 + u^2)^{3/2}} = \frac{2\|f\|_\infty}{\gamma r_\perp} \longrightarrow 0.$$

Consequently

$$\boxed{\mathbf{E} = \mathbf{D}_\perp\delta(ct - z), \qquad \mathbf{B} = \hat{\mathbf{z}} \times \mathbf{D}_\perp\delta(ct - z), \quad \mathbf{D}_\perp = \frac{2q\mathbf{r}_\perp}{r_\perp^2}.}$$

Here $\mathbf{D}_\perp$ is only an auxiliary vector, not a material displacement field. Its $1/r_\perp$ singularity is locally integrable in the transverse plane. **(b)** A small transverse circle gives $\oint(\mathbf{r}_\perp/r_\perp^2)\cdot d\mathbf{a} = 2\pi$; therefore $\nabla_\perp \cdot \mathbf{D}_\perp = 4\pi q\delta^{(2)}(\mathbf{r}_\perp)$. Also $\nabla_\perp \times \mathbf{D}_\perp = 0$ distributionally, since $\mathbf{D}_\perp = 2q\nabla_\perp \ln r_\perp$. Put $\delta = \delta(ct - z)$ and let δ' mean differentiation with respect to its argument. Then

$$\begin{aligned} \nabla\cdot\mathbf{E} &= 4\pi q\delta^{(2)}(\mathbf{r}_\perp)\delta, & \nabla\cdot\mathbf{B} &= 0,\\ \nabla\times\mathbf{E} &= -\hat{\mathbf{z}}\times\mathbf{D}_\perp\delta', & \frac{1}{c}\partial_t\mathbf{B} &= \hat{\mathbf{z}}\times\mathbf{D}_\perp\delta',\\ \nabla\times\mathbf{B} &= \hat{\mathbf{z}}(\nabla_\perp\cdot\mathbf{D}_\perp)\delta + \mathbf{D}_\perp\delta', & \frac{1}{c}\partial_t\mathbf{E} &= \mathbf{D}_\perp\delta'. \end{aligned}$$

The derivative terms cancel in both curl equations. The surviving sources are $\rho = q\delta^{(2)}(\mathbf{r}_\perp)\delta$ and $\mathbf{J} = c\rho\hat{\mathbf{z}}$, or

$$\boxed{J^\alpha = qc(1,\hat{\mathbf{z}})\,\delta^{(2)}(\mathbf{r}_\perp)\delta(ct-z).}$$

They satisfy $\partial_t\rho + \nabla\cdot\mathbf{J} = 0$. **(c)** Set $L = \ln(\lambda r_\perp)$. For the first potential, $\Phi = A_z = -2q\delta L$, $\mathbf{A}_\perp = 0$. Then $\mathbf{E}_\perp = -\nabla_\perp\Phi = 2q\delta\nabla_\perp L$, while $E_z = -\partial_z\Phi - c^{-1}\partial_t A_z = -2q\delta' L + 2q\delta' L = 0$. The curl of $A_z\hat{\mathbf{z}}$ is $2q\delta\hat{\mathbf{z}}\times\nabla_\perp L$, as required.

For the second potential, $\Phi = A_z = 0$ and $\mathbf{A}_\perp = -2q\Theta(ct-z)\nabla_\perp L$. Its time derivative gives the same $\mathbf{E}$, and its z derivative gives the same $\mathbf{B}$; its transverse curl is zero. In the convention $\mathbf{A}_2 = \mathbf{A}_1 + \nabla\chi$, $\Phi_2 = \Phi_1 - c^{-1}\partial_t\chi$, the gauge function is

$$\boxed{\chi = -2q\Theta(ct-z)\ln(\lambda r_\perp).}$$

Indeed $c^{-1}\partial_t\chi = -2q\delta L = \Phi_1$, $\partial_z\chi = 2q\delta L = -A_{1z}$, and its transverse gradient is precisely $\mathbf{A}_{2\perp}$. Changing λ leaves the fields unchanged.

11.19 A particle of mass M and 4-momentum P decays into two particles of masses m_1 and m_2.

(a) Use the conservation of energy and momentum in the form, $p_2 = P - p_1$, and the invariance of scalar products of 4-vectors to show that the total energy of the first particle in the rest frame of the decaying particle is

$$E_1 = \frac{M^2 + m_1^2 - m_2^2}{2M}$$

and that E_2 is obtained by interchanging m_1 and m_2.

(b) Show that the *kinetic energy* T_i of the ith particle in the same frame is

$$T_i = \Delta M\left(1 - \frac{m_i}{M} - \frac{\Delta M}{2M}\right)$$

where $\Delta M = M - m_1 - m_2$ is the mass excess or Q value of the process.

(c) The charged pi-meson ($M = 139.6$ MeV) decays into a mu-meson ($m_1 = 105.7$ MeV) and a neutrino ($m_2 = 0$). Calculate the kinetic energies of the mu-meson and the neutrino in the pi-meson's rest frame. The unique kinetic energy of the muon is the signature of a two-body decay. It entered importantly in the discovery of the pi-meson in photographic emulsions by Powell and coworkers in 1947.

Solution.

(a) In energy units the mass-shell conditions are $P^2 = M^2$ and $p_i^2 = m_i^2$. Square $p_2 = P - p_1$:

$$m_2^2 = M^2 + m_1^2 - 2P \cdot p_1.$$

In the parent rest frame $P = (M, \mathbf{0})$, so $P \cdot p_1 = ME_1$. Therefore

$$\boxed{E_1 = \frac{M^2 + m_1^2 - m_2^2}{2M}, \qquad E_2 = \frac{M^2 + m_2^2 - m_1^2}{2M}.}$$

These satisfy $E_1 + E_2 = M$. Momentum conservation makes the momenta equal and opposite, with common magnitude

$$p_*^2 = E_1^2 - m_1^2 = \frac{[M^2 - (m_1 + m_2)^2][M^2 - (m_1 - m_2)^2]}{4M^2}.$$

The decay is kinematically possible if $M \geq m_1 + m_2$. **(b)** Subtract m_i and use $m_j = M - m_i - \Delta M$:

$$\begin{aligned} T_i &= \frac{(M - m_i)^2 - m_j^2}{2M} \\ &= \frac{(M - m_i - m_j)(M - m_i + m_j)}{2M} \\ &= \boxed{\Delta M \left(1 - \frac{m_i}{M} - \frac{\Delta M}{2M}\right).} \end{aligned}$$

In particular $T_1 + T_2 = \Delta M$, as energy conservation requires. **(c)** With the stated masses, $\Delta M = 33.9$ MeV and

$$E_\nu = p_* = \frac{(139.6)^2 - (105.7)^2}{2(139.6)} = 29.7839183 \text{ MeV}.$$

The neutrino has zero rest energy, while $E_\mu = 139.6 - E_\nu = 109.8160817$ MeV. Hence

$$\boxed{T_\mu = 4.11608 \text{ MeV}, \qquad T_\nu = 29.78392 \text{ MeV}.}$$

Their sum is 33.9 MeV, and their momentum magnitudes are both 29.78392 MeV/c.

11.20 The lambda particle (Λ) is a neutral baryon of mass $M = 1115$ MeV that decays with a lifetime of $\tau = 2.9 \times 10^{-10}$ s into a nucleon of mass $m_1 \simeq 939$ MeV and a pi-meson of mass $m_2 \simeq 140$ MeV. It was first observed in flight by its charged decay mode $\Lambda \to p + \pi^-$ in cloud chambers. The charged tracks originate from a single point and have the appearance of an inverted vee or lambda. The particles' identities and momenta can be inferred from their ranges and curvature in the magnetic field of the chamber.

(a) Using conservation of momentum and energy and the invariance of scalar products of 4-vectors show that, if the opening angle θ between the two tracks is measured, the mass of the decaying particle can be found from the formula

$$M^2 = m_1^2 + m_2^2 + 2E_1E_2 - 2p_1p_2 \cos\theta$$

where here p_1 and p_2 are the magnitudes of the 3-momenta.

(b) A lambda particle is created with a total energy of 10 GeV in a collision in the top plate of a cloud chamber. How far on the average will it travel in the chamber before decaying? What range of opening angles will occur for a 10 GeV lambda if the decay is more or less isotropic in the lambda's rest frame?

Solution.

(a) The parent four-momentum is $P = p_1 + p_2$ in any frame. Squaring it and using the on-shell conditions gives

$$\boxed{M^2 = m_1^2 + m_2^2 + 2(E_1E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2) = m_1^2 + m_2^2 + 2E_1E_2 - 2p_1p_2\cos\theta.}$$

The measured momenta determine $E_i = \sqrt{p_i^2 + m_i^2}$, so this is a mass reconstruction from the tracks, without knowing the parent's original energy. **(b)** For $E_\Lambda = 10^4$ MeV,

$$\gamma = \frac{10^4}{1115} = 8.96861, \qquad \beta = \sqrt{1 - \gamma^{-2}} = 0.993764.$$

The proper decay law is $\exp(-t'/\tau)$, and $t = \gamma t'$, so the mean distance traveled is

$$\boxed{\langle \ell \rangle = \beta\gamma c\tau = \sqrt{(10^4/1115)^2 - 1}\,(299792458)(2.9 \times 10^{-10}) = 0.77487 \text{ m}.}$$

For the angular range, first solve the two-body problem in the parent rest frame:

$$E_1^* = 944.10135 \text{ MeV}, \quad E_2^* = 170.89865 \text{ MeV}, \quad p_* = \sqrt{(E_1^*)^2 - 939^2} = 98.01199 \text{ MeV}.$$

Choose the laboratory parent velocity along z and put $u = \cos\theta_*$ for the proton's rest-frame emission angle. The transverse momenta remain opposite, while the longitudinal momenta become

$$p_{1z} = \gamma(p_*u + \beta E_1^*), \qquad p_{2z} = \gamma(-p_*u + \beta E_2^*), \qquad |\mathbf{p}_{1\perp}| = |\mathbf{p}_{2\perp}| = p_*\sqrt{1-u^2}.$$

Both longitudinal momenta are positive for every $-1 \le u \le 1$, since $\beta E_i^* > p_*$. Thus the opening angle is the sum

$$\theta(u) = \tan^{-1}\frac{\sqrt{1-u^2}}{\gamma(u+a)} + \tan^{-1}\frac{\sqrt{1-u^2}}{\gamma(b-u)}, \quad a = \frac{\beta E_1^*}{p_*}, \quad b = \frac{\beta E_2^*}{p_*}.$$

It is zero at $u = \pm 1$. To locate its maximum, set $k = 1/\gamma$. Differentiating the two arctangents gives

$$\frac{d\theta}{du} = \frac{k}{\sqrt{1-u^2}}\left[\frac{1-bu}{(b-u)^2 + k^2(1-u^2)} - \frac{1+au}{(a+u)^2 + k^2(1-u^2)}\right].$$

For an explicit algebraic reduction, the numerator after cross multiplication factors as

$$\begin{aligned}&(1-bu)[(a+u)^2 + k^2(1-u^2)] - (1+au)[(b-u)^2 + k^2(1-u^2)]\\ &\quad = (a+b)[(k^2-1)u^3 + (2-ab-k^2)u + a - b].\end{aligned}$$

Since $a + b > 0$, the stationary condition is

$$(k^2-1)u^3 + (2-ab-k^2)u + a - b = 0.$$

The sole root in $(-1, 1)$ is $u = 0.527082$; substitution gives $\theta_{\max} = 0.08780948$ rad. Hence

$$\boxed{0 \le \theta \le 5.03111^\circ.}$$

For an isotropic decay u is uniform on $[-1, 1]$, so the full interval is populated; its endpoints are limiting configurations of zero measure. Equal transverse emission ($u = 0$) need not maximize the opening angle when the daughter masses are unequal.

11.21 If a system of mass M decays or transforms at rest into a number of particles, the sum of whose masses is less than M by an amount ΔM,

(a) show that the maximum kinetic energy of the ith particle (mass m_i) is

$$(T_i)_{\max} = \Delta M\left(1 - \frac{m_i}{M} - \frac{\Delta M}{2M}\right)$$

(b) determine the maximum kinetic energies in MeV and also the ratios to ΔM for each of the particles in the following decays or transformations of particles at rest:

$$\begin{aligned}
\mu &\to e + \nu + \bar{\nu} \\
K^+ &\to \pi^+ + \pi^- + \pi^+ \\
K^\pm &\to e^\pm + \pi^0 + \nu \\
K^\pm &\to \mu^\pm + \pi^0 + \nu \\
p + \bar{p} &\to 2\pi^+ + 2\pi^- + \pi^0 \\
p + \bar{p} &\to K^+ + K^- + 3\pi^0
\end{aligned}$$

Solution.

(a) Combine every daughter except i into a four-vector $Q = \sum_{j\neq i} p_j$ with invariant mass $M_R = \sqrt{Q^2}$. When $M_R > 0$, work in its rest frame: each constituent energy is at least its rest mass, so $M_R \geq \sum_{j\neq i} m_j$. If every remaining particle is massless, a collinear set can instead have $Q^2 = 0$; it has no rest frame, but obeys the same bound $M_R \geq 0$. Massive constituents attain their bound when they share a common velocity. If the remainder contains both massive and massless particles, equality may require the massless energies to tend to zero. The maximum below then denotes a kinematic supremum. Treating i and the remainder as the two bodies in Problem 11.19 gives

$$E_i = \frac{M^2 + m_i^2 - M_R^2}{2M}.$$

This decreases with M_R^2, and therefore its upper endpoint uses $M_R = \sum_{j\neq i} m_j = M - m_i - \Delta M$. Subtracting m_i and factoring the difference of squares gives

$$\boxed{(T_i)_{\max} = \frac{(M - m_i)^2 - (M - m_i - \Delta M)^2}{2M} = \Delta M\left(1 - \frac{m_i}{M} - \frac{\Delta M}{2M}\right).}$$

This is a kinematic endpoint; a matrix element or phase-space factor can make the decay rate vanish at that exact endpoint. It does not assert that every daughter simultaneously has its maximum energy. **(b)** Use the rounded masses in the neighboring problems,

$$\begin{gathered}
m_e = 0.511, \qquad m_\mu = 105.7, \qquad m_{\pi^\pm} = 139.6, \\
m_{\pi^0} = 135.0, \qquad m_p = 938.5, \qquad m_{K^\pm} = 493.7 \qquad (\text{MeV}),
\end{gathered}$$

and zero neutrino masses. The kaon mass is the rounded value of 493.677 MeV in the Particle Data Group's charged-kaon mass review (2025 update). For annihilation at rest $M = 2m_p = 1877$ MeV. For each row first compute $\Delta M = M - \sum_j m_j$, then the fraction $f_i = 1 - m_i/M - \Delta M/(2M)$, and

finally $(T_i)_{\max} = \Delta M f_i$. For example the three-pion kaon decay has $\Delta M = 493.7 - 3(139.6) = 74.9$ MeV and

$$f_\pi = 1 - \frac{139.6}{493.7} - \frac{74.9}{2(493.7)} = 0.6413814, \qquad (T_\pi)_{\max} = 48.0395 \text{ MeV}.$$

Applying exactly this calculation to each distinct daughter species gives the complete table; identical or charge-conjugate daughters in the same row have the same endpoint.

Parent process	Daughter	ΔM (MeV)	$(T_i)_{\max}$ (MeV)	f_i
$\mu \to e\nu\bar{\nu}$	e	105.189	52.3402	0.497583
	each $\nu, \bar{\nu}$	105.189	52.8488	0.502417
$K^+ \to \pi^+\pi^-\pi^+$	each π	74.900	48.0395	0.641381
$K^\pm \to e^\pm\pi^0\nu$	$e^\pm$	358.189	227.8817	0.636205
	π^0	358.189	130.3073	0.363795
	ν	358.189	228.2524	0.637240
$K^\pm \to \mu^\pm\pi^0\nu$	$\mu^\pm$	253.000	134.0075	0.529674
	π^0	253.000	118.9925	0.470326
	ν	253.000	188.1742	0.743772
$p\bar{p} \to 2\pi^+2\pi^-\pi^0$	each $\pi^\pm$	1183.600	722.3933	0.610336
	π^0	1183.600	725.2939	0.612786
$p\bar{p} \to K^+K^-3\pi^0$	each $K^\pm$	484.600	294.5810	0.607885
	each π^0	484.600	387.1895	0.798988

The displayed precision documents the calculation with the stated rounded inputs, rather than implying more precise mass measurements. [6]

11.22 The presence in the universe of an apparently uniform "sea" of blackbody radiation at a temperature of roughly 3K gives one mechanism for an upper limit on the energies of photons that have traveled an appreciable distance since their creation. Photon-photon collisions can result in the creation of a charged particle and its antiparticle ("pair creation") if there is sufficient energy in the center of "mass" of the two photons. The lowest threshold and also the largest cross section occurs for an electron-positron pair.

(a) Taking the energy of a typical 3K photon to be $E = 2.5 \times 10^{-4}$ eV, calculate the energy for an incident photon such that there is energy just sufficient to make an electron-positron pair. For photons with energies larger than this threshold value, the cross section increases to a maximum of the order of $(e^2/mc^2)^2$ and then decreases slowly at higher energies. This interaction is one mechanism for the disappearance of such photons as they travel cosmological distances.

(b) There is some evidence for a diffuse x-ray background with photons having energies of several hundred electron volts or more. Beyond 1 keV the spectrum falls as E^{-n} with $n \simeq 1.5$. Repeat the calculation of the threshold incident energy, assuming that the energy of the photon in the "sea" is 500 eV.

Solution.

For photon energies E_γ and ε and the angle ψ between their propagation directions, their total four-momentum satisfies

$$s = (k_1 + k_2)^2 = 2E_\gamma\varepsilon(1 - \cos\psi).$$

[6] Mass input: Particle Data Group, "Charged Kaon Mass," PDG charged-kaon mass review (2025).

At the electron-positron threshold the two particles have no relative momentum in their center-of-momentum frame, so $s_{\rm th} = (2m_e c^2)^2$. Consequently

$$E_{\gamma,\rm th}(\psi) = \frac{2(m_e c^2)^2}{\varepsilon(1-\cos\psi)}.$$

The lowest threshold is for a head-on encounter, $\psi = \pi$; parallel photons cannot create a pair at any finite energy. With $m_e c^2 = 0.511 \times 10^6$ eV the minimum becomes

$$\boxed{E_{\gamma,\rm th} = \frac{(0.511 \times 10^6 \text{ eV})^2}{\varepsilon}.}$$

(a) The specified $\varepsilon = 2.5 \times 10^{-4}$ eV gives

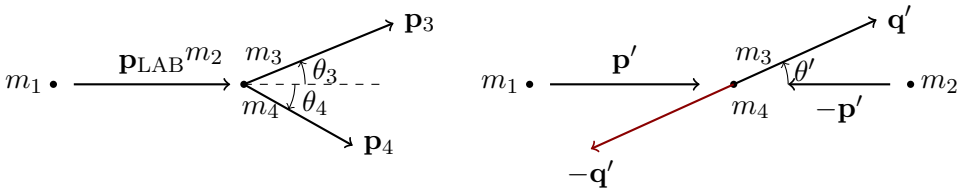
$$\boxed{E_{\gamma,\rm th} = 1.044484 \times 10^{15} \text{ eV} \simeq 1.04 \text{ PeV}.}$$

(b) At $\varepsilon = 500$ eV the corresponding value is

$$\boxed{E_{\gamma,\rm th} = 5.22242 \times 10^{8} \text{ eV} \simeq 522 \text{ MeV}.}$$

These are minimum kinematic thresholds for the selected background photon energies. A distribution of energies and incidence angles produces an attenuation band, rather than a sharp universal cutoff; no cross-section or mean-free-path estimate is needed to determine these thresholds.

11.23 In a collision process a particle of mass m_2, at rest in the laboratory, is struck by a particle of mass m_1, momentum $\mathbf{p}_{\rm LAB}$ and total energy $E_{\rm LAB}$. In the collision the two initial particles are transformed into two others of mass m_3 and m_4. The configurations of the momentum vectors in the center of momentum (cm) frame (traditionally called the center-of-mass frame) and the laboratory frame are shown in the figure.



Laboratory frame cm frame

Problem 11.23

(a) Use invariant scalar products to show that the total energy W in the cm frame has its square given by

$$W^2 = m_1^2 + m_2^2 + 2m_2 E_{\rm LAB}$$

and that the cms 3-momentum $\mathbf{p}'$ is

$$\mathbf{p}' = \frac{m_2 \mathbf{p}_{\rm LAB}}{W}$$

(b) Show that the Lorentz transformation parameters $\boldsymbol{\beta}_{\rm cm}$ and $\gamma_{\rm cm}$ describing the velocity of the cm frame in the laboratory are

$$\boldsymbol{\beta}_{\rm cm} = \frac{\mathbf{p}_{\rm LAB}}{m_2 + E_{\rm LAB}}, \qquad \gamma_{\rm cm} = \frac{m_2 + E_{\rm LAB}}{W}$$

(c) Show that the results of parts a and b reduce in the nonrelativistic limit to the familiar expressions,

$$W \simeq m_1 + m_2 + \left(\frac{m_2}{m_1 + m_2}\right)\frac{p_{\mathrm{LAB}}^2}{2m_1}$$
$$\mathbf{p}' \simeq \left(\frac{m_2}{m_1 + m_2}\right)\mathbf{p}_{\mathrm{LAB}}, \qquad \boldsymbol{\beta}_{\mathrm{cm}} \simeq \frac{\mathbf{p}_{\mathrm{LAB}}}{m_1 + m_2}$$

Solution.

The cm diagram's outgoing arrows are collinear and equal in length, as required by $\mathbf{q}' + (-\mathbf{q}') = 0$ and as drawn in the source figure (p. 575). Let the incident momentum define the positive z axis and write $p = |\mathbf{p}_{\mathrm{LAB}}|$, $E = E_{\mathrm{LAB}}$. **(a)** The initial four-momenta are $p_1 = (E, \mathbf{p}_{\mathrm{LAB}})$ and $p_2 = (m_2, \mathbf{0})$. Their sum has invariant square

$$\boxed{W^2 = (p_1 + p_2)^2 = m_1^2 + m_2^2 + 2m_2E.}$$

This equals its energy squared in the frame where total momentum vanishes. The incident energy in that frame follows from $P \cdot p_1 = WE_1' = m_1^2 + m_2E$:

$$(p')^2 = (E_1')^2 - m_1^2 = \frac{(m_1^2 + m_2E)^2 - m_1^2(m_1^2 + m_2^2 + 2m_2E)}{W^2} = \frac{m_2^2(E^2 - m_1^2)}{W^2}.$$

A collinear boost does not change the direction of the incident momentum in this case, and therefore

$$\boxed{\mathbf{p}' = \frac{m_2}{W}\mathbf{p}_{\mathrm{LAB}}.}$$

The outgoing masses constrain whether the reaction can occur, $W \geq m_3 + m_4$, but do not alter these initial-state formulas. **(b)** Transform the total four-momentum $P = (E + m_2, \mathbf{p}_{\mathrm{LAB}})$. Requiring $\mathbf{P}' = 0$ gives $p - \beta_{\mathrm{cm}}(E + m_2) = 0$. The invariant $W^2 = (E + m_2)^2 - p^2$ then fixes gamma:

$$\boxed{\boldsymbol{\beta}_{\mathrm{cm}} = \frac{\mathbf{p}_{\mathrm{LAB}}}{E + m_2}, \qquad \gamma_{\mathrm{cm}} = \frac{E + m_2}{W}.}$$

For example the transformed incident momentum is $\gamma_{\mathrm{cm}}(p - \beta_{\mathrm{cm}}E) = m_2p/W$, an independent check of part (a). **(c)** At $p \ll m_1$ expand $E = m_1 + p^2/(2m_1) + O(p^4/m_1^3)$, whence

$$W^2 = (m_1 + m_2)^2 + \frac{m_2}{m_1}p^2 + O(p^4).$$

Taking the square root with $\sqrt{A^2 + \delta} = A + \delta/(2A) + O(\delta^2)$ gives

$$\boxed{W = m_1 + m_2 + \frac{m_2}{m_1 + m_2}\frac{p^2}{2m_1} + O(p^4).}$$

Keeping the leading terms in the exact momentum and velocity relations similarly gives

$$\boxed{\mathbf{p}' \simeq \frac{m_2}{m_1 + m_2}\mathbf{p}_{\mathrm{LAB}}, \qquad \boldsymbol{\beta}_{\mathrm{cm}} \simeq \frac{\mathbf{p}_{\mathrm{LAB}}}{m_1 + m_2}.}$$

Restoring ordinary mass units in the last expression supplies the factor $1/c$ appropriate to $\boldsymbol{\beta}$; the actual center of mass velocity is the total momentum divided by the total mass.

11.24 The threshold kinetic energy T_{th} in the laboratory for a given reaction is the kinetic energy of the incident particle on a stationary target just sufficient to make the center of mass energy W equal to the sum of the rest energies of the particles in the final state. Calculate the threshold kinetic energies for the following processes. Express your answers in MeV or GeV and also in units of the rest energy of the incident particle (unless it is a massless particle).

(a) Pi-meson photoproduction, $\gamma p \to \pi^0 p$

$$(m_p = 938.5 \text{ MeV}, \qquad m_{\pi^0} = 135.0 \text{ MeV})$$

(b) Nucleon-antinucleon pair production in nucleon-nucleon collisions, for example, $pp \to ppp\bar{p}$.

(c) Nucleon-antinucleon pair production in electron-electron collisions, $e^-e^- \to e^-e^-p\bar{p}$ and $e^+e^- \to p\bar{p}$ ($m_e = 0.511$ MeV).

Solution.

For a projectile of mass m_1, a stationary target of mass m_2, and total final rest mass M_f, set the invariant from Problem 11.23 equal to M_f^2:

$$M_f^2 = m_1^2 + m_2^2 + 2m_2(m_1 + T_{\text{th}}).$$

Solving for the kinetic energy gives, for the endothermic processes considered here,

$$\boxed{T_{\text{th}} = \frac{M_f^2 - (m_1 + m_2)^2}{2m_2}.}$$

For a photon $m_1 = 0$ and its kinetic energy means its entire energy. The threshold final particles share a common laboratory velocity; they are all at rest relative to one another in the cm frame. **(a)** Here $M_f = m_p + m_{\pi^0}$, and hence

$$E_{\gamma,\text{th}} = \frac{(m_p + m_{\pi^0})^2 - m_p^2}{2m_p} = m_{\pi^0} + \frac{m_{\pi^0}^2}{2m_p} = \boxed{144.710 \text{ MeV.}}$$

The 9.710 MeV above the pion rest energy accounts for the unavoidable laboratory recoil. **(b)** For $pp \to ppp\bar{p}$, $M_f = 4m_p$ and $m_1 = m_2 = m_p$. Thus

$$\boxed{T_{\text{th}} = \frac{16m_p^2 - 4m_p^2}{2m_p} = 6m_p = 5631.0 \text{ MeV} = 5.6310 \text{ GeV.}}$$

In incident-particle rest-energy units the answer is exactly 6. **(c)** For $e^-e^- \to e^-e^-p\bar{p}$, $M_f = 2m_e + 2m_p$, so

$$T_{\text{th}} = \frac{(2m_e + 2m_p)^2 - 4m_e^2}{2m_e} = 4m_p + \frac{2m_p^2}{m_e}$$
$$= \boxed{3.451043 \times 10^6 \text{ MeV} = 3451.043 \text{ GeV} = 6.753508 \times 10^6 m_e.}$$

For $e^+e^- \to p\bar{p}$, the final mass is only $2m_p$, giving

$$T_{\text{th}} = \frac{4m_p^2 - 4m_e^2}{2m_e} = \frac{2m_p^2}{m_e} - 2m_e$$
$$= \boxed{3.447288 \times 10^6 \text{ MeV} = 3447.288 \text{ GeV} = 6.746160 \times 10^6 m_e.}$$

All numbers use the question's $m_p = 938.5$ MeV and $m_e = 0.511$ MeV. The very large fixed-target energies reflect the small mass of the stationary electron in the denominator of the invariant; they are not thresholds for a collider with both beams moving.

11.25 In colliding beam machines such as the Tevatron at Fermilab or the numerous e^+e^- storage rings, counterrotating relativistic beams of particles are stored and made to collide more or less head-on in one or more interaction regions. Let the particles in the two beams have masses m_1 and m_2 and momenta p_1 and p_2, respectively, and let them intersect with a crossing angle θ, defined as the departure from head-on incidence, so the angle between their momentum vectors is $\pi - \theta$.

(a) Show that, to order $(m/p)^2$ inclusive, the square of the total energy in the cm frame is

$$W^2 = 4p_1p_2\cos^2\frac{\theta}{2} + (p_1 + p_2)\left(\frac{m_1^2}{p_1} + \frac{m_2^2}{p_2}\right)$$

(b) Show that the cm inertial frame has a velocity in the laboratory given by

$$\beta_{\text{cm}} = \frac{(p_1 + p_2)\sin(\theta/2)}{(E_1 + E_2)\sin\alpha}$$

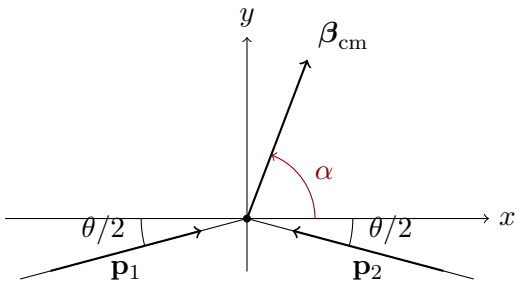
where

$$\tan\alpha = \left(\frac{p_1 + p_2}{p_1 - p_2}\right)\tan\frac{\theta}{2}$$

The angle α is defined in the figure.

(c) Check that the results of part b agree with those of Problem 11.23b.

(d) If the crossing angle is $\theta = 20^\circ$ and the colliding protons have $p_1 = p_2 = 100$ GeV/c, is the laboratory frame a reasonable approximation to the cm frame? Consider, for example, a proton-proton inelastic collision involving pion production and examine the collinearity of two pions produced with equal and opposite momenta of 10 GeV/c in the cm frame.



Problem 11.25

Solution.

The source figure on printed p. 576 measures α from the horizontal axis. The corrected diagram and angle convention in the statement make this explicit. Put $h = \theta/2$ and choose

$$\mathbf{p}_1 = p_1(\cos h, \sin h, 0), \qquad \mathbf{p}_2 = p_2(-\cos h, \sin h, 0).$$

Their included angle is $\pi - \theta$, so head-on collision is $\theta = 0$. **(a)** Their scalar product is $\mathbf{p}_1 \cdot \mathbf{p}_2 = -p_1p_2\cos\theta$. Thus exactly

$$W^2 = m_1^2 + m_2^2 + 2E_1E_2 + 2p_1p_2\cos\theta.$$

Expand $E_i = p_i + m_i^2/(2p_i) + O(m_i^4/p_i^3)$:

$$2E_1E_2 = 2p_1p_2 + \frac{p_2m_1^2}{p_1} + \frac{p_1m_2^2}{p_2} + O(m^4/p^2).$$

Combining the explicit rest-mass terms with these corrections and using $1 + \cos\theta = 2\cos^2 h$ yields

$$\boxed{W^2 = 4p_1p_2\cos^2 h + (p_1 + p_2)\left(\frac{m_1^2}{p_1} + \frac{m_2^2}{p_2}\right) + O(m^4/p^2).}$$

The expansion assumes both beams are relativistic; the exact formula preceding it has no such restriction. **(b)** The total momentum components and boost vector are

$$P_x = (p_1 - p_2)\cos h, \quad P_y = (p_1 + p_2)\sin h, \qquad \boldsymbol{\beta}_{\rm cm} = \frac{(P_x, P_y, 0)}{E_1 + E_2}.$$

Consequently $\alpha = \text{atan2}(P_y, P_x)$ satisfies

$$\boxed{\tan\alpha = \frac{p_1 + p_2}{p_1 - p_2}\tan h, \qquad \beta_{\rm cm} = \frac{(p_1 + p_2)\sin h}{(E_1 + E_2)\sin\alpha}.}$$

Using atan2 supplies the correct quadrant when $p_2 > p_1$. If $P_y = 0$, use the nonsingular equivalent $\beta_{\rm cm} = \sqrt{P_x^2 + P_y^2}/(E_1 + E_2)$. If the total momentum vanishes, the cm frame is the laboratory and its direction α is undefined. **(c)** Set $p_2 = 0$ in these *exact* part (b) formulas. Then $E_2 = m_2$, $\mathbf{P} = \mathbf{p}_1$, and $\boldsymbol{\beta}_{\rm cm} = \mathbf{p}_1/(E_1 + m_2)$. Also $\gamma_{\rm cm} = (E_1 + m_2)/W$, since $W^2 = (E_1 + m_2)^2 - p_1^2$. These are Problem 11.23b. The ultrarelativistic expansion in part (a), containing $1/p_2$, should not be used for this limit. **(d)** Equal beams give $\alpha = \pi/2$ and

$$\beta_{\rm cm} = \frac{p}{\sqrt{p^2 + m_p^2}}\sin 10^\circ = 0.173641, \qquad \gamma_{\rm cm} = 1.015425,$$

using $p = 100$ GeV and $m_p = 0.9385$ GeV. The invariant energy is $W = 2\sqrt{m_p^2 + p^2\cos^2 10^\circ} = 196.9705$ GeV. To assess direction changes, take a back-to-back pion pair of momenta $\pm q\hat{\mathbf{n}}$ in the cm frame, with $q = 10$ GeV and $E_\pi^* = \sqrt{q^2 + (0.1396)^2} = 10.000974$ GeV. Decompose $\hat{\mathbf{n}}$ relative to the cm velocity, with $u = \hat{\mathbf{n}} \cdot \hat{\boldsymbol{\beta}}_{\rm cm}$. The laboratory components are

$$p_{\pm,\parallel} = \gamma_{\rm cm}(\pm qu + \beta_{\rm cm}E_\pi^*), \quad \mathbf{p}_{\pm,\perp} = \pm q\mathbf{n}_\perp.$$

For emission perpendicular to the boost ($u = 0$), both pions gain the same longitudinal momentum $\gamma_{\rm cm}\beta_{\rm cm}E_\pi^* = 1.76336$ GeV. Their opening angle is therefore

$$\boxed{\Theta = 2\tan^{-1}\frac{q}{\gamma_{\rm cm}\beta_{\rm cm}E_\pi^*} = 159.9990^\circ,}$$

about 20° away from collinearity. Along the boost ($u = \pm 1$) the momenta remain opposite because $q > \beta_{\rm cm}E_\pi^*$; intermediate orientations give intermediate deviations. Thus the laboratory is close to the cm frame for percent-level total-energy estimates, but does not approximate back-to-back directions well when angular correlations at the degree level matter.

11.26 In an elastic scattering process the incident particle imparts energy to the stationary target. The energy ΔE lost by the incident particle appears as recoil kinetic energy of the target. In the notation of Problem 11.23, $m_3 = m_1$ and $m_4 = m_2$, while $\Delta E = T_4 = E_4 - m_4$.

(a) Show that ΔE can be expressed in the following different ways,

$$\Delta E = \frac{m_2}{W^2}\, p_{\mathrm{LAB}}^2 (1 - \cos\theta')$$

$$\Delta E = \frac{2m_2 p_{\mathrm{LAB}}^2 \cos^2\theta_4}{W^2 + p_{\mathrm{LAB}}^2 \sin^2\theta_4}$$

$$\Delta E = \frac{Q^2}{2m_2}$$

where $Q^2 = -(p_1 - p_3)^2 = (\mathbf{p}_1 - \mathbf{p}_3)^2 - (E_1 - E_3)^2$ is the Lorentz invariant momentum transfer (squared).

(b) Show that for charged particles other than electrons incident on stationary electrons ($m_1 \gg m_2$) the maximum energy loss is approximately

$$\Delta E_{\max} \simeq 2\gamma^2\beta^2 m_e$$

where γ and β are characteristic of the incident particle and $\gamma \ll (m_1/m_e)$. Give this result a simple interpretation by considering the relevant collision in the rest frame of the incident particle and then transforming back to the laboratory.

(c) For electron-electron collisions, show that the maximum energy transfer is

$$\Delta E_{\max}^{(e)} = (\gamma - 1) m_e$$

Solution.

(a) Use the cm boost of Problem 11.23 and let $p_* = m_2 p_{\mathrm{LAB}}/W$. Elasticity leaves this momentum magnitude unchanged. The target initially has cm longitudinal momentum $-p_*$ and finally has longitudinal momentum $-p_* \cos\theta'$, with the same cm energy E_2^*. Transforming its energy back gives

$$m_2 = \gamma_{\mathrm{cm}}(E_2^* - \beta_{\mathrm{cm}} p_*), \quad E_4 = \gamma_{\mathrm{cm}}(E_2^* - \beta_{\mathrm{cm}} p_* \cos\theta').$$

Subtract the equations and use $\gamma_{\mathrm{cm}}\beta_{\mathrm{cm}} = p_{\mathrm{LAB}}/W$:

$$\boxed{\Delta E = \frac{m_2 p_{\mathrm{LAB}}^2}{W^2}(1 - \cos\theta').}$$

For the laboratory-angle form, write $P = p_1 + p_2$ and $p_3 = P - p_4$. The relation $p_3^2 = m_1^2$ implies $2P \cdot p_4 = W^2 + m_2^2 - m_1^2 = 2m_2(E_{\mathrm{LAB}} + m_2)$. Set $T = E_4 - m_2$ and $p_4 = \sqrt{T^2 + 2m_2 T}$; cancellation of the rest terms leaves

$$(E_{\mathrm{LAB}} + m_2)T = p_{\mathrm{LAB}} \sqrt{T^2 + 2m_2 T}\, \cos\theta_4.$$

For $T > 0$, square and divide by T to get

$$T[(E_{\mathrm{LAB}} + m_2)^2 - p_{\mathrm{LAB}}^2 \cos^2\theta_4] = 2m_2 p_{\mathrm{LAB}}^2 \cos^2\theta_4.$$

Since $(E_{\rm LAB}+m_2)^2 = W^2 + p_{\rm LAB}^2$,

$$\boxed{\Delta E = \frac{2m_2 p_{\rm LAB}^2 \cos^2\theta_4}{W^2 + p_{\rm LAB}^2 \sin^2\theta_4}.}$$

The original unsquared equation restricts a recoiling target to $0 \le \theta_4 \le \pi/2$; $T = 0$ follows by continuity. Finally momentum conservation gives $p_1 - p_3 = p_4 - p_2$, hence

$$Q^2 = -(p_4 - p_2)^2 = -[2m_2^2 - 2m_2 E_4] = 2m_2 T, \qquad \boxed{\Delta E = Q^2/(2m_2).}$$

(b) The first form is largest at $\theta' = \pi$. With $m_2 = m_e$, $E_{\rm LAB} = \gamma m_1$, and $p_{\rm LAB} = \gamma\beta m_1$, its exact value is

$$\Delta E_{\max} = \frac{2m_e\gamma^2\beta^2}{1 + 2\gamma m_e/m_1 + (m_e/m_1)^2} \simeq \boxed{2m_e\gamma^2\beta^2},$$

provided $m_e/m_1 \ll 1$ and $\gamma m_e/m_1 \ll 1$. In the heavy projectile's rest frame the initial electron has energy γm_e and momentum $-\gamma\beta m_e$. Neglecting projectile recoil, maximal transfer reverses this momentum to $+\gamma\beta m_e$. Transforming back to the laboratory gives

$$E_{e,f} = \gamma(\gamma m_e + \beta\gamma\beta m_e) = \gamma^2 m_e(1+\beta^2),$$

so $E_{e,f} - m_e = 2\gamma^2\beta^2 m_e$, the same result. **(c)** For equal masses the exact denominator is $2(1+\gamma)$:

$$\boxed{\Delta E_{\max}^{(e)} = \frac{2m_e\gamma^2\beta^2}{2(1+\gamma)} = m_e\frac{\gamma^2-1}{\gamma+1} = (\gamma-1)m_e.}$$

This is complete transfer of the incoming kinetic energy to the initially stationary, labeled electron. Because the final electrons are identical, an experimental convention that calls the more energetic outgoing electron the "incident" electron instead restricts the labeled loss to half this value; it does not change the kinematic endpoint asked for here.

11.27

(a) A charge density ρ' of zero total charge, but with a dipole moment $\mathbf{p}$, exists in reference frame K'. There is no current density in K'. The frame K' moves with a velocity $\mathbf{v} = \boldsymbol{\beta}c$ in the frame K. Find the charge and current densities ρ and $\mathbf{J}$ in the frame K and show that there is a magnetic dipole moment, $\mathbf{m} = (\mathbf{p}\times\boldsymbol{\beta})/2$, correct to first order in β. What is the electric dipole moment in K to the same order in β?

(b) Instead of the charge density, but no current density, in K', consider no charge density, but a localized, time-independent current density $\mathbf{J}'$ that has a magnetic dipole moment $\mathbf{m}$. Find the charge and current densities in K and show that to first order in β there is an electric dipole moment $\mathbf{p} = \boldsymbol{\beta}\times\mathbf{m}$ in addition to the magnetic dipole moment.

Solution.

Take localized, stationary rest-frame sources, and let $\mathbf{R} = \mathbf{x} - \mathbf{v}t$ measure position from their moving center. Rest and laboratory simultaneous positions are related by $\mathbf{r}' = \mathbf{R}_\perp + \gamma\mathbf{R}_\parallel$, with $d^3x = d^3r'/\gamma$. **(a)** The inverse boost of $(c\rho', \mathbf{0})$ gives exactly

$$\boxed{\rho(\mathbf{x},t) = \gamma\rho'(\mathbf{r}'), \qquad \mathbf{J}(\mathbf{x},t) = \gamma\rho'(\mathbf{r}')\mathbf{v}.}$$

The total charge stays zero. The dipole integral transforms as

$$\mathbf{p}_K = \int \mathbf{x}\rho\, d^3x = \int (\mathbf{v}t + \mathbf{r}'_\perp + \mathbf{r}'_\parallel/\gamma)\rho'(\mathbf{r}')\, d^3r' = \mathbf{p}_\perp + \mathbf{p}_\parallel/\gamma.$$

The translation term vanishes by neutrality. Thus $\mathbf{p}_K = \mathbf{p} + O(\beta^2)$. With the Gaussian definition of the magnetic moment,

$$\mathbf{m}_K = \frac{1}{2c}\int \mathbf{x}\times\mathbf{J}\, d^3x = \frac{1}{2}\mathbf{p}_K \times \boldsymbol{\beta} = \boxed{\tfrac{1}{2}\mathbf{p}\times\boldsymbol{\beta} + O(\beta^3).}$$

The factor 1/2 is the current-moment definition, rather than a field transformation factor. **(b)** For the steady rest current $\mathbf{J}'(\mathbf{r}')$ and $\rho' = 0$, the exact transformed densities are

$$\boxed{\rho = \frac{\gamma}{c}\boldsymbol{\beta}\cdot\mathbf{J}'(\mathbf{r}'), \qquad \mathbf{J} = \mathbf{J}'_\perp(\mathbf{r}') + \gamma\mathbf{J}'_\parallel(\mathbf{r}').}$$

To first order the spatial distortion and $\gamma - 1$ are quadratic, so $\mathbf{J} = \mathbf{J}'(\mathbf{R}) + O(\beta^2)$ and $\rho = \boldsymbol{\beta}\cdot\mathbf{J}'(\mathbf{R})/c + O(\beta^3)$. For a localized divergence-free current, integration of $\partial_k(r_i J'_k)$ gives $\int J'_i\, d^3r' = 0$, so the induced total charge vanishes. Integrating the divergence of $r_i r_j \mathbf{J}'$ similarly gives

$$0 = \int (r_i J'_j + r_j J'_i)\, d^3r'.$$

Thus $M_{ij} = \int r_i J'_j\, d^3r'$ is antisymmetric. Since $m_k = (2c)^{-1}\varepsilon_{kij}M_{ij}$, its inverse is $M_{ij} = c\varepsilon_{ijk}m_k$. The electric moment is therefore

$$(p_K)_i = \frac{\beta_j}{c}M_{ij} = \varepsilon_{ijk}\beta_j m_k, \qquad \boxed{\mathbf{p}_K = \boldsymbol{\beta}\times\mathbf{m} + O(\beta^2).}$$

At this order the magnetic moment integral retains its rest value, $\mathbf{m}_K = \mathbf{m} + O(\beta^2)$. Steadiness is needed for that last claim: a time-dependent rest current is sampled at different rest times on a laboratory simultaneity slice. The statement now explicitly supplies this assumption, as appropriate for a transported permanent dipole.

11.28 Revisit Problems 6.21 and 6.22 from the viewpoint of Lorentz transformations. An electric dipole instantaneously at rest at the origin in the frame K' has potentials, $\Phi' = \mathbf{p}\cdot\mathbf{r}'/r'^3$, and $\mathbf{A}' = 0$ (and thus only an electric field). The frame K' moves with uniform velocity $\mathbf{v} = \boldsymbol{\beta}c$ in the frame K.

(a) Show that in frame K to first order in β the potentials are

$$\Phi = \frac{\mathbf{p}\cdot\mathbf{R}}{R^3}, \qquad \mathbf{A} = \boldsymbol{\beta}\,\frac{(\mathbf{p}\cdot\mathbf{R})}{R^3}$$

where $\mathbf{R} = \mathbf{x} - \mathbf{x}_0(t)$, with $\mathbf{v} = d\mathbf{x}_0/dt$ at time t.

(b) Show explicitly that the potentials in K satisfy the Lorentz condition.

(c) Show that to first order in β the electric field $\mathbf{E}$ in K is just the electric dipole field (centered at $\mathbf{x}_0$), or a dipole field plus time-dependent higher multipoles, if viewed from a fixed origin, and the magnetic field is $\mathbf{B} = \boldsymbol{\beta}\times\mathbf{E}$. Where is the effective magnetic dipole moment of Problem 6.21 or 11.27a?

Solution.

Assume a permanent dipole $\mathbf{p}$ and constant $\boldsymbol{\beta}$, with coincident origins at $t = 0$. **(a)** The inverse four-potential transformation gives $\Phi = \gamma\Phi'$ and $\mathbf{A} = \gamma\boldsymbol{\beta}\Phi'$ because $\mathbf{A}' = 0$. At the same observation event, $\mathbf{r}' = \mathbf{R}_\perp + \gamma\mathbf{R}_\parallel = \mathbf{R} + O(\beta^2)$. Therefore, through first order,

$$\boxed{\Phi = \frac{\mathbf{p}\cdot\mathbf{R}}{R^3}, \qquad \mathbf{A} = \boldsymbol{\beta}\frac{\mathbf{p}\cdot\mathbf{R}}{R^3}.}$$

The time dependence is entirely through $\mathbf{R} = \mathbf{x} - \mathbf{x}_0(t)$, with $\dot{\mathbf{x}}_0 = c\boldsymbol{\beta}$. **(b)** For any function of $\mathbf{R}$, $c^{-1}\partial_t = -\boldsymbol{\beta}\cdot\nabla$. Hence the Lorenz condition is satisfied directly:

$$\boxed{\nabla\cdot\mathbf{A} + \frac{1}{c}\partial_t\Phi = \boldsymbol{\beta}\cdot\nabla\Phi - \boldsymbol{\beta}\cdot\nabla\Phi = 0.}$$

(c) Since $\mathbf{A}$ already contains one power of β, its time derivative contributes only at order β^2. For $R > 0$, with $\mathbf{n} = \mathbf{R}/R$,

$$\mathbf{E} = -\nabla\Phi + O(\beta^2) = \boxed{\frac{3\mathbf{n}(\mathbf{n}\cdot\mathbf{p}) - \mathbf{p}}{R^3} + O(\beta^2)},$$
$$\mathbf{B} = \nabla\times(\boldsymbol{\beta}\Phi) = \nabla\Phi\times\boldsymbol{\beta} = \boxed{\boldsymbol{\beta}\times\mathbf{E} + O(\beta^3)}.$$

If expanded about the fixed origin for $r > |\mathbf{x}_0|$, the potential is

$$\Phi = \frac{\mathbf{p}\cdot\mathbf{x}}{r^3} - (\mathbf{x}_0\cdot\nabla)\frac{\mathbf{p}\cdot\mathbf{x}}{r^3} + \cdots = \frac{\mathbf{p}\cdot\mathbf{n}_r}{r^2} + \frac{3(\mathbf{x}_0\cdot\mathbf{n}_r)(\mathbf{p}\cdot\mathbf{n}_r) - \mathbf{x}_0\cdot\mathbf{p}}{r^3} + \cdots.$$

The second term is the electric quadrupole potential, with $Q_{ij} = 3(x_{0i}p_j + x_{0j}p_i) - 2\delta_{ij}\mathbf{x}_0\cdot\mathbf{p}$, and is time dependent even though the intrinsic dipole is fixed.

To locate the magnetic moment explicitly, use $(\mathbf{p}\times\boldsymbol{\beta})\times\mathbf{R} = \boldsymbol{\beta}(\mathbf{p}\cdot\mathbf{R}) - \mathbf{p}(\boldsymbol{\beta}\cdot\mathbf{R})$ to split

$$\mathbf{A} = \underbrace{\frac{\mathbf{m}\times\mathbf{R}}{R^3}}_{\mathbf{A}_m} + \underbrace{\frac{\mathbf{p}(\boldsymbol{\beta}\cdot\mathbf{R}) + \boldsymbol{\beta}(\mathbf{p}\cdot\mathbf{R})}{2R^3}}_{\mathbf{A}_{\rm sym}}, \qquad \mathbf{m} = \tfrac{1}{2}\mathbf{p}\times\boldsymbol{\beta}.$$

The numerator of $\mathbf{A}_{\rm sym}$ has zero curl. Its radial factor nevertheless gives the nonzero field

$$\mathbf{B}_{\rm sym} = -\frac{3}{2R^3}\mathbf{n}\times[\mathbf{p}(\boldsymbol{\beta}\cdot\mathbf{n}) + \boldsymbol{\beta}(\mathbf{p}\cdot\mathbf{n})].$$

Thus the magnetic dipole field of $\mathbf{m}$ is present but must be added to this symmetric contribution. To check its displacement current, let $\mathbf{E}_p(\mathbf{x})$ be the static dipole field at the fixed origin. The first translation term is $\mathbf{E}_Q = -(\mathbf{x}_0\cdot\nabla)\mathbf{E}_p$, so

$$\frac{1}{c}\partial_t\mathbf{E}_Q = -(\boldsymbol{\beta}\cdot\nabla)\mathbf{E}_p.$$

At this same fixed-origin multipole order, the curl of the static magnetic dipole term vanishes away from the source, while

$$\nabla\times\mathbf{B}_{\rm sym} = \nabla\times(\boldsymbol{\beta}\times\mathbf{E}_p) = \boldsymbol{\beta}(\nabla\cdot\mathbf{E}_p) - (\boldsymbol{\beta}\cdot\nabla)\mathbf{E}_p = -(\boldsymbol{\beta}\cdot\nabla)\mathbf{E}_p.$$

Thus it is precisely the magnetic contribution needed for the translating quadrupole's displacement current, as in Problem 6.22. Adding both magnetic terms gives $\boldsymbol{\beta}\times\mathbf{E}$; interpreting that entire field as the field of a stationary magnetic dipole would discard the second term. Point-source contact terms can be obtained by taking the displayed potentials distributionally; the ordinary formulas above are for $R > 0$.

11.29 Instead of the electric dipole potential of Problem 11.28, consider a point magnetic moment $\mathbf{m}$ in the moving frame K', with its potentials, $\Phi' = 0$, $\mathbf{A}' = \mathbf{m} \times \mathbf{r}'/r'^3$ (and so only a magnetic field).

(a) Show that to first order in β the potentials in K are

$$\Phi = \frac{(\boldsymbol{\beta} \times \mathbf{m}) \cdot \mathbf{R}}{R^3}, \qquad \mathbf{A} = \frac{(\mathbf{m} \times \mathbf{R})}{R^3}$$

Note that the scalar potential is the same as the static potential of the electric dipole moment of Problem 11.27b. [But this gives only the irrotational part of the electric field.]

(b) Calculate the electric and magnetic fields in K from the potentials and show that the electric field can be expressed alternatively as

$$\mathbf{E} = \mathbf{E}_{\text{dipole}}(\mathbf{p}_{\text{eff}} = \boldsymbol{\beta} \times \mathbf{m}) - \mathbf{m} \times \frac{[3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}]}{R^3}$$

$$\mathbf{E} = \mathbf{E}_{\text{dipole}}\left(\mathbf{p}_{\text{eff}} = \frac{\boldsymbol{\beta}}{2} \times \mathbf{m}\right) + \frac{3}{2}\,\mathbf{n} \times \frac{[\mathbf{m}(\mathbf{n} \cdot \boldsymbol{\beta}) + \boldsymbol{\beta}(\mathbf{n} \cdot \mathbf{m})]}{R^3}$$

$$\mathbf{E} = \mathbf{B} \times \boldsymbol{\beta}$$

where $\mathbf{B}$ is the magnetic dipole field. In light of Problem 6.22, comment on the interpretation of the different forms.

Solution.

(a) The inverse potential transformation is $\Phi = \gamma(\Phi' + \boldsymbol{\beta} \cdot \mathbf{A}')$, $\mathbf{A}_\parallel = \gamma(\mathbf{A}'_\parallel + \boldsymbol{\beta}\Phi')$, and $\mathbf{A}_\perp = \mathbf{A}'_\perp$. With $\Phi' = 0$, and $\mathbf{r}' = \mathbf{R} + O(\beta^2)$, this gives

$$\Phi = \frac{\boldsymbol{\beta} \cdot (\mathbf{m} \times \mathbf{R})}{R^3} = \boxed{\frac{(\boldsymbol{\beta} \times \mathbf{m}) \cdot \mathbf{R}}{R^3}}, \qquad \boxed{\mathbf{A} = \frac{\mathbf{m} \times \mathbf{R}}{R^3}}$$

to the requested order. The potential is in Lorenz gauge: $\nabla \cdot \mathbf{A} = 0$ and $c^{-1}\partial_t \Phi = O(\beta^2)$. **(b)** Put $\mathbf{p} = \boldsymbol{\beta} \times \mathbf{m}$ and $\mathbf{n} = \mathbf{R}/R$. For $R > 0$ the curl gives

$$\boxed{\mathbf{B} = \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{R^3} + O(\beta^2).}$$

The electric field has both a gradient and a time-derivative part:

$$-\nabla\Phi = \mathbf{E}_{\text{dipole}}(\mathbf{p}), \qquad -\frac{1}{c}\partial_t \mathbf{A} = (\boldsymbol{\beta} \cdot \nabla)\frac{\mathbf{m} \times \mathbf{R}}{R^3}.$$

Use $(\boldsymbol{\beta} \cdot \nabla)(\mathbf{R}/R^3) = [\boldsymbol{\beta} - 3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta})]/R^3$ to obtain the first requested form:

$$\boxed{\mathbf{E} = \mathbf{E}_{\text{dipole}}(\boldsymbol{\beta} \times \mathbf{m}) - \frac{\mathbf{m} \times [3\mathbf{n}(\mathbf{n} \cdot \boldsymbol{\beta}) - \boldsymbol{\beta}]}{R^3}.}$$

To rearrange it explicitly, use the vector identity

$$\mathbf{p} = \mathbf{n}(\mathbf{n} \cdot \mathbf{p}) + (\mathbf{n} \cdot \mathbf{m})\boldsymbol{\beta} \times \mathbf{n} - (\mathbf{n} \cdot \boldsymbol{\beta})\mathbf{m} \times \mathbf{n}.$$

It follows by resolving both $\boldsymbol{\beta}$ and $\mathbf{m}$ into components parallel and perpendicular to $\mathbf{n}$; the cross product of the two transverse components is $\mathbf{n}(\mathbf{n}\cdot\mathbf{p})$. Subtracting half the dipole field from the first expression and using this identity leaves

$$\boxed{\mathbf{E} = \mathbf{E}_{\text{dipole}}(\tfrac{1}{2}\boldsymbol{\beta}\times\mathbf{m}) + \frac{3}{2R^3}\mathbf{n}\times[\mathbf{m}(\mathbf{n}\cdot\boldsymbol{\beta}) + \boldsymbol{\beta}(\mathbf{n}\cdot\mathbf{m})].}$$

Alternatively substitute the identity for $\mathbf{n}(\mathbf{n}\cdot\mathbf{p})$ in the first expression. The terms proportional to $\mathbf{m}\times\mathbf{n}$ cancel, leaving

$$\mathbf{E} = \frac{3(\mathbf{n}\cdot\mathbf{m})\mathbf{n}\times\boldsymbol{\beta} - \mathbf{m}\times\boldsymbol{\beta}}{R^3} = \boxed{\mathbf{B}\times\boldsymbol{\beta}}.$$

This agrees with the inverse field transformation from a purely magnetic rest field. The physical electric moment inferred from the charge density is the full $\boldsymbol{\beta}\times\mathbf{m}$ of 11.27b. The first expression separates the irrotational field of that moment from the induction field. The second is a useful symmetric/antisymmetric rearrangement, dual to 6.22; its half-size dipole coefficient does not change the charge moment. The third makes the Lorentz transformation manifest. None of these partial terms by itself is the entire electric field.

For a point moment the displayed formulas are ordinary fields away from $R = 0$. The contact terms, if needed, follow by testing the derivative of R_j/R^3 over a small sphere: $\int n_i n_j\, d\Omega = (4\pi/3)\delta_{ij}$ gives

$$\partial_i \frac{R_j}{R^3} = \text{PV}\,\frac{\delta_{ij} - 3n_i n_j}{R^3} + \frac{4\pi}{3}\delta_{ij}\delta^{(3)}(\mathbf{R}).$$

Since $\nabla\times(\mathbf{m}\times\mathbf{R}/R^3) = \mathbf{m}\nabla\cdot(\mathbf{R}/R^3) - (\mathbf{m}\cdot\nabla)(\mathbf{R}/R^3)$, its contact coefficient is $(4\pi - 4\pi/3)\mathbf{m} = (8\pi/3)\mathbf{m}$. The two electric contributions give respectively $-(4\pi/3)\mathbf{p}\delta^{(3)}(\mathbf{R})$ and $(4\pi/3)(\mathbf{m}\times\boldsymbol{\beta})\delta^{(3)}(\mathbf{R})$. Their sum is $-(8\pi/3)\mathbf{p}\delta^{(3)}(\mathbf{R})$, so $\mathbf{E} = \mathbf{B}\times\boldsymbol{\beta}$ holds also as a distribution.

11.30 An isotropic linear material medium, characterized by the constitutive relations (in its rest frame K'), $\mathbf{D}' = \epsilon\mathbf{E}'$ and $\mu\mathbf{H}' = \mathbf{B}'$, is in uniform translation with velocity $\mathbf{v}$ in the inertial frame K. By exploiting the fact that $F_{\mu\nu} = (\mathbf{E}, \mathbf{B})$ and $G_{\mu\nu} = (\mathbf{D}, \mathbf{H})$ transform as second rank 4-tensors under Lorentz transformations, show that the macroscopic fields $\mathbf{D}$ and $\mathbf{H}$ are given in terms of $\mathbf{E}$ and $\mathbf{B}$ by

$$\mathbf{D} = \epsilon\mathbf{E} + \gamma^2\left(\epsilon - \frac{1}{\mu}\right)[\beta^2\mathbf{E}_\perp + \boldsymbol{\beta}\times\mathbf{B}]$$

$$\mathbf{H} = \frac{1}{\mu}\mathbf{B} + \gamma^2\left(\epsilon - \frac{1}{\mu}\right)[-\beta^2\mathbf{B}_\perp + \boldsymbol{\beta}\times\mathbf{E}]$$

where $\mathbf{E}_\perp$ and $\mathbf{B}_\perp$ are components perpendicular to $\mathbf{v}$.

Solution.

Decompose every field parallel and perpendicular to $\boldsymbol{\beta}$. Boosting the laboratory field tensor into the medium rest frame gives

$$\mathbf{E}'_\parallel = \mathbf{E}_\parallel, \quad \mathbf{B}'_\parallel = \mathbf{B}_\parallel, \quad \mathbf{E}'_\perp = \gamma(\mathbf{E}_\perp + \boldsymbol{\beta}\times\mathbf{B}), \quad \mathbf{B}'_\perp = \gamma(\mathbf{B}_\perp - \boldsymbol{\beta}\times\mathbf{E}).$$

Apply $\mathbf{D}' = \epsilon\mathbf{E}'$ and $\mathbf{H}' = \mathbf{B}'/\mu$, then transform G back to the laboratory. Its inverse boost has the same structure as that of $(\mathbf{E}, \mathbf{B})$, with velocity sign reversed:

$$\mathbf{D}_\parallel = \epsilon\mathbf{E}_\parallel, \qquad \mathbf{H}_\parallel = \mathbf{B}_\parallel/\mu,$$
$$\mathbf{D}_\perp = \gamma(\mathbf{D}'_\perp - \boldsymbol{\beta}\times\mathbf{H}'), \qquad \mathbf{H}_\perp = \gamma(\mathbf{H}'_\perp + \boldsymbol{\beta}\times\mathbf{D}').$$

Substituting the rest constitutive laws and then the first boost,

$$\begin{aligned}\mathbf{D}_\perp &= \gamma^2\left[\epsilon(\mathbf{E}_\perp+\boldsymbol{\beta}\times\mathbf{B})-\frac{1}{\mu}\boldsymbol{\beta}\times(\mathbf{B}_\perp-\boldsymbol{\beta}\times\mathbf{E})\right]\\ &= \gamma^2\left[(\epsilon-\beta^2/\mu)\mathbf{E}_\perp+(\epsilon-1/\mu)\boldsymbol{\beta}\times\mathbf{B}\right],\\ \mathbf{H}_\perp &= \gamma^2\left[\frac{1}{\mu}(\mathbf{B}_\perp-\boldsymbol{\beta}\times\mathbf{E})+\epsilon\boldsymbol{\beta}\times(\mathbf{E}_\perp+\boldsymbol{\beta}\times\mathbf{B})\right]\\ &= \gamma^2\left[(1/\mu-\epsilon\beta^2)\mathbf{B}_\perp+(\epsilon-1/\mu)\boldsymbol{\beta}\times\mathbf{E}\right].\end{aligned}$$

The second equalities use $\boldsymbol{\beta}\times(\boldsymbol{\beta}\times\mathbf{E})=-\beta^2\mathbf{E}_\perp$, and the same identity for $\mathbf{B}$. Finally $\gamma^2(1-\beta^2)=1$ implies

$$\gamma^2(\epsilon-\beta^2/\mu)=\epsilon+\gamma^2\beta^2(\epsilon-1/\mu),\qquad \gamma^2(1/\mu-\epsilon\beta^2)=1/\mu-\gamma^2\beta^2(\epsilon-1/\mu).$$

Adding the parallel components now gives

$$\boxed{\mathbf{D}=\epsilon\mathbf{E}+\gamma^2(\epsilon-1/\mu)[\beta^2\mathbf{E}_\perp+\boldsymbol{\beta}\times\mathbf{B}],}$$

$$\boxed{\mathbf{H}=\mathbf{B}/\mu+\gamma^2(\epsilon-1/\mu)[-\beta^2\mathbf{B}_\perp+\boldsymbol{\beta}\times\mathbf{E}].}$$

They reduce to the rest constitutive laws at $\beta=0$ and to $\mathbf{D}=\mathbf{E}$, $\mathbf{H}=\mathbf{B}$ for vacuum. The derivation assumes the stated local, isotropic constitutive law in the material rest frame; a dispersive nonlocal law would also require transforming its frequency and wave-vector arguments.

11.31 Consider a hollow right-circular cylinder of magnetic insulator (relative permittivity ϵ and permeability μ and inner and outer radii a and b) set in rotation about its axis at angular speed ω in a uniform axial magnetic field B_0. Neglect end effects and assume no free charge or current on either cylindrical surface. In 1913 the Wilsons measured the voltage difference between its inner and outer surfaces caused by a radial internal electric field. Assuming that locally the relations of Problem 11.30 hold, that the velocity $\mathbf{v}=\omega\rho\hat{\boldsymbol{\phi}}$, and that there are only the field components E_ρ and B_z, which are independent of z and ϕ, solve the equations $\nabla\cdot\mathbf{D}=0$ and $\nabla\times\mathbf{H}=0$ within the cylinder and show that the internal fields are

$$E_\rho=-\frac{\mu\omega\rho B_0}{c(1-\omega^2\rho^2/c^2)}\left(1-\frac{1}{\mu\epsilon}\right);\qquad B_z=\mu B_0\left[\frac{1-\omega^2\rho^2/(c^2\mu\epsilon)}{1-\omega^2\rho^2/c^2}\right]$$

and that for nonrelativistic motion ($\omega b/c\ll 1$) the voltage difference is

$$V=\frac{\mu\omega}{2c}B_0(b^2-a^2)\left(1-\frac{1}{\mu\epsilon}\right)$$

This experiment was historically regarded as an early validation of special relativity and Minkowski's electrodynamics of material media in motion; its composite-material limitation is discussed in the solution. If you are curious about how the Wilsons made a magnetic insulator, look up the paper.

Reference: M. Wilson and H. A. Wilson, *Proc. Roy. Soc. London* **A89**, 99–106 (1913).

Solution.

Use the axially uniform, effectively infinite-cylinder model specified in the question, with no free charge on either surface, no free surface current, and $|\omega|b<c$. The exterior magnetic field is $B_0\hat{\mathbf{z}}$.

Define $\beta = \omega\rho/c$, $\gamma^2 = (1-\beta^2)^{-1}$, $E = E_\rho$, and $B = B_z$. Because $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\boldsymbol{\rho}}$ and $\hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\rho}} = -\hat{\mathbf{z}}$, the formulas of 11.30 give

$$\begin{aligned} D_\rho &= \gamma^2[(\epsilon - \beta^2/\mu)E + (\epsilon - 1/\mu)\beta B], \\ H_z &= \gamma^2[(1/\mu - \epsilon\beta^2)B - (\epsilon - 1/\mu)\beta E]. \end{aligned}$$

The differential equations integrate once to

$$\frac{1}{\rho}\frac{d(\rho D_\rho)}{d\rho} = 0 \implies D_\rho = C/\rho, \qquad \frac{dH_z}{d\rho} = 0 \implies H_z = H_*.$$

In the empty bore a radial, divergence-free field has the same C/ρ form. Regularity at the axis makes its coefficient zero. Continuity of normal D without free surface charge then sets $C = 0$ inside the material as well. Continuity of tangential H at the outer wall, with vacuum outside, sets $H_* = B_0$. These boundary conditions fix the integration constants; the bulk differential equations alone do not.

Multiply the two constitutive equations by $1-\beta^2$ and set $A = \epsilon - 1/\mu$. The resulting linear system is

$$(\epsilon - \beta^2/\mu)E + A\beta B = 0, \qquad -A\beta E + (1/\mu - \epsilon\beta^2)B = (1-\beta^2)B_0.$$

Its determinant is

$$\begin{aligned} &(\epsilon - \beta^2/\mu)(1/\mu - \epsilon\beta^2) + A^2\beta^2 \\ &\qquad = \frac{\epsilon}{\mu}(1 - 2\beta^2 + \beta^4) = \frac{\epsilon}{\mu}(1-\beta^2)^2. \end{aligned}$$

For nonzero ϵ, μ it is nonzero at subluminal speed. Cramer's rule therefore gives

$$\boxed{E_\rho = -\frac{\mu\beta B_0}{1-\beta^2}\left(1 - \frac{1}{\mu\epsilon}\right), \qquad B_z = \mu B_0 \frac{1 - \beta^2/(\mu\epsilon)}{1-\beta^2}.}$$

Restoring $\beta = \omega\rho/c$ reproduces the first field in the question and fixes the ambiguous denominator of the second: its numerator is $1 - \omega^2\rho^2/(c^2\mu\epsilon)$. At $\omega = 0$, $E_\rho = 0$ and $B_z = \mu B_0$.

The outer-minus-inner potential difference is

$$\begin{aligned} V = \Phi(b) - \Phi(a) &= -\int_a^b E_\rho \, d\rho \\ &= \frac{\mu\omega B_0}{c}\left(1 - \frac{1}{\mu\epsilon}\right)\int_a^b \frac{\rho\, d\rho}{1 - \omega^2\rho^2/c^2} \\ &= \boxed{-\frac{\mu c B_0}{2\omega}\left(1 - \frac{1}{\mu\epsilon}\right)\ln\frac{1 - \omega^2 b^2/c^2}{1 - \omega^2 a^2/c^2}.} \end{aligned}$$

Use $\ln(1-x) = -x + O(x^2)$ to obtain

$$\boxed{V = \frac{\mu\omega B_0}{2c}(b^2 - a^2)\left(1 - \frac{1}{\mu\epsilon}\right) + O\!\left(\frac{\mu\omega^3 B_0 b^4}{c^3}\left|1 - \frac{1}{\mu\epsilon}\right|\right).}$$

The exact logarithmic expression has the continuous limit $V = 0$ at $\omega = 0$. For positive ωB_0 and $\mu\epsilon > 1$, the radial field is inward and the outer surface is at higher potential. The calculation is

conditional on the local moving-medium law stipulated in the problem; it does not assume a global inertial rest frame for the rotating material.

For the historical construction asked about, the Wilsons embedded steel balls in wax: the composite was insulating as a whole while the steel supplied its magnetic response. This is described and reexamined in the primary experimental report by Hertzberg et al. (2001). They explain that the original composite could not distinguish competing predictions for a homogeneous rotating magnetic insulator, because its magnetization lay in conducting steel. Their subsequent measurement using a homogeneous magnetic insulator, yttrium iron garnet, supports the locally inertial constitutive treatment assumed in this problem. This material qualification motivates the marked refinement of the historical claim; it does not alter the conditional field calculation above. [7]

[7] J. B. Hertzberg et al., "Measurement of the Relativistic Potential Difference across a Rotating Magnetic Dielectric Cylinder," primary experimental report, 2001, abstract and introduction, `https://arxiv.org/abs/gr-qc/0104101`.

12 Chapter 12. Dynamics of Relativistic Particles and Electromagnetic Fields

Problems

12.1

(a) Show that the Lorentz invariant Lagrangian (in the sense of Section 12.1B)

$$L = -\frac{mU_\alpha U^\alpha}{2} - \frac{q}{c} U_\alpha A^\alpha$$

gives the correct relativistic equations of motion for a particle of mass m and charge q interacting with an external field described by the 4-vector potential $A^\alpha(x)$.

(b) Define the canonical momenta and write out the effective Hamiltonian in both covariant and space-time form. The effective Hamiltonian is a Lorentz invariant. What is its value?

Solution.

(a) Use $g_{\alpha\beta} = \mathrm{diag}(1,-1,-1,-1)$, $x^0 = ct$, $U^\alpha = dx^\alpha/d\tau$ and Gaussian units. As in Section 12.1B, vary the four coordinates before imposing $U_\alpha U^\alpha = c^2$; substituting that constraint into the Lagrangian before variation would incorrectly remove its kinetic term. The derivatives entering the Euler–Lagrange equation are

$$\frac{\partial L}{\partial U^\alpha} = -mU_\alpha - \frac{q}{c}A_\alpha, \qquad \frac{\partial L}{\partial x^\alpha} = -\frac{q}{c}U^\beta \partial_\alpha A_\beta.$$

Consequently

$$-m\frac{dU_\alpha}{d\tau} - \frac{q}{c}U^\beta \partial_\beta A_\alpha + \frac{q}{c}U^\beta \partial_\alpha A_\beta = 0,$$

or

$$\boxed{m\frac{dU^\alpha}{d\tau} = \frac{q}{c}F^{\alpha\beta}U_\beta.}$$

Antisymmetry of F gives $d(U^2)/d\tau = 0$, so the proper-time normalization, once imposed initially, is preserved. Its spatial and time components are respectively $d\mathbf{p}/dt = q(\mathbf{E} + \mathbf{v}\times\mathbf{B}/c)$ and $d(\gamma mc^2)/dt = q\mathbf{E}\cdot\mathbf{v}$.

(b) For consistency with Jackson's (12.33)–(12.34), define the physical canonical four-momentum with the minus sign used there:

$$P_\alpha = -\frac{\partial L}{\partial U^\alpha} = mU_\alpha + \frac{q}{c}A_\alpha, \qquad \widetilde{H} = P_\alpha U^\alpha + L.$$

Eliminating U gives

$$\boxed{\widetilde{H} = \frac{1}{2m}\left(P_\alpha - \frac{q}{c}A_\alpha\right)\left(P^\alpha - \frac{q}{c}A^\alpha\right).}$$

In space-time notation $P^\alpha = (W/c, \mathbf{P})$, $W = \gamma mc^2 + q\Phi$, $\mathbf{P} = \gamma m\mathbf{v} + q\mathbf{A}/c$, so

$$\boxed{\widetilde{H} = \frac{1}{2m}\left[\frac{(W-q\Phi)^2}{c^2} - \left|\mathbf{P} - \frac{q}{c}\mathbf{A}\right|^2\right] = \frac{1}{2}mc^2.}$$

This invariant generates proper-time motion; it is not the laboratory energy W. Hamilton's equations reproduce the force equation when applied before substituting its on-shell value: explicitly,

$$\frac{dx^\alpha}{d\tau} = \frac{\partial \widetilde{H}}{\partial P_\alpha} = \frac{1}{m}\left(P^\alpha - \frac{q}{c}A^\alpha\right) = U^\alpha, \qquad \frac{dP_\alpha}{d\tau} = -\frac{\partial \widetilde{H}}{\partial x^\alpha} = \frac{q}{c}U^\beta \partial_\alpha A_\beta.$$

Differentiating $P_\alpha = mU_\alpha + (q/c)A_\alpha$ in the second equation supplies the antisymmetric field tensor in part (a). If instead one uses the ordinary Legendre definition $\pi_\alpha = \partial L/\partial U^\alpha$ for this negative-sign action, then $\pi = -P$ and $K = \pi_\alpha U^\alpha - L = -\widetilde{H} = -mc^2/2$. The sign difference is entirely a convention; the boxed positive value uses the convention explicitly requested by Section 12.1B.

12.2

(a) Show from Hamilton's principle that Lagrangians that differ only by a total time derivative of some function of the coordinates and time are equivalent in the sense that they yield the same Euler–Lagrange equations of motion.

(b) Show explicitly that the gauge transformation $A^\alpha \to A^\alpha + \partial^\alpha \Lambda$ of the potentials in the charged-particle Lagrangian (12.12) merely generates another equivalent Lagrangian.

Solution.

(a) Let $L' = L + dF(\mathbf{q}, t)/dt$, with F differentiable. For paths with fixed endpoints,

$$S'[\mathbf{q}] - S[\mathbf{q}] = F(\mathbf{q}_f, t_f) - F(\mathbf{q}_i, t_i), \qquad \delta(S' - S) = 0.$$

Thus stationary paths coincide. The actions need not have identical numerical values: it is their variations that agree. One can also check the Euler–Lagrange expression directly. Since

$$\frac{dF}{dt} = F_t + \sum_j F_{q_j}\dot{q}_j,$$

its contribution for coordinate q_i is

$$\frac{d}{dt}F_{q_i} - \left(F_{tq_i} + \sum_j F_{q_j q_i}\dot{q}_j\right) = 0.$$

(b) In the metric used here $A^\alpha = (\Phi, \mathbf{A})$, so the stated transformation means

$$\Phi' = \Phi + \frac{1}{c}\partial_t \Lambda, \qquad \mathbf{A}' = \mathbf{A} - \nabla\Lambda.$$

Substitute into (12.12), $L = -mc^2\sqrt{1 - v^2/c^2} - q\Phi + (q/c)\mathbf{v}\cdot\mathbf{A}$:

$$L' - L = -\frac{q}{c}\left(\partial_t \Lambda + \mathbf{v}\cdot\nabla\Lambda\right) = \boxed{-\frac{q}{c}\frac{d\Lambda(\mathbf{x}(t), t)}{dt}.}$$

This is part (a) with $F = -q\Lambda/c$. The equivalent covariant statement is $\Delta L_\tau = -(q/c)d\Lambda/d\tau$. Canonical energy and momentum change with the gauge, but the kinetic momentum and electromagnetic force do not.

12.3 A particle with mass m and charge e moves in a uniform, static, electric field $\mathbf{E}_0$.

(a) Solve for the velocity and position of the particle as explicit functions of time, assuming that the initial velocity $\mathbf{v}_0$ was perpendicular to the electric field.

(b) Eliminate the time to obtain the trajectory of the particle in space. Discuss the shape of the path for short and long times (define "short" and "long" times).

Solution.

(a) Choose $\mathbf{E}_0 = E_0\hat{\mathbf{x}}$ and $\mathbf{v}_0 = v_0\hat{\mathbf{y}}$, and set the initial position to zero. Write $F = eE_0$, $p_0 = \gamma_0 m v_0$, $\gamma_0 = (1 - v_0^2/c^2)^{-1/2}$. The force equation integrates to

$$p_x = Ft, \quad p_y = p_0, \quad p_z = 0, \qquad \gamma mc = \sqrt{(\gamma_0 mc)^2 + F^2t^2}.$$

The last equality uses $m^2c^2 + p_0^2 = \gamma_0^2 m^2 c^2$. Therefore

$$\boxed{v_x = \frac{cFt}{\sqrt{(\gamma_0 mc)^2 + F^2t^2}}, \qquad v_y = \frac{cp_0}{\sqrt{(\gamma_0 mc)^2 + F^2t^2}}, \quad v_z = 0.}$$

For the position integrals put $\xi = Ft/(\gamma_0 mc)$. The antiderivatives are $\int \xi\, d\xi/\sqrt{1+\xi^2} = \sqrt{1+\xi^2}$ and $\int d\xi/\sqrt{1+\xi^2} = \operatorname{arsinh}\xi$. Subtracting their values at $t = 0$ gives, for $F \neq 0$,

$$\boxed{x(t) = \frac{\gamma_0 mc^2}{F}\left[\sqrt{1 + \left(\frac{Ft}{\gamma_0 mc}\right)^2} - 1\right], \quad y(t) = \frac{p_0 c}{F}\operatorname{arsinh}\left(\frac{Ft}{\gamma_0 mc}\right).}$$

(b) The signs also cover a negative charge. Eliminating t when $p_0 \neq 0$ yields a catenary turned on its side:

$$\boxed{x = \frac{\gamma_0 mc^2}{F}\left[\cosh\left(\frac{Fy}{p_0 c}\right) - 1\right].}$$

The crossover time is $t_* = \gamma_0 mc/|F|$, when the gained longitudinal momentum is comparable with the initial total energy in units of c. For $|t| \ll t_*$,

$$x = \frac{Ft^2}{2\gamma_0 m} + O(t^4), \qquad y = v_0 t + O(t^3), \qquad x \simeq \frac{F}{2\gamma_0 m v_0^2}y^2.$$

For $t \gg t_*$,

$$x \simeq \operatorname{sgn}(F)c(t - t_*), \qquad y \simeq \frac{p_0 c}{|F|}\log(2t/t_*).$$

The velocity tends to $\operatorname{sgn}(F)c\hat{\mathbf{x}}$; the transverse displacement still grows logarithmically. If $v_0 = 0$, the trajectory is simply the field line $y = z = 0$, and if $F = 0$ the continuous limit is uniform straight motion.

12.4 It is desired to make an $E \times B$ velocity selector with uniform, static, crossed, electric and magnetic fields over a length L. If the entrance and exit slit widths are Δx, discuss the interval Δu of velocities, around the mean value $u = cE/B$, that is transmitted by the device as a function of the mass, the momentum or energy of the incident particles, the field strengths, the length of the selector, and any other relevant variables. Neglect fringing effects at the ends. Base your

discussion on the practical facts that $L \sim$ few meters, $E_{\max} \sim 3 \times 10^6$ V/m, $\Delta x \sim 10^{-3}$–10^{-4} m, $u \sim 0.5$–$0.995c$. (It is instructive to consider the equation of motion in a frame moving at the mean speed u along the beam direction, as well as in the laboratory.)

References: C. A. Coombes et al., *Phys. Rev.* **112**, 1303 (1958); P. Eberhard, M. L. Good, and H. K. Ticho, *Rev. Sci. Instrum.* **31**, 1054 (1960).

Solution.

Take the beam along z, $\mathbf{E} = E\widehat{\mathbf{x}}$ and $\mathbf{B} = B\widehat{\mathbf{y}}$, with $E, B > 0$. Let $u = cE/B$, $\Gamma = (1-u^2/c^2)^{-1/2}$ and write the incident speed as $u + \delta u$. The following bandwidth calculation is a linearization about the selected orbit; it is reliable only when $|\Gamma^2 \delta u| \ll c$ and angular deviations are small.

In the frame moving at u, $E' = 0$, $B' = B/\Gamma$, and the small incident velocity is $v'_z = \Gamma^2 \delta u$. The cyclotron frequency there is $\omega' = qB/(\Gamma mc)$. To first order in the velocity mismatch,

$$x' = -\frac{\Gamma^2 \delta u}{\omega'}(1 - \cos \omega' t'), \qquad z' = \frac{\Gamma^2 \delta u}{\omega'} \sin \omega' t'.$$

At the exit plane $t' = L/(\Gamma u)$ to zeroth order, sufficient in the already first-order transverse displacement. Hence

$$\boxed{d \equiv x(L) - x(0) = D\,\delta u, \qquad D = -\frac{\Gamma^3 mc}{qB}(1 - \cos\varphi), \quad \varphi = \frac{qBL}{\Gamma^2 mc\,u} = \frac{qEL}{\Gamma^2 mu^2}.}$$

The SI version replaces qB/c by qB and has $B = E/u$. The same result follows in the laboratory from

$$\dot{v}_x = -\frac{qB}{\Gamma mc}\delta v_z, \qquad \delta\dot{v}_z = \frac{qB}{\Gamma^3 mc} v_x,$$

whose oscillation frequency is $qB/(\Gamma^2 mc)$. For an initial small angle $\eta = v_x(0)/u$, the displacement gains $\Gamma u \eta \sin\varphi/\omega'$; thus the beam emittance and collimation also determine resolution.

For two equal slits of full width Δx and a parallel beam uniformly illuminating the entrance slit, the transmitted fraction at a given speed is

$$T(\delta u) = \max\left(0, 1 - \frac{|D\delta u|}{\Delta x}\right).$$

Thus the full interval with nonzero geometrical transmission is

$$\boxed{\Delta u_{\text{support}} = \frac{2\Delta x}{|D|}.}$$

The full width at half maximum is half this value. A pencil beam starting at the center of the entrance slit also has full acceptance width $\Delta x/|D|$. Stating this convention avoids an arbitrary factor of two in the bandwidth.

When $|\varphi| \ll 1$,

$$D \simeq -\frac{qEL^2}{2\Gamma mu^3}, \qquad \boxed{\frac{\Delta u_{\text{support}}}{u} \simeq \frac{4\Gamma mu^2 \Delta x}{|q|EL^2} = \frac{4pu\Delta x}{|q|EL^2}.}$$

Here $p = \Gamma mu$, and $pu = \mathcal{E}\,u^2/c^2$ for total particle energy $\mathcal{E}$. Narrower slits, longer guides and larger electric fields improve this small-phase resolution, while large momentum worsens it. Magnetic field and electric field cannot be varied independently at fixed selected speed. At $\varphi = 2\pi j$ the linear

transverse displacement vanishes: refocusing defeats the selector. If the electric field is adjustable, the first-lobe sensitivity is proportional to $(1-\cos\varphi)/\varphi$; its maximum satisfies $\tan(\varphi/2)=\varphi$ and occurs at $\varphi = 2.33112237$.

For illustration use $L = 2$ m, $E = 3$ MV/m and $\Delta x = 0.1$ mm. The magnetic field is 0.0200–0.0101 T over the stated speed range. The linear formula gives

Particle	u/c	φ	$\Delta u_{\text{support}}/u$
electron	0.5	35.2251	1.71×10^{-3}
electron	0.9	2.75423	6.23×10^{-5}
electron	0.995	0.118304	1.69×10^{-4}
proton	0.5	0.0191842	9.03×10^{-3}
proton	0.9	0.0015000	5.81×10^{-2}
proton	0.995	6.4430×10^{-5}	0.310 (extrapolation)

The last entry violates the narrow-band assumption and is not a physical interval centered on $0.995c$. Integrating the exact relativistic force equation gives the connected interval $0.9500180c < v < c$ for nonzero slit overlap, with all speeds between the mean and c accepted; the device has poor velocity discrimination there. For the electrons at $0.5c$, reducing the field to the first-lobe optimum improves resolution and avoids the high-order oscillatory response. Widening the slits to 1 mm multiplies the local linear bandwidth by ten, until the linear approximation itself fails. For a broad incident spectrum one must use the exact trajectories, since additional disconnected transmission bands can occur.

12.5 A particle of mass m and charge e moves in the laboratory in crossed, static, uniform, electric and magnetic fields. $\mathbf{E}$ is parallel to the x axis; $\mathbf{B}$ is parallel to the y axis.

(a) For $|\mathbf{E}| < |\mathbf{B}|$ make the necessary Lorentz transformation described in Section 12.3 to obtain explicitly parametric equations for the particle's trajectory.

(b) Repeat the calculation of part a for $|\mathbf{E}| > |\mathbf{B}|$.

Solution.

Use Gaussian units, choose the initial event as the origin, and write the signed components as $\mathbf{E} = E\hat{\mathbf{x}}$, $\mathbf{B} = B\hat{\mathbf{y}}$. A boost along z with speed w acts on coordinates by

$$x = x', \quad y = y', \quad z = \Gamma_w(z' + wt'), \quad t = \Gamma_w(t' + wz'/c^2).$$

It transforms the initial four-velocity according to $U_0'^0 = \Gamma_w(U_0^0 - wU_0^z/c)$, $U_0'^z = \Gamma_w(U_0^z - wU_0^0/c)$, with $U_0'^x = U_0^x$, $U_0'^y = U_0^y$. Thus the following formulas accommodate arbitrary initial velocity, not just a particle initially at rest.

(a) For $|E| < |B|$, choose $w = cE/B$. Then $E' = 0$, $B' = B/\Gamma_w$ and the primed energy $\gamma' mc^2$ is constant. Define the signed frequency $\Omega' = eB'/(\gamma' mc)$ and $v'_{i0} = U_0'^i/\gamma'$. The two coupled equations are $\dot{v}'_x = -\Omega' v'_z$, $\dot{v}'_z = \Omega' v'_x$; v'_y is constant. Integration gives

$$\begin{aligned} x' &= \frac{v'_{x0}}{\Omega'}\sin\Omega' t' + \frac{v'_{z0}}{\Omega'}(\cos\Omega' t' - 1), \\ y' &= v'_{y0} t', \\ z' &= \frac{v'_{z0}}{\Omega'}\sin\Omega' t' + \frac{v'_{x0}}{\Omega'}(1 - \cos\Omega' t'). \end{aligned}$$

Together with the inverse boost these are explicit parametric laboratory trajectories, with parameter t'. They describe a helical motion in the magnetic frame and its $\mathbf{E}\times\mathbf{B}$ drift in the laboratory.

(b) For $|E|>|B|$, choose $w=cB/E$. Now $B'=0$, $E'=E/\Gamma_w$. Proper time is the convenient parameter; set $\kappa=eE'/(mc)$. The primed equations are

$$\frac{dU'^0}{d\tau}=\kappa U'^x, \qquad \frac{dU'^x}{d\tau}=\kappa U'^0, \qquad U'^y=U_0'^y, \quad U'^z=U_0'^z.$$

Their integrated trajectory is

$$\begin{aligned} ct' &= \frac{U_0'^0\sinh\kappa\tau + U_0'^x(\cosh\kappa\tau-1)}{\kappa},\\ x' &= \frac{U_0'^x\sinh\kappa\tau + U_0'^0(\cosh\kappa\tau-1)}{\kappa},\\ y' &= U_0'^y\tau, \qquad z' = U_0'^z\tau. \end{aligned}$$

Applying the same inverse boost gives $x(\tau),y(\tau),z(\tau),t(\tau)$. Here the particle gains unbounded energy rather than gyrating at constant energy. The formulas with zero frequency are interpreted by their sine or hyperbolic-sine limits. The separate null-field case $|E|=|B|$ is excluded by the question's strict inequalities; it is included in the general solution of Problem 12.6.

12.6 Static, uniform electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$, make an angle of θ with respect to each other.

(a) By a suitable choice of axes, solve the force equation for the motion of a particle of charge e and mass m in rectangular coordinates.

(b) For $\mathbf{E}$ and $\mathbf{B}$ parallel, show that with appropriate constants of integration, etc., the parametric solution can be written

$$x=AR\sin\phi, \qquad y=AR\cos\phi, \qquad z=\frac{R}{\rho}\sqrt{1+A^2}\,\cosh(\rho\phi)$$

$$ct=\frac{R}{\rho}\sqrt{1+A^2}\,\sinh(\rho\phi)$$

where $R=(mc^2/eB)$, $\rho=(E/B)$, A is an arbitrary constant, and ϕ is the parameter [actually c/R times the proper time].

Solution.

(a) For the general initial-value problem choose $\mathbf{B}=B\widehat{\mathbf{z}}$ and $\mathbf{E}=E\sin\theta\,\widehat{\mathbf{x}}+E\cos\theta\,\widehat{\mathbf{z}}$. Define signed inverse times

$$f=\frac{eE\sin\theta}{mc}, \qquad h=\frac{eE\cos\theta}{mc}, \qquad b=\frac{eB}{mc}.$$

With $X=(ct,x,y,z)^{\mathsf T}$ and $U=dX/d\tau$, the four rectangular equations are the constant-coefficient system

$$\frac{dU}{d\tau}=MU, \qquad M=\begin{pmatrix} 0 & f & 0 & h\\ f & 0 & b & 0\\ 0 & -b & 0 & 0\\ h & 0 & 0 & 0\end{pmatrix}.$$

This form includes the energy equation and avoids time-dependent γ coefficients. To evaluate its solution explicitly, put

$$S = f^2 + h^2 - b^2, \quad D = \sqrt{S^2 + 4h^2b^2}, \quad \kappa^2 = (D+S)/2, \quad \Omega^2 = (D-S)/2.$$

The characteristic identity is $M^4 - SM^2 - h^2b^2\mathbf{1} = 0$: expansion of $\det(\lambda\mathbf{1} - M)$ gives $\lambda^4 - (f^2 + h^2 - b^2)\lambda^2 - h^2b^2 = (\lambda^2 - \kappa^2)(\lambda^2 + \Omega^2)$, and Cayley–Hamilton replaces λ by M. For $D \neq 0$ define

$$P_e = \frac{M^2 + \Omega^2\mathbf{1}}{D}, \qquad P_b = \frac{\kappa^2\mathbf{1} - M^2}{D}.$$

These complementary projectors obey $M^2P_e = \kappa^2P_e$ and $M^2P_b = -\Omega^2P_b$. Thus no unevaluated matrix exponential is needed:

$$U(\tau) = \left\{P_e\left[\cosh\kappa\tau\,\mathbf{1} + \frac{\sinh\kappa\tau}{\kappa}M\right] + P_b\left[\cos\Omega\tau\,\mathbf{1} + \frac{\sin\Omega\tau}{\Omega}M\right]\right\}U_0,$$

$$X(\tau) - X_0 = \left\{P_e\left[\frac{\sinh\kappa\tau}{\kappa}\mathbf{1} + \frac{\cosh\kappa\tau - 1}{\kappa^2}M\right] + P_b\left[\frac{\sin\Omega\tau}{\Omega}\mathbf{1} + \frac{1 - \cos\Omega\tau}{\Omega^2}M\right]\right\}U_0.$$

The four entries give the requested rectangular coordinates for any initial four-velocity U_0 satisfying $U_0^2 = c^2$. Terms with a zero κ or Ω have their continuous limits, e.g. $(1 - \cos\Omega\tau)/\Omega^2 \to \tau^2/2$. If $D = 0$, the field is zero or null ($E = B$, $\mathbf{E} \perp \mathbf{B}$). Then $M^3 = 0$ and the complete nonsingular solution is

$$U = (\mathbf{1} + \tau M + \tfrac{1}{2}\tau^2M^2)U_0, \qquad X = X_0 + (\tau\mathbf{1} + \tfrac{1}{2}\tau^2M + \tfrac{1}{6}\tau^3M^2)U_0.$$

In all cases $M^{\mathsf{T}}g + gM = 0$, which verifies that the four-velocity normalization is conserved.

(b) For parallel fields $f = 0$, the (U^0, U^z) pair is hyperbolic and the (U^x, U^y) pair rotates. Put $R = mc^2/(eB)$, $\rho = E/B$, $\phi = c\tau/R$. Choose integration constants so that

$$U^0 = c\sqrt{1 + A^2}\cosh(\rho\phi), \quad U^z = c\sqrt{1 + A^2}\sinh(\rho\phi), \quad U^x = Ac\cos\phi, \quad U^y = -Ac\sin\phi.$$

These satisfy $U^2 = c^2$ and all four equations directly. Integrating, with the permitted coordinate offsets, gives

$$\boxed{x = AR\sin\phi, \quad y = AR\cos\phi, \quad z = \frac{R}{\rho}\sqrt{1 + A^2}\cosh(\rho\phi), \quad ct = \frac{R}{\rho}\sqrt{1 + A^2}\sinh(\rho\phi).}$$

The apparently divergent constant in z as $E \to 0$ is removed by shifting its origin before taking the limit; the physical motion then becomes the usual helix or circle.

12.7 A constant uniform magnetic induction B in the negative z direction exists in a region limited by the planes $x = 0$ and $x = a$. For $x < 0$ and $x > a$, there is no magnetic induction.

(a) Determine the total electromagnetic momentum $\mathbf{G}$ in magnitude and direction of the combination of a particle with point charge q at (x_0, y_0, z_0) in the presence of this magnetic induction. Find $\mathbf{G}$ for the charge located on either side of and within the region occupied by the magnetic field. Assume the particle is at rest or in nonrelativistic motion. For the ideal infinite slab, take symmetric circular limits in each transverse plane when evaluating the field momentum.

(b) The particle is normally incident on the field region from $x < 0$ with nonrelativistic momentum p. Assuming that $p > |q|Ba/c$, determine the components of momentum after the particle has emerged into the field-free region, $x > a$. Compare the components of the sum of mechanical (particle) and electromagnetic momenta initially and finally. Why are some components of the sum conserved and some not?

(c) Assume that $p < |q|Ba/2c$ and that the initial conditions are such that the particle's motion is confined within the region of the magnetic induction at fixed z. Discuss the conservation, or lack of it, of the components of the sum of mechanical and electromagnetic momentum as the particle moves in its path. Comment.

Solution.

(a) Compute the interaction field momentum $\mathbf{G} = (4\pi c)^{-1} \int \mathbf{E}_q \times \mathbf{B}\, d^3x$; the point charge's divergent self-energy is not part of this cross term. The ideal infinite slab is interpreted with symmetric transverse integration. This prescription sets integrals of the odd transverse Coulomb components to zero; without a prescription some all-space component integrals are only conditionally convergent. At a fixed x,

$$\int_{\mathbb{R}^2} E_x \, dy\, dz = 2\pi q \int_0^\infty \frac{(x - x_0)s\, ds}{[(x - x_0)^2 + s^2]^{3/2}} = 2\pi q \, \operatorname{sgn}(x - x_0).$$

As $\mathbf{B} = -B\widehat{\mathbf{z}}$, $\mathbf{E} \times \mathbf{B} = B(-E_y, E_x, 0)$, and therefore

$$\boxed{\mathbf{G} = \frac{qB}{2c}\widehat{\mathbf{y}} \begin{cases} a, & x_0 < 0, \\ a - 2x_0, & 0 \le x_0 \le a, \\ -a, & x_0 > a. \end{cases}}$$

Its magnitude is the absolute value of the displayed coefficient; its direction reverses with that coefficient. The result is continuous at both walls and independent of y_0, z_0.

(b) For normal incidence set $p_x = p > 0$, $p_y = 0$ before the slab. The force components in the slab obey

$$\dot{p}_x = -\frac{qB}{c} v_y, \qquad \dot{p}_y = \frac{qB}{c} v_x, \qquad p_x^2 + p_y^2 = p^2.$$

Consequently $p_y = qBx/c$ on the first traversal. If $p > |q|Ba/c$, the particle exits through $x = a$ with

$$\boxed{p_{x,f} = \sqrt{p^2 - (qBa/c)^2}, \qquad p_{y,f} = qBa/c, \qquad p_{z,f} = 0.}$$

The initial and final sums of particle and cross-field momentum are

$$\mathbf{P}_i = (p, qBa/(2c), 0), \qquad \mathbf{P}_f = (\sqrt{p^2 - (qBa/c)^2}, qBa/(2c), 0).$$

The y, z components are conserved, but the x component changes. The external currents and supports maintaining the slab exchange momentum with the particle-field subsystem. The slab is translation invariant along y, z, but its two interfaces break translation invariance along x.

(c) For an orbit entirely inside the slab, let its center be (x_c, y_c) and its radius $r_g = pc/(|q|B) < a/2$, with $r_g < x_c < a - r_g$. Integrating the same equations gives

$$p_y = \frac{qB}{c}(x - x_c), \qquad p_x = -\frac{qB}{c}(y - y_c).$$

It follows that

$$\boxed{P_y = p_y + G_y = \frac{qB}{c}(a/2 - x_c), \qquad P_z = 0,}$$

whereas $P_x = p_x$ oscillates. Even though the orbit never touches the interfaces, the source apparatus still exchanges x momentum through its interaction with the charge's field. The result does not describe an isolated mechanical system. The charge-sign corrections in the inequalities make explicit the intended radius conditions; the originally printed inequalities assumed $q > 0$.

12.8 In Problems 6.5 and 6.6 a nonvanishing momentum of the electromagnetic fields was found for a charge and a current-carrying toroid at rest. This paradox is among situations involving "hidden momentum." Since the field momentum is proportional to $1/c^2$, you may infer that relativistic effects may enter the considerations.

(a) Consider the charge carriers in the toroid (or other current-carrying systems) of mass m, charge e, and individual mechanical momentum $\mathbf{p} = \gamma m\mathbf{v}$, and the current density $\mathbf{J} = en\mathbf{v}$, where n is the number density of carriers and γ is the relativistic Lorentz factor. Use conservation of energy for each charge carrier to show that, for the "static" field situation of Problem 6.5, the total mechanical momentum of the charge carriers,

$$\mathbf{P}_{\text{mech}} = \int d^3x\, \gamma nm\mathbf{v} = -\frac{1}{c^2}\int d^3x\, \Phi\mathbf{J}$$

just opposite to the field momentum of Problem 6.5a.

(b) Consider the toroid of Problem 6.6 to be of rectangular cross section, with width w and height h both small compared to a, and hollow tubes of uniform cross section A_w. Show that the electrostatic potential energy difference between the inner and outer vertical segments of each tube yields the change in γmc^2 necessary to generate a net vertical mechanical momentum equal and opposite to the result of Problem 6.6a, with due regard to differences in units.

Reference: Vaidman, *op. cit.*

Solution.

(a) For a charge carrier on a stationary closed stream, energy conservation in the electrostatic field gives

$$\gamma mc^2 + e\Phi = \mathcal{E}_{\text{stream}}.$$

Since $\mathbf{J} = en\mathbf{v}$, its mechanical momentum density is

$$\gamma nm\mathbf{v} = \frac{\mathcal{E}_{\text{stream}}}{ec^2}\mathbf{J} - \frac{\Phi\mathbf{J}}{c^2}.$$

A localized steady current has $\nabla\cdot\mathbf{J} = 0$ and $\int \mathbf{J}\, d^3x = 0$, by integrating $\partial_j(x_i J_j)$ over space. Apply this separately to each closed stream if their energy constants differ. The first term then integrates to zero and

$$\boxed{\mathbf{P}_{\text{mech}} = -\frac{1}{c^2}\int \Phi\mathbf{J}\, d^3x = -\mathbf{P}_{\text{field}}.}$$

The density n need not be constant: conservation of current requires its product with the local speed to adjust as the carriers accelerate and decelerate. Replacing γ by a common constant would incorrectly discard this order-$1/c^2$ momentum.

(b) For the small rectangular turn let the inner and outer vertical segments have potentials Φ_i, Φ_o, and take conventional current I upward on the inner segment. This orientation produces positive azimuthal B and positive axial field momentum for a positive central charge Q. The energy difference gives

$$mc^2(\gamma_i - \gamma_o) = -e(\Phi_i - \Phi_o).$$

Each segment has length h and tube area A_w; because $en_j v_j A_w = I$, its signed vertical momentum is $\pm\gamma_j mIh/e$. Hence per turn

$$P_{z,\text{turn}} = \frac{mIh}{e}(\gamma_i - \gamma_o) = -\frac{Ih}{c^2}(\Phi_i - \Phi_o).$$

In Gaussian units $\Phi_i - \Phi_o \simeq Qw/a^2$, so N turns have

$$\boxed{P_{z,\text{mech}} \simeq -\frac{NIQwh}{c^2 a^2}.}$$

In SI units the potential difference is instead $Qw/(4\pi\epsilon_0 a^2)$, giving, with $A = wh$,

$$\boxed{P_{z,\text{mech}} \simeq -\frac{\mu_0 QINA}{4\pi a^2},}$$

exactly opposite to Problem 6.6(a). Reversing the current reverses both momenta. The transverse contributions vanish by symmetry around the toroid; the tube area cancels because it fixes carrier number and speed through the same current constraint.

12.9 The magnetic field of the earth can be represented approximately by a magnetic dipole of magnetic moment $M = 8.1 \times 10^{25}$ gauss-cm^3. Consider the motion of energetic electrons in the neighborhood of the earth under the action of this dipole field (Van Allen electron belts). [Note that **M** points south.]

(a) Show that the equation for a line of magnetic force is $r = r_0 \sin^2\theta$, where θ is the usual polar angle (colatitude) measured from the axis of the dipole, and find an expression for the magnitude of B along any line of force as a function of θ.

(b) A positively charged particle circles around a line of force in the equatorial plane with a gyration radius a and a mean radius R ($a \ll R$). Show that the particle's azimuthal position (east longitude) changes approximately linearly in time according to

$$\phi(t) = \phi_0 - \frac{3}{2}\left(\frac{a}{R}\right)^2 \omega_B (t - t_0)$$

where ω_B is the frequency of gyration at radius R.

(c) If, in addition to its circular motion of part b, the particle has a small component of velocity parallel to the lines of force, show that it undergoes small oscillations in θ around $\theta = \pi/2$ with a frequency $\Omega = (3/\sqrt{2})(a/R)\omega_B$. Find the change in longitude per cycle of oscillation in latitude.

(d) For an electron of 10 MeV kinetic energy at a mean radius $R = 3 \times 10^7$ m, find ω_B and a, and so determine how long it takes to drift once around the earth and how long it takes to execute one cycle of oscillation in latitude. Calculate the same quantities for an electron of 10 keV at the same radius.

Solution.

(a) Let M denote the magnitude of the dipole moment. In spherical coordinates about its axis,

$$B_r = \frac{2M\cos\theta}{r^3}, \qquad B_\theta = \frac{M\sin\theta}{r^3}.$$

A field line has $dr/(r\,d\theta) = B_r/B_\theta = 2\cot\theta$. Integration gives $\log r = 2\log\sin\theta + \text{constant}$, and therefore

$$\boxed{r = r_0\sin^2\theta, \qquad B(\theta) = \frac{M}{r_0^3}\frac{\sqrt{1+3\cos^2\theta}}{\sin^6\theta}.}$$

Here r_0 is the equatorial radius of that line.

(b) At the geographic equator Earth's southward moment produces a northward $\mathbf{B}$. The field magnitude is $B = M/R^3$ and $\nabla B = -(3B/R)\widehat{\mathbf{R}}$. Equation (12.55), or averaging the force on the small gyrating current loop, gives

$$\mathbf{v}_D = \frac{c\mu_g}{qB^2}\mathbf{B}\times\nabla B, \qquad \mu_g = \frac{p_\perp^2}{2\gamma mB} = \frac{q\omega_B a^2}{2c}.$$

Indeed, the small current loop experiences $\mathbf{F}_{\nabla B} = -\mu_g\nabla B$. Its slowly drifting center balances this force with $q\mathbf{v}_D\times\mathbf{B}/c$. Crossing that balance with $\mathbf{B}$ and taking $\mathbf{v}_D\perp\mathbf{B}$ gives the displayed drift velocity. For a positive charge $\omega_B = qB/(\gamma mc) > 0$, so

$$v_{D\phi} = -\frac{3a^2\omega_B}{2R}, \qquad \boxed{\phi(t) = \phi_0 - \frac{3}{2}(a/R)^2\omega_B(t-t_0).}$$

The direction is westward for positive charges and eastward for electrons. The approximation requires $a/R \ll 1$; a and R refer to the unperturbed local gyration and guiding center.

(c) For the small parallel motion put $\lambda = \pi/2 - \theta$. Along the same field line $r = R\cos^2\lambda$, and expansion gives

$$\frac{B(\lambda)}{B_0} = \frac{\sqrt{1+3\sin^2\lambda}}{\cos^6\lambda} = 1 + \frac{9}{2}\lambda^2 + O(\lambda^4).$$

Arc length from the equator is $s = R\lambda + O(\lambda^3)$. Conservation of $p_\perp^2/B$, with constant total γ, gives the mirror force $\dot{p}_\parallel = -\mu_g\partial_s B$. Thus

$$\gamma m\ddot{s} = -\frac{9\mu_g B_0}{R^2}s, \qquad \boxed{\Omega^2 = \frac{9v_\perp^2}{2R^2}, \quad \Omega = \frac{3}{\sqrt{2}}\frac{a}{R}\omega_B.}$$

This is a small-amplitude bounce frequency. The extra curvature drift from the small parallel velocity is second order in that velocity and is omitted consistently. Over a full bounce,

$$\boxed{\Delta\phi = -\frac{3}{2}\left(\frac{a}{R}\right)^2\omega_B\frac{2\pi}{\Omega} = -\sqrt{2}\pi\frac{a}{R}}$$

for a positive charge; the electron sign is opposite.

(d) For the numerical values convert the given moment to $M = 8.1\times10^{15}\ \mathrm{T\,m^3}$, giving $B = M/R^3 = 3.00\times10^{-7}$ T at $R = 3.00\times10^7$ m. In SI units use

$$\gamma = 1 + \frac{K}{m_e c^2}, \quad v = c\sqrt{1-\gamma^{-2}}, \quad |\omega_B| = \frac{eB}{\gamma m_e}, \quad a = \frac{v}{|\omega_B|},$$

$$T_D = \frac{2\pi}{(3/2)(a/R)^2|\omega_B|}, \qquad T_{\text{bounce}} = \frac{2\pi\sqrt{2}R}{3v}.$$

These correspond to nearly equatorial pitch, as in parts (b),(c).

Quantity	$K = 10$ MeV	$K = 10$ keV
γ	20.5695	1.01957
$\|\omega_B\|$ (rad/s)	2565.18	51751.8
a (m)	1.16732×10^5	1129.53
T_D (s)	107.854	57096.6
T_{bounce} (s)	0.296748	1.52010

Thus the drifts take about 1.80 min and 15.9 h, respectively. The small-radius ratios are 0.00389 and 3.77×10^{-5}, consistent with the expansion used here.

12.10 A charged particle finds itself instantaneously in the equatorial plane of the earth's magnetic field (assumed to be a dipole field) at a distance R from the center of the earth. Its velocity vector at that instant makes an angle α with the equatorial plane ($v_\parallel/v_\perp = \tan\alpha$). Assuming that the particle spirals along the lines of force with a gyration radius $a \ll R$, and that the flux linked by the orbit is a constant of the motion, find an equation for the maximum magnetic latitude λ reached by the particle as a function of the angle α. Plot a graph (*not a sketch*) of λ versus α. Mark parametrically along the curve the values of α for which a particle at radius R in the equatorial plane will hit the earth (radius R_0) for $R/R_0 = 1.2$, 1.5, 2.0, 2.5, 3, 4, 5.

Solution.

In a static magnetic field the total momentum magnitude p is constant. The linked-flux invariant is equivalent to $p_\perp^2/B = $ constant. At the equator $p_{\perp 0} = p\cos\alpha$ because α is measured from the equatorial plane, rather than from the field direction. At the turning point all momentum is transverse, so

$$\frac{p^2\cos^2\alpha}{B_0} = \frac{p^2}{B_m}, \qquad \frac{B_m}{B_0} = \sec^2\alpha.$$

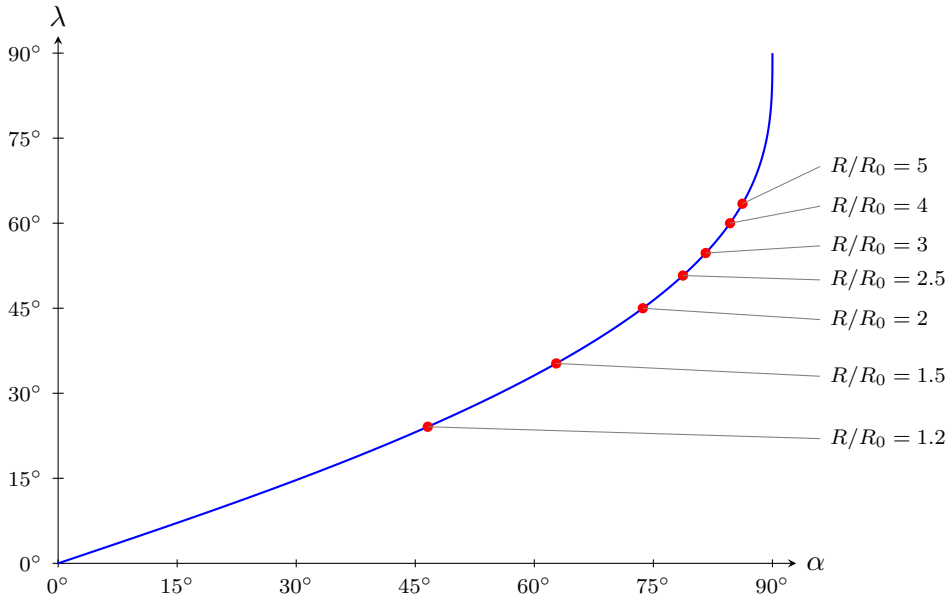
Using the field-line equation from Problem 12.9 gives

$$\boxed{\cos^2\alpha = \frac{\cos^6\lambda}{\sqrt{1+3\sin^2\lambda}}.}$$

For $f(\lambda) = \cos^6\lambda/\sqrt{1+3\sin^2\lambda}$,

$$\frac{f'}{f} = -6\tan\lambda - \frac{3\sin\lambda\cos\lambda}{1+3\sin^2\lambda} < 0 \qquad (0 < \lambda < \pi/2).$$

Thus on $0 \le \alpha, \lambda \le \pi/2$ the right side decreases from one to zero, so there is exactly one turning latitude for each launch angle. The sign of the initial parallel velocity chooses the northern or southern turning point; the formula gives its magnitude. The curve below is computed from this equation, with the limiting endpoints included.



A field line of equatorial radius R intersects Earth when $R\cos^2\lambda_E = R_0$. Consequently

$$\boxed{\lambda_E = \arccos\sqrt{R_0/R}, \qquad \alpha_E = \arccos\left[\frac{(R_0/R)^{3/2}}{(4-3R_0/R)^{1/4}}\right].}$$

The marked points have the following values:

R/R_0	α_E (degrees)	λ_E (degrees)
1.2	46.5759	24.0948
1.5	62.7595	35.2644
2.0	73.6699	45.0000
2.5	78.7220	50.7685
3.0	81.5915	54.7356
4.0	84.6582	60.0000
5.0	86.2233	63.4349

For $|\alpha| < \alpha_E$ the particle mirrors above the surface; for $|\alpha| \geq \alpha_E$ it reaches Earth before, or exactly at, its formal mirror point. This conclusion uses the ideal dipole and guiding-center approximation, neglecting the finite gyration radius in the surface-intersection condition.

12.11 Consider the precession of the spin of a muon, initially longitudinally polarized, as the muon moves in a circular orbit in a plane perpendicular to a uniform magnetic field $\mathbf{B}$.

(a) Show that the difference Ω of the spin precession frequency and the orbital gyration frequency is

$$\Omega = \frac{eBa}{m_\mu c}$$

independent of the muon's energy, where $a = (g-2)/2$ is the magnetic moment anomaly. (Find equations of motion for the components of spin along the mutually perpendicular directions defined by the particle's velocity, the radius vector from the center of the circle to the particle, and the magnetic field.)

(b) For the CERN Muon Storage Ring, the orbit radius is $R = 2.5$ meters and $B = 17 \times 10^3$ gauss. What is the momentum of the muon? What is the time dilatation factor γ? How many periods of precession $T = 2\pi/\Omega$ occur per observed laboratory mean lifetime of the muons? [$m_\mu = 105.66$ MeV, $\tau_0 = 2.2 \times 10^{-6}$ s, $a \simeq \alpha/2\pi$].

(c) Express the difference frequency Ω in units of the orbital rotation frequency and compute how many precessional periods (at the difference frequency) occur per rotation for a 300 MeV muon, a 300 MeV electron, a 5 GeV electron (this last typical of the e^+e^- storage ring at Cornell).

Solution.

(a) Let $a = (g-2)/2$ and choose $\widehat{\mathbf{b}}$ along the field. For $\mathbf{E} = 0$ and $\boldsymbol{\beta} \cdot \mathbf{B} = 0$, the BMT rest-spin equation (11.170) reduces to

$$\frac{d\mathbf{s}}{dt} = \boldsymbol{\omega}_s \times \mathbf{s}, \qquad \boldsymbol{\omega}_s = -\frac{e}{m_\mu c}\left(a + \frac{1}{\gamma}\right)\mathbf{B}.$$

The velocity and radius vectors rotate with $\boldsymbol{\omega}_c = -e\mathbf{B}/(\gamma m_\mu c)$. Thus in the frame rotating with the orbit,

$$\left(\frac{d\mathbf{s}}{dt}\right)_{\text{orbit}} = (\boldsymbol{\omega}_s - \boldsymbol{\omega}_c) \times \mathbf{s} = -\frac{ea}{m_\mu c}\mathbf{B} \times \mathbf{s}.$$

For a positive charge let $\widehat{\mathbf{t}} = \mathbf{v}/v$ and $\widehat{\mathbf{r}} = \widehat{\mathbf{b}} \times \widehat{\mathbf{t}}$ be the outward radius direction. With $\Omega = eaB/(m_\mu c)$ the component equations are

$$\dot{s}_t = \Omega s_r, \qquad \dot{s}_r = -\Omega s_t, \qquad \dot{s}_b = 0.$$

Initially longitudinal spin has $s_t = s\cos\Omega t$, $s_r = -s\sin\Omega t$, $s_b = 0$. Hence the difference-frequency magnitude

$$\boxed{|\Omega| = \frac{|e|Ba}{m_\mu c}}$$

is independent of energy. Reversing the charge reverses the precession direction without changing this magnitude.

(b) For the given ring, $B = 1.7$ T and $R = 2.5$ m. Converting the Gaussian orbit relation $pc = eBR$ to SI gives $p = eBR$, or

$$p[\text{GeV}/c] = 0.299792458\, B[\text{T}]R[\text{m}].$$

With the specified rest energy and $a = \alpha/(2\pi)$,

$$\boxed{p = 1.27412 \text{ GeV}/c, \quad \gamma = \sqrt{1 + (pc/m_\mu c^2)^2} = 12.1001,}$$

$$|\Omega| = 1.67944 \times 10^6 \text{ rad/s}, \quad T = 3.74123\ \mu\text{s}, \quad \boxed{\frac{\gamma\tau_0}{T} = 7.1153.}$$

This uses the approximation to the anomaly requested by the problem, not a fitted modern value.

(c) In any such orbit,

$$\boxed{\frac{|\Omega|}{|\omega_c|} = a\gamma,}$$

which is also the number of spin-orbit difference cycles per orbital revolution. Interpreting the quoted MeV/GeV values as total energies,

Particle and total energy	Cycles per orbit
300 MeV muon	0.00329759
300 MeV electron	0.681847
5 GeV electron	11.3641

If "300 MeV" is instead taken as kinetic energy, one must use $\gamma = 1 + K/(mc^2)$: the three numbers become 0.00445900, 0.683008 and 11.3653, respectively. The distinction matters most for the muon, so the energy convention has been made explicit.

12.12 In Section 11.11 the BMT equation of motion for the spin of a particle of charge e and a magnetic moment with an arbitrary g factor was obtained.

(a) Verify that (11.171) is the correct equation for the time derivative of the longitudinal component of the rest-frame spin vector $\mathbf{s}$.

(b) Let $\hat{\mathbf{n}}$ be a unit 3-vector perpendicular to $\hat{\boldsymbol{\beta}}$ and coplanar with $\hat{\boldsymbol{\beta}}$ and $\mathbf{s}$ ($\hat{\mathbf{n}}$ is generally time dependent). Let θ be the angle between $\hat{\boldsymbol{\beta}}$ and $\mathbf{s}$. Show that the time rate of change of θ can be written as

$$\frac{d\theta}{dt} = \frac{e}{mc}\left[\left(\frac{g}{2} - 1\right)\hat{\mathbf{n}} \cdot (\hat{\boldsymbol{\beta}} \times \mathbf{B}) + \left(\frac{g\beta}{2} - \frac{1}{\beta}\right)\hat{\mathbf{n}} \cdot \mathbf{E}\right]$$

where $\mathbf{E}$ and $\mathbf{B}$ are the fields in the laboratory and $c\boldsymbol{\beta} = c\beta\hat{\boldsymbol{\beta}}$ is the particle's instantaneous velocity in the laboratory.

(c) For a particle moving undeflected through an $E \times B$ velocity selector and with $(\hat{\mathbf{n}} \times \hat{\boldsymbol{\beta}}) \cdot \mathbf{B} = B$, find $d\theta/dt$ in terms of the gyration frequency $eB/\gamma mc$.

(d) By defining the two 4-vectors, $L^\alpha = (\gamma\beta, \gamma\hat{\boldsymbol{\beta}})$ and $N^\alpha = (0, \hat{\mathbf{n}})$, show that $d\theta/d\tau$ can be written in the quasi-covariant form

$$\frac{d\theta}{d\tau} = \frac{e}{mc}\left[\frac{g}{2}\, L_\alpha - \frac{1}{v}\, U_\alpha\right] F^{\alpha\beta} N_\beta$$

where U^α is the particle's 4-velocity.

Solution.

(a) Write $a_g = g/2 - 1$, $\mathbf{b} = \hat{\boldsymbol{\beta}}$ and $\mathbf{E}_\perp = \mathbf{E} - \mathbf{b}(\mathbf{b} \cdot \mathbf{E})$. Equations (11.168) and (11.170), in the notation needed here, are

$$\dot{\mathbf{b}} = \frac{e}{mc\gamma\beta}[\mathbf{E}_\perp + \beta\mathbf{b} \times \mathbf{B}], \qquad \dot{\mathbf{s}} = \boldsymbol{\omega}_s \times \mathbf{s},$$

$$\boldsymbol{\omega}_s = -\frac{e}{mc}\left[(a_g + \gamma^{-1})\mathbf{B} - \frac{a_g\gamma}{\gamma+1}(\boldsymbol{\beta} \cdot \mathbf{B})\boldsymbol{\beta} - (a_g + (\gamma+1)^{-1})\boldsymbol{\beta} \times \mathbf{E}\right].$$

Differentiate the longitudinal component $h = \mathbf{b} \cdot \mathbf{s}$:

$$\dot{h} = \mathbf{s} \cdot (\dot{\mathbf{b}} + \mathbf{b} \times \boldsymbol{\omega}_s).$$

The term in $\boldsymbol{\omega}_s$ parallel to $\mathbf{b}$ drops out. Use $\mathbf{b}\times(\boldsymbol{\beta}\times\mathbf{E}) = -\beta\mathbf{E}_\perp$. The remaining coefficients combine as

$$\dot{\mathbf{b}} + \mathbf{b}\times\boldsymbol{\omega}_s = \frac{e}{mc}\Bigg[-a_g \mathbf{b}\times\mathbf{B} + \left(\frac{1}{\gamma\beta} - \frac{\beta}{\gamma+1} - a_g\beta\right)\mathbf{E}_\perp\Bigg].$$

Since $1/(\gamma\beta) - \beta/(\gamma+1) = 1/\beta - \beta$,

$$\boxed{\dot{h} = -\frac{e}{mc}\mathbf{s}_\perp \cdot \left[a_g \mathbf{b}\times\mathbf{B} + \left(\frac{g\beta}{2} - \frac{1}{\beta}\right)\mathbf{E}\right].}$$

This verifies the precise form of (11.171), including the derivative of the velocity direction, which cannot be omitted.

(b) The spin length is constant because its derivative is a cross product. Write $h = s\cos\theta$, $\mathbf{s}_\perp = s\sin\theta\,\hat{\mathbf{n}}$. Dividing $\dot{h} = -s\sin\theta\,\dot{\theta}$ by the same factor therefore gives

$$\boxed{\dot{\theta} = \frac{e}{mc}\left[\left(\frac{g}{2}-1\right)\hat{\mathbf{n}}\cdot(\mathbf{b}\times\mathbf{B}) + \left(\frac{g\beta}{2} - \frac{1}{\beta}\right)\hat{\mathbf{n}}\cdot\mathbf{E}\right].}$$

At exactly parallel spin, the direction $\hat{\mathbf{n}}$ is fixed by a continuous limit from the ensuing rotation plane.

(c) For the undeflected selector $\mathbf{E} = -\boldsymbol{\beta}\times\mathbf{B}$, and the stated orientation gives $\hat{\mathbf{n}}\cdot(\mathbf{b}\times\mathbf{B}) = B$, $\hat{\mathbf{n}}\cdot\mathbf{E} = -\beta B$. Thus

$$\dot{\theta} = \frac{eB}{mc}\left[\frac{g}{2} - 1 - \beta\left(\frac{g\beta}{2} - \frac{1}{\beta}\right)\right] = \boxed{\frac{g}{2\gamma}\frac{eB}{\gamma mc}.}$$

There is spin rotation even though the reference trajectory is straight.

(d) Finally $L_\alpha = (\gamma\beta, -\gamma\mathbf{b})$, $U_\alpha = (\gamma c, -\gamma c\beta\mathbf{b})$, $N_\alpha = (0, -\hat{\mathbf{n}})$, so, with $v = c\beta$,

$$\frac{g}{2}L_\alpha - \frac{1}{v}U_\alpha = \gamma\left(\frac{g\beta}{2} - \frac{1}{\beta}, -\left(\frac{g}{2}-1\right)\mathbf{b}\right).$$

Using $F^{0i} = -E_i$, $F^{ij} = -\epsilon_{ijk}B_k$, its contraction with $F^{\alpha\beta}N_\beta$ is

$$\gamma\left[\left(\frac{g\beta}{2} - \frac{1}{\beta}\right)\hat{\mathbf{n}}\cdot\mathbf{E} + \left(\frac{g}{2}-1\right)\hat{\mathbf{n}}\cdot(\mathbf{b}\times\mathbf{B})\right].$$

Because $d\theta/d\tau = \gamma\, d\theta/dt$, this is precisely the claimed quasi-covariant expression. "Quasi" is appropriate because the laboratory-defined directions and speed enter L, N and v.

12.13

(a) Specialize the Darwin Lagrangian (12.82) to the interaction of two charged particles (m_1, q_1) and (m_2, q_2). Introduce reduced particle coordinates, $\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$, $\mathbf{v} = \mathbf{v}_1 - \mathbf{v}_2$ and also center of mass coordinates. Write out the Lagrangian in the reference frame in which the velocity of the center of mass vanishes and evaluate the canonical momentum components, $p_x = \partial L/\partial v_x$, etc.

(b) Calculate the Hamiltonian to first order in $1/c^2$ and show that it is

$$H = \frac{p^2}{2}\left(\frac{1}{m_1}+\frac{1}{m_2}\right) + \frac{q_1 q_2}{r} - \frac{p^4}{8c^2}\left(\frac{1}{m_1^3}+\frac{1}{m_2^3}\right) + \frac{q_1 q_2}{2m_1 m_2 c^2}\left(\frac{p^2+(\mathbf{p}\cdot\hat{\mathbf{r}})^2}{r}\right)$$

Compare with the various terms in (42.1) of Bethe and Salpeter [*op. cit.* (Section 12.6), p. 193]. Discuss the agreements and disagreements.

Solution.

(a) Drop the constant rest energies and retain terms through c^{-2}. The two-particle form of the Darwin Lagrangian is

$$L = \sum_{i=1}^{2}\left(\frac{m_i v_i^2}{2} + \frac{m_i v_i^4}{8c^2}\right) - \frac{Q}{r} + \frac{Q}{2c^2 r}[\mathbf{v}_1\cdot\mathbf{v}_2 + (\mathbf{v}_1\cdot\hat{\mathbf{r}})(\mathbf{v}_2\cdot\hat{\mathbf{r}})], \quad Q = q_1 q_2.$$

Introduce $M = m_1 + m_2$, $\mu = m_1 m_2/M$ and $\mathbf{R} = (m_1\mathbf{x}_1 + m_2\mathbf{x}_2)/M$. Then

$$\mathbf{v}_1 = \mathbf{V} + \frac{m_2}{M}\mathbf{v}, \qquad \mathbf{v}_2 = \mathbf{V} - \frac{m_1}{M}\mathbf{v}.$$

At $\mathbf{V} = 0$ define $C = m_1(m_2/M)^4 + m_2(m_1/M)^4 = \mu^4(m_1^{-3} + m_2^{-3})$ and $v_r = \mathbf{v}\cdot\hat{\mathbf{r}}$. Substitution term by term gives

$$\boxed{L = \frac{1}{2}\mu v^2 - \frac{Q}{r} + \frac{Cv^4}{8c^2} - \frac{Q\mu}{2Mc^2 r}(v^2 + v_r^2).}$$

For example the interaction sign follows from $\mathbf{v}_1\cdot\mathbf{v}_2 = -(\mu/M)v^2$. The three canonical momentum components are

$$\boxed{p_i = \mu v_i + \frac{C}{2c^2}v^2 v_i - \frac{Q\mu}{Mc^2 r}(v_i + v_r\hat{r}_i), \qquad i = x, y, z.}$$

The Newtonian center-of-mass velocity and the exact center-of-energy velocity differ at order c^{-2}. Eliminating the latter at zero total canonical momentum changes this reduced Lagrangian only at order c^{-4}, since its leading center term is $MV^2/2$.

(b) Forming $H = \mathbf{p}\cdot\mathbf{v} - L$ before inversion gives

$$H = \frac{1}{2}\mu v^2 + \frac{Q}{r} + \frac{3Cv^4}{8c^2} - \frac{Q\mu}{2Mc^2 r}(v^2 + v_r^2).$$

Iterate the momentum equation once, substituting $\mathbf{p}/\mu$ only in its already order-c^{-2} terms:

$$\mathbf{v} = \frac{\mathbf{p}}{\mu} - \frac{Cp^2\mathbf{p}}{2\mu^4 c^2} + \frac{Q}{M\mu c^2 r}[\mathbf{p} + (\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}] + O(c^{-4}).$$

Consequently the correction from $\mu v^2/2$ is $-Cp^4/(2\mu^4 c^2) + Q[p^2 + p_r^2]/(M\mu c^2 r)$. Combining it with the two explicit corrections above yields

$$\begin{aligned} H &= \frac{p^2}{2\mu} + \frac{Q}{r} - \frac{Cp^4}{8\mu^4 c^2} + \frac{Q}{2M\mu c^2 r}(p^2 + p_r^2) + O(c^{-4}) \\ &= \boxed{\frac{p^2}{2}\left(\frac{1}{m_1}+\frac{1}{m_2}\right) + \frac{Q}{r} - \frac{p^4}{8c^2}\left(\frac{1}{m_1^3}+\frac{1}{m_2^3}\right) + \frac{Q[p^2+(\mathbf{p}\cdot\hat{\mathbf{r}})^2]}{2m_1 m_2 c^2 r}.} \end{aligned}$$

In Bethe and Salpeter's actual (42.1), p. 193, these are H_0 (nonrelativistic Coulomb and kinetic energy), H_1 (the quartic kinetic correction) and the classical limit of H_2 (orbit–orbit interaction). Their $p_r^2 = r^{-2}\mathbf{r}\cdot(\mathbf{r}\cdot\mathbf{p})\mathbf{p}$ has a specified operator ordering; the classical momenta commute. Their H_3 contains spin–orbit and spin–other-orbit interactions, H_4 is an explicitly $\hbar$-dependent derivative term, and H_5 contains dipole and contact spin–spin terms. None is produced by classical spinless point charges. In particular the name "Darwin Lagrangian" does not imply that the quantum contact correction is already present. These differences reflect the quantum and spin degrees of freedom, rather than a discrepancy in the classical order-c^{-2} interaction.

12.14 An alternative Lagrangian density for the electromagnetic field is

$$\mathcal{L} = -\frac{1}{8\pi}\,\partial_\alpha A_\beta \partial^\alpha A^\beta - \frac{1}{c}\,J_\alpha A^\alpha$$

(a) Derive the Euler–Lagrange equations of motion. Are they the Maxwell equations? Under what assumptions?

(b) Show explicitly, and with what assumptions, that this Lagrangian density differs from (12.85) by a 4-divergence. Does this added 4-divergence affect the action or the equations of motion?

Solution.

(a) Vary A_β as the field coordinates. The derivatives are

$$\frac{\partial\mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -\frac{1}{4\pi}\partial^\alpha A^\beta, \qquad \frac{\partial\mathcal{L}}{\partial A_\beta} = -\frac{1}{c}J^\beta.$$

Thus the four Euler–Lagrange equations are

$$\boxed{\Box A^\beta = \frac{4\pi}{c}J^\beta.}$$

In contrast, the inhomogeneous Maxwell equation in potentials is

$$\partial_\alpha F^{\alpha\beta} = \Box A^\beta - \partial^\beta D = \frac{4\pi}{c}J^\beta, \qquad D = \partial_\alpha A^\alpha.$$

They agree in Lorenz gauge $D = 0$ (or more generally if D is a space-time constant). The homogeneous equations hold identically because $F = dA$. Conservation of the source current gives $\Box D = 0$ from the alternative equations, but does not by itself set D to zero: one also needs Lorenz-compatible initial and boundary data. In particular $D = \partial_t D = 0$ initially eliminates the unwanted homogeneous solution.

(b) To compare the actions, introduce

$$I = \partial_\alpha A_\beta \partial^\alpha A^\beta, \qquad K = \partial_\alpha A_\beta \partial^\beta A^\alpha.$$

Since $F_{\alpha\beta}F^{\alpha\beta} = 2(I - K)$, $\mathcal{L}_{\text{alt}} - \mathcal{L}_{\text{Maxwell}} = -K/(8\pi)$. The product rule twice gives

$$\begin{aligned} K &= \partial_\alpha(A_\beta\partial^\beta A^\alpha) - A_\beta\partial^\beta D \\ &= \partial_\alpha(A_\beta\partial^\beta A^\alpha - A^\alpha D) + D^2. \end{aligned}$$

Therefore the exact off-gauge identity is

$$\boxed{\mathcal{L}_{\rm alt} = \mathcal{L}_{\rm Maxwell} - \frac{D^2}{8\pi} - \frac{1}{8\pi}\partial_\alpha(A_\beta \partial^\beta A^\alpha - A^\alpha D).}$$

Under the Lorenz condition the D^2 term vanishes and the difference is a four-divergence. Its action integral is a boundary integral; it can shift the numerical action, but not the bulk equations for compactly supported variations or boundary conditions that make its variation vanish. Off Lorenz gauge there is also a gauge-fixing term, so claiming unconditional equivalence by a divergence would be incorrect. Imposing the gauge consistently on the variations, or on the initial-value solution of part (a), is essential.

12.15 Consider the Proca equations for a localized steady-state distribution of current that has only a static magnetic moment. This model can be used to study the observable effects of a finite photon mass on the earth's magnetic field. Note that if the magnetization is $\mathcal{M}(\mathbf{x})$ the current density can be written as $\mathbf{J} = c(\nabla \times \mathcal{M})$.

(a) Show that if $\mathcal{M} = \mathbf{m}f(\mathbf{x})$, where $\mathbf{m}$ is a fixed vector and $f(\mathbf{x})$ is a localized scalar function, the vector potential is

$$\mathbf{A}(\mathbf{x}) = -\mathbf{m} \times \nabla \int f(\mathbf{x}') \, \frac{e^{-\mu|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|} \, d^3x'$$

(b) If the magnetic dipole is a point dipole at the origin [$f(\mathbf{x}) = \delta(\mathbf{x})$], show that the magnetic field away from the origin is

$$\mathbf{B}(\mathbf{x}) = [3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}]\left(1 + \mu r + \frac{\mu^2 r^2}{3}\right)\frac{e^{-\mu r}}{r^3} - \tfrac{2}{3}\mu^2 \mathbf{m} \, \frac{e^{-\mu r}}{r}$$

(c) The result of part b shows that at fixed $r = R$ (on the surface of the earth), the earth's magnetic field will appear as a dipole angular distribution, plus an added constant magnetic field (an apparently external field) antiparallel to $\mathbf{m}$. Satellite and surface observations lead to the conclusion that this "external" field is less than 4×10^{-3} times the dipole field at the magnetic equator. Estimate a lower limit on μ^{-1} in earth radii and an upper limit on the photon mass in grams from this datum.

This method of estimating μ is due to E. Schrödinger, *Proc. R. Irish Acad.* **A49**, 135 (1943). See A. S. Goldhaber and M. M. Nieto, *Phys. Rev. Lett.* **21**, 567 (1968).

Solution.

(a) For a stationary conserved current the Proca equation in Gaussian units is

$$(\nabla^2 - \mu^2)\mathbf{A} = -\frac{4\pi}{c}\mathbf{J}, \qquad \mathbf{A}(\mathbf{x}) = \frac{1}{c}\int \frac{e^{-\mu|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|}\mathbf{J}(\mathbf{x}') \, d^3x'.$$

Insert $\mathbf{J} = c\nabla' \times [\mathbf{m}f(\mathbf{x}')]$ and integrate by parts. The surface term vanishes for localized f, and $\nabla' G(\mathbf{x}-\mathbf{x}') = -\nabla G$ gives

$$\begin{aligned}\mathbf{A} &= \int f(\mathbf{x}')\mathbf{m} \times \nabla' G \, d^3x' \\ &= -\mathbf{m} \times \nabla \int f(\mathbf{x}') \frac{e^{-\mu|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|} \, d^3x'.\end{aligned}$$

This proves part (a), and $\nabla \cdot \mathbf{A} = 0$ as required.

(b) For a point dipole put $Y(r) = e^{-\mu r}/r$. The vector identity for the curl of a cross product gives, away from the origin,

$$\mathbf{B} = \nabla \times (-\mathbf{m} \times \nabla Y) = (\mathbf{m} \cdot \nabla)\nabla Y - \mathbf{m}\nabla^2 Y.$$

Here

$$Y' = -\frac{e^{-\mu r}}{r^2}(1+\mu r), \quad Y'' = \frac{e^{-\mu r}}{r^3}(2+2\mu r+\mu^2 r^2), \quad \nabla^2 Y = \mu^2 Y,$$

and $\partial_i \partial_j Y = (Y'' - Y'/r)\hat{r}_i \hat{r}_j + (Y'/r)\delta_{ij}$. Therefore, writing $s = \mu r$,

$$\begin{aligned}\mathbf{B} &= \frac{e^{-s}}{r^3}\{(3+3s+s^2)\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{m}) - (1+s+s^2)\mathbf{m}\} \\ &= \boxed{[3\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{m}) - \mathbf{m}]\left(1+s+\frac{s^2}{3}\right)\frac{e^{-s}}{r^3} - \frac{2}{3}\mu^2\mathbf{m}\frac{e^{-s}}{r}.}\end{aligned}$$

A distributional point-dipole field also has its usual $(8\pi/3)\mathbf{m}\delta^{(3)}(\mathbf{x})$ contact term; it is excluded by the question's "away from the origin" qualification.

(c) On a sphere of radius R, the amplitude of the dipole-shaped part at its equator is $B_d = (m/R^3)(1+s+s^2/3)e^{-s}$, while the apparent uniform external component has magnitude $B_e = (2m/3R^3)s^2 e^{-s}$ and points opposite to $\mathbf{m}$. The given observational ratio imposes

$$\frac{B_e}{B_d} = \frac{(2/3)s^2}{1+s+s^2/3} < 0.004.$$

To leading order, $s < \sqrt{0.006} = 0.07746$, hence

$$\boxed{\mu^{-1} \gtrsim 13R.}$$

Retaining the displayed expression gives the positive root $s < 0.0806015$, or $\mu^{-1} > 12.4067R$. With $R = 6.371 \times 10^8$ cm and $m_\gamma = \hbar\mu/c$,

$$\boxed{m_\gamma < 4.45 \times 10^{-48}\ \text{g}}$$

for the full ratio, about 4.3×10^{-48} g at leading order. These are the bounds from the datum and model supplied in the problem, rather than a claim about the current best photon-mass limit.

12.16

(a) Starting with the Proca Lagrangian density (12.91) and following the same procedure as for the electromagnetic fields, show that the symmetric stress-energy-momentum tensor for the Proca *fields* is

$$\Theta^{\alpha\beta} = \frac{1}{4\pi}\left[g^{\alpha\gamma}F_{\gamma\lambda}F^{\lambda\beta} + \frac{1}{4}\,g^{\alpha\beta}F_{\lambda\nu}F^{\lambda\nu} + \mu^2\left(A^\alpha A^\beta - \frac{1}{2}\,g^{\alpha\beta}A_\lambda A^\lambda\right)\right]$$

(b) For these fields in interaction with the external source J^β, as in (12.91), show that the differential conservation laws take the same form as for the electromagnetic fields, namely,

$$\partial_\alpha \Theta^{\alpha\beta} = \frac{J_\lambda F^{\lambda\beta}}{c}$$

(c) Show explicitly that the time-time and space-time components of $\Theta^{\alpha\beta}$ are

$$\Theta^{00} = \frac{1}{8\pi}\left[E^2 + B^2 + \mu^2(A^0A^0 + \mathbf{A}\cdot\mathbf{A})\right]$$
$$\Theta^{i0} = \frac{1}{4\pi}\left[(\mathbf{E}\times\mathbf{B})_i + \mu^2 A^i A^0\right]$$

Solution.

(a) Use the free-field part of (12.91),

$$\mathcal{L}_f = -\frac{F_{\lambda\nu}F^{\lambda\nu}}{16\pi} + \frac{\mu^2 A_\lambda A^\lambda}{8\pi}, \qquad \frac{\partial\mathcal{L}_f}{\partial(\partial_\alpha A_\lambda)} = -\frac{F^{\alpha\lambda}}{4\pi}.$$

The canonical tensor is

$$T^{\alpha\beta} = -\frac{1}{4\pi}F^{\alpha\lambda}\partial^\beta A_\lambda - g^{\alpha\beta}\mathcal{L}_f.$$

Use $\partial^\beta A_\lambda = F^\beta{}_\lambda + \partial_\lambda A^\beta$ and expand the latter term by the product rule:

$$T^{\alpha\beta} = \frac{1}{4\pi}F^\alpha{}_\lambda F^{\lambda\beta} - g^{\alpha\beta}\mathcal{L}_f - \frac{1}{4\pi}\partial_\lambda(F^{\alpha\lambda}A^\beta) + \frac{1}{4\pi}(\partial_\lambda F^{\alpha\lambda})A^\beta.$$

For a free Proca field, $\partial_\lambda F^{\lambda\alpha} + \mu^2 A^\alpha = 0$, so the last term is $\mu^2 A^\alpha A^\beta/(4\pi)$. Adding the divergence improvement cancels the penultimate term and gives

$$\boxed{\Theta^{\alpha\beta} = \frac{1}{4\pi}\left[g^{\alpha\gamma}F_{\gamma\lambda}F^{\lambda\beta} + \frac{1}{4}g^{\alpha\beta}F_{\lambda\nu}F^{\lambda\nu} + \mu^2\left(A^\alpha A^\beta - \frac{1}{2}g^{\alpha\beta}A_\lambda A^\lambda\right)\right].}$$

It is symmetric. The same field tensor, which can also be obtained by varying the free-field action with respect to the metric, is used in the presence of prescribed sources. The interaction energy is not included as an extra term in this field tensor.

(b) For a conserved external current the Proca equations imply

$$\partial_\alpha F^{\alpha\beta} + \mu^2 A^\beta = \frac{4\pi}{c}J^\beta, \qquad \partial_\alpha A^\alpha = 0 \quad (\mu \neq 0).$$

The divergence of the Maxwell part of Θ follows from the Bianchi identity. In explicit contracted form it gives

$$F^{\alpha\lambda}\partial_\alpha F_\lambda{}^\beta = -\tfrac{1}{4}\partial^\beta(F_{\lambda\nu}F^{\lambda\nu}),$$

so the derivatives acting on the second F cancel those of $F^2/4$:

$$\partial_\alpha\Theta^{\alpha\beta}_M = \frac{1}{4\pi}(\partial_\alpha F^{\alpha\lambda})F_\lambda{}^\beta = \frac{1}{c}J_\lambda F^{\lambda\beta} - \frac{\mu^2}{4\pi}A_\lambda F^{\lambda\beta}.$$

For clarity, the mass contribution has divergence

$$\begin{aligned}\partial_\alpha\Theta^{\alpha\beta}_\mu &= \frac{\mu^2}{4\pi}[(\partial_\alpha A^\alpha)A^\beta + A^\alpha\partial_\alpha A^\beta - A_\lambda\partial^\beta A^\lambda] \\ &= \frac{\mu^2}{4\pi}A_\lambda F^{\lambda\beta}.\end{aligned}$$

The two mass terms cancel, proving

$$\boxed{\partial_\alpha \Theta^{\alpha\beta} = \frac{1}{c} J_\lambda F^{\lambda\beta}.}$$

Current conservation is essential here; without it an additional $\mu^2 A^\beta (\partial \cdot A)/(4\pi)$ term remains.

(c) Finally $F^{0i} = -E_i$, $F^{ij} = -\epsilon_{ijk} B_k$ give $F_{\lambda\nu} F^{\lambda\nu} = 2(B^2 - E^2)$. For the 00 component the Maxwell terms combine to $(E^2 + B^2)/2$, and the mass term is $\mu^2[(A^0)^2 + \mathbf{A}^2]/2$ inside the factor $1/(4\pi)$. For $i0$, $g^{i0} = 0$ and the field product is $(\mathbf{E} \times \mathbf{B})_i$. Thus

$$\boxed{\Theta^{00} = \frac{E^2 + B^2 + \mu^2[(A^0)^2 + \mathbf{A}^2]}{8\pi}, \qquad \Theta^{i0} = \frac{(\mathbf{E} \times \mathbf{B})_i + \mu^2 A^i A^0}{4\pi}.}$$

Both the energy density and its mass contribution are nonnegative.

12.17 Consider the "Thomson" scattering of Proca waves (photons with mass) by a free electron.

(a) As a preliminary, show that for an incident plane wave of unit amplitude, $\mathbf{A} = \boldsymbol{\epsilon}_0 \cos(kz - \omega t)$, where $\boldsymbol{\epsilon}_0$ is a polarization vector of unit magnitude describing either longitudinal (l) or transverse (t) fields, the time-averaged energy fluxes (measured by Θ^{30}) are $F_t = \omega k/8\pi$ and $F_l = (\mu/\omega)^2 F_t$. Show also for arbitrary polarization that the ratio of time-averaged energy flux to energy density is

$$\frac{\langle \Theta^{30} \rangle}{\langle \Theta^{00} \rangle} = \beta = \frac{k}{\omega} = \sqrt{1 - \frac{\mu^2}{\omega^2}}$$

as expected for particles of mass μ.

(b) For polarization $\boldsymbol{\epsilon}_0$ initially and polarization $\boldsymbol{\epsilon}$ finally, show that the "Thomson" cross section for scattering is

$$\frac{d\sigma}{d\Omega}(\boldsymbol{\epsilon}, \boldsymbol{\epsilon}_0) = r_0^2 E_0 \left| \boldsymbol{\epsilon}^* \cdot \boldsymbol{\epsilon}_0 \right|^2 \frac{F_{\text{out}}}{F_{\text{in}}}$$

where r_0 is the classical electron radius, E_0 is a factor for the efficiency of the incident Proca field in exciting the electron, and the final factor is a ratio of the outgoing to incident fluxes. What is the value of E_0?

(c) For an unpolarized *transverse* wave incident, show that the scattering cross section is

$$\left(\frac{d\sigma}{d\Omega} \right)_t = \frac{r_0^2}{2} \left[1 + \cos^2\theta + \left(\frac{\mu}{\omega} \right)^2 \sin^2\theta \right]$$

where the first term is the familiar transverse to transverse scattering and the second is transverse to longitudinal.

(d) For a *longitudinally* polarized wave incident, show that the cross section, summed over outgoing polarizations, is

$$\left(\frac{d\sigma}{d\Omega} \right)_l = \left(\frac{\mu}{\omega} \right)^2 r_0^2 \left[\sin^2\theta + \left(\frac{\mu}{\omega} \right)^2 \cos^2\theta \right]$$

where the first term is the longitudinal to transverse scattering and the second is longitudinal to longitudinal.

Note that in the limit $\mu/\omega \to 0$, the longitudinal fields decouple and we recover the standard Thomson cross section.

Solution.

(a) The formulas in this problem use $c = 1$, so $\omega^2 = k^2 + \mu^2$; retain Gaussian normalization of fields. For a plane wave write $\mathbf{A} = \boldsymbol{\epsilon}\cos\phi$, $\phi = kz - \omega t$, and decompose $\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_t + \epsilon_l\widehat{\mathbf{z}}$. The Lorenz condition requires $A^0 = (k/\omega)\epsilon_l\cos\phi$. Differentiation gives

$$\mathbf{E} = -\omega\left(\boldsymbol{\epsilon}_t + \frac{\mu^2}{\omega^2}\epsilon_l\widehat{\mathbf{z}}\right)\sin\phi, \qquad \mathbf{B} = -k\widehat{\mathbf{z}}\times\boldsymbol{\epsilon}_t\sin\phi.$$

For transverse unit amplitude, averaging $\sin^2\phi$ and $\cos^2\phi$ gives

$$\langle\Theta^{00}\rangle_t = \frac{\omega^2 + k^2 + \mu^2}{16\pi} = \frac{\omega^2}{8\pi}, \qquad F_t = \frac{\omega k}{8\pi}.$$

For longitudinal unit amplitude $\mathbf{B} = 0$, and the two averages instead are

$$\langle\Theta^{00}\rangle_l = \frac{\mu^4/\omega^2 + \mu^2(1 + k^2/\omega^2)}{16\pi} = \frac{\mu^2}{8\pi}, \qquad F_l = \frac{\mu^2 k}{8\pi\omega}.$$

Consequently $F_l = (\mu/\omega)^2 F_t$. For arbitrary polarization the transverse and longitudinal contributions are orthogonal in the quadratic energy and flux:

$$\langle\Theta^{00}\rangle = \frac{\omega^2|\boldsymbol{\epsilon}_t|^2 + \mu^2|\epsilon_l|^2}{8\pi}, \qquad \boxed{\frac{\langle\Theta^{30}\rangle}{\langle\Theta^{00}\rangle} = \frac{k}{\omega} = \sqrt{1 - \mu^2/\omega^2}.}$$

The physical energy-transport speed with c restored is ck/ω when ω is expressed in inverse-length units.

(b) In the Thomson limit the electron displacement is small, its motion nonrelativistic, and recoil can be neglected. Let $s = \mu^2/\omega^2$ and let $d_0 = 1$ for incident transverse polarization, $d_0 = s$ for incident longitudinal polarization. The electric-field amplitude is ωd_0 per unit incident vector-potential amplitude. With $e^{-i\omega t}$ phasors, $\mathbf{E}_0 = i\omega d_0\boldsymbol{\epsilon}_0$ and $-i\omega m\mathbf{v}_0 = e\mathbf{E}_0$, so $\mathbf{v}_0 = -ed_0\boldsymbol{\epsilon}_0/m$. To first order in the displacement the current is $\mathbf{J}_0(\mathbf{x}) = e\mathbf{v}_0\delta^{(3)}(\mathbf{x})$. Inserting it into the outgoing Helmholtz integral $\mathbf{A}_{\rm sc} = \int e^{ik|\mathbf{x}-\mathbf{x}'|}\mathbf{J}_0(\mathbf{x}')/|\mathbf{x}-\mathbf{x}'|\,d^3x'$ gives

$$\mathbf{A}_{\rm sc} \simeq -\frac{e^2}{m}\frac{e^{ikr}}{r}d_0\boldsymbol{\epsilon}_0.$$

The overall minus sign is the driven electron's phase reversal; it disappears from the cross section. Projection onto a unit outgoing polarization $\boldsymbol{\epsilon}$ multiplies this by $\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0$. Power per solid angle is amplitude squared times the corresponding unit-amplitude flux. With $r_0 = e^2/m$ in these units,

$$\boxed{\frac{d\sigma}{d\Omega}(\boldsymbol{\epsilon},\boldsymbol{\epsilon}_0) = r_0^2 E_0|\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0|^2\frac{F_{\rm out}}{F_{\rm in}}, \quad E_0 = d_0^2 = \begin{cases}1, & t \text{ incident},\\ (\mu/\omega)^4, & l \text{ incident}.\end{cases}}$$

For a coherent mixture of incident transverse and longitudinal polarizations the force acts on $\boldsymbol{\epsilon}_{0t} + s\boldsymbol{\epsilon}_{0l}$, rather than multiplying the entire vector by one scalar; the displayed E_0 applies to the pure t, l channels requested here.

(c) Choose scattering in the xz plane at angle θ from the incident z direction. Convenient outgoing polarizations are $\boldsymbol{\epsilon}_\theta = (\cos\theta, 0, -\sin\theta)$, $\boldsymbol{\epsilon}_\phi = (0,1,0)$ and $\boldsymbol{\epsilon}_l = (\sin\theta, 0, \cos\theta)$. For equally weighted

incident x, y polarizations the transverse sum of squared overlaps is $(1+\cos^2\theta)/2$; the longitudinal sum is $\sin^2\theta/2$. Their flux ratios are one and s. Therefore

$$\boxed{\left(\frac{d\sigma}{d\Omega}\right)_t = \frac{r_0^2}{2}[1+\cos^2\theta + s\sin^2\theta].}$$

(d) For incident longitudinal polarization $\boldsymbol{\epsilon}_0 = \hat{\mathbf{z}}$, the corresponding overlaps are $\sin^2\theta$ and $\cos^2\theta$. Multiplying by $E_0 = s^2$ and the flux ratios $1/s$ and one gives

$$\boxed{\left(\frac{d\sigma}{d\Omega}\right)_l = sr_0^2[\sin^2\theta + s\cos^2\theta].}$$

As an integral check, $\sigma_t = (8\pi/3)r_0^2(1+s/2)$ and $\sigma_l = s\sigma_t$. The massless limit leaves the usual Thomson result and removes both incident and outgoing longitudinal contributions.

12.18 Prove, by means of the divergence theorem in four dimensions or otherwise, that for source-free electromagnetic fields confined to a finite region of space, the 3-space integrals of Θ^{00} and Θ^{0i} transform as the components of a constant 4-vector, as implied by (12.106).

Solution.

Let Σ be any spacelike hypersurface with future-directed surface element $d\Sigma_\alpha$, and define

$$P^\beta[\Sigma] = \frac{1}{c}\int_\Sigma \Theta^{\alpha\beta} d\Sigma_\alpha.$$

The tensor and directed surface element transform covariantly; contracting their first index leaves a four-vector index β. This fact alone does not yet prove that equal-time integrals in different frames are the same four-vector, because the equal-time surfaces are different. Surface independence supplies the missing step.

Take two such surfaces and close the intervening four-volume with a distant timelike boundary. The divergence theorem and $\partial_\alpha \Theta^{\alpha\beta} = 0$ give

$$0 = \int_{\mathcal{V}} \partial_\alpha \Theta^{\alpha\beta}\, d^4x = \int_{\Sigma_2} \Theta^{\alpha\beta} d\Sigma_\alpha - \int_{\Sigma_1} \Theta^{\alpha\beta} d\Sigma_\alpha + \int_{\text{side}} \Theta^{\alpha\beta} d\Sigma_\alpha.$$

Localization makes the side term vanish as the boundary is removed. Thus $P^\beta[\Sigma_2] = P^\beta[\Sigma_1]$. For a compact wave packet this comparison remains valid even for a tilted hyperplane: its slope is subluminal, so sufficiently distant parts of the closing surface lie outside the packet's causal support. Suitable decaying fields work as well if their boundary flux tends to zero.

On a laboratory $t =$ constant plane, $d\Sigma_\alpha = (d^3x, 0, 0, 0)$, so

$$P^0 = \frac{1}{c}\int \Theta^{00}\, d^3x = E_{\text{em}}/c, \qquad P^i = \frac{1}{c}\int \Theta^{0i}\, d^3x.$$

A Lorentz transformation maps this plane to a generally tilted plane in the primed frame, on which the integral is $\Lambda^\beta{}_\gamma P^\gamma$. By surface independence one may replace it by $t' =$ constant. Hence

$$\boxed{P'^\beta = \Lambda^\beta{}_\gamma P^\gamma.}$$

Taking Σ_1, Σ_2 as two parallel constant-time planes also proves $dP^\beta/dt = 0$. The integrals of Θ^{00}, Θ^{0i} therefore form the constant four-vector $(E_{\text{em}}, c\mathbf{P}_{\text{em}})$. The premise is a free localized field; if confining walls exert forces, their momentum exchange must be included and the field subsystem alone does not satisfy this proof's source-free boundary conditions.

12.19 Source-free electromagnetic fields exist in a localized region of space. Consider the various conservation laws that are contained in the integral of $\partial_\alpha M^{\alpha\beta\gamma} = 0$ over all space, where $M^{\alpha\beta\gamma}$ is defined by (12.117).

(a) Show that when β and γ are both space indices conservation of the total field angular momentum follows.

(b) Show that when $\beta = 0$ the conservation law is

$$\frac{d\mathbf{X}}{dt} = \frac{c^2 \mathbf{P}_{em}}{E_{em}}$$

where $\mathbf{X}$ is the coordinate of the center of mass of the electromagnetic fields, defined by

$$\mathbf{X} \int u \, d^3x = \int \mathbf{x} u \, d^3x$$

where u is the electromagnetic energy density and E_{em} and $\mathbf{P}_{em}$ are the total energy and momentum of the fields.

Solution.

(a) Equation (12.117) defines

$$M^{\alpha\beta\gamma} = x^\beta \Theta^{\alpha\gamma} - x^\gamma \Theta^{\alpha\beta}.$$

Its divergence is

$$\partial_\alpha M^{\alpha\beta\gamma} = x^\beta \partial_\alpha \Theta^{\alpha\gamma} - x^\gamma \partial_\alpha \Theta^{\alpha\beta} + \Theta^{\beta\gamma} - \Theta^{\gamma\beta} = 0.$$

Both the source-free conservation law and the symmetry of Θ are used. Integrate over space, use $\partial_0 = c^{-1}\partial_t$, and discard the vanishing flux at infinity. The six antisymmetric quantities

$$\mathcal{J}^{\beta\gamma} = \frac{1}{c} \int M^{0\beta\gamma} \, d^3x$$

are constants.

For spatial indices,

$$\mathcal{J}^{ij} = \int (x^i g^j - x^j g^i) \, d^3x, \qquad g^i = \Theta^{0i}/c.$$

The three independent components are equivalent to

$$\boxed{\mathbf{L}_{\text{em}} = \int \mathbf{x} \times \mathbf{g} \, d^3x = \frac{1}{4\pi c} \int \mathbf{x} \times (\mathbf{E} \times \mathbf{B}) \, d^3x, \quad \frac{d\mathbf{L}_{\text{em}}}{dt} = 0.}$$

(b) For $\beta = 0, \gamma = i$, with $u = \Theta^{00}$,

$$\begin{aligned} \mathcal{J}^{0i} &= \frac{1}{c} \int (ct\Theta^{0i} - x^i u) \, d^3x \\ &= ctP^i_{\text{em}} - \frac{E_{\text{em}}}{c} X^i. \end{aligned}$$

Energy and momentum are separately constant. Differentiating the last equation therefore yields

$$0 = cP^i_{\text{em}} - \frac{E_{\text{em}}}{c} \frac{dX^i}{dt}, \qquad \boxed{\frac{d\mathbf{X}}{dt} = \frac{c^2 \mathbf{P}_{\text{em}}}{E_{\text{em}}}.}$$

Here the defined "center of mass" is the center of energy. It moves uniformly even though the local field energy changes and propagates. The result requires $E_{\text{em}} > 0$; for the zero-field solution the centroid is undefined, rather than an additional physical degree of freedom.

12.20 A uniform superconductor with London penetration depth λ_L fills the half-space $x > 0$. The vector potential is tangential and for $x < 0$ is given by

$$A_y = (ae^{ikx} + be^{-ikx})e^{-i\omega t}$$

Find the vector potential inside the superconductor. Determine expressions for the electric and magnetic fields at the surface. Evaluate the surface impedance Z_s (in Gaussian units, $4\pi/c$ times the ratio of tangential electric field to tangential magnetic field). Show that in the appropriate limit your result for Z_s reduces to that given in Section 12.9.

Solution.

Use the stated $e^{-i\omega t}$ convention, $k = \omega/c > 0$, and the transverse gauge with $\Phi = 0$. Equation (12.99), $J_y = -cA_y/(4\pi\lambda_L^2)$, turns the wave equation inside the material into

$$\frac{d^2A_y}{dx^2} + \left(k^2 - \lambda_L^{-2}\right) A_y = 0.$$

Define $q = \sqrt{k^2 - \lambda_L^{-2}}$, choosing $\operatorname{Im} q \geq 0$ and positive real q above threshold. The decaying or outgoing solution in the half-space is

$$A_y = Ce^{iqx - i\omega t}, \qquad x > 0.$$

The tangential electric and magnetic fields are $E_y = ikA_y$ and $B_z = \partial_x A_y$. There is no delta-function sheet current in this finite-penetration London model, so both are continuous at $x = 0$. Thus

$$a + b = C, \qquad k(a - b) = qC,$$

and solution of these two equations gives

$$\boxed{b = \frac{k-q}{k+q}a, \qquad A_y(x>0) = \frac{2ka}{k+q}e^{iqx-i\omega t}.}$$

In particular the surface fields are

$$\boxed{E_y(0) = \frac{2ik^2a}{k+q}e^{-i\omega t}, \qquad B_z(0) = \frac{2ikqa}{k+q}e^{-i\omega t}.}$$

For $\omega < c/\lambda_L$ put $\kappa = \sqrt{\lambda_L^{-2} - \omega^2/c^2}$, so $q = i\kappa$. The defined Gaussian surface impedance is

$$\boxed{Z_s = \frac{4\pi}{c}\frac{E_y(0)}{B_z(0)} = \frac{4\pi k}{cq} = -\frac{4\pi i\omega}{c^2\kappa}.}$$

It is purely inductive with this time convention. When $k\lambda_L \ll 1$,

$$\boxed{Z_s \simeq -\frac{4\pi i\omega\lambda_L}{c^2} = -\frac{8\pi^2 i}{c}\frac{\lambda_L}{\lambda},}$$

where $\lambda = 2\pi c/\omega$. This is Section 12.9's result; in SI units it is $Z_s = -i\omega\mu_0\lambda_L$. For $k > \lambda_L^{-1}$ the same ideal model has real q and a positive real impedance representing energy propagation into the bulk. At $q = 0$ use the continuous field solution; its magnetic surface field vanishes and the impedance ratio diverges.

12.21 A two-fluid model for the electrodynamics of superconductors posits two types of electron, normal and superconducting, with number densities, charges, masses, and collisional damping constants, n_j, e_j, m_j, and γ_j, respectively ($j = \mathrm{N}, \mathrm{S}$). The electrical conductivity consists of the sum of two terms of the Drude form (7.58) with $f_0 N \to n_j$, $e \to e_j$, $m \to m_j$, $\gamma_0 \to \gamma_j$. The normal (superconducting) electrons are distinguished by $\gamma_\mathrm{N} \neq 0$ ($\gamma_\mathrm{S} = 0$).

(a) Show that the conductivity of the superconductor at very low frequencies is largely imaginary (inductive) with a small resistive component from the normal electrons.

(b) Show that use of Ohm's law with the conductivity of part a in the Maxwell equations results in the static London equation for the electric field in the limit $\omega \to 0$, with the penetration depth (12.100), provided the carriers are identified with the superconducting component of the electric fluid.

Solution.

(a) Let $D_j = n_j e_j^2/m_j$ and use $e^{-i\omega t}$. The Drude equation for each component is $m_j(\gamma_j - i\omega)\mathbf{v}_j = e_j\mathbf{E}$, so

$$\sigma(\omega) = \frac{D_N}{\gamma_N - i\omega} + \frac{D_S}{-i\omega}.$$

For $\omega \ll \gamma_N$ and a nonzero superconducting density,

$$\boxed{\sigma(\omega) = \frac{iD_S}{\omega} + \sigma_n + i\frac{D_N\omega}{\gamma_N^2} + O(\omega^2), \qquad \sigma_n = \frac{D_N}{\gamma_N} > 0.}$$

The imaginary term dominates when $\omega\sigma_n \ll D_S$. To see its inductive sign directly,

$$\frac{1}{\sigma} = -\frac{i\omega}{D_S} + \frac{\sigma_n\omega^2}{D_S^2} + O(\omega^3).$$

Thus the leading voltage-to-current response is inductive and the small positive real part is dissipative.

(b) In the homogeneous transverse problem, Maxwell's equations are

$$\nabla \times \mathbf{E} = \frac{i\omega}{c}\mathbf{B}, \qquad \nabla \times \mathbf{B} = \left(-\frac{i\omega}{c} + \frac{4\pi\sigma}{c}\right)\mathbf{E}.$$

Take the curl of the first equation and use $\nabla \cdot \mathbf{E} = 0$ in the bulk. This gives

$$\nabla^2\mathbf{E} + \left(\frac{\omega^2}{c^2} + \frac{4\pi i\omega\sigma}{c^2}\right)\mathbf{E} = 0.$$

Substitution of the low-frequency conductivity makes the bracket

$$-\frac{4\pi D_S}{c^2} + \frac{4\pi i\omega\sigma_n}{c^2} + O(\omega^2).$$

It therefore tends to the static London operator:

$$\boxed{\nabla^2\mathbf{E} - \lambda_L^{-2}\mathbf{E} = 0, \qquad \lambda_L = \sqrt{\frac{m_S c^2}{4\pi n_S e_S^2}}.}$$

This is (12.100) with the carriers identified as the superconducting component. The normal component contributes the vanishing order-ω loss term, not the finite penetration scale. The statement concerns the limit of a time-dependent transverse response. A finite stationary electric field would, by $\partial_t \mathbf{J}_S = D_S\mathbf{E}$, continually accelerate the supercurrent; a static equilibrium does not sustain such a bulk field. If $n_S = 0$ the finite London length disappears, as expected for a normal conductor.

13 Chapter 13. Energy Loss, Scattering; Cherenkov and Transition Radiation

Problems

13.1 If the light particle (electron) in the Coulomb scattering of Section 13.1 is treated classically, scattering through an angle θ is correlated uniquely to an incident trajectory of impact parameter b according to

$$b = \frac{ze^2}{pv}\cot\frac{\theta}{2}$$

where $p = \gamma mv$ and the differential scattering cross section is $\dfrac{d\sigma}{d\Omega} = \dfrac{b}{\sin\theta}\left|\dfrac{db}{d\theta}\right|$.

(a) Express the invariant momentum transfer squared in terms of impact parameter and show that the energy transfer $T(b)$ is

$$T(b) = \frac{2z^2e^4}{mv^2}\cdot\frac{1}{b^2 + b_{\min}^{(c)2}}$$

where $b_{\min}^{(c)} = ze^2/pv$ and $T(0) = T_{\max} = 2\gamma^2\beta^2mc^2$.

(b) Calculate the small transverse impulse Δp given to the (nearly stationary) light particle by the transverse electric field (11.152) of the heavy particle $q = ze$ as it passes by at large impact parameter b in a (nearly) straight line path at speed v. Find the energy transfer $T \approx (\Delta p)^2/2m$ in terms of b. Compare with the exact classical result of part a. Comment.

Solution.

We use Gaussian units in this chapter, write $S = -dE_{\text{projectile}}/dx > 0$ for a stopping power, and take $e > 0$ as the magnitude of the electron charge. The heavy projectile is treated as infinitely massive in its rest frame. Consequently the magnitude $p = \gamma mv$ of the electron momentum is unchanged in that frame. The stipulated impact-parameter law is the Rutherford model; the word exact below refers to that model, whereas the straight-path impulse is its large-b approximation.

(a) Put $b_c = ze^2/(pv)$. The given relation implies

$$\cot^2\frac{\theta}{2} = \frac{b^2}{b_c^2}, \qquad \sin^2\frac{\theta}{2} = \frac{b_c^2}{b^2 + b_c^2}.$$

The incoming and outgoing four-momenta have equal energy in the projectile rest frame. Hence their invariant difference is

$$Q^2 \equiv -(p_e' - p_e)^2 = |\mathbf{p}_e' - \mathbf{p}_e|^2 = 2p^2(1 - \cos\theta) = \frac{4p^2b_c^2}{b^2 + b_c^2}.$$

Here four-momenta have units of momentum, so Q^2 has units of momentum squared, as in Jackson's (13.2). In the laboratory the electron starts at rest. If it gains kinetic energy T, its final energy and momentum obey

$$E_e' = mc^2 + T, \qquad p_e'^2c^2 = T^2 + 2mc^2T.$$

Therefore the same invariant is

$$Q^2 = p_e'^2 - (T/c)^2 = 2mT.$$

Combining the two evaluations gives

$$\boxed{T(b) = \frac{2p^2 b_c^2}{m(b^2 + b_c^2)} = \frac{2z^2 e^4}{mv^2} \frac{1}{b^2 + b_c^2}.}$$

In particular,

$$T(0) = \frac{2p^2}{m} = 2\gamma^2 mv^2 = 2\gamma^2 \beta^2 mc^2 = T_{\max}.$$

A finite projectile mass changes this maximum by the recoil denominator given immediately after Jackson's (13.4); it is omitted consistently here.

(b) Equation (11.152), with closest approach at $t = 0$, gives

$$E_\perp(t) = \frac{ze\gamma b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}, \qquad E_\parallel(t) = -\frac{ze\gamma vt}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}.$$

The parallel impulse vanishes by oddness. The magnitude of the transverse impulse, using $u = \gamma vt/b$, is

$$|\Delta p_\perp| = e \int_{-\infty}^{\infty} E_\perp \, dt = \frac{ze^2}{bv} \int_{-\infty}^{\infty} \frac{du}{(1+u^2)^{3/2}} = \frac{ze^2}{bv} \left[\frac{u}{\sqrt{1+u^2}} \right]_{-\infty}^{\infty} = \frac{2ze^2}{bv}.$$

The electron's actual impulse points opposite to $\mathbf{E}$; the energy depends on its square. For a nearly stationary, nonrelativistic electron,

$$\boxed{T_{\text{imp}}(b) = \frac{(\Delta p_\perp)^2}{2m} = \frac{2z^2 e^4}{mv^2 b^2}.}$$

Indeed

$$T(b) = T_{\text{imp}}(b) \left[1 - \frac{b_c^2}{b^2} + O\left(\frac{b_c^4}{b^4} \right) \right].$$

Besides $b \gg b_c$, using $(\Delta p)^2/(2m)$ requires $2ze^2/(bvmc) \ll 1$. The cancellation of γ in the impulse is physical: the transverse field is enhanced by γ, but its duration is shortened by the same factor. Extrapolating the impulse result to $b = 0$ would violate its assumptions and miss the finite $T_{\max}$.

13.2 Time-varying electromagnetic fields $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$ of finite duration act on a charged particle of charge e and mass m bound harmonically to the origin with natural frequency ω_0 and small damping constant Γ. The fields may be caused by a passing charged particle or some other external source. The charge's motion in response to the fields is nonrelativistic and small in amplitude compared to the scale of spatial variation of the fields (dipole approximation). Take the oscillator initially unexcited at the origin, and retain the leading energy transfer in linear response to the applied fields. Show that the energy transferred to the oscillator in the limit of very small damping is

$$\Delta E = \frac{\pi e^2}{m} |\mathbf{E}(\omega_0)|^2$$

where $\mathbf{E}(\omega)$ is the symmetric Fourier transform of $\mathbf{E}(0, t)$:

$$\mathbf{E}(0,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{E}(\omega) e^{-i\omega t} \, d\omega; \qquad \mathbf{E}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{E}(0,t) e^{i\omega t} \, dt$$

Solution.

Work to first order in the applied fields when computing the displacement, and to second order when computing the deposited energy. In the dipole approximation the equation of motion is

$$m(\ddot{\mathbf{x}} + \Gamma\dot{\mathbf{x}} + \omega_0^2\mathbf{x}) = e\mathbf{E}(0,t).$$

The magnetic force is proportional to $\dot{\mathbf{x}} \times \mathbf{B}$, so it changes the first-order motion only at second order in the applied field. It also does no instantaneous work. Spatial gradients of $\mathbf{E}$ are beyond the stated dipole approximation. Assume the oscillator is at the origin with zero velocity before the pulse; an initially excited oscillator would have an additional phase-dependent interference term. The marked assumptions make explicit the scope needed for the formula on printed p. 655; the nonrelativistic and dipole conditions alone do not specify the initial state.

First set $\Gamma = 0$. The retarded solution is

$$\mathbf{x}(t) = \frac{e}{m\omega_0}\int_{-\infty}^{t} \mathbf{E}(0,t')\sin[\omega_0(t-t')]\,dt'.$$

After the pulse, define the real vectors

$$\mathbf{C} = \int_{-\infty}^{\infty} \mathbf{E}(0,t')\cos\omega_0 t'\,dt', \qquad \mathbf{S} = \int_{-\infty}^{\infty} \mathbf{E}(0,t')\sin\omega_0 t'\,dt'.$$

Expansion of the sine gives

$$\mathbf{x} = \frac{e}{m\omega_0}(\mathbf{C}\sin\omega_0 t - \mathbf{S}\cos\omega_0 t), \qquad \dot{\mathbf{x}} = \frac{e}{m}(\mathbf{C}\cos\omega_0 t + \mathbf{S}\sin\omega_0 t).$$

The cross terms cancel between kinetic and potential energies:

$$\Delta E = \frac{m}{2}\left(|\dot{\mathbf{x}}|^2 + \omega_0^2|\mathbf{x}|^2\right) = \frac{e^2}{2m}(|\mathbf{C}|^2 + |\mathbf{S}|^2).$$

By the stated Fourier convention, $\mathbf{E}(\omega_0) = (\mathbf{C} + i\mathbf{S})/\sqrt{2\pi}$. Thus

$$\boxed{\Delta E = \frac{\pi e^2}{m}|\mathbf{E}(\omega_0)|^2.}$$

Even with $\mathbf{B} = 0$, the initial-state condition matters. Let the unforced motion before the pulse be $\mathbf{a}\cos\omega_0 t + \mathbf{b}\sin\omega_0 t$. The pulse changes these two coefficients to $\mathbf{a} - e\mathbf{S}/(m\omega_0)$ and $\mathbf{b} + e\mathbf{C}/(m\omega_0)$, respectively. Subtracting the initial oscillator energy from the final one gives

$$\Delta E = \frac{e^2}{2m}(|\mathbf{C}|^2 + |\mathbf{S}|^2) + e\omega_0(\mathbf{b}\cdot\mathbf{C} - \mathbf{a}\cdot\mathbf{S}).$$

The second term can have either sign and vanishes for an initially unexcited oscillator (or after averaging uniformly over its initial phase).

One can also keep the damping until the final step. Fourier transformation gives

$$\mathbf{x}(\omega) = \frac{e\mathbf{E}(\omega)}{m(\omega_0^2 - \omega^2 - i\Gamma\omega)}.$$

After all the energy has been dissipated, the work done by the external field equals $m\Gamma\int|\dot{\mathbf{x}}|^2 dt$. Parseval's identity and $\mathbf{E}(-\omega) = \mathbf{E}(\omega)^*$ then give

$$W_\Gamma = \frac{2e^2}{m}\int_0^\infty \frac{\Gamma\omega^2|\mathbf{E}(\omega)|^2}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2}\,d\omega.$$

Near the positive resonance, $\omega = \omega_0 + s$,

$$\frac{2\Gamma\omega^2}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2} \simeq \frac{\Gamma/2}{s^2 + (\Gamma/2)^2} \longrightarrow \pi\delta(s).$$

This reproduces the boxed result. Thus the small-damping limit can mean either the mechanical energy just after the pulse, before appreciable decay, or the total energy subsequently dissipated.

13.3 The external fields of Problem 13.2 are caused by a charge ze passing the origin in a straight-line path at speed v and impact parameter b. The fields are given by (11.152).

(a) Evaluate the Fourier transforms for the perpendicular and parallel components of the electric field at the origin and show that

$$E_\perp(\omega) = \frac{ze}{bv}\left(\frac{2}{\pi}\right)^{1/2}\xi K_1(\xi); \qquad E_\parallel(\omega) = -i\,\frac{ze}{\gamma bv}\left(\frac{2}{\pi}\right)^{1/2}\xi K_0(\xi)$$

where $\xi = \omega b/\gamma v$, and $K_\nu(\xi)$ is the modified Bessel function of the second kind and order ν. [See references to tables of Fourier transforms in Section 13.3.]

(b) Using the result of Problem 13.2, write down the energy transfer ΔE to a harmonically bound charged particle. From the limiting forms of the modified Bessel functions for small and large argument, show that your result agrees with the appropriate limit of $T(b)$ in Problem 13.1 on the one hand and the arguments at the end of Section 13.1 on the adiabatic behavior for $b \gg \gamma v/\omega_0$ on the other.

Solution.

Take the perpendicular axis from the trajectory toward the observer, so that the fields are those displayed in Problem 13.1(b). We first take $\omega > 0$, as is appropriate when inserting ω_0 in the energy formula.

(a) Set $u = \gamma vt/b$ and $\xi = \omega b/(\gamma v)$. The two transforms become

$$E_\perp(\omega) = \frac{ze}{bv\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{(1+u^2)^{3/2}}\,du, \qquad E_\parallel(\omega) = -\frac{ze}{\gamma bv\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{ue^{i\xi u}}{(1+u^2)^{3/2}}\,du.$$

For completeness, the basic transform can be derived from

$$(1+u^2)^{-1/2} = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2}e^{-s(1+u^2)}\,ds.$$

The Gaussian u integration gives

$$\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{\sqrt{1+u^2}}\,du = \int_0^\infty \frac{ds}{s}e^{-s-\xi^2/(4s)} = 2K_0(\xi).$$

A convergence factor justifies this conditionally convergent transform and can be removed for $\xi > 0$. More generally the denominator $\sqrt{u^2+a^2}$ gives $2K_0(a\xi)$. Differentiating in a and setting $a = 1$ yields

$$\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{(1+u^2)^{3/2}}\,du = 2\xi K_1(\xi).$$

Also,

$$\frac{u}{(1+u^2)^{3/2}} = -\frac{d}{du}(1+u^2)^{-1/2},$$

so integration by parts gives the second integral as $2i\xi K_0(\xi)$. Consequently,

$$\boxed{E_\perp(\omega) = \frac{ze}{bv}\sqrt{\frac{2}{\pi}}\,\xi K_1(\xi), \qquad E_\parallel(\omega) = -i\frac{ze}{\gamma bv}\sqrt{\frac{2}{\pi}}\,\xi K_0(\xi).}$$

For negative frequencies the real-field condition determines the continuation: replace each Bessel argument by $|\xi|$, replace the transverse prefactor ξ by $|\xi|$, and retain the signed ξ in the longitudinal prefactor. This also makes the even/odd Fourier symmetries explicit.

(b) Orthogonality of the parallel and perpendicular directions eliminates cross terms in $|\mathbf{E}|^2$. With $\xi_0 = \omega_0 b/(\gamma v)$, Problem 13.2 gives

$$\boxed{\Delta E(b) = \frac{2z^2e^4}{mv^2b^2}\left[\xi_0^2K_1^2(\xi_0) + \frac{\xi_0^2}{\gamma^2}K_0^2(\xi_0)\right].}$$

At small argument,

$$K_1(\xi) = \xi^{-1} + O(\xi\ln\xi), \qquad K_0(\xi) = -\ln(\xi/2) - \gamma_{\rm E} + O(\xi^2\ln\xi).$$

Thus $\xi^2K_1^2 \to 1$, $\xi^2K_0^2 \to 0$, and

$$\Delta E \longrightarrow \frac{2z^2e^4}{mv^2b^2}.$$

This is the large-b impulse limit of Problem 13.1, with the additional condition $b \ll \gamma v/\omega_0$ that the pulse be sudden on the oscillator time scale. The useful overlap is $b_c \ll b \ll \gamma v/\omega_0$, together with a nonrelativistic response.

For $\xi \gg 1$,

$$K_\nu(\xi) = \sqrt{\frac{\pi}{2\xi}}e^{-\xi}\left[1 + \frac{4\nu^2-1}{8\xi} + O(\xi^{-2})\right],$$

and therefore

$$\Delta E \simeq \frac{\pi z^2e^4}{mv^2b^2}\,\xi_0(1+\gamma^{-2})e^{-2\xi_0}\left[1+O(\xi_0^{-1})\right].$$

For $b \gg \gamma v/\omega_0$ the oscillator follows the slowly varying force and returns almost to its initial state. The remaining excitation is exponentially small, quantitatively establishing adiabatic suppression.

13.4

(a) Taking $\hbar\langle\omega\rangle = 12Z$ eV in the quantum-mechanical energy-loss formula, calculate the rate of energy loss (in MeV/cm) in air at NTP, aluminum, copper, and lead for a proton and a mu meson, each with kinetic energies of 10, 100, 1000 MeV.

(b) Convert your results to energy loss in units of MeV·(cm^2/g) and compare the values obtained in different materials. Explain why, for each fixed projectile and kinetic energy, the energy losses in MeV-(cm^2/g) are within about a factor of 2.3 of each other, whereas the values in MeV/cm differ greatly.

Solution.

(a) Use Jackson's (13.14)–(13.15), with unit projectile charge, exactly as requested:

$$S = \frac{4\pi N Z e^4}{mc^2\beta^2}\left[\ln\frac{2mc^2\beta^2\gamma^2}{I} - \beta^2\right], \qquad I = \hbar\langle\omega\rangle = 12Z \text{ eV}.$$

Here m is the electron mass and $N = \rho N_{\mathrm{A}}/A$ is the atomic number density. For projectile rest energy Mc^2 and kinetic energy T,

$$\gamma = 1 + \frac{T}{Mc^2}, \qquad \beta^2 = 1 - \gamma^{-2}, \qquad \beta^2\gamma^2 = \gamma^2 - 1 = \frac{T(T + 2Mc^2)}{M^2c^4}.$$

Writing $r_0 = e^2/(mc^2)$ for the classical electron radius gives the convenient mass stopping power

$$\boxed{\frac{S}{\rho} = \frac{K}{\beta^2}\frac{Z}{A}\left[\ln\frac{2(0.51099895\ \text{MeV})(\gamma^2-1)}{I\ (\text{MeV})} - \beta^2\right], \quad K = 0.30707493\ \text{MeV}\,\text{cm}^2\,\text{mol}^{-1}.}$$

The coefficient is $K = 4\pi N_{\mathrm{A}} r_0^2 mc^2$, and A is expressed in $\mathrm{g\,mol^{-1}}$. We use $M_p c^2 = 938.27209$ MeV and $M_\mu c^2 = 105.65838$ MeV.

The adopted material parameters are

	air	Al	Cu	Pb
ρ ($\mathrm{g\,cm^{-3}}$)	0.0012931	2.699	8.960	11.35
Z/A ($\mathrm{mol\,g^{-1}}$)	0.499185	0.481811	0.456362	0.395753
I (eV)	87.596	156	348	984

NTP here means 0°C and one atmosphere; this convention matters only for the air column in MeV/cm. The densities and atomic weights are from the Particle Data Group atomic/nuclear-property tables.[8] For air we apply $I_j = 12Z_j$ to each constituent and add their stopping powers. This is equivalent to

$$\left\langle\frac{Z}{A}\right\rangle = \sum_j w_j\frac{Z_j}{A_j}, \qquad \ln I_{\text{air}} = \frac{\sum_j w_j (Z_j/A_j)\ln(12Z_j\ \text{eV})}{\sum_j w_j Z_j/A_j}.$$

The dry-air mass fractions for C, N, O, Ar are respectively 0.000124, 0.755267, 0.231781, 0.012827, normalized to sum to one. Thus the supplied $12Z$ prescription is maintained for the mixture instead of inserting the tabulated experimental excitation energy.

For example, a 10 MeV proton has $\gamma = 1.0106579$, $\beta^2 = 0.0209798$, and $2mc^2(\gamma^2-1) = 0.0219008$ MeV. For aluminum the substitution is

$$\frac{S}{\rho} = \frac{0.30707493(13/26.9815385)}{0.0209798}\left[\ln\frac{0.0219008}{0.000156} - 0.0209798\right] = 34.7207\ \text{MeV}\,\text{cm}^2/\text{g},$$

and $S = 2.699(34.7207) = 93.7112$ MeV/cm. Applying the same calculation to every case gives

		S (MeV/cm)			
particle	T (MeV)	air	Al	Cu	Pb
p	10	0.051969	93.711	246.646	202.605
p	100	0.008311	15.485	43.198	39.634
p	1000	0.002533	4.801	13.781	13.266
μ	10	0.009082	16.904	47.064	43.035
μ	100	0.002600	4.924	14.116	13.558
μ	1000	0.002613	5.031	14.802	14.813

[8] PDG, 2005 Table 6.1 gives the air density at 0°C. The dry-air table gives mass fractions; the aluminum, copper, and lead tables give elemental densities and molar masses.

These are the model's local collisional stopping powers, not an average over a slab in which the particle slows appreciably. Density, shell, radiative, and finite-projectile-mass corrections have not been added to the explicitly requested (13.14).

(b) Dividing each column by its density produces

		S/ρ (MeV cm^2/g)			
particle	T (MeV)	air	Al	Cu	Pb
p	10	40.189	34.721	27.527	17.851
p	100	6.427	5.737	4.821	3.492
p	1000	1.959	1.779	1.538	1.169
μ	10	7.024	6.263	5.253	3.792
μ	100	2.011	1.825	1.575	1.195
μ	1000	2.020	1.864	1.652	1.305

At a fixed projectile speed the material dependence is through Z/A, which changes only from about 0.50 to 0.40, and through a logarithm of I. The density factor, which differs by almost four orders of magnitude between air and lead, has disappeared. This explains the similarity of mass stopping powers and the large differences in MeV/cm.

The original assertion of a factor strictly less than two has one exception within the problem's own model:

$$\left.\frac{(S/\rho)_{\text{air}}}{(S/\rho)_{\text{Pb}}}\right|_{p,\ 10\,\text{MeV}} = \frac{40.1894}{17.8506} = 2.25143.$$

The other five row-wise ratios lie between 1.55 and 1.85. Thus about a factor of two, or the marked bound of 2.3, is accurate; a factor of two for all 24 entries irrespective of projectile energy would also be incorrect.

13.5 Consider the energy loss by close collisions of a fast, but nonrelativistic, heavy particle of charge ze passing through an electronic plasma. Assume that the screened Coulomb interaction $V(r) = ze^2 \exp(-k_{\text{D}} r)/r$, where k_{D} is the Debye screening parameter, acts between the electrons and the incident particle.

(a) Show that the energy transfer in a collision at impact parameter b is given approximately by

$$\Delta E(b) \simeq \frac{2(ze^2)^2}{mv^2}\, k_{\text{D}}^2 K_1^2(k_{\text{D}} b)$$

where m is the electron mass and v is the velocity of the incident particle.

(b) Determine the energy loss per unit distance traveled for collisions with impact parameter greater than $b_{\min}$. Assuming $k_{\text{D}} b_{\min} \ll 1$, show that

$$\left(\frac{dE}{dx}\right)_{k_{\text{D}} b<1} \simeq \frac{(ze)^2}{v^2}\, \omega_p^2 \ln\left(\frac{1}{1.47 k_{\text{D}} b_{\min}}\right)$$

where $b_{\min}$ is given by the larger of the classical and quantum minimum impact parameters [(13.16) and above].

Solution.

(a) During a weak collision use the undeflected trajectory $r(t) = \sqrt{b^2 + v^2 t^2}$. Integrating the interaction along that path gives

$$\mathcal{V}(b) = \int_{-\infty}^{\infty} V(r(t))\, dt = \frac{ze^2}{v} \int_{-\infty}^{\infty} \frac{e^{-k_{\text{D}}\sqrt{b^2+z^2}}}{\sqrt{b^2+z^2}}\, dz.$$

Set $z = b\sinh u$, so that $r = b\cosh u$ and $dz/r = du$. The integral representation $K_0(x) = \int_0^\infty e^{-x\cosh u}\,du$ gives

$$\mathcal{V}(b) = \frac{2ze^2}{v} K_0(k_{\mathrm{D}} b).$$

The transverse force is $-\partial V/\partial b$, and differentiation commutes with the convergent integral. Therefore

$$|\Delta p_\perp| = \left| -\frac{d\mathcal{V}}{db} \right| = \frac{2ze^2 k_{\mathrm{D}}}{v} K_1(k_{\mathrm{D}} b).$$

The longitudinal force is odd in t and integrates to zero. With a nonrelativistic electron recoil,

$$\boxed{\Delta E(b) \simeq \frac{(\Delta p_\perp)^2}{2m} = \frac{2(ze^2)^2}{mv^2} k_{\mathrm{D}}^2 K_1^2(k_{\mathrm{D}} b).}$$

For $k_{\mathrm{D}} b \ll 1$ the factor $k_{\mathrm{D}} K_1(k_{\mathrm{D}} b)$ tends to $1/b$, recovering the unscreened impulse.

(b) Let n_e be the electron density, so $\omega_p^2 = 4\pi n_e e^2/m$. The number of electrons encountered per unit path length in an annulus is $2\pi n_e b\,db$. Consequently, putting $u = k_{\mathrm{D}} b_{\min}$,

$$S_{\mathrm{close}} = 2\pi n_e \int_{b_{\min}}^\infty b\,\Delta E(b)\,db = \frac{(ze)^2 \omega_p^2}{v^2} \int_u^\infty x K_1^2(x)\,dx.$$

The upper limit may be infinite because screening suppresses the large-b integrand exponentially. To evaluate the last integral, use

$$K_0' = -K_1, \qquad K_1' = -K_0 - K_1/x, \qquad K_2 = K_0 + 2K_1/x.$$

Differentiating the expression below with these identities gives $-uK_1^2(u)$, and it vanishes at infinity. Hence

$$\int_u^\infty x K_1^2(x)\,dx = \frac{u^2}{2}\left[K_0(u)K_2(u) - K_1^2(u)\right] = uK_0K_1 + \frac{u^2}{2}(K_0^2 - K_1^2).$$

As $u \to 0$, $uK_1 \to 1$, $u^2 K_0^2 \to 0$, and $K_0 = -\ln(u/2) - \gamma_{\mathrm{E}} + o(1)$, giving

$$\int_u^\infty x K_1^2(x)\,dx = \ln\frac{2}{u} - \gamma_{\mathrm{E}} - \frac{1}{2} + O(u^2 \ln^2 u) = \ln\frac{1}{C_{\mathrm{c}} u} + O(u^2 \ln^2 u), \quad C_{\mathrm{c}} = \frac{e^{\gamma_{\mathrm{E}}+1/2}}{2} = 1.468\ldots.$$

Thus

$$\boxed{S_{\mathrm{close}} \simeq \frac{(ze)^2\omega_p^2}{v^2} \ln\frac{1}{1.47\,k_{\mathrm{D}} b_{\min}}.}$$

The labels close and $k_{\mathrm{D}} b < 1$ refer to the dominant range; the screened integral itself has not been sharply truncated at $k_{\mathrm{D}} b = 1$. In this nonrelativistic, heavy-projectile approximation the cutoff is

$$b_{\min} \sim \max\left(\frac{ze^2}{mv^2}, \frac{\hbar}{mv}\right).$$

It excludes either large classical deflections or distances below the electron's reduced de Broglie wavelength. The logarithmic formula requires $k_{\mathrm{D}} b_{\min} \ll 1$ and a speed well above the electron thermal speed.

13.6 The energy loss in a plasma from distant collisions can be found with Fermi's method for the density effect. Consider the nonrelativistic limit of (13.36) with the relative dielectric constant of a plasma given by (7.59) augmented by some damping,

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\Gamma}$$

Assume that the arguments of the Bessel functions are small (corresponding to a speed of the incident particle large compared to thermal speeds in the plasma).

(a) Show that the energy loss (13.36) for $k_{\rm D}b > 1$ becomes

$$\left(\frac{dE}{dx}\right)_{k_{\rm D}b>1} \approx \frac{2z^2e^2}{\pi v^2}\int_0^\infty \mathrm{Re}\left(\frac{i\omega}{\epsilon(\omega)}\right)\ln\left(\frac{1.123k_{\rm D}v}{\omega}\right)d\omega$$

(b) With the assumption that $\Gamma \ll \omega_p$ in $\epsilon(\omega)$, show that the formula of part a yields

$$\left(\frac{dE}{dx}\right)_{k_{\rm D}b>1} \approx \frac{z^2e^2}{v^2}\,\omega_p^2\ln\left(\frac{1.123k_{\rm D}v}{\omega_p}\right)$$

Combine with the close-collision result of Problem 13.5 to find the total energy loss of a nonrelativistic particle passing through a plasma,

$$\left(\frac{dE}{dx}\right) \approx \frac{z^2e^2}{v^2}\,\omega_p^2\ln\left(\frac{\Lambda v}{\omega_p b_{\rm min}}\right)$$

where Λ is a number of order unity. The presence of ω_p in the logarithm suggests that the energy loss may be quantized in units of $\hbar\omega_p$. In fact, electrons passing through thin metallic foils do show this discreteness in energy loss, allowing determination of effective plasma frequencies in metals. [See H. Raether, *Springer Tracts in Modern Physics*, Vol. 38, ed. G. Höhler, Springer-Verlag, Berlin (1965), pp. 84–157.]

Solution.

(a) With a cylindrical separation radius a, Jackson's (13.36), on printed p. 634, is

$$S_{b>a} = \frac{2z^2e^2}{\pi v^2}\,\mathrm{Re}\int_0^\infty i\omega\,\lambda^* aK_1(\lambda^* a)K_0(\lambda a)\left(\frac{1}{\epsilon} - \beta^2\right)d\omega, \qquad \lambda^2 = \frac{\omega^2}{v^2}(1-\beta^2\epsilon).$$

For $\beta \ll 1$ set $\lambda = \omega/v$ to leading order and drop β^2. Put $a = k_{\rm D}^{-1}$. At the frequencies responsible for energy deposition, $u = \omega/(k_{\rm D}v) \ll 1$, and

$$uK_1(u) = 1 + O(u^2\ln u), \qquad K_0(u) = \ln(2/u) - \gamma_{\rm E} + O(u^2\ln u).$$

The Bessel factors are then real and

$$\boxed{S_{b>k_{\rm D}^{-1}} \simeq \frac{2z^2e^2}{\pi v^2}\int_0^\infty \mathrm{Re}\left(\frac{i\omega}{\epsilon(\omega)}\right)\ln\frac{2e^{-\gamma_{\rm E}}k_{\rm D}v}{\omega}\,d\omega.}$$

Since $2e^{-\gamma_{\rm E}} = 1.12291897$, this is the stated 1.123.

(b) Algebraically,

$$\frac{1}{\epsilon} = \frac{\omega^2 + i\Gamma\omega}{\omega^2 - \omega_p^2 + i\Gamma\omega} = 1 + \frac{\omega_p^2}{\omega^2 - \omega_p^2 + i\Gamma\omega}.$$

Multiplication by the conjugate denominator therefore gives the positive loss function

$$\operatorname{Re}\frac{i\omega}{\epsilon} = \frac{\Gamma\omega_p^2\omega^2}{(\omega^2-\omega_p^2)^2+\Gamma^2\omega^2}.$$

For $\omega = \omega_p + s$ with $|s|, \Gamma \ll \omega_p$, this becomes

$$\frac{\Gamma\omega_p^2}{4s^2+\Gamma^2} = \frac{\omega_p^2}{2}\frac{\Gamma/2}{s^2+(\Gamma/2)^2} \longrightarrow \frac{\pi\omega_p^2}{2}\delta(\omega-\omega_p).$$

The logarithm varies slowly across this narrow resonance. Its value at ω_p can be taken outside the integral, yielding

$$\boxed{S_{\text{distant}} \simeq \frac{z^2e^2\omega_p^2}{v^2}\ln\frac{1.12291897\,k_{\text{D}}v}{\omega_p}.}$$

The small-argument condition used above is therefore specifically $\omega_p/(k_{\text{D}}v) \ll 1$. For a thermal Debye plasma ω_p/k_{D} is its electron thermal-speed scale.

Adding Problem 13.5 makes the arbitrary matching scale k_{D} cancel inside the logarithm:

$$S = S_{\text{close}} + S_{\text{distant}} = \frac{z^2e^2\omega_p^2}{v^2}\ln\left[\frac{2e^{-\gamma_{\text{E}}}}{C_{\text{c}}}\,\frac{v}{\omega_p b_{\min}}\right].$$

Thus, retaining the unrounded constants,

$$\boxed{S \simeq \frac{z^2e^2\omega_p^2}{v^2}\ln\frac{\Lambda v}{\omega_p b_{\min}}, \qquad \Lambda = 4e^{-2\gamma_{\text{E}}-1/2} = 0.764803\ldots.}$$

Using 1.123 and 1.47 literally instead gives 0.76395, a difference from rounding. The occurrence of ω_p here arises from the collective pole in $1/\epsilon$; classical theory establishes that frequency, while the stated discrete losses $\hbar\omega_p$ require quantization of the collective excitation.

13.7 With the same approximations as were used to discuss multiple scattering, show that the *projected* transverse displacement y (see Fig. 13.7) of an incident particle is described approximately by a Gaussian distribution,

$$P(y)\,dy = A\,\exp\left[\frac{-y^2}{2\langle y^2\rangle}\right]dy$$

where the mean square displacement is $\langle y^2\rangle = (x^2/6)\langle\Theta^2\rangle$, x being the thickness of the material traversed and $\langle\Theta^2\rangle$ the mean square angle of scattering.

Solution.

Let s measure distance from the entrance and let $\delta\theta_y$ be the projected angular kick in a collision at s. In the small-angle approximation this kick contributes $(x-s)\delta\theta_y$ to the final displacement. With independent, azimuthally symmetric collisions,

$$y = \sum_j (x-s_j)\delta\theta_{yj}, \qquad \langle\delta\theta_{yj}\rangle = 0, \qquad \langle\delta\theta_{yj}^2\rangle = \frac{1}{2}\langle\theta^2\rangle.$$

Write $N\sigma\,ds$ for the mean number of collisions in ds. The mean projected angular variance added per unit length is

$$D_\theta = N\sigma\,\frac{\langle\theta^2\rangle}{2}.$$

All cross terms in $\langle y^2\rangle$ vanish because kicks at distinct positions are independent with zero mean. Thus

$$\langle y^2\rangle = D_\theta \int_0^x (x-s)^2\, ds = D_\theta \left[-\frac{(x-s)^3}{3}\right]_0^x = \frac{D_\theta x^3}{3}.$$

The total, unprojected exit-angle variance is

$$\langle \Theta^2\rangle = N\sigma x \langle \theta^2\rangle = 2D_\theta x.$$

Eliminating D_θ gives

$$\boxed{\langle y^2\rangle = \frac{x^2}{6}\langle \Theta^2\rangle.}$$

To establish the Gaussian shape rather than just its variance, take the characteristic function of the sum:

$$\langle e^{iky}\rangle = \prod_j \left\langle e^{ik(x-s_j)\delta\theta_{yj}}\right\rangle.$$

Keeping the second cumulant in the same many-small-kicks approximation used for the angular distribution,

$$\ln\langle e^{iky}\rangle \simeq -\frac{k^2}{2}\sum_j (x-s_j)^2\langle \delta\theta_{yj}^2\rangle = -\frac{k^2}{2}\langle y^2\rangle.$$

The inverse Fourier transform is a normalized Gaussian:

$$\boxed{P(y) = \frac{1}{\sqrt{2\pi\langle y^2\rangle}}\exp\left[-\frac{y^2}{2\langle y^2\rangle}\right].}$$

This is the central part of the distribution. Rare, unusually large single kicks cause non-Gaussian tails, just as in the angular problem. Constant D_θ also assumes negligible slowing over x; if the momentum changes, the general variance is $\int_0^x D_\theta(s)(x-s)^2\, ds$ instead.

13.8 If the finite size of the nucleus is taken into account in the "single-scattering" tail of the multiple-scattering distribution, there is a critical thickness x_c beyond which the single-scattering tail is effectively suppressed within the angular-cutoff approximation.

(a) Define x_c in a reasonable way and calculate its value (in cm) for aluminum and lead, assuming that the incident particle is relativistic and singly charged.

(b) For these thicknesses calculate the number of collisions that occur and determine whether the Gaussian approximation is valid.

Solution.

(a) A sharp disappearance is a feature of an effective angular cutoff, not of the exact nuclear form factor. We define a conservative critical thickness by the criterion suggested at the end of Section 13.6:

$$\boxed{\langle \Theta^2\rangle(x_c) = \theta_{\max}^2.}$$

At this thickness the typical cumulative deflection is already as large as the largest effective individual deflection, so a distinct single-scattering tail cannot extend outside the central distribution.

Use the source's estimates, on printed pp. 641–644,

$$a = 1.4a_0 Z^{-1/3}, \qquad R = 1.4A^{1/3}\ \text{fm}, \qquad \theta_{\min} = \frac{\hbar}{pa}, \qquad \theta_{\max} = \frac{\hbar}{pR}, \qquad L = \ln\frac{\theta_{\max}}{\theta_{\min}} = \ln\frac{a}{R}.$$

For a singly charged relativistic projectile ($z = 1$, $v \simeq c$), (13.65) is

$$\langle\Theta^2\rangle = 2\pi N x \left(\frac{2Ze^2}{pc}\right)^2 L.$$

Equating this to $\hbar^2/(p^2R^2)$ cancels the incident momentum:

$$\frac{\hbar^2}{p^2R^2} = \frac{8\pi N Z^2 e^4 L}{p^2c^2} x_c, \qquad \boxed{x_c = \frac{1}{8\pi N Z^2 \alpha^2 R^2 L}.}$$

For charge ze and speed βc, the corresponding expression has an additional factor β^2/z^2. The assumed relativistic singly charged case is the one tabulated here.

Use $a_0 = 5.2917721 \times 10^{-9}$ cm and $N = \rho N_{\text{A}}/A$, with the material data of Problem 13.4:

	Al	Pb
Z	13	82
A	26.9815	207.2
N (cm^{-3})	6.024×10^{22}	3.299×10^{22}
a (cm)	3.15076×10^{-9}	1.70526×10^{-9}
R (cm)	4.19904×10^{-13}	8.28434×10^{-13}
L	8.92313	7.62969
x_c (cm)	46.65	0.6433

These are estimates within the source's screening and nuclear-radius model, not precision material thresholds.

The definition is not unique. In Fig. 13.8 the tail starts near $\theta_{\text{proj}} \simeq 2.5\sqrt{\langle\Theta^2\rangle}$, because the dimensionless projected angle is $\alpha_{\text{ang}} = \theta_{\text{proj}}/\sqrt{\langle\Theta^2\rangle}$. Equating that crossover to $\theta_{\max}$ instead gives the earlier onset of strong tail suppression at $x_c/2.5^2 \simeq 7.46$ cm in aluminum and 0.103 cm in lead. Both definitions express the same comparison of cumulative and single-event deflections. The adopted boxed definition specifies which one is used for the requested numbers.

(b) The total cross section (13.59) and mean number of collisions are

$$\sigma = \pi\left(\frac{2Ze^2}{pc}\right)^2 \theta_{\min}^{-2} = \pi a^2 (2Z\alpha)^2, \qquad n_c = N\sigma x_c.$$

Substitution of the expression for x_c cancels N, $Z^2\alpha^2$:

$$\boxed{n_c = \frac{a^2}{2R^2L} = \begin{cases} 3.15 \times 10^6, & \text{Al}, \\ 2.78 \times 10^5, & \text{Pb}. \end{cases}}$$

Equivalently, $\langle\Theta^2\rangle = n_c\langle\theta^2\rangle$ and $\langle\theta^2\rangle = 2\theta_{\min}^2 L$ give the same result. Even the earlier crossover definition gives about 5.05×10^5 and 4.44×10^4 collisions. Both are much larger than the $n \lesssim 200$ regime identified in the source as problematic for a Gaussian approximation. The many-collision approximation is therefore well justified. The incident energy must also be large enough that slowing and large total deflections over these thicknesses can be neglected; relativistic speed by itself does not guarantee negligible energy loss through 47 cm of aluminum.

13.9 Assuming that Plexiglas or Lucite has an index of refraction of 1.50 in the visible region, compute the angle of emission of visible Cherenkov radiation for electrons and protons as a function of their kinetic energies in MeV. Determine how many quanta with wavelengths between 4000 and 6000 Å are emitted per centimeter of path in Lucite by a 1 MeV electron, a 500 MeV proton, and a 5 GeV proton.

Solution.

For refractive index $n = 1.50$, (13.50) gives $\cos\theta_C = 1/(n\beta)$. The velocity at kinetic energy T is

$$\beta(T) = \frac{\sqrt{T(T + 2Mc^2)}}{T + Mc^2}.$$

The requested functions of energy, with T and Mc^2 in MeV, are therefore

$$\boxed{\theta_C(T; M) = \cos^{-1}\left[\frac{T + Mc^2}{1.50\sqrt{T(T + 2Mc^2)}}\right], \qquad T > T_{\text{th}}.}$$

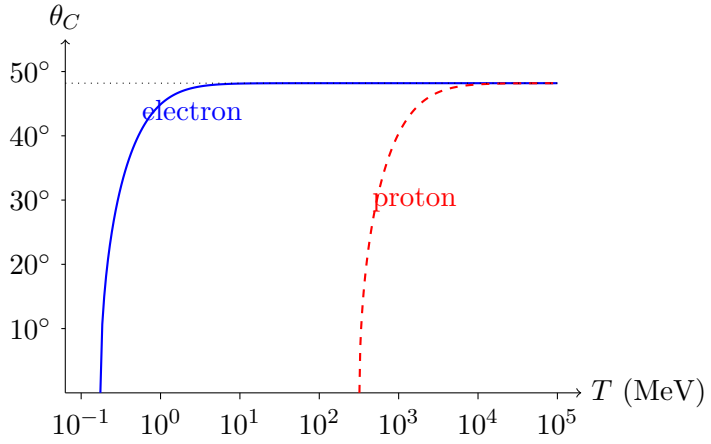
There is no Cherenkov emission below threshold. Setting $n\beta = 1$ gives

$$\gamma_{\text{th}} = \frac{1}{\sqrt{1 - n^{-2}}} = \frac{3}{\sqrt{5}}, \qquad T_{\text{th}} = Mc^2\left(\frac{3}{\sqrt{5}} - 1\right).$$

With the masses used in Problem 13.4,

$$T_{\text{th},e} = 0.174578 \text{ MeV}, \qquad T_{\text{th},p} = 320.552 \text{ MeV}.$$

Both curves approach the common limit $\theta_C(\infty) = \cos^{-1}(2/3) = 48.1897°$:



The horizontal energy axis is logarithmic; each curve starts at its own threshold.

To count photons, divide the Frank–Tamm spectral energy (13.48) by $\hbar\omega$:

$$\frac{d^2N_\gamma}{dx\,d\omega} = \frac{z^2e^2}{\hbar c^2}\left(1 - \frac{1}{n^2\beta^2}\right).$$

The specified wavelengths are vacuum wavelengths, so $\omega = 2\pi c/\lambda$ and $|d\omega/d\lambda| = 2\pi c/\lambda^2$. Thus

$$\frac{d^2N_\gamma}{dx\,d\lambda} = \frac{2\pi z^2\alpha}{\lambda^2}\left(1 - \frac{1}{n^2\beta^2}\right),$$

and for singly charged particles

$$\frac{N_\gamma}{x} = 2\pi\alpha\left(\frac{1}{4.000\times10^{-5}\text{ cm}} - \frac{1}{6.000\times10^{-5}\text{ cm}}\right)\left(1-\frac{1}{2.25\beta^2}\right)$$
$$= \boxed{382.088\left(1-\frac{1}{2.25\beta^2}\right)\text{cm}^{-1}.}$$

The resulting angles and yields are

particle and kinetic energy	β	θ_C	N_γ/x (cm^{-1})
e, 1 MeV	0.94108	44.8946°	190.34
p, 500 MeV	0.75791	28.4042°	86.46
p, 5000 MeV	0.98744	47.5344°	207.92

These are emitted photon numbers. Absorption, interface escape losses, and detector efficiency would be additional experimental factors.

13.10 A particle of charge ze moves along the z axis with constant speed v, passing $z = 0$ at $t = 0$. The medium through which the particle moves is described by a dielectric constant $\epsilon(\omega)$.

(a) Beginning with the potential $\Phi(\mathbf{k},\omega)$ of (13.25), show that the potential of frequency ω is given as a function of spatial coordinate $\mathbf{x}$ by

$$\Phi(\omega,\mathbf{x}) = \frac{ze}{v\epsilon(\omega)}\sqrt{\frac{2}{\pi}}\,K_0\left(\frac{|\omega|\,\rho}{v}\sqrt{1-\beta^2\epsilon}\right)e^{i\omega z/v}$$

where z and $\rho = \sqrt{x^2+y^2}$ are the cylindrical coordinates of the observation point.

(b) Assuming that ϵ is independent of frequency and that $\beta^2\epsilon < 1$, take the Fourier transform with respect to ω of the expression in part a and obtain $\Phi(\mathbf{x},t)$. Calculate the electric and magnetic fields and compare them to the vacuum fields (11.152). Show that, among other things, the vacuum factor γ is replaced by $\Gamma = (1-\beta^2\epsilon)^{-1/2}$.

(c) Repeat the calculations of parts a and b with $\beta^2\epsilon > 1$. Show that now

$$\Phi(\omega,\mathbf{x}) = \frac{ze}{v\epsilon(\omega)}\sqrt{\frac{\pi}{2}}\,e^{i\omega z/v}\left[-N_0\left(\frac{|\omega|\,\rho}{v}\sqrt{\beta^2\epsilon-1}\right) \pm iJ_0\left(\frac{|\omega|\,\rho}{v}\sqrt{\beta^2\epsilon-1}\right)\right]$$

for $\omega \gtrless 0$. Calculate the remaining Fourier transform to obtain $\Phi(\mathbf{x},t)$. Relate your answer to the result given in Section 13.4 for $\mathbf{A}(\mathbf{x},t)$.

Solution.

Write $q = ze$, let ρ denote cylindrical radius, and assume a nonmagnetic medium. The retarded prescription is essential, especially above the Cherenkov threshold: take $\text{Im}\,\epsilon(\omega) > 0$ for $\omega > 0$ and then approach the lossless limit.

(a) The symmetric four-dimensional transform in (13.25) is

$$\Phi(\mathbf{k},\omega) = \frac{2q}{\epsilon(\omega)}\frac{\delta(\omega - vk_z)}{k_\perp^2 + k_z^2 - \epsilon(\omega)\omega^2/c^2}.$$

Performing the k_z integral in the three-dimensional inverse transform gives

$$\Phi(\omega,\mathbf{x}) = \frac{2q\,e^{i\omega z/v}}{(2\pi)^{3/2}v\epsilon}\int \frac{d^2k_\perp\, e^{i\mathbf{k}_\perp\cdot\boldsymbol{\rho}}}{k_\perp^2+\lambda^2}, \qquad \lambda^2 = \frac{\omega^2}{v^2}(1-\beta^2\epsilon).$$

The azimuthal integral is $2\pi J_0(k_\perp\rho)$. Initially take $\lambda > 0$ real, so that the auxiliary exponential integral converges. A useful evaluation of the radial integral is

$$\begin{aligned}\int_0^\infty \frac{kJ_0(k\rho)}{k^2+\lambda^2}\,dk &= \int_0^\infty e^{-s\lambda^2}ds\int_0^\infty kJ_0(k\rho)e^{-sk^2}\,dk \\ &= \frac{1}{2}\int_0^\infty \frac{ds}{s}e^{-s\lambda^2-\rho^2/(4s)} = K_0(\lambda\rho).\end{aligned}$$

Analytic continuation to $\operatorname{Re}\lambda > 0$ then implements the retarded prescription. The intermediate real-s representation requires $\operatorname{Re}(\lambda^2) > 0$; the continued Bessel identity has the wider domain. Combining the normalization factors gives

$$\boxed{\Phi(\omega,\mathbf{x}) = \frac{q}{v\epsilon(\omega)}\sqrt{\frac{2}{\pi}}\,K_0\!\left(\frac{|\omega|\rho}{v}\sqrt{1-\beta^2\epsilon(\omega)}\right)e^{i\omega z/v}.}$$

The square-root branch is continued from positive real part, and the negative-frequency potential is the complex conjugate of the positive-frequency potential. Equation (13.25) also gives, at each frequency, $\mathbf{A} = \beta\epsilon\Phi\,\hat{\mathbf{z}}$.

(b) Now let ϵ be a real positive constant and $\beta^2\epsilon < 1$. Define

$$\zeta = vt - z, \qquad \Gamma = (1-\beta^2\epsilon)^{-1/2}, \qquad a = \frac{\rho}{\Gamma v}, \qquad \tau = \frac{\zeta}{v}.$$

Pairing positive and negative frequencies gives

$$\Phi(\mathbf{x},t) = \frac{2q}{\pi v\epsilon}\int_0^\infty K_0(a\omega)\cos(\tau\omega)\,d\omega.$$

For clarity, using $K_0(a\omega) = \int_0^\infty e^{-a\omega\cosh u}du$ evaluates the integral as

$$\begin{aligned}\int_0^\infty K_0(a\omega)\cos(\tau\omega)d\omega &= \int_0^\infty \frac{a\cosh u}{a^2\cosh^2 u+\tau^2}\,du \\ &= a\int_0^\infty \frac{ds}{a^2s^2+a^2+\tau^2} = \frac{\pi}{2\sqrt{a^2+\tau^2}},\end{aligned}$$

where $s = \sinh u$. Consequently

$$\boxed{\Phi(\mathbf{x},t) = \frac{q}{\epsilon\sqrt{\zeta^2+\rho^2/\Gamma^2}} = \frac{q\Gamma}{\epsilon\sqrt{\rho^2+\Gamma^2\zeta^2}}, \qquad A_z = \beta\epsilon\Phi.}$$

Set $D = \rho^2+\Gamma^2\zeta^2$. Direct differentiation yields

$$E_\rho = -\partial_\rho\Phi = \frac{q\Gamma\rho}{\epsilon D^{3/2}}.$$

For the longitudinal field, $\partial_z = -\partial_\zeta$ and $\partial_t = v\partial_\zeta$, so both potentials contribute:

$$\begin{aligned}E_z &= -\partial_z\Phi - \frac{1}{c}\partial_t A_z = (1-\beta^2\epsilon)\partial_\zeta\Phi \\ &= -\Gamma^{-2}\frac{q\Gamma^3\zeta}{\epsilon D^{3/2}} = -\frac{q\Gamma\zeta}{\epsilon D^{3/2}}.\end{aligned}$$

Finally,

$$\boxed{\mathbf{E} = \frac{q\Gamma}{\epsilon[\rho^2 + \Gamma^2(z-vt)^2]^{3/2}}[\rho\,\hat{\boldsymbol{\rho}} + (z-vt)\hat{\mathbf{z}}], \qquad \mathbf{B} = \beta\epsilon E_\rho\,\hat{\boldsymbol{\phi}}.}$$

For $\epsilon = 1$, $\Gamma = \gamma$, and these are precisely the uniform-motion vacuum fields (11.152). In a medium the electric field also carries a factor $1/\epsilon$, and the magnetic-to-transverse-electric ratio becomes $\beta\epsilon$.

(c) Let $\kappa = \sqrt{\beta^2\epsilon - 1}$ and now put $a = \kappa\rho/v$. The retarded continuation gives $\lambda\rho = -ia\omega$ for $\omega > 0$. The Bessel continuation

$$K_0(-ix) = \frac{\pi}{2}[-N_0(x) + iJ_0(x)], \qquad x > 0,$$

and its complex conjugate for $\omega < 0$ give

$$\boxed{\Phi(\omega, \mathbf{x}) = \frac{q}{v\epsilon}\sqrt{\frac{\pi}{2}}\,e^{i\omega z/v}[-N_0(a|\omega|) + i\ \mathrm{sgn}(\omega)J_0(a|\omega|)].}$$

Here N_0 is Jackson's Neumann function. Pairing the two frequency halves now yields

$$\Phi(\mathbf{x}, t) = \frac{q}{v\epsilon}\int_0^\infty [-N_0(a\omega)\cos(\tau\omega) + J_0(a\omega)\sin(\tau\omega)]\,d\omega.$$

These are retarded, regulated integrals. Their values follow by setting $s = 0^+ - i\tau$ in the Laplace transforms

$$\int_0^\infty e^{-s\omega}J_0(a\omega)d\omega = \frac{1}{\sqrt{a^2+s^2}}, \qquad \int_0^\infty e^{-s\omega}N_0(a\omega)d\omega = -\frac{2}{\pi}\frac{\mathrm{arsinh}(s/a)}{\sqrt{a^2+s^2}}.$$

Specifically,

$$\int_0^\infty J_0(a\omega)\sin(\tau\omega)d\omega = \frac{\mathrm{sgn}\,\tau\,H(|\tau|-a)}{\sqrt{\tau^2-a^2}}, \qquad \int_0^\infty N_0(a\omega)\cos(\tau\omega)d\omega = -\frac{H(|\tau|-a)}{\sqrt{\tau^2-a^2}}.$$

They cancel for $\tau < -a$, both have zero relevant part for $|\tau| < a$, and add for $\tau > a$. Thus

$$\boxed{\Phi(\mathbf{x}, t) = \frac{2q}{\epsilon}\frac{H(\zeta - \kappa\rho)}{\sqrt{\zeta^2 - \kappa^2\rho^2}}, \qquad \mathbf{A} = \frac{2q\beta\,H(\zeta-\kappa\rho)}{\sqrt{\zeta^2-\kappa^2\rho^2}}\hat{\mathbf{z}}.}$$

The vector potential agrees with (13.51), after rotating Jackson's direction of motion from x to z. It is nonzero only inside the wake behind the particle. The factor of two reflects the two retarded source positions contributing inside a superluminal cone.

For points strictly inside the cone, put $S = \zeta^2 - \kappa^2\rho^2$. The regular parts of the fields are

$$\boxed{E_\rho = -\frac{2q\kappa^2\rho}{\epsilon S^{3/2}}, \qquad E_z = \frac{2q\kappa^2\zeta}{\epsilon S^{3/2}}, \qquad B_\phi = \beta\epsilon E_\rho.}$$

They vanish strictly outside the cone. At the front one must differentiate the whole regulated potential, including its support, before removing the regulator; the divergent front is not represented by these interior formulas alone. The assumption of a frequency-independent dielectric constant creates this singularity, as discussed immediately after (13.51). Dispersion and absorption regularize a physical front.

13.11 A magnetic monopole with magnetic charge g passes through matter and loses energy by collisions with electrons, just as does a particle with electric charge ze.

(a) In the same approximation as presented in Section 13.1, show that the energy loss per unit distance is given approximately by (13.14), but with $ze \to \beta g$ and with only the leading logarithm retained, yielding

$$\left(\frac{dE}{dx}\right)_{\substack{\text{magnetic}\\ \text{monopole}}} \simeq 4\pi N Z \frac{g^2 e^2}{mc^2} \ln\left(\frac{2\gamma^2 m v^2}{\hbar\langle\omega\rangle}\right)$$

(b) With the Dirac quantization condition (6.153) determining the magnetic charge, what z value is necessary for an ordinary charged particle in order that it lose energy at relativistic speeds at the same rate as a monopole? Sketch for the magnetic monopole a curve of dE/dx equivalent to Fig. 13.1 and comment on the differences.

Solution.

(a) In Gaussian units the static monopole field is $\mathbf{B}' = g\mathbf{r}'/r'^3$. A boost along its path gives

$$B_\perp(t) = \frac{g\gamma b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}, \qquad \mathbf{E} = -\boldsymbol{\beta} \times \mathbf{B}.$$

An initially slow electron is accelerated by this transverse electric field. Its magnitude is exactly the electric-charge expression of Problem 13.1(b) with ze replaced by βg, although its direction is rotated by 90° around the trajectory. Thus

$$|\Delta p| = \frac{2e\beta g}{bv} = \frac{2eg}{bc}, \qquad \Delta E(b) = \frac{2e^2 g^2}{mc^2 b^2}.$$

For electron density $n_e = NZ$, integrating over impact parameters gives

$$S_g = 2\pi n_e \int b\,\Delta E(b)\,db = \frac{4\pi N Z g^2 e^2}{mc^2} \ln\frac{b_{\max}}{b_{\min}}.$$

The bound-electron and quantum matching used in Section 13.1 sets the logarithm to $\ln B_q$, with $B_q = 2\gamma^2 m v^2/(\hbar\langle\omega\rangle)$, to the accuracy retained here. Therefore

$$\boxed{S_g \simeq \frac{4\pi N Z g^2 e^2}{mc^2} \ln\frac{2\gamma^2 m v^2}{\hbar\langle\omega\rangle}.}$$

There is no $1/\beta^2$ prefactor: the extra β in the induced electric field cancels it.

The displayed result is a *leading-logarithm* formula. A literal algebraic substitution in the full electric-charge (13.14) would leave $\ln B_q - \beta^2$ rather than $\ln B_q$. The omitted nonlogarithmic term, and monopole-specific corrections beyond this impulse treatment, cannot be inferred from the replacement rule. The marked qualification in the statement makes the approximation consistent.

(b) Converting the Dirac condition (6.153) to the Gaussian normalization used here gives

$$eg = \frac{n\hbar c}{2}, \qquad \frac{g}{e} = \frac{n}{2\alpha}.$$

Since $\beta \simeq 1$ at relativistic speed, the equivalent ordinary charge is

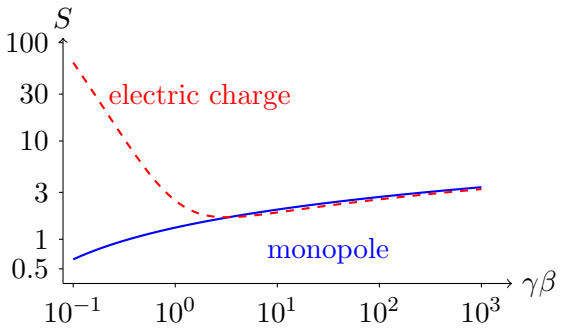
$$\boxed{z_{\text{equivalent}} \simeq \frac{|g|}{e} = 68.518\,|n|.}$$

For the minimum monopole, the stopping power at such speeds is therefore roughly $(68.518)^2 \simeq 4695$ times that of a singly charged particle, at the same leading-logarithm accuracy.

To compare shapes with Fig. 13.1, put $u = \gamma\beta$ and use the figure's $I = 160$ eV. Divide out the squared effective charge:

$$S_{\text{plot,g}} = 0.15 \ln \frac{2mc^2u^2}{I}, \qquad S_{\text{plot,e}} = \frac{0.15}{\beta^2}\left[\ln \frac{2mc^2u^2}{I} - \beta^2\right], \qquad \beta^2 = \frac{u^2}{1+u^2}.$$

The resulting sketch uses logarithmic axes:



Its ordinate S is the same approximate aluminum scale as Fig. 13.1, after dividing the physical stopping power by $(g/e)^2$ for the monopole or by z^2 for the electric particle. The monopole curve rises logarithmically with u, whereas the electric curve has a large low-speed rise and a minimum. Both rise logarithmically at large u. In particular, the monopole lacks the electric particle's $1/\beta^2$ increase near the end of its range. Neither this high-speed approximation nor a negative extrapolation of its logarithm describes the actual stopping law arbitrarily close to rest.

13.12 A relativistic particle of charge ze moves along the z axis with a constant speed βc. The half-space $z \le 0$ is filled with a uniform isotropic dielectric medium with plasma frequency ω_1, and the space $z > 0$ with a similar medium whose plasma frequency is ω_2. Discuss the emission of transition radiation as the particle traverses the interface, using the approximation of Section 13.7.

(a) Show that the radiation intensity per unit circular frequency interval and per unit solid angle is given approximately by

$$\frac{d^2I}{d\omega\, d\Omega} \simeq \frac{z^2e^2\theta^2}{\pi^2 c}\left|\frac{1}{\dfrac{1}{\gamma^2}+\dfrac{\omega_1^2}{\omega^2}+\theta^2} - \frac{1}{\dfrac{1}{\gamma^2}+\dfrac{\omega_2^2}{\omega^2}+\theta^2}\right|^2$$

where θ is the angle of emission relative to the velocity of the particle and $\gamma = (1-\beta^2)^{-1/2}$.

(b) Show that the total energy radiated is

$$I \simeq \frac{z^2e^2}{3c} \cdot \frac{(\omega_1-\omega_2)^2}{(\omega_1+\omega_2)} \cdot \gamma$$

Solution.

(a) Adopt the assumptions of Section 13.7: $\gamma \gg 1$, $\theta \ll 1$, $\omega \gg \omega_j$, negligible absorption and reflection, and $\epsilon_j = 1 - \omega_j^2/\omega^2$. Define

$$A = \gamma^{-2} + \theta^2, \qquad q_j = \frac{\omega}{v} - k_j \cos\theta \simeq \frac{\omega}{2c}\left(A + \frac{\omega_j^2}{\omega^2}\right).$$

The polarization driving the radiation is proportional to $\omega_j^2 \mathbf{E}_i$, while its longitudinal phase is $e^{iq_j z}$. With convergence factors at infinity the two half-space integrals have opposite signs:

$$\int_{-\infty}^{0} e^{iq_1 z + 0^+ z} dz = \frac{1}{iq_1}, \qquad \int_{0}^{\infty} e^{iq_2 z - 0^+ z} dz = -\frac{1}{iq_2}.$$

The transverse integral and its normalization can be kept explicit. Write $q = ze$, $\ell = \omega/(\gamma v)$, $h = k \sin\theta$, and $C = \sqrt{2/\pi}\, q\omega/(\gamma v^2)$. The incident fields are $E_\rho = CK_1(\ell\rho)$ and $E_z = -iCK_0(\ell\rho)/\gamma$ after removing their common z phase. The two-dimensional transform and its derivative give

$$\int d^2\rho\, e^{-ihx} K_0(\ell\rho) = \frac{2\pi}{\ell^2 + h^2}, \qquad \int d^2\rho\, e^{-ihx} \cos\phi K_1(\ell\rho) = -\frac{2\pi i h}{\ell(\ell^2 + h^2)}.$$

For the second identity use $\cos\phi K_1(\ell\rho) = -\ell^{-1}\partial_x K_0(\ell\rho)$ and integrate by parts. Projection onto the radiation polarization in the observation plane gives

$$T = \cos\theta\, T_\rho - \sin\theta\, T_z = -\frac{2\pi i C}{\ell^2 + h^2}\left(\frac{h\cos\theta}{\ell} - \frac{\sin\theta}{\gamma}\right) \simeq -\frac{2i\sqrt{2\pi}\, q}{\omega}\frac{\theta}{A}.$$

Here $v \simeq c$, $h \simeq \omega\theta/c$, and the retained denominator is $\ell^2 + h^2 \simeq (\omega^2/c^2)A$. Generalizing (13.79) to the two media therefore gives

$$\frac{d^2 I}{d\omega\, d\Omega} = \frac{1}{32\pi^3 c^3}\left| T\left(\frac{\omega_1^2}{iq_1} - \frac{\omega_2^2}{iq_2}\right)\right|^2.$$

Inserting q_j leaves a common amplitude factor $4\sqrt{2\pi}\, qc$ multiplying

$$\begin{aligned}\frac{\theta}{A}\left[\frac{\omega_2^2}{A\omega^2 + \omega_2^2} - \frac{\omega_1^2}{A\omega^2 + \omega_1^2}\right] &= \frac{\theta}{A}\left[\left(1 - \frac{A\omega^2}{A\omega^2 + \omega_2^2}\right) - \left(1 - \frac{A\omega^2}{A\omega^2 + \omega_1^2}\right)\right] \\ &= \theta\left[\frac{1}{A + \omega_1^2/\omega^2} - \frac{1}{A + \omega_2^2/\omega^2}\right].\end{aligned}$$

Squaring and using $(4\sqrt{2\pi}\, qc)^2/(32\pi^3 c^3) = q^2/(\pi^2 c)$ gives

$$\boxed{\frac{d^2 I}{d\omega\, d\Omega} = \frac{z^2 e^2 \theta^2}{\pi^2 c}\left|\frac{1}{\gamma^{-2} + \theta^2 + \omega_1^2/\omega^2} - \frac{1}{\gamma^{-2} + \theta^2 + \omega_2^2/\omega^2}\right|^2.}$$

The two sides contribute coherently; adding their intensities separately would miss the cancellation when $\omega_1 = \omega_2$.

(b) Put $a = \omega_1$, $b = \omega_2$. For real, lossless media, the squared difference is

$$\left[\frac{1}{A + a^2/\omega^2} - \frac{1}{A + b^2/\omega^2}\right]^2 = \frac{(a^2 - b^2)^2 \omega^4}{(A\omega^2 + a^2)^2 (A\omega^2 + b^2)^2}.$$

We need its frequency integral. One convenient derivation starts with the partial-fraction integral

$$J(a, b) = \int_0^\infty \frac{s^2\, ds}{(s^2 + a^2)(s^2 + b^2)} = \frac{\pi}{2(a + b)}.$$

Indeed, its integrand is $[b^2/(s^2 + b^2) - a^2/(s^2 + a^2)]/(b^2 - a^2)$. Differentiating J gives

$$\begin{aligned}\int_0^\infty \frac{s^4\, ds}{(s^2 + a^2)^2 (s^2 + b^2)^2} &= -\frac{1}{2a}\partial_a J - \frac{b}{4a}\partial_a \partial_b J \\ &= \frac{\pi}{4a(a + b)^2} - \frac{\pi b}{4a(a + b)^3} = \frac{\pi}{4(a + b)^3}.\end{aligned}$$

The continuous limit covers $a = b$ or one vanishing frequency. Setting $s = \sqrt{A}\,\omega$ therefore gives

$$\int_0^\infty d\omega \left[\frac{1}{A + a^2/\omega^2} - \frac{1}{A + b^2/\omega^2}\right]^2 = \frac{\pi(a-b)^2}{4A^{5/2}(a+b)}.$$

Now $d\Omega \simeq 2\pi\theta\, d\theta$, so

$$I = \frac{z^2e^2}{2c}\frac{(a-b)^2}{a+b}\int_0^\infty \frac{\theta^3\, d\theta}{(\theta^2+\gamma^{-2})^{5/2}}.$$

To show the last step explicitly, set $u = \theta^2 + \gamma^{-2}$:

$$\begin{aligned}\int_0^\infty \frac{\theta^3\, d\theta}{(\theta^2+\gamma^{-2})^{5/2}} &= \frac{1}{2}\int_{\gamma^{-2}}^\infty (u - \gamma^{-2})u^{-5/2}\, du \\ &= \frac{1}{2}\left[-2u^{-1/2} + \frac{2}{3}\gamma^{-2}u^{-3/2}\right]_{\gamma^{-2}}^\infty = \frac{2\gamma}{3}.\end{aligned}$$

Hence

$$\boxed{I \simeq \frac{z^2e^2}{3c}\frac{(\omega_1-\omega_2)^2}{\omega_1+\omega_2}\,\gamma.}$$

Equal media give zero; setting one plasma frequency to zero recovers (13.87). Extending the high-frequency, small-angle approximation over the full integration range gives the leading term proportional to γ; low-frequency details of a real dielectric affect subleading terms.

13.13 Consider the transition radiation emitted by a relativistic particle traversing a dielectric foil of thickness a perpendicular to its path. Assuming that reflections can be ignored because

$$|[n(\omega) - 1]/[n(\omega) + 1]|$$

is very small, show that the differential angular and frequency spectrum is given by the single-interface result (13.84) times the factor,

$$\mathcal{F} = 4\sin^2\Theta, \qquad \text{with } \Theta = \nu\left(1 + \frac{1}{\nu^2} + \eta\right)\frac{a}{4D}$$

Here $D = \gamma c/\omega_p$ is the formation length, $\nu = \omega/\gamma\omega_p$, and $\eta = (\gamma\theta)^2$. Provided $a \gg D$, the factor $\mathcal{F}$ oscillates extremely rapidly in angle or frequency, averaging to $\langle\mathcal{F}\rangle = 2$. For such foils the smoothed intensity distribution is just twice that for a single interface. Frequency distributions for different values of $\Gamma = 2D/a$ are displayed in Fig. 1 of G. B. Yodh, X. Artru, and R. Ramaty, *Astrophys. J.* **181**, 725 (1973).

Solution.

Use the same forward-radiation and high-frequency approximation as in Section 13.7. At fixed frequency and observation angle, the polarization source at depth z carries the driving phase $e^{i\omega z/v}$ and the radiation propagator carries $e^{-ikz\cos\theta}$. Define

$$q_f = \frac{\omega}{v} - k\cos\theta.$$

The entire change from a semi-infinite medium to a foil is then in the longitudinal integral. A half-space, with a vanishing convergence factor, has integral i/q_f, whereas a foil from 0 to a has

$$\int_0^a e^{iq_f z}dz = \frac{e^{iq_f a} - 1}{iq_f} = \frac{2e^{iq_f a/2}}{q_f}\sin\frac{q_f a}{2}.$$

Their squared ratio is

$$\boxed{\mathcal{F} = |1 - e^{iq_f a}|^2 = 2 - 2\cos(q_f a) = 4\sin^2\frac{q_f a}{2}.}$$

This is also the interference of the two interface amplitudes. The entrance and exit discontinuities have opposite sign, and their phase difference is $q_f a$.

To express the phase in the requested variables, expand each factor:

$$\beta^{-1} = 1 + \frac{1}{2\gamma^2} + O(\gamma^{-4}), \qquad n = 1 - \frac{\omega_p^2}{2\omega^2} + O(\omega_p^4/\omega^4), \qquad \cos\theta = 1 - \frac{\theta^2}{2} + O(\theta^4).$$

Consequently,

$$q_f = \frac{\omega}{c}(\beta^{-1} - n\cos\theta) \simeq \frac{\omega}{2c}\left(\gamma^{-2} + \frac{\omega_p^2}{\omega^2} + \theta^2\right).$$

With $\nu = \omega/(\gamma\omega_p)$, $\eta = \gamma^2\theta^2$, and $D = \gamma c/\omega_p$,

$$\boxed{\Theta = \frac{q_f a}{2} = \frac{\nu a}{4D}(1 + \nu^{-2} + \eta), \qquad \mathcal{F} = 4\sin^2\Theta.}$$

For a very thin foil, $q_f a \ll 1$, the factor is $\mathcal{F} \simeq (q_f a)^2$ and the two interfaces almost cancel. For a thick foil the phase oscillates through many cycles within an ordinary frequency or angular bin. Phase averaging gives explicitly

$$\langle\mathcal{F}\rangle = \frac{1}{\pi}\int_{\Theta_0}^{\Theta_0+\pi} 4\sin^2\Theta\, d\Theta = 2.$$

Thus the smoothed thick-foil intensity is twice the single-interface intensity. The criterion is loss of phase resolution, not a pointwise limit of $\sin^2\Theta$ as $a \to \infty$.

13.14 Transition radiation is emitted by a relativistic particle traversing normally a uniform array of N dielectric foils, each of thickness a, separated by air gaps (effectively vacuum), each of length b. Assume that multiple reflections can be neglected for the whole stack. This requires

$$\left|\frac{n(\omega) - 1}{n(\omega) + 1}\right| \simeq \frac{\omega_p^2}{4\omega^2} \ll \frac{1}{N}$$

(a) Show that if the dielectric constant of the medium varies in the z direction as $\epsilon(\omega, z) = 1 - (\omega_p^2/\omega^2)\rho(z)$, the differential spectrum of transition radiation is given approximately by the single-interface result (13.84) times

$$\mathcal{F} = \left|\mu\int dz\ \rho(z)e^{i\omega z/v}\exp\left(-i\cos\theta\int^z k(z')\,dz'\right)\right|^2$$

where $\rho(0) = 1$ by convention, $\mu = \omega/v - k(0)\cos\theta$, and $k(z) = (\omega/c)\sqrt{\epsilon(\omega, z)}$.

(b) Show that for the stack of N foils

$$\mathcal{F} = 4\sin^2\Theta\,\frac{\sin^2[N(\Theta + \Psi)]}{\sin^2[\Theta + \Psi]}$$

where Θ is defined in Problem 13.13 and $\Psi = \nu(1 + \eta)(b/4D)$. Compare G. M. Garibyan, *Zh. Eksp. Teor. Fiz.* **60**, 39 (1970) [transl. *Sov. Phys. JETP* **33**, 23 (1971)].

The practical theory of multilayered transition radiation detectors is treated in great detail by X. Artru, G. B. Yodh, and G. Mennessier, *Phys. Rev. D* **12**, 1289 (1975).

Solution.

(a) In the weak-susceptibility approximation used in (13.76),

$$\mathbf{P}(\mathbf{x},\omega) = -\frac{\omega_p^2}{4\pi\omega^2}\rho(z)\,\mathbf{E}_i(\mathbf{x},\omega).$$

The exciting field has z dependence $e^{i\omega z/v}$. Neglecting reflected waves, propagation from a source at z adds the phase $\exp[-i\cos\theta\int^z k(z')dz']$. Corrections to the transverse radiation kernel from the small change of index are discarded in amplitude, while these accumulated phases are retained. Define

$$\varphi(z) = \frac{\omega z}{v} - \cos\theta\int^z k(z')\,dz', \qquad \varphi'(z) = \frac{\omega}{v} - k(z)\cos\theta.$$

The transverse integral in (13.79) is common to all z. The new longitudinal amplitude is therefore proportional to

$$L[\rho] = \int dz\,\rho(z)e^{i\varphi(z)}.$$

For the reference half-space with $\rho = 1$ and $k = k(0)$, the same integral is

$$L_{\text{ref}} = \lim_{\delta\downarrow 0}\int_0^\infty e^{i\mu z-\delta z}\,dz = \frac{i}{\mu}, \qquad \mu = \frac{\omega}{v} - k(0)\cos\theta.$$

Taking the ratio of squared amplitudes gives

$$\boxed{\mathcal{F} = \left|\mu\int dz\,\rho(z)e^{i\omega z/v}\exp\left[-i\cos\theta\int^z k(z')\,dz'\right]\right|^2.}$$

A change in the lower limit of the phase integral gives an overall phase and does not affect $\mathcal{F}$. The convention $\rho(0) = 1$ sets the reference single-interface normalization. This formula is an approximation for propagation in weakly refracting material; for strongly varying indices the transverse propagation and reflections cannot in general be represented by this phase alone.

(b) Number the foils $j = 0,\ldots,N-1$ and place the jth entrance at $z_j = j(a+b)$. Let

$$q_f = \frac{\omega}{v} - k_f\cos\theta, \qquad q_g = \frac{\omega}{v} - \frac{\omega}{c}\cos\theta, \qquad 2\Theta = q_f a, \quad 2\Psi = q_g b.$$

At $z = z_j + s$ inside foil j the accumulated phase is

$$\varphi(z) = j(q_f a + q_g b) + q_f s, \qquad 0 < s < a.$$

The gaps have $\rho = 0$ and contribute only to phase, not to the source integral. Since $\mu = q_f$, direct integration yields

$$\begin{aligned}\mu L[\rho] &= q_f\sum_{j=0}^{N-1} e^{ij(q_f a+q_g b)}\int_0^a e^{iq_f s}\,ds \\ &= \frac{e^{2i\Theta}-1}{i}\sum_{j=0}^{N-1} e^{2ij(\Theta+\Psi)}.\end{aligned}$$

The finite geometric series, with $\chi = \Theta + \Psi$, is

$$\sum_{j=0}^{N-1} e^{2ij\chi} = \frac{1-e^{2iN\chi}}{1-e^{2i\chi}} = e^{i(N-1)\chi}\frac{\sin(N\chi)}{\sin\chi}.$$

The squared modulus gives

$$\boxed{\mathcal{F} = 4\sin^2\Theta\,\frac{\sin^2[N(\Theta+\Psi)]}{\sin^2(\Theta+\Psi)}.}$$

The vacuum gap phase is, to the same order as in Problem 13.13,

$$q_g \simeq \frac{\omega}{2c}(\gamma^{-2}+\theta^2), \qquad \boxed{\Psi = \frac{\nu b}{4D}(1+\eta).}$$

At $\chi = \ell\pi$ the apparent $0/0$ is removable: the original series equals N, up to its common phase, and the array factor is N^2. At $N = 1$ this reduces to the foil factor. At $b = 0$ the product reduces to $4\sin^2(N\Theta)$, exactly the result for a single contiguous slab of thickness Na.

Garibyan's cited paper, printed p. 24, Eq. (5), has the same foil sine factor and squared array factor. Its Eq. (3) requires $|r|N \ll 1$, not merely a small reflection coefficient at one surface, in agreement with the condition in this problem.[9] The finite-N expression here retains the peak widths, of order $1/N$ in χ. Replacing these peaks by delta functions needs the remaining spectral and angular factors to vary little over those widths; no such additional approximation is needed for the displayed answer.

13.15

(a) Find the number N_γ of transition radiation quanta with frequencies greater than ω_p emitted per interface, starting from the energy spectrum (13.85). Show that for $\gamma \gg 1$,

$$N_\gamma = \frac{z^2e^2}{\pi\hbar c}\left[(\ln\gamma - 1)^2 + \frac{\pi^2}{12}\right]$$

where terms of order $\gamma^{-2}\ln\gamma$ have been neglected.

(b) Using the result from part a for the number of photons and the value $\hbar\omega_p = 20$ eV, find the mean energy of the photons (in keV) for $\gamma = 10^3$, 10^4, 10^5.

Solution.

(a) Write the checked source spectrum (13.85) as

$$\frac{dI}{d\nu} = \frac{z^2e^2\gamma\omega_p}{\pi c}f(\nu), \qquad f(\nu) = (1+2\nu^2)\ln(1+\nu^{-2}) - 2, \qquad \nu = \frac{\omega}{\gamma\omega_p}.$$

A photon has energy $\hbar\omega = \hbar\gamma\omega_p\nu$. The cutoff $\omega > \omega_p$ is therefore $\nu > 1/\gamma$, giving

$$N_\gamma = \frac{z^2\alpha}{\pi}\mathcal{N}(\gamma), \qquad \mathcal{N}(\gamma) = \int_{1/\gamma}^{\infty}\frac{f(\nu)}{\nu}\,d\nu.$$

We evaluate the integral rather than integrate only the low-frequency asymptote, which would lose the constant term. With $y = \nu^{-2}$, $d\nu/\nu = -dy/(2y)$, so

$$\mathcal{N}(\gamma) = \frac{1}{2}\int_0^{\gamma^2}\left[\frac{\ln(1+y)}{y} + \frac{2\ln(1+y)}{y^2} - \frac{2}{y}\right]dy.$$

[9] G. M. Garibyan, *Sov. Phys. JETP* **33**, 23–29 (1971), publisher PDF, p. 24, Eqs. (3)–(5).

The combination of the last two terms has a finite lower limit. Using

$$\frac{d}{dy}\,\mathrm{Li}_2(-y) = -\frac{\ln(1+y)}{y}, \qquad \frac{d}{dy}\left[\frac{\ln(1+y)}{y}\right] = \frac{1}{y(1+y)} - \frac{\ln(1+y)}{y^2},$$

an antiderivative of half the displayed integrand is

$$-\frac{1}{2}\,\mathrm{Li}_2(-y) - \frac{\ln(1+y)}{y} - \ln(1+y).$$

Its value tends to -1 as $y \to 0$. Thus the exact result within the approximate spectrum is

$$\mathcal{N}(\gamma) = 1 - \frac{1}{2}\,\mathrm{Li}_2(-\gamma^2) - (1+\gamma^{-2})\ln(1+\gamma^2).$$

To expand it at large γ, use the real dilogarithm identity

$$\mathrm{Li}_2(-x) + \mathrm{Li}_2(-1/x) = -\frac{\pi^2}{6} - \frac{1}{2}\ln^2 x, \qquad x > 0.$$

This follows by differentiation; the integration constant is fixed at $x = 1$ by $\mathrm{Li}_2(-1) = -\pi^2/12$. With $r = \gamma^{-2}$, it gives

$$\begin{aligned}\mathcal{N}(\gamma) &= (\ln\gamma - 1)^2 + \frac{\pi^2}{12} + \frac{1}{2}\,\mathrm{Li}_2(-r) - 2r\ln\gamma - (1+r)\ln(1+r) \\ &= (\ln\gamma - 1)^2 + \frac{\pi^2}{12} - \frac{2\ln\gamma + 3/2}{\gamma^2} + O(\gamma^{-4}).\end{aligned}$$

Therefore

$$\boxed{N_\gamma \simeq \frac{z^2e^2}{\pi\hbar c}\left[(\ln\gamma - 1)^2 + \frac{\pi^2}{12}\right].}$$

The marked remainder in the question records the logarithm accompanying the first omitted inverse-square term.

(b) Equation (13.87) gives $I_{\mathrm{all}} = z^2\alpha\hbar\omega_p\gamma/3$. If the energy below ω_p is neglected consistently at leading order in $1/\gamma$, division by part (a) gives

$$\boxed{\langle\hbar\omega\rangle_{\mathrm{leading}} = \frac{\pi\gamma\hbar\omega_p}{3[(\ln\gamma - 1)^2 + \pi^2/12]}.}$$

For $\hbar\omega_p = 0.020$ keV the numerical values are

γ	10^3	10^4	10^5
N_γ/z^2	0.0829805	0.158491	0.258632
$\langle\hbar\omega\rangle_{\mathrm{leading}}$ (keV)	0.58627	3.06951	18.81012

There is a small distinction between this conventional use of the total energy and the mean energy of precisely the photons counted in part (a). For the latter the numerator must have the same cutoff. To quantify it, integration by parts gives

$$J_0(u) = \int_0^u f(\nu)d\nu = \left(u + \frac{2u^3}{3}\right)\ln(1+u^{-2}) - \frac{2u}{3} + \frac{2}{3}\arctan u.$$

Indeed the logarithmic derivative is $-2/[u(1+u^2)]$, and the remaining rational integral is $2\int(1+2u^2/3)/(1+u^2)\,du - 2u$; it gives the last two terms. The limit $J_0(\infty) = \pi/3$ also checks (13.87). Therefore

$$\langle\hbar\omega\rangle_{\omega>\omega_p} = \gamma\hbar\omega_p\,\frac{\pi/3 - J_0(1/\gamma)}{\mathcal{N}(\gamma)} = \begin{cases} 0.57854\text{ keV}, & \gamma = 10^3, \\ 3.06411\text{ keV}, & \gamma = 10^4, \\ 18.80598\text{ keV}, & \gamma = 10^5. \end{cases}$$

Both tables use the supplied high-frequency model down to its nominal cutoff. The second values are the internally consistent cutoff means; the first are the leading large-γ answer obtained directly from (13.87). Their difference is of relative order $\ln\gamma/\gamma$, so it is larger than the inverse-square correction to photon number alone.

13.16 A highly relativistic neutral particle of mass m possessing a magnetic moment μ parallel to its direction of motion emits transition radiation as it crosses at right angles a plane interface from vacuum into a dielectric medium characterized at high frequencies by a plasma frequency ω_p. The magnetic moment μ is defined in the particle's rest frame. [The particle could be a neutron or, of more potential interest, a neutrino with a small mass.]

(a) Show that the intensity of transition radiation is given by (13.79), provided the electric field of the incident particle $\mathbf{E}_i$, is given by $(\beta\mu/\gamma ze)$ times the partial derivative in the z direction of E_ρ in (13.80). Note that the electric field actually points azimuthally, but in the leading ultrarelativistic, small-angle approximation this affects only the polarization of the radiation, not its intensity.

(b) Show that in the combined limit of $\gamma \gg 1$ and $\omega \gg \omega_p$, the intensity distributions in angle and frequency are given by (13.84) and (13.85), each multiplied by $(\mu\omega/ze\gamma c)^2$.

(c) By expressing μ in units of the Bohr magneton $\mu_\mathrm{B} = e\hbar/2m_ec$ and the plasma frequency in atomic units ($\hbar\omega_0 = e^2/a_0 = 27.2$ eV), show that the ratio of frequency distributions of transition radiation emitted by the magnetic moment to that emitted by an electron with the same speed is

$$\frac{dI_\mu(\nu)}{dI_e(\nu)} = \frac{\alpha^4}{4}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2\left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2\cdot\nu^2$$

where $\alpha = 1/137$ is the fine structure constant and $\nu = \omega/\gamma\omega_p$ is the dimensionless frequency variable.

(d) Calculate the total energy of transition radiation, imposing conservation of energy, that is, $\nu \le \nu_\mathrm{max} = mc^2/\hbar\omega_p$. [This constraint will give only a crude estimate of the energy in the quantum regime where $\nu_\mathrm{max} < 1$ because the derivation is classical throughout.] Show that the ratio of total energies for the magnetic moment and an electron of the same speed can be written as

$$\frac{I_\mu}{I_e} = \frac{\alpha^4}{20}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2\left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2\cdot G(\nu_\mathrm{max})$$

where $G \approx 1$ for $\nu_\mathrm{max} \gg 1$ and $G \approx (10\nu_\mathrm{max}^3/\pi)\cdot[\ln(1/\nu_\mathrm{max}) - 2/3]$ for $\nu_\mathrm{max} \ll 1$. For fixed particle energy and magnetic moment, how does the actual amount of radiated energy vary with the particle's mass for very small mass?

Hint: the integrals of Section 2.7 of Gradshteyn and Ryzhik may be of use, although integration by parts is effective.

Solution.

(a) In the particle rest frame, with $\boldsymbol{\mu} = \mu\hat{\mathbf{z}}$,

$$\mathbf{E}' = 0, \qquad \mathbf{B}' = \frac{3\mathbf{r}'(\boldsymbol{\mu}\cdot\mathbf{r}')}{r'^5} - \frac{\boldsymbol{\mu}}{r'^3}.$$

The component required for a boost parallel to the moment is $B'_\rho = 3\mu\rho z'/(\rho^2 + z'^2)^{5/2}$. At a fixed laboratory event $z' = \gamma(z - vt)$ and $\rho' = \rho$. The field transformation gives $\mathbf{E}_\perp = -\gamma\boldsymbol{\beta}\times\mathbf{B}'_\perp$, hence

$$E_{\phi,\mu}(\rho, z, t) = -\frac{3\beta\gamma^2\mu\rho(z - vt)}{[\rho^2 + \gamma^2(z - vt)^2]^{5/2}}.$$

For comparison, the radial field of charge $q = ze$ at the same speed is

$$E_{\rho,q} = \frac{q\gamma\rho}{[\rho^2 + \gamma^2(z - vt)^2]^{3/2}}, \qquad \partial_z E_{\rho,q} = -\frac{3q\gamma^3\rho(z - vt)}{[\rho^2 + \gamma^2(z - vt)^2]^{5/2}}.$$

Thus the asserted relation, including its sign, is

$$\boxed{E_{\phi,\mu} = \frac{\beta\mu}{\gamma ze}\,\partial_z E_{\rho,q}.}$$

The induced polarization and its radiation are still calculated by (13.76)–(13.79), now with this azimuthal incident electric field. Axisymmetry makes the radiation polarized perpendicular to the plane containing the trajectory and observer. Charge radiation is polarized in that plane. Equality of their angular shapes after a scalar multiplication is the leading ultrarelativistic, small-angle result, as established next.

(b) A Fourier component of either driving field contains $e^{i\omega z/v}$. Therefore

$$E_{\phi,\mu}(\omega) = \frac{\beta\mu}{\gamma ze}\frac{i\omega}{v}E_{\rho,q}(\omega) = \frac{i\mu\omega}{ze\gamma c}E_{\rho,q}(\omega).$$

The common longitudinal radiation integral is unchanged. To check the polarization projection explicitly, observe in the x–z plane, put $h = k\sin\theta$ and $\ell = \omega/(\gamma v)$, and write

$$E_{\rho,q} = CK_1(\ell\rho)e^{i\omega z/v}, \qquad E_{z,q} = -\frac{iC}{\gamma}K_0(\ell\rho)e^{i\omega z/v}, \quad C = \frac{ze\omega}{\gamma v^2}\sqrt{\frac{2}{\pi}}.$$

After removing the common z phase, the transverse integrals are

$$T_\rho = -2\pi iC\int_0^\infty \rho K_1(\ell\rho)J_1(h\rho)d\rho = -\frac{2\pi iCh}{\ell(\ell^2 + h^2)},$$

$$T_z = -\frac{2\pi iC}{\gamma}\int_0^\infty \rho K_0(\ell\rho)J_0(h\rho)d\rho = -\frac{2\pi iC}{\gamma(\ell^2 + h^2)}.$$

The second Bessel integral is the radial transform used in (13.81); the first follows from integrating $K_1(\ell\rho) = -\ell^{-1}\partial_\rho K_0(\ell\rho)$ by parts. The charge's transverse radiation component is

$$T_q = \cos\theta\, T_\rho - \sin\theta\, T_z = T_\rho\left(\cos\theta - \frac{1}{\gamma^2\beta n}\right).$$

For the azimuthal field the other polarization has component $T_\mu = i\mu\omega T_\rho/(ze\gamma c)$. In the limit $\gamma \gg 1$, $\theta \ll 1$, $\omega \gg \omega_p$, the parenthesis in T_q tends to one. Hence the amplitudes differ only by $i\mu\omega/(ze\gamma c)$ to the retained order, and

$$\boxed{\left(\frac{d^2I}{d\omega\, d\Omega}\right)_\mu = \left(\frac{\mu\omega}{ze\gamma c}\right)^2 \left(\frac{d^2I}{d\omega\, d\Omega}\right)_q.}$$

The multiplier does not depend on angle, so the same factor multiplies the frequency spectrum (13.85). At finite angle or finite γ the polarization projection above explains the higher-order difference; no exact equality outside the stated approximation is needed.

(c) Set $z = 1$ for comparison with an electron and use $\omega = \gamma\omega_p\nu$. Then

$$\left(\frac{\mu\omega}{e\gamma c}\right)^2 = \left(\frac{\mu}{\mu_\mathrm{B}}\right)^2 \left(\frac{\hbar\omega_p}{2m_ec^2}\right)^2 \nu^2.$$

The atomic relations $a_0 = \hbar/(\alpha m_e c)$ and $e^2 = \alpha\hbar c$ give

$$\hbar\omega_0 = \frac{e^2}{a_0} = \alpha^2 m_e c^2.$$

Thus

$$\boxed{\frac{dI_\mu}{dI_e} = \frac{\alpha^4}{4}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2 \left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2 \nu^2.}$$

The incident neutral-particle mass is m; the electron mass m_e appearing in μ_B and atomic units is a separate quantity.

(d) A photon cannot carry more than the particle energy. For $\gamma \gg 1$, $\hbar\omega \le \gamma mc^2$ gives $u \equiv \nu_\mathrm{max} = mc^2/(\hbar\omega_p)$ as in the question. Using the function $f(\nu)$ defined in Problem 13.15, the cutoff energy is

$$I_\mu(u) = \frac{\mu^2\gamma\omega_p^3}{\pi c^3} J_2(u), \qquad J_2(u) = \int_0^u \nu^2 f(\nu)\, d\nu.$$

Integrate the logarithmic term by parts with

$$P(\nu) = \frac{\nu^3}{3} + \frac{2\nu^5}{5}, \qquad \frac{d}{d\nu}\ln(1+\nu^{-2}) = -\frac{2}{\nu(1+\nu^2)}.$$

This gives

$$J_2(u) = P(u)\ln(1+u^{-2}) + 2\int_0^u \frac{\nu^2/3 + 2\nu^4/5}{1+\nu^2}\, d\nu - \frac{2u^3}{3}.$$

The rational function has the explicit division

$$\frac{\nu^2/3 + 2\nu^4/5}{1+\nu^2} = \frac{2\nu^2}{5} - \frac{1}{15} + \frac{1}{15(1+\nu^2)}.$$

Consequently

$$\boxed{J_2(u) = \left(\frac{u^3}{3} + \frac{2u^5}{5}\right)\ln(1+u^{-2}) - \frac{2u^3}{5} - \frac{2u}{15} + \frac{2}{15}\arctan u.}$$

Its lower-limit value is zero. At large u the polynomial terms cancel against the first terms of the logarithm and $J_2(\infty) = \pi/15$.

For an electron of the same speed, (13.87) gives $I_e = e^2\gamma\omega_p/(3c)$. Division yields

$$\frac{I_\mu(u)}{I_e} = \frac{3\mu^2\omega_p^2}{\pi e^2 c^2} J_2(u) = \frac{\mu^2\omega_p^2}{5e^2c^2}\, G(u), \qquad \boxed{G(u) = \frac{15}{\pi} J_2(u).}$$

Using the conversion in part (c),

$$\boxed{\frac{I_\mu}{I_e} = \frac{\alpha^4}{20}\left(\frac{\mu}{\mu_{\rm B}}\right)^2 \left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2 G(u).}$$

The reference electron energy here is the untruncated classical result (13.87), consistent with the question's normalization.

For $u \gg 1$, $f(u) = 1/(6u^4) + O(u^{-6})$, so

$$G(u) = 1 - \frac{15}{\pi}\int_u^\infty \nu^2 f(\nu) d\nu = 1 - \frac{5}{2\pi u} + O(u^{-3}) \longrightarrow 1.$$

For $u \ll 1$, $f(\nu) = 2\ln(1/\nu) - 2 + O(\nu^2 \ln(1/\nu))$. Direct integration gives

$$\begin{aligned} J_2(u) &= 2\left[\frac{\nu^3}{3}\ln\frac{1}{\nu} + \frac{\nu^3}{9}\right]_0^u - \frac{2u^3}{3} + O(u^5\ln(1/u)) \\ &= \frac{2u^3}{3}\left[\ln\frac{1}{u} - \frac{2}{3}\right] + O(u^5\ln(1/u)), \end{aligned}$$

and hence

$$\boxed{G(u) \simeq \frac{10u^3}{\pi}\left[\ln\frac{1}{u} - \frac{2}{3}\right] \quad (u \ll 1).}$$

Finally, hold the total particle energy $\mathcal{E} = \gamma mc^2$ and μ fixed. Then $\gamma \propto m^{-1}$, $u \propto m$, and

$$I_\mu \;\propto\; \gamma J_2(u) \;\propto\; m^2\left[\ln\frac{\hbar\omega_p}{mc^2} - \frac{2}{3}\right] \longrightarrow 0 \qquad (m \to 0).$$

The growth of γ therefore does not produce a divergence once the allowed photon energy is imposed. As stipulated in the question, this mass dependence is only the classical spectrum with a kinematic cutoff; quantum recoil and the radiation matrix element require a quantum calculation when $u < 1$.

14 Chapter 14. Radiation by Moving Charges

Problems

14.1 Verify by explicit calculation that the Liénard–Wiechert expressions for *all* components of **E** and **B** for a particle moving with constant velocity agree with the ones obtained in the text by means of a Lorentz transformation. Follow the general method at the end of Section 14.1.

Solution.

We use Gaussian units throughout this chapter. Let the observation time be t, let $\mathbf{r} = \mathbf{x} - \mathbf{v}t$ point from the *present* position of the charge, and write $\mathbf{R} = R\mathbf{n}$ for the vector from its retarded position. Uniform motion and $t - t' = R/c$ imply

$$\mathbf{R} = \mathbf{r} + \boldsymbol{\beta} R, \qquad \mathbf{r} = R(\mathbf{n} - \boldsymbol{\beta}).$$

Squaring the first equation gives $(1-\beta^2)R^2 - 2(\boldsymbol{\beta}\cdot\mathbf{r})R - r^2 = 0$. The positive retarded-distance root is

$$R = \frac{\boldsymbol{\beta}\cdot\mathbf{r} + s}{1-\beta^2}, \quad s = \sqrt{(\boldsymbol{\beta}\cdot\mathbf{r})^2 + (1-\beta^2)r^2} = \sqrt{r^2 - |\boldsymbol{\beta}\times\mathbf{r}|^2}.$$

Taking the scalar product of $\mathbf{R} = \mathbf{r} + \boldsymbol{\beta}R$ with $\boldsymbol{\beta}$ shows $R(1 - \mathbf{n}\cdot\boldsymbol{\beta}) = (1-\beta^2)R - \boldsymbol{\beta}\cdot\mathbf{r} = s$. Since the acceleration term is zero, the Liénard–Wiechert fields reduce to

$$\boxed{\mathbf{E} = \frac{e(1-\beta^2)\mathbf{r}}{s^3}, \qquad \mathbf{B} = \mathbf{n}\times\mathbf{E} = \boldsymbol{\beta}\times\mathbf{E}.}$$

The second equality uses $\mathbf{n} = \boldsymbol{\beta} + \mathbf{r}/R$ and $\mathbf{E} \parallel \mathbf{r}$.

For an explicit component comparison, take $\mathbf{v} = v\hat{\mathbf{z}}$, $z_* = z - vt$, and $D_* = [x^2 + y^2 + \gamma^2 z_*^2]^{3/2}$. Then

$$(E_x, E_y, E_z) = \frac{e\gamma}{D_*}(x, y, z_*), \qquad (B_x, B_y, B_z) = \frac{e\gamma\beta}{D_*}(-y, x, 0).$$

In the rest frame $x' = x$, $y' = y$, $z' = \gamma z_*$, $r'^2 = x^2 + y^2 + \gamma^2 z_*^2$, and $\mathbf{E}' = e(x', y', z')/r'^3$, $\mathbf{B}' = 0$. The transformations $E_z = E'_z$, $\mathbf{E}_\perp = \gamma\mathbf{E}'_\perp$, and $\mathbf{B} = \boldsymbol{\beta}\times\mathbf{E}$ reproduce every component above, including both vanishing longitudinal magnetic and transverse special cases.

14.2 A particle of charge e is moving in nearly uniform nonrelativistic motion. For times near $t = t_0$, its vectorial position can be expanded in a Taylor series with fixed vector coefficients multiplying powers of $(t - t_0)$.

(a) Show that, in an inertial frame where the particle is instantaneously at rest at the origin but has a small acceleration $\mathbf{a}$, the Liénard–Wiechert electric field, correct to order $1/c^2$ inclusive, at that instant is $\mathbf{E} = \mathbf{E}_v + \mathbf{E}_a$, where the velocity and acceleration fields are

$$\mathbf{E}_v = e\frac{\hat{\mathbf{r}}}{r^2} + \frac{e}{2c^2 r}\left[\mathbf{a} - 3\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{a})\right]; \qquad \mathbf{E}_a = -\frac{e}{c^2 r}\left[\mathbf{a} - \hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{a})\right]$$

and that the total electric field to this order is

$$\mathbf{E} = e\frac{\hat{\mathbf{r}}}{r^2} - \frac{e}{2c^2 r}\left[\mathbf{a} + \hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{a})\right]$$

The unit vector $\hat{\mathbf{r}}$ points from the origin to the observation point and r is the magnitude of the distance. Comment on the r dependences of the velocity and acceleration fields. Where is the expansion likely to be valid?

(b) What is the result for the instantaneous magnetic induction $\mathbf{B}$ to the same order? Comment.

(c) Show that the $1/c^2$ term in the electric field has zero divergence and that the curl of the electric field is $\nabla \times \mathbf{E} = e(\hat{\mathbf{r}} \times \mathbf{a})/c^2 r^2$. From Faraday's law, find the magnetic induction $\mathbf{B}$ at times near $t = 0$. Compare with the familiar elementary expression.

Solution.

Set the instant in question to $t = 0$, so $\mathbf{r}_q(t') = \frac{1}{2}\mathbf{a}t'^2 + O(t'^3)$ and $\boldsymbol{\beta}(t') = \mathbf{a}t'/c + O(t'^2/c)$. All expansions below hold at fixed r as c becomes large, retaining terms through c^{-2}.

(a) The retarded equation gives $t' = -r/c + O(c^{-3})$. Consequently

$$\mathbf{r}_q(t') = \frac{r^2\mathbf{a}}{2c^2} + O(c^{-3}), \quad \boldsymbol{\beta}_{\rm ret} = -\frac{r\mathbf{a}}{c^2} + O(c^{-3}),$$

$$R = r - \frac{r^2\hat{\mathbf{r}}\cdot\mathbf{a}}{2c^2} + O(c^{-3}), \quad \mathbf{n} = \hat{\mathbf{r}} - \frac{r}{2c^2}[\mathbf{a} - \hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{a})] + O(c^{-3}).$$

Put $A = \hat{\mathbf{r}}\cdot\mathbf{a}$ and $\kappa = 1 - \mathbf{n}\cdot\boldsymbol{\beta}$. The factors in the velocity field become

$$\kappa^{-3} = 1 - 3rA/c^2 + O(c^{-3}), \quad R^{-2} = r^{-2}(1 + rA/c^2) + O(c^{-3}), \quad \mathbf{n} - \boldsymbol{\beta} = \hat{\mathbf{r}} + \frac{r}{2c^2}(\mathbf{a} + A\hat{\mathbf{r}}) + O(c^{-3}).$$

Also $1 - \beta^2 = 1 + O(c^{-4})$. Multiplication gives

$$\mathbf{E}_v = \frac{e}{r^2}\left[\hat{\mathbf{r}} + \frac{r}{2c^2}(\mathbf{a} + A\hat{\mathbf{r}}) - \frac{2rA}{c^2}\hat{\mathbf{r}}\right] = \frac{e\hat{\mathbf{r}}}{r^2} + \frac{e}{2c^2 r}(\mathbf{a} - 3A\hat{\mathbf{r}}).$$

In the acceleration term it suffices at this order to set $R = r$, $\mathbf{n} = \hat{\mathbf{r}}$, $\boldsymbol{\beta} = 0$:

$$\mathbf{E}_a = \frac{e}{c^2 r}\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{a}) = -\frac{e}{c^2 r}(\mathbf{a} - A\hat{\mathbf{r}}).$$

Adding proves $\mathbf{E} = e\hat{\mathbf{r}}/r^2 - e(\mathbf{a} + A\hat{\mathbf{r}})/(2c^2 r)$. The exact velocity and acceleration fields scale as R^{-2} and R^{-1} when written in retarded variables. Expanding the retarded position and velocity about the present instant introduces r^{-1} terms into the former too. This is a local slow-motion expansion, requiring $ar/c^2 \ll 1$ and higher terms such as $|\dot{\mathbf{a}}|r^2/c^3 \ll 1$; it cannot be extended without restriction to the distant radiation zone.

(b) Use the retarded $\mathbf{n}$, not $\hat{\mathbf{r}}$, in $\mathbf{B} = \mathbf{n} \times \mathbf{E}$. The two contributions are

$$\hat{\mathbf{r}} \times \delta\mathbf{E} = -\frac{e}{2c^2 r}\hat{\mathbf{r}} \times \mathbf{a}, \qquad \delta\mathbf{n} \times \frac{e\hat{\mathbf{r}}}{r^2} = +\frac{e}{2c^2 r}\hat{\mathbf{r}} \times \mathbf{a}.$$

They cancel: $\boxed{\mathbf{B}(\mathbf{x}, 0) = 0 + O(c^{-3})}$. An instantaneous rest frame does not remove the acceleration electric field, but it does remove the magnetic field through this order.

(c) Write the correction as $\delta\mathbf{E} = -e[\mathbf{a}/r + \mathbf{r}(\mathbf{a}\cdot\mathbf{r})/r^3]/(2c^2)$. For $r > 0$,

$$\nabla \cdot (\mathbf{a}/r) = -\frac{\mathbf{a}\cdot\mathbf{r}}{r^3}, \qquad \nabla \cdot \left[\frac{\mathbf{r}(\mathbf{a}\cdot\mathbf{r})}{r^3}\right] = \frac{\mathbf{a}\cdot\mathbf{r}}{r^3},$$

so its divergence is zero. There is no added delta function at the origin because its flux through a shrinking sphere tends to zero. Furthermore

$$\nabla \times (\mathbf{a}/r) = -\frac{\hat{\mathbf{r}} \times \mathbf{a}}{r^2}, \quad \nabla \times \left[\frac{\mathbf{r}(\mathbf{a}\cdot\mathbf{r})}{r^3}\right] = -\frac{\hat{\mathbf{r}} \times \mathbf{a}}{r^2}.$$

Thus $\nabla \times \mathbf{E} = e\hat{\mathbf{r}} \times \mathbf{a}/(c^2 r^2)$. Faraday's law gives $\partial_t \mathbf{B} = e\mathbf{a} \times \hat{\mathbf{r}}/(cr^2)$, and integration about $t = 0$ yields

$$\boxed{\mathbf{B}(\mathbf{x}, t) = \frac{e}{cr^2}(t\mathbf{a}) \times \hat{\mathbf{r}} + \cdots = \frac{e\mathbf{v}(t) \times \hat{\mathbf{r}}}{cr^2} + \cdots .}$$

This is precisely the elementary nonrelativistic moving-charge field.

14.3 The Heaviside–Feynman expression for the electric field of a particle of charge e in arbitrary motion, an alternative to the Liénard–Wiechert expression (14.14), is

$$\mathbf{E} = e\left[\frac{\mathbf{n}}{R^2}\right]_{\text{ret}} + e\left[\frac{R}{c}\right]_{\text{ret}} \frac{d}{dt}\left[\frac{\mathbf{n}}{R^2}\right]_{\text{ret}} + e\frac{d^2}{c^2 dt^2}[\mathbf{n}]_{\text{ret}}$$

where the time derivatives are with respect to the time at the observation point. The magnetic field is given by (14.13).

Using the fact that the retarded time is $t' = t - R(t')/c$ and that, as a result,

$$\frac{dt}{dt'} = 1 - \boldsymbol{\beta}(t') \cdot \mathbf{n}(t')$$

show that the form above yields (14.14) when the time differentiations are performed.

Solution.

All unbracketed quantities in this derivation are evaluated at t'. Define $b = \mathbf{n} \cdot \boldsymbol{\beta}$, $\kappa = 1 - b$, and $\mathbf{u} = b\mathbf{n} - \boldsymbol{\beta}$. At a fixed observation point,

$$\frac{d}{dt} = \kappa^{-1}\frac{d}{dt'}, \quad \dot{R} = -cb, \quad \dot{\mathbf{n}} = \frac{c}{R}\mathbf{u}, \quad \dot{b} = \frac{c}{R}(b^2 - \beta^2) + \mathbf{n} \cdot \dot{\boldsymbol{\beta}},$$

where a dot here means d/dt'. The first differentiated expression is

$$\frac{d}{dt}\frac{\mathbf{n}}{R^2} = \frac{c}{\kappa R^3}(3b\mathbf{n} - \boldsymbol{\beta}).$$

For the second, differentiate $d\mathbf{n}/dt = c\mathbf{u}/(R\kappa)$ and use $\dot{\mathbf{u}} = \dot{b}\mathbf{n} + cb\mathbf{u}/R - \dot{\boldsymbol{\beta}}$. This gives

$$\frac{d^2\mathbf{n}}{dt^2} = \frac{c}{R\kappa^2}\left[\dot{b}\mathbf{n} - \dot{\boldsymbol{\beta}} + \left(\frac{2cb}{R} + \frac{\dot{b}}{\kappa}\right)\mathbf{u}\right].$$

Substitute these in the proposed field and split $\dot{b}$ into its velocity and acceleration parts. The coefficient of $\mathbf{n}/R^2$ has numerator

$$(1 + 2b)\kappa^2 + (b^2 - \beta^2)\kappa + b[2b\kappa + b^2 - \beta^2] = 1 - \beta^2,$$

and that of $\boldsymbol{\beta}/R^2$ has numerator $-\kappa^2 - 2b\kappa - b^2 + \beta^2 = -(1 - \beta^2)$, both over κ^3. Writing $q = \mathbf{n} \cdot \dot{\boldsymbol{\beta}}$, the remaining terms are

$$\frac{e}{cR\kappa^3}[q(\mathbf{n} - \boldsymbol{\beta}) - \kappa\dot{\boldsymbol{\beta}}] = \frac{e}{cR\kappa^3}\mathbf{n} \times [(\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}].$$

Therefore

$$\boxed{\mathbf{E} = e\left[\frac{(1 - \beta^2)(\mathbf{n} - \boldsymbol{\beta})}{\kappa^3 R^2} + \frac{\mathbf{n} \times [(\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]}{c\kappa^3 R}\right]_{\text{ret}},}$$

which is (14.14), with $\mathbf{B} = \mathbf{n} \times \mathbf{E}$. The factors of κ^{-1} from observation-time differentiation are essential to the result.

14.4 Using the Liénard–Wiechert fields, discuss the time-averaged power radiated per unit solid angle in nonrelativistic motion of a particle with charge e, moving

(a) along the z axis with instantaneous position $z(t) = a\cos\omega_0 t$,

(b) in a circle of radius R in the x-y plane with constant angular frequency ω_0.

Sketch the angular distribution of the radiation and determine the total power radiated in each case.

Solution.

To leading nonrelativistic order the acceleration field is $\mathbf{E}_a = e\mathbf{n}\times(\mathbf{n}\times\mathbf{a})/(c^2R)$, $\mathbf{B}_a = \mathbf{n}\times\mathbf{E}_a$. The Gaussian Poynting flux $c|\mathbf{E}_a|^2/(4\pi)$ therefore gives

$$\frac{d\overline{P}}{d\Omega} = \frac{e^2}{4\pi c^3}\left[\overline{a^2} - \overline{(\mathbf{n}\cdot\mathbf{a})^2}\right].$$

At this order the distinction between source-time and arrival-time averages is beyond the retained accuracy.

(a) For $z = a\cos\omega_0 t$, $\mathbf{a} = -a\omega_0^2\cos\omega_0 t\,\hat{\mathbf{z}}$. Using $\overline{\cos^2\omega_0 t} = 1/2$ gives

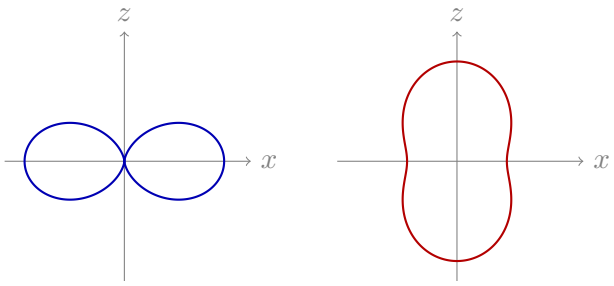
$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{e^2a^2\omega_0^4}{8\pi c^3}\sin^2\theta, \qquad \overline{P} = \frac{e^2a^2\omega_0^4}{3c^3}.}$$

The angular integral is $2\pi\int_{-1}^{1}(1-u^2)du = 8\pi/3$.

(b) For circular motion, $\mathbf{a} = -R\omega_0^2(\hat{\mathbf{x}}\cos\omega_0 t + \hat{\mathbf{y}}\sin\omega_0 t)$. Here $\overline{a^2} = R^2\omega_0^4$ and $\overline{(\mathbf{n}\cdot\mathbf{a})^2} = R^2\omega_0^4\sin^2\theta/2$. Hence

$$\boxed{\frac{d\overline{P}}{d\Omega} = \frac{e^2R^2\omega_0^4}{8\pi c^3}(1+\cos^2\theta), \qquad \overline{P} = \frac{2e^2R^2\omega_0^4}{3c^3}.}$$

This time $2\pi\int_{-1}^{1}(1+u^2)du = 16\pi/3$. The meridional polar plots below use normalized intensity as radius; rotation about z gives the full distributions.



Linear oscillation Circular motion

14.5 A *nonrelativistic* particle of charge ze, mass m, and kinetic energy E makes a *head-on* collision with a fixed central force field of finite range. The interaction is repulsive and described by a potential $V(r)$, which becomes greater than E at close distances.

(a) Show that the total energy radiated is given by

$$\Delta W = \frac{4}{3}\frac{z^2e^2}{m^2c^3}\sqrt{\frac{m}{2}}\int_{r_{\min}}^{\infty}\left|\frac{dV}{dr}\right|^2\frac{dr}{\sqrt{V(r_{\min})-V(r)}}$$

where $r_{\min}$ is the closest distance of approach in the collision.

(b) If the interaction is a Coulomb potential $V(r) = zZe^2/r$, show that the total energy radiated is

$$\Delta W = \frac{8}{45}\frac{zmv_0^5}{Zc^3}$$

where v_0 is the velocity of the charge at infinity.

Solution.

Radiation is evaluated to first order in its small effect on the orbit; thus mechanical energy conservation determines the unperturbed trajectory. Take $V(\infty) = 0$, so $E = mv_0^2/2$ and $V(r_{\min}) = E$.

(a) Newton's law and the Larmor formula give

$$|\mathbf{a}| = |V'|/m, \qquad P = \frac{2z^2e^2}{3m^2c^3}(V')^2, \qquad |\dot{r}| = \sqrt{\frac{2}{m}[E - V(r)]}.$$

The incoming and outgoing branches have equal radiation. Changing time to radius on each branch therefore gives

$$\boxed{\Delta W = 2\int_{r_{\min}}^{\infty}\frac{P}{|\dot{r}|}dr = \frac{4z^2e^2}{3m^2c^3}\sqrt{\frac{m}{2}}\int_{r_{\min}}^{\infty}\frac{[V'(r)]^2}{\sqrt{E - V(r)}}dr.}$$

The turning-point square-root singularity is integrable for a simple turning point. A Coulomb tail, although not of finite range literally, also makes this radiation integral converge at infinity.

(b) Put $A = zZe^2 > 0$. Then $r_{\min} = A/E$ and $[V']^2 = A^2/r^4$. With $u = r_{\min}/r$,

$$\int_{r_{\min}}^{\infty}\frac{A^2dr}{r^4\sqrt{E - A/r}} = \frac{E^{5/2}}{A}\int_0^1\frac{u^2}{\sqrt{1-u}}du.$$

The substitution $u = 1 - s^2$ evaluates the last integral as $2\int_0^1(1-s^2)^2ds = 2(1 - 2/3 + 1/5) = 16/15$. Substitution into part (a), followed by $E = mv_0^2/2$, gives

$$\Delta W = \frac{64z^2e^2E^{5/2}}{45\sqrt{2}\,m^{3/2}c^3A} = \boxed{\frac{8}{45}\frac{zmv_0^5}{Zc^3}}.$$

The consistency condition is $\Delta W \ll E$, in addition to nonrelativistic motion.

14.6

(a) Generalize the circumstances of the collision of Problem 14.5 to nonzero angular momentum (impact parameter) and show that the total energy radiated is given by

$$\Delta W = \frac{4z^2e^2}{3m^2c^3}\left(\frac{m}{2}\right)^{1/2}\int_{r_{\min}}^{\infty}\left(\frac{dV}{dr}\right)^2\left(E - V(r) - \frac{L^2}{2mr^2}\right)^{-1/2}dr$$

where $r_{\min}$ is the closest distance of approach (root of $E - V - L^2/2mr^2$), $L = mbv_0$, where b is the impact parameter, and v_0 is the incident speed ($E = mv_0^2/2$).

(b) Specialize to a repulsive Coulomb potential $V(r) = zZe^2/r$. Show that ΔW can be written in terms of impact parameter as

$$\Delta W = \frac{2zmv_0^5}{Zc^3}\left[-t^{-4} + t^{-5}\left(1 + \frac{t^2}{3}\right)\tan^{-1}t\right]$$

where $t = bmv_0^2/zZe^2$ is the ratio of twice the impact parameter to the distance of closest approach in a head-on collision.

Show that in the limit of t going to zero the result of Problem 14.5b is recovered, while for $t \gg 1$ one obtains the approximate result of Problem 14.7a.

(c) Using the relation between the scattering angle θ and t ($= \cot\theta/2$), show that ΔW can be expressed as

$$\Delta W = \frac{2zmv_0^5}{Zc^3}\tan^3\frac{\theta}{2}\left[\frac{1}{6}(\pi-\theta)\left(1+3\tan^2\frac{\theta}{2}\right)-\tan\frac{\theta}{2}\right]$$

(d) What changes occur for an *attractive* Coulomb potential?

Solution.

(a) Angular momentum conservation gives $r^2\dot{\phi} = L/m$ and the radial energy equation is

$$\frac{m}{2}\dot{r}^2 + V(r) + \frac{L^2}{2mr^2} = E.$$

The actual acceleration remains $-V'(r)\hat{\mathbf{r}}/m$; the centrifugal term is part of the radial-coordinate equation and is not an extra force in the Larmor formula. On each branch,

$$dt = \frac{|dr|}{\sqrt{(2/m)[E - V - L^2/(2mr^2)]}}.$$

Doubling their equal contributions gives

$$\Delta W = 2\int_{r_{\min}}^{\infty} \frac{2z^2e^2[V'(r)]^2}{3m^2c^3}\frac{dr}{\sqrt{(2/m)[E-V(r)-L^2/(2mr^2)]}},$$

which is the stated integral after taking $\sqrt{m/2}$ outside. Here $L = mbv_0$ and $r_{\min}$ is its positive turning-point root.

(b) Let $A = zZe^2 > 0$, $a_C = A/(mv_0^2)$, $t = b/a_C$, and $u = a_C/r$. Then

$$\dot{r}^2 = v_0^2(1-2u-t^2u^2), \quad u_+ = \frac{\sqrt{1+t^2}-1}{t^2}, \quad \Delta W = \frac{4zmv_0^5}{3Zc^3}J(t),$$

where $J(t) = \int_0^{u_+} u^2(1-2u-t^2u^2)^{-1/2}du$. To do the integral put $s = (1+t^2u)/\sqrt{1+t^2} = \cos\chi$. The endpoints are $\chi = \tan^{-1}t$ and 0, and therefore

$$\begin{aligned} J(t) &= \frac{1}{t^5}\int_0^{\tan^{-1}t}[(1+t^2)\cos^2\chi - 2\sqrt{1+t^2}\cos\chi + 1]d\chi \\ &= \frac{1}{t^5}\left[\frac{t^2+3}{2}\chi + \frac{1+t^2}{4}\sin 2\chi - 2\sqrt{1+t^2}\sin\chi\right]_0^{\tan^{-1}t} \\ &= \frac{(t^2+3)\tan^{-1}t - 3t}{2t^5}. \end{aligned}$$

In the final line use $\sin\chi = t/\sqrt{1+t^2}$ and $\sin 2\chi = 2t/(1+t^2)$. Hence

$$\boxed{\Delta W = \frac{2zmv_0^5}{Zc^3}\left[-t^{-4} + t^{-5}\left(1+\frac{t^2}{3}\right)\tan^{-1}t\right].}$$

At small t the bracket is $4/45 - 8t^2/105 + O(t^4)$, obtained by multiplying the series for $\tan^{-1} t$ before subtracting t^{-4}. This recovers 14.5. At large t it is $\pi/(6t^3) + O(t^{-4})$; insertion of $t = bmv_0^2/A$ gives $\pi z^4 Z^2 e^6/(3m^2c^3 v_0 b^3)$, as in 14.7.

(c) For repulsive Coulomb scattering, $t = \cot(\theta/2)$ and $\tan^{-1} t = (\pi - \theta)/2$. Writing $s = \tan(\theta/2) = 1/t$, the bracket becomes

$$-s^4 + \frac{\pi - \theta}{2}(s^5 + s^3/3) = s^3 \left[\frac{\pi - \theta}{6}(1 + 3s^2) - s \right],$$

which is the requested angular expression.

(d) For attraction define $A = |zZe^2|$, $a_C = A/(mv_0^2)$ and $t = b/a_C > 0$. The square root is now $\sqrt{1 + 2u - t^2 u^2}$, with $u_+ = (1 + \sqrt{1 + t^2})/t^2$. Use $s = (t^2 u - 1)/\sqrt{1 + t^2} = \cos\chi$. The upper angular endpoint becomes $\pi - \tan^{-1} t$ and the numerator is $[(1 + t^2)\cos^2\chi + 2\sqrt{1 + t^2}\cos\chi + 1]$. Integrating the three terms gives

$$J_{\rm att} = \frac{(t^2 + 3)(\pi - \tan^{-1} t) + 3t}{2t^5},$$

so

$$\boxed{\Delta W_{\rm att} = \frac{2|z/Z| m v_0^5}{c^3} \left[t^{-4} + t^{-5} \left(1 + \frac{t^2}{3} \right) (\pi - \tan^{-1} t) \right].}$$

The magnitude of the deflection still obeys $t = \cot(\theta/2)$, with the bending toward the center. The large-t limit is the same as for repulsion. At small t the result instead grows as $2\pi|z/Z| m v_0^5/(c^3 t^5)$: a point attractive center has no finite head-on turning radius. Before taking that limit, the assumptions fail when the periapsis speed or fractional radiative loss is large; the formula is valid only for trajectories remaining nonrelativistic with small loss throughout the encounter.

14.7 A nonrelativistic particle of charge ze, mass m, and initial speed v_0 is incident on a fixed charge Ze at an impact parameter b that is large enough to ensure that the particle's deflection in the course of the collision is very small.

(a) Using the Larmor power formula and Newton's second law, calculate the total energy radiated, assuming (after you have computed the acceleration) that the particle's trajectory is a straight line at constant speed:

$$\Delta W = \frac{\pi z^4 Z^2 e^6}{3m^2 c^3 v_0} \frac{1}{b^3}$$

(b) The expression found in part a is an approximation that fails at small enough impact parameter. For a repulsive potential the closest distance of approach at zero impact parameter, $r_c = 2zZe^2/mv_0^2$, serves as a length against which to measure b. The approximation will be valid for $b \gg r_c$. Compare the result of replacing b by r_c in part a with the answer of Problem 14.5 for a *head-on* collision.

(c) A radiation cross section χ (with dimensions of energy times area) can be defined classically by multiplying $\Delta W(b)$ by $2\pi b\, db$ and integrating over all impact parameters. Because of the divergence of the expression at small b, one must cut off the integration at some $b = b_{\rm min}$. If, as in Chapter 13, the uncertainty principle is used to specify the minimum impact parameter, one may expect to obtain an approximation to the quantum-mechanical result. Compute such a cross section with the expression from part a. Compare your result with the Bethe–Heitler formula [N^{-1} times (15.30)].

Solution.

(a) Put the straight reference trajectory at $\mathbf{r}(t) = (b, 0, v_0 t)$. First compute its acceleration from the Coulomb force, rather than differentiating the straight reference path:

$$\mathbf{a}(t) = \frac{zZe^2}{m} \frac{(b, 0, v_0 t)}{(b^2 + v_0^2 t^2)^{3/2}}, \qquad |\mathbf{a}|^2 = \frac{z^2 Z^2 e^4}{m^2 (b^2 + v_0^2 t^2)^2}.$$

The straight line is used only to evaluate this first-order force. With $u = v_0 t/b$,

$$\Delta W = \frac{2z^4 Z^2 e^6}{3m^2 c^3 v_0 b^3} \int_{-\infty}^{\infty} \frac{du}{(1+u^2)^2}.$$

Set $u = \tan\chi$: the integral becomes $\int_{-\pi/2}^{\pi/2} \cos^2\chi \, d\chi = \pi/2$, proving the stated result.

(b) Since $r_c = 2zZe^2/(mv_0^2)$,

$$\Delta W_{\text{straight}}(r_c) = \frac{\pi}{24} \frac{zmv_0^5}{Zc^3}, \qquad \frac{\Delta W_{\text{straight}}(r_c)}{\Delta W_{\text{head-on}}} = \frac{15\pi}{64} \simeq 0.7363.$$

Thus that uncontrolled substitution happens to give the same scale, but underestimates the actual head-on result by about 26%. It does not extend the small-deflection approximation to small b.

(c) Let $C = \pi z^4 Z^2 e^6/(3m^2 c^3 v_0)$, so that $\Delta W = C/b^3$. Then

$$\chi = \int_{b_{\min}}^{\infty} 2\pi b \Delta W \, db = 2\pi C \int_{b_{\min}}^{\infty} b^{-2} db = \frac{2\pi^2 z^4 Z^2 e^6}{3m^2 c^3 v_0 b_{\min}}.$$

Using the reduced de Broglie scale $b_{\min} = \hbar/(mv_0)$ gives

$$\boxed{\chi_{\text{cutoff}} = \frac{2\pi^2}{3} z^4 Z^2 \alpha r_0^2 mc^2, \qquad r_0 = \frac{e^2}{mc^2}, \quad \alpha = \frac{e^2}{\hbar c}.}$$

The nonrelativistic Bethe–Heitler result in (15.30), divided by target density, contains the dimensionless integral

$$\mathcal{I} = \int_0^1 \ln \frac{1 + \sqrt{1-x}}{\sqrt{x}} \, dx.$$

Set $x = \sin^2\vartheta$ and use $(1 + \cos\vartheta)/\sin\vartheta = \cot(\vartheta/2)$. Integration by parts gives

$$\mathcal{I} = \left[\sin^2\vartheta \ln\cot\frac{\vartheta}{2} \right]_0^{\pi/2} + \int_0^{\pi/2} \sin\vartheta \, d\vartheta = 1;$$

the boundary term vanishes at both ends. Thus $\chi_{\text{BH}} = (16/3) z^4 Z^2 \alpha r_0^2 mc^2$, with m the projectile mass and $|z| = 1$ for an electron. The cutoff estimate is larger by $\pi^2/8 \simeq 1.2337$. The common energy independence and scale are reproduced, but a sharp uncertainty cutoff cannot fix the quantum coefficient. This comparison uses the Born regime $|zZ|\alpha c/v_0 \ll 1$, in which the de Broglie cutoff also lies outside the strong-deflection scale.

14.8 A swiftly moving particle of charge ze and mass m passes a fixed point charge Ze in an approximately straight-line path at impact parameter b and nearly constant speed v. Show that the total energy radiated in the encounter is

$$\Delta W = \frac{\pi z^4 Z^2 e^6}{4m^2 c^4 \beta}\left(\gamma^2 + \frac{1}{3}\right)\frac{1}{b^3}$$

This is the relativistic generalization of the result of Problem 14.7.

Solution.

Let $q = ze$ and $A = zZe^2$, and again evaluate the external force on $\mathbf{r} = (b, 0, vt)$:

$$F_\perp = \frac{Ab}{(b^2 + v^2t^2)^{3/2}}, \qquad F_\parallel = \frac{Avt}{(b^2 + v^2t^2)^{3/2}}.$$

At fixed zeroth-order speed, $a_\perp = F_\perp/(\gamma m)$ and $a_\parallel = F_\parallel/(\gamma^3 m)$. The Liénard power becomes

$$P = \frac{2q^2}{3c^3}(\gamma^4 a_\perp^2 + \gamma^6 a_\parallel^2) = \frac{2q^2A^2}{3m^2c^3}\frac{\gamma^2 b^2 + v^2t^2}{(b^2 + v^2t^2)^3}.$$

With $u = vt/b$, the two needed integrals are

$$\int_{-\infty}^{\infty}\frac{du}{(1+u^2)^3} = \frac{3\pi}{8}, \qquad \int_{-\infty}^{\infty}\frac{u^2 du}{(1+u^2)^3} = \frac{\pi}{8}.$$

For example $u = \tan\chi$ turns them into integrals of $\cos^4\chi$ and $\sin^2\chi\cos^2\chi$ on $(-\pi/2, \pi/2)$, whose double-angle expansions give the displayed values. Hence

$$\boxed{\Delta W = \frac{\pi z^4 Z^2 e^6}{4m^2 c^4 \beta b^3}\left(\gamma^2 + \frac{1}{3}\right).}$$

As $\gamma \to 1$, the bracket becomes $4/3$, reproducing 14.7. The trajectory approximation requires a small transverse impulse $2|A|/(bv) \ll \gamma m v$ and a small energy loss relative to the incoming energy.

14.9 A particle of mass m, charge q, moves in a plane perpendicular to a uniform, static, magnetic induction B.

(a) Calculate the total energy radiated per unit time, expressing it in terms of the constants already defined and the ratio γ of the particles's total energy to its rest energy.

(b) If at time $t = 0$ the particle has a total energy $E_0 = \gamma_0 mc^2$, show that it will have energy $E = \gamma mc^2 < E_0$ at a time t, where

$$t \simeq \frac{3m^3c^5}{2q^4B^2}\left(\frac{1}{\gamma} - \frac{1}{\gamma_0}\right)$$

provided $\gamma \gg 1$.

(c) If the particle is initially nonrelativistic and has a *kinetic* energy T_0 at $t = 0$, what is its kinetic energy at time t?

(d) If the particle is actually trapped in the magnetic dipole field of the earth and is spiraling back and forth along a line of force, does it radiate more energy while near the equator, or while near its turning points? Why? Make quantitative statements if you can.

Solution.

(a) The magnetic force is perpendicular to the velocity: $\gamma ma = |q|vB/c$. Inserting this in the Liénard formula gives

$$\boxed{P = \frac{2q^2\gamma^4 a^2}{3c^3} = \frac{2q^4B^2}{3m^2c^3}(\gamma^2 - 1).}$$

Define $C = 2q^4B^2/(3m^3c^5)$. The slow energy loss obeys $\dot{\gamma} = -C(\gamma^2 - 1)$.

(b) Separation of variables gives the useful exact relation

$$Ct = \int_{\gamma}^{\gamma_0} \frac{dg}{g^2 - 1} = \frac{1}{2}\ln\frac{(\gamma_0 - 1)(\gamma + 1)}{(\gamma_0 + 1)(\gamma - 1)}.$$

Equivalently $(\gamma - 1)/(\gamma + 1) = [(\gamma_0 - 1)/(\gamma_0 + 1)]e^{-2Ct}$. For both Lorentz factors much larger than one, $(g^2 - 1)^{-1} = g^{-2} + O(g^{-4})$, so

$$\boxed{t \simeq \frac{1}{C}\left(\frac{1}{\gamma} - \frac{1}{\gamma_0}\right) = \frac{3m^3c^5}{2q^4B^2}\left(\frac{1}{\gamma} - \frac{1}{\gamma_0}\right).}$$

(c) For $T = (\gamma - 1)mc^2 \ll mc^2$, $\gamma^2 - 1 = 2T/(mc^2) + O(T^2)$, hence

$$\boxed{T(t) = T_0 e^{-2Ct} = T_0 \exp\left[-\frac{4q^4B^2}{3m^3c^5}t\right].}$$

The rest energy does not decay according to this nonrelativistic law.

(d) Over one weakly damped bounce the adiabatic invariant $p_\perp^2/B$ is constant and the total momentum is approximately constant. If ϑ is the local angle between velocity and field, $\sin^2\vartheta = B/B_m$, where B_m is the mirror-point field. The local radiation power is proportional to $B^2\gamma^2\beta^2\sin^2\vartheta$, and therefore to B^3 along that orbit. Thus

$$\boxed{\frac{P_m}{P_{\rm eq}} = \left(\frac{B_m}{B_{\rm eq}}\right)^3 = \sin^{-6}\vartheta_{\rm eq}.}$$

Radiation is strongest near the turning points, where the field and perpendicular fraction are largest; the slow parallel motion there also increases the time spent in that region. For a dipole field line at magnetic latitude λ, $B/B_{\rm eq} = \sqrt{1 + 3\sin^2\lambda}/\cos^6\lambda$, which can be inserted in the cubic law for a quantitative spatial comparison.

14.10 A particle of charge e moves at constant velocity $\boldsymbol{\beta}c$ for $t < 0$. During the short time interval, $0 < t < \Delta t$, its velocity remains in the same direction, but its speed decreases linearly in time to zero. For $t > \Delta t$, the particle remains at rest.

(a) Show that the radiant energy emitted per unit solid angle is

$$\frac{dE}{d\Omega} = \frac{e^2\beta^2}{16\pi c\,\Delta t}\frac{(2 - \beta\cos\theta)\,[1 + (1 - \beta\cos\theta)^2]\sin^2\theta}{(1 - \beta\cos\theta)^4}$$

where θ is the polar angle relative to the direction of the initial velocity.

(b) In the limit of $\gamma \gg 1$, show that the angular distribution can be expressed as

$$\frac{dE}{d\xi} \approx \frac{e^2\beta^2\gamma^4}{c\,\Delta t}\frac{\xi}{(1+\xi)^4}$$

where $\xi = (\gamma\theta)^2$. Show that $\langle\theta^2\rangle^{1/2} \approx \sqrt{2}/\gamma$ and that the expression for the total energy radiated is in agreement with the result from (14.43) in the same limit.

Solution.

(a) The velocity and its derivative remain parallel to the initial axis, so the angular power per unit *source* time is

$$\frac{dP}{d\Omega} = \frac{e^2}{4\pi c}\frac{\dot\beta^2\sin^2\theta}{(1-\beta(t')\cos\theta)^5}.$$

Here $\beta(t') = \beta(1 - t'/\Delta t)$ and $\dot\beta = -\beta/\Delta t$. With $u = \cos\theta$ and $s = 1 - t'/\Delta t$,

$$\frac{dE}{d\Omega} = \frac{e^2\beta^2\sin^2\theta}{4\pi c\Delta t}\int_0^1 \frac{ds}{(1-\beta u s)^5} = \frac{e^2\beta^2\sin^2\theta}{16\pi c\Delta t}\frac{(1-\beta u)^{-4}-1}{\beta u}.$$

Set $D = 1 - \beta u$. Since $(1 - D^4)/(1 - D) = (1 + D)(1 + D^2)$, this is the stated expression. The apparent quotient at $u = 0$ is removable and equals 4 before the factor 1/16; it reproduces the elementary nonrelativistic limit.

(b) For $\theta = O(\gamma^{-1})$ set $\xi = \gamma^2\theta^2$. Then $D \simeq (1 + \xi)/(2\gamma^2)$, $\sin^2\theta \simeq \xi/\gamma^2$, and $d\Omega \simeq \pi d\xi/\gamma^2$. Keeping the leading factors yields

$$\boxed{\frac{dE}{d\xi} \simeq \frac{e^2\beta^2\gamma^4}{c\Delta t}\frac{\xi}{(1+\xi)^4}.}$$

With $y = 1/(1 + \xi)$, the normalization and first-moment integrals are

$$\int_0^\infty \frac{\xi\,d\xi}{(1+\xi)^4} = \int_0^1 y(1-y)dy = \frac{1}{6}, \quad \int_0^\infty \frac{\xi^2 d\xi}{(1+\xi)^4} = \int_0^1 (1-y)^2 dy = \frac{1}{3}.$$

Hence $E \simeq e^2\beta^2\gamma^4/(6c\Delta t)$ and $\langle\theta^2\rangle = 2/\gamma^2$. As an independent total-power check, (14.43) for parallel acceleration integrates to

$$E = \frac{2e^2\beta^2}{3c\Delta t}\int_0^1 \frac{ds}{(1-\beta^2 s^2)^3} = \frac{2e^2\beta^2}{3c\Delta t}\left[\frac{\gamma^4}{4} + \frac{3\gamma^2}{8} + \frac{3\,\mathrm{artanh}\,\beta}{8\beta}\right].$$

An antiderivative after $w = \beta s$ is $w/[4(1 - w^2)^2] + 3w/[8(1 - w^2)] + 3\,\mathrm{artanh}\,w/8$. Its leading term agrees with the angular integral.

14.11 A particle of charge ze and mass m moves in external electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$.

(a) Show that the classical relativistic result for the instantaneous energy radiated per unit time can be written

$$P = \frac{2}{3}\frac{z^4e^4}{m^2c^3}\gamma^2[(\mathbf{E} + \boldsymbol{\beta}\times\mathbf{B})^2 - (\boldsymbol{\beta}\cdot\mathbf{E})^2]$$

where $\mathbf{E}$ and $\mathbf{B}$ are evaluated at the position of the particle and γ is the particle's instantaneous Lorentz factor.

(b) Show that the expression in part a can be put into the manifestly Lorentz-invariant form,

$$P = \frac{2z^4 r_0^2}{3m^2 c} \cdot F^{\mu\nu} p_\nu p^\lambda F_{\lambda\mu}$$

where $r_0 = e^2/mc^2$ is the classical charged particle radius.

Solution.

(a) Put $q = ze$ and $\mathbf{F} = q(\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B})$. Differentiating $\mathbf{p} = \gamma m \mathbf{v}$, with $\dot{\gamma} = \gamma^3 \boldsymbol{\beta} \cdot \mathbf{a}/c$, gives

$$\mathbf{F} = \gamma m[\mathbf{a} + \gamma^2 \boldsymbol{\beta}(\boldsymbol{\beta} \cdot \mathbf{a})], \quad \mathbf{a} = \frac{\mathbf{F} - \boldsymbol{\beta}(\boldsymbol{\beta} \cdot \mathbf{F})}{\gamma m}.$$

The parallel and perpendicular components are $a_\parallel = F_\parallel/(\gamma^3 m)$, $a_\perp = F_\perp/(\gamma m)$. Thus

$$P = \frac{2q^2}{3c^3}(\gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2) = \frac{2q^2\gamma^2}{3m^2c^3}[\mathbf{F}^2 - (\boldsymbol{\beta} \cdot \mathbf{F})^2]$$
$$= \boxed{\frac{2z^4 e^4 \gamma^2}{3m^2 c^3}[(\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B})^2 - (\boldsymbol{\beta} \cdot \mathbf{E})^2].}$$

In the last step $\boldsymbol{\beta} \cdot (\boldsymbol{\beta} \times \mathbf{B}) = 0$.

(b) Use metric $(+,-,-,-)$ and $p^\mu = \gamma mc(1, \boldsymbol{\beta})$. Define $w^\mu = F^{\mu\nu} p_\nu$. Jackson's (11.137) has $F^{0i} = -E_i$ and $F^{ij} = -\epsilon_{ijk} B_k$. Since $p_\nu = \gamma mc(1, -\boldsymbol{\beta})$, direct contraction gives $w^0 = \gamma mc\, \boldsymbol{\beta} \cdot \mathbf{E}$ and $w^i = \gamma mc(E_i + \epsilon_{ijk}\beta_j B_k)$. Hence

$$w^\mu = \gamma mc\big(\boldsymbol{\beta} \cdot \mathbf{E}, \mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}\big).$$

Antisymmetry gives $p^\lambda F_{\lambda\mu} = -w_\mu$, so

$$F^{\mu\nu} p_\nu p^\lambda F_{\lambda\mu} = -w^\mu w_\mu = m^2 c^2 \gamma^2 [(\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B})^2 - (\boldsymbol{\beta} \cdot \mathbf{E})^2].$$

Substitute $r_0^2 = e^4/(m^2 c^4)$ to obtain exactly the stated invariant formula. It is nonnegative because the Lorentz four-force is spacelike or zero. This invariance concerns total emitted energy per unit source time; it does not assert invariance of an angle-resolved received power.

14.12 As in Problem 14.4a a charge e moves in simple harmonic motion along the z axis, $z(t') = a\cos(\omega_0 t')$.

(a) Show that the instantaneous power radiated per unit solid angle is

$$\frac{dP(t')}{d\Omega} = \frac{e^2 c \beta^4}{4\pi a^2} \frac{\sin^2\theta\, \cos^2(\omega_0 t')}{(1 + \beta\cos\theta\, \sin\omega_0 t')^5}$$

where $\beta = a\omega_0/c$.

(b) By performing a time averaging, show that the average power per unit solid angle is

$$\frac{dP}{d\Omega} = \frac{e^2 c \beta^4}{32\pi a^2} \left[\frac{4 + \beta^2 \cos^2\theta}{(1 - \beta^2\cos^2\theta)^{7/2}}\right] \sin^2\theta$$

(c) Make rough sketches of the angular distribution for nonrelativistic and relativistic motion.

Solution.

(a) Let $u = \omega_0 t'$. The axial velocity and acceleration are $\beta_z = -\beta \sin u$ and $\dot{\beta}_z = -\beta\omega_0 \cos u$. The cross product in the angular Liénard formula has squared norm $\dot{\beta}_z^2 \sin^2\theta$, since its velocity-parallel terms cancel. Consequently

$$\frac{dP(t')}{d\Omega} = \frac{e^2\beta^2\omega_0^2}{4\pi c}\,\frac{\sin^2\theta\cos^2 u}{(1+\beta\cos\theta\sin u)^5}.$$

Using $\omega_0 = c\beta/a$ gives the proposed result.

(b) Put $q = \beta\cos\theta$ and $I_j(q) = (2\pi)^{-1}\int_0^{2\pi}(1+q\sin u)^{-j}du$. For the starting integral use $t = \tan(u/2)$, splitting the period at $u = \pi$ so the two pieces cover the real line:

$$I_1 = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2\,dt}{t^2+2qt+1} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2\,dt}{(t+q)^2+1-q^2} = \frac{1}{\sqrt{1-q^2}}.$$

Differentiation gives $I'_{j-1} = -(j-1)\langle \sin u/(1+q\sin u)^j\rangle$, whereas $I_{j-1} = I_j + q\langle \sin u/(1+q\sin u)^j\rangle$. Eliminating the last average gives $I_j = I_{j-1} + qI'_{j-1}/(j-1)$. Starting with $I_2 = (1-q^2)^{-3/2}$, successive differentiation yields

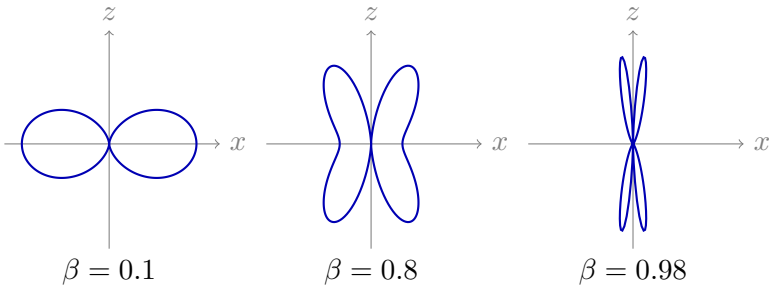
$$I_3 = \frac{2+q^2}{2(1-q^2)^{5/2}},\quad I_4 = \frac{2+3q^2}{2(1-q^2)^{7/2}},\quad I_5 = \frac{8+24q^2+3q^4}{8(1-q^2)^{9/2}}.$$

Since $q^2\sin^2 u = (1+q\sin u)^2 - 2(1+q\sin u) + 1$,

$$\begin{aligned}\left\langle \frac{\cos^2 u}{(1+q\sin u)^5}\right\rangle &= I_5 - \frac{I_3 - 2I_4 + I_5}{q^2}\\ &= \frac{(q^2-1)I_5 + 2I_4 - I_3}{q^2} = \frac{4+q^2}{8(1-q^2)^{7/2}}.\end{aligned}$$

The $q = 0$ value is the continuous limit $1/2$. Substitution proves the stated average. Because the position repeats after a full period, the total observation-time interval is also that period; averaging emitted energy per cycle therefore gives the same mean power for a fixed distant direction.

(c) For $\beta \ll 1$ the pattern is $\sin^2\theta$. For $\gamma = (1-\beta^2)^{-1/2} \gg 1$, near either pole it is proportional to $\theta^2/(\gamma^{-2}+\theta^2)^{7/2}$. Differentiation with respect to θ^2 gives the maximum at $\theta^2 = 2/(5\gamma^2)$; the axis itself remains a node. The normalized plots show the two narrowing lobes for increasing β.



$\beta = 0.1$ $\beta = 0.8$ $\beta = 0.98$

14.13 Show explicitly by use of the Poisson sum formula or other means that, if the motion of a radiating particle repeats itself with periodicity T, the continuous frequency spectrum becomes a discrete spectrum containing frequencies that are integral multiples of the fundamental. Show that a general expression for the time-averaged power radiated per unit solid angle in each multiple m of the fundamental frequency $\omega_0 = 2\pi/T$ is:

$$\frac{dP_m}{d\Omega} = \frac{e^2\omega_0^4 m^2}{(2\pi c)^3}\left|\int_0^{2\pi/\omega_0} \mathbf{v}(t)\times\mathbf{n}\,\exp\left[im\omega_0\left(t-\frac{\mathbf{n}\cdot\mathbf{x}(t)}{c}\right)\right]dt\right|^2$$

Solution.

Let $\omega > 0$ and write the one-period vector integral

$$\mathbf{F}_T(\omega) = \int_0^T \mathbf{v}(t)\times\mathbf{n}\exp\{i\omega[t-\mathbf{n}\cdot\mathbf{x}(t)/c]\}\,dt.$$

For M repeated periods, periodicity of both $\mathbf{x}$ and $\mathbf{v}$ makes the full transform $\mathbf{F}_T(\omega)\sum_{j=0}^{M-1} e^{ij\omega T}$. The spectral energy formula is $e^2\omega^2|\mathbf{F}|^2/(4\pi^2c^3)$. Divide it by the observation duration MT and use the distributional limit of the squared geometric sum:

$$\frac{1}{MT}\left|\sum_{j=0}^{M-1} e^{ij\omega T}\right|^2 \longrightarrow \frac{2\pi}{T^2}\sum_{m=-\infty}^{\infty}\delta(\omega-m\omega_0).$$

To check the normalization, expand the square as $\sum_{s=-(M-1)}^{M-1}(M-|s|)e^{is\omega T}$. After division by MT, its limit is $T^{-1}\sum_s e^{is\omega T}$; Poisson summation gives $\sum_s e^{is\omega T} = (2\pi/T)\sum_m \delta(\omega - 2\pi m/T)$. Thus the weight of the positive line $\omega = m\omega_0$ is

$$\boxed{\frac{dP_m}{d\Omega} = \frac{e^2(m\omega_0)^2}{2\pi c^3 T^2}|\mathbf{F}_T(m\omega_0)|^2 = \frac{e^2\omega_0^4 m^2}{(2\pi c)^3}|\mathbf{F}_T(m\omega_0)|^2.}$$

The $m = 0$ line does not radiate because of the explicit frequency square. Finite M replaces each delta by a narrow peak of width of order $1/(MT)$; the infinite-time limit is taken after dividing by duration, so no undefined square of a delta function is used.

14.14

(a) Show that for the simple harmonic motion of a charge discussed in Problem 14.12 the average power radiated per unit solid angle in the mth harmonic is

$$\frac{dP_m}{d\Omega} = \frac{e^2 c\beta^2}{2\pi a^2}\,m^2\tan^2\theta\,J_m^2(m\beta\cos\theta)$$

(b) Show that in the nonrelativistic limit the total power radiated is all in the fundamental and has the value

$$P \simeq \frac{2}{3}\frac{e^2}{c^3}\,\omega_0^4\,\overline{a^2}$$

where $\overline{a^2}$ is the mean square amplitude of oscillation.

Solution.

(a) Set $u = \omega_0 t$ and $b = \beta\cos\theta$ in 14.13. The phase is $m(u - b\cos u)$ and $v_z dt = -a\sin u\, du$. The needed Bessel representation is

$$I_m(b) = \int_0^{2\pi} e^{im(u-b\cos u)} du = 2\pi(-i)^m J_m(mb).$$

Integrating the derivative of the exponential over its full period gives

$$0 = im\int_0^{2\pi} (1 + b\sin u) e^{im(u-b\cos u)} du.$$

Thus $\int \sin u\, e^{im(u-b\cos u)} du = -I_m/b$, and the magnitude of the transverse velocity integral is $2\pi a|\tan\theta\, J_m(m\beta\cos\theta)|/\beta$. Its squared magnitude in 14.13 gives

$$\boxed{\frac{dP_m}{d\Omega} = \frac{e^2 c\beta^2}{2\pi a^2} m^2 \tan^2\theta J_m(m\beta\cos\theta)^2.}$$

The apparent singularity at $\cos\theta = 0$ is removable. There the fundamental has a finite limit because $J_1(x) \sim x/2$, and every $m > 1$ term vanishes because $J_m(x) = O(x^m)$.

(b) For fixed m and small β, $J_m(m\beta\cos\theta) \simeq (m\beta\cos\theta/2)^m/m!$. The mth power is thus of order β^{2m+2} in the displayed normalization; the fundamental dominates. For $m = 1$,

$$\frac{dP_1}{d\Omega} \simeq \frac{e^2 c\beta^4}{8\pi a^2}\sin^2\theta, \quad P_1 \simeq \frac{e^2 c\beta^4}{3a^2} = \frac{e^2\omega_0^4 a^2}{3c^3}.$$

With $\overline{z^2} = a^2/2$, the last expression is $(2e^2/3c^3)\omega_0^4\overline{z^2}$, the stated formula with its mean-square-amplitude notation $\overline{a^2}$.

14.15 A particle of charge e moves in a circular path of radius R in the x-y plane with a constant angular velocity ω_0.

(a) Show that the exact expression for the angular distribution of power radiated into the mth multiple of ω_0 is

$$\frac{dP_m}{d\Omega} = \frac{e^2\omega_0^4 R^2}{2\pi c^3} m^2 \left\{ \left[\frac{dJ_m(m\beta\sin\theta)}{d(m\beta\sin\theta)}\right]^2 + \frac{\cot^2\theta}{\beta^2} J_m^2(m\beta\sin\theta) \right\}$$

where $\beta = \omega_0 R/c$, and $J_m(x)$ is the Bessel function of order m.

(b) Assume nonrelativistic motion and obtain an approximate result for $dP_m/d\Omega$. Show that the results of Problem 14.4b are obtained in this limit.

(c) Assume extreme relativistic motion and obtain the results found in the text for a relativistic particle in instantaneously circular motion. [*Watson* (pp. 79, 249) may be of assistance to you.]

Solution.

(a) By axial symmetry take the observation azimuth to be zero, $\mathbf{n} = (\sin\theta, 0, \cos\theta)$, with transverse basis $\boldsymbol{\epsilon}_\theta = (\cos\theta, 0, -\sin\theta)$ and $\boldsymbol{\epsilon}_\phi = (0, 1, 0)$. Put $u = \omega_0 t$ and $b = \beta\sin\theta$. The position and velocity integrals are

$$\mathbf{r} = R(\cos u, \sin u, 0), \qquad \mathbf{v}dt = R(-\sin u, \cos u, 0)du,$$

with phase $m(u - b\cos u)$. From 14.14, $\int \sin u\, e^{i\text{phase}} du = -I_m/b$. Differentiating $I_m(b)$ also gives $\int \cos u\, e^{i\text{phase}} du = (i/m)I'_m(b) = 2\pi i(-i)^m J'_m(mb)$. Therefore the two transverse components of the velocity integral are

$$V_\theta = 2\pi R(-i)^m \frac{\cot\theta}{\beta} J_m(mb), \qquad V_\phi = 2\pi i R(-i)^m J'_m(mb).$$

Their orthogonality makes the squared modulus a sum. Substitution in 14.13 yields the expression in the question, including its relative β^{-2} factor.

(b) For small β, $J_m(x) \simeq (x/2)^m/m!$ and $J'_m(x) \simeq mJ_m(x)/x$. At $x = m\beta\sin\theta$ this gives

$$\boxed{\frac{dP_m}{d\Omega} \simeq \frac{e^2\omega_0^2}{2\pi c}\frac{m^{2m+2}\beta^{2m}}{2^{2m}(m!)^2}\sin^{2m-2}\theta(1+\cos^2\theta).}$$

Only $m = 1$ survives to leading order; it equals $e^2R^2\omega_0^4(1+\cos^2\theta)/(8\pi c^3)$, and its angular integral is $2e^2R^2\omega_0^4/(3c^3)$, exactly 14.4(b).

(c) Write $\psi = \theta - \pi/2$ and $A = \gamma^{-2} + \psi^2$. For $\gamma \gg 1$, $|\psi| \ll 1$, $\beta\sin\theta = 1 - A/2 + \cdots$. Use $J_m(x) = \pi^{-1}\int_0^\pi \cos(mu - x\sin u)du$ for integer m. In the large-m region contributing to radiation, the phase near zero is

$$mu - m\beta\sin\theta\sin u \simeq \frac{m}{2}(Au + u^3/3).$$

Put $u = \sqrt{A}\,s$ and $\xi = mA^{3/2}/3$. The Airy-integral identities

$$\int_0^\infty \cos\left[\frac{3\xi}{2}(s + s^3/3)\right] ds = \frac{1}{\sqrt{3}}K_{1/3}(\xi), \quad \int_0^\infty s\sin\left[\frac{3\xi}{2}(s + s^3/3)\right] ds = \frac{1}{\sqrt{3}}K_{2/3}(\xi)$$

then give

$$J_m(m\beta\sin\theta) \simeq \frac{\sqrt{A}}{\pi\sqrt{3}}K_{1/3}(\xi), \quad J'_m(m\beta\sin\theta) \simeq \frac{A}{\pi\sqrt{3}}K_{2/3}(\xi).$$

The second relation follows by differentiating the integral with respect to its argument before making the same small-u expansion. Part (a) consequently becomes

$$\frac{dP_m}{d\Omega} \simeq \frac{e^2\omega_0^2 m^2}{6\pi^3 c}A^2\left[K_{2/3}^2(\xi) + \frac{\psi^2}{A}K_{1/3}^2(\xi)\right].$$

For many closely spaced harmonics, divide by the spacing ω_0 to get power per unit frequency. Multiplication by the orbit period $2\pi/\omega_0$ converts this to the single-pulse energy spectrum:

$$\boxed{\frac{d^2I}{d\omega d\Omega} \simeq \frac{e^2}{3\pi^2 c}\left(\frac{\omega\rho}{c}\right)^2 A^2\left[K_{2/3}^2(\xi) + \frac{\psi^2}{A}K_{1/3}^2(\xi)\right],}$$

where $\rho = R \simeq c/\omega_0$, $\omega_c = 3\gamma^3 c/(2\rho)$, and $\xi = (\omega/2\omega_c)(1+\gamma^2\psi^2)^{3/2}$. This is the instantaneously circular result (14.79), with its two orthogonal polarization contributions retained.

14.16 Exploiting the fact that $k_0 d^3N/d^3k$, the number of quanta per invariant phase-space element d^3k/k_0, is a Lorentz-invariant quantity, show that the energy radiated per unit frequency interval per unit solid angle, (14.79), can be written in the invariant and coordinate-free form

$$\hbar\omega\frac{d^3N}{d^3k} = \frac{4e^2}{3\pi^2m^2}\left[\frac{(p\cdot k)^2}{[d^2(p\cdot k)/d\tau^2]^2}\left(\frac{d(\boldsymbol{\epsilon}_1\cdot p)}{d\tau}\right)^2 K_{2/3}^2(\xi) + \frac{(p\cdot k)(\boldsymbol{\epsilon}_2\cdot p)^2}{2[d^2(p\cdot k)/d\tau^2]}K_{1/3}^2(\xi)\right]$$

where the same ultrarelativistic local-curvature approximation as in (14.79) is understood, $d\tau$ is the proper time interval of the particle of mass m, p^μ is the 4-momentum of the particle, k^μ is the 4-wave vector of the radiation, and $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$ are polarization vectors parallel to the acceleration and in the direction $\boldsymbol{\epsilon}_1 \times \mathbf{k}$, respectively. The parameter is

$$\xi = \frac{2\sqrt{2}}{3m} \cdot \frac{(p \cdot k)^{3/2}}{(|d^2(p \cdot k)/d\tau^2|)^{1/2}}$$

This expression can be used to obtain the results of Problem 14.17 in an alternative manner. *Hint*: In proceeding with a solution, it is useful to expand $\mathbf{k} \cdot \mathbf{r}(t)$ around $t = 0$ in terms of the velocity, acceleration, etc. and compare with (14.72). One finds, for example, that $\omega c^2/\rho^2 = -\mathbf{k} \cdot d^2\mathbf{v}(0)/dt^2$, and, because the energy is constant,

$$\rho^2 = \frac{\omega E^3}{m^2 c^4 \, |d^2(p \cdot k)/d\tau^2|}$$

Solution.

The result has the same local-curvature and ultrarelativistic accuracy as (14.79); it is not an exact formula for arbitrary motion. Use metric $(+,-,-,-)$. At the emission point choose fixed polarization four-vectors $\epsilon_i^\mu = (0, \boldsymbol{\epsilon}_i)$ in the frame used for the local circular calculation, and transform those four-vectors along with p and k when changing frames.

Since $d^3k = (\omega^2/c^3) d\omega d\Omega$,

$$\hbar\omega \frac{d^3N}{d^3k} = \frac{c^3}{\omega^2} \frac{d^2I}{d\omega d\Omega}.$$

The left side is invariant because the number of quanta and d^3k/k^0 are invariant and $k^0 = \omega/c$. Let $A = \gamma^{-2} + \psi^2$, $E = \gamma mc^2$, $Q = p \cdot k$, and $D = d^2(p \cdot k)/d\tau^2$. In the local circular frame, the following reductions keep the same leading terms as the spectral formula:

$$Q = \frac{E\omega}{c^2}(1 - \beta\cos\psi) \simeq \frac{E\omega}{2c^2} A, \qquad \epsilon_2 \cdot p \simeq \frac{E}{c}\psi \quad \text{up to its unimportant sign.}$$

For constant energy, $d/d\tau = \gamma d/dt$ and $\mathbf{p} = \gamma m\mathbf{v}$. Expanding the phase gives

$$\omega\left(t - \frac{\mathbf{n} \cdot \mathbf{r}(t)}{c}\right) = \omega(1 - \mathbf{n} \cdot \boldsymbol{\beta})t - \frac{\omega}{2c}\mathbf{n} \cdot \mathbf{a}\, t^2 - \frac{\omega}{6c}\mathbf{n} \cdot \dot{\mathbf{a}}\, t^3 + \cdots .$$

At closest alignment $\mathbf{n} \cdot \mathbf{a} = 0$ and $-\mathbf{k} \cdot \dot{\mathbf{a}} \simeq \omega c^2/\rho^2$. Consequently

$$D = -\gamma^3 m\mathbf{k} \cdot \dot{\mathbf{a}} \simeq \frac{\omega E^3}{m^2 c^4 \rho^2} > 0, \qquad \left(\frac{d(\epsilon_1 \cdot p)}{d\tau}\right)^2 \simeq \left(\gamma^2 m \frac{c^2}{\rho}\right)^2 = \frac{E^4}{m^2 c^4 \rho^2}.$$

These establish the hint's expression for ρ^2. Substitution in the two proposed invariant coefficients gives, separately,

$$\frac{4}{m^2}\frac{Q^2}{D^2}\left(\frac{d(\epsilon_1 \cdot p)}{d\tau}\right)^2 = \rho^2 A^2, \qquad \frac{4}{m^2}\frac{Q(\epsilon_2 \cdot p)^2}{2D} = \rho^2 A\psi^2.$$

Also

$$\frac{2\sqrt{2}}{3m}\frac{Q^{3/2}}{\sqrt{|D|}} = \frac{\omega\rho}{3c} A^{3/2} = \xi.$$

Multiplying (14.79) by c^3/ω^2 now gives exactly the expression in the question. Every factor $Q, D, \epsilon_i \cdot p$ and its proper-time derivative is a Lorentz scalar under the stated simultaneous transformation, so equality established in this convenient frame holds in every frame, to the same approximation order.

14.17 A particle of charge e and mass m moves relativistically in a helical path in a uniform magnetic field $\mathbf{B}$. The pitch angle of the helix is α ($\alpha = 0$ corresponds to circular motion).

(a) By arguments similar to those of Section 14.4, show that an observer far from the helix within the beaming cone, for $\gamma \gg 1$ and $\gamma \cos\alpha \gg 1$, would detect radiation with a fundamental frequency

$$\omega_0 = \frac{\omega_B}{\cos^2\alpha}$$

and that the spectrum would extend up to frequencies in the order of

$$\omega_c = \frac{3}{2}\gamma^3 \omega_B \cos\alpha$$

where $\omega_B = eB/\gamma mc$. (Take care in determining the radius of curvature ρ of the helical path.)

(b) From part a and the results of Section 14.6, show that the *power* received by the observer per unit solid angle and per unit circular frequency interval is

$$\frac{d^2P}{d\omega\, d\Omega} \simeq \frac{3e^2\gamma^2}{8\pi^3 c}\frac{\omega_B}{\cos^2\alpha}\left(\frac{\omega}{\omega_c}\right)^2 (1+\gamma^2\psi^2)^2\left[K_{2/3}^2(\xi) + \frac{\gamma^2\psi^2}{1+\gamma^2\psi^2}\,K_{1/3}^2(\xi)\right]$$

where ω_B and ω_c are defined above, $\xi = (\omega/2\omega_c)(1+\gamma^2\psi^2)^{3/2}$, and ψ is the angle of observation measured relative to the particle's velocity vector, as in Fig. 14.9.

Solution.

Let the magnetic field define the z axis. Then $v_\perp = v\cos\alpha$, $v_\parallel = v\sin\alpha$, and the rotation period in source time is $T = 2\pi/\omega_B$.

(a) During one turn the particle advances $\Delta z = v\sin\alpha T$. For an observer at polar angle θ, subtracting the path-length change from the source period gives the exact far-observer period and its corresponding fundamental:

$$T_{\rm obs} = T(1-\beta\sin\alpha\cos\theta), \qquad \boxed{\omega_{\rm obs} = \frac{\omega_B}{1-\beta\sin\alpha\cos\theta}.}$$

The strong radiation is seen near $\theta_0 = \pi/2 - \alpha$, where the velocity points toward the observer. For $\gamma \gg 1$ and $\gamma\cos\alpha \gg 1$, its denominator is $\cos^2\alpha$ to leading order; thus $\omega_0 \simeq \omega_B/\cos^2\alpha$ as stated. The viewing and pitch qualifications are necessary: a general viewing direction has the Doppler factor displayed above.

The helix's cylindrical radius is $r_L = v\cos\alpha/\omega_B$, but its acceleration magnitude is $a = v\omega_B\cos\alpha$. Since speed is constant, the curvature radius is

$$\rho = \frac{v^2}{a} = \frac{v}{\omega_B\cos\alpha} = \frac{r_L}{\cos^2\alpha},$$

not r_L. The synchrotron cutoff scale is therefore

$$\boxed{\omega_c = \frac{3\gamma^3 c}{2\rho} \simeq \frac{3}{2}\gamma^3\omega_B\cos\alpha.}$$

(b) In terms of $x = \omega/\omega_c$, the single-pulse spectrum in 14.15 is

$$\frac{d^2I}{d\omega d\Omega} = \frac{3e^2\gamma^2}{4\pi^2 c}x^2(1+\gamma^2\psi^2)^2\left[K_{2/3}^2(\xi) + \frac{\gamma^2\psi^2}{1+\gamma^2\psi^2}K_{1/3}^2(\xi)\right].$$

This follows directly by replacing $(\omega\rho/c)^2(\gamma^{-2}+\psi^2)^2$ by $(9/4)\gamma^2x^2(1+\gamma^2\psi^2)^2$. The received mean power is the pulse energy divided by $T_{\rm obs}\simeq 2\pi\cos^2\alpha/\omega_B$; this produces the coefficient $3e^2\gamma^2\omega_B/(8\pi^3c\cos^2\alpha)$ and exactly the expression requested in part (b). Here ψ is the small displacement from closest alignment, as in the local circular radiation formula. The approximation presumes that the beaming cone does not collapse onto the field axis.

14.18

(a) By comparison of (14.91) with (14.79), show that the frequency spectrum of the received power for the situation in Problem 14.17 is

$$\frac{dP}{d\omega}=\left(\frac{\sqrt{3}e^2\gamma\omega_B}{2\pi c\,\cos\alpha}\right)G\left(\frac{\omega}{\omega_c}\right)$$

where $G(x)=x\displaystyle\int_x^\infty K_{5/3}(t)\,dt$ and the other symbols are as in Problem 14.17. This expression shows that the *shape* of the power spectrum in units of ω/ω_c is unchanged by the spiraling.

(b) Show that the integral over frequencies yields

$$P=\frac{2e^2\omega_B^2\gamma^4}{3c}$$

Comparison with (14.31) shows that the total received power is independent of the pitch angle of motion.

[In doing the integration over solid angles in part a, note that $\psi=0$ corresponds to $\theta=\pi/2-\alpha$.]

Solution.

(a) For the circular orbit with the same curvature ρ, write $\omega_\rho=c/\rho=\omega_B\cos\alpha$ to leading order. Its spectral power, from (14.91), is $\sqrt{3}e^2\gamma\omega_\rho G(\omega/\omega_c)/(2\pi c)$. The helix has the same local pulse spectrum, but its received repetition rate is $\omega_B/\cos^2\alpha$ rather than ω_ρ: this multiplies the angle-resolved power by $\cos^{-3}\alpha$. The angular integration has a compensating geometric factor, because

$$d\Omega\simeq 2\pi\sin\theta_0\,d\psi=2\pi\cos\alpha\,d\psi$$

instead of $2\pi d\psi$ for the circular orbit. Combining these factors with $\omega_\rho=\omega_B\cos\alpha$ gives

$$\boxed{\frac{dP_{\rm rec}}{d\omega}=\frac{\sqrt{3}e^2\gamma\omega_B}{2\pi c\cos\alpha}G\left(\frac{\omega}{\omega_c}\right),\quad G(x)=x\int_x^\infty K_{5/3}(t)dt.}$$

(b) Interchange the positive integrals:

$$\int_0^\infty G(x)dx=\int_0^\infty K_{5/3}(t)\left(\int_0^t x\,dx\right)dt=\frac{1}{2}\int_0^\infty t^2K_{5/3}(t)dt.$$

The Bessel integral $\int_0^\infty t^{s-1}K_\nu(t)dt=2^{s-2}\Gamma[(s+\nu)/2]\Gamma[(s-\nu)/2]$ with $s=3$ gives

$$\int_0^\infty G(x)dx=\Gamma(7/3)\Gamma(2/3)=\frac{4}{9}\Gamma(1/3)\Gamma(2/3)=\frac{8\pi}{9\sqrt{3}}.$$

Multiplication by the coefficient in (a) and $\omega_c = 3\gamma^3\omega_B \cos\alpha/2$ gives

$$\boxed{P_{\rm rec} = \frac{2e^2\omega_B^2\gamma^4}{3c}.}$$

This is the stated *received* power in the beaming approximation. The emitted energy per unit source time is instead $P_{\rm emit} = 2e^2\omega_B^2\gamma^4\beta^2\cos^2\alpha/(3c)$, from the Liénard formula. The leading ratio $P_{\rm rec}/P_{\rm emit} \simeq \cos^{-2}\alpha$ is the shortening of the arrival interval, not extra energy generation. The small-$\cos\alpha$ limit must respect the qualifications of 14.17.

14.19 Consider the angular and frequency spectrum of radiation produced by a magnetic moment in nonrelativistic motion, using (14.70) and the fact that a magnetization density $\mathcal{M}$ produces an effective current density $\mathbf{J}_{\rm eff} = c\nabla\times\mathcal{M}$.

(a) Show that a magnetic moment $\boldsymbol{\mu}$ with magnetization, $\mathcal{M} = \boldsymbol{\mu}(t)\,\delta[\mathbf{x}-\mathbf{r}(t)]$, in nonrelativistic motion gives a radiation intensity (energy radiated per unit solid angle per unit frequency interval),

$$\frac{d^2I_{\rm mag}}{d\omega\,d\Omega} = \frac{\omega^4}{4\pi^2c^3}\left|\int dt\;\mathbf{n}\times\boldsymbol{\mu}(t)e^{i\omega(t-\mathbf{n}\cdot\mathbf{r}(t)/c)}\right|^2$$

(b) The magnetic moment is located at the origin and is caused to precess by an external torque such that $\mu_x = \mu_0\sin\omega_0 t$ and $\mu_z = \mu_0\cos\omega_0 t$ for the time interval $t = -T/2$ to $t = T/2$, where $\omega_0 T/2\pi \gg 1$, with smooth switching outside the long periodic interval. Show that the frequency distribution of the radiation is very strongly peaked at $\omega = \omega_0$, that the angular distribution of radiation is proportional to $(1+\sin^2\theta\sin^2\phi)$, and that for $T\to\infty$, the total time-averaged power radiated is

$$\langle P\rangle = \frac{2\omega_0^4}{3c^3}\,\mu_0^2$$

Compare the result with the power calculated by the method of Section 9.3.

Solution.

(a) The Gaussian current spectral formula is

$$\frac{d^2I}{d\omega d\Omega} = \frac{\omega^2}{4\pi^2c^3}\left|\int dt\,d^3x\;\mathbf{n}\times\mathbf{J}(\mathbf{x},t)e^{i\omega(t-\mathbf{n}\cdot\mathbf{x}/c)}\right|^2.$$

For $\mathbf{J} = c\nabla\times\boldsymbol{\mathcal{M}}$, integration by parts puts the curl on the spatial exponential and gives

$$\int\mathbf{J}e^{-i(\omega/c)\mathbf{n}\cdot\mathbf{x}}d^3x = i\omega\mathbf{n}\times\boldsymbol{\mu}(t)e^{-i(\omega/c)\mathbf{n}\cdot\mathbf{r}(t)}.$$

The outer cross product projects on the transverse plane. Since $|\mathbf{n}\times(\mathbf{n}\times\mathbf{u})| = |\mathbf{n}\times\mathbf{u}|$, its squared norm produces

$$\boxed{\frac{d^2I_{\rm mag}}{d\omega d\Omega} = \frac{\omega^4}{4\pi^2c^3}\left|\int\mathbf{n}\times\boldsymbol{\mu}(t)e^{i\omega[t-\mathbf{n}\cdot\mathbf{r}(t)/c]}dt\right|^2.}$$

This uses the prescribed magnetization current of the problem.

(b) Define $S_T(\Omega) = 2\sin(\Omega T/2)/\Omega$. A rectangular observation window of the long sinusoidal motion has

$$\widetilde{\mu}_x = \frac{\mu_0}{2i}[S_T(\omega+\omega_0) - S_T(\omega-\omega_0)], \qquad \widetilde{\mu}_z = \frac{\mu_0}{2}[S_T(\omega+\omega_0) + S_T(\omega-\omega_0)].$$

Near the positive-frequency line, $\omega_0 T \gg 1$ makes the resonant $S_T(\omega-\omega_0)$ dominate. The vector is then $(\mu_0/2)S_T(\omega-\omega_0)(i\hat{\mathbf{x}}+\hat{\mathbf{z}})$, whose transverse squared norm is

$$\frac{\mu_0^2}{4}S_T^2[2-|in_x+n_z|^2]=\frac{\mu_0^2}{4}S_T^2(1+n_y^2).$$

Thus the angular pattern is $1+\sin^2\theta\sin^2\phi$. The identity $S_T(\Omega)^2/T\to 2\pi\delta(\Omega)$ gives

$$\frac{d\overline{P}}{d\Omega}=\frac{\mu_0^2\omega_0^4}{8\pi c^3}(1+n_y^2).$$

Since $\int(1+n_y^2)d\Omega=16\pi/3$,

$$\boxed{\overline{P}=\frac{2\mu_0^2\omega_0^4}{3c^3}.}$$

It agrees with the magnetic dipole Larmor formula $P=2|\ddot{\boldsymbol{\mu}}|^2/(3c^3)$, because the precessing vector has $|\ddot{\boldsymbol{\mu}}|^2=\mu_0^2\omega_0^4$. The physical turn-on and turn-off must be smooth. Interpreting a sharp rectangular window as an actual instantaneous creation or destruction of the dipole would give divergent high-frequency energy. The window above extracts the long-duration spectral line; smooth switching gives finite transient energy whose contribution to mean power vanishes as $T\to\infty$.

14.20 Apply part a of Problem 14.19 to the radiation emitted by a magnetic moment at the origin flipping from pointing down to pointing up, with components,

$$\mu_z=\mu_0\tanh(\nu t),\qquad \mu_x=\mu_0\,\mathrm{sech}(\nu t),\qquad \mu_y=0$$

where ν^{-1} is characteristic of the time taken to flip.

(a) Find the angular distribution of radiation and show that the intensity per unit frequency interval is

$$\frac{dI_{\mathrm{mag}}}{dx}=\frac{4}{3}\left(\frac{\nu}{c}\right)^3\mu_0^2\,\{16(x/\pi)^4[\mathrm{cosech}^2x+\mathrm{sech}^2x]\}$$

where $x=\pi\omega/2\nu$ is a dimensionless frequency variable and the quantity in curly brackets is the normalized frequency distribution in x. Make a plot of this distribution and find the mean value of ω in units of ν.

(b) Apply the method of Problem 9.7 to calculate the instantaneous power and total energy radiated by the flipping dipole. Compare with the answer in part a.

Solution.

(a) Use the Fourier convention $\widetilde{f}(\omega)=\int f(t)e^{i\omega t}dt$, and let $x=\pi\omega/(2\nu)>0$. To evaluate the transforms, put $u=e^{2\nu t}$ and $y=\omega/(2\nu)$, so $dt=du/(2\nu u)$. The beta integral $\int_0^\infty u^{a-1}(1+u)^{-a-b}du=\Gamma(a)\Gamma(b)/\Gamma(a+b)$ gives

$$\int_{-\infty}^{\infty}\mathrm{sech}(\nu t)e^{i\omega t}dt=\frac{1}{\nu}\int_0^\infty\frac{u^{-1/2+iy}}{1+u}du=\frac{\Gamma(1/2+iy)\Gamma(1/2-iy)}{\nu}=\frac{\pi}{\nu\cosh\pi y},$$
$$\int_{-\infty}^{\infty}\mathrm{sech}^2(\nu t)e^{i\omega t}dt=\frac{2}{\nu}\int_0^\infty\frac{u^{iy}}{(1+u)^2}du=\frac{2\Gamma(1+iy)\Gamma(1-iy)}{\nu}=\frac{2\pi y}{\nu\sinh\pi y}.$$

The final equalities use $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$ and $\Gamma(1+iy)=iy\Gamma(iy)$. Thus the transforms needed are

$$\widetilde{\mu}_x=\frac{\pi\mu_0}{\nu}\,\mathrm{sech}\,x,\qquad \widetilde{\mu}_z=\frac{i\pi\mu_0}{\nu}\,\mathrm{csch}\,x.$$

The second follows without assigning an ordinary convergent integral to $t \mapsto \tanh \nu t$: transform $\dot{\mu}_z = \mu_0 \nu \operatorname{sech}^2 \nu t$, whose transform is $\pi \mu_0 \omega \operatorname{csch} x/\nu$, and divide by $-i\omega$. Any zero-frequency distribution disappears after multiplication by ω^4 in the radiation formula. The two components are in quadrature, so their interference has zero real part. With $n_x = \sin\theta\cos\phi$, $n_z = \cos\theta$,

$$\boxed{\frac{d^2 I}{d\omega d\Omega} = \frac{\mu_0^2 \omega^4}{4c^3 \nu^2}[(1-n_x^2)\operatorname{sech}^2 x + (1-n_z^2)\operatorname{csch}^2 x].}$$

Both angular factors integrate to $8\pi/3$. Multiplication by $d\omega/dx = 2\nu/\pi$ then gives

$$\frac{dI}{dx} = \frac{4}{3}\left(\frac{\nu}{c}\right)^3 \mu_0^2 F(x), \qquad F(x) = \frac{16x^4}{\pi^4}[\operatorname{csch}^2 x + \operatorname{sech}^2 x].$$

For explicit normalization and the mean frequency, expand

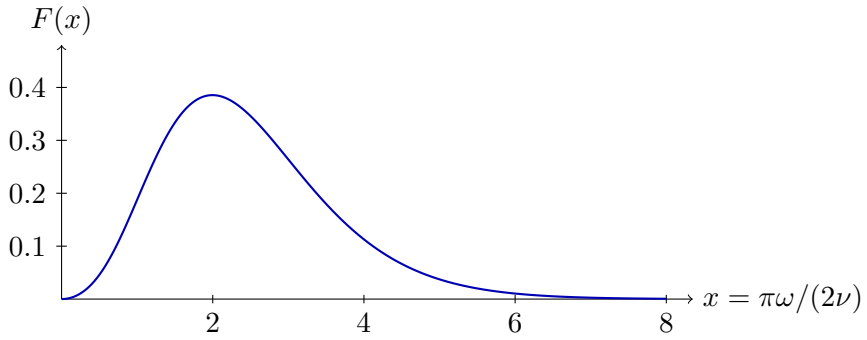
$$\operatorname{csch}^2 x + \operatorname{sech}^2 x = 8 \sum_{\substack{j \geq 1 \\ j \text{ odd}}} j e^{-2jx}.$$

For $s > 1$, termwise integration, using $\int_0^\infty x^s e^{-2jx} dx = \Gamma(s+1)/(2j)^{s+1}$, gives

$$\int_0^\infty x^s [\operatorname{csch}^2 x + \operatorname{sech}^2 x] dx = 2^{2-s}\Gamma(s+1)(1-2^{-s})\zeta(s).$$

For $s = 4$ this is $\pi^4/16$, hence $\int F dx = 1$; for $s = 5$ it is $465\zeta(5)/32$. Therefore

$$\boxed{\frac{\langle\omega\rangle}{\nu} = \frac{2}{\pi}\int_0^\infty x F(x) dx = \frac{465\zeta(5)}{\pi^5} = 1.57562219.}$$



The frequency-integrated angular distribution, obtained either from these transforms or part (b), is

$$\boxed{\frac{dI}{d\Omega} = \frac{\mu_0^2 \nu^3}{60\pi c^3}[14(1-n_x^2) + 16(1-n_z^2)].}$$

(b) Put $s = \operatorname{sech}(\nu t)$ and $h = \tanh(\nu t)$, so $s^2 + h^2 = 1$. Differentiating the specified components twice gives

$$\ddot{\mu}_x = \mu_0 \nu^2 s(1-2s^2), \qquad \ddot{\mu}_z = -2\mu_0 \nu^2 s^2 h.$$

Their squared sum simplifies without a time average:

$$|\ddot{\boldsymbol{\mu}}|^2 = \mu_0^2 \nu^4 [s^2(1-2s^2)^2 + 4s^4(1-s^2)] = \mu_0^2 \nu^4 s^2.$$

The instantaneous and integrated powers are thus

$$\boxed{P(t) = \frac{2\mu_0^2\nu^4}{3c^3}\,\mathrm{sech}^2\,\nu t, \qquad I = \frac{4\mu_0^2\nu^3}{3c^3}.}$$

For the angular integral quoted above, the cross term is odd in t. The two squared-derivative integrals are

$$\int \ddot{\mu}_x^2 dt = \mu_0^2\nu^3\left(2 - 4\frac{4}{3} + 4\frac{16}{15}\right) = \frac{14}{15}\mu_0^2\nu^3,$$

$$\int \ddot{\mu}_z^2 dt = 4\mu_0^2\nu^3\left(\frac{4}{3} - \frac{16}{15}\right) = \frac{16}{15}\mu_0^2\nu^3.$$

These use $h = \tanh u$, $dh = \mathrm{sech}^2\,u\,du$:

$$\int_{-\infty}^{\infty} \mathrm{sech}^{2j}\,u\,du = \int_{-1}^{1}(1-h^2)^{j-1}dh = \begin{cases} 2, & j=1,\\ 2(1-1/3) = 4/3, & j=2,\\ 2(1-2/3+1/5) = 16/15, & j=3. \end{cases}$$

Substitution in $dP/d\Omega = |\mathbf{n}\times\ddot{\boldsymbol{\mu}}|^2/(4\pi c^3)$ proves the angular result and its agreement with the spectral integral.

14.21 Bohr's correspondence principle states that in the limit of large quantum numbers the classical power radiated in the fundamental is equal to the product of the quantum energy ($\hbar\omega_0$) and the reciprocal mean lifetime of the transition from principal quantum number n to $(n-1)$.

(a) Using nonrelativistic approximations, show that in a hydrogen-like atom the transition probability (reciprocal mean lifetime) for a transition from a circular orbit of principal quantum number n to $(n-1)$ is given classically by

$$\frac{1}{\tau} = \frac{2}{3}\frac{e^2}{\hbar c}\left(\frac{Ze^2}{\hbar c}\right)^4 \frac{mc^2}{\hbar}\frac{1}{n^5}$$

(b) For hydrogen compare the classical value from part a with the correct quantum-mechanical results for the mean lives of the transitions $2p \to 1s$ (1.6×10^{-9} s), $4f \to 3d$ (7.3×10^{-8} s), $6h \to 5g$ (6.1×10^{-7} s).

Solution.

(a) In a circular Coulomb orbit, $mvr = n\hbar$ and $mv^2/r = Ze^2/r^2$. Solving these two equations gives

$$r_n = \frac{n^2\hbar^2}{mZe^2}, \quad v_n = \frac{Ze^2}{n\hbar} = \frac{Z\alpha c}{n}, \quad \omega_0 = \frac{v_n}{r_n} = \frac{mZ^2\alpha^2c^2}{\hbar n^3}.$$

The nonrelativistic circular power is $P = 2e^2\omega_0^2v_n^2/(3c^3)$ and is in the fundamental to this accuracy. Dividing by $\hbar\omega_0$ yields

$$\boxed{\tau^{-1} = \frac{P}{\hbar\omega_0} = \frac{2}{3}\alpha(Z\alpha)^4\frac{mc^2}{\hbar n^5}.}$$

The use of the classical orbital frequency for the quantum transition energy is the large-n correspondence approximation.

(b) For hydrogen, $mc^2/\hbar = c/(\alpha a_0)$, hence $\tau = 3n^5a_0/(2\alpha^4c)$. With the same constants as 9.10:

Transition	Classical τ (s)	Given quantum τ (s)	Ratio
$2p \to 1s$	2.9879×10^{-9}	1.6×10^{-9}	1.8674
$4f \to 3d$	9.5612×10^{-8}	7.3×10^{-8}	1.3097
$6h \to 5g$	7.2605×10^{-7}	6.1×10^{-7}	1.1902

The relative discrepancy decreases with n, consistent with the domain of the correspondence principle. This comparison uses the rounded quantum lifetimes supplied in the question, not an asserted exact small-n equivalence.

14.22 Periodic motion of charges gives rise to a discrete frequency spectrum in multiples of the basic frequency of the motion. Appreciable radiation in multiples of the fundamental can occur because of relativistic effects (Problems 14.14 and 14.15) even though the components of velocity are truly *sinusoidal*, or it can occur if the components of velocity are not sinusoidal, even though periodic. An example of this latter motion is an electron undergoing nonrelativistic elliptic motion in a hydrogen atom.

The orbit can be specified by the parametric equations

$$x = a(\cos u - \epsilon)$$
$$y = a\sqrt{1-\epsilon^2}\sin u$$

where

$$\omega_0 t = u - \epsilon \sin u$$

a is the semimajor axis, ϵ is the eccentricity, ω_0 is the orbital frequency, and u is an angle related to the polar angle θ of the particle by $\tan(u/2) = \sqrt{(1-\epsilon)/(1+\epsilon)}\tan(\theta/2)$. In terms of the binding energy B and the angular momentum L, the various constants are

$$a = \frac{e^2}{2B}, \qquad \epsilon = \sqrt{1 - \frac{2BL^2}{me^4}}, \qquad \omega_0^2 = \frac{8B^3}{me^4}$$

(a) Show that the power radiated in the kth multiple of ω_0 is

$$P_k = \frac{4e^2}{3c^3}(k\omega_0)^4 a^2 \left\{ \frac{1}{k^2}\left[(J_k'(k\epsilon))^2 + \left(\frac{1-\epsilon^2}{\epsilon^2}\right) J_k^2(k\epsilon) \right] \right\}$$

where $J_k(x)$ is a Bessel function of order k.

(b) Verify that for circular orbits the general result above agrees with part a of Problem 14.21.

Solution.

(a) Let $M = \omega_0 t = u - \epsilon \sin u$. For a real coordinate the positive-frequency phasor is $\mathbf{r}_k = \pi^{-1}\int_0^{2\pi} \mathbf{r}(u)(1-\epsilon\cos u)e^{ik(u-\epsilon\sin u)}du$. Integration by parts transfers the factor $(1-\epsilon\cos u)e^{ikM} = (ik)^{-1}d(e^{ikM})/du$ to $\mathbf{r}$; the endpoint term vanishes by periodicity. The basic integral is

$$I_k = \int_0^{2\pi} e^{ik(u-\epsilon\sin u)}du = 2\pi J_k(k\epsilon).$$

Differentiation with respect to ϵ gives $\int \sin u\, e^{ikM}du = 2\pi i J_k'(k\epsilon)$. The integral of $d(e^{ikM})/du$ also gives $\int \cos u\, e^{ikM}du = I_k/\epsilon$. Using $x' = -a\sin u$ and $y' = a\sqrt{1-\epsilon^2}\cos u$ therefore yields

$$\boxed{x_k = \frac{2a}{k}J_k'(k\epsilon), \qquad y_k = \frac{2ia\sqrt{1-\epsilon^2}}{k\epsilon}J_k(k\epsilon).}$$

The constant displacement in x contributes only a nonradiating mean. The time-averaged Larmor power of each phasor is $P_k = e^2(k\omega_0)^4(|x_k|^2 + |y_k|^2)/(3c^3)$. Inserting the two amplitudes gives exactly the requested expression.

(b) As $\epsilon \to 0$, $J_1(\epsilon)/\epsilon \to 1/2$ and $J_1'(\epsilon) \to 1/2$, so $(x_1, y_1) = (a, ia)$. For $k > 1$, both terms tend to zero with positive powers of ϵ. Thus $P_1 = 2e^2\omega_0^4 a^2/(3c^3)$ and higher harmonics vanish. With the circular Bohr radius and frequency of 14.21, $P_1/(\hbar\omega_0)$ is precisely its transition rate for hydrogen.

14.23 Instead of a single charge e moving with constant velocity $\omega_0 R$ in a circular path of radius R, as in Problem 14.15, N charges q_j move with fixed relative positions θ_j around the same circle.

(a) Show that the power radiated into the mth multiple of ω_0 is

$$\frac{dP_m(N)}{d\Omega} = \frac{dP_m(1)}{d\Omega} F_m(N)$$

where $dP_m(1)/d\Omega$ is the result of part a in Problem 14.15 with $e \to 1$, and

$$F_m(N) = \left| \sum_{j=1}^{N} q_j e^{im\theta_j} \right|^2$$

(b) Show that, if the charges are all equal in magnitude and sign and uniformly spaced around the circle, energy is radiated only into multiples of $N\omega_0$, but with an intensity N^2 times that for a single charge. Give a qualitative explanation of these facts.

(c) For the situation of part b, without detailed calculations show that for nonrelativistic motion the dependence on N of the total power radiated is dominantly as β^{2N}, so that in the limit $N \to \infty$ no radiation is emitted.

(d) By arguments like those of part c show that for N relativistic particles of equal charge and symmetrically arrayed, the radiated power varies with N mainly as $e^{-2N/3\gamma^3}$ for $N \gg \gamma^3$, to leading ultrarelativistic order in the exponential decay rate, so that again in the limit $N \to \infty$ no radiation is emitted.

(e) What relevance have the results of parts c and d to the radiation properties of a steady current in a loop?

Solution.

(a) The jth trajectory is a time shift of the reference one, $\mathbf{r}_j(t) = \mathbf{r}_1(t + \theta_j/\omega_0)$. Changing integration variable in the one-period amplitude of 14.13 therefore multiplies it by $q_j e^{-im\theta_j}$. Coherent addition before squaring gives

$$\frac{dP_m(N)}{d\Omega} = \frac{dP_m(1)}{d\Omega}\bigg|_{e=1} \left| \sum_j q_j e^{-im\theta_j} \right|^2 .$$

For real charges the modulus is unchanged by complex conjugation, so this is the stated $F_m(N)$ with its positive phase convention.

(b) The charges must have the same sign as well as magnitude; otherwise the asserted selection rule is false. For example two opposite charges half a circle apart radiate in odd harmonics. For $q_j = q$ and $\theta_j = \theta_* + 2\pi j/N$,

$$\sum_{j=0}^{N-1} e^{2\pi i m j/N} = \begin{cases} N, & m \equiv 0 \pmod N, \\ (1 - e^{2\pi i m})/(1 - e^{2\pi i m/N}) = 0, & \text{otherwise.} \end{cases}$$

Thus $F_m = N^2 q^2$ for $m = N, 2N, \ldots$ and is zero otherwise. After time T/N the charge distribution is identical, which explains the missing lower harmonics. At an allowed harmonic all amplitudes add in phase, explaining the square N^2.

(c) At fixed harmonic order, the nonrelativistic expansion in 14.15 gives $P_m = O(\beta^{2m})$ for fixed orbital frequency. The first allowed $m = N$ therefore starts at β^{2N}; higher allowed orders carry further powers. Applying Stirling's formula to this leading small-β term gives $(e\beta/2)^{2N}$ times algebraic factors, but that expansion is not uniform when $N\beta^2$ becomes large. For the actual limit $N \to \infty$ at fixed $0 < \beta < 1$, put $\beta = \operatorname{sech}\eta$ with $\eta > 0$. In the Bessel integral the saddle is at $u = i\eta$: $1 - \beta\cos u = 0$, and

$$iN(u - \beta\sin u) = -N(\eta - \tanh\eta) - \tfrac{1}{2}N\tanh\eta\,(u - i\eta)^2 + \cdots .$$

Its Gaussian integral gives $J_N(N\beta) \sim e^{-N(\eta - \tanh\eta)}/\sqrt{2\pi N\tanh\eta}$. The exponent is strictly negative because $\eta - \tanh\eta = \int_0^\eta \tanh^2 s\, ds > 0$. Arguments $N\beta\sin\theta$ away from the orbital plane only increase the suppression; derivatives of J_N change algebraic prefactors. The statement "dominantly as β^{2N}" describes the leading small-β power; it is not an exact large-N formula at arbitrary fixed speed. For small fixed β the exponential suppression beats the N^2 coherence enhancement and the radiation tends to zero.

(d) For a fixed finite $\gamma > 1$, the large-N result of part (c) gives the exact leading exponential factor in the power,

$$\exp[-2N(\eta - \tanh\eta)] = \exp\left[-2N\left(\operatorname{artanh}\frac{1}{\gamma} - \frac{1}{\gamma}\right)\right],$$

because $\tanh\eta = \sqrt{1 - \beta^2} = 1/\gamma$. Angular integration around the orbital plane and the N^2 coherence factor change algebraic prefactors, not this exponential rate. Expanding $\operatorname{artanh} u = u + u^3/3 + u^5/5 + \cdots$ gives

$$2N\left(\operatorname{artanh}\frac{1}{\gamma} - \frac{1}{\gamma}\right) = \frac{2N}{3\gamma^3} + \frac{2N}{5\gamma^5} + O(N/\gamma^7).$$

The first term is the requested ultrarelativistic suppression,

$$\boxed{\exp[-2N/(3\gamma^3)] \qquad (N \gg \gamma^3),}$$

consistent with $\omega_c/\omega_0 = 3\gamma^3/2$ and the $e^{-\omega/\omega_c}$ synchrotron tail. It approximates the *exponential rate* at large γ; its relative error in the power need not remain small for arbitrarily large N at fixed γ. In particular the omitted exponential correction is small only if $N/\gamma^5 \ll 1$. For the strict $N \to \infty$ limit keep the full positive rate above. The next allowed harmonic has twice that exponent, so the first allowed harmonic dominates and the total again vanishes.

(e) A continuum of uniformly distributed moving charge is a steady current and a time-independent charge distribution. Its fields are static and it has no radiation. The discrete calculations approach that result through coherent cancellation, even though each particle separately accelerates and radiates. Adding individual particle powers would discard the interference essential to a steady current.

14.24 As an idealization of steady-state currents flowing in a circuit, consider a system of N identical charges q moving with constant *speed* v (but subject to accelerations) in an arbitrary closed path. Successive charges are separated by a constant small interval Δ.

Starting with the Liénard–Wiechert potentials for each particle, and making no assumptions concerning the speed v relative to the velocity of light show that, in the limit $N \to \infty$, $q \to 0$, and $\Delta \to 0$, but $Nq =$ constant and $q/\Delta =$ constant, no radiation is emitted by the system and the electric and magnetic fields of the system are the usual static values.

(In a real wire there are the stationary positive ions, possible static surface charge, and even a tiny volume charge density from an internal Hall effect.)

Solution.

Let s be arc length around the closed curve of length L, let $\hat{\mathbf{t}}(s)$ be its tangent, and let $\lambda = q/\Delta = Nq/L$ be the limiting uniform line charge density. The current is $I = \lambda v$. We work at an observation point off the ideal filament, so all source distances are nonzero.

At fixed observation time label each particle by the arc coordinate s_0 it would have at that time. Its retarded coordinate s satisfies

$$s_0 = s + vR(s)/c \pmod{L}, \qquad R(s) = |\mathbf{x} - \mathbf{r}(s)|.$$

As $dR/ds = -\mathbf{n} \cdot \hat{\mathbf{t}}$,

$$\boxed{ds_0 = [1 - \boldsymbol{\beta}(s) \cdot \mathbf{n}(s)]ds = \kappa(s)ds, \qquad \boldsymbol{\beta}(s) = \frac{v}{c}\hat{\mathbf{t}}(s).}$$

For $v < c$, $\kappa > 0$, so this is a nonsingular change of variable around the entire curve. The sum of Liénard–Wiechert potentials becomes a Riemann integral over s_0:

$$\Phi(\mathbf{x}, t) = \lim \sum_j \frac{q}{\kappa_j R_j} = \lambda \oint \frac{ds_0}{\kappa R} = \boxed{\lambda \oint \frac{ds}{|\mathbf{x} - \mathbf{r}(s)|}},$$

$$\mathbf{A}(\mathbf{x}, t) = \lim \sum_j \frac{q\boldsymbol{\beta}_j}{\kappa_j R_j} = \lambda \oint \frac{\boldsymbol{\beta}(s)ds}{R(s)} = \boxed{\frac{I}{c} \oint \frac{d\boldsymbol{\ell}}{|\mathbf{x} - \mathbf{r}(s)|}}.$$

The retardation Jacobian cancels the apparent velocity enhancement exactly, without a small-v/c expansion. Both integrals are independent of time. Differentiating them gives

$$\boxed{\mathbf{E} = \lambda \oint \frac{\mathbf{x} - \mathbf{r}(s)}{|\mathbf{x} - \mathbf{r}(s)|^3} ds, \quad \mathbf{B} = \frac{I}{c} \oint \frac{d\boldsymbol{\ell} \times [\mathbf{x} - \mathbf{r}(s)]}{|\mathbf{x} - \mathbf{r}(s)|^3}.}$$

These are the ordinary electrostatic line-charge and magnetostatic Biot–Savart fields. Their sources and fields are stationary, so no outgoing radiation term remains. Stationary background charges can be added by superposition; they do not change this cancellation.

14.25

(a) Within the framework of approximations of Section 14.6, show that, for a relativistic particle moving in a path with instantaneous radius of curvature ρ, the frequency-angle spectra of radiations with positive and negative helicity are

$$\frac{d^2 I_\pm}{d\omega\, d\Omega} = \frac{e^2}{6\pi^2 c} \left(\frac{\omega\rho}{c}\right)^2 \left(\frac{1}{\gamma^2} + \theta^2\right)^2 \left| K_{2/3}(\xi) \pm \frac{\theta}{\left(\frac{1}{\gamma^2} + \theta^2\right)^{1/2}} K_{1/3}(\xi) \right|^2$$

(b) From the formulas of Section 14.6 and part a above, discuss the polarization of the total radiation emitted as a function of frequency and angle. In particular, determine the state of polarization at (1) high frequencies ($\omega > \omega_c$) for all angles, (2) intermediate and low frequencies ($\omega < \omega_c$) for large angles, (3) intermediate and low frequencies at very small angles.

(c) See the paper by P. Joos, *Phys. Rev. Letters*, **4**, 558 (1960), for experimental comparison. See also *Handbook on Synchrotron Radiation*, (*op. cit.*), Vol. 1A, p. 139.

Solution.

(a) Let $A = \gamma^{-2} + \theta^2$ and $q = \theta/\sqrt{A}$. Choose the ordered linear basis of (14.71), $\boldsymbol{\epsilon}_1 = \boldsymbol{\epsilon}_\parallel$ and $\boldsymbol{\epsilon}_2 = \boldsymbol{\epsilon}_\perp = \mathbf{n} \times \boldsymbol{\epsilon}_\parallel$; thus $\boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2 = \mathbf{n}$. The odd and even Airy integrals in (14.77)–(14.78), with $C = 2\rho A/(\sqrt{3}c)$, give

$$A_\parallel = iCK_{2/3}(\xi), \qquad A_\perp = CqK_{1/3}(\xi),$$

so the vector in (14.74) is

$$-\boldsymbol{\epsilon}_1 A_\parallel + \boldsymbol{\epsilon}_2 A_\perp = -iC[\boldsymbol{\epsilon}_1 K_{2/3}(\xi) + i\boldsymbol{\epsilon}_2 q K_{1/3}(\xi)].$$

Removing the common phase $-i$ therefore gives the Jones vector $(K_{2/3}(\xi),\, iqK_{1/3}(\xi))$. Projection on the helicity basis $\boldsymbol{\epsilon}_\pm = (\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)/\sqrt{2}$ gives amplitudes $[K_{2/3} \pm qK_{1/3}]/\sqrt{2}$. The sum of their squared moduli is $K_{2/3}^2 + q^2 K_{1/3}^2$, the polarization-summed spectrum (14.79). The prefactor from (14.74) and the projection's $1/2$ is $e^2\omega^2 C^2/(8\pi^2 c) = e^2(\omega\rho/c)^2 A^2/(6\pi^2 c)$, giving

$$\boxed{\frac{d^2 I_\pm}{d\omega\, d\Omega} = \frac{e^2}{6\pi^2 c}\left(\frac{\omega\rho}{c}\right)^2 A^2 \left| K_{2/3}(\xi) \pm \frac{\theta}{\sqrt{A}} K_{1/3}(\xi)\right|^2 .}$$

Interchanging the names of the circular basis vectors interchanges $+$ and $-$; the measurable helicity asymmetry changes sign above and below the orbital plane.

(b) An exact description within this approximation is the signed ellipse ratio. Use Jackson's Stokes parameters (7.27): $s_0 = |E_1|^2 + |E_2|^2$, $s_1 = |E_1|^2 - |E_2|^2$, $s_2 = 2\,\mathrm{Re}(E_1^* E_2)$, and $s_3 = 2\,\mathrm{Im}(E_1^* E_2)$, where the same $e^{-i\omega t}$ convention and ordered linear basis are used as in part (a). Setting $E_2 = irE_1$ gives

$$r = \frac{\theta}{\sqrt{\gamma^{-2} + \theta^2}} \frac{K_{1/3}(\xi)}{K_{2/3}(\xi)}, \qquad \frac{s_1}{s_0} = \frac{1 - r^2}{1 + r^2}, \quad \frac{s_2}{s_0} = 0, \quad \frac{s_3}{s_0} = \frac{2r}{1 + r^2}.$$

Each resolved frequency and direction is fully polarized; the major axis is in the orbital plane, and $\theta = 0$ is exactly linear. For $\omega \gg \omega_c$, $\xi \gg 1$ for every angle and $K_{1/3}/K_{2/3} = 1 - 1/(6\xi) + O(\xi^{-2})$. Thus $r \simeq \theta/\sqrt{\gamma^{-2} + \theta^2}$: linear near the plane, elliptical for $|\theta| \sim 1/\gamma$, and nearly circular at $|\theta| \gg 1/\gamma$, where the intensity is exponentially weak.

At lower frequencies, "large angle" must be compared with the frequency-dependent cone, not only with $1/\gamma$. For $\xi \ll 1$,

$$\frac{K_{1/3}(\xi)}{K_{2/3}(\xi)} \simeq \frac{\Gamma(1/3)}{\Gamma(2/3)}(\xi/2)^{1/3}.$$

When $|\theta| \gg 1/\gamma$, $q \simeq \mathrm{sgn}\,\theta$: inside the broader low-frequency cone the polarization can still be predominantly linear; it becomes more elliptical as ξ grows, and nearly circular only in its $\xi \gg 1$ angular tail. The crossover angular scale is of order $(c/\omega\rho)^{1/3}$. For very small angles $|\theta| \ll 1/\gamma$ and $\omega \ll \omega_c$,

$$r \simeq \gamma\theta \frac{\Gamma(1/3)}{\Gamma(2/3)} \left(\frac{\omega}{4\omega_c}\right)^{1/3},$$

so the polarization is especially close to linear. These expressions also specify the intermediate-frequency cases without asserting a circular polarization where ξ is actually small.

(c) The cited experimental paper is P. Joos, "Measurement of the Polarization of Synchrotron Radiation," *Phys. Rev. Lett.* **4**, 558 (1960), DOI: 10.1103/PhysRevLett.4.558. Its publisher record verifies the reference. The direct comparison quantities from this solution are the two linear intensities in (b), or their ratio r^2, averaged over an experiment's angular and frequency acceptance. The full experimental curves were not available from the publisher's accessible record, so no numerical agreement with those curves is asserted here.

14.26 Consider the synchrotron radiation from the Crab nebula. Electrons with energies up to 10^{13} eV move in a magnetic field of the order of 10^{-4} gauss.

(a) For $E = 10^{13}$ eV, $B = 3 \times 10^{-4}$ gauss, calculate the orbit radius ρ, the fundamental frequency $\omega_0 = c/\rho$, and the critical frequency ω_c. What is the energy $\hbar\omega_c$ in keV?

(b) Show that for a relativistic electron of energy E in a constant magnetic field the power spectrum of synchrotron radiation can be written

$$P(E, \omega) = \text{const}\left(\frac{\omega}{E^2}\right)^{1/3} f\left(\frac{\omega}{\omega_c}\right)$$

where $f(x)$ is a cutoff function having the value unity at $x = 0$ and vanishing rapidly for $x \gg 1$ [e.g., $f \simeq \exp(-\omega/\omega_c)$], and $\omega_c = (3/2)(eB/mc)(E/mc^2)^2 \cos\theta$, where θ is the pitch angle of the helical path. Cf. Problem 14.17a.

(c) If electrons are distributed in energy according to the spectrum $N(E)\, dE \propto E^{-n}\, dE$, show that the synchrotron radiation has the power spectrum

$$\langle P(\omega) \rangle\, d\omega \propto \omega^{-\alpha}\, d\omega$$

where $\alpha = (n - 1)/2$.

(d) Observations on the radiofrequency and optical continuous spectrum from the Crab nebula show that on the frequency interval from $\omega \sim 10^8$ s^{-1} to $\omega \sim 6 \times 10^{15}$ s^{-1} the constant $\alpha \simeq 0.35$. At frequencies above 10^{18} s^{-1} the spectrum of radiation falls steeply with $\alpha \gtrsim 1.5$. Determine the index n for the electron-energy spectrum, as well as an upper cutoff for that spectrum. Is this cutoff consistent with the numbers of part a?

(e) The half-life of a particle emitting synchrotron radiation is defined as the time taken for it to lose one half of its initial energy. From the result of Problem 14.9b, find a formula for the half-life of an electron in years when B is given in milligauss and E in GeV. What is the half-life using the numbers from part a? How does this compare with the known lifetime of the Crab nebula? Must the energetic electrons be continually replenished? From what source?

Solution.

The numerical assumptions and spectral indices in this problem are used as the specified model inputs, rather than as a current broadband model of the Crab nebula.

(a) For a perpendicular ultrarelativistic orbit, $\rho = pc/(eB) \simeq E/(eB)$ in Gaussian units. Equivalently in SI, $\rho = E/(eBc)$, with $E = 10^{13}$ eV and $B = 3 \times 10^{-8}$ T. These give

$$\gamma = 1.95695 \times 10^7, \quad \boxed{\rho = 1.11188 \times 10^{14} \text{ cm}, \qquad \omega_0 = 2.69627 \times 10^{-4} \text{ s}^{-1},}$$

$$\boxed{\omega_c = \tfrac{3}{2}\gamma^3\omega_0 = 3.03106 \times 10^{18}\ \mathrm{s}^{-1}, \qquad \hbar\omega_c = 1.99508\ \mathrm{keV}.}$$

Angular frequencies are used throughout; no extra factor 2π is inserted when multiplying by $\hbar$.

(b) At fixed field and pitch angle the spectrum is a constant amplitude times $G(x)$, $x = \omega/\omega_c$, with $\omega_c = C_B E^2$ and $C_B = 3eB\cos\theta/(2m^3c^5)$ in Gaussian units. For $x \ll 1$, insert $K_{5/3}(t) \sim 2^{2/3}\Gamma(5/3)t^{-5/3}$ in the defining integral:

$$G(x) \sim x\, 2^{2/3}\Gamma(5/3)\frac{3}{2}x^{-2/3} = 2^{2/3}\Gamma(2/3)x^{1/3}.$$

Define $f(x) = G(x)/[2^{2/3}\Gamma(2/3)x^{1/3}]$. Then $f(0) = 1$, and $G(x) \sim \sqrt{\pi x/2}e^{-x}$ at large x gives $f(x) \propto x^{1/6}e^{-x}$. Absorbing fixed powers of C_B into a constant proves $P(E,\omega) = \mathrm{const}\,(\omega/E^2)^{1/3}f(\omega/\omega_c)$.

(c) Over the power-law electron interval,

$$\langle P(\omega)\rangle \propto \omega^{1/3}\int E^{-n-2/3}f[\omega/(C_B E^2)]dE.$$

With $x = \omega/(C_B E^2)$, $E = (\omega/C_B x)^{1/2}$ and $|dE| = \frac{1}{2}(\omega/C_B)^{1/2}x^{-3/2}dx$, this becomes

$$\langle P(\omega)\rangle \propto \omega^{(1-n)/2}\int_0^\infty x^{n/2-7/6}f(x)dx.$$

The integral is frequency independent and converges at zero for $n > 1/3$, while the high-x end is cut off exponentially. For a finite electron-energy interval the same slope holds away from its corresponding spectral endpoints. Thus $\boxed{\alpha = (n-1)/2}$.

(d) The supplied low-frequency index gives $n = 1 + 2(0.35) = 1.70$. If the high-frequency tail is itself a power law, its electron index would be $n \gtrsim 4$. A steepening alone does not determine a unique hard upper cutoff. If it is interpreted as a characteristic cutoff beginning at $\omega_c \sim 10^{18}\ \mathrm{s}^{-1}$, the field in (a) implies

$$\boxed{E_{\mathrm{cut}} \sim 10^{13}\sqrt{\frac{10^{18}}{3.03106 \times 10^{18}}}\ \mathrm{eV} = 5.74 \times 10^{12}\ \mathrm{eV}.}$$

This is consistent in scale with electrons reaching 10^{13} eV, whose critical frequency is a few times higher. The exponential kernel, pitch distribution, and an actual electron cutoff shape would be needed to infer a more precise upper energy from data.

(e) Write the ultrarelativistic loss law as $\dot{E} = -A_B E^2$. Its solution is $1/E(t) = 1/E_0 + A_B t$, so $E(t_{1/2}) = E_0/2$ gives

$$t_{1/2} = \frac{1}{A_B E_0} = \frac{3m_e^4c^7}{2e^4B^2E_0} \quad \text{(Gaussian, perpendicular orbit).}$$

Converting one milligauss to 10^{-7} T and one GeV to $10^9 e$ joules gives

$$\boxed{t_{1/2} = \frac{8.3532 \times 10^3}{B_{\mathrm{mG}}^2 E_{\mathrm{GeV}}}\ \mathrm{yr}.}$$

At $B_{\mathrm{mG}} = 0.3$, $E_{\mathrm{GeV}} = 10^4$, this is 9.2813 yr. For a fixed nonzero pitch the loss rate has the additional perpendicular fraction $\cos^2\theta$, so the lifetime is longer by its reciprocal. The Crab supernova was observed in 1054; its age is about 10^3 yr, much longer than this loss time. The energetic radiating

population therefore requires continuing acceleration or replenishment. The central rotation-powered pulsar supplies a relativistic wind, with particle acceleration in the nebula/wind termination region. The historical date and pulsar-wind interpretation are supported by NASA's "The Crab Nebula: Observations Through Time" (5 May 2020) and "NASA Satellites Find High-Energy Surprises in 'Constant' Crab Nebula" (2011). The acceleration interpretation is also described in NASA/CXC's "Historic Nebula Seen Like Never Before With NASA's IXPE" (2023). Source URLs are recorded with the supplementary calculation.

14.27 Consider the radiation emitted at twice the fundamental frequency in the average rest frame of an electron in the sinusoidal undulator of Sections 14.7.C and 14.7.D. The radiation is a coherent sum of $E1$ radiation from the $z'(t')$ motion and $E2$ radiation from the $x'(t')$ motion.

(a) Using the techniques and notation of Chapter 9, show that the radiation-zone magnetic induction is given to sufficient accuracy by

$$\mathbf{B} = \frac{-iek'^2 a}{8} \frac{K}{\sqrt{1+K^2/2}}\, \mathbf{n} \times [\hat{\mathbf{z}} - 4\hat{\mathbf{x}}(\mathbf{n}\cdot\hat{\mathbf{x}})]$$

where $k' = 2\bar{\gamma}k_0$, $\mathbf{n}$ is a unit vector in the direction of $\mathbf{k}'$, and a factor of $\exp[ik'(r'-ct')]/r'$ is understood.

(b) Show that the time-averaged radiated power in the average rest frame, summed over outgoing polarizations, can be written

$$\frac{dP'}{d\Omega'} = \frac{e^2 c}{8\pi} \frac{K^2}{(1+K^2/2)} \frac{a^2}{64} \cdot S'$$

where

$$S' = k_x'^4 + k_y'^4 + 18k_x'^2 k_y'^2 + 17k_x'^2 k_z'^2 + k_y'^2 k_z'^2 + 8k' k_x'^2 k_z'$$

(c) Using the invariance arguments in the text in going from (14.111) to (14.118), show that the laboratory frequency spectrum of the second harmonic is

$$\frac{dP_2}{d\nu} = \frac{3}{16} P_1 \frac{K^2}{(1+K^2/2)^2} \cdot \nu^2(10 - 21\nu + 20\nu^2 - 6\nu^3)$$

where $\nu = k/2\bar{\gamma}^2 k_0$ and P_1 is the power in the fundamental, (14.117). For the angular range $\eta_1 < \eta < \eta_2$, the minimum and maximum ν values are $\nu_{\min} = 2/(1+\eta_2)$ and $\nu_{\max} = 2/(1+\eta_1)$. What is the total power radiated in the second harmonic?

Solution.

Put $D = 1 + K^2/2$, $\bar{\gamma} = \gamma/\sqrt{D}$, and $\Omega' = \bar{\gamma} k_0 c$. In the weak-undulator expansion the orbit from Section 14.7.C is

$$x'(t') = a \sin \Omega' t', \qquad z'(t') = -b \sin 2\Omega' t', \quad b = \frac{Ka}{8\sqrt{D}}, \quad a = \frac{K}{\gamma k_0}, \quad k' = 2\bar{\gamma} k_0.$$

Only terms needed through this second-harmonic amplitude are retained; in particular $k'a = 2K/\sqrt{D}$.

(a) At frequency $2\Omega'$, the electric dipole phasor is $\mathbf{p} = -ieb\hat{\mathbf{z}}$. Since $x'^2 = (a^2/2)(1 - \cos 2\Omega' t')$, the electric quadrupole phasor is

$$Q = \frac{ea^2}{2} \operatorname{diag}(-2, 1, 1) = \frac{ea^2}{2}(\mathbf{1} - 3\hat{\mathbf{x}}\hat{\mathbf{x}}).$$

The Gaussian radiation fields of these moments are

$$\mathbf{B}_{E1} = k'^2 \mathbf{n}\times\mathbf{p} = -iek'^2 b\,\mathbf{n}\times\hat{\mathbf{z}}, \qquad \mathbf{B}_{E2} = -\frac{ik'^3}{6}\mathbf{n}\times(Q\mathbf{n}) = \frac{iek'^3 a^2}{4} n_x\,\mathbf{n}\times\hat{\mathbf{x}}.$$

A common $e^{ik'(r'-ct')}/r'$ is understood. Because $k'a^2/(4b) = 4$, their coherent sum is

$$\boxed{\mathbf{B} = -\frac{iek'^2 a}{8}\frac{K}{\sqrt{D}}\mathbf{n}\times(\hat{\mathbf{z}} - 4n_x\hat{\mathbf{x}}).}$$

The magnetic dipole and mixed xz quadrupole terms first occur at other harmonics at the neglected order for this calculation.

(b) Let $u = n_z$. Expanding the squared cross product gives

$$|\mathbf{n}\times(\hat{\mathbf{z}} - 4n_x\hat{\mathbf{x}})|^2 = 1 + 16n_x^2 - (u - 4n_x^2)^2 = 1 - u^2 + 16n_x^2 - 16n_x^4 + 8un_x^2.$$

Multiply by k'^4 and replace $1 = n_x^2 + n_y^2 + n_z^2$ in the quadratic terms. The result is

$$S' = k_x'^4 + k_y'^4 + 18k_x'^2k_y'^2 + 17k_x'^2k_z'^2 + k_y'^2k_z'^2 + 8k'k_x'^2k_z',$$

including the odd-in-k_z' interference term. The time average $c|\mathbf{B}|^2 r'^2/(8\pi)$ gives the stated $dP'/d\Omega' = e^2cK^2a^2S'/(512\pi D)$.

(c) For the boost to the laboratory, let $\delta = \bar{\gamma}(1+\bar{\beta}u)$. Photon energies multiply by δ, solid angles obey $d\Omega = \delta^{-2}d\Omega'$, and a fixed number of undulator periods occupies times related by $\Delta t = \bar{\gamma}\Delta t'$. Thus $dP/d\Omega = (\delta^3/\bar{\gamma})dP'/d\Omega'$. At second-harmonic rest frequency $k' = 2\bar{\gamma}k_0$,

$$\nu = \frac{k}{2\bar{\gamma}^2 k_0} = 1 + \bar{\beta}u \simeq 1 + u, \quad \frac{dP_2}{d\nu} \simeq \nu\int_0^{2\pi}\frac{dP'}{d\Omega'}(u = \nu - 1, \phi)d\phi.$$

This last relation includes both aberration and the frequency Jacobian $du = d\nu/\bar{\beta}$, with $\bar{\beta}$ set to one at the stated accuracy. For the angular factor in (b), $\langle n_x^2\rangle_\phi = (1-u^2)/2$ and $\langle n_x^4\rangle_\phi = 3(1-u^2)^2/8$. Hence

$$\langle S'/k'^4\rangle_\phi = (1-u^2)(3+4u+6u^2) = \nu(10 - 21\nu + 20\nu^2 - 6\nu^3).$$

Using $a^2 = K^2/(D\bar{\gamma}^2k_0^2)$ and $P_1 = e^2c\bar{\gamma}^2K^2k_0^2/3$, the remaining coefficient is

$$\frac{2\pi e^2cK^2a^2k'^4}{512\pi D} = \frac{3}{16}P_1\frac{K^2}{D^2}.$$

Therefore the requested spectrum follows for $0 < \nu < 2$. The aberration relation $u \simeq (1-\eta)/(1+\eta)$ gives $\nu = 2/(1+\eta)$ and the stated acceptance endpoints. For full angular acceptance,

$$\int_0^2 \nu^2(10 - 21\nu + 20\nu^2 - 6\nu^3)d\nu = \left[\frac{10\nu^3}{3} - \frac{21\nu^4}{4} + 4\nu^5 - \nu^6\right]_0^2 = \frac{20}{3}.$$

Thus

$$\boxed{P_2 = \frac{5}{4}P_1\frac{K^2}{(1+K^2/2)^2}.}$$

This is the retained-order second-harmonic result; the rational factors in the chosen notation do not turn a weak-undulator expansion into an all-K formula.

15 Chapter 15. Bremsstrahlung, Method of Virtual Quanta, Radiative Beta Processes

Problems

15.1 In radiative collision problems it is useful to have the radiation amplitude expressed explicitly as an integral involving the accelerations of the particles, as in (14.65), for example. In the nonrelativistic limit, particles do not move rapidly or far during the period of acceleration; only the lowest order velocity and retardation effects need be kept in an approximate description.

(a) Show that the integral (times e) in (14.65), which is, apart from an inessential phase, $\sqrt{2\pi}\,cR$ times the Fourier transform with the normalization of (14.54) of the Liénard–Wiechert electric field at distance R, can be expanded in inverse powers of c (remembering that $\beta = v/c$ and $k = \omega/c$) as follows:

$$\mathbf{J} = e\int dt\; e^{i\omega t}(\dot{\boldsymbol{\beta}}_\perp + \beta_\parallel\dot{\boldsymbol{\beta}}_\perp + \dot{\beta}_\parallel\boldsymbol{\beta}_\perp - ikr_\parallel\boldsymbol{\beta}_\perp + \cdots)$$

or

$$\mathbf{J} = e\int dt\; e^{i\omega t}\left(\dot{\boldsymbol{\beta}}_\perp + \frac{d}{dt}\,(\beta_\parallel\boldsymbol{\beta}_\perp) - ikr_\parallel\boldsymbol{\beta}_\perp + \cdots\right)$$

where $\beta_\parallel = \boldsymbol{\beta}\cdot\mathbf{n}$, $r_\parallel = \mathbf{r}\cdot\mathbf{n}$, and $\boldsymbol{\beta}_\perp = (\mathbf{n}\times\boldsymbol{\beta})\times\mathbf{n}$. Neglected terms are of order $1/c^3$ and higher.

(b) Show that the first term in part a corresponds to the electric dipole approximation, while the next terms are the magnetic dipole and the electric quadrupole contributions. [Some integrations by parts are required, with a convergence factor $e^{-\epsilon|t|}$ to give meaning to the integrals, as discussed following (14.67).] For a group of charges, show that the generalization of part a can be written as

$$\mathbf{J} = \frac{1}{c}\int dt\; e^{i\omega t}\left(\frac{d^2\mathbf{p}_\perp(t)}{dt^2} + \frac{d^2\mathbf{m}(t)}{dt^2}\times\mathbf{n} + \frac{1}{6c}\frac{d^3\mathbf{Q}_\perp}{dt^3}(\mathbf{n},t) + \cdots\right)$$

and the radiated intensity per unit solid angle and per unit frequency as

$$\frac{d^2I}{d\omega\;d\Omega} = \frac{1}{4\pi^2c^3}\left|\int dt\; e^{i\omega t}\left(\frac{d^2\mathbf{p}_\perp(t)}{dt^2} + \frac{d^2\mathbf{m}(t)}{dt^2}\times\mathbf{n} + \frac{1}{6c}\frac{d^3\mathbf{Q}_\perp}{dt^3}(\mathbf{n},t) + \cdots\right)\right|^2$$

where

$$\mathbf{p} = \sum_j q_j\mathbf{r}_j, \qquad \mathbf{m} = \sum_j q_j\frac{\mathbf{r}_j\times\boldsymbol{\beta}_j}{2}, \qquad Q_{\alpha\beta} = \sum_j q_j(3x_{j\alpha}x_{j\beta} - r_j^2\,\delta_{\alpha\beta})$$

and the vector $\mathbf{Q}(\mathbf{n})$ has components, $Q_\alpha = \sum_\beta Q_{\alpha\beta}n_\beta$. Relate to the treatment of multipole radiation in Sections 9.2 and 9.3 and Problem 9.7.

Solution.

Use Gaussian units and let $P = \mathbf{1} - \mathbf{n}\mathbf{n}^\mathsf{T}$ be the transverse projector. Frequencies below are positive.

(a) The identity (14.66), checked against (14.65), is

$$\frac{\mathbf{n}\times[(\mathbf{n}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}}]}{(1-\beta_\parallel)^2}=-\frac{d}{dt}\frac{\boldsymbol{\beta}_\perp}{1-\beta_\parallel}.$$

Thus the problem's $\mathbf{J}$ chooses an overall minus sign relative to the integral printed in (14.65); this is the stated inessential phase and has no effect on intensity. There is also a normalization qualification in the introductory description: Jackson's transform on p. 674, (14.54), contains $1/\sqrt{2\pi}$. Indeed the far-zone field and the observation-time Jacobian are

$$\mathbf{E}_{\rm rad}(t_{\rm obs})=\frac{e}{cR}\left[\frac{\mathbf{n}\times[(\mathbf{n}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}}]}{(1-\mathbf{n}\cdot\boldsymbol{\beta})^3}\right]_{\rm ret},\qquad dt_{\rm obs}=(1-\mathbf{n}\cdot\boldsymbol{\beta})\,dt.$$

Using $t_{\rm obs}=t+(R-\mathbf{n}\cdot\mathbf{r})/c$ therefore gives

$$\widetilde{\mathbf{E}}_{\rm rad}(\omega)=-\frac{e^{i\omega R/c}}{\sqrt{2\pi}\,cR}\mathbf{J}(\omega).$$

This explains the marked factor in the statement; the displayed definition of $\mathbf{J}$ and its intensity formula need no change. Expand the positive version of the acceleration integral:

$$\begin{aligned}\frac{d}{dt}\frac{\boldsymbol{\beta}_\perp}{1-\beta_\parallel}&=\frac{\dot{\boldsymbol{\beta}}_\perp}{1-\beta_\parallel}+\frac{\boldsymbol{\beta}_\perp\dot\beta_\parallel}{(1-\beta_\parallel)^2}\\&=\dot{\boldsymbol{\beta}}_\perp+\beta_\parallel\dot{\boldsymbol{\beta}}_\perp+\dot\beta_\parallel\boldsymbol{\beta}_\perp+O(c^{-3}),\\ e^{-ikr_\parallel}&=1-ikr_\parallel+O(c^{-2}).\end{aligned}$$

The leading acceleration is already $O(c^{-1})$. Multiplication therefore gives

$$\boxed{\mathbf{J}=e\int e^{i\omega t}[\dot{\boldsymbol{\beta}}_\perp+(\beta_\parallel\boldsymbol{\beta}_\perp)^{\cdot}-ikr_\parallel\dot{\boldsymbol{\beta}}_\perp]dt+O(c^{-3}).}$$

The expansion requires both $v/c\ll1$ and small source displacement in units of c/ω during the acceleration. It does not require an eternally unbounded straight trajectory to fit inside a wavelength.

(b) Put $b=r_\parallel$ and $\mathbf{a}=\mathbf{r}_\perp$ temporarily. The second-order integral is

$$\frac{e}{c}^2\int e^{i\omega t}[(\dot b\dot{\mathbf{a}})^{\cdot}-i\omega b\ddot{\mathbf{a}}]dt.$$

Insert $e^{-\epsilon|t|}$ and integrate by parts before taking $\epsilon\downarrow0$. For $\omega>0$ the boundary terms vanish in this regulated sense, and

$$\int e^{i\omega t}(-i\omega)b\ddot{\mathbf{a}}\,dt=\int e^{i\omega t}(\dot b\ddot{\mathbf{a}}+b\,\dddot{\mathbf{a}})dt.$$

The resulting integrand is $e[\ddot b\dot{\mathbf{a}}+2\dot b\ddot{\mathbf{a}}+b\,\dddot{\mathbf{a}}]/c^2$. For one charge the definitions in the question give

$$\mathbf{m}\times\mathbf{n}=\frac{e}{2c}(b\dot{\mathbf{a}}-\dot b\mathbf{a}),\qquad\mathbf{Q}_\perp=3eb\mathbf{a}.$$

Differentiate both products explicitly:

$$\begin{aligned}(b\dot{\mathbf{a}}-\dot b\mathbf{a})''&=b\,\dddot{\mathbf{a}}+\dot b\ddot{\mathbf{a}}-\ddot b\dot{\mathbf{a}}-\dddot b\,\mathbf{a},\\(b\mathbf{a})'''&=b\,\dddot{\mathbf{a}}+3\dot b\ddot{\mathbf{a}}+3\ddot b\dot{\mathbf{a}}+\dddot b\,\mathbf{a}.\end{aligned}$$

Their sum proves

$$\frac{1}{c}\left(\ddot{\mathbf{m}}\times\mathbf{n}+\frac{\dddot{\mathbf{Q}}_\perp}{6c}\right)=\frac{e}{c^2}(b\,\ddddot{\mathbf{a}}+2\dot{b}\dddot{\mathbf{a}}+\ddot{b}\ddot{\mathbf{a}}).$$

Also $e\dot{\boldsymbol{\beta}}_\perp=\ddot{\mathbf{p}}_\perp/c$. Adding the amplitudes of all charges, before squaring, yields

$$\boxed{\mathbf{J}=\frac{1}{c}\int e^{i\omega t}\left(\ddot{\mathbf{p}}_\perp+\ddot{\mathbf{m}}\times\mathbf{n}+\frac{\dddot{\mathbf{Q}}_\perp}{6c}\right)dt+O(c^{-3}),}$$

and $d^2I/(d\omega\,d\Omega)=|\mathbf{J}|^2/(4\pi^2c)$ gives the stated intensity. These are precisely the electric dipole, magnetic dipole and traceless electric quadrupole moments of Sections 9.2–9.3. For a Fourier component their derivatives supply $-\omega^2,-\omega^2,i\omega^3$, respectively; the last coefficient includes the $1/6$ appropriate to the definition $Q_{ij}=\sum q(3x_ix_j-r^2\delta_{ij})$. Their coherent sum also contains the interference terms discussed in Problem 9.7. No magnetic-monopole term occurs because the current is conserved.

15.2 A nonrelativistic particle of charge e and mass m collides with a fixed, smooth, hard sphere of radius R. Assuming that the collision is elastic, show that in the dipole approximation (neglecting retardation effects) the classical differential cross section for the emission of photons per unit solid angle per unit energy interval is

$$\frac{d^2\sigma}{d\Omega\,d(\hbar\omega)}=\frac{R^2}{12\pi}\frac{e^2}{\hbar c}\left(\frac{v}{c}\right)^2\frac{1}{\hbar\omega}\,(2+3\sin^2\theta)$$

where θ is measured relative to the incident direction. Sketch the angular distribution. Integrate over angles to get the total bremsstrahlung cross section. Qualitatively, what factor (or factors) govern the upper limit to the frequency spectrum?

Solution.

Let $\varepsilon=\hbar\omega$, $\alpha=e^2/(\hbar c)$, $\boldsymbol{\beta}=\beta\hat{\mathbf{z}}$ and let $\mathbf{u}$ be the outgoing velocity direction. If χ is the particle's scattering angle, specular reflection gives $b=R\cos(\chi/2)$. Consequently

$$d\sigma_{\rm el}=b|db|\,d\phi=\frac{R^2}{4}\sin\chi\,d\chi\,d\phi=\frac{R^2}{4}d\Omega_{\mathbf{u}}.$$

The soft dipole photon probability for this one collision is

$$\frac{d^2N}{d\varepsilon\,d\Omega}=\frac{\alpha\beta^2}{4\pi^2\varepsilon}|P(\mathbf{u}-\hat{\mathbf{z}})|^2,\qquad P=\mathbf{1}-\mathbf{n}\mathbf{n}^{\mathsf T}.$$

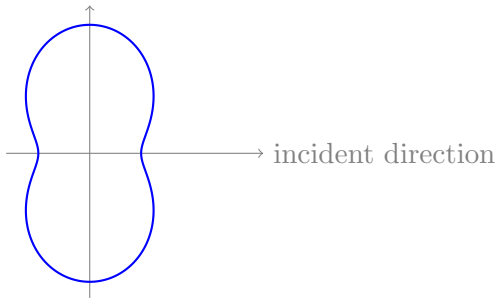
The average over outgoing directions uses $\langle u_i\rangle=0$, $\langle u_iu_j\rangle=\delta_{ij}/3$. It follows that

$$\langle|P\mathbf{u}|^2\rangle=\tfrac{2}{3},\qquad|P\hat{\mathbf{z}}|^2=\sin^2\theta,\qquad\langle P\mathbf{u}\cdot P\hat{\mathbf{z}}\rangle=0.$$

Multiplication by the total elastic area πR^2 gives

$$\boxed{\frac{d^2\sigma}{d\Omega\,d\varepsilon}=\frac{\alpha\beta^2R^2}{12\pi\varepsilon}(2+3\sin^2\theta).}$$

The distribution is independent of azimuth, symmetric forward and backward, and has maximum-to-minimum ratio $5/2$. The following polar sketch uses radius proportional to $2+3\sin^2\theta$ and a horizontal incident direction.



$r(\theta) \propto 2 + 3\sin^2\theta$

Because $\int d\Omega = 4\pi$ and $\int \sin^2\theta\, d\Omega = 8\pi/3$,

$$\boxed{\frac{d\sigma}{d\varepsilon} = \frac{4\alpha R^2\beta^2}{3\varepsilon}.}$$

On a photon-energy band $[\varepsilon_1, \varepsilon_2]$ this gives $\sigma = (4/3)\alpha R^2\beta^2 \ln(\varepsilon_2/\varepsilon_1)$. Its number integral has the familiar infrared divergence if the lower limit is set to zero, whereas its energy integral is finite. An ideal instantaneous hard collision has no classical high-frequency cutoff. For a real interaction, a collision duration τ produces loss of coherence for $\omega\tau \gtrsim 1$; the range over which the force changes sets τ. Energy conservation additionally requires $\varepsilon \le mv^2/2$ (a massive fixed scatterer absorbs momentum with negligible recoil energy). The dipole and soft formulas should be used below both scales, not extrapolated as exact spectra at their endpoint.

15.3 Treat Problem 15.2 without the assumption of nonrelativistic motion, using (15.2) and assuming the elastic impact is of negligible duration. Show that the cross section for photon emission is now

$$\frac{d^2\sigma}{d\Omega\; d(\hbar\omega)} = \frac{R^2}{4\pi}\frac{e^2}{\hbar c}\frac{\beta^2}{\hbar\omega}\left[\frac{\sin^2\theta}{(1-\beta\cos\theta)^2} + \frac{1}{\beta^3}\ln\left(\frac{1+\beta}{1-\beta}\right) - \frac{2}{\beta^2}\right]$$

Solution.

The elastic angular distribution remains $d\sigma_{\rm el} = (R^2/4)d\Omega_{\mathbf{u}}$: it is a geometrical consequence of specular reflection, independent of speed. Equation (15.2) gives

$$\frac{d^2\sigma}{d\Omega\, d\varepsilon} = \frac{\alpha R^2}{16\pi^2\varepsilon}\int d\Omega_{\mathbf{u}}\left|\frac{\beta P\mathbf{u}}{1-\beta\mathbf{n}\cdot\mathbf{u}} - \frac{\beta P\hat{\mathbf{z}}}{1-\beta\cos\theta}\right|^2.$$

The angular average of the first vector is zero: rotation about $\mathbf{n}$ leaves only a vector parallel to $\mathbf{n}$, and P removes that component. Thus the cross term vanishes. Set $t = \mathbf{n}\cdot\mathbf{u}$. The first squared term is

$$2\pi\beta^2\int_{-1}^{1}\frac{1-t^2}{(1-\beta t)^2}\,dt.$$

With $y = 1-\beta t$ its integral becomes

$$\begin{aligned}\int_{-1}^{1}\frac{1-t^2}{(1-\beta t)^2}\,dt &= \frac{1}{\beta^3}\int_{1-\beta}^{1+\beta}\left[\frac{\beta^2-1}{y^2} + \frac{2}{y} - 1\right]dy \\ &= \frac{2}{\beta^3}\ln\frac{1+\beta}{1-\beta} - \frac{4}{\beta^2}.\end{aligned}$$

The second squared term integrates to $4\pi\beta^2\sin^2\theta/(1-\beta\cos\theta)^2$. Therefore

$$\boxed{\frac{d^2\sigma}{d\Omega\,d\varepsilon} = \frac{\alpha R^2\beta^2}{4\pi\varepsilon}\left[\frac{\sin^2\theta}{(1-\beta\cos\theta)^2} + \frac{1}{\beta^3}\ln\frac{1+\beta}{1-\beta} - \frac{2}{\beta^2}\right].}$$

As a check, the apparent small-β singularity cancels because $\ln[(1+\beta)/(1-\beta)] = 2\beta + 2\beta^3/3 + \cdots$. The bracket becomes $\sin^2\theta + 2/3$, recovering Problem 15.2. Integration over the photon direction gives the useful further check

$$\frac{d\sigma}{d\varepsilon} = \frac{2\alpha R^2}{\varepsilon}\left[\frac{1}{\beta}\ln\frac{1+\beta}{1-\beta} - 2\right].$$

The instantaneous-impact assumption removes the classical duration cutoff; it does not remove the soft-photon restriction implicit in using an elastic trajectory at unchanged particle energy.

15.4 A group of charged particles with charges e_j and coordinates $\mathbf{r}_j(t)$ undergo interactions and are accelerated only during a time $-\tau/2 < t < \tau/2$, during which their velocities change from $c\boldsymbol{\beta}_j$ to $c\boldsymbol{\beta}'_j$.

(a) Show that for $\omega\tau \ll 1$ the intensity of radiation emitted with polarization $\boldsymbol{\epsilon}$ per unit solid angle and unit frequency interval is

$$\frac{d^2I}{d\omega\,d\Omega} = \frac{1}{4\pi^2 c}\,|\boldsymbol{\epsilon}^*\cdot\mathbf{E}|^2$$

where

$$\mathbf{E} = \sum_j e_j\left(\frac{\boldsymbol{\beta}'_j}{1-\mathbf{n}\cdot\boldsymbol{\beta}'_j} - \frac{\boldsymbol{\beta}_j}{1-\mathbf{n}\cdot\boldsymbol{\beta}_j}\right)e^{-i\omega\mathbf{n}\cdot\mathbf{r}_j(0)/c}$$

(b) An ω^0 meson of mass 784 MeV decays into $\pi^+\pi^-$ and e^+e^- with branching ratios of 1.3×10^{-2} and 8×10^{-5}, respectively. Show that for both decay modes the frequency spectrum of radiated energy at low frequencies is

$$\frac{dI}{d\omega} = \frac{e^2}{\pi c}\left[\left(\frac{1+\beta^2}{\beta}\right)\ln\left(\frac{1+\beta}{1-\beta}\right) - 2\right] \simeq \frac{4e^2}{\pi c}\left[\ln\left(\frac{M_\omega}{m}\right) - \frac{1}{2}\right]$$

where M_ω is the mass of the ω^0 meson and m is the mass of one of the decay products. Evaluate approximately the *total* energy radiated in each decay by integrating the spectrum up to the maximum frequency allowed kinematically. What fraction of the rest energy of the ω^0 is it in each decay?

Solution.

(a) For each charge use the derivative identity (14.66), with the common overall sign chosen as in Problem 15.1. During acceleration,

$$\omega[t - \mathbf{n}\cdot\mathbf{r}_j(t)/c] = -\omega\mathbf{n}\cdot\mathbf{r}_j(0)/c + O(\omega\tau),$$

since $|\mathbf{r}_j(t) - \mathbf{r}_j(0)| \le c|t|$. Thus the phase can be pulled out of its acceleration integral and

$$\int dt\,\frac{d}{dt}\frac{\boldsymbol{\beta}_j}{1-\mathbf{n}\cdot\boldsymbol{\beta}_j} = \frac{\boldsymbol{\beta}'_j}{1-\mathbf{n}\cdot\boldsymbol{\beta}'_j} - \frac{\boldsymbol{\beta}_j}{1-\mathbf{n}\cdot\boldsymbol{\beta}_j}.$$

Summing the amplitudes proves exactly the stated $\mathbf{E}$ and $d^2I/(d\omega\, d\Omega) = |\boldsymbol{\epsilon}^* \cdot \mathbf{E}|^2/(4\pi^2 c)$, to leading order in $\omega\tau$. The fixed positions need not be mutually close compared with a wavelength; their relative phases have deliberately been retained. Polarization vectors satisfy $\boldsymbol{\epsilon} \cdot \mathbf{n} = 0$.

(b) In the parent rest frame the two opposite charges are created at the same event with velocities $\pm c\beta\widehat{\mathbf{z}}$, where $\beta = \sqrt{1 - 4m^2/M_\omega^2}$. The neutral parent contributes no endpoint term, so

$$\mathbf{E} = \frac{2e\beta\widehat{\mathbf{z}}}{1 - \beta^2\cos^2\theta}, \qquad \frac{d^2I}{d\omega\, d\Omega} = \frac{e^2\beta^2}{\pi^2 c}\frac{1 - \cos^2\theta}{(1 - \beta^2\cos^2\theta)^2}.$$

To integrate, write $L = \ln[(1+\beta)/(1-\beta)]$ and use

$$\frac{1 - x^2}{(1 - \beta^2 x^2)^2} = \frac{\beta^2 - 1}{2\beta^2}\frac{d}{dx}\frac{x}{1 - \beta^2 x^2} + \frac{\beta^2 + 1}{2\beta^2}\frac{1}{1 - \beta^2 x^2}.$$

The two elementary endpoint integrals give

$$\int_{-1}^{1} \frac{1 - x^2}{(1 - \beta^2 x^2)^2} dx = -\frac{1}{\beta^2} + \frac{1 + \beta^2}{2\beta^3} L.$$

Multiplying by $2\pi e^2\beta^2/(\pi^2 c)$ proves

$$\boxed{\frac{dI}{d\omega} = \frac{e^2}{\pi c}\left[\frac{1 + \beta^2}{\beta} L - 2\right].}$$

For $m \ll M_\omega$, $1 - \beta \simeq 2m^2/M_\omega^2$, $L \simeq 2\ln(M_\omega/m)$ and $(1 + \beta^2)/\beta \simeq 2$, which gives the stated approximation.

For the requested endpoint estimate set $C = (1/\hbar)dI/d\omega = dI/d\varepsilon$. The remaining pair has invariant mass at least $2m$, and hence

$$(M_\omega c^2 - \varepsilon)^2 - \varepsilon^2 \geq (2mc^2)^2, \qquad \varepsilon_{\max} = \frac{M_\omega^2 - 4m^2}{2M_\omega} c^2.$$

Using this endpoint rather than $(M_\omega - 2m)c^2$ is necessary because the pair must balance the photon momentum. Holding the soft coefficient fixed gives $I_{\rm est} = C\varepsilon_{\max}$:

Mode	β	$\varepsilon_{\max}$ (MeV)	$I_{\rm est}$ (MeV)	$I_{\rm est}/M_\omega c^2$
$\pi^+\pi^-$	0.934439	342.285	3.8042	4.8523×10^{-3}
e^+e^-	0.99999915	391.999	24.897	3.1757×10^{-2}

Here $m_\pi c^2 = 139.6$ MeV, $m_e c^2 = 0.510999$ MeV and $\alpha^{-1} = 137.036$ were used. These are energies per decay in the indicated channel. Multiplication by the stated branching ratios gives contributions 0.04945 and 0.001992 MeV per parent decay averaged over all channels. The endpoint extrapolation is only the rough estimate requested: recoil changes the daughter velocities and phase space, so a constant soft spectrum is not a controlled prediction at $\varepsilon \sim \varepsilon_{\max}$.

15.5 A situation closely related to that of Problem 15.4b is the emission of radiation caused by the disappearance of charges and magnetic moments in the annihilation of electrons and positrons to form hadrons in high-energy storage ring experiments. If the differential cross section for the process $e^+e^- \to$ hadrons is $d\sigma_0$, *without* the emission of photons, calculate the cross section for the same process accompanied by a *soft* photon ($\hbar\omega \to 0$) in the energy interval $d(\hbar\omega)$ around $\hbar\omega$. Compare your results with the quantum-mechanical expressions:

$$\frac{d^2\sigma}{d\Omega\, d(\hbar\omega)} = \frac{\alpha}{\pi^2}\frac{d\sigma_0(s')}{\hbar\omega}\sqrt{\frac{s'^2-4m^2s'}{s^2-4m^2s}}\cdot\frac{\left[\beta^2\sin^2\theta+\dfrac{\hbar^2\omega^2}{s'+2m^2}\left(1-\beta^4\cos^4\theta\right)\right]}{(1-\beta^2\cos^2\theta)^2}$$

$$\frac{d\sigma}{d(\hbar\omega)} = \frac{4\alpha}{\pi}\frac{d\sigma_0(s')}{\hbar\omega}\sqrt{\frac{s'^2-4m^2s'}{s^2-4m^2s}}\left\{\frac{1}{2}\left[\frac{1+\beta^2}{2\beta}\ln\left(\frac{1+\beta}{1-\beta}\right)-1\right]\right.$$
$$\left.+\frac{\hbar^2\omega^2}{s'+2m^2}\left[\frac{1}{\beta}\ln\left(\frac{1+\beta}{1-\beta}\right)-1\right]\right\}$$

where $s=(p_1+p_2)^2$, $s'=(p_1+p_2-k)^2$, $\beta=$ electron v/c in c.m. frame. Neglect the emission of radiation by any of the hadrons, all assumed to be much heavier than the electrons.

The factors proportional to ω^2 in the numerators of these expressions can be attributed to the disappearance of the magnetic moments. If you have not included such contributions in your semiclassical calculation, you may consider doing so.

Solution.

Write $\varepsilon=\hbar\omega$ and use the center-of-mass frame. The incoming electron and positron have opposite velocities $\pm c\beta\hat{\mathbf{z}}$; their disappearance reverses the amplitude in Problem 15.4 but not its square. Under the stipulated neglect of hadron radiation, the probability of a soft photon factorizes from the nonradiative process:

$$\boxed{\frac{d^2\sigma}{d\Omega\, d\varepsilon} = d\sigma_0(s)\frac{\alpha}{\pi^2\varepsilon}\frac{\beta^2\sin^2\theta}{(1-\beta^2\cos^2\theta)^2}.}$$

This is a cross section still differential in whatever hadronic variables are represented by $d\sigma_0$. The angular integral already evaluated in Problem 15.4 gives

$$\boxed{\frac{d\sigma}{d\varepsilon} = d\sigma_0(s)\frac{\alpha}{\pi\varepsilon}\left[\frac{1+\beta^2}{\beta}\ln\frac{1+\beta}{1-\beta}-2\right].}$$

For ultrarelativistic incident beams this becomes $(4\alpha/\pi\varepsilon)d\sigma_0(s)[\ln(\sqrt{s}/m)-1/2]$, with energies and masses in common units. A finite photon-energy band gives this coefficient times $\ln(\varepsilon_2/\varepsilon_1)$.

To compare term by term with the supplied quantum result, note that $s'=s-2\sqrt{s}\varepsilon$ in units $c=1$. Thus $d\sigma_0(s')\to d\sigma_0(s)$ and

$$\sqrt{\frac{s'^2-4m^2s'}{s^2-4m^2s}}\longrightarrow 1, \qquad \frac{\varepsilon^2}{s'+2m^2}\longrightarrow 0.$$

The entire coefficient of $1/\varepsilon$, both angular and integrated, agrees. In particular the first term in the braces of the quantum integrated expression equals $[(1+\beta^2)L/\beta-2]/4$, where $L=\ln[(1+\beta)/(1-\beta)]$.

A sudden disappearance of a magnetic moment gives a Fourier amplitude proportional to $\omega\mathbf{m}$, in contrast to the constant electric-charge endpoint amplitude. Its intensity is therefore $O(\omega^2)$ and its

photon spectrum is $O(\varepsilon)$. It does not modify the soft answer. The angular function of the quoted magnetic contribution can be checked without assuming an unprovided classical spin ensemble:

$$\begin{aligned}\int d\Omega\,\frac{1-\beta^4\cos^4\theta}{(1-\beta^2\cos^2\theta)^2} &= 2\pi\int_{-1}^{1}\left[\frac{2}{1-\beta^2x^2}-1\right]dx\\ &= 4\pi\left(\frac{L}{\beta}-1\right).\end{aligned}$$

This reproduces the second term of the supplied integrated formula. Its precise coefficient $1/(s'+2m^2)$ describes spin and the annihilation process beyond the universal soft approximation; the spinless endpoint calculation by itself does not determine it. Likewise recoil and the variation of $d\sigma_0(s')$ cannot be inferred by extending a strictly soft classical trajectory.

15.6 For the soft-photon limit of radiation emitted when there is a small change $\Delta\boldsymbol{\beta}$ of velocity, (15.9) applies, with convenient polarization vectors shown in Fig. 15.2.

(a) Show that

$$\begin{aligned}\boldsymbol{\epsilon}_\parallel\cdot[\Delta\boldsymbol{\beta}+\mathbf{n}\times(\boldsymbol{\beta}\times\Delta\boldsymbol{\beta})] &= |\Delta\boldsymbol{\beta}|\,(\beta-\cos\theta)\cos\phi\\ \boldsymbol{\epsilon}_\perp\cdot[\Delta\boldsymbol{\beta}+\mathbf{n}\times(\boldsymbol{\beta}\times\Delta\boldsymbol{\beta})] &= |\Delta\boldsymbol{\beta}|\,(1-\beta\cos\theta)\sin\phi\end{aligned}$$

leading to (15.10) after averaging over ϕ.

(b) Show that in the limit $\gamma\gg1$ and $\theta\ll1$

$$\lim_{\omega\to0}\frac{d^2I}{d\omega\,d\Omega}\approx\frac{z^2e^2\gamma^4|\Delta\boldsymbol{\beta}|^2}{2\pi^2c}\cdot\left[\frac{(\gamma^2\theta^2-1)^2}{(1+\gamma^2\theta^2)^4}+\frac{1}{(1+\gamma^2\theta^2)^2}\right]$$

where the first (second) term in square brackets corresponds to the parallel (perpendicular) polarization. This expression leads immediately to (15.11).

(c) Show that the result of part b gives the expression for $P(\theta)$ given following (15.10).

(d) Show that the angular integral of (15.11) or the answer in part b can be written

$$\lim_{\omega\to0}\frac{dI}{d\omega}\approx\frac{z^2e^2\gamma^2|\Delta\boldsymbol{\beta}|^2}{\pi c}\cdot\int_1^\infty\frac{dy}{y^4}\,(2-2y+y^2)=\frac{2}{3}\,\frac{z^2e^2\gamma^2|\Delta\boldsymbol{\beta}|^2}{\pi c}$$

Solution.

(a) Choose the axes of Fig. 15.2:

$$\boldsymbol{\beta}=\beta(0,0,1),\quad \Delta\boldsymbol{\beta}=D(\cos\phi,\sin\phi,0),\quad \mathbf{n}=(\sin\theta,0,\cos\theta),$$

with $D=|\Delta\boldsymbol{\beta}|$ and convenient polarizations $\boldsymbol{\epsilon}_\parallel=(-\cos\theta,0,\sin\theta)$, $\boldsymbol{\epsilon}_\perp=(0,1,0)$. The vector triple product gives

$$\begin{aligned}\mathbf{A} &= \Delta\boldsymbol{\beta}+\mathbf{n}\times(\boldsymbol{\beta}\times\Delta\boldsymbol{\beta})\\ &= (1-\beta\cos\theta)D(\cos\phi,\sin\phi,0)+\beta D\sin\theta\cos\phi(0,0,1).\end{aligned}$$

Consequently

$$\epsilon_\parallel \cdot \mathbf{A} = D[-\cos\theta + \beta\cos^2\theta + \beta\sin^2\theta]\cos\phi = D(\beta - \cos\theta)\cos\phi,$$

while $\epsilon_\perp \cdot \mathbf{A} = D(1 - \beta\cos\theta)\sin\phi$. Since the averages of $\cos^2\phi$ and $\sin^2\phi$ are both $1/2$, (15.9) gives

$$I'_\parallel = \frac{z^2e^2D^2}{8\pi^2c}\frac{(\beta - \cos\theta)^2}{(1 - \beta\cos\theta)^4}, \qquad I'_\perp = \frac{z^2e^2D^2}{8\pi^2c}\frac{1}{(1 - \beta\cos\theta)^2},$$

where I' denotes $d^2I/(d\omega\, d\Omega)$.

(b) For $\gamma \gg 1$, $\theta \ll 1$, retain $x = \gamma^2\theta^2$ without assuming it small:

$$1 - \beta\cos\theta \simeq \frac{1 + x}{2\gamma^2}, \qquad \beta - \cos\theta \simeq \frac{x - 1}{2\gamma^2}.$$

Substitution yields

$$\boxed{I'_\parallel + I'_\perp = \frac{z^2e^2D^2\gamma^4}{2\pi^2c}\left[\frac{(x-1)^2}{(1+x)^4} + \frac{1}{(1+x)^2}\right] = \frac{z^2e^2D^2\gamma^4}{\pi^2c}\frac{1+x^2}{(1+x)^4}.}$$

The last equality is (15.11).

(c) The text defines positive polarization as the excess perpendicular intensity:

$$P = \frac{I'_\perp - I'_\parallel}{I'_\perp + I'_\parallel} = \frac{(1+x)^2 - (x-1)^2}{(1+x)^2 + (x-1)^2} = \boxed{\frac{2x}{1+x^2} = \frac{2\gamma^2\theta^2}{1+\gamma^4\theta^4}.}$$

It vanishes on axis and tends to one at $\theta = 1/\gamma$. The exact expression from part (a) has its maximum at $\cos\theta = \beta$, consistent with this limit.

(d) The forward cone dominates, so $d\Omega \simeq 2\pi\theta d\theta = \pi dx/\gamma^2$. The upper integration limit can be extended to infinity at this leading order. Therefore, on setting $y = 1 + x$,

$$\begin{aligned}\frac{dI}{d\omega} &\simeq \frac{z^2e^2D^2\gamma^2}{\pi c}\int_0^\infty \frac{1+x^2}{(1+x)^4}dx \\ &= \frac{z^2e^2D^2\gamma^2}{\pi c}\int_1^\infty (y^{-2} - 2y^{-3} + 2y^{-4})dy \\ &= \boxed{\frac{2z^2e^2D^2\gamma^2}{3\pi c}.}\end{aligned}$$

The small-deflection expansion itself requires $\gamma D \ll 1$; taking the high-energy limit at fixed large deflection would violate (15.9), even though the displayed cone formula is algebraically finite.

15.7 Consider the radiation emitted in nuclear fission by the sudden creation of two fragments of charge and mass $(Z_1e,\ A_1m)$ and $(Z_2e,\ A_2m)$ recoiling in opposite directions with total c.m.s. kinetic energy E. Treat the nuclei as point charges and their motion after the very short initial period of acceleration is nonrelativistic, but keep terms up to second order in $1/c$, as in Problem 15.1. For simplicity, assume that the fragments move with constant speeds in opposite directions away from the origin for $t > 0$. The relative speed is $c\beta$.

(a) Using the appropriate generalization of Problem 15.1a, show that the intensity of radiation per unit solid angle and per unit photon energy in the c.m. system is

$$\frac{d^2I}{d(\hbar\omega)\,d\Omega} = \frac{\alpha\beta^2\sin^2\theta}{4\pi^2}\,|p + q\beta\cos\theta|^2$$

where θ is the angle between the line of recoil and the direction of observation, and

$$\alpha = \frac{e^2}{\hbar c}; \qquad p = \frac{Z_1A_2 - Z_2A_1}{A_1 + A_2}; \qquad q = \frac{Z_1A_2^2 + Z_2A_1^2}{(A_1 + A_2)^2}$$

Show that the radiated energy per unit photon energy is

$$\frac{dI}{d(\hbar\omega)} = \frac{2\alpha\beta^2}{3\pi}\left(p^2 + \frac{\beta^2q^2}{5}\right)$$

where the first term is the electric dipole and the second the quadrupole radiation.

(b) As an example of the asymmetric fission of ^{235}U by thermal neutrons, take $Z_1 = 36$, $A_1 = 95$ (krypton), $Z_2 = 56$, $A_2 = 138$ (barium) (three neutrons are emitted during fission), with $E = 170$ MeV and $mc^2 = 931.5$ MeV. What are the values of p^2 and q^2? Determine the total amount of energy (in MeV) radiated by this "inner bremsstrahlung" process, with the substitution, $\beta^2 \to \beta^2(1 - \hbar\omega/E)$, as a crude way to incorporate conservation of energy. What are the relative amounts of energy radiated in the dipole and quadrupole modes? In actual fission, roughly 7 MeV of electromagnetic energy is radiated within 10^{-8} s. How does your estimate compare? If it is much smaller or larger, attempt to explain.

Solution.

(a) Write $A = A_1 + A_2$, and take fragment 1 along $+z$. Momentum conservation gives $\beta_1 = (A_2/A)\beta$, $\beta_2 = -(A_1/A)\beta$. The initial parent is at rest, so its transverse soft endpoint amplitude vanishes. Expanding the two final endpoint factors gives

$$\begin{aligned}\frac{\mathbf{E}_\perp}{e} &= P\widehat{\mathbf{z}}\left[Z_1\frac{\beta_1}{1 - \beta_1\cos\theta} + Z_2\frac{\beta_2}{1 - \beta_2\cos\theta}\right] \\ &= P\widehat{\mathbf{z}}\,[p\beta + q\beta^2\cos\theta + O(\beta^3)],\end{aligned}$$

where

$$p = \frac{Z_1A_2 - Z_2A_1}{A}, \qquad q = \frac{Z_1A_2^2 + Z_2A_1^2}{A^2}.$$

Hence the square of the amplitude retained through c^{-2} is

$$\boxed{\frac{d^2I}{d\varepsilon\,d\Omega} = \frac{\alpha\beta^2\sin^2\theta}{4\pi^2}|p + q\beta\cos\theta|^2.}$$

The mixed term is odd in $\cos\theta$ and integrates to zero. The remaining integrals are

$$\int \sin^2\theta\,d\Omega = \frac{8\pi}{3}, \qquad \int \sin^2\theta\cos^2\theta\,d\Omega = \frac{8\pi}{15},$$

so

$$\boxed{\frac{dI}{d\varepsilon} = \frac{2\alpha\beta^2}{3\pi}\left(p^2 + \frac{q^2\beta^2}{5}\right).}$$

The leading charge-weighted dipole motion is proportional to p; the quadrupole amplitude is proportional to q. Straight radial recoil has zero orbital magnetic moment. As specified, this is a multipole truncation of the amplitude: if a complete relativistic intensity through β^4 were wanted, the interference of the $O(\beta)$ and omitted $O(\beta^3)$ amplitudes would also be needed.

(b) The relative kinetic energy is $E = \mu c^2\beta^2/2$, $\mu = mA_1A_2/A$, so the numerical inputs give

$$p = -\frac{352}{233}, \quad p^2 = 2.28230397, \qquad q = \frac{1190984}{54289}, \quad q^2 = 481.269312, \quad \beta^2 = 0.00648708050.$$

Replace every β^2 by $\beta^2(1-\varepsilon/E)$ and integrate from 0 to E. In the dipole term $\int_0^E(1-\varepsilon/E)d\varepsilon = E/2$; in the quadrupole term the squared factor integrates to $E/3$. Thus

$$I_D = \frac{\alpha p^2\beta^2 E}{3\pi} = 0.00194879 \text{ MeV}, \qquad I_Q = \frac{2\alpha q^2\beta^4 E}{45\pi} = 0.000355442 \text{ MeV}.$$

The total is

$$\boxed{I_D + I_Q = 0.00230424 \text{ MeV} = 2.30424 \text{ keV},}$$

with fractions 84.57% dipole and 15.43% quadrupole. It is about 3.29×10^{-4} of the quoted 7 MeV. The model includes only radiation caused by the translational recoil of rigid point fragments; it contains no internal excited nuclear states. Thus it cannot account for the much larger prompt electromagnetic energy released by de-excitation of the fragments. The near equality of the two charge-to-mass ratios additionally suppresses the dipole coefficient. The comparison does not justify extrapolating the point-particle spectrum as a calculation of nuclear de-excitation.

15.8 Two identical point particles of charge q and mass m interact by means of a short-range repulsive interaction that is equivalent to a hard sphere of radius R in their relative separation. Neglecting the electromagnetic *interaction* between the two particles, determine the radiation cross section in the center-of-mass system for a collision between these identical particles to the lowest nonvanishing approximation. Show that the differential cross section for emission of photons per unit solid angle per unit energy interval is

$$\frac{d^2\sigma}{d(\hbar\omega)\,d\Omega} = \left(\frac{q^2}{\hbar c}\right)\cdot\frac{\beta^2R^2}{48\pi}\cdot\frac{1}{\hbar\omega}\left[\beta^2\left(\frac{2}{5}+3\sin^2\theta\cos^2\theta\right)+\left(\frac{\omega R}{c}\right)^2\left(1-\frac{3}{8}\sin^4\theta\right)\right]$$

where θ is measured relative to the incident direction and $c\beta$ is the relative speed. By integration over the angles of emission, show that the total cross section for radiation per unit photon energy $\hbar\omega$ is

$$\frac{d\sigma}{d(\hbar\omega)} = \left(\frac{q^2}{\hbar c}\right)\cdot\frac{\beta^2R^2}{15}\cdot\frac{1}{\hbar\omega}\left[\beta^2+\left(\frac{\omega R}{c}\right)^2\right]$$

Compare these results with that of Problem 15.2 as to frequency dependence, relative magnitude, etc.

Solution.

Work in the center-of-mass frame and let the incident relative velocity be $c\beta\hat{\mathbf{z}}$, so that each particle has speed $c\beta/2$. At contact write the separation as $R\mathbf{t}$, with

$$\mathbf{t} = (\sqrt{1-u^2}\cos\phi, \sqrt{1-u^2}\sin\phi, -u), \qquad u = \sqrt{1-b^2/R^2} \in [0,1].$$

Specular reflection of the relative velocity gives $\mathbf{w} = \widehat{\mathbf{z}} + 2u\mathbf{t}$ after collision. The two charge positions are $\pm R\mathbf{t}/2$. Apply the expansion of Problem 15.1 to each instantaneous change, retaining both the velocity correction and the contact-position phase:

$$\begin{aligned}\frac{\mathbf{J}}{q} &= \frac{\beta^2}{2}P[(\mathbf{n}\cdot\mathbf{w})\mathbf{w} - (\mathbf{n}\cdot\widehat{\mathbf{z}})\widehat{\mathbf{z}}] - \frac{i\beta kR}{2}(\mathbf{n}\cdot\mathbf{t})P(\mathbf{w}-\widehat{\mathbf{z}}) \\ &= \beta^2 P\mathbf{K} - i\beta kR\,uNP\mathbf{t}, \\ \mathbf{K} &= uN\widehat{\mathbf{z}} + u(C+2uN)\mathbf{t}, \quad C=\cos\theta, \quad N=\mathbf{n}\cdot\mathbf{t}, \quad P=\mathbf{1}-\mathbf{n}\mathbf{n}^{\mathsf{T}}.\end{aligned}$$

The leading electric dipoles cancel between the identical charges. Their orbital magnetic moment is constant in this central elastic collision and therefore does not radiate. The displayed terms are the first quadrupole contributions, including the impulsive term associated with the finite separation at contact. The approximation requires $\beta \ll 1$ and $kR \ll 1$. The real and imaginary terms have no interference in $|\mathbf{J}|^2$.

Here are the impact-parameter integrals explicitly. Set $S^2 = 1 - C^2$, choose $\mathbf{n} = (S, 0, C)$, and average over ϕ:

$$\begin{aligned}\langle N\rangle &= -Cu, \\ \langle N^2\rangle &= \tfrac{1}{2}S^2(1-u^2) + C^2u^2, \\ \langle N^3\rangle &= -Cu[\tfrac{3}{2}S^2(1-u^2) + C^2u^2], \\ \langle N^4\rangle &= \tfrac{3}{8}S^4(1-u^2)^2 + 3S^2C^2u^2(1-u^2) + C^4u^4.\end{aligned}$$

Using $\widehat{\mathbf{z}}\cdot\mathbf{t} = -u$, direct scalar products give

$$|P\mathbf{K}|^2 = u^2[C^2 + (1-4C^2)N^2 + 2uCN - 8uCN^3 - 4u^2N^4], \qquad u^2N^2|P\mathbf{t}|^2 = u^2(N^2 - N^4).$$

Since $b|db|d\phi = R^2 u\,du\,d\phi$, substitution of the four polynomials above gives

$$\begin{aligned}\int |P\mathbf{K}|^2\, b\,db\,d\phi &= 2\pi R^2 \int_0^1 u^3[C^2 + (1-4C^2)\langle N^2\rangle + 2uC\langle N\rangle - 8uC\langle N^3\rangle - 4u^2\langle N^4\rangle]du \\ &= \frac{\pi R^2}{12}\left(\frac{2}{5} + 3S^2C^2\right), \\ \int u^2N^2|P\mathbf{t}|^2\, b\,db\,d\phi &= 2\pi R^2\int_0^1 u^3[\langle N^2\rangle - \langle N^4\rangle]du = \frac{\pi R^2}{12}\left(1 - \frac{3}{8}S^4\right).\end{aligned}$$

All integrations use only $\int_0^1 u^j du = 1/(j+1)$. Multiplication by $q^2/(4\pi^2\hbar c\varepsilon)$ gives

$$\boxed{\frac{d^2\sigma}{d\varepsilon\,d\Omega} = \frac{q^2}{\hbar c}\frac{\beta^2R^2}{48\pi\varepsilon}\left[\beta^2\left(\frac{2}{5} + 3\sin^2\theta\cos^2\theta\right) + (kR)^2\left(1 - \frac{3}{8}\sin^4\theta\right)\right].}$$

The two angular brackets have the same integral:

$$\frac{2}{5}(4\pi) + 3\frac{8\pi}{15} = \frac{16\pi}{5}, \qquad 4\pi - \frac{3}{8}\frac{32\pi}{15} = \frac{16\pi}{5}.$$

Therefore

$$\boxed{\frac{d\sigma}{d\varepsilon} = \frac{q^2}{\hbar c}\frac{\beta^2R^2}{15\varepsilon}[\beta^2 + (\omega R/c)^2].}$$

No extra factor 1/2 is inserted: the incoming beam particles are specified and their classical impact parameters are integrated once. For the same charge, radius and relative speed, division by the fixed-sphere result of Problem 15.2 gives $[\beta^2 + (\omega R/c)^2]/20$. The soft term is suppressed by an additional β^2 because the dipole cancels. The second photon spectrum increases as ω rather than $1/\omega$, but this truncated result is valid only while $\omega R/c \ll 1$ and while photon recoil and the finite collision duration can be ignored.

15.9 A particle of charge ze, mass m, and nonrelativistic velocity v is deflected in a screened Coulomb field, $V(r) = Zze^2e^{-\alpha r}/r$, and consequently emits radiation. Discuss the radiation with the approximation that the particle moves in an almost straight-line trajectory past the force center.

(a) Show that, if the impact parameter is b, the energy radiated per unit frequency interval is

$$\frac{dI}{d\omega}(\omega, b) = \frac{8}{3\pi}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\alpha^2K_1^2(\alpha b)$$

for $\omega \ll v/b$, and negligible for $\omega \gg v/b$.

(b) Show that the radiation cross section is

$$\frac{d\chi(\omega)}{d\omega} \simeq -\frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\left(\frac{c}{v}\right)^2\left\{\frac{x^2}{2}\left[K_0^2(x) - K_1^2(x) + \frac{2K_0(x)K_1(x)}{x}\right]\right\}_{x_1}^{x_2}$$

where $x_1 = \alpha b_{\min}$, $x_2 = \alpha b_{\max}$.

(c) With $b_{\min} = \hbar/mv$, $b_{\max} = v/\omega$, and $\alpha^{-1} = 1.4a_0Z^{-1/3}$, determine the radiation cross section in the two limits, $x_2 \ll 1$ and $x_2 \gg 1$. Compare your results with the "screening" and "no screening" limits of the text.

Solution.

Write $g = Zze^2$, and approximate the orbit by $\mathbf{r}(t) = (b, 0, vt)$. This requires a small deflection on the impact parameters retained. To distinguish the screening parameter from the fine-structure constant, retain the question's α only in the potential and write $\alpha_{\rm fs} = e^2/(\hbar c)$ below.

(a) At zero frequency the longitudinal impulse vanishes by symmetry. The integrated potential is

$$\int_{-\infty}^{\infty} V(\sqrt{b^2 + v^2t^2})dt = \frac{2g}{v}\int_0^{\infty} e^{-\alpha b\cosh u}du = \frac{2g}{v}K_0(\alpha b),$$

where $vt = b\sinh u$. Differentiation with respect to b gives

$$\Delta v_x = -\frac{1}{m}\partial_b\int V dt = \frac{2g\alpha}{mv}K_1(\alpha b).$$

The dipole formula $dI/d\omega = 2(ze)^2|\Delta\mathbf{v}|^2/(3\pi c^3)$ therefore yields

$$\boxed{\frac{dI}{d\omega}(\omega, b) \simeq \frac{8}{3\pi}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\frac{c^2}{v^2}\alpha^2K_1^2(\alpha b).}$$

For completeness, the straight-path Fourier transform at arbitrary frequency is

$$\int Ve^{i\omega t}dt = \frac{2g}{v}K_0(b\lambda), \quad \lambda = \sqrt{\alpha^2 + \omega^2/v^2},$$

To see the transform explicitly, put $q = \omega/v$ and $\tan\delta = q/\alpha$, with $0 \le \delta < \pi/2$. Then $\alpha = \lambda\cos\delta$, $q = \lambda\sin\delta$ and the hyperbolic substitution $vt = b\sinh u$ gives

$$\int V e^{i\omega t}dt = \frac{g}{v}\int_{-\infty}^{\infty} e^{-b(\alpha\cosh u - iq\sinh u)}du = \frac{g}{v}\int_{-\infty}^{\infty} e^{-b\lambda\cosh(u-i\delta)}du.$$

The integrand is analytic between this line and the real u axis and decays at both ends of the joining contours, since $\cos(\mathrm{Im}\,u) > 0$ in this strip. Moving the line to the real axis gives $2gK_0(b\lambda)/v$. The unscreened case $\alpha = 0$ follows by the limit $\alpha \downarrow 0$ at $\omega > 0$. For the force components, $\widetilde{a}_x = -\partial_b\widetilde{V}/m$, while $dV/dt = v\,\partial_z V$ implies $\widetilde{a}_z = i\omega\widetilde{V}/(mv)$ after integration by parts. Consequently

$$\widetilde{a}_x = \frac{2g\lambda}{mv}K_1(b\lambda), \qquad \widetilde{a}_z = \frac{2ig\omega}{mv^2}K_0(b\lambda).$$

Thus the exact straight-path dipole answer replaces $\alpha^2K_1^2(\alpha b)$ by $\lambda^2K_1^2(b\lambda) + (\omega/v)^2K_0^2(b\lambda)$. For $\omega b/v \ll 1$ its leading limit is the boxed result; for $\omega b/v \gg 1$, $K_\nu(b\lambda)$ has the exponentially small factor $e^{-b\lambda}$. This also verifies the stated cutoff.

(b) Integrate over $2\pi b\,db$. Define

$$C_0 = \frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{z^2e^2}{mc^2}\right)^2\frac{c^2}{v^2}, \qquad G(x) = \frac{x^2}{2}[K_0^2 - K_1^2] + xK_0K_1.$$

Use $K_0' = -K_1$, $K_1' = -K_0 - K_1/x$ to differentiate:

$$\begin{aligned} G' &= x(K_0^2 - K_1^2) + x^2(K_0K_0' - K_1K_1') + K_0K_1 + x(K_0'K_1 + K_0K_1') \\ &= -xK_1^2. \end{aligned}$$

It follows that

$$\boxed{\frac{d\chi}{d\omega} = C_0\int_{x_1}^{x_2} xK_1^2(x)dx = -C_0[G(x)]_{x_1}^{x_2},}$$

which is the stated expression. It is nonnegative for $x_2 \ge x_1$.

(c) For small x, $K_0(x) = \ln(2/x) - \gamma_E + O(x^2\ln x)$ and $K_1(x) = x^{-1} + O(x\ln x)$, giving

$$G(x) = \ln(2/x) - \gamma_E - \tfrac{1}{2} + O(x^2\ln^2 x).$$

If $x_2 \ll 1$ and $b_{\max} > b_{\min}$, both endpoints are small:

$$\boxed{\frac{d\chi}{d\omega} \simeq C_0\ln\frac{b_{\max}}{b_{\min}} = C_0\ln\frac{mv^2}{\hbar\omega}.}$$

This is the unscreened nonrelativistic logarithm, agreeing with the soft limit of (15.29) up to the order-unity factor inherent in the sharp impact-parameter cutoffs. That exact Born result contains $\ln(2mv^2/\hbar\omega)$.

If $x_2 \gg 1$, $G(x_2)$ is exponentially small, so the general result is $C_0G(x_1)$, independent of frequency. If also $x_1 \ll 1$,

$$\begin{aligned} \frac{d\chi}{d\omega} &\simeq C_0\left[\ln\frac{2}{\alpha b_{\min}} - \gamma_E - \frac{1}{2}\right] \\ &= \boxed{C_0\ln\left[2.8e^{-\gamma_E - 1/2}\frac{m}{m_e}\frac{v/c}{\alpha_{\mathrm{fs}}Z^{1/3}}\right].} \end{aligned}$$

Here $a_0 = \hbar/(m_e c\alpha_{\rm fs})$ has been used; the m_e in the atomic screening length is not necessarily the projectile mass m. The logarithm is approximately $\ln[130.67(m/m_e)(v/c)Z^{-1/3}]$. This is the complete-screening plateau; it is the nonrelativistic counterpart of the constant screened spectrum in the text. The transition occurs near $\omega \sim \alpha v$. If x_1 is not small one must retain $G(x_1)$ rather than use a possibly negative logarithmic approximation. Likewise $b_{\rm max} < b_{\rm min}$ means that there is no interval for this cutoff model, not a negative cross section.

15.10 A particle of charge ze, mass m, and velocity v is deflected in a hyperbolic path by a fixed repulsive Coulomb potential, $V(r) = Zze^2/r$. Assume the nonrelativistic dipole approximation (but no further approximations).

(a) Show that the energy radiated per unit frequency interval by the particle when initially incident at impact parameter b is

$$\frac{d}{d\omega}I(\omega,b) = \frac{8}{3\pi}\frac{(zea\omega)^2}{c^3}\,e^{-(\pi\omega/\omega_0)}\left\{\left[K'_{i\omega/\omega_0}\left(\frac{\omega\epsilon}{\omega_0}\right)\right]^2 + \frac{\epsilon^2-1}{\epsilon^2}\left[K_{i\omega/\omega_0}\left(\frac{\omega\epsilon}{\omega_0}\right)\right]^2\right\}$$

(b) Show that the radiation cross section is

$$\frac{d}{d\omega}\chi(\omega) = \frac{16}{3}\frac{(zeav)^2}{c^3}\,e^{-(\pi\omega/\omega_0)}\frac{\omega}{\omega_0}\,K_{i\omega/\omega_0}\left(\frac{\omega}{\omega_0}\right)\left[-K'_{i\omega/\omega_0}\left(\frac{\omega}{\omega_0}\right)\right]$$

(c) Prove that the radiation cross section reduces to that obtained in the text for classical bremsstrahlung for $\omega \ll \omega_0$. What is the limiting form for $\omega \gg \omega_0$?

(d) What modifications occur for an attractive Coulomb interaction?

The hyperbolic path may be described by

$$x = a(\epsilon + \cosh\xi), \qquad y = -b\,\sinh\xi, \qquad \omega_0 t = (\xi + \epsilon\,\sinh\xi)$$

where $a = Zze^2/mv^2$, $\epsilon = \sqrt{1+(b/a)^2}$, $\omega_0 = v/a$.

Solution.

Put $s = \omega/\omega_0 > 0$ and $z = s\epsilon$. A prime on K_{is} will mean derivative with respect to its real argument, holding its order fixed.

(a) The dipole spectral formula is

$$\frac{dI}{d\omega} = \frac{2(ze)^2}{3\pi c^3}\left|\int \ddot{\mathbf{r}}(t)e^{i\omega t}dt\right|^2.$$

Introduce the master integral

$$H(s,\epsilon) = \int_{-\infty}^{\infty} e^{is(\xi+\epsilon\sinh\xi)}d\xi.$$

A convergence factor justifies shifting the contour to $\xi = u + i\pi/2$. Since $\sinh(u+i\pi/2) = i\cosh u$,

$$H = e^{-\pi s/2}\int_{-\infty}^{\infty} e^{-s\epsilon\cosh u + isu}du = 2e^{-\pi s/2}K_{is}(s\epsilon).$$

Its derivative and an integration by parts give

$$\int \sinh\xi\, e^{is(\xi+\epsilon\sinh\xi)}d\xi = \frac{\partial_\epsilon H}{is} = -2ie^{-\pi s/2}K'_{is}(z),$$

$$0 = \int \frac{d}{d\xi} e^{is(\xi+\epsilon\sinh\xi)} d\xi = is\left(H + \epsilon\int \cosh\xi\, e^{is(\xi+\epsilon\sinh\xi)} d\xi\right).$$

Finally integrate the acceleration once by parts and use $dx/d\xi = a\sinh\xi$, $dy/d\xi = -b\cosh\xi$:

$$\widetilde{a}_x = -2a\omega e^{-\pi s/2} K'_{is}(z), \qquad \widetilde{a}_y = -\frac{2ib\omega}{\epsilon} e^{-\pi s/2} K_{is}(z).$$

K_{is} is real for real positive argument. Since $b^2 = a^2(\epsilon^2 - 1)$, squaring these orthogonal components gives

$$\boxed{\frac{dI}{d\omega} = \frac{8}{3\pi}\frac{(zea\omega)^2}{c^3} e^{-\pi s}\left[K'_{is}(s\epsilon)^2 + \frac{\epsilon^2-1}{\epsilon^2} K_{is}(s\epsilon)^2\right].}$$

(b) The impact measure is $2\pi b\, db = 2\pi a^2 \epsilon\, d\epsilon$. The modified Bessel equation with order is reads

$$z^2 K'' + zK' - (z^2 - s^2)K = 0.$$

Multiplication by K/z proves

$$\frac{d}{dz}(zKK') = z(K')^2 + (z - s^2/z)K^2.$$

With $z = s\epsilon$, the impact integral therefore is

$$\int_1^\infty \epsilon\left[(K')^2 + (1-\epsilon^{-2})K^2\right] d\epsilon = \frac{1}{s^2}[zKK']_s^\infty = -\frac{1}{s} K_{is}(s) K'_{is}(s).$$

The term at infinity vanishes exponentially. Substituting this integral and $a\omega = vs$ gives

$$\boxed{\frac{d\chi}{d\omega} = \frac{16}{3}\frac{(zeav)^2}{c^3} e^{-\pi s} s K_{is}(s)[-K'_{is}(s)].}$$

(c) As $s \downarrow 0$, $K_{is}(s) = \ln(2/s) - \gamma_E + o(1)$ and $-sK'_{is}(s) = 1 + o(1)$. Hence

$$\boxed{\frac{d\chi}{d\omega} \sim \frac{16}{3}\frac{Z^2 z^4 e^6}{m^2 c^3 v^2} \ln\frac{2e^{-\gamma_E} m v^3}{Z z e^2 \omega}.}$$

This is (15.25), and fixes its undetermined constant to $2e^{-\gamma_E} = 1.12291897$ for the repulsive exact orbit.

For large s, both the order and argument grow together; the fixed-order large-argument expansion is inapplicable. Shift the integral representation in the other direction to obtain its turning-point form, with the oscillatory integrals understood with a convergence factor,

$$K_{is}(z) = e^{-\pi s/2}\int_0^\infty \cos(z\sinh u - su)\, du, \qquad -K'_{is}(z) = e^{-\pi s/2}\int_0^\infty \sinh u\, \sin(z\sinh u - su)\, du.$$

The derivative holds s fixed. Only afterward set $z = s$. The phase is $s(\sinh u - u) = su^3/6 + O(su^5)$, so the dominant region has $u = O(s^{-1/3})$ and $\sinh u = u + O(u^3)$. With $y = su^3/6$, the two leading integrals become

$$\int_0^\infty \cos(su^3/6)\, du = \frac{(6/s)^{1/3}}{3}\int_0^\infty y^{-2/3}\cos y\, dy,$$

$$\int_0^\infty u\sin(su^3/6)\,du = \frac{(6/s)^{2/3}}{3}\int_0^\infty y^{-1/3}\sin y\,dy.$$

Thus

$$K_{is}(s) \sim e^{-\pi s/2}\frac{\Gamma(1/3)\cos(\pi/6)}{3}(6/s)^{1/3},$$
$$-K'_{is}(s) \sim e^{-\pi s/2}\frac{\Gamma(2/3)\sin(\pi/3)}{3}(6/s)^{2/3}.$$

These follow from $\int_0^\infty x^{a-1}e^{ix}dx = e^{i\pi a/2}\Gamma(a)$ with a convergence factor. Using $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$ yields $sK(-K') \sim (\pi/\sqrt{3})e^{-\pi s}$, and consequently

$$\boxed{\frac{d\chi}{d\omega} \sim \frac{16\pi}{3\sqrt{3}}\frac{(zeav)^2}{c^3}e^{-2\pi\omega/\omega_0}.}$$

(d) For attraction let $a = |Zze^2|/(mv^2) > 0$, $\omega_0 = v/a$ and $\epsilon = \sqrt{1+b^2/a^2}$. A hyperbolic parametrization has $t = (\epsilon\sinh\xi - \xi)/\omega_0$ and, up to axis reversals, $x = a(\epsilon - \cosh\xi)$, $y = b\sinh\xi$. The master integral is now

$$\int e^{is(\epsilon\sinh\xi - \xi)}d\xi = 2e^{+\pi s/2}K_{is}(s\epsilon).$$

Thus every spectrum above changes $e^{-\pi s}$ to $e^{+\pi s}$, with the same squared Bessel bracket and the same integrated product. The leading low-frequency logarithm is unchanged; the high-frequency cross section instead tends to

$$\boxed{\left(\frac{d\chi}{d\omega}\right)_{\text{attractive}} \longrightarrow \frac{16\pi}{3\sqrt{3}}\frac{(zeav)^2}{c^3}.}$$

This formal plateau comes from ever smaller impact parameters near an attractive point center. At any fixed $b > 0$ the spectrum still decays at high frequency. The integrated ideal point-Coulomb model allows arbitrarily large speed near the center, so eventually the nonrelativistic approximation, negligible radiation reaction, finite nuclear size, or quantum mechanics must cut it off. The exactly head-on attractive orbit is a singular zero-measure case and is understood as the limit $b \downarrow 0$, not as a regular passage through the center.

15.11 Using the method of virtual quanta, discuss the relationship between the cross section for photodisintegration of a nucleus and electrodisintegration of a nucleus.

(a) Show that, for electrons of energy $E = \gamma mc^2 \gg mc^2$, the electron disintegration cross section is approximately:

$$\sigma_{\text{el}}(E) \simeq \frac{2}{\pi}\frac{e^2}{\hbar c}\int_{\omega_T}^{E/\hbar}\sigma_{\text{photo}}(\omega)\,\ln\left(\frac{k\gamma^2 mc^2}{\hbar\omega}\right)\frac{d\omega}{\omega}$$

where $\hbar\omega_T$ is the threshold energy for the process and k is a constant of order unity.

(b) Assuming that $\sigma_{\text{photo}}(\omega)$ has the resonance shape:

$$\sigma_{\text{photo}}(\omega) \simeq \frac{A}{2\pi}\frac{e^2}{Mc}\frac{\Gamma}{(\omega-\omega_0)^2 + (\Gamma/2)^2}$$

where the width Γ is small compared to $(\omega_0 - \omega_T)$, sketch the behavior of $\sigma_{\text{el}}(E)$ as a function of E and show that for $E \gg \hbar\omega_0$,

$$\sigma_{\text{el}}(E) \simeq \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\frac{Ae^2}{Mc}\frac{1}{\omega_0}\ln\left(\frac{kE^2}{mc^2\hbar\omega_0}\right)$$

(c) In the limit of a very narrow resonance, the photonuclear cross section can be written as $\sigma_{\rm photo}(\omega) = (Ae^2/Mc)\,\delta(\omega - \omega_0)$. Then the result of part b would represent the electrodisintegration cross section for $E > \hbar\omega_0$. The corresponding bremsstrahlung-induced cross section is given in the same approximation by (15.47), multiplied by $(Ae^2/Mc\hbar\omega_0)$, where Z is the atomic number of the radiator. Comparisons of the electron- and bremsstrahlung-induced disintegration cross sections of a number of nuclei are given by E. Wolynec et al. *Phys. Rev.* C **11**, 1083 (1975). Calculate the quantity called F as a function of E (with a giant dipole resonance energy $\hbar\omega_0 \approx 20$ MeV) and compare its magnitude and energy dependence (at the high energy end) with the data in Figures 1–5 of Wolynec et al. The comparison is only qualitative at $E \approx \hbar\omega_0$ because of the breadth of the dipole resonance. [F is the ratio of the bremsstrahlung-induced cross section in units of $Z^2 r_0^2$ to the electrodisintegration cross section.]

Solution.

(a) The virtual-photon number spectrum is (15.58), with $q = e$:

$$N(\varepsilon) \simeq \frac{2\alpha}{\pi\beta^2\varepsilon}\left[\ln\frac{2e^{-\gamma_E}\hbar\gamma v}{\varepsilon b_{\rm min}} - \frac{\beta^2}{2}\right].$$

For an ultrarelativistic electron use $b_{\rm min} \sim \hbar/(\gamma mc)$, as in Table 15.1, unless nuclear size supplies the larger cutoff. The square bracket is then $\ln(k\gamma^2 mc^2/\varepsilon)$, where k absorbs the order-unity cutoff convention. The particular sharp-cutoff choice just written gives $k = 2e^{-\gamma_E - 1/2} = 0.681085$ for $\beta \to 1$. Since one virtual photon of energy ε has the same absorption probability as a real photon in this approximation,

$$\sigma_{\rm el}(E) = \int_{\hbar\omega_T}^{E} \sigma_{\rm photo}(\varepsilon/\hbar) N(\varepsilon) d\varepsilon.$$

Set $\varepsilon = \hbar\omega$, $E = \gamma mc^2$ to obtain

$$\boxed{\sigma_{\rm el}(E) \simeq \frac{2\alpha}{\pi}\int_{\omega_T}^{E/\hbar} \sigma_{\rm photo}(\omega) \ln\frac{k\gamma^2 mc^2}{\hbar\omega}\,\frac{d\omega}{\omega}.}$$

Replacing the actual endpoint $E - mc^2$ by E uses $E \gg mc^2$. The same assumption and negligible nuclear-size corrections delimit this logarithmic equivalent-photon estimate.

(b) Define $S = Ae^2/(Mc)$. The resonance has area

$$\int_{-\infty}^{\infty} \sigma_{\rm photo}(\omega) d\omega = \frac{S}{2\pi}\int_{-\infty}^{\infty} \frac{\Gamma\, d\omega}{(\omega - \omega_0)^2 + (\Gamma/2)^2} = S.$$

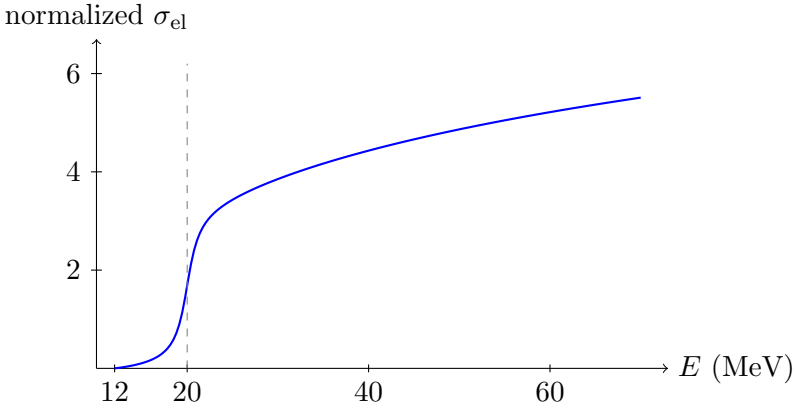
For $E/\hbar - \omega_0 \gg \Gamma$ and $\omega_0 - \omega_T \gg \Gamma$, the slowly varying factor $\omega^{-1}\ln[kE^2/(mc^2\hbar\omega)]$ can be evaluated at ω_0, so

$$\boxed{\sigma_{\rm el}(E) \simeq \frac{2\alpha S}{\pi\omega_0}\ln\frac{kE^2}{mc^2\hbar\omega_0}.}$$

Near the onset, keeping the finite integration limits gives the useful approximation

$$\sigma_{\rm el}(E) \simeq \frac{2\alpha S}{\pi\omega_0}\ln\frac{kE^2}{mc^2\hbar\omega_0}\,\frac{1}{\pi}\left[\arctan\frac{2(E/\hbar - \omega_0)}{\Gamma} - \arctan\frac{2(\omega_T - \omega_0)}{\Gamma}\right].$$

It rises from zero at threshold, rounds upward across the resonance, and subsequently grows logarithmically. In the zero-width limit it becomes a step at $E = \hbar\omega_0$ followed by the logarithm. The sketch below evaluates the original finite-width integral for $\hbar\omega_T = 12$ MeV, $\hbar\omega_0 = 20$ MeV, $\hbar\Gamma = 2$ MeV, and the stated k; the vertical unit is $2\alpha S/(\pi\omega_0)$.



(c) For an electron radiator, (15.47) is

$$\frac{d\chi}{d\omega} = \frac{16}{3} Z^2 r_0^2 \frac{e^2}{c} \ln \frac{233}{Z^{1/3}}.$$

Its product with $S/(\hbar\omega_0)$ is the bremsstrahlung-induced quantity σ_B in the question. It has two factors of area before the radiator factor $Z^2 r_0^2$ is divided out. Thus

$$\frac{\sigma_B}{Z^2 r_0^2} = \frac{16}{3} \alpha \frac{S}{\omega_0} \ln \frac{233}{Z^{1/3}}, \qquad \boxed{F(E;Z) = \frac{8\pi}{3} \frac{\ln(233/Z^{1/3})}{\ln[kE^2/(mc^2\hbar\omega_0)]}.}$$

The resonance strength cancels. Its high-energy slope is $dF/d\ln E = -2F/\ln[kE^2/(mc^2\hbar\omega_0)] < 0$. With $mc^2 = 0.510999$ MeV, $\hbar\omega_0 = 20$ MeV and $k = 0.681085$, representative single-element radiator values are

E (MeV)	$Z = 6$	$Z = 9$	$Z = 17$	$Z = 29$
25	10.904	10.600	10.124	9.724
30	9.932	9.656	9.222	8.858
35	9.237	8.980	8.576	8.237
40	8.708	8.466	8.086	7.766

The original Figures 1–5 of Wolynec et al. were checked in their reproductions in NBS IR 83-2742, volumes III, V, VII and IX, data sheets 368, 84, 186 and 425, respectively. Near the upper end of the measured range, their plotted F values are approximately 8–10 for carbon, 8 for fluorine, 8–9 for chlorine, 7–8 for ^{65}Cu, and 9 for ^{63}Cu. All show a decreasing high-energy trend. The simple expression therefore gives the observed scale and trend. These are readings of the plotted ranges, not a fit or a claim of precision agreement; the two copper reactions already demonstrate that one universal 20 MeV delta resonance cannot describe every channel.

At threshold the actual broad resonances produce the rises and maxima seen in the figures; the delta model has a discontinuous onset and cannot reproduce them. Moreover (15.47) assumes complete screening, whereas 20–40 MeV electrons need not satisfy its asymptotic condition for all

radiator elements and photon energies. Mixtures, incomplete screening, Coulomb distortion and the real photonuclear line shapes must be retained for a quantitative comparison. The problem's prescribed formula is consequently a qualitative estimate even at the upper end of these data, rather than a precision prediction.

15.12 A fast particle of charge ze, mass M, and velocity v, collides with a hydrogen-like atom with one electron of charge $-e$, mass m, bound to a nuclear center of charge Ze. The collisions can be divided into two kinds: close collisions where the particle passes through the atom ($b < d$), and distant collisions where the particle passes by outside the atom ($b > d$). The atomic "radius" d can be taken as a_0/Z. For the close collisions the interaction of the incident particle and the electron can be treated as a two-body collision and the energy transfer calculated from the Rutherford cross section. For the distant collisions the excitation and ionization of the atom can be considered the result of the photoelectric effect by the virtual quanta of the incident particle's fields.

For simplicity assume that for photon energies Q greater than the ionization potential I the photoelectric cross section is

$$\sigma_\gamma(Q) = \frac{8\pi^2}{137}\left(\frac{a_0}{Z}\right)^2\left(\frac{I}{Q}\right)^3$$

(This obeys the empirical $Z^4\lambda^3$ law for x-ray absorption and has a coefficient adjusted to satisfy the dipole sum rule, $\int \sigma_\gamma(Q)\, dQ = 2\pi^2 e^2\hbar/mc$.)

(a) Calculate the differential cross sections $d\sigma/dQ$ for energy transfer Q for close and distant collisions (write them as functions of Q/I as far as possible and in units of $2\pi z^2 e^4/mv^2I^2$). *Plot* the two distributions for $Q/I > 1$ for nonrelativistic motion of the incident particle and $\frac{1}{2}mv^2 = 10^3 I$.

(b) Show that the number of distant collisions measured by the integrated cross section is much larger than the number of close collisions, but that the energy transfer per collision is much smaller. Show that the energy loss is divided approximately equally between the two kinds of collisions, and verify that your total energy loss is in essential agreement with Bethe's result (13.14).

Solution.

Put $d = a_0/Z$, $I = \hbar^2/(2md^2)$, $y = Q/I$, $\beta = v/c$, and $\gamma = (1-\beta^2)^{-1/2}$. Use the stated dipole-photoabsorption model and a nearly straight projectile path for the distant collisions. Define the requested unit

$$S_0 = \frac{2\pi z^2 e^4}{mv^2 I^2}.$$

It has dimensions area per energy, as required for $d\sigma/dQ$.

(a) In a nonrelativistic two-body collision let $\mu = mM/(m+M)$ be the reduced mass and χ the relative scattering angle. The Rutherford cross section and the electron recoil are

$$\frac{d\sigma}{d\Omega_*} = \frac{z^2e^4}{4\mu^2v^4\sin^4(\chi/2)}, \qquad Q = \frac{2\mu^2v^2}{m}\sin^2(\chi/2).$$

Since $d\Omega_* = 4\pi d[\sin^2(\chi/2)]$,

$$\boxed{\frac{1}{S_0}\frac{d\sigma_{\text{close}}}{dQ} = \frac{1}{y^2}, \quad 1 \lesssim y < Y, \quad Y = \frac{2\mu^2v^2}{mI}.}$$

The lower limit $Q \sim I$ marks failure of the free-electron approximation; this is a division of collision mechanisms to logarithmic accuracy, not an exact treatment of the atomic boundary. For a relativistic heavy projectile the corresponding classical spinless upper limit is $Q_{\max} = 2mc^2\beta^2\gamma^2/[1 + 2\gamma m/M + (m/M)^2]$, with the same leading Q^{-2} law. Electron spin and identical-projectile exchange are outside the Rutherford model specified in the problem.

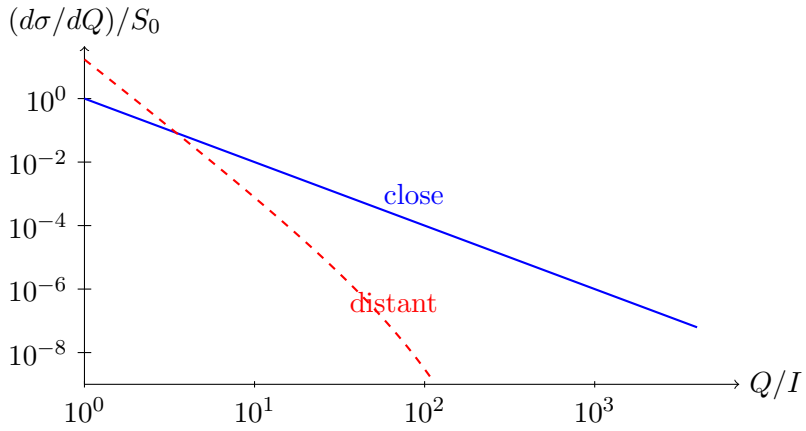
For distant encounters use (15.54) with minimum impact parameter d. The integrated number of virtual photons per unit photon energy is

$$N(Q) = \frac{2z^2\alpha_{\rm fs}}{\pi\beta^2 Q}B(x), \quad x = \frac{Qd}{\hbar\gamma v}, \qquad B(x) = xK_0(x)K_1(x) - \frac{\beta^2 x^2}{2}[K_1^2(x) - K_0^2(x)].$$

Multiplying by $\sigma_\gamma(Q) = 8\pi^2\alpha_{\rm fs}d^2/y^3$ and using $md^2 = \hbar^2/(2I)$ gives

$$\boxed{\frac{1}{S_0}\frac{d\sigma_{\rm distant}}{dQ} = \frac{4}{y^4}B(x), \qquad x = \frac{y}{2\gamma\sqrt{S}}, \quad S = \frac{mv^2}{2I}.}$$

For nonrelativistic motion $\gamma \simeq 1$ and the β^2 term in B is omitted. At the requested $S = 1000$ the distributions are therefore y^{-2} and $4y^{-4}xK_0(x)K_1(x)$, $x = y/(2\sqrt{1000})$.



The close curve is shown up to its heavy-projectile endpoint $Y = 4S = 4000$. For finite M replace it by $Y = 4000[M/(M + m)]^2$. The distant curve falls exponentially when $Q \gtrsim \hbar v/d$, in addition to its initial Q^{-4} dependence.

(b) Divide collision numbers by $S_0 I$ and energy transfers by $S_0 I^2$. For close collisions,

$$n_c = \int_1^Y y^{-2}dy = 1 - Y^{-1}, \qquad \ell_c = \int_1^Y y^{-1}dy = \ln Y.$$

For the distant spectrum the exact model integrals are

$$n_d = 4\int_1^\infty y^{-4}B(y/2\sqrt{S})dy, \qquad \ell_d = 4\int_1^\infty y^{-3}B(y/2\sqrt{S})dy.$$

To expose the physics, use the low-x expansion $B(x) = \ln(2/x) - \gamma_E + O(x^2\ln^2 x)$ in the nonrelativistic limit. Write $L_0 = \ln(4\sqrt{S}) - \gamma_E$. Then

$$n_d \simeq 4\int_1^\infty y^{-4}(L_0 - \ln y)dy = \frac{4}{3}(L_0 - \tfrac{1}{3}),$$
$$\ell_d \simeq 4\int_1^\infty y^{-3}(L_0 - \ln y)dy = 2(L_0 - \tfrac{1}{2}).$$

Here $\int_1^\infty y^{-p}dy = (p-1)^{-1}$ and differentiating it with respect to p gives the logarithmic integral $(p-1)^{-2}$. The tail beyond the virtual-photon cutoff changes only small corrections for $S \gg 1$. At $S = 1000$, direct Bessel quadrature gives the values in the supplementary calculation; rounded to the accuracy of the model,

$$n_c = 0.99975, \quad n_d = 5.23393, \qquad \ell_c = 8.29405, \quad \ell_d = 7.51412.$$

Thus distant events are about five times more numerous, and their mean transfer $I\ell_d/n_d \simeq 1.44I$ is much smaller than the close mean $I\ell_c/n_c \simeq 8.30I$. Their energy losses are nevertheless comparable, about 48% and 52% of the sum.

For a number density N_a of these one-electron atoms, the leading nonrelativistic heavy-projectile stopping power is

$$\begin{aligned} -\frac{dE}{dx} &= N_a S_0 I^2(\ell_c + \ell_d) \\ &\simeq \frac{2\pi N_a z^2 e^4}{mv^2}[\ln(64S^2) - 2\gamma_E - 1] \\ &= \frac{4\pi N_a z^2 e^4}{mv^2} \ln\frac{2mv^2}{I_{\text{eff}}}, \qquad I_{\text{eff}} = \frac{1}{2}e^{\gamma_E + 1/2} I \simeq 1.47I. \end{aligned}$$

This is Bethe's logarithmic form (13.14), with an effective mean excitation energy of the order of the ionization energy. The photoabsorption model omits discrete excitation lines and uses a crude separation of close and distant events, so it cannot fix the exact atomic mean-excitation constant. If one simply sets that constant to I, Bethe's dimensionless sum is $2\ln(4S) = 16.5881$, versus approximately 15.8082 here, a difference of less than 5% at the stipulated speed. For finite projectile mass the recoil factor in Y reduces the close contribution in the way already shown.

15.13 In the decay of a pi meson at rest a mu meson and a neutrino are created. The total kinetic energy available is $(m_\pi - m_\mu)c^2 = 34$ MeV. The mu meson has a kinetic energy of 4.1 MeV. Determine the number of quanta emitted per unit energy interval because of the sudden creation of the moving mu meson. Assuming that the photons are emitted perpendicular to the direction of motion of the mu meson (actually it is a $\sin^2\theta$ distribution), show that the maximum photon energy is 17 MeV. Find how many quanta are emitted with energies greater than one-tenth of the maximum, and compare your result with the observed ratio of radiative pi-mu decays. [$1.24 \pm 0.25 \times 10^{-4}$ for muons with kinetic energy less than 3.4 MeV. See also H. Primakoff, *Phys. Rev.*, **84**, 1255 (1951).]

Solution.

Let $\varepsilon = \hbar\omega$ and neglect radiation from the stationary parent pion and the neutral neutrino. For a suddenly created muon with speed $c\beta$, (15.65) gives

$$\frac{d^2N}{d\varepsilon\, d\Omega} = \frac{\alpha}{4\pi^2\varepsilon} \frac{\beta^2 \sin^2\theta}{(1 - \beta\cos\theta)^2}.$$

Put $x = \cos\theta$. The same elementary substitution $y = 1 - \beta x$ as in Problem 15.3 gives

$$\int_{-1}^{1} \frac{\beta^2(1-x^2)}{(1-\beta x)^2} dx = \frac{2}{\beta}\ln\frac{1+\beta}{1-\beta} - 4.$$

Therefore

$$\boxed{\frac{dN}{d\varepsilon} = \frac{C(\beta)}{\varepsilon}, \qquad C(\beta) = \frac{\alpha}{\pi}\left[\frac{1}{\beta}\ln\frac{1+\beta}{1-\beta} - 2\right].}$$

For $m_\mu c^2 = 105.66$ MeV and $T_\mu = 4.1$ MeV, $\beta = \sqrt{T_\mu(T_\mu + 2m_\mu c^2)}/(T_\mu + m_\mu c^2) = 0.270764$ and $C = 1.18800 \times 10^{-4}$. The nonrelativistic limit is $C \simeq 2\alpha\beta^2/(3\pi)$.

To implement the stipulated perpendicular-photon approximation, use $c = 1$ temporarily, let $\Delta = m_\pi - m_\mu = 34$ MeV, and call the muon momentum p. The neutrino momentum is then $\sqrt{p^2 + \varepsilon^2}$, so energy conservation reads

$$\Delta = T_\mu + \varepsilon + \sqrt{T_\mu^2 + 2m_\mu T_\mu + \varepsilon^2}.$$

Squaring and cancelling the common T_μ^2 and ε^2 terms gives

$$\boxed{T_\mu(\varepsilon) = \frac{\Delta(\Delta - 2\varepsilon)}{2(m_\pi - \varepsilon)}, \qquad 0 \le \varepsilon \le \Delta/2 = 17 \text{ MeV}.}$$

The maximum occurs at vanishing muon momentum. With the stated rounded masses the zero-photon energy from this relation is 4.139 MeV, consistent with the quoted 4.1 MeV accuracy. A frozen soft coefficient gives the requested count

$$\boxed{N(\varepsilon > 1.7 \text{ MeV}) \simeq C \ln(17/1.7) = 2.7355 \times 10^{-4}.}$$

This is about 2.2 times the measured ratio. There are two distinct limitations to that direct comparison. First, emission of an energetic photon reduces the muon speed, so the constant soft coefficient overestimates the high-energy portion. As a simple kinematic improvement, insert the $\beta[T_\mu(\varepsilon)]$ just derived into C and integrate:

$$\int_{1.7}^{17} \frac{C(\beta[T_\mu(\varepsilon)])}{\varepsilon} d\varepsilon = 1.7495 \times 10^{-4}.$$

Second, the experimental cut $T_\mu < 3.4$ MeV does not correspond to $\varepsilon > 1.7$ MeV. In this same perpendicular model it is

$$\varepsilon > \frac{\Delta^2 - 2(3.4)m_\pi}{2(\Delta - 3.4)} = 3.3711 \text{ MeV}.$$

With this matched cut the improved estimate is 1.0329×10^{-4}, comparable to $(1.24 \pm 0.25) \times 10^{-4}$. This last agreement is at the level of the model: the real angular distribution, decay phase space and non-soft amplitude are not replaced exactly by perpendicular emission and a velocity-dependent soft coefficient.

15.14 In internal conversion, the nucleus makes a transition from one state to another and an orbital electron is ejected. The electron has a kinetic energy equal to the transition energy minus its binding energy. For a conversion line of 1 MeV determine the number of quanta emitted per unit energy because of the sudden ejection of the electron. What fraction of the electrons will have energies less than 99% of the total energy? Will this low-energy tail on the conversion line be experimentally observable?

Solution.

Interpret the 1 MeV conversion line as the electron's kinetic energy T_0, as specified in the question. Initially bound electronic motion, nuclear recoil and atomic relaxation are neglected in the sudden-ejection approximation. Then

$$\gamma = 1 + \frac{T_0}{m_e c^2} = 2.956951, \qquad \beta = \sqrt{1 - \gamma^{-2}} = 0.941079.$$

The single-charge endpoint formula (15.67), or the angular integral in Problem 15.13, gives

$$\boxed{\frac{dN}{d\varepsilon} = \frac{\alpha}{\pi\varepsilon}\left[\frac{1}{\beta}\ln\frac{1+\beta}{1-\beta} - 2\right] = \frac{0.00398042}{\varepsilon}.}$$

If ε is measured in MeV, the displayed density is per MeV. The spectrum has this form for soft photons; the electron's kinetic-energy budget bounds its endpoint by $\varepsilon \le 1$ MeV.

An electron below $0.99T_0$ has lost more than $0.01T_0 = 10$ keV. At leading order in the emission probability, the requested fraction is the one-photon number on this band:

$$\boxed{f \simeq 0.00398042\int_{0.01}^{1}\frac{d\varepsilon}{\varepsilon} = 0.00398042\ln 100 = 0.01833,}$$

about 1.8%. Multiple-photon corrections are higher order in this small probability. Using the full relativistic electron energy $E_0 = T_0 + m_e c^2$ in the phrase "99% of the total energy" instead would impose $\varepsilon > 0.01E_0 = 15.11$ keV and give $f = 0.01669$; the usual line-energy convention refers to the measured kinetic energy and gives the first value.

This is an intrinsically observable, percent-level continuum below the conversion peak: a measurement resolving 10 keV and collecting adequate counts can distinguish such a tail. No detector resolution or background is specified, so its visibility in any particular apparatus cannot be asserted. External bremsstrahlung and electron energy loss in the source also make low-energy tails and must be controlled to isolate this intrinsic contribution. The estimated fraction is only logarithmically sensitive to the endpoint, but holding β fixed all the way to 1 MeV is still an extrapolation of the soft spectrum, not an exact conversion-line shape.

15.15 One of the decay modes of a K^+ meson is the three-pion decay, $K^+ \to \pi^+\pi^+\pi^-$. The energy release is 75 MeV, small enough that the pions can be treated nonrelativistically in rough approximation.

(a) Show that the differential spectrum of radiated intensity at low frequencies in the K meson rest frame is approximately

$$\frac{d^2I}{d\omega\, d\Omega} \simeq \frac{2e^2}{\pi^2 c}\cdot\frac{T_-}{m_\pi c^2}\cdot\sin^2\theta$$

where T_- is the kinetic energy of the negative pion and θ is the angle of emission of the photon relative to the momentum of the negative pion.

(b) Estimate the branching ratio for emission of a photon of energy greater than Δ relative to the nonradiative three-pion decay. What is its numerical value for $\Delta = 1$ MeV? 10 MeV? Compare with experiment ($\sim 2\times10^{-3}$ for $\Delta = 11$ MeV).

Solution.

(a) In the kaon rest frame momentum conservation, to leading nonrelativistic order for equal pion masses, is

$$\mathbf{v}_{+1} + \mathbf{v}_{+2} + \mathbf{v}_- = 0.$$

The parent's initial velocity is zero. The electric-dipole endpoint amplitude of the three created pions is consequently

$$\mathbf{J} = \frac{e}{c}P(\mathbf{v}_{+1} + \mathbf{v}_{+2} - \mathbf{v}_-) = -\frac{2e}{c}P\mathbf{v}_-.$$

Using $v_-^2 = 2T_-/m_\pi$ in $|\mathbf{J}|^2/(4\pi^2 c)$ gives

$$\boxed{\frac{d^2I}{d\omega\, d\Omega} = \frac{2e^2}{\pi^2 c}\frac{T_-}{m_\pi c^2}\sin^2\theta.}$$

The direction is measured from the negative pion because its velocity alone expresses the charge-weighted total after using momentum conservation. Integrating $\sin^2\theta$ yields

$$\frac{dN}{d\varepsilon} = \frac{16\alpha}{3\pi}\frac{T_-}{m_\pi c^2}\frac{1}{\varepsilon}.$$

(b) The nonradiative Dalitz distribution is not specified, so a numerical branching ratio requires an assumption about its average T_-. For a constant decay amplitude and the symmetric nonrelativistic three-body phase space, all three pions have the same mean energy, $\langle T_-\rangle = Q/3 = 25$ MeV, where $Q = 75$ MeV. This also follows by permutation symmetry from $\sum T_i = Q$. With $m_\pi c^2 = 139.6$ MeV,

$$C_3 = \frac{16\alpha}{3\pi}\frac{25}{139.6} = 0.00221855.$$

Extrapolating a frozen soft coefficient up to the available energy gives $B_0(\Delta) = C_3 \ln(Q/\Delta)$. A more useful rough estimate accounts for the reduced kinetic budget by $\langle T_-(\varepsilon)\rangle \simeq (Q-\varepsilon)/3$. It gives the elementary integral

$$\begin{aligned} B(\Delta) &\simeq C_3 \int_\Delta^Q \left(1 - \frac{\varepsilon}{Q}\right)\frac{d\varepsilon}{\varepsilon} \\ &= \boxed{C_3\left[\ln\frac{Q}{\Delta} - 1 + \frac{\Delta}{Q}\right].} \end{aligned}$$

Here B is relative to nonradiative three-pion decay, as requested. The resulting estimates are

Δ (MeV)	Frozen soft B_0	Reduced-budget B
1	9.579×10^{-3}	7.390×10^{-3}
10	4.470×10^{-3}	2.547×10^{-3}
11	4.259×10^{-3}	2.366×10^{-3}

The last improved value is close in scale to the stated 2×10^{-3} measurement. It is not a precision prediction: the available three-pion phase space and the actual decay amplitude also change when a photon is emitted. Even the exact endpoint is $[M_K^2 - 9m_\pi^2]c^2/(2M_K) \simeq 69.3$ MeV rather than Q, because the pion system recoils; Q is its nonrelativistic approximation. The coefficient of the soft logarithm follows from part (a), while the nonlogarithmic correction and hard-photon tail depend on the additional decay model. The table states explicitly which approximation supplies those estimates.

15.16 One of the decay modes of the charged K meson ($M_{\mathrm{K}} = 493.7$ MeV) is $\mathrm{K}^+ \to \pi^+\pi^0$ ($M_{\pi^+} = 139.6$ MeV, $M_{\pi^0} = 135.0$ MeV). Inner bremsstrahlung is emitted by the creation of the positive pion. A study of this radiative decay mode was made by Edwards et al. [*Phys. Rev.* **D5**, 2720 (1972)].

(a) Calculate the classical distribution in angle and frequency of soft photons and compare with the data of Fig. 6 of Edwards et al. Compute the classical distribution also for $\beta = 0.71$, corresponding to a charged pion of kinetic energy 58 MeV, and compare.

(b) Estimate the number of radiative decays for charged pion kinetic energies on the interval, 55 MeV $\le T_\pi \le$ 90 MeV, as a fraction of *all* K^+ decays (the $\pi^+\pi^0$ decay mode is 21% of all decays). You can treat the kinematics, including the photon, correctly, or you can approximate reality with an idealization that has the neutral pion always with the same momentum and the photon and the charged pion with parallel momenta (see part a for justification of this assumption). This idealization permits you to correlate directly the limits on the charged pion's kinetic energy with that of the photon. At fixed neutral momentum equal to its nonradiative value, this shortcut cannot also conserve energy exactly for a nonzero photon. Use it only as a momentum-based estimate, or use the full kinematics. Compare your estimate with the experimental value for the branching ratio for $\pi^+\pi^0\gamma$ (with the limited range of π^+ energies) of $(2.75 \pm 0.15) \times 10^{-4}$.

Solution.

(a) Use $c = 1$ for the kinematics, with all masses in MeV. Two-body energy and momentum conservation give

$$E_* = \frac{M_K^2 + m_+^2 - m_0^2}{2M_K} = 248.129279 \text{ MeV}, \quad p_* = \sqrt{E_*^2 - m_+^2} = 205.134051 \text{ MeV},$$

$$T_* = 108.529279 \text{ MeV}, \qquad \beta_* = p_*/E_* = 0.82672247.$$

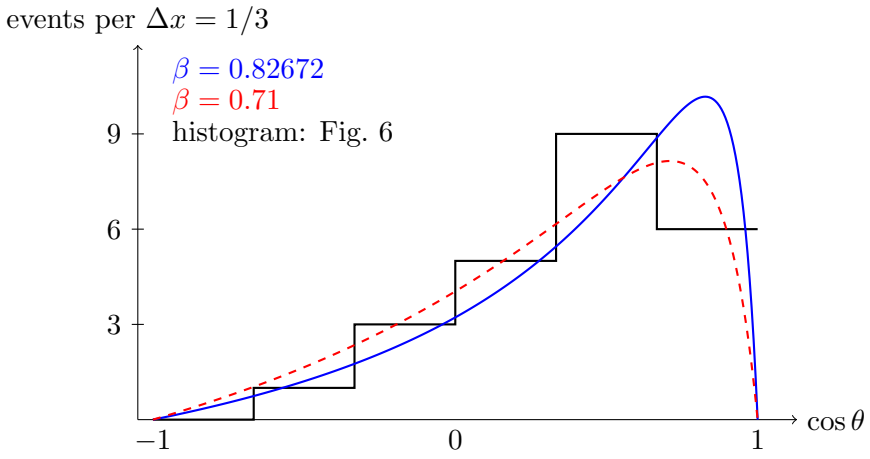
The initial kaon is stationary and the neutral pion has no charge. The soft radiation is therefore the endpoint radiation of the charged pion alone. Restoring c in the spectral intensity gives

$$\boxed{\frac{d^2I}{d\omega\, d\Omega} = \frac{e^2}{4\pi^2 c}\frac{\beta^2 \sin^2\theta}{(1-\beta\cos\theta)^2}, \qquad \frac{d^2N}{d\varepsilon\, dx} = \frac{\alpha}{2\pi\varepsilon}\frac{\beta^2(1-x^2)}{(1-\beta x)^2}, \quad x = \cos\theta.}$$

The intensity is independent of frequency in the soft limit, while the photon number is proportional to $1/\varepsilon$. Differentiating $(1-x^2)/(1-\beta x)^2$ gives $2(\beta - x)/(1-\beta x)^3$, so its maximum is at $x = \beta$: $\theta_{\max} = 34.24°$ for β_* and $44.77°$ for $\beta = 0.71$.

Figure 6 was checked in Edwards et al.'s original preprint, COO-1195-230 (December 1971), Fig. 6, p. 26 of the OSTI scan. It has 24 events in six equal x bins and imposes $58 \le T_\pi \le 90$ MeV. Normalizing either classical curve to 24 events gives the comparison

x bin	$[-1,-2/3]$	$[-2/3,-1/3]$	$[-1/3,0]$	$[0,1/3]$	$[1/3,2/3]$	$[2/3,1]$
Observed	0	1	3	5	9	6
$\beta = 0.82672$	0.35	1.22	2.44	4.25	7.07	8.66
$\beta = 0.71$	0.50	1.66	3.16	5.07	7.24	6.38



The lower-speed curve moves probability from the last bin toward the middle and describes the broad forward preference of these few events. The comparison is qualitative: their finite energy cut changes the pion speed and the angular acceptance, whereas each curve above has one fixed speed. In particular the original plot's 58 MeV lower cut is distinct from the 55 MeV cut requested in part (b).

(b) Exact three-body kinematics can be used without the inconsistent fixed-neutral-momentum idealization. Put $E = m_+ + T$, $p = \sqrt{E^2 - m_+^2}$ and $x = \hat{\mathbf{p}} \cdot \hat{\mathbf{k}}$. The neutral pion has momentum $-\mathbf{p} - \varepsilon\mathbf{n}$. Squaring $M_K - E - \varepsilon = \sqrt{m_0^2 + p^2 + \varepsilon^2 + 2p\varepsilon x}$ gives

$$\boxed{\varepsilon(T, x) = \frac{M_K(E_* - E)}{M_K - E + px}.}$$

At fixed x, increasing T decreases ε. In the parallel case the interval $55 \le T \le 90$ corresponds to $20.4933 \le \varepsilon \le 60.7979$ MeV. A fixed $\beta = 0.71$ and this parallel endpoint estimate give

$$0.21\,\frac{\alpha}{\pi}\left[\frac{1}{0.71}\ln\frac{1.71}{0.29} - 2\right]\ln\frac{60.7979}{20.4933} = 2.6475 \times 10^{-4}.$$

Using the full angular dependence of the endpoints, still at fixed β, instead gives the reproducible one-dimensional integral

$$B_\beta = 0.21\,\frac{\alpha}{2\pi}\int_{-1}^{1}\frac{\beta^2(1-x^2)}{(1-\beta x)^2}\ln\frac{\varepsilon(55,x)}{\varepsilon(90,x)}dx.$$

Its values are 4.1461×10^{-4} at β_* and 2.3943×10^{-4} at $\beta = 0.71$. The variation illustrates the uncertainty in extrapolating a fixed-speed soft formula to this rather hard photon interval.

A more consistent estimate retains both the outgoing speed and the three-body phase-space Jacobian, while assuming a constant underlying nonradiative decay amplitude and only its charge-radiation factor. The following derivation specifies that extra model rather than silently changing the soft approximation. In units $\hbar = c = 1$, let the nonradiative amplitude be $\mathcal{M}_0$. The endpoint factor, summed over photon polarizations, is

$$|\mathcal{M}_\gamma|^2 = 4\pi\alpha|\mathcal{M}_0|^2\frac{p^2(1-x^2)}{\varepsilon^2(E-px)^2}.$$

To retain the normalization explicitly, start from

$$d\Gamma_\gamma = \frac{|\mathcal{M}_\gamma|^2}{2M_K}(2\pi)^4\delta^{(4)}(P-p-p_0-k)\frac{d^3p}{(2\pi)^3 2E}\frac{d^3p_0}{(2\pi)^3 2E_0}\frac{d^3k}{(2\pi)^3 2\varepsilon}.$$

Integrating the spatial delta function fixes $\mathbf{p}_0 = -\mathbf{p} - \varepsilon\mathbf{n}$. The remaining measure is

$$\frac{|\mathcal{M}_\gamma|^2}{16M_K(2\pi)^5}\frac{p^2\varepsilon}{EE_0}\,dp\,d\varepsilon\,d\Omega_p\,d\Omega_k\,\delta(M_K - E - E_0 - \varepsilon).$$

Rotational symmetry supplies $\int d\Omega_p = 4\pi$ and a further 2π from the photon's azimuth about $\mathbf{p}$. The energy delta function's absolute derivative with respect to p is

$$\frac{p}{E} + \frac{p+\varepsilon x}{E_0} = \frac{p(M_K-\varepsilon)+E\varepsilon x}{EE_0}, \qquad E_0 = M_K - E - \varepsilon.$$

Thus the p integral gives

$$\frac{d\Gamma_\gamma}{d\varepsilon\,dx} = \frac{|\mathcal{M}_\gamma|^2}{64\pi^3 M_K}\frac{p^2\varepsilon}{p(M_K-\varepsilon)+E\varepsilon x}.$$

Dividing by $\Gamma_0 = |\mathcal{M}_0|^2 p_*/(8\pi M_K^2)$ gives

$$\frac{d^2N}{d\varepsilon\,dx} = \frac{\alpha}{2\pi\varepsilon}\frac{\beta(T)^2(1-x^2)}{[1-\beta(T)x]^2}\,W(T,x), \qquad W = \frac{M_K p^2}{p_*[p(M_K-\varepsilon)+E\varepsilon x]}.$$

$W \to 1$ as $\varepsilon \to 0$, independently checking the soft normalization. To integrate directly over the stated T interval, use

$$-\frac{\partial \ln\varepsilon}{\partial T} = \frac{1}{E_* - E} + \frac{-1+xE/p}{M_K - E + px}.$$

Consequently the numerical integral, with energies in MeV, is

$$\begin{aligned} B_{\text{model}} &= 0.21\,\frac{\alpha}{2\pi}\int_{55}^{90} dT \int_{-1}^{1} dx\,\frac{\beta(T)^2(1-x^2)}{[1-\beta(T)x]^2}\,W(T,x) \\ &\quad\times\left[\frac{1}{E_*-E} + \frac{-1+xE/p}{M_K-E+px}\right] \\ &= \boxed{2.5928\times10^{-4}} \quad \text{as a fraction of all } K^+ \text{ decays.} \end{aligned}$$

The supplementary code independently integrates in T and in photon energy, using the exact kinematic transformation, to check this value. It is close to the stated $(2.75 \pm 0.15)\times10^{-4}$, at the accuracy warranted by this inner-bremsstrahlung model; additional direct radiative amplitudes are not determined by a classical calculation.

Finally, the fixed-neutral-momentum option needs a qualification. If its momentum is held at the nonradiative value p_* and the photon is parallel to the charged pion, momentum conservation gives $p + \varepsilon = p_*$. Exact energy conservation at that same fixed neutral energy gives $E + \varepsilon = E_*$. Subtracting implies $E - p = E_* - p_*$; because $E - p = m_+^2/(E+p)$ is strictly decreasing with p, this requires $p = p_*$ and $\varepsilon = 0$. Thus a nonzero radiative event cannot satisfy all those constraints; the proposed shortcut can only be a momentum-based estimate. The calculations above use consistent photon kinematics instead.

16 Chapter 16. Radiation Damping, Classical Models of Charged Particles

Problems

16.1 A nonrelativistic particle of charge e and mass m is bound by a linear, isotropic, restoring force with force constant $m\omega_0^2$.

Using (16.13) and (16.16) of Section 16.2, show that the energy and angular momentum of the particle both decrease exponentially from their initial values as $e^{-\Gamma t}$, where $\Gamma = \omega_0^2\tau$. Quantum mechanically, the mean excitation energy of an oscillator decays in exactly the same way because the total radiative transition probability for a state with quantum numbers n_0, l_0 is $\Gamma(n_0, l_0) = n_0\Gamma$. The decay of the angular momentum approaches the classical law only for $l_0 \gg 1$.

Solution.

We use Gaussian units and write $\tau_r = 2e^2/(3mc^3)$ for the radiation-reaction time (the τ of this question). The equations (16.13) and (16.16) describe *secular*, orbit-averaged changes; they require $\omega_0\tau_r \ll 1$. For $V = \frac{1}{2}m\omega_0^2 r^2$, their explicit forms are

$$\frac{d\mathcal{E}}{dt} = -\frac{\tau_r}{m}\left\langle (V')^2 \right\rangle = -m\tau_r\omega_0^4\langle r^2\rangle, \qquad \frac{d\mathbf{L}}{dt} = -\frac{\tau_r}{m}\left\langle \frac{V'}{r} \right\rangle \mathbf{L} = -\tau_r\omega_0^2\mathbf{L}.$$

To express the first in terms of energy, write the unperturbed orbit $\mathbf{r} = \mathbf{A}\cos\omega_0 t + \mathbf{B}\sin\omega_0 t$. A full-period average gives

$$\langle r^2\rangle = \frac{A^2 + B^2}{2}, \quad \langle v^2\rangle = \omega_0^2\frac{A^2 + B^2}{2}, \quad \mathcal{E} = \frac{m}{2}\langle v^2 + \omega_0^2 r^2\rangle = m\omega_0^2\langle r^2\rangle.$$

Thus both differential equations have the same decay constant:

$$\boxed{\mathcal{E}(t) = \mathcal{E}(0)e^{-\Gamma t}, \qquad \mathbf{L}(t) = \mathbf{L}(0)e^{-\Gamma t}, \qquad \Gamma = \omega_0^2\tau_r.}$$

The angular-momentum direction is unchanged. The energy here is the slowly varying orbital energy; small within-cycle corrections are outside the averaging approximation. In the reduced-order oscillator $\ddot{\mathbf{r}} + \Gamma\dot{\mathbf{r}} + \omega_0^2\mathbf{r} = 0$, the coordinate amplitude decays as $e^{-\Gamma t/2}$, giving the same energy envelope. The quantum comparison in the question concerns excitation energy above the ground state, whose zero-point energy is not radiated away.

16.2 A nonrelativistic electron of charge $-e$ and mass m bound in an attractive Coulomb potential $(-Ze^2/r)$ moves in a circular orbit in the absence of radiation reaction.

(a) Show that both the energy and angular-momentum equations (16.13) and (16.16) lead to the solution for the slowly changing orbit radius,

$$r^3(t) = r_0^3 - 9Z(c\tau)^3\frac{t}{\tau}$$

where r_0 is the value of $r(t)$ at $t = 0$.

(b) For circular orbits in a Bohr atom the orbit radius and the principal quantum number n are related by $r = n^2 a_0/Z$. If the transition probability for transitions from $n \to (n-1)$ is defined as $-dn/dt$, show that the result of part a agrees with that found in Problem 14.21.

(c) From part a calculate the numerical value of the times taken for a mu meson of mass $m = 207m_e$ to fall from a circular orbit with principal quantum number $n_1 = 10$ to one with $n_2 = 4$, and $n_2 = 1$. These are reasonable estimates of the time taken for a mu meson to cascade down to its lowest orbit after capture by an isolated atom.

Solution.

Put $A = Ze^2$ and $\tau_r = 2e^2/(3mc^3)$. The orbit remains nearly circular when its radiative evolution is slow compared with a period.

(a) Circular force balance and the orbital energy give

$$v^2 = \frac{A}{mr}, \qquad \mathcal{E} = -\frac{A}{2r}, \qquad L = \sqrt{mAr}.$$

The energy equation is

$$\frac{A}{2r^2}\dot{r} = -\frac{\tau_r}{m}\frac{A^2}{r^4}, \quad \text{hence} \quad \dot{r} = -\frac{2\tau_r A}{mr^2}.$$

Independently, the angular-momentum equation gives

$$\dot{L} = -\frac{\tau_r A}{mr^3}L, \qquad \dot{L} = \frac{L}{2r}\dot{r},$$

which yields the same $\dot{r}$. Multiply it by $3r^2$ and use $e^2/m = 3c^3\tau_r/2$:

$$\frac{d(r^3)}{dt} = -\frac{6\tau_r Ze^2}{m} = -9Zc^3\tau_r^2.$$

Integration from $t = 0$ gives

$$\boxed{r^3(t) = r_0^3 - 9Z(c\tau_r)^3\frac{t}{\tau_r}.}$$

This formal shrinking-orbit law ceases to apply when the orbit is no longer nonrelativistic and slowly varying; it does not describe the final singular collapse of a point-particle model reliably.

(b) For a particle of mass m, define its Bohr length $a_m = \hbar^2/(me^2)$, so $r = n^2a_m/Z$. Since $r^3 = n^6a_m^3/Z^3$, differentiation gives

$$6n^5\frac{a_m^3}{Z^3}\dot{n} = -9Zc^3\tau_r^2, \qquad -\dot{n} = \frac{3Z^4c^3\tau_r^2}{2a_m^3n^5}.$$

Substitute $a_m = \hbar/(mc\alpha)$ and $\tau_r = 2\alpha\hbar/(3mc^2)$:

$$\boxed{-\dot{n} = \frac{2}{3}\alpha(Z\alpha)^4\frac{mc^2}{\hbar}\frac{1}{n^5},}$$

exactly the classical transition-rate estimate in 14.21.

(c) Integrate $dt = dn/\dot{n}$ from n_1 down to n_2. Since $\dot{n} < 0$, reversing the integration limits gives the positive time

$$\Delta t = \frac{3\hbar}{2Z^4\alpha^5mc^2}\int_{n_2}^{n_1} n^5dn = \boxed{\frac{\hbar(n_1^6 - n_2^6)}{4Z^4\alpha^5mc^2}.}$$

For the specified $m = 207m_e$, $\hbar/(4\alpha^5mc^2) = 7.51776552 \times 10^{-14}$ s. Thus

$$\boxed{\Delta t_{10\to4} = \frac{7.48697 \times 10^{-8}}{Z^4}\ \text{s}, \qquad \Delta t_{10\to1} = \frac{7.51776 \times 10^{-8}}{Z^4}\ \text{s}.}$$

No nuclear charge was specified, so the Z^{-4} dependence is retained; for hydrogen set $Z = 1$. Most of the classical cascade time is spent at large n. These are the requested classical estimates, with a fixed point nucleus, no screening or recoil, and the supplied mass ratio.

16.3 An electron moving in an attractive Coulomb field ($-Ze^2/r$) with binding energy ϵ and angular momentum L has an elliptic orbit,

$$\frac{1}{r} = \frac{Ze^2 m}{L^2}\left[1 + \sqrt{1 - \frac{2\epsilon L^2}{Z^2 e^4 m}}\,\cos(\theta - \theta_0)\right]$$

The eccentricity of the ellipse is given by the square root multiplying the cosine.

(a) By performing the appropriate time averages over the orbit, show that the secular changes in energy and angular momentum are

$$\frac{d\epsilon}{dt} = \frac{2^{3/2}}{3}\frac{Z^3 e^8 m^{1/2}}{c^3}\frac{\epsilon^{3/2}}{L^5}\left(3 - \frac{2\epsilon L^2}{Z^2 e^4 m}\right)$$

$$\frac{dL}{dt} = -\frac{2^{5/2}}{3}\frac{Ze^4}{m^{1/2}c^3}\frac{\epsilon^{3/2}}{L^2}$$

(b) If the initial values of ϵ and L are ϵ_0 and L_0, show that

$$\epsilon(L) = \frac{Z^2 e^4 m}{2L^2}\left[1 - \left(\frac{L}{L_0}\right)^3\right] + \frac{\epsilon_0}{L_0}\,L$$

Calculate the eccentricity of the ellipse, and show that it decreases from its initial value as $(L/L_0)^{3/2}$, showing that the orbit tends to become circular as time goes on.

(c) Compare your results here to the special case of a circular orbit of Problem 16.2.

Hint: In performing the time averages make use of Kepler's law of equal areas ($dt = mr^2\,d\theta/L$) to convert time integrals to angular integrals.

Solution.

Let $A = Ze^2$, let $\varepsilon > 0$ denote the binding energy, and write the eccentricity as η to distinguish it from the charge. Set the periapsis at $\theta = 0$. The orbit and its parameters are

$$r = \frac{p}{1 + \eta\cos\theta}, \quad p = \frac{L^2}{mA}, \quad \eta^2 = 1 - \frac{2\varepsilon L^2}{mA^2}, \quad T = 2\pi\sqrt{\frac{m[A/(2\varepsilon)]^3}{A}} = \frac{2\pi A\sqrt{m}}{(2\varepsilon)^{3/2}}.$$

(a) Since $dt = mr^2 d\theta/L$, the two needed averages are

$$\langle r^{-4}\rangle = \frac{m}{TLp^2}\int_0^{2\pi}(1 + \eta\cos\theta)^2 d\theta = \frac{2\pi m}{TLp^2}(1 + \eta^2/2),$$

$$\langle r^{-3}\rangle = \frac{m}{TLp}\int_0^{2\pi}(1 + \eta\cos\theta)d\theta = \frac{2\pi m}{TLp}.$$

Here $\int\cos\theta\,d\theta = 0$ and $\int\cos^2\theta\,d\theta = \pi$ over the full period. The energy radiated makes the binding energy increase:

$$\dot\varepsilon = \frac{2e^2A^2}{3m^2c^3}\langle r^{-4}\rangle, \qquad \dot L = -\frac{2e^2AL}{3m^2c^3}\langle r^{-3}\rangle.$$

Insert $p = L^2/(mA)$ and $2\pi/T = (2\varepsilon)^{3/2}/(A\sqrt{m})$:

$$\dot\varepsilon = \frac{2^{5/2}e^2A^3\sqrt{m}}{3c^3}\frac{\varepsilon^{3/2}}{L^5}(1 + \eta^2/2), \qquad \dot L = -\frac{2^{5/2}e^2A}{3\sqrt{m}c^3}\frac{\varepsilon^{3/2}}{L^2}.$$

Finally $1+\eta^2/2 = [3 - 2\varepsilon L^2/(mA^2)]/2$ and $e^2A^3 = Z^3e^8$, $e^2A = Ze^4$ give both formulas in the question.

(b) Put $C = mA^2$. Dividing the two rates gives

$$\frac{d\varepsilon}{dL} = -\frac{C}{2L^3}\left(3 - \frac{2\varepsilon L^2}{C}\right) = \frac{\varepsilon}{L} - \frac{3C}{2L^3}.$$

Multiplication by $1/L$ makes the left side a total derivative:

$$\frac{d}{dL}\left(\frac{\varepsilon}{L}\right) = -\frac{3C}{2L^4}, \qquad \frac{\varepsilon}{L} - \frac{\varepsilon_0}{L_0} = \frac{C}{2}(L^{-3} - L_0^{-3}).$$

Therefore

$$\boxed{\varepsilon(L) = \frac{mZ^2e^4}{2L^2}\left[1 - (L/L_0)^3\right] + \frac{\varepsilon_0}{L_0}L.}$$

Substitution in the eccentricity relation yields

$$\eta^2 = \left(1 - \frac{2\varepsilon_0 L_0^2}{C}\right)(L/L_0)^3, \qquad \boxed{\eta = \eta_0 (L/L_0)^{3/2}.}$$

Since $\dot{L} < 0$, a bound ellipse becomes progressively more circular within this secular approximation.

(c) For an initially circular orbit $\eta_0 = 0$, hence $\eta = 0$ and $\varepsilon = C/(2L^2) = A/(2r)$ at every later secular time. With $L^2 = mAr$, the angular-momentum rate in (a) becomes $\dot{L} = -\tau_r AL/(mr^3)$. Using $\dot{L} = L\dot{r}/(2r)$ gives $\dot{r} = -2\tau_r A/(mr^2)$ and the law in 16.2, with no additional assumption about a finite eccentricity.

16.4 A classical model of an electron is a spherical shell of charge of radius a and total charge e.

(a) Using (16.30) for the "mass" $M(\omega)$ and the angular average of $e^{i\omega R/c}/R$, show that

$$M(\omega) = m + \frac{2}{3}\frac{e^2}{ac^2}\cdot\left(\frac{e^{i\xi} - 1 - i\xi}{i\xi}\right)$$

where $\xi = 2\omega a/c$, and $m = m_0 + 2e^2/3ac^2$ is the physical mass of the electron.

(b) Expand in powers of the frequency (ξ) and show that, to lowest nontrivial order, $M(\omega)$ has a zero in the upper half-plane at $\omega\tau = i$, where $\tau = 2e^2/3mc^3$. What is the physical significance of such a zero?

(c) For the exact result of part a, show that the zeros of $\omega M(\omega)$, if any, are defined by the two simultaneous equations, proportional to the real and imaginary parts of $i\omega M(\omega)$,

$$e^{-y}\cos x - 1 + y(1 - a/c\tau) = 0$$
$$e^{-y}\sin x - x(1 - a/c\tau) = 0$$

where $x = \mathrm{Re}\,\xi$ and $y = \mathrm{Im}\,\xi$. Find the condition on the radius a such that $\omega M(\omega)$ has no zeros in the upper half-ω-plane. Express the condition also in terms of the mechanical mass m_0 for fixed physical mass. What about zeros and/or singularities in the lower half-plane?

Solution.

Write $\tau_r = 2e^2/(3mc^3)$ and $h = a/(c\tau_r) > 0$. Here m is the fixed positive physical mass; m_0 may have either sign.

(a) Equation (16.30) is

$$M(\omega) = m_0 + \frac{2}{3c^2}\int d^3x\, d^3x'\, \rho(\mathbf{x})\rho(\mathbf{x}')\frac{e^{i\omega R/c}}{R}.$$

Fix one point on the spherical shell and let u be the cosine of the angle to the other point. Their separation satisfies $R^2 = 2a^2(1-u)$, so $du = -R\,dR/a^2$. Its normalized angular average is

$$\frac{1}{2}\int_{-1}^{1}\frac{e^{i\omega R/c}}{R}du = \frac{1}{2a^2}\int_0^{2a} e^{i\omega R/c}dR = \frac{1}{a}\frac{e^{i\xi}-1}{i\xi}, \qquad \xi = \frac{2\omega a}{c}.$$

The remaining shell integral multiplies this average by e^2. Thus, with $m = m_0 + 2e^2/(3ac^2)$,

$$\boxed{M = m_0 + \frac{2e^2}{3ac^2}\frac{e^{i\xi}-1}{i\xi} = m + \frac{2e^2}{3ac^2}\frac{e^{i\xi}-1-i\xi}{i\xi}.}$$

The value at $\xi = 0$ is its continuous limit, $M(0) = m$.

(b) Expanding the exponential first gives

$$\frac{e^{i\xi}-1-i\xi}{i\xi} = \frac{i\xi}{2} - \frac{\xi^2}{6} - \frac{i\xi^3}{24} + \cdots, \qquad M = m[1 + i\omega\tau_r + O(\omega^2 a\tau_r/c)].$$

The truncated expression vanishes at $\omega\tau_r = i$. Because the response is $v(\omega) = F(\omega)/[-i\omega M(\omega)]$, this is a pole associated with a homogeneous e^{t/τ_r} runaway. It is a zero of the *truncated* mass function, not necessarily of the exact finite-radius model. For $a \gtrsim c\tau_r$, its frequency is outside the small-$|\xi|$ expansion used to obtain it.

(c) Multiplying M/m by $i\xi h$ gives the equation

$$e^{i\xi} - 1 - i(1-h)\xi = 0.$$

It describes zeros of ωM, including the free-particle zero at $\omega = 0$, which is not a zero of M itself. With $r = 1-h$, $\xi = x + iy$, its real and imaginary parts are

$$e^{-y}\cos x - 1 + ry = 0, \qquad e^{-y}\sin x - rx = 0,$$

as asserted. If $h \geq 1$, then $r \leq 0$: for $y > 0$ the first equation would require $e^{-y}\cos x = 1 - ry \geq 1$, which is impossible. For $0 < h < 1$, set $x = 0$ and consider $f(y) = e^{-y} - 1 + ry$. Here $f(0) = 0$, $f'(0) = -h < 0$, $f''(y) > 0$, and $f(y) \to +\infty$ as $y \to +\infty$. There is exactly one additional positive root. Therefore

$$\boxed{\text{No upper-half-plane zero} \iff a \geq c\tau_r \iff m_0 \geq 0.}$$

There are no off-axis upper-half-plane roots even when $m_0 < 0$. For $x \neq 0$, eliminate r between the two equations to get $e^y = \cos x + y\sin x/x \leq 1 + y$, contradicting $e^y > 1 + y$ for $y > 0$.

For completeness, the zeros can be parameterized for $r \neq 0$ by the branches of Lambert's function:

$$\xi_k = i\left[\frac{1}{r} + W_k\left(-\frac{e^{-1/r}}{r}\right)\right].$$

This follows by putting $z = -i\xi$ and $(z - 1/r)e^{z-1/r} = -e^{-1/r}/r$. One branch gives the already identified root at zero. For $h > 1$ all other roots lie strictly in the lower half-plane; for $0 < h < 1$ one is on the positive imaginary axis and the remaining nonzero roots lie in the lower half-plane. They occur in pairs $\xi, -\xi^*$ off the imaginary axis. At the borderline $h = 1$, $e^{i\xi} = 1$ and the nonzero roots are real, $\xi = 2\pi n$: the model then has undamped modes rather than strictly damped ones. The exact M is entire: its apparent singularity at zero is removable, and it has no finite poles. Its zeros give poles of the response; lower-half-plane poles describe decaying transients. The exponential also means that infinity is not a removable singularity of M.

16.5 The particle of Problem 16.4, with $a > c\tau$ (positive mechanical mass), is initially at rest in a spatially uniform, but time-varying electric field $E(t) = E_0\Theta(t)$.

(a) Show that its speed in the direction of the field is given by the integral

$$v(t) = -\frac{eE_0 a}{\pi c}\int_{-\infty}^{\infty} d\xi\, \frac{e^{-iT\xi}}{\xi^2 M(\xi)}$$

where $T = ct/2a$.

(b) From the analytic properties of $M(\xi)$ established in part c of Problem 16.4, show that $v(t) = 0$ for $t \le 0$ (no preacceleration).

Solution.

Let $F_0 = eE_0$, $\tau_r = 2e^2/(3mc^3)$, and assume the stable finite-radius case $a > c\tau_r$, equivalently $m_0 > 0$. The contour prescription at real-axis poles is part of the initial rest condition.

(a) Use Jackson's symmetric Fourier pair (printed p. 753),

$$\widetilde{f}(\omega) = \frac{1}{\sqrt{2\pi}}\int f(t)e^{i\omega t}dt, \qquad f(t) = \frac{1}{\sqrt{2\pi}}\int \widetilde{f}(\omega)e^{-i\omega t}d\omega.$$

Regularizing the step by $e^{-\epsilon t}$ gives

$$\widetilde{F}(\omega) = \lim_{\epsilon\downarrow 0}\frac{F_0}{\sqrt{2\pi}}\int_0^{\infty} e^{(i\omega-\epsilon)t}dt = \frac{iF_0}{\sqrt{2\pi}(\omega + i0)}.$$

Equation (16.29), $-i\omega M\widetilde{v} = \widetilde{F}$, then gives $\widetilde{v} = -F_0/[\sqrt{2\pi}\,\omega^2 M(\omega)]$, with the integration line passing above the zero-frequency pole. The inverse transform supplies the second factor $1/\sqrt{2\pi}$. Consequently

$$v(t) = -\frac{F_0}{2\pi}\int_{\mathbb{R}+i0}\frac{e^{-i\omega t}}{\omega^2 M(\omega)}d\omega.$$

Set $\xi = 2a\omega/c$, $T = ct/(2a)$ and $d\omega/\omega^2 = (2a/c)d\xi/\xi^2$. This yields

$$\boxed{v(t) = -\frac{eE_0 a}{\pi c}\int_{\mathbb{R}+i0}\frac{e^{-iT\xi}}{\xi^2 M(\xi)}d\xi.}$$

The real-line integral in the question must be understood with this retarded prescription, rather than as an ordinary integral through a double pole.

(b) For $t < 0$, close the contour in the upper half-plane. There are no enclosed poles by 16.4 and the contour passes above $\xi = 0$. In this half-plane $e^{i\xi}$ is bounded and $M(\xi) \to m_0 \neq 0$ as $|\xi| \to \infty$.

The factor $e^{-iT\xi}$ decays for $T<0$, and even at $T=0$ the integrand falls as ξ^{-2}, making the arc contribution vanish. Cauchy's theorem therefore gives

$$\boxed{v(t)=0 \quad (t\le 0).}$$

One can check the initial behavior directly. Let $d=2a/c$ and $k=2e^2/(3ac^2d)$. The exact mass function gives the causal delay equation

$$m_0\dot{v}(t)+k[v(t)-v(t-d)]=F(t).$$

For $0<t<d$ its history is zero, so $v(t)=(F_0/k)(1-e^{-kt/m_0})$; in particular $v(0)=0$. At $m_0=0$ a causal solution still exists, but it has an instantaneous right-hand jump $v(0^+)=F_0/k$ and real undamped response poles. For $m_0<0$ the upper-plane runaway pole invalidates the stated real-line contour argument: a prescription excluding runaway behavior then produces preacceleration. A causal contour above that pole instead retains the runaway. The positive-mass qualification separates these different cases.

16.6 A particle of bare mass m_0 and charge e has a charge density, $\rho(\mathbf{x})=e\,e^{-r/a}/4\pi a^2 r$.

(a) Show that the charge form factor is $f(\mathbf{k})=(1+k^2a^2)^{-1}$.

(b) Show that the mass, (16.33), is

$$m=m_0+\frac{mc\tau}{2a}$$

(c) Show that the zeros of $M(\omega)$, (16.34), in the complex ω plane, are given by

$$\omega\tau=-i(c\tau/a)\left[1\pm(1-2a/c\tau)^{-1/2}\right]$$

(d) Find the trajectories of the roots in the complex ω plane for $m_0>0$ and $m_0<0$. Find the limiting form for the roots when $a/c\tau\ll 1$ and $a/c\tau\gg 1$. Discuss.

Solution.

Set $h=a/(c\tau_r)>0$, $\tau_r=2e^2/(3mc^3)$, and use the retarded frequency convention of 16.4–16.5.

(a) Spherical symmetry gives

$$f(k)=\frac{1}{e}\int\rho(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}}d^3x=\frac{1}{a^2k}\int_0^\infty e^{-r/a}\sin(kr)dr.$$

The integral is the imaginary part of $\int_0^\infty e^{-(a^{-1}-ik)r}dr=(a^{-1}-ik)^{-1}$; its imaginary part is $k/(a^{-2}+k^2)$. Therefore

$$\boxed{f(k)=\frac{1}{1+a^2k^2},\qquad f(0)=1,}$$

which also checks the total-charge normalization.

(b) The radial version of (16.33) is

$$m-m_0=\frac{4e^2}{3\pi c^2}\int_0^\infty f(k)^2dk=\frac{4e^2}{3\pi ac^2}\int_0^\infty\frac{dq}{(1+q^2)^2}.$$

Putting $q = \tan\theta$ makes the last integral $\int_0^{\pi/2} \cos^2\theta\, d\theta = \pi/4$. Hence

$$\boxed{m = m_0 + \frac{e^2}{3ac^2} = m_0 + \frac{mc\tau_r}{2a}, \qquad \frac{m_0}{m} = 1 - \frac{1}{2h}.}$$

(c) To evaluate the frequency dependence put $s = \omega a/c$. The radial mass integral becomes

$$M = m_0 + \frac{4e^2}{3\pi ac^2}\int_0^\infty \frac{q^2\, dq}{(1+q^2)^2(q^2 - s^2)}, \qquad \mathrm{Im}\, s > 0.$$

The partial fractions are

$$\frac{q^2}{(1+q^2)^2(q^2-s^2)} = \frac{s^2}{(1+s^2)^2}\left(\frac{1}{q^2-s^2} - \frac{1}{q^2+1}\right) + \frac{1}{1+s^2}\frac{1}{(q^2+1)^2}.$$

The three integrals are $i\pi/(2s)$, $\pi/2$, and $\pi/4$. For the first, closing the even full-line integrand above the real axis encloses the pole $q = s$ with residue $1/(2s)$; half the full-line integral is $i\pi/(2s)$. Combining the terms gives

$$\frac{\pi}{4}\frac{1 - s^2 + 2is}{(1+s^2)^2} = \frac{\pi}{4(1-is)^2}, \qquad \boxed{M = m_0 + \frac{m}{2h}(1-is)^{-2}.}$$

The apparent exception at $s = i$ in the partial fractions is removed by continuity. A zero of M for $m_0 \neq 0$ obeys $(1-is)^2 = (1-2h)^{-1}$, hence

$$\boxed{\omega\tau_r = -\frac{i}{h}\left[1 \pm (1-2h)^{-1/2}\right].}$$

There is also a double pole of M at $\omega = -ic/a$; it is a zero, not a pole, of the mechanical response $1/[-i\omega M]$.

(d) For $m_0 > 0$ one has $h > 1/2$. The roots are

$$\omega\tau_r = \pm\frac{1}{h\sqrt{2h-1}} - \frac{i}{h}.$$

Both are damped. In coordinates $X = \mathrm{Re}(\omega\tau_r)$, $Y = \mathrm{Im}(\omega\tau_r)$, their trajectories satisfy

$$X^2 = -\frac{Y^3}{Y+2}, \qquad -2 < Y < 0.$$

As $h \downarrow 1/2$ their real parts tend to opposite infinities and $Y \to -2$; as $h \to \infty$ both approach the origin from below, with

$$\omega\tau_r = \pm\frac{1}{\sqrt{2}h^{3/2}}[1 + O(h^{-1})] - \frac{i}{h}.$$

For $m_0 < 0$, $0 < h < 1/2$ and both roots are imaginary:

$$\omega_+\tau_r = \frac{i}{h}[(1-2h)^{-1/2} - 1], \qquad \omega_-\tau_r = -\frac{i}{h}[(1-2h)^{-1/2} + 1].$$

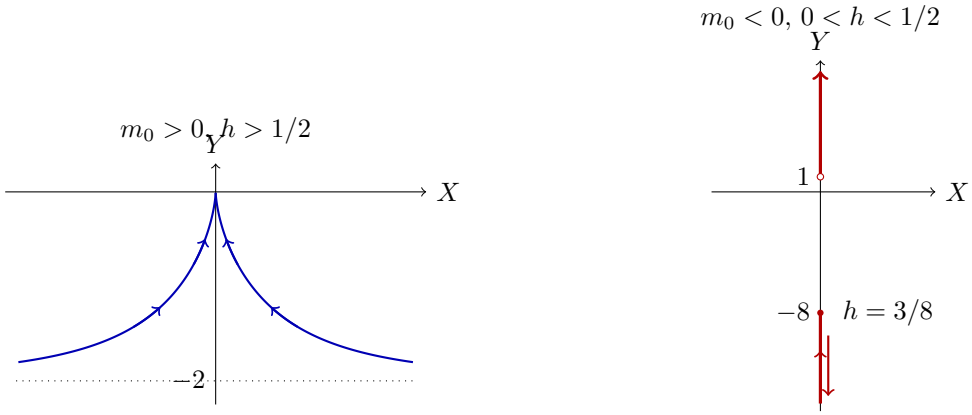
The first is a runaway, the second a decaying mode. The upper root rises monotonically from $Y = 1$ to $+\infty$. The lower root first rises from $-\infty$ to $Y = -8$, then returns to $-\infty$. To locate that turning point put $s = (1-2h)^{-1/2}$:

$$\frac{d}{dh}\frac{1+s}{h} = \frac{hs^3 - 1 - s}{h^2}, \qquad hs^3 - 1 - s = \tfrac{1}{2}(s^3 - 3s - 2) = \tfrac{1}{2}(s-2)(s+1)^2.$$

Thus $s = 2$, $h = 3/8$, and $Y_- = -(1+2)/(3/8) = -8$. Expanding $(1-2h)^{-1/2} = 1 + h + 3h^2/2 + \cdots$ gives

$$\omega_+ \tau_r = i[1 + 3h/2 + O(h^2)], \qquad \omega_- \tau_r = -i[2/h + 1 + 3h/2 + O(h^2)].$$

The point limit retains the Abraham–Lorentz runaway at i/τ_r; the other root goes to negative imaginary infinity. As $h \uparrow 1/2$ the two roots go to opposite imaginary infinities. At exactly $h = 1/2$, $m_0 = 0$ and $M = m/(1 - is)^2$ has no finite zeros: the displayed root formula is then interpreted through its limits, not evaluated at its singular denominator. The plot shows these branches; arrows indicate increasing h.



16.7 The Dirac (1938) relativistic theory of classical point electrons has as its equation of motion,

$$\frac{dp_\mu}{d\tau} = F_\mu^{\text{ext}} + F_\mu^{\text{rad}}$$

where p_μ is the particle's 4-momentum, τ is the particle's proper time, and F_μ^{rad} is the covariant generalization of the radiative reaction force (16.8).

Using the requirement that any force must satisfy $F_\mu p^\mu = 0$, show that

$$F_\mu^{\text{rad}} = \frac{2e^2}{3mc^3}\left[\frac{d^2 p_\mu}{d\tau^2} + \frac{p_\mu}{m^2c^2}\left(\frac{dp_\nu}{d\tau}\frac{dp^\nu}{d\tau}\right)\right]$$

Solution.

Use metric $(+,-,-,-)$, with $p_\mu p^\mu = m^2c^2$. In this problem a dot denotes a derivative with respect to proper time. Differentiating the mass-shell relation once and twice gives

$$p \cdot \dot{p} = 0, \qquad p \cdot \ddot{p} = -\dot{p} \cdot \dot{p}.$$

The covariant local term corresponding to the nonrelativistic jerk is $\ddot{p}_\mu$. Its component along p_μ must be removed because a four-force cannot change the rest mass. The orthogonal projection is

$$\ddot{p}_\mu - \frac{p_\mu}{m^2c^2}(p \cdot \ddot{p}) = \ddot{p}_\mu + \frac{p_\mu}{m^2c^2}(\dot{p} \cdot \dot{p}).$$

After raising the index, the contravariant spatial component in the instantaneous nonrelativistic limit is $m\, d^2\mathbf{v}/dt^2$; the covariant spatial components of both momentum and force have the opposite sign. The known force $2e^2\, d^2\mathbf{v}/(3c^3 dt^2)$ fixes its coefficient. Thus

$$\boxed{F_\mu^{\text{rad}} = \frac{2e^2}{3mc^3}\left[\ddot{p}_\mu + \frac{p_\mu}{m^2c^2}(\dot{p} \cdot \dot{p})\right].}$$

Contraction with p^μ explicitly gives zero. This construction uses the same local jerk-order radiation-reaction model as the stated Dirac equation; orthogonality alone would not rule out arbitrary additional transverse forces of other orders.

16.8

(a) Show that for relativistic motion in one dimension the equation of motion of Problem 16.7 can be written in the form,

$$\dot{p} - \frac{2e^2}{3mc^3}\left(\ddot{p} - \frac{p\dot{p}^2}{p^2 + m^2c^2}\right) = \sqrt{1 + \frac{p^2}{m^2c^2}}\, f(\tau)$$

where p is the momentum in the direction of motion, a dot means differentiation with respect to proper time, and $f(\tau)$ is the ordinary Newtonian force as a function of proper time.

(b) Show that the substitution of $p = mc\sinh y$ reduces the relativistic equation to the Abraham–Lorentz form (16.9) in y and τ. Write down the general solution for $p(\tau)$, with the initial condition that

$$p(\tau) = p_0 \qquad \text{at} \qquad \tau = 0$$

Solution.

To avoid confusing two different times, retain τ for proper time and write $\tau_r = 2e^2/(3mc^3)$ for the damping constant.

(a) In one-dimensional motion $p^\mu = (\sqrt{p^2 + m^2c^2}, p, 0, 0)$, so

$$\dot{p}^0 = \frac{p\dot{p}}{\sqrt{p^2 + m^2c^2}}, \qquad \dot{p}_\mu \dot{p}^\mu = -\frac{m^2c^2\dot{p}^2}{p^2 + m^2c^2}.$$

The spatial external four-force is $F^1_{\text{ext}} = \gamma f(\tau)$, with $\gamma = \sqrt{1 + p^2/(m^2c^2)}$. The spatial Lorentz–Dirac equation from 16.7 is therefore

$$\boxed{\dot{p} - \tau_r\left(\ddot{p} - \frac{p\dot{p}^2}{p^2 + m^2c^2}\right) = \gamma f(\tau).}$$

(b) Set $p = mc\sinh y$. Then

$$\dot{p} = mc\cosh y\, \dot{y}, \qquad \ddot{p} = mc(\cosh y\, \ddot{y} + \sinh y\, \dot{y}^2),$$

$$\frac{p\dot{p}^2}{p^2 + m^2c^2} = mc\sinh y\, \dot{y}^2.$$

The quadratic terms cancel and division by $\gamma = \cosh y$ gives $\dot{y} - \tau_r\ddot{y} = f/(mc)$, the Abraham–Lorentz equation in rapidity and proper time. Put $u = \dot{y}$. Multiplication by $e^{-\tau/\tau_r}$ gives

$$\frac{d}{d\tau}(ue^{-\tau/\tau_r}) = -\frac{f(\tau)}{mc\tau_r}e^{-\tau/\tau_r}, \quad u(\tau) = e^{\tau/\tau_r}\left[C - \frac{1}{mc\tau_r}\int_0^\tau e^{-s/\tau_r} f(s)ds\right].$$

Here $C = u(0)$ remains an arbitrary initial rapidity derivative. Integrating once more and interchanging the finite integrals yields

$$y(\tau) = \operatorname{arsinh}\frac{p_0}{mc} + C\tau_r(e^{\tau/\tau_r} - 1) - \frac{1}{mc}\int_0^\tau [e^{(\tau-s)/\tau_r} - 1]f(s)ds, \qquad \boxed{p(\tau) = mc\sinh y(\tau).}$$

The one specified momentum fixes $y(0)$, but leaves C free; a second-order equation for y needs another condition. If runaway behavior is excluded and the integral converges, set

$$C = \frac{1}{mc\tau_r}\int_0^\infty e^{-s/\tau_r} f(s)ds.$$

Then $\dot{y} = (mc)^{-1}\int_0^\infty e^{-s} f(\tau + \tau_r s)ds$, which exhibits the future-force dependence of the nonrunaway solution.

16.9

(a) Show that the radiation reaction force in the Lorentz–Dirac equation of Problem 16.7 can be expressed alternatively as

$$F_\mu^{\text{rad}} = \frac{2e^2}{3mc^3}\left[\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2c^2}\right)\frac{d^2p^\nu}{d\tau^2}\right]$$

(b) The relativistic generalization of (16.10) can be obtained by replacing $d^2p^\nu/d\tau^2$ by $g^{\nu\alpha}dF_\alpha^{\text{ext}}/d\tau$ in the expression for F_μ^{rad}. Show that the spatial part of the generalization of (16.10) becomes

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} + \tau\left[\gamma\frac{d\mathbf{F}}{dt} - \frac{\gamma^3}{c^2}\frac{d\mathbf{v}}{dt}\times(\mathbf{v}\times\mathbf{F})\right]$$

where $\mathbf{F}$ is the spatial part of $\mathbf{F}_{\text{ext}}^\mu/\gamma$. For a charged particle in external electric and magnetic fields $\mathbf{F}$ is the Lorentz force.

Reference: G. W. Ford and R. F. O'Connell, *Phys. Lett.* A **174**, 182 (1993).

Solution.

Write $\tau_r = 2e^2/(3mc^3)$, reserving τ for proper time.

(a) The differentiated mass-shell identity from 16.7 gives $p_\nu\ddot{p}^\nu = -\dot{p}_\nu\dot{p}^\nu$. Consequently

$$\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2c^2}\right)\ddot{p}^\nu = \ddot{p}_\mu + \frac{p_\mu}{m^2c^2}\dot{p}_\nu\dot{p}^\nu,$$

which proves the projected expression for F_μ^{rad}.

(b) Replace $\ddot{p}^\mu$ in this projected term by $dF_{\text{ext}}^\mu/d\tau$. This is reduction of order, valid through first order in τ_r. Write ordinary laboratory time derivatives explicitly and put $\mathbf{a} = d\mathbf{v}/dt$:

$$p^\mu = m\gamma(c,\mathbf{v}), \qquad F_{\text{ext}}^\mu = \gamma(\mathbf{v}\cdot\mathbf{F}/c, \mathbf{F}), \qquad \frac{d}{d\tau} = \gamma\frac{d}{dt}.$$

For $W^\mu = dF_{\text{ext}}^\mu/d\tau$ the spatial component is $\mathbf{W} = \gamma(d\gamma/dt)\mathbf{F} + \gamma^2 d\mathbf{F}/dt$. Since $p\cdot F_{\text{ext}} = 0$,

$$p\cdot W = -\frac{dp}{d\tau}\cdot F_{\text{ext}} = m\gamma^3\mathbf{a}\cdot\mathbf{F}.$$

For example the terms containing $d\gamma/dt$ cancel in this scalar product between its time and spatial components; the remaining spatial term is $-m\gamma^3\mathbf{a}\cdot\mathbf{F}$ before the minus sign. Projection, followed by division by γ, therefore gives

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} + \tau_r\left[\gamma\frac{d\mathbf{F}}{dt} + \frac{d\gamma}{dt}\mathbf{F} - \frac{\gamma^3}{c^2}\mathbf{v}(\mathbf{a}\cdot\mathbf{F})\right].$$

Use $d\gamma/dt = \gamma^3 \mathbf{v}\cdot\mathbf{a}/c^2$ and $\mathbf{a}\times(\mathbf{v}\times\mathbf{F}) = \mathbf{v}(\mathbf{a}\cdot\mathbf{F}) - \mathbf{F}(\mathbf{a}\cdot\mathbf{v})$:

$$\boxed{\frac{d\mathbf{p}}{dt} = \mathbf{F} + \tau_r\left[\gamma\frac{d\mathbf{F}}{dt} - \frac{\gamma^3}{c^2}\frac{d\mathbf{v}}{dt}\times(\mathbf{v}\times\mathbf{F})\right].}$$

The transcription had omitted the multiplier τ_r before the bracket; the textbook's printed page 772 includes it, and its units are required to convert a force derivative to a force. For a consistently explicit first-order equation, evaluate the acceleration and force derivatives inside the bracket using the zeroth-order Lorentz-force motion. Changes from substituting the corrected motion there start at order τ_r^2.

16.10 The Abraham–Lorentz equation of motion (16.9) can be replaced by an integro-differential equation if the external force is considered a function of time.

(a) Show that a first integral of (16.9) that eliminates the possibility of "runaway" solutions is

$$m\dot{\mathbf{v}}(t) = \int_0^\infty e^{-s}\mathbf{F}(t+\tau s)\,ds$$

(b) Show that a Taylor expansion of the force for small τ leads to the formal, generally asymptotic series

$$m\dot{\mathbf{v}}(t) \sim \sum_{n=0}^{\infty}\tau^n\frac{d^n\mathbf{F}(t)}{dt^n}$$

The approximate equation (16.10) contains the first two terms of the infinite series.

(c) For a step-function force in one dimension, $F(t) = F_0\Theta(t)$, solve the integro-differential equation of part a for the acceleration and velocity for $t<0$ and $t>0$ for a particle at rest at $t=-\infty$. Plot ma/F_0 and $mv/F_0\tau$ in units of t/τ. Compare with the solution from (16.10). Comment.

Solution.

(a) Write the Abraham–Lorentz equation as $\mathbf{a} - \tau_r\dot{\mathbf{a}} = \mathbf{F}/m$, where $\tau_r = 2e^2/(3mc^3)$. An integrating factor gives

$$\frac{d}{dt}(e^{-t/\tau_r}\mathbf{a}) = -\frac{e^{-t/\tau_r}}{m\tau_r}\mathbf{F}(t).$$

Impose the nonrunaway boundary condition $e^{-t/\tau_r}\mathbf{a}(t)\to 0$ as $t\to+\infty$ and integrate from t to infinity. With $t' = t+\tau_r s$, the result is

$$\boxed{m\mathbf{a}(t) = \frac{1}{\tau_r}\int_t^\infty e^{-(t'-t)/\tau_r}\mathbf{F}(t')dt' = \int_0^\infty e^{-s}\mathbf{F}(t+\tau_r s)ds.}$$

The force must not grow so rapidly that this integral fails to converge.

(b) At any fixed truncation order N, Taylor's formula gives

$$\mathbf{F}(t+\tau_r s) = \sum_{n=0}^{N}\frac{(\tau_r s)^n}{n!}\mathbf{F}^{(n)}(t) + R_{N+1}(t,s,\tau_r).$$

Since $\int_0^\infty e^{-s}s^n ds = n!$, integration yields

$$m\mathbf{a}(t) = \sum_{n=0}^{N} \tau_r^n \mathbf{F}^{(n)}(t) + \int_0^\infty e^{-s} R_{N+1}\, ds.$$

Under the corresponding derivative bounds the last term is $O(\tau_r^{N+1})$. Thus the infinite expression is generally an *asymptotic* series, written with $\sim$; equality requires convergence and justification of the interchange. For example $F(t) = F_0/[1 + (t/T)^2]$ gives coefficients $F^{(2n)}(0) = (-1)^n(2n)!F_0/T^{2n}$, so the series diverges for every nonzero τ_r, although the force integral is finite. Keeping only $n = 0, 1$ gives $m\mathbf{a} = \mathbf{F} + \tau_r\dot{\mathbf{F}}$, the reduced-order approximation (16.10).

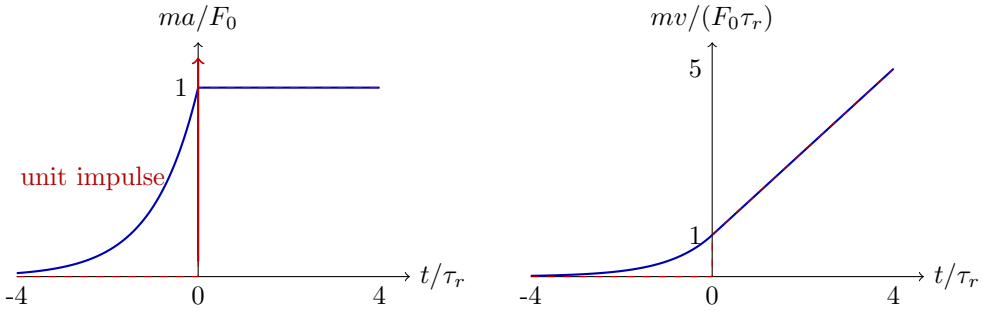
(c) For $t < 0$ the step contributes only for $s > -t/\tau_r$; for $t > 0$ it contributes on the entire positive axis. Hence

$$\boxed{\frac{ma}{F_0} = \begin{cases} e^{t/\tau_r}, & t < 0, \\ 1, & t > 0, \end{cases} \qquad \frac{mv}{F_0\tau_r} = \begin{cases} e^{t/\tau_r}, & t < 0, \\ 1 + t/\tau_r, & t > 0. \end{cases}}$$

The velocity follows by integration from rest at $-\infty$; its two branches and the acceleration are continuous at zero. The nonrunaway solution accelerates before the force turns on. For the reduced-order equation,

$$ma_{\text{red}} = F_0[\Theta(t) + \tau_r\delta(t)], \qquad mv_{\text{red}} = F_0[t\Theta(t) + \tau_r\Theta(t)].$$

It has no preacceleration but an impulsive acceleration and a velocity jump at the step. An ideal discontinuity does not meet the slowly varying-force condition needed for a uniform small-τ_r expansion. The two velocities agree after turn-on, but their onset behavior differs.



16.11 A nonrelativistic particle of charge e and mass m is accelerated in one-dimensional motion across a gap of width d by a constant electric field. The mathematical idealization is that the particle has applied to it an external force $m\alpha$ while its coordinate lies in the interval $(0, d)$. Without radiation damping the particle, having initial velocity v_0, is accelerated uniformly for a time $T = (-v_0/\alpha) + \sqrt{(v_0^2/\alpha^2) + (2d/\alpha)}$, emerging at $x = d$ with a final velocity $v_1 = \sqrt{v_0^2 + 2\alpha d}$.

With radiation damping the motion is altered so that the particle takes a time T' to cross the gap and emerges with a velocity v_1'.

(a) Solve the integro-differential equation of motion, including damping, assuming T and T' large compared to τ. Sketch a velocity-versus-time diagram for the motion with and without damping.

(b) Show that to lowest order in τ,

$$T' = T - \tau\left(1 - \frac{v_0}{v_1}\right)$$

$$v_1' = v_1 - \frac{\alpha^2 \tau}{v_1} T$$

(c) Verify that the sum of the energy radiated and the change in the particle's kinetic energy is equal to the work done by the applied field.

Solution.

Let $\tau_r = 2e^2/(3mc^3)$, choose entry at $x = 0, t = 0$, and write T' for the exit time. The prescribed v_0 is the incoming velocity before the preacceleration region. Take $\alpha > 0$, $d > 0$, $v_0 \geq 0$.

(a) Along the monotone crossing trajectory the force is $m\alpha$ for $0 < t < T'$ and zero otherwise. Inserting this finite pulse in the nonrunaway force integral of 16.10 gives

$$a(t) = \begin{cases} \alpha(1 - e^{-T'/\tau_r})e^{t/\tau_r}, & t < 0, \\ \alpha[1 - e^{(t-T')/\tau_r}], & 0 < t < T', \\ 0, & t > T'. \end{cases}$$

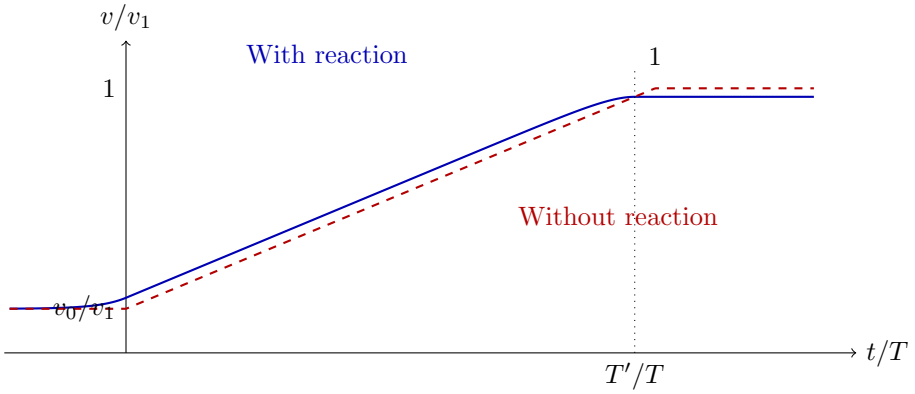
For example in the middle interval the allowed integration range is $0 < s < (T' - t)/\tau_r$, whose exponential integral is $1 - e^{(t-T')/\tau_r}$. Integrating a and imposing $v(-\infty) = v_0$ gives

$$\boxed{v(t) = \begin{cases} v_0 + \alpha\tau_r(1 - e^{-T'/\tau_r})e^{t/\tau_r}, & t < 0, \\ v_0 + \alpha[t + \tau_r - \tau_r e^{(t-T')/\tau_r}], & 0 < t < T', \\ v_0 + \alpha T', & t > T'. \end{cases}}$$

Both transitions are continuous. In particular $v(0)$ is slightly larger than v_0 and $v_1' = v_0 + \alpha T'$. The spatial crossing condition determines T' self-consistently:

$$d = \int_0^{T'} v(t)dt = v_0 T' + \frac{\alpha}{2} T'^2 + \alpha\tau_r T' - \alpha\tau_r^2(1 - e^{-T'/\tau_r}).$$

Its derivative with respect to T' is $v_0 + \alpha T' + \alpha\tau_r(1 - e^{-T'/\tau_r}) > 0$ for $T' > 0$, so the positive crossing time is unique. For $T' \gg \tau_r$ the exponential is negligible except in the short entry and exit layers. The plot compares this solution with the undamped piecewise linear velocity for one representative $T/\tau_r \gg 1$ case.



$T/\tau_r = 20$, $v_0/v_1 = 1/6$

(b) Write $T' = T + \delta T$ in the crossing condition and use $d = v_0 T + \alpha T^2/2$. Keeping terms linear in τ_r gives

$$0 = (v_0 + \alpha T)\delta T + \alpha \tau_r T + O(\tau_r^2), \qquad \delta T = -\frac{\alpha \tau_r T}{v_1}, \quad v_1 = v_0 + \alpha T.$$

Therefore

$$\boxed{T' = T - \tau_r(1 - v_0/v_1) + O(\tau_r^2), \qquad v_1' = v_1 - \frac{\alpha^2 \tau_r T}{v_1} + O(\tau_r^2).}$$

The earlier entry preacceleration shortens the time spent in the gap, while the final kinetic energy is lower because energy is radiated.

(c) With $E = e^{-T'/\tau_r}$, integrate the squared acceleration on the two nonzero intervals:

$$\int_{-\infty}^{0} a^2 dt = \frac{\alpha^2 \tau_r}{2}(1 - E)^2, \quad \int_0^{T'} a^2 dt = \alpha^2 \left[T' - 2\tau_r(1 - E) + \frac{\tau_r}{2}(1 - E^2)\right].$$

Their sum is $\alpha^2[T' - \tau_r(1 - E)]$. Since the Larmor power is $m\tau_r a^2$,

$$W_{\rm rad} = m\tau_r \alpha^2 [T' - \tau_r(1 - E)], \qquad \Delta K = m\alpha(v_0 T' + \alpha T'^2/2).$$

Adding and using the exact crossing condition gives

$$\boxed{\Delta K + W_{\rm rad} = m\alpha d = W_{\rm ext}.}$$

This equality includes the radiation before entry. In the local Abraham–Lorentz energy equation the extra total derivative is the Schott term $m\tau_r v a$; it vanishes at both asymptotic endpoints, which is why it does not alter the integrated balance.

16.12 A classical model for the description of collision broadening of spectral lines is that the oscillator is interrupted by a collision after oscillating for a time T so that the coherence of the wave train is lost.

(a) Taking the oscillator used in Section 16.7 and assuming that the probability that a collision will occur between time T and $(T + dT)$ is $(\nu e^{-\nu T}\, dT)$, where ν is the mean collision frequency, show that the averaged spectral distribution is

$$\frac{dI(\omega)}{d\omega} = \frac{I_0}{2\pi} \frac{\Gamma + 2\nu}{(\omega - \omega_0)^2 + \left(\dfrac{\Gamma}{2} + \nu\right)^2}$$

so that the breadth of the line is $(2\nu + \Gamma)$.

(b) For the sodium doublet at 5893 Å the oscillator strength is $f = 0.975$, so that the natural width is essentially the classical value, $\Delta\lambda = 1.2 \times 10^{-4}$ Å. Estimate the Doppler width of the line, assuming the sodium atoms are in thermal equilibrium at a temperature of 500K, and compare it with the natural width. Assuming a collision cross section of 10^{-16} cm^2, determine the collision breadth of the sodium doublet as a function of the pressure of the sodium vapor. For what pressure is the collision breadth equal to the natural breadth? The Doppler breadth?

Solution.

(a) Treat a collision as destroying phase coherence, with uncorrelated phases after it, while retaining the specified radiative amplitude decay $e^{-\Gamma t/2}$. This is the phase-interruption model in the question. Let I_0 be the total radiated energy, as in (16.72), and put $\Delta = \omega - \omega_0$, absorbing any small fixed line shift in ω_0. For two times separated by s, the probability of no intervening collision is obtained directly from the given waiting-time density:

$$\Pr(T > s) = \int_s^\infty \nu e^{-\nu T} dT = e^{-\nu s}.$$

Products of amplitudes before and after an intervening collision average to zero. Thus the averaged squared Fourier amplitude is proportional to

$$\begin{aligned}\mathcal{S}(\Delta) &= \int_0^\infty dt \int_0^\infty dt'\, e^{-\Gamma(t+t')/2} e^{-\nu|t-t'|} e^{i\Delta(t-t')} \\ &= 2\,\mathrm{Re} \int_0^\infty du\, e^{-\Gamma u} \int_0^\infty ds\, e^{-(\Gamma/2+\nu-i\Delta)s} \\ &= \frac{2}{\Gamma}\,\mathrm{Re}\,\frac{1}{\Gamma/2+\nu-i\Delta} = \frac{\Gamma+2\nu}{\Gamma[\Delta^2+(\Gamma/2+\nu)^2]}.\end{aligned}$$

Here the two triangles $t > t'$ and $t < t'$ use $u = \min(t, t')$ and $s = |t - t'|$; their diagonal has zero integration measure. The same normalization as the uncollided wave train is $I_0\Gamma/(2\pi)$, giving

$$\boxed{\frac{dI}{d\omega} = \frac{I_0}{2\pi}\frac{\Gamma+2\nu}{(\omega-\omega_0)^2+(\Gamma/2+\nu)^2}.}$$

Its area over the narrow-line frequency approximation is I_0. At half maximum $|\Delta| = \Gamma/2 + \nu$, so the full width at half maximum is $\Gamma + 2\nu$. Permanently terminating radiation at the first collision would also reduce the total energy; it is not the phase-loss model just averaged.

(b) Use full widths at half maximum throughout, consistent with the line-breadth convention of Section 16.7. The line-of-sight Maxwell distribution is proportional to $e^{-Mv_z^2/(2k_BT)}$. With $\delta\omega = \omega_0 v_z/c$, its half-maximum points obey $Mv_z^2/(2k_BT) = \ln 2$. Thus

$$\Delta\lambda_D = \lambda_0\sqrt{\frac{8k_BT\ln 2}{Mc^2}}.$$

For $M \simeq 23\,u$, $T = 500$ K, and $\lambda_0 = 5893$ Å,

$$\boxed{\Delta\lambda_D = 1.96794 \times 10^{-2}\ \text{Å},}$$

about 164 times the supplied natural width $\Delta\lambda_N = 1.2 \times 10^{-4}$ Å. The one-sided $1/e$ Gaussian width is smaller by $2\sqrt{\ln 2}$; it must not be compared as if it were the full width.

For collisions of identical sodium atoms, the relative Maxwell speed has reduced mass $M/2$, and its mean is $\bar{v}_{\rm rel} = \sqrt{16k_BT/(\pi M)} = 959.45$ m/s. Using ideal-gas density $n = P/(k_BT)$ and the given $\sigma = 10^{-16}$ cm^2 $= 10^{-20}$ m^2,

$$\nu = n\sigma\bar{v}_{\rm rel} = (1.38986 \times 10^3\ \text{s}^{-1}\text{Pa}^{-1})P.$$

The collision contribution to angular-frequency FWHM is 2ν. Since $|d\lambda/d\omega| = \lambda_0^2/(2\pi c)$,

$$\boxed{\Delta\lambda_C = \frac{\lambda_0^2\nu}{\pi c} = (5.12474 \times 10^{-9}\ \text{Å/Pa})P = (6.83243 \times 10^{-7}\ \text{Å/Torr})P_{\rm Torr}.}$$

Consequently

$$\boxed{P_{C=N} = 2.34158 \times 10^4 \text{ Pa} \simeq 176 \text{ Torr},} \qquad \boxed{P_{C=D} = 3.84007 \times 10^6 \text{ Pa} \simeq 28{,}800 \text{ Torr}.}$$

These are estimates within the specified constant-cross-section, ideal-gas collision model. The comparisons are between individual broadening contributions; a Gaussian Doppler profile convolved with a Lorentzian natural/collision profile is a Voigt profile, whose total width is not obtained by simply adding all three widths.

16.13 A single particle oscillator under the action of an applied electric field $\mathbf{E}_0 e^{-i\omega t}$ has a dipole moment given by

$$\mathbf{p} = \alpha(\omega)\mathbf{E}_0 e^{-i\omega t}$$

(a) Show that the total dipole cross section can be written as

$$\sigma_t(\omega) = \frac{2\pi}{c}\left[-i\omega\alpha(\omega) + \text{c.c.}\right]$$

(b) Using only the facts that all the normal modes of oscillation must have some damping and that the polarizability $\alpha(\omega)$ must approach the free-particle value $(-e^2/m\omega^2)$ at high frequencies, show that the cross section satisfies the dipole sum rule,

$$\int_0^\infty \sigma_t(\omega)\, d\omega = \frac{2\pi^2 e^2}{mc}$$

(The discussion of Kramers–Kronig dispersion relations in Chapter 7 is clearly relevant.)

Solution.

(a) The total cross section means extinction, namely scattering plus any absorption. For phasor $\mathbf{p} = \alpha\mathbf{E}_0$, the mean work taken from the incident field is

$$\overline{P} = \frac{1}{2}\,\mathrm{Re}[(-i\omega\mathbf{p})\cdot\mathbf{E}_0^*] = \frac{\omega}{2}\,\mathrm{Im}\,\alpha\,|\mathbf{E}_0|^2.$$

The incident plane-wave flux in Gaussian units is $c|\mathbf{E}_0|^2/(8\pi)$. Dividing gives

$$\boxed{\sigma_t = \frac{4\pi\omega}{c}\,\mathrm{Im}\,\alpha = \frac{2\pi}{c}[-i\omega\alpha + \text{c.c.}].}$$

If the only damping is radiative this equals the total scattered power cross section; otherwise it also contains the absorbed part.

(b) The causal response of a stable damped oscillator is analytic for $\mathrm{Im}\,z > 0$; its normal-mode poles lie below the real axis. A real time-domain response also gives $\alpha(-\omega) = \alpha(\omega)^*$. Consider $z\alpha(z)$ on a semicircle of radius R in the upper half-plane, traversed counterclockwise, with the real segment going from $-R$ to R. The specified high-frequency limit is $z\alpha(z) \sim -e^2/(mz)$. On the arc $z = Re^{i\theta}$, $dz = iz\,d\theta$, hence

$$\lim_{R\to\infty}\int_{\text{arc}} z\alpha(z)dz = -\frac{ie^2}{m}\int_0^\pi d\theta = -\frac{i\pi e^2}{m}.$$

There are no enclosed poles, so the real-axis integral is $i\pi e^2/m$. Pair its negative and positive parts using the reality relation:

$$\int_{-\infty}^{\infty} \omega\alpha(\omega)d\omega = \int_0^{\infty} \omega[\alpha(\omega) - \alpha(\omega)^*]d\omega = 2i\int_0^{\infty} \omega \,\mathrm{Im}\,\alpha(\omega)d\omega.$$

It follows that $\int_0^{\infty} \omega \,\mathrm{Im}\,\alpha \, d\omega = \pi e^2/(2m)$. Multiplication by the factor in (a) proves

$$\boxed{\int_0^{\infty} \sigma_t(\omega)d\omega = \frac{2\pi^2 e^2}{mc}.}$$

The sum does not depend on the resonance frequencies or widths. The analyticity and high-frequency behavior are essential: extending a low-frequency radiation-reaction approximation beyond its validity can introduce an upper-half-plane runaway pole or the wrong asymptotic mass, in which case that approximate response cannot be used for this contour proof.